lyche-numerical-linear-algebra: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE's — a formula has none of its own. A line's confidence is not a formula's.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00001	7	■1.000	`\boldsymbol{A}`	$\boldsymbol{A}$	• *System of linear equations*. Given a (square) matrix $\boldsymbol{A}$ and a vector $\boldsymbol{b}$. Find a
lyche-numerical-linear-algebra_F00002	7	■1.000	`\boldsymbol{b}`	$\boldsymbol{b}$	• *System of linear equations*. Given a (square) matrix $\boldsymbol{A}$ and a vector $\boldsymbol{b}$. Find a
lyche-numerical-linear-algebra_F00003	7	■0.993	`\boldsymbol{x}`	$\boldsymbol{x}$	vector $\boldsymbol{x}$ such that $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$.
lyche-numerical-linear-algebra_F00004	7	■0.993	`\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}`	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}$	vector $\boldsymbol{x}$ such that $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$.
lyche-numerical-linear-algebra_F00005	7	■1.000	`\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}`	$\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}$	such that the sum of squares of the components of $\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}$ is as small as possible.
lyche-numerical-linear-algebra_F00006	7	■1.000	`\lambda`	λ	• *Eigenvalues and eigenvectors*. Given a (square) matrix $\boldsymbol{A}$. Find a number λ
lyche-numerical-linear-algebra_F00007	7	■0.997	`\boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{x}`	$\boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{x}$	and/or a nonzero vector $\boldsymbol{x}$ such that $\boldsymbol{A}\boldsymbol{x}=\lambda\boldsymbol{x}$.
lyche-numerical-linear-algebra_F00008	10	■1.000	`\mathbb{R}^{n}`	$\mathbb{R}^{n}$	1.2.3 The Vector Spaces $\mathbb{R}^n$ and $\mathbb{C}^n$
lyche-numerical-linear-algebra_F00009	10	■1.000	`\mathbb{C}^{n}`	$\mathbb{C}^{n}$	1.2.3 The Vector Spaces $\mathbb{R}^n$ and $\mathbb{C}^n$
lyche-numerical-linear-algebra_F00010	13	■1.000	`\boldsymbol{A}^{*} \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^{*}`	$\boldsymbol{A}^{*} \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^{*}$	7.1.1 The Matrices $\boldsymbol{A}^*\boldsymbol{A}$, $\boldsymbol{A}\boldsymbol{A}^*$
lyche-numerical-linear-algebra_F00011	13	■1.000	`p`	p	8.2.3 The Operator p-Norms
lyche-numerical-linear-algebra_F00012	15	■1.000	`\omega`	ω	12.5 The Optimal SOR Parameter ω
lyche-numerical-linear-algebra_F00013	16	■1.000	`\lambda_{m}`	λ_{m}	14.5.2 Approximating λ_m
lyche-numerical-linear-algebra_F00014	17	■1.000	`T`	T	Fig. 1.1 The triangle T defined by the three points P_1, P_2 and P_3
lyche-numerical-linear-algebra_F00015	17	■1.000	`P_{1}, P_{2}`	P_{1}, P_{2}	Fig. 1.1 The triangle T defined by the three points P_1, P_2 and P_3
lyche-numerical-linear-algebra_F00016	17	■1.000	`P_{3}`	P_{3}	Fig. 1.1 The triangle T defined by the three points P_1, P_2 and P_3
lyche-numerical-linear-algebra_F00017	17	■1.000	`f(x)=\arctan (10 x)+\pi / 2`	$f(x)=\arctan (10 x)+\pi / 2$	$f(x) = \arctan(10x) + \pi/2$ on $[-1, 1]$. See text
lyche-numerical-linear-algebra_F00018	17	■1.000	`[-1,1]`	$[-1,1]$	$f(x) = \arctan(10x) + \pi/2$ on $[-1, 1]$. See text
lyche-numerical-linear-algebra_F00019	17	■1.000	`n=14`	$n=14$	$f(x) = \arctan(10x) + \pi/2$ at $n = 14$ uniform points on
lyche-numerical-linear-algebra_F00020	17	■1.000	`f(x)=x^{4}`	$f(x)=x^{4}$	Fig. 2.3 A cubic spline with one knot interpolating $f(x) = x^4$ on
lyche-numerical-linear-algebra_F00021	17	■0.994	`5 \times 5`	5×5	Fig. 3.2 Lower triangular 5×5 band matrices: $d = 1$
lyche-numerical-linear-algebra_F00022	17	■0.994	`d=1`	$d=1$	Fig. 3.2 Lower triangular 5×5 band matrices: $d = 1$
lyche-numerical-linear-algebra_F00023	17	■1.000	`d=2`	$d=2$	(left) and $d = 2$ right
lyche-numerical-linear-algebra_F00024	17	■0.997	`\boldsymbol{v}_{1}`	$\boldsymbol{v}_{1}$	Fig. 5.1 The construction of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ in Gram-Schmidt. The

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00025	17	■ 0.997	`\boldsymbol{v}_{2}`	$\boldsymbol{v}_{2}$	Fig. 5.1 The construction of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ in Gram-Schmidt. The
lyche-numerical-linear-algebra_F00026	17	■ 1.000	`c`	c	constant c is given by $c := \langle \boldsymbol{s}_2, \boldsymbol{v}_1 \rangle / \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle$
lyche-numerical-linear-algebra_F00027	17	■ 1.000	`c:=\left\langle\boldsymbol{s}_{2}, \boldsymbol{v}_{1} \right\rangle /\left\langle\boldsymbol{v}_{1}, \boldsymbol{v}_{1}\right\rangle`	$c:=\left\langle\boldsymbol{s}_{2}, \boldsymbol{v}_{1} \right\rangle /\left\langle\boldsymbol{v}_{1}, \boldsymbol{v}_{1}\right\rangle$	constant c is given by $c := \langle \boldsymbol{s}_2, \boldsymbol{v}_1 \rangle / \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle$
lyche-numerical-linear-algebra_F00028	17	■ 1.000	`\boldsymbol{s}+\boldsymbol{t}`	$\boldsymbol{s}+\boldsymbol{t}$	Fig. 5.2 The orthogonal projections of $\boldsymbol{s}+\boldsymbol{t}$ into $\mathcal{S}$ and $\mathcal{T}$
lyche-numerical-linear-algebra_F00029	17	■ 1.000	`\mathcal{S}`	$\mathcal{S}$	Fig. 5.2 The orthogonal projections of $\boldsymbol{s}+\boldsymbol{t}$ into $\mathcal{S}$ and $\mathcal{T}$
lyche-numerical-linear-algebra_F00030	17	■ 1.000	`\mathcal{T}`	$\mathcal{T}$	Fig. 5.2 The orthogonal projections of $\boldsymbol{s}+\boldsymbol{t}$ into $\mathcal{S}$ and $\mathcal{T}$
lyche-numerical-linear-algebra_F00031	17	■ 0.683	`y_{1}^{2} / 9+y_{2}^{2}=1`	$y_{1}^{2} / 9+y_{2}^{2}=1$	Fig. 7.1 The ellipse $y_1^2/9 + y_2^2 = 1$ (left) and the rotated ellipse $\boldsymbol{A}\mathcal{S}$
lyche-numerical-linear-algebra_F00032	17	■ 0.683	`\boldsymbol{A} \boldsymbol{\mathcal { S }}`	$\boldsymbol{A} \boldsymbol{\mathcal { S }}$	Fig. 7.1 The ellipse $y_1^2/9 + y_2^2 = 1$ (left) and the rotated ellipse $\boldsymbol{A}\mathcal{S}$
lyche-numerical-linear-algebra_F00033	18	■ 0.995	`n=100`	$n=100$	$(n = 100)$
lyche-numerical-linear-algebra_F00034	18	■ 1.000	`\alpha \rightarrow\left|1-\alpha \lambda_{1}\right|`	$\alpha \rightarrow\left\|1-\alpha \lambda_{1}\right\|$	Fig. 12.1 The functions $\alpha \rightarrow \|1-\alpha\lambda_1\|$ and $\alpha \rightarrow \|1-\alpha\lambda_n\|$
lyche-numerical-linear-algebra_F00035	18	■ 1.000	`\alpha \rightarrow\left|1-\alpha \lambda_{n}\right|`	$\alpha \rightarrow\left\|1-\alpha \lambda_{n}\right\|$	Fig. 12.1 The functions $\alpha \rightarrow \|1-\alpha\lambda_1\|$ and $\alpha \rightarrow \|1-\alpha\lambda_n\|$
lyche-numerical-linear-algebra_F00036	18	■ 0.737	`12.2 \rho\left(\boldsymbol{G}_{\omega}\right)`	$12.2 \rho\left(\boldsymbol{G}_{\omega}\right)$	Fig. 12.2 $\rho(\boldsymbol{G}_\omega)$ with $\omega \in [0, 2]$ for $n = 100$, (lower curve) and
lyche-numerical-linear-algebra_F00037	18	■ 0.737	`\omega \in[0,2]`	$\omega \in[0,2]$	Fig. 12.2 $\rho(\boldsymbol{G}_\omega)$ with $\omega \in [0, 2]$ for $n = 100$, (lower curve) and
lyche-numerical-linear-algebra_F00038	18	■ 1.000	`n=2500`	$n=2500$	$n = 2500$ (upper curve)
lyche-numerical-linear-algebra_F00039	18	■ 0.987	`Q(x, y)`	$Q(x, y)$	Fig. 13.1 Level curves for $Q(x, y)$ given by (13.4). Also shown
lyche-numerical-linear-algebra_F00040	18	■ 1.000	`Q`	Q	gradient iteration (right) to find the minimum of Q (cf
lyche-numerical-linear-algebra_F00041	18	■ 1.000	`\boldsymbol{x}-\boldsymbol{x}_{0}`	$\boldsymbol{x}-\boldsymbol{x}_{0}$	Fig. 13.2 The orthogonal projection of $\boldsymbol{x}-\boldsymbol{x}_0$ into $\mathbb{W}_k$
lyche-numerical-linear-algebra_F00042	18	■ 1.000	`\mathbb{W}_{k}`	$\mathbb{W}_{k}$	Fig. 13.2 The orthogonal projection of $\boldsymbol{x}-\boldsymbol{x}_0$ into $\mathbb{W}_k$
lyche-numerical-linear-algebra_F00043	18	■ 0.999	`k=3 . f \equiv Q-Q^{*}`	$k=3 . f \equiv Q-Q^{*}$	$k = 3$. $f \equiv Q - Q^*$ has a double zero at μ_1 and one zero
lyche-numerical-linear-algebra_F00044	18	■ 0.999	`\mu_{1}`	μ_{1}	$k = 3$. $f \equiv Q - Q^*$ has a double zero at μ_1 and one zero
lyche-numerical-linear-algebra_F00045	18	■ 1.000	`\mu_{2}`	μ_{2}	between μ_2 and μ_3
lyche-numerical-linear-algebra_F00046	18	■ 1.000	`\mu_{3}`	μ_{3}	between μ_2 and μ_3
lyche-numerical-linear-algebra_F00047	18	■ 0.991	`R_{i}`	R_{i}	Fig. 14.1 The Gershgorin disk R_i
lyche-numerical-linear-algebra_F00048	19	■ 1.000	`k_{n}`	k_{n}	Table 12.1 The number of iterations k_n to solve the discrete Poisson
lyche-numerical-linear-algebra_F00049	19	■ 1.000	`n`	n	problem with n unknowns using the methods of Jacobi,
lyche-numerical-linear-algebra_F00050	19	■ 1.000	`10^{-8}`	10^{-8}	Gauss-Seidel, and SOR (see text) with a tolerance 10^{-8}
lyche-numerical-linear-algebra_F00051	19	■ 1.000	`\boldsymbol{G}_{J}, \boldsymbol{G}_{1}, \boldsymbol{G}_{\omega^{*}}`	$\boldsymbol{G}_{J}, \boldsymbol{G}_{1}, \boldsymbol{G}_{\omega^{*}}$	Table 12.2 Spectral radial for $\boldsymbol{G}_J$, $\boldsymbol{G}_1$, $\boldsymbol{G}_{\omega^*}$ and the smallest integer
lyche-numerical-linear-algebra_F00052	19	■ 1.000	`\rho(\boldsymbol{G})^{k_{n}} \leq 10^{-8}`	$\rho(\boldsymbol{G})^{k_{n}} \leq 10^{-8}$	k_n such that $\rho(\boldsymbol{G})^{k_n} \leq 10^{-8}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00053	19	1.000	`K`	K	Table 13.1 The number of iterations K for the averaging problem on a
lyche-numerical-linear-algebra_F00054	19	1.000	`\sqrt{n} \times \sqrt{n}`	$\sqrt{n} \times \sqrt{n}$	$\sqrt{n} \times \sqrt{n}$ grid for various n
lyche-numerical-linear-algebra_F00055	19	0.926	`K_{p r e}`	$K_{p r e}$	Table 13.3 The number of iterations K (no preconditioning) and K_{pre}
lyche-numerical-linear-algebra_F00056	21	1.000	`\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}`	$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	complex numbers are denoted by $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$, respectively.
lyche-numerical-linear-algebra_F00057	21	1.000	`v:=e`	$v:=e$	2. We use the "colon equal" symbol $v := e$ to indicate that the symbol v is defined
lyche-numerical-linear-algebra_F00058	21	1.000	`v`	v	2. We use the "colon equal" symbol $v := e$ to indicate that the symbol v is defined
lyche-numerical-linear-algebra_F00059	21	1.000	`e`	e	by the expression e.
lyche-numerical-linear-algebra_F00060	21	1.000	`\boldsymbol{x} \in \mathbb{R}^{n}`	$\boldsymbol{x} \in \mathbb{R}^{n}$	column vectors. Thus $\boldsymbol{x} \in \mathbb{R}^n$ means
lyche-numerical-linear-algebra_F00061	21	1.000	`x_{i} \in \mathbb{R}`	$x_{i} \in \mathbb{R}$	where $x_i \in \mathbb{R}$ for $i = 1, \ldots, n$. Row vectors are normally identified using the
lyche-numerical-linear-algebra_F00062	21	1.000	`i=1, \ldots, n`	$i=1, \ldots, n$	where $x_i \in \mathbb{R}$ for $i = 1, \ldots, n$. Row vectors are normally identified using the
lyche-numerical-linear-algebra_F00063	21	1.000	`\boldsymbol{x}^{T}`	$\boldsymbol{x}^{T}$	transpose operation. Thus if $\boldsymbol{x} \in \mathbb{R}^n$ then $\boldsymbol{x}$ is a column vector and $\boldsymbol{x}^T$ is a row
lyche-numerical-linear-algebra_F00064	22	1.000	`\mathbb{R}^{m \times n}`	$\mathbb{R}^{m \times n}$	5. $\mathbb{R}^{m \times n}$ is the set of matrices $\boldsymbol{A}$ with real elements. The integers m and n are the
lyche-numerical-linear-algebra_F00065	22	1.000	`m`	m	5. $\mathbb{R}^{m \times n}$ is the set of matrices $\boldsymbol{A}$ with real elements. The integers m and n are the
lyche-numerical-linear-algebra_F00066	22	0.998	`i`	i	The element in the ith row and jth column of $\boldsymbol{A}$ will be denoted by $a_{i,j}$, a_{ij},
lyche-numerical-linear-algebra_F00067	22	0.998	`j`	j	The element in the ith row and jth column of $\boldsymbol{A}$ will be denoted by $a_{i,j}$, a_{ij},
lyche-numerical-linear-algebra_F00068	22	0.998	`a_{i, j}, a_{i j}`	$a_{i, j}, a_{i j}$	The element in the ith row and jth column of $\boldsymbol{A}$ will be denoted by $a_{i,j}$, a_{ij},
lyche-numerical-linear-algebra_F00069	22	1.000	`\boldsymbol{A}(i, j)`	$\boldsymbol{A}(i, j)$	$\boldsymbol{A}(i, j)$ or $(\boldsymbol{A})_{i,j}$. We use the notations
lyche-numerical-linear-algebra_F00070	22	1.000	`(\boldsymbol{A})_{i, j}`	$(\boldsymbol{A})_{i, j}$	$\boldsymbol{A}(i, j)$ or $(\boldsymbol{A})_{i,j}$. We use the notations
lyche-numerical-linear-algebra_F00071	22	1.000	`\boldsymbol{a}_{: j}`	$\boldsymbol{a}_{: j}$	for the columns $\boldsymbol{a}_{:j}$ and rows $\boldsymbol{a}_{i:}^T$ of $\boldsymbol{A}$. We often drop the colon and write $\boldsymbol{a}_j$
lyche-numerical-linear-algebra_F00072	22	1.000	`\boldsymbol{a}_{i:}^{T}`	$\boldsymbol{a}_{i:}^{T}$	for the columns $\boldsymbol{a}_{:j}$ and rows $\boldsymbol{a}_{i:}^T$ of $\boldsymbol{A}$. We often drop the colon and write $\boldsymbol{a}_j$
lyche-numerical-linear-algebra_F00073	22	1.000	`\boldsymbol{a}_{j}`	$\boldsymbol{a}_{j}$	for the columns $\boldsymbol{a}_{:j}$ and rows $\boldsymbol{a}_{i:}^T$ of $\boldsymbol{A}$. We often drop the colon and write $\boldsymbol{a}_j$
lyche-numerical-linear-algebra_F00074	22	1.000	`\boldsymbol{a}_{i}^{T}`	$\boldsymbol{a}_{i}^{T}$	and $\boldsymbol{a}_i^T$ with the risk of some confusion. If $m = 1$ then $\boldsymbol{A}$ is a row vector, if
lyche-numerical-linear-algebra_F00075	22	1.000	`m=1`	$m=1$	and $\boldsymbol{a}_i^T$ with the risk of some confusion. If $m = 1$ then $\boldsymbol{A}$ is a row vector, if
lyche-numerical-linear-algebra_F00076	22	1.000	`n=1`	$n=1$	$n = 1$ then $\boldsymbol{A}$ is a column vector, while if $m = n$ then $\boldsymbol{A}$ is a square matrix.
lyche-numerical-linear-algebra_F00077	22	1.000	`m=n`	$m=n$	$n = 1$ then $\boldsymbol{A}$ is a column vector, while if $m = n$ then $\boldsymbol{A}$ is a square matrix.
lyche-numerical-linear-algebra_F00078	22	0.999	`\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \cdots`	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \cdots$	In this text we will denote matrices by boldface capital letters $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \cdots$ and
lyche-numerical-linear-algebra_F00079	22	1.000	`\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \cdots`	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \cdots$	vectors most often by boldface lower case letters $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \cdots$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00080	22	1.000	`x=a+i b`	$x=a+i b$	6. A **complex number** is a number written in the form $x = a+ib$, where a, b
lyche-numerical-linear-algebra_F00081	22	1.000	`a, b`	a, b	6. A **complex number** is a number written in the form $x = a+ib$, where a, b
lyche-numerical-linear-algebra_F00082	22	1.000	`i^{2}=-1`	$i^{2}=-1$	are real numbers and i, **the imaginary unit**, satisfies $i^2 = -1$. The set of all
lyche-numerical-linear-algebra_F00083	22	1.000	`\mathbb{C}`	$\mathbb{C}$	such numbers is denoted by $\mathbb{C}$. The numbers $a = \operatorname{Re} x$ and $b = \operatorname{Im} x$ are the
lyche-numerical-linear-algebra_F00084	22	1.000	`a=\operatorname{Re} x`	$a=\operatorname{Re} x$	such numbers is denoted by $\mathbb{C}$. The numbers $a = \operatorname{Re} x$ and $b = \operatorname{Im} x$ are the
lyche-numerical-linear-algebra_F00085	22	1.000	`b=\operatorname{Im} x`	$b=\operatorname{Im} x$	such numbers is denoted by $\mathbb{C}$. The numbers $a = \operatorname{Re} x$ and $b = \operatorname{Im} x$ are the
lyche-numerical-linear-algebra_F00086	22	1.000	`x`	x	**real and imaginary part** of x. The number $\overline{x} := a - ib$ is called the **complex**
lyche-numerical-linear-algebra_F00087	22	1.000	`\bar{x}:=a-i b`	$\bar{x}:=a-i b$	**real and imaginary part** of x. The number $\overline{x} := a - ib$ is called the **complex**
lyche-numerical-linear-algebra_F00088	22	1.000	`|x|:=\sqrt{\bar{x} x}=\sqrt{a^{2}+b^{2}}`	$\|x\|:=\sqrt{\bar{x} x}=\sqrt{a^{2}+b^{2}}$	**conjugate** of $x = a+ib$, and $\|x\| := \sqrt{\overline{x}x} = \sqrt{a^2+b^2}$ the **absolute value** or
lyche-numerical-linear-algebra_F00089	23	1.000	`e^{x+y}=e^{x} e^{y}`	$e^{x+y}=e^{x} e^{y}$	We have $e^{x+y} = e^x e^y$ for all $x, y \in \mathbb{C}$. The **polar form** of a complex number is
lyche-numerical-linear-algebra_F00090	23	1.000	`x, y \in \mathbb{C}`	$x, y \in \mathbb{C}$	We have $e^{x+y} = e^x e^y$ for all $x, y \in \mathbb{C}$. The **polar form** of a complex number is
lyche-numerical-linear-algebra_F00091	23	1.000	`\boldsymbol{A} \in \mathbb{C}^{m \times n}`	$\boldsymbol{A} \in \mathbb{C}^{m \times n}$	7. For matrices and vectors with complex elements we use the notation $\boldsymbol{A} \in \mathbb{C}^{m\times n}$
lyche-numerical-linear-algebra_F00092	23	1.000	`\boldsymbol{x} \in \mathbb{C}^{n}`	$\boldsymbol{x} \in \mathbb{C}^{n}$	and $\boldsymbol{x} \in \mathbb{C}^n$. We define complex row vectors using either the transpose $\boldsymbol{x}^T$ or
lyche-numerical-linear-algebra_F00093	23	1.000	`\boldsymbol{x}^{*}:=\overline{\boldsymbol{x}}^{T}=\left[ \bar{x}_{1}, \ldots, \bar{x}_{n}\right]`	$\boldsymbol{x}^{*}:=\overline{\boldsymbol{x}}^{T}=\left[\bar{x}_{1}, \ldots, \bar{x}_{n}\right]$	the conjugate transpose operation $\boldsymbol{x}^* := \overline{\boldsymbol{x}}^T = [\overline{x}_1, \ldots, \overline{x}_n]$. If $\boldsymbol{x} \in \mathbb{R}^n$ then
lyche-numerical-linear-algebra_F00094	23	0.809	`x^{*}=x^{T}`	$x^{*}=x^{T}$	$\boldsymbol{x}^* = \boldsymbol{x}^T$.
lyche-numerical-linear-algebra_F00095	23	1.000	`\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n}`	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n}$	8. For $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ and $a \in \mathbb{C}$ the operations of vector addition and scalar
lyche-numerical-linear-algebra_F00096	23	1.000	`a \in \mathbb{C}`	$a \in \mathbb{C}$	8. For $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ and $a \in \mathbb{C}$ the operations of vector addition and scalar
lyche-numerical-linear-algebra_F00097	23	1.000	`\boldsymbol{C}:=\boldsymbol{A}+\boldsymbol{B}`	$\boldsymbol{C}:=\boldsymbol{A}+\boldsymbol{B}$	• **matrix addition** $\boldsymbol{C} := \boldsymbol{A} + \boldsymbol{B}$ if $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ are matrices of the same size, i.e.,
lyche-numerical-linear-algebra_F00098	23	1.000	`\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}`	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$	• **matrix addition** $\boldsymbol{C} := \boldsymbol{A} + \boldsymbol{B}$ if $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ are matrices of the same size, i.e.,
lyche-numerical-linear-algebra_F00099	23	1.000	`c_{i j}:=a_{i j}+b_{i j}`	$c_{i j}:=a_{i j}+b_{i j}$	with the same number of rows and columns, and $c_{ij} := a_{ij} + b_{ij}$ for all i, j.
lyche-numerical-linear-algebra_F00100	23	1.000	`i, j`	i, j	with the same number of rows and columns, and $c_{ij} := a_{ij} + b_{ij}$ for all i, j.
lyche-numerical-linear-algebra_F00101	23	1.000	`\boldsymbol{C}:=\alpha \boldsymbol{A}`	$\boldsymbol{C}:=\alpha \boldsymbol{A}$	• **multiplication by a scalar** $\boldsymbol{C} := \alpha\boldsymbol{A}$, where $c_{ij} := \alpha a_{ij}$ for all i, j.
lyche-numerical-linear-algebra_F00102	23	1.000	`c_{i j}:=\alpha a_{i j}`	$c_{i j}:=\alpha a_{i j}$	• **multiplication by a scalar** $\boldsymbol{C} := \alpha\boldsymbol{A}$, where $c_{ij} := \alpha a_{ij}$ for all i, j.
lyche-numerical-linear-algebra_F00103	23	1.000	`\boldsymbol{C}:=\boldsymbol{A} \boldsymbol{B}, \boldsymbol{C}=\boldsymbol{A} \cdot \boldsymbol{B}`	$\boldsymbol{C}:=\boldsymbol{A} \boldsymbol{B}, \boldsymbol{C}=\boldsymbol{A} \cdot \boldsymbol{B}$	• **matrix multiplication** $\boldsymbol{C} := \boldsymbol{A}\boldsymbol{B}$, $\boldsymbol{C} = \boldsymbol{A} \cdot \boldsymbol{B}$ or $\boldsymbol{C} = \boldsymbol{A} * \boldsymbol{B}$, where $A \in$
lyche-numerical-linear-algebra_F00104	23	1.000	`\boldsymbol{C}=\boldsymbol{A} * \boldsymbol{B}`	$\boldsymbol{C}=\boldsymbol{A} * \boldsymbol{B}$	• **matrix multiplication** $\boldsymbol{C} := \boldsymbol{A}\boldsymbol{B}$, $\boldsymbol{C} = \boldsymbol{A} \cdot \boldsymbol{B}$ or $\boldsymbol{C} = \boldsymbol{A} * \boldsymbol{B}$, where $A \in$
lyche-numerical-linear-algebra_F00105	23	1.000	`\boldsymbol{A} \in`	$\boldsymbol{A} \in$	• **matrix multiplication** $\boldsymbol{C} := \boldsymbol{A}\boldsymbol{B}$, $\boldsymbol{C} = \boldsymbol{A} \cdot \boldsymbol{B}$ or $\boldsymbol{C} = \boldsymbol{A} * \boldsymbol{B}$, where $A \in$
lyche-numerical-linear-algebra_F00106	23	1.000	`\mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}, \boldsymbol{C} \in \mathbb{C}^{m \times n}`	$\mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}, \boldsymbol{C} \in \mathbb{C}^{m \times n}$	$\mathbb{C}^{m\times p}$, $\boldsymbol{B} \in \mathbb{C}^{p\times n}$, $\boldsymbol{C} \in \mathbb{C}^{m\times n}$, and $c_{ij} := \sum_{k=1}^{p} a_{ik}b_{kj}$ for $i = 1, \ldots, m$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00107	23	■ 1.000	`c_{i j}:=\sum_{k=1}^{p} a_{i k} b_{k j}`	$c_{i j}:=\sum_{k=1}^{p} a_{i k} b_{k j}$	$\mathbb{C}^{m\times p}$, $\boldsymbol{B}\in\mathbb{C}^{p\times n}$, $\boldsymbol{C}\in\mathbb{C}^{m\times n}$, and $c_{ij}:=\sum_{k=1}^{p}a_{ik}b_{kj}$ for $i=1,\ldots,m$,
lyche-numerical-linear-algebra_F00108	23	■ 1.000	`i=1, \ldots, m`	$i=1, \ldots, m$	$\mathbb{C}^{m\times p}$, $\boldsymbol{B}\in\mathbb{C}^{p\times n}$, $\boldsymbol{C}\in\mathbb{C}^{m\times n}$, and $c_{ij}:=\sum_{k=1}^{p}a_{ik}b_{kj}$ for $i=1,\ldots,m$,
lyche-numerical-linear-algebra_F00109	23	■ 1.000	`j=1, \ldots, n`	$j=1, \ldots, n$	$j=1,\ldots,n$.
lyche-numerical-linear-algebra_F00110	23	■ 1.000	`\boldsymbol{C}:=\boldsymbol{A} \times \boldsymbol{B}, \boldsymbol{D}:=\boldsymbol{A} / \boldsymbol{B}`	$\boldsymbol{C}:=\boldsymbol{A} \times \boldsymbol{B}, \boldsymbol{D}:=\boldsymbol{A} / \boldsymbol{B}$	• **element-by-element matrix operations** $\boldsymbol{C}:=\boldsymbol{A}\times\boldsymbol{B}$, $\boldsymbol{D}:=\boldsymbol{A}/\boldsymbol{B}$, and $\boldsymbol{E}:=$
lyche-numerical-linear-algebra_F00111	23	■ 1.000	`\boldsymbol{E}:=`	$\boldsymbol{E}:=$	• **element-by-element matrix operations** $\boldsymbol{C}:=\boldsymbol{A}\times\boldsymbol{B}$, $\boldsymbol{D}:=\boldsymbol{A}/\boldsymbol{B}$, and $\boldsymbol{E}:=$
lyche-numerical-linear-algebra_F00112	23	■ 1.000	`\boldsymbol{A} \wedge r`	$\boldsymbol{A} \wedge r$	$\boldsymbol{A}\wedge r$ where all matrices are of the same size and $c_{ij}:=a_{ij}b_{ij}$, $d_{ij}:=a_{ij}/b_{ij}$
lyche-numerical-linear-algebra_F00113	23	■ 1.000	`c_{i j}:=a_{i j} b_{i j}, d_{i j}:=a_{i j} / b_{i j}`	$c_{i j}:=a_{i j} b_{i j}, d_{i j}:=a_{i j} / b_{i j}$	$\boldsymbol{A}\wedge r$ where all matrices are of the same size and $c_{ij}:=a_{ij}b_{ij}$, $d_{ij}:=a_{ij}/b_{ij}$
lyche-numerical-linear-algebra_F00114	23	■ 1.000	`e_{i j}:=a_{i j}^{r}`	$e_{i j}:=a_{i j}^{r}$	and $e_{ij}:=a_{ij}^{r}$ for all i, j and suitable r. For the division $\boldsymbol{A}/\boldsymbol{B}$ we assume that
lyche-numerical-linear-algebra_F00115	23	■ 1.000	`r`	r	and $e_{ij}:=a_{ij}^{r}$ for all i, j and suitable r. For the division $\boldsymbol{A}/\boldsymbol{B}$ we assume that
lyche-numerical-linear-algebra_F00116	23	■ 1.000	`\boldsymbol{A} / \boldsymbol{B}`	$\boldsymbol{A} / \boldsymbol{B}$	and $e_{ij}:=a_{ij}^{r}$ for all i, j and suitable r. For the division $\boldsymbol{A}/\boldsymbol{B}$ we assume that
lyche-numerical-linear-algebra_F00117	23	■ 1.000	`\boldsymbol{B}`	$\boldsymbol{B}$	all elements of $\boldsymbol{B}$ are nonzero. The element-by-element product $\boldsymbol{C}=\boldsymbol{A}\times\boldsymbol{B}$
lyche-numerical-linear-algebra_F00118	23	■ 1.000	`\boldsymbol{C}=\boldsymbol{A} \times \boldsymbol{B}`	$\boldsymbol{C}=\boldsymbol{A} \times \boldsymbol{B}$	all elements of $\boldsymbol{B}$ are nonzero. The element-by-element product $\boldsymbol{C}=\boldsymbol{A}\times\boldsymbol{B}$
lyche-numerical-linear-algebra_F00119	23	■ 1.000	`\boldsymbol{A} \in \mathbb{R}^{m \times n}`	$\boldsymbol{A} \in \mathbb{R}^{m \times n}$	10. Let $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ or $\boldsymbol{A}\in\mathbb{C}^{m\times n}$. The **transpose** $\boldsymbol{A}^T$ and **conjugate transpose**
lyche-numerical-linear-algebra_F00120	23	■ 1.000	`\boldsymbol{A}^{T}`	$\boldsymbol{A}^{T}$	10. Let $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ or $\boldsymbol{A}\in\mathbb{C}^{m\times n}$. The **transpose** $\boldsymbol{A}^T$ and **conjugate transpose**
lyche-numerical-linear-algebra_F00121	23	■ 1.000	`\boldsymbol{A}^{*}`	$\boldsymbol{A}^{*}$	$\boldsymbol{A}^*$ are $n\times m$ matrices with elements $a_{ij}^T:=a_{ji}$ and $a_{ij}^*:=\overline{a}_{ji}$, respectively.
lyche-numerical-linear-algebra_F00122	23	■ 1.000	`n \times m`	$n \times m$	$\boldsymbol{A}^*$ are $n\times m$ matrices with elements $a_{ij}^T:=a_{ji}$ and $a_{ij}^*:=\overline{a}_{ji}$, respectively.
lyche-numerical-linear-algebra_F00123	23	■ 1.000	`a_{i j}^{T}:=a_{j i}`	$a_{i j}^{T}:=a_{j i}$	$\boldsymbol{A}^*$ are $n\times m$ matrices with elements $a_{ij}^T:=a_{ji}$ and $a_{ij}^*:=\overline{a}_{ji}$, respectively.
lyche-numerical-linear-algebra_F00124	23	■ 1.000	`a_{i j}^{*}:=\bar{a}_{j i}`	$a_{i j}^{*}:=\bar{a}_{j i}$	$\boldsymbol{A}^*$ are $n\times m$ matrices with elements $a_{ij}^T:=a_{ji}$ and $a_{ij}^*:=\overline{a}_{ji}$, respectively.
lyche-numerical-linear-algebra_F00125	23	■ 1.000	`n, p`	n, p	If $\boldsymbol{B}$ is an n, p matrix then $(\boldsymbol{AB})^T=\boldsymbol{B}^T\boldsymbol{A}^T$ and $(\boldsymbol{AB})^*=\boldsymbol{B}^*\boldsymbol{A}^*$. A matrix
lyche-numerical-linear-algebra_F00126	23	■ 1.000	`(\boldsymbol{A} \boldsymbol{B})^{T}=\boldsymbol{B}^{T} \boldsymbol{A}^{T}`	$(\boldsymbol{A} \boldsymbol{B})^{T}=\boldsymbol{B}^{T} \boldsymbol{A}^{T}$	If $\boldsymbol{B}$ is an n, p matrix then $(\boldsymbol{AB})^T=\boldsymbol{B}^T\boldsymbol{A}^T$ and $(\boldsymbol{AB})^*=\boldsymbol{B}^*\boldsymbol{A}^*$. A matrix
lyche-numerical-linear-algebra_F00127	23	■ 1.000	`(\boldsymbol{A} \boldsymbol{B})^{*}=\boldsymbol{B}^{*} \boldsymbol{A}^{*}`	$(\boldsymbol{A} \boldsymbol{B})^{*}=\boldsymbol{B}^{*} \boldsymbol{A}^{*}$	If $\boldsymbol{B}$ is an n, p matrix then $(\boldsymbol{AB})^T=\boldsymbol{B}^T\boldsymbol{A}^T$ and $(\boldsymbol{AB})^*=\boldsymbol{B}^*\boldsymbol{A}^*$. A matrix
lyche-numerical-linear-algebra_F00128	23	■ 1.000	`\boldsymbol{A} \in \mathbb{C}^{n \times n}`	$\boldsymbol{A} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}\in\mathbb{C}^{n\times n}$ is **symmetric** if $\boldsymbol{A}^T=\boldsymbol{A}$ and **Hermitian** if $\boldsymbol{A}^*=\boldsymbol{A}$.
lyche-numerical-linear-algebra_F00129	23	■ 1.000	`\boldsymbol{A}^{T}=\boldsymbol{A}`	$\boldsymbol{A}^{T}=\boldsymbol{A}$	$\boldsymbol{A}\in\mathbb{C}^{n\times n}$ is **symmetric** if $\boldsymbol{A}^T=\boldsymbol{A}$ and **Hermitian** if $\boldsymbol{A}^*=\boldsymbol{A}$.
lyche-numerical-linear-algebra_F00130	23	■ 1.000	`\boldsymbol{A}^{*}=\boldsymbol{A}`	$\boldsymbol{A}^{*}=\boldsymbol{A}$	$\boldsymbol{A}\in\mathbb{C}^{n\times n}$ is **symmetric** if $\boldsymbol{A}^T=\boldsymbol{A}$ and **Hermitian** if $\boldsymbol{A}^*=\boldsymbol{A}$.
lyche-numerical-linear-algebra_F00131	23	■ 1.000	`\boldsymbol{I}_{n}=\boldsymbol{I}:=\left[\delta_{i j} \right]_{i, j=1}^{n}`	$\boldsymbol{I}_{n}=\boldsymbol{I}:=\left[\delta_{i j}\right]_{i, j=1}^{n}$	while $\boldsymbol{I}_n=\boldsymbol{I}:=[\delta_{ij}]_{i,j=1}^{n}$, where
lyche-numerical-linear-algebra_F00132	23	■ 1.000	`\boldsymbol{I}`	$\boldsymbol{I}$	rows of $\boldsymbol{I}$ are the unit vectors $\boldsymbol{e}_1,\boldsymbol{e}_2,\ldots,\boldsymbol{e}_n$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00133	23	1.000	`\boldsymbol{e}_{1}, \boldsymbol{e}_{2}, \ldots, \boldsymbol{e}_{n}`	$\boldsymbol{e}_{1}, \boldsymbol{e}_{2}, \ldots, \boldsymbol{e}_{n}$	rows of $\boldsymbol{I}$ are the unit vectors $\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n$.
lyche-numerical-linear-algebra_F00134	24	1.000	`\boldsymbol{A} \in \mathbb{R}^{n \times n}`	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$	$\boldsymbol{A} \in \mathbb{R}^{n\times n}$ or $\boldsymbol{A} \in \mathbb{C}^{n\times n}$. Then $\boldsymbol{A}$ is
lyche-numerical-linear-algebra_F00135	24	1.000	`a_{i j}=0`	$a_{i j}=0$	• **diagonal** if $a_{ij} = 0$ for $i \neq j$.
lyche-numerical-linear-algebra_F00136	24	1.000	`i \neq j`	$i \neq j$	• **diagonal** if $a_{ij} = 0$ for $i \neq j$.
lyche-numerical-linear-algebra_F00137	24	1.000	`i>j`	$i>j$	• **upper triangular** or **right triangular** if $a_{ij} = 0$ for $i > j$.
lyche-numerical-linear-algebra_F00138	24	1.000	`i<j`	$i<j$	• **lower triangular** or **left triangular** if $a_{ij} = 0$ for $i < j$.
lyche-numerical-linear-algebra_F00139	24	1.000	`i>j+1`	$i>j+1$	• **upper Hessenberg** if $a_{ij} = 0$ for $i > j+1$.
lyche-numerical-linear-algebra_F00140	24	1.000	`i<j+1`	$i<j+1$	• **lower Hessenberg** if $a_{ij} = 0$ for $i < j+1$.
lyche-numerical-linear-algebra_F00141	24	1.000	`|i-j|>1`	$\lvert i-j\rvert>1$	• **tridiagonal** if $a_{ij} = 0$ for $\lvert i-j\rvert > 1$.
lyche-numerical-linear-algebra_F00142	24	0.999	`d`	d	• d-**banded** if $a_{ij} = 0$ for $\lvert i-j\rvert > d$.
lyche-numerical-linear-algebra_F00143	24	0.999	`|i-j|>d`	$\lvert i-j\rvert>d$	• d-**banded** if $a_{ij} = 0$ for $\lvert i-j\rvert > d$.
lyche-numerical-linear-algebra_F00144	24	1.000	`n \times n`	$n \times n$	13. We use the following notations for diagonal- and tridiagonal $n \times n$ matrices
lyche-numerical-linear-algebra_F00145	24	1.000	`b_{i i}:=d_{i}`	$b_{i i}:=d_{i}$	Here $b_{ii} := d_i$ for $i = 1, \ldots, n$, $b_{i+1,i} := a_i$, $b_{i,i+1} := c_i$ for $i = 1, \ldots, n-1$,
lyche-numerical-linear-algebra_F00146	24	1.000	`i=1, \ldots, n, b_{i+1, i}:=a_{i}, b_{i, i+1}:=c_{i}`	$i=1, \ldots, n, b_{i+1, i}:=a_{i}, b_{i, i+1}:=c_{i}$	Here $b_{ii} := d_i$ for $i = 1, \ldots, n$, $b_{i+1,i} := a_i$, $b_{i,i+1} := c_i$ for $i = 1, \ldots, n-1$,
lyche-numerical-linear-algebra_F00147	24	1.000	`i=1, \ldots, n-1`	$i=1, \ldots, n-1$	Here $b_{ii} := d_i$ for $i = 1, \ldots, n$, $b_{i+1,i} := a_i$, $b_{i,i+1} := c_i$ for $i = 1, \ldots, n-1$,
lyche-numerical-linear-algebra_F00148	24	1.000	`b_{i j}:=0`	$b_{i j}:=0$	and $b_{ij} := 0$ otherwise.
lyche-numerical-linear-algebra_F00149	24	1.000	`1 \leq i_{1}<i_{2}<\cdots<i_{r} \leq m, 1 \leq j_{1}<j_{2}<\cdots<`	$1 \leq i_{1}<i_{2}<\cdots<i_{r} \leq m, 1 \leq j_{1}<j_{2}<\cdots<$	14. Suppose $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ and $1 \leq i_1 < i_2 < \cdots < i_r \leq m$, $1 \leq j_1 < j_2 < \cdots <$
lyche-numerical-linear-algebra_F00150	24	1.000	`j_{c} \leq n`	$j_{c} \leq n$	$j_c \leq n$. The matrix $\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) \in \mathbb{C}^{r\times c}$ is the submatrix of $\boldsymbol{A}$ consisting of rows
lyche-numerical-linear-algebra_F00151	24	1.000	`\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) \in \mathbb{C}^{r \times c}`	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) \in \mathbb{C}^{r \times c}$	$j_c \leq n$. The matrix $\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) \in \mathbb{C}^{r\times c}$ is the submatrix of $\boldsymbol{A}$ consisting of rows
lyche-numerical-linear-algebra_F00152	24	1.000	`\boldsymbol{i}:=\left[i_{1}, \ldots, i_{r}\right]`	$\boldsymbol{i}:=\left[i_{1}, \ldots, i_{r}\right]$	$\boldsymbol{i} := [i_1, \ldots, i_r]$ and columns $\boldsymbol{j} := [j_1, \ldots, j_c]$
lyche-numerical-linear-algebra_F00153	24	1.000	`\boldsymbol{j}:=\left[j_{1}, \ldots, j_{c}\right]`	$\boldsymbol{j}:=\left[j_{1}, \ldots, j_{c}\right]$	$\boldsymbol{i} := [i_1, \ldots, i_r]$ and columns $\boldsymbol{j} := [j_1, \ldots, j_c]$
lyche-numerical-linear-algebra_F00154	25	1.000	`\mathbb{R}^{2}`	$\mathbb{R}^{2}$	Many mathematical systems have analogous properties to vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$.
lyche-numerical-linear-algebra_F00155	25	1.000	`\mathbb{R}^{3}`	$\mathbb{R}^{3}$	Many mathematical systems have analogous properties to vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$.
lyche-numerical-linear-algebra_F00156	25	1.000	`\mathcal{V}`	$\mathcal{V}$	**Definition 1.1 (Real Vector Space)** A **real vector space** is a nonempty set $\mathcal{V}$,
lyche-numerical-linear-algebra_F00157	25	1.000	`+: \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V}`	$+: \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V}$	whose objects are called **vectors**, together with two operations $+ : \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V}$
lyche-numerical-linear-algebra_F00158	25	0.992	`\cdot: \mathbb{R} \times \mathcal{V} \longrightarrow \mathcal{V}`	$\cdot: \mathbb{R} \times \mathcal{V} \longrightarrow \mathcal{V}$	and $\cdot : \mathbb{R} \times \mathcal{V} \longrightarrow \mathcal{V}$, called **addition** and **scalar multiplication**, satisfying the
lyche-numerical-linear-algebra_F00159	25	1.000	`\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}`	$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$	following axioms for all vectors $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{w}$ in $\mathcal{V}$ and scalars c, d in $\mathbb{R}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00160	25	1.000	`c, d`	c, d	following axioms for all vectors $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{w}$ in $\mathcal{V}$ and scalars c, d in $\mathbb{R}$.
lyche-numerical-linear-algebra_F00161	25	1.000	`\mathbb{R}`	$\mathbb{R}$	following axioms for all vectors $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{w}$ in $\mathcal{V}$ and scalars c, d in $\mathbb{R}$.
lyche-numerical-linear-algebra_F00162	25	1.000	`\boldsymbol{u}+\boldsymbol{v}`	$\boldsymbol{u}+\boldsymbol{v}$	(V1) The sum $\boldsymbol{u}+\boldsymbol{v}$ is in $\mathcal{V}$,
lyche-numerical-linear-algebra_F00163	25	0.960	`\boldsymbol{u}+\boldsymbol{v}=\boldsymbol{v}+\boldsymbol{u}`	$\boldsymbol{u}+\boldsymbol{v}=\boldsymbol{v}+\boldsymbol{u}$	(V2) $\boldsymbol{u}+\boldsymbol{v}=\boldsymbol{v}+\boldsymbol{u}$,
lyche-numerical-linear-algebra_F00164	25	0.994	`\boldsymbol{u}+(\boldsymbol{v}+\boldsymbol{w})=(\boldsymbol{u}+\boldsymbol{v})+\boldsymbol{w}`	$\boldsymbol{u}+(\boldsymbol{v}+\boldsymbol{w})=(\boldsymbol{u}+\boldsymbol{v})+\boldsymbol{w}$	(V3) $\boldsymbol{u}+(\boldsymbol{v}+\boldsymbol{w})=(\boldsymbol{u}+\boldsymbol{v})+\boldsymbol{w}$,
lyche-numerical-linear-algebra_F00165	25	0.999	`\mathbf{0}`	$\mathbf{0}$	(V4) There is a **zero vector** $\mathbf{0}$ such that $\boldsymbol{u}+\mathbf{0}=\boldsymbol{u}$,
lyche-numerical-linear-algebra_F00166	25	0.999	`\boldsymbol{u}+\mathbf{0}=\boldsymbol{u}`	$\boldsymbol{u}+\mathbf{0}=\boldsymbol{u}$	(V4) There is a **zero vector** $\mathbf{0}$ such that $\boldsymbol{u}+\mathbf{0}=\boldsymbol{u}$,
lyche-numerical-linear-algebra_F00167	25	1.000	`\boldsymbol{u}`	$\boldsymbol{u}$	(V5) For each $\boldsymbol{u}$ in $\mathcal{V}$ there is a vector $-\boldsymbol{u}$ in $\mathcal{V}$ such that $\boldsymbol{u}+(-\boldsymbol{u})=\mathbf{0}$,
lyche-numerical-linear-algebra_F00168	25	1.000	`-\boldsymbol{u}`	$-\boldsymbol{u}$	(V5) For each $\boldsymbol{u}$ in $\mathcal{V}$ there is a vector $-\boldsymbol{u}$ in $\mathcal{V}$ such that $\boldsymbol{u}+(-\boldsymbol{u})=\mathbf{0}$,
lyche-numerical-linear-algebra_F00169	25	1.000	`\boldsymbol{u}+(-\boldsymbol{u})=\mathbf{0}`	$\boldsymbol{u}+(-\boldsymbol{u})=\mathbf{0}$	(V5) For each $\boldsymbol{u}$ in $\mathcal{V}$ there is a vector $-\boldsymbol{u}$ in $\mathcal{V}$ such that $\boldsymbol{u}+(-\boldsymbol{u})=\mathbf{0}$,
lyche-numerical-linear-algebra_F00170	25	1.000	`c \cdot \boldsymbol{u}`	$c \cdot \boldsymbol{u}$	(S1) The scalar multiple $c \cdot \boldsymbol{u}$ is in $\mathcal{V}$,
lyche-numerical-linear-algebra_F00171	25	1.000	`c \cdot(\boldsymbol{u}+\boldsymbol{v})=c \cdot \boldsymbol{u}+c \cdot \boldsymbol{v}`	$c \cdot(\boldsymbol{u}+\boldsymbol{v})=c \cdot \boldsymbol{u}+c \cdot \boldsymbol{v}$	(S2) $c \cdot(\boldsymbol{u}+\boldsymbol{v})=c \cdot \boldsymbol{u}+c \cdot \boldsymbol{v}$,
lyche-numerical-linear-algebra_F00172	25	1.000	`(c+d) \cdot \boldsymbol{u}=c \cdot \boldsymbol{u}+d \cdot \boldsymbol{u}`	$(c+d) \cdot \boldsymbol{u}=c \cdot \boldsymbol{u}+d \cdot \boldsymbol{u}$	(S3) $(c+d) \cdot \boldsymbol{u}=c \cdot \boldsymbol{u}+d \cdot \boldsymbol{u}$,
lyche-numerical-linear-algebra_F00173	25	1.000	`c \cdot(d \cdot \boldsymbol{u})=(c d) \cdot \boldsymbol{u}`	$c \cdot(d \cdot \boldsymbol{u})=(c d) \cdot \boldsymbol{u}$	(S4) $c \cdot(d \cdot \boldsymbol{u})=(c d) \cdot \boldsymbol{u}$,
lyche-numerical-linear-algebra_F00174	25	1.000	`1 \cdot \boldsymbol{u}=\boldsymbol{u}`	$1 \cdot \boldsymbol{u}=\boldsymbol{u}$	(S5) $1 \cdot \boldsymbol{u}=\boldsymbol{u}$.
lyche-numerical-linear-algebra_F00175	25	0.589	`c \boldsymbol{v}`	$c \boldsymbol{v}$	The scalar multiplication symbol $\cdot$ is often omitted, writing $c\boldsymbol{v}$ instead of $c \cdot \boldsymbol{v}$. We
lyche-numerical-linear-algebra_F00176	25	0.589	`c \cdot \boldsymbol{v}`	$c \cdot \boldsymbol{v}$	The scalar multiplication symbol $\cdot$ is often omitted, writing $c\boldsymbol{v}$ instead of $c \cdot \boldsymbol{v}$. We
lyche-numerical-linear-algebra_F00177	25	1.000	`\boldsymbol{u}-\boldsymbol{v}:=\boldsymbol{u}+(-\boldsymbol{v})`	$\boldsymbol{u}-\boldsymbol{v}:=\boldsymbol{u}+(-\boldsymbol{v})$	define $\boldsymbol{u}-\boldsymbol{v}:=\boldsymbol{u}+(-\boldsymbol{v})$. We call $\mathcal{V}$ a **complex vector space** if the scalars consist
lyche-numerical-linear-algebra_F00178	25	0.974	`\boldsymbol{u} \in \mathcal{V}`	$\boldsymbol{u} \in \mathcal{V}$	2. For each $\boldsymbol{u} \in \mathcal{V}$ the **negative** $-\boldsymbol{u}$ of $\boldsymbol{u}$ is unique.
lyche-numerical-linear-algebra_F00179	25	1.000	`0 \boldsymbol{u}=\mathbf{0}, c \mathbf{0}=\mathbf{0}`	$0 \boldsymbol{u}=\mathbf{0}, c \mathbf{0}=\mathbf{0}$	3. $0\boldsymbol{u}=\mathbf{0}$, $c\mathbf{0}=\mathbf{0}$, and $-\boldsymbol{u}=(-1)\boldsymbol{u}$.
lyche-numerical-linear-algebra_F00180	25	1.000	`-\boldsymbol{u}=(-1) \boldsymbol{u}`	$-\boldsymbol{u}=(-1) \boldsymbol{u}$	3. $0\boldsymbol{u}=\mathbf{0}$, $c\mathbf{0}=\mathbf{0}$, and $-\boldsymbol{u}=(-1)\boldsymbol{u}$.
lyche-numerical-linear-algebra_F00181	25	1.000	`n \in \mathbb{N}`	$n \in \mathbb{N}$	1. The spaces $\mathbb{R}^n$ and $\mathbb{C}^n$, where $n \in \mathbb{N}$, are real and complex vector spaces,
lyche-numerical-linear-algebra_F00182	25	1.000	`\mathcal{D}`	$\mathcal{D}$	2. Let $\mathcal{D}$ be a subset of $\mathbb{R}$ and $d \in \mathbb{N}$. The set $\mathcal{V}$ of all functions $\boldsymbol{f}, \boldsymbol{g}: \mathcal{D} \rightarrow \mathbb{R}^d$ is a
lyche-numerical-linear-algebra_F00183	25	1.000	`d \in \mathbb{N}`	$d \in \mathbb{N}$	2. Let $\mathcal{D}$ be a subset of $\mathbb{R}$ and $d \in \mathbb{N}$. The set $\mathcal{V}$ of all functions $\boldsymbol{f}, \boldsymbol{g}: \mathcal{D} \rightarrow \mathbb{R}^d$ is a
lyche-numerical-linear-algebra_F00184	25	1.000	`\boldsymbol{f}, \boldsymbol{g}: \mathcal{D} \rightarrow \mathbb{R}^{d}`	$\boldsymbol{f}, \boldsymbol{g}: \mathcal{D} \rightarrow \mathbb{R}^{d}$	2. Let $\mathcal{D}$ be a subset of $\mathbb{R}$ and $d \in \mathbb{N}$. The set $\mathcal{V}$ of all functions $\boldsymbol{f}, \boldsymbol{g}: \mathcal{D} \rightarrow \mathbb{R}^d$ is a
lyche-numerical-linear-algebra_F00185	25	1.000	`\boldsymbol{f}, \boldsymbol{g}`	$\boldsymbol{f}, \boldsymbol{g}$	Two functions $\boldsymbol{f}, \boldsymbol{g}$ in $\mathcal{V}$ are equal if $\boldsymbol{f}(t)=\boldsymbol{g}(t)$ for all $t \in \mathcal{D}$. The zero
lyche-numerical-linear-algebra_F00186	25	1.000	`\boldsymbol{f}(t)=\boldsymbol{g}(t)`	$\boldsymbol{f}(t)=\boldsymbol{g}(t)$	Two functions $\boldsymbol{f}, \boldsymbol{g}$ in $\mathcal{V}$ are equal if $\boldsymbol{f}(t)=\boldsymbol{g}(t)$ for all $t \in \mathcal{D}$. The zero
lyche-numerical-linear-algebra_F00187	25	1.000	`t \in \mathcal{D}`	$t \in \mathcal{D}$	Two functions $\boldsymbol{f}, \boldsymbol{g}$ in $\mathcal{V}$ are equal if $\boldsymbol{f}(t)=\boldsymbol{g}(t)$ for all $t \in \mathcal{D}$. The zero

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00188	25	0.998	`\boldsymbol{f}(t)=\mathbf{0}`	$\boldsymbol{f}(t)=\mathbf{0}$	element is the **zero function** given by $\boldsymbol{f}(t)=\mathbf{0}$ for all $t\in\mathcal{D}$ and the negative
lyche-numerical-linear-algebra_F00189	25	1.000	`\boldsymbol{f}`	$\boldsymbol{f}$	of $\boldsymbol{f}$ is given by $-\boldsymbol{f}=(-1)\boldsymbol{f}$. In the following we will use boldface letters for
lyche-numerical-linear-algebra_F00190	25	1.000	`-\boldsymbol{f}=(-1) \boldsymbol{f}`	$-\boldsymbol{f}=(-1) \boldsymbol{f}$	of $\boldsymbol{f}$ is given by $-\boldsymbol{f}=(-1)\boldsymbol{f}$. In the following we will use boldface letters for
lyche-numerical-linear-algebra_F00191	25	1.000	`d>1`	$d>1$	functions only if $d>1$.
lyche-numerical-linear-algebra_F00192	25	0.997	`n \geq 0`	$n \geq 0$	3. For $n\geq 0$ the space Π_n of polynomials of degree at most n consists of all
lyche-numerical-linear-algebra_F00193	25	0.997	`\Pi_{n}`	Π_{n}	3. For $n\geq 0$ the space Π_n of polynomials of degree at most n consists of all
lyche-numerical-linear-algebra_F00194	25	1.000	`p: \mathbb{R} \rightarrow \mathbb{R}, p: \mathbb{R} \rightarrow \mathbb{C}`	$p: \mathbb{R} \rightarrow \mathbb{R}, p: \mathbb{R} \rightarrow \mathbb{C}$	polynomials $p:\mathbb{R}\rightarrow\mathbb{R}$, $p:\mathbb{R}\rightarrow\mathbb{C}$, or $p:\mathbb{C}\rightarrow\mathbb{C}$ of the form
lyche-numerical-linear-algebra_F00195	25	1.000	`p: \mathbb{C} \rightarrow \mathbb{C}`	$p: \mathbb{C} \rightarrow \mathbb{C}$	polynomials $p:\mathbb{R}\rightarrow\mathbb{R}$, $p:\mathbb{R}\rightarrow\mathbb{C}$, or $p:\mathbb{C}\rightarrow\mathbb{C}$ of the form
lyche-numerical-linear-algebra_F00196	26	1.000	`a_{0}, \ldots, a_{n}`	$a_{0}, \ldots, a_{n}$	where the coefficients $a_0,\ldots,a_n$ are real or complex numbers. p is called the
lyche-numerical-linear-algebra_F00197	26	1.000	`0 \leq k \leq n`	$0 \leq k \leq n$	integer $0\leq k\leq n$ such that $p(t)=a_0+\cdots+a_kt^k$ with $a_k\neq 0$. The degree of
lyche-numerical-linear-algebra_F00198	26	1.000	`p(t)=a_{0}+\cdots+a_{k} t^{k}`	$p(t)=a_{0}+\cdots+a_{k} t^{k}$	integer $0\leq k\leq n$ such that $p(t)=a_0+\cdots+a_kt^k$ with $a_k\neq 0$. The degree of
lyche-numerical-linear-algebra_F00199	26	1.000	`a_{k} \neq 0`	$a_{k} \neq 0$	integer $0\leq k\leq n$ such that $p(t)=a_0+\cdots+a_kt^k$ with $a_k\neq 0$. The degree of
lyche-numerical-linear-algebra_F00200	26	1.000	`n \geq 1`	$n \geq 1$	**Definition 1.2 (Linear Combination)** For $n\geq 1$ let $\mathcal{X}:=\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\}$ be a set of
lyche-numerical-linear-algebra_F00201	26	1.000	`\mathcal{X}:=\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right\}`	$\mathcal{X}:=\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right\}$	**Definition 1.2 (Linear Combination)** For $n\geq 1$ let $\mathcal{X}:=\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\}$ be a set of
lyche-numerical-linear-algebra_F00202	26	1.000	`c_{1}, \ldots, c_{n}`	$c_{1}, \ldots, c_{n}$	vectors in a vector space $\mathcal{V}$ and let $c_1,\ldots,c_n$ be scalars.
lyche-numerical-linear-algebra_F00203	26	1.000	`c_{1} \boldsymbol{x}_{1}+\cdots+c_{n} \boldsymbol{x}_{n}`	$c_{1} \boldsymbol{x}_{1}+\cdots+c_{n} \boldsymbol{x}_{n}$	1. The sum $c_1\boldsymbol{x}_1+\cdots+c_n\boldsymbol{x}_n$ is called a **linear combination** of $\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n$.
lyche-numerical-linear-algebra_F00204	26	1.000	`\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}`	$\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}$	1. The sum $c_1\boldsymbol{x}_1+\cdots+c_n\boldsymbol{x}_n$ is called a **linear combination** of $\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n$.
lyche-numerical-linear-algebra_F00205	26	1.000	`c_{j} \boldsymbol{x}_{j} \neq \mathbf{0}`	$c_{j} \boldsymbol{x}_{j} \neq \mathbf{0}$	2. The linear combination is **nontrivial** if $c_j\boldsymbol{x}_j\neq\mathbf{0}$ for at least one j.
lyche-numerical-linear-algebra_F00206	26	0.999	`\mathcal{X}`	$\mathcal{X}$	3. The set of all linear combinations of elements in $\mathcal{X}$ is denoted $\operatorname{span}(\mathcal{X})$.
lyche-numerical-linear-algebra_F00207	26	0.999	`\operatorname{span}(\mathcal{X})`	$\operatorname{span}(\mathcal{X})$	3. The set of all linear combinations of elements in $\mathcal{X}$ is denoted $\operatorname{span}(\mathcal{X})$.
lyche-numerical-linear-algebra_F00208	26	1.000	`\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right\}`	$\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right\}$	exists $n\in\mathbb{N}$ and $\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\}$ in $\mathcal{V}$ such that $\mathcal{V}=\operatorname{span}(\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\})$.
lyche-numerical-linear-algebra_F00209	26	1.000	`\mathcal{V}=\operatorname{span}\left(\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right\}\right)`	$\mathcal{V}=\operatorname{span}\left(\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right\}\right)$	exists $n\in\mathbb{N}$ and $\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\}$ in $\mathcal{V}$ such that $\mathcal{V}=\operatorname{span}(\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\})$.
lyche-numerical-linear-algebra_F00210	26	1.000	`\boldsymbol{x}=\left[x_{1}, \ldots, x_{m}\right]^{T}`	$\boldsymbol{x}=\left[x_{1}, \ldots, x_{m}\right]^{T}$	1. Any $\boldsymbol{x}=[x_1,\ldots,x_m]^T$ in $\mathbb{C}^m$ can be written as a linear combination of the unit
lyche-numerical-linear-algebra_F00211	26	1.000	`\mathbb{C}^{m}`	$\mathbb{C}^{m}$	1. Any $\boldsymbol{x}=[x_1,\ldots,x_m]^T$ in $\mathbb{C}^m$ can be written as a linear combination of the unit
lyche-numerical-linear-algebra_F00212	26	0.991	`\boldsymbol{x}=x_{1} \boldsymbol{e}_{1}+x_{2} \boldsymbol{e}_{2}+\cdots+x_{m} \boldsymbol{e}_{m}`	$\boldsymbol{x}=x_{1} \boldsymbol{e}_{1}+x_{2} \boldsymbol{e}_{2}+\cdots+x_{m} \boldsymbol{e}_{m}$	vectors as $\boldsymbol{x}=x_1\boldsymbol{e}_1+x_2\boldsymbol{e}_2+\cdots+x_m\boldsymbol{e}_m$. Thus, $\mathbb{C}^m=\operatorname{span}(\{\boldsymbol{e}_1,\ldots,\boldsymbol{e}_m\})$ and
lyche-numerical-linear-algebra_F00213	26	0.991	`\mathbb{C}^{m}=\operatorname{span}\left(\left\{\boldsymbol{e}_{1}, \ldots, \boldsymbol{e}_{m}\right\}\right)`	$\mathbb{C}^{m}=\operatorname{span}\left(\left\{\boldsymbol{e}_{1}, \ldots, \boldsymbol{e}_{m}\right\}\right)$	vectors as $\boldsymbol{x}=x_1\boldsymbol{e}_1+x_2\boldsymbol{e}_2+\cdots+x_m\boldsymbol{e}_m$. Thus, $\mathbb{C}^m=\operatorname{span}(\{\boldsymbol{e}_1,\ldots,\boldsymbol{e}_m\})$ and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00214	26	1.000	`\mathbb{R}^{m}`	$\mathbb{R}^{m}$	$\mathbb{C}^m$ is finite dimensional. Similarly $\mathbb{R}^m$ is finite dimensional.
lyche-numerical-linear-algebra_F00215	26	1.000	`\Pi=\cup_{n} \Pi_{n}`	$\Pi=\cup_{n} \Pi_{n}$	2. Let $\Pi = \cup_n \Pi_n$ be the space of all polynomials. Π is a vector space that
lyche-numerical-linear-algebra_F00216	26	1.000	`\Pi`	Π	2. Let $\Pi = \cup_n \Pi_n$ be the space of all polynomials. Π is a vector space that
lyche-numerical-linear-algebra_F00217	26	1.000	`\Pi=`	$\Pi=$	is not finite dimensional. For suppose Π is finite dimensional. Then $\Pi =$
lyche-numerical-linear-algebra_F00218	26	0.986	`\operatorname{span}\left(\left\{p_{1}, \ldots, p_{m} \right\}\right)`	$\operatorname{span}\left(\left\{p_{1}, \ldots, p_{m}\right\}\right)$	$\operatorname{span}(\{p_1, \ldots, p_m\})$ for some polynomials $p_1, \ldots, p_m$. Let d be an integer such
lyche-numerical-linear-algebra_F00219	26	0.986	`p_{1}, \ldots, p_{m}`	$p_{1}, \ldots, p_{m}$	$\operatorname{span}(\{p_1, \ldots, p_m\})$ for some polynomials $p_1, \ldots, p_m$. Let d be an integer such
lyche-numerical-linear-algebra_F00220	26	1.000	`p_{j}`	p_{j}	that the degree of p_j is less than d for $j = 1, \ldots, m$. A polynomial of degree d
lyche-numerical-linear-algebra_F00221	26	1.000	`j=1, \ldots, m`	$j=1, \ldots, m$	that the degree of p_j is less than d for $j = 1, \ldots, m$. A polynomial of degree d
lyche-numerical-linear-algebra_F00222	26	0.958	`\mathcal{X}=\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right\}`	$\mathcal{X}=\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right\}$	**Definition 1.3 (Linear Independence)** A set $\mathcal{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ of nonzero
lyche-numerical-linear-algebra_F00223	26	1.000	`\boldsymbol{x} \in \operatorname{span}(\mathcal{X})`	$\boldsymbol{x} \in \operatorname{span}(\mathcal{X})$	1. If $\boldsymbol{x} \in \operatorname{span}(\mathcal{X})$ then the scalars $c_1, \ldots, c_n$ in the representation $\boldsymbol{x} = c_1\boldsymbol{x}_1 + \cdots +$
lyche-numerical-linear-algebra_F00224	26	1.000	`\boldsymbol{x}=c_{1} \boldsymbol{x}_{1}+\cdots+`	$\boldsymbol{x}=c_{1} \boldsymbol{x}_{1}+\cdots+$	1. If $\boldsymbol{x} \in \operatorname{span}(\mathcal{X})$ then the scalars $c_1, \ldots, c_n$ in the representation $\boldsymbol{x} = c_1\boldsymbol{x}_1 + \cdots +$
lyche-numerical-linear-algebra_F00225	26	0.995	`c_{n} \boldsymbol{x}_{n}`	$c_{n} \boldsymbol{x}_{n}$	$c_n\boldsymbol{x}_n$ are unique.
lyche-numerical-linear-algebra_F00226	26	0.995	`\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}`	$\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}$	**Lemma 1.1 (Linear Independence and Span)** *Suppose* $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ *span a vector*
lyche-numerical-linear-algebra_F00227	26	1.000	`\boldsymbol{w}_{1}, \ldots, \boldsymbol{w}_{k}`	$\boldsymbol{w}_{1}, \ldots, \boldsymbol{w}_{k}$	*space* $\mathcal{V}$ *and that* $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ *are linearly independent vectors in* $\mathcal{V}$. *Then* $k \leq n$.
lyche-numerical-linear-algebra_F00228	26	1.000	`k \leq n`	$k \leq n$	*space* $\mathcal{V}$ *and that* $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ *are linearly independent vectors in* $\mathcal{V}$. *Then* $k \leq n$.
lyche-numerical-linear-algebra_F00229	27	1.000	`k>n`	$k>n$	***Proof*** Suppose $k > n$. Write $\boldsymbol{w}_1$ as a linear combination of elements from
lyche-numerical-linear-algebra_F00230	27	1.000	`\boldsymbol{w}_{1}`	$\boldsymbol{w}_{1}$	***Proof*** Suppose $k > n$. Write $\boldsymbol{w}_1$ as a linear combination of elements from
lyche-numerical-linear-algebra_F00231	27	1.000	`\mathcal{X}_{0}:=\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right\}`	$\mathcal{X}_{0}:=\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right\}$	the set $\mathcal{X}_0 := \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$, say $\boldsymbol{w}_1 = c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n$. Since $\boldsymbol{w}_1 \neq \mathbf{0}$
lyche-numerical-linear-algebra_F00232	27	1.000	`\boldsymbol{w}_{1}=c_{1} \boldsymbol{v}_{1}+\cdots+c_{n} \boldsymbol{v}_{n}`	$\boldsymbol{w}_{1}=c_{1} \boldsymbol{v}_{1}+\cdots+c_{n} \boldsymbol{v}_{n}$	the set $\mathcal{X}_0 := \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$, say $\boldsymbol{w}_1 = c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n$. Since $\boldsymbol{w}_1 \neq \mathbf{0}$
lyche-numerical-linear-algebra_F00233	27	1.000	`\boldsymbol{w}_{1} \neq \mathbf{0}`	$\boldsymbol{w}_{1} \neq \mathbf{0}$	the set $\mathcal{X}_0 := \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$, say $\boldsymbol{w}_1 = c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n$. Since $\boldsymbol{w}_1 \neq \mathbf{0}$
lyche-numerical-linear-algebra_F00234	27	1.000	`c_{i_{1}}`	$c_{i_{1}}$	not all the c's are equal to zero. Pick a nonzero c, say c_{i_1}. Then $\boldsymbol{v}_{i_1}$ can be
lyche-numerical-linear-algebra_F00235	27	1.000	`\boldsymbol{v}_{i_{1}}`	$\boldsymbol{v}_{i_{1}}$	not all the c's are equal to zero. Pick a nonzero c, say c_{i_1}. Then $\boldsymbol{v}_{i_1}$ can be
lyche-numerical-linear-algebra_F00236	27	1.000	`\boldsymbol{v}`	$\boldsymbol{v}$	expressed as a linear combination of $\boldsymbol{w}_1$ and the remaining $\boldsymbol{v}$'s. So the set $\mathcal{X}_1 :=$
lyche-numerical-linear-algebra_F00237	27	1.000	`\mathcal{X}_{1}:=`	$\mathcal{X}_{1}:=$	expressed as a linear combination of $\boldsymbol{w}_1$ and the remaining $\boldsymbol{v}$'s. So the set $\mathcal{X}_1 :=$
lyche-numerical-linear-algebra_F00238	27	1.000	`\left\{\boldsymbol{w}_{1}, \boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{i_{1}-1}, \boldsymbol{v}_{i_{1}+1}, \ldots, \boldsymbol{v}_{n}\right\}`	$\left\{\boldsymbol{w}_{1}, \boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{i_{1}-1}, \boldsymbol{v}_{i_{1}+1}, \ldots, \boldsymbol{v}_{n}\right\}$	$\{\boldsymbol{w}_1, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_{i_1-1}, \boldsymbol{v}_{i_1+1}, \ldots, \boldsymbol{v}_n\}$ must also be a spanning set for $\mathcal{V}$. We repeat this
lyche-numerical-linear-algebra_F00239	27	1.000	`\boldsymbol{w}_{2}`	$\boldsymbol{w}_{2}$	for $\boldsymbol{w}_2$ and $\mathcal{X}_1$. In the linear combination $\boldsymbol{w}_2 = d_{i_1}\boldsymbol{w}_1 + \sum_{j \neq i_1} d_j\boldsymbol{v}_j$, we must
lyche-numerical-linear-algebra_F00240	27	1.000	`\mathcal{X}_{1}`	$\mathcal{X}_{1}$	for $\boldsymbol{w}_2$ and $\mathcal{X}_1$. In the linear combination $\boldsymbol{w}_2 = d_{i_1}\boldsymbol{w}_1 + \sum_{j \neq i_1} d_j\boldsymbol{v}_j$, we must

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F00241	27	■1.000	\boldsymbol{w}_{2}=d_{i_{1}} \boldsymbol{w}_{1}+\sum_{j \neq i_{1}} d_{j} \boldsymbol{v}_{j}	$\boldsymbol{w}_{2}=d_{i_{1}} \boldsymbol{w}_{1}+\sum_{j \neq i_{1}} d_{j} \boldsymbol{v}_{j}$	for $\boldsymbol{w}_2$ and $\mathcal{X}_1$. In the linear combination $\boldsymbol{w}_2 = d_{i_1}\boldsymbol{w}_1 + \sum_{j\neq i_1} d_j \boldsymbol{v}_j$, we must
lyche-numerical-linear-algebra_ F00242	27	■0.900	d_{i_{2}} \neq 0	$d_{i_{2}} \neq 0$	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\boldsymbol{w}_2 = d_1\boldsymbol{w}_1$ contradicting
lyche-numerical-linear-algebra_ F00243	27	■0.900	i_{2}	i_{2}	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\boldsymbol{w}_2 = d_1\boldsymbol{w}_1$ contradicting
lyche-numerical-linear-algebra_ F00244	27	■0.900	i_{2} \neq i_{1}	$i_{2} \neq i_{1}$	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\boldsymbol{w}_2 = d_1\boldsymbol{w}_1$ contradicting
lyche-numerical-linear-algebra_ F00245	27	■0.900	\boldsymbol{w}_{2}=d_{1} \boldsymbol{w}_{1}	$\boldsymbol{w}_{2}=d_{1} \boldsymbol{w}_{1}$	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\boldsymbol{w}_2 = d_1\boldsymbol{w}_1$ contradicting
lyche-numerical-linear-algebra_ F00246	27	■1.000	\boldsymbol{w}	$\boldsymbol{w}$	the linear independence of the $\boldsymbol{w}$'s. So the set $\mathcal{X}_2$ consisting of the $\boldsymbol{v}$'s with $\boldsymbol{v}_{i_1}$
lyche-numerical-linear-algebra_ F00247	27	■1.000	\mathcal{X}_{2}	$\mathcal{X}_{2}$	the linear independence of the $\boldsymbol{w}$'s. So the set $\mathcal{X}_2$ consisting of the $\boldsymbol{v}$'s with $\boldsymbol{v}_{i_1}$
lyche-numerical-linear-algebra_ F00248	27	■0.998	\boldsymbol{v}_{i_{2}}	$\boldsymbol{v}_{i_{2}}$	replaced by $\boldsymbol{w}_1$ and $\boldsymbol{v}_{i_2}$ replaced by $\boldsymbol{w}_2$ is again a spanning set for $\mathcal{V}$. Repeating this
lyche-numerical-linear-algebra_ F00249	27	■1.000	n-2	$n-2$	process $n-2$ more times we obtain a spanning set $\mathcal{X}_n$ where $\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n$ have been
lyche-numerical-linear-algebra_ F00250	27	■1.000	\mathcal{X}_{n}	$\mathcal{X}_{n}$	process $n-2$ more times we obtain a spanning set $\mathcal{X}_n$ where $\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n$ have been
lyche-numerical-linear-algebra_ F00251	27	■1.000	\boldsymbol{w}_{1}, \ldots, \boldsymbol{w}_{n}	$\boldsymbol{w}_{1}, \ldots, \boldsymbol{w}_{n}$	replaced by $\boldsymbol{w}_1,\ldots,\boldsymbol{w}_n$. Since $k > n$ we can then write $\boldsymbol{w}_k$ as a linear combination
lyche-numerical-linear-algebra_ F00252	27	■1.000	\boldsymbol{w}_{k}	$\boldsymbol{w}_{k}$	replaced by $\boldsymbol{w}_1,\ldots,\boldsymbol{w}_n$. Since $k > n$ we can then write $\boldsymbol{w}_k$ as a linear combination
lyche-numerical-linear-algebra_ F00253	27	—	\square	$\square$	—
lyche-numerical-linear-algebra_ F00254	27	■1.000	\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n} \right\}	$\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n} \right\}$	**Definition 1.4 (Basis)** A finite set of vectors $\{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n\}$ in a vector space $\mathcal{V}$ is a
lyche-numerical-linear-algebra_ F00255	27	■1.000	\operatorname{span}\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right\}=\mathcal{V}	$\operatorname{span}\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right\}=\mathcal{V}$	1. $\operatorname{span}\{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n\} = \mathcal{V}$.
lyche-numerical-linear-algebra_ F00256	27	■0.999	\left\{\boldsymbol{v}_{i_{1}}, \ldots, \boldsymbol{v}_{i_{k}}\right\}	$\left\{\boldsymbol{v}_{i_{1}}, \ldots, \boldsymbol{v}_{i_{k}}\right\}$	*that* $\{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n\}$ *is a spanning set for* $\mathcal{V}$. *Then we can find a subset* $\{\boldsymbol{v}_{i_1},\ldots,\boldsymbol{v}_{i_k}\}$
lyche-numerical-linear-algebra_ F00257	27	■1.000	\mathcal{V}=\operatorname{span}\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right\}	$\mathcal{V}=\operatorname{span}\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right\}$	***Proof*** Let $\mathcal{V} = \operatorname{span}\{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n\}$ be a finite dimensional vector space. By Theo-
lyche-numerical-linear-algebra_ F00258	27	■1.000	\operatorname{dim} \mathcal{V}	$\operatorname{dim} \mathcal{V}$	*space and denoted* $\dim \mathcal{V}$.
lyche-numerical-linear-algebra_ F00259	27	■1.000	\mathcal{X}=\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right\}	$\mathcal{X}=\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right\}$	***Proof*** Suppose $\mathcal{X} = \{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n\}$ and $\mathcal{Y} = \{\boldsymbol{w}_1,\ldots,\boldsymbol{w}_k\}$ are two bases for $\mathcal{V}$. By
lyche-numerical-linear-algebra_ F00260	27	■1.000	\mathcal{Y}=\left\{\boldsymbol{w}_{1}, \ldots, \boldsymbol{w}_{k}\right\}	$\mathcal{Y}=\left\{\boldsymbol{w}_{1}, \ldots, \boldsymbol{w}_{k}\right\}$	***Proof*** Suppose $\mathcal{X} = \{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n\}$ and $\mathcal{Y} = \{\boldsymbol{w}_1,\ldots,\boldsymbol{w}_k\}$ are two bases for $\mathcal{V}$. By
lyche-numerical-linear-algebra_ F00261	27	■1.000	\mathcal{Y}	$\mathcal{Y}$	Lemma 1.1 we have $k \leq n$. Using the same Lemma with $\mathcal{X}$ and $\mathcal{Y}$ switched we
lyche-numerical-linear-algebra_ F00262	27	■1.000	n \leq k	$n \leq k$	obtain $n \leq k$. We conclude that $n = k$.
lyche-numerical-linear-algebra_ F00263	27	■1.000	n=k	$n=k$	obtain $n \leq k$. We conclude that $n = k$.
lyche-numerical-linear-algebra_ F00264	27	■1.000	\left\{\boldsymbol{e}_{1}, \ldots, \boldsymbol{e}_{n} \right\}	$\left\{\boldsymbol{e}_{1}, \ldots, \boldsymbol{e}_{n} \right\}$	The set of unit vectors $\{\boldsymbol{e}_1,\ldots,\boldsymbol{e}_n\}$ form a basis for both $\mathbb{R}^n$ and $\mathbb{C}^n$.
lyche-numerical-linear-algebra_ F00265	27	■0.996	\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{k} \right\}	$\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{k} \right\}$	*vectors* $\{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_k\}$ *in a finite dimensional vector space* $\mathcal{V}$ *can be enlarged to a*
lyche-numerical-linear-algebra_ F00266	28	■1.000	\boldsymbol{v}_{k+1}	$\boldsymbol{v}_{k+1}$	***Proof*** If $\{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_k\}$ does not span $\mathcal{V}$ we can enlarge the set by one vector $\boldsymbol{v}_{k+1}$
lyche-numerical-linear-algebra_ F00267	28	■1.000	\boldsymbol{u}, \boldsymbol{v} \in \mathcal{S}	$\boldsymbol{u}, \boldsymbol{v} \in \mathcal{S}$	(V1) The sum $\boldsymbol{u}+\boldsymbol{v}$ is in $\mathcal{S}$ for any $\boldsymbol{u},\boldsymbol{v} \in \mathcal{S}$.
lyche-numerical-linear-algebra_ F00268	28	■1.000	c \boldsymbol{u}	$c \boldsymbol{u}$	(S1) The scalar multiple $c\boldsymbol{u}$ is in $\mathcal{S}$ for any scalar c and any $\boldsymbol{u} \in \mathcal{S}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F00269	28	■ 1.000	`\boldsymbol{u} \in \mathcal{S}`	$\boldsymbol{u} \in \mathcal{S}$	(S1) The scalar multiple $c\boldsymbol{u}$ is in $\mathcal{S}$ for any scalar c and any $\boldsymbol{u} \in \mathcal{S}$.
lyche-numerical-linear-algebra_ F00270	28	■ 1.000	`V 1-V 5`	$V 1-V 5$	axioms $V1 - V5$ and $S1 - S5$ are satisfied for $\mathcal{S}$. In particular, $\mathcal{S}$ must contain the
lyche-numerical-linear-algebra_ F00271	28	■ 1.000	`S 1-S 5`	$S 1-S 5$	axioms $V1 - V5$ and $S1 - S5$ are satisfied for $\mathcal{S}$. In particular, $\mathcal{S}$ must contain the
lyche-numerical-linear-algebra_ F00272	28	■ 1.000	`\{\mathbf{0}\}`	$\{\mathbf{0}\}$	1. $\{\mathbf{0}\}$, where $\mathbf{0}$ is the zero vector is a subspace, the **trivial subspace**. The dimension
lyche-numerical-linear-algebra_ F00273	28	■ 1.000	`\mathcal{X}=\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right\} \subseteq \mathcal{V}`	$\mathcal{X}=\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right\} \subseteq \mathcal{V}$	3. span$(\mathcal{X})$ is a subspace of $\mathcal{V}$ for any $\mathcal{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\} \subseteq \mathcal{V}$. Indeed, it is easy to
lyche-numerical-linear-algebra_ F00274	28	■ 1.000	`\mathcal{S} \oplus \mathcal{T}`	$\mathcal{S} \oplus \mathcal{T}$	denoted $\mathcal{S} \oplus \mathcal{T}$ if $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$.
lyche-numerical-linear-algebra_ F00275	28	■ 1.000	`\mathcal{S} \cap \mathcal{T}=\{\mathbf{0}\}`	$\mathcal{S} \cap \mathcal{T}=\{\mathbf{0}\}$	denoted $\mathcal{S} \oplus \mathcal{T}$ if $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$.
lyche-numerical-linear-algebra_ F00276	29	■ 1.000	`\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p} \right\}`	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p} \right\}$	***Proof*** Let $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$ be a basis for $\mathcal{S} \cap \mathcal{T}$, where $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\} = \emptyset$, the empty
lyche-numerical-linear-algebra_ F00277	29	■ 1.000	`\mathcal{S} \cap \mathcal{T}`	$\mathcal{S} \cap \mathcal{T}$	***Proof*** Let $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$ be a basis for $\mathcal{S} \cap \mathcal{T}$, where $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\} = \emptyset$, the empty
lyche-numerical-linear-algebra_ F00278	29	■ 1.000	`\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p} \right\}=\emptyset`	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p} \right\}=\emptyset$	***Proof*** Let $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$ be a basis for $\mathcal{S} \cap \mathcal{T}$, where $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\} = \emptyset$, the empty
lyche-numerical-linear-algebra_ F00279	29	■ 1.000	`\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}, \boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{q}\right\}`	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}, \boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{q}\right\}$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_q\}$ for $\mathcal{S}$ and a basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{t}_1, \ldots, \boldsymbol{t}_r\}$ for $\mathcal{T}$. Every
lyche-numerical-linear-algebra_ F00280	29	■ 1.000	`\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}, \boldsymbol{t}_{1}, \ldots, \boldsymbol{t}_{r}\right\}`	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}, \boldsymbol{t}_{1}, \ldots, \boldsymbol{t}_{r}\right\}$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_q\}$ for $\mathcal{S}$ and a basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{t}_1, \ldots, \boldsymbol{t}_r\}$ for $\mathcal{T}$. Every
lyche-numerical-linear-algebra_ F00281	29	■ 1.000	`\boldsymbol{x} \in \mathcal{S}+\mathcal{T}`	$\boldsymbol{x} \in \mathcal{S}+\mathcal{T}$	$\boldsymbol{x} \in \mathcal{S} + \mathcal{T}$ can be written as a linear combination of
lyche-numerical-linear-algebra_ F00282	29	■ 1.000	`\mathcal{S}+\mathcal{T}`	$\mathcal{S}+\mathcal{T}$	so these vectors span $\mathcal{S} + \mathcal{T}$. We show that they are linearly independent and hence
lyche-numerical-linear-algebra_ F00283	29	■ 1.000	`\boldsymbol{u}+\boldsymbol{s}+\boldsymbol{t}=\mathbf{0}`	$\boldsymbol{u}+\boldsymbol{s}+\boldsymbol{t}=\mathbf{0}$	a basis. Suppose $\boldsymbol{u} + \boldsymbol{s} + \boldsymbol{t} = \mathbf{0}$, where $\boldsymbol{u} := \sum_{j=1}^p \alpha_j \boldsymbol{u}_j$, $\boldsymbol{s} := \sum_{j=1}^q \rho_j \boldsymbol{s}_j$, and
lyche-numerical-linear-algebra_ F00284	29	■ 1.000	`\boldsymbol{u}:=\sum_{j=1}^{p} \alpha_{j} \boldsymbol{u}_{j}, \boldsymbol{s}:=\sum_{j=1}^{q} \rho_{j} \boldsymbol{s}_{j}`	$\boldsymbol{u}:=\sum_{j=1}^{p} \alpha_{j} \boldsymbol{u}_{j}, \boldsymbol{s}:=\sum_{j=1}^{q} \rho_{j} \boldsymbol{s}_{j}$	a basis. Suppose $\boldsymbol{u} + \boldsymbol{s} + \boldsymbol{t} = \mathbf{0}$, where $\boldsymbol{u} := \sum_{j=1}^p \alpha_j \boldsymbol{u}_j$, $\boldsymbol{s} := \sum_{j=1}^q \rho_j \boldsymbol{s}_j$, and
lyche-numerical-linear-algebra_ F00285	29	■ 1.000	`\boldsymbol{t}:=\sum_{j=1}^{r} \sigma_{j} \boldsymbol{t}_{j}`	$\boldsymbol{t}:=\sum_{j=1}^{r} \sigma_{j} \boldsymbol{t}_{j}$	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$. Now $\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$ belongs to both $\mathcal{S}$ and to $\mathcal{T}$ and hence $\boldsymbol{s} \in$
lyche-numerical-linear-algebra_ F00286	29	■ 1.000	`\boldsymbol{s}=-(\boldsymbol{u}+\boldsymbol{t})`	$\boldsymbol{s}=-(\boldsymbol{u}+\boldsymbol{t})$	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$. Now $\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$ belongs to both $\mathcal{S}$ and to $\mathcal{T}$ and hence $\boldsymbol{s} \in$
lyche-numerical-linear-algebra_ F00287	29	■ 1.000	`\boldsymbol{s} \in`	$\boldsymbol{s} \in$	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$. Now $\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$ belongs to both $\mathcal{S}$ and to $\mathcal{T}$ and hence $\boldsymbol{s} \in$
lyche-numerical-linear-algebra_ F00288	29	■ 1.000	`\boldsymbol{s}`	$\boldsymbol{s}$	$\mathcal{S} \cap \mathcal{T}$. Therefore $\boldsymbol{s}$ can be written as a linear combination of $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p$ say $\boldsymbol{s} :=$
lyche-numerical-linear-algebra_ F00289	29	■ 1.000	`\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}`	$\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}$	$\mathcal{S} \cap \mathcal{T}$. Therefore $\boldsymbol{s}$ can be written as a linear combination of $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p$ say $\boldsymbol{s} :=$
lyche-numerical-linear-algebra_ F00290	29	■ 1.000	`\boldsymbol{s}:=`	$\boldsymbol{s}:=$	$\mathcal{S} \cap \mathcal{T}$. Therefore $\boldsymbol{s}$ can be written as a linear combination of $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p$ say $\boldsymbol{s} :=$
lyche-numerical-linear-algebra_ F00291	29	■ 1.000	`\sum_{j=1}^{p} \beta_{j} \boldsymbol{u}_{j}`	$\sum_{j=1}^{p} \beta_{j} \boldsymbol{u}_{j}$	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$. But then $\mathbf{0} = \sum_{j=1}^p \beta_j \boldsymbol{u}_j - \sum_{j=1}^q \rho_j \boldsymbol{s}_j$ and since
lyche-numerical-linear-algebra_ F00292	29	■ 1.000	`\mathbf{0}=\sum_{j=1}^{p} \beta_{j} \boldsymbol{u}_{j}-\sum_{j=1}^{q} \rho_{j} \boldsymbol{s}_{j}`	$\mathbf{0}=\sum_{j=1}^{p} \beta_{j} \boldsymbol{u}_{j}-\sum_{j=1}^{q} \rho_{j} \boldsymbol{s}_{j}$	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$. But then $\mathbf{0} = \sum_{j=1}^p \beta_j \boldsymbol{u}_j - \sum_{j=1}^q \rho_j \boldsymbol{s}_j$ and since
lyche-numerical-linear-algebra_ F00293	29	■ 1.000	`\beta_{1}=\cdots=\beta_{p}=\rho_{1}=\cdots=\rho_{q}=0`	$\beta_{1}=\cdots=\beta_{p}=\rho_{1}=\cdots=\rho_{q}=0$	is linearly independent we must have $\beta_1 = \cdots = \beta_p = \rho_1 = \cdots = \rho_q = 0$
lyche-numerical-linear-algebra_ F00294	29	■ 1.000	`\boldsymbol{s}=\mathbf{0}`	$\boldsymbol{s}=\mathbf{0}$	and hence $\boldsymbol{s} = \mathbf{0}$. We then have $\boldsymbol{u} + \boldsymbol{t} = \mathbf{0}$ and by linear independence of

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F00295	29	1.000	`\boldsymbol{u}+\boldsymbol{t}=\mathbf{0}`	$\boldsymbol{u}+\boldsymbol{t}=\mathbf{0}$	and hence $\boldsymbol{s} = \mathbf{0}$. We then have $\boldsymbol{u} + \boldsymbol{t} = \mathbf{0}$ and by linear independence of
lyche-numerical-linear-algebra_ F00296	29	1.000	`\alpha_{1}=\cdots=\alpha_{p}=\sigma_{1}=\cdots=\sigma_{r}=0`	$\alpha_{1}=\cdots=\alpha_{p}=\sigma_{1}=\cdots=\sigma_{r}=0$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{t}_1, \ldots, \boldsymbol{t}_r\}$ we obtain $\alpha_1 = \cdots = \alpha_p = \sigma_1 = \cdots = \sigma_r = 0$. We
lyche-numerical-linear-algebra_ F00297	29	1.000	`\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}, \boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{q}, \boldsymbol{t}_{1}, \ldots, \boldsymbol{t}_{r}\right \}`	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}, \boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{q}, \boldsymbol{t}_{1}, \ldots, \boldsymbol{t}_{r}\right\}$	have shown that the vectors $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_q, \boldsymbol{t}_1, \ldots, \boldsymbol{t}_r\}$ constitute a basis
lyche-numerical-linear-algebra_ F00298	29	1.000	`\operatorname{dim}\{\mathbf{0}\}=0`	$\operatorname{dim}\{\mathbf{0}\}=0$	and (1.7) follows. Equation (1.7) implies (1.8) since $\dim\{\mathbf{0}\} = 0$.
lyche-numerical-linear-algebra_ F00299	29	1.000	`\left\{\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{n} \right\}`	$\left\{\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{n}\right\}$	*dimensional vector space* $\mathcal{V}$ *and let* $\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n\}$ *be a basis for* $\mathcal{S}$ *and* $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$
lyche-numerical-linear-algebra_ F00300	29	1.000	`\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{m} \right\}`	$\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{m}\right\}$	*dimensional vector space* $\mathcal{V}$ *and let* $\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n\}$ *be a basis for* $\mathcal{S}$ *and* $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$
lyche-numerical-linear-algebra_ F00301	29	1.000	`\boldsymbol{s}_{j}`	$\boldsymbol{s}_{j}$	*a basis for* $\mathcal{V}$. *Then each* $\boldsymbol{s}_j$ *can be expressed as a linear combination of* $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$,
lyche-numerical-linear-algebra_ F00302	29	1.000	`\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{m}`	$\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{m}$	*a basis for* $\mathcal{V}$. *Then each* $\boldsymbol{s}_j$ *can be expressed as a linear combination of* $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$,
lyche-numerical-linear-algebra_ F00303	30	1.000	`\boldsymbol{x} \in \mathcal{S}`	$\boldsymbol{x} \in \mathcal{S}$	*If* $\boldsymbol{x} \in \mathcal{S}$ *then* $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ *for some coefficients* $\boldsymbol{b} :=$
lyche-numerical-linear-algebra_ F00304	30	1.000	`\boldsymbol{x}=\sum_{j=1}^{n} c_{j} \boldsymbol{s}_{j}=\sum_{i=1}^{m} b_{i} \boldsymbol{v}_{i}`	$\boldsymbol{x}=\sum_{j=1}^{n} c_{j} \boldsymbol{s}_{j}=\sum_{i=1}^{m} b_{i} \boldsymbol{v}_{i}$	*If* $\boldsymbol{x} \in \mathcal{S}$ *then* $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ *for some coefficients* $\boldsymbol{b} :=$
lyche-numerical-linear-algebra_ F00305	30	1.000	`\boldsymbol{b}:=`	$\boldsymbol{b}:=$	*If* $\boldsymbol{x} \in \mathcal{S}$ *then* $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ *for some coefficients* $\boldsymbol{b} :=$
lyche-numerical-linear-algebra_ F00306	30	1.000	`\left[b_{1}, \ldots, \boldsymbol{b}_{m}\right]^{T}, \boldsymbol{c}:=\left[c_{1}, \ldots, c_{n}\right]^{T}`	$\left[b_{1}, \ldots, \boldsymbol{b}_{m}\right]^{T}, \boldsymbol{c}:=\left[c_{1}, \ldots, c_{n}\right]^{T}$	$[b_1, \ldots, \boldsymbol{b}_m]^T$, $\boldsymbol{c} := [c_1, \ldots, c_n]^T$. *Moreover* $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{c}$, *where* $\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{m\times n}$ *is*
lyche-numerical-linear-algebra_ F00307	30	1.000	`\boldsymbol{b}=\boldsymbol{A} \boldsymbol{c}`	$\boldsymbol{b}=\boldsymbol{A} \boldsymbol{c}$	$[b_1, \ldots, \boldsymbol{b}_m]^T$, $\boldsymbol{c} := [c_1, \ldots, c_n]^T$. *Moreover* $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{c}$, *where* $\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{m\times n}$ *is*
lyche-numerical-linear-algebra_ F00308	30	1.000	`\boldsymbol{A}=\left[a_{i j}\right] \in \mathbb{C}^{m \times n}`	$\boldsymbol{A}=\left[a_{i j}\right] \in \mathbb{C}^{m \times n}$	$[b_1, \ldots, \boldsymbol{b}_m]^T$, $\boldsymbol{c} := [c_1, \ldots, c_n]^T$. *Moreover* $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{c}$, *where* $\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{m\times n}$ *is*
lyche-numerical-linear-algebra_ F00309	30	0.999	`a_{i j}`	$a_{i j}$	***Proof*** Equation (1.9) holds for some a_{ij} since $\boldsymbol{s}_j \in \mathcal{V}$ and $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$ spans $\mathcal{V}$.
lyche-numerical-linear-algebra_ F00310	30	0.999	`\boldsymbol{s}_{j} \in \mathcal{V}`	$\boldsymbol{s}_{j} \in \mathcal{V}$	***Proof*** Equation (1.9) holds for some a_{ij} since $\boldsymbol{s}_j \in \mathcal{V}$ and $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$ spans $\mathcal{V}$.
lyche-numerical-linear-algebra_ F00311	30	0.980	`\left(c_{j}\right)`	$\left(c_{j}\right)$	be written $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some scalars (c_j) and (b_i). But then
lyche-numerical-linear-algebra_ F00312	30	0.980	`\left(b_{i}\right)`	$\left(b_{i}\right)$	be written $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some scalars (c_j) and (b_i). But then
lyche-numerical-linear-algebra_ F00313	30	1.000	`b_{i}=\sum_{j=1}^{n} a_{i j} c_{j}`	$b_{i}=\sum_{j=1}^{n} a_{i j} c_{j}$	Since $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$ is linearly independent it follows that $b_i = \sum_{j=1}^n a_{ij} c_j$ for $i =$
lyche-numerical-linear-algebra_ F00314	30	1.000	`i=`	$i=$	Since $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$ is linearly independent it follows that $b_i = \sum_{j=1}^n a_{ij} c_j$ for $i =$
lyche-numerical-linear-algebra_ F00315	30	1.000	`1, \ldots, m`	$1, \ldots, m$	$1, \ldots, m$ or $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{c}$. Finally, to show that $\boldsymbol{A}$ has linearly independent columns
lyche-numerical-linear-algebra_ F00316	30	1.000	`\boldsymbol{b}:=\boldsymbol{A} \boldsymbol{c}=\mathbf{0}`	$\boldsymbol{b}:=\boldsymbol{A} \boldsymbol{c}=\mathbf{0}$	suppose $\boldsymbol{b} := \boldsymbol{A}\boldsymbol{c} = \mathbf{0}$ for some $\boldsymbol{c} = [c_1, \ldots, c_n]^T$. Define $\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$. Then
lyche-numerical-linear-algebra_ F00317	30	1.000	`\boldsymbol{c}=\left[c_{1}, \ldots, c_{n}\right]^{T}`	$\boldsymbol{c}=\left[c_{1}, \ldots, c_{n}\right]^{T}$	suppose $\boldsymbol{b} := \boldsymbol{A}\boldsymbol{c} = \mathbf{0}$ for some $\boldsymbol{c} = [c_1, \ldots, c_n]^T$. Define $\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$. Then
lyche-numerical-linear-algebra_ F00318	30	1.000	`\boldsymbol{x}:=\sum_{j=1}^{n} c_{j} \boldsymbol{s}_{j}`	$\boldsymbol{x}:=\sum_{j=1}^{n} c_{j} \boldsymbol{s}_{j}$	suppose $\boldsymbol{b} := \boldsymbol{A}\boldsymbol{c} = \mathbf{0}$ for some $\boldsymbol{c} = [c_1, \ldots, c_n]^T$. Define $\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$. Then
lyche-numerical-linear-algebra_ F00319	30	0.995	`\boldsymbol{x}=\sum_{i=1}^{m} b_{i} \boldsymbol{v}_{i}`	$\boldsymbol{x}=\sum_{i=1}^{m} b_{i} \boldsymbol{v}_{i}$	$\boldsymbol{x} = \sum_{i=1}^m b_i \boldsymbol{v}_i$ and since $\boldsymbol{b} = \mathbf{0}$ we have $\boldsymbol{x} = \mathbf{0}$. But since $\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n\}$ is linearly

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00320	30	■ 0.995	`\boldsymbol{b}=\mathbf{0}`	$\boldsymbol{b}=\mathbf{0}$	$\boldsymbol{x} = \sum_{i=1}^{m} b_i \boldsymbol{v}_i$ and since $\boldsymbol{b} = \mathbf{0}$ we have $\boldsymbol{x} = \mathbf{0}$. But since $\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n\}$ is linearly
lyche-numerical-linear-algebra_F00321	30	■ 0.995	`\boldsymbol{x}=\mathbf{0}`	$\boldsymbol{x}=\mathbf{0}$	$\boldsymbol{x} = \sum_{i=1}^{m} b_i \boldsymbol{v}_i$ and since $\boldsymbol{b} = \mathbf{0}$ we have $\boldsymbol{x} = \mathbf{0}$. But since $\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n\}$ is linearly
lyche-numerical-linear-algebra_F00322	30	■ 1.000	`\boldsymbol{c}=\mathbf{0}`	$\boldsymbol{c}=\mathbf{0}$	independent it follows that $\boldsymbol{c} = \mathbf{0}$.
lyche-numerical-linear-algebra_F00323	30	■ 1.000	`\mathcal{V}=\mathbb{R}^{m}`	$\mathcal{V}=\mathbb{R}^{m}$	When $\mathcal{V} = \mathbb{R}^m$ or $\mathbb{C}^m$ we can think of n vectors in $\mathcal{V}$, say $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$, as a set
lyche-numerical-linear-algebra_F00324	30	■ 1.000	`m \times n`	$m \times n$	$\mathcal{X} := \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ or as the columns of an $m \times n$ matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n]$. A
lyche-numerical-linear-algebra_F00325	30	■ 1.000	`\boldsymbol{X}=\left[\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right]`	$\boldsymbol{X}=\left[\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right]$	$\mathcal{X} := \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ or as the columns of an $m \times n$ matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n]$. A
lyche-numerical-linear-algebra_F00326	30	■ 0.998	`\boldsymbol{X} \boldsymbol{c}`	$\boldsymbol{X} \boldsymbol{c}$	linear combination can then be written as a matrix times vector $\boldsymbol{X}\boldsymbol{c}$, where $\boldsymbol{c} =$
lyche-numerical-linear-algebra_F00327	30	■ 0.998	`\boldsymbol{c}=`	$\boldsymbol{c}=$	linear combination can then be written as a matrix times vector $\boldsymbol{X}\boldsymbol{c}$, where $\boldsymbol{c} =$
lyche-numerical-linear-algebra_F00328	30	■ 1.000	`\left[c_{1}, \ldots, c_{n}\right]^{T}`	$\left[c_{1}, \ldots, c_{n}\right]^{T}$	$[c_1, \ldots, c_n]^T$ is the vector of scalars. Thus
lyche-numerical-linear-algebra_F00329	30	■ 1.000	`\boldsymbol{x}_{j} \in \mathcal{V}, j=1, \ldots, n`	$\boldsymbol{x}_{j} \in \mathcal{V}, j=1, \ldots, n$	ated with an $m \times n$ matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n]$, where $\boldsymbol{x}_j \in \mathcal{V}$, $j = 1, \ldots, n$ are the
lyche-numerical-linear-algebra_F00330	30	■ 1.000	`\mathcal{R}(\boldsymbol{X})`	$\mathcal{R}(\boldsymbol{X})$	1. The subspace $\mathcal{R}(\boldsymbol{X})$ is called the **column space** of $\boldsymbol{X}$. It is the smallest subspace
lyche-numerical-linear-algebra_F00331	30	■ 1.000	`\boldsymbol{X}`	$\boldsymbol{X}$	1. The subspace $\mathcal{R}(\boldsymbol{X})$ is called the **column space** of $\boldsymbol{X}$. It is the smallest subspace
lyche-numerical-linear-algebra_F00332	30	■ 1.000	`\mathcal{R}\left(\boldsymbol{X}^{T}\right)`	$\mathcal{R}\left(\boldsymbol{X}^{T}\right)$	2. $\mathcal{R}(\boldsymbol{X}^T)$ is called the **row space** of $\boldsymbol{X}$. It is generated by the rows of $\boldsymbol{X}$ written as
lyche-numerical-linear-algebra_F00333	30	■ 1.000	`\mathcal{N}(\boldsymbol{X}):=\left\{\boldsymbol{y} \in \mathbb{R}^{n}: \boldsymbol{X} \boldsymbol{y}=\mathbf{0} \right\}`	$\mathcal{N}(\boldsymbol{X}):=\left\{\boldsymbol{y} \in \mathbb{R}^{n}: \boldsymbol{X} \boldsymbol{y}=\mathbf{0}\right\}$	3. The subspace $\mathcal{N}(\boldsymbol{X}) := \{\boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{X}\boldsymbol{y} = \mathbf{0}\}$ is called the **null space** or **kernel**
lyche-numerical-linear-algebra_F00334	30	■ 1.000	`\mathcal{N}(\boldsymbol{X})`	$\mathcal{N}(\boldsymbol{X})$	**space** of $\boldsymbol{X}$. The dimension of $\mathcal{N}(\boldsymbol{X})$ is called the **nullity** of $\boldsymbol{X}$ and denoted
lyche-numerical-linear-algebra_F00335	30	■ 1.000	`\operatorname{null}(\boldsymbol{X})`	$\operatorname{null}(\boldsymbol{X})$	$\operatorname{null}(\boldsymbol{X})$.
lyche-numerical-linear-algebra_F00336	31	■ 0.922	`\boldsymbol{X} \in`	$\boldsymbol{X} \in$	**Theorem 1.5 (Counting Dimensions of Fundamental Subspaces)** *Suppose* $\boldsymbol{X} \in$
lyche-numerical-linear-algebra_F00337	31	■ 1.000	`\mathbb{C}^{m \times n}`	$\mathbb{C}^{m \times n}$	$\mathbb{C}^{m \times n}$. *Then*
lyche-numerical-linear-algebra_F00338	31	■ 1.000	`\operatorname{rank}(\boldsymbol{X})=\operatorname{rank} \left(\boldsymbol{X}^{*}\right)`	$\operatorname{rank}(\boldsymbol{X})=\operatorname{rank}\left(\boldsymbol{X}^{*}\right)$	*1.* $\operatorname{rank}(\boldsymbol{X}) = \operatorname{rank}(\boldsymbol{X}^*)$.
lyche-numerical-linear-algebra_F00339	31	■ 1.000	`\operatorname{rank}(\boldsymbol{X})+\operatorname{null}( \boldsymbol{X})=n`	$\operatorname{rank}(\boldsymbol{X})+\operatorname{null}(\boldsymbol{X})=n$	*2.* $\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}) = n$,
lyche-numerical-linear-algebra_F00340	31	■ 1.000	`\operatorname{rank}(\boldsymbol{X})+\operatorname{null} \left(\boldsymbol{X}^{*}\right)=m`	$\operatorname{rank}(\boldsymbol{X})+\operatorname{null}\left(\boldsymbol{X}^{*}\right)=m$	*3.* $\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}^*) = m$,
lyche-numerical-linear-algebra_F00341	31	■ 1.000	`x_{j}`	x_{j}	x_j, and the components b_i of the right hand side are real or complex numbers. The
lyche-numerical-linear-algebra_F00342	31	■ 1.000	`b_{i}`	b_{i}	x_j, and the components b_i of the right hand side are real or complex numbers. The
lyche-numerical-linear-algebra_F00343	32	■ 0.992	`\boldsymbol{a}_{j}=\left[a_{1 j}, \ldots, \boldsymbol{a}_{m j}\right]^{T} \in \mathbb{C}^{m}`	$\boldsymbol{a}_{j}=\left[a_{1 j}, \ldots, \boldsymbol{a}_{m j}\right]^{T} \in \mathbb{C}^{m}$	where $\boldsymbol{a}_j = \left[a_{1j}, \ldots, \boldsymbol{a}_{mj}\right]^T \in \mathbb{C}^m$ for $j = 1, \ldots, n$ and $\boldsymbol{b} = \left[b_1, \ldots, b_m\right]^T \in \mathbb{C}^m$.
lyche-numerical-linear-algebra_F00344	32	■ 0.992	`\boldsymbol{b}=\left[b_{1}, \ldots, b_{m}\right]^{T} \in \mathbb{C}^{m}`	$\boldsymbol{b}=\left[b_{1}, \ldots, b_{m}\right]^{T} \in \mathbb{C}^{m}$	where $\boldsymbol{a}_j = \left[a_{1j}, \ldots, \boldsymbol{a}_{mj}\right]^T \in \mathbb{C}^m$ for $j = 1, \ldots, n$ and $\boldsymbol{b} = \left[b_1, \ldots, b_m\right]^T \in \mathbb{C}^m$.
lyche-numerical-linear-algebra_F00345	32	■ 1.000	`m<n, m=n`	$m<n, m=n$	**square**, or **overdetermined** if $m < n$, $m = n$, or $m > n$, respectively.
lyche-numerical-linear-algebra_F00346	32	■ 1.000	`m>n`	$m>n$	**square**, or **overdetermined** if $m < n$, $m = n$, or $m > n$, respectively.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00347	32	1.000	`m<n`	$m<n$	**Lemma 1.3 (Underdetermined System)** *Suppose* $A \in \mathbb{R}^{m\times n}$ *with* $m < n$. *Then*
lyche-numerical-linear-algebra_F00348	32	0.570	`\boldsymbol{A x}=\mathbf{0}`	$\boldsymbol{A x}=\mathbf{0}$	*there is a nonzero* $\boldsymbol{x} \in \mathbb{R}^n$ *such that* $\boldsymbol{Ax} = \mathbf{0}$.
lyche-numerical-linear-algebra_F00349	32	1.000	`\boldsymbol{A} \boldsymbol{x}=\mathbf{0}`	$\boldsymbol{A} \boldsymbol{x}=\mathbf{0}$	nonzero $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{Ax} = \mathbf{0}$.
lyche-numerical-linear-algebra_F00350	32	1.000	`\mathbb{R}^{n \times n}`	$\mathbb{R}^{n \times n}$	$\mathbb{R}^{n\times n}$ is said to be **nonsingular** if the only real solution of the homogeneous system
lyche-numerical-linear-algebra_F00351	32	0.658	`\boldsymbol{b} \in \mathbb{R}^{n}`	$\boldsymbol{b} \in \mathbb{R}^{n}$	*The linear system* $\boldsymbol{Ax} = \boldsymbol{b}$ *has a unique solution* $\boldsymbol{x} \in \mathbb{R}^n$ *for any* $\boldsymbol{b} \in \mathbb{R}^n$ *if and only*
lyche-numerical-linear-algebra_F00352	32	1.000	`\boldsymbol{B}=[\boldsymbol{A} \boldsymbol{b}] \in \mathbb{R}^{n \times(n+1)}`	$\boldsymbol{B}=[\boldsymbol{A} \boldsymbol{b}] \in \mathbb{R}^{n \times(n+1)}$	***Proof*** Suppose $\boldsymbol{A}$ is nonsingular. We define $\boldsymbol{B} = [\boldsymbol{A}\ \boldsymbol{b}] \in \mathbb{R}^{n\times(n+1)}$ by adding a
lyche-numerical-linear-algebra_F00353	32	0.997	`\boldsymbol{z} \in \mathbb{R}^{n+1}`	$\boldsymbol{z} \in \mathbb{R}^{n+1}$	column to $\boldsymbol{A}$. By Lemma 1.3 there is a nonzero $\boldsymbol{z} \in \mathbb{R}^{n+1}$ such that $\boldsymbol{Bz} = \mathbf{0}$. If we
lyche-numerical-linear-algebra_F00354	32	0.997	`\boldsymbol{B} \boldsymbol{z}=\mathbf{0}`	$\boldsymbol{B} \boldsymbol{z}=\mathbf{0}$	column to $\boldsymbol{A}$. By Lemma 1.3 there is a nonzero $\boldsymbol{z} \in \mathbb{R}^{n+1}$ such that $\boldsymbol{Bz} = \mathbf{0}$. If we
lyche-numerical-linear-algebra_F00355	32	0.896	`\boldsymbol{z}=\left[\begin{array}{c}\tilde{\boldsymbol{z}} \\ z_{n+1}\end{array}\right]`	$\boldsymbol{z}=\left[\begin{array}{c}\tilde{\boldsymbol{z}} \\ z_{n+1}\end{array}\right]$	write $\boldsymbol{z} = \left[\begin{array}{c}\tilde{\boldsymbol{z}} \\ z_{n+1}\end{array}\right]$ where $\tilde{\boldsymbol{z}} = [z_1, \ldots, z_n]^T \in \mathbb{R}^n$ and $z_{n+1} \in \mathbb{R}$, then
lyche-numerical-linear-algebra_F00356	32	0.896	`\tilde{\boldsymbol{z}}=\left[z_{1}, \ldots, z_{n} \right]^{T} \in \mathbb{R}^{n}`	$\tilde{\boldsymbol{z}}=\left[z_{1}, \ldots, z_{n} \right]^{T} \in \mathbb{R}^{n}$	write $\boldsymbol{z} = \left[\begin{array}{c}\tilde{\boldsymbol{z}} \\ z_{n+1}\end{array}\right]$ where $\tilde{\boldsymbol{z}} = [z_1, \ldots, z_n]^T \in \mathbb{R}^n$ and $z_{n+1} \in \mathbb{R}$, then
lyche-numerical-linear-algebra_F00357	32	0.896	`z_{n+1} \in \mathbb{R}`	$z_{n+1} \in \mathbb{R}$	write $\boldsymbol{z} = \left[\begin{array}{c}\tilde{\boldsymbol{z}} \\ z_{n+1}\end{array}\right]$ where $\tilde{\boldsymbol{z}} = [z_1, \ldots, z_n]^T \in \mathbb{R}^n$ and $z_{n+1} \in \mathbb{R}$, then
lyche-numerical-linear-algebra_F00358	33	0.737	`z_{n+1}=0`	$z_{n+1}=0$	We cannot have $z_{n+1} = 0$ for then $\boldsymbol{A}\tilde{\boldsymbol{z}} = \mathbf{0}$ for a nonzero $\tilde{\boldsymbol{z}}$, contradicting the
lyche-numerical-linear-algebra_F00359	33	0.737	`\boldsymbol{A} \tilde{\boldsymbol{z}}=\mathbf{0}`	$\boldsymbol{A} \tilde{\boldsymbol{z}}=\mathbf{0}$	We cannot have $z_{n+1} = 0$ for then $\boldsymbol{A}\tilde{\boldsymbol{z}} = \mathbf{0}$ for a nonzero $\tilde{\boldsymbol{z}}$, contradicting the
lyche-numerical-linear-algebra_F00360	33	0.737	`\tilde{\boldsymbol{z}}`	$\tilde{\boldsymbol{z}}$	We cannot have $z_{n+1} = 0$ for then $\boldsymbol{A}\tilde{\boldsymbol{z}} = \mathbf{0}$ for a nonzero $\tilde{\boldsymbol{z}}$, contradicting the
lyche-numerical-linear-algebra_F00361	33	0.997	`\boldsymbol{x}:=-\tilde{\boldsymbol{z}} / z_{n+1}`	$\boldsymbol{x}:=-\tilde{\boldsymbol{z}} / z_{n+1}$	nonsingularity of $\boldsymbol{A}$. Define $\boldsymbol{x} := -\tilde{\boldsymbol{z}}/z_{n+1}$. Then
lyche-numerical-linear-algebra_F00362	33	1.000	`\boldsymbol{A} \boldsymbol{y}=\boldsymbol{b}`	$\boldsymbol{A} \boldsymbol{y}=\boldsymbol{b}$	Suppose $\boldsymbol{Ax} = \boldsymbol{b}$ and $\boldsymbol{Ay} = \boldsymbol{b}$ for $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Then $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \mathbf{0}$ and since $\boldsymbol{A}$ is
lyche-numerical-linear-algebra_F00363	33	1.000	`\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n}`	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n}$	Suppose $\boldsymbol{Ax} = \boldsymbol{b}$ and $\boldsymbol{Ay} = \boldsymbol{b}$ for $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Then $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \mathbf{0}$ and since $\boldsymbol{A}$ is
lyche-numerical-linear-algebra_F00364	33	1.000	`\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\mathbf{0}`	$\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\mathbf{0}$	Suppose $\boldsymbol{Ax} = \boldsymbol{b}$ and $\boldsymbol{Ay} = \boldsymbol{b}$ for $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Then $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \mathbf{0}$ and since $\boldsymbol{A}$ is
lyche-numerical-linear-algebra_F00365	33	1.000	`\boldsymbol{x}-\boldsymbol{y}=\mathbf{0}`	$\boldsymbol{x}-\boldsymbol{y}=\mathbf{0}$	nonsingular we conclude that $\boldsymbol{x} - \boldsymbol{y} = \mathbf{0}$ or $\boldsymbol{x} = \boldsymbol{y}$. Thus the solution is unique.
lyche-numerical-linear-algebra_F00366	33	1.000	`\boldsymbol{x}=\boldsymbol{y}`	$\boldsymbol{x}=\boldsymbol{y}$	nonsingular we conclude that $\boldsymbol{x} - \boldsymbol{y} = \mathbf{0}$ or $\boldsymbol{x} = \boldsymbol{y}$. Thus the solution is unique.
lyche-numerical-linear-algebra_F00367	33	1.000	`m<`	$m<$	**Lemma 1.4 (Complex Underdetermined System)** *Suppose* $A \in \mathbb{C}^{m\times n}$ *with* $m <$
lyche-numerical-linear-algebra_F00368	33	1.000	`\boldsymbol{b} \in \mathbb{C}^{n}`	$\boldsymbol{b} \in \mathbb{C}^{n}$	$\boldsymbol{b} \in \mathbb{C}^n$ *if and only if the matrix* $\boldsymbol{A}$ *is nonsingular.*
lyche-numerical-linear-algebra_F00369	33	1.000	`\boldsymbol{B} \in \mathbb{C}^{n \times n}`	$\boldsymbol{B} \in \mathbb{C}^{n \times n}$	Suppose $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ is a square matrix. A matrix $\boldsymbol{B} \in \mathbb{C}^{n\times n}$ is called a **right inverse**
lyche-numerical-linear-algebra_F00370	33	1.000	`\boldsymbol{A} \boldsymbol{B}=\boldsymbol{I}`	$\boldsymbol{A} \boldsymbol{B}=\boldsymbol{I}$	of $\boldsymbol{A}$ if $\boldsymbol{AB} = \boldsymbol{I}$. A matrix $\boldsymbol{C} \in \mathbb{C}^{n\times n}$ is said to be a **left inverse** of $\boldsymbol{A}$ if $\boldsymbol{CA} = \boldsymbol{I}$.
lyche-numerical-linear-algebra_F00371	33	1.000	`\boldsymbol{C} \in \mathbb{C}^{n \times n}`	$\boldsymbol{C} \in \mathbb{C}^{n \times n}$	of $\boldsymbol{A}$ if $\boldsymbol{AB} = \boldsymbol{I}$. A matrix $\boldsymbol{C} \in \mathbb{C}^{n\times n}$ is said to be a **left inverse** of $\boldsymbol{A}$ if $\boldsymbol{CA} = \boldsymbol{I}$.
lyche-numerical-linear-algebra_F00372	33	1.000	`\boldsymbol{C} \boldsymbol{A}=\boldsymbol{I}`	$\boldsymbol{C} \boldsymbol{A}=\boldsymbol{I}$	of $\boldsymbol{A}$ if $\boldsymbol{AB} = \boldsymbol{I}$. A matrix $\boldsymbol{C} \in \mathbb{C}^{n\times n}$ is said to be a **left inverse** of $\boldsymbol{A}$ if $\boldsymbol{CA} = \boldsymbol{I}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F00373	33	1.000	`\boldsymbol{C}`	$\boldsymbol{C}$	inverse $\boldsymbol{B}$ and a left inverse $\boldsymbol{C}$ then
lyche-numerical-linear-algebra_ F00374	33	1.000	`\boldsymbol{A}^{-1}`	$\boldsymbol{A}^{-1}$	and this common inverse is called the **inverse** of $\boldsymbol{A}$ and denoted by $\boldsymbol{A}^{-1}$. Thus the
lyche-numerical-linear-algebra_ F00375	33	1.000	`\boldsymbol{A}^{-1} \boldsymbol{A}=\boldsymbol{A} \boldsymbol{A}^{-1}=\boldsymbol{I}`	$\boldsymbol{A}^{-1} \boldsymbol{A}=\boldsymbol{A} \boldsymbol{A}^{-1}=\boldsymbol{I}$	inverse satisfies $\boldsymbol{A}^{-1}\boldsymbol{A}=\boldsymbol{A}\boldsymbol{A}^{-1}=\boldsymbol{I}$.
lyche-numerical-linear-algebra_ F00376	33	0.998	`\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times n}`	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$	**Theorem 1.8 (Product of Nonsingular Matrices)** *If* $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n\times n}$ *with* $\boldsymbol{A}\boldsymbol{B} =$
lyche-numerical-linear-algebra_ F00377	33	0.998	`\boldsymbol{A} \boldsymbol{B}=`	$\boldsymbol{A} \boldsymbol{B}=$	**Theorem 1.8 (Product of Nonsingular Matrices)** *If* $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n\times n}$ *with* $\boldsymbol{A}\boldsymbol{B} =$
lyche-numerical-linear-algebra_ F00378	33	1.000	`\boldsymbol{B} \boldsymbol{A}=\boldsymbol{I}`	$\boldsymbol{B} \boldsymbol{A}=\boldsymbol{I}$	*if either* $\boldsymbol{A}\boldsymbol{B}=\boldsymbol{I}$ *or* $\boldsymbol{B}\boldsymbol{A}=\boldsymbol{I}$ *then* $\boldsymbol{A}$ *is nonsingular and* $\boldsymbol{A}^{-1}=\boldsymbol{B}$.
lyche-numerical-linear-algebra_ F00379	33	1.000	`\boldsymbol{A}^{-1}=\boldsymbol{B}`	$\boldsymbol{A}^{-1}=\boldsymbol{B}$	*if either* $\boldsymbol{A}\boldsymbol{B}=\boldsymbol{I}$ *or* $\boldsymbol{B}\boldsymbol{A}=\boldsymbol{I}$ *then* $\boldsymbol{A}$ *is nonsingular and* $\boldsymbol{A}^{-1}=\boldsymbol{B}$.
lyche-numerical-linear-algebra_ F00380	33	1.000	`\boldsymbol{C} \boldsymbol{x}=\mathbf{0}`	$\boldsymbol{C} \boldsymbol{x}=\mathbf{0}$	***Proof*** Suppose both $\boldsymbol{A}$ and $\boldsymbol{B}$ are nonsingular and let $\boldsymbol{C}\boldsymbol{x}=\mathbf{0}$. Then $\boldsymbol{A}\boldsymbol{B}\boldsymbol{x}=\mathbf{0}$ and
lyche-numerical-linear-algebra_ F00381	33	1.000	`\boldsymbol{A} \boldsymbol{B} \boldsymbol{x}=\mathbf{0}`	$\boldsymbol{A} \boldsymbol{B} \boldsymbol{x}=\mathbf{0}$	***Proof*** Suppose both $\boldsymbol{A}$ and $\boldsymbol{B}$ are nonsingular and let $\boldsymbol{C}\boldsymbol{x}=\mathbf{0}$. Then $\boldsymbol{A}\boldsymbol{B}\boldsymbol{x}=\mathbf{0}$ and
lyche-numerical-linear-algebra_ F00382	33	1.000	`\boldsymbol{B} \boldsymbol{x}=\mathbf{0}`	$\boldsymbol{B} \boldsymbol{x}=\mathbf{0}$	since $\boldsymbol{A}$ is nonsingular we see that $\boldsymbol{B}\boldsymbol{x}=\mathbf{0}$. Since $\boldsymbol{B}$ is nonsingular we have $\boldsymbol{x}=\mathbf{0}$.
lyche-numerical-linear-algebra_ F00383	34	1.000	`\boldsymbol{C} \boldsymbol{x}=(\boldsymbol{A} \boldsymbol{B}) \boldsymbol{x}=\boldsymbol{A}( \boldsymbol{B} \boldsymbol{x})=\boldsymbol{A} \mathbf{0}= \mathbf{0}`	$\boldsymbol{C} \boldsymbol{x}=(\boldsymbol{A} \boldsymbol{B}) \boldsymbol{x}=\boldsymbol{A}(\boldsymbol{B} \boldsymbol{x})=\boldsymbol{A} \mathbf{0}=\mathbf{0}$	vector so that $\boldsymbol{B}\boldsymbol{x}=\mathbf{0}$. But then $\boldsymbol{C}\boldsymbol{x}=(\boldsymbol{A}\boldsymbol{B})\boldsymbol{x}=\boldsymbol{A}(\boldsymbol{B}\boldsymbol{x})=\boldsymbol{A}\mathbf{0}=\mathbf{0}$ so $\boldsymbol{C}$ is
lyche-numerical-linear-algebra_ F00384	34	1.000	`\tilde{\boldsymbol{x}}`	$\tilde{\boldsymbol{x}}$	singular. Finally suppose $\boldsymbol{B}$ is nonsingular, but $\boldsymbol{A}$ is singular. Let $\tilde{\boldsymbol{x}}$ be a nonzero
lyche-numerical-linear-algebra_ F00385	34	1.000	`\boldsymbol{A} \tilde{\boldsymbol{x}}=\mathbf{0}`	$\boldsymbol{A} \tilde{\boldsymbol{x}}=\mathbf{0}$	vector such that $\boldsymbol{A}\tilde{\boldsymbol{x}}=\mathbf{0}$. By Theorem 1.7 there is a vector $\boldsymbol{x}$ such that $\boldsymbol{B}\boldsymbol{x}=\tilde{\boldsymbol{x}}$ and
lyche-numerical-linear-algebra_ F00386	34	1.000	`\boldsymbol{B} \boldsymbol{x}=\tilde{\boldsymbol{x}}`	$\boldsymbol{B} \boldsymbol{x}=\tilde{\boldsymbol{x}}$	vector such that $\boldsymbol{A}\tilde{\boldsymbol{x}}=\mathbf{0}$. By Theorem 1.7 there is a vector $\boldsymbol{x}$ such that $\boldsymbol{B}\boldsymbol{x}=\tilde{\boldsymbol{x}}$ and
lyche-numerical-linear-algebra_ F00387	34	1.000	`\boldsymbol{C} \boldsymbol{x}=(\boldsymbol{A} \boldsymbol{B}) \boldsymbol{x}=\boldsymbol{A}( \boldsymbol{B} \boldsymbol{x})=\boldsymbol{A} \tilde{ \boldsymbol{x}}=\mathbf{0}`	$\boldsymbol{C} \boldsymbol{x}=(\boldsymbol{A} \boldsymbol{B}) \boldsymbol{x}=\boldsymbol{A}(\boldsymbol{B} \boldsymbol{x})=\boldsymbol{A} \tilde{\boldsymbol{x}}=\mathbf{0}$	$\boldsymbol{x}$ is nonzero since $\tilde{\boldsymbol{x}}$ is nonzero. But then $\boldsymbol{C}\boldsymbol{x}=(\boldsymbol{A}\boldsymbol{B})\boldsymbol{x}=\boldsymbol{A}(\boldsymbol{B}\boldsymbol{x})=\boldsymbol{A}\tilde{\boldsymbol{x}}=\mathbf{0}$ for a
lyche-numerical-linear-algebra_ F00388	34	1.000	`\boldsymbol{A} \boldsymbol{b}_{i}=\boldsymbol{e}_{i}`	$\boldsymbol{A} \boldsymbol{b}_{i}=\boldsymbol{e}_{i}$	systems $\boldsymbol{A}\boldsymbol{b}_i=\boldsymbol{e}_i$ has a unique solution $\boldsymbol{b}_i$ for $i=1,\ldots,n$. Let $\boldsymbol{B}=[\boldsymbol{b}_1,\ldots,\boldsymbol{b}_n]$.
lyche-numerical-linear-algebra_ F00389	34	1.000	`\boldsymbol{b}_{i}`	$\boldsymbol{b}_{i}$	systems $\boldsymbol{A}\boldsymbol{b}_i=\boldsymbol{e}_i$ has a unique solution $\boldsymbol{b}_i$ for $i=1,\ldots,n$. Let $\boldsymbol{B}=[\boldsymbol{b}_1,\ldots,\boldsymbol{b}_n]$.
lyche-numerical-linear-algebra_ F00390	34	1.000	`\boldsymbol{B}=\left[\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{n}\right]`	$\boldsymbol{B}=\left[\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{n}\right]$	systems $\boldsymbol{A}\boldsymbol{b}_i=\boldsymbol{e}_i$ has a unique solution $\boldsymbol{b}_i$ for $i=1,\ldots,n$. Let $\boldsymbol{B}=[\boldsymbol{b}_1,\ldots,\boldsymbol{b}_n]$.
lyche-numerical-linear-algebra_ F00391	34	0.968	`\boldsymbol{A} \boldsymbol{B}=\left[\boldsymbol{A} \boldsymbol{b}_{1}, \ldots, \boldsymbol{A} \boldsymbol{b}_{n}\right]=\left[\boldsymbol{e}_{1}, \ldots, \boldsymbol{e}_{n}\right]=\boldsymbol{I}`	$\boldsymbol{A} \boldsymbol{B}=\left[\boldsymbol{A} \boldsymbol{b}_{1}, \ldots, \boldsymbol{A} \boldsymbol{b}_{n}\right]=\left[\boldsymbol{e}_{1}, \ldots, \boldsymbol{e}_{n}\right]=\boldsymbol{I}$	Then $\boldsymbol{A}\boldsymbol{B}=[\boldsymbol{A}\boldsymbol{b}_1,\ldots,\boldsymbol{A}\boldsymbol{b}_n]=[\boldsymbol{e}_1,\ldots,\boldsymbol{e}_n]=\boldsymbol{I}$ so that $\boldsymbol{A}$ has a right inverse $\boldsymbol{B}$.
lyche-numerical-linear-algebra_ F00392	34	1.000	`1.8 \boldsymbol{B}`	$1.8 \boldsymbol{B}$	By Theorem 1.8 $\boldsymbol{B}$ is nonsingular since $\boldsymbol{I}$ is nonsingular and $\boldsymbol{A}\boldsymbol{B}=\boldsymbol{I}$. Since $\boldsymbol{B}$ is
lyche-numerical-linear-algebra_ F00393	34	1.000	`\boldsymbol{B} \boldsymbol{C}=\boldsymbol{I}`	$\boldsymbol{B} \boldsymbol{C}=\boldsymbol{I}$	inverse $\boldsymbol{C}$, i.e. $\boldsymbol{B}\boldsymbol{C}=\boldsymbol{I}$. But then $\boldsymbol{A}\boldsymbol{B}=\boldsymbol{B}\boldsymbol{C}=\boldsymbol{I}$ so $\boldsymbol{B}$ has both a right inverse and
lyche-numerical-linear-algebra_ F00394	34	1.000	`\boldsymbol{A} \boldsymbol{B}=\boldsymbol{B} \boldsymbol{C}=\boldsymbol{I}`	$\boldsymbol{A} \boldsymbol{B}=\boldsymbol{B} \boldsymbol{C}=\boldsymbol{I}$	inverse $\boldsymbol{C}$, i.e. $\boldsymbol{B}\boldsymbol{C}=\boldsymbol{I}$. But then $\boldsymbol{A}\boldsymbol{B}=\boldsymbol{B}\boldsymbol{C}=\boldsymbol{I}$ so $\boldsymbol{B}$ has both a right inverse and
lyche-numerical-linear-algebra_ F00395	34	1.000	`\boldsymbol{A}=\boldsymbol{C}`	$\boldsymbol{A}=\boldsymbol{C}$	a left inverse which must be equal so $\boldsymbol{A}=\boldsymbol{C}$. Since $\boldsymbol{B}\boldsymbol{C}=\boldsymbol{I}$ we have $\boldsymbol{B}\boldsymbol{A}=\boldsymbol{I}$, so
lyche-numerical-linear-algebra_ F00396	34	0.995	`\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n \times n}`	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n \times n}$	**Corollary 1.2 (Basic Properties of the Inverse Matrix)** *Suppose* $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n\times n}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00397	34	■ 1.000	`\left(\boldsymbol{A}^{-1}\right)^{-1}=\boldsymbol{A}`	$\left(\boldsymbol{A}^{-1}\right)^{-1}=\boldsymbol{A}$	1. $\boldsymbol{A}^{-1}$ *is nonsingular and* $(\boldsymbol{A}^{-1})^{-1}=\boldsymbol{A}$.
lyche-numerical-linear-algebra_F00398	34	■ 1.000	`\boldsymbol{C}=\boldsymbol{A} \boldsymbol{B}`	$\boldsymbol{C}=\boldsymbol{A} \boldsymbol{B}$	2. $\boldsymbol{C}=\boldsymbol{AB}$ *is nonsingular and* $\boldsymbol{C}^{-1}=\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.
lyche-numerical-linear-algebra_F00399	34	■ 1.000	`\boldsymbol{C}^{-1}=\boldsymbol{B}^{-1} \boldsymbol{A}^{-1}`	$\boldsymbol{C}^{-1}=\boldsymbol{B}^{-1} \boldsymbol{A}^{-1}$	2. $\boldsymbol{C}=\boldsymbol{AB}$ *is nonsingular and* $\boldsymbol{C}^{-1}=\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.
lyche-numerical-linear-algebra_F00400	34	■ 1.000	`\left(\boldsymbol{A}^{T}\right)^{-1}=\left(\boldsymbol{A}^{-1}\right)^{T}=: \boldsymbol{A}^{-T}`	$\left(\boldsymbol{A}^{T}\right)^{-1}=\left(\boldsymbol{A}^{-1}\right)^{T}=: \boldsymbol{A}^{-T}$	3. $\boldsymbol{A}^{T}$ *is nonsingular and* $(\boldsymbol{A}^{T})^{-1}=(\boldsymbol{A}^{-1})^{T}=:\boldsymbol{A}^{-T}$.
lyche-numerical-linear-algebra_F00401	34	■ 1.000	`\left(\boldsymbol{A}^{*}\right)^{-1}=\left(\boldsymbol{A}^{-1}\right)^{*}=: \boldsymbol{A}^{-*}`	$\left(\boldsymbol{A}^{*}\right)^{-1}=\left(\boldsymbol{A}^{-1}\right)^{*}=: \boldsymbol{A}^{-*}$	4. $\boldsymbol{A}^{*}$ *is nonsingular and* $(\boldsymbol{A}^{*})^{-1}=(\boldsymbol{A}^{-1})^{*}=:\boldsymbol{A}^{-*}$.
lyche-numerical-linear-algebra_F00402	34	■ 1.000	`c \boldsymbol{A}`	$c \boldsymbol{A}$	5. $c\boldsymbol{A}$ *is nonsingular and* $(c\boldsymbol{A})^{-1}=\frac{1}{c}\boldsymbol{A}^{-1}$.
lyche-numerical-linear-algebra_F00403	34	■ 1.000	`(c \boldsymbol{A})^{-1}=\frac{1}{c} \boldsymbol{A}^{-1}`	$(c \boldsymbol{A})^{-1}=\frac{1}{c} \boldsymbol{A}^{-1}$	5. $c\boldsymbol{A}$ *is nonsingular and* $(c\boldsymbol{A})^{-1}=\frac{1}{c}\boldsymbol{A}^{-1}$.
lyche-numerical-linear-algebra_F00404	34	■ 1.000	`\boldsymbol{A}^{-1} \boldsymbol{A}=\boldsymbol{I}`	$\boldsymbol{A}^{-1} \boldsymbol{A}=\boldsymbol{I}$	1. Since $\boldsymbol{A}^{-1}\boldsymbol{A}=\boldsymbol{I}$ the matrix $\boldsymbol{A}$ is a right inverse of $\boldsymbol{A}^{-1}$. Thus $\boldsymbol{A}^{-1}$ is nonsingular
lyche-numerical-linear-algebra_F00405	34	■ 1.000	`\left(\boldsymbol{B}^{-1} \boldsymbol{A}^{-1}\right)(\boldsymbol{A} \boldsymbol{B})=\boldsymbol{B}^{-1}\left(\boldsymbol{A}^{-1} \boldsymbol{A}\right) \boldsymbol{B}=\boldsymbol{B}^{-1} \boldsymbol{B}=\boldsymbol{I}`	$\left(\boldsymbol{B}^{-1} \boldsymbol{A}^{-1}\right)(\boldsymbol{A} \boldsymbol{B})=\boldsymbol{B}^{-1}\left(\boldsymbol{A}^{-1} \boldsymbol{A}\right) \boldsymbol{B}=\boldsymbol{B}^{-1} \boldsymbol{B}=\boldsymbol{I}$	2. We note that $(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB})=\boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B}=\boldsymbol{B}^{-1}\boldsymbol{B}=\boldsymbol{I}$. Thus $\boldsymbol{AB}$ is
lyche-numerical-linear-algebra_F00406	34	■ 1.000	`\boldsymbol{A} \boldsymbol{B}`	$\boldsymbol{A} \boldsymbol{B}$	2. We note that $(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB})=\boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B}=\boldsymbol{B}^{-1}\boldsymbol{B}=\boldsymbol{I}$. Thus $\boldsymbol{AB}$ is
lyche-numerical-linear-algebra_F00407	34	■ 1.000	`\boldsymbol{I}=\boldsymbol{I}^{T}=\left(\boldsymbol{A}^{-1} \boldsymbol{A}\right)^{T}=\boldsymbol{A}^{T}\left(\boldsymbol{A}^{-1}\right)^{T}`	$\boldsymbol{I}=\boldsymbol{I}^{T}=\left(\boldsymbol{A}^{-1} \boldsymbol{A}\right)^{T}=\boldsymbol{A}^{T}\left(\boldsymbol{A}^{-1}\right)^{T}$	3. Now $\boldsymbol{I}=\boldsymbol{I}^{T}=(\boldsymbol{A}^{-1}\boldsymbol{A})^{T}=\boldsymbol{A}^{T}(\boldsymbol{A}^{-1})^{T}$ showing that $(\boldsymbol{A}^{-1})^{T}$ is a right inverse
lyche-numerical-linear-algebra_F00408	34	■ 1.000	`\left(\boldsymbol{A}^{-1}\right)^{T}`	$\left(\boldsymbol{A}^{-1}\right)^{T}$	3. Now $\boldsymbol{I}=\boldsymbol{I}^{T}=(\boldsymbol{A}^{-1}\boldsymbol{A})^{T}=\boldsymbol{A}^{T}(\boldsymbol{A}^{-1})^{T}$ showing that $(\boldsymbol{A}^{-1})^{T}$ is a right inverse
lyche-numerical-linear-algebra_F00409	34	■ 1.000	`\frac{1}{c} \boldsymbol{A}^{-1}`	$\frac{1}{c} \boldsymbol{A}^{-1}$	4. The matrix $\frac{1}{c}\boldsymbol{A}^{-1}$ is a one sided inverse of $c\boldsymbol{A}$.
lyche-numerical-linear-algebra_F00410	35	■ 1.000	`n!`	$n!$	This sum ranges of all $n!$ permutations of $\{1,2,\ldots,n\}$. Moreover, $\operatorname{sign}(\sigma)$ equals
lyche-numerical-linear-algebra_F00411	35	■ 1.000	`\{1,2, \ldots, n\}`	$\{1,2, \ldots, n\}$	This sum ranges of all $n!$ permutations of $\{1,2,\ldots,n\}$. Moreover, $\operatorname{sign}(\sigma)$ equals
lyche-numerical-linear-algebra_F00412	35	■ 1.000	`\operatorname{sign}(\sigma)`	$\operatorname{sign}(\sigma)$	This sum ranges of all $n!$ permutations of $\{1,2,\ldots,n\}$. Moreover, $\operatorname{sign}(\sigma)$ equals
lyche-numerical-linear-algebra_F00413	35	■ 1.000	`\sigma`	σ	the number of times a bigger integer precedes a smaller one in σ. We also denote
lyche-numerical-linear-algebra_F00414	35	■ 1.000	`\epsilon`	ϵ	The first term on the right corresponds to the identity permutation ϵ given by $\epsilon(i)=$
lyche-numerical-linear-algebra_F00415	35	■ 1.000	`\epsilon(i)=`	$\epsilon(i)=$	The first term on the right corresponds to the identity permutation ϵ given by $\epsilon(i)=$
lyche-numerical-linear-algebra_F00416	35	■ 1.000	`i, i=1,2`	$i, i=1,2$	i, $i=1,2$. The second term comes from the permutation $\sigma=\{2,1\}$. For $n=3$
lyche-numerical-linear-algebra_F00417	35	■ 1.000	`\sigma=\{2,1\}`	$\sigma=\{2,1\}$	i, $i=1,2$. The second term comes from the permutation $\sigma=\{2,1\}$. For $n=3$
lyche-numerical-linear-algebra_F00418	35	■ 1.000	`n=3`	$n=3$	i, $i=1,2$. The second term comes from the permutation $\sigma=\{2,1\}$. For $n=3$
lyche-numerical-linear-algebra_F00419	35	■ 0.719	`\{1,2,3\}`	$\{1,2,3\}$	there are six permutations of $\{1,2,3\}$. Then
lyche-numerical-linear-algebra_F00420	35	■ 0.999	`\operatorname{sign}(\{1,2,3\})=\operatorname{sign}(\{2,3,1\})=\operatorname{sign}(\{3,1,2\})=1`	$\operatorname{sign}(\{1,2,3\})=\operatorname{sign}(\{2,3,1\})=\operatorname{sign}(\{3,1,2\})=1$	This follows since $\operatorname{sign}(\{1,2,3\})=\operatorname{sign}(\{2,3,1\})=\operatorname{sign}(\{3,1,2\})=1$, and
lyche-numerical-linear-algebra_F00421	35	■ 0.749	`\operatorname{sign}(\{2,1,3\})=\operatorname{sign}(\{3,2,1\})=\operatorname{sign}(\{1,3,2\})=-1`	$\operatorname{sign}(\{2,1,3\})=\operatorname{sign}(\{3,2,1\})=\operatorname{sign}(\{1,3,2\})=-1$	$\operatorname{sign}(\{2,1,3\})=\operatorname{sign}(\{3,2,1\})=\operatorname{sign}(\{1,3,2\})=-1$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00422	35	1.000	`\operatorname{det}(\boldsymbol{A})=`	$\operatorname{det}(\boldsymbol{A})=$	to reduce it to a simpler form. For example, if $\boldsymbol{A}$ is triangular then $\det(\boldsymbol{A}) =$
lyche-numerical-linear-algebra_F00423	36	1.000	`a_{11} a_{22} \cdots a_{n n}`	$a_{11} a_{22} \cdots a_{n n}$	$a_{11}a_{22}\cdots a_{nn}$, the product of the diagonal elements. In particular, for the identity
lyche-numerical-linear-algebra_F00424	36	1.000	`\operatorname{det}(\boldsymbol{I})=1`	$\operatorname{det}(\boldsymbol{I})=1$	matrix $\det(\boldsymbol{I}) = 1$. The elementary operations using either rows or columns are
lyche-numerical-linear-algebra_F00425	36	1.000	`\operatorname{det}(\boldsymbol{B})=-\operatorname{det}( \boldsymbol{A})`	$\operatorname{det}(\boldsymbol{B})=-\operatorname{det}(\boldsymbol{A})$	1. Interchanging two rows(columns): $\det(\boldsymbol{B}) = -\det(\boldsymbol{A})$,
lyche-numerical-linear-algebra_F00426	36	1.000	`\alpha, \operatorname{det}(\boldsymbol{B})=\alpha \operatorname{det}(\boldsymbol{A})`	$\alpha, \operatorname{det}(\boldsymbol{B})=\alpha \operatorname{det}(\boldsymbol{A})$	2. Multiply a row(column) by a scalar: α, $\det(\boldsymbol{B}) = \alpha\det(\boldsymbol{A})$,
lyche-numerical-linear-algebra_F00427	36	1.000	`\boldsymbol{A}_{i, j}`	$\boldsymbol{A}_{i, j}$	Here $\boldsymbol{A}_{i,j}$ denotes the submatrix of $\boldsymbol{A}$ obtained by deleting the ith row and jth
lyche-numerical-linear-algebra_F00428	36	1.000	`1 \leq i, j \leq n`	$1 \leq i, j \leq n$	column of $\boldsymbol{A}$. For $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ and $1 \leq i, j \leq n$ the determinant $\det(\boldsymbol{A}_{ij})$ is called the
lyche-numerical-linear-algebra_F00429	36	1.000	`\operatorname{det}\left(\boldsymbol{A}_{i j}\right)`	$\operatorname{det}\left(\boldsymbol{A}_{i j}\right)$	column of $\boldsymbol{A}$. For $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ and $1 \leq i, j \leq n$ the determinant $\det(\boldsymbol{A}_{ij})$ is called the
lyche-numerical-linear-algebra_F00430	36	0.969	`\left(x_{1}, y_{1}\right)`	$\left(x_{1}, y_{1}\right)$	line through two points (x_1, y_1) and (x_2, y_2) in the plane can be written as the
lyche-numerical-linear-algebra_F00431	36	0.969	`\left(x_{2}, y_{2}\right)`	$\left(x_{2}, y_{2}\right)$	line through two points (x_1, y_1) and (x_2, y_2) in the plane can be written as the
lyche-numerical-linear-algebra_F00432	36	1.000	`\operatorname{det}(\boldsymbol{A})=0`	$\operatorname{det}(\boldsymbol{A})=0$	Rearranging the equation $\det(\boldsymbol{A}) = 0$ we obtain
lyche-numerical-linear-algebra_F00433	37	1.000	`\boldsymbol{A}, \boldsymbol{B}`	$\boldsymbol{A}, \boldsymbol{B}$	$\boldsymbol{A}, \boldsymbol{B}$ are square matrices of order n with real or complex elements, then
lyche-numerical-linear-algebra_F00434	37	0.984	`\operatorname{det}(\boldsymbol{A B})=\operatorname{det}( \boldsymbol{A}) \operatorname{det}(\boldsymbol{B})`	$\operatorname{det}(\boldsymbol{A B})=\operatorname{det}(\boldsymbol{A}) \operatorname{det}(\boldsymbol{B})$	1. $\det(\boldsymbol{AB}) = \det(\boldsymbol{A})\det(\boldsymbol{B})$.
lyche-numerical-linear-algebra_F00435	37	1.000	`\operatorname{det}\left(\boldsymbol{A}^{T}\right)= \operatorname{det}(\boldsymbol{A})`	$\operatorname{det}\left(\boldsymbol{A}^{T}\right)=\operatorname{det}(\boldsymbol{A})$	2. $\det(\boldsymbol{A}^T) = \det(\boldsymbol{A})$, and $\det(\boldsymbol{A}^*) = \overline{\det(\boldsymbol{A})}$, (complex conjugate).
lyche-numerical-linear-algebra_F00436	37	1.000	`\operatorname{det}\left(\boldsymbol{A}^{*}\right)= \overline{\operatorname{det}(\boldsymbol{A})}`	$\operatorname{det}\left(\boldsymbol{A}^{*}\right)=\overline{\operatorname{det}(\boldsymbol{A})}$	2. $\det(\boldsymbol{A}^T) = \det(\boldsymbol{A})$, and $\det(\boldsymbol{A}^*) = \overline{\det(\boldsymbol{A})}$, (complex conjugate).
lyche-numerical-linear-algebra_F00437	37	1.000	`\operatorname{det}(a \boldsymbol{A})=a^{n} \operatorname{det}(\boldsymbol{A})`	$\operatorname{det}(a \boldsymbol{A})=a^{n} \operatorname{det}(\boldsymbol{A})$	3. $\det(a\boldsymbol{A}) = a^n\det(\boldsymbol{A})$, for $a \in \mathbb{C}$.
lyche-numerical-linear-algebra_F00438	37	0.943	`\boldsymbol{A}=\left[\begin{array}{ll}\boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E}\end{array} \right]`	$\boldsymbol{A}=\left[\begin{array}{ll}\boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E}\end{array}\right]$	5. If $\boldsymbol{A} = \begin{bmatrix}\boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E}\end{bmatrix}$ for some square matrices $\boldsymbol{C}, \boldsymbol{E}$ then $\det(\boldsymbol{A}) = \det(\boldsymbol{C})\det(\boldsymbol{E})$.
lyche-numerical-linear-algebra_F00439	37	0.943	`\boldsymbol{C}, \boldsymbol{E}`	$\boldsymbol{C}, \boldsymbol{E}$	5. If $\boldsymbol{A} = \begin{bmatrix}\boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E}\end{bmatrix}$ for some square matrices $\boldsymbol{C}, \boldsymbol{E}$ then $\det(\boldsymbol{A}) = \det(\boldsymbol{C})\det(\boldsymbol{E})$.
lyche-numerical-linear-algebra_F00440	37	0.943	`\operatorname{det}(\boldsymbol{A})=\operatorname{det}( \boldsymbol{C}) \operatorname{det}(\boldsymbol{E})`	$\operatorname{det}(\boldsymbol{A})=\operatorname{det}(\boldsymbol{C}) \operatorname{det}(\boldsymbol{E})$	5. If $\boldsymbol{A} = \begin{bmatrix}\boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E}\end{bmatrix}$ for some square matrices $\boldsymbol{C}, \boldsymbol{E}$ then $\det(\boldsymbol{A}) = \det(\boldsymbol{C})\det(\boldsymbol{E})$.
lyche-numerical-linear-algebra_F00441	37	1.000	`\boldsymbol{x}=`	$\boldsymbol{x}=$	6. **Cramer's rule** Suppose $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ is nonsingular and $\boldsymbol{b} \in \mathbb{C}^n$. Let $\boldsymbol{x} =$
lyche-numerical-linear-algebra_F00442	37	0.982	`\left[x_{1}, x_{2}, \ldots, x_{n}\right]^{T}`	$\left[x_{1}, x_{2}, \ldots, x_{n}\right]^{T}$	$[x_1, x_2, \ldots, x_n]^T$ be the unique solution of $\boldsymbol{Ax} = \boldsymbol{b}$. Then
lyche-numerical-linear-algebra_F00443	37	1.000	`\boldsymbol{A}_{j}(\boldsymbol{b})`	$\boldsymbol{A}_{j}(\boldsymbol{b})$	where $\boldsymbol{A}_j(\boldsymbol{b})$ denote the matrix obtained from $\boldsymbol{A}$ by replacing the jth column of
lyche-numerical-linear-algebra_F00444	37	0.999	`\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}`	$\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$	where the matrix $\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n\times n}$ with elements $\operatorname{adj}(\boldsymbol{A})_{i,j} = (-1)^{i+j}\det(\boldsymbol{A}_{j,i})$
lyche-numerical-linear-algebra_F00445	37	0.999	`\operatorname{adj}(\boldsymbol{A})_{i, j}=(-1)^{i+j} \operatorname{det}\left(\boldsymbol{A}_{j, i}\right)`	$\operatorname{adj}(\boldsymbol{A})_{i, j}=(-1)^{i+j} \operatorname{det}\left(\boldsymbol{A}_{j, i}\right)$	where the matrix $\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n\times n}$ with elements $\operatorname{adj}(\boldsymbol{A})_{i,j} = (-1)^{i+j}\det(\boldsymbol{A}_{j,i})$
lyche-numerical-linear-algebra_F00446	37	1.000	`\boldsymbol{A}_{j, i}`	$\boldsymbol{A}_{j, i}$	is called the **adjoint** of $\boldsymbol{A}$. Moreover, $\boldsymbol{A}_{j,i}$ denotes the submatrix of $\boldsymbol{A}$ obtained

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F00447	37	1.000	`\boldsymbol{A} \in \mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}`	$\boldsymbol{A} \in \mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}$	8. **Cauchy-Binet formula**: Let $\boldsymbol{A} \in \mathbb{C}^{m \times p}$, $\boldsymbol{B} \in \mathbb{C}^{p \times n}$ and $\boldsymbol{C} = \boldsymbol{AB}$. Suppose
lyche-numerical-linear-algebra_ F00448	37	1.000	`1 \leq r \leq \min \{m, n, p\}`	$1 \leq r \leq \min \{m, n, p\}$	$1 \leq r \leq \min\{m, n, p\}$ and let $\boldsymbol{i} = \{i_1, \ldots, i_r\}$ and $\boldsymbol{j} = \{j_1, \ldots, j_r\}$ be integers
lyche-numerical-linear-algebra_ F00449	37	1.000	`\boldsymbol{i}=\left\{i_{1}, \ldots, i_{r}\right\}`	$\boldsymbol{i}=\left\{i_{1}, \ldots, i_{r}\right\}$	$1 \leq r \leq \min\{m, n, p\}$ and let $\boldsymbol{i} = \{i_1, \ldots, i_r\}$ and $\boldsymbol{j} = \{j_1, \ldots, j_r\}$ be integers
lyche-numerical-linear-algebra_ F00450	37	1.000	`\boldsymbol{j}=\left\{j_{1}, \ldots, j_{r}\right\}`	$\boldsymbol{j}=\left\{j_{1}, \ldots, j_{r}\right\}$	$1 \leq r \leq \min\{m, n, p\}$ and let $\boldsymbol{i} = \{i_1, \ldots, i_r\}$ and $\boldsymbol{j} = \{j_1, \ldots, j_r\}$ be integers
lyche-numerical-linear-algebra_ F00451	37	1.000	`1 \leq i_{1}<i_{2}<\cdots<i_{r} \leq m`	$1 \leq i_{1}<i_{2}<\cdots<i_{r} \leq m$	with $1 \leq i_1 < i_2 < \cdots < i_r \leq m$ and $1 \leq j_1 < j_2 < \cdots < j_r \leq n$. Then
lyche-numerical-linear-algebra_ F00452	37	1.000	`1 \leq j_{1}<j_{2}<\cdots<j_{r} \leq n`	$1 \leq j_{1}<j_{2}<\cdots<j_{r} \leq n$	with $1 \leq i_1 < i_2 < \cdots < i_r \leq m$ and $1 \leq j_1 < j_2 < \cdots < j_r \leq n$. Then
lyche-numerical-linear-algebra_ F00453	37	0.999	`\boldsymbol{k}=\left\{k_{1}, \ldots, k_{r}\right\}`	$\boldsymbol{k}=\left\{k_{1}, \ldots, k_{r}\right\}$	where we sum over all $\boldsymbol{k} = \{k_1, \ldots, k_r\}$ with $1 \leq k_1 < k_2 < \cdots < k_r \leq p$.
lyche-numerical-linear-algebra_ F00454	37	0.999	`1 \leq k_{1}<k_{2}<\cdots<k_{r} \leq p`	$1 \leq k_{1}<k_{2}<\cdots<k_{r} \leq p$	where we sum over all $\boldsymbol{k} = \{k_1, \ldots, k_r\}$ with $1 \leq k_1 < k_2 < \cdots < k_r \leq p$.
lyche-numerical-linear-algebra_ F00455	38	0.937	`\lambda \in \mathbb{C}`	$\lambda \in \mathbb{C}$	Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is a square matrix, $\lambda \in \mathbb{C}$ and $\boldsymbol{x} \in \mathbb{C}^n$. We say that $(\lambda, \boldsymbol{x})$ is an
lyche-numerical-linear-algebra_ F00456	38	0.937	`\lambda, \boldsymbol{x}`	$\lambda, \boldsymbol{x}$	Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is a square matrix, $\lambda \in \mathbb{C}$ and $\boldsymbol{x} \in \mathbb{C}^n$. We say that $(\lambda, \boldsymbol{x})$ is an
lyche-numerical-linear-algebra_ F00457	38	1.000	`\sigma(\boldsymbol{A})`	$\sigma(\boldsymbol{A})$	of $\boldsymbol{A}$ and is denoted by $\sigma(\boldsymbol{A})$. For example, $\sigma(\boldsymbol{I}) = \{1, \ldots, 1\} = \{1\}$.
lyche-numerical-linear-algebra_ F00458	38	1.000	`\sigma(\boldsymbol{I})=\{1, \ldots, 1\}=\{1\}`	$\sigma(\boldsymbol{I})=\{1, \ldots, 1\}=\{1\}$	of $\boldsymbol{A}$ and is denoted by $\sigma(\boldsymbol{A})$. For example, $\sigma(\boldsymbol{I}) = \{1, \ldots, 1\} = \{1\}$.
lyche-numerical-linear-algebra_ F00459	38	1.000	`\lambda \in`	$\lambda \in$	**Lemma 1.5 (Characteristic Equation)** *For any* $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ *we have* $\lambda \in$
lyche-numerical-linear-algebra_ F00460	38	0.995	`\sigma(\boldsymbol{A}) \Longleftrightarrow \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=0`	$\sigma(\boldsymbol{A}) \Longleftrightarrow \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=0$	$\sigma(\boldsymbol{A}) \Longleftrightarrow \det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$.
lyche-numerical-linear-algebra_ F00461	38	0.699	`(\lambda, \boldsymbol{x})`	$(\lambda, \boldsymbol{x})$	***Proof*** Suppose $(\lambda, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}$. The equation $\boldsymbol{Ax} = \lambda \boldsymbol{x}$ can be written
lyche-numerical-linear-algebra_ F00462	38	1.000	`(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{x}= \mathbf{0}`	$(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{x}= \mathbf{0}$	$(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{x} = \mathbf{0}$. Since $\boldsymbol{x}$ is nonzero the matrix $\boldsymbol{A} - \lambda \boldsymbol{I}$ must be singular with a
lyche-numerical-linear-algebra_ F00463	38	1.000	`\boldsymbol{A}-\lambda \boldsymbol{I}`	$\boldsymbol{A}-\lambda \boldsymbol{I}$	$(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{x} = \mathbf{0}$. Since $\boldsymbol{x}$ is nonzero the matrix $\boldsymbol{A} - \lambda \boldsymbol{I}$ must be singular with a
lyche-numerical-linear-algebra_ F00464	38	1.000	`\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=0`	$\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=0$	zero determinant. Conversely, if $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$ then $\boldsymbol{A} - \lambda \boldsymbol{I}$ is singular and
lyche-numerical-linear-algebra_ F00465	38	1.000	`\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})`	$\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})$	The expression $\det(\boldsymbol{A} - \lambda \boldsymbol{I})$ is a polynomial of exact degree n in λ. For $n = 3$
lyche-numerical-linear-algebra_ F00466	38	1.000	`r(\lambda)`	$r(\lambda)$	where each term in $r(\lambda)$ has at most $n - 2$ factors containing λ. It follows that r is
lyche-numerical-linear-algebra_ F00467	38	0.998	`n-2, \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})`	$n-2, \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})$	a polynomial of degree at most $n - 2$, $\det(\boldsymbol{A} - \lambda \boldsymbol{I})$ is a polynomial of exact degree
lyche-numerical-linear-algebra_ F00468	38	1.000	`\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=(-1)^{n} \operatorname{det}(\lambda \boldsymbol{I}-\boldsymbol{A})`	$\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=(-1)^{n} \operatorname{det}(\lambda \boldsymbol{I}-\boldsymbol{A})$	We observe that $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (-1)^n \det(\lambda \boldsymbol{I} - \boldsymbol{A})$ so $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$ if and
lyche-numerical-linear-algebra_ F00469	38	1.000	`\operatorname{det}(\lambda \boldsymbol{I}- \boldsymbol{A})=0`	$\operatorname{det}(\lambda \boldsymbol{I}- \boldsymbol{A})=0$	only if $\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = 0$.
lyche-numerical-linear-algebra_ F00470	39	0.958	`\pi_{\boldsymbol{A}}: \mathbb{C} \rightarrow`	$\pi_{\boldsymbol{A}}: \mathbb{C} \rightarrow$	**Definition 1.9 (Characteristic Polynomial of a Matrix)** The function $\pi_{\boldsymbol{A}} : \mathbb{C} \rightarrow$
lyche-numerical-linear-algebra_ F00471	39	1.000	`\pi_{\boldsymbol{A}}(\lambda)=\operatorname{det}( \boldsymbol{A}-\lambda \boldsymbol{I})`	$\pi_{\boldsymbol{A}}(\lambda)=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})$	$\mathbb{C}$ given by $\pi_{\boldsymbol{A}}(\lambda) = \det(\boldsymbol{A} - \lambda \boldsymbol{I})$ is called the **characteristic polynomial** of $\boldsymbol{A}$.
lyche-numerical-linear-algebra_ F00472	39	0.994	`\lambda_{1}, \ldots, \lambda_{n}`	$\lambda_{1}, \ldots, \lambda_{n}$	ties, precisely n eigenvalues $\lambda_1, \ldots, \lambda_n$ some of which might be complex even if $\boldsymbol{A}$
lyche-numerical-linear-algebra_ F00473	39	1.000	`\boldsymbol{A} \overline{\boldsymbol{x}}=\bar{\lambda} \overline{\boldsymbol{x}}`	$\boldsymbol{A} \overline{\boldsymbol{x}}=\bar{\lambda} \overline{\boldsymbol{x}}$	$\boldsymbol{A}$ real gives $\boldsymbol{A}\overline{\boldsymbol{x}} = \bar{\lambda}\overline{\boldsymbol{x}}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F00474	39	0.945	\pi_{\boldsymbol{A}}	$\pi_{\boldsymbol{A}}$	***Proof*** We compare two different expansions of $\pi_{\boldsymbol{A}}$. On the one hand from (1.16)
lyche-numerical-linear-algebra_ F00475	39	1.000	c_{n-1}=(-1)^{n-1} \operatorname{trace}(\boldsymbol{A})	$c_{n-1}=(-1)^{n-1} \operatorname{trace}(\boldsymbol{A})$	where $c_{n-1}=(-1)^{n-1} \operatorname{trace}(\boldsymbol{A})$ and $c_0=\pi_{\boldsymbol{A}}(0)=\det(\boldsymbol{A})$. On the other hand
lyche-numerical-linear-algebra_ F00476	39	1.000	c_{0}=\pi_{\boldsymbol{A}}(0)=\operatorname{det}(\boldsymbol{A})	$c_{0}=\pi_{\boldsymbol{A}}(0)=\operatorname{det}(\boldsymbol{A})$	where $c_{n-1}=(-1)^{n-1} \operatorname{trace}(\boldsymbol{A})$ and $c_0=\pi_{\boldsymbol{A}}(0)=\det(\boldsymbol{A})$. On the other hand
lyche-numerical-linear-algebra_ F00477	39	1.000	d_{n-1}=(-1)^{n-1}\left(\lambda_{1}+\cdots+\lambda_{n} \right)	$d_{n-1}=(-1)^{n-1}\left(\lambda_{1}+\cdots+\lambda_{n}\right)$	where $d_{n-1}=(-1)^{n-1}(\lambda_1+\cdots+\lambda_n)$ and $d_0=\lambda_1\cdots\lambda_n$. Since $c_j=d_j$ for all j
lyche-numerical-linear-algebra_ F00478	39	1.000	d_{0}=\lambda_{1} \cdots \lambda_{n}	$d_{0}=\lambda_{1} \cdots \lambda_{n}$	where $d_{n-1}=(-1)^{n-1}(\lambda_1+\cdots+\lambda_n)$ and $d_0=\lambda_1\cdots\lambda_n$. Since $c_j=d_j$ for all j
lyche-numerical-linear-algebra_ F00479	39	1.000	c_{j}=d_{j}	$c_{j}=d_{j}$	where $d_{n-1}=(-1)^{n-1}(\lambda_1+\cdots+\lambda_n)$ and $d_0=\lambda_1\cdots\lambda_n$. Since $c_j=d_j$ for all j
lyche-numerical-linear-algebra_ F00480	39	1.000	\boldsymbol{A}=\left[\begin{array}{ll}2 & 1 \\ 1 & 2 \end{array}\right]	$\boldsymbol{A}=\left[\begin{array}{ll}2 & 1 \\ 1 & 2\end{array}\right]$	Thus, if $\boldsymbol{A}=\left[\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right]$ then $\operatorname{trace}(\boldsymbol{A})=4$, $\det(\boldsymbol{A})=3$ so that $\pi_{\boldsymbol{A}}(\lambda)=\lambda^2-4\lambda+3$.
lyche-numerical-linear-algebra_ F00481	39	1.000	\operatorname{trace}(\boldsymbol{A})=4, \operatorname{det}(\boldsymbol{A})=3	$\operatorname{trace}(\boldsymbol{A})=4, \operatorname{det}(\boldsymbol{A})=3$	Thus, if $\boldsymbol{A}=\left[\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right]$ then $\operatorname{trace}(\boldsymbol{A})=4$, $\det(\boldsymbol{A})=3$ so that $\pi_{\boldsymbol{A}}(\lambda)=\lambda^2-4\lambda+3$.
lyche-numerical-linear-algebra_ F00482	39	1.000	\pi_{\boldsymbol{A}}(\lambda)=\lambda^{2}-4 \lambda+3	$\pi_{\boldsymbol{A}}(\lambda)=\lambda^{2}-4 \lambda+3$	Thus, if $\boldsymbol{A}=\left[\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right]$ then $\operatorname{trace}(\boldsymbol{A})=4$, $\det(\boldsymbol{A})=3$ so that $\pi_{\boldsymbol{A}}(\lambda)=\lambda^2-4\lambda+3$.
lyche-numerical-linear-algebra_ F00483	39	1.000	\Longleftrightarrow \boldsymbol{A} \boldsymbol{x}= \mathbf{0}	$\Longleftrightarrow \boldsymbol{A} \boldsymbol{x}=\mathbf{0}$	Since $\boldsymbol{A}$ is singular $\Longleftrightarrow$ $\boldsymbol{A}\boldsymbol{x}=\mathbf{0}$, some $\boldsymbol{x}\neq 0$ $\Longleftrightarrow$ $\boldsymbol{A}\boldsymbol{x}=\mathbf{0}\boldsymbol{x}$, some $\boldsymbol{x}\neq$
lyche-numerical-linear-algebra_ F00484	39	1.000	\boldsymbol{x} \neq 0 \Longleftrightarrow \boldsymbol{A} \boldsymbol{x}=\mathbf{0} \boldsymbol{x}	$\boldsymbol{x} \neq 0 \Longleftrightarrow \boldsymbol{A} \boldsymbol{x}=\mathbf{0} \boldsymbol{x}$	Since $\boldsymbol{A}$ is singular $\Longleftrightarrow$ $\boldsymbol{A}\boldsymbol{x}=\mathbf{0}$, some $\boldsymbol{x}\neq 0$ $\Longleftrightarrow$ $\boldsymbol{A}\boldsymbol{x}=\mathbf{0}\boldsymbol{x}$, some $\boldsymbol{x}\neq$
lyche-numerical-linear-algebra_ F00485	39	1.000	\boldsymbol{x} \neq	$\boldsymbol{x} \neq$	Since $\boldsymbol{A}$ is singular $\Longleftrightarrow$ $\boldsymbol{A}\boldsymbol{x}=\mathbf{0}$, some $\boldsymbol{x}\neq 0$ $\Longleftrightarrow$ $\boldsymbol{A}\boldsymbol{x}=\mathbf{0}\boldsymbol{x}$, some $\boldsymbol{x}\neq$
lyche-numerical-linear-algebra_ F00486	39	0.981	0 \Longleftrightarrow	$0 \Longleftrightarrow$	0 $\Longleftrightarrow$ zero is an eigenvalue of $\boldsymbol{A}$, we obtain
lyche-numerical-linear-algebra_ F00487	40	0.848	\boldsymbol{A}=\left[\begin{array}{cc}\boldsymbol{B} & \boldsymbol{D} \\ 0 & \boldsymbol{C}\end{array}\right]	$\boldsymbol{A}=\left[\begin{array}{cc}\boldsymbol{B} & \boldsymbol{D} \\ 0 & \boldsymbol{C}\end{array}\right]$	**Theorem 1.12 (Eigenvalues of a Block Triangular Matrix)** *If* $\boldsymbol{A}=\left[\begin{smallmatrix}\boldsymbol{B}&\boldsymbol{D}\\0&\boldsymbol{C}\end{smallmatrix}\right]$ *is*
lyche-numerical-linear-algebra_ F00488	40	0.998	\pi_{\boldsymbol{A}}=\pi_{\boldsymbol{B}} \cdot \pi_{ \boldsymbol{C}}	$\pi_{\boldsymbol{A}}=\pi_{\boldsymbol{B}} \cdot \pi_{\boldsymbol{C}}$	*block triangular then* $\pi_{\boldsymbol{A}}=\pi_{\boldsymbol{B}}\cdot\pi_{\boldsymbol{C}}$.
lyche-numerical-linear-algebra_ F00489	40	1.000	\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n \times n}	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n \times n}$	**a)** With $\boldsymbol{A},\boldsymbol{B}\in\mathbb{R}^{n\times n}$, how many arithmetic operations are required to form the
lyche-numerical-linear-algebra_ F00490	40	1.000	2 n \times 2 n	$2 n \times 2 n$	**b)** Consider the $2n\times 2n$ block matrix
lyche-numerical-linear-algebra_ F00491	40	1.000	\boldsymbol{A}, \ldots, \boldsymbol{Z}	$\boldsymbol{A}, \ldots, \boldsymbol{Z}$	where all matrices $\boldsymbol{A},\ldots,\boldsymbol{Z}$ are in $\mathbb{R}^{n\times n}$. How many operations does it take to
lyche-numerical-linear-algebra_ F00492	40	1.000	\boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}	$\boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}$	compute $\boldsymbol{W}$, $\boldsymbol{X}$, $\boldsymbol{Y}$ and $\boldsymbol{Z}$ by the obvious algorithm?
lyche-numerical-linear-algebra_ F00493	40	1.000	\boldsymbol{Z}	$\boldsymbol{Z}$	compute $\boldsymbol{W}$, $\boldsymbol{X}$, $\boldsymbol{Y}$ and $\boldsymbol{Z}$ by the obvious algorithm?
lyche-numerical-linear-algebra_ F00494	40	1.000	m \times m	$m \times m$	matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ of size $m\times m$, with $m=2^k$ for some $k\geq 0$, calculates the
lyche-numerical-linear-algebra_ F00495	40	1.000	m=2^{k}	$m=2^{k}$	matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ of size $m\times m$, with $m=2^k$ for some $k\geq 0$, calculates the
lyche-numerical-linear-algebra_ F00496	40	1.000	k \geq 0	$k \geq 0$	matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ of size $m\times m$, with $m=2^k$ for some $k\geq 0$, calculates the
lyche-numerical-linear-algebra_ F00497	40	1.000	\mathcal{O}\left(m^{\log _{2}(7)}\right)	$\mathcal{O}\left(m^{\log _{2}(7)}\right)$	**e)** Show that the operation count of the recursive algorithm is $\mathcal{O}\left(m^{\log_2(7)}\right)$. Note
lyche-numerical-linear-algebra_ F00498	40	1.000	\log _{2}(7) \approx 2.8<3	$\log _{2}(7) \approx 2.8<3$	that $\log_2(7)\approx 2.8<3$, so this is less costly than straightforward matrix
lyche-numerical-linear-algebra_ F00499	41	1.000	2 \times 2	2×2	**Exercise 1.2 (The Inverse of a General 2×2 Matrix)** Show that

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00500	41	1.000	`a, b, c, d`	a, b, c, d	for any a, b, c, d such that $ad - bc \neq 0$.
lyche-numerical-linear-algebra_F00501	41	1.000	`a d-b c \neq 0`	$a d-b c \neq 0$	for any a, b, c, d such that $ad - bc \neq 0$.
lyche-numerical-linear-algebra_F00502	41	1.000	`\boldsymbol{B}, \boldsymbol{C} \in`	$\boldsymbol{B}, \boldsymbol{C} \in$	**Exercise 1.4 (Sherman-Morrison Formula)** Suppose $\boldsymbol{A} \in \mathbb{C}^{n\times n}$, and $\boldsymbol{B}, \boldsymbol{C} \in$
lyche-numerical-linear-algebra_F00503	41	1.000	`\mathbb{R}^{n \times m}`	$\mathbb{R}^{n \times m}$	$\mathbb{R}^{n\times m}$ for some $n, m \in \mathbb{N}$. If $(\boldsymbol{I}+\boldsymbol{C}^T\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}$ exists then
lyche-numerical-linear-algebra_F00504	41	1.000	`n, m \in \mathbb{N}`	$n, m \in \mathbb{N}$	$\mathbb{R}^{n\times m}$ for some $n, m \in \mathbb{N}$. If $(\boldsymbol{I}+\boldsymbol{C}^T\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}$ exists then
lyche-numerical-linear-algebra_F00505	41	1.000	`\left(\boldsymbol{I}+\boldsymbol{C}^{T} \boldsymbol{A}^{-1} \boldsymbol{B}\right)^{-1}`	$\left(\boldsymbol{I}+\boldsymbol{C}^{T} \boldsymbol{A}^{-1} \boldsymbol{B}\right)^{-1}$	$\mathbb{R}^{n\times m}$ for some $n, m \in \mathbb{N}$. If $(\boldsymbol{I}+\boldsymbol{C}^T\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}$ exists then
lyche-numerical-linear-algebra_F00506	41	1.000	`\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^{n}`	$\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^{n}$	**a)** Let $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ and suppose $\boldsymbol{v}^T\boldsymbol{u} \neq 1$. Show that $\boldsymbol{I}-\boldsymbol{u}\boldsymbol{v}^T$ has an inverse given
lyche-numerical-linear-algebra_F00507	41	1.000	`\boldsymbol{v}^{T} \boldsymbol{u} \neq 1`	$\boldsymbol{v}^{T} \boldsymbol{u} \neq 1$	**a)** Let $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ and suppose $\boldsymbol{v}^T\boldsymbol{u} \neq 1$. Show that $\boldsymbol{I}-\boldsymbol{u}\boldsymbol{v}^T$ has an inverse given
lyche-numerical-linear-algebra_F00508	41	1.000	`\boldsymbol{I}-\boldsymbol{u} \boldsymbol{v}^{T}`	$\boldsymbol{I}-\boldsymbol{u} \boldsymbol{v}^{T}$	**a)** Let $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ and suppose $\boldsymbol{v}^T\boldsymbol{u} \neq 1$. Show that $\boldsymbol{I}-\boldsymbol{u}\boldsymbol{v}^T$ has an inverse given
lyche-numerical-linear-algebra_F00509	41	0.813	`\boldsymbol{I}-\tau \boldsymbol{u} \boldsymbol{v}^{T}`	$\boldsymbol{I}-\tau \boldsymbol{u} \boldsymbol{v}^{T}$	by $\boldsymbol{I}-\tau\boldsymbol{u}\boldsymbol{v}^T$, where $\tau = 1/(\boldsymbol{v}^T\boldsymbol{u}-1)$.
lyche-numerical-linear-algebra_F00510	41	0.813	`\tau=1 /\left(\boldsymbol{v}^{T} \boldsymbol{u}-1\right)`	$\tau=1 /\left(\boldsymbol{v}^{T} \boldsymbol{u}-1\right)$	by $\boldsymbol{I}-\tau\boldsymbol{u}\boldsymbol{v}^T$, where $\tau = 1/(\boldsymbol{v}^T\boldsymbol{u}-1)$.
lyche-numerical-linear-algebra_F00511	41	1.000	`\boldsymbol{C}:=\boldsymbol{A}^{-1}`	$\boldsymbol{C}:=\boldsymbol{A}^{-1}$	**b)** Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be nonsingular with inverse $\boldsymbol{C} := \boldsymbol{A}^{-1}$, and let $\boldsymbol{a} \in \mathbb{R}^n$. Let $\overline{\boldsymbol{A}}$ be
lyche-numerical-linear-algebra_F00512	41	1.000	`\boldsymbol{a} \in \mathbb{R}^{n}`	$\boldsymbol{a} \in \mathbb{R}^{n}$	**b)** Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be nonsingular with inverse $\boldsymbol{C} := \boldsymbol{A}^{-1}$, and let $\boldsymbol{a} \in \mathbb{R}^n$. Let $\overline{\boldsymbol{A}}$ be
lyche-numerical-linear-algebra_F00513	41	1.000	`\overline{\boldsymbol{A}}`	$\overline{\boldsymbol{A}}$	**b)** Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be nonsingular with inverse $\boldsymbol{C} := \boldsymbol{A}^{-1}$, and let $\boldsymbol{a} \in \mathbb{R}^n$. Let $\overline{\boldsymbol{A}}$ be
lyche-numerical-linear-algebra_F00514	41	1.000	`\boldsymbol{a}^{T}`	$\boldsymbol{a}^{T}$	the matrix which differs from $\boldsymbol{A}$ by exchanging the ith row of $\boldsymbol{A}$ with $\boldsymbol{a}^T$, i.e.,
lyche-numerical-linear-algebra_F00515	41	0.995	`\overline{\boldsymbol{A}}=\boldsymbol{A}-\boldsymbol{e}_{i}\left(\boldsymbol{e}_{i}^{T} \boldsymbol{A}-\boldsymbol{a}^{T}\right)`	$\overline{\boldsymbol{A}}=\boldsymbol{A}-\boldsymbol{e}_{i}\left(\boldsymbol{e}_{i}^{T} \boldsymbol{A}-\boldsymbol{a}^{T}\right)$	$\overline{\boldsymbol{A}} = \boldsymbol{A}-\boldsymbol{e}_i(\boldsymbol{e}_i^T\boldsymbol{A}-\boldsymbol{a}^T)$, where $\boldsymbol{e}_i$ is the ith column in the identity matrix $\boldsymbol{I}$.
lyche-numerical-linear-algebra_F00516	41	0.995	`\boldsymbol{e}_{i}`	$\boldsymbol{e}_{i}$	$\overline{\boldsymbol{A}} = \boldsymbol{A}-\boldsymbol{e}_i(\boldsymbol{e}_i^T\boldsymbol{A}-\boldsymbol{a}^T)$, where $\boldsymbol{e}_i$ is the ith column in the identity matrix $\boldsymbol{I}$.
lyche-numerical-linear-algebra_F00517	41	1.000	`\overline{\boldsymbol{C}}=\overline{\boldsymbol{A}}^{-1}`	$\overline{\boldsymbol{C}}=\overline{\boldsymbol{A}}^{-1}$	then $\overline{\boldsymbol{A}}$ has an inverse $\overline{\boldsymbol{C}} = \overline{\boldsymbol{A}}^{-1}$ given by
lyche-numerical-linear-algebra_F00518	41	1.000	`\boldsymbol{a}`	$\boldsymbol{a}$	**c)** Write an algorithm which to given $\boldsymbol{C}$ and $\boldsymbol{a}$ checks if (1.20) holds and computes
lyche-numerical-linear-algebra_F00519	41	0.999	`\overline{\boldsymbol{C}}`	$\overline{\boldsymbol{C}}$	$\overline{\boldsymbol{C}}$ provided $\lambda \neq 0$. (hint: Use (1.21) to find formulas for computing each column
lyche-numerical-linear-algebra_F00520	41	0.999	`\lambda \neq 0`	$\lambda \neq 0$	$\overline{\boldsymbol{C}}$ provided $\lambda \neq 0$. (hint: Use (1.21) to find formulas for computing each column
lyche-numerical-linear-algebra_F00521	41	1.000	`\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{E} \in`	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{E} \in$	**Exercise 1.6 (Matrix Products (Exam Exercise 2009-1))** Let $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{E} \in$
lyche-numerical-linear-algebra_F00522	41	1.000	`\boldsymbol{B} \boldsymbol{C}`	$\boldsymbol{B} \boldsymbol{C}$	**a)** How many operations are required to compute the matrix product $\boldsymbol{B}\boldsymbol{C}$? How
lyche-numerical-linear-algebra_F00523	41	1.000	`\boldsymbol{L} \in \mathbb{R}^{n \times n}`	$\boldsymbol{L} \in \mathbb{R}^{n \times n}$	**b)** Show that there exists a lower triangular matrix $\boldsymbol{L} \in \mathbb{R}^{n\times n}$ such that $\boldsymbol{A} = \boldsymbol{L}+\boldsymbol{L}^T$.
lyche-numerical-linear-algebra_F00524	41	1.000	`\boldsymbol{A}=\boldsymbol{L}+\boldsymbol{L}^{T}`	$\boldsymbol{A}=\boldsymbol{L}+\boldsymbol{L}^{T}$	**b)** Show that there exists a lower triangular matrix $\boldsymbol{L} \in \mathbb{R}^{n\times n}$ such that $\boldsymbol{A} = \boldsymbol{L}+\boldsymbol{L}^T$.
lyche-numerical-linear-algebra_F00525	42	1.000	`\boldsymbol{E}^{T} \boldsymbol{A} \boldsymbol{E}=\boldsymbol{S}+\boldsymbol{S}^{T}`	$\boldsymbol{E}^{T} \boldsymbol{A} \boldsymbol{E}=\boldsymbol{S}+\boldsymbol{S}^{T}$	**c)** We have $\boldsymbol{E}^T\boldsymbol{A}\boldsymbol{E} = \boldsymbol{S}+\boldsymbol{S}^T$ where $\boldsymbol{S} = \boldsymbol{E}^T\boldsymbol{L}\boldsymbol{E}$. How many operations are

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00526	42	1.000	`\boldsymbol{S}=\boldsymbol{E}^{T} \boldsymbol{L} \boldsymbol{E}`	$\boldsymbol{S}=\boldsymbol{E}^{T} \boldsymbol{L} \boldsymbol{E}$	**c)** We have $\boldsymbol{E}^T\boldsymbol{A}\boldsymbol{E} = \boldsymbol{S}+\boldsymbol{S}^T$ where $\boldsymbol{S} = \boldsymbol{E}^T\boldsymbol{L}\boldsymbol{E}$. How many operations are
lyche-numerical-linear-algebra_F00527	42	1.000	`\boldsymbol{E}^{T} \boldsymbol{A} \boldsymbol{E}`	$\boldsymbol{E}^{T} \boldsymbol{A} \boldsymbol{E}$	required to compute $\boldsymbol{E}^T\boldsymbol{A}\boldsymbol{E}$ in this way?
lyche-numerical-linear-algebra_F00528	42	1.000	`\left(x_{1}, y_{1}, z_{1}\right),\left(x_{2}, y_{2}, z_{2}\right),\left(x_{3}, y_{3}, z_{3}\right)`	$\left(x_{1}, y_{1}, z_{1}\right),\left(x_{2}, y_{2}, z_{2}\right),\left(x_{3}, y_{3}, z_{3}\right)$	is the equation for a plane through three points (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)
lyche-numerical-linear-algebra_F00529	43	1.000	`P_{i}=\left(x_{i}, y_{i}\right), i=1,2,3`	$P_{i}=\left(x_{i}, y_{i}\right), i=1,2,3$	**Exercise 1.10 (Signed Area of a Triangle)** Let $P_i = (x_i, y_i)$, $i = 1, 2, 3$, be three
lyche-numerical-linear-algebra_F00530	43	0.816	`T \mathrm{is}^{2}`	$T \mathrm{is}^{2}$	points in the plane defining a triangle T. Show that the area of T is[2]
lyche-numerical-linear-algebra_F00531	43	0.999	`\prod_{i>j}\left(x_{i}-x_{j}\right)=\prod_{i=2}^{n} \left(x_{i}-x_{1}\right)\left(x_{i}-x_{2}\right) \cdots \left(x_{i}-x_{i-1}\right)`	$\prod_{i>j}\left(x_{i}-x_{j}\right)=\prod_{i=2}^{n} \left(x_{i}-x_{1}\right)\left(x_{i}-x_{2}\right) \cdots \left(x_{i}-x_{i-1}\right)$	where $\prod_{i>j}(x_i - x_j) = \prod_{i=2}^n (x_i - x_1)(x_i - x_2)\cdots(x_i - x_{i-1})$. This determinant
lyche-numerical-linear-algebra_F00532	43	1.000	`\boldsymbol{\alpha}=\left[\alpha_{1}, \ldots, \alpha_{n} \right]^{T}, \boldsymbol{\beta}=`	$\boldsymbol{\alpha}=\left[\alpha_{1}, \ldots, \alpha_{n} \right]^{T}, \boldsymbol{\beta}=$	**Exercise 1.12 (Cauchy Determinant (1842))** Let $\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_n]^T$, $\boldsymbol{\beta} =$
lyche-numerical-linear-algebra_F00533	43	1.000	`\left[\beta_{1}, \ldots, \beta_{n}\right]^{T}`	$\left[\beta_{1}, \ldots, \beta_{n}\right]^{T}$	$[\beta_1, \ldots, \beta_n]^T$ be in $\mathbb{R}^n$.
lyche-numerical-linear-algebra_F00534	43	0.996	`a_{i, j}=1 /\left(\alpha_{i}+\beta_{j}\right), i, j=`	$a_{i, j}=1 /\left(\alpha_{i}+\beta_{j}\right), i, j=$	**a)** Consider the matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ with elements $a_{i,j} = 1/(\alpha_i + \beta_j)$, $i, j =$
lyche-numerical-linear-algebra_F00535	43	1.000	`1,2, \ldots, n`	$1,2, \ldots, n$	$1, 2, \ldots, n$. Show that
lyche-numerical-linear-algebra_F00536	43	0.997	`A(T)=A\left(A B P_{3} P_{1}\right)+A\left(P_{3} B C P_{2}\right)-A\left(P_{1} A C P_{2}\right)`	$A(T)=A\left(A B P_{3} P_{1}\right)+A\left(P_{3} B C P_{2}\right)-A\left(P_{1} A C P_{2}\right)$	[2]Hint: $A(T) = A(ABP_3P_1) + A(P_3BCP_2) - A(P_1ACP_2)$, c.f. Fig. 1.1.
lyche-numerical-linear-algebra_F00537	43	0.997	`x_{n}^{k}`	x_{n}^{k}	[3]Hint: subtract x_n^k times column k from column $k+1$ for $k = n-1, n-2, \ldots, 1$.
lyche-numerical-linear-algebra_F00538	43	0.997	`k`	k	[3]Hint: subtract x_n^k times column k from column $k+1$ for $k = n-1, n-2, \ldots, 1$.
lyche-numerical-linear-algebra_F00539	43	0.997	`k+1`	$k+1$	[3]Hint: subtract x_n^k times column k from column $k+1$ for $k = n-1, n-2, \ldots, 1$.
lyche-numerical-linear-algebra_F00540	43	0.997	`k=n-1, n-2, \ldots, 1`	$k=n-1, n-2, \ldots, 1$	[3]Hint: subtract x_n^k times column k from column $k+1$ for $k = n-1, n-2, \ldots, 1$.
lyche-numerical-linear-algebra_F00541	44	1.000	`P=\prod_{i=1}^{n} \prod_{j=1}^{n} a_{i j}`	$P=\prod_{i=1}^{n} \prod_{j=1}^{n} a_{i j}$	where $P = \prod_{i=1}^n \prod_{j=1}^n a_{ij}$, and for $\boldsymbol{\gamma} = [\gamma_1, \ldots, \gamma_n]^T$
lyche-numerical-linear-algebra_F00542	44	1.000	`\boldsymbol{\gamma}=\left[\gamma_{1}, \ldots, \gamma_{n} \right]^{T}`	$\boldsymbol{\gamma}=\left[\gamma_{1}, \ldots, \gamma_{n} \right]^{T}$	where $P = \prod_{i=1}^n \prod_{j=1}^n a_{ij}$, and for $\boldsymbol{\gamma} = [\gamma_1, \ldots, \gamma_n]^T$
lyche-numerical-linear-algebra_F00543	44	1.000	`\prod_{j=1}^{n}\left(\alpha_{i}+\beta_{j}\right)`	$\prod_{j=1}^{n}\left(\alpha_{i}+\beta_{j}\right)$	Hint: Multiply the ith row of $\boldsymbol{A}$ by $\prod_{j=1}^n(\alpha_i + \beta_j)$ for $i = 1, 2, \ldots, n$. Call the
lyche-numerical-linear-algebra_F00544	44	1.000	`i=1,2, \ldots, n`	$i=1,2, \ldots, n$	Hint: Multiply the ith row of $\boldsymbol{A}$ by $\prod_{j=1}^n(\alpha_i + \beta_j)$ for $i = 1, 2, \ldots, n$. Call the
lyche-numerical-linear-algebra_F00545	44	0.999	`n-1`	$n-1$	resulting matrix $\boldsymbol{C}$. Each element of $\boldsymbol{C}$ is a product of $n-1$ factors $\alpha_r + \beta_s$. Hence
lyche-numerical-linear-algebra_F00546	44	0.999	`\alpha_{r}+\beta_{s}`	$\alpha_{r}+\beta_{s}$	resulting matrix $\boldsymbol{C}$. Each element of $\boldsymbol{C}$ is a product of $n-1$ factors $\alpha_r + \beta_s$. Hence
lyche-numerical-linear-algebra_F00547	44	1.000	`\operatorname{det}(\boldsymbol{C})`	$\operatorname{det}(\boldsymbol{C})$	$\det(\boldsymbol{C})$ is a sum of terms where each term contain precisely $n(n-1)$ factors
lyche-numerical-linear-algebra_F00548	44	1.000	`n(n-1)`	$n(n-1)$	$\det(\boldsymbol{C})$ is a sum of terms where each term contain precisely $n(n-1)$ factors
lyche-numerical-linear-algebra_F00549	44	0.997	`\operatorname{det}(\boldsymbol{C})=q(\alpha, \beta)`	$\operatorname{det}(\boldsymbol{C})=q(\alpha, \beta)$	$\alpha_r + \beta_s$. Thus $\det(\boldsymbol{C}) = q(\alpha, \beta)$ where q is a polynomial of degree at most
lyche-numerical-linear-algebra_F00550	44	0.997	`q`	q	$\alpha_r + \beta_s$. Thus $\det(\boldsymbol{C}) = q(\alpha, \beta)$ where q is a polynomial of degree at most
lyche-numerical-linear-algebra_F00551	44	1.000	`\alpha_{i}`	α_{i}	$n(n-1)$ in α_i and β_j. Since $\det(\boldsymbol{A})$ and therefore $\det(\boldsymbol{C})$ vanishes if $\alpha_i = \alpha_j$ for

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00552	44	1.000	`\beta_{j}`	β_{j}	$n(n-1)$ in α_i and β_j. Since $\det(\boldsymbol{A})$ and therefore $\det(\boldsymbol{C})$ vanishes if $\alpha_i=\alpha_j$ for
lyche-numerical-linear-algebra_F00553	44	1.000	`\operatorname{det}(\boldsymbol{A})`	$\operatorname{det}(\boldsymbol{A})$	$n(n-1)$ in α_i and β_j. Since $\det(\boldsymbol{A})$ and therefore $\det(\boldsymbol{C})$ vanishes if $\alpha_i=\alpha_j$ for
lyche-numerical-linear-algebra_F00554	44	1.000	`\alpha_{i}=\alpha_{j}`	$\alpha_{i}=\alpha_{j}$	$n(n-1)$ in α_i and β_j. Since $\det(\boldsymbol{A})$ and therefore $\det(\boldsymbol{C})$ vanishes if $\alpha_i=\alpha_j$ for
lyche-numerical-linear-algebra_F00555	44	1.000	`\beta_{r}=\beta_{s}`	$\beta_{r}=\beta_{s}$	some $i \neq j$ or $\beta_r=\beta_s$ for some $r \neq s$, we have that $q(\boldsymbol{\alpha}, \boldsymbol{\beta})$ must be divisible
lyche-numerical-linear-algebra_F00556	44	1.000	`r \neq s`	$r \neq s$	some $i \neq j$ or $\beta_r=\beta_s$ for some $r \neq s$, we have that $q(\boldsymbol{\alpha}, \boldsymbol{\beta})$ must be divisible
lyche-numerical-linear-algebra_F00557	44	1.000	`q(\boldsymbol{\alpha}, \boldsymbol{\beta})`	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})$	some $i \neq j$ or $\beta_r=\beta_s$ for some $r \neq s$, we have that $q(\boldsymbol{\alpha}, \boldsymbol{\beta})$ must be divisible
lyche-numerical-linear-algebra_F00558	44	1.000	`g(\boldsymbol{\alpha})`	$g(\boldsymbol{\alpha})$	by each factor in $g(\boldsymbol{\alpha})$ and $g(\boldsymbol{\beta})$. Since $g(\boldsymbol{\alpha})$ and $g(\boldsymbol{\beta})$ is a polynomial of degree
lyche-numerical-linear-algebra_F00559	44	1.000	`g(\boldsymbol{\beta})`	$g(\boldsymbol{\beta})$	by each factor in $g(\boldsymbol{\alpha})$ and $g(\boldsymbol{\beta})$. Since $g(\boldsymbol{\alpha})$ and $g(\boldsymbol{\beta})$ is a polynomial of degree
lyche-numerical-linear-algebra_F00560	44	1.000	`\boldsymbol{\alpha}`	$\boldsymbol{\alpha}$	for some constant k independent of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Show that $k=1$ by choosing
lyche-numerical-linear-algebra_F00561	44	1.000	`\boldsymbol{\beta}`	$\boldsymbol{\beta}$	for some constant k independent of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Show that $k=1$ by choosing
lyche-numerical-linear-algebra_F00562	44	1.000	`k=1`	$k=1$	for some constant k independent of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Show that $k=1$ by choosing
lyche-numerical-linear-algebra_F00563	44	1.000	`\beta_{i}+\alpha_{i}=0, i=1,2, \ldots, n`	$\beta_{i}+\alpha_{i}=0, i=1,2, \ldots, n$	$\beta_i+\alpha_i=0, i=1,2,\ldots,n$.
lyche-numerical-linear-algebra_F00564	44	1.000	`\boldsymbol{A}^{-1}=\left(b_{j, k}\right)`	$\boldsymbol{A}^{-1}=\left(b_{j, k}\right)$	the elements of $\boldsymbol{A}^{-1}=(b_{j,k})$. Answer:
lyche-numerical-linear-algebra_F00565	44	1.000	`\boldsymbol{H}_{n}=\left(h_{i, j}\right)`	$\boldsymbol{H}_{n}=\left(h_{i, j}\right)$	**Exercise 1.13 (Inverse of the Hilbert Matrix)** Let $\boldsymbol{H}_n=(h_{i,j})$ be the $n \times n$
lyche-numerical-linear-algebra_F00566	44	1.000	`h_{i, j}=1 /(i+j-1)`	$h_{i, j}=1 /(i+j-1)$	matrix with elements $h_{i,j}=1/(i+j-1)$. Use Exercise 1.12 to show that the elements
lyche-numerical-linear-algebra_F00567	44	1.000	`t_{i, j}^{n}`	$t_{i, j}^{n}$	$t_{i,j}^n$ in $\boldsymbol{T}_n=\boldsymbol{H}_n^{-1}$ are given by
lyche-numerical-linear-algebra_F00568	44	1.000	`\boldsymbol{T}_{n}=\boldsymbol{H}_{n}^{-1}`	$\boldsymbol{T}_{n}=\boldsymbol{H}_{n}^{-1}$	$t_{i,j}^n$ in $\boldsymbol{T}_n=\boldsymbol{H}_n^{-1}$ are given by
lyche-numerical-linear-algebra_F00569	45	1.000	`\boldsymbol{L}`	$\boldsymbol{L}$	triangular matrix $\boldsymbol{L}$ and an upper triangular matrix $\boldsymbol{U}$ resulting in the product $\boldsymbol{A}=$
lyche-numerical-linear-algebra_F00570	45	1.000	`\boldsymbol{U}`	$\boldsymbol{U}$	triangular matrix $\boldsymbol{L}$ and an upper triangular matrix $\boldsymbol{U}$ resulting in the product $\boldsymbol{A}=$
lyche-numerical-linear-algebra_F00571	45	1.000	`\boldsymbol{A}=`	$\boldsymbol{A}=$	triangular matrix $\boldsymbol{L}$ and an upper triangular matrix $\boldsymbol{U}$ resulting in the product $\boldsymbol{A}=$
lyche-numerical-linear-algebra_F00572	45	1.000	`\boldsymbol{L} \boldsymbol{U}`	$\boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{LU}$. We also consider the factorization $\boldsymbol{A}=\boldsymbol{LDU}$, where $\boldsymbol{L}$ is lower triangular,
lyche-numerical-linear-algebra_F00573	45	1.000	`\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U}`	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U}$	$\boldsymbol{LU}$. We also consider the factorization $\boldsymbol{A}=\boldsymbol{LDU}$, where $\boldsymbol{L}$ is lower triangular,
lyche-numerical-linear-algebra_F00574	45	1.000	`\boldsymbol{D}`	$\boldsymbol{D}$	$\boldsymbol{D}$ is diagonal and $\boldsymbol{U}$ is upper triangular. Moreover, $\boldsymbol{L}$ and $\boldsymbol{U}$ have ones on their
lyche-numerical-linear-algebra_F00575	45	1.000	`\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{R}`	$\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{R}$	on QR factorization. Here $\boldsymbol{A}=\boldsymbol{QR}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^*\boldsymbol{Q}=\boldsymbol{I}$, and $\boldsymbol{R}$
lyche-numerical-linear-algebra_F00576	45	1.000	`\boldsymbol{Q}`	$\boldsymbol{Q}$	on QR factorization. Here $\boldsymbol{A}=\boldsymbol{QR}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^*\boldsymbol{Q}=\boldsymbol{I}$, and $\boldsymbol{R}$
lyche-numerical-linear-algebra_F00577	45	1.000	`\boldsymbol{Q}^{*} \boldsymbol{Q}=\boldsymbol{I}`	$\boldsymbol{Q}^{*} \boldsymbol{Q}=\boldsymbol{I}$	on QR factorization. Here $\boldsymbol{A}=\boldsymbol{QR}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^*\boldsymbol{Q}=\boldsymbol{I}$, and $\boldsymbol{R}$
lyche-numerical-linear-algebra_F00578	45	1.000	`\boldsymbol{R}`	$\boldsymbol{R}$	on QR factorization. Here $\boldsymbol{A}=\boldsymbol{QR}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^*\boldsymbol{Q}=\boldsymbol{I}$, and $\boldsymbol{R}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F00579	46	0.996	`[a, b], n+1 \geq 2`	$[a, b], n+1 \geq 2$	Given an interval $[a, b]$, $n+1 \geq 2$ equidistant sites in $[a, b]$
lyche-numerical-linear-algebra_ F00580	46	0.996	`[a, b]`	$[a, b]$	Given an interval $[a, b]$, $n+1 \geq 2$ equidistant sites in $[a, b]$
lyche-numerical-linear-algebra_ F00581	46	1.000	`y`	y	and y values $\boldsymbol{y}:=[y_1, \ldots, y_{n+1}]^T \in \mathbb{R}^{n+1}$. We seek a function $g:[a, b] \rightarrow \mathbb{R}$ such
lyche-numerical-linear-algebra_ F00582	46	1.000	`\boldsymbol{y}:=\left[y_{1}, \ldots, y_{n+1}\right]^{T} \in \mathbb{R}^{n+1}`	$\boldsymbol{y}:=\left[y_{1}, \ldots, y_{n+1}\right]^{T} \in \mathbb{R}^{n+1}$	and y values $\boldsymbol{y}:=[y_1, \ldots, y_{n+1}]^T \in \mathbb{R}^{n+1}$. We seek a function $g:[a, b] \rightarrow \mathbb{R}$ such
lyche-numerical-linear-algebra_ F00583	46	1.000	`g:[a, b] \rightarrow \mathbb{R}`	$g:[a, b] \rightarrow \mathbb{R}$	and y values $\boldsymbol{y}:=[y_1, \ldots, y_{n+1}]^T \in \mathbb{R}^{n+1}$. We seek a function $g:[a, b] \rightarrow \mathbb{R}$ such
lyche-numerical-linear-algebra_ F00584	47	1.000	`a \leq x_{1}<x_{2}<\cdots<x_{n+1} \leq b`	$a \leq x_{1}<x_{2}<\cdots<x_{n+1} \leq b$	$a \leq x_1 < x_2 < \cdots < x_{n+1} \leq b$.
lyche-numerical-linear-algebra_ F00585	47	1.000	`n+1`	$n+1$	Since there are $n+1$ interpolation conditions in (2.2) a natural choice for a function
lyche-numerical-linear-algebra_ F00586	47	1.000	`g`	g	g is a polynomial of degree n. As shown in most books on numerical methods such
lyche-numerical-linear-algebra_ F00587	47	1.000	`n=1, g`	$n=1, g$	when $n=1$, g is the straight line
lyche-numerical-linear-algebra_ F00588	47	1.000	`f`	f	The function f and the polynomial g of degree at most 13 satisfying (2.2) with
lyche-numerical-linear-algebra_ F00589	47	1.000	`[a, b]=[-1,1]`	$[a, b]=[-1,1]$	$[a, b]=[-1, 1]$ and $y_i = f(x_i)$, $i=1, \ldots, 14$ is shown in Fig. 2.1. The interpolant
lyche-numerical-linear-algebra_ F00590	47	1.000	`y_{i}=f\left(x_{i}\right), i=1, \ldots, 14`	$y_{i}=f\left(x_{i}\right), i=1, \ldots, 14$	$[a, b]=[-1, 1]$ and $y_i = f(x_i)$, $i=1, \ldots, 14$ is shown in Fig. 2.1. The interpolant
lyche-numerical-linear-algebra_ F00591	48	1.000	`p_{i}`	p_{i}	where each p_i is a polynomial of degree ≤ 1. In particular, p_1 is given in (2.3) and
lyche-numerical-linear-algebra_ F00592	48	1.000	`\leq 1`	≤ 1	where each p_i is a polynomial of degree ≤ 1. In particular, p_1 is given in (2.3) and
lyche-numerical-linear-algebra_ F00593	48	1.000	`p_{1}`	p_{1}	where each p_i is a polynomial of degree ≤ 1. In particular, p_1 is given in (2.3) and
lyche-numerical-linear-algebra_ F00594	48	0.656	`\leq 2`	≤ 2	continuous derivatives of order ≤ 2 (C^2). We consider here the following functions
lyche-numerical-linear-algebra_ F00595	48	0.656	`C^{2}`	C^{2}	continuous derivatives of order ≤ 2 (C^2). We consider here the following functions
lyche-numerical-linear-algebra_ F00596	49	0.950	`D_{2}`	D_{2}	**Definition 2.1 (The D_2-Spline Problem)** Given $n \in \mathbb{N}$, an interval $[a, b]$, $\boldsymbol{y} \in$
lyche-numerical-linear-algebra_ F00597	49	0.950	`\boldsymbol{y} \in`	$\boldsymbol{y} \in$	**Definition 2.1 (The D_2-Spline Problem)** Given $n \in \mathbb{N}$, an interval $[a, b]$, $\boldsymbol{y} \in$
lyche-numerical-linear-algebra_ F00598	49	1.000	`\mathbb{R}^{n+1}`	$\mathbb{R}^{n+1}$	$\mathbb{R}^{n+1}$, knots (sites) $x_1, \ldots, x_{n+1}$ given by (2.1) and numbers μ_1, μ_{n+1}. The problem
lyche-numerical-linear-algebra_ F00599	49	1.000	`x_{1}, \ldots, x_{n+1}`	$x_{1}, \ldots, x_{n+1}$	$\mathbb{R}^{n+1}$, knots (sites) $x_1, \ldots, x_{n+1}$ given by (2.1) and numbers μ_1, μ_{n+1}. The problem
lyche-numerical-linear-algebra_ F00600	49	1.000	`\mu_{1}, \mu_{n+1}`	μ_{1}, μ_{n+1}	$\mathbb{R}^{n+1}$, knots (sites) $x_1, \ldots, x_{n+1}$ given by (2.1) and numbers μ_1, μ_{n+1}. The problem
lyche-numerical-linear-algebra_ F00601	49	1.000	`g \in C^{2}[a, b]`	$g \in C^{2}[a, b]$	• **smoothness**: $g \in C^2[a, b]$, i.e., derivatives of order ≤ 2 are continuous on $\mathbb{R}$,
lyche-numerical-linear-algebra_ F00602	49	1.000	`g\left(x_{i}\right)=y_{i}, \quad i=1,2, \ldots, n+1`	$g\left(x_{i}\right)=y_{i}, \quad i=1,2, \ldots, n+1$	• **interpolation**: $g(x_i) = y_i, \quad i=1, 2, \ldots, n+1$,
lyche-numerical-linear-algebra_ F00603	49	1.000	`\boldsymbol{D}_{\mathbf{2}}`	$\boldsymbol{D}_{\mathbf{2}}$	• **D_2 boundary conditions**: $g''(a) = \mu_1, \quad g''(b) = \mu_{n+1}$.
lyche-numerical-linear-algebra_ F00604	49	1.000	`g^{\prime \prime}(a)=\mu_{1}, \quad g^{\prime \prime}(b)=\mu_{n+1}`	$g^{\prime \prime}(a)=\mu_{1}, \quad g^{\prime \prime}(b)=\mu_{n+1}$	• **D_2 boundary conditions**: $g''(a) = \mu_1, \quad g''(b) = \mu_{n+1}$.
lyche-numerical-linear-algebra_ F00605	49	1.000	`N`	N	We call g a D_2-**spline**. It is called an N-**spline** or **natural spline** if $\mu_1 = \mu_{n+1} = 0$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00606	49	1.000	\mu_{1}=\mu_{n+1}=0	$\mu_{1}=\mu_{n+1}=0$	We call g a D_2-**spline**. It is called an N-**spline** or **natural spline** if $\mu_1 = \mu_{n+1} = 0$.
lyche-numerical-linear-algebra_F00607	49	1.000	n=2	$n=2$	*Example 2.1 (A D_2-Spline)* Suppose we choose $n = 2$ and sample data from the
lyche-numerical-linear-algebra_F00608	49	1.000	f:[0,2] \rightarrow \mathbb{R}	$f:[0,2] \rightarrow \mathbb{R}$	function $f : [0, 2] \to \mathbb{R}$ given by $f(x) = x^4$. Thus we consider the D_2-spline
lyche-numerical-linear-algebra_F00609	49	1.000	[a, b]=[0,2], \boldsymbol{y}:=[0,1,16]^{T}	$[a, b]=[0,2], \boldsymbol{y}:=[0,1,16]^{T}$	problem with $[a, b] = [0, 2]$, $\boldsymbol{y} := [0, 1, 16]^T$ and $\mu_1 = g''(0) = 0$, $\mu_3 = g''(2) =$
lyche-numerical-linear-algebra_F00610	49	1.000	\mu_{1}=g^{\prime \prime}(0)=0, \mu_{3}=g^{\prime \prime}(2)=	$\mu_{1}=g^{\prime \prime}(0)=0, \mu_{3}=g^{\prime \prime}(2)=$	problem with $[a, b] = [0, 2]$, $\boldsymbol{y} := [0, 1, 16]^T$ and $\mu_1 = g''(0) = 0$, $\mu_3 = g''(2) =$
lyche-numerical-linear-algebra_F00611	49	1.000	x_{1}=0, x_{2}=1	$x_{1}=0, x_{2}=1$	48. The knots are $x_1 = 0$, $x_2 = 1$ and $x_3 = 2$. The function g given by
lyche-numerical-linear-algebra_F00612	49	1.000	x_{3}=2	$x_{3}=2$	48. The knots are $x_1 = 0$, $x_2 = 1$ and $x_3 = 2$. The function g given by
lyche-numerical-linear-algebra_F00613	49	1.000	p_{2}	p_{2}	is a D_2-spline solving this problem. Indeed, p_1 and p_2 are cubic polynomials. For
lyche-numerical-linear-algebra_F00614	49	1.000	p_{1}(1)=p_{2}(1)=1, p_{1}^{\prime}(1)=p_{2}^{\prime}(1)=4, p_{1}^{\prime \prime}(1)=p_{2}^{\prime \prime}(1)=	$p_{1}(1)=p_{2}(1)=1, p_{1}^{\prime}(1)=p_{2}^{\prime}(1)=4, p_{1}^{\prime \prime}(1)=p_{2}^{\prime \prime}(1)=$	smoothness we find $p_1(1) = p_2(1) = 1$, $p_1'(1) = p_2'(1) = 4$, $p_1''(1) = p_2''(1) =$
lyche-numerical-linear-algebra_F00615	49	1.000	g \in C^{2}[0,2]	$g \in C^{2}[0,2]$	9 which implies that $g \in C^2[0, 2]$. Finally we check that the interpolation and
lyche-numerical-linear-algebra_F00616	49	1.000	g(0)=p_{1}(0)=0, g(1)=p_{2}(1)=1, g(2)=	$g(0)=p_{1}(0)=0, g(1)=p_{2}(1)=1, g(2)=$	boundary conditions hold. Indeed, $g(0) = p_1(0) = 0$, $g(1) = p_2(1) = 1$, $g(2) =$
lyche-numerical-linear-algebra_F00617	49	1.000	p_{2}(2)=16, g^{\prime \prime}(0)=p_{1}^{\prime \prime}(0)=0	$p_{2}(2)=16, g^{\prime \prime}(0)=p_{1}^{\prime \prime}(0)=0$	$p_2(2) = 16$, $g''(0) = p_1''(0) = 0$ and $g''(2) = p_2''(2) = 48$. Note that $p_1'''(x) = 9 \neq$
lyche-numerical-linear-algebra_F00618	49	1.000	g^{\prime \prime}(2)=p_{2}^{\prime \prime}(2)=48	$g^{\prime \prime}(2)=p_{2}^{\prime \prime}(2)=48$	$p_2(2) = 16$, $g''(0) = p_1''(0) = 0$ and $g''(2) = p_2''(2) = 48$. Note that $p_1'''(x) = 9 \neq$
lyche-numerical-linear-algebra_F00619	49	1.000	p_{1}^{\prime \prime \prime}(x)=9 \neq	$p_{1}^{\prime \prime \prime}(x)=9 \neq$	$p_2(2) = 16$, $g''(0) = p_1''(0) = 0$ and $g''(2) = p_2''(2) = 48$. Note that $p_1'''(x) = 9 \neq$
lyche-numerical-linear-algebra_F00620	49	1.000	39=p_{2}^{\prime \prime \prime}(x)	$39=p_{2}^{\prime \prime \prime}(x)$	$39 = p_2'''(x)$ showing that the third derivative of g is piecewise constant with a jump
lyche-numerical-linear-algebra_F00621	49	1.000	3(n-1) C^{2}	$3(n-1) C^{2}$	interpolant. Indeed counting requirements we have $3(n - 1)$ C^2 conditions,
lyche-numerical-linear-algebra_F00622	49	1.000	4 n	$4 n$	$n + 1$ conditions (2.2), and two boundary conditions, adding to $4n$. Since a
lyche-numerical-linear-algebra_F00623	49	1.000	p_{1}, \ldots, p_{n}	$p_{1}, \ldots, p_{n}$	coefficients of the n polynomials $p_1, \ldots, p_n$ and give hope for uniqueness of the
lyche-numerical-linear-algebra_F00624	50	0.829	[0,2]	$[0,2]$	**Fig. 2.3** A cubic spline with one knot interpolating $f(x) = x^4$ on $[0, 2]$
lyche-numerical-linear-algebra_F00625	50	1.000	(0,2)	$(0,2)$	**Lemma 2.1 (Representing Each p_i Using $(0, 2)$ Interpolation)** *Given $a < b$,*
lyche-numerical-linear-algebra_F00626	50	1.000	a<b	$a<b$	**Lemma 2.1 (Representing Each p_i Using $(0, 2)$ Interpolation)** *Given $a < b$,*
lyche-numerical-linear-algebra_F00627	50	1.000	h=(b-a) / n	$h=(b-a) / n$	*$h = (b-a)/n$ with $n \geq 2$, $x_i = a+(i-1)h$, and numbers y_i, μ_i for $i = 1, \ldots, n+1$.*
lyche-numerical-linear-algebra_F00628	50	1.000	n \geq 2, x_{i}=a+(i-1) h	$n \geq 2, x_{i}=a+(i-1) h$	*$h = (b-a)/n$ with $n \geq 2$, $x_i = a+(i-1)h$, and numbers y_i, μ_i for $i = 1, \ldots, n+1$.*
lyche-numerical-linear-algebra_F00629	50	1.000	y_{i}, \mu_{i}	y_{i}, μ_{i}	*$h = (b-a)/n$ with $n \geq 2$, $x_i = a+(i-1)h$, and numbers y_i, μ_i for $i = 1, \ldots, n+1$.*
lyche-numerical-linear-algebra_F00630	50	1.000	i=1, \ldots, n+1	$i=1, \ldots, n+1$	*$h = (b-a)/n$ with $n \geq 2$, $x_i = a+(i-1)h$, and numbers y_i, μ_i for $i = 1, \ldots, n+1$.*
lyche-numerical-linear-algebra_F00631	51	1.000	1 \leq i \leq n	$1 \leq i \leq n$	***Proof*** Consider p_i in the form (2.7) for some $1 \leq i \leq n$. Evoking (2.6) we find
lyche-numerical-linear-algebra_F00632	51	1.000	p_{i}\left(x_{i}\right)=c_{i, 1}=y_{i}	$p_{i}\left(x_{i}\right)=c_{i, 1}=y_{i}$	$p_i(x_i) = c_{i,1} = y_i$. Since $p_i''(x) = 2c_{i,3}+6c_{i,4}(x-x_i)$ we obtain $c_{i,3}$ from $p_i''(x_i) =$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F00633	51	1.000	`p_{i}^{\prime \prime}(x)=2 c_{i, 3}+6 c_{i, 4} \left(x-x_{i}\right)`	$p_{i}^{\prime \prime}(x)=2 c_{i, 3}+6 c_{i, 4} \left(x-x_{i}\right)$	$p_i(x_i) = c_{i,1} = y_i$. Since $p_i''(x) = 2c_{i,3}+6c_{i,4}(x-x_i)$ we obtain $c_{i,3}$ from $p_i''(x_i) =$
lyche-numerical-linear-algebra_ F00634	51	1.000	`c_{i, 3}`	$c_{i, 3}$	$p_i(x_i) = c_{i,1} = y_i$. Since $p_i''(x) = 2c_{i,3}+6c_{i,4}(x-x_i)$ we obtain $c_{i,3}$ from $p_i''(x_i) =$
lyche-numerical-linear-algebra_ F00635	51	1.000	`p_{i}^{\prime \prime}\left(x_{i}\right)=`	$p_{i}^{\prime \prime}\left(x_{i}\right)=$	$p_i(x_i) = c_{i,1} = y_i$. Since $p_i''(x) = 2c_{i,3}+6c_{i,4}(x-x_i)$ we obtain $c_{i,3}$ from $p_i''(x_i) =$
lyche-numerical-linear-algebra_ F00636	51	1.000	`2 c_{i, 3}=\mu_{i}`	$2 c_{i, 3}=\mu_{i}$	$2c_{i,3} = \mu_i$ (a moment), and then $c_{i,4}$ from $p_i''(x_{i+1}) = \mu_i + 6hc_{i,4} = \mu_{i+1}$. Finally
lyche-numerical-linear-algebra_ F00637	51	1.000	`c_{i, 4}`	$c_{i, 4}$	$2c_{i,3} = \mu_i$ (a moment), and then $c_{i,4}$ from $p_i''(x_{i+1}) = \mu_i + 6hc_{i,4} = \mu_{i+1}$. Finally
lyche-numerical-linear-algebra_ F00638	51	1.000	`p_{i}^{\prime \prime}\left(x_{i+1}\right)=\mu_{i}+6 h c_{i, 4}=\mu_{i+1}`	$p_{i}^{\prime \prime}\left(x_{i+1}\right)=\mu_{i}+6 h c_{i, 4}=\mu_{i+1}$	$2c_{i,3} = \mu_i$ (a moment), and then $c_{i,4}$ from $p_i''(x_{i+1}) = \mu_i + 6hc_{i,4} = \mu_{i+1}$. Finally
lyche-numerical-linear-algebra_ F00639	51	1.000	`c_{i, 2}`	$c_{i, 2}$	we find $c_{i,2}$ by solving $p_i(x_{i+1}) = y_i + c_{i,2}h + \frac{\mu_i}{2}h^2 + \frac{\mu_{i+1}-\mu_i}{6h}h^3 = y_{i+1}$. For
lyche-numerical-linear-algebra_ F00640	51	1.000	`p_{i}\left(x_{i+1}\right)=y_{i}+c_{i, 2} h+\frac{ \mu_{i}}{2} h^{2}+\frac{\mu_{i+1}-\mu_{i}}{6 h} h^{3}=y_{i+1}`	$p_{i}\left(x_{i+1}\right)=y_{i}+c_{i, 2} h+\frac{ \mu_{i}}{2} h^{2}+\frac{\mu_{i+1}-\mu_{i}}{6 h} h^{3}=y_{i+1}$	we find $c_{i,2}$ by solving $p_i(x_{i+1}) = y_i + c_{i,2}h + \frac{\mu_i}{2}h^2 + \frac{\mu_{i+1}-\mu_i}{6h}h^3 = y_{i+1}$. For
lyche-numerical-linear-algebra_ F00641	51	1.000	`j=0,1,2,3`	$j=0,1,2,3$	$j = 0, 1, 2, 3$ the shifted powers $(x - x_i)^j$ constitute a basis for cubic polynomials
lyche-numerical-linear-algebra_ F00642	51	1.000	`\left(x-x_{i}\right)^{j}`	$\left(x-x_{i}\right)^{j}$	$j = 0, 1, 2, 3$ the shifted powers $(x - x_i)^j$ constitute a basis for cubic polynomials
lyche-numerical-linear-algebra_ F00643	51	0.999	`\mu_{1}, \ldots`	$\mu_{1}, \ldots$	**Theorem 2.1 (Constructing a D_2-Spline)** *Suppose for some moments* $\mu_1, \ldots,$
lyche-numerical-linear-algebra_ F00644	51	1.000	`\mu_{n+1}`	μ_{n+1}	μ_{n+1} *that each p_i is given as in Lemma 2.1 for* $i = 1, \ldots, n$. *If in addition*
lyche-numerical-linear-algebra_ F00645	51	0.973	`\mu_{1}, \ldots, \mu_{n+1}`	$\mu_{1}, \ldots, \mu_{n+1}$	$\mu_1, \ldots, \mu_{n+1}$. Consider the C^2 requirement. Since $p_{i-1}(x_i) = p_i(x_i) = y_i$ and
lyche-numerical-linear-algebra_ F00646	51	0.973	`p_{i-1}\left(x_{i}\right)=p_{i}\left(x_{i}\right)=y_{i}`	$p_{i-1}\left(x_{i}\right)=p_{i}\left(x_{i}\right)=y_{i}$	$\mu_1, \ldots, \mu_{n+1}$. Consider the C^2 requirement. Since $p_{i-1}(x_i) = p_i(x_i) = y_i$ and
lyche-numerical-linear-algebra_ F00647	51	1.000	`p_{i-1}^{\prime \prime}\left(x_{i}\right)=p_{i}^{\prime \prime}\left(x_{i}\right)=\mu_{i}`	$p_{i-1}^{\prime \prime}\left(x_{i}\right)=p_{i}^{\prime \prime}\left(x_{i}\right)=\mu_{i}$	$p_{i-1}''(x_i) = p_i''(x_i) = \mu_i$ for $i = 2, \ldots, n$ it follows that $g \in C^2$ if and only if
lyche-numerical-linear-algebra_ F00648	51	1.000	`i=2, \ldots, n`	$i=2, \ldots, n$	$p_{i-1}''(x_i) = p_i''(x_i) = \mu_i$ for $i = 2, \ldots, n$ it follows that $g \in C^2$ if and only if
lyche-numerical-linear-algebra_ F00649	51	1.000	`g \in C^{2}`	$g \in C^{2}$	$p_{i-1}''(x_i) = p_i''(x_i) = \mu_i$ for $i = 2, \ldots, n$ it follows that $g \in C^2$ if and only if
lyche-numerical-linear-algebra_ F00650	51	1.000	`p_{i-1}^{\prime}\left(x_{i}\right)=p_{i}^{\prime} \left(x_{i}\right)`	$p_{i-1}^{\prime}\left(x_{i}\right)=p_{i}^{\prime} \left(x_{i}\right)$	$p_{i-1}'(x_i) = p_i'(x_i)$ for $i = 2, \ldots, n$. By (2.7)
lyche-numerical-linear-algebra_ F00651	51	1.000	`g\left(x_{i}\right)=y_{i}, i=1, \ldots, n+1, g^{\prime \prime}(a)=p_{1}^{\prime \prime}\left(x_{1}\right)= \mu_{1}`	$g\left(x_{i}\right)=y_{i}, i=1, \ldots, n+1, g^{\prime \prime}(a)=p_{1}^{\prime \prime}\left(x_{1}\right)= \mu_{1}$	By construction $g(x_i) = y_i, i = 1, \ldots, n+1, g''(a) = p_1''(x_1) = \mu_1$ and $g''(b) =$
lyche-numerical-linear-algebra_ F00652	51	1.000	`g^{\prime \prime}(b)=`	$g^{\prime \prime}(b)=$	By construction $g(x_i) = y_i, i = 1, \ldots, n+1, g''(a) = p_1''(x_1) = \mu_1$ and $g''(b) =$
lyche-numerical-linear-algebra_ F00653	51	1.000	`p_{n}^{\prime \prime}\left(x_{n+1}\right)=\mu_{n+1}`	$p_{n}^{\prime \prime}\left(x_{n+1}\right)=\mu_{n+1}$	$p_n''(x_{n+1}) = \mu_{n+1}$. It follows that g solves the D_2-spline problem.
lyche-numerical-linear-algebra_ F00654	51	1.000	`\mu_{2}, \ldots, \mu_{n}`	$\mu_{2}, \ldots, \mu_{n}$	In order for the D_2-spline to exist we need to show that $\mu_2, \ldots, \mu_n$ always can be
lyche-numerical-linear-algebra_ F00655	51	1.000	`n \geq 3`	$n \geq 3$	determined from (2.9). For $n \geq 3$ and with μ_1 and μ_{n+1} given (2.9) can be written
lyche-numerical-linear-algebra_ F00656	52	1.000	`\boldsymbol{A}=\left[a_{i j}\right] \in \mathbb{C}^{n \times n}`	$\boldsymbol{A}=\left[a_{i j}\right] \in \mathbb{C}^{n \times n}$	**Definition 2.2 (Strict Diagonal Dominance)** The matrix $\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{n \times n}$ is
lyche-numerical-linear-algebra_ F00657	52	1.000	`\left|x_{k}\right|=\max _{i}\left|x_{i}\right|`	$\left\|x_{k}\right\|=\max _{i}\left\|x_{i}\right\|$	that $\|x_k\| = \max_i \|x_i\|$. Then

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00658	52	0.999	`\max _{1 \leq i \leq n}\left|x_{i}\right|=\left|x_{k}\right| \leq \frac{\left|b_{k}\right|}{\sigma_{k}} \leq \max _{1 \leq i \leq n}\left(\frac{\left|b_{i}\right|}{\sigma_{i}}\right)`	$\max _{1 \leq i \leq n}\left\|x_{i}\right\|=\left\|x_{k}\right\| \leq \frac{\left\|b_{k}\right\|}{\sigma_{k}} \leq \max _{1 \leq i \leq n}\left(\frac{\left\|b_{i}\right\|}{\sigma_{i}}\right)$	and this implies $\max_{1\leq i\leq n}\|x_i\| = \|x_k\| \leq \frac{\|b_k\|}{\sigma_k} \leq \max_{1\leq i\leq n}\left(\frac{\|b_i\|}{\sigma_i}\right)$. For the
lyche-numerical-linear-algebra_F00659	52	0.999	`\max _{1 \leq i \leq n}\left|x_{i}\right| \leq 0`	$\max _{1 \leq i \leq n}\left\|x_{i}\right\| \leq 0$	nonsingularity, if $\boldsymbol{A}\boldsymbol{x} = \mathbf{0}$, then $\max_{1\leq i\leq n}\|x_i\| \leq 0$ by (2.13), and so $\boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra_F00660	52	1.000	`g_{1}`	g_{1}	uniqueness suppose g_1 and g_2 are two D_2-splines interpolating the same data. Then
lyche-numerical-linear-algebra_F00661	52	1.000	`g_{2}`	g_{2}	uniqueness suppose g_1 and g_2 are two D_2-splines interpolating the same data. Then
lyche-numerical-linear-algebra_F00662	52	1.000	`g:=g_{1}-g_{2}`	$g:=g_{1}-g_{2}$	$g := g_1 - g_2$ is an N-spline satisfying (2.2) with $\boldsymbol{y} = \mathbf{0}$. The solution $[\mu_2, \ldots, \mu_n]^T$
lyche-numerical-linear-algebra_F00663	52	1.000	`\boldsymbol{y}=\mathbf{0}`	$\boldsymbol{y}=\mathbf{0}$	$g := g_1 - g_2$ is an N-spline satisfying (2.2) with $\boldsymbol{y} = \mathbf{0}$. The solution $[\mu_2, \ldots, \mu_n]^T$
lyche-numerical-linear-algebra_F00664	52	1.000	`\left[\mu_{2}, \ldots, \mu_{n}\right]^{T}`	$\left[\mu_{2}, \ldots, \mu_{n}\right]^{T}$	$g := g_1 - g_2$ is an N-spline satisfying (2.2) with $\boldsymbol{y} = \mathbf{0}$. The solution $[\mu_2, \ldots, \mu_n]^T$
lyche-numerical-linear-algebra_F00665	52	1.000	`\mu_{1}=\mu_{n+1}`	$\mu_{1}=\mu_{n+1}$	of (2.11) and also $\mu_1 = \mu_{n+1}$ are zero. It follows from (2.8) that all coefficients $c_{i,j}$
lyche-numerical-linear-algebra_F00666	52	1.000	`c_{i, j}`	$c_{i, j}$	of (2.11) and also $\mu_1 = \mu_{n+1}$ are zero. It follows from (2.8) that all coefficients $c_{i,j}$
lyche-numerical-linear-algebra_F00667	52	1.000	`g=0`	$g=0$	are zero. We conclude that $g = 0$ and $g_1 = g_2$.
lyche-numerical-linear-algebra_F00668	52	1.000	`g_{1}=g_{2}`	$g_{1}=g_{2}$	are zero. We conclude that $g = 0$ and $g_1 = g_2$.
lyche-numerical-linear-algebra_F00669	52	1.000	`B`	B	*Example 2.2 (Cubic B-Spline)* For the N-spline with $[a, b] = [0, 4]$ and $\boldsymbol{y} =$
lyche-numerical-linear-algebra_F00670	52	1.000	`[a, b]=[0,4]`	$[a, b]=[0,4]$	*Example 2.2 (Cubic B-Spline)* For the N-spline with $[a, b] = [0, 4]$ and $\boldsymbol{y} =$
lyche-numerical-linear-algebra_F00671	52	1.000	`\boldsymbol{y}=`	$\boldsymbol{y}=$	*Example 2.2 (Cubic B-Spline)* For the N-spline with $[a, b] = [0, 4]$ and $\boldsymbol{y} =$
lyche-numerical-linear-algebra_F00672	52	0.997	`\left[0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}, 0\right]`	$\left[0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}, 0\right]$	$[0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}, 0]$ the linear system (2.9) takes the form
lyche-numerical-linear-algebra_F00673	53	1.000	`\mu_{2}=\mu_{4}=1, \mu_{3}=-2`	$\mu_{2}=\mu_{4}=1, \mu_{3}=-2$	The solution is $\mu_2 = \mu_4 = 1$, $\mu_3 = -2$. The knotset is $\{0, 1, 2, 3, 4\}$. Using (2.8)
lyche-numerical-linear-algebra_F00674	53	1.000	`\{0,1,2,3,4\}`	$\{0,1,2,3,4\}$	The solution is $\mu_2 = \mu_4 = 1$, $\mu_3 = -2$. The knotset is $\{0, 1, 2, 3, 4\}$. Using (2.8)
lyche-numerical-linear-algebra_F00675	53	1.000	`(0,4)`	$(0,4)$	A plot of this spline is shown in Fig. 2.4. On $(0, 4)$ the function g equals the nonzero
lyche-numerical-linear-algebra_F00676	53	1.000	`D^{2}`	D^{2}	To find the D^2-spline g we have to solve the triangular system (2.11). Consider
lyche-numerical-linear-algebra_F00677	53	1.000	`\boldsymbol{A}=\operatorname{tridiag}\left(a_{i}, d_{i}, c_{i}\right) \in`	$\boldsymbol{A}=\operatorname{tridiag}\left(a_{i}, d_{i}, c_{i}\right) \in$	solving a general tridiagonal linear system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ where $\boldsymbol{A} = \operatorname{tridiag}(a_i, d_i, c_i) \in$
lyche-numerical-linear-algebra_F00678	54	1.000	`\mathbb{C}^{n \times n}`	$\mathbb{C}^{n \times n}$	$\mathbb{C}^{n\times n}$. Instead of using Gaussian elimination directly, we can construct two matrices
lyche-numerical-linear-algebra_F00679	54	1.000	`\boldsymbol{A}=\boldsymbol{L} \boldsymbol{U}`	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{L}$ and $\boldsymbol{U}$ such that $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}$. Since $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{L}\boldsymbol{U}\boldsymbol{x} = \boldsymbol{b}$ we can find $\boldsymbol{x}$ by solving
lyche-numerical-linear-algebra_F00680	54	1.000	`\boldsymbol{A} \boldsymbol{x}=\boldsymbol{L} \boldsymbol{U} \boldsymbol{x}=\boldsymbol{b}`	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{L} \boldsymbol{U} \boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{L}$ and $\boldsymbol{U}$ such that $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}$. Since $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{L}\boldsymbol{U}\boldsymbol{x} = \boldsymbol{b}$ we can find $\boldsymbol{x}$ by solving
lyche-numerical-linear-algebra_F00681	54	1.000	`\boldsymbol{L} \boldsymbol{z}=\boldsymbol{b}`	$\boldsymbol{L} \boldsymbol{z}=\boldsymbol{b}$	two systems $\boldsymbol{L}\boldsymbol{z} = \boldsymbol{b}$ and $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{z}$. Moreover $\boldsymbol{L}$ and $\boldsymbol{U}$ are both triangular and
lyche-numerical-linear-algebra_F00682	54	1.000	`\boldsymbol{U} \boldsymbol{x}=\boldsymbol{z}`	$\boldsymbol{U} \boldsymbol{x}=\boldsymbol{z}$	two systems $\boldsymbol{L}\boldsymbol{z} = \boldsymbol{b}$ and $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{z}$. Moreover $\boldsymbol{L}$ and $\boldsymbol{U}$ are both triangular and
lyche-numerical-linear-algebra_F00683	54	1.000	`u_{1}, u_{2}, \ldots, u_{n-1}`	$u_{1}, u_{2}, \ldots, u_{n-1}$	then $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}$. If $u_1, u_2, \ldots, u_{n-1}$ are nonzero then (2.16) is well defined. If in

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00684	54	0.999	`u_{n} \neq 0`	$u_{n} \neq 0$	addition $u_n \neq 0$ then we can solve $\boldsymbol{L}\boldsymbol{z} = \boldsymbol{b}$ and $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{z}$ for $\boldsymbol{z}$ and $\boldsymbol{x}$. We formulate
lyche-numerical-linear-algebra_F00685	54	0.999	`\boldsymbol{z}`	$\boldsymbol{z}$	addition $u_n \neq 0$ then we can solve $\boldsymbol{L}\boldsymbol{z} = \boldsymbol{b}$ and $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{z}$ for $\boldsymbol{z}$ and $\boldsymbol{x}$. We formulate
lyche-numerical-linear-algebra_F00686	54	1.000	`\boldsymbol{l} \in \mathbb{C}^{n-1}, \boldsymbol{u} \in \mathbb{C}^{n}`	$\boldsymbol{l} \in \mathbb{C}^{n-1}, \boldsymbol{u} \in \mathbb{C}^{n}$	this as two algorithms. In `trifactor`, vectors $\boldsymbol{l} \in \mathbb{C}^{n-1}$, $\boldsymbol{u} \in \mathbb{C}^n$ are computed
lyche-numerical-linear-algebra_F00687	54	0.852	`\boldsymbol{a}, \boldsymbol{c} \in \mathbb{C}^{n-1}, \boldsymbol{d} \in \mathbb{C}^{n}`	$\boldsymbol{a}, \boldsymbol{c} \in \mathbb{C}^{n-1}, \boldsymbol{d} \in \mathbb{C}^{n}$	from $\boldsymbol{a}, \boldsymbol{c} \in \mathbb{C}^{n-1}$, $\boldsymbol{d} \in \mathbb{C}^n$. This implements the LU factorization of a tridiagonal
lyche-numerical-linear-algebra_F00688	55	1.000	`\boldsymbol{l}, \in \mathbb{C}^{n-1}`	$\boldsymbol{l}, \in \mathbb{C}^{n-1}$	computed from a previous call to `trifactor`. Here $\boldsymbol{l}, \in \mathbb{C}^{n-1}$ and $\boldsymbol{u} \in \mathbb{C}^n$ are
lyche-numerical-linear-algebra_F00689	55	1.000	`\boldsymbol{u} \in \mathbb{C}^{n}`	$\boldsymbol{u} \in \mathbb{C}^{n}$	computed from a previous call to `trifactor`. Here $\boldsymbol{l}, \in \mathbb{C}^{n-1}$ and $\boldsymbol{u} \in \mathbb{C}^n$ are
lyche-numerical-linear-algebra_F00690	55	1.000	`\boldsymbol{b} \in \mathbb{C}^{n, r}`	$\boldsymbol{b} \in \mathbb{C}^{n, r}$	output from `trifactor` and $\boldsymbol{b} \in \mathbb{C}^{n,r}$ for some $r \in \mathbb{N}$:
lyche-numerical-linear-algebra_F00691	55	1.000	`r \in \mathbb{N}`	$r \in \mathbb{N}$	output from `trifactor` and $\boldsymbol{b} \in \mathbb{C}^{n,r}$ for some $r \in \mathbb{N}$:
lyche-numerical-linear-algebra_F00692	55	1.000	`u_{k}`	u_{k}	***Proof*** We show that the u_k's in (2.16) are nonzero for $k = 1, \ldots, n$. For this it is
lyche-numerical-linear-algebra_F00693	55	1.000	`k=1, \ldots, n`	$k=1, \ldots, n$	***Proof*** We show that the u_k's in (2.16) are nonzero for $k = 1, \ldots, n$. For this it is
lyche-numerical-linear-algebra_F00694	55	1.000	`a_{0}:=c_{n}:=0`	$a_{0}:=c_{n}:=0$	and where $a_0 := c_n := 0$. By assumption $\|u_1\| = \|d_1\| = \sigma_1 + \|c_1\|$. Suppose $\|u_k\| \geq$
lyche-numerical-linear-algebra_F00695	55	1.000	`\left|u_{1}\right|=\left|d_{1}\right|=\sigma_{1}+ \left|c_{1}\right|`	$\left\|u_{1}\right\|=\left\|d_{1}\right\|=\sigma_{1}+\left\|c_{1}\right\|$	and where $a_0 := c_n := 0$. By assumption $\|u_1\| = \|d_1\| = \sigma_1 + \|c_1\|$. Suppose $\|u_k\| \geq$
lyche-numerical-linear-algebra_F00696	55	1.000	`\left|u_{k}\right| \geq`	$\left\|u_{k}\right\| \geq$	and where $a_0 := c_n := 0$. By assumption $\|u_1\| = \|d_1\| = \sigma_1 + \|c_1\|$. Suppose $\|u_k\| \geq$
lyche-numerical-linear-algebra_F00697	55	0.999	`\sigma_{k}+\left|c_{k}\right|`	$\sigma_{k}+\left\|c_{k}\right\|$	$\sigma_k + \|c_k\|$ for some $1 \leq k \leq n-1$. Then $\|c_k\|/\|u_k\| < 1$ and by (2.16) and strict
lyche-numerical-linear-algebra_F00698	55	0.999	`1 \leq k \leq n-1`	$1 \leq k \leq n-1$	$\sigma_k + \|c_k\|$ for some $1 \leq k \leq n-1$. Then $\|c_k\|/\|u_k\| < 1$ and by (2.16) and strict
lyche-numerical-linear-algebra_F00699	55	0.999	`\left|c_{k}\right| /\left|u_{k}\right|<1`	$\left\|c_{k}\right\| /\left\|u_{k}\right\|<1$	$\sigma_k + \|c_k\|$ for some $1 \leq k \leq n-1$. Then $\|c_k\|/\|u_k\| < 1$ and by (2.16) and strict
lyche-numerical-linear-algebra_F00700	56	1.000	`u_{1}=d_{1}`	$u_{1}=d_{1}$	*given by* (2.16)*. Then* (2.17) *holds,* $u_1 = d_1$ *and*
lyche-numerical-linear-algebra_F00701	56	1.000	`l_{k}`	l_{k}	we cannot have severe growth in the computed elements u_k and l_k.
lyche-numerical-linear-algebra_F00702	56	1.000	`3 n-3`	$3 n-3$	onal matrix of order n using (2.16) is $3n-3$, while the number of arithmetic oper-
lyche-numerical-linear-algebra_F00703	56	1.000	`r(5 n-4)`	$r(5 n-4)$	ations for Algorithm `trisolve` is $r(5n-4)$, where r is the number of right-hand
lyche-numerical-linear-algebra_F00704	56	1.000	`O(n)`	$O(n)$	sides. This means that the complexity to solve a tridiagonal system is $O(n)$, or
lyche-numerical-linear-algebra_F00705	56	1.000	`8 n-7`	$8 n-7$	more precisely $8n-7$ when $r = 1$, and this number only grows linearly[2] with n.
lyche-numerical-linear-algebra_F00706	56	1.000	`r=1`	$r=1$	more precisely $8n-7$ when $r = 1$, and this number only grows linearly[2] with n.
lyche-numerical-linear-algebra_F00707	56	0.939	`[0,1]`	$[0,1]$	where f is a given continuous function on $[0, 1]$ and u is an unknown function. This
lyche-numerical-linear-algebra_F00708	56	0.939	`u`	u	where f is a given continuous function on $[0, 1]$ and u is an unknown function. This
lyche-numerical-linear-algebra_F00709	56	1.000	`u(0)=u(1)=`	$u(0)=u(1)=$	integration constants so that the homogeneous boundary conditions $u(0) = u(1) =$
lyche-numerical-linear-algebra_F00710	56	1.000	`f(x)=1`	$f(x)=1$	0 are satisfied. For example, if $f(x) = 1$ then $u(x) = x(x-1)/2$ is the solution.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00711	56	1.000	`u(x)=x(x-1) / 2`	$u(x)=x(x-1)/2$	0 are satisfied. For example, if $f(x)=1$ then $u(x)=x(x-1)/2$ is the solution.
lyche-numerical-linear-algebra_F00712	56	1.000	`O\left(n^{3}\right)`	$O\left(n^{3}\right)$	[2]We show in Sect. 3.3.2 that Gaussian elimination on a full $n \times n$ system is an $O(n^3)$ process.
lyche-numerical-linear-algebra_F00713	57	1.000	`h:=1 /(m+1)`	$h:=1/(m+1)$	positive integer m, let $h := 1/(m+1)$ be the discretization parameter, and replace
lyche-numerical-linear-algebra_F00714	57	0.816	`x_{j}:=j h`	$x_{j}:=jh$	the interval [0, 1] by grid points $x_j := jh$ for $j = 0, 1, \ldots, m+1$. We then obtain
lyche-numerical-linear-algebra_F00715	57	0.816	`j=0,1, \ldots, m+1`	$j=0,1,\ldots,m+1$	the interval [0, 1] by grid points $x_j := jh$ for $j = 0, 1, \ldots, m+1$. We then obtain
lyche-numerical-linear-algebra_F00716	57	0.958	`v_{j}`	v_{j}	approximations v_j to the exact solution $u(x_j)$ for $j = 1, \ldots, m$ by replacing the
lyche-numerical-linear-algebra_F00717	57	0.958	`u\left(x_{j}\right)`	$u\left(x_{j}\right)$	approximations v_j to the exact solution $u(x_j)$ for $j = 1, \ldots, m$ by replacing the
lyche-numerical-linear-algebra_F00718	57	1.000	`h^{2}`	h^{2}	Moving the h^2 factor to the right hand side this can be written as an $m \times m$ linear
lyche-numerical-linear-algebra_F00719	57	1.000	`\boldsymbol{T}`	$\boldsymbol{T}$	The matrix $\boldsymbol{T}$ is called the **second derivative matrix** and will occur frequently in
lyche-numerical-linear-algebra_F00720	57	0.999	`\boldsymbol{T}=\operatorname{tridiag}\left(a_{i}, d_{i}, c_{i}\right) \in`	$\boldsymbol{T}=\operatorname{tridiag}\left(a_{i}, d_{i}, c_{i}\right) \in$	this book. It is our second example of a tridiagonal matrix, $\boldsymbol{T} = \operatorname{tridiag}(a_i, d_i, c_i) \in$
lyche-numerical-linear-algebra_F00721	57	1.000	`\mathbb{R}^{m \times m}`	$\mathbb{R}^{m \times m}$	$\mathbb{R}^{m\times m}$, where in this case $a_i = c_i = -1$ and $d_i = 2$ for all i.
lyche-numerical-linear-algebra_F00722	57	1.000	`a_{i}=c_{i}=-1`	$a_{i}=c_{i}=-1$	$\mathbb{R}^{m\times m}$, where in this case $a_i = c_i = -1$ and $d_i = 2$ for all i.
lyche-numerical-linear-algebra_F00723	57	1.000	`d_{i}=2`	$d_{i}=2$	$\mathbb{R}^{m\times m}$, where in this case $a_i = c_i = -1$ and $d_i = 2$ for all i.
lyche-numerical-linear-algebra_F00724	58	1.000	`\boldsymbol{A}_{1}`	$\boldsymbol{A}_{1}$	They are all weakly diagonally dominant, but $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are singular, while $\boldsymbol{A}_3$ is
lyche-numerical-linear-algebra_F00725	58	1.000	`\boldsymbol{A}_{2}`	$\boldsymbol{A}_{2}$	They are all weakly diagonally dominant, but $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are singular, while $\boldsymbol{A}_3$ is
lyche-numerical-linear-algebra_F00726	58	1.000	`\boldsymbol{A}_{3}`	$\boldsymbol{A}_{3}$	They are all weakly diagonally dominant, but $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are singular, while $\boldsymbol{A}_3$ is
lyche-numerical-linear-algebra_F00727	58	1.000	`\operatorname{det}\left(\boldsymbol{A}_{3}\right)=4 \neq 0`	$\operatorname{det}\left(\boldsymbol{A}_{3}\right)=4 \neq 0$	has a zero row, and $\det(\boldsymbol{A}_3) = 4 \neq 0$. It follows that for the nonsingularity and
lyche-numerical-linear-algebra_F00728	58	1.000	`\left|d_{1}\right|>\left|c_{1}\right|`	$\left\lvert d_{1}\right\rvert>\left\lvert c_{1}\right\rvert$	$\mathbb{C}^{n\times n}$ *is tridiagonal and weakly diagonally dominant. If in addition* $\lvert d_1\rvert > \lvert c_1\rvert$ *and*
lyche-numerical-linear-algebra_F00729	58	0.574	`a_{i} \neq 0`	$a_{i} \neq 0$	$a_i \neq 0$ *for* $i = 1, \ldots, n-2$, *then* $\boldsymbol{A}$ *has a unique LU factorization* (2.15). *If in*
lyche-numerical-linear-algebra_F00730	58	0.574	`i=1, \ldots, n-2`	$i=1,\ldots,n-2$	$a_i \neq 0$ *for* $i = 1, \ldots, n-2$, *then* $\boldsymbol{A}$ *has a unique LU factorization* (2.15). *If in*
lyche-numerical-linear-algebra_F00731	58	0.574	`L U`	LU	$a_i \neq 0$ *for* $i = 1, \ldots, n-2$, *then* $\boldsymbol{A}$ *has a unique LU factorization* (2.15). *If in*
lyche-numerical-linear-algebra_F00732	58	1.000	`d_{n} \neq 0`	$d_{n} \neq 0$	*addition* $d_n \neq 0$, *then* $\boldsymbol{A}$ *is nonsingular.*
lyche-numerical-linear-algebra_F00733	58	0.999	`k=1, \ldots, n-1`	$k=1,\ldots,n-1$	factorization if the u_k's in (2.16) are nonzero for $k = 1, \ldots, n-1$. For this it is
lyche-numerical-linear-algebra_F00734	58	0.991	`\left|u_{k}\right|>\left|c_{k}\right|`	$\left\lvert u_{k}\right\rvert>\left\lvert c_{k}\right\rvert$	sufficient to show by induction that $\lvert u_k\rvert > \lvert c_k\rvert$ for $k = 1, \ldots, n-1$. By assumption
lyche-numerical-linear-algebra_F00735	58	0.798	`\left|u_{1}\right|=\left|d_{1}\right|>\left|c_{1}\right|`	$\left\lvert u_{1}\right\rvert=\left\lvert d_{1}\right\rvert>\left\lvert c_{1}\right\rvert$	$\lvert u_1\rvert = \lvert d_1\rvert > \lvert c_1\rvert$. Suppose $\lvert u_k\rvert > \lvert c_k\rvert$ for some $1 \leq k \leq n-2$. Then $\lvert c_k\rvert/\lvert u_k\rvert < 1$
lyche-numerical-linear-algebra_F00736	58	0.798	`1 \leq k \leq n-2`	$1 \leq k \leq n-2$	$\lvert u_1\rvert = \lvert d_1\rvert > \lvert c_1\rvert$. Suppose $\lvert u_k\rvert > \lvert c_k\rvert$ for some $1 \leq k \leq n-2$. Then $\lvert c_k\rvert/\lvert u_k\rvert < 1$
lyche-numerical-linear-algebra_F00737	58	1.000	`k=n-1`	$k=n-1$	This also holds for $k = n-1$ if $a_{n-1} \neq 0$. By (2.23) and weak diagonal
lyche-numerical-linear-algebra_F00738	58	1.000	`a_{n-1} \neq 0`	$a_{n-1} \neq 0$	This also holds for $k = n-1$ if $a_{n-1} \neq 0$. By (2.23) and weak diagonal

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00739	58	■ 1.000	`\left|u_{k+1}\right|>\left|d_{k+1}\right|-\left|a_{k}\right| \geq\left|c_{k+1}\right|`	$\left\|u_{k+1}\right\|>\left\|d_{k+1}\right\|-\left\|a_{k}\right\| \geq\left\|c_{k+1}\right\|$	dominance $\|u_{k+1}\| > \|d_{k+1}\| - \|a_k\| \geq \|c_{k+1}\|$ and it follows by induction that an
lyche-numerical-linear-algebra_F00740	58	■ 1.000	`\left|u_{n}\right|>\left|d_{n}\right|-\left|a_{n-1}\right| \geq 0`	$\left\|u_{n}\right\|>\left\|d_{n}\right\|-\left\|a_{n-1}\right\| \geq 0$	$a_{n-1} \neq 0$ then by (2.16) $\|u_n\| > \|d_n\| - \|a_{n-1}\| \geq 0$ by weak diagonal dominance,
lyche-numerical-linear-algebra_F00741	58	■ 1.000	`a_{n-1}=0`	$a_{n-1}=0$	while if $a_{n-1} = 0$ then again by (2.23) $\|u_n\| \geq \|d_n\| > 0$.
lyche-numerical-linear-algebra_F00742	58	■ 1.000	`\left|u_{n}\right| \geq\left|d_{n}\right|>0`	$\left\|u_{n}\right\| \geq\left\|d_{n}\right\|>0$	while if $a_{n-1} = 0$ then again by (2.23) $\|u_n\| \geq \|d_n\| > 0$.
lyche-numerical-linear-algebra_F00743	58	■ 1.000	`\boldsymbol{T} \boldsymbol{v}=\boldsymbol{b}`	$\boldsymbol{T} \boldsymbol{v}=\boldsymbol{b}$	Consider now the special system $\boldsymbol{T}\boldsymbol{v} = \boldsymbol{b}$ given by (2.21). The matrix $\boldsymbol{T}$ is weakly
lyche-numerical-linear-algebra_F00744	58	■ 1.000	`\boldsymbol{v}=\boldsymbol{T}^{-1} \boldsymbol{b}`	$\boldsymbol{v}=\boldsymbol{T}^{-1} \boldsymbol{b}$	solution of $\boldsymbol{T}\boldsymbol{v} = \boldsymbol{b}$ as $\boldsymbol{v} = \boldsymbol{T}^{-1}\boldsymbol{b}$. However, this is not a good idea. In fact, all
lyche-numerical-linear-algebra_F00745	58	■ 1.000	`\boldsymbol{T}^{-1}`	$\boldsymbol{T}^{-1}$	elements in $\boldsymbol{T}^{-1}$ are nonzero and the calculation of $\boldsymbol{T}^{-1}\boldsymbol{b}$ requires $O(n^2)$ operations.
lyche-numerical-linear-algebra_F00746	58	■ 1.000	`\boldsymbol{T}^{-1} \boldsymbol{b}`	$\boldsymbol{T}^{-1} \boldsymbol{b}$	elements in $\boldsymbol{T}^{-1}$ are nonzero and the calculation of $\boldsymbol{T}^{-1}\boldsymbol{b}$ requires $O(n^2)$ operations.
lyche-numerical-linear-algebra_F00747	58	■ 1.000	`O\left(n^{2}\right)`	$O\left(n^{2}\right)$	elements in $\boldsymbol{T}^{-1}$ are nonzero and the calculation of $\boldsymbol{T}^{-1}\boldsymbol{b}$ requires $O(n^2)$ operations.
lyche-numerical-linear-algebra_F00748	59	■ 1.000	`\boldsymbol{x} \in`	$\boldsymbol{x} \in$	Recall that if $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ is a square matrix and $\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ for some nonzero $\boldsymbol{x} \in$
lyche-numerical-linear-algebra_F00749	59	■ 1.000	`L`	L	Consider a horizontal beam of length L located between 0 and L on the x-axis of
lyche-numerical-linear-algebra_F00750	59	■ 1.000	`x=0`	$x=0$	the plane. We assume that the beam is fixed at $x = 0$ and $x = L$ and that a force F is
lyche-numerical-linear-algebra_F00751	59	■ 1.000	`x=L`	$x=L$	the plane. We assume that the beam is fixed at $x = 0$ and $x = L$ and that a force F is
lyche-numerical-linear-algebra_F00752	59	■ 1.000	`F`	F	the plane. We assume that the beam is fixed at $x = 0$ and $x = L$ and that a force F is
lyche-numerical-linear-algebra_F00753	59	■ 1.000	`(L, 0)`	$(L, 0)$	applied at $(L, 0)$ in the direction towards the origin. This situation can be modeled
lyche-numerical-linear-algebra_F00754	59	■ 1.000	`y(x)`	$y(x)$	where $y(x)$ is the vertical displacement of the beam at x, and R is a constant defined
lyche-numerical-linear-algebra_F00755	59	■ 1.000	`R`	R	where $y(x)$ is the vertical displacement of the beam at x, and R is a constant defined
lyche-numerical-linear-algebra_F00756	59	■ 1.000	`u:[0,1] \rightarrow \mathbb{R}`	$u:[0,1] \rightarrow \mathbb{R}$	by considering the function $u : [0, 1] \rightarrow \mathbb{R}$ given by $u(t) := y(tL)$. Since $u''(t) =$
lyche-numerical-linear-algebra_F00757	59	■ 1.000	`u(t):=y(t L)`	$u(t):=y(t L)$	by considering the function $u : [0, 1] \rightarrow \mathbb{R}$ given by $u(t) := y(tL)$. Since $u''(t) =$
lyche-numerical-linear-algebra_F00758	59	■ 1.000	`u^{\prime \prime}(t)=`	$u^{\prime \prime}(t)=$	by considering the function $u : [0, 1] \rightarrow \mathbb{R}$ given by $u(t) := y(tL)$. Since $u''(t) =$
lyche-numerical-linear-algebra_F00759	59	■ 1.000	`L^{2} y^{\prime \prime}(t L)`	$L^{2} y^{\prime \prime}(t L)$	$L^2 y''(tL)$, the problem (2.24) then becomes
lyche-numerical-linear-algebra_F00760	59	■ 1.000	`u=0`	$u=0$	Clearly $u = 0$ is a solution, but we can have nonzero solutions corresponding
lyche-numerical-linear-algebra_F00761	59	■ 1.000	`F=0`	$F=0$	called an eigenfunction. If $F = 0$ then $K = 0$ and $u = 0$ is the only solution, but
lyche-numerical-linear-algebra_F00762	59	■ 1.000	`K=0`	$K=0$	called an eigenfunction. If $F = 0$ then $K = 0$ and $u = 0$ is the only solution, but
lyche-numerical-linear-algebra_F00763	59	■ 1.000	`u(t)=\sin (\pi t)`	$u(t)=\sin (\pi t)$	With $u(t) = \sin(\pi t)$ we find $u''(t) = -\pi^2 u(t)$ and this u is a solution if $K = \pi^2$.
lyche-numerical-linear-algebra_F00764	59	■ 1.000	`u^{\prime \prime}(t)=-\pi^{2} u(t)`	$u^{\prime \prime}(t)=-\pi^{2} u(t)$	With $u(t) = \sin(\pi t)$ we find $u''(t) = -\pi^2 u(t)$ and this u is a solution if $K = \pi^2$.
lyche-numerical-linear-algebra_F00765	59	■ 1.000	`K=\pi^{2}`	$K=\pi^{2}$	With $u(t) = \sin(\pi t)$ we find $u''(t) = -\pi^2 u(t)$ and this u is a solution if $K = \pi^2$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F00766	59	1.000	`F=\frac{\pi^{2} R}{L^{2}}`	$F=\frac{\pi^{2} R}{L^{2}}$	find $F = \frac{\pi^2 R}{L^2}$.
lyche-numerical-linear-algebra_ F00767	59	1.000	`m \in \mathbb{N}, h:=1 /(m+`	$m \in \mathbb{N}, h:=1 /(m+$	We can approximate this eigenvalue numerically. Choosing $m \in \mathbb{N}$, $h := 1/(m +$
lyche-numerical-linear-algebra_ F00768	60	1.000	`v_{j} \approx u(j h)`	$v_{j} \approx u(j h)$	where $v_j \approx u(jh)$ for $j = 0, \ldots, m+1$. If we define $\lambda := h^2 K$ then we obtain the
lyche-numerical-linear-algebra_ F00769	60	1.000	`j=0, \ldots, m+1`	$j=0, \ldots, m+1$	where $v_j \approx u(jh)$ for $j = 0, \ldots, m+1$. If we define $\lambda := h^2 K$ then we obtain the
lyche-numerical-linear-algebra_ F00770	60	1.000	`\lambda:=h^{2} K`	$\lambda:=h^{2} K$	where $v_j \approx u(jh)$ for $j = 0, \ldots, m+1$. If we define $\lambda := h^2 K$ then we obtain the
lyche-numerical-linear-algebra_ F00771	60	1.000	`\lambda=4 \sin ^{2}(\pi h / 2)`	$\lambda=4 \sin ^{2}(\pi h / 2)$	that the smallest eigenvalue of (2.26) is given by $\lambda = 4\sin^2(\pi h/2)$. Since $\lambda =$
lyche-numerical-linear-algebra_ F00772	60	1.000	`\lambda=`	$\lambda=$	that the smallest eigenvalue of (2.26) is given by $\lambda = 4\sin^2(\pi h/2)$. Since $\lambda =$
lyche-numerical-linear-algebra_ F00773	60	1.000	`h^{2} K=\frac{h^{2} F L^{2}}{R}`	$h^{2} K=\frac{h^{2} F L^{2}}{R}$	$h^2 K = \frac{h^2 F L^2}{R}$ we can solve for F to obtain
lyche-numerical-linear-algebra_ F00774	60	1.000	`h`	h	For small h this is a good approximation to the value $\frac{\pi^2 R}{L^2}$ we computed above.
lyche-numerical-linear-algebra_ F00775	60	1.000	`\frac{\pi^{2} R}{L^{2}}`	$\frac{\pi^{2} R}{L^{2}}$	For small h this is a good approximation to the value $\frac{\pi^2 R}{L^2}$ we computed above.
lyche-numerical-linear-algebra_ F00776	60	0.997	`\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1)`	$\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1)$	The second derivative matrix $\boldsymbol{T} = \operatorname{tridiag}(-1, 2, -1)$ is a special case of the
lyche-numerical-linear-algebra_ F00777	60	0.990	`a, d \in \mathbb{R}`	$a, d \in \mathbb{R}$	where $a, d \in \mathbb{R}$. We call this the **1D test matrix**. It is symmetric and strictly
lyche-numerical-linear-algebra_ F00778	60	0.990	`\mathbf{1 D}`	$\mathbf{1 D}$	where $a, d \in \mathbb{R}$. We call this the **1D test matrix**. It is symmetric and strictly
lyche-numerical-linear-algebra_ F00779	60	1.000	`|d|>2|a|`	$\|d\|>2\|a\|$	diagonally dominant if $\|d\| > 2\|a\|$.
lyche-numerical-linear-algebra_ F00780	61	1.000	`m=3`	$m=3$	For $m = 3$,
lyche-numerical-linear-algebra_ F00781	61	1.000	`\boldsymbol{T}_{1}=\left(t_{k j}\right)_{k, j}=`	$\boldsymbol{T}_{1}=\left(t_{k j}\right)_{k, j}=$	**Lemma 2.2 (Eigenpairs of 1D Test Matrix)** *Suppose* $\boldsymbol{T}_1 = (t_{kj})_{k,j} =$
lyche-numerical-linear-algebra_ F00782	61	0.999	`\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}`	$\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$	$\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$ *with* $m \geq 2$, $a, d \in \mathbb{R}$, *and let* $h = 1/(m+1)$.
lyche-numerical-linear-algebra_ F00783	61	0.999	`m \geq 2, a, d \in \mathbb{R}`	$m \geq 2, a, d \in \mathbb{R}$	$\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$ *with* $m \geq 2$, $a, d \in \mathbb{R}$, *and let* $h = 1/(m+1)$.
lyche-numerical-linear-algebra_ F00784	61	0.999	`h=1 /(m+1)`	$h=1 /(m+1)$	$\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$ *with* $m \geq 2$, $a, d \in \mathbb{R}$, *and let* $h = 1/(m+1)$.
lyche-numerical-linear-algebra_ F00785	61	1.000	`\boldsymbol{T}_{1} \boldsymbol{s}_{j}=\lambda_{j} \boldsymbol{s}_{j}`	$\boldsymbol{T}_{1} \boldsymbol{s}_{j}=\lambda_{j} \boldsymbol{s}_{j}$	*1. We have* $\boldsymbol{T}_1 \boldsymbol{s}_j = \lambda_j \boldsymbol{s}_j$ *for* $j = 1, \ldots, m$, *where*
lyche-numerical-linear-algebra_ F00786	61	1.000	`1<k<m`	$1<k<m$	***Proof*** We find for $1 < k < m$
lyche-numerical-linear-algebra_ F00787	61	1.000	`k=1, m`	$k=1, m$	This also holds for $k = 1, m$, and part 1 follows. Since $j\pi h = j\pi/(m+1) \in (0, \pi)$
lyche-numerical-linear-algebra_ F00788	61	1.000	`j \pi h=j \pi /(m+1) \in(0, \pi)`	$j \pi h=j \pi /(m+1) \in(0, \pi)$	This also holds for $k = 1, m$, and part 1 follows. Since $j\pi h = j\pi/(m+1) \in (0, \pi)$
lyche-numerical-linear-algebra_ F00789	61	0.988	`(0, \pi)`	$(0, \pi)$	for $j = 1, \ldots, m$ and the cosine function is strictly monotone decreasing on $(0, \pi)$
lyche-numerical-linear-algebra_ F00790	61	1.000	`\boldsymbol{T}_{1}`	$\boldsymbol{T}_{1}$	the eigenvalues are distinct, and since $\boldsymbol{T}_1$ is symmetric it follows from Lemma 2.3
lyche-numerical-linear-algebra_ F00791	62	1.000	`i=\sqrt{-1}`	$i=\sqrt{-1}$	complex exponentials. With $i = \sqrt{-1}$ we find

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_F00792</td><td>62</td><td>1.000</td><td>\boldsymbol{A}^{*}:=\overline{\boldsymbol{A}}^{T}</td><td>$\boldsymbol{A}^{*}:=\overline{\boldsymbol{A}}^{T}$</td><td>Recall that the conjugate transpose of a matrix is defined by $\boldsymbol{A}^{*}:=\overline{\boldsymbol{A}}^{T}$, where</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00793</td><td>62</td><td>1.000</td><td>\boldsymbol{x} \neq 0</td><td>$\boldsymbol{x} \neq 0$</td><td>Proof Suppose $\boldsymbol{A}^{*}=\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{x}=\lambda\boldsymbol{x}$ with $\boldsymbol{x} \neq 0$. We multiply both sides of</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00794</td><td>62</td><td>0.998</td><td>\boldsymbol{x}^{*}</td><td>$\boldsymbol{x}^{*}$</td><td>$\boldsymbol{A}\boldsymbol{x}=\lambda\boldsymbol{x}$ from the left by $\boldsymbol{x}^{*}$ and divide by $\boldsymbol{x}^{*}\boldsymbol{x}$ to obtain $\lambda=\frac{\boldsymbol{x}^{*}\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^{*}\boldsymbol{x}}$. Taking</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00795</td><td>62</td><td>0.998</td><td>\boldsymbol{x}^{*} \boldsymbol{x}</td><td>$\boldsymbol{x}^{*} \boldsymbol{x}$</td><td>$\boldsymbol{A}\boldsymbol{x}=\lambda\boldsymbol{x}$ from the left by $\boldsymbol{x}^{*}$ and divide by $\boldsymbol{x}^{*}\boldsymbol{x}$ to obtain $\lambda=\frac{\boldsymbol{x}^{*}\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^{*}\boldsymbol{x}}$. Taking</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00796</td><td>62</td><td>0.998</td><td>\lambda=\frac{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{*} \boldsymbol{x}}</td><td>$\lambda=\frac{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{*} \boldsymbol{x}}$</td><td>$\boldsymbol{A}\boldsymbol{x}=\lambda\boldsymbol{x}$ from the left by $\boldsymbol{x}^{*}$ and divide by $\boldsymbol{x}^{*}\boldsymbol{x}$ to obtain $\lambda=\frac{\boldsymbol{x}^{*}\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^{*}\boldsymbol{x}}$. Taking</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00797</td><td>62</td><td>0.996</td><td>\bar{\lambda}=\lambda^{*}=\frac{\left(\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}\right)^{*}}{\left(\boldsymbol{x}^{*} \boldsymbol{x}\right)^{*}}= \frac{\boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{x}}{\boldsymbol{x}^{*} \boldsymbol{x}}= \frac{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}{ \boldsymbol{x}^{*} \boldsymbol{x}}=\lambda</td><td>$\bar{\lambda}=\lambda^{*}=\frac{\left(\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}\right)^{*}}{\left(\boldsymbol{x}^{*} \boldsymbol{x}\right)^{*}}= \frac{\boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{x}}{\boldsymbol{x}^{*} \boldsymbol{x}}= \frac{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}{ \boldsymbol{x}^{*} \boldsymbol{x}}=\lambda$</td><td>complex conjugates we find $\bar{\lambda}=\lambda^{*}=\frac{(\boldsymbol{x}^{*}\boldsymbol{A}\boldsymbol{x})^{*}}{(\boldsymbol{x}^{*}\boldsymbol{x})^{*}}=\frac{\boldsymbol{x}^{*}\boldsymbol{A}^{*}\boldsymbol{x}}{\boldsymbol{x}^{*}\boldsymbol{x}}=\frac{\boldsymbol{x}^{*}\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^{*}\boldsymbol{x}}=\lambda$, and λ is real.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00798</td><td>62</td><td>0.725</td><td>(\mu, \boldsymbol{y})</td><td>$(\mu, \boldsymbol{y})$</td><td>Suppose that $(\lambda, \boldsymbol{x})$ and $(\mu, \boldsymbol{y})$ are two eigenpairs for $\boldsymbol{A}$ with $\mu \neq \lambda$. Multiplying</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00799</td><td>62</td><td>0.725</td><td>\mu \neq \lambda</td><td>$\mu \neq \lambda$</td><td>Suppose that $(\lambda, \boldsymbol{x})$ and $(\mu, \boldsymbol{y})$ are two eigenpairs for $\boldsymbol{A}$ with $\mu \neq \lambda$. Multiplying</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00800</td><td>62</td><td>1.000</td><td>\boldsymbol{y}^{*}</td><td>$\boldsymbol{y}^{*}$</td><td>$\boldsymbol{A}\boldsymbol{x}=\lambda\boldsymbol{x}$ by $\boldsymbol{y}^{*}$ gives</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00801</td><td>62</td><td>1.000</td><td>\mu</td><td>μ</td><td>using that μ is real. Since $\lambda \neq \mu$ it follows that $\boldsymbol{y}^{*}\boldsymbol{x}=0$, which means that $\boldsymbol{x}$ and $\boldsymbol{y}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00802</td><td>62</td><td>1.000</td><td>\lambda \neq \mu</td><td>$\lambda \neq \mu$</td><td>using that μ is real. Since $\lambda \neq \mu$ it follows that $\boldsymbol{y}^{*}\boldsymbol{x}=0$, which means that $\boldsymbol{x}$ and $\boldsymbol{y}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00803</td><td>62</td><td>1.000</td><td>\boldsymbol{y}^{*} \boldsymbol{x}=0</td><td>$\boldsymbol{y}^{*} \boldsymbol{x}=0$</td><td>using that μ is real. Since $\lambda \neq \mu$ it follows that $\boldsymbol{y}^{*}\boldsymbol{x}=0$, which means that $\boldsymbol{x}$ and $\boldsymbol{y}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00804</td><td>62</td><td>1.000</td><td>\boldsymbol{y}</td><td>$\boldsymbol{y}$</td><td>using that μ is real. Since $\lambda \neq \mu$ it follows that $\boldsymbol{y}^{*}\boldsymbol{x}=0$, which means that $\boldsymbol{x}$ and $\boldsymbol{y}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00805</td><td>63</td><td>0.997</td><td>\left[\begin{array}{ll}\boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22}\end{array}\right]=\left[\begin{array}{l|ll}1 & 2 & 3 \\ \hline 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right]</td><td>$\left[\begin{array}{ll}\boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22}\end{array}\right]=\left[\begin{array}{l|ll}1 & 2 & 3 \\ \hline 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right]$</td><td>$(i)\ \left[\begin{array}{ll}\boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22}\end{array}\right]=\left[\begin{array}{l|ll}1 & 2 & 3 \\ \hline 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right],$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00806</td><td>63</td><td>0.949</td><td>\left[\boldsymbol{a}_{: 1}, \boldsymbol{a}_{: 2}, \boldsymbol{a}_{: 3}\right]=\left[\begin{array}{c|c|c}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right]</td><td>$\left[\boldsymbol{a}_{: 1}, \boldsymbol{a}_{: 2}, \boldsymbol{a}_{: 3}\right]=\left[\begin{array}{c|c|c}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right]$</td><td>$(ii)\ \left[\boldsymbol{a}_{:1}, \boldsymbol{a}_{:2}, \boldsymbol{a}_{:3}\right]=\left[\begin{array}{c|c|c}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right],$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00807</td><td>63</td><td>0.125</td><td>\left[\begin{array}{c}\boldsymbol{a}_{1}^{T} \\ \boldsymbol{a}_{2:}^{T} \\ \boldsymbol{a}_{3:}^{T} \end{array}\right]=\left[\begin{array}{l}\frac{123}{45} 6 \\ \hline \frac{789}{788}\end{array}\right]</td><td>$\left[\begin{array}{c}\boldsymbol{a}_{1}^{T} \\ \boldsymbol{a}_{2:}^{T} \\ \boldsymbol{a}_{3:}^{T} \end{array}\right]=\left[\begin{array}{l}\frac{123}{45} 6 \\ \hline \frac{789}{788}\end{array}\right]$</td><td>$(iii)\ \left[\begin{array}{c}\boldsymbol{a}_{1:}^{T} \\ \boldsymbol{a}_{2:}^{T} \\ \boldsymbol{a}_{3:}^{T}\end{array}\right]=\left[\begin{array}{ccc}1 & 2 & 3 \\ \hline 4 & 5 & 6 \\ \hline 7 & 8 & 9\end{array}\right],$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F00808</td><td>63</td><td>0.673</td><td>\left[\boldsymbol{A}_{11}, \boldsymbol{A}_{12}\right]= \left[\begin{array}{c|cc}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right]</td><td>$\left[\boldsymbol{A}_{11}, \boldsymbol{A}_{12}\right]= \left[\begin{array}{c|cc}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right]$</td><td>$(iv)\ \left[\boldsymbol{A}_{11}, \boldsymbol{A}_{12}\right]=\left[\begin{array}{c|cc}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right].$</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F00809	63	0.902	`(i i)`	(ii)	while in (ii) and (iii) $\boldsymbol{A}$ has been partitioned into columns and rows, respectively.
lyche-numerical-linear-algebra_ F00810	63	0.902	`(i i i) \boldsymbol{A}`	$(iii)\boldsymbol{A}$	while in (ii) and (iii) $\boldsymbol{A}$ has been partitioned into columns and rows, respectively.
lyche-numerical-linear-algebra_ F00811	63	1.000	`\boldsymbol{A} \in \mathbb{C}^{m \times p}`	$\boldsymbol{A} \in \mathbb{C}^{m \times p}$	In the following we assume that $\boldsymbol{A} \in \mathbb{C}^{m \times p}$ and $\boldsymbol{B} \in \mathbb{C}^{p \times n}$. Here are some rules
lyche-numerical-linear-algebra_ F00812	63	1.000	`\boldsymbol{B} \in \mathbb{C}^{p \times n}`	$\boldsymbol{B} \in \mathbb{C}^{p \times n}$	In the following we assume that $\boldsymbol{A} \in \mathbb{C}^{m \times p}$ and $\boldsymbol{B} \in \mathbb{C}^{p \times n}$. Here are some rules
lyche-numerical-linear-algebra_ F00813	63	1.000	`\boldsymbol{B}=\left[\boldsymbol{b}_{: 1}, \ldots, \boldsymbol{b}_{: n}\right]`	$\boldsymbol{B}=\left[\boldsymbol{b}_{: 1}, \ldots, \boldsymbol{b}_{: n}\right]$	1. If $\boldsymbol{B}=\left[\boldsymbol{b}_{: 1}, \ldots, \boldsymbol{b}_{: n}\right]$ is partitioned into columns then the partition of the product
lyche-numerical-linear-algebra_ F00814	63	1.000	`\boldsymbol{A} \boldsymbol{e}_{j}`	$\boldsymbol{A} \boldsymbol{e}_{j}$	and we see that column j of $\boldsymbol{A}$ can be written $\boldsymbol{A} \boldsymbol{e}_{j}$ for $j=1, \ldots, p$.
lyche-numerical-linear-algebra_ F00815	63	1.000	`j=1, \ldots, p`	$j=1, \ldots, p$	and we see that column j of $\boldsymbol{A}$ can be written $\boldsymbol{A} \boldsymbol{e}_{j}$ for $j=1, \ldots, p$.
lyche-numerical-linear-algebra_ F00816	63	1.000	`\boldsymbol{A}=\boldsymbol{I}`	$\boldsymbol{A}=\boldsymbol{I}$	and taking $\boldsymbol{A}=\boldsymbol{I}$ it follows that row i of $\boldsymbol{B}$ can be written $\boldsymbol{e}_{i}^{T} \boldsymbol{B}$ for $i=1, \ldots, m$.
lyche-numerical-linear-algebra_ F00817	63	1.000	`\boldsymbol{e}_{i}^{T} \boldsymbol{B}`	$\boldsymbol{e}_{i}^{T} \boldsymbol{B}$	and taking $\boldsymbol{A}=\boldsymbol{I}$ it follows that row i of $\boldsymbol{B}$ can be written $\boldsymbol{e}_{i}^{T} \boldsymbol{B}$ for $i=1, \ldots, m$.
lyche-numerical-linear-algebra_ F00818	63	1.000	`\boldsymbol{A} \boldsymbol{x}`	$\boldsymbol{A} \boldsymbol{x}$	3. It is often useful to write the matrix-vector product $\boldsymbol{A}\boldsymbol{x}$ as a linear combination
lyche-numerical-linear-algebra_ F00819	64	1.000	`\boldsymbol{B}=\left[\boldsymbol{B}_{1}, \boldsymbol{B}_{2}\right]`	$\boldsymbol{B}=\left[\boldsymbol{B}_{1}, \boldsymbol{B}_{2}\right]$	4. If $\boldsymbol{B}=\left[\boldsymbol{B}_{1}, \boldsymbol{B}_{2}\right]$, where $\boldsymbol{B}_{1} \in \mathbb{C}^{p \times r}$ and $\boldsymbol{B}_{2} \in \mathbb{C}^{p \times(n-r)}$ then
lyche-numerical-linear-algebra_ F00820	64	1.000	`\boldsymbol{B}_{1} \in \mathbb{C}^{p \times r}`	$\boldsymbol{B}_{1} \in \mathbb{C}^{p \times r}$	4. If $\boldsymbol{B}=\left[\boldsymbol{B}_{1}, \boldsymbol{B}_{2}\right]$, where $\boldsymbol{B}_{1} \in \mathbb{C}^{p \times r}$ and $\boldsymbol{B}_{2} \in \mathbb{C}^{p \times(n-r)}$ then
lyche-numerical-linear-algebra_ F00821	64	1.000	`\boldsymbol{B}_{2} \in \mathbb{C}^{p \times(n-r)}`	$\boldsymbol{B}_{2} \in \mathbb{C}^{p \times(n-r)}$	4. If $\boldsymbol{B}=\left[\boldsymbol{B}_{1}, \boldsymbol{B}_{2}\right]$, where $\boldsymbol{B}_{1} \in \mathbb{C}^{p \times r}$ and $\boldsymbol{B}_{2} \in \mathbb{C}^{p \times(n-r)}$ then
lyche-numerical-linear-algebra_ F00822	64	0.993	`\boldsymbol{A}=\left[\begin{array}{l}\boldsymbol{A}_{1} \\ \boldsymbol{A}_{2}\end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{l}\boldsymbol{A}_{1} \\ \boldsymbol{A}_{2}\end{array}\right]$	5. If $\boldsymbol{A}=\begin{bmatrix}\boldsymbol{A}_{1} \\ \boldsymbol{A}_{2}\end{bmatrix}$, where $\boldsymbol{A}_{1} \in \mathbb{C}^{k \times p}$ and $\boldsymbol{A}_{2} \in \mathbb{C}^{(m-k) \times p}$ then
lyche-numerical-linear-algebra_ F00823	64	0.993	`\boldsymbol{A}_{1} \in \mathbb{C}^{k \times p}`	$\boldsymbol{A}_{1} \in \mathbb{C}^{k \times p}$	5. If $\boldsymbol{A}=\begin{bmatrix}\boldsymbol{A}_{1} \\ \boldsymbol{A}_{2}\end{bmatrix}$, where $\boldsymbol{A}_{1} \in \mathbb{C}^{k \times p}$ and $\boldsymbol{A}_{2} \in \mathbb{C}^{(m-k) \times p}$ then
lyche-numerical-linear-algebra_ F00824	64	0.993	`\boldsymbol{A}_{2} \in \mathbb{C}^{(m-k) \times p}`	$\boldsymbol{A}_{2} \in \mathbb{C}^{(m-k) \times p}$	5. If $\boldsymbol{A}=\begin{bmatrix}\boldsymbol{A}_{1} \\ \boldsymbol{A}_{2}\end{bmatrix}$, where $\boldsymbol{A}_{1} \in \mathbb{C}^{k \times p}$ and $\boldsymbol{A}_{2} \in \mathbb{C}^{(m-k) \times p}$ then
lyche-numerical-linear-algebra_ F00825	64	0.985	`\boldsymbol{A}=\left[\boldsymbol{A}_{1}, \boldsymbol{A}_{2}\right]`	$\boldsymbol{A}=\left[\boldsymbol{A}_{1}, \boldsymbol{A}_{2}\right]$	6. If $\boldsymbol{A}=\left[\boldsymbol{A}_{1}, \boldsymbol{A}_{2}\right]$ and $\boldsymbol{B}=\begin{bmatrix}\boldsymbol{B}_{1} \\ \boldsymbol{B}_{2}\end{bmatrix}$, where $\boldsymbol{A}_{1} \in \mathbb{C}^{m \times s}$, $\boldsymbol{A}_{2} \in \mathbb{C}^{m \times(p-s)}$, $\boldsymbol{B}_{1} \in$
lyche-numerical-linear-algebra_ F00826	64	0.985	`\boldsymbol{B}=\left[\begin{array}{l}\boldsymbol{B}_{1} \\ \boldsymbol{B}_{2}\end{array}\right]`	$\boldsymbol{B}=\left[\begin{array}{l}\boldsymbol{B}_{1} \\ \boldsymbol{B}_{2}\end{array}\right]$	6. If $\boldsymbol{A}=\left[\boldsymbol{A}_{1}, \boldsymbol{A}_{2}\right]$ and $\boldsymbol{B}=\begin{bmatrix}\boldsymbol{B}_{1} \\ \boldsymbol{B}_{2}\end{bmatrix}$, where $\boldsymbol{A}_{1} \in \mathbb{C}^{m \times s}$, $\boldsymbol{A}_{2} \in \mathbb{C}^{m \times(p-s)}$, $\boldsymbol{B}_{1} \in$
lyche-numerical-linear-algebra_ F00827	64	0.985	`\boldsymbol{A}_{1} \in \mathbb{C}^{m \times s}, \boldsymbol{A}_{2} \in \mathbb{C}^{m \times(p-s)}, \boldsymbol{B}_{1} \in`	$\boldsymbol{A}_{1} \in \mathbb{C}^{m \times s}, \boldsymbol{A}_{2} \in \mathbb{C}^{m \times(p-s)}, \boldsymbol{B}_{1} \in$	6. If $\boldsymbol{A}=\left[\boldsymbol{A}_{1}, \boldsymbol{A}_{2}\right]$ and $\boldsymbol{B}=\begin{bmatrix}\boldsymbol{B}_{1} \\ \boldsymbol{B}_{2}\end{bmatrix}$, where $\boldsymbol{A}_{1} \in \mathbb{C}^{m \times s}$, $\boldsymbol{A}_{2} \in \mathbb{C}^{m \times(p-s)}$, $\boldsymbol{B}_{1} \in$
lyche-numerical-linear-algebra_ F00828	64	1.000	`\mathbb{C}^{s \times n}`	$\mathbb{C}^{s \times n}$	$\mathbb{C}^{s \times n}$ and $\boldsymbol{B}_{2} \in \mathbb{C}^{(p-s) \times n}$ then
lyche-numerical-linear-algebra_ F00829	64	1.000	`\boldsymbol{B}_{2} \in \mathbb{C}^{(p-s) \times n}`	$\boldsymbol{B}_{2} \in \mathbb{C}^{(p-s) \times n}$	$\mathbb{C}^{s \times n}$ and $\boldsymbol{B}_{2} \in \mathbb{C}^{(p-s) \times n}$ then
lyche-numerical-linear-algebra_ F00830	64	1.000	`\boldsymbol{A}=\left[\begin{array}{ll}\boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22}\end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{ll}\boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22}\end{array}\right]$	7. If $\boldsymbol{A}=\begin{bmatrix}\boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22}\end{bmatrix}$ and $\boldsymbol{B}=\begin{bmatrix}\boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22}\end{bmatrix}$ then

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00831	64	1.000	`\boldsymbol{B}=\left[\begin{array}{ll}\boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22}\end{array}\right]`	$\boldsymbol{B}=\left[\begin{array}{ll}\boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22}\end{array}\right]$	7. If $\boldsymbol{A}=\begin{bmatrix}\boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22}\end{bmatrix}$ and $\boldsymbol{B}=\begin{bmatrix}\boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22}\end{bmatrix}$ then
lyche-numerical-linear-algebra_F00832	64	1.000	`\boldsymbol{A}_{11}`	$\boldsymbol{A}_{11}$	number of columns in $\boldsymbol{A}_{11}$ and $\boldsymbol{A}_{21}$ equals the number of rows in $\boldsymbol{B}_{11}$ and $\boldsymbol{B}_{12}$
lyche-numerical-linear-algebra_F00833	64	1.000	`\boldsymbol{A}_{21}`	$\boldsymbol{A}_{21}$	number of columns in $\boldsymbol{A}_{11}$ and $\boldsymbol{A}_{21}$ equals the number of rows in $\boldsymbol{B}_{11}$ and $\boldsymbol{B}_{12}$
lyche-numerical-linear-algebra_F00834	64	1.000	`\boldsymbol{B}_{11}`	$\boldsymbol{B}_{11}$	number of columns in $\boldsymbol{A}_{11}$ and $\boldsymbol{A}_{21}$ equals the number of rows in $\boldsymbol{B}_{11}$ and $\boldsymbol{B}_{12}$
lyche-numerical-linear-algebra_F00835	64	1.000	`\boldsymbol{B}_{12}`	$\boldsymbol{B}_{12}$	number of columns in $\boldsymbol{A}_{11}$ and $\boldsymbol{A}_{21}$ equals the number of rows in $\boldsymbol{B}_{11}$ and $\boldsymbol{B}_{12}$
lyche-numerical-linear-algebra_F00836	65	1.000	`\boldsymbol{A}_{i k} \boldsymbol{B}_{k j}`	$\boldsymbol{A}_{i k} \boldsymbol{B}_{k j}$	8. Consider finally the general case. If all the matrix products $\boldsymbol{A}_{ik}\boldsymbol{B}_{kj}$ in
lyche-numerical-linear-algebra_F00837	65	1.000	`\boldsymbol{A}, \boldsymbol{A}_{11}`	$\boldsymbol{A}, \boldsymbol{A}_{11}$	*where* $\boldsymbol{A}, \boldsymbol{A}_{11}$ *and* $\boldsymbol{A}_{22}$ *are square matrices. Then* $\boldsymbol{A}$ *is nonsingular if and only if*
lyche-numerical-linear-algebra_F00838	65	1.000	`\boldsymbol{A}_{22}`	$\boldsymbol{A}_{22}$	*where* $\boldsymbol{A}, \boldsymbol{A}_{11}$ *and* $\boldsymbol{A}_{22}$ *are square matrices. Then* $\boldsymbol{A}$ *is nonsingular if and only if*
lyche-numerical-linear-algebra_F00839	65	1.000	`\boldsymbol{B}:=\boldsymbol{A}^{-1}`	$\boldsymbol{B}:=\boldsymbol{A}^{-1}$	***Proof*** Suppose $\boldsymbol{A}$ is nonsingular. We partition $\boldsymbol{B}:=\boldsymbol{A}^{-1}$ conformally with $\boldsymbol{A}$ and
lyche-numerical-linear-algebra_F00840	66	1.000	`\boldsymbol{B}_{21}=`	$\boldsymbol{B}_{21}=$	The first equation implies that $\boldsymbol{A}_{11}$ is nonsingular, this in turn implies that $\boldsymbol{B}_{21}=$
lyche-numerical-linear-algebra_F00841	66	0.577	`\mathbf{0 A}_{11}^{-1}=\mathbf{0}`	$\mathbf{0 A}_{11}^{-1}=\mathbf{0}$	$\mathbf{0}\boldsymbol{A}_{11}^{-1}=\mathbf{0}$ in the second equation, and then the third equation simplifies to
lyche-numerical-linear-algebra_F00842	66	1.000	`\boldsymbol{B}_{22} \boldsymbol{A}_{22}=\boldsymbol{I}`	$\boldsymbol{B}_{22} \boldsymbol{A}_{22}=\boldsymbol{I}$	$\boldsymbol{B}_{22}\boldsymbol{A}_{22}=\boldsymbol{I}$. We conclude that also $\boldsymbol{A}_{22}$ is nonsingular. From the fourth equation
lyche-numerical-linear-algebra_F00843	66	1.000	`a_{i i}`	$a_{i i}$	$\boldsymbol{A}=[a_{ij}]\in\mathbb{C}^{n\times n}$ *is nonsingular if and only if the diagonal elements* a_{ii},
lyche-numerical-linear-algebra_F00844	66	1.000	`a_{i i}^{-1}, i=1, \ldots, n`	$a_{i i}^{-1}, i=1, \ldots, n$	*diagonal elements* $a_{ii}^{-1}, i=1,\ldots,n$.
lyche-numerical-linear-algebra_F00845	66	0.785	`a_{11}`	a_{11}	$[a_{11}]$ is nonsingular if and only if $a_{11}\neq\mathbf{0}$ and in that case $\boldsymbol{A}^{-1}=[a_{11}^{-1}]$. Suppose
lyche-numerical-linear-algebra_F00846	66	0.785	`a_{11} \neq \mathbf{0}`	$a_{11} \neq \mathbf{0}$	$[a_{11}]$ is nonsingular if and only if $a_{11}\neq\mathbf{0}$ and in that case $\boldsymbol{A}^{-1}=[a_{11}^{-1}]$. Suppose
lyche-numerical-linear-algebra_F00847	66	0.785	`\boldsymbol{A}^{-1}=\left[a_{11}^{-1}\right]`	$\boldsymbol{A}^{-1}=\left[a_{11}^{-1}\right]$	$[a_{11}]$ is nonsingular if and only if $a_{11}\neq\mathbf{0}$ and in that case $\boldsymbol{A}^{-1}=[a_{11}^{-1}]$. Suppose
lyche-numerical-linear-algebra_F00848	66	1.000	`\boldsymbol{A} \in \mathbb{C}^{(k+1) \times(k+1)}`	$\boldsymbol{A} \in \mathbb{C}^{(k+1) \times(k+1)}$	the result holds for $n=k$ and let $\boldsymbol{A}\in\mathbb{C}^{(k+1)\times(k+1)}$ be upper triangular. We partition
lyche-numerical-linear-algebra_F00849	66	1.000	`\boldsymbol{A}_{k} \in \mathbb{C}^{k \times k}`	$\boldsymbol{A}_{k} \in \mathbb{C}^{k \times k}$	and note that $\boldsymbol{A}_k\in\mathbb{C}^{k\times k}$ is upper triangular. By Lemma 2.4 $\boldsymbol{A}$ is nonsingular if and
lyche-numerical-linear-algebra_F00850	66	1.000	`2.4 \boldsymbol{A}`	$2.4 \boldsymbol{A}$	and note that $\boldsymbol{A}_k\in\mathbb{C}^{k\times k}$ is upper triangular. By Lemma 2.4 $\boldsymbol{A}$ is nonsingular if and
lyche-numerical-linear-algebra_F00851	66	1.000	`\boldsymbol{A}_{k}`	$\boldsymbol{A}_{k}$	only if $\boldsymbol{A}_k$ and $(a_{k+1,k+1})$ are nonsingular and in that case
lyche-numerical-linear-algebra_F00852	66	1.000	`\left(a_{k+1, k+1}\right)`	$\left(a_{k+1, k+1}\right)$	only if $\boldsymbol{A}_k$ and $(a_{k+1,k+1})$ are nonsingular and in that case
lyche-numerical-linear-algebra_F00853	66	1.000	`\boldsymbol{c} \in \mathbb{C}^{n}`	$\boldsymbol{c} \in \mathbb{C}^{n}$	for some $\boldsymbol{c}\in\mathbb{C}^n$. By the induction hypothesis $\boldsymbol{A}_k$ is nonsingular if and only if the
lyche-numerical-linear-algebra_F00854	66	1.000	`a_{11}, \ldots, a_{k k}`	$a_{11}, \ldots, a_{k k}$	diagonal elements $a_{11},\ldots,a_{kk}$ of $\boldsymbol{A}_k$ are nonzero and in that case $\boldsymbol{A}_k^{-1}$ is upper
lyche-numerical-linear-algebra_F00855	66	1.000	`\boldsymbol{A}_{k}^{-1}`	$\boldsymbol{A}_{k}^{-1}$	diagonal elements $a_{11},\ldots,a_{kk}$ of $\boldsymbol{A}_k$ are nonzero and in that case $\boldsymbol{A}_k^{-1}$ is upper

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F00856	66	1.000	`a_{i i}^{-1}, i=1, \ldots, k`	$a_{i i}^{-1}, i=1, \ldots, k$	triangular with diagonal elements a_{ii}^{-1}, $i = 1, \ldots, k$. The result for $\boldsymbol{A}$ follows.
lyche-numerical-linear-algebra_ F00857	66	0.515	`\boldsymbol{C}=\boldsymbol{A B}=\left(c_{i j}\right)`	$\boldsymbol{C}=\boldsymbol{A B}=\left(c_{i j}\right)$	**Lemma 2.6 (Product of Triangular Matrices)** *The product* $\boldsymbol{C} = \boldsymbol{AB} = (c_{ij})$ *of*
lyche-numerical-linear-algebra_ F00858	66	1.000	`\boldsymbol{A}=\left(a_{i j}\right)`	$\boldsymbol{A}=\left(a_{i j}\right)$	*two upper (lower) triangular matrices* $\boldsymbol{A} = (a_{ij})$ *and* $\boldsymbol{B} = (b_{ij})$ *is upper (lower)*
lyche-numerical-linear-algebra_ F00859	66	1.000	`\boldsymbol{B}=\left(b_{i j}\right)`	$\boldsymbol{B}=\left(b_{i j}\right)$	*two upper (lower) triangular matrices* $\boldsymbol{A} = (a_{ij})$ *and* $\boldsymbol{B} = (b_{ij})$ *is upper (lower)*
lyche-numerical-linear-algebra_ F00860	66	1.000	`c_{i i}=a_{i i} b_{i i}`	$c_{i i}=a_{i i} b_{i i}$	*triangular with diagonal elements* $c_{ii} = a_{ii}b_{ii}$ *for all* i.
lyche-numerical-linear-algebra_ F00861	67	0.822	`\left\{\left(x-x_{i}\right)^{j}\right\}_{0 \leq j \leq n}`	$\left\{\left(x-x_{i}\right)^{j}\right\}_{0 \leq j \leq n}$	$\{(x - x_i)^j\}_{0 \le j \le n}$ is a basis for polynomials of degree n.[3]
lyche-numerical-linear-algebra_ F00862	67	0.822	`n .^{3}`	$n .^{3}$	$\{(x - x_i)^j\}_{0 \le j \le n}$ is a basis for polynomials of degree n.[3]
lyche-numerical-linear-algebra_ F00863	67	1.000	`D_{1}`	D_{1}	pose instead of the D_2 boundary conditions we use D_1 conditions given by $g'(a) =$
lyche-numerical-linear-algebra_ F00864	67	1.000	`g^{\prime}(a)=`	$g^{\prime}(a)=$	pose instead of the D_2 boundary conditions we use D_1 conditions given by $g'(a) =$
lyche-numerical-linear-algebra_ F00865	67	1.000	`s_{1}`	s_{1}	s_1 and $g'(b) = s_{n+1}$ for some given numbers s_1 and s_{n+1}. Show that the linear
lyche-numerical-linear-algebra_ F00866	67	1.000	`g^{\prime}(b)=s_{n+1}`	$g^{\prime}(b)=s_{n+1}$	s_1 and $g'(b) = s_{n+1}$ for some given numbers s_1 and s_{n+1}. Show that the linear
lyche-numerical-linear-algebra_ F00867	67	1.000	`s_{n+1}`	s_{n+1}	s_1 and $g'(b) = s_{n+1}$ for some given numbers s_1 and s_{n+1}. Show that the linear
lyche-numerical-linear-algebra_ F00868	67	1.000	`\delta^{2} y_{i}:=y_{i+1}-2 y_{i}+y_{i-1}, i=2, \ldots, n`	$\delta^{2} y_{i}:=y_{i+1}-2 y_{i}+y_{i-1}, i=2, \ldots, n$	where $\delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}$, $i = 2, \ldots, n$. Hint: Use (2.10) to compute $g'(x_1)$
lyche-numerical-linear-algebra_ F00869	67	1.000	`g^{\prime}\left(x_{1}\right)`	$g^{\prime}\left(x_{1}\right)$	where $\delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}$, $i = 2, \ldots, n$. Hint: Use (2.10) to compute $g'(x_1)$
lyche-numerical-linear-algebra_ F00870	67	1.000	`g^{\prime}\left(x_{n+1}\right)`	$g^{\prime}\left(x_{n+1}\right)$	and $g'(x_{n+1})$, Is g unique?
lyche-numerical-linear-algebra_ F00871	67	1.000	`x_{i}`	x_{i}	[3]Hint: consider an arbitrary polynomial of degree n and expanded it in Taylor series around x_i.
lyche-numerical-linear-algebra_ F00872	68	0.999	`h \in C^{2}[a, b]`	$h \in C^{2}[a, b]$	*for all* $h \in C^2[a, b]$ *such that* $h(x_i) = g(x_i)$, $i = 1, \ldots, n + 1$.
lyche-numerical-linear-algebra_ F00873	68	0.999	`h\left(x_{i}\right)=g\left(x_{i}\right), i=1, \ldots, n+1`	$h\left(x_{i}\right)=g\left(x_{i}\right), i=1, \ldots, n+1$	*for all* $h \in C^2[a, b]$ *such that* $h(x_i) = g(x_i)$, $i = 1, \ldots, n + 1$.
lyche-numerical-linear-algebra_ F00874	68	1.000	`\int_{a}^{b} g^{\prime \prime} e^{\prime \prime}= \left[g^{\prime \prime} e^{\prime}\right]_{a}^{b}- \int_{a}^{b} g^{\prime \prime \prime} e^{\prime}`	$\int_{a}^{b} g^{\prime \prime} e^{\prime \prime}=\left[g^{\prime \prime} e^{\prime}\right]_{a}^{b}-\int_{a}^{b} g^{\prime \prime \prime} e^{\prime}$	Integration by parts gives $\int_a^b g''e'' = [g''e']_a^b - \int_a^b g'''e'$. The first term is zero since
lyche-numerical-linear-algebra_ F00875	68	1.000	`g^{\prime \prime}`	$g^{\prime \prime}$	g'' is continuous and $g''(b) = g''(a) = 0$. For the second term, since g''' is equal to
lyche-numerical-linear-algebra_ F00876	68	1.000	`g^{\prime \prime}(b)=g^{\prime \prime}(a)=0`	$g^{\prime \prime}(b)=g^{\prime \prime}(a)=0$	g'' is continuous and $g''(b) = g''(a) = 0$. For the second term, since g''' is equal to
lyche-numerical-linear-algebra_ F00877	68	1.000	`g^{\prime \prime \prime}`	$g^{\prime \prime \prime}$	g'' is continuous and $g''(b) = g''(a) = 0$. For the second term, since g''' is equal to
lyche-numerical-linear-algebra_ F00878	68	0.998	`v_{i}`	v_{i}	a constant v_i on each subinterval (x_i, x_{i+1}) and $e(x_i) = 0$, for $i = 1, \ldots, n + 1$
lyche-numerical-linear-algebra_ F00879	68	0.998	`x_{i}, x_{i+1}`	x_{i}, x_{i+1}	a constant v_i on each subinterval (x_i, x_{i+1}) and $e(x_i) = 0$, for $i = 1, \ldots, n + 1$
lyche-numerical-linear-algebra_ F00880	68	0.998	`e\left(x_{i}\right)=0`	$e\left(x_{i}\right)=0$	a constant v_i on each subinterval (x_i, x_{i+1}) and $e(x_i) = 0$, for $i = 1, \ldots, n + 1$
lyche-numerical-linear-algebra_ F00881	68	1.000	`h=g+e`	$h=g+e$	Writing $h = g + e$ and using (2.35)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00882	68	1.000	\boldsymbol{y} \in \mathbb{R}^{n+1}	$\boldsymbol{y} \in \mathbb{R}^{n+1}$	an interval $[a, b]$, a vector $\boldsymbol{y} \in \mathbb{R}^{n+1}$ and μ_1, μ_{n+1}. The vector $\boldsymbol{x} = [x_1, \ldots, x_n]$ and
lyche-numerical-linear-algebra_F00883	68	1.000	\boldsymbol{x}=\left[x_{1}, \ldots, x_{n}\right]	$\boldsymbol{x}=\left[x_{1}, \ldots, x_{n}\right]$	an interval $[a, b]$, a vector $\boldsymbol{y} \in \mathbb{R}^{n+1}$ and μ_1, μ_{n+1}. The vector $\boldsymbol{x} = [x_1, \ldots, x_n]$ and
lyche-numerical-linear-algebra_F00884	69	1.000	\boldsymbol{C} \in \mathbb{R}^{n \times 4}	$\boldsymbol{C} \in \mathbb{R}^{n \times 4}$	the coefficient matrix $\boldsymbol{C} \in \mathbb{R}^{n\times 4}$ in (2.7) are returned in the following algorithm. It
lyche-numerical-linear-algebra_F00885	69	1.000	g\left(r_{j}\right)	$g\left(r_{j}\right)$	form (2.4) we need to compute values $g(r_j)$ at a number of sites $\boldsymbol{r} = [r_1, \ldots, r_m] \in$
lyche-numerical-linear-algebra_F00886	69	1.000	\boldsymbol{r}=\left[r_{1}, \ldots, r_{m}\right] \in	$\boldsymbol{r}=\left[r_{1}, \ldots, r_{m}\right] \in$	form (2.4) we need to compute values $g(r_j)$ at a number of sites $\boldsymbol{r} = [r_1, \ldots, r_m] \in$
lyche-numerical-linear-algebra_F00887	69	1.000	i_{j}	i_{j}	find an integer i_j so that $g(r_j) = p_{i_j}(r_j)$.
lyche-numerical-linear-algebra_F00888	69	1.000	g\left(r_{j}\right)=p_{i_{j}}\left(r_{j}\right)	$g\left(r_{j}\right)=p_{i_{j}}\left(r_{j}\right)$	find an integer i_j so that $g(r_j) = p_{i_j}(r_j)$.
lyche-numerical-linear-algebra_F00889	69	1.000	k \in \mathbb{N}, \boldsymbol{t}=\left[t_{1}, \ldots, t_{k}\right]	$k \in \mathbb{N}, \boldsymbol{t}=\left[t_{1}, \ldots, t_{k}\right]$	Given $k \in \mathbb{N}$, $\boldsymbol{t} = [t_1, \ldots, t_k]$ and a real number x. We consider the problem
lyche-numerical-linear-algebra_F00890	69	1.000	i=1	$i=1$	of computing an integer i so that $i = 1$ if $x < t_2$, $i = k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F00891	69	1.000	x<t_{2}, i=k	$x<t_{2}, i=k$	of computing an integer i so that $i = 1$ if $x < t_2$, $i = k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F00892	69	1.000	x \geq t_{k}	$x \geq t_{k}$	of computing an integer i so that $i = 1$ if $x < t_2$, $i = k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F00893	69	1.000	t_{i} \leq	$t_{i} \leq$	of computing an integer i so that $i = 1$ if $x < t_2$, $i = k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F00894	69	1.000	x<t_{i+1}	$x<t_{i+1}$	$x < t_{i+1}$ otherwise. If $\boldsymbol{x} \in \mathbb{R}^m$ is a vector then an m-vector $\boldsymbol{i}$ should be computed,
lyche-numerical-linear-algebra_F00895	69	1.000	\boldsymbol{x} \in \mathbb{R}^{m}	$\boldsymbol{x} \in \mathbb{R}^{m}$	$x < t_{i+1}$ otherwise. If $\boldsymbol{x} \in \mathbb{R}^m$ is a vector then an m-vector $\boldsymbol{i}$ should be computed,
lyche-numerical-linear-algebra_F00896	69	1.000	\boldsymbol{i}	$\boldsymbol{i}$	$x < t_{i+1}$ otherwise. If $\boldsymbol{x} \in \mathbb{R}^m$ is a vector then an m-vector $\boldsymbol{i}$ should be computed,
lyche-numerical-linear-algebra_F00897	69	1.000	\boldsymbol{i}=\left[i_{1}, \ldots, i_{m}\right]	$\boldsymbol{i}=\left[i_{1}, \ldots, i_{m}\right]$	The following MATLAB function determines $\boldsymbol{i} = [i_1, \ldots, i_m]$. It uses the built in
lyche-numerical-linear-algebra_F00898	70	1.000	\boldsymbol{x}, \boldsymbol{C}	$\boldsymbol{x}, \boldsymbol{C}$	Given output $\boldsymbol{x}, \boldsymbol{C}$ of splineint, defining a cubic spline g, and a vector $\boldsymbol{X}$,
lyche-numerical-linear-algebra_F00899	70	1.000	\boldsymbol{G}=g(\boldsymbol{X})	$\boldsymbol{G}=g(\boldsymbol{X})$	splinevalăcomputes the vector $\boldsymbol{G} = g(\boldsymbol{X})$.
lyche-numerical-linear-algebra_F00900	71	1.000	f \in C^{2}[x-h, x+h]	$f \in C^{2}[x-h, x+h]$	**a)** Show using Taylor expansion that if $f \in C^2[x-h, x+h]$ then for some η_2
lyche-numerical-linear-algebra_F00901	71	1.000	\eta_{2}	η_{2}	**a)** Show using Taylor expansion that if $f \in C^2[x-h, x+h]$ then for some η_2
lyche-numerical-linear-algebra_F00902	71	1.000	f \in C^{4}[x-h, x+h]	$f \in C^{4}[x-h, x+h]$	**b)** Show that if $f \in C^4[x-h, x+h]$ then for some η_4
lyche-numerical-linear-algebra_F00903	71	1.000	\eta_{4}	η_{4}	**b)** Show that if $f \in C^4[x-h, x+h]$ then for some η_4
lyche-numerical-linear-algebra_F00904	71	1.000	\delta^{2} f(x)	$\delta^{2} f(x)$	$\delta^2 f(x)$ is known as the **central difference approximation** to the second derivative
lyche-numerical-linear-algebra_F00905	71	1.000	f, q	f, q	We assume that the given functions f, q and r are continuous on $[a, b]$ and that
lyche-numerical-linear-algebra_F00906	71	1.000	q(x) \geq 0	$q(x) \geq 0$	$q(x) \geq 0$ for $x \in [a, b]$. It can then be shown that (2.36) has a unique solution u.
lyche-numerical-linear-algebra_F00907	71	1.000	x \in[a, b]	$x \in[a, b]$	$q(x) \geq 0$ for $x \in [a, b]$. It can then be shown that (2.36) has a unique solution u.
lyche-numerical-linear-algebra_F00908	71	1.000	m \in \mathbb{N}, h=(b-a) /(m+1), x_{j}=a+j h	$m \in \mathbb{N}, h=(b-a) /(m+1), x_{j}=a+j h$	To solve (2.36) numerically we choose $m \in \mathbb{N}$, $h = (b-a)/(m+1)$, $x_j = a + jh$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00909	71	0.993	`v_{0}=g_{0}`	$v_{0}=g_{0}$	with $v_0 = g_0$ and $v_{m+1} = g_1$.
lyche-numerical-linear-algebra_F00910	71	0.993	`v_{m+1}=g_{1}`	$v_{m+1}=g_{1}$	with $v_0 = g_0$ and $v_{m+1} = g_1$.
lyche-numerical-linear-algebra_F00911	71	1.000	`\boldsymbol{A} \boldsymbol{v}=\boldsymbol{b}`	$\boldsymbol{A} \boldsymbol{v}=\boldsymbol{b}$	**a)** Show that (2.37) leads to a tridiagonal linear system $\boldsymbol{Av} = \boldsymbol{b}$, where $\boldsymbol{A} =$
lyche-numerical-linear-algebra_F00912	71	0.523	`\operatorname{tridiag}\left(a_{j}, d_{j} . c_{j}\right) \in \mathbb{R}^{m \times m}`	$\operatorname{tridiag}\left(a_{j}, d_{j} . c_{j}\right) \in \mathbb{R}^{m \times m}$	$\operatorname{tridiag}(a_j, d_j.c_j) \in \mathbb{R}^{m\times m}$ has elements
lyche-numerical-linear-algebra_F00913	72	1.000	`\frac{h}{2}|r(x)|<1`	$\frac{h}{2}\|r(x)\|<1$	h is so small that $\frac{h}{2}\|r(x)\| < 1$ for all $x \in [a, b]$.
lyche-numerical-linear-algebra_F00914	72	1.000	`v_{1}, \ldots, v_{m}`	$v_{1}, \ldots, v_{m}$	**(c)** Propose a method to find $v_1, \ldots, v_m$.
lyche-numerical-linear-algebra_F00915	72	1.000	`r=0, f=q=1`	$r=0, f=q=1$	**a)** Consider the problem (2.36) with $r = 0$, $f = q = 1$ and boundary conditions
lyche-numerical-linear-algebra_F00916	72	1.000	`u(0)=1, u(1)=0`	$u(0)=1, u(1)=0$	$u(0) = 1$, $u(1) = 0$. The exact solution is $u(x) = 1 - \sinh x/\sinh 1$. Write a
lyche-numerical-linear-algebra_F00917	72	1.000	`u(x)=1-\sinh x / \sinh 1`	$u(x)=1-\sinh x / \sinh 1$	$u(0) = 1$, $u(1) = 0$. The exact solution is $u(x) = 1 - \sinh x/\sinh 1$. Write a
lyche-numerical-linear-algebra_F00918	72	1.000	`h=0.1,0.05,0.025,0.0125`	$h=0.1,0.05,0.025,0.0125$	computer program to solve (2.37) for $h = 0.1, 0.05, 0.025, 0.0125$, and compute
lyche-numerical-linear-algebra_F00919	72	1.000	`\max _{1 \leq j \leq m}\left|u\left(x_{j}\right)-v_{j}\right|`	$\max _{1 \leq j \leq m}\left\|u\left(x_{j}\right)-v_{j}\right\|$	the "error" $\max_{1\le j\le m}\|u(x_j) - v_j\|$ for each h.
lyche-numerical-linear-algebra_F00920	72	1.000	`v_{j}, j=`	$v_{j}, j=$	**b)** Make a combined plot of the solution u and the computed points v_j, $j =$
lyche-numerical-linear-algebra_F00921	72	1.000	`0, \ldots, m+1`	$0, \ldots, m+1$	$0, \ldots, m + 1$ for $h = 0.1$.
lyche-numerical-linear-algebra_F00922	72	1.000	`h=0.1`	$h=0.1$	$0, \ldots, m + 1$ for $h = 0.1$.
lyche-numerical-linear-algebra_F00923	72	1.000	`h^{p}`	h^{p}	**c)** One can show that the error is proportional to h^p for some integer p. Estimate p
lyche-numerical-linear-algebra_F00924	72	1.000	`a_{i, i+1} a_{i+1, i}>0`	$a_{i, i+1} a_{i+1, i}>0$	tridiagonal and suppose $a_{i,i+1}a_{i+1,i} > 0$ for $i = 1, \ldots, n-1$. Show that there
lyche-numerical-linear-algebra_F00925	72	1.000	`\boldsymbol{D}=\operatorname{diag}\left(d_{1}, \ldots, d_{n}\right)`	$\boldsymbol{D}=\operatorname{diag}\left(d_{1}, \ldots, d_{n}\right)$	exists a diagonal matrix $\boldsymbol{D} = \operatorname{diag}(d_1, \ldots, d_n)$ with $d_i > 0$ for all i such that
lyche-numerical-linear-algebra_F00926	72	1.000	`d_{i}>0`	$d_{i}>0$	exists a diagonal matrix $\boldsymbol{D} = \operatorname{diag}(d_1, \ldots, d_n)$ with $d_i > 0$ for all i such that
lyche-numerical-linear-algebra_F00927	72	1.000	`\boldsymbol{B}:=\boldsymbol{D} \boldsymbol{A} \boldsymbol{D}^{-1}`	$\boldsymbol{B}:=\boldsymbol{D} \boldsymbol{A} \boldsymbol{D}^{-1}$	$\boldsymbol{B} := \boldsymbol{DAD}^{-1}$ is symmetric.
lyche-numerical-linear-algebra_F00928	73	1.000	`\boldsymbol{T}=\boldsymbol{L} \boldsymbol{U}`	$\boldsymbol{T}=\boldsymbol{L} \boldsymbol{U}$	**Exercise 2.14 (LU Factorization of 2. Derivative Matrix)** Show that $\boldsymbol{T} = \boldsymbol{LU}$,
lyche-numerical-linear-algebra_F00929	73	0.998	`\boldsymbol{S} \in \mathbb{R}^{m \times m}`	$\boldsymbol{S} \in \mathbb{R}^{m \times m}$	**Exercise 2.15 (Inverse of the 2. Derivative Matrix)** Let $\boldsymbol{S} \in \mathbb{R}^{m\times m}$ have elements
lyche-numerical-linear-algebra_F00930	73	1.000	`s_{i j}`	$s_{i j}$	s_{ij} given by
lyche-numerical-linear-algebra_F00931	73	1.000	`\boldsymbol{S} \boldsymbol{T}=\boldsymbol{I}`	$\boldsymbol{S} \boldsymbol{T}=\boldsymbol{I}$	Show that $\boldsymbol{ST} = \boldsymbol{I}$ and conclude that $\boldsymbol{T}^{-1} = \boldsymbol{S}$.
lyche-numerical-linear-algebra_F00932	73	1.000	`\boldsymbol{T}^{-1}=\boldsymbol{S}`	$\boldsymbol{T}^{-1}=\boldsymbol{S}$	Show that $\boldsymbol{ST} = \boldsymbol{I}$ and conclude that $\boldsymbol{T}^{-1} = \boldsymbol{S}$.
lyche-numerical-linear-algebra_F00933	73	1.000	`a_{i j}=\boldsymbol{e}_{i}^{T} \boldsymbol{A} \boldsymbol{e}_{j}`	$a_{i j}=\boldsymbol{e}_{i}^{T} \boldsymbol{A} \boldsymbol{e}_{j}$	that $a_{ij} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j$ for all i, j.
lyche-numerical-linear-algebra_F00934	73	1.000	`\boldsymbol{A}=\sum_{i=1}^{m} \sum_{j=1}^{n} a_{i j} \boldsymbol{e}_{i} \boldsymbol{e}_{j}^{T}`	$\boldsymbol{A}=\sum_{i=1}^{m} \sum_{j=1}^{n} a_{i j} \boldsymbol{e}_{i} \boldsymbol{e}_{j}^{T}$	show that $\boldsymbol{A} = \sum_{i=1}^m \sum_{j=1}^n a_{ij}\boldsymbol{e}_i\boldsymbol{e}_j^T$.
lyche-numerical-linear-algebra_F00935	73	0.942	`\boldsymbol{A}^{T} \boldsymbol{A}`	$\boldsymbol{A}^{T} \boldsymbol{A}$	**Exercise 2.18 (The Product $\boldsymbol{A}^T\boldsymbol{A}$)** Let $\boldsymbol{B} = \boldsymbol{A}^T\boldsymbol{A}$. Explain why this product is

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00936	73	■0.942	`\boldsymbol{B}=\boldsymbol{A}^{T} \boldsymbol{A}`	$\boldsymbol{B}=\boldsymbol{A}^{T} \boldsymbol{A}$	**Exercise 2.18 (The Product $A^T A$)** Let $\boldsymbol{B} = \boldsymbol{A}^T \boldsymbol{A}$. Explain why this product is
lyche-numerical-linear-algebra_F00937	73	■1.000	`b_{i j}=\boldsymbol{a}_{: i}^{T} \boldsymbol{a}_{: j}`	$b_{i j}=\boldsymbol{a}_{: i}^{T} \boldsymbol{a}_{: j}$	defined for any matrix $\boldsymbol{A}$. Show that $b_{ij} = \boldsymbol{a}_{:i}^T \boldsymbol{a}_{:j}$ for all i, j.
lyche-numerical-linear-algebra_F00938	73	■1.000	`\boldsymbol{B} \in \mathbb{R}^{p \times n}`	$\boldsymbol{B} \in \mathbb{R}^{p \times n}$	**Exercise 2.19 (Outer Product Expansion)** For $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ and $\boldsymbol{B} \in \mathbb{R}^{p\times n}$ show
lyche-numerical-linear-algebra_F00939	73	■1.000	`\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}`	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}$	$\boldsymbol{A} \in \mathbb{R}^{m\times n}$, $\boldsymbol{B} \in \mathbb{R}^{m\times p}$, and $\boldsymbol{X} \in \mathbb{R}^{n\times p}$. Show that
lyche-numerical-linear-algebra_F00940	73	■1.000	`\boldsymbol{X} \in \mathbb{R}^{n \times p}`	$\boldsymbol{X} \in \mathbb{R}^{n \times p}$	$\boldsymbol{A} \in \mathbb{R}^{m\times n}$, $\boldsymbol{B} \in \mathbb{R}^{m\times p}$, and $\boldsymbol{X} \in \mathbb{R}^{n\times p}$. Show that
lyche-numerical-linear-algebra_F00941	74	■0.989	`\boldsymbol{B}=`	$\boldsymbol{B}=$	**Exercise 2.21 (Block Multiplication Example)** Suppose $\boldsymbol{A} = [\boldsymbol{A}_1, \boldsymbol{A}_2]$ and $\boldsymbol{B} =$
lyche-numerical-linear-algebra_F00942	74	■1.000	`\left[\begin{array}{c}\boldsymbol{B}_{1} \\ \mathbf{0} \end{array}\right]`	$\left[\begin{array}{c}\boldsymbol{B}_{1} \\ \mathbf{0} \end{array}\right]$	$\begin{bmatrix}\boldsymbol{B}_1\\ \mathbf{0}\end{bmatrix}$. When is $\boldsymbol{A}\boldsymbol{B} = \boldsymbol{A}_1\boldsymbol{B}_1$?
lyche-numerical-linear-algebra_F00943	74	■1.000	`\boldsymbol{A} \boldsymbol{B}=\boldsymbol{A}_{1} \boldsymbol{B}_{1}`	$\boldsymbol{A} \boldsymbol{B}=\boldsymbol{A}_{1} \boldsymbol{B}_{1}$	$\begin{bmatrix}\boldsymbol{B}_1\\ \mathbf{0}\end{bmatrix}$. When is $\boldsymbol{A}\boldsymbol{B} = \boldsymbol{A}_1\boldsymbol{B}_1$?
lyche-numerical-linear-algebra_F00944	74	■0.999	`\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in`	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in$	**Exercise 2.22 (Another Block Multiplication Example)** Suppose $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in$
lyche-numerical-linear-algebra_F00945	74	■1.000	`\boldsymbol{A}_{1}, \boldsymbol{B}_{1}, \boldsymbol{C}_{1} \in \mathbb{R}^{(n-1) \times(n-1)}`	$\boldsymbol{A}_{1}, \boldsymbol{B}_{1}, \boldsymbol{C}_{1} \in \mathbb{R}^{(n-1) \times(n-1)}$	where $\boldsymbol{A}_1, \boldsymbol{B}_1, \boldsymbol{C}_1 \in \mathbb{R}^{(n-1)\times(n-1)}$. Show that
lyche-numerical-linear-algebra_F00946	74	■0.921	`\boldsymbol{T} \boldsymbol{x}=\boldsymbol{b} ?`	$\boldsymbol{T} \boldsymbol{x}=\boldsymbol{b} ?$	$\boldsymbol{T}\boldsymbol{x} = \boldsymbol{b}$?
lyche-numerical-linear-algebra_F00947	74	■1.000	`\left[\begin{array}{cc}1 & 1+i \\ 1+i & 2\end{array} \right]`	$\left[\begin{array}{cc}1 & 1+i \\ 1+i & 2\end{array} \right]$	2.7.6 Is the matrix $\left[\begin{smallmatrix}1 & 1+i\\ 1+i & 2\end{smallmatrix}\right]$ Hermitian? Symmetric?
lyche-numerical-linear-algebra_F00948	75	■1.000	`\boldsymbol{P}`	$\boldsymbol{P}$	factorization, if row interchanges are introduced. Here $\boldsymbol{P}$ is a permutation matrix, $\boldsymbol{L}$
lyche-numerical-linear-algebra_F00949	75	■0.951	`3 \times 3`	3×3	linear equations in n unknowns.[1] We first recall how it works on a 3×3 system.
lyche-numerical-linear-algebra_F00950	76	■1.000	`a_{11}^{(1)} \neq 0`	$a_{11}^{(1)} \neq 0$	To solve this system by Gaussian elimination suppose $a_{11}^{(1)} \neq 0$. We subtract $l_{21}^{(1)} :=$
lyche-numerical-linear-algebra_F00951	76	■1.000	`l_{21}^{(1)}:=`	$l_{21}^{(1)}:=$	To solve this system by Gaussian elimination suppose $a_{11}^{(1)} \neq 0$. We subtract $l_{21}^{(1)} :=$
lyche-numerical-linear-algebra_F00952	76	■0.997	`a_{21}^{(1)} / a_{11}^{(1)}`	$a_{21}^{(1)} / a_{11}^{(1)}$	$a_{21}^{(1)}/a_{11}^{(1)}$ times equation I from equation II and $l_{31}^{(1)} := a_{31}^{(1)}/a_{11}^{(1)}$ times equation I
lyche-numerical-linear-algebra_F00953	76	■0.997	`l_{31}^{(1)}:=a_{31}^{(1)} / a_{11}^{(1)}`	$l_{31}^{(1)}:=a_{31}^{(1)} / a_{11}^{(1)}$	$a_{21}^{(1)}/a_{11}^{(1)}$ times equation I from equation II and $l_{31}^{(1)} := a_{31}^{(1)}/a_{11}^{(1)}$ times equation I
lyche-numerical-linear-algebra_F00954	76	■1.000	`b_{i}^{(2)}=b_{i}^{(1)}-l_{i 1}^{(1)} b_{i}^{(1)}`	$b_{i}^{(2)}=b_{i}^{(1)}-l_{i 1}^{(1)} b_{i}^{(1)}$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_i^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i,1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00955	76	■1.000	`i=2,3`	$i=2,3$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_i^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i,1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00956	76	■1.000	`a_{i j}^{(2)}=a_{i j}^{(1)}-l_{i, 1}^{(1)} a_{1 j}^{(1)}`	$a_{i j}^{(2)}=a_{i j}^{(1)}-l_{i, 1}^{(1)} a_{1 j}^{(1)}$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_i^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i,1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00957	76	■1.000	`i, j=2,3`	$i, j=2,3$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_i^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i,1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00958	76	■1.000	`a_{11}^{(1)}=0`	$a_{11}^{(1)}=0$	If $a_{11}^{(1)} = 0$ and $a_{21}^{(1)} \neq 0$ we first interchange equation I and equation II. If $a_{11}^{(1)} =$
lyche-numerical-linear-algebra_F00959	76	■1.000	`a_{21}^{(1)} \neq 0`	$a_{21}^{(1)} \neq 0$	If $a_{11}^{(1)} = 0$ and $a_{21}^{(1)} \neq 0$ we first interchange equation I and equation II. If $a_{11}^{(1)} =$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00960	76	1.000	a_{11}^{(1)}=	$a_{11}^{(1)}=$	If $a_{11}^{(1)}=0$ and $a_{21}^{(1)}\neq 0$ we first interchange equation I and equation II. If $a_{11}^{(1)}=$
lyche-numerical-linear-algebra_F00961	76	0.997	a_{21}^{(1)}=0	$a_{21}^{(1)}=0$	$a_{21}^{(1)}=0$ we interchange equation I and III. Since the system is nonsingular the first
lyche-numerical-linear-algebra_F00962	76	0.705	a_{22}^{(2)} \neq 0	$a_{22}^{(2)} \neq 0$	If $a_{22}^{(2)}\neq 0$ we subtract $l_{32}^{(2)}:=a_{32}^{(2)}/a_{22}^{(2)}$ times equation II′ from equation III′ to
lyche-numerical-linear-algebra_F00963	76	0.705	l_{32}^{(2)}:=a_{32}^{(2)} / a_{22}^{(2)}	$l_{32}^{(2)}:=a_{32}^{(2)} / a_{22}^{(2)}$	If $a_{22}^{(2)}\neq 0$ we subtract $l_{32}^{(2)}:=a_{32}^{(2)}/a_{22}^{(2)}$ times equation II′ from equation III′ to
lyche-numerical-linear-algebra_F00964	76	0.705	{ }^{\prime}	${ }^{\prime}$	If $a_{22}^{(2)}\neq 0$ we subtract $l_{32}^{(2)}:=a_{32}^{(2)}/a_{22}^{(2)}$ times equation II′ from equation III′ to
lyche-numerical-linear-algebra_F00965	76	1.000	a_{33}^{(3)}=a_{33}^{(2)}-l_{32}^{(2)} a_{23}^{(2)}	$a_{33}^{(3)}=a_{33}^{(2)}-l_{32}^{(2)} a_{23}^{(2)}$	where $a_{33}^{(3)}=a_{33}^{(2)}-l_{32}^{(2)}a_{23}^{(2)}$ and $b_3^{(3)}=b_3^{(2)}-l_{32}^{(2)}b_2^{(2)}$. If $a_{22}^{(2)}=0$ then $a_{32}^{(2)}\neq 0$
lyche-numerical-linear-algebra_F00966	76	1.000	b_{3}^{(3)}=b_{3}^{(2)}-l_{32}^{(2)} b_{2}^{(2)}	$b_{3}^{(3)}=b_{3}^{(2)}-l_{32}^{(2)} b_{2}^{(2)}$	where $a_{33}^{(3)}=a_{33}^{(2)}-l_{32}^{(2)}a_{23}^{(2)}$ and $b_3^{(3)}=b_3^{(2)}-l_{32}^{(2)}b_2^{(2)}$. If $a_{22}^{(2)}=0$ then $a_{32}^{(2)}\neq 0$
lyche-numerical-linear-algebra_F00967	76	1.000	a_{22}^{(2)}=0	$a_{22}^{(2)}=0$	where $a_{33}^{(3)}=a_{33}^{(2)}-l_{32}^{(2)}a_{23}^{(2)}$ and $b_3^{(3)}=b_3^{(2)}-l_{32}^{(2)}b_2^{(2)}$. If $a_{22}^{(2)}=0$ then $a_{32}^{(2)}\neq 0$
lyche-numerical-linear-algebra_F00968	76	1.000	a_{32}^{(2)} \neq 0	$a_{32}^{(2)} \neq 0$	where $a_{33}^{(3)}=a_{33}^{(2)}-l_{32}^{(2)}a_{23}^{(2)}$ and $b_3^{(3)}=b_3^{(2)}-l_{32}^{(2)}b_2^{(2)}$. If $a_{22}^{(2)}=0$ then $a_{32}^{(2)}\neq 0$
lyche-numerical-linear-algebra_F00969	76	1.000	a_{k k}^{(k)} \neq 0, k=1,2	$a_{k k}^{(k)} \neq 0, k=1,2$	Indeed, if $a_{kk}^{(k)}\neq 0, k=1,2$ then
lyche-numerical-linear-algebra_F00970	77	1.000	\boldsymbol{A}^{(1)}	$\boldsymbol{A}^{(1)}$	$\boldsymbol{A}^{(1)}$ (cf. the proof of Theorem 3.2).
lyche-numerical-linear-algebra_F00971	77	1.000	\boldsymbol{A}^{(k)} \boldsymbol{x}=\boldsymbol{b}^{(k)}	$\boldsymbol{A}^{(k)} \boldsymbol{x}=\boldsymbol{b}^{(k)}$	$\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$ and generate a sequence of equivalent systems $\boldsymbol{A}^{(k)}\boldsymbol{x}=\boldsymbol{b}^{(k)}$ for $k=$
lyche-numerical-linear-algebra_F00972	77	1.000	k=	$k=$	$\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$ and generate a sequence of equivalent systems $\boldsymbol{A}^{(k)}\boldsymbol{x}=\boldsymbol{b}^{(k)}$ for $k=$
lyche-numerical-linear-algebra_F00973	77	1.000	1, \ldots, n	$1, \ldots, n$	$1,\ldots,n$, where $\boldsymbol{A}^{(1)}=\boldsymbol{A}$, $\boldsymbol{b}^{(1)}=\boldsymbol{b}$, and $\boldsymbol{A}^{(k)}$ has zeros under the diagonal in its
lyche-numerical-linear-algebra_F00974	77	1.000	\boldsymbol{A}^{(1)}=\boldsymbol{A}, \boldsymbol{b}^{(1)}=\boldsymbol{b}	$\boldsymbol{A}^{(1)}=\boldsymbol{A}, \boldsymbol{b}^{(1)}=\boldsymbol{b}$	$1,\ldots,n$, where $\boldsymbol{A}^{(1)}=\boldsymbol{A}$, $\boldsymbol{b}^{(1)}=\boldsymbol{b}$, and $\boldsymbol{A}^{(k)}$ has zeros under the diagonal in its
lyche-numerical-linear-algebra_F00975	77	1.000	\boldsymbol{A}^{(k)}	$\boldsymbol{A}^{(k)}$	$1,\ldots,n$, where $\boldsymbol{A}^{(1)}=\boldsymbol{A}$, $\boldsymbol{b}^{(1)}=\boldsymbol{b}$, and $\boldsymbol{A}^{(k)}$ has zeros under the diagonal in its
lyche-numerical-linear-algebra_F00976	77	1.000	k-1	$k-1$	first $k-1$ columns. Thus $\boldsymbol{A}^{(n)}$ is upper triangular and the system $\boldsymbol{A}^{(n)}\boldsymbol{x}=\boldsymbol{b}^{(n)}$ is
lyche-numerical-linear-algebra_F00977	77	1.000	\boldsymbol{A}^{(n)}	$\boldsymbol{A}^{(n)}$	first $k-1$ columns. Thus $\boldsymbol{A}^{(n)}$ is upper triangular and the system $\boldsymbol{A}^{(n)}\boldsymbol{x}=\boldsymbol{b}^{(n)}$ is
lyche-numerical-linear-algebra_F00978	77	1.000	\boldsymbol{A}^{(n)} \boldsymbol{x}=\boldsymbol{b}^{(n)}	$\boldsymbol{A}^{(n)} \boldsymbol{x}=\boldsymbol{b}^{(n)}$	first $k-1$ columns. Thus $\boldsymbol{A}^{(n)}$ is upper triangular and the system $\boldsymbol{A}^{(n)}\boldsymbol{x}=\boldsymbol{b}^{(n)}$ is
lyche-numerical-linear-algebra_F00979	77	1.000	\boldsymbol{A}^{(k+1)}	$\boldsymbol{A}^{(k+1)}$	The process transforming $\boldsymbol{A}^{(k)}$ into $\boldsymbol{A}^{(k+1)}$ for $k=1,\ldots,n-1$ can be described
lyche-numerical-linear-algebra_F00980	78	1.000	j=k	$j=k$	For $j=k$ it follows from (3.2) that $a_{ik}^{(k+1)}=a_{ik}^{(k)}-\frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}a_{kk}^{(k)}=0$ for $i=$
lyche-numerical-linear-algebra_F00981	78	1.000	a_{i k}^{(k+1)}=a_{i k}^{(k)}-\frac{a_{i k}^{(k)}}{a_{k k}^{(k)}} a_{k k}^{(k)}=0	$a_{i k}^{(k+1)}=a_{i k}^{(k)}-\frac{a_{i k}^{(k)}}{a_{k k}^{(k)}} a_{k k}^{(k)}=0$	For $j=k$ it follows from (3.2) that $a_{ik}^{(k+1)}=a_{ik}^{(k)}-\frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}a_{kk}^{(k)}=0$ for $i=$
lyche-numerical-linear-algebra_F00982	78	1.000	k+1, \ldots, n	$k+1, \ldots, n$	$k+1,\ldots,n$. Thus $\boldsymbol{A}^{(k+1)}$ will have zeros under the diagonal in its first k columns
lyche-numerical-linear-algebra_F00983	78	1.000	l_{i k}^{(k)}	$l_{i k}^{(k)}$	and the elimination is carried one step further. The numbers $l_{ik}^{(k)}$ in (3.2) are called

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00984	78	1.000	`a_{k k}^{(k)}`	$a_{k k}^{(k)}$	$a_{kk}^{(k)}$ are nonzero for $k = 1, \ldots, n-1$. This depends on certain submatrices of $\boldsymbol{A}$
lyche-numerical-linear-algebra_F00985	78	1.000	`\boldsymbol{A}_{[k]} \in \mathbb{C}^{k \times k}`	$\boldsymbol{A}_{[k]} \in \mathbb{C}^{k \times k}$	**Definition 3.1 (Principal Submatrix)** For $k = 1, \ldots, n$ the matrices $\boldsymbol{A}_{[k]} \in \mathbb{C}^{k \times k}$
lyche-numerical-linear-algebra_F00986	78	0.997	`\boldsymbol{B} \in \mathbb{C}^{k \times k}`	$\boldsymbol{B} \in \mathbb{C}^{k \times k}$	matrix $\boldsymbol{B} \in \mathbb{C}^{k \times k}$ is called a **principal submatrix** of $\boldsymbol{A}$ if $\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$, where
lyche-numerical-linear-algebra_F00987	78	0.997	`\boldsymbol{B}=\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})`	$\boldsymbol{B}=\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$	matrix $\boldsymbol{B} \in \mathbb{C}^{k \times k}$ is called a **principal submatrix** of $\boldsymbol{A}$ if $\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$, where
lyche-numerical-linear-algebra_F00988	78	1.000	`\boldsymbol{r}=\left[r_{1}, \ldots, r_{k}\right]`	$\boldsymbol{r}=\left[r_{1}, \ldots, r_{k}\right]$	$\boldsymbol{r} = [r_1, \ldots, r_k]$ for some $1 \leq r_1 < \cdots < r_k \leq n$. Thus,
lyche-numerical-linear-algebra_F00989	78	1.000	`1 \leq r_{1}<\cdots<r_{k} \leq n`	$1 \leq r_{1}<\cdots<r_{k} \leq n$	$\boldsymbol{r} = [r_1, \ldots, r_k]$ for some $1 \leq r_1 < \cdots < r_k \leq n$. Thus,
lyche-numerical-linear-algebra_F00990	78	1.000	`r_{j}=j`	$r_{j}=j$	A principal submatrix is leading if $r_j = j$ for $j = 1, \ldots, k$. Also a principal
lyche-numerical-linear-algebra_F00991	78	1.000	`j=1, \ldots, k`	$j=1, \ldots, k$	A principal submatrix is leading if $r_j = j$ for $j = 1, \ldots, k$. Also a principal
lyche-numerical-linear-algebra_F00992	78	1.000	`\boldsymbol{A}=\left[\begin{array}{lll}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{lll}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right]$	*Example 3.2 (Principal Submatrices)* The principal submatrices of $\boldsymbol{A} = \left[\begin{smallmatrix}1&2&3\\4&5&6\\7&8&9\end{smallmatrix}\right]$
lyche-numerical-linear-algebra_F00993	78	1.000	`a_{k, k}^{(k)} \neq 0`	$a_{k, k}^{(k)} \neq 0$	**Theorem 3.1** *We have $a_{k,k}^{(k)} \neq 0$ for $k = 1, \ldots, n-1$ if and only if the leading*
lyche-numerical-linear-algebra_F00994	78	1.000	`\boldsymbol{A}_{[k]}`	$\boldsymbol{A}_{[k]}$	*principal submatrices $\boldsymbol{A}_{[k]}$ of $\boldsymbol{A}$ are nonsingular for $k = 1, \ldots, n-1$. Moreover*
lyche-numerical-linear-algebra_F00995	79	1.000	`\boldsymbol{B}_{k}=\boldsymbol{A}_{k-1}^{(k)}`	$\boldsymbol{B}_{k}=\boldsymbol{A}_{k-1}^{(k)}$	***Proof*** Let $\boldsymbol{B}_k = \boldsymbol{A}_{k-1}^{(k)}$ be the upper left $k-1$ corner of $\boldsymbol{A}^{(k)}$ given by (3.1). Observe
lyche-numerical-linear-algebra_F00996	79	1.000	`\boldsymbol{B}_{k}`	$\boldsymbol{B}_{k}$	that the elements of $\boldsymbol{B}_k$ are computed from $\boldsymbol{A}$ by using only elements from $\boldsymbol{A}_{[k-1]}$.
lyche-numerical-linear-algebra_F00997	79	1.000	`\boldsymbol{A}_{[k-1]}`	$\boldsymbol{A}_{[k-1]}$	that the elements of $\boldsymbol{B}_k$ are computed from $\boldsymbol{A}$ by using only elements from $\boldsymbol{A}_{[k-1]}$.
lyche-numerical-linear-algebra_F00998	79	1.000	`\boldsymbol{B}_{k+1}`	$\boldsymbol{B}_{k+1}$	of diagonal elements of $\boldsymbol{B}_{k+1}$ and (3.3) follows. But then $a_{11}^{(1)} \cdots a_{kk}^{(k)} \neq 0$ for $k =$
lyche-numerical-linear-algebra_F00999	79	1.000	`a_{11}^{(1)} \cdots a_{k k}^{(k)} \neq 0`	$a_{11}^{(1)} \cdots a_{k k}^{(k)} \neq 0$	of diagonal elements of $\boldsymbol{B}_{k+1}$ and (3.3) follows. But then $a_{11}^{(1)} \cdots a_{kk}^{(k)} \neq 0$ for $k =$
lyche-numerical-linear-algebra_F01000	79	1.000	`1, \ldots, n-1`	$1, \ldots, n-1$	$1, \ldots, n-1$ if and only if $\det(\boldsymbol{A}_{[k]}) \neq 0$ for $k = 1, \ldots, n-1$, or equivalently $\boldsymbol{A}_{[k]}$
lyche-numerical-linear-algebra_F01001	79	1.000	`\operatorname{det}\left(\boldsymbol{A}_{[k]}\right) \neq 0`	$\operatorname{det}\left(\boldsymbol{A}_{[k]}\right) \neq 0$	$1, \ldots, n-1$ if and only if $\det(\boldsymbol{A}_{[k]}) \neq 0$ for $k = 1, \ldots, n-1$, or equivalently $\boldsymbol{A}_{[k]}$
lyche-numerical-linear-algebra_F01002	79	1.000	`l_{i j}^{(j)}`	$l_{i j}^{(j)}$	*where the $l_{ij}^{(j)}$ and $a_{ij}^{(i)}$ are given by* (3.2).
lyche-numerical-linear-algebra_F01003	79	1.000	`a_{i j}^{(i)}`	$a_{i j}^{(i)}$	*where the $l_{ij}^{(j)}$ and $a_{ij}^{(i)}$ are given by* (3.2).
lyche-numerical-linear-algebra_F01004	79	1.000	`i \leq j`	$i \leq j$	Thus for $i \leq j$ we find
lyche-numerical-linear-algebra_F01005	79	1.000	`a_{n n}^{(n)}=0`	$a_{n n}^{(n)}=0$	matrix $\boldsymbol{U}$ is then singular, and we must have $a_{nn}^{(n)} = 0$ when $\boldsymbol{A}$ is singular.
lyche-numerical-linear-algebra_F01006	80	1.000	`\boldsymbol{L} \boldsymbol{U} \boldsymbol{x}=\boldsymbol{b}`	$\boldsymbol{L} \boldsymbol{U} \boldsymbol{x}=\boldsymbol{b}$	Since $\boldsymbol{LUx} = \boldsymbol{b}$ we have $\boldsymbol{Ly} = \boldsymbol{b}$, where $\boldsymbol{y} := \boldsymbol{Ux}$. We first solve $\boldsymbol{Ly} = \boldsymbol{b}$, for $\boldsymbol{y}$
lyche-numerical-linear-algebra_F01007	80	1.000	`\boldsymbol{L} \boldsymbol{y}=\boldsymbol{b}`	$\boldsymbol{L} \boldsymbol{y}=\boldsymbol{b}$	Since $\boldsymbol{LUx} = \boldsymbol{b}$ we have $\boldsymbol{Ly} = \boldsymbol{b}$, where $\boldsymbol{y} := \boldsymbol{Ux}$. We first solve $\boldsymbol{Ly} = \boldsymbol{b}$, for $\boldsymbol{y}$
lyche-numerical-linear-algebra_F01008	80	1.000	`\boldsymbol{y}:=\boldsymbol{U} \boldsymbol{x}`	$\boldsymbol{y}:=\boldsymbol{U} \boldsymbol{x}$	Since $\boldsymbol{LUx} = \boldsymbol{b}$ we have $\boldsymbol{Ly} = \boldsymbol{b}$, where $\boldsymbol{y} := \boldsymbol{Ux}$. We first solve $\boldsymbol{Ly} = \boldsymbol{b}$, for $\boldsymbol{y}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01009	80	1.000	`\boldsymbol{U} \boldsymbol{x}=\boldsymbol{y}`	$\boldsymbol{U} \boldsymbol{x}=\boldsymbol{y}$	and then $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{y}$ for $\boldsymbol{x}$.
lyche-numerical-linear-algebra_F01010	80	0.999	`2.5 \boldsymbol{A}`	$2.5 \boldsymbol{A}$	A nonsingular triangular linear system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ is easy to solve. By Lemma 2.5 $\boldsymbol{A}$
lyche-numerical-linear-algebra_F01011	80	1.000	`x_{1}=b_{1} / a_{11}`	$x_{1}=b_{1} / a_{11}$	From the first equation we find $x_1 = b_1/a_{11}$. Solving the second equation for x_2 we
lyche-numerical-linear-algebra_F01012	80	1.000	`x_{2}`	x_{2}	From the first equation we find $x_1 = b_1/a_{11}$. Solving the second equation for x_2 we
lyche-numerical-linear-algebra_F01013	80	0.999	`x_{2}=\left(b_{2}-a_{21} x_{1}\right) / a_{22}`	$x_{2}=\left(b_{2}-a_{21} x_{1}\right) / a_{22}$	obtain $x_2 = (b_2 - a_{21}x_1)/a_{22}$. Finally the third equation gives $x_3 = (b_3 - a_{31}x_1 -$
lyche-numerical-linear-algebra_F01014	80	0.999	`x_{3}=\left(b_{3}-a_{31} x_{1}-\right.`	$x_{3}=\left(b_{3}-a_{31} x_{1}-\right.$	obtain $x_2 = (b_2 - a_{21}x_1)/a_{22}$. Finally the third equation gives $x_3 = (b_3 - a_{31}x_1 -$
lyche-numerical-linear-algebra_F01015	80	1.000	`\left.a_{32} x_{2}\right) / a_{33}`	$\left.a_{32} x_{2}\right) / a_{33}$	$a_{32}x_2)/a_{33}$. This process is known as forward substitution. In general
lyche-numerical-linear-algebra_F01016	80	1.000	`a_{k, j}=0`	$a_{k, j}=0$	so that $a_{k,j} = 0$ for $j \notin \{l_k, l_k+1, \ldots, k$ for $k = 1, 2, \ldots, n$, and where
lyche-numerical-linear-algebra_F01017	80	1.000	`j \notin\left\{l_{k}, l_{k}+1, \ldots, k\right.`	$j \notin\left\{l_{k}, l_{k}+1, \ldots, k\right.$	so that $a_{k,j} = 0$ for $j \notin \{l_k, l_k+1, \ldots, k$ for $k = 1, 2, \ldots, n$, and where
lyche-numerical-linear-algebra_F01018	80	1.000	`k=1,2, \ldots, n`	$k=1,2, \ldots, n$	so that $a_{k,j} = 0$ for $j \notin \{l_k, l_k+1, \ldots, k$ for $k = 1, 2, \ldots, n$, and where
lyche-numerical-linear-algebra_F01019	80	1.000	`l_{k}:=\max (1, k-d)`	$l_{k}:=\max (1, k-d)$	$l_k := \max(1, k-d)$, see Fig. 3.2. For a lower triangular d-band matrix the
lyche-numerical-linear-algebra_F01020	80	1.000	`d=n`	$d=n$	Note that (3.8) reduces to (3.7) if $d = n$. Letting $A(k, l_k : (k-1)) * x(l_k : (k-1))$
lyche-numerical-linear-algebra_F01021	80	1.000	`A\left(k, l_{k}:(k-1)\right) * x\left(l_{k}:(k-1)\right)`	$A\left(k, l_{k}:(k-1)\right) * x\left(l_{k}:(k-1)\right)$	Note that (3.8) reduces to (3.7) if $d = n$. Letting $A(k, l_k : (k-1)) * x(l_k : (k-1))$
lyche-numerical-linear-algebra_F01022	80	0.995	`\sum_{j=l_{k}}^{k-1} a_{k j} x_{j}`	$\sum_{j=l_{k}}^{k-1} a_{k j} x_{j}$	denote the sum $\sum_{j=l_k}^{k-1} a_{kj}x_j$ we arrive at the following algorithm, where the initial
lyche-numerical-linear-algebra_F01023	81	1.000	`x_{n}`	x_{n}	substitution or 'bottom-up'. We first find x_n from the last equation and then move
lyche-numerical-linear-algebra_F01024	81	1.000	`\boldsymbol{A} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^{n}`	$\boldsymbol{A} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^{n}$	banded matrix $\boldsymbol{A} \in \mathbb{C}^{n \times n}$, $\boldsymbol{b} \in \mathbb{C}^n$ and d, as input, and returns an $\boldsymbol{x} \in \mathbb{C}^n$ so
lyche-numerical-linear-algebra_F01025	81	1.000	`n-k+1`	$n-k+1$	This system is of order $n-k+1$. The unknowns are $x_k, \ldots, x_n$.
lyche-numerical-linear-algebra_F01026	81	1.000	`x_{k}, \ldots, x_{n}`	$x_{k}, \ldots, x_{n}$	This system is of order $n-k+1$. The unknowns are $x_k, \ldots, x_n$.
lyche-numerical-linear-algebra_F01027	82	1.000	`x_{k}=b_{k} / a_{k, k}`	$x_{k}=b_{k} / a_{k, k}$	We see that $x_k = b_k/a_{k,k}$ and eliminating x_k from the remaining equations we
lyche-numerical-linear-algebra_F01028	82	1.000	`x_{k}`	x_{k}	We see that $x_k = b_k/a_{k,k}$ and eliminating x_k from the remaining equations we
lyche-numerical-linear-algebra_F01029	82	1.000	`n-k`	$n-k$	obtain a system of order $n-k$ with unknowns $x_{k+1}, \ldots, x_n$
lyche-numerical-linear-algebra_F01030	82	1.000	`x_{k+1}, \ldots, x_{n}`	$x_{k+1}, \ldots, x_{n}$	obtain a system of order $n-k$ with unknowns $x_{k+1}, \ldots, x_n$
lyche-numerical-linear-algebra_F01031	82	1.000	`k=1,2, \ldots n`	$k=1,2, \ldots n$	Thus at the kth step, $k = 1, 2, \ldots n$ we set $x_k = b_k/A(k,k)$ and update b as follows:
lyche-numerical-linear-algebra_F01032	82	1.000	`x_{k}=b_{k} / A(k, k)`	$x_{k}=b_{k} / A(k, k)$	Thus at the kth step, $k = 1, 2, \ldots n$ we set $x_k = b_k/A(k,k)$ and update b as follows:
lyche-numerical-linear-algebra_F01033	82	1.000	`b`	b	Thus at the kth step, $k = 1, 2, \ldots n$ we set $x_k = b_k/A(k,k)$ and update b as follows:
lyche-numerical-linear-algebra_F01034	82	1.000	`M, D, A, S`	M, D, A, S	operations. Let M, D, A, S be the number of (complex) multiplications, divisions,
lyche-numerical-linear-algebra_F01035	82	1.000	`a_{i j}^{k+1}=a_{i j}^{(k)}-l_{i k}^{(k)} a_{k j}^{(k)}`	$a_{i j}^{k+1}=a_{i j}^{(k)}-l_{i k}^{(k)} a_{k j}^{(k)}$	calculation of $a_{ij}^{k+1} = a_{ij}^{(k)} - l_{ik}^{(k)} a_{kj}^{(k)}$ which is carried out $(n-k)^2$ times. Moreover,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01036	82	1.000	`(n-k)^{2}`	$(n-k)^{2}$	calculation of $a_{ij}^{k+1} = a_{ij}^{(k)} - l_{ik}^{(k)} a_{kj}^{(k)}$ which is carried out $(n-k)^2$ times. Moreover,
lyche-numerical-linear-algebra_F01037	83	1.000	`M+`	$M+$	each calculation involves one subtraction and one multiplication. Thus we find $M+$
lyche-numerical-linear-algebra_F01038	83	1.000	`S=2 \sum_{k=1}^{n-1}(n-k)^{2}=2 \sum_{m=1}^{n-1} m^{2}= \frac{2}{3} n(n-1)\left(n-\frac{1}{2}\right)`	$S=2 \sum_{k=1}^{n-1}(n-k)^{2}=2 \sum_{m=1}^{n-1} m^{2}= \frac{2}{3} n(n-1)\left(n-\frac{1}{2}\right)$	$S = 2\sum_{k=1}^{n-1}(n-k)^2 = 2\sum_{m=1}^{n-1} m^2 = \frac{2}{3}n(n-1)(n-\frac{1}{2})$. For each k there are $n-k$
lyche-numerical-linear-algebra_F01039	83	1.000	`\sum_{k=1}^{n-1}(n-k)=\frac{1}{2} n(n-1)`	$\sum_{k=1}^{n-1}(n-k)=\frac{1}{2} n(n-1)$	divisions giving a sum of $\sum_{k=1}^{n-1}(n-k) = \frac{1}{2}n(n-1)$. Since there are no additions
lyche-numerical-linear-algebra_F01040	83	1.000	`N_{L U}`	$N_{L U}$	We are only interested in N_{LU} when n is large and for such n the term $\frac{2}{3}n^3$
lyche-numerical-linear-algebra_F01041	83	1.000	`\frac{2}{3} n^{3}`	$\frac{2}{3} n^{3}$	We are only interested in N_{LU} when n is large and for such n the term $\frac{2}{3}n^3$
lyche-numerical-linear-algebra_F01042	83	1.000	`G_{n}`	G_{n}	the number G_n as follows:
lyche-numerical-linear-algebra_F01043	83	0.997	`3.2\left(G_{n}:=\frac{2}{3} n^{3}\right)`	$3.2\left(G_{n}:=\frac{2}{3} n^{3}\right)$	**Definition 3.2** ($G_n := \frac{2}{3}n^3$) We define $G_n := \frac{2}{3}n^3$.
lyche-numerical-linear-algebra_F01044	83	0.997	`G_{n}:=\frac{2}{3} n^{3}`	$G_{n}:=\frac{2}{3} n^{3}$	**Definition 3.2** ($G_n := \frac{2}{3}n^3$) We define $G_n := \frac{2}{3}n^3$.
lyche-numerical-linear-algebra_F01045	83	1.000	`2 n^{3} / 3`	$2 n^{3} / 3$	There is a quick way to arrive at the leading term $2n^3/3$. We only consider the
lyche-numerical-linear-algebra_F01046	83	1.000	`M+S`	$M+S$	loop contributing to $M+S$. Then we replace sums by integrals letting the summation
lyche-numerical-linear-algebra_F01047	83	1.000	`N_{S}`	N_{S}	Consider next N_S, the number of forward plus backward substitutions. By (3.7)
lyche-numerical-linear-algebra_F01048	83	1.000	`n=10^{6}`	$n=10^{6}$	system requires $O(n^2)$ arithmetic operations. Thus, if $n = 10^6$ and one arith-
lyche-numerical-linear-algebra_F01049	83	1.000	`c=10^{-14}`	$c=10^{-14}$	metic operation requires $c = 10^{-14}$ seconds of computing time then $cn^3 =$
lyche-numerical-linear-algebra_F01050	83	1.000	`c n^{3}=`	$c n^{3}=$	metic operation requires $c = 10^{-14}$ seconds of computing time then $cn^3 =$
lyche-numerical-linear-algebra_F01051	83	1.000	`10^{4}`	10^{4}	10^4 seconds ≈ 3 hours and $cn^2 = 0.01$ second, giving dramatic differences in
lyche-numerical-linear-algebra_F01052	83	1.000	`\approx 3`	≈ 3	10^4 seconds ≈ 3 hours and $cn^2 = 0.01$ second, giving dramatic differences in
lyche-numerical-linear-algebra_F01053	83	1.000	`c n^{2}=0.01`	$c n^{2}=0.01$	10^4 seconds ≈ 3 hours and $cn^2 = 0.01$ second, giving dramatic differences in
lyche-numerical-linear-algebra_F01054	84	1.000	`\left[\begin{array}{ll}0 & 1 \\ 1 & 1\end{array}\right] \left[\begin{array}{l}x_{1} \\ x_{2}\end{array}\right]= \left[\begin{array}{l}1 \\ 1\end{array}\right]`	$\left[\begin{array}{ll}0 & 1 \\ 1 & 1\end{array}\right] \left[\begin{array}{l}x_{1} \\ x_{2}\end{array}\right]= \left[\begin{array}{l}1 \\ 1\end{array}\right]$	nonsingular system. A simple example is $\left[\begin{smallmatrix}0 & 1 \\ 1 & 1\end{smallmatrix}\right]\left[\begin{smallmatrix}x_1 \\ x_2\end{smallmatrix}\right] = \left[\begin{smallmatrix}1 \\ 1\end{smallmatrix}\right]$. We show here that any
lyche-numerical-linear-algebra_F01055	84	0.980	`(k, k)`	(k, k)	as **pivoting**. The element which is moved to the diagonal position (k,k) is called
lyche-numerical-linear-algebra_F01056	84	1.000	`r_{k} \geq k`	$r_{k} \geq k$	1. **Choose** $r_k \geq k$ so that $a_{r_k,k}^{(k)} \neq 0$.
lyche-numerical-linear-algebra_F01057	84	1.000	`a_{r_{k}, k}^{(k)} \neq 0`	$a_{r_{k}, k}^{(k)} \neq 0$	1. **Choose** $r_k \geq k$ so that $a_{r_k,k}^{(k)} \neq 0$.
lyche-numerical-linear-algebra_F01058	84	1.000	`r_{k}`	r_{k}	2. **Interchange** rows r_k and k of $\boldsymbol{A}^{(k)}$.
lyche-numerical-linear-algebra_F01059	84	1.000	`a_{i j}^{(k+1)}`	$a_{i j}^{(k+1)}$	3. **Eliminate** by computing $l_{ik}^{(k)}$ and $a_{ij}^{(k+1)}$ using (3.2).

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F01060	84	1.000	`\boldsymbol{A}^{(1)}=\boldsymbol{A}`	$\boldsymbol{A}^{(1)}=\boldsymbol{A}$	$\boldsymbol{A}^{(1)}=\boldsymbol{A}$ this holds for $k=1$. By Lemma 2.4 the lower right diagonal block in
lyche-numerical-linear-algebra_ F01061	84	1.000	`\boldsymbol{e}_{i_{1}}, \ldots, \boldsymbol{e}_{i_{n}}`	$\boldsymbol{e}_{i_{1}}, \ldots, \boldsymbol{e}_{i_{n}}$	where $\boldsymbol{e}_{i_1},\dots,\boldsymbol{e}_{i_n}$ is a permutation of the unit vectors $\boldsymbol{e}_1,\dots,\boldsymbol{e}_n\in\mathbb{R}^n$.
lyche-numerical-linear-algebra_ F01062	84	1.000	`\boldsymbol{e}_{1}, \ldots, \boldsymbol{e}_{n} \in \mathbb{R}^{n}`	$\boldsymbol{e}_{1}, \ldots, \boldsymbol{e}_{n} \in \mathbb{R}^{n}$	where $\boldsymbol{e}_{i_1},\dots,\boldsymbol{e}_{i_n}$ is a permutation of the unit vectors $\boldsymbol{e}_1,\dots,\boldsymbol{e}_n\in\mathbb{R}^n$.
lyche-numerical-linear-algebra_ F01063	84	1.000	`\boldsymbol{p}=\left[i_{1}, \ldots, i_{n}\right]^{T}`	$\boldsymbol{p}=\left[i_{1}, \ldots, i_{n}\right]^{T}$	Every permutation $\boldsymbol{p}=[i_1,\dots,i_n]^T$ of the integers $1,2,\dots,n$ gives rise to a
lyche-numerical-linear-algebra_ F01064	85	0.997	`\boldsymbol{A} \boldsymbol{P}=\left(\boldsymbol{A} \boldsymbol{e}_{i_{1}}, \ldots, \boldsymbol{A} \boldsymbol{e}_{i_{n}}\right)=\boldsymbol{A}(:, \boldsymbol{p})`	$\boldsymbol{A} \boldsymbol{P}=\left(\boldsymbol{A} \boldsymbol{e}_{i_{1}}, \ldots, \boldsymbol{A} \boldsymbol{e}_{i_{n}}\right)=\boldsymbol{A}(:, \boldsymbol{p})$	Indeed, $\boldsymbol{A}\boldsymbol{P}=(\boldsymbol{A}\boldsymbol{e}_{i_1},\dots,\boldsymbol{A}\boldsymbol{e}_{i_n})=\boldsymbol{A}(:,\boldsymbol{p})$ and $\boldsymbol{P}^T\boldsymbol{A}=(\boldsymbol{A}^T\boldsymbol{P})^T=(\boldsymbol{A}^T($:
lyche-numerical-linear-algebra_ F01065	85	0.997	`\boldsymbol{P}^{T} \boldsymbol{A}=\left( \boldsymbol{A}^{T} \boldsymbol{P}\right)^{T}=\left( \boldsymbol{A}^{T}(\right.`	$\boldsymbol{P}^{T} \boldsymbol{A}=\left(\boldsymbol{A}^{T} \boldsymbol{P}\right)^{T}=\left(\boldsymbol{A}^{T}(\right.$	Indeed, $\boldsymbol{A}\boldsymbol{P}=(\boldsymbol{A}\boldsymbol{e}_{i_1},\dots,\boldsymbol{A}\boldsymbol{e}_{i_n})=\boldsymbol{A}(:,\boldsymbol{p})$ and $\boldsymbol{P}^T\boldsymbol{A}=(\boldsymbol{A}^T\boldsymbol{P})^T=(\boldsymbol{A}^T($:
lyche-numerical-linear-algebra_ F01066	85	0.773	`, \boldsymbol{p}))^{T}=\boldsymbol{A}(\boldsymbol{p},:)`	$, \boldsymbol{p}))^{T}=\boldsymbol{A}(\boldsymbol{p},:)$	$,\boldsymbol{p}))^T=\boldsymbol{A}(\boldsymbol{p},:)$.
lyche-numerical-linear-algebra_ F01067	85	1.000	`\boldsymbol{P}^{T} \boldsymbol{P}=\boldsymbol{I}`	$\boldsymbol{P}^{T} \boldsymbol{P}=\boldsymbol{I}$	Since $\boldsymbol{P}^T\boldsymbol{P}=\boldsymbol{I}$ the inverse of $\boldsymbol{P}$ is equal to its transpose, $\boldsymbol{P}^{-1}=\boldsymbol{P}^T$ and
lyche-numerical-linear-algebra_ F01068	85	1.000	`\boldsymbol{P}^{-1}=\boldsymbol{P}^{T}`	$\boldsymbol{P}^{-1}=\boldsymbol{P}^{T}$	Since $\boldsymbol{P}^T\boldsymbol{P}=\boldsymbol{I}$ the inverse of $\boldsymbol{P}$ is equal to its transpose, $\boldsymbol{P}^{-1}=\boldsymbol{P}^T$ and
lyche-numerical-linear-algebra_ F01069	85	1.000	`\boldsymbol{P} \boldsymbol{P}^{T}=\boldsymbol{I}`	$\boldsymbol{P} \boldsymbol{P}^{T}=\boldsymbol{I}$	$\boldsymbol{P}\boldsymbol{P}^T=\boldsymbol{I}$ as well. We will use a particularly simple permutation matrix.
lyche-numerical-linear-algebra_ F01070	85	0.997	`(j, k)`	(j, k)	**Definition 3.4** We define a (j,k)**-Interchange matrix** $\boldsymbol{I}_{jk}$ by interchanging col-
lyche-numerical-linear-algebra_ F01071	85	0.997	`\boldsymbol{I}_{j k}`	$\boldsymbol{I}_{j k}$	**Definition 3.4** We define a (j,k)**-Interchange matrix** $\boldsymbol{I}_{jk}$ by interchanging col-
lyche-numerical-linear-algebra_ F01072	85	0.999	`\boldsymbol{I}_{j k}=\boldsymbol{I}_{k j}`	$\boldsymbol{I}_{j k}=\boldsymbol{I}_{k j}$	Since $\boldsymbol{I}_{jk}=\boldsymbol{I}_{kj}$, and we obtain the identity by applying $\boldsymbol{I}_{jk}$ twice, we see that
lyche-numerical-linear-algebra_ F01073	85	1.000	`\boldsymbol{I}_{j k}^{2}=\boldsymbol{I}`	$\boldsymbol{I}_{j k}^{2}=\boldsymbol{I}$	$\boldsymbol{I}_{jk}^2=\boldsymbol{I}$ and an interchange matrix is symmetric and equal to its own inverse. Pre-
lyche-numerical-linear-algebra_ F01074	85	1.000	`\boldsymbol{p}_{k}`	$\boldsymbol{p}_{k}$	We can keep track of the row interchanges using **pivot vectors** $\boldsymbol{p}_k$. We define
lyche-numerical-linear-algebra_ F01075	85	1.000	`\boldsymbol{p}_{k+1}`	$\boldsymbol{p}_{k+1}$	We obtain $\boldsymbol{p}_{k+1}$ from $\boldsymbol{p}_k$ by interchanging the entries r_k and k in $\boldsymbol{p}_k$. In particular,
lyche-numerical-linear-algebra_ F01076	85	0.686	`\boldsymbol{P}_{k}:=\boldsymbol{I}_{r_{k}, k}`	$\boldsymbol{P}_{k}:=\boldsymbol{I}_{r_{k}, k}$	interchange matrices $\boldsymbol{P}_k:=\boldsymbol{I}_{r_k,k}$. Since $\boldsymbol{P}_k\boldsymbol{I}(\boldsymbol{p}_k,:)=\boldsymbol{I}(\boldsymbol{P}_k\boldsymbol{p}_k,:)=\boldsymbol{I}(\boldsymbol{p}_{k+1},:)$
lyche-numerical-linear-algebra_ F01077	85	0.686	`\boldsymbol{P}_{k} \boldsymbol{I}\left( \boldsymbol{p}_{k},:\right)=\boldsymbol{I}\left( \boldsymbol{P}_{k} \boldsymbol{p}_{k},:\right)= \boldsymbol{I}\left(\boldsymbol{p}_{k+1},:\right)`	$\boldsymbol{P}_{k} \boldsymbol{I}\left(\boldsymbol{p}_{k},:\right)=\boldsymbol{I}\left(\boldsymbol{P}_{k} \boldsymbol{p}_{k},:\right)= \boldsymbol{I}\left(\boldsymbol{p}_{k+1},:\right)$	interchange matrices $\boldsymbol{P}_k:=\boldsymbol{I}_{r_k,k}$. Since $\boldsymbol{P}_k\boldsymbol{I}(\boldsymbol{p}_k,:)=\boldsymbol{I}(\boldsymbol{P}_k\boldsymbol{p}_k,:)=\boldsymbol{I}(\boldsymbol{p}_{k+1},:)$
lyche-numerical-linear-algebra_ F01078	85	1.000	`a_{i j}^{(1)}=a_{i j}`	$a_{i j}^{(1)}=a_{i j}$	pivoting starting with $a_{ij}^{(1)}=a_{ij}$ can be described as follows:
lyche-numerical-linear-algebra_ F01079	86	0.994	`\boldsymbol{A}=\boldsymbol{P} \boldsymbol{L} \boldsymbol{U}`	$\boldsymbol{A}=\boldsymbol{P} \boldsymbol{L} \boldsymbol{U}$	$\mathbb{C}^{n\times n}$ *leads to the factorization* $\boldsymbol{A}=\boldsymbol{P}\boldsymbol{L}\boldsymbol{U}$, *where* $\boldsymbol{P}$ *is a permutation matrix,* $\boldsymbol{L}$
lyche-numerical-linear-algebra_ F01080	86	0.976	`\boldsymbol{P}=\boldsymbol{I}(:, \boldsymbol{p})`	$\boldsymbol{P}=\boldsymbol{I}(:, \boldsymbol{p})$	*explicitly,* $\boldsymbol{P}=\boldsymbol{I}(:,\boldsymbol{p})$, *where* $\boldsymbol{p}=\boldsymbol{I}_{r_{n-1},n-1}\cdots\boldsymbol{I}_{r_1,1}[1,\dots,n]^T$, *and*
lyche-numerical-linear-algebra_ F01081	86	0.976	`\boldsymbol{p}=\boldsymbol{I}_{r_{n-1}, n-1} \cdots \boldsymbol{I}_{r_{1}, 1}[1, \ldots, n]^{T}`	$\boldsymbol{p}=\boldsymbol{I}_{r_{n-1}, n-1} \cdots \boldsymbol{I}_{r_{1}, 1}[1, \ldots, n]^{T}$	*explicitly,* $\boldsymbol{P}=\boldsymbol{I}(:,\boldsymbol{p})$, *where* $\boldsymbol{p}=\boldsymbol{I}_{r_{n-1},n-1}\cdots\boldsymbol{I}_{r_1,1}[1,\dots,n]^T$, *and*
lyche-numerical-linear-algebra_ F01082	86	1.000	`\boldsymbol{P}^{T} \boldsymbol{A}=\boldsymbol{L} \boldsymbol{U}`	$\boldsymbol{P}^{T} \boldsymbol{A}=\boldsymbol{L} \boldsymbol{U}$	The PLU factorization can also be written $\boldsymbol{P}^T\boldsymbol{A}=\boldsymbol{L}\boldsymbol{U}$. This shows that for a
lyche-numerical-linear-algebra_ F01083	87	0.998	`\mathrm{fl}\left(x_{1}\right)`	$\mathrm{fl}\left(x_{1}\right)$	solutions $\mathrm{fl}(x_1)$ and $\mathrm{fl}(x_2)$ computed in this way is

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F01084	87	0.998	`\mathrm{fl}\left(x_{2}\right)`	$\mathrm{fl}\left(x_{2}\right)$	solutions $\mathrm{fl}(x_1)$ and $\mathrm{fl}(x_2)$ computed in this way is
lyche-numerical-linear-algebra_ F01085	87	1.000	`x_{1}`	x_{1}	The computed value 0 of x_1 is completely wrong. Suppose instead we apply
lyche-numerical-linear-algebra_ F01086	88	1.000	`\mathrm{fl}\left(x_{1}\right)=1`	$\mathrm{fl}\left(x_{1}\right)=1$	In this case rounding each calculation to three digits produces $\mathrm{fl}(x_1) = 1$ and
lyche-numerical-linear-algebra_ F01087	88	1.000	`\mathrm{fl}\left(x_{2}\right)=2`	$\mathrm{fl}\left(x_{2}\right)=2$	$\mathrm{fl}(x_2) = 2$ which is quite satisfactory since it is the exact solution rounded to three
lyche-numerical-linear-algebra_ F01088	88	1.000	`s_{k}`	s_{k}	coefficients of wildly different sizes. Note that the scaling factors s_k are computed
lyche-numerical-linear-algebra_ F01089	88	1.000	`r_{k}, s_{k}`	r_{k}, s_{k}	with r_k, s_k the smallest such indices in case of a tie, is known as **complete pivoting**.
lyche-numerical-linear-algebra_ F01090	88	0.967	`\mathbf{L U}`	$\mathbf{L U}$	that $\boldsymbol{A} = \boldsymbol{LU}$ is an **LU factorization** of $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ if $\boldsymbol{L} \in \mathbb{C}^{n\times n}$ is lower triangular
lyche-numerical-linear-algebra_ F01091	88	0.967	`\boldsymbol{L} \in \mathbb{C}^{n \times n}`	$\boldsymbol{L} \in \mathbb{C}^{n \times n}$	that $\boldsymbol{A} = \boldsymbol{LU}$ is an **LU factorization** of $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ if $\boldsymbol{L} \in \mathbb{C}^{n\times n}$ is lower triangular
lyche-numerical-linear-algebra_ F01092	88	1.000	`\boldsymbol{U} \in \mathbb{C}^{n \times n}`	$\boldsymbol{U} \in \mathbb{C}^{n \times n}$	and $\boldsymbol{U} \in \mathbb{C}^{n\times n}$ is upper triangular , i.e.,
lyche-numerical-linear-algebra_ F01093	88	0.963	`n^{2}`	n^{2}	To find an LU factorization there is one equation for each of the n^2 elements in $\boldsymbol{A}$,
lyche-numerical-linear-algebra_ F01094	88	1.000	`n^{2}+n`	$n^{2}+n$	and $\boldsymbol{L}$ and $\boldsymbol{U}$ contain a total of n^2+n unknown elements. There are several ways to
lyche-numerical-linear-algebra_ F01095	89	0.701	`\quad l_{i i}=1`	$\quad l_{i i}=1$	L1U: $l_{ii} = 1$ all i,
lyche-numerical-linear-algebra_ F01096	89	0.992	`\quad u_{i i}=1`	$\quad u_{i i}=1$	LU1: $u_{ii} = 1$ all i,
lyche-numerical-linear-algebra_ F01097	89	0.998	`\quad \boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U}, l_{i i}=u_{i i}=1`	$\quad \boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U}, l_{i i}=u_{i i}=1$	LDU: $\boldsymbol{A} = \boldsymbol{LDU}, l_{ii} = u_{ii} = 1$ all i, $\boldsymbol{D} = \mathrm{diag}(d_{11},\ldots,d_{nn})$.
lyche-numerical-linear-algebra_ F01098	89	0.998	`i, \boldsymbol{D}=\operatorname{diag}\left(d_{11}, \ldots, d_{n n}\right)`	$i, \boldsymbol{D}=\operatorname{diag}\left(d_{11}, \ldots, d_{n n}\right)$	LDU: $\boldsymbol{A} = \boldsymbol{LDU}, l_{ii} = u_{ii} = 1$ all i, $\boldsymbol{D} = \mathrm{diag}(d_{11},\ldots,d_{nn})$.
lyche-numerical-linear-algebra_ F01099	89	0.889	`a, b, c, d \in \mathbb{C}`	$a, b, c, d \in \mathbb{C}$	*Example 3.5 (L1U of 2 × 2 Matrix)* Let $a,b,c,d \in \mathbb{C}$. An L1U factorization of
lyche-numerical-linear-algebra_ F01100	89	0.855	`\boldsymbol{A}=\left[\begin{array}{ll}a & b \\ c & d \end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{ll}a & b \\ c & d \end{array}\right]$	$\boldsymbol{A} = [\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}]$ must satisfy the equations
lyche-numerical-linear-algebra_ F01101	89	0.984	`l_{1}`	l_{1}	for the unknowns l_1 in $\boldsymbol{L}$ and u_1, u_2, u_3 in $\boldsymbol{U}$. The equations are
lyche-numerical-linear-algebra_ F01102	89	0.984	`u_{1}, u_{2}, u_{3}`	u_{1}, u_{2}, u_{3}	for the unknowns l_1 in $\boldsymbol{L}$ and u_1, u_2, u_3 in $\boldsymbol{U}$. The equations are
lyche-numerical-linear-algebra_ F01103	89	0.999	`a l_{1}=c`	$a l_{1}=c$	equation $al_1 = c$. There are essentially three cases
lyche-numerical-linear-algebra_ F01104	89	1.000	`a \neq 0`	$a \neq 0$	1. $a \neq 0$: The matrix has a unique L1U factorization.
lyche-numerical-linear-algebra_ F01105	89	1.000	`a=c=0`	$a=c=0$	2. $a = c = 0$: The L1U factorization exists, but it is not unique. Any value for l_1
lyche-numerical-linear-algebra_ F01106	89	1.000	`a=0, c \neq 0`	$a=0, c \neq 0$	3. $a = 0, c \neq 0$: No L1U factorization exists.
lyche-numerical-linear-algebra_ F01107	89	1.000	`\boldsymbol{A}_{4}`	$\boldsymbol{A}_{4}$	has no L1U factorization, $\boldsymbol{A}_3$ has an L1U factorization but it is not unique, and $\boldsymbol{A}_4$
lyche-numerical-linear-algebra_ F01108	90	1.000	`\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}, \boldsymbol{U}_{[k]}`	$\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}, \boldsymbol{U}_{[k]}$	*L1U factorization of $\boldsymbol{A} \in \mathbb{C}^{n\times n}$. For $k = 1,\ldots,n$ let $\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}, \boldsymbol{U}_{[k]}$ be the leading*
lyche-numerical-linear-algebra_ F01109	90	0.986	`\boldsymbol{A}, \boldsymbol{L}, \boldsymbol{U}`	$\boldsymbol{A}, \boldsymbol{L}, \boldsymbol{U}$	*principal submatrices of $\boldsymbol{A}, \boldsymbol{L}, \boldsymbol{U}$, respectively. Then $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]}\boldsymbol{U}_{[k]}$ is an L1U*
lyche-numerical-linear-algebra_ F01110	90	0.986	`\boldsymbol{A}_{[k]}=\boldsymbol{L}_{[k]} \boldsymbol{U}_{[k]}`	$\boldsymbol{A}_{[k]}=\boldsymbol{L}_{[k]} \boldsymbol{U}_{[k]}$	*principal submatrices of $\boldsymbol{A}, \boldsymbol{L}, \boldsymbol{U}$, respectively. Then $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]}\boldsymbol{U}_{[k]}$ is an L1U*

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01111	90	1.000	`\boldsymbol{F}_{k}, \boldsymbol{N}_{k}, \boldsymbol{T}_{k} \in \mathbb{C}^{n-k, n-k}`	$\boldsymbol{F}_{k}, \boldsymbol{N}_{k}, \boldsymbol{T}_{k} \in \mathbb{C}^{n-k, n-k}$	where $\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{T}_k \in \mathbb{C}^{n-k,n-k}$. Comparing blocks we find $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]}\boldsymbol{U}_{[k]}$. Since
lyche-numerical-linear-algebra_F01112	90	1.000	`\boldsymbol{L}_{[k]}`	$\boldsymbol{L}_{[k]}$	$\boldsymbol{L}_{[k]}$ is unit lower triangular and $\boldsymbol{U}_{[k]}$ is upper triangular this is an L1U factorization
lyche-numerical-linear-algebra_F01113	90	1.000	`\boldsymbol{U}_{[k]}`	$\boldsymbol{U}_{[k]}$	$\boldsymbol{L}_{[k]}$ is unit lower triangular and $\boldsymbol{U}_{[k]}$ is upper triangular this is an L1U factorization
lyche-numerical-linear-algebra_F01114	90	0.597	`1 \times 1`	1×1	is clearly true for $n = 1$, since the unique L1U factorization of a 1×1 matrix is
lyche-numerical-linear-algebra_F01115	90	1.000	`\left[a_{11}\right]=[1]\left[a_{11}\right]`	$\left[a_{11}\right]=[1]\left[a_{11}\right]$	$[a_{11}] = [1][a_{11}]$. Suppose that $\boldsymbol{A}_{[n-1]}$ has a unique L1U factorization $\boldsymbol{A}_{[n-1]} =$
lyche-numerical-linear-algebra_F01116	90	1.000	`\boldsymbol{A}_{[n-1]}`	$\boldsymbol{A}_{[n-1]}$	$[a_{11}] = [1][a_{11}]$. Suppose that $\boldsymbol{A}_{[n-1]}$ has a unique L1U factorization $\boldsymbol{A}_{[n-1]} =$
lyche-numerical-linear-algebra_F01117	90	1.000	`\boldsymbol{A}_{[n-1]}=`	$\boldsymbol{A}_{[n-1]}=$	$[a_{11}] = [1][a_{11}]$. Suppose that $\boldsymbol{A}_{[n-1]}$ has a unique L1U factorization $\boldsymbol{A}_{[n-1]} =$
lyche-numerical-linear-algebra_F01118	90	0.992	`\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}`	$\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	$\boldsymbol{L}_{n-1}\boldsymbol{U}_{n-1}$, and that $\boldsymbol{A}_{[1]}, \ldots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By block multiplication
lyche-numerical-linear-algebra_F01119	90	0.992	`\boldsymbol{A}_{[1]}, \ldots, \boldsymbol{A}_{[n-1]}`	$\boldsymbol{A}_{[1]}, \ldots, \boldsymbol{A}_{[n-1]}$	$\boldsymbol{L}_{n-1}\boldsymbol{U}_{n-1}$, and that $\boldsymbol{A}_{[1]}, \ldots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By block multiplication
lyche-numerical-linear-algebra_F01120	90	1.000	`\boldsymbol{A}_{[n-1]}=\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}`	$\boldsymbol{A}_{[n-1]}=\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	if and only if $\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1}\boldsymbol{U}_{n-1}$ and $\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$ and $u_{nn} \in \mathbb{C}$ are determined
lyche-numerical-linear-algebra_F01121	90	1.000	`\boldsymbol{l}_{n}, \boldsymbol{u}_{n} \in \mathbb{C}^{n-1}`	$\boldsymbol{l}_{n}, \boldsymbol{u}_{n} \in \mathbb{C}^{n-1}$	if and only if $\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1}\boldsymbol{U}_{n-1}$ and $\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$ and $u_{nn} \in \mathbb{C}$ are determined
lyche-numerical-linear-algebra_F01122	90	1.000	`u_{n n} \in \mathbb{C}`	$u_{n n} \in \mathbb{C}$	if and only if $\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1}\boldsymbol{U}_{n-1}$ and $\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$ and $u_{nn} \in \mathbb{C}$ are determined
lyche-numerical-linear-algebra_F01123	90	1.000	`\boldsymbol{L}_{n-1}`	$\boldsymbol{L}_{n-1}$	Since $\boldsymbol{A}_{[n-1]}$ is nonsingular it follows that $\boldsymbol{L}_{n-1}$ and $\boldsymbol{U}_{n-1}$ are nonsingular and
lyche-numerical-linear-algebra_F01124	90	1.000	`\boldsymbol{U}_{n-1}`	$\boldsymbol{U}_{n-1}$	Since $\boldsymbol{A}_{[n-1]}$ is nonsingular it follows that $\boldsymbol{L}_{n-1}$ and $\boldsymbol{U}_{n-1}$ are nonsingular and
lyche-numerical-linear-algebra_F01125	90	1.000	`\boldsymbol{l}_{n}, \boldsymbol{u}_{n}`	$\boldsymbol{l}_{n}, \boldsymbol{u}_{n}$	therefore $\boldsymbol{l}_n, \boldsymbol{u}_n$, and u_{nn} are uniquely given. Thus (3.17) gives a unique L1U
lyche-numerical-linear-algebra_F01126	90	1.000	`u_{n n}`	$u_{n n}$	therefore $\boldsymbol{l}_n, \boldsymbol{u}_n$, and u_{nn} are uniquely given. Thus (3.17) gives a unique L1U
lyche-numerical-linear-algebra_F01127	91	1.000	`k \leq n-1`	$k \leq n-1$	$\boldsymbol{A}_{[k]}$ is singular for some $k \leq n-1$. We will show that this leads to a contradiction.
lyche-numerical-linear-algebra_F01128	91	1.000	`\boldsymbol{A}_{[j]}`	$\boldsymbol{A}_{[j]}$	Let k be the smallest integer so that $\boldsymbol{A}_{[k]}$ is singular. Since $\boldsymbol{A}_{[j]}$ is nonsingular for
lyche-numerical-linear-algebra_F01129	91	0.916	`\boldsymbol{j} \leq k-1`	$\boldsymbol{j} \leq k-1$	$j \leq k-1$ it follows from what we have already shown that $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]}\boldsymbol{U}_{[k]}$ is the
lyche-numerical-linear-algebra_F01130	91	1.000	`\boldsymbol{U}_{[k]}^{T} \boldsymbol{M}_{k}^{T}= \boldsymbol{C}_{k}^{T}`	$\boldsymbol{U}_{[k]}^{T} \boldsymbol{M}_{k}^{T}= \boldsymbol{C}_{k}^{T}$	and $\boldsymbol{L}_{[k]}$ is nonsingular. By (3.16) we have $\boldsymbol{U}_{[k]}^T\boldsymbol{M}_k^T = \boldsymbol{C}_k^T$. This can be written as
lyche-numerical-linear-algebra_F01131	91	1.000	`\boldsymbol{M}_{k}^{T}`	$\boldsymbol{M}_{k}^{T}$	$n-k$ linear systems for the columns of $\boldsymbol{M}_k^T$. By assumption $\boldsymbol{M}_k^T$ exists, but since
lyche-numerical-linear-algebra_F01132	91	1.000	`\boldsymbol{U}_{[k]}^{T}`	$\boldsymbol{U}_{[k]}^{T}$	$\boldsymbol{U}_{[k]}^T$ is singular $\boldsymbol{M}_k$ is not unique, a contradiction.
lyche-numerical-linear-algebra_F01133	91	1.000	`\boldsymbol{M}_{k}`	$\boldsymbol{M}_{k}$	$\boldsymbol{U}_{[k]}^T$ is singular $\boldsymbol{M}_k$ is not unique, a contradiction.
lyche-numerical-linear-algebra_F01134	91	1.000	`k=n`	$k=n$	By combining the last two equations in (3.18) we obtain with $k = n$
lyche-numerical-linear-algebra_F01135	91	1.000	`k-d`	$k-d$	is d-banded then the first $k-d$ components in $\boldsymbol{r}_k$ and $\boldsymbol{c}_k$ are zero so both $\boldsymbol{L}$ and $\boldsymbol{U}$
lyche-numerical-linear-algebra_F01136	91	1.000	`\boldsymbol{r}_{k}`	$\boldsymbol{r}_{k}$	is d-banded then the first $k-d$ components in $\boldsymbol{r}_k$ and $\boldsymbol{c}_k$ are zero so both $\boldsymbol{L}$ and $\boldsymbol{U}$
lyche-numerical-linear-algebra_F01137	91	1.000	`\boldsymbol{c}_{k}`	$\boldsymbol{c}_{k}$	is d-banded then the first $k-d$ components in $\boldsymbol{r}_k$ and $\boldsymbol{c}_k$ are zero so both $\boldsymbol{L}$ and $\boldsymbol{U}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01138	91	1.000	`\boldsymbol{U}_{k-1}^{T} \boldsymbol{l}_{k}= \boldsymbol{r}_{k}`	$\boldsymbol{U}_{k-1}^{T} \boldsymbol{l}_{k}= \boldsymbol{r}_{k}$	to solve the lower triangular system $\boldsymbol{U}_{k-1}^{T}\boldsymbol{l}_k = \boldsymbol{r}_k$ for the kth row $\boldsymbol{l}_k^T$ in $\boldsymbol{L}$ and the
lyche-numerical-linear-algebra_F01139	91	1.000	`\boldsymbol{l}_{k}^{T}`	$\boldsymbol{l}_{k}^{T}$	to solve the lower triangular system $\boldsymbol{U}_{k-1}^{T}\boldsymbol{l}_k = \boldsymbol{r}_k$ for the kth row $\boldsymbol{l}_k^T$ in $\boldsymbol{L}$ and the
lyche-numerical-linear-algebra_F01140	91	0.699	`\left[\begin{array}{c}\boldsymbol{u}_{k} \\ u_{k} \end{array}\right]`	$\left[\begin{array}{c}\boldsymbol{u}_{k} \\ u_{k} \end{array}\right]$	kth column $\left[\begin{smallmatrix}\boldsymbol{u}_k \\ u_{kk}\end{smallmatrix}\right]$ in $\boldsymbol{U}$ for $k = 2, \ldots, n$. This leads to the following algorithm to
lyche-numerical-linear-algebra_F01141	91	0.699	`k=2, \ldots, n`	$k=2, \ldots, n$	kth column $\left[\begin{smallmatrix}\boldsymbol{u}_k \\ u_{kk}\end{smallmatrix}\right]$ in $\boldsymbol{U}$ for $k = 2, \ldots, n$. This leads to the following algorithm to
lyche-numerical-linear-algebra_F01142	91	1.000	`d \geq 1`	$d \geq 1$	compute the L1U factorization of a d-banded matrix $\boldsymbol{A}$ with $d \geq 1$. The algorithm
lyche-numerical-linear-algebra_F01143	91	1.000	`O\left(d^{2} n\right)`	$O\left(d^{2} n\right)$	the number of arithmetic operation for this algorithm is $O(d^2n)$.
lyche-numerical-linear-algebra_F01144	91	0.996	`k<n`	$k<n$	factorization even if $\boldsymbol{A}_{[k]}$ is singular for some $k < n$. By Theorem 4.1 such an LU
lyche-numerical-linear-algebra_F01145	91	1.000	`\boldsymbol{A}=\boldsymbol{I} \boldsymbol{A}`	$\boldsymbol{A}=\boldsymbol{I} \boldsymbol{A}$	$\boldsymbol{A}$ is $\boldsymbol{A} = \boldsymbol{I}\boldsymbol{A}$ so it always exists even if $\boldsymbol{A}$ has zeros somewhere on the diagonal.
lyche-numerical-linear-algebra_F01146	91	1.000	`a_{k k}`	$a_{k k}$	By Lemma 2.5, if some a_{kk} is zero then $\boldsymbol{A}_{[k]}$ is singular and the L1U factorization
lyche-numerical-linear-algebra_F01147	92	1.000	`\boldsymbol{A}_{i i}`	$\boldsymbol{A}_{i i}$	where each diagonal block $\boldsymbol{A}_{ii}$ is square. We call the factorization
lyche-numerical-linear-algebra_F01148	92	0.434	`\mathbf{L}`	$\mathbf{L}$	a **block L1U factorization of** $\boldsymbol{A}$. Here the ith diagonal blocks $\boldsymbol{I}$ and $\boldsymbol{U}_{ii}$ in $\boldsymbol{L}$ and
lyche-numerical-linear-algebra_F01149	92	0.434	`\boldsymbol{U}_{i i}`	$\boldsymbol{U}_{i i}$	a **block L1U factorization of** $\boldsymbol{A}$. Here the ith diagonal blocks $\boldsymbol{I}$ and $\boldsymbol{U}_{ii}$ in $\boldsymbol{L}$ and
lyche-numerical-linear-algebra_F01150	92	0.999	`k=1, \ldots, m-1`	$k=1, \ldots, m-1$	*are nonsingular for* $k = 1, \ldots, m-1$.
lyche-numerical-linear-algebra_F01151	92	1.000	`\boldsymbol{A}_{\{k\}}`	$\boldsymbol{A}_{\{k\}}$	***Proof*** Suppose $\boldsymbol{A}_{\{k\}}$ is nonsingular for $k = 1, \ldots, m-1$. Following the proof
lyche-numerical-linear-algebra_F01152	92	0.993	`\boldsymbol{A}_{\{m-1\}}`	$\boldsymbol{A}_{\{m-1\}}$	in Theorem 3.4 suppose $\boldsymbol{A}_{\{m-1\}}$ has a unique block LU factorization $\boldsymbol{A}_{\{m-1\}} =$
lyche-numerical-linear-algebra_F01153	92	0.993	`\boldsymbol{A}_{\{m-1\}}=`	$\boldsymbol{A}_{\{m-1\}}=$	in Theorem 3.4 suppose $\boldsymbol{A}_{\{m-1\}}$ has a unique block LU factorization $\boldsymbol{A}_{\{m-1\}} =$
lyche-numerical-linear-algebra_F01154	92	0.999	`\boldsymbol{L}_{\{m-1\}} \boldsymbol{U}_{\{m-1\}}`	$\boldsymbol{L}_{\{m-1\}} \boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}}\boldsymbol{U}_{\{m-1\}}$, and that $\boldsymbol{A}_{\{1\}}, \ldots, \boldsymbol{A}_{\{m-1\}}$ are nonsingular. Then $\boldsymbol{L}_{\{m-1\}}$ and
lyche-numerical-linear-algebra_F01155	92	0.999	`\boldsymbol{A}_{\{1\}}, \ldots, \boldsymbol{A}_{\{m-1\}}`	$\boldsymbol{A}_{\{1\}}, \ldots, \boldsymbol{A}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}}\boldsymbol{U}_{\{m-1\}}$, and that $\boldsymbol{A}_{\{1\}}, \ldots, \boldsymbol{A}_{\{m-1\}}$ are nonsingular. Then $\boldsymbol{L}_{\{m-1\}}$ and
lyche-numerical-linear-algebra_F01156	92	0.999	`\boldsymbol{L}_{\{m-1\}}`	$\boldsymbol{L}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}}\boldsymbol{U}_{\{m-1\}}$, and that $\boldsymbol{A}_{\{1\}}, \ldots, \boldsymbol{A}_{\{m-1\}}$ are nonsingular. Then $\boldsymbol{L}_{\{m-1\}}$ and
lyche-numerical-linear-algebra_F01157	92	0.523	`\boldsymbol{U}_{\{m-1\}}`	$\boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{U}_{\{m-1\}}$ are nonsingular and
lyche-numerical-linear-algebra_F01158	93	0.934	`\boldsymbol{A}_{\{k\}}=\boldsymbol{L}_{\{k\}} \boldsymbol{U}_{\{k\}}`	$\boldsymbol{A}_{\{k\}}=\boldsymbol{L}_{\{k\}} \boldsymbol{U}_{\{k\}}$	that $\boldsymbol{A}_{\{k\}} = \boldsymbol{L}_{\{k\}}\boldsymbol{U}_{\{k\}}$ is the unique block LU factorization of $\boldsymbol{A}_{[k]}$ for $k = 1, \ldots, m$.
lyche-numerical-linear-algebra_F01159	93	0.934	`k=1, \ldots, m`	$k=1, \ldots, m$	that $\boldsymbol{A}_{\{k\}} = \boldsymbol{L}_{\{k\}}\boldsymbol{U}_{\{k\}}$ is the unique block LU factorization of $\boldsymbol{A}_{[k]}$ for $k = 1, \ldots, m$.
lyche-numerical-linear-algebra_F01160	93	1.000	`\boldsymbol{U}_{i i}=\tilde{\boldsymbol{L}}_{i i} \tilde{ \boldsymbol{U}}_{i i}`	$\boldsymbol{U}_{i i}=\tilde{\boldsymbol{L}}_{i i} \tilde{ \boldsymbol{U}}_{i i}$	factorization $\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii}\tilde{\boldsymbol{U}}_{ii}$. Then $\boldsymbol{A} = \hat{\boldsymbol{L}}\hat{\boldsymbol{U}}$, where $\hat{\boldsymbol{L}} := \boldsymbol{L}\,\mathrm{diag}(\tilde{\boldsymbol{L}}_{ii})$ and
lyche-numerical-linear-algebra_F01161	93	1.000	`\boldsymbol{A}=\hat{\boldsymbol{L}} \hat{\boldsymbol{U}}`	$\boldsymbol{A}=\hat{\boldsymbol{L}} \hat{\boldsymbol{U}}$	factorization $\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii}\tilde{\boldsymbol{U}}_{ii}$. Then $\boldsymbol{A} = \hat{\boldsymbol{L}}\hat{\boldsymbol{U}}$, where $\hat{\boldsymbol{L}} := \boldsymbol{L}\,\mathrm{diag}(\tilde{\boldsymbol{L}}_{ii})$ and
lyche-numerical-linear-algebra_F01162	93	1.000	`\hat{\boldsymbol{L}}:=\boldsymbol{L} \operatorname{diag} \left(\tilde{\boldsymbol{L}}_{i i}\right)`	$\hat{\boldsymbol{L}}:=\boldsymbol{L} \operatorname{diag} \left(\tilde{\boldsymbol{L}}_{i i}\right)$	factorization $\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii}\tilde{\boldsymbol{U}}_{ii}$. Then $\boldsymbol{A} = \hat{\boldsymbol{L}}\hat{\boldsymbol{U}}$, where $\hat{\boldsymbol{L}} := \boldsymbol{L}\,\mathrm{diag}(\tilde{\boldsymbol{L}}_{ii})$ and
lyche-numerical-linear-algebra_F01163	93	0.987	`\hat{\boldsymbol{U}}:=\operatorname{diag}\left(\tilde{ \boldsymbol{L}}_{i i}^{-1}\right) \boldsymbol{U}`	$\hat{\boldsymbol{U}}:=\operatorname{diag}\left(\tilde{ \boldsymbol{L}}_{i i}^{-1}\right) \boldsymbol{U}$	$\hat{\boldsymbol{U}} := \mathrm{diag}(\tilde{\boldsymbol{L}}_{ii}^{-1})\boldsymbol{U}$, and this is an ordinary LU factorization of $\boldsymbol{A}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01164	93	0.997	`\left[\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{n} \right]`	$\left[\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{n} \right]$	$[\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n]$ is also lower triangular. The kth column $\boldsymbol{b}_k$ of $\boldsymbol{B}$ is the solution of the
lyche-numerical-linear-algebra_F01165	93	0.997	`\boldsymbol{b}_{k}`	$\boldsymbol{b}_{k}$	$[\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n]$ is also lower triangular. The kth column $\boldsymbol{b}_k$ of $\boldsymbol{B}$ is the solution of the
lyche-numerical-linear-algebra_F01166	93	1.000	`\boldsymbol{A} \boldsymbol{b}_{k}=\boldsymbol{e}_{k}`	$\boldsymbol{A} \boldsymbol{b}_{k}=\boldsymbol{e}_{k}$	linear systems $\boldsymbol{A}\boldsymbol{b}_k = \boldsymbol{e}_k$. Show that $b_k(k) = 1/a(k,k)$ for $k = 1, \ldots, n$, and
lyche-numerical-linear-algebra_F01167	93	1.000	`b_{k}(k)=1 / a(k, k)`	$b_{k}(k)=1 / a(k, k)$	linear systems $\boldsymbol{A}\boldsymbol{b}_k = \boldsymbol{e}_k$. Show that $b_k(k) = 1/a(k,k)$ for $k = 1, \ldots, n$, and
lyche-numerical-linear-algebra_F01168	94	1.000	`k=n, n-1, \ldots, 1`	$k=n, n-1, \ldots, 1$	for $k = n, n-1, \ldots, 1$.
lyche-numerical-linear-algebra_F01169	94	1.000	`\frac{1}{3} n\left(2 n^{2}+1\right) \approx G_{n}`	$\frac{1}{3} n\left(2 n^{2}+1\right) \approx G_{n}$	cation $\boldsymbol{A}\boldsymbol{B}$ can be done in $\frac{1}{3}n(2n^2+1) \approx G_n$ arithmetic operations when $\boldsymbol{A} \in \mathbb{R}^{n\times n}$
lyche-numerical-linear-algebra_F01170	94	1.000	`\boldsymbol{B} \in \mathbb{R}^{n \times n}`	$\boldsymbol{B} \in \mathbb{R}^{n \times n}$	is lower triangular and $\boldsymbol{B} \in \mathbb{R}^{n\times n}$ is upper triangular. What about $\boldsymbol{B}\boldsymbol{A}$?
lyche-numerical-linear-algebra_F01171	94	1.000	`\boldsymbol{B} \boldsymbol{A}`	$\boldsymbol{B} \boldsymbol{A}$	is lower triangular and $\boldsymbol{B} \in \mathbb{R}^{n\times n}$ is upper triangular. What about $\boldsymbol{B}\boldsymbol{A}$?
lyche-numerical-linear-algebra_F01172	94	1.000	`\boldsymbol{P}, \boldsymbol{L}, \boldsymbol{U}`	$\boldsymbol{P}, \boldsymbol{L}, \boldsymbol{U}$	**Exercise 3.5 (Using PLU for A^*)** Suppose we know the PLU factors $\boldsymbol{P}, \boldsymbol{L}, \boldsymbol{U}$ in a
lyche-numerical-linear-algebra_F01173	94	1.000	`\boldsymbol{A}^{*} \boldsymbol{x}=\boldsymbol{b}`	$\boldsymbol{A}^{*} \boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{A}^*\boldsymbol{x} = \boldsymbol{b}$ economically.
lyche-numerical-linear-algebra_F01174	94	1.000	`2 G_{n}`	$2 G_{n}$	$2G_n$ arithmetic operations to compute $\boldsymbol{A}^{-1} = \boldsymbol{U}^{-1}\boldsymbol{L}^{-1}\boldsymbol{P}^T$. Here we have not
lyche-numerical-linear-algebra_F01175	94	1.000	`\boldsymbol{A}^{-1}=\boldsymbol{U}^{-1} \boldsymbol{L}^{-1} \boldsymbol{P}^{T}`	$\boldsymbol{A}^{-1}=\boldsymbol{U}^{-1} \boldsymbol{L}^{-1} \boldsymbol{P}^{T}$	$2G_n$ arithmetic operations to compute $\boldsymbol{A}^{-1} = \boldsymbol{U}^{-1}\boldsymbol{L}^{-1}\boldsymbol{P}^T$. Here we have not
lyche-numerical-linear-algebra_F01176	94	1.000	`\boldsymbol{P}^{T}`	$\boldsymbol{P}^{T}$	counted the final multiplication with $\boldsymbol{P}^T$ which amounts to n row interchanges.
lyche-numerical-linear-algebra_F01177	95	0.993	`k=1,2, \ldots, n-1`	$k=1,2, \ldots, n-1$	1. for $k = 1, 2, \ldots, n-1$
lyche-numerical-linear-algebra_F01178	95	1.000	`a_{k, j}`	$a_{k, j}$	(a) exchange $a_{k,j}$ and $a_{k+1,j}$ for $j = k, k+1, \ldots, n$
lyche-numerical-linear-algebra_F01179	95	1.000	`a_{k+1, j}`	$a_{k+1, j}$	(a) exchange $a_{k,j}$ and $a_{k+1,j}$ for $j = k, k+1, \ldots, n$
lyche-numerical-linear-algebra_F01180	95	1.000	`j=k, k+1, \ldots, n`	$j=k, k+1, \ldots, n$	(a) exchange $a_{k,j}$ and $a_{k+1,j}$ for $j = k, k+1, \ldots, n$
lyche-numerical-linear-algebra_F01181	95	1.000	`i=k+1, k+2, \ldots, n`	$i=k+1, k+2, \ldots, n$	(b) for $i = k+1, k+2, \ldots, n$
lyche-numerical-linear-algebra_F01182	95	1.000	`a_{i, k}=m_{i, k}=a_{i, k} / a_{k, k}`	$a_{i, k}=m_{i, k}=a_{i, k} / a_{k, k}$	i. $a_{i,k} = m_{i,k} = a_{i,k}/a_{k,k}$
lyche-numerical-linear-algebra_F01183	95	1.000	`a_{i, j}=a_{i, j}-m_{i, k} a_{k, j}`	$a_{i, j}=a_{i, j}-m_{i, k} a_{k, j}$	ii. $a_{i,j} = a_{i,j} - m_{i,k}a_{k,j}$ for $j = k+1, k+2, \ldots, n$
lyche-numerical-linear-algebra_F01184	95	1.000	`j=k+1, k+2, \ldots, n`	$j=k+1, k+2, \ldots, n$	ii. $a_{i,j} = a_{i,j} - m_{i,k}a_{k,j}$ for $j = k+1, k+2, \ldots, n$
lyche-numerical-linear-algebra_F01185	95	1.000	`b_{k}`	b_{k}	(a) exchange b_k and b_{k+1}
lyche-numerical-linear-algebra_F01186	95	1.000	`b_{k+1}`	b_{k+1}	(a) exchange b_k and b_{k+1}
lyche-numerical-linear-algebra_F01187	95	1.000	`b_{i}=b_{i}-a_{i, k} b_{k}`	$b_{i}=b_{i}-a_{i, k} b_{k}$	(b) $b_i = b_i - a_{i,k}b_k$ for $i = k+1, k+2, \ldots, n$
lyche-numerical-linear-algebra_F01188	95	1.000	`x_{n}=b_{n} / a_{n, n}`	$x_{n}=b_{n} / a_{n, n}$	2. $x_n = b_n/a_{n,n}$
lyche-numerical-linear-algebra_F01189	95	0.981	`=0`	$=0$	(a) $sum = 0$
lyche-numerical-linear-algebra_F01190	95	0.958	`+a_{k, j} x_{j}`	$+a_{k, j} x_{j}$	(b) $sum = sum + a_{k,j}x_j$ for $j = k+1, k+2, \ldots, n$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01191	95	1.000	`x_{k}=\left(b_{k}-\right.`	$x_k = (b_k -$	(c) $x_k = (b_k - sum)/a_{k,k}$
lyche-numerical-linear-algebra_F01192	95	1.000	`) / a_{k, k}`	$)/a_{k,k}$	(c) $x_k = (b_k - sum)/a_{k,k}$
lyche-numerical-linear-algebra_F01193	95	1.000	`\boldsymbol{H} \in \mathbb{R}^{n \times n}`	$\boldsymbol{H} \in \mathbb{R}^{n \times n}$	We say that $\boldsymbol{H} \in \mathbb{R}^{n\times n}$ is **unreduced upper Hessenberg** if it is upper Hessenberg
lyche-numerical-linear-algebra_F01194	95	1.000	`h_{i, i-1} \neq 0`	$h_{i,i-1} \neq 0$	and the subdiagonal elements $h_{i,i-1} \neq 0$ for $i = 2, \ldots, n$.
lyche-numerical-linear-algebra_F01195	95	0.983	`\boldsymbol{H} \boldsymbol{x}=\boldsymbol{b}`	$\boldsymbol{H}\boldsymbol{x}=\boldsymbol{b}$	solving the linear system $\boldsymbol{H}\boldsymbol{x} = \boldsymbol{b}$ using suitable specializations of Algorithms 1
lyche-numerical-linear-algebra_F01196	95	1.000	`\boldsymbol{U} \in \mathbb{R}^{n \times n}`	$\boldsymbol{U} \in \mathbb{R}^{n \times n}$	**c)** Let $\boldsymbol{U} \in \mathbb{R}^{n\times n}$ be upper triangular and nonsingular. We define
lyche-numerical-linear-algebra_F01197	95	1.000	`\boldsymbol{v} \in \mathbb{R}^{n}`	$\boldsymbol{v} \in \mathbb{R}^{n}$	where $\boldsymbol{v} \in \mathbb{R}^n$ and $\boldsymbol{e}_1$ is the first unit vector in $\mathbb{R}^n$. We also let
lyche-numerical-linear-algebra_F01198	95	1.000	`\boldsymbol{e}_{1}`	$\boldsymbol{e}_1$	where $\boldsymbol{v} \in \mathbb{R}^n$ and $\boldsymbol{e}_1$ is the first unit vector in $\mathbb{R}^n$. We also let
lyche-numerical-linear-algebra_F01199	95	0.999	`\boldsymbol{I}_{i, j}`	$\boldsymbol{I}_{i,j}$	where the $\boldsymbol{I}_{i,j}$ are obtained from the identity matrix by interchanging rows i and
lyche-numerical-linear-algebra_F01200	95	0.822	`\boldsymbol{E}:=\boldsymbol{C} \boldsymbol{P}`	$\boldsymbol{E} := \boldsymbol{C}\boldsymbol{P}$	j. Explain why the matrix $\boldsymbol{E} := \boldsymbol{C}\boldsymbol{P}$ is unreduced upper Hessenberg.
lyche-numerical-linear-algebra_F01201	96	1.000	`\boldsymbol{W} \in \mathbb{R}^{n}`	$\boldsymbol{W} \in \mathbb{R}^{n}$	$\boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}$. To a given $\boldsymbol{W} \in \mathbb{R}^n$ we define a rank one modification of $\boldsymbol{A}$ by
lyche-numerical-linear-algebra_F01202	96	1.000	`\boldsymbol{B}=\boldsymbol{L} \boldsymbol{H} \boldsymbol{P}^{T}`	$\boldsymbol{B} = \boldsymbol{L}\boldsymbol{H}\boldsymbol{P}^T$	Show that $\boldsymbol{B}$ has the factorization $\boldsymbol{B} = \boldsymbol{L}\boldsymbol{H}\boldsymbol{P}^T$, where $\boldsymbol{L}$ is unit lower triangular,
lyche-numerical-linear-algebra_F01203	96	1.000	`\boldsymbol{H}`	$\boldsymbol{H}$	$\boldsymbol{P}$ is given by (3.29) and $\boldsymbol{H}$ is unreduced upper Hessenberg.
lyche-numerical-linear-algebra_F01204	96	1.000	`\boldsymbol{B} \boldsymbol{x}=\boldsymbol{b}`	$\boldsymbol{B}\boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{B}\boldsymbol{x} = \boldsymbol{b}$, where $\boldsymbol{B}$ is given by (3.30). We assume that the matrices $\boldsymbol{L}$ and $\boldsymbol{U}$ in
lyche-numerical-linear-algebra_F01205	96	0.999	`1 \leq`	$1 \leq$	**Exercise 3.9 (# Operations for Banded Triangular Systems)** Show that for $1 \leq$
lyche-numerical-linear-algebra_F01206	96	1.000	`d \leq n`	$d \leq n$	$d \leq n$ Algorithm 3.4, with $A(k,k) = 1$ for $k = 1, \ldots, n$ in Algorithm 3.1, requires
lyche-numerical-linear-algebra_F01207	96	1.000	`A(k, k)=1`	$A(k,k)=1$	$d \leq n$ Algorithm 3.4, with $A(k,k) = 1$ for $k = 1, \ldots, n$ in Algorithm 3.1, requires
lyche-numerical-linear-algebra_F01208	96	1.000	`N_{L U}(n, d):=\left(2 d^{2}+d\right) n-\left(d^{2}+d \right)(8 d+1) / 6=O\left(d^{2} n\right)`	$N_{LU}(n,d) := \left(2d^2+d\right)n - \left(d^2+d\right)(8d+1)/6 = O\left(d^2 n\right)$	exactly $N_{LU}(n,d) := (2d^2+d)n - (d^2+d)(8d+1)/6 = O(d^2 n)$ operations.[2] In
lyche-numerical-linear-algebra_F01209	96	1.000	`d=n-1`	$d=n-1$	particular, for a full matrix $d = n-1$ and we find $N_{LU}(n,n) = \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \approx$
lyche-numerical-linear-algebra_F01210	96	1.000	`N_{L U}(n, n)=\frac{2}{3} n^{3}-\frac{1}{2} n^{2}- \frac{1}{6} n \approx`	$N_{LU}(n,n) = \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \approx$	particular, for a full matrix $d = n-1$ and we find $N_{LU}(n,n) = \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \approx$
lyche-numerical-linear-algebra_F01211	96	1.000	`N_{L U}(n, 1)=3 n-3=O(n)`	$N_{LU}(n,1) = 3n-3 = O(n)$	tridiagonal matrix $N_{LU}(n,1) = 3n-3 = O(n)$.
lyche-numerical-linear-algebra_F01212	96	0.985	`\boldsymbol{A}=\left[\left[\begin{array}{ll}1 & 1 \\ 0 & 1\end{array}\right]\right.`	$\boldsymbol{A} = \left[\left[\begin{array}{ll}1 & 1 \\ 0 & 1\end{array}\right]\right]$	**Exercise 3.12 (Row Interchange)** Show that $\boldsymbol{A} = \left[\left[\begin{smallmatrix}1 & 1 \\ 0 & 1\end{smallmatrix}\right]\right]$ has a unique L1U
lyche-numerical-linear-algebra_F01213	96	0.999	`2 \leq k \leq d`	$2 \leq k \leq d$	[2]Hint: Consider the cases $2 \leq k \leq d$ and $d+1 \leq k \leq n$ separately.
lyche-numerical-linear-algebra_F01214	96	0.999	`d+1 \leq k \leq n`	$d+1 \leq k \leq n$	[2]Hint: Consider the cases $2 \leq k \leq d$ and $d+1 \leq k \leq n$ separately.
lyche-numerical-linear-algebra_F01215	97	1.000	`u_{k k}`	u_{kk}	u_{kk} in the L1U factorization are
lyche-numerical-linear-algebra_F01216	97	1.000	`\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1) \times(n-1)}`	$\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1)\times(n-1)}$	where $\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1)\times(n-1)}$. Define $\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2}\boldsymbol{U}_{2,2}$ and $\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$.
lyche-numerical-linear-algebra_F01217	97	1.000	`\boldsymbol{A}_{2,2}:=\boldsymbol{L}_{2,2} \boldsymbol{U}_{2,2}`	$\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2}\boldsymbol{U}_{2,2}$	where $\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1)\times(n-1)}$. Define $\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2}\boldsymbol{U}_{2,2}$ and $\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F01218	97	1.000	\boldsymbol{B}_{2,2}:=\boldsymbol{A}_{2,2}^{-1}	$\boldsymbol{B}_{2,2}:=\boldsymbol{A}_{2,2}^{-1}$	where $\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1)\times(n-1)}$. Define $\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2}\boldsymbol{U}_{2,2}$ and $\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$.
lyche-numerical-linear-algebra_ F01219	97	1.000	l_{i, j}, i>j	$l_{i, j}, i>j$	**b)** Suppose that the elements $l_{i,j}$, $i > j$ in $\boldsymbol{L}$ and $u_{i,j}$, $j \geq i$ in $\boldsymbol{U}$ are stored in
lyche-numerical-linear-algebra_ F01220	97	1.000	u_{i, j}, j \geq i	$u_{i, j}, j \geq i$	**b)** Suppose that the elements $l_{i,j}$, $i > j$ in $\boldsymbol{L}$ and $u_{i,j}$, $j \geq i$ in $\boldsymbol{U}$ are stored in
lyche-numerical-linear-algebra_ F01221	97	1.000	a_{i, j}	$a_{i, j}$	$\boldsymbol{A}$ with elements $a_{i,j}$. Write an algorithm that overwrites the elements in $\boldsymbol{A}$ with
lyche-numerical-linear-algebra_ F01222	97	1.000	\boldsymbol{s} \in \mathbb{R}^{n}	$\boldsymbol{s} \in \mathbb{R}^{n}$	ones in $\boldsymbol{A}^{-1}$. Only one extra vector $\boldsymbol{s} \in \mathbb{R}^n$ should be used.
lyche-numerical-linear-algebra_ F01223	97	1.000	\boldsymbol{T} \boldsymbol{H} \boldsymbol{x}= \boldsymbol{b}	$\boldsymbol{T} \boldsymbol{H} \boldsymbol{x}=\boldsymbol{b}$	**Exercise 3.18 (Solving $\boldsymbol{T}\boldsymbol{H}\boldsymbol{x} = \boldsymbol{b}$ (Exam Exercise 1981-3))** In this exercise we
lyche-numerical-linear-algebra_ F01224	97	1.000	\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n \times n}	$\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n \times n}$	consider nonsingular matrices $\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n\times n}$ with $\boldsymbol{T} = (t_{ij})$ upper triangular,
lyche-numerical-linear-algebra_ F01225	97	1.000	\boldsymbol{T}=\left(t_{i j}\right)	$\boldsymbol{T}=\left(t_{i j}\right)$	consider nonsingular matrices $\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n\times n}$ with $\boldsymbol{T} = (t_{ij})$ upper triangular,
lyche-numerical-linear-algebra_ F01226	97	1.000	\boldsymbol{H}=\left(h_{i j}\right)	$\boldsymbol{H}=\left(h_{i j}\right)$	$\boldsymbol{H} = (h_{ij})$ upper Hessenberg and $\boldsymbol{S} := \boldsymbol{T}\boldsymbol{H}$. We assume that $\boldsymbol{H}$ has a unique
lyche-numerical-linear-algebra_ F01227	97	1.000	\boldsymbol{S}:=\boldsymbol{T} \boldsymbol{H}	$\boldsymbol{S}:=\boldsymbol{T} \boldsymbol{H}$	$\boldsymbol{H} = (h_{ij})$ upper Hessenberg and $\boldsymbol{S} := \boldsymbol{T}\boldsymbol{H}$. We assume that $\boldsymbol{H}$ has a unique
lyche-numerical-linear-algebra_ F01228	97	1.000	\boldsymbol{H}=\boldsymbol{L} \boldsymbol{U}	$\boldsymbol{H}=\boldsymbol{L} \boldsymbol{U}$	LU1 factorization $\boldsymbol{H} = \boldsymbol{L}\boldsymbol{U}$ with $\Vert\boldsymbol{L}\Vert_\infty\Vert\boldsymbol{U}\Vert_\infty \leq K\Vert\boldsymbol{H}\Vert_\infty$ for a constant K not
lyche-numerical-linear-algebra_ F01229	97	1.000	\\|\boldsymbol{L}\\|_{\infty}\\|\boldsymbol{U}\\|_{\infty} \leq K\\|\boldsymbol{H}\\|_{\infty}	$\Vert\boldsymbol{L}\Vert_{\infty}\Vert\boldsymbol{U}\Vert_{\infty} \leq K\Vert\boldsymbol{H}\Vert_{\infty}$	LU1 factorization $\boldsymbol{H} = \boldsymbol{L}\boldsymbol{U}$ with $\Vert\boldsymbol{L}\Vert_\infty\Vert\boldsymbol{U}\Vert_\infty \leq K\Vert\boldsymbol{H}\Vert_\infty$ for a constant K not
lyche-numerical-linear-algebra_ F01230	97	1.000	\boldsymbol{S}	$\boldsymbol{S}$	**a)** Give an algorithm which computes $\boldsymbol{S}$ from $\boldsymbol{T}$ and $\boldsymbol{H}$ without using the lower parts
lyche-numerical-linear-algebra_ F01231	97	0.999	\left(t_{i j}, i>j\right)	$\left(t_{i j}, i>j\right)$	$(t_{ij}, i > j)$ of $\boldsymbol{T}$ and $(h_{ij}, i > j+1)$ of $\boldsymbol{H}$. In what order should the elements
lyche-numerical-linear-algebra_ F01232	97	0.999	\left(h_{i j}, i>j+1\right)	$\left(h_{i j}, i>j+1\right)$	$(t_{ij}, i > j)$ of $\boldsymbol{T}$ and $(h_{ij}, i > j+1)$ of $\boldsymbol{H}$. In what order should the elements
lyche-numerical-linear-algebra_ F01233	98	0.987	\boldsymbol{T}, \boldsymbol{H}	$\boldsymbol{T}, \boldsymbol{H}$	**d)** Given $\boldsymbol{b} \in \mathbb{R}^n$ and $\boldsymbol{T}$,$\boldsymbol{H}$ as before. Suppose $\boldsymbol{S}$ and the LU1-factorization are not
lyche-numerical-linear-algebra_ F01234	98	1.000	\boldsymbol{S} \boldsymbol{x}=\boldsymbol{b}	$\boldsymbol{S} \boldsymbol{x}=\boldsymbol{b}$	computed. We want to find $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{S}\boldsymbol{x} = \boldsymbol{b}$. We have the 2 following
lyche-numerical-linear-algebra_ F01235	98	0.779	\boldsymbol{S}=\boldsymbol{T} \boldsymbol{H}	$\boldsymbol{S}=\boldsymbol{T} \boldsymbol{H}$	1. $\boldsymbol{S} = \boldsymbol{T}\boldsymbol{H}$
lyche-numerical-linear-algebra_ F01236	98	0.991	\boldsymbol{T} \boldsymbol{z}=\boldsymbol{b}	$\boldsymbol{T} \boldsymbol{z}=\boldsymbol{b}$	1. Solve $\boldsymbol{T}\boldsymbol{z} = \boldsymbol{b}$
lyche-numerical-linear-algebra_ F01237	98	0.894	\boldsymbol{H} \boldsymbol{x}=\boldsymbol{z}	$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{z}$	2. Solve $\boldsymbol{H}\boldsymbol{x} = \boldsymbol{z}$
lyche-numerical-linear-algebra_ F01238	98	1.000	\boldsymbol{a}_{1}, \boldsymbol{a}_{2}, \ldots, \boldsymbol{a}_{n}	$\boldsymbol{a}_{1}, \boldsymbol{a}_{2}, \ldots, \boldsymbol{a}_{n}$	$\mathbb{R}^{n\times n}$ be nonsingular with columns $\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_n$. We assume that $\boldsymbol{A}$ has a unique
lyche-numerical-linear-algebra_ F01239	98	1.000	p \leq n	$p \leq n$	For a positive integer $p \leq n$ and $\boldsymbol{b} \in \mathbb{R}^n$ we define
lyche-numerical-linear-algebra_ F01240	98	1.000	\boldsymbol{H}:=\boldsymbol{L}^{-1} \boldsymbol{B}	$\boldsymbol{H}:=\boldsymbol{L}^{-1} \boldsymbol{B}$	**a)** Show that $\boldsymbol{H} := \boldsymbol{L}^{-1}\boldsymbol{B}$ is upper Hessenberg. We assume that $\boldsymbol{H}$ has a unique
lyche-numerical-linear-algebra_ F01241	98	1.000	\boldsymbol{H}=\boldsymbol{L}_{H} \boldsymbol{U}_{H}	$\boldsymbol{H}=\boldsymbol{L}_{H} \boldsymbol{U}_{H}$	L1U factorization. $\boldsymbol{H} = \boldsymbol{L}_H\boldsymbol{U}_H$.
lyche-numerical-linear-algebra_ F01242	98	1.000	\boldsymbol{H}:=\boldsymbol{L}_{H} \boldsymbol{U}_{H}	$\boldsymbol{H}:=\boldsymbol{L}_{H} \boldsymbol{U}_{H}$	**c)** Suppose we have found the L1U factorization $\boldsymbol{H} := \boldsymbol{L}_H\boldsymbol{U}_H$ of $\boldsymbol{H}$. Explain how
lyche-numerical-linear-algebra_ F01243	98	1.000	\boldsymbol{L}_{H}	$\boldsymbol{L}_{H}$	we can find the L1U factorization of $\boldsymbol{B}$ from $\boldsymbol{L}_H$ and $\boldsymbol{U}_H$.
lyche-numerical-linear-algebra_ F01244	98	1.000	\boldsymbol{U}_{H}	$\boldsymbol{U}_{H}$	we can find the L1U factorization of $\boldsymbol{B}$ from $\boldsymbol{L}_H$ and $\boldsymbol{U}_H$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01245	98	0.999	`\boldsymbol{A}=\boldsymbol{U} \boldsymbol{L}`	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{L}$	$\mathbb{R}^{n\times n}$ has a U1L factorization if $\boldsymbol{A}=\boldsymbol{U}\boldsymbol{L}$ for an upper triangular matrix $\boldsymbol{U}\in\mathbb{R}^{n\times n}$
lyche-numerical-linear-algebra_F01246	98	1.000	`\boldsymbol{P} \in \mathbb{R}^{n \times n}`	$\boldsymbol{P} \in \mathbb{R}^{n \times n}$	**b)** Let the columns of $\boldsymbol{P}\in\mathbb{R}^{n\times n}$ be the unit vectors in reverse order, i.e.,
lyche-numerical-linear-algebra_F01247	99	1.000	`\boldsymbol{P}^{T}=\boldsymbol{P}`	$\boldsymbol{P}^{T}=\boldsymbol{P}$	Show that $\boldsymbol{P}^T=\boldsymbol{P}$ and $\boldsymbol{P}^2=\boldsymbol{I}$. What is the connection between the elements
lyche-numerical-linear-algebra_F01248	99	1.000	`\boldsymbol{P}^{2}=\boldsymbol{I}`	$\boldsymbol{P}^{2}=\boldsymbol{I}$	Show that $\boldsymbol{P}^T=\boldsymbol{P}$ and $\boldsymbol{P}^2=\boldsymbol{I}$. What is the connection between the elements
lyche-numerical-linear-algebra_F01249	99	1.000	`\boldsymbol{P} \boldsymbol{A}`	$\boldsymbol{P} \boldsymbol{A}$	in $\boldsymbol{A}$ and $\boldsymbol{P}\boldsymbol{A}$?
lyche-numerical-linear-algebra_F01250	99	1.000	`\boldsymbol{B}:=\boldsymbol{P} \boldsymbol{A} \boldsymbol{P}`	$\boldsymbol{B}:=\boldsymbol{P} \boldsymbol{A} \boldsymbol{P}$	**c)** Let $\boldsymbol{B}:=\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$. Find integers r,s, depending on i,j,n, such that $b_{i,j}=a_{r,s}$.
lyche-numerical-linear-algebra_F01251	99	1.000	`r, s`	r, s	**c)** Let $\boldsymbol{B}:=\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$. Find integers r,s, depending on i,j,n, such that $b_{i,j}=a_{r,s}$.
lyche-numerical-linear-algebra_F01252	99	1.000	`i, j, n`	i, j, n	**c)** Let $\boldsymbol{B}:=\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$. Find integers r,s, depending on i,j,n, such that $b_{i,j}=a_{r,s}$.
lyche-numerical-linear-algebra_F01253	99	1.000	`b_{i, j}=a_{r, s}`	$b_{i, j}=a_{r, s}$	**c)** Let $\boldsymbol{B}:=\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$. Find integers r,s, depending on i,j,n, such that $b_{i,j}=a_{r,s}$.
lyche-numerical-linear-algebra_F01254	99	1.000	`b_{i, j}`	$b_{i, j}$	The elements $b_{i,j}$ in $\boldsymbol{B}$ should be stored in position i,j in $\boldsymbol{A}$. You should not use
lyche-numerical-linear-algebra_F01255	99	1.000	`w \in \mathbb{R}`	$w \in \mathbb{R}$	other matrices than $\boldsymbol{A}$ and a scalar $w\in\mathbb{R}$.
lyche-numerical-linear-algebra_F01256	99	1.000	`\boldsymbol{P} \boldsymbol{A} \boldsymbol{P}= \boldsymbol{M} \boldsymbol{R}`	$\boldsymbol{P} \boldsymbol{A} \boldsymbol{P}= \boldsymbol{M} \boldsymbol{R}$	**e)** Let $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}=\boldsymbol{M}\boldsymbol{R}$ be an L1U factorization of $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$, i.e., $\boldsymbol{M}$ is lower triangular
lyche-numerical-linear-algebra_F01257	99	1.000	`\boldsymbol{P} \boldsymbol{A} \boldsymbol{P}`	$\boldsymbol{P} \boldsymbol{A} \boldsymbol{P}$	**e)** Let $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}=\boldsymbol{M}\boldsymbol{R}$ be an L1U factorization of $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$, i.e., $\boldsymbol{M}$ is lower triangular
lyche-numerical-linear-algebra_F01258	99	1.000	`\boldsymbol{M}`	$\boldsymbol{M}$	**e)** Let $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}=\boldsymbol{M}\boldsymbol{R}$ be an L1U factorization of $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$, i.e., $\boldsymbol{M}$ is lower triangular
lyche-numerical-linear-algebra_F01259	99	1.000	`\boldsymbol{M}, \boldsymbol{R}`	$\boldsymbol{M}, \boldsymbol{R}$	$\boldsymbol{L}$ in a U1L factorization of $\boldsymbol{A}$ in terms of $\boldsymbol{M}$, $\boldsymbol{R}$ and $\boldsymbol{P}$.
lyche-numerical-linear-algebra_F01260	99	0.504	`\hat{\boldsymbol{L}}`	$\hat{\boldsymbol{L}}$	**Exercise 3.21 (Making Block LU into LU)** Show that $\hat{\boldsymbol{L}}$ is unit lower triangular
lyche-numerical-linear-algebra_F01261	99	1.000	`\hat{\boldsymbol{U}}`	$\hat{\boldsymbol{U}}$	and $\hat{\boldsymbol{U}}$ is upper triangular.
lyche-numerical-linear-algebra_F01262	100	0.997	`a_{j i}=\bar{a}_{i j}`	$a_{j i}=\bar{a}_{i j}$	matrices. Recall that a matrix $\boldsymbol{A}\in\mathbb{C}^{n\times n}$ is **Hermitian** if $\boldsymbol{A}^*=\boldsymbol{A}$, i.e., $a_{ji}=\overline{a}_{ij}$
lyche-numerical-linear-algebra_F01263	100	1.000	`a_{i i}=\bar{a}_{i i}`	$a_{i i}=\bar{a}_{i i}$	for all i,j. A real Hermitian matrix is symmetric. Since $a_{ii}=\overline{a}_{ii}$ the diagonal
lyche-numerical-linear-algebra_F01264	100	1.000	`\boldsymbol{U}=\boldsymbol{L}^{*}`	$\boldsymbol{U}=\boldsymbol{L}^{*}$	1. the LDL* factorization which is an LDU factorization with $\boldsymbol{U}=\boldsymbol{L}^*$ and $\boldsymbol{D}$ a
lyche-numerical-linear-algebra_F01265	100	1.000	`l_{i i}>0`	$l_{i i}>0$	2. the LL* factorization which is an LU factorization with $\boldsymbol{U}=\boldsymbol{L}^*$ and $l_{ii}>0$
lyche-numerical-linear-algebra_F01266	100	0.999	`\boldsymbol{A}^{*}=\left(\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^{*}\right)^{*}=\boldsymbol{L} \boldsymbol{D}^{*} \boldsymbol{L}^{*}=\boldsymbol{A}`	$\boldsymbol{A}^{*}=\left(\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^{*}\right)^{*}=\boldsymbol{L} \boldsymbol{D}^{*} \boldsymbol{L}^{*}=\boldsymbol{A}$	that $\boldsymbol{A}^*=(\boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^*)^*=\boldsymbol{L}\boldsymbol{D}^*\boldsymbol{L}^*=\boldsymbol{A}$. The LL* factorization is called a **Cholesky**
lyche-numerical-linear-algebra_F01267	100	0.966	`b \in \mathbb{C}`	$b \in \mathbb{C}$	*Example 4.1 (LDL* of 2×2 Hermitian Matrix)* Let $a,d\in\mathbb{R}$ and $b\in\mathbb{C}$. An LDL*
lyche-numerical-linear-algebra_F01268	101	1.000	`d_{1}, d_{2}`	d_{1}, d_{2}	for the unknowns l_1 in $\boldsymbol{L}$ and d_1,d_2 in $\boldsymbol{D}$. They are determined from
lyche-numerical-linear-algebra_F01269	101	1.000	`d_{1}`	d_{1}	1. $a\neq 0$: The matrix has a unique LDL* factorization. Note that d_1 and d_2 are real.
lyche-numerical-linear-algebra_F01270	101	1.000	`d_{2}`	d_{2}	1. $a\neq 0$: The matrix has a unique LDL* factorization. Note that d_1 and d_2 are real.
lyche-numerical-linear-algebra_F01271	101	1.000	`a=b=0`	$a=b=0$	2. $a=b=0$: The LDL* factorization exists, but it is not unique. Any value for l_1

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01272	101	1.000	`a=0, b \neq 0`	$a=0, b \neq 0$	3. $a=0, b \neq 0$: No LDL* factorization exists.
lyche-numerical-linear-algebra_F01273	101	0.830	`\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D L}^{*}`	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D L}^{*}$	**Lemma 4.1 (LDL* of Leading Principal Sub Matrices)** *Suppose* $\boldsymbol{A}=\boldsymbol{L D L}^{*}$
lyche-numerical-linear-algebra_F01274	101	0.954	`L D L^{*}`	$L D L^{*}$	*is an LDL* factorization of* $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. *For* $k=1, \ldots, n$ *let* $\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}$ *and* $\boldsymbol{D}_{[k]}$
lyche-numerical-linear-algebra_F01275	101	0.954	`\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}`	$\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}$	*is an LDL* factorization of* $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. *For* $k=1, \ldots, n$ *let* $\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}$ *and* $\boldsymbol{D}_{[k]}$
lyche-numerical-linear-algebra_F01276	101	0.954	`\boldsymbol{D}_{[k]}`	$\boldsymbol{D}_{[k]}$	*is an LDL* factorization of* $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. *For* $k=1, \ldots, n$ *let* $\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}$ *and* $\boldsymbol{D}_{[k]}$
lyche-numerical-linear-algebra_F01277	101	1.000	`\boldsymbol{A}, \boldsymbol{L}`	$\boldsymbol{A}, \boldsymbol{L}$	*be the leading principal submatrices of* $\boldsymbol{A}, \boldsymbol{L}$ *and* $\boldsymbol{D}$, *respectively. Then* $\boldsymbol{A}_{[k]}=$
lyche-numerical-linear-algebra_F01278	101	1.000	`\boldsymbol{A}_{[k]}=`	$\boldsymbol{A}_{[k]}=$	*be the leading principal submatrices of* $\boldsymbol{A}, \boldsymbol{L}$ *and* $\boldsymbol{D}$, *respectively. Then* $\boldsymbol{A}_{[k]}=$
lyche-numerical-linear-algebra_F01279	101	1.000	`\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^{*}`	$\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^{*}$	$\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^{*}$ *is an LDL* factorization of* $\boldsymbol{A}_{[k]}$ *for* $k=1, \ldots, n$.
lyche-numerical-linear-algebra_F01280	101	1.000	`\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^{*}`	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^{*}$	***Proof*** For $k=1, \ldots, n-1$ we partition $\boldsymbol{A}=\boldsymbol{L D L}^{*}$ as follows:
lyche-numerical-linear-algebra_F01281	101	1.000	`\boldsymbol{F}_{k}, \boldsymbol{N}_{k}, \boldsymbol{E}_{k} \in \mathbb{C}^{n-k, n-k}`	$\boldsymbol{F}_{k}, \boldsymbol{N}_{k}, \boldsymbol{E}_{k} \in \mathbb{C}^{n-k, n-k}$	where $\boldsymbol{F}_{k}, \boldsymbol{N}_{k}, \boldsymbol{E}_{k} \in \mathbb{C}^{n-k, n-k}$. Block multiplication gives $\boldsymbol{A}_{[k]}=\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^{*}$.
lyche-numerical-linear-algebra_F01282	101	1.000	`\boldsymbol{A}_{[k]}=\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^{*}`	$\boldsymbol{A}_{[k]}=\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^{*}$	where $\boldsymbol{F}_{k}, \boldsymbol{N}_{k}, \boldsymbol{E}_{k} \in \mathbb{C}^{n-k, n-k}$. Block multiplication gives $\boldsymbol{A}_{[k]}=\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^{*}$.
lyche-numerical-linear-algebra_F01283	101	1.000	`\boldsymbol{A}=\boldsymbol{A}^{*}`	$\boldsymbol{A}=\boldsymbol{A}^{*}$	*factorization if and only if* $\boldsymbol{A}=\boldsymbol{A}^{*}$ *and* $\boldsymbol{A}_{[k]}$ *is nonsingular for* $k=1, \ldots, n-1$.
lyche-numerical-linear-algebra_F01284	101	1.000	`\boldsymbol{A}_{[k]}^{*}=\boldsymbol{A}_{[k]}`	$\boldsymbol{A}_{[k]}^{*}=\boldsymbol{A}_{[k]}$	Note that $\boldsymbol{A}_{[k]}^{*}=\boldsymbol{A}_{[k]}$ for $k=1, \ldots, n$. We use induction on n to show that $\boldsymbol{A}$
lyche-numerical-linear-algebra_F01285	101	0.983	`\left[a_{11}\right]=[1]\left[a_{11}\right][1]`	$\left[a_{11}\right]=[1]\left[a_{11}\right][1]$	unique LDL* factorization of a 1-by-1 matrix is $[a_{11}]=[1][a_{11}][1]$ and a_{11} is real
lyche-numerical-linear-algebra_F01286	101	0.997	`\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*}`	$\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*}$	$\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*}$, and that $\boldsymbol{A}_{[1]}, \ldots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By definition $\boldsymbol{D}_{n-1}$ is
lyche-numerical-linear-algebra_F01287	101	0.997	`\boldsymbol{D}_{n-1}`	$\boldsymbol{D}_{n-1}$	$\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*}$, and that $\boldsymbol{A}_{[1]}, \ldots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By definition $\boldsymbol{D}_{n-1}$ is
lyche-numerical-linear-algebra_F01288	102	1.000	`\boldsymbol{A}_{[n-1]}=\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*}`	$\boldsymbol{A}_{[n-1]}=\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*}$	if and only if $\boldsymbol{A}_{[n-1]}=\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*}$, and
lyche-numerical-linear-algebra_F01289	102	1.000	`d_{n n}`	$d_{n n}$	nonsingular. Also d_{nn} is real since a_{nn} and $\boldsymbol{D}_{n-1}$ are real.
lyche-numerical-linear-algebra_F01290	102	1.000	`a_{n n}`	$a_{n n}$	nonsingular. Also d_{nn} is real since a_{nn} and $\boldsymbol{D}_{n-1}$ are real.
lyche-numerical-linear-algebra_F01291	102	1.000	`\frac{1}{2} G_{n}`	$\frac{1}{2} G_{n}$	$\frac{1}{2} G_{n}$, half the number of operations needed for the LU factorization. Indeed, in the
lyche-numerical-linear-algebra_F01292	102	1.000	`\boldsymbol{m}`	$\boldsymbol{m}$	and $\boldsymbol{m}$, while only one such system is needed to find $\boldsymbol{m}$ in the Hermitian case (4.3).
lyche-numerical-linear-algebra_F01293	102	1.000	`f: \mathbb{C}^{n} \rightarrow \mathbb{R}`	$f: \mathbb{C}^{n} \rightarrow \mathbb{R}$	Given $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. The function $f: \mathbb{C}^{n} \rightarrow \mathbb{R}$ given by
lyche-numerical-linear-algebra_F01294	103	0.998	`\overline{f(\boldsymbol{x})}=\overline{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}=\left(\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}\right)^{*}= \boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{x}=f( \boldsymbol{x})`	$\overline{f(\boldsymbol{x})}=\overline{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}=\left(\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}\right)^{*}=\boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{x}=f(\boldsymbol{x})$	$\overline{f(\boldsymbol{x})}=\overline{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}=(\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x})^{*}=\boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{x}=f(\boldsymbol{x})$.
lyche-numerical-linear-algebra_F01295	103	1.000	`\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}>0`	$\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}>0$	(i) **positive definite** if $A^{*}=\boldsymbol{A}$ and $\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}>0$ for all nonzero $\boldsymbol{x} \in \mathbb{C}^{n}$;
lyche-numerical-linear-algebra_F01296	103	1.000	`\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x} \geq 0`	$\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x} \geq 0$	(ii) **positive semidefinite** if $A^{*}=\boldsymbol{A}$ and $\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x} \geq 0$ for all $\boldsymbol{x} \in \mathbb{C}^{n}$;
lyche-numerical-linear-algebra_F01297	103	1.000	`-\boldsymbol{A}`	$-\boldsymbol{A}$	(iii) **negative (semi)definite** if $-\boldsymbol{A}$ is positive (semi)definite.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01298	103	1.000	\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=0 \Longrightarrow \boldsymbol{x}=\mathbf{0}	$\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=0 \Longrightarrow \boldsymbol{x}=\mathbf{0}$	$\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} = 0 \implies \boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra F01299	103	0.999	\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=0	$\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=0$	3. A positive definite matrix $\boldsymbol{A}$ is nonsingular. For if $\boldsymbol{A}\boldsymbol{x} = \mathbf{0}$ then $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} = 0$ and
lyche-numerical-linear-algebra F01300	103	1.000	\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^{T}=\boldsymbol{A}	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^{T}=\boldsymbol{A}$	$\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\boldsymbol{A}^T = \boldsymbol{A}$ and $\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x} \in \mathbb{R}^n$ then $\boldsymbol{z}^*\boldsymbol{A}\boldsymbol{z} > 0$ for
lyche-numerical-linear-algebra F01301	103	1.000	\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}>0	$\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}>0$	$\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\boldsymbol{A}^T = \boldsymbol{A}$ and $\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x} \in \mathbb{R}^n$ then $\boldsymbol{z}^*\boldsymbol{A}\boldsymbol{z} > 0$ for
lyche-numerical-linear-algebra F01302	103	1.000	\boldsymbol{z}^{*} \boldsymbol{A} \boldsymbol{z}>0	$\boldsymbol{z}^{*} \boldsymbol{A} \boldsymbol{z}>0$	$\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\boldsymbol{A}^T = \boldsymbol{A}$ and $\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x} \in \mathbb{R}^n$ then $\boldsymbol{z}^*\boldsymbol{A}\boldsymbol{z} > 0$ for
lyche-numerical-linear-algebra F01303	103	0.999	\boldsymbol{z} \in \mathbb{C}^{n}	$\boldsymbol{z} \in \mathbb{C}^{n}$	all nonzero $\boldsymbol{z} \in \mathbb{C}^n$. For if $\boldsymbol{z} = \boldsymbol{x} + i\,\boldsymbol{y} \neq \mathbf{0}$ with $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ then
lyche-numerical-linear-algebra F01304	103	0.999	\boldsymbol{z}=\boldsymbol{x}+i \boldsymbol{y} \neq \mathbf{0}	$\boldsymbol{z}=\boldsymbol{x}+i \boldsymbol{y} \neq \mathbf{0}$	all nonzero $\boldsymbol{z} \in \mathbb{C}^n$. For if $\boldsymbol{z} = \boldsymbol{x} + i\,\boldsymbol{y} \neq \mathbf{0}$ with $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ then
lyche-numerical-linear-algebra F01305	103	0.991	\boldsymbol{x}, \boldsymbol{y}	$\boldsymbol{x}, \boldsymbol{y}$	and this is positive since at least one of the real vectors $\boldsymbol{x}, \boldsymbol{y}$ is nonzero.
lyche-numerical-linear-algebra F01306	103	1.000	\nabla f	∇f	important in nonlinear optimization. Consider (cf. (16.1)) the gradient ∇f and
lyche-numerical-linear-algebra F01307	103	0.658	H f	$H f$	hessian Hf of a function $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$
lyche-numerical-linear-algebra F01308	103	0.658	f: \Omega \subset \mathbb{R}^{n} \rightarrow \mathbb{R}	$f: \Omega \subset \mathbb{R}^{n} \rightarrow \mathbb{R}$	hessian Hf of a function $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$
lyche-numerical-linear-algebra F01309	103	1.000	\Omega	Ω	We assume that f has continuous first and second order partial derivatives on Ω.
lyche-numerical-linear-algebra F01310	103	1.000	\nabla f(\boldsymbol{x})=\mathbf{0}	$\nabla f(\boldsymbol{x})=\mathbf{0}$	that if $\nabla f(\boldsymbol{x}) = \mathbf{0}$ and $Hf(\boldsymbol{x})$ is positive definite then $\boldsymbol{x}$ is a local minimum for f.
lyche-numerical-linear-algebra F01311	103	1.000	H f(\boldsymbol{x})	$H f(\boldsymbol{x})$	that if $\nabla f(\boldsymbol{x}) = \mathbf{0}$ and $Hf(\boldsymbol{x})$ is positive definite then $\boldsymbol{x}$ is a local minimum for f.
lyche-numerical-linear-algebra F01312	103	0.933	\boldsymbol{A}^{*} \boldsymbol{A}	$\boldsymbol{A}^{*} \boldsymbol{A}$	**Lemma 4.2 (The Matrix $\boldsymbol{A}^*\boldsymbol{A}$)** *The matrix $\boldsymbol{A}^*\boldsymbol{A}$ is positive semidefinite for any*
lyche-numerical-linear-algebra F01313	103	1.000	m, n \in \mathbb{N}	$m, n \in \mathbb{N}$	$m, n \in \mathbb{N}$ *and* $\boldsymbol{A} \in \mathbb{C}^{m\times n}$. *It is positive definite if and only if* $\boldsymbol{A}$ *has linearly*
lyche-numerical-linear-algebra F01314	104	0.986	\boldsymbol{z}:=\boldsymbol{A} \boldsymbol{x}	$\boldsymbol{z}:=\boldsymbol{A} \boldsymbol{x}$	***Proof*** Clearly $\boldsymbol{A}^*\boldsymbol{A}$ is Hermitian. Let $\boldsymbol{x} \in \mathbb{C}^n$ and set $\boldsymbol{z} := \boldsymbol{A}\boldsymbol{x}$. By the definition
lyche-numerical-linear-algebra F01315	104	1.000	\boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{z}^{*} \boldsymbol{z}=\\|\boldsymbol{z}\\|_{2}^{2}=\\|\boldsymbol{A} \boldsymbol{x}\\|_{2}^{2} \geq 0	$\boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{z}^{*} \boldsymbol{z}=\Vert\boldsymbol{z}\Vert_{2}^{2}=\Vert\boldsymbol{A} \boldsymbol{x}\Vert_{2}^{2} \geq 0$	(1.11) of the Euclidean norm we have $\boldsymbol{x}^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x} = \boldsymbol{z}^*\boldsymbol{z} = \Vert\boldsymbol{z}\Vert_2^2 = \Vert\boldsymbol{A}\boldsymbol{x}\Vert_2^2 \geq 0$
lyche-numerical-linear-algebra F01316	104	0.515	\boldsymbol{T}=	$\boldsymbol{T}=$	**Lemma 4.3 ($\boldsymbol{T}$ Is Positive Definite)** *The second derivative matrix* $\boldsymbol{T} =$
lyche-numerical-linear-algebra F01317	104	0.983	\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{n \times n}	$\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{n \times n}$	$\operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{n\times n}$ *is positive definite.*
lyche-numerical-linear-algebra F01318	104	0.994	\boldsymbol{x}^{T} \boldsymbol{T} \boldsymbol{x} \geq 0	$\boldsymbol{x}^{T} \boldsymbol{T} \boldsymbol{x} \geq 0$	Thus $\boldsymbol{x}^T\boldsymbol{T}\boldsymbol{x} \geq 0$ and if $\boldsymbol{x}^T\boldsymbol{T}\boldsymbol{x} = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra F01319	104	0.994	\boldsymbol{x}^{T} \boldsymbol{T} \boldsymbol{x}=0	$\boldsymbol{x}^{T} \boldsymbol{T} \boldsymbol{x}=0$	Thus $\boldsymbol{x}^T\boldsymbol{T}\boldsymbol{x} \geq 0$ and if $\boldsymbol{x}^T\boldsymbol{T}\boldsymbol{x} = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra F01320	104	0.994	x_{1}=x_{n}=0	$x_{1}=x_{n}=0$	Thus $\boldsymbol{x}^T\boldsymbol{T}\boldsymbol{x} \geq 0$ and if $\boldsymbol{x}^T\boldsymbol{T}\boldsymbol{x} = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra F01321	104	0.994	x_{i}=x_{i+1}	$x_{i}=x_{i+1}$	Thus $\boldsymbol{x}^T\boldsymbol{T}\boldsymbol{x} \geq 0$ and if $\boldsymbol{x}^T\boldsymbol{T}\boldsymbol{x} = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra F01322	104	0.999	\boldsymbol{x}=0	$\boldsymbol{x}=0$	$1, \ldots, n-1$ which implies that $\boldsymbol{x} = 0$. Hence $\boldsymbol{T}$ is positive definite.
lyche-numerical-linear-algebra F01323	104	1.000	\boldsymbol{B}=\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r}) \in \mathbb{C}^{k \times k}	$\boldsymbol{B}=\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r}) \in \mathbb{C}^{k \times k}$	Recall that a **principal submatrix** $\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r}) \in \mathbb{C}^{k\times k}$ of a matrix $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ has

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01324	104	■1.000	`b_{i, j}=a_{r_{i}, r_{j}}`	$b_{i, j}=a_{r_{i}, r_{j}}$	elements $b_{i,j} = a_{r_i,r_j}$ for $i, j = 1, \ldots, k$, where $1 \le r_1 < \cdots < r_k \le n$. It is a
lyche-numerical-linear-algebra F01325	104	■1.000	`i, j=1, \ldots, k`	$i, j=1, \ldots, k$	elements $b_{i,j} = a_{r_i,r_j}$ for $i, j = 1, \ldots, k$, where $1 \le r_1 < \cdots < r_k \le n$. It is a
lyche-numerical-linear-algebra F01326	104	■1.000	`\boldsymbol{r}=[1,2, \ldots, k]^{T}`	$\boldsymbol{r}=[1,2, \ldots, k]^{T}$	**leading principal submatrix**, denoted $\boldsymbol{A}_{[k]}$ if $\boldsymbol{r} = [1, 2, \ldots, k]^T$. We have
lyche-numerical-linear-algebra F01327	104	■1.000	`\boldsymbol{B}:=\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})`	$\boldsymbol{B}:=\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$	***Proof*** Let $\boldsymbol{X}$ and $\boldsymbol{B} := \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$ be given by (4.5). If $\boldsymbol{A}$ is positive semidefinite then
lyche-numerical-linear-algebra F01328	104	■1.000	`\boldsymbol{y}^{*} \boldsymbol{B} \boldsymbol{y}=0`	$\boldsymbol{y}^{*} \boldsymbol{B} \boldsymbol{y}=0$	Suppose $\boldsymbol{A}$ is positive definite and $\boldsymbol{y}^*\boldsymbol{B}\boldsymbol{y} = 0$. By (4.6) we have $\boldsymbol{x} = \boldsymbol{0}$ and since
lyche-numerical-linear-algebra F01329	105	■1.000	`\boldsymbol{A}^{-*}:=\left(\boldsymbol{A}^{-1}\right)^{*}=\left(\boldsymbol{A}^{*}\right)^{-1}`	$\boldsymbol{A}^{-*}:=\left(\boldsymbol{A}^{-1}\right)^{*}=\left(\boldsymbol{A}^{*}\right)^{-1}$	***Proof*** Recall that $\boldsymbol{A}^{-*} := (\boldsymbol{A}^{-1})^* = (\boldsymbol{A}^*)^{-1}$.
lyche-numerical-linear-algebra F01330	105	■1.000	`1 \Longrightarrow 2 \Longrightarrow 3 \Longrightarrow 1`	$1 \Longrightarrow 2 \Longrightarrow 3 \Longrightarrow 1$	We show that $1 \implies 2 \implies 3 \implies 1$.
lyche-numerical-linear-algebra F01331	105	■0.999	`1 \Longrightarrow`	$1 \Longrightarrow$	$1 \implies 2$: Suppose $\boldsymbol{A}$ is positive definite. By Lemma 4.4 the leading principal
lyche-numerical-linear-algebra F01332	105	■1.000	`\boldsymbol{x}_{i}:=\boldsymbol{L}^{-*} \boldsymbol{e}_{i}`	$\boldsymbol{x}_{i}:=\boldsymbol{L}^{-*} \boldsymbol{e}_{i}$	is positive we note that $\boldsymbol{x}_i := \boldsymbol{L}^{-*}\boldsymbol{e}_i$ is nonzero since $\boldsymbol{L}^{-*}$ is nonsingular. But
lyche-numerical-linear-algebra F01333	105	■1.000	`\boldsymbol{L}^{-*}`	$\boldsymbol{L}^{-*}$	is positive we note that $\boldsymbol{x}_i := \boldsymbol{L}^{-*}\boldsymbol{e}_i$ is nonzero since $\boldsymbol{L}^{-*}$ is nonsingular. But
lyche-numerical-linear-algebra F01334	105	■1.000	`d_{i i}=\boldsymbol{e}_{i}^{*} \boldsymbol{D} \boldsymbol{e}_{i}=\boldsymbol{e}_{i}^{*} \boldsymbol{L}^{-1} \boldsymbol{A} \boldsymbol{L}^{-*} \boldsymbol{e}_{i}=\boldsymbol{x}_{i}^{*} \boldsymbol{A} \boldsymbol{x}_{i}>0`	$d_{i i}=\boldsymbol{e}_{i}^{*} \boldsymbol{D} \boldsymbol{e}_{i}=\boldsymbol{e}_{i}^{*} \boldsymbol{L}^{-1} \boldsymbol{A} \boldsymbol{L}^{-*} \boldsymbol{e}_{i}=\boldsymbol{x}_{i}^{*} \boldsymbol{A} \boldsymbol{x}_{i}>0$	then $d_{ii} = \boldsymbol{e}_i^*\boldsymbol{D}\boldsymbol{e}_i = \boldsymbol{e}_i^*\boldsymbol{L}^{-1}\boldsymbol{A}\boldsymbol{L}^{-*}\boldsymbol{e}_i = \boldsymbol{x}_i^*\boldsymbol{A}\boldsymbol{x}_i > 0$ since $\boldsymbol{A}$ is positive definite.
lyche-numerical-linear-algebra F01335	105	■0.913	`2 \Longrightarrow`	$2 \Longrightarrow$	$2 \implies 3$: Suppose $\boldsymbol{A}$ has an LDL* factorization $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^*$ with positive
lyche-numerical-linear-algebra F01336	105	■0.913	`\mathrm{LDL}^{*}`	LDL^{*}	$2 \implies 3$: Suppose $\boldsymbol{A}$ has an LDL* factorization $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^*$ with positive
lyche-numerical-linear-algebra F01337	105	■1.000	`d_{i i}`	$d_{i i}$	diagonal elements d_{ii} in $\boldsymbol{D}$. Then $\boldsymbol{A} = \boldsymbol{S}\boldsymbol{S}^*$, where $\boldsymbol{S} := \boldsymbol{L}\boldsymbol{D}^{1/2}$ and $\boldsymbol{D}^{1/2} :=$
lyche-numerical-linear-algebra F01338	105	■1.000	`\boldsymbol{A}=\boldsymbol{S} \boldsymbol{S}^{*}`	$\boldsymbol{A}=\boldsymbol{S} \boldsymbol{S}^{*}$	diagonal elements d_{ii} in $\boldsymbol{D}$. Then $\boldsymbol{A} = \boldsymbol{S}\boldsymbol{S}^*$, where $\boldsymbol{S} := \boldsymbol{L}\boldsymbol{D}^{1/2}$ and $\boldsymbol{D}^{1/2} :=$
lyche-numerical-linear-algebra F01339	105	■1.000	`\boldsymbol{S}:=\boldsymbol{L} \boldsymbol{D}^{1 / 2}`	$\boldsymbol{S}:=\boldsymbol{L} \boldsymbol{D}^{1 / 2}$	diagonal elements d_{ii} in $\boldsymbol{D}$. Then $\boldsymbol{A} = \boldsymbol{S}\boldsymbol{S}^*$, where $\boldsymbol{S} := \boldsymbol{L}\boldsymbol{D}^{1/2}$ and $\boldsymbol{D}^{1/2} :=$
lyche-numerical-linear-algebra F01340	105	■1.000	`\boldsymbol{D}^{1 / 2}:=`	$\boldsymbol{D}^{1 / 2}:=$	diagonal elements d_{ii} in $\boldsymbol{D}$. Then $\boldsymbol{A} = \boldsymbol{S}\boldsymbol{S}^*$, where $\boldsymbol{S} := \boldsymbol{L}\boldsymbol{D}^{1/2}$ and $\boldsymbol{D}^{1/2} :=$
lyche-numerical-linear-algebra F01341	105	■1.000	`\operatorname{diag}\left(\sqrt{d_{11}}, \ldots, \sqrt{d_{n n}}\right)`	$\operatorname{diag}\left(\sqrt{d_{11}}, \ldots, \sqrt{d_{n n}}\right)$	$\operatorname{diag}(\sqrt{d_{11}}, \ldots, \sqrt{d_{nn}})$, and this is a Cholesky factorization of $\boldsymbol{A}$.
lyche-numerical-linear-algebra F01342	105	■1.000	`3 \Longrightarrow`	$3 \Longrightarrow$	$3 \implies 1$: Suppose $\boldsymbol{A}$ has a Cholesky factorization $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{L}^*$. Clearly $\boldsymbol{A}^* = \boldsymbol{A}$.
lyche-numerical-linear-algebra F01343	105	■1.000	`\boldsymbol{A}=\boldsymbol{L} \boldsymbol{L}^{*}`	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{L}^{*}$	$3 \implies 1$: Suppose $\boldsymbol{A}$ has a Cholesky factorization $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{L}^*$. Clearly $\boldsymbol{A}^* = \boldsymbol{A}$.
lyche-numerical-linear-algebra F01344	105	■1.000	`\boldsymbol{L} \boldsymbol{L}^{*}=\boldsymbol{S} \boldsymbol{S}^{*}`	$\boldsymbol{L} \boldsymbol{L}^{*}=\boldsymbol{S} \boldsymbol{S}^{*}$	For uniqueness suppose $\boldsymbol{L}\boldsymbol{L}^* = \boldsymbol{S}\boldsymbol{S}^*$ are two Cholesky factorizations of the
lyche-numerical-linear-algebra F01345	105	■1.000	`\boldsymbol{S}^{-1} \boldsymbol{L}=\boldsymbol{S}^{*} \boldsymbol{L}^{-*}`	$\boldsymbol{S}^{-1} \boldsymbol{L}=\boldsymbol{S}^{*} \boldsymbol{L}^{-*}$	Then $\boldsymbol{S}^{-1}\boldsymbol{L} = \boldsymbol{S}^*\boldsymbol{L}^{-*}$, where by Lemma 2.5 $\boldsymbol{S}^{-1}\boldsymbol{L}$ is lower triangular and $\boldsymbol{S}^*\boldsymbol{L}^{-*}$
lyche-numerical-linear-algebra F01346	105	■1.000	`2.5 \boldsymbol{S}^{-1} \boldsymbol{L}`	$2.5 \boldsymbol{S}^{-1} \boldsymbol{L}$	Then $\boldsymbol{S}^{-1}\boldsymbol{L} = \boldsymbol{S}^*\boldsymbol{L}^{-*}$, where by Lemma 2.5 $\boldsymbol{S}^{-1}\boldsymbol{L}$ is lower triangular and $\boldsymbol{S}^*\boldsymbol{L}^{-*}$
lyche-numerical-linear-algebra F01347	105	■1.000	`\boldsymbol{S}^{*} \boldsymbol{L}^{-*}`	$\boldsymbol{S}^{*} \boldsymbol{L}^{-*}$	Then $\boldsymbol{S}^{-1}\boldsymbol{L} = \boldsymbol{S}^*\boldsymbol{L}^{-*}$, where by Lemma 2.5 $\boldsymbol{S}^{-1}\boldsymbol{L}$ is lower triangular and $\boldsymbol{S}^*\boldsymbol{L}^{-*}$
lyche-numerical-linear-algebra F01348	105	■1.000	`\ell_{i i} / s_{i i}`	$\ell_{i i} / s_{i i}$	is upper triangular, with diagonal elements ℓ_{ii}/s_{ii} and s_{ii}/ℓ_{ii}, respectively. But then

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01349	105	■ 1.000	`s_{i i} / \ell_{i i}`	s_{ii}/ℓ_{ii}	is upper triangular, with diagonal elements ℓ_{ii}/s_{ii} and s_{ii}/ℓ_{ii}, respectively. But then
lyche-numerical-linear-algebra F01350	105	■ 1.000	`\ell_{i i}^{2}=s_{i i}^{2}`	$\ell_{ii}^{2}=s_{ii}^{2}$	both matrices must be equal to the same diagonal matrix and $\ell_{ii}^2 = s_{ii}^2$. By positivity
lyche-numerical-linear-algebra F01351	105	■ 1.000	`\ell_{i i}=s_{i i}`	$\ell_{ii}=s_{ii}$	$\ell_{ii} = s_{ii}$ and we conclude that $\boldsymbol{S}^{-1}\boldsymbol{L} = \boldsymbol{I} = \boldsymbol{S}^*\boldsymbol{L}^{-*}$ which means that $\boldsymbol{L} = \boldsymbol{S}$.
lyche-numerical-linear-algebra F01352	105	■ 1.000	`\boldsymbol{S}^{-1} \boldsymbol{L}=\boldsymbol{I}= \boldsymbol{S}^{*} \boldsymbol{L}^{-*}`	$\boldsymbol{S}^{-1}\boldsymbol{L}=\boldsymbol{I}=\boldsymbol{S}^{*}\boldsymbol{L}^{-*}$	$\ell_{ii} = s_{ii}$ and we conclude that $\boldsymbol{S}^{-1}\boldsymbol{L} = \boldsymbol{I} = \boldsymbol{S}^*\boldsymbol{L}^{-*}$ which means that $\boldsymbol{L} = \boldsymbol{S}$.
lyche-numerical-linear-algebra F01353	105	■ 1.000	`\boldsymbol{L}=\boldsymbol{S}`	$\boldsymbol{L}=\boldsymbol{S}$	$\ell_{ii} = s_{ii}$ and we conclude that $\boldsymbol{S}^{-1}\boldsymbol{L} = \boldsymbol{I} = \boldsymbol{S}^*\boldsymbol{L}^{-*}$ which means that $\boldsymbol{L} = \boldsymbol{S}$.
lyche-numerical-linear-algebra F01354	105	■ 0.993	`\boldsymbol{A}=\boldsymbol{R}^{*} \boldsymbol{R}`	$\boldsymbol{A}=\boldsymbol{R}^{*}\boldsymbol{R}$	A Cholesky factorization can also be written in the equivalent form $\boldsymbol{A} = \boldsymbol{R}^*\boldsymbol{R}$,
lyche-numerical-linear-algebra F01355	105	■ 1.000	`\boldsymbol{R}=\boldsymbol{L}^{*}`	$\boldsymbol{R}=\boldsymbol{L}^{*}$	where $\boldsymbol{R} = \boldsymbol{L}^*$ is upper triangular with positive diagonal elements.
lyche-numerical-linear-algebra F01356	105	■ 0.993	`4.4(2 \times 2)`	$4.4(2 \times 2)$	*Example 4.4 (2 × 2)* The matrix $\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ has an LDL* and a Cholesky-
lyche-numerical-linear-algebra F01357	105	■ 0.993	`\boldsymbol{A}=\left[\begin{array}{cc}2 & -1 \\ -1 & 2 \end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right]$	*Example 4.4 (2 × 2)* The matrix $\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ has an LDL* and a Cholesky-
lyche-numerical-linear-algebra F01358	106	■ 1.000	`\frac{1}{2} G_{n}=n^{3} / 3`	$\frac{1}{2} G_{n}=n^{3}/3$	count $\frac{1}{2}G_n = n^3/3$ for a full matrix.
lyche-numerical-linear-algebra F01359	106	■ 1.000	`a_{i i}>0,\left(a_{i i} \geq 0\right)`	$a_{ii}>0,\left(a_{ii} \geq 0\right)$	*1.* $a_{ii} > 0$, $(a_{ii} \geq 0)$,
lyche-numerical-linear-algebra F01360	106	■ 1.000	`\left|\operatorname{Re}\left(a_{i j}\right)\right|< \left(a_{i i}+a_{j j}\right) / 2,\left(\left| \operatorname{Re}\left(a_{i j}\right)\right| \leq \left(a_{i i}+a_{j j}\right) / 2\right)`	$\left\vert\operatorname{Re}\left(a_{ij}\right)\right\vert<\left(a_{ii}+a_{jj}\right)/2,\left(\left\vert\operatorname{Re}\left(a_{ij}\right)\right\vert \leq \left(a_{ii}+a_{jj}\right)/2\right)$	*2.* $\vert Re\,(a_{ij})\vert < (a_{ii} + a_{jj})/2$, $(\vert Re\,(a_{ij})\vert \leq (a_{ii} + a_{jj})/2)$,
lyche-numerical-linear-algebra F01361	106	■ 1.000	`\left|a_{i j}\right|<\sqrt{a_{i i} a_{j j}},\left( \left|a_{i j}\right| \leq \sqrt{a_{i i} a_{j j}}\right)`	$\left\vert a_{ij}\right\vert<\sqrt{a_{ii} a_{jj}},\left(\left\vert a_{ij}\right\vert \leq \sqrt{a_{ii} a_{jj}}\right)$	*3.* $\vert a_{ij}\vert < \sqrt{a_{ii}a_{jj}}$, $(\vert a_{ij}\vert \leq \sqrt{a_{ii}a_{jj}})$,
lyche-numerical-linear-algebra F01362	106	■ 1.000	`a_{i i}=0`	$a_{ii}=0$	*4. If $\boldsymbol{A}$ is positive semidefinite and $a_{ii} = 0$ for some i then $a_{ij} = a_{ji} = 0$ for*
lyche-numerical-linear-algebra F01363	106	■ 1.000	`a_{i j}=a_{j i}=0`	$a_{ij}=a_{ji}=0$	*4. If $\boldsymbol{A}$ is positive semidefinite and $a_{ii} = 0$ for some i then $a_{ij} = a_{ji} = 0$ for*
lyche-numerical-linear-algebra F01364	106	■ 1.000	`a_{i i}=\boldsymbol{e}_{i}^{T} \boldsymbol{A} \boldsymbol{e}_{i}>(\geq) 0`	$a_{ii}=\boldsymbol{e}_{i}^{T}\boldsymbol{A}\boldsymbol{e}_{i}>(\geq) 0$	***Proof*** Clearly $a_{ii} = \boldsymbol{e}_i^T\boldsymbol{A}\boldsymbol{e}_i > (\geq) 0$ and Part 1 follows. If $\alpha, \beta \in \mathbb{C}$ and $\alpha\boldsymbol{e}_i + \beta\boldsymbol{e}_j \neq$
lyche-numerical-linear-algebra F01365	106	■ 1.000	`\alpha, \beta \in \mathbb{C}`	$\alpha, \beta \in \mathbb{C}$	***Proof*** Clearly $a_{ii} = \boldsymbol{e}_i^T\boldsymbol{A}\boldsymbol{e}_i > (\geq) 0$ and Part 1 follows. If $\alpha, \beta \in \mathbb{C}$ and $\alpha\boldsymbol{e}_i + \beta\boldsymbol{e}_j \neq$
lyche-numerical-linear-algebra F01366	106	■ 1.000	`\alpha \boldsymbol{e}_{i}+\beta \boldsymbol{e}_{j} \neq`	$\alpha \boldsymbol{e}_{i}+\beta \boldsymbol{e}_{j} \neq$	***Proof*** Clearly $a_{ii} = \boldsymbol{e}_i^T\boldsymbol{A}\boldsymbol{e}_i > (\geq) 0$ and Part 1 follows. If $\alpha, \beta \in \mathbb{C}$ and $\alpha\boldsymbol{e}_i + \beta\boldsymbol{e}_j \neq$
lyche-numerical-linear-algebra F01367	106	■ 0.999	`\alpha=1, \beta= \pm 1`	$\alpha=1, \beta= \pm 1$	Taking $\alpha = 1$, $\beta = \pm 1$ we obtain $a_{ii} + a_{jj} \pm 2\mathrm{Re}\, a_{ij} > 0$ and this implies Part 2.
lyche-numerical-linear-algebra F01368	106	■ 0.999	`a_{i i}+a_{j j} \pm 2 \operatorname{Re} a_{i j}>0`	$a_{ii}+a_{jj} \pm 2 \operatorname{Re} a_{ij}>0$	Taking $\alpha = 1$, $\beta = \pm 1$ we obtain $a_{ii} + a_{jj} \pm 2\mathrm{Re}\, a_{ij} > 0$ and this implies Part 2.
lyche-numerical-linear-algebra F01369	106	■ 0.834	`\alpha=-a_{i j}, \beta=a_{i i}`	$\alpha=-a_{ij}, \beta=a_{ii}$	We first show 3. when $\boldsymbol{A}$ is positive definite. Taking $\alpha = -a_{ij}$, $\beta = a_{ii}$ in (4.7) we
lyche-numerical-linear-algebra F01370	106	■ 1.000	`a_{i i}>0`	$a_{ii}>0$	Since $a_{ii} > 0$ Part 3 follows in the positive definite case.
lyche-numerical-linear-algebra F01371	106	■ 1.000	`\varepsilon>0`	$\varepsilon>0$	Suppose now $\boldsymbol{A}$ is positive semidefinite. For $\varepsilon > 0$ we define $\boldsymbol{B} := \boldsymbol{A} + \varepsilon\boldsymbol{I}$. The
lyche-numerical-linear-algebra F01372	106	■ 1.000	`\boldsymbol{B}:=\boldsymbol{A}+\varepsilon \boldsymbol{I}`	$\boldsymbol{B}:=\boldsymbol{A}+\varepsilon \boldsymbol{I}$	Suppose now $\boldsymbol{A}$ is positive semidefinite. For $\varepsilon > 0$ we define $\boldsymbol{B} := \boldsymbol{A} + \varepsilon\boldsymbol{I}$. The

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01373	106	1.000	\boldsymbol{x}^{*} \boldsymbol{B} \boldsymbol{x} \geq \varepsilon|\boldsymbol{x}|_{2}^{2}>0	$\boldsymbol{x}^{*} \boldsymbol{B} \boldsymbol{x} \geq \varepsilon\|\boldsymbol{x}\|_{2}^{2}>0$	matrix $\boldsymbol{B}$ is positive definite since it is Hermitian and $\boldsymbol{x}^*\boldsymbol{B}\boldsymbol{x} \geq \varepsilon\|\boldsymbol{x}\|_2^2 > 0$ for any
lyche-numerical-linear-algebra F01374	107	1.000	\left[\begin{array}{ll}2 & 1 \\ 1 & 2\end{array}\right]	$\left[\begin{array}{ll}2 & 1 \\ 1 & 2\end{array}\right]$	The matrix $\left[\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right]$ enjoys all the necessary conditions in Theorem 4.3. But to
lyche-numerical-linear-algebra F01375	107	1.000	\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{x}^{*} \boldsymbol{x}	$\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{x}^{*} \boldsymbol{x}$	if $\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ and $\boldsymbol{x}$ is nonzero, then $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}^*\boldsymbol{x}$. This implies that $\lambda > 0(\geq 0)$
lyche-numerical-linear-algebra F01376	107	1.000	\lambda>0(\geq 0)	$\lambda>0(\geq 0)$	if $\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ and $\boldsymbol{x}$ is nonzero, then $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}^*\boldsymbol{x}$. This implies that $\lambda > 0(\geq 0)$
lyche-numerical-linear-algebra F01377	107	1.000	\boldsymbol{x}^{*} \boldsymbol{x}=|\boldsymbol{x}|_{2}^{2}>0	$\boldsymbol{x}^{*} \boldsymbol{x}=\|\boldsymbol{x}\|_{2}^{2}>0$	since $\boldsymbol{A}$ is positive (semi)definite and $\boldsymbol{x}^*\boldsymbol{x} = \|\boldsymbol{x}\|_2^2 > 0$. Conversely, suppose
lyche-numerical-linear-algebra F01378	107	1.000	\boldsymbol{U}^{*} \boldsymbol{U}=	$\boldsymbol{U}^{*} \boldsymbol{U}=$	Theorem 6.9 (the spectral theorem) there is a matrix $\boldsymbol{U} \in \mathbb{C}^{n\times n}$ with $\boldsymbol{U}^*\boldsymbol{U} =$
lyche-numerical-linear-algebra F01379	107	1.000	\boldsymbol{U} \boldsymbol{U}^{*}=\boldsymbol{I}	$\boldsymbol{U} \boldsymbol{U}^{*}=\boldsymbol{I}$	$\boldsymbol{U}\boldsymbol{U}^* = \boldsymbol{I}$ such that $\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \operatorname{diag}(\lambda_1,\ldots,\lambda_n)$. Let $\boldsymbol{x} \in \mathbb{C}^n$ and define $\boldsymbol{z} :=$
lyche-numerical-linear-algebra F01380	107	1.000	\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)	$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)$	$\boldsymbol{U}\boldsymbol{U}^* = \boldsymbol{I}$ such that $\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \operatorname{diag}(\lambda_1,\ldots,\lambda_n)$. Let $\boldsymbol{x} \in \mathbb{C}^n$ and define $\boldsymbol{z} :=$
lyche-numerical-linear-algebra F01381	107	1.000	\boldsymbol{z}:=	$\boldsymbol{z}:=$	$\boldsymbol{U}\boldsymbol{U}^* = \boldsymbol{I}$ such that $\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \operatorname{diag}(\lambda_1,\ldots,\lambda_n)$. Let $\boldsymbol{x} \in \mathbb{C}^n$ and define $\boldsymbol{z} :=$
lyche-numerical-linear-algebra F01382	107	1.000	\boldsymbol{U}^{*} \boldsymbol{x}=\left[z_{1}, \ldots, z_{n}\right]^{T} \in \mathbb{C}^{n}	$\boldsymbol{U}^{*} \boldsymbol{x}=\left[z_{1}, \ldots, z_{n}\right]^{T} \in \mathbb{C}^{n}$	$\boldsymbol{U}^*\boldsymbol{x} = [z_1,\ldots,z_n]^T \in \mathbb{C}^n$. Then $\boldsymbol{x} = \boldsymbol{U}\boldsymbol{U}^*\boldsymbol{x} = \boldsymbol{U}\boldsymbol{z}$ and by the spectral theorem
lyche-numerical-linear-algebra F01383	107	1.000	\boldsymbol{x}=\boldsymbol{U} \boldsymbol{U}^{*} \boldsymbol{x}=\boldsymbol{U} \boldsymbol{z}	$\boldsymbol{x}=\boldsymbol{U} \boldsymbol{U}^{*} \boldsymbol{x}=\boldsymbol{U} \boldsymbol{z}$	$\boldsymbol{U}^*\boldsymbol{x} = [z_1,\ldots,z_n]^T \in \mathbb{C}^n$. Then $\boldsymbol{x} = \boldsymbol{U}\boldsymbol{U}^*\boldsymbol{x} = \boldsymbol{U}\boldsymbol{z}$ and by the spectral theorem
lyche-numerical-linear-algebra F01384	107	0.982	\boldsymbol{U}^{*}	$\boldsymbol{U}^{*}$	It follows that $\boldsymbol{A}$ is positive semidefinite. Since $\boldsymbol{U}^*$ is nonsingular we see that $\boldsymbol{z} =$
lyche-numerical-linear-algebra F01385	107	0.982	\boldsymbol{z}=	$\boldsymbol{z}=$	It follows that $\boldsymbol{A}$ is positive semidefinite. Since $\boldsymbol{U}^*$ is nonsingular we see that $\boldsymbol{z} =$
lyche-numerical-linear-algebra F01386	107	1.000	\boldsymbol{U}^{*} \boldsymbol{x}	$\boldsymbol{U}^{*} \boldsymbol{x}$	$\boldsymbol{U}^*\boldsymbol{x}$ is nonzero if $\boldsymbol{x}$ is nonzero, and therefore $\boldsymbol{A}$ is positive definite.
lyche-numerical-linear-algebra F01387	108	1.000	\boldsymbol{A}=\boldsymbol{B} \boldsymbol{B}^{*}	$\boldsymbol{A}=\boldsymbol{B} \boldsymbol{B}^{*}$	*4.* $\boldsymbol{A} = \boldsymbol{B}\boldsymbol{B}^*$ *for a nonsingular* $\boldsymbol{B} \in \mathbb{C}^{n\times n}$.
lyche-numerical-linear-algebra F01388	108	0.793	\boldsymbol{B}=\boldsymbol{L}	$\boldsymbol{B}=\boldsymbol{L}$	and by Theorem 4.2 a unique Cholesky factorization $\boldsymbol{A} = \boldsymbol{B}\boldsymbol{B}^*$ with $\boldsymbol{B} = \boldsymbol{L}$.
lyche-numerical-linear-algebra F01389	108	0.973	4 \Longrightarrow	$4 \Longrightarrow$	$4 \implies 1$: This follows from Lemma 4.2.
lyche-numerical-linear-algebra F01390	108	1.000	\boldsymbol{A}:=\left[\begin{array}{ll}3 & 1 \\ 1 & 3 \end{array}\right]	$\boldsymbol{A}:=\left[\begin{array}{ll}3 & 1 \\ 1 & 3 \end{array}\right]$	$\boldsymbol{A} := \begin{bmatrix}3 & 1\\ 1 & 3\end{bmatrix}$.
lyche-numerical-linear-algebra F01391	108	0.991	\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}=2 x_{1}^{2}+2 x_{2}^{2}+\left(x_{1}+x_{2}\right)^{2}>0	$\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}=2 x_{1}^{2}+2 x_{2}^{2}+\left(x_{1}+x_{2}\right)^{2}>0$	1. We have $\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} = 2x_1^2 + 2x_2^2 + (x_1+x_2)^2 > 0$ for all nonzero $\boldsymbol{x}$ showing that $\boldsymbol{A}$
lyche-numerical-linear-algebra F01392	108	1.000	\lambda_{1}=2	$\lambda_{1}=2$	2. The eigenvalues of $\boldsymbol{A}$ are $\lambda_1 = 2$ and $\lambda_2 = 4$. They are positive showing that $\boldsymbol{A}$
lyche-numerical-linear-algebra F01393	108	1.000	\lambda_{2}=4	$\lambda_{2}=4$	2. The eigenvalues of $\boldsymbol{A}$ are $\lambda_1 = 2$ and $\lambda_2 = 4$. They are positive showing that $\boldsymbol{A}$
lyche-numerical-linear-algebra F01394	108	0.998	\operatorname{det}\left(\boldsymbol{A}_{[1]}\right)=3	$\operatorname{det}\left(\boldsymbol{A}_{[1]}\right)=3$	3. We find $\det(\boldsymbol{A}_{[1]}) = 3$ and $\det(\boldsymbol{A}_{[2]}) = 8$ showing again that $\boldsymbol{A}$ is positive definite
lyche-numerical-linear-algebra F01395	108	0.998	\operatorname{det}\left(\boldsymbol{A}_{[2]}\right)=8	$\operatorname{det}\left(\boldsymbol{A}_{[2]}\right)=8$	3. We find $\det(\boldsymbol{A}_{[1]}) = 3$ and $\det(\boldsymbol{A}_{[2]}) = 8$ showing again that $\boldsymbol{A}$ is positive definite
lyche-numerical-linear-algebra F01396	109	1.000	\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=\left|\boldsymbol{L}^{*} \boldsymbol{x}\right|_{2}^{2} \geq 0	$\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=\left\|\boldsymbol{L}^{*} \boldsymbol{x}\right\|_{2}^{2} \geq 0$	$\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} = \|\boldsymbol{L}^*\boldsymbol{x}\|_2^2 \geq 0$ and $\boldsymbol{A}$ is positive semidefinite. For the converse we use
lyche-numerical-linear-algebra F01397	109	1.000	\alpha=\boldsymbol{e}_{1}^{*} \boldsymbol{A} \boldsymbol{e}_{1}>0	$\alpha=\boldsymbol{e}_{1}^{*} \boldsymbol{A} \boldsymbol{e}_{1}>0$	There are two cases. Suppose first $\alpha = \boldsymbol{e}_1^*\boldsymbol{A}\boldsymbol{e}_1 > 0$. We claim that $\boldsymbol{C} := \boldsymbol{B} -$
lyche-numerical-linear-algebra F01398	109	1.000	\boldsymbol{C}:=\boldsymbol{B}-	$\boldsymbol{C}:=\boldsymbol{B}-$	There are two cases. Suppose first $\alpha = \boldsymbol{e}_1^*\boldsymbol{A}\boldsymbol{e}_1 > 0$. We claim that $\boldsymbol{C} := \boldsymbol{B} -$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01399	109	1.000	`\boldsymbol{v} \boldsymbol{v}^{*} / \alpha`	$\boldsymbol{v} \boldsymbol{v}^{*} / \alpha$	$\boldsymbol{v}\boldsymbol{v}^*/\alpha$ is positive semidefinite. $\boldsymbol{C}$ is Hermitian since $\boldsymbol{B}$ is. To show that $\boldsymbol{C}$ is positive
lyche-numerical-linear-algebra F01400	109	1.000	`\boldsymbol{y} \in \mathbb{C}^{n-1}`	$\boldsymbol{y} \in \mathbb{C}^{n-1}$	semidefinite we consider any $\boldsymbol{y} \in \mathbb{C}^{n-1}$ and define $\boldsymbol{x}^* := [-\boldsymbol{y}^*\boldsymbol{v}/\alpha, \boldsymbol{y}^*] \in \mathbb{C}^n$. Then
lyche-numerical-linear-algebra F01401	109	1.000	`\boldsymbol{x}^{*}:=\left[-\boldsymbol{y}^{*} \boldsymbol{v} / \alpha, \boldsymbol{y}^{*}\right] \in \mathbb{C}^{n}`	$\boldsymbol{x}^{*}:=\left[-\boldsymbol{y}^{*} \boldsymbol{v} / \alpha, \boldsymbol{y}^{*}\right] \in \mathbb{C}^{n}$	semidefinite we consider any $\boldsymbol{y} \in \mathbb{C}^{n-1}$ and define $\boldsymbol{x}^* := [-\boldsymbol{y}^*\boldsymbol{v}/\alpha, \boldsymbol{y}^*] \in \mathbb{C}^n$. Then
lyche-numerical-linear-algebra F01402	109	1.000	`\boldsymbol{C} \in \mathbb{C}^{(n-1) \times(n-1)}`	$\boldsymbol{C} \in \mathbb{C}^{(n-1) \times(n-1)}$	So $\boldsymbol{C} \in \mathbb{C}^{(n-1)\times(n-1)}$ is positive semidefinite and by the induction hypothesis it has
lyche-numerical-linear-algebra F01403	109	1.000	`\boldsymbol{C}=\boldsymbol{L}_{1} \boldsymbol{L}_{1}^{*}`	$\boldsymbol{C}=\boldsymbol{L}_{1} \boldsymbol{L}_{1}^{*}$	a semi-Cholesky factorization $\boldsymbol{C} = \boldsymbol{L}_1\boldsymbol{L}_1^*$. The matrix
lyche-numerical-linear-algebra F01404	109	1.000	`\alpha=0`	$\alpha=0$	If $\alpha = 0$ then part 4 of Theorem 4.3 implies that $\boldsymbol{v} = \mathbf{0}$. Moreover, $\boldsymbol{B} \in$
lyche-numerical-linear-algebra F01405	109	1.000	`\boldsymbol{v}=\mathbf{0}`	$\boldsymbol{v}=\mathbf{0}$	If $\alpha = 0$ then part 4 of Theorem 4.3 implies that $\boldsymbol{v} = \mathbf{0}$. Moreover, $\boldsymbol{B} \in$
lyche-numerical-linear-algebra F01406	109	1.000	`\boldsymbol{B} \in`	$\boldsymbol{B} \in$	If $\alpha = 0$ then part 4 of Theorem 4.3 implies that $\boldsymbol{v} = \mathbf{0}$. Moreover, $\boldsymbol{B} \in$
lyche-numerical-linear-algebra F01407	109	1.000	`\mathbb{C}^{(n-1) \times(n-1)}`	$\mathbb{C}^{(n-1) \times(n-1)}$	$\mathbb{C}^{(n-1)\times(n-1)}$ in (4.8) is positive semidefinite and therefore has a semi-Cholesky
lyche-numerical-linear-algebra F01408	110	0.554	`\boldsymbol{B}=\boldsymbol{L}_{1} \boldsymbol{L}_{1}^{*}`	$\boldsymbol{B}=\boldsymbol{L}_{1} \boldsymbol{L}_{1}^{*}$	factorization $\boldsymbol{B} = \boldsymbol{L}_1\boldsymbol{L}_1^*$. But then $\boldsymbol{L}\boldsymbol{L}^*$, where $\boldsymbol{L} = \begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & \boldsymbol{L}_1 \end{bmatrix}$ is a semi-Cholesky
lyche-numerical-linear-algebra F01409	110	0.554	`\boldsymbol{L}^{*}`	$\boldsymbol{L}^{*}$	factorization $\boldsymbol{B} = \boldsymbol{L}_1\boldsymbol{L}_1^*$. But then $\boldsymbol{L}\boldsymbol{L}^*$, where $\boldsymbol{L} = \begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & \boldsymbol{L}_1 \end{bmatrix}$ is a semi-Cholesky
lyche-numerical-linear-algebra F01410	110	0.554	`\boldsymbol{L}=\left[\begin{array}{cc}0 & \mathbf{0}^{*} \\ \mathbf{0} & \boldsymbol{L}_{1}\end{array}\right]`	$\boldsymbol{L}=\left[\begin{array}{cc}0 & \mathbf{0}^{*} \\ \mathbf{0} & \boldsymbol{L}_{1}\end{array}\right]$	factorization $\boldsymbol{B} = \boldsymbol{L}_1\boldsymbol{L}_1^*$. But then $\boldsymbol{L}\boldsymbol{L}^*$, where $\boldsymbol{L} = \begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & \boldsymbol{L}_1 \end{bmatrix}$ is a semi-Cholesky
lyche-numerical-linear-algebra F01411	110	1.000	`\boldsymbol{v}^{*}=\left[\boldsymbol{u}^{*}, \mathbf{0}^{*}\right]`	$\boldsymbol{v}^{*}=\left[\boldsymbol{u}^{*}, \mathbf{0}^{*}\right]$	***Proof*** Suppose $A \in \mathbb{C}^{n\times n}$ is d-banded. Then $\boldsymbol{v}^* = [\boldsymbol{u}^*, \mathbf{0}^*]$ in (4.8), where $\boldsymbol{u} \in \mathbb{C}^d$,
lyche-numerical-linear-algebra F01412	110	1.000	`\boldsymbol{u} \in \mathbb{C}^{d}`	$\boldsymbol{u} \in \mathbb{C}^{d}$	***Proof*** Suppose $A \in \mathbb{C}^{n\times n}$ is d-banded. Then $\boldsymbol{v}^* = [\boldsymbol{u}^*, \mathbf{0}^*]$ in (4.8), where $\boldsymbol{u} \in \mathbb{C}^d$,
lyche-numerical-linear-algebra F01413	110	1.000	`\boldsymbol{C}:=\boldsymbol{B}-\boldsymbol{v} \boldsymbol{v}^{*} / \alpha`	$\boldsymbol{C}:=\boldsymbol{B}-\boldsymbol{v} \boldsymbol{v}^{*} / \alpha$	and therefore $\boldsymbol{C} := \boldsymbol{B} - \boldsymbol{v}\boldsymbol{v}^*/\alpha$ differs from $\boldsymbol{B}$ only in the upper left $d \times d$ corner. It
lyche-numerical-linear-algebra F01414	110	1.000	`d \times d`	$d \times d$	and therefore $\boldsymbol{C} := \boldsymbol{B} - \boldsymbol{v}\boldsymbol{v}^*/\alpha$ differs from $\boldsymbol{B}$ only in the upper left $d \times d$ corner. It
lyche-numerical-linear-algebra F01415	110	1.000	`n, \boldsymbol{C}=\boldsymbol{L}_{1} \boldsymbol{L}_{1}^{*}`	$n, \boldsymbol{C}=\boldsymbol{L}_{1} \boldsymbol{L}_{1}^{*}$	follows that $\boldsymbol{C}$ has the same bandwidth as $\boldsymbol{B}$ and $\boldsymbol{A}$. By induction on n, $\boldsymbol{C} = \boldsymbol{L}_1\boldsymbol{L}_1^*$,
lyche-numerical-linear-algebra F01416	110	0.999	`\boldsymbol{L}_{1}^{*}`	$\boldsymbol{L}_{1}^{*}$	where $\boldsymbol{L}_1^*$ has the same bandwidth as $\boldsymbol{C}$. But then $\boldsymbol{L}$ in (4.10) has the same bandwidth
lyche-numerical-linear-algebra F01417	110	1.000	`\left[\beta, \boldsymbol{v}^{*} / \beta\right]^{*}`	$\left[\beta, \boldsymbol{v}^{*} / \beta\right]^{*}$	of $\boldsymbol{L}$ is $[\beta, \boldsymbol{v}^*/\beta]^*$ if $\alpha > 0$. If $\alpha = 0$ then by 4 in Theorem 4.3 the first column of
lyche-numerical-linear-algebra F01418	110	1.000	`\alpha>0`	$\alpha>0$	of $\boldsymbol{L}$ is $[\beta, \boldsymbol{v}^*/\beta]^*$ if $\alpha > 0$. If $\alpha = 0$ then by 4 in Theorem 4.3 the first column of
lyche-numerical-linear-algebra F01419	110	0.947	`\boldsymbol{C}=\boldsymbol{B}-\boldsymbol{v} \boldsymbol{v}^{*} / \alpha`	$\boldsymbol{C}=\boldsymbol{B}-\boldsymbol{v} \boldsymbol{v}^{*} / \alpha$	$\boldsymbol{C} = \boldsymbol{B} - \boldsymbol{v}\boldsymbol{v}^*/\alpha$ in the lower part of $A(2:n, 2:n)$.
lyche-numerical-linear-algebra F01420	110	0.947	`A(2: n, 2: n)`	$A(2: n, 2: n)$	$\boldsymbol{C} = \boldsymbol{B} - \boldsymbol{v}\boldsymbol{v}^*/\alpha$ in the lower part of $A(2:n, 2:n)$.
lyche-numerical-linear-algebra F01421	110	1.000	`\min (i+d, n)`	$\min (i+d, n)$	replace all occurrences of n by $\min(i+d, n)$. Continuing the reduction we arrive
lyche-numerical-linear-algebra F01422	111	1.000	`\ell_{k k}=0`	$\ell_{k k}=0$	Column k of $\boldsymbol{L}$ is zero for those k where $\ell_{kk} = 0$. We reduce round-off noise by
lyche-numerical-linear-algebra F01423	111	0.910	`1 \Longleftrightarrow 2`	$1 \Longleftrightarrow 2$	$1 \Longleftrightarrow 2$: This follows from Lemma 4.5.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F01424	111	0.858	`1 \Longleftrightarrow`	$1 \Longleftrightarrow$	$1 \iff 4$: This follows from Theorem 4.5
lyche-numerical-linear-algebra-F01425	111	0.933	`1 \Longleftrightarrow 3`	$1 \Longleftrightarrow 3$	$1 \iff 3$: We refer to page 567 of [15], where it is shown that $4 \implies 3$ (and
lyche-numerical-linear-algebra-F01426	111	0.933	`4 \Longrightarrow 3`	$4 \Longrightarrow 3$	$1 \iff 3$: We refer to page 567 of [15], where it is shown that $4 \implies 3$ (and
lyche-numerical-linear-algebra-F01427	111	0.835	`1 \Longrightarrow 3`	$1 \Longrightarrow 3$	therefore $1 \implies 3$ since $1 \iff 4$) and $3 \implies 1$.
lyche-numerical-linear-algebra-F01428	111	0.835	`1 \Longleftrightarrow 4`	$1 \Longleftrightarrow 4$	therefore $1 \implies 3$ since $1 \iff 4$) and $3 \implies 1$.
lyche-numerical-linear-algebra-F01429	111	0.835	`3 \Longrightarrow 1`	$3 \Longrightarrow 1$	therefore $1 \implies 3$ since $1 \iff 4$) and $3 \implies 1$.
lyche-numerical-linear-algebra-F01430	112	0.997	`\boldsymbol{A}:=\left[\begin{array}{ll}1 & 1 \\ 1 & 1 \end{array}\right]`	$\boldsymbol{A}:=\left[\begin{array}{ll}1 & 1 \\ 1 & 1 \end{array}\right]$	matrix $\boldsymbol{A} := \begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$.
lyche-numerical-linear-algebra-F01431	112	0.996	`\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=x_{1}^{2}+x_{2}^{2}+x_{1} x_{2}+x_{2} x_{1}=\left(x_{1}+x_{2}\right)^{2} \geq 0`	$\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=x_{1}^{2}+x_{2}^{2}+x_{1} x_{2}+x_{2} x_{1}=\left(x_{1}+x_{2}\right)^{2} \geq 0$	1. We have $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} = x_1^2 + x_2^2 + x_1x_2 + x_2x_1 = (x_1+x_2)^2 \geq 0$ for all $\boldsymbol{x} \in \mathbb{R}^2$
lyche-numerical-linear-algebra-F01432	112	0.996	`\boldsymbol{x} \in \mathbb{R}^{2}`	$\boldsymbol{x} \in \mathbb{R}^{2}$	1. We have $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} = x_1^2 + x_2^2 + x_1x_2 + x_2x_1 = (x_1+x_2)^2 \geq 0$ for all $\boldsymbol{x} \in \mathbb{R}^2$
lyche-numerical-linear-algebra-F01433	112	1.000	`\lambda_{2}=0`	$\lambda_{2}=0$	2. The eigenvalues of $\boldsymbol{A}$ are $\lambda_1 = 2$ and $\lambda_2 = 0$ and they are nonnegative showing
lyche-numerical-linear-algebra-F01434	112	0.998	`\operatorname{det}\left(\left[a_{11}\right]\right)=`	$\operatorname{det}\left(\left[a_{11}\right]\right)=$	3. There are three principal sub matrices, and they have determinants $\det([a_{11}]) =$
lyche-numerical-linear-algebra-F01435	112	0.971	`\operatorname{det}\left(\left[a_{22}\right]\right)=1`	$\operatorname{det}\left(\left[a_{22}\right]\right)=1$	1, $\det([a_{22}]) = 1$ and $\det(\boldsymbol{A}) = 0$ and showing again that $\boldsymbol{A}$ is positive
lyche-numerical-linear-algebra-F01436	112	0.575	`\boldsymbol{A}=\boldsymbol{B B}^{*}`	$\boldsymbol{A}=\boldsymbol{B B}^{*}$	4. Finally $\boldsymbol{A}$ is positive semidefinite since $\boldsymbol{A} = \boldsymbol{B}\boldsymbol{B}^*$, where $\boldsymbol{B} = \begin{bmatrix}1 & 0\\ 1 & 0\end{bmatrix}$.
lyche-numerical-linear-algebra-F01437	112	0.575	`\boldsymbol{B}=\left[\begin{array}{ll}1 & 0 \\ 1 & 0 \end{array}\right]`	$\boldsymbol{B}=\left[\begin{array}{ll}1 & 0 \\ 1 & 0 \end{array}\right]$	4. Finally $\boldsymbol{A}$ is positive semidefinite since $\boldsymbol{A} = \boldsymbol{B}\boldsymbol{B}^*$, where $\boldsymbol{B} = \begin{bmatrix}1 & 0\\ 1 & 0\end{bmatrix}$.
lyche-numerical-linear-algebra-F01438	112	0.546	`\boldsymbol{A}:=\left[\begin{array}{rr}0 & 0 \\ 0 & -1 \end{array}\right]`	$\boldsymbol{A}:=\left[\begin{array}{rr}0 & 0 \\ 0 & -1 \end{array}\right]$	is not enough consider the matrix $\boldsymbol{A} := \begin{bmatrix}0 & 0\\ 0 & -1\end{bmatrix}$. The leading principal minors are
lyche-numerical-linear-algebra-F01439	112	1.000	`\operatorname{det}(\boldsymbol{A})>0`	$\operatorname{det}(\boldsymbol{A})>0$	4. $\det(\boldsymbol{A}) > 0$,
lyche-numerical-linear-algebra-F01440	112	1.000	`a_{i i} a_{j j}>a_{i j} a_{j i}`	$a_{i i} a_{j j}>a_{i j} a_{j i}$	5. $a_{ii}a_{jj} > a_{ij}a_{ji}$, for $i \neq j$.
lyche-numerical-linear-algebra-F01441	113	0.752	`\lambda=\frac{\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{T} \boldsymbol{x}}>0`	$\lambda=\frac{\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{T} \boldsymbol{x}}>0$	$\lambda = \frac{\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}} > 0$.
lyche-numerical-linear-algebra-F01442	113	0.981	`\left[\begin{array}{cc}a_{i i} & a_{i j} \\ a_{j i} & a_{j j}\end{array}\right]`	$\left[\begin{array}{cc}a_{i i} & a_{i j} \\ a_{j i} & a_{j j}\end{array}\right]$	5. The principal submatrix $\begin{bmatrix}a_{ii} & a_{ij}\\ a_{ji} & a_{jj}\end{bmatrix}$ has a positive determinant.
lyche-numerical-linear-algebra-F01443	113	1.000	`a \in \mathbb{R}`	$a \in \mathbb{R}$	is positive definite for any $a \in \mathbb{R}$. Indeed, for any nonzero $\boldsymbol{x} \in \mathbb{R}^2$
lyche-numerical-linear-algebra-F01444	113	1.000	`\boldsymbol{A}[a]`	$\boldsymbol{A}[a]$	The eigenvalues of $\boldsymbol{A}[a]$ are positive for $a \in [1-\frac{\sqrt{5}}{2}, 1+\frac{\sqrt{5}}{2}]$ and complex for
lyche-numerical-linear-algebra-F01445	113	1.000	`a \in\left[1-\frac{\sqrt{5}}{2}, 1+\frac{\sqrt{5}}{2} \right]`	$a \in\left[1-\frac{\sqrt{5}}{2}, 1+\frac{\sqrt{5}}{2} \right]$	The eigenvalues of $\boldsymbol{A}[a]$ are positive for $a \in [1-\frac{\sqrt{5}}{2}, 1+\frac{\sqrt{5}}{2}]$ and complex for
lyche-numerical-linear-algebra-F01446	113	1.000	`a`	a	other values of a.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01447	114	1.000	`\boldsymbol{A}:=\left[\begin{array}{cc}1 & 0 \\ -2 & 1 \end{array}\right]`	$\boldsymbol{A}:=\left[\begin{array}{cc}1 & 0 \\ -2 & 1\end{array}\right]$	the matrix $\boldsymbol{A}:=\left[\begin{array}{cc}1 & 0 \\ -2 & 1\end{array}\right]$.
lyche-numerical-linear-algebra F01448	114	1.000	`\boldsymbol{E} \in \mathbb{R}^{n \times n}`	$\boldsymbol{E} \in \mathbb{R}^{n \times n}$	**a)** Let $\boldsymbol{E} \in \mathbb{R}^{n\times n}$ be of the form $\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{u}\in\mathbb{R}^n$. Show that $\boldsymbol{E}$ is
lyche-numerical-linear-algebra F01449	114	1.000	`\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u} \boldsymbol{u}^{T}`	$\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^{T}$	**a)** Let $\boldsymbol{E} \in \mathbb{R}^{n\times n}$ be of the form $\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{u}\in\mathbb{R}^n$. Show that $\boldsymbol{E}$ is
lyche-numerical-linear-algebra F01450	114	1.000	`\boldsymbol{u} \in \mathbb{R}^{n}`	$\boldsymbol{u} \in \mathbb{R}^{n}$	**a)** Let $\boldsymbol{E} \in \mathbb{R}^{n\times n}$ be of the form $\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{u}\in\mathbb{R}^n$. Show that $\boldsymbol{E}$ is
lyche-numerical-linear-algebra F01451	114	1.000	`\boldsymbol{E}`	$\boldsymbol{E}$	**a)** Let $\boldsymbol{E} \in \mathbb{R}^{n\times n}$ be of the form $\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{u}\in\mathbb{R}^n$. Show that $\boldsymbol{E}$ is
lyche-numerical-linear-algebra F01452	114	0.834	`\boldsymbol{E}^{-1} .1`	$\boldsymbol{E}^{-1}.1$	symmetric and positive definite, and find an expression for $\boldsymbol{E}^{-1}$.[1]
lyche-numerical-linear-algebra F01453	114	1.000	`\boldsymbol{A}=\boldsymbol{B}+\boldsymbol{u} \boldsymbol{u}^{T}`	$\boldsymbol{A}=\boldsymbol{B}+\boldsymbol{u}\boldsymbol{u}^{T}$	**b)** Let $\boldsymbol{A}\in\mathbb{R}^{n\times n}$ be of the form $\boldsymbol{A}=\boldsymbol{B}+\boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{B}\in\mathbb{R}^{n\times n}$ is symmetric and
lyche-numerical-linear-algebra F01454	114	1.000	`\boldsymbol{A}=\boldsymbol{L} \boldsymbol{L}^{T}`	$\boldsymbol{A}=\boldsymbol{L}\boldsymbol{L}^{T}$	symmetric positive definite matrix with a known Cholesky factorization $\boldsymbol{A}=\boldsymbol{L}\boldsymbol{L}^T$.
lyche-numerical-linear-algebra F01455	114	0.998	`\boldsymbol{A}_{+}`	$\boldsymbol{A}_{+}$	Furthermore, let $\boldsymbol{A}_+$ be a corresponding $(n+1)\times(n+1)$ matrix of the form
lyche-numerical-linear-algebra F01456	114	0.998	`(n+1) \times(n+1)`	$(n+1)\times(n+1)$	Furthermore, let $\boldsymbol{A}_+$ be a corresponding $(n+1)\times(n+1)$ matrix of the form
lyche-numerical-linear-algebra F01457	114	1.000	`\alpha`	α	where $\boldsymbol{a}$ is a vector in $\mathbb{R}^n$, and α is a real number. We assume that the matrix $\boldsymbol{A}_+$ is
lyche-numerical-linear-algebra F01458	114	1.000	`\boldsymbol{A}_{+}=\boldsymbol{L}_{+} \boldsymbol{L}_{+}^{T}`	$\boldsymbol{A}_{+}=\boldsymbol{L}_{+}\boldsymbol{L}_{+}^{T}$	**a)** Show that if $\boldsymbol{A}_+=\boldsymbol{L}_+\boldsymbol{L}_+^T$ is the Cholesky factorization of $\boldsymbol{A}_+$, then $\boldsymbol{L}_+$ is of the
lyche-numerical-linear-algebra F01459	114	1.000	`\boldsymbol{L}_{+}`	$\boldsymbol{L}_{+}$	**a)** Show that if $\boldsymbol{A}_+=\boldsymbol{L}_+\boldsymbol{L}_+^T$ is the Cholesky factorization of $\boldsymbol{A}_+$, then $\boldsymbol{L}_+$ is of the
lyche-numerical-linear-algebra F01460	114	1.000	`\alpha>\left|\boldsymbol{L}^{-1} \boldsymbol{a}\right |_{2}^{2}`	$\alpha>\left\|\boldsymbol{L}^{-1}\boldsymbol{a}\right\|_{2}^{2}$	**b)** Explain why $\alpha>\|\boldsymbol{L}^{-1}\boldsymbol{a}\|_2^2$.
lyche-numerical-linear-algebra F01461	114	1.000	`\boldsymbol{E}^{-1}`	$\boldsymbol{E}^{-1}$	[1]Hint: The matrix $\boldsymbol{E}^{-1}$ is of the form $\boldsymbol{E}^{-1}=\boldsymbol{I}+a\boldsymbol{u}\boldsymbol{u}^T$ for some $a\in\mathbb{R}$.
lyche-numerical-linear-algebra F01462	114	1.000	`\boldsymbol{E}^{-1}=\boldsymbol{I}+a \boldsymbol{u} \boldsymbol{u}^{T}`	$\boldsymbol{E}^{-1}=\boldsymbol{I}+a\boldsymbol{u}\boldsymbol{u}^{T}$	[1]Hint: The matrix $\boldsymbol{E}^{-1}$ is of the form $\boldsymbol{E}^{-1}=\boldsymbol{I}+a\boldsymbol{u}\boldsymbol{u}^T$ for some $a\in\mathbb{R}$.
lyche-numerical-linear-algebra F01463	115	0.493	`\left[\begin{array}{lll}10 & 4 & 3 \\ 4 & 0 & 2 \\ 3 & 2 & 5\end{array}\right]`	$\left[\begin{array}{lll}10 & 4 & 3 \\ 4 & 0 & 2 \\ 3 & 2 & 5\end{array}\right]$	4.6.3 Is the matrix $\left[\begin{smallmatrix}10 & 4 & 3 \\ 4 & 0 & 2 \\ 3 & 2 & 5\end{smallmatrix}\right]$ positive definite?
lyche-numerical-linear-algebra F01464	117	1.000	`\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}`	$\mathcal{V}\times\mathcal{V}\rightarrow\mathbb{C}$	a function $\mathcal{V}\times\mathcal{V}\to\mathbb{C}$ satisfying for all $\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}\in\mathcal{V}$ and all $a,b\in\mathbb{C}$ the following
lyche-numerical-linear-algebra F01465	117	1.000	`\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}`	$\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}\in\mathcal{V}$	a function $\mathcal{V}\times\mathcal{V}\to\mathbb{C}$ satisfying for all $\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}\in\mathcal{V}$ and all $a,b\in\mathbb{C}$ the following
lyche-numerical-linear-algebra F01466	117	1.000	`a, b \in \mathbb{C}`	$a,b\in\mathbb{C}$	a function $\mathcal{V}\times\mathcal{V}\to\mathbb{C}$ satisfying for all $\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}\in\mathcal{V}$ and all $a,b\in\mathbb{C}$ the following
lyche-numerical-linear-algebra F01467	117	1.000	`\langle\boldsymbol{x}, \boldsymbol{x}\rangle \geq 0`	$\langle\boldsymbol{x},\boldsymbol{x}\rangle\geq 0$	1. $\langle\boldsymbol{x},\boldsymbol{x}\rangle\geq 0$ with equality if and only if $\boldsymbol{x}=\mathbf{0}$.
lyche-numerical-linear-algebra F01468	117	1.000	`\langle\boldsymbol{x}, \boldsymbol{y}\rangle=\overline{ \langle\boldsymbol{y}, \boldsymbol{x}\rangle}`	$\langle\boldsymbol{x},\boldsymbol{y}\rangle=\overline{\langle\boldsymbol{y},\boldsymbol{x}\rangle}$	2. $\langle\boldsymbol{x},\boldsymbol{y}\rangle=\overline{\langle\boldsymbol{y},\boldsymbol{x}\rangle}$
lyche-numerical-linear-algebra F01469	117	1.000	`\langle a \boldsymbol{x}+b \boldsymbol{y}, \boldsymbol{z} \rangle=a\langle\boldsymbol{x}, \boldsymbol{z}\rangle+b \langle\boldsymbol{y}, \boldsymbol{z}\rangle`	$\langle a\boldsymbol{x}+b\boldsymbol{y},\boldsymbol{z}\rangle=a\langle\boldsymbol{x},\boldsymbol{z}\rangle+b\langle\boldsymbol{y},\boldsymbol{z}\rangle$	3. $\langle a\boldsymbol{x}+b\boldsymbol{y},\boldsymbol{z}\rangle=a\langle\boldsymbol{x},\boldsymbol{z}\rangle+b\langle\boldsymbol{y},\boldsymbol{z}\rangle$.
lyche-numerical-linear-algebra F01470	117	0.996	`(\mathcal{V},\langle\cdot, \cdot\rangle)`	$(\mathcal{V},\langle\cdot,\cdot\rangle)$	The pair $(\mathcal{V},\langle\cdot,\cdot\rangle)$ is called an **inner product space**.
lyche-numerical-linear-algebra F01471	117	0.975	`|\boldsymbol{x}|=|\boldsymbol{x}|_{2}=\sqrt{ \boldsymbol{x}^{*} \boldsymbol{x}}`	$\\|\boldsymbol{x}\\|=\\|\boldsymbol{x}\\|_{2}=\sqrt{\boldsymbol{x}^{*}\boldsymbol{x}}$	$\\|\boldsymbol{x}\\|=\\|\boldsymbol{x}\\|_2=\sqrt{\boldsymbol{x}^*\boldsymbol{x}}$.
lyche-numerical-linear-algebra F01472	118	1.000	`0 \boldsymbol{x}+\boldsymbol{y}=\mathbf{0}`	$0\boldsymbol{x}+\boldsymbol{y}=\mathbf{0}$	***Proof*** If $\boldsymbol{y}=\mathbf{0}$ then $0\boldsymbol{x}+\boldsymbol{y}=\mathbf{0}$ and $\boldsymbol{x}$ and $\boldsymbol{y}$ are linearly dependent. Moreover the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01473	118	1.000	`\langle\boldsymbol{x}, \boldsymbol{y}\rangle=\langle \boldsymbol{x}, 0 \boldsymbol{y}\rangle=0\langle \boldsymbol{x}, \boldsymbol{y}\rangle=0`	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle=\langle \boldsymbol{x}, 0 \boldsymbol{y}\rangle=0\langle \boldsymbol{x}, \boldsymbol{y}\rangle=0$	inequality holds with equality since $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \langle \boldsymbol{x}, 0\boldsymbol{y}\rangle = 0\langle \boldsymbol{x}, \boldsymbol{y}\rangle = 0$ and $\|\boldsymbol{y}\| = 0$.
lyche-numerical-linear-algebra F01474	118	1.000	`|\boldsymbol{y}|=0`	$\|\boldsymbol{y}\|=0$	inequality holds with equality since $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \langle \boldsymbol{x}, 0\boldsymbol{y}\rangle = 0\langle \boldsymbol{x}, \boldsymbol{y}\rangle = 0$ and $\|\boldsymbol{y}\| = 0$.
lyche-numerical-linear-algebra F01475	118	1.000	`\boldsymbol{y} \neq \mathbf{0}`	$\boldsymbol{y} \neq \mathbf{0}$	So assume $\boldsymbol{y} \neq \mathbf{0}$. Define
lyche-numerical-linear-algebra F01476	118	0.862	`\langle\boldsymbol{z}, \boldsymbol{y}\rangle=\langle \boldsymbol{x}, \boldsymbol{y}\rangle-a\langle \boldsymbol{y}, \boldsymbol{y}\rangle=0`	$\langle\boldsymbol{z}, \boldsymbol{y}\rangle=\langle \boldsymbol{x}, \boldsymbol{y}\rangle-a\langle \boldsymbol{y}, \boldsymbol{y}\rangle=0$	By linearity $\langle \boldsymbol{z}, \boldsymbol{y}\rangle = \langle \boldsymbol{x}, \boldsymbol{y}\rangle - a\langle \boldsymbol{y}, \boldsymbol{y}\rangle = 0$ so that by 2. and (5.1)
lyche-numerical-linear-algebra F01477	118	1.000	`|\boldsymbol{y}|^{2}`	$\|\boldsymbol{y}\|^{2}$	Multiplying by $\|\boldsymbol{y}\|^2$ gives (5.3). We have equality if and only if $\boldsymbol{z} = \mathbf{0}$, which means
lyche-numerical-linear-algebra F01478	118	1.000	`\boldsymbol{z}=\mathbf{0}`	$\boldsymbol{z}=\mathbf{0}$	Multiplying by $\|\boldsymbol{y}\|^2$ gives (5.3). We have equality if and only if $\boldsymbol{z} = \mathbf{0}$, which means
lyche-numerical-linear-algebra F01479	118	1.000	`|\boldsymbol{x}| \geq 0`	$\|\boldsymbol{x}\| \geq 0$	*1.* $\|\boldsymbol{x}\| \geq 0$ *with equality if and only if* $\boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra F01480	118	1.000	`|a \boldsymbol{x}|=|a||\boldsymbol{x}|`	$\|a \boldsymbol{x}\|=\|a\|\|\boldsymbol{x}\|$	*2.* $\|a\boldsymbol{x}\| = \|a\|\,\|\boldsymbol{x}\|$.
lyche-numerical-linear-algebra F01481	118	1.000	`|\boldsymbol{x}+\boldsymbol{y}| \leq|\boldsymbol{x}|+|\boldsymbol{y}|`	$\|\boldsymbol{x}+\boldsymbol{y}\| \leq\|\boldsymbol{x}\|+\|\boldsymbol{y}\|$	*3.* $\|\boldsymbol{x}+\boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$,
lyche-numerical-linear-algebra F01482	118	1.000	`|\boldsymbol{x}|:=\sqrt{\langle\boldsymbol{x}, \boldsymbol{x}\rangle}`	$\|\boldsymbol{x}\|:=\sqrt{\langle\boldsymbol{x}, \boldsymbol{x}\rangle}$	*where* $\|\boldsymbol{x}\| := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x}\rangle}$.
lyche-numerical-linear-algebra F01483	118	1.000	`\left||: \mathbb{C}^{n} \rightarrow \mathbb{R}\right.`	$\left\|\|: \mathbb{C}^{n} \rightarrow \mathbb{R}\right.$	In general a function $\|\ \| : \mathbb{C}^n \to \mathbb{R}$ that satisfies these three properties is called a
lyche-numerical-linear-algebra F01484	119	1.000	`|\boldsymbol{x}+a \boldsymbol{y}|^{2}=\langle \boldsymbol{x}+a \boldsymbol{y}, \boldsymbol{x}+a \boldsymbol{y}\rangle`	$\|\boldsymbol{x}+a \boldsymbol{y}\|^{2}=\langle \boldsymbol{x}+a \boldsymbol{y}, \boldsymbol{x}+a \boldsymbol{y}\rangle$	second one follows from (5.1). Expanding $\|\boldsymbol{x} + a\boldsymbol{y}\|^2 = \langle \boldsymbol{x} + a\boldsymbol{y}, \boldsymbol{x} + a\boldsymbol{y}\rangle$ using
lyche-numerical-linear-algebra F01485	119	1.000	`a=1`	$a=1$	Now (5.5) with $a = 1$ and the Cauchy-Schwarz inequality implies
lyche-numerical-linear-algebra F01486	119	0.999	`-1 \leq \frac{\langle\boldsymbol{x}, \boldsymbol{y}\rangle}{|\boldsymbol{x}||\boldsymbol{y}|} \leq 1`	$-1 \leq \frac{\langle\boldsymbol{x}, \boldsymbol{y}\rangle}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|} \leq 1$	In the real case the Cauchy-Schwarz inequality implies that $-1 \leq \frac{\langle \boldsymbol{x}, \boldsymbol{y}\rangle}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|} \leq 1$
lyche-numerical-linear-algebra F01487	119	1.000	`\theta`	θ	for nonzero $\boldsymbol{x}$ and $\boldsymbol{y}$, so there is a unique angle θ in $[0, \pi]$ such that
lyche-numerical-linear-algebra F01488	119	1.000	`[0, \pi]`	$[0, \pi]$	for nonzero $\boldsymbol{x}$ and $\boldsymbol{y}$, so there is a unique angle θ in $[0, \pi]$ such that
lyche-numerical-linear-algebra F01489	119	1.000	`\boldsymbol{x} \perp \boldsymbol{y}`	$\boldsymbol{x} \perp \boldsymbol{y}$	uct space are **orthogonal** or **perpendicular**, denoted as $\boldsymbol{x} \perp \boldsymbol{y}$, if $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = 0$. The
lyche-numerical-linear-algebra F01490	119	1.000	`\langle\boldsymbol{x}, \boldsymbol{y}\rangle=0`	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle=0$	uct space are **orthogonal** or **perpendicular**, denoted as $\boldsymbol{x} \perp \boldsymbol{y}$, if $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = 0$. The
lyche-numerical-linear-algebra F01491	119	1.000	`|\boldsymbol{x}|=|\boldsymbol{y}|=1`	$\|\boldsymbol{x}\|=\|\boldsymbol{y}\|=1$	vectors are **orthonormal** if in addition $\|\boldsymbol{x}\| = \|\boldsymbol{y}\| = 1$.
lyche-numerical-linear-algebra F01492	119	1.000	`\theta=\pi / 2`	$\theta=\pi / 2$	or $\mathbb{C}^n$ it follows that $\boldsymbol{x} \perp \boldsymbol{y}$ if and only if $\theta = \pi/2$.
lyche-numerical-linear-algebra F01493	119	0.997	`\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{j}\right\rangle=0`	$\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{j}\right\rangle=0$	**orthogonal basis** for $\mathcal{S}$ if it is a basis for $\mathcal{S}$ and $\langle \boldsymbol{v}_i, \boldsymbol{v}_j\rangle = 0$ for $i \neq j$. It is an
lyche-numerical-linear-algebra F01494	119	0.734	`\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{j}\right\rangle=\delta_{i j}`	$\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{j}\right\rangle=\delta_{i j}$	**orthonormal basis** for $\mathcal{S}$ if it is a basis for $\mathcal{S}$ and $\langle \boldsymbol{v}_i, \boldsymbol{v}_j\rangle = \delta_{ij}$ for all i, j.
lyche-numerical-linear-algebra F01495	120	0.999	`\left\{\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{k}\right\}`	$\left\{\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{k}\right\}$	**Theorem 5.4 (Gram-Schmidt)** *Let* $\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k\}$ *be a basis for a real or complex*
lyche-numerical-linear-algebra F01496	120	0.833	`(\mathcal{S},\langle\cdot, \cdot\rangle)`	$(\mathcal{S},\langle\cdot, \cdot\rangle)$	*inner product space* $(\mathcal{S}, \langle\cdot, \cdot\rangle)$. *Define*
lyche-numerical-linear-algebra F01497	120	0.999	`S_{j}:=\operatorname{span}\left\{\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{j}\right\}`	$S_{j}:=\operatorname{span}\left\{\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{j}\right\}$	k. Define subspaces $S_j := \operatorname{span}\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_j\}$ for $j = 1, \ldots, k$. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$
lyche-numerical-linear-algebra F01498	120	0.999	`\boldsymbol{v}_{1}=\boldsymbol{s}_{1}`	$\boldsymbol{v}_{1}=\boldsymbol{s}_{1}$	k. Define subspaces $S_j := \operatorname{span}\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_j\}$ for $j = 1, \ldots, k$. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01499	120	1.000	`S_{1}`	S_{1}	is an orthogonal basis for S_1. Suppose for some $j \geq 2$ that $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{j-1}$ is an
lyche-numerical-linear-algebra_F01500	120	1.000	`j \geq 2`	$j \geq 2$	is an orthogonal basis for S_1. Suppose for some $j \geq 2$ that $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{j-1}$ is an
lyche-numerical-linear-algebra_F01501	120	1.000	`\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{j-1}`	$\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{j-1}$	is an orthogonal basis for S_1. Suppose for some $j \geq 2$ that $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{j-1}$ is an
lyche-numerical-linear-algebra_F01502	120	1.000	`S_{j-1}`	S_{j-1}	orthogonal basis for S_{j-1} and let $\boldsymbol{v}_j$ be given by (5.8) as a linear combination of
lyche-numerical-linear-algebra_F01503	120	1.000	`\boldsymbol{v}_{j}`	$\boldsymbol{v}_{j}$	orthogonal basis for S_{j-1} and let $\boldsymbol{v}_j$ be given by (5.8) as a linear combination of
lyche-numerical-linear-algebra_F01504	120	1.000	`\boldsymbol{v}_{i}`	$\boldsymbol{v}_{i}$	$\boldsymbol{s}_j$ and $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{j-1}$. Now each of these $\boldsymbol{v}_i$ is a linear combination of $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_i$,
lyche-numerical-linear-algebra_F01505	120	1.000	`\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{i}`	$\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{i}$	$\boldsymbol{s}_j$ and $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{j-1}$. Now each of these $\boldsymbol{v}_i$ is a linear combination of $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_i$,
lyche-numerical-linear-algebra_F01506	120	1.000	`\boldsymbol{v}_{j}=\sum_{i=1}^{j} a_{i} \boldsymbol{s}_{i}`	$\boldsymbol{v}_{j}=\sum_{i=1}^{j} a_{i} \boldsymbol{s}_{i}$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^{j} a_i \boldsymbol{s}_i$ for some $a_0, \ldots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra_F01507	120	1.000	`a_{0}, \ldots, a_{j}`	$a_{0}, \ldots, a_{j}$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^{j} a_i \boldsymbol{s}_i$ for some $a_0, \ldots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra_F01508	120	1.000	`a_{j}=1`	$a_{j}=1$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^{j} a_i \boldsymbol{s}_i$ for some $a_0, \ldots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra_F01509	120	1.000	`\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{j}`	$\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{j}$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^{j} a_i \boldsymbol{s}_i$ for some $a_0, \ldots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra_F01510	120	1.000	`a_{j} \neq 0`	$a_{j} \neq 0$	are linearly independent and $a_j \neq 0$ we deduce that $\boldsymbol{v}_j \neq 0$. By the induction
lyche-numerical-linear-algebra_F01511	120	1.000	`\boldsymbol{v}_{j} \neq 0`	$\boldsymbol{v}_{j} \neq 0$	are linearly independent and $a_j \neq 0$ we deduce that $\boldsymbol{v}_j \neq 0$. By the induction
lyche-numerical-linear-algebra_F01512	120	1.000	`l=1, \ldots, j-1`	$l=1, \ldots, j-1$	for $l = 1, \ldots, j-1$. Thus $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j$ is an orthogonal basis for S_j.
lyche-numerical-linear-algebra_F01513	120	1.000	`\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{j}`	$\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{j}$	for $l = 1, \ldots, j-1$. Thus $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j$ is an orthogonal basis for S_j.
lyche-numerical-linear-algebra_F01514	120	1.000	`S_{j}`	S_{j}	for $l = 1, \ldots, j-1$. Thus $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j$ is an orthogonal basis for S_j.
lyche-numerical-linear-algebra_F01515	120	1.000	`\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{k} \right\}`	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{k} \right\}$	If $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ is an orthogonal basis for $\mathcal{S}$ then clearly $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\}$ is an
lyche-numerical-linear-algebra_F01516	121	1.000	`\mathcal{S} \subset \mathcal{T}`	$\mathcal{S} \subset \mathcal{T}$	**Theorem 5.5 (Orthogonal Extension of Basis)** *Suppose $\mathcal{S} \subset \mathcal{T}$ are finite dimen-*
lyche-numerical-linear-algebra_F01517	121	1.000	`\operatorname{dim} \mathcal{S}:=k<\operatorname{dim} \mathcal{T}=n`	$\operatorname{dim} \mathcal{S}:=k<\operatorname{dim} \mathcal{T}=n$	***Proof*** Suppose $\dim \mathcal{S} := k < \dim \mathcal{T} = n$. Using Theorem 1.3 we first extend
lyche-numerical-linear-algebra_F01518	121	1.000	`\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{k}`	$\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{k}$	an orthogonal basis $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k$ for $\mathcal{S}$ to a basis $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k, \boldsymbol{s}_{k+1}, \ldots, \boldsymbol{s}_n$ for $\mathcal{T}$, and
lyche-numerical-linear-algebra_F01519	121	1.000	`\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{k}, \boldsymbol{s}_{k+1}, \ldots, \boldsymbol{s}_{n}`	$\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{k}, \boldsymbol{s}_{k+1}, \ldots, \boldsymbol{s}_{n}$	an orthogonal basis $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k$ for $\mathcal{S}$ to a basis $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k, \boldsymbol{s}_{k+1}, \ldots, \boldsymbol{s}_n$ for $\mathcal{T}$, and
lyche-numerical-linear-algebra_F01520	121	0.955	`\boldsymbol{v}_{i}=\boldsymbol{s}_{i}`	$\boldsymbol{v}_{i}=\boldsymbol{s}_{i}$	$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ for $\mathcal{T}$. This is an extension of the basis for $\mathcal{S}$ since $\boldsymbol{v}_i = \boldsymbol{s}_i$ for $i =$
lyche-numerical-linear-algebra_F01521	121	1.000	`1, \ldots, k`	$1, \ldots, k$	$1, \ldots, k$. We show this by induction. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$. Suppose for some $2 \leq r < k$
lyche-numerical-linear-algebra_F01522	121	1.000	`2 \leq r<k`	$2 \leq r<k$	$1, \ldots, k$. We show this by induction. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$. Suppose for some $2 \leq r < k$
lyche-numerical-linear-algebra_F01523	121	1.000	`\boldsymbol{v}_{j}=\boldsymbol{s}_{j}`	$\boldsymbol{v}_{j}=\boldsymbol{s}_{j}$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \ldots, r-1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
lyche-numerical-linear-algebra_F01524	121	1.000	`j=1, \ldots, r-1`	$j=1, \ldots, r-1$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \ldots, r-1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
lyche-numerical-linear-algebra_F01525	121	1.000	`j=r`	$j=r$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \ldots, r-1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01526	121	1.000	`\left\langle\boldsymbol{s}_{r}, \boldsymbol{v}_{i}\right \rangle=`	$\left\langle\boldsymbol{s}_{r}, \boldsymbol{v}_{i}\right\rangle=$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \ldots, r-1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
lyche-numerical-linear-algebra_F01527	121	0.999	`\left\langle\boldsymbol{s}_{r}, \boldsymbol{s}_{i}\right \rangle=0`	$\left\langle\boldsymbol{s}_{r}, \boldsymbol{s}_{i}\right\rangle=0$	$\langle \boldsymbol{s}_r, \boldsymbol{s}_i \rangle = 0$ for $i < r$ we obtain $\boldsymbol{v}_r = \boldsymbol{s}_r$.
lyche-numerical-linear-algebra_F01528	121	0.999	`i<r`	$i<r$	$\langle \boldsymbol{s}_r, \boldsymbol{s}_i \rangle = 0$ for $i < r$ we obtain $\boldsymbol{v}_r = \boldsymbol{s}_r$.
lyche-numerical-linear-algebra_F01529	121	0.999	`\boldsymbol{v}_{r}=\boldsymbol{s}_{r}`	$\boldsymbol{v}_{r}=\boldsymbol{s}_{r}$	$\langle \boldsymbol{s}_r, \boldsymbol{s}_i \rangle = 0$ for $i < r$ we obtain $\boldsymbol{v}_r = \boldsymbol{s}_r$.
lyche-numerical-linear-algebra_F01530	121	0.999	`\mathcal{S}=\operatorname{span}\left(\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{k}\right)`	$\mathcal{S}=\operatorname{span}\left(\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{k}\right)$	Letting $\mathcal{S} = \operatorname{span}(\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k)$ and $\mathcal{T}$ be $\mathbb{R}^n$ or $\mathbb{C}^n$ we obtain
lyche-numerical-linear-algebra_F01531	121	1.000	`1 \leq k<n`	$1 \leq k<n$	**Corollary 5.1 (Extending Orthogonal Vectors to a Basis)** *For* $1 \leq k < n$ *a*
lyche-numerical-linear-algebra_F01532	121	1.000	`\langle\boldsymbol{x}, \boldsymbol{y}\rangle`	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle$	an inner product $\langle \boldsymbol{x}, \boldsymbol{y} \rangle$. We define
lyche-numerical-linear-algebra_F01533	121	1.000	`\mathcal{S}+\mathcal{T}:=\{\boldsymbol{s}+\boldsymbol{t}: \boldsymbol{s} \in \mathcal{S}`	$\mathcal{S}+\mathcal{T}:=\{\boldsymbol{s}+\boldsymbol{t}: \boldsymbol{s} \in \mathcal{S}$	• **Sum**: $\mathcal{S} + \mathcal{T} := \{\boldsymbol{s} + \boldsymbol{t} : \boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}\}$,
lyche-numerical-linear-algebra_F01534	121	1.000	`\boldsymbol{t} \in \mathcal{T}\}`	$\boldsymbol{t} \in \mathcal{T}\}$	• **Sum**: $\mathcal{S} + \mathcal{T} := \{\boldsymbol{s} + \boldsymbol{t} : \boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}\}$,
lyche-numerical-linear-algebra_F01535	121	0.967	`\boldsymbol{\operatorname { s u m }} \mathcal{S} \oplus \mathcal{T}`	*(not rendered)*	• **direct sum** $\mathcal{S} \oplus \mathcal{T}$: a sum where $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$,
lyche-numerical-linear-algebra_F01536	121	0.998	`\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}`	$\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	• **orthogonal sum** $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$ for all $\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$.
lyche-numerical-linear-algebra_F01537	121	0.998	`\langle\boldsymbol{s}, \boldsymbol{t}\rangle=0`	$\langle\boldsymbol{s}, \boldsymbol{t}\rangle=0$	• **orthogonal sum** $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$ for all $\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$.
lyche-numerical-linear-algebra_F01538	121	0.998	`\boldsymbol{s} \in \mathcal{S}`	$\boldsymbol{s} \in \mathcal{S}$	• **orthogonal sum** $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$ for all $\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$.
lyche-numerical-linear-algebra_F01539	121	0.998	`\boldsymbol{t} \in \mathcal{T}`	$\boldsymbol{t} \in \mathcal{T}$	• **orthogonal sum** $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$ for all $\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$.
lyche-numerical-linear-algebra_F01540	121	1.000	`\boldsymbol{v} \in \mathcal{S} \oplus \mathcal{T}`	$\boldsymbol{v} \in \mathcal{S} \oplus \mathcal{T}$	• Every $\boldsymbol{v} \in \mathcal{S} \oplus \mathcal{T}$ can be decomposed uniquely in the form $\boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}$, where
lyche-numerical-linear-algebra_F01541	121	1.000	`\boldsymbol{v}=\boldsymbol{s}+\boldsymbol{t}`	$\boldsymbol{v}=\boldsymbol{s}+\boldsymbol{t}$	• Every $\boldsymbol{v} \in \mathcal{S} \oplus \mathcal{T}$ can be decomposed uniquely in the form $\boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}$, where
lyche-numerical-linear-algebra_F01542	121	1.000	`\boldsymbol{v}=\boldsymbol{s}_{1}+\boldsymbol{t}_{1}= \boldsymbol{s}_{2}+\boldsymbol{t}_{2}`	$\boldsymbol{v}=\boldsymbol{s}_{1}+\boldsymbol{t}_{1}=\boldsymbol{s}_{2}+\boldsymbol{t}_{2}$	$\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$. For if $\boldsymbol{v} = \boldsymbol{s}_1 + \boldsymbol{t}_1 = \boldsymbol{s}_2 + \boldsymbol{t}_2$ for $\boldsymbol{s}_1, \boldsymbol{s}_2 \in \mathcal{S}$ and $\boldsymbol{t}_1, \boldsymbol{t}_2 \in \mathcal{T}$,
lyche-numerical-linear-algebra_F01543	121	1.000	`\boldsymbol{s}_{1}, \boldsymbol{s}_{2} \in \mathcal{S}`	$\boldsymbol{s}_{1}, \boldsymbol{s}_{2} \in \mathcal{S}$	$\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$. For if $\boldsymbol{v} = \boldsymbol{s}_1 + \boldsymbol{t}_1 = \boldsymbol{s}_2 + \boldsymbol{t}_2$ for $\boldsymbol{s}_1, \boldsymbol{s}_2 \in \mathcal{S}$ and $\boldsymbol{t}_1, \boldsymbol{t}_2 \in \mathcal{T}$,
lyche-numerical-linear-algebra_F01544	121	1.000	`\boldsymbol{t}_{1}, \boldsymbol{t}_{2} \in \mathcal{T}`	$\boldsymbol{t}_{1}, \boldsymbol{t}_{2} \in \mathcal{T}$	$\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$. For if $\boldsymbol{v} = \boldsymbol{s}_1 + \boldsymbol{t}_1 = \boldsymbol{s}_2 + \boldsymbol{t}_2$ for $\boldsymbol{s}_1, \boldsymbol{s}_2 \in \mathcal{S}$ and $\boldsymbol{t}_1, \boldsymbol{t}_2 \in \mathcal{T}$,
lyche-numerical-linear-algebra_F01545	121	0.969	`\mathbf{0}=\boldsymbol{s}_{1}-\boldsymbol{s}_{2}+ \boldsymbol{t}_{1}-\boldsymbol{t}_{2}`	$\mathbf{0}=\boldsymbol{s}_{1}-\boldsymbol{s}_{2}+\boldsymbol{t}_{1}-\boldsymbol{t}_{2}$	then $\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$ or $\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$. It follows that $\boldsymbol{s}_1 - \boldsymbol{s}_2$ and $\boldsymbol{t}_2 - \boldsymbol{t}_1$
lyche-numerical-linear-algebra_F01546	121	0.969	`\boldsymbol{s}_{1}-\boldsymbol{s}_{2}=\boldsymbol{t}_{2}- \boldsymbol{t}_{1}`	$\boldsymbol{s}_{1}-\boldsymbol{s}_{2}=\boldsymbol{t}_{2}-\boldsymbol{t}_{1}$	then $\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$ or $\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$. It follows that $\boldsymbol{s}_1 - \boldsymbol{s}_2$ and $\boldsymbol{t}_2 - \boldsymbol{t}_1$
lyche-numerical-linear-algebra_F01547	121	0.969	`\boldsymbol{s}_{1}-\boldsymbol{s}_{2}`	$\boldsymbol{s}_{1}-\boldsymbol{s}_{2}$	then $\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$ or $\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$. It follows that $\boldsymbol{s}_1 - \boldsymbol{s}_2$ and $\boldsymbol{t}_2 - \boldsymbol{t}_1$
lyche-numerical-linear-algebra_F01548	121	0.969	`\boldsymbol{t}_{2}-\boldsymbol{t}_{1}`	$\boldsymbol{t}_{2}-\boldsymbol{t}_{1}$	then $\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$ or $\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$. It follows that $\boldsymbol{s}_1 - \boldsymbol{s}_2$ and $\boldsymbol{t}_2 - \boldsymbol{t}_1$
lyche-numerical-linear-algebra_F01549	121	1.000	`\boldsymbol{s}_{1}-\boldsymbol{s}_{2}=\boldsymbol{t}_{2}- \boldsymbol{t}_{1}=\mathbf{0}`	$\boldsymbol{s}_{1}-\boldsymbol{s}_{2}=\boldsymbol{t}_{2}-\boldsymbol{t}_{1}=\mathbf{0}$	belong to both $\mathcal{S}$ and $\mathcal{T}$ and hence to $\mathcal{S} \cap \mathcal{T}$. But then $\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1 = \mathbf{0}$ so
lyche-numerical-linear-algebra_F01550	121	1.000	`\boldsymbol{s}_{1}=\boldsymbol{s}_{2}`	$\boldsymbol{s}_{1}=\boldsymbol{s}_{2}$	$\boldsymbol{s}_1 = \boldsymbol{s}_2$ and $\boldsymbol{t}_2 = \boldsymbol{t}_1$.
lyche-numerical-linear-algebra_F01551	121	1.000	`\boldsymbol{t}_{2}=\boldsymbol{t}_{1}`	$\boldsymbol{t}_{2}=\boldsymbol{t}_{1}$	$\boldsymbol{s}_1 = \boldsymbol{s}_2$ and $\boldsymbol{t}_2 = \boldsymbol{t}_1$.
lyche-numerical-linear-algebra_F01552	121	1.000	`\boldsymbol{v} \in \mathcal{S} \cap \mathcal{T}`	$\boldsymbol{v} \in \mathcal{S} \cap \mathcal{T}$	• An orthogonal sum is a direct sum. For if $\boldsymbol{v} \in \mathcal{S} \cap \mathcal{T}$ then $\boldsymbol{v}$ is orthogonal to itself,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01553	121	1.000	`\langle\boldsymbol{v}, \boldsymbol{v}\rangle=0`	$\langle\boldsymbol{v}, \boldsymbol{v}\rangle=0$	$\langle \boldsymbol{v}, \boldsymbol{v}\rangle = 0$, which implies that $\boldsymbol{v} = 0$. We often write $\mathcal{T} := \mathcal{S}^{\perp}$.
lyche-numerical-linear-algebra_F01554	121	1.000	`\boldsymbol{v}=0`	$\boldsymbol{v}=0$	$\langle \boldsymbol{v}, \boldsymbol{v}\rangle = 0$, which implies that $\boldsymbol{v} = 0$. We often write $\mathcal{T} := \mathcal{S}^{\perp}$.
lyche-numerical-linear-algebra_F01555	121	1.000	`\mathcal{T}:=\mathcal{S}^{\perp}`	$\mathcal{T}:=\mathcal{S}^{\perp}$	$\langle \boldsymbol{v}, \boldsymbol{v}\rangle = 0$, which implies that $\boldsymbol{v} = 0$. We often write $\mathcal{T} := \mathcal{S}^{\perp}$.
lyche-numerical-linear-algebra_F01556	121	1.000	`\boldsymbol{v}=\boldsymbol{s}_{0}+\boldsymbol{t}_{0} \in \mathcal{S} \oplus \mathcal{T}`	$\boldsymbol{v}=\boldsymbol{s}_{0}+\boldsymbol{t}_{0} \in \mathcal{S} \oplus \mathcal{T}$	• Suppose $\boldsymbol{v} = \boldsymbol{s}_0 + \boldsymbol{t}_0 \in \mathcal{S} \oplus \mathcal{T}$, where $\boldsymbol{s}_0 \in \mathcal{S}$ and $\boldsymbol{t}_0 \in \mathcal{T}$. The vector $\boldsymbol{s}_0$ is
lyche-numerical-linear-algebra_F01557	121	1.000	`\boldsymbol{s}_{0} \in \mathcal{S}`	$\boldsymbol{s}_{0} \in \mathcal{S}$	• Suppose $\boldsymbol{v} = \boldsymbol{s}_0 + \boldsymbol{t}_0 \in \mathcal{S} \oplus \mathcal{T}$, where $\boldsymbol{s}_0 \in \mathcal{S}$ and $\boldsymbol{t}_0 \in \mathcal{T}$. The vector $\boldsymbol{s}_0$ is
lyche-numerical-linear-algebra_F01558	121	1.000	`\boldsymbol{t}_{0} \in \mathcal{T}`	$\boldsymbol{t}_{0} \in \mathcal{T}$	• Suppose $\boldsymbol{v} = \boldsymbol{s}_0 + \boldsymbol{t}_0 \in \mathcal{S} \oplus \mathcal{T}$, where $\boldsymbol{s}_0 \in \mathcal{S}$ and $\boldsymbol{t}_0 \in \mathcal{T}$. The vector $\boldsymbol{s}_0$ is
lyche-numerical-linear-algebra_F01559	121	1.000	`\boldsymbol{s}_{0}`	$\boldsymbol{s}_{0}$	• Suppose $\boldsymbol{v} = \boldsymbol{s}_0 + \boldsymbol{t}_0 \in \mathcal{S} \oplus \mathcal{T}$, where $\boldsymbol{s}_0 \in \mathcal{S}$ and $\boldsymbol{t}_0 \in \mathcal{T}$. The vector $\boldsymbol{s}_0$ is
lyche-numerical-linear-algebra_F01560	121	1.000	`\boldsymbol{t}_{0}`	$\boldsymbol{t}_{0}$	called the **oblique projection of $\boldsymbol{v}$ into $\mathcal{S}$ along $\mathcal{T}$**. Similarly, The vector $\boldsymbol{t}_0$ is
lyche-numerical-linear-algebra_F01561	122	1.000	`\mathcal{T}=\mathcal{S}^{\perp}`	$\mathcal{T}=\mathcal{S}^{\perp}$	called the **orthogonal projection** of $\boldsymbol{v}$ in $\mathcal{T} = \mathcal{S}^{\perp}$. The orthogonal projections
lyche-numerical-linear-algebra_F01562	122	1.000	`\langle\cdot, \cdot\rangle`	$\langle\cdot, \cdot\rangle$	*dimensional real or complex vector space $\mathcal{V}$ with an inner product $\langle\cdot,\cdot\rangle$. The*
lyche-numerical-linear-algebra_F01563	122	0.984	`\boldsymbol{v} \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}`	$\boldsymbol{v} \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	*orthogonal projections $\boldsymbol{s}_0$ of $\boldsymbol{v} \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$ into $\mathcal{S}$ and $\boldsymbol{t}_0$ of $\boldsymbol{v} \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$ into $\mathcal{T}$ satisfy*
lyche-numerical-linear-algebra_F01564	122	1.000	`\boldsymbol{v}=\boldsymbol{s}_{0}+\boldsymbol{t}_{0}`	$\boldsymbol{v}=\boldsymbol{s}_{0}+\boldsymbol{t}_{0}$	$\boldsymbol{v} = \boldsymbol{s}_0 + \boldsymbol{t}_0$, *and*
lyche-numerical-linear-algebra_F01565	122	1.000	`\left\langle\boldsymbol{s}_{0}, \boldsymbol{s}\right\rangle=\left\langle\boldsymbol{v}-\boldsymbol{t}_{0}, \boldsymbol{s}\right\rangle=\langle\boldsymbol{v}, \boldsymbol{s}\rangle`	$\left\langle\boldsymbol{s}_{0}, \boldsymbol{s}\right\rangle=\left\langle\boldsymbol{v}-\boldsymbol{t}_{0}, \boldsymbol{s}\right\rangle=\langle\boldsymbol{v}, \boldsymbol{s}\rangle$	***Proof*** We have $\langle \boldsymbol{s}_0, \boldsymbol{s}\rangle = \langle \boldsymbol{v} - \boldsymbol{t}_0, \boldsymbol{s}\rangle = \langle \boldsymbol{v}, \boldsymbol{s}\rangle$, since $\langle \boldsymbol{t}_0, \boldsymbol{s}\rangle = 0$ for all $\boldsymbol{s} \in \mathcal{S}$ and
lyche-numerical-linear-algebra_F01566	122	1.000	`\left\langle\boldsymbol{t}_{0}, \boldsymbol{s}\right\rangle=0`	$\left\langle\boldsymbol{t}_{0}, \boldsymbol{s}\right\rangle=0$	***Proof*** We have $\langle \boldsymbol{s}_0, \boldsymbol{s}\rangle = \langle \boldsymbol{v} - \boldsymbol{t}_0, \boldsymbol{s}\rangle = \langle \boldsymbol{v}, \boldsymbol{s}\rangle$, since $\langle \boldsymbol{t}_0, \boldsymbol{s}\rangle = 0$ for all $\boldsymbol{s} \in \mathcal{S}$ and
lyche-numerical-linear-algebra_F01567	123	1.000	`|\boldsymbol{v}|:=\sqrt{\langle\boldsymbol{v}, \boldsymbol{v}\rangle}`	$\|\boldsymbol{v}\|:=\sqrt{\langle\boldsymbol{v}, \boldsymbol{v}\rangle}$	$\|\boldsymbol{v}\| := \sqrt{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}$. *If $\boldsymbol{s}_0 \in \mathcal{S}$ is the orthogonal projection of $\boldsymbol{v} \in \mathcal{V}$ then*
lyche-numerical-linear-algebra_F01568	123	1.000	`\boldsymbol{v} \in \mathcal{V}`	$\boldsymbol{v} \in \mathcal{V}$	$\|\boldsymbol{v}\| := \sqrt{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}$. *If $\boldsymbol{s}_0 \in \mathcal{S}$ is the orthogonal projection of $\boldsymbol{v} \in \mathcal{V}$ then*
lyche-numerical-linear-algebra_F01569	123	0.975	`s_{0} \neq s \in \mathcal{S}`	$s_{0} \neq s \in \mathcal{S}$	***Proof*** Let $\boldsymbol{s}_0 \neq \boldsymbol{s} \in \mathcal{S}$ and $0 \neq \boldsymbol{u} := \boldsymbol{s}_0 - \boldsymbol{s} \in \mathcal{S}$. It follows from (5.9) that
lyche-numerical-linear-algebra_F01570	123	0.975	`0 \neq \boldsymbol{u}:=s_{0}-s \in \mathcal{S}`	$0 \neq \boldsymbol{u}:=s_{0}-s \in \mathcal{S}$	***Proof*** Let $\boldsymbol{s}_0 \neq \boldsymbol{s} \in \mathcal{S}$ and $0 \neq \boldsymbol{u} := \boldsymbol{s}_0 - \boldsymbol{s} \in \mathcal{S}$. It follows from (5.9) that
lyche-numerical-linear-algebra_F01571	123	1.000	`\left\langle\boldsymbol{v}-\boldsymbol{s}_{0}, \boldsymbol{u}\right\rangle=0`	$\left\langle\boldsymbol{v}-\boldsymbol{s}_{0}, \boldsymbol{u}\right\rangle=0$	$\langle \boldsymbol{v} - \boldsymbol{s}_0, \boldsymbol{u}\rangle = 0$. By (5.7) (Pythagoras) we obtain
lyche-numerical-linear-algebra_F01572	123	1.000	`\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\boldsymbol{y}^{*} \boldsymbol{x}=\sum_{j=1}^{n} x_{j} \overline{y_{j}}`	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\boldsymbol{y}^{*} \boldsymbol{x}=\sum_{j=1}^{n} x_{j} \overline{y_{j}}$	$\mathbb{C}^n$ given by $\langle \boldsymbol{x}, \boldsymbol{y}\rangle := \boldsymbol{y}^*\boldsymbol{x} = \sum_{j=1}^{n} x_j \overline{y_j}$. For symmetric and Hermitian matrices
lyche-numerical-linear-algebra_F01573	123	1.000	`\boldsymbol{A}^{T}=\boldsymbol{A} \Longleftrightarrow \langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{y}\rangle=\langle\boldsymbol{x}, \overline{\boldsymbol{A}} \boldsymbol{y}\rangle`	$\boldsymbol{A}^{T}=\boldsymbol{A} \Longleftrightarrow \langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{y}\rangle=\langle\boldsymbol{x}, \overline{\boldsymbol{A}} \boldsymbol{y}\rangle$	*1.* $\boldsymbol{A}^T = \boldsymbol{A} \iff \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle = \langle \boldsymbol{x}, \overline{\boldsymbol{A}}\boldsymbol{y}\rangle$ *for all* $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra_F01574	123	1.000	`\boldsymbol{A}^{*}=\boldsymbol{A} \Longleftrightarrow \langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{y}\rangle=\langle\boldsymbol{x}, \boldsymbol{A} \boldsymbol{y}\rangle`	$\boldsymbol{A}^{*}=\boldsymbol{A} \Longleftrightarrow \langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{y}\rangle=\langle\boldsymbol{x}, \boldsymbol{A} \boldsymbol{y}\rangle$	2. $\boldsymbol{A}^* = \boldsymbol{A} \iff \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle = \langle \boldsymbol{x}, \boldsymbol{A}\boldsymbol{y}\rangle$ *for all* $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra_F01575	123	1.000	`\boldsymbol{x}=\boldsymbol{e}_{j}`	$\boldsymbol{x}=\boldsymbol{e}_{j}$	For the converse we take $\boldsymbol{x} = \boldsymbol{e}_j$ and $\boldsymbol{y} = \boldsymbol{e}_i$ for some i, j and obtain
lyche-numerical-linear-algebra_F01576	123	1.000	`\boldsymbol{y}=\boldsymbol{e}_{i}`	$\boldsymbol{y}=\boldsymbol{e}_{i}$	For the converse we take $\boldsymbol{x} = \boldsymbol{e}_j$ and $\boldsymbol{y} = \boldsymbol{e}_i$ for some i, j and obtain

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01577	123	0.815	\boldsymbol{A}=\boldsymbol{A}^{T}	$\boldsymbol{A}=\boldsymbol{A}^{T}$	Thus, $\boldsymbol{A} = \boldsymbol{A}^T$ since they have the same i, j element for all i, j. The proof of 2. is
lyche-numerical-linear-algebra F01578	123	1.000	\boldsymbol{U}^{*} \boldsymbol{U}=\boldsymbol{I}	$\boldsymbol{U}^{*} \boldsymbol{U}=\boldsymbol{I}$	A square matrix $\boldsymbol{U} \in \mathbb{C}^{n\times n}$ is **unitary** if $\boldsymbol{U}^*\boldsymbol{U} = \boldsymbol{I}$. If $\boldsymbol{U}$ is real then $\boldsymbol{U}^T\boldsymbol{U} =$
lyche-numerical-linear-algebra F01579	123	1.000	\boldsymbol{U}^{T} \boldsymbol{U}=	$\boldsymbol{U}^{T} \boldsymbol{U}=$	A square matrix $\boldsymbol{U} \in \mathbb{C}^{n\times n}$ is **unitary** if $\boldsymbol{U}^*\boldsymbol{U} = \boldsymbol{I}$. If $\boldsymbol{U}$ is real then $\boldsymbol{U}^T\boldsymbol{U} =$
lyche-numerical-linear-algebra F01580	123	1.000	\boldsymbol{U}^{-1}=\boldsymbol{U}^{*}	$\boldsymbol{U}^{-1}=\boldsymbol{U}^{*}$	If $\boldsymbol{U}^*\boldsymbol{U} = \boldsymbol{I}$ the matrix $\boldsymbol{U}$ is nonsingular, $\boldsymbol{U}^{-1} = \boldsymbol{U}^*$ and therefore $\boldsymbol{U}\boldsymbol{U}^* =$
lyche-numerical-linear-algebra F01581	123	1.000	\boldsymbol{U} \boldsymbol{U}^{*}=	$\boldsymbol{U} \boldsymbol{U}^{*}=$	If $\boldsymbol{U}^*\boldsymbol{U} = \boldsymbol{I}$ the matrix $\boldsymbol{U}$ is nonsingular, $\boldsymbol{U}^{-1} = \boldsymbol{U}^*$ and therefore $\boldsymbol{U}\boldsymbol{U}^* =$
lyche-numerical-linear-algebra F01582	123	1.000	\boldsymbol{U} \boldsymbol{U}^{-1}=\boldsymbol{I}	$\boldsymbol{U} \boldsymbol{U}^{-1}=\boldsymbol{I}$	$\boldsymbol{U}\boldsymbol{U}^{-1} = \boldsymbol{I}$ as well. Moreover, both the columns and rows of a unitary matrix of
lyche-numerical-linear-algebra F01583	123	1.000	\boldsymbol{U}_{1}^{*} \boldsymbol{U}_{1}=\boldsymbol{I}	$\boldsymbol{U}_{1}^{*} \boldsymbol{U}_{1}=\boldsymbol{I}$	matrices is unitary. Indeed, if $\boldsymbol{U}_1^*\boldsymbol{U}_1 = \boldsymbol{I}$ and $\boldsymbol{U}_2^*\boldsymbol{U}_2 = \boldsymbol{I}$ then $(\boldsymbol{U}_1\boldsymbol{U}_2)^*(\boldsymbol{U}_1\boldsymbol{U}_2) =$
lyche-numerical-linear-algebra F01584	123	1.000	\boldsymbol{U}_{2}^{*} \boldsymbol{U}_{2}=\boldsymbol{I}	$\boldsymbol{U}_{2}^{*} \boldsymbol{U}_{2}=\boldsymbol{I}$	matrices is unitary. Indeed, if $\boldsymbol{U}_1^*\boldsymbol{U}_1 = \boldsymbol{I}$ and $\boldsymbol{U}_2^*\boldsymbol{U}_2 = \boldsymbol{I}$ then $(\boldsymbol{U}_1\boldsymbol{U}_2)^*(\boldsymbol{U}_1\boldsymbol{U}_2) =$
lyche-numerical-linear-algebra F01585	123	1.000	\left(\boldsymbol{U}_{1} \boldsymbol{U}_{2}\right)^{*} \left(\boldsymbol{U}_{1} \boldsymbol{U}_{2}\right)=	$\left(\boldsymbol{U}_{1} \boldsymbol{U}_{2}\right)^{*} \left(\boldsymbol{U}_{1} \boldsymbol{U}_{2}\right)=$	matrices is unitary. Indeed, if $\boldsymbol{U}_1^*\boldsymbol{U}_1 = \boldsymbol{I}$ and $\boldsymbol{U}_2^*\boldsymbol{U}_2 = \boldsymbol{I}$ then $(\boldsymbol{U}_1\boldsymbol{U}_2)^*(\boldsymbol{U}_1\boldsymbol{U}_2) =$
lyche-numerical-linear-algebra F01586	123	0.984	\boldsymbol{U}_{2}^{*} \boldsymbol{U}_{1}^{*} \boldsymbol{U}_{1} \boldsymbol{U}_{2}=\boldsymbol{I}	$\boldsymbol{U}_{2}^{*} \boldsymbol{U}_{1}^{*} \boldsymbol{U}_{1} \boldsymbol{U}_{2}=\boldsymbol{I}$	$\boldsymbol{U}_2^*\boldsymbol{U}_1^*\boldsymbol{U}_1\boldsymbol{U}_2 = \boldsymbol{I}$.
lyche-numerical-linear-algebra F01587	124	0.997	\langle\boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y}\rangle=\langle\boldsymbol{x}, \boldsymbol{y}\rangle	$\langle\boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y}\rangle=\langle\boldsymbol{x}, \boldsymbol{y}\rangle$	$\langle \boldsymbol{U}\boldsymbol{x}, \boldsymbol{U}\boldsymbol{y}\rangle = \langle \boldsymbol{x}, \boldsymbol{y}\rangle$ *for all* $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$. *In particular, if* $\boldsymbol{U}$ *is unitary then* $\\|\boldsymbol{U}\boldsymbol{x}\\|_2 =$
lyche-numerical-linear-algebra F01588	124	0.997	\\|\boldsymbol{U} \boldsymbol{x}\\|_{2}=	$\\|\boldsymbol{U} \boldsymbol{x}\\|_{2}=$	$\langle \boldsymbol{U}\boldsymbol{x}, \boldsymbol{U}\boldsymbol{y}\rangle = \langle \boldsymbol{x}, \boldsymbol{y}\rangle$ *for all* $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$. *In particular, if* $\boldsymbol{U}$ *is unitary then* $\\|\boldsymbol{U}\boldsymbol{x}\\|_2 =$
lyche-numerical-linear-algebra F01589	124	1.000	\\|\boldsymbol{x}\\|_{2}	$\\|\boldsymbol{x}\\|_{2}$	$\\|\boldsymbol{x}\\|_2$ *for all* $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra F01590	124	1.000	i, j=1, \ldots, n	$i, j=1, \ldots, n$	$i, j = 1, \ldots, n$
lyche-numerical-linear-algebra F01591	124	1.000	\left(\boldsymbol{U}^{*} \boldsymbol{U}\right)_{i, j}=\delta_{i, j}	$\left(\boldsymbol{U}^{*} \boldsymbol{U}\right)_{i, j}=\delta_{i, j}$	so that $(\boldsymbol{U}^*\boldsymbol{U})_{i,j} = \delta_{i,j}$ for all i, j. The last part of the theorem follows immediately
lyche-numerical-linear-algebra F01592	124	1.000	\boldsymbol{y}=\boldsymbol{x}	$\boldsymbol{y}=\boldsymbol{x}$	by taking $\boldsymbol{y} = \boldsymbol{x}$:
lyche-numerical-linear-algebra F01593	124	1.000	\boldsymbol{H} \in \mathbb{C}^{n \times n}	$\boldsymbol{H} \in \mathbb{C}^{n \times n}$	**Definition 5.4 (Householder Transformation)** A matrix $\boldsymbol{H} \in \mathbb{C}^{n\times n}$ of the form
lyche-numerical-linear-algebra F01594	124	1.000	\boldsymbol{H}^{*}=(\boldsymbol{I}-	$\boldsymbol{H}^{*}=(\boldsymbol{I}-$	A Householder transformation is Hermitian and unitary. Indeed, $\boldsymbol{H}^* = (\boldsymbol{I} -$
lyche-numerical-linear-algebra F01595	124	1.000	\left.\boldsymbol{u} \boldsymbol{u}^{*}\right)^{*}=\boldsymbol{H}	$\left.\boldsymbol{u} \boldsymbol{u}^{*}\right)^{*}=\boldsymbol{H}$	$\boldsymbol{u}\boldsymbol{u}^*)^* = \boldsymbol{H}$ and
lyche-numerical-linear-algebra F01596	124	1.000	\boldsymbol{I}-2 \boldsymbol{u} \boldsymbol{u}^{*}	$\boldsymbol{I}-2 \boldsymbol{u} \boldsymbol{u}^{*}$	used $\boldsymbol{I} - 2\boldsymbol{u}\boldsymbol{u}^*$, where $\boldsymbol{u}^*\boldsymbol{u} = 1$. For any nonzero $\boldsymbol{v} \in \mathbb{C}^n$ the matrix
lyche-numerical-linear-algebra F01597	124	1.000	\boldsymbol{u}^{*} \boldsymbol{u}=1	$\boldsymbol{u}^{*} \boldsymbol{u}=1$	used $\boldsymbol{I} - 2\boldsymbol{u}\boldsymbol{u}^*$, where $\boldsymbol{u}^*\boldsymbol{u} = 1$. For any nonzero $\boldsymbol{v} \in \mathbb{C}^n$ the matrix
lyche-numerical-linear-algebra F01598	124	1.000	\boldsymbol{v} \in \mathbb{C}^{n}	$\boldsymbol{v} \in \mathbb{C}^{n}$	used $\boldsymbol{I} - 2\boldsymbol{u}\boldsymbol{u}^*$, where $\boldsymbol{u}^*\boldsymbol{u} = 1$. For any nonzero $\boldsymbol{v} \in \mathbb{C}^n$ the matrix
lyche-numerical-linear-algebra F01599	125	1.000	\boldsymbol{H}=\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*}	$\boldsymbol{H}=\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*}$	is a Householder transformation. Indeed, $\boldsymbol{H} = \boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*$, where $\boldsymbol{u} := \sqrt{2}\frac{\boldsymbol{v}}{\\|\boldsymbol{v}\\|_2}$ has
lyche-numerical-linear-algebra F01600	125	1.000	\boldsymbol{u}:=\sqrt{2} \frac{\boldsymbol{v}}{\\|\boldsymbol{v}\\|_{2}}	$\boldsymbol{u}:=\sqrt{2} \frac{\boldsymbol{v}}{\\|\boldsymbol{v}\\|_{2}}$	is a Householder transformation. Indeed, $\boldsymbol{H} = \boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*$, where $\boldsymbol{u} := \sqrt{2}\frac{\boldsymbol{v}}{\\|\boldsymbol{v}\\|_2}$ has
lyche-numerical-linear-algebra F01601	125	1.000	\sqrt{2}	$\sqrt{2}$	length $\sqrt{2}$.
lyche-numerical-linear-algebra F01602	125	1.000	\\|\boldsymbol{x}\\|_{2}=\\|\boldsymbol{y}\\|_{2}, \boldsymbol{y}^{*} \boldsymbol{x}	$\\|\boldsymbol{x}\\|_{2}=\\|\boldsymbol{y}\\|_{2}, \boldsymbol{y}^{*} \boldsymbol{x}$	**Lemma 5.2** *Suppose* $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ *are two vectors such that* $\\|\boldsymbol{x}\\|_2 = \\|\boldsymbol{y}\\|_2$, $\boldsymbol{y}^*\boldsymbol{x}$ *is*

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01603	125	0.993	\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y} \neq \mathbf{0}	$\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y} \neq \mathbf{0}$	*real and* $\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \mathbf{0}$. *Then* $\left(\boldsymbol{I} - 2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x} = \boldsymbol{y}$.
lyche-numerical-linear-algebra F01604	125	0.993	\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}} \right) \boldsymbol{x}=\boldsymbol{y}	$\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}} \right) \boldsymbol{x}=\boldsymbol{y}$	*real and* $\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \mathbf{0}$. *Then* $\left(\boldsymbol{I} - 2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x} = \boldsymbol{y}$.
lyche-numerical-linear-algebra F01605	125	1.000	\boldsymbol{x}^{*} \boldsymbol{x}=\boldsymbol{y}^{*} \boldsymbol{y}	$\boldsymbol{x}^{*} \boldsymbol{x}=\boldsymbol{y}^{*} \boldsymbol{y}$	***Proof*** Since $\boldsymbol{x}^*\boldsymbol{x} = \boldsymbol{y}^*\boldsymbol{y}$ and $\operatorname{Re}(\boldsymbol{y}^*\boldsymbol{x}) = \boldsymbol{y}^*\boldsymbol{x}$ we have
lyche-numerical-linear-algebra F01606	125	1.000	\operatorname{Re}\left(\boldsymbol{y}^{*} \boldsymbol{x} \right)=\boldsymbol{y}^{*} \boldsymbol{x}	$\operatorname{Re}\left(\boldsymbol{y}^{*} \boldsymbol{x} \right)=\boldsymbol{y}^{*} \boldsymbol{x}$	***Proof*** Since $\boldsymbol{x}^*\boldsymbol{x} = \boldsymbol{y}^*\boldsymbol{y}$ and $\operatorname{Re}(\boldsymbol{y}^*\boldsymbol{x}) = \boldsymbol{y}^*\boldsymbol{x}$ we have
lyche-numerical-linear-algebra F01607	125	0.998	\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}} \right) \boldsymbol{x}=\boldsymbol{x}-\frac{2 \boldsymbol{v}^{*} \boldsymbol{x}}{\boldsymbol{v}^{*} \boldsymbol{v}} \boldsymbol{v}=\boldsymbol{x}-\boldsymbol{v}=\boldsymbol{y}	$\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}} \right) \boldsymbol{x}=\boldsymbol{x}-\frac{2 \boldsymbol{v}^{*} \boldsymbol{x}}{\boldsymbol{v}^{*} \boldsymbol{v}} \boldsymbol{v}=\boldsymbol{x}-\boldsymbol{v}=\boldsymbol{y}$	But then $\left(\boldsymbol{I} - 2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x} = \boldsymbol{x} - \frac{2\boldsymbol{v}^*\boldsymbol{x}}{\boldsymbol{v}^*\boldsymbol{v}}\boldsymbol{v} = \boldsymbol{x} - \boldsymbol{v} = \boldsymbol{y}$.
lyche-numerical-linear-algebra F01608	125	1.000	\boldsymbol{H} \boldsymbol{x}	$\boldsymbol{H} \boldsymbol{x}$	If $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ it follows that $\boldsymbol{H}\boldsymbol{x}$ is the reflected image of $\boldsymbol{x}$. The "mirror" $\mathcal{M} :=$
lyche-numerical-linear-algebra F01609	125	1.000	\mathcal{M}:=	$\mathcal{M}:=$	If $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ it follows that $\boldsymbol{H}\boldsymbol{x}$ is the reflected image of $\boldsymbol{x}$. The "mirror" $\mathcal{M} :=$
lyche-numerical-linear-algebra F01610	125	1.000	\left\{\boldsymbol{w} \in \mathbb{R}^{n}: \boldsymbol{w}^{*} \boldsymbol{v}=0\right\}	$\left\{\boldsymbol{w} \in \mathbb{R}^{n}: \boldsymbol{w}^{*} \boldsymbol{v}=0\right\}$	$\{\boldsymbol{w} \in \mathbb{R}^n : \boldsymbol{w}^*\boldsymbol{v} = 0\}$ contains the vector $(\boldsymbol{x}+\boldsymbol{y})/2$ and has normal $\boldsymbol{x}-\boldsymbol{y}$. This is
lyche-numerical-linear-algebra F01611	125	1.000	(\boldsymbol{x}+\boldsymbol{y}) / 2	$(\boldsymbol{x}+\boldsymbol{y}) / 2$	$\{\boldsymbol{w} \in \mathbb{R}^n : \boldsymbol{w}^*\boldsymbol{v} = 0\}$ contains the vector $(\boldsymbol{x}+\boldsymbol{y})/2$ and has normal $\boldsymbol{x}-\boldsymbol{y}$. This is
lyche-numerical-linear-algebra F01612	125	1.000	\boldsymbol{x}-\boldsymbol{y}	$\boldsymbol{x}-\boldsymbol{y}$	$\{\boldsymbol{w} \in \mathbb{R}^n : \boldsymbol{w}^*\boldsymbol{v} = 0\}$ contains the vector $(\boldsymbol{x}+\boldsymbol{y})/2$ and has normal $\boldsymbol{x}-\boldsymbol{y}$. This is
lyche-numerical-linear-algebra F01613	126	1.000	\boldsymbol{x}:=[1,0,1]^{T}	$\boldsymbol{x}:=[1,0,1]^{T}$	*Example 5.1 (Reflector)* Suppose $\boldsymbol{x} := [1,0,1]^T$ and $\boldsymbol{y} := [-1,0,1]^T$. Then $\boldsymbol{v} =$
lyche-numerical-linear-algebra F01614	126	1.000	\boldsymbol{y}:=[-1,0,1]^{T}	$\boldsymbol{y}:=[-1,0,1]^{T}$	*Example 5.1 (Reflector)* Suppose $\boldsymbol{x} := [1,0,1]^T$ and $\boldsymbol{y} := [-1,0,1]^T$. Then $\boldsymbol{v} =$
lyche-numerical-linear-algebra F01615	126	1.000	\boldsymbol{v}=	$\boldsymbol{v}=$	*Example 5.1 (Reflector)* Suppose $\boldsymbol{x} := [1,0,1]^T$ and $\boldsymbol{y} := [-1,0,1]^T$. Then $\boldsymbol{v} =$
lyche-numerical-linear-algebra F01616	126	1.000	\boldsymbol{x}-\boldsymbol{y}=[2,0,0]^{T}	$\boldsymbol{x}-\boldsymbol{y}=[2,0,0]^{T}$	$\boldsymbol{x}-\boldsymbol{y} = [2,0,0]^T$ and
lyche-numerical-linear-algebra F01617	126	0.999	y z	$y z$	is the yz plane (cf. Fig. 5.3), $\boldsymbol{H}\boldsymbol{x} = [-1,0,1]^T = \boldsymbol{y}$, and $\boldsymbol{P}\boldsymbol{x} = [0,0,1]^T =$
lyche-numerical-linear-algebra F01618	126	0.999	\boldsymbol{H} \boldsymbol{x}=[-1,0,1]^{T}=\boldsymbol{y}	$\boldsymbol{H} \boldsymbol{x}=[-1,0,1]^{T}=\boldsymbol{y}$	is the yz plane (cf. Fig. 5.3), $\boldsymbol{H}\boldsymbol{x} = [-1,0,1]^T = \boldsymbol{y}$, and $\boldsymbol{P}\boldsymbol{x} = [0,0,1]^T =$
lyche-numerical-linear-algebra F01619	126	0.999	\boldsymbol{P} \boldsymbol{x}=[0,0,1]^{T}=	$\boldsymbol{P} \boldsymbol{x}=[0,0,1]^{T}=$	is the yz plane (cf. Fig. 5.3), $\boldsymbol{H}\boldsymbol{x} = [-1,0,1]^T = \boldsymbol{y}$, and $\boldsymbol{P}\boldsymbol{x} = [0,0,1]^T =$
lyche-numerical-linear-algebra F01620	126	1.000	(\boldsymbol{x}+\boldsymbol{y}) / 2 \in \mathcal{M}	$(\boldsymbol{x}+\boldsymbol{y}) / 2 \in \mathcal{M}$	$(\boldsymbol{x}+\boldsymbol{y})/2 \in \mathcal{M}$.
lyche-numerical-linear-algebra F01621	126	1.000	\boldsymbol{H} \boldsymbol{x}=\mathbf{0}	$\boldsymbol{H} \boldsymbol{x}=\mathbf{0}$	***Proof*** If $\boldsymbol{x} = 0$ then $\boldsymbol{H}\boldsymbol{x} = \mathbf{0}$ and $a = 0$. Any $\boldsymbol{u}$ with $\\|\boldsymbol{u}\\|_2 = \sqrt{2}$ will work, and
lyche-numerical-linear-algebra F01622	126	1.000	a=0	$a=0$	***Proof*** If $\boldsymbol{x} = 0$ then $\boldsymbol{H}\boldsymbol{x} = \mathbf{0}$ and $a = 0$. Any $\boldsymbol{u}$ with $\\|\boldsymbol{u}\\|_2 = \sqrt{2}$ will work, and
lyche-numerical-linear-algebra F01623	126	1.000	\\|\boldsymbol{u}\\|_{2}=\sqrt{2}	$\\|\boldsymbol{u}\\|_{2}=\sqrt{2}$	***Proof*** If $\boldsymbol{x} = 0$ then $\boldsymbol{H}\boldsymbol{x} = \mathbf{0}$ and $a = 0$. Any $\boldsymbol{u}$ with $\\|\boldsymbol{u}\\|_2 = \sqrt{2}$ will work, and
lyche-numerical-linear-algebra F01624	126	1.000	\boldsymbol{u}:=\sqrt{2} \boldsymbol{e}_{1}	$\boldsymbol{u}:=\sqrt{2} \boldsymbol{e}_{1}$	we choose $\boldsymbol{u} := \sqrt{2}\boldsymbol{e}_1$ in this case. For $\boldsymbol{x} \neq \mathbf{0}$ we define
lyche-numerical-linear-algebra F01625	126	1.000	\boldsymbol{x} \neq \mathbf{0}	$\boldsymbol{x} \neq \mathbf{0}$	we choose $\boldsymbol{u} := \sqrt{2}\boldsymbol{e}_1$ in this case. For $\boldsymbol{x} \neq \mathbf{0}$ we define
lyche-numerical-linear-algebra F01626	126	0.971	|\rho|=1	$\lvert\rho\rvert=1$	Since $\lvert\rho\rvert = 1$ we have $\rho\\|\boldsymbol{x}\\|_2 z = \lvert\rho\rvert^2\boldsymbol{x} = \boldsymbol{x}$. Moreover, $\\|z\\|_2 = 1$ and $z_1 =$

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-al F01627</td><td>126</td><td>0.971</td><td><code>\rho\|\boldsymbol{x}\|_{2} \boldsymbol{z}=|\rho|^{2} \boldsymbol{x}=\boldsymbol{x}</code></td><td>$\rho\|\boldsymbol{x}\|_{2} \boldsymbol{z}=|\rho|^{2} \boldsymbol{x}=\boldsymbol{x}$</td><td>Since $|\rho| = 1$ we have $\rho\|\boldsymbol{x}\|_2 z = |\rho|^2\boldsymbol{x} = \boldsymbol{x}$. Moreover, $\|z\|_2 = 1$ and $z_1 =$</td></tr>
<tr><td>lyche-numerical-linear-al F01628</td><td>126</td><td>0.971</td><td><code>\|\boldsymbol{z}\|_{2}=1</code></td><td>$\|\boldsymbol{z}\|_{2}=1$</td><td>Since $|\rho| = 1$ we have $\rho\|\boldsymbol{x}\|_2 z = |\rho|^2\boldsymbol{x} = \boldsymbol{x}$. Moreover, $\|z\|_2 = 1$ and $z_1 =$</td></tr>
<tr><td>lyche-numerical-linear-al F01629</td><td>126</td><td>0.971</td><td><code>z_{1}=</code></td><td>$z_{1}=$</td><td>Since $|\rho| = 1$ we have $\rho\|\boldsymbol{x}\|_2 z = |\rho|^2\boldsymbol{x} = \boldsymbol{x}$. Moreover, $\|z\|_2 = 1$ and $z_1 =$</td></tr>
<tr><td>lyche-numerical-linear-al F01630</td><td>126</td><td>0.969</td><td><code>\left|x_{1}\right| /\|\boldsymbol{x}\|_{2}</code></td><td>$\left|x_{1}\right| /\|\boldsymbol{x}\|_{2}$</td><td>$|x_1|/\|\boldsymbol{x}\|_2$ is real so that $\boldsymbol{u}^*\boldsymbol{u} = \frac{(z+e_1)^*(z+e_1)}{1+z_1} = \frac{2+2z_1}{1+z_1} = 2$. Finally,</td></tr>
<tr><td>lyche-numerical-linear-al F01631</td><td>126</td><td>0.969</td><td><code>\boldsymbol{u}^{*} \boldsymbol{u}=\frac{\left(z+\boldsymbol{e}_{1}\right)^{*}\left(z+\boldsymbol{e}_{1}\right)}{1+z_{1}}=\frac{2+2 z_{1}}{1+z_{1}}=2</code></td><td>$\boldsymbol{u}^{*} \boldsymbol{u}=\frac{\left(z+\boldsymbol{e}_{1}\right)^{*}\left(z+\boldsymbol{e}_{1}\right)}{1+z_{1}}=\frac{2+2 z_{1}}{1+z_{1}}=2$</td><td>$|x_1|/\|\boldsymbol{x}\|_2$ is real so that $\boldsymbol{u}^*\boldsymbol{u} = \frac{(z+e_1)^*(z+e_1)}{1+z_1} = \frac{2+2z_1}{1+z_1} = 2$. Finally,</td></tr>
<tr><td>lyche-numerical-linear-al F01632</td><td>127</td><td>1.000</td><td><code>\boldsymbol{u}^{*} \boldsymbol{u}=2</code></td><td>$\boldsymbol{u}^{*} \boldsymbol{u}=2$</td><td>adapted from [17]. To any given $\boldsymbol{x} \in \mathbb{C}^n$, a number a and a vector $\boldsymbol{u}$ with $\boldsymbol{u}^*\boldsymbol{u} = 2$</td></tr>
<tr><td>lyche-numerical-linear-al F01633</td><td>127</td><td>0.986</td><td><code>\left(\boldsymbol{I}-\boldsymbol{u u}^{*}\right) \boldsymbol{x}=a \boldsymbol{e}_{1}</code></td><td>$\left(\boldsymbol{I}-\boldsymbol{u u}^{*}\right) \boldsymbol{x}=a \boldsymbol{e}_{1}$</td><td>is computed so that $(\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*)\boldsymbol{x} = a\boldsymbol{e}_1$:</td></tr>
<tr><td>lyche-numerical-linear-al F01634</td><td>127</td><td>0.999</td><td><code>z_{1}=\left|x_{1}\right| /\|\boldsymbol{x}\|_{2} \geq 0</code></td><td>$z_{1}=\left|x_{1}\right| /\|\boldsymbol{x}\|_{2} \geq 0$</td><td>• In Theorem 5.8 the first component of z is $z_1 = |x_1|/\|\boldsymbol{x}\|_2 \ge 0$. Since $\|z\|_2 =$</td></tr>
<tr><td>lyche-numerical-linear-al F01635</td><td>127</td><td>0.999</td><td><code>\|\boldsymbol{z}\|_{2}=</code></td><td>$\|\boldsymbol{z}\|_{2}=$</td><td>• In Theorem 5.8 the first component of z is $z_1 = |x_1|/\|\boldsymbol{x}\|_2 \ge 0$. Since $\|z\|_2 =$</td></tr>
<tr><td>lyche-numerical-linear-al F01636</td><td>127</td><td>1.000</td><td><code>1 \leq 1+z_{1} \leq 2</code></td><td>$1 \leq 1+z_{1} \leq 2$</td><td>1 we have $1 \le 1 + z_1 \le 2$. It follows that we avoid cancelation error when</td></tr>
<tr><td>lyche-numerical-linear-al F01637</td><td>127</td><td>1.000</td><td><code>1+z_{1}</code></td><td>$1+z_{1}$</td><td>computing $1+z_1$ and $\boldsymbol{u}$ and a are computed in a numerically stable way.</td></tr>
<tr><td>lyche-numerical-linear-al F01638</td><td>127</td><td>0.887</td><td><code>\boldsymbol{H} \boldsymbol{x}=\left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*}\right) \boldsymbol{x}=\boldsymbol{x}-\left(\boldsymbol{u}^{*} \boldsymbol{x}\right) \boldsymbol{u}</code></td><td>$\boldsymbol{H} \boldsymbol{x}=\left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*}\right) \boldsymbol{x}=\boldsymbol{x}-\left(\boldsymbol{u}^{*} \boldsymbol{x}\right) \boldsymbol{u}$</td><td>Indeed, $\boldsymbol{Hx} = (\boldsymbol{I} - \boldsymbol{uu}^*)\boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^*\boldsymbol{x})\boldsymbol{u}$. If $\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$ this requires $2m$</td></tr>
<tr><td>lyche-numerical-linear-al F01639</td><td>127</td><td>0.887</td><td><code>\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^{m}</code></td><td>$\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^{m}$</td><td>Indeed, $\boldsymbol{Hx} = (\boldsymbol{I} - \boldsymbol{uu}^*)\boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^*\boldsymbol{x})\boldsymbol{u}$. If $\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$ this requires $2m$</td></tr>
<tr><td>lyche-numerical-linear-al F01640</td><td>127</td><td>0.887</td><td><code>2 m</code></td><td>$2 m$</td><td>Indeed, $\boldsymbol{Hx} = (\boldsymbol{I} - \boldsymbol{uu}^*)\boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^*\boldsymbol{x})\boldsymbol{u}$. If $\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$ this requires $2m$</td></tr>
<tr><td>lyche-numerical-linear-al F01641</td><td>127</td><td>0.994</td><td><code>\boldsymbol{u}^{T} \boldsymbol{x}, m</code></td><td>$\boldsymbol{u}^{T} \boldsymbol{x}, m$</td><td>operations to find $\boldsymbol{u}^T\boldsymbol{x}$, m operations for $(\boldsymbol{u}^T\boldsymbol{x})\boldsymbol{u}$ and m operations for the final</td></tr>
<tr><td>lyche-numerical-linear-al F01642</td><td>127</td><td>0.994</td><td><code>\left(\boldsymbol{u}^{T} \boldsymbol{x}\right) \boldsymbol{u}</code></td><td>$\left(\boldsymbol{u}^{T} \boldsymbol{x}\right) \boldsymbol{u}$</td><td>operations to find $\boldsymbol{u}^T\boldsymbol{x}$, m operations for $(\boldsymbol{u}^T\boldsymbol{x})\boldsymbol{u}$ and m operations for the final</td></tr>
<tr><td>lyche-numerical-linear-al F01643</td><td>127</td><td>1.000</td><td><code>4 m</code></td><td>$4 m$</td><td>subtraction of the two vectors, a total of $4m$ arithmetic operations. If $\boldsymbol{A} \in \mathbb{R}^{m\times n}$</td></tr>
<tr><td>lyche-numerical-linear-al F01644</td><td>127</td><td>1.000</td><td><code>4 m n</code></td><td>$4 m n$</td><td>then $4mn$ operations are required for $\boldsymbol{HA} = \boldsymbol{A} - (\boldsymbol{u}^T\boldsymbol{A})\boldsymbol{u}$, i.e., $4m$ operations</td></tr>
<tr><td>lyche-numerical-linear-al F01645</td><td>127</td><td>1.000</td><td><code>\boldsymbol{H} \boldsymbol{A}=\boldsymbol{A}-\left(\boldsymbol{u}^{T} \boldsymbol{A}\right) \boldsymbol{u}</code></td><td>$\boldsymbol{H} \boldsymbol{A}=\boldsymbol{A}-\left(\boldsymbol{u}^{T} \boldsymbol{A}\right) \boldsymbol{u}$</td><td>then $4mn$ operations are required for $\boldsymbol{HA} = \boldsymbol{A} - (\boldsymbol{u}^T\boldsymbol{A})\boldsymbol{u}$, i.e., $4m$ operations</td></tr>
<tr><td>lyche-numerical-linear-al F01646</td><td>127</td><td>1.000</td><td><code>\boldsymbol{x}^{T}:=[\boldsymbol{y}, \boldsymbol{z}]^{T}</code></td><td>$\boldsymbol{x}^{T}:=[\boldsymbol{y}, \boldsymbol{z}]^{T}$</td><td>a vector. Suppose $\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$, where $\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$ for some $1 \le k <$</td></tr>
<tr><td>lyche-numerical-linear-al F01647</td><td>127</td><td>1.000</td><td><code>\boldsymbol{y} \in \mathbb{C}^{k}, \boldsymbol{z} \in \mathbb{C}^{n-k}</code></td><td>$\boldsymbol{y} \in \mathbb{C}^{k}, \boldsymbol{z} \in \mathbb{C}^{n-k}$</td><td>a vector. Suppose $\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$, where $\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$ for some $1 \le k <$</td></tr>
<tr><td>lyche-numerical-linear-al F01648</td><td>127</td><td>1.000</td><td><code>1 \leq k<</code></td><td>$1 \leq k<$</td><td>a vector. Suppose $\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$, where $\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$ for some $1 \le k <$</td></tr>
<tr><td>lyche-numerical-linear-al F01649</td><td>127</td><td>0.915</td><td><code>[\hat{\boldsymbol{u}}, a]:=\operatorname{housegen}(\boldsymbol{z})</code></td><td>$[\hat{\boldsymbol{u}}, a]:=\operatorname{housegen}(\boldsymbol{z})$</td><td>$n$. The command $[\hat{\boldsymbol{u}}, a] := \operatorname{housegen}(\boldsymbol{z})$ defines a Householder transformation</td></tr>
<tr><td>lyche-numerical-linear-al F01650</td><td>127</td><td>0.384</td><td><code>\hat{\boldsymbol{H}}=\boldsymbol{I}-\hat{\boldsymbol{u}} \hat{\boldsymbol{u}}^{*}</code></td><td>$\hat{\boldsymbol{H}}=\boldsymbol{I}-\hat{\boldsymbol{u}} \hat{\boldsymbol{u}}^{*}$</td><td>$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^*$ so that $\hat{\boldsymbol{H}}\boldsymbol{z} = a\boldsymbol{e}_1 \in \mathbb{C}^{n-k}$. With $\boldsymbol{u} := \left[\begin{smallmatrix}\boldsymbol{0}\\ \hat{\boldsymbol{u}}\end{smallmatrix}\right] \in \mathbb{C}^n$ we see that</td></tr>
<tr><td>lyche-numerical-linear-al F01651</td><td>127</td><td>0.384</td><td><code>\hat{\boldsymbol{H}} z=a \boldsymbol{e}_{1} \in \mathbb{C}^{n-k}</code></td><td>$\hat{\boldsymbol{H}} z=a \boldsymbol{e}_{1} \in \mathbb{C}^{n-k}$</td><td>$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^*$ so that $\hat{\boldsymbol{H}}\boldsymbol{z} = a\boldsymbol{e}_1 \in \mathbb{C}^{n-k}$. With $\boldsymbol{u} := \left[\begin{smallmatrix}\boldsymbol{0}\\ \hat{\boldsymbol{u}}\end{smallmatrix}\right] \in \mathbb{C}^n$ we see that</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01652	127	0.384	`\boldsymbol{u}:=\left[\begin{array}{c}\mathbf{0} \\ \hat{\boldsymbol{u}}\end{array}\right] \in \mathbb{C}^{n}`	$\boldsymbol{u}:=\left[\begin{array}{c}\mathbf{0} \\ \hat{\boldsymbol{u}}\end{array}\right] \in \mathbb{C}^{n}$	$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^*$ so that $\hat{\boldsymbol{H}}\boldsymbol{z} = a\boldsymbol{e}_1 \in \mathbb{C}^{n-k}$. With $\boldsymbol{u} := \left[\begin{smallmatrix}\mathbf{0}\\ \hat{\boldsymbol{u}}\end{smallmatrix}\right] \in \mathbb{C}^n$ we see that
lyche-numerical-linear-algebra F01653	127	1.000	`\boldsymbol{u}^{*} \boldsymbol{u}=\hat{\boldsymbol{u}}^{*} \hat{\boldsymbol{u}}=2`	$\boldsymbol{u}^{*} \boldsymbol{u}=\hat{\boldsymbol{u}}^{*} \hat{\boldsymbol{u}}=2$	$\boldsymbol{u}^*\boldsymbol{u} = \hat{\boldsymbol{u}}^*\hat{\boldsymbol{u}} = 2$, and
lyche-numerical-linear-algebra F01654	128	1.000	`\boldsymbol{R} \in \mathbb{C}^{m \times n}`	$\boldsymbol{R} \in \mathbb{C}^{m \times n}$	We say that a matrix $\boldsymbol{R} \in \mathbb{C}^{m\times n}$ is **upper trapezoidal**, if $r_{i,j} = 0$ for $j < i$ and
lyche-numerical-linear-algebra F01655	128	1.000	`r_{i, j}=0`	$r_{i, j}=0$	We say that a matrix $\boldsymbol{R} \in \mathbb{C}^{m\times n}$ is **upper trapezoidal**, if $r_{i,j} = 0$ for $j < i$ and
lyche-numerical-linear-algebra F01656	128	1.000	`j<i`	$j<i$	We say that a matrix $\boldsymbol{R} \in \mathbb{C}^{m\times n}$ is **upper trapezoidal**, if $r_{i,j} = 0$ for $j < i$ and
lyche-numerical-linear-algebra F01657	128	1.000	`i=2,3 \ldots, m`	$i=2,3 \ldots, m$	$i = 2, 3 \ldots, m$. Upper trapezoidal matrices corresponding to $m < n$, $m = n$, and
lyche-numerical-linear-algebra F01658	128	1.000	`m \leq n`	$m \leq n$	We treat the cases $m > n$ and $m \le n$ separately and consider first $m > n$. We
lyche-numerical-linear-algebra F01659	128	0.996	`\boldsymbol{H}_{1}, \ldots, \boldsymbol{H}_{n}`	$\boldsymbol{H}_{1}, \ldots, \boldsymbol{H}_{n}$	describe how to find a sequence $\boldsymbol{H}_1, \ldots, \boldsymbol{H}_n$ of Householder transformations such
lyche-numerical-linear-algebra F01660	128	1.000	`\boldsymbol{R}_{1}`	$\boldsymbol{R}_{1}$	and where $\boldsymbol{R}_1$ is square and upper triangular. We define
lyche-numerical-linear-algebra F01661	129	1.000	`\hat{\boldsymbol{H}}_{k}:=\boldsymbol{I}-\hat{\boldsymbol{u}}_{k} \hat{\boldsymbol{u}}_{k}^{*}`	$\hat{\boldsymbol{H}}_{k}:=\boldsymbol{I}-\hat{\boldsymbol{u}}_{k} \hat{\boldsymbol{u}}_{k}^{*}$	Let $\hat{\boldsymbol{H}}_k := \boldsymbol{I} - \hat{\boldsymbol{u}}_k\hat{\boldsymbol{u}}_k^*$ be a Householder transformation that maps the first column
lyche-numerical-linear-algebra F01662	129	1.000	`\left[a_{k, k}^{(k)}, \ldots, a_{m, k}^{(k)}\right]^{T}`	$\left[a_{k, k}^{(k)}, \ldots, a_{m, k}^{(k)}\right]^{T}$	$[a_{k,k}^{(k)}, \ldots, a_{m,k}^{(k)}]^T$ of $\boldsymbol{D}_k$ to a multiple of $\boldsymbol{e}_1$, $\hat{\boldsymbol{H}}_k(\boldsymbol{D}_k\boldsymbol{e}_1) = a_k\boldsymbol{e}_1$. Using Algorithm 5.1
lyche-numerical-linear-algebra F01663	129	1.000	`\boldsymbol{D}_{k}`	$\boldsymbol{D}_{k}$	$[a_{k,k}^{(k)}, \ldots, a_{m,k}^{(k)}]^T$ of $\boldsymbol{D}_k$ to a multiple of $\boldsymbol{e}_1$, $\hat{\boldsymbol{H}}_k(\boldsymbol{D}_k\boldsymbol{e}_1) = a_k\boldsymbol{e}_1$. Using Algorithm 5.1
lyche-numerical-linear-algebra F01664	129	1.000	`\boldsymbol{e}_{1}, \hat{\boldsymbol{H}}_{k}\left(\boldsymbol{D}_{k} \boldsymbol{e}_{1}\right)=a_{k} \boldsymbol{e}_{1}`	$\boldsymbol{e}_{1}, \hat{\boldsymbol{H}}_{k}\left(\boldsymbol{D}_{k} \boldsymbol{e}_{1}\right)=a_{k} \boldsymbol{e}_{1}$	$[a_{k,k}^{(k)}, \ldots, a_{m,k}^{(k)}]^T$ of $\boldsymbol{D}_k$ to a multiple of $\boldsymbol{e}_1$, $\hat{\boldsymbol{H}}_k(\boldsymbol{D}_k\boldsymbol{e}_1) = a_k\boldsymbol{e}_1$. Using Algorithm 5.1
lyche-numerical-linear-algebra F01665	129	0.865	`\left[\hat{\boldsymbol{u}}_{k}, a_{k}\right]=\operatorname{housegen}\left(\boldsymbol{D}_{k} \boldsymbol{e}_{1}\right)`	$\left[\hat{\boldsymbol{u}}_{k}, a_{k}\right]=\operatorname{housegen}\left(\boldsymbol{D}_{k} \boldsymbol{e}_{1}\right)$	we have $[\hat{\boldsymbol{u}}_k, a_k] = \mathtt{housegen}(\boldsymbol{D}_k\boldsymbol{e}_1)$. Then $\boldsymbol{H}_k := \left[\begin{smallmatrix}\boldsymbol{I}_{k-1} & \mathbf{0}\\ \mathbf{0} & \hat{\boldsymbol{H}}_k\end{smallmatrix}\right]$ is a Householder
lyche-numerical-linear-algebra F01666	129	0.865	`\boldsymbol{H}_{k}:=\left[\begin{array}{cc}\boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_{k}\end{array}\right]`	$\boldsymbol{H}_{k}:=\left[\begin{array}{cc}\boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_{k}\end{array}\right]$	we have $[\hat{\boldsymbol{u}}_k, a_k] = \mathtt{housegen}(\boldsymbol{D}_k\boldsymbol{e}_1)$. Then $\boldsymbol{H}_k := \left[\begin{smallmatrix}\boldsymbol{I}_{k-1} & \mathbf{0}\\ \mathbf{0} & \hat{\boldsymbol{H}}_k\end{smallmatrix}\right]$ is a Householder
lyche-numerical-linear-algebra F01667	129	1.000	`\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}`	$\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$	where $\boldsymbol{B}_{k+1} \in \mathbb{C}^{k\times k}$ is upper triangular and $\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k)\times(n-k)}$. Thus $\boldsymbol{A}_{k+1}$ is
lyche-numerical-linear-algebra F01668	129	1.000	`\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times(n-k)}`	$\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times(n-k)}$	where $\boldsymbol{B}_{k+1} \in \mathbb{C}^{k\times k}$ is upper triangular and $\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k)\times(n-k)}$. Thus $\boldsymbol{A}_{k+1}$ is
lyche-numerical-linear-algebra F01669	129	1.000	`\boldsymbol{A}_{k+1}`	$\boldsymbol{A}_{k+1}$	where $\boldsymbol{B}_{k+1} \in \mathbb{C}^{k\times k}$ is upper triangular and $\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k)\times(n-k)}$. Thus $\boldsymbol{A}_{k+1}$ is
lyche-numerical-linear-algebra F01670	129	1.000	`\boldsymbol{R}:=\boldsymbol{A}_{n+1}=\left[\begin{array}{c}\boldsymbol{R}_{1} \\ \mathbf{0}\end{array}\right]`	$\boldsymbol{R}:=\boldsymbol{A}_{n+1}=\left[\begin{array}{c}\boldsymbol{R}_{1} \\ \mathbf{0}\end{array}\right]$	further. At the end $\boldsymbol{R} := \boldsymbol{A}_{n+1} = \left[\begin{smallmatrix}\boldsymbol{R}_1\\ \mathbf{0}\end{smallmatrix}\right]$, where $\boldsymbol{R}_1$ is upper triangular.
lyche-numerical-linear-algebra F01671	129	1.000	`m>1`	$m>1$	already in upper trapezoidal form. Suppose $m > 1$. In this case $m - 1$ Householder
lyche-numerical-linear-algebra F01672	129	1.000	`m-1`	$m-1$	already in upper trapezoidal form. Suppose $m > 1$. In this case $m - 1$ Householder
lyche-numerical-linear-algebra F01673	129	0.999	`\boldsymbol{H}_{m-1} \cdots \boldsymbol{H}_{1} \boldsymbol{A}`	$\boldsymbol{H}_{m-1} \cdots \boldsymbol{H}_{1} \boldsymbol{A}$	transformations will suffice and $\boldsymbol{H}_{m-1}\cdots\boldsymbol{H}_1\boldsymbol{A}$ is upper trapezoidal.
lyche-numerical-linear-algebra F01674	129	1.000	`\hat{\boldsymbol{u}}_{k}=\left[u_{k k}, \ldots, u_{m k}\right]^{T}`	$\hat{\boldsymbol{u}}_{k}=\left[u_{k k}, \ldots, u_{m k}\right]^{T}$	In an algorithm we can store most of the vector $\hat{\boldsymbol{u}}_k = [u_{kk}, \ldots, u_{mk}]^T$ and the
lyche-numerical-linear-algebra F01675	129	1.000	`u_{k, k}`	$u_{k, k}$	matrix and $\boldsymbol{A}_k$ in $\boldsymbol{A}$. However, the elements $u_{k,k}$ and $a_k = r_{k,k}$ have to compete for

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01676	129	1.000	`a_{k}=r_{k, k}`	$a_{k}=r_{k, k}$	matrix and $\boldsymbol{A}_k$ in $\boldsymbol{A}$. However, the elements $u_{k,k}$ and $a_k = r_{k,k}$ have to compete for
lyche-numerical-linear-algebra_F01677	129	1.000	`m=4`	$m=4$	the diagonal in $\boldsymbol{A}$. For $m = 4$ and $n = 3$ the two possibilities look as follows:
lyche-numerical-linear-algebra_F01678	129	1.000	`\boldsymbol{B} \in \mathbb{C}^{m \times r}`	$\boldsymbol{B} \in \mathbb{C}^{m \times r}$	$\boldsymbol{B} \in \mathbb{C}^{m\times r}$ as input, and uses housegen to compute Householder transformations
lyche-numerical-linear-algebra_F01679	129	1.000	`\boldsymbol{H}_{1}, \ldots, \boldsymbol{H}_{s}`	$\boldsymbol{H}_{1}, \ldots, \boldsymbol{H}_{s}$	$\boldsymbol{H}_1, \ldots, \boldsymbol{H}_s$ so that $\boldsymbol{R} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1\boldsymbol{A}$ is upper trapezoidal, and $\boldsymbol{C} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1\boldsymbol{B}$.
lyche-numerical-linear-algebra_F01680	129	1.000	`\boldsymbol{R}=\boldsymbol{H}_{s} \cdots \boldsymbol{H}_{1} \boldsymbol{A}`	$\boldsymbol{R}=\boldsymbol{H}_{s} \cdots \boldsymbol{H}_{1} \boldsymbol{A}$	$\boldsymbol{H}_1, \ldots, \boldsymbol{H}_s$ so that $\boldsymbol{R} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1\boldsymbol{A}$ is upper trapezoidal, and $\boldsymbol{C} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1\boldsymbol{B}$.
lyche-numerical-linear-algebra_F01681	129	1.000	`\boldsymbol{C}=\boldsymbol{H}_{s} \cdots \boldsymbol{H}_{1} \boldsymbol{B}`	$\boldsymbol{C}=\boldsymbol{H}_{s} \cdots \boldsymbol{H}_{1} \boldsymbol{B}$	$\boldsymbol{H}_1, \ldots, \boldsymbol{H}_s$ so that $\boldsymbol{R} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1\boldsymbol{A}$ is upper trapezoidal, and $\boldsymbol{C} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1\boldsymbol{B}$.
lyche-numerical-linear-algebra_F01682	129	1.000	`r_{k, k}`	$r_{k, k}$	matrix with m rows and 0 columns. $r_{k,k}$ is stored in $\boldsymbol{A}$, and $u_{k,k}$ is stored in a
lyche-numerical-linear-algebra_F01683	129	0.961	`\operatorname{side}(\mathrm{s}) \boldsymbol{B}`	$\operatorname{side}(\mathrm{s}) \boldsymbol{B}$	and least squares problems with right hand side(s) $\boldsymbol{B}$, and to compute the product of
lyche-numerical-linear-algebra_F01684	129	0.999	`\boldsymbol{B}=\boldsymbol{I}`	$\boldsymbol{B}=\boldsymbol{I}$	the Householder transformations by choosing $\boldsymbol{B} = \boldsymbol{I}$.
lyche-numerical-linear-algebra_F01685	130	0.998	`v=\hat{\boldsymbol{u}}_{k}`	$v=\hat{\boldsymbol{u}}_{k}$	Here $v = \hat{\boldsymbol{u}}_k$ and the update is computed as $\hat{\boldsymbol{H}}_k\boldsymbol{C} = (\boldsymbol{I} - \boldsymbol{v}\boldsymbol{v}^*)\boldsymbol{C} = \boldsymbol{C} - \boldsymbol{v}(\boldsymbol{v}^*\boldsymbol{C})$.
lyche-numerical-linear-algebra_F01686	130	0.998	`\hat{\boldsymbol{H}}_{k} \boldsymbol{C}=\left( \boldsymbol{I}-\boldsymbol{v} \boldsymbol{v}^{*}\right) \boldsymbol{C}=\boldsymbol{C}-\boldsymbol{v}\left( \boldsymbol{v}^{*} \boldsymbol{C}\right)`	$\hat{\boldsymbol{H}}_{k} \boldsymbol{C}=\left(\boldsymbol{I}-\boldsymbol{v} \boldsymbol{v}^{*}\right) \boldsymbol{C}=\boldsymbol{C}-\boldsymbol{v}\left(\boldsymbol{v}^{*} \boldsymbol{C}\right)$	Here $v = \hat{\boldsymbol{u}}_k$ and the update is computed as $\hat{\boldsymbol{H}}_k\boldsymbol{C} = (\boldsymbol{I} - \boldsymbol{v}\boldsymbol{v}^*)\boldsymbol{C} = \boldsymbol{C} - \boldsymbol{v}(\boldsymbol{v}^*\boldsymbol{C})$.
lyche-numerical-linear-algebra_F01687	130	1.000	`n+1, \ldots, m`	$n+1, \ldots, m$	zeros in rows $n+1, \ldots, m$.
lyche-numerical-linear-algebra_F01688	130	1.000	`\boldsymbol{C}-\boldsymbol{v} *\left(\boldsymbol{v}^{*} * \boldsymbol{C}\right)`	$\boldsymbol{C}-\boldsymbol{v} *\left(\boldsymbol{v}^{*} * \boldsymbol{C}\right)$	The bulk of the work in Algorithm 5.2 is the computation of $\boldsymbol{C}-\boldsymbol{v}*(\boldsymbol{v}^* * \boldsymbol{C})$ for each
lyche-numerical-linear-algebra_F01689	130	1.000	`\boldsymbol{C} \in \mathbb{C}^{(m-k+1) \times(n+r-k)}`	$\boldsymbol{C} \in \mathbb{C}^{(m-k+1) \times(n+r-k)}$	k. Since in Algorithm 5.2, $\boldsymbol{C} \in \mathbb{C}^{(m-k+1)\times(n+r-k)}$ and $m \geq n$ the cost of computing
lyche-numerical-linear-algebra_F01690	130	1.000	`m \geq n`	$m \geq n$	k. Since in Algorithm 5.2, $\boldsymbol{C} \in \mathbb{C}^{(m-k+1)\times(n+r-k)}$ and $m \geq n$ the cost of computing
lyche-numerical-linear-algebra_F01691	130	1.000	`\boldsymbol{C}-\boldsymbol{v} *\left(\boldsymbol{v}^{T} * \boldsymbol{C}\right)`	$\boldsymbol{C}-\boldsymbol{v} *\left(\boldsymbol{v}^{T} * \boldsymbol{C}\right)$	the update $\boldsymbol{C} - \boldsymbol{v} * (\boldsymbol{v}^T * \boldsymbol{C})$ in the real case is $4(m-k+1)(n+r-k)$ arithmetic
lyche-numerical-linear-algebra_F01692	130	1.000	`4(m-k+1)(n+r-k)`	$4(m-k+1)(n+r-k)$	the update $\boldsymbol{C} - \boldsymbol{v} * (\boldsymbol{v}^T * \boldsymbol{C})$ in the real case is $4(m-k+1)(n+r-k)$ arithmetic
lyche-numerical-linear-algebra_F01693	130	1.000	`r=0`	$r=0$	For $m = n$ and $r = 0$ this gives $4n^3/3 = 2G_n$ for the number of arithmetic
lyche-numerical-linear-algebra_F01694	130	1.000	`4 n^{3} / 3=2 G_{n}`	$4 n^{3} / 3=2 G_{n}$	For $m = n$ and $r = 0$ this gives $4n^3/3 = 2G_n$ for the number of arithmetic
lyche-numerical-linear-algebra_F01695	130	1.000	`\boldsymbol{R} \boldsymbol{x}=\boldsymbol{c}`	$\boldsymbol{R} \boldsymbol{x}=\boldsymbol{c}$	obtain an upper triangular system $\boldsymbol{R}\boldsymbol{x} = \boldsymbol{c}$ that is upper triangular and nonsingular if
lyche-numerical-linear-algebra_F01696	131	1.000	`4 n^{3} / 3=`	$4 n^{3} / 3=$	reason is that the number of arithmetic operations in (5.16) when $m = n$ is $4n^3/3 =$
lyche-numerical-linear-algebra_F01697	131	0.998	`\boldsymbol{R}=\boldsymbol{H}_{n-1} \cdots \boldsymbol{H}_{1} \boldsymbol{A}`	$\boldsymbol{R}=\boldsymbol{H}_{n-1} \cdots \boldsymbol{H}_{1} \boldsymbol{A}$	Algorithm 5.2 gives $\boldsymbol{R} = \boldsymbol{H}_{n-1} \cdots \boldsymbol{H}_1\boldsymbol{A}$ implying the factorization $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$,
lyche-numerical-linear-algebra_F01698	131	1.000	`\boldsymbol{Q}=\boldsymbol{H}_{1} \cdots \boldsymbol{H}_{n-1}`	$\boldsymbol{Q}=\boldsymbol{H}_{1} \cdots \boldsymbol{H}_{n-1}$	where $\boldsymbol{Q} = \boldsymbol{H}_1 \cdots \boldsymbol{H}_{n-1}$ is unitary and $\boldsymbol{R}$ is upper triangular. This is known as a
lyche-numerical-linear-algebra_F01699	131	0.982	`\mathbf{Q R}`	$\mathbf{Q R}$	$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ is a **QR decomposition** of $\boldsymbol{A}$ if $\boldsymbol{Q} \in \mathbb{C}^{m,m}$ is square and unitary and
lyche-numerical-linear-algebra_F01700	131	0.982	`\boldsymbol{Q} \in \mathbb{C}^{m, m}`	$\boldsymbol{Q} \in \mathbb{C}^{m, m}$	$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ is a **QR decomposition** of $\boldsymbol{A}$ if $\boldsymbol{Q} \in \mathbb{C}^{m,m}$ is square and unitary and
lyche-numerical-linear-algebra_F01701	131	1.000	`\boldsymbol{R}_{1} \in \mathbb{C}^{n \times n}`	$\boldsymbol{R}_{1} \in \mathbb{C}^{n \times n}$	where $\boldsymbol{R}_1 \in \mathbb{C}^{n\times n}$ is upper triangular and $\boldsymbol{0}_{m-n,n}$ is the zero matrix with $m - n$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01702	131	1.000	`\mathbf{0}_{m-n, n}`	$\mathbf{0}_{m-n, n}$	where $\boldsymbol{R}_1 \in \mathbb{C}^{n \times n}$ is upper triangular and $\mathbf{0}_{m-n,n}$ is the zero matrix with $m-n$
lyche-numerical-linear-algebra_F01703	131	1.000	`m-n`	$m-n$	where $\boldsymbol{R}_1 \in \mathbb{C}^{n \times n}$ is upper triangular and $\mathbf{0}_{m-n,n}$ is the zero matrix with $m-n$
lyche-numerical-linear-algebra_F01704	131	0.961	`\boldsymbol{A}=\boldsymbol{Q}_{1} \boldsymbol{R}_{1}`	$\boldsymbol{A}=\boldsymbol{Q}_{1} \boldsymbol{R}_{1}$	rows and n columns. For $m \geq n$ we call $\boldsymbol{A} = \boldsymbol{Q}_1\boldsymbol{R}_1$ a **QR factorization** of $\boldsymbol{A}$ if
lyche-numerical-linear-algebra_F01705	131	1.000	`\boldsymbol{Q}_{1} \in \mathbb{C}^{m \times n}`	$\boldsymbol{Q}_{1} \in \mathbb{C}^{m \times n}$	$\boldsymbol{Q}_1 \in \mathbb{C}^{m \times n}$ has orthonormal columns and $\boldsymbol{R}_1 \in \mathbb{C}^{n \times n}$ is upper triangular.
lyche-numerical-linear-algebra_F01706	131	1.000	`\boldsymbol{Q} \boldsymbol{R}`	$\boldsymbol{Q} \boldsymbol{R}$	$\boldsymbol{Q}\boldsymbol{R}$ by simply using the first n columns of $\boldsymbol{Q}$ and the first n rows of $\boldsymbol{R}$. Indeed, if
lyche-numerical-linear-algebra_F01707	131	1.000	`\left[\boldsymbol{Q}_{1}, \boldsymbol{Q}_{2}\right]`	$\left[\boldsymbol{Q}_{1}, \boldsymbol{Q}_{2}\right]$	we partition $\boldsymbol{Q}$ as $[\boldsymbol{Q}_1, \boldsymbol{Q}_2]$ and $\boldsymbol{R} = \left[\begin{smallmatrix}\boldsymbol{R}_1\\ \mathbf{0}\end{smallmatrix}\right]$, where $\boldsymbol{Q}_1 \in \mathbb{R}^{m \times n}$ and $\boldsymbol{R}_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra_F01708	131	1.000	`\boldsymbol{R}=\left[\begin{array}{c}\boldsymbol{R}_{1} \\ \mathbf{0}\end{array}\right]`	$\boldsymbol{R}=\left[\begin{array}{c}\boldsymbol{R}_{1} \\ \mathbf{0}\end{array}\right]$	we partition $\boldsymbol{Q}$ as $[\boldsymbol{Q}_1, \boldsymbol{Q}_2]$ and $\boldsymbol{R} = \left[\begin{smallmatrix}\boldsymbol{R}_1\\ \mathbf{0}\end{smallmatrix}\right]$, where $\boldsymbol{Q}_1 \in \mathbb{R}^{m \times n}$ and $\boldsymbol{R}_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra_F01709	131	1.000	`\boldsymbol{Q}_{1} \in \mathbb{R}^{m \times n}`	$\boldsymbol{Q}_{1} \in \mathbb{R}^{m \times n}$	we partition $\boldsymbol{Q}$ as $[\boldsymbol{Q}_1, \boldsymbol{Q}_2]$ and $\boldsymbol{R} = \left[\begin{smallmatrix}\boldsymbol{R}_1\\ \mathbf{0}\end{smallmatrix}\right]$, where $\boldsymbol{Q}_1 \in \mathbb{R}^{m \times n}$ and $\boldsymbol{R}_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra_F01710	131	1.000	`\boldsymbol{R}_{1} \in \mathbb{R}^{n \times n}`	$\boldsymbol{R}_{1} \in \mathbb{R}^{n \times n}$	we partition $\boldsymbol{Q}$ as $[\boldsymbol{Q}_1, \boldsymbol{Q}_2]$ and $\boldsymbol{R} = \left[\begin{smallmatrix}\boldsymbol{R}_1\\ \mathbf{0}\end{smallmatrix}\right]$, where $\boldsymbol{Q}_1 \in \mathbb{R}^{m \times n}$ and $\boldsymbol{R}_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra_F01711	131	1.000	`\left\{\boldsymbol{q}_{1}, \ldots, \boldsymbol{q}_{n} \right\}`	$\left\{\boldsymbol{q}_{1}, \ldots, \boldsymbol{q}_{n} \right\}$	columns $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\}$ of $\boldsymbol{Q}_1$ into an orthonormal basis $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n, \boldsymbol{q}_{n+1}, \ldots, \boldsymbol{q}_m\}$
lyche-numerical-linear-algebra_F01712	131	1.000	`\boldsymbol{Q}_{1}`	$\boldsymbol{Q}_{1}$	columns $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\}$ of $\boldsymbol{Q}_1$ into an orthonormal basis $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n, \boldsymbol{q}_{n+1}, \ldots, \boldsymbol{q}_m\}$
lyche-numerical-linear-algebra_F01713	131	1.000	`\left\{\boldsymbol{q}_{1}, \ldots, \boldsymbol{q}_{n}, \boldsymbol{q}_{n+1}, \ldots, \boldsymbol{q}_{m}\right\}`	$\left\{\boldsymbol{q}_{1}, \ldots, \boldsymbol{q}_{n}, \boldsymbol{q}_{n+1}, \ldots, \boldsymbol{q}_{m}\right\}$	columns $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\}$ of $\boldsymbol{Q}_1$ into an orthonormal basis $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n, \boldsymbol{q}_{n+1}, \ldots, \boldsymbol{q}_m\}$
lyche-numerical-linear-algebra_F01714	131	1.000	`\boldsymbol{Q}=\left[\boldsymbol{q}_{1}, \ldots, \boldsymbol{q}_{m}\right]`	$\boldsymbol{Q}=\left[\boldsymbol{q}_{1}, \ldots, \boldsymbol{q}_{m}\right]$	$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$, where $\boldsymbol{Q} = [\boldsymbol{q}_1, \ldots, \boldsymbol{q}_m]$ and $\boldsymbol{R} = \left[\begin{smallmatrix}\boldsymbol{R}_1\\ \mathbf{0}\end{smallmatrix}\right]$.
lyche-numerical-linear-algebra_F01715	132	0.743	`\boldsymbol{Q}^{T} \boldsymbol{Q}=\boldsymbol{I}`	$\boldsymbol{Q}^{T} \boldsymbol{Q}=\boldsymbol{I}$	Since $\boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}$ and $\boldsymbol{R}$ is upper trapezoidal, this is a QR decomposition of $\boldsymbol{A}$. A
lyche-numerical-linear-algebra_F01716	132	1.000	`Q R`	$Q R$	*$m, n \in \mathbb{N}$ has a QR decomposition.*
lyche-numerical-linear-algebra_F01717	132	1.000	`\boldsymbol{A}=[1] \boldsymbol{A}`	$\boldsymbol{A}=[1] \boldsymbol{A}$	***Proof*** If $m = 1$ then $\boldsymbol{A}$ is already in upper trapezoidal form and $\boldsymbol{A} = [1]\boldsymbol{A}$ is a
lyche-numerical-linear-algebra_F01718	132	1.000	`s:=\min (m-1, n)`	$s:=\min (m-1, n)$	QR decomposition of $\boldsymbol{A}$. Suppose $m > 1$ and set $s := \min(m-1, n)$. Note that
lyche-numerical-linear-algebra_F01719	132	0.999	`\boldsymbol{R}=\boldsymbol{C} \boldsymbol{A}`	$\boldsymbol{R}=\boldsymbol{C} \boldsymbol{A}$	for any vector $\boldsymbol{x}$. With $\boldsymbol{B} = \boldsymbol{I}$ in Algorithm 5.2 we obtain $\boldsymbol{R} = \boldsymbol{C}\boldsymbol{A}$ and $\boldsymbol{C} =$
lyche-numerical-linear-algebra_F01720	132	0.999	`\boldsymbol{C}=`	$\boldsymbol{C}=$	for any vector $\boldsymbol{x}$. With $\boldsymbol{B} = \boldsymbol{I}$ in Algorithm 5.2 we obtain $\boldsymbol{R} = \boldsymbol{C}\boldsymbol{A}$ and $\boldsymbol{C} =$
lyche-numerical-linear-algebra_F01721	132	0.615	`\boldsymbol{H}_{s} \cdots \boldsymbol{H}_{2} \boldsymbol{H}_{1}`	$\boldsymbol{H}_{s} \cdots \boldsymbol{H}_{2} \boldsymbol{H}_{1}$	$\boldsymbol{H}_s \cdots \boldsymbol{H}_2\boldsymbol{H}_1$. Thus $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ is a QR decomposition of $\boldsymbol{A}$ since $\boldsymbol{Q} := \boldsymbol{C}^* =$
lyche-numerical-linear-algebra_F01722	132	0.615	`\boldsymbol{Q}:=\boldsymbol{C}^{*}=`	$\boldsymbol{Q}:=\boldsymbol{C}^{*}=$	$\boldsymbol{H}_s \cdots \boldsymbol{H}_2\boldsymbol{H}_1$. Thus $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ is a QR decomposition of $\boldsymbol{A}$ since $\boldsymbol{Q} := \boldsymbol{C}^* =$
lyche-numerical-linear-algebra_F01723	132	1.000	`\boldsymbol{H}_{1} \cdots \boldsymbol{H}_{s}`	$\boldsymbol{H}_{1} \cdots \boldsymbol{H}_{s}$	$\boldsymbol{H}_1 \cdots \boldsymbol{H}_s$ is a product of unitary matrices and therefore unitary.
lyche-numerical-linear-algebra_F01724	132	1.000	`\boldsymbol{A}^{*} \boldsymbol{A}=`	$\boldsymbol{A}^{*} \boldsymbol{A}=$	***Proof*** Let $\boldsymbol{A} = \boldsymbol{Q}_1\boldsymbol{R}_1$ be a QR factorization of $\boldsymbol{A} \in \mathbb{C}^{m \times n}$. Now $\boldsymbol{A}^*\boldsymbol{A} =$
lyche-numerical-linear-algebra_F01725	132	1.000	`\boldsymbol{R}_{1}^{*} \boldsymbol{Q}_{1}^{*} \boldsymbol{Q}_{1} \boldsymbol{R}_{1}= \boldsymbol{R}_{1}^{*} \boldsymbol{R}_{1}`	$\boldsymbol{R}_{1}^{*} \boldsymbol{Q}_{1}^{*} \boldsymbol{Q}_{1} \boldsymbol{R}_{1}= \boldsymbol{R}_{1}^{*} \boldsymbol{R}_{1}$	$\boldsymbol{R}_1^*\boldsymbol{Q}_1^*\boldsymbol{Q}_1\boldsymbol{R}_1 = \boldsymbol{R}_1^*\boldsymbol{R}_1$. By Lemma 4.2 the matrix $\boldsymbol{A}^*\boldsymbol{A}$ is positive definite, the
lyche-numerical-linear-algebra_F01726	132	1.000	`\boldsymbol{R}_{1}^{*} \boldsymbol{R}_{1}`	$\boldsymbol{R}_{1}^{*} \boldsymbol{R}_{1}$	matrix $\boldsymbol{R}_1$ is nonsingular, and if its diagonal elements are positive then $\boldsymbol{R}_1^*\boldsymbol{R}_1$ is the
lyche-numerical-linear-algebra_F01727	132	1.000	`\boldsymbol{Q}_{1}=\boldsymbol{A} \boldsymbol{R}_{1}^{-1}`	$\boldsymbol{Q}_{1}=\boldsymbol{A} \boldsymbol{R}_{1}^{-1}$	that $\boldsymbol{R}_1$ is unique and since necessarily $\boldsymbol{Q}_1 = \boldsymbol{A}\boldsymbol{R}_1^{-1}$, it must also be unique.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F01728	132	1.000	`\boldsymbol{B}:=\boldsymbol{A}^{T} \boldsymbol{A}=\left[ \begin{array}{cc}5 & -4 \\ -4 & 5\end{array}\right]`	$\boldsymbol{B}:=\boldsymbol{A}^{T} \boldsymbol{A}=\left[\begin{array}{cc}5 & -4 \\ -4 & 5\end{array}\right]$	uniqueness proof of Theorem 5.10. We find $\boldsymbol{B} := \boldsymbol{A}^T\boldsymbol{A} = \left[\begin{smallmatrix}5 & -4\\ -4 & 5\end{smallmatrix}\right]$. The Cholesky
lyche-numerical-linear-al F01729	132	1.000	`\boldsymbol{B}=\boldsymbol{R}^{T} \boldsymbol{R}`	$\boldsymbol{B}=\boldsymbol{R}^{T} \boldsymbol{R}$	factorization of $\boldsymbol{B} = \boldsymbol{R}^T\boldsymbol{R}$ is given by $\boldsymbol{R} = \frac{1}{\sqrt{5}}\left[\begin{smallmatrix}5 & -4\\ 0 & 3\end{smallmatrix}\right]$. Now $\boldsymbol{R}^{-1} = \frac{1}{3\sqrt{5}}\left[\begin{smallmatrix}3 & 4\\ 0 & 5\end{smallmatrix}\right]$
lyche-numerical-linear-al F01730	132	1.000	`\boldsymbol{R}=\frac{1}{\sqrt{5}}\left[\begin{array}{cc}5 & -4 \\ 0 & 3\end{array}\right]`	$\boldsymbol{R}=\frac{1}{\sqrt{5}}\left[\begin{array}{cc}5 & -4 \\ 0 & 3\end{array}\right]$	factorization of $\boldsymbol{B} = \boldsymbol{R}^T\boldsymbol{R}$ is given by $\boldsymbol{R} = \frac{1}{\sqrt{5}}\left[\begin{smallmatrix}5 & -4\\ 0 & 3\end{smallmatrix}\right]$. Now $\boldsymbol{R}^{-1} = \frac{1}{3\sqrt{5}}\left[\begin{smallmatrix}3 & 4\\ 0 & 5\end{smallmatrix}\right]$
lyche-numerical-linear-al F01731	132	1.000	`\boldsymbol{R}^{-1}=\frac{1}{3 \sqrt{5}}\left[ \begin{array}{ll}3 & 4 \\ 0 & 5\end{array}\right]`	$\boldsymbol{R}^{-1}=\frac{1}{3 \sqrt{5}}\left[\begin{array}{ll}3 & 4 \\ 0 & 5\end{array}\right]$	factorization of $\boldsymbol{B} = \boldsymbol{R}^T\boldsymbol{R}$ is given by $\boldsymbol{R} = \frac{1}{\sqrt{5}}\left[\begin{smallmatrix}5 & -4\\ 0 & 3\end{smallmatrix}\right]$. Now $\boldsymbol{R}^{-1} = \frac{1}{3\sqrt{5}}\left[\begin{smallmatrix}3 & 4\\ 0 & 5\end{smallmatrix}\right]$
lyche-numerical-linear-al F01732	132	0.889	`\boldsymbol{Q}=\boldsymbol{A} \boldsymbol{R}^{-1}= \frac{1}{\sqrt{5}}\left[\begin{array}{cc}2 & 1 \\ -1 & 2\end{array}\right]`	$\boldsymbol{Q}=\boldsymbol{A} \boldsymbol{R}^{-1}= \frac{1}{\sqrt{5}}\left[\begin{array}{cc}2 & 1 \\ -1 & 2\end{array}\right]$	so $\boldsymbol{Q} = \boldsymbol{A}\boldsymbol{R}^{-1} = \frac{1}{\sqrt{5}}\left[\begin{smallmatrix}2 & 1\\ -1 & 2\end{smallmatrix}\right]$. Since $\boldsymbol{A}$ is square $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ is both the QR
lyche-numerical-linear-al F01733	133	0.992	`\boldsymbol{a}_{1}, \ldots, \boldsymbol{a}_{n}`	$\boldsymbol{a}_{1}, \ldots, \boldsymbol{a}_{n}$	$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ *be the result of applying Gram Schmidt to the columns* $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ *of* $\boldsymbol{A}$,
lyche-numerical-linear-al F01734	133	1.000	`\mathcal{R}(\boldsymbol{A})`	$\mathcal{R}(\boldsymbol{A})$	has orthonormal columns, since $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\}$ is an orthonormal basis for $\mathcal{R}(\boldsymbol{A})$ by
lyche-numerical-linear-al F01735	133	1.000	`\boldsymbol{A}=\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right]=\left[\boldsymbol{a}_{1}, \boldsymbol{a}_{2}\right]`	$\boldsymbol{A}=\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right]=\left[\boldsymbol{a}_{1}, \boldsymbol{a}_{2}\right]$	and factorization of the matrix $\boldsymbol{A} = \left[\begin{smallmatrix}2 & -1\\ -1 & 2\end{smallmatrix}\right] = [\boldsymbol{a}_1, \boldsymbol{a}_2]$ using Gram-Schmidt.
lyche-numerical-linear-al F01736	133	0.999	`\boldsymbol{v}_{1}=\boldsymbol{a}_{1}`	$\boldsymbol{v}_{1}=\boldsymbol{a}_{1}$	Using (5.17) we find $\boldsymbol{v}_1 = \boldsymbol{a}_1$ and $\boldsymbol{v}_2 = \boldsymbol{a}_2 - \frac{\boldsymbol{a}_2^T\boldsymbol{v}_1}{\boldsymbol{v}_1^T\boldsymbol{v}_1}\boldsymbol{v}_1 = \frac{3}{5}\left[\begin{smallmatrix}1\\ 2\end{smallmatrix}\right]$. Thus $\boldsymbol{Q} = [\boldsymbol{q}_1, \boldsymbol{q}_2]$,
lyche-numerical-linear-al F01737	133	0.999	`\boldsymbol{v}_{2}=\boldsymbol{a}_{2}-\frac{ \boldsymbol{a}_{2}^{T} \boldsymbol{v}_{1}}{ \boldsymbol{v}_{1}^{T} \boldsymbol{v}_{1}} \boldsymbol{v}_{1}=\frac{3}{5}\left[\begin{array}{l}1 \\ 2\end{array}\right]`	$\boldsymbol{v}_{2}=\boldsymbol{a}_{2}-\frac{ \boldsymbol{a}_{2}^{T} \boldsymbol{v}_{1}}{ \boldsymbol{v}_{1}^{T} \boldsymbol{v}_{1}} \boldsymbol{v}_{1}=\frac{3}{5}\left[\begin{array}{l}1 \\ 2\end{array}\right]$	Using (5.17) we find $\boldsymbol{v}_1 = \boldsymbol{a}_1$ and $\boldsymbol{v}_2 = \boldsymbol{a}_2 - \frac{\boldsymbol{a}_2^T\boldsymbol{v}_1}{\boldsymbol{v}_1^T\boldsymbol{v}_1}\boldsymbol{v}_1 = \frac{3}{5}\left[\begin{smallmatrix}1\\ 2\end{smallmatrix}\right]$. Thus $\boldsymbol{Q} = [\boldsymbol{q}_1, \boldsymbol{q}_2]$,
lyche-numerical-linear-al F01738	133	0.999	`\boldsymbol{Q}=\left[\boldsymbol{q}_{1}, \boldsymbol{q}_{2}\right]`	$\boldsymbol{Q}=\left[\boldsymbol{q}_{1}, \boldsymbol{q}_{2}\right]$	Using (5.17) we find $\boldsymbol{v}_1 = \boldsymbol{a}_1$ and $\boldsymbol{v}_2 = \boldsymbol{a}_2 - \frac{\boldsymbol{a}_2^T\boldsymbol{v}_1}{\boldsymbol{v}_1^T\boldsymbol{v}_1}\boldsymbol{v}_1 = \frac{3}{5}\left[\begin{smallmatrix}1\\ 2\end{smallmatrix}\right]$. Thus $\boldsymbol{Q} = [\boldsymbol{q}_1, \boldsymbol{q}_2]$,
lyche-numerical-linear-al F01739	133	1.000	`\boldsymbol{q}_{1}=\frac{1}{\sqrt{5}}\left[ \begin{array}{c}2 \\ -1\end{array}\right]`	$\boldsymbol{q}_{1}=\frac{1}{\sqrt{5}}\left[\begin{array}{c}2 \\ -1\end{array}\right]$	where $\boldsymbol{q}_1 = \frac{1}{\sqrt{5}}\left[\begin{smallmatrix}2\\ -1\end{smallmatrix}\right]$ and $\boldsymbol{q}_2 = \frac{1}{\sqrt{5}}\left[\begin{smallmatrix}1\\ 2\end{smallmatrix}\right]$. By (5.18) we find
lyche-numerical-linear-al F01740	133	1.000	`\boldsymbol{q}_{2}=\frac{1}{\sqrt{5}}\left[ \begin{array}{l}1 \\ 2\end{array}\right]`	$\boldsymbol{q}_{2}=\frac{1}{\sqrt{5}}\left[\begin{array}{l}1 \\ 2\end{array}\right]$	where $\boldsymbol{q}_1 = \frac{1}{\sqrt{5}}\left[\begin{smallmatrix}2\\ -1\end{smallmatrix}\right]$ and $\boldsymbol{q}_2 = \frac{1}{\sqrt{5}}\left[\begin{smallmatrix}1\\ 2\end{smallmatrix}\right]$. By (5.18) we find
lyche-numerical-linear-al F01741	134	1.000	`n=4`	$n=4$	by a **Wilkinson diagram** for $n = 4$
lyche-numerical-linear-al F01742	134	1.000	`\boldsymbol{P} \in \mathbb{R}^{2,2}`	$\boldsymbol{P} \in \mathbb{R}^{2,2}$	a **Given's rotation**) is a matrix $\boldsymbol{P} \in \mathbb{R}^{2,2}$ of the form
lyche-numerical-linear-al F01743	134	1.000	`\theta \in[0,2 \pi)`	$\theta \in[0,2 \pi)$	A plane rotation is an orthogonal matrix and there is a unique angle $\theta \in [0, 2\pi)$
lyche-numerical-linear-al F01744	134	1.000	`c=\cos \theta`	$c=\cos \theta$	such that $c = \cos\theta$ and $s = \sin\theta$. Moreover, the identity matrix is a plane rotation
lyche-numerical-linear-al F01745	134	1.000	`s=\sin \theta`	$s=\sin \theta$	such that $c = \cos\theta$ and $s = \sin\theta$. Moreover, the identity matrix is a plane rotation
lyche-numerical-linear-al F01746	134	1.000	`\theta=0`	$\theta=0$	corresponding to $\theta = 0$. A vector $\boldsymbol{x}$ in the plane is rotated an angle θ clockwise by
lyche-numerical-linear-al F01747	134	1.000	`\boldsymbol{P}=\boldsymbol{R}`	$\boldsymbol{P}=\boldsymbol{R}$	$\boldsymbol{P} = \boldsymbol{R}$. See Exercise 5.16 and Fig. 5.4.
lyche-numerical-linear-al F01748	135	1.000	`\boldsymbol{P}=\boldsymbol{I}`	$\boldsymbol{P}=\boldsymbol{I}$	and we have introduced a zero in $\boldsymbol{x}$. We can take $\boldsymbol{P} = \boldsymbol{I}$ when $\boldsymbol{x} = \boldsymbol{0}$.
lyche-numerical-linear-al F01749	135	1.000	`1 \leq i<j \leq n`	$1 \leq i<j \leq n$	For an n-vector $\boldsymbol{x} \in \mathbb{R}^n$ and $1 \le i < j \le n$ we define a **rotation in the** i, j**-plane**

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01750	135	0.998	\boldsymbol{P}_{i j}=\left(p_{k l}\right) \in \mathbb{R}^{n \times n}	$\boldsymbol{P}_{i j}=\left(p_{k l}\right) \in \mathbb{R}^{n \times n}$	as a matrix $\boldsymbol{P}_{ij} = (p_{kl}) \in \mathbb{R}^{n\times n}$ by $p_{kl} = \delta_{kl}$ except for positions ii, jj, ij, ji,
lyche-numerical-linear-algebra F01751	135	0.998	p_{k l}=\delta_{k l}	$p_{k l}=\delta_{k l}$	as a matrix $\boldsymbol{P}_{ij} = (p_{kl}) \in \mathbb{R}^{n\times n}$ by $p_{kl} = \delta_{kl}$ except for positions ii, jj, ij, ji,
lyche-numerical-linear-algebra F01752	135	0.998	i i, j j, i j, j i	$i i, j j, i j, j i$	as a matrix $\boldsymbol{P}_{ij} = (p_{kl}) \in \mathbb{R}^{n\times n}$ by $p_{kl} = \delta_{kl}$ except for positions ii, jj, ij, ji,
lyche-numerical-linear-algebra F01753	135	1.000	\boldsymbol{B}=\boldsymbol{P}_{i j} \boldsymbol{A}	$\boldsymbol{B}=\boldsymbol{P}_{i j} \boldsymbol{A}$	column i and j. In particular, if $\boldsymbol{B} = \boldsymbol{P}_{ij}\boldsymbol{A}$ and $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{P}_{ij}$ then $\boldsymbol{B}(k,:) = \boldsymbol{A}(k,:)$,
lyche-numerical-linear-algebra F01754	135	1.000	\boldsymbol{C}=\boldsymbol{A} \boldsymbol{P}_{i j}	$\boldsymbol{C}=\boldsymbol{A} \boldsymbol{P}_{i j}$	column i and j. In particular, if $\boldsymbol{B} = \boldsymbol{P}_{ij}\boldsymbol{A}$ and $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{P}_{ij}$ then $\boldsymbol{B}(k,:) = \boldsymbol{A}(k,:)$,
lyche-numerical-linear-algebra F01755	135	1.000	\boldsymbol{B}(k,:)=\boldsymbol{A}(k,:)	$\boldsymbol{B}(k,:)=\boldsymbol{A}(k,:)$	column i and j. In particular, if $\boldsymbol{B} = \boldsymbol{P}_{ij}\boldsymbol{A}$ and $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{P}_{ij}$ then $\boldsymbol{B}(k,:) = \boldsymbol{A}(k,:)$,
lyche-numerical-linear-algebra F01756	135	1.000	\boldsymbol{C}(:, k)=\boldsymbol{A}(:, k)	$\boldsymbol{C}(:, k)=\boldsymbol{A}(:, k)$	$\boldsymbol{C}(:,k) = \boldsymbol{A}(:,k)$ for all $k \neq i, j$ and
lyche-numerical-linear-algebra F01757	135	1.000	k \neq i, j	$k \neq i, j$	$\boldsymbol{C}(:,k) = \boldsymbol{A}(:,k)$ for all $k \neq i, j$ and
lyche-numerical-linear-algebra F01758	135	0.999	2 n^{3}	$2 n^{3}$	number of arithmetic operations is asymptotically $2n^3$, corresponding to the work
lyche-numerical-linear-algebra F01759	135	1.000	\boldsymbol{P}_{i, i+1}	$\boldsymbol{P}_{i, i+1}$	using rotations $\boldsymbol{P}_{i,i+1}$ for $i = 1, \ldots, n-1$. For $n = 4$ the process can be illustrated
lyche-numerical-linear-algebra F01760	136	1.000	\langle\boldsymbol{x}, \boldsymbol{y}\rangle:= \boldsymbol{y}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:= \boldsymbol{y}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}$	dent columns. Show that $\langle \boldsymbol{x}, \boldsymbol{y}\rangle := \boldsymbol{y}^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x}$ defines an inner product on $\mathbb{C}^n$.
lyche-numerical-linear-algebra F01761	136	0.998	[0, \pi / 2]	$[0, \pi / 2]$	plex case there is a unique angle θ in $[0, \pi/2]$ such that
lyche-numerical-linear-algebra F01762	136	1.000	\boldsymbol{x}=\boldsymbol{e}_{1}	$\boldsymbol{x}=\boldsymbol{e}_{1}$	**Exercise 5.4 (What Does Algorithm Housegen Do When $\boldsymbol{x} = \boldsymbol{e}_1$?)** Determine
lyche-numerical-linear-algebra F01763	136	1.000	\\|\boldsymbol{x}\\|_{2}=\\|\boldsymbol{y}\\|_{2}	$\\|\boldsymbol{x}\\|_{2}=\\|\boldsymbol{y}\\|_{2}$	$\\|\boldsymbol{x}\\|_2 = \\|\boldsymbol{y}\\|_2$ and $\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \boldsymbol{0}$ then it follows from Example 5.1 that
lyche-numerical-linear-algebra F01764	136	0.998	\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^{T}}{\boldsymbol{v}^{T} \boldsymbol{v}}\right) \boldsymbol{x}=\boldsymbol{y}	$\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^{T}}{\boldsymbol{v}^{T} \boldsymbol{v}}\right) \boldsymbol{x}=\boldsymbol{y}$	$(\boldsymbol{I} - 2\frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}})\boldsymbol{x} = \boldsymbol{y}$. Use this to construct a Householder transformation $\boldsymbol{H}$ such
lyche-numerical-linear-algebra F01765	136	1.000	\boldsymbol{H} \boldsymbol{x}=\boldsymbol{y}	$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{y}$	that $\boldsymbol{H}\boldsymbol{x} = \boldsymbol{y}$ in the following cases.
lyche-numerical-linear-algebra F01766	136	0.995	\boldsymbol{x}=\left[\begin{array}{l}3 \\\\ 4\end{array}\right], \quad \boldsymbol{y}=\left[\begin{array}{l}5 \\\\ 0\end{array}\right]	$\boldsymbol{x}=\left[\begin{array}{l}3 \\\\ 4\end{array}\right], \quad \boldsymbol{y}=\left[\begin{array}{l}5 \\\\ 0\end{array}\right]$	**a)** $\boldsymbol{x} = \begin{bmatrix}3\\\\4\end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix}5\\\\0\end{bmatrix}$.
lyche-numerical-linear-algebra F01767	136	1.000	\boldsymbol{x}=\left[\begin{array}{l}2 \\\\ 2 \\\\ 1 \end{array}\right], \quad \boldsymbol{y}=\left[\begin{array}{l}0 \\\\ 3 \\\\ 0\end{array}\right]	$\boldsymbol{x}=\left[\begin{array}{l}2 \\\\ 2 \\\\ 1 \end{array}\right], \quad \boldsymbol{y}=\left[\begin{array}{l}0 \\\\ 3 \\\\ 0\end{array}\right]$	**b)** $\boldsymbol{x} = \begin{bmatrix}2\\\\2\\\\1\end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix}0\\\\3\\\\0\end{bmatrix}$.
lyche-numerical-linear-algebra F01768	137	1.000	\boldsymbol{x}=[\cos \phi, \sin \phi]^{T}	$\boldsymbol{x}=[\cos \phi, \sin \phi]^{T}$	Find $\boldsymbol{H}\boldsymbol{x}$ if $\boldsymbol{x} = [\cos\phi, \sin\phi]^T$.
lyche-numerical-linear-algebra F01769	137	0.997	\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y} \neq 0	$\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y} \neq 0$	**a)** Suppose $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ with $\\|\boldsymbol{x}\\|_2 = \\|\boldsymbol{y}\\|_2$ and $\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq 0$. Show that
lyche-numerical-linear-algebra F01770	137	1.000	\boldsymbol{B} \in \mathbb{R}^{4,4}	$\boldsymbol{B} \in \mathbb{R}^{4,4}$	**b)** Let $\boldsymbol{B} \in \mathbb{R}^{4,4}$ be given by
lyche-numerical-linear-algebra F01771	137	1.000	0<\epsilon<1	$0<\epsilon<1$	where $0 < \epsilon < 1$. Compute a Householder transformation $\boldsymbol{H}$ and a matrix $\boldsymbol{B}_1$
lyche-numerical-linear-algebra F01772	137	1.000	\boldsymbol{B}_{1}	$\boldsymbol{B}_{1}$	where $0 < \epsilon < 1$. Compute a Householder transformation $\boldsymbol{H}$ and a matrix $\boldsymbol{B}_1$
lyche-numerical-linear-algebra F01773	137	0.857	\boldsymbol{B}_{1}:=\boldsymbol{H} \boldsymbol{B} \boldsymbol{H}	$\boldsymbol{B}_{1}:=\boldsymbol{H} \boldsymbol{B} \boldsymbol{H}$	such that the first column of $\boldsymbol{B}_1 := \boldsymbol{H}\boldsymbol{B}\boldsymbol{H}$ has a zero in the last two positions.

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-al F01774</td><td>138</td><td>1.000</td><td>\boldsymbol{H}_{1}, \boldsymbol{H}_{2} \in \mathbb{R}^{3 \times 3}</td><td>$\boldsymbol{H}_{1}, \boldsymbol{H}_{2} \in \mathbb{R}^{3 \times 3}$</td><td>Find Householder transformations $\boldsymbol{H}_1, \boldsymbol{H}_2 \in \mathbb{R}^{3\times 3}$ such that $\boldsymbol{H}_2\boldsymbol{H}_1\boldsymbol{A}$ is</td></tr>
<tr><td>lyche-numerical-linear-al F01775</td><td>138</td><td>1.000</td><td>\boldsymbol{H}_{2} \boldsymbol{H}_{1} \boldsymbol{A}</td><td>$\boldsymbol{H}_{2} \boldsymbol{H}_{1} \boldsymbol{A}$</td><td>Find Householder transformations $\boldsymbol{H}_1, \boldsymbol{H}_2 \in \mathbb{R}^{3\times 3}$ such that $\boldsymbol{H}_2\boldsymbol{H}_1\boldsymbol{A}$ is</td></tr>
<tr><td>lyche-numerical-linear-al F01776</td><td>138</td><td>0.999</td><td>\boldsymbol{A}=\left[\boldsymbol{a}_{1}, \ldots, \boldsymbol{a}_{n}\right] \in \mathbb{C}^{n \times n}</td><td>$\boldsymbol{A}=\left[\boldsymbol{a}_{1}, \ldots, \boldsymbol{a}_{n}\right] \in \mathbb{C}^{n \times n}$</td><td>tion to prove a classical determinant inequality. For any $\boldsymbol{A} = [\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n] \in \mathbb{C}^{n\times n}$</td></tr>
<tr><td>lyche-numerical-linear-al F01777</td><td>138</td><td>1.000</td><td>|\operatorname{det}(\boldsymbol{Q})|=1</td><td>$|\operatorname{det}(\boldsymbol{Q})|=1$</td><td>a) Show that if $\boldsymbol{Q}$ is unitary then $|\det(\boldsymbol{Q})| = 1$.</td></tr>
<tr><td>lyche-numerical-linear-al F01778</td><td>138</td><td>1.000</td><td>\boldsymbol{R}=\left[\boldsymbol{r}_{1}, \ldots, \boldsymbol{r}_{n}\right]</td><td>$\boldsymbol{R}=\left[\boldsymbol{r}_{1}, \ldots, \boldsymbol{r}_{n}\right]$</td><td>b) Let $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ be a QR factorization of $\boldsymbol{A}$ and let $\boldsymbol{R} = [\boldsymbol{r}_1, \ldots, \boldsymbol{r}_n]$. Show that</td></tr>
<tr><td>lyche-numerical-linear-al F01779</td><td>138</td><td>0.986</td><td>\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)_{j j}=\left\|\boldsymbol{a}_{j}\right\|_{2}^{2}=\left(\boldsymbol{R}^{*} \boldsymbol{R}\right)_{j j}=\left\|\boldsymbol{r}_{j}\right\|_{2}^{2}</td><td>$\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)_{j j}=\left\|\boldsymbol{a}_{j}\right\|_{2}^{2}=\left(\boldsymbol{R}^{*} \boldsymbol{R}\right)_{j j}=\left\|\boldsymbol{r}_{j}\right\|_{2}^{2}$</td><td>$(\boldsymbol{A}^*\boldsymbol{A})_{jj} = \|\boldsymbol{a}_j\|_2^2 = (\boldsymbol{R}^*\boldsymbol{R})_{jj} = \|\boldsymbol{r}_j\|_2^2$.</td></tr>
<tr><td>lyche-numerical-linear-al F01780</td><td>138</td><td>1.000</td><td>|\operatorname{det}(\boldsymbol{A})|=\prod_{j=1}^{n} \left|r_{j j}\right| \leq \prod_{j=1}^{n}\left\| \boldsymbol{a}_{j}\right\|_{2}</td><td>$|\operatorname{det}(\boldsymbol{A})|=\prod_{j=1}^{n} \left|r_{j j}\right| \leq \prod_{j=1}^{n}\left\| \boldsymbol{a}_{j}\right\|_{2}$</td><td>c) Show that $|\det(\boldsymbol{A})| = \prod_{j=1}^n |r_{jj}| \le \prod_{j=1}^n \|\boldsymbol{a}_j\|_2$.</td></tr>
<tr><td>lyche-numerical-linear-al F01781</td><td>138</td><td>0.949</td><td>\boldsymbol{B}=\boldsymbol{L}^{T} \boldsymbol{L}</td><td>$\boldsymbol{B}=\boldsymbol{L}^{T} \boldsymbol{L}$</td><td>form $\boldsymbol{B} = \boldsymbol{L}^T\boldsymbol{L}$, where $\boldsymbol{L}$ is lower triangular with positive diagonal elements (you</td></tr>
<tr><td>lyche-numerical-linear-al F01782</td><td>138</td><td>0.995</td><td>\boldsymbol{B}=\boldsymbol{L} \boldsymbol{L}^{T}</td><td>$\boldsymbol{B}=\boldsymbol{L} \boldsymbol{L}^{T}$</td><td>$\boldsymbol{B} = \boldsymbol{L}\boldsymbol{L}^T$.</td></tr>
<tr><td>lyche-numerical-linear-al F01783</td><td>138</td><td>1.000</td><td>l_{i, j}</td><td>$l_{i, j}$</td><td>a) Suppose $\boldsymbol{B} = \boldsymbol{L}^T\boldsymbol{L}$. Write down the equations to determine the elements $l_{i,j}$ of</td></tr>
<tr><td>lyche-numerical-linear-al F01784</td><td>138</td><td>1.000</td><td>i=n, n-1, \ldots, 1</td><td>$i=n, n-1, \ldots, 1$</td><td>$\boldsymbol{L}$, in the order $i = n, n-1, \ldots, 1$ and $j = i, 1, 2 \ldots, i-1$.</td></tr>
<tr><td>lyche-numerical-linear-al F01785</td><td>138</td><td>1.000</td><td>j=i, 1,2 \ldots, i-1</td><td>$j=i, 1,2 \ldots, i-1$</td><td>$\boldsymbol{L}$, in the order $i = n, n-1, \ldots, 1$ and $j = i, 1, 2 \ldots, i-1$.</td></tr>
<tr><td>lyche-numerical-linear-al F01786</td><td>138</td><td>1.000</td><td>\boldsymbol{L}^{T} \boldsymbol{L}</td><td>$\boldsymbol{L}^{T} \boldsymbol{L}$</td><td>b) Explain (without making a detailed algorithm) how the $\boldsymbol{L}^T\boldsymbol{L}$ factorization can be</td></tr>
<tr><td>lyche-numerical-linear-al F01787</td><td>138</td><td>1.000</td><td>\boldsymbol{B} \boldsymbol{x}=\boldsymbol{c}</td><td>$\boldsymbol{B} \boldsymbol{x}=\boldsymbol{c}$</td><td>used to solve the linear system $\boldsymbol{B}\boldsymbol{x} = \boldsymbol{c}$. Compute $\|\boldsymbol{L}\|_F$. Is the algorithm stable?</td></tr>
<tr><td>lyche-numerical-linear-al F01788</td><td>138</td><td>1.000</td><td>\|\boldsymbol{L}\|_{F}</td><td>$\|\boldsymbol{L}\|_{F}$</td><td>used to solve the linear system $\boldsymbol{B}\boldsymbol{x} = \boldsymbol{c}$. Compute $\|\boldsymbol{L}\|_F$. Is the algorithm stable?</td></tr>
<tr><td>lyche-numerical-linear-al F01789</td><td>138</td><td>1.000</td><td>\boldsymbol{Q} \boldsymbol{L}</td><td>$\boldsymbol{Q} \boldsymbol{L}$</td><td>$\boldsymbol{Q}\boldsymbol{L}$, where $\boldsymbol{Q} \in \mathbb{R}^{n\times n}$ is orthogonal and $\boldsymbol{L} \in \mathbb{R}^{n\times n}$ is lower triangular with</td></tr>
<tr><td>lyche-numerical-linear-al F01790</td><td>138</td><td>1.000</td><td>\boldsymbol{Q} \in \mathbb{R}^{n \times n}</td><td>$\boldsymbol{Q} \in \mathbb{R}^{n \times n}$</td><td>$\boldsymbol{Q}\boldsymbol{L}$, where $\boldsymbol{Q} \in \mathbb{R}^{n\times n}$ is orthogonal and $\boldsymbol{L} \in \mathbb{R}^{n\times n}$ is lower triangular with</td></tr>
<tr><td>lyche-numerical-linear-al F01791</td><td>138</td><td>1.000</td><td>\boldsymbol{a}:=\left[a_{1}, \ldots, a_{n}\right]^{T} \in \mathbb{R}^{n}</td><td>$\boldsymbol{a}:=\left[a_{1}, \ldots, a_{n}\right]^{T} \in \mathbb{R}^{n}$</td><td>a) Given vectors $\boldsymbol{a} := [a_1, \ldots, a_n]^T \in \mathbb{R}^n$ and $\boldsymbol{e}_n := [0, \ldots, 0, 1]^T$. Find $\boldsymbol{v} \in \mathbb{R}^n$</td></tr>
<tr><td>lyche-numerical-linear-al F01792</td><td>138</td><td>1.000</td><td>\boldsymbol{e}_{n}:=[0, \ldots, 0,1]^{T}</td><td>$\boldsymbol{e}_{n}:=[0, \ldots, 0,1]^{T}$</td><td>a) Given vectors $\boldsymbol{a} := [a_1, \ldots, a_n]^T \in \mathbb{R}^n$ and $\boldsymbol{e}_n := [0, \ldots, 0, 1]^T$. Find $\boldsymbol{v} \in \mathbb{R}^n$</td></tr>
<tr><td>lyche-numerical-linear-al F01793</td><td>138</td><td>0.724</td><td>\boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}}</td><td>$\boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}}$</td><td>such that the Householder transformation $\boldsymbol{H} := \boldsymbol{I} - 2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}$ satisfies $\boldsymbol{H}\boldsymbol{a} = -s\boldsymbol{e}_n$,</td></tr>
<tr><td>lyche-numerical-linear-al F01794</td><td>138</td><td>0.724</td><td>\boldsymbol{H} \boldsymbol{a}=-s \boldsymbol{e}_{n}</td><td>$\boldsymbol{H} \boldsymbol{a}=-s \boldsymbol{e}_{n}$</td><td>such that the Householder transformation $\boldsymbol{H} := \boldsymbol{I} - 2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}$ satisfies $\boldsymbol{H}\boldsymbol{a} = -s\boldsymbol{e}_n$,</td></tr>
<tr><td>lyche-numerical-linear-al F01795</td><td>138</td><td>1.000</td><td>|s|=\|\boldsymbol{a}\|_{2}</td><td>$|s|=\|\boldsymbol{a}\|_{2}$</td><td>where $|s| = \|\boldsymbol{a}\|_2$. How should we choose the sign of s?</td></tr>
<tr><td>lyche-numerical-linear-al F01796</td><td>138</td><td>1.000</td><td>s</td><td>s</td><td>where $|s| = \|\boldsymbol{a}\|_2$. How should we choose the sign of s?</td></tr>
<tr><td>lyche-numerical-linear-al F01797</td><td>138</td><td>0.630</td><td>\Longleftrightarrow \boldsymbol{R}</td><td>$\Longleftrightarrow \boldsymbol{R}$</td><td>[1]Show that we have equality $\iff$ $\boldsymbol{R}$ is diagonal $\iff$ $\boldsymbol{A}^*\boldsymbol{A}$ is diagonal.</td></tr>
<tr><td>lyche-numerical-linear-al F01798</td><td>138</td><td>0.630</td><td>\Longleftrightarrow \boldsymbol{A}^{*} \boldsymbol{A}</td><td>$\Longleftrightarrow \boldsymbol{A}^{*} \boldsymbol{A}$</td><td>[1]Show that we have equality $\iff$ $\boldsymbol{R}$ is diagonal $\iff$ $\boldsymbol{A}^*\boldsymbol{A}$ is diagonal.</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01799	139	1.000	`1 \leq r \leq n, \boldsymbol{v}_{r} \in \mathbb{R}^{r}, \boldsymbol{v}_{r} \neq \mathbf{0}`	$1 \leq r \leq n, \boldsymbol{v}_{r} \in \mathbb{R}^{r}, \boldsymbol{v}_{r} \neq \mathbf{0}$	**b)** Let $1 \le r \le n$, $\boldsymbol{v}_r \in \mathbb{R}^r$, $\boldsymbol{v}_r \neq \mathbf{0}$, and
lyche-numerical-linear-algebra F01800	139	1.000	`\boldsymbol{H}:=\left[\begin{array}{cc}\boldsymbol{V}_{r} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}_{n-r} \end{array}\right]`	$\boldsymbol{H}:=\left[\begin{array}{cc}\boldsymbol{V}_{r} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}_{n-r} \end{array}\right]$	Show that $\boldsymbol{H} := \left[\begin{smallmatrix}\boldsymbol{V}_r & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}_{n-r}\end{smallmatrix}\right]$ is a Householder transformation. Show also that if
lyche-numerical-linear-algebra F01801	139	1.000	`a_{i, j}=0`	$a_{i, j}=0$	$a_{i,j} = 0$ for $i = 1,\ldots,r$ and $j = r+1,\ldots,n$ then the last r columns of $\boldsymbol{A}$ and
lyche-numerical-linear-algebra F01802	139	1.000	`i=1, \ldots, r`	$i=1, \ldots, r$	$a_{i,j} = 0$ for $i = 1,\ldots,r$ and $j = r+1,\ldots,n$ then the last r columns of $\boldsymbol{A}$ and
lyche-numerical-linear-algebra F01803	139	1.000	`j=r+1, \ldots, n`	$j=r+1, \ldots, n$	$a_{i,j} = 0$ for $i = 1,\ldots,r$ and $j = r+1,\ldots,n$ then the last r columns of $\boldsymbol{A}$ and
lyche-numerical-linear-algebra F01804	139	1.000	`\boldsymbol{H} \boldsymbol{A}`	$\boldsymbol{H} \boldsymbol{A}$	$\boldsymbol{H}\boldsymbol{A}$ are the same.
lyche-numerical-linear-algebra F01805	139	1.000	`\boldsymbol{H}_{1}, \ldots, \boldsymbol{H}_{n-1}`	$\boldsymbol{H}_{1}, \ldots, \boldsymbol{H}_{n-1}$	$\boldsymbol{A} \in \mathbb{R}^{n\times n}$ can find Householder transformations $\boldsymbol{H}_1,\ldots,\boldsymbol{H}_{n-1}$ such that
lyche-numerical-linear-algebra F01806	139	1.000	`\boldsymbol{H}_{n-1}, \ldots, \boldsymbol{H}_{1} \boldsymbol{A}`	$\boldsymbol{H}_{n-1}, \ldots, \boldsymbol{H}_{1} \boldsymbol{A}$	$\boldsymbol{H}_{n-1},\ldots,\boldsymbol{H}_1\boldsymbol{A}$ is lower triangular. Give a $\boldsymbol{Q}\boldsymbol{L}$ factorization of $\boldsymbol{A}$.
lyche-numerical-linear-algebra F01807	139	1.000	`d \leq n-1`	$d \leq n-1$	$\mathbb{R}^{n\times n}$ be a nonsingular symmetric band matrix with bandwidth $d \le n-1$, so that
lyche-numerical-linear-algebra F01808	139	1.000	`\boldsymbol{B}:=\boldsymbol{A}^{T} \boldsymbol{A}`	$\boldsymbol{B}:=\boldsymbol{A}^{T} \boldsymbol{A}$	$a_{ij} = 0$ for all i, j with $\lvert i-j\rvert > d$. We define $\boldsymbol{B} := \boldsymbol{A}^T\boldsymbol{A}$ and let $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ be the
lyche-numerical-linear-algebra F01809	139	1.000	`\leq 2 d`	$\leq 2 d$	**b)** Show that $\boldsymbol{B}$ has bandwidth $\le 2d$.
lyche-numerical-linear-algebra F01810	139	0.426	`\mathrm{B}=`	$\mathrm{B}=$	**c)** Write a MATLAB `function B=ata(A,d)` which computes $\boldsymbol{B}$. You shall
lyche-numerical-linear-algebra F01811	139	0.426	`(\mathrm{A}, \mathrm{d})`	(A, d)	**c)** Write a MATLAB `function B=ata(A,d)` which computes $\boldsymbol{B}$. You shall
lyche-numerical-linear-algebra F01812	139	1.000	`\mathcal{O}\left(c n^{2}\right)`	$\mathcal{O}\left(c n^{2}\right)$	exploit the symmetry and the function should only use $\mathcal{O}(cn^2)$ flops, where c
lyche-numerical-linear-algebra F01813	139	1.000	`\boldsymbol{A}^{T} \boldsymbol{A}=\boldsymbol{R}^{T} \boldsymbol{R}`	$\boldsymbol{A}^{T} \boldsymbol{A}=\boldsymbol{R}^{T} \boldsymbol{R}$	**e)** Show that $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{R}^T\boldsymbol{R}$.
lyche-numerical-linear-algebra F01814	139	1.000	`2 d`	$2 d$	**f)** Explain why $\boldsymbol{R}$ has upper bandwidth $2d$.
lyche-numerical-linear-algebra F01815	140	0.928	`\boldsymbol{x}=\left[\begin{array}{c}r \cos \alpha \\ r \sin \alpha\end{array}\right]`	$\boldsymbol{x}=\left[\begin{array}{c}r \cos \alpha \\ r \sin \alpha\end{array}\right]$	**Exercise 5.16 (Plane Rotation)** Show that if $\boldsymbol{x} = \left[\begin{smallmatrix}r\cos\alpha \\ r\sin\alpha\end{smallmatrix}\right]$ then $\boldsymbol{P}\boldsymbol{x} =$
lyche-numerical-linear-algebra F01816	140	0.928	`\boldsymbol{P} \boldsymbol{x}=`	$\boldsymbol{P} \boldsymbol{x}=$	**Exercise 5.16 (Plane Rotation)** Show that if $\boldsymbol{x} = \left[\begin{smallmatrix}r\cos\alpha \\ r\sin\alpha\end{smallmatrix}\right]$ then $\boldsymbol{P}\boldsymbol{x} =$
lyche-numerical-linear-algebra F01817	140	0.711	`\left[\begin{array}{l}r \cos (\alpha-\theta) \\ r \sin ( \alpha-\theta)\end{array}\right]`	$\left[\begin{array}{l}r \cos (\alpha-\theta) \\ r \sin (\alpha-\theta)\end{array}\right]$	$\left[\begin{smallmatrix}r\cos(\alpha-\theta) \\ r\sin(\alpha-\theta)\end{smallmatrix}\right]$.
lyche-numerical-linear-algebra F01818	140	0.988	`\boldsymbol{H} \in \mathbb{R}^{4,4}`	$\boldsymbol{H} \in \mathbb{R}^{4,4}$	**Exercise 5.17 (Updating the QR Decomposition)** Let $\boldsymbol{H} \in \mathbb{R}^{4,4}$ be upper
lyche-numerical-linear-algebra F01819	140	1.000	`\boldsymbol{G}_{1}, \boldsymbol{G}_{2}, \boldsymbol{G}_{3}`	$\boldsymbol{G}_{1}, \boldsymbol{G}_{2}, \boldsymbol{G}_{3}$	Hessenberg. Find Givens rotation matrices $\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3$ such that
lyche-numerical-linear-algebra F01820	140	1.000	`\boldsymbol{G}_{k}=\boldsymbol{P}_{i, j}`	$\boldsymbol{G}_{k}=\boldsymbol{P}_{i, j}$	is upper triangular. (Here each $\boldsymbol{G}_k = \boldsymbol{P}_{i,j}$ for suitable i, j, c og s, and for each k
lyche-numerical-linear-algebra F01821	140	1.000	`i, j, c`	i, j, c	is upper triangular. (Here each $\boldsymbol{G}_k = \boldsymbol{P}_{i,j}$ for suitable i, j, c og s, and for each k
lyche-numerical-linear-algebra F01822	140	0.999	`\boldsymbol{P}_{k, k+1}`	$\boldsymbol{P}_{k, k+1}$	solves the linear system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ using rotations $\boldsymbol{P}_{k,k+1}$ for $k = 1,\ldots,n-1$. It
lyche-numerical-linear-algebra F01823	140	0.838	`\boldsymbol{G}:=\left[\begin{array}{rc}c & s \\ -s & c \end{array}\right] \in \mathbb{R}^{2 \times 2}`	$\boldsymbol{G}:=\left[\begin{array}{rc}c & s \\ -s & c \end{array}\right] \in \mathbb{R}^{2 \times 2}$	rotation of order 2 has the form $\boldsymbol{G} := \left[\begin{smallmatrix}c & s \\ -s & c\end{smallmatrix}\right] \in \mathbb{R}^{2\times 2}$, where $s^2 + c^2 = 1$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01824	140	0.838	s^{2}+c^{2}=1	$s^{2}+c^{2}=1$	rotation of order 2 has the form $\boldsymbol{G}:=\left[\begin{array}{cc}c & s\\ -s & c\end{array}\right]\in\mathbb{R}^{2\times 2}$, where $s^2+c^2=1$.
lyche-numerical-linear-algebra_F01825	141	1.000	\boldsymbol{G}	$\boldsymbol{G}$	**a)** Is $\boldsymbol{G}$ symmetric and unitary?
lyche-numerical-linear-algebra_F01826	141	1.000	x_{1}, x_{2} \in \mathbb{R}	$x_{1}, x_{2} \in \mathbb{R}$	**b)** Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2+x_2^2}$. Find $\boldsymbol{G}$ and y_1, y_2 so that $y_1 = y_2$,
lyche-numerical-linear-algebra_F01827	141	1.000	r:=\sqrt{x_{1}^{2}+x_{2}^{2}}	$r:=\sqrt{x_{1}^{2}+x_{2}^{2}}$	**b)** Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2+x_2^2}$. Find $\boldsymbol{G}$ and y_1, y_2 so that $y_1 = y_2$,
lyche-numerical-linear-algebra_F01828	141	1.000	y_{1}, y_{2}	y_{1}, y_{2}	**b)** Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2+x_2^2}$. Find $\boldsymbol{G}$ and y_1, y_2 so that $y_1 = y_2$,
lyche-numerical-linear-algebra_F01829	141	1.000	y_{1}=y_{2}	$y_{1}=y_{2}$	**b)** Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2+x_2^2}$. Find $\boldsymbol{G}$ and y_1, y_2 so that $y_1 = y_2$,
lyche-numerical-linear-algebra_F01830	141	1.000	\left[\begin{array}{l}y_{1} \\\\ y_{2}\end{array}\right]= \boldsymbol{G}\left[\begin{array}{l}x_{1} \\\\ x_{2} \end{array}\right]	$\left[\begin{array}{l}y_{1} \\ y_{2}\end{array}\right]=\boldsymbol{G}\left[\begin{array}{l}x_{1} \\ x_{2}\end{array}\right]$	where $\left[\begin{array}{l}y_1\\ y_2\end{array}\right]=\boldsymbol{G}\left[\begin{array}{l}x_1\\ x_2\end{array}\right]$.
lyche-numerical-linear-algebra_F01831	141	1.000	i j	$i j$	rotation in the ij-plane is an $m\times m$-matrix, denoted $\boldsymbol{P}_{i,j}$, which differs from the
lyche-numerical-linear-algebra_F01832	141	1.000	\boldsymbol{P}_{i, j}	$\boldsymbol{P}_{i, j}$	rotation in the ij-plane is an $m\times m$-matrix, denoted $\boldsymbol{P}_{i,j}$, which differs from the
lyche-numerical-linear-algebra_F01833	141	0.938	i i, i j, j i, j j	$i i, i j, j i, j j$	identity matrix only in the entries ii, ij, ji, jj, which equal
lyche-numerical-linear-algebra_F01834	141	1.000	\theta \in \mathbb{R}	$\theta \in \mathbb{R}$	**a)** For $\theta\in\mathbb{R}$, let $\boldsymbol{P}$ be a Givens rotation of the form
lyche-numerical-linear-algebra_F01835	141	1.000	\theta \in(-\pi / 2, \pi / 2]	$\theta \in(-\pi / 2, \pi / 2]$	and let $\boldsymbol{x}$ be a fixed vector in $\mathbb{R}^2$. Show that there exists a unique $\theta\in(-\pi/2,\pi/2]$
lyche-numerical-linear-algebra_F01836	141	0.910	\boldsymbol{P} \boldsymbol{x}= \pm\\|\boldsymbol{x}\\|_{2} \boldsymbol{e}_{1}	$\boldsymbol{P} \boldsymbol{x}= \pm\|\boldsymbol{x}\|_{2} \boldsymbol{e}_{1}$	so that $\boldsymbol{P}\boldsymbol{x}=\pm\|\boldsymbol{x}\|_2\boldsymbol{e}_1$, where $\boldsymbol{e}_1=(1,0)^T$.
lyche-numerical-linear-algebra_F01837	141	0.910	\boldsymbol{e}_{1}=(1,0)^{T}	$\boldsymbol{e}_{1}=(1,0)^{T}$	so that $\boldsymbol{P}\boldsymbol{x}=\pm\|\boldsymbol{x}\|_2\boldsymbol{e}_1$, where $\boldsymbol{e}_1=(1,0)^T$.
lyche-numerical-linear-algebra_F01838	141	1.000	\boldsymbol{w} \in \mathbb{R}^{m}	$\boldsymbol{w} \in \mathbb{R}^{m}$	**b)** Show that, for any vector $\boldsymbol{w}\in\mathbb{R}^m$, one can find rotations in the 12-plane, 23-
lyche-numerical-linear-algebra_F01839	141	0.481	(m-1) m	$(m-1) m$	plane, ..., $(m-1)m$-plane, so that
lyche-numerical-linear-algebra_F01840	141	0.989	\alpha= \pm\\|\boldsymbol{w}\\|_{2}	$\alpha= \pm\|\boldsymbol{w}\|_{2}$	where $\alpha=\pm\|\boldsymbol{w}\|_2$.
lyche-numerical-linear-algebra_F01841	142	1.000	\boldsymbol{A}_{-}	$\boldsymbol{A}_{-}$	**d)** Let again $\boldsymbol{A}$ be an $m\times n$-matrix with $m\geq n$, and let $\boldsymbol{A}_-$ be the matrix obtained
lyche-numerical-linear-algebra_F01842	142	0.916	\boldsymbol{A} .{ }^{2}	$\boldsymbol{A} .{ }^{2}$	$\boldsymbol{A}_-$, when we already have a QR decomposition $\boldsymbol{A}=\boldsymbol{Q}\boldsymbol{R}$ of $\boldsymbol{A}$.[2]
lyche-numerical-linear-algebra_F01843	142	0.936	z	z	Assume also that $\boldsymbol{z}$ is a given column vector of length n.
lyche-numerical-linear-algebra_F01844	142	0.999	\boldsymbol{A}+\boldsymbol{z} \boldsymbol{z}^{*}	$\boldsymbol{A}+\boldsymbol{z} \boldsymbol{z}^{*}$	**a)** Explain why $\boldsymbol{A}+\boldsymbol{z}\boldsymbol{z}^*$ has a unique Cholesky factorization.
lyche-numerical-linear-algebra_F01845	142	1.000	\boldsymbol{R}^{*} \boldsymbol{R}	$\boldsymbol{R}^{*} \boldsymbol{R}$	factorization $\boldsymbol{R}^*\boldsymbol{R}$.
lyche-numerical-linear-algebra_F01846	142	0.987	P_{i_{1}, n+1}, P_{i_{2}, n+1}, \ldots, P_{i_{n}, n+1}	$P_{i_{1}, n+1}, P_{i_{2}, n+1}, \ldots, P_{i_{n}, n+1}$	**c)** Explain how one can find plane rotations $P_{i_1,n+1}, P_{i_2,n+1},\ldots,P_{i_n,n+1}$ so that
lyche-numerical-linear-algebra_F01847	142	1.000	\boldsymbol{R}^{\prime}	$\boldsymbol{R}^{\prime}$	with $\boldsymbol{R}'$ upper triangular, and explain how to obtain a QR decomposition of

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01848	142	0.996	`\left[\begin{array}{c}\boldsymbol{L}^{*} \\ \boldsymbol{z}^{*}\end{array}\right]`	$\left[\begin{array}{c}\boldsymbol{L}^{*} \\ \boldsymbol{z}^{*}\end{array}\right]$	$\begin{bmatrix}\boldsymbol{L}^*\\ z^*\end{bmatrix}$ from this. In particular you should write down the numbers $i_1,\ldots,i_n$.
lyche-numerical-linear-algebra F01849	142	0.996	`i_{1}, \ldots, i_{n}`	$i_{1}, \ldots, i_{n}$	$\begin{bmatrix}\boldsymbol{L}^*\\ z^*\end{bmatrix}$ from this. In particular you should write down the numbers $i_1,\ldots,i_n$.
lyche-numerical-linear-algebra F01850	142	1.000	`\boldsymbol{Q}^{T} \boldsymbol{A}_{-}`	$\boldsymbol{Q}^{T} \boldsymbol{A}_{-}$	[2]Consider the matrix $\boldsymbol{Q}^T\boldsymbol{A}_-$.
lyche-numerical-linear-algebra F01851	146	0.999	`\left(\lambda_{k}, \boldsymbol{x}_{k}\right), k=`	$\left(\lambda_{k}, \boldsymbol{x}_{k}\right), k=$	***Proof*** The proof is by contradiction. Suppose $\boldsymbol{A}$ has eigenpairs $(\lambda_k,\boldsymbol{x}_k)$, k =
lyche-numerical-linear-algebra F01852	146	0.999	`\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{m} \right\}`	$\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{m} \right\}$	that $\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_m\}$ is linearly dependent. Thus $\sum_{j=1}^m c_j\boldsymbol{x}_j=\boldsymbol{0}$, where at least one
lyche-numerical-linear-algebra F01853	146	0.999	`\sum_{j=1}^{m} c_{j} \boldsymbol{x}_{j}=\mathbf{0}`	$\sum_{j=1}^{m} c_{j} \boldsymbol{x}_{j}=\mathbf{0}$	that $\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_m\}$ is linearly dependent. Thus $\sum_{j=1}^m c_j\boldsymbol{x}_j=\boldsymbol{0}$, where at least one
lyche-numerical-linear-algebra F01854	146	1.000	`c_{j}`	c_{j}	c_j is nonzero. We must have $m\geq 2$ since eigenvectors are nonzero. We find
lyche-numerical-linear-algebra F01855	146	1.000	`m \geq 2`	$m \geq 2$	c_j is nonzero. We must have $m\geq 2$ since eigenvectors are nonzero. We find
lyche-numerical-linear-algebra F01856	146	1.000	`\sum_{j=1}^{m} c_{j} \lambda_{m} \boldsymbol{x}_{j}= \mathbf{0}`	$\sum_{j=1}^{m} c_{j} \lambda_{m} \boldsymbol{x}_{j}= \mathbf{0}$	From the last relation we subtract $\sum_{j=1}^m c_j\lambda_m\boldsymbol{x}_j=\boldsymbol{0}$ and find $\sum_{j=1}^{m-1} c_j(\lambda_j -$
lyche-numerical-linear-algebra F01857	146	1.000	`\sum_{j=1}^{m-1} c_{j}\left(\lambda_{j}-\right.`	$\sum_{j=1}^{m-1} c_{j}\left(\lambda_{j}-\right.$	From the last relation we subtract $\sum_{j=1}^m c_j\lambda_m\boldsymbol{x}_j=\boldsymbol{0}$ and find $\sum_{j=1}^{m-1} c_j(\lambda_j -$
lyche-numerical-linear-algebra F01858	146	1.000	`\left.\lambda_{m}\right) \boldsymbol{x}_{j}=\mathbf{0}`	$\left.\lambda_{m}\right) \boldsymbol{x}_{j}=\mathbf{0}$	$\lambda_m)\boldsymbol{x}_j=\boldsymbol{0}$. But since $\lambda_j-\lambda_m\neq 0$ for $j=1,\ldots,m-1$ and at least one
lyche-numerical-linear-algebra F01859	146	1.000	`\lambda_{j}-\lambda_{m} \neq 0`	$\lambda_{j}-\lambda_{m} \neq 0$	$\lambda_m)\boldsymbol{x}_j=\boldsymbol{0}$. But since $\lambda_j-\lambda_m\neq 0$ for $j=1,\ldots,m-1$ and at least one
lyche-numerical-linear-algebra F01860	146	1.000	`j=1, \ldots, m-1`	$j=1, \ldots, m-1$	$\lambda_m)\boldsymbol{x}_j=\boldsymbol{0}$. But since $\lambda_j-\lambda_m\neq 0$ for $j=1,\ldots,m-1$ and at least one
lyche-numerical-linear-algebra F01861	146	1.000	`c_{j} \neq 0`	$c_{j} \neq 0$	$c_j\neq 0$ for $j<m$ we see that $\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_{m-1}\}$ is linearly dependent, contradicting
lyche-numerical-linear-algebra F01862	146	1.000	`j<m`	$j<m$	$c_j\neq 0$ for $j<m$ we see that $\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_{m-1}\}$ is linearly dependent, contradicting
lyche-numerical-linear-algebra F01863	146	1.000	`\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{m-1} \right\}`	$\left\{\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{m-1} \right\}$	$c_j\neq 0$ for $j<m$ we see that $\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_{m-1}\}$ is linearly dependent, contradicting
lyche-numerical-linear-algebra F01864	146	1.000	`\boldsymbol{I} \boldsymbol{x}=\boldsymbol{x}`	$\boldsymbol{I} \boldsymbol{x}=\boldsymbol{x}$	Since $\boldsymbol{I}\boldsymbol{x}=\boldsymbol{x}$ and $\lambda_1=\lambda_2=1$ any vector $\boldsymbol{x}\in\mathbb{C}^2$ is an eigenvector for $\boldsymbol{I}$. In
lyche-numerical-linear-algebra F01865	146	1.000	`\lambda_{1}=\lambda_{2}=1`	$\lambda_{1}=\lambda_{2}=1$	Since $\boldsymbol{I}\boldsymbol{x}=\boldsymbol{x}$ and $\lambda_1=\lambda_2=1$ any vector $\boldsymbol{x}\in\mathbb{C}^2$ is an eigenvector for $\boldsymbol{I}$. In
lyche-numerical-linear-algebra F01866	146	1.000	`\boldsymbol{x} \in \mathbb{C}^{2}`	$\boldsymbol{x} \in \mathbb{C}^{2}$	Since $\boldsymbol{I}\boldsymbol{x}=\boldsymbol{x}$ and $\lambda_1=\lambda_2=1$ any vector $\boldsymbol{x}\in\mathbb{C}^2$ is an eigenvector for $\boldsymbol{I}$. In
lyche-numerical-linear-algebra F01867	146	1.000	`\boldsymbol{e}_{2}`	$\boldsymbol{e}_{2}$	particular the two unit vectors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ are eigenvectors and form an orthonormal
lyche-numerical-linear-algebra F01868	146	1.000	`\mathbb{C}^{2}`	$\mathbb{C}^{2}$	basis for $\mathbb{C}^2$. We conclude that the identity matrix is nondefective. The matrix $\boldsymbol{J}$
lyche-numerical-linear-algebra F01869	146	1.000	`\boldsymbol{J}`	$\boldsymbol{J}$	basis for $\mathbb{C}^2$. We conclude that the identity matrix is nondefective. The matrix $\boldsymbol{J}$
lyche-numerical-linear-algebra F01870	146	1.000	`\boldsymbol{J} \boldsymbol{x}=\boldsymbol{x}`	$\boldsymbol{J} \boldsymbol{x}=\boldsymbol{x}$	also has the eigenvalue one with multiplicity two, but since $\boldsymbol{J}\boldsymbol{x}=\boldsymbol{x}$ if and only if
lyche-numerical-linear-algebra F01871	146	1.000	`x_{2}=0`	$x_{2}=0$	$x_2=0$, any eigenvector must be a multiple of $\boldsymbol{e}_1$. Thus $\boldsymbol{J}$ is defective.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01872	146	1.000	`\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right]`	$\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right]$	*Example 6.2 (Eigenvector Expansion Example)* Eigenpairs of $\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$ are
lyche-numerical-linear-algebra_F01873	146	1.000	`\left(1,[1,1]^{T}\right)`	$\left(1,[1,1]^{T}\right)$	$(1,[1,1]^T)$ and $(3,[1,-1]^T)$. Any $\boldsymbol{x} = [x_1,x_2]^T \in \mathbb{C}^2$ has the eigenvector
lyche-numerical-linear-algebra_F01874	146	1.000	`\left(3,[1,-1]^{T}\right)`	$\left(3,[1,-1]^{T}\right)$	$(1,[1,1]^T)$ and $(3,[1,-1]^T)$. Any $\boldsymbol{x} = [x_1,x_2]^T \in \mathbb{C}^2$ has the eigenvector
lyche-numerical-linear-algebra_F01875	146	1.000	`\boldsymbol{x}=\left[x_{1}, x_{2}\right]^{T} \in \mathbb{C}^{2}`	$\boldsymbol{x}=\left[x_{1}, x_{2}\right]^{T} \in \mathbb{C}^{2}$	$(1,[1,1]^T)$ and $(3,[1,-1]^T)$. Any $\boldsymbol{x} = [x_1,x_2]^T \in \mathbb{C}^2$ has the eigenvector
lyche-numerical-linear-algebra_F01876	147	1.000	`\boldsymbol{S} \in \mathbb{C}^{n \times n}`	$\boldsymbol{S} \in \mathbb{C}^{n \times n}$	**similar** if there is a nonsingular matrix $\boldsymbol{S} \in \mathbb{C}^{n\times n}$ such that $\boldsymbol{B} = \boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}$. The
lyche-numerical-linear-algebra_F01877	147	1.000	`\boldsymbol{B}=\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}`	$\boldsymbol{B}=\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$	**similar** if there is a nonsingular matrix $\boldsymbol{S} \in \mathbb{C}^{n\times n}$ such that $\boldsymbol{B} = \boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}$. The
lyche-numerical-linear-algebra_F01878	147	1.000	`\boldsymbol{A} \rightarrow \boldsymbol{B}`	$\boldsymbol{A} \rightarrow \boldsymbol{B}$	transformation $\boldsymbol{A} \rightarrow \boldsymbol{B}$ is called a **similarity transformation**. The columns of $\boldsymbol{S}$
lyche-numerical-linear-algebra_F01879	147	1.000	`\boldsymbol{s}_{1}, \boldsymbol{s}_{2}, \ldots, \boldsymbol{s}_{n}`	$\boldsymbol{s}_{1}, \boldsymbol{s}_{2}, \ldots, \boldsymbol{s}_{n}$	are denoted by $\boldsymbol{s}_1, \boldsymbol{s}_2, \ldots, \boldsymbol{s}_n$.
lyche-numerical-linear-algebra_F01880	147	1.000	`\operatorname{det}(\boldsymbol{A} \boldsymbol{C})=`	$\operatorname{det}(\boldsymbol{A} \boldsymbol{C})=$	teristic polynomial. Indeed, by the product rule for determinants $\det(\boldsymbol{A}\boldsymbol{C}) =$
lyche-numerical-linear-algebra_F01881	147	1.000	`\operatorname{det}(\boldsymbol{A}) \operatorname{det}(\boldsymbol{C})`	$\operatorname{det}(\boldsymbol{A}) \operatorname{det}(\boldsymbol{C})$	$\det(\boldsymbol{A})\det(\boldsymbol{C})$ so that
lyche-numerical-linear-algebra_F01882	147	0.866	`\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}`	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$	2. $(\lambda, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}$ if and only if $(\lambda, \boldsymbol{S}\boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}$. In
lyche-numerical-linear-algebra_F01883	147	0.866	`\lambda, \boldsymbol{S} \boldsymbol{x}`	$\lambda, \boldsymbol{S} \boldsymbol{x}$	2. $(\lambda, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}$ if and only if $(\lambda, \boldsymbol{S}\boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}$. In
lyche-numerical-linear-algebra_F01884	147	1.000	`\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}\right) \boldsymbol{x}=\lambda \boldsymbol{x}`	$\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}\right) \boldsymbol{x}=\lambda \boldsymbol{x}$	fact $(\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S})\boldsymbol{x} = \lambda \boldsymbol{x}$ if and only if $\boldsymbol{A}(\boldsymbol{S}\boldsymbol{x}) = \lambda(\boldsymbol{S}\boldsymbol{x})$.
lyche-numerical-linear-algebra_F01885	147	1.000	`\boldsymbol{A}(\boldsymbol{S} \boldsymbol{x})=\lambda(\boldsymbol{S} \boldsymbol{x})`	$\boldsymbol{A}(\boldsymbol{S} \boldsymbol{x})=\lambda(\boldsymbol{S} \boldsymbol{x})$	fact $(\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S})\boldsymbol{x} = \lambda \boldsymbol{x}$ if and only if $\boldsymbol{A}(\boldsymbol{S}\boldsymbol{x}) = \lambda(\boldsymbol{S}\boldsymbol{x})$.
lyche-numerical-linear-algebra_F01886	147	1.000	`\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}=\boldsymbol{D}=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)`	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}=\boldsymbol{D}=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)$	3. If $\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S} = \boldsymbol{D} = \operatorname{diag}(\lambda_1,\ldots,\lambda_n)$ we can partition $\boldsymbol{A}\boldsymbol{S} = \boldsymbol{S}\boldsymbol{D}$ by
lyche-numerical-linear-algebra_F01887	147	1.000	`\boldsymbol{A} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{D}`	$\boldsymbol{A} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{D}$	3. If $\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S} = \boldsymbol{D} = \operatorname{diag}(\lambda_1,\ldots,\lambda_n)$ we can partition $\boldsymbol{A}\boldsymbol{S} = \boldsymbol{S}\boldsymbol{D}$ by
lyche-numerical-linear-algebra_F01888	147	1.000	`\left[\boldsymbol{A} \boldsymbol{s}_{1}, \ldots, \boldsymbol{A} \boldsymbol{s}_{n}\right]=\left[\lambda_{1} \boldsymbol{s}_{1}, \ldots, \lambda_{n} \boldsymbol{s}_{n}\right]`	$\left[\boldsymbol{A} \boldsymbol{s}_{1}, \ldots, \boldsymbol{A} \boldsymbol{s}_{n}\right]=\left[\lambda_{1} \boldsymbol{s}_{1}, \ldots, \lambda_{n} \boldsymbol{s}_{n}\right]$	columns to obtain $[\boldsymbol{A}\boldsymbol{s}_1,\ldots,\boldsymbol{A}\boldsymbol{s}_n] = [\lambda_1 \boldsymbol{s}_1,\ldots,\lambda_n \boldsymbol{s}_n]$. Thus the columns of
lyche-numerical-linear-algebra_F01889	147	1.000	`\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n \times n}`	$\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$	4. For any square matrices $\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n\times n}$ the two products $\boldsymbol{A}\boldsymbol{C}$ and $\boldsymbol{C}\boldsymbol{A}$ have the
lyche-numerical-linear-algebra_F01890	147	1.000	`\boldsymbol{A} \boldsymbol{C}`	$\boldsymbol{A} \boldsymbol{C}$	4. For any square matrices $\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n\times n}$ the two products $\boldsymbol{A}\boldsymbol{C}$ and $\boldsymbol{C}\boldsymbol{A}$ have the
lyche-numerical-linear-algebra_F01891	147	1.000	`\boldsymbol{C} \boldsymbol{A}`	$\boldsymbol{C} \boldsymbol{A}$	4. For any square matrices $\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n\times n}$ the two products $\boldsymbol{A}\boldsymbol{C}$ and $\boldsymbol{C}\boldsymbol{A}$ have the
lyche-numerical-linear-algebra_F01892	147	1.000	`\boldsymbol{C} \in \mathbb{C}^{n \times m}`	$\boldsymbol{C} \in \mathbb{C}^{n \times m}$	$\mathbb{C}^{m\times n}$ and $\boldsymbol{C} \in \mathbb{C}^{n\times m}$, with say $m > n$, the bigger matrix has $m-n$ extra zero
lyche-numerical-linear-algebra_F01893	147	1.000	`n+m`	$n+m$	To show this define for any $m, n \in \mathbb{N}$ block triangular matrices of order $n+m$ by
lyche-numerical-linear-algebra_F01894	147	0.997	`\boldsymbol{S}^{-1}=\left[\begin{array}{cc}\boldsymbol{I} & -\boldsymbol{A} \\ \mathbf{0} & \boldsymbol{I} \end{array}\right]`	$\boldsymbol{S}^{-1}=\left[\begin{array}{cc}\boldsymbol{I} & -\boldsymbol{A} \\ \mathbf{0} & \boldsymbol{I} \end{array}\right]$	The matrix $\boldsymbol{S}$ is nonsingular with $\boldsymbol{S}^{-1} = \begin{bmatrix}\boldsymbol{I} & -\boldsymbol{A}\\ \mathbf{0} & \boldsymbol{I}\end{bmatrix}$. Moreover, $\boldsymbol{E}\boldsymbol{S} = \boldsymbol{S}\boldsymbol{F}$ so

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F01895	147	0.997	`\boldsymbol{E} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{F}`	$\boldsymbol{E} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{F}$	The matrix $\boldsymbol{S}$ is nonsingular with $\boldsymbol{S}^{-1} = \begin{bmatrix} \boldsymbol{I} & -\boldsymbol{A} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix}$. Moreover, $\boldsymbol{ES} = \boldsymbol{SF}$ so
lyche-numerical-linear-algebra_F01896	147	1.000	`\boldsymbol{F}`	$\boldsymbol{F}$	$\boldsymbol{E}$ and $\boldsymbol{F}$ are similar and therefore have the same characteristic polynomials.
lyche-numerical-linear-algebra_F01897	147	1.000	`\pi_{\boldsymbol{E}}(\lambda)=\lambda^{n} \pi_{ \boldsymbol{A} \boldsymbol{C}}(\lambda)=\pi_{ \boldsymbol{F}}(\lambda)=\lambda^{m} \pi_{\boldsymbol{C} \boldsymbol{A}}(\lambda)`	$\pi_{\boldsymbol{E}}(\lambda)=\lambda^{n} \pi_{\boldsymbol{A} \boldsymbol{C}}(\lambda)=\pi_{\boldsymbol{F}}(\lambda)=\lambda^{m} \pi_{\boldsymbol{C} \boldsymbol{A}}(\lambda)$	diagonal blocks. But then $\pi_{\boldsymbol{E}}(\lambda) = \lambda^n\pi_{\boldsymbol{AC}}(\lambda) = \pi_{\boldsymbol{F}}(\lambda) = \lambda^m\pi_{\boldsymbol{CA}}(\lambda)$. This
lyche-numerical-linear-algebra_F01898	148	1.000	`\lambda_{1}, \ldots, \lambda_{k}`	$\lambda_{1}, \ldots, \lambda_{k}$	Suppose $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ has k distinct eigenvalues $\lambda_1, \ldots, \lambda_k$ with multiplicities
lyche-numerical-linear-algebra_F01899	148	1.000	`a_{1}, \ldots, a_{k}`	$a_{1}, \ldots, a_{k}$	$a_1, \ldots, a_k$ so that
lyche-numerical-linear-algebra_F01900	148	0.890	`a_{i}=a\left(\lambda_{i}\right)=a_{\boldsymbol{A}}\left( \lambda_{i}\right)`	$a_{i}=a\left(\lambda_{i}\right)=a_{\boldsymbol{A}}\left(\lambda_{i}\right)$	The positive integer $a_i = a(\lambda_i) = a_{\boldsymbol{A}}(\lambda_i)$ is called the **multiplicity**, or more
lyche-numerical-linear-algebra_F01901	148	1.000	`\lambda_{i}`	λ_{i}	precisely the **algebraic multiplicity** of the eigenvalue λ_i. The multiplicity of an
lyche-numerical-linear-algebra_F01902	148	1.000	`a_{i}`	a_{i}	eigenvalue is simple (double, triple) if a_i is equal to one (two, three).
lyche-numerical-linear-algebra_F01903	148	1.000	`\lambda \in \sigma(\boldsymbol{A})`	$\lambda \in \sigma(\boldsymbol{A})$	To define a second kind of multiplicity we consider for each $\lambda \in \sigma(\boldsymbol{A})$ the
lyche-numerical-linear-algebra_F01904	148	1.000	`g=g(\lambda)=`	$g=g(\lambda)=$	**Definition 6.2 (Geometric Multiplicity)** The **geometric multiplicity** $g = g(\lambda) =$
lyche-numerical-linear-algebra_F01905	148	1.000	`g_{\boldsymbol{A}}(\lambda)`	$g_{\boldsymbol{A}}(\lambda)$	$g_{\boldsymbol{A}}(\lambda)$ of an eigenvalue λ of $\boldsymbol{A}$ is the dimension of the nullspace $\mathcal{N}(\boldsymbol{A} - \lambda\boldsymbol{I})$.
lyche-numerical-linear-algebra_F01906	148	1.000	`\mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I})`	$\mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I})$	$g_{\boldsymbol{A}}(\lambda)$ of an eigenvalue λ of $\boldsymbol{A}$ is the dimension of the nullspace $\mathcal{N}(\boldsymbol{A} - \lambda\boldsymbol{I})$.
lyche-numerical-linear-algebra_F01907	148	1.000	`\lambda=1`	$\lambda=1$	value $\lambda = 1$ with $\pi_{\boldsymbol{I}}(\lambda) = (1-\lambda)^n$. Since $\boldsymbol{I} - \lambda\boldsymbol{I}$ is the zero matrix when $\lambda = 1$,
lyche-numerical-linear-algebra_F01908	148	1.000	`\pi_{\boldsymbol{I}}(\lambda)=(1-\lambda)^{n}`	$\pi_{\boldsymbol{I}}(\lambda)=(1-\lambda)^{n}$	value $\lambda = 1$ with $\pi_{\boldsymbol{I}}(\lambda) = (1-\lambda)^n$. Since $\boldsymbol{I} - \lambda\boldsymbol{I}$ is the zero matrix when $\lambda = 1$,
lyche-numerical-linear-algebra_F01909	148	1.000	`\boldsymbol{I}-\lambda \boldsymbol{I}`	$\boldsymbol{I}-\lambda \boldsymbol{I}$	value $\lambda = 1$ with $\pi_{\boldsymbol{I}}(\lambda) = (1-\lambda)^n$. Since $\boldsymbol{I} - \lambda\boldsymbol{I}$ is the zero matrix when $\lambda = 1$,
lyche-numerical-linear-algebra_F01910	148	1.000	`a=g=n`	$a=g=n$	the nullspace of $\boldsymbol{I} - \lambda\boldsymbol{I}$ is all of n-space and it follows that $a = g = n$. On the other
lyche-numerical-linear-algebra_F01911	148	0.985	`\boldsymbol{J}:=\left[\begin{array}{ll}1 & 1 \\ 0 & 1 \end{array}\right]`	$\boldsymbol{J}:=\left[\begin{array}{ll}1 & 1 \\ 0 & 1\end{array}\right]$	hand we saw in Example 6.1 that the matrix $\boldsymbol{J} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has the eigenvalue $\lambda = 1$
lyche-numerical-linear-algebra_F01912	148	1.000	`a=2`	$a=2$	with $a = 2$ and any eigenvector is a multiple of $\boldsymbol{e}_1$. Thus $g = 1$.
lyche-numerical-linear-algebra_F01913	148	1.000	`g=1`	$g=1$	with $a = 2$ and any eigenvector is a multiple of $\boldsymbol{e}_1$. Thus $g = 1$.
lyche-numerical-linear-algebra_F01914	148	0.991	`\operatorname{dim} \mathcal{N}\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I} \right)=k`	$\operatorname{dim} \mathcal{N}\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I}\right)=k$	$\dim\mathcal{N}(\boldsymbol{S}^{-1}\boldsymbol{AS} - \lambda\boldsymbol{I}) = k$, and $\dim\mathcal{N}(\boldsymbol{A} - \lambda\boldsymbol{I}) = \ell$. We need to show that
lyche-numerical-linear-algebra_F01915	148	0.991	`\operatorname{dim} \mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I})=\ell`	$\operatorname{dim} \mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I})=\ell$	$\dim\mathcal{N}(\boldsymbol{S}^{-1}\boldsymbol{AS} - \lambda\boldsymbol{I}) = k$, and $\dim\mathcal{N}(\boldsymbol{A} - \lambda\boldsymbol{I}) = \ell$. We need to show that
lyche-numerical-linear-algebra_F01916	148	1.000	`k=\ell`	$k=\ell$	$k = \ell$. Suppose $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ is a basis for $\mathcal{N}(\boldsymbol{S}^{-1}\boldsymbol{AS} - \lambda\boldsymbol{I})$. Then $\boldsymbol{S}^{-1}\boldsymbol{AS}\boldsymbol{v}_i = \lambda\boldsymbol{v}_i$
lyche-numerical-linear-algebra_F01917	148	1.000	`\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{k}`	$\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{k}$	$k = \ell$. Suppose $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ is a basis for $\mathcal{N}(\boldsymbol{S}^{-1}\boldsymbol{AS} - \lambda\boldsymbol{I})$. Then $\boldsymbol{S}^{-1}\boldsymbol{AS}\boldsymbol{v}_i = \lambda\boldsymbol{v}_i$
lyche-numerical-linear-algebra_F01918	148	1.000	`\mathcal{N}\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I}\right)`	$\mathcal{N}\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I}\right)$	$k = \ell$. Suppose $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ is a basis for $\mathcal{N}(\boldsymbol{S}^{-1}\boldsymbol{AS} - \lambda\boldsymbol{I})$. Then $\boldsymbol{S}^{-1}\boldsymbol{AS}\boldsymbol{v}_i = \lambda\boldsymbol{v}_i$
lyche-numerical-linear-algebra_F01919	148	1.000	`\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} \boldsymbol{v}_{i}=\lambda \boldsymbol{v}_{i}`	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} \boldsymbol{v}_{i}=\lambda \boldsymbol{v}_{i}$	$k = \ell$. Suppose $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ is a basis for $\mathcal{N}(\boldsymbol{S}^{-1}\boldsymbol{AS} - \lambda\boldsymbol{I})$. Then $\boldsymbol{S}^{-1}\boldsymbol{AS}\boldsymbol{v}_i = \lambda\boldsymbol{v}_i$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01920	148	1.000	`\boldsymbol{A} \boldsymbol{S} \boldsymbol{v}_{i}=\lambda \boldsymbol{S} \boldsymbol{v}_{i}, i=1, \ldots, k`	$\boldsymbol{A} \boldsymbol{S} \boldsymbol{v}_{i}=\lambda \boldsymbol{S} \boldsymbol{v}_{i}, i=1, \ldots, k$	or $\boldsymbol{A}\boldsymbol{S}\boldsymbol{v}_i = \lambda\boldsymbol{S}\boldsymbol{v}_i$, $i = 1, \ldots, k$. But then $\{\boldsymbol{S}\boldsymbol{v}_1, \ldots, \boldsymbol{S}\boldsymbol{v}_k\} \subset \mathcal{N}(\boldsymbol{A} - \lambda\boldsymbol{I})$, which
lyche-numerical-linear-algebra F01921	148	1.000	`\left\{\boldsymbol{S} \boldsymbol{v}_{1}, \ldots, \boldsymbol{S} \boldsymbol{v}_{k}\right\} \subset \mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I})`	$\left\{\boldsymbol{S} \boldsymbol{v}_{1}, \ldots, \boldsymbol{S} \boldsymbol{v}_{k}\right\} \subset \mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I})$	or $\boldsymbol{A}\boldsymbol{S}\boldsymbol{v}_i = \lambda\boldsymbol{S}\boldsymbol{v}_i$, $i = 1, \ldots, k$. But then $\{\boldsymbol{S}\boldsymbol{v}_1, \ldots, \boldsymbol{S}\boldsymbol{v}_k\} \subset \mathcal{N}(\boldsymbol{A} - \lambda\boldsymbol{I})$, which
lyche-numerical-linear-algebra F01922	148	1.000	`k \leq \ell`	$k \leq \ell$	implies that $k \leq \ell$. Similarly, if $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_\ell$ is a basis for $\mathcal{N}(\boldsymbol{A} - \lambda\boldsymbol{I})$ then
lyche-numerical-linear-algebra F01923	148	1.000	`\boldsymbol{w}_{1}, \ldots, \boldsymbol{w}_{\ell}`	$\boldsymbol{w}_{1}, \ldots, \boldsymbol{w}_{\ell}$	implies that $k \leq \ell$. Similarly, if $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_\ell$ is a basis for $\mathcal{N}(\boldsymbol{A} - \lambda\boldsymbol{I})$ then
lyche-numerical-linear-algebra F01924	148	1.000	`\left\{\boldsymbol{S}^{-1} \boldsymbol{w}_{1}, \ldots, \boldsymbol{S}^{-1} \boldsymbol{w}_{\ell}\right\} \subset \mathcal{N}\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I}\right)`	$\left\{\boldsymbol{S}^{-1} \boldsymbol{w}_{1}, \ldots, \boldsymbol{S}^{-1} \boldsymbol{w}_{\ell}\right\} \subset \mathcal{N}\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I}\right)$	$\{\boldsymbol{S}^{-1}\boldsymbol{w}_1, \ldots, \boldsymbol{S}^{-1}\boldsymbol{w}_\ell\} \subset \mathcal{N}(\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S} - \lambda\boldsymbol{I})$. which implies that $k \geq \ell$. We conclude
lyche-numerical-linear-algebra F01925	148	1.000	`k \geq \ell`	$k \geq \ell$	$\{\boldsymbol{S}^{-1}\boldsymbol{w}_1, \ldots, \boldsymbol{S}^{-1}\boldsymbol{w}_\ell\} \subset \mathcal{N}(\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S} - \lambda\boldsymbol{I})$. which implies that $k \geq \ell$. We conclude
lyche-numerical-linear-algebra F01926	149	1.000	`\boldsymbol{J}_{m}(\lambda)`	$\boldsymbol{J}_{m}(\lambda)$	A **Jordan block** of order m, denoted $\boldsymbol{J}_m(\lambda)$ is an $m \times m$ matrix of the form
lyche-numerical-linear-algebra F01927	149	0.764	`\boldsymbol{J}_{3}(\lambda)=\left[\begin{array}{ccc} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{array}\right]`	$\boldsymbol{J}_{3}(\lambda)=\left[\begin{array}{ccc} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{array}\right]$	A 3×3 Jordan block has the form $\boldsymbol{J}_3(\lambda) = \left[\begin{smallmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{smallmatrix}\right]$. Since a Jordan block is upper
lyche-numerical-linear-algebra F01928	149	1.000	`\boldsymbol{J}_{m}(\lambda) \boldsymbol{v}=\lambda \boldsymbol{v}`	$\boldsymbol{J}_{m}(\lambda) \boldsymbol{v}=\lambda \boldsymbol{v}$	$\boldsymbol{e}_1$. Indeed, if $\boldsymbol{J}_m(\lambda)\boldsymbol{v} = \lambda\boldsymbol{v}$ for some $\boldsymbol{v} = [v_1, \ldots, v_m]$ then $\lambda v_{i-1} + v_i = \lambda v_{i-1}$,
lyche-numerical-linear-algebra F01929	149	1.000	`\boldsymbol{v}=\left[v_{1}, \ldots, v_{m}\right]`	$\boldsymbol{v}=\left[v_{1}, \ldots, v_{m}\right]$	$\boldsymbol{e}_1$. Indeed, if $\boldsymbol{J}_m(\lambda)\boldsymbol{v} = \lambda\boldsymbol{v}$ for some $\boldsymbol{v} = [v_1, \ldots, v_m]$ then $\lambda v_{i-1} + v_i = \lambda v_{i-1}$,
lyche-numerical-linear-algebra F01930	149	1.000	`\lambda v_{i-1}+v_{i}=\lambda v_{i-1}`	$\lambda v_{i-1}+v_{i}=\lambda v_{i-1}$	$\boldsymbol{e}_1$. Indeed, if $\boldsymbol{J}_m(\lambda)\boldsymbol{v} = \lambda\boldsymbol{v}$ for some $\boldsymbol{v} = [v_1, \ldots, v_m]$ then $\lambda v_{i-1} + v_i = \lambda v_{i-1}$,
lyche-numerical-linear-algebra F01931	149	1.000	`i=2, \ldots, m`	$i=2, \ldots, m$	$i = 2, \ldots, m$ which shows that $v_2 = \cdots = v_m = 0$. Thus, the eigenvalue λ of
lyche-numerical-linear-algebra F01932	149	1.000	`v_{2}=\cdots=v_{m}=0`	$v_{2}=\cdots=v_{m}=0$	$i = 2, \ldots, m$ which shows that $v_2 = \cdots = v_m = 0$. Thus, the eigenvalue λ of
lyche-numerical-linear-algebra F01933	149	0.997	`a=m`	$a=m$	$\boldsymbol{J}_m(\lambda)$ has algebraic multiplicity $a = m$ and geometric multiplicity $g = 1$.
lyche-numerical-linear-algebra F01934	149	1.000	`g_{1}, \ldots, g_{k}`	$g_{1}, \ldots, g_{k}$	*$a_1, \ldots, a_k$ and geometric multiplicities $g_1, \ldots, g_k$. There is a nonsingular matrix*
lyche-numerical-linear-algebra F01935	150	1.000	`\boldsymbol{U}_{i}`	$\boldsymbol{U}_{i}$	*where each $\boldsymbol{U}_i$ is block diagonal having g_i Jordan blocks along the diagonal*
lyche-numerical-linear-algebra F01936	150	1.000	`g_{i}`	g_{i}	*where each $\boldsymbol{U}_i$ is block diagonal having g_i Jordan blocks along the diagonal*
lyche-numerical-linear-algebra F01937	150	1.000	`m_{i, 1}, \ldots, m_{i, g_{i}}`	$m_{i, 1}, \ldots, m_{i, g_{i}}$	*Here $m_{i,1}, \ldots, m_{i,g_i}$ are positive integers and they are unique if they are ordered so*
lyche-numerical-linear-algebra F01938	150	1.000	`m_{i, 1} \geq m_{i, 2} \geq \cdots \geq m_{i, g_{i}}`	$m_{i, 1} \geq m_{i, 2} \geq \cdots \geq m_{i, g_{i}}$	*that $m_{i,1} \geq m_{i,2} \geq \cdots \geq m_{i,g_i}$. Moreover, $a_i = \sum_{j=1}^{g_i} m_{i,j}$ for all i.*
lyche-numerical-linear-algebra F01939	150	1.000	`a_{i}=\sum_{j=1}^{g_{i}} m_{i, j}`	$a_{i}=\sum_{j=1}^{g_{i}} m_{i, j}$	*that $m_{i,1} \geq m_{i,2} \geq \cdots \geq m_{i,g_i}$. Moreover, $a_i = \sum_{j=1}^{g_i} m_{i,j}$ for all i.*
lyche-numerical-linear-algebra F01940	150	1.000	`\boldsymbol{A} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{J}`	$\boldsymbol{A} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{J}$	They satisfy the matrix equation $\boldsymbol{A}\boldsymbol{S} = \boldsymbol{S}\boldsymbol{J}$.
lyche-numerical-linear-algebra F01941	150	1.000	`\boldsymbol{U}_{1}=\operatorname{diag}\left( \boldsymbol{J}_{3}(2), \boldsymbol{J}_{2}(2), \boldsymbol{J}_{1}(2)\right)`	$\boldsymbol{U}_{1}=\operatorname{diag}\left(\boldsymbol{J}_{3}(2), \boldsymbol{J}_{2}(2), \boldsymbol{J}_{1}(2)\right)$	• $\boldsymbol{U}_1 = \operatorname{diag}(\boldsymbol{J}_3(2), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2))$ and $\boldsymbol{U}_2 = \boldsymbol{J}_2(3)$.
lyche-numerical-linear-algebra F01942	150	1.000	`\boldsymbol{U}_{2}=\boldsymbol{J}_{2}(3)`	$\boldsymbol{U}_{2}=\boldsymbol{J}_{2}(3)$	• $\boldsymbol{U}_1 = \operatorname{diag}(\boldsymbol{J}_3(2), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2))$ and $\boldsymbol{U}_2 = \boldsymbol{J}_2(3)$.
lyche-numerical-linear-algebra F01943	150	1.000	`\lambda=2`	$\lambda=2$	number of Jordan blocks corresponding to $\lambda = 2$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F01944	150	■1.000	`\boldsymbol{S}=\left[\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{8}\right]`	$\boldsymbol{S}=\left[\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{8}\right]$	The columns of $\boldsymbol{S}=[\boldsymbol{s}_1,\ldots,\boldsymbol{s}_8]$ are determined from the columns of $\boldsymbol{J}$ as
lyche-numerical-linear-al F01945	151	■1.000	`\boldsymbol{J}_{2}(3), \boldsymbol{J}_{2}(2), \boldsymbol{J}_{1}(2), \boldsymbol{J}_{3}(2)`	$\boldsymbol{J}_{2}(3), \boldsymbol{J}_{2}(2), \boldsymbol{J}_{1}(2), \boldsymbol{J}_{3}(2)$	the 4 Jordan blocks $\boldsymbol{J}_2(3), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2), \boldsymbol{J}_3(2)$ are uniquely given.
lyche-numerical-linear-al F01946	151	■1.000	`m_{i, j} \geq 1`	$m_{i, j} \geq 1$	corresponding $\boldsymbol{U}_i$. Moreover each $\boldsymbol{U}_i$ contains g_i Jordan blocks of size $m_{i,j}\geq 1$.
lyche-numerical-linear-al F01947	151	■1.000	`g_{i} \leq a_{i}`	$g_{i} \leq a_{i}$	Thus $g_i \leq a_i$.
lyche-numerical-linear-al F01948	151	■1.000	`\boldsymbol{e}_{1}, \boldsymbol{e}_{4}, \boldsymbol{e}_{6}, \boldsymbol{e}_{7}`	$\boldsymbol{e}_{1}, \boldsymbol{e}_{4}, \boldsymbol{e}_{6}, \boldsymbol{e}_{7}$	should then be clear. There are only 4 eigenvectors of $\boldsymbol{J}$, namely $\boldsymbol{e}_1,\boldsymbol{e}_4,\boldsymbol{e}_6,\boldsymbol{e}_7$
lyche-numerical-linear-al F01949	151	■1.000	`k=2`	$k=2$	independent. Moreover there are $k=2$ distinct eigenvalues and $g_1+g_2=$
lyche-numerical-linear-al F01950	151	■1.000	`g_{1}+g_{2}=`	$g_{1}+g_{2}=$	independent. Moreover there are $k=2$ distinct eigenvalues and $g_1+g_2=$
lyche-numerical-linear-al F01951	151	■1.000	`3+1=4`	$3+1=4$	$3+1=4$.
lyche-numerical-linear-al F01952	151	■1.000	`\sum_{i} a_{i}=n`	$\sum_{i} a_{i}=n$	3. Since $g_i \leq a_i$ for all i and $\sum_i a_i = n$ we have $\sum_i g_i = n$ if and only if $a_i = g_i$
lyche-numerical-linear-al F01953	151	■1.000	`\sum_{i} g_{i}=n`	$\sum_{i} g_{i}=n$	3. Since $g_i \leq a_i$ for all i and $\sum_i a_i = n$ we have $\sum_i g_i = n$ if and only if $a_i = g_i$
lyche-numerical-linear-al F01954	151	■1.000	`a_{i}=g_{i}`	$a_{i}=g_{i}$	3. Since $g_i \leq a_i$ for all i and $\sum_i a_i = n$ we have $\sum_i g_i = n$ if and only if $a_i = g_i$
lyche-numerical-linear-al F01955	151	■1.000	`i=1, \ldots, k`	$i=1, \ldots, k$	for $i=1,\ldots,k$.
lyche-numerical-linear-al F01956	151	■1.000	`\boldsymbol{S}=\boldsymbol{U}`	$\boldsymbol{S}=\boldsymbol{U}$	We turn now to **unitary similarity transformations** $\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}$, where $\boldsymbol{S}=\boldsymbol{U}$ is
lyche-numerical-linear-al F01957	151	■1.000	`\boldsymbol{S}^{-1}=\boldsymbol{U}^{*}`	$\boldsymbol{S}^{-1}=\boldsymbol{U}^{*}$	unitary. Thus $\boldsymbol{S}^{-1}=\boldsymbol{U}^*$ and a unitary similarity transformation takes the form
lyche-numerical-linear-al F01958	151	■1.000	`\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}`	$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}$.
lyche-numerical-linear-al F01959	152	■1.000	`\boldsymbol{R}:=\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}`	$\boldsymbol{R}:=\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}$	*matrix $\boldsymbol{U}\in\mathbb{C}^{n\times n}$ such that $\boldsymbol{R}:=\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}$ is upper triangular.*
lyche-numerical-linear-al F01960	152	■1.000	`\boldsymbol{A}=\boldsymbol{U} \boldsymbol{R} \boldsymbol{U}^{*}`	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{R} \boldsymbol{U}^{*}$	call $\boldsymbol{A}=\boldsymbol{U}\boldsymbol{R}\boldsymbol{U}^*$ the **Schur factorization** of $\boldsymbol{A}$.
lyche-numerical-linear-al F01961	152	■1.000	`k \times k`	$k \times k$	Assume that the theorem is true for all $k\times k$ matrices, and suppose $\boldsymbol{A}\in\mathbb{C}^{n\times n}$, where
lyche-numerical-linear-al F01962	152	■0.993	`n:=k+1`	$n:=k+1$	$n:=k+1$. Let $(\lambda_1,\boldsymbol{v}_1)$ be an eigenpair for $\boldsymbol{A}$ with $\\|\boldsymbol{v}_1\\|_2=1$. By Theorem 5.5
lyche-numerical-linear-al F01963	152	■0.993	`\left(\lambda_{1}, \boldsymbol{v}_{1}\right)`	$\left(\lambda_{1}, \boldsymbol{v}_{1}\right)$	$n:=k+1$. Let $(\lambda_1,\boldsymbol{v}_1)$ be an eigenpair for $\boldsymbol{A}$ with $\\|\boldsymbol{v}_1\\|_2=1$. By Theorem 5.5
lyche-numerical-linear-al F01964	152	■0.993	`\left|\boldsymbol{v}_{1}\right|_{2}=1`	$\left\\|\boldsymbol{v}_{1}\right\\|_{2}=1$	$n:=k+1$. Let $(\lambda_1,\boldsymbol{v}_1)$ be an eigenpair for $\boldsymbol{A}$ with $\\|\boldsymbol{v}_1\\|_2=1$. By Theorem 5.5
lyche-numerical-linear-al F01965	152	■1.000	`\left\{\boldsymbol{v}_{1}, \boldsymbol{v}_{2}, \ldots, \boldsymbol{v}_{n}\right\}`	$\left\{\boldsymbol{v}_{1}, \boldsymbol{v}_{2}, \ldots, \boldsymbol{v}_{n}\right\}$	we can extend $\boldsymbol{v}_1$ to an orthonormal basis $\{\boldsymbol{v}_1,\boldsymbol{v}_2,\ldots,\boldsymbol{v}_n\}$ for $\mathbb{C}^n$. The matrix $\boldsymbol{V}:=$
lyche-numerical-linear-al F01966	152	■1.000	`\boldsymbol{V}:=`	$\boldsymbol{V}:=$	we can extend $\boldsymbol{v}_1$ to an orthonormal basis $\{\boldsymbol{v}_1,\boldsymbol{v}_2,\ldots,\boldsymbol{v}_n\}$ for $\mathbb{C}^n$. The matrix $\boldsymbol{V}:=$
lyche-numerical-linear-al F01967	152	■1.000	`\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right] \in \mathbb{C}^{n \times n}`	$\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right] \in \mathbb{C}^{n \times n}$	$[\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n]\in\mathbb{C}^{n\times n}$ is unitary, and
lyche-numerical-linear-al F01968	152	■1.000	`\boldsymbol{W}_{1} \in \mathbb{C}^{(n-1) \times(n-1)}`	$\boldsymbol{W}_{1} \in \mathbb{C}^{(n-1) \times(n-1)}$	By the induction hypothesis there is a unitary matrix $\boldsymbol{W}_1\in\mathbb{C}^{(n-1)\times(n-1)}$ such that
lyche-numerical-linear-al F01969	152	■1.000	`\boldsymbol{W}_{1}^{*} \boldsymbol{M} \boldsymbol{W}_{1}`	$\boldsymbol{W}_{1}^{*} \boldsymbol{M} \boldsymbol{W}_{1}$	$\boldsymbol{W}_1^*\boldsymbol{M}\boldsymbol{W}_1$ is upper triangular. Define
lyche-numerical-linear-al F01970	152	■1.000	`\boldsymbol{W}`	$\boldsymbol{W}$	Then $\boldsymbol{W}$ and $\boldsymbol{U}$ are unitary and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01971	152	0.570	\boldsymbol{U}^{T} \boldsymbol{A U}	$\boldsymbol{U}^{T} \boldsymbol{A U}$	*eigenvalues there exists an orthogonal matrix* $\boldsymbol{U} \in \mathbb{R}^{n\times n}$ *such that* $\boldsymbol{U}^T\boldsymbol{AU}$ *is upper*
lyche-numerical-linear-algebra F01972	152	1.000	\lambda_{1}	λ_{1}	***Proof*** Consider the proof of Theorem 6.5. Since $\boldsymbol{A}$ and λ_1 are real the eigenvector
lyche-numerical-linear-algebra F01973	152	1.000	\boldsymbol{W}^{T} \boldsymbol{W}=\boldsymbol{I}	$\boldsymbol{W}^{T} \boldsymbol{W}=\boldsymbol{I}$	$\boldsymbol{v}_1$ is real and the matrix $\boldsymbol{W}$ is real and $\boldsymbol{W}^T\boldsymbol{W} = \boldsymbol{I}$. By the induction hypothesis $\boldsymbol{V}$
lyche-numerical-linear-algebra F01974	152	1.000	\boldsymbol{V}	$\boldsymbol{V}$	$\boldsymbol{v}_1$ is real and the matrix $\boldsymbol{W}$ is real and $\boldsymbol{W}^T\boldsymbol{W} = \boldsymbol{I}$. By the induction hypothesis $\boldsymbol{V}$
lyche-numerical-linear-algebra F01975	152	1.000	\boldsymbol{V}^{T} \boldsymbol{V}=\boldsymbol{I}	$\boldsymbol{V}^{T} \boldsymbol{V}=\boldsymbol{I}$	is real and $\boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}$. But then also $\boldsymbol{U} = \boldsymbol{VW}$ is real and $\boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}$.
lyche-numerical-linear-algebra F01976	152	1.000	\boldsymbol{U}=\boldsymbol{V} \boldsymbol{W}	$\boldsymbol{U}=\boldsymbol{V} \boldsymbol{W}$	is real and $\boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}$. But then also $\boldsymbol{U} = \boldsymbol{VW}$ is real and $\boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}$.
lyche-numerical-linear-algebra F01977	152	1.000	\boldsymbol{U}^{T} \boldsymbol{U}=\boldsymbol{I}	$\boldsymbol{U}^{T} \boldsymbol{U}=\boldsymbol{I}$	is real and $\boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}$. But then also $\boldsymbol{U} = \boldsymbol{VW}$ is real and $\boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}$.
lyche-numerical-linear-algebra F01978	153	1.000	n-1 . \boldsymbol{M}	$n-1 . \boldsymbol{M}$	$n \times n$ matrix $\boldsymbol{A}$, we obtain a matrix $\boldsymbol{M}$ of order $n-1$. $\boldsymbol{M}$ has the same eigenvalues
lyche-numerical-linear-algebra F01979	153	1.000	\boldsymbol{T}:=\left[\begin{array}{ccc}2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2\end{array}\right]	$\boldsymbol{T}:=\left[\begin{array}{ccc}2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2\end{array}\right]$	numerical work. The second derivative matrix $\boldsymbol{T} := \left[\begin{smallmatrix}2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2\end{smallmatrix}\right]$ has an eigenpair
lyche-numerical-linear-algebra F01980	153	0.675	\left(2, \boldsymbol{x}_{1}\right)	$\left(2, \boldsymbol{x}_{1}\right)$	$(2, \boldsymbol{x}_1)$, where $\boldsymbol{x}_1 = [-1, 0, 1]^T$. Find the remaining eigenvalues using deflation.
lyche-numerical-linear-algebra F01981	153	0.675	\boldsymbol{x}_{1}=[-1,0,1]^{T}	$\boldsymbol{x}_{1}=[-1,0,1]^{T}$	$(2, \boldsymbol{x}_1)$, where $\boldsymbol{x}_1 = [-1, 0, 1]^T$. Find the remaining eigenvalues using deflation.
lyche-numerical-linear-algebra F01982	153	1.000	\boldsymbol{x}_{1}	$\boldsymbol{x}_{1}$	For this we extend $\boldsymbol{x}_1$ to a basis $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\}$ for $\mathbb{R}^3$ by defining $\boldsymbol{x}_2 = [0, 1, 0]^T$,
lyche-numerical-linear-algebra F01983	153	1.000	\left\{\boldsymbol{x}_{1}, \boldsymbol{x}_{2}, \boldsymbol{x}_{3}\right\}	$\left\{\boldsymbol{x}_{1}, \boldsymbol{x}_{2}, \boldsymbol{x}_{3}\right\}$	For this we extend $\boldsymbol{x}_1$ to a basis $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\}$ for $\mathbb{R}^3$ by defining $\boldsymbol{x}_2 = [0, 1, 0]^T$,
lyche-numerical-linear-algebra F01984	153	1.000	\boldsymbol{x}_{2}=[0,1,0]^{T}	$\boldsymbol{x}_{2}=[0,1,0]^{T}$	For this we extend $\boldsymbol{x}_1$ to a basis $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\}$ for $\mathbb{R}^3$ by defining $\boldsymbol{x}_2 = [0, 1, 0]^T$,
lyche-numerical-linear-algebra F01985	153	1.000	\boldsymbol{x}_{3}=[1,0,1]^{T}	$\boldsymbol{x}_{3}=[1,0,1]^{T}$	$\boldsymbol{x}_3 = [1, 0, 1]^T$. This is already an orthogonal basis and normalizing we obtain the
lyche-numerical-linear-algebra F01986	153	1.000	\boldsymbol{A}^{*} \boldsymbol{A}=\boldsymbol{A} \boldsymbol{A}^{*}	$\boldsymbol{A}^{*} \boldsymbol{A}=\boldsymbol{A} \boldsymbol{A}^{*}$	A matrix $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ is **normal** if $\boldsymbol{A}^*\boldsymbol{A} = \boldsymbol{A}\boldsymbol{A}^*$. In this section we show that a matrix
lyche-numerical-linear-algebra F01987	153	1.000	\boldsymbol{A}^{*}=-\boldsymbol{A}	$\boldsymbol{A}^{*}=-\boldsymbol{A}$	2. $\boldsymbol{A}^* = -\boldsymbol{A}$,
lyche-numerical-linear-algebra F01988	153	1.000	\boldsymbol{A}^{*}=\boldsymbol{A}^{-1}	$\boldsymbol{A}^{*}=\boldsymbol{A}^{-1}$	3. $\boldsymbol{A}^* = \boldsymbol{A}^{-1}$,
lyche-numerical-linear-algebra F01989	153	0.999	\boldsymbol{A}=\operatorname{diag}\left(d_{1}, \ldots, d_{n}\right)	$\boldsymbol{A}=\operatorname{diag}\left(d_{1}, \ldots, d_{n}\right)$	4. $\boldsymbol{A} = \operatorname{diag}(d_1, \ldots, d_n)$.
lyche-numerical-linear-algebra F01990	154	0.989	\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}=\boldsymbol{D}	$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}=\boldsymbol{D}$	*normal if and only if there exists a unitary matrix* $\boldsymbol{U} \in \mathbb{C}^{n\times n}$ *such that* $\boldsymbol{U}^*\boldsymbol{AU} = \boldsymbol{D}$
lyche-numerical-linear-algebra F01991	154	1.000	\boldsymbol{D}=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)	$\boldsymbol{D}=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)$	*is diagonal. If* $\boldsymbol{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ *and* $\boldsymbol{U} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n]$ *then* $(\lambda_j, \boldsymbol{u}_j)$, $j =$
lyche-numerical-linear-algebra F01992	154	1.000	\boldsymbol{U}=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{n}\right]	$\boldsymbol{U}=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{n}\right]$	*is diagonal. If* $\boldsymbol{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ *and* $\boldsymbol{U} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n]$ *then* $(\lambda_j, \boldsymbol{u}_j)$, $j =$
lyche-numerical-linear-algebra F01993	154	1.000	\left(\lambda_{j}, \boldsymbol{u}_{j}\right), j=	$\left(\lambda_{j}, \boldsymbol{u}_{j}\right), j=$	*is diagonal. If* $\boldsymbol{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ *and* $\boldsymbol{U} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n]$ *then* $(\lambda_j, \boldsymbol{u}_j)$, $j =$
lyche-numerical-linear-algebra F01994	154	0.996	\boldsymbol{B}=\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}	$\boldsymbol{B}=\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}$	***Proof*** If $\boldsymbol{B} = \boldsymbol{U}^*\boldsymbol{AU}$, with $\boldsymbol{B}$ diagonal, and $\boldsymbol{U}^*\boldsymbol{U} = \boldsymbol{I}$, then $\boldsymbol{A} = \boldsymbol{UBU}^*$ and
lyche-numerical-linear-algebra F01995	154	0.996	\boldsymbol{A}=\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^{*}	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^{*}$	***Proof*** If $\boldsymbol{B} = \boldsymbol{U}^*\boldsymbol{AU}$, with $\boldsymbol{B}$ diagonal, and $\boldsymbol{U}^*\boldsymbol{U} = \boldsymbol{I}$, then $\boldsymbol{A} = \boldsymbol{UBU}^*$ and
lyche-numerical-linear-algebra F01996	154	1.000	\boldsymbol{B} \boldsymbol{B}^{*}=\boldsymbol{B}^{*} \boldsymbol{B}	$\boldsymbol{B} \boldsymbol{B}^{*}=\boldsymbol{B}^{*} \boldsymbol{B}$	Now $\boldsymbol{BB}^* = \boldsymbol{B}^*\boldsymbol{B}$ since $\boldsymbol{B}$ is diagonal, and $\boldsymbol{A}$ is normal.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01997	154	1.000	`\boldsymbol{B}:=\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}`	$\boldsymbol{B}:=\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{I}$ such that $\boldsymbol{B} := \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}$ is upper triangular. Since $\boldsymbol{A}$ is normal $\boldsymbol{B}$ is normal. Indeed,
lyche-numerical-linear-algebra F01998	154	0.458	`e_{i i}`	$e_{i i}$	must be diagonal. The diagonal elements e_{ii} in $\boldsymbol{E} := \boldsymbol{B}^*\boldsymbol{B}$ and f_{ii} in $\boldsymbol{F} := \boldsymbol{B}\boldsymbol{B}^*$
lyche-numerical-linear-algebra F01999	154	0.458	`\boldsymbol{E}:=\boldsymbol{B}^{*} \boldsymbol{B}`	$\boldsymbol{E}:=\boldsymbol{B}^{*} \boldsymbol{B}$	must be diagonal. The diagonal elements e_{ii} in $\boldsymbol{E} := \boldsymbol{B}^*\boldsymbol{B}$ and f_{ii} in $\boldsymbol{F} := \boldsymbol{B}\boldsymbol{B}^*$
lyche-numerical-linear-algebra F02000	154	0.458	`f_{i i}`	$f_{i i}$	must be diagonal. The diagonal elements e_{ii} in $\boldsymbol{E} := \boldsymbol{B}^*\boldsymbol{B}$ and f_{ii} in $\boldsymbol{F} := \boldsymbol{B}\boldsymbol{B}^*$
lyche-numerical-linear-algebra F02001	154	0.458	`\boldsymbol{F}:=\boldsymbol{B B}^{*}`	$\boldsymbol{F}:=\boldsymbol{B B}^{*}$	must be diagonal. The diagonal elements e_{ii} in $\boldsymbol{E} := \boldsymbol{B}^*\boldsymbol{B}$ and f_{ii} in $\boldsymbol{F} := \boldsymbol{B}\boldsymbol{B}^*$
lyche-numerical-linear-algebra F02002	154	1.000	`\left|b_{11}\right|^{2}=\left|b_{11}\right|^{2}+ \left|b_{12}\right|^{2}+\cdots+\left|b_{1 n}\right|^{2}`	$\left\|b_{11}\right\|^{2}=\left\|b_{11}\right\|^{2}+\left\|b_{12}\right\|^{2}+\cdots+\left\|b_{1 n}\right\|^{2}$	$i = 1$ we have $\|b_{11}\|^2 = \|b_{11}\|^2+\|b_{12}\|^2+\cdots+\|b_{1n}\|^2$, so $b_{1k} = 0$ for $k = 2, 3, \ldots, n$.
lyche-numerical-linear-algebra F02003	154	1.000	`b_{1 k}=0`	$b_{1 k}=0$	$i = 1$ we have $\|b_{11}\|^2 = \|b_{11}\|^2+\|b_{12}\|^2+\cdots+\|b_{1n}\|^2$, so $b_{1k} = 0$ for $k = 2, 3, \ldots, n$.
lyche-numerical-linear-algebra F02004	154	1.000	`k=2,3, \ldots, n`	$k=2,3, \ldots, n$	$i = 1$ we have $\|b_{11}\|^2 = \|b_{11}\|^2+\|b_{12}\|^2+\cdots+\|b_{1n}\|^2$, so $b_{1k} = 0$ for $k = 2, 3, \ldots, n$.
lyche-numerical-linear-algebra F02005	154	1.000	`i-1`	$i-1$	Suppose $\boldsymbol{B}$ is diagonal in its first $i-1$ rows so that $b_{jk} = 0$ for $j = 1, \ldots, i-1$,
lyche-numerical-linear-algebra F02006	154	1.000	`b_{j k}=0`	$b_{j k}=0$	Suppose $\boldsymbol{B}$ is diagonal in its first $i-1$ rows so that $b_{jk} = 0$ for $j = 1, \ldots, i-1$,
lyche-numerical-linear-algebra F02007	154	1.000	`j=1, \ldots, i-1`	$j=1, \ldots, i-1$	Suppose $\boldsymbol{B}$ is diagonal in its first $i-1$ rows so that $b_{jk} = 0$ for $j = 1, \ldots, i-1$,
lyche-numerical-linear-algebra F02008	154	0.999	`k=j+1, \ldots, n`	$k=j+1, \ldots, n$	$k = j+1, \ldots, n$. Then
lyche-numerical-linear-algebra F02009	154	1.000	`b_{i k}=0, k=i+1, \ldots, n`	$b_{i k}=0, k=i+1, \ldots, n$	and it follows that $b_{ik} = 0, k = i+1, \ldots, n$. By induction on the rows we see that
lyche-numerical-linear-algebra F02010	154	0.877	`\boldsymbol{U}^{T} \boldsymbol{A} \boldsymbol{U}=`	$\boldsymbol{U}^{T} \boldsymbol{A} \boldsymbol{U}=$	*Example 6.6* The orthogonal diagonalization of $\boldsymbol{A} = \left[\begin{smallmatrix}2 & -1\\ -1 & 2\end{smallmatrix}\right]$ is $\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U} =$
lyche-numerical-linear-algebra F02011	154	0.987	`\operatorname{diag}(1,3)`	$\operatorname{diag}(1,3)$	$\operatorname{diag}(1, 3)$, where $\boldsymbol{U} = \frac{1}{\sqrt{2}}\left[\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right]$.
lyche-numerical-linear-algebra F02012	154	0.987	`\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc}1 & 1 \\ 1 & -1\end{array}\right]`	$\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc}1 & 1 \\ 1 & -1\end{array}\right]$	$\operatorname{diag}(1, 3)$, where $\boldsymbol{U} = \frac{1}{\sqrt{2}}\left[\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right]$.
lyche-numerical-linear-algebra F02013	155	0.835	`R(\boldsymbol{x})=\frac{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{*} \boldsymbol{x}}= \lambda`	$R(\boldsymbol{x})=\frac{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{*} \boldsymbol{x}}=\lambda$	If $(\lambda, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}$ then $R(\boldsymbol{x}) = \frac{\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^*\boldsymbol{x}} = \lambda$.
lyche-numerical-linear-algebra F02014	155	0.761	`\lambda_{j}, \boldsymbol{u}_{j}`	$\lambda_{j}, \boldsymbol{u}_{j}$	*is normal with orthonormal eigenpairs* $(\lambda_j, \boldsymbol{u}_j)$, *for* $j = 1, 2, \ldots, n$. *Then the*
lyche-numerical-linear-algebra F02015	155	0.761	`j=1,2, \ldots, n`	$j=1,2, \ldots, n$	*is normal with orthonormal eigenpairs* $(\lambda_j, \boldsymbol{u}_j)$, *for* $j = 1, 2, \ldots, n$. *Then the*
lyche-numerical-linear-algebra F02016	155	1.000	`\boldsymbol{x}^{*} \boldsymbol{x}=\sum_{i=1}^{n} \sum_{j=1}^{n} \bar{c}_{i} \overline{\boldsymbol{u}}_{i} c_{j} \boldsymbol{u}_{j}=`	$\boldsymbol{x}^{*} \boldsymbol{x}=\sum_{i=1}^{n} \sum_{j=1}^{n} \bar{c}_{i} \overline{\boldsymbol{u}}_{i} c_{j} \boldsymbol{u}_{j}=$	***Proof*** By orthonormality of the eigenvectors $\boldsymbol{x}^*\boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\boldsymbol{u}}_i c_j \boldsymbol{u}_j =$
lyche-numerical-linear-algebra F02017	155	1.000	`\sum_{j=1}^{n}\left|c_{j}\right|^{2}`	$\sum_{j=1}^{n}\left\|c_{j}\right\|^{2}$	$\sum_{j=1}^n \|c_j\|^2$. Similarly, $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\boldsymbol{u}}_i c_j \lambda_j \boldsymbol{u}_j = \sum_{i=1}^n \lambda_i \|c_i\|^2$. and
lyche-numerical-linear-algebra F02018	155	1.000	`\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}= \sum_{i=1}^{n} \sum_{j=1}^{n} \bar{c}_{i} \overline{ \boldsymbol{u}}_{i} c_{j} \lambda_{j} \boldsymbol{u}_{j}= \sum_{i=1}^{n} \lambda_{i}\left|c_{i}\right|^{2}`	$\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=\sum_{i=1}^{n} \sum_{j=1}^{n} \bar{c}_{i} \overline{\boldsymbol{u}}_{i} c_{j} \lambda_{j} \boldsymbol{u}_{j}=\sum_{i=1}^{n} \lambda_{i}\left\|c_{i}\right\|^{2}$	$\sum_{j=1}^n \|c_j\|^2$. Similarly, $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\boldsymbol{u}}_i c_j \lambda_j \boldsymbol{u}_j = \sum_{i=1}^n \lambda_i \|c_i\|^2$. and
lyche-numerical-linear-algebra F02019	155	1.000	`\sum_{i=1}^{n}\left|c_{i}\right|^{2} / \sum_{j=1}^{n} \left|c_{j}\right|^{2}=1`	$\sum_{i=1}^{n}\left\|c_{i}\right\|^{2} / \sum_{j=1}^{n}\left\|c_{j}\right\|^{2}=1$	combination since $\sum_{i=1}^n \|c_i\|^2 / \sum_{j=1}^n \|c_j\|^2 = 1$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F02020	155	0.998	`\lambda=\mu+i \nu, \bar{\lambda}=\mu-i \nu`	$\lambda=\mu+i \nu, \bar{\lambda}=\mu-i \nu$	eigenvalues of a real matrix occur in conjugate pairs, $\lambda = \mu + i\nu$, $\bar{\lambda} = \mu - i\nu$,
lyche-numerical-linear-algebra_F02021	155	0.758	`\mu, \nu`	μ, ν	where μ, ν are real. The real 2×2 matrix
lyche-numerical-linear-algebra_F02022	155	0.999	`\lambda=\mu+i \nu`	$\lambda=\mu+i \nu$	has eigenvalues $\lambda = \mu + i\nu$ and $\bar{\lambda} = \mu - i\nu$.
lyche-numerical-linear-algebra_F02023	155	0.999	`\bar{\lambda}=\mu-i \nu`	$\bar{\lambda}=\mu-i \nu$	has eigenvalues $\lambda = \mu + i\nu$ and $\bar{\lambda} = \mu - i\nu$.
lyche-numerical-linear-algebra_F02024	156	1.000	`\pi_{\boldsymbol{R}}=`	$\pi_{\boldsymbol{R}}=$	Since $\boldsymbol{R}$ is block triangular the characteristic polynomial of $\boldsymbol{R}$ is given by $\pi_{\boldsymbol{R}} =$
lyche-numerical-linear-algebra_F02025	156	0.831	`\boldsymbol{\pi}_{\boldsymbol{D}_{1}} \boldsymbol{\pi}_{ \boldsymbol{D}_{2}} \boldsymbol{\pi}_{\boldsymbol{D}_{3}}`	$\boldsymbol{\pi}_{\boldsymbol{D}_{1}} \boldsymbol{\pi}_{\boldsymbol{D}_{2}} \boldsymbol{\pi}_{\boldsymbol{D}_{3}}$	$\pi_{\boldsymbol{D}_1}\pi_{\boldsymbol{D}_2}\pi_{\boldsymbol{D}_3}$. We find
lyche-numerical-linear-algebra_F02026	156	1.000	`\boldsymbol{D}_{1}`	$\boldsymbol{D}_{1}$	and the eigenvalues $\boldsymbol{D}_1$ and $\boldsymbol{D}_3$ are $\lambda_1 = 2+i$, $\lambda_2 = 2-i$, while $\boldsymbol{D}_2$ obviously has
lyche-numerical-linear-algebra_F02027	156	1.000	`\boldsymbol{D}_{3}`	$\boldsymbol{D}_{3}$	and the eigenvalues $\boldsymbol{D}_1$ and $\boldsymbol{D}_3$ are $\lambda_1 = 2+i$, $\lambda_2 = 2-i$, while $\boldsymbol{D}_2$ obviously has
lyche-numerical-linear-algebra_F02028	156	1.000	`\lambda_{1}=2+i, \lambda_{2}=2-i`	$\lambda_{1}=2+i, \lambda_{2}=2-i$	and the eigenvalues $\boldsymbol{D}_1$ and $\boldsymbol{D}_3$ are $\lambda_1 = 2+i$, $\lambda_2 = 2-i$, while $\boldsymbol{D}_2$ obviously has
lyche-numerical-linear-algebra_F02029	156	1.000	`\boldsymbol{D}_{2}`	$\boldsymbol{D}_{2}$	and the eigenvalues $\boldsymbol{D}_1$ and $\boldsymbol{D}_3$ are $\lambda_1 = 2+i$, $\lambda_2 = 2-i$, while $\boldsymbol{D}_2$ obviously has
lyche-numerical-linear-algebra_F02030	156	1.000	`\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{n} \right\}`	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{n}\right\}$	*For any such* $\boldsymbol{U}$ *the columns* $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$ *of* $\boldsymbol{U}$ *are orthonormal eigenvectors of* $\boldsymbol{A}$
lyche-numerical-linear-algebra_F02031	156	1.000	`\boldsymbol{A} \boldsymbol{u}_{j}=\lambda_{j} \boldsymbol{u}_{j}`	$\boldsymbol{A} \boldsymbol{u}_{j}=\lambda_{j} \boldsymbol{u}_{j}$	*and* $\boldsymbol{A}\boldsymbol{u}_j = \lambda_j \boldsymbol{u}_j$ *for* $j = 1, \ldots, n$.
lyche-numerical-linear-algebra_F02032	156	0.995	`\lambda_{1}, \lambda_{2}, \ldots, \lambda_{n}`	$\lambda_{1}, \lambda_{2}, \ldots, \lambda_{n}$	*ric. Then* $\boldsymbol{A}$ *has real eigenvalues* $\lambda_1, \lambda_2, \ldots, \lambda_n$. *Moreover, there is an orthogonal*
lyche-numerical-linear-algebra_F02033	157	0.999	`R(\boldsymbol{x})=R_{\boldsymbol{A}}(\boldsymbol{x}):= \frac{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}{ \boldsymbol{x}^{*} \boldsymbol{x}}`	$R(\boldsymbol{x})=R_{\boldsymbol{A}}(\boldsymbol{x}):=\frac{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{*} \boldsymbol{x}}$	terms of the Rayleigh quotient $R(\boldsymbol{x}) = R_{\boldsymbol{A}}(\boldsymbol{x}) := \frac{\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^*\boldsymbol{x}}$. First we show
lyche-numerical-linear-algebra_F02034	157	0.924	`\left(\lambda_{j}, \boldsymbol{u}_{j}\right), 1 \leq j \leq n`	$\left(\lambda_{j}, \boldsymbol{u}_{j}\right), 1 \leq j \leq n$	*eigenpairs* $(\lambda_j, \boldsymbol{u}_j)$, $1 \leq j \leq n$, *ordered so that* $\lambda_1 \geq \cdots \geq \lambda_n$. *Let* $1 \leq k \leq n$. *For*
lyche-numerical-linear-algebra_F02035	157	0.924	`\lambda_{1} \geq \cdots \geq \lambda_{n}`	$\lambda_{1} \geq \cdots \geq \lambda_{n}$	*eigenpairs* $(\lambda_j, \boldsymbol{u}_j)$, $1 \leq j \leq n$, *ordered so that* $\lambda_1 \geq \cdots \geq \lambda_n$. *Let* $1 \leq k \leq n$. *For*
lyche-numerical-linear-algebra_F02036	157	0.924	`1 \leq k \leq n`	$1 \leq k \leq n$	*eigenpairs* $(\lambda_j, \boldsymbol{u}_j)$, $1 \leq j \leq n$, *ordered so that* $\lambda_1 \geq \cdots \geq \lambda_n$. *Let* $1 \leq k \leq n$. *For*
lyche-numerical-linear-algebra_F02037	157	1.000	`\mathcal{S}=\tilde{\mathcal{S}}:=\operatorname{span} \left(\boldsymbol{u}_{k}, \ldots, \boldsymbol{u}_{n} \right)`	$\mathcal{S}=\tilde{\mathcal{S}}:=\operatorname{span}\left(\boldsymbol{u}_{k}, \ldots, \boldsymbol{u}_{n}\right)$	*with equality for* $\mathcal{S} = \tilde{\mathcal{S}} := \operatorname{span}(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n)$ *and* $\boldsymbol{x} = \boldsymbol{u}_k$.
lyche-numerical-linear-algebra_F02038	157	1.000	`\boldsymbol{x}=\boldsymbol{u}_{k}`	$\boldsymbol{x}=\boldsymbol{u}_{k}$	*with equality for* $\mathcal{S} = \tilde{\mathcal{S}} := \operatorname{span}(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n)$ *and* $\boldsymbol{x} = \boldsymbol{u}_k$.
lyche-numerical-linear-algebra_F02039	157	1.000	`\mathcal{S}^{\prime}:=`	$\mathcal{S}^{\prime}:=$	***Proof*** Let $\mathcal{S}$ be any subspace of $\mathbb{C}^n$ of dimension $n-k+1$ and define $\mathcal{S}' :=$
lyche-numerical-linear-algebra_F02040	157	1.000	`\operatorname{span}\left(\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{k}\right)`	$\operatorname{span}\left(\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{k}\right)$	$\operatorname{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S} + \mathcal{S}' :=$
lyche-numerical-linear-algebra_F02041	157	1.000	`\boldsymbol{y} \in \mathcal{S}`	$\boldsymbol{y} \in \mathcal{S}$	$\operatorname{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S} + \mathcal{S}' :=$
lyche-numerical-linear-algebra_F02042	157	1.000	`R(\boldsymbol{y}) \geq \lambda_{k}`	$R(\boldsymbol{y}) \geq \lambda_{k}$	$\operatorname{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S} + \mathcal{S}' :=$
lyche-numerical-linear-algebra_F02043	157	1.000	`\mathcal{S}+\mathcal{S}^{\prime}:=`	$\mathcal{S}+\mathcal{S}^{\prime}:=$	$\operatorname{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S} + \mathcal{S}' :=$
lyche-numerical-linear-algebra_F02044	157	1.000	`\left\{\boldsymbol{s}+\boldsymbol{s}^{\prime}: \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s}^{\prime} \in \mathcal{S}^{\prime}\right\}`	$\left\{\boldsymbol{s}+\boldsymbol{s}^{\prime}: \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s}^{\prime} \in \mathcal{S}^{\prime}\right\}$	$\{\boldsymbol{s} + \boldsymbol{s}' : \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s}' \in \mathcal{S}'\}$ is a subspace of $\mathbb{C}^n$ and by (1.7)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F02045	157	■0.996	`\mathcal{S} \cap \mathcal{S}^{\prime}`	$\mathcal{S} \cap \mathcal{S}^{\prime}$	It follows that $\mathcal{S} \cap \mathcal{S}'$ is nonempty. Let $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}' = \sum_{j=1}^{k} c_j \boldsymbol{u}_j$ with $\sum_{j=1}^{k} \lvert c_j \rvert^2 =$
lyche-numerical-linear-al F02046	157	■0.996	`\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}^{\prime}= \sum_{j=1}^{k} c_{j} \boldsymbol{u}_{j}`	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}^{\prime}= \sum_{j=1}^{k} c_{j} \boldsymbol{u}_{j}$	It follows that $\mathcal{S} \cap \mathcal{S}'$ is nonempty. Let $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}' = \sum_{j=1}^{k} c_j \boldsymbol{u}_j$ with $\sum_{j=1}^{k} \lvert c_j \rvert^2 =$
lyche-numerical-linear-al F02047	157	■0.996	`\sum_{j=1}^{k}\left|c_{j}\right|^{2}=`	$\sum_{j=1}^{k}\left\lvert c_{j}\right\rvert^{2}=$	It follows that $\mathcal{S} \cap \mathcal{S}'$ is nonempty. Let $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}' = \sum_{j=1}^{k} c_j \boldsymbol{u}_j$ with $\sum_{j=1}^{k} \lvert c_j \rvert^2 =$
lyche-numerical-linear-al F02048	157	■0.997	`c_{j}=0`	$c_{j}=0$	1. Defining $c_j = 0$ for $k+1 \le j \le n$, we obtain by Theorem 6.8
lyche-numerical-linear-al F02049	157	■0.997	`k+1 \leq j \leq n`	$k+1 \leq j \leq n$	1. Defining $c_j = 0$ for $k+1 \le j \le n$, we obtain by Theorem 6.8
lyche-numerical-linear-al F02050	157	■0.877	`\boldsymbol{z} \in \mathcal{S}=\tilde{\mathcal{S}}`	$\boldsymbol{z} \in \mathcal{S}=\tilde{\mathcal{S}}$	and (6.11) follows. To show equality suppose $\boldsymbol{z} \in \mathcal{S} = \tilde{\mathcal{S}}$. Now $\boldsymbol{z} = \sum_{j=k}^{n} d_j \boldsymbol{u}_j$ for
lyche-numerical-linear-al F02051	157	■0.877	`\boldsymbol{z}=\sum_{j=k}^{n} d_{j} \boldsymbol{u}_{j}`	$\boldsymbol{z}=\sum_{j=k}^{n} d_{j} \boldsymbol{u}_{j}$	and (6.11) follows. To show equality suppose $\boldsymbol{z} \in \mathcal{S} = \tilde{\mathcal{S}}$. Now $\boldsymbol{z} = \sum_{j=k}^{n} d_j \boldsymbol{u}_j$ for
lyche-numerical-linear-al F02052	157	■0.999	`d_{k}, \ldots, d_{n}`	$d_{k}, \ldots, d_{n}$	some $d_k, \ldots, d_n$ with $\sum_{j=k}^{n} \lvert d_j \rvert^2 = 1$ and by Lemma 6.8 $R(z) = \sum_{j=k}^{n} \lambda_j \lvert d_j \rvert^2 \le$
lyche-numerical-linear-al F02053	157	■0.999	`\sum_{j=k}^{n}\left|d_{j}\right|^{2}=1`	$\sum_{j=k}^{n}\left\lvert d_{j}\right\rvert^{2}=1$	some $d_k, \ldots, d_n$ with $\sum_{j=k}^{n} \lvert d_j \rvert^2 = 1$ and by Lemma 6.8 $R(z) = \sum_{j=k}^{n} \lambda_j \lvert d_j \rvert^2 \le$
lyche-numerical-linear-al F02054	157	■0.999	`6.8 R(z)=\sum_{j=k}^{n} \lambda_{j}\left|d_{j}\right|^{2} \leq`	$6.8 R(z)=\sum_{j=k}^{n} \lambda_{j}\left\lvert d_{j}\right\rvert^{2} \leq$	some $d_k, \ldots, d_n$ with $\sum_{j=k}^{n} \lvert d_j \rvert^2 = 1$ and by Lemma 6.8 $R(z) = \sum_{j=k}^{n} \lambda_j \lvert d_j \rvert^2 \le$
lyche-numerical-linear-al F02055	157	■0.959	`\lambda_{k}`	λ_{k}	λ_k. Since $\boldsymbol{z} \in \tilde{\mathcal{S}}$ is arbitrary we have $\max_{\substack{\boldsymbol{x} \in \tilde{\mathcal{S}} \\ \boldsymbol{x} \neq \boldsymbol{0}}} R(\boldsymbol{x}) \le \lambda_k$ and equality in (6.11)
lyche-numerical-linear-al F02056	157	■0.959	`\boldsymbol{z} \in \tilde{\mathcal{S}}`	$\boldsymbol{z} \in \tilde{\mathcal{S}}$	λ_k. Since $\boldsymbol{z} \in \tilde{\mathcal{S}}$ is arbitrary we have $\max_{\substack{\boldsymbol{x} \in \tilde{\mathcal{S}} \\ \boldsymbol{x} \neq \boldsymbol{0}}} R(\boldsymbol{x}) \le \lambda_k$ and equality in (6.11)
lyche-numerical-linear-al F02057	157	■0.959	`\max _{\substack{\boldsymbol{x} \in \tilde{\mathcal{S}} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) \leq \lambda_{k}`	$\max _{\substack{\boldsymbol{x} \in \tilde{\mathcal{S}} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) \leq \lambda_{k}$	λ_k. Since $\boldsymbol{z} \in \tilde{\mathcal{S}}$ is arbitrary we have $\max_{\substack{\boldsymbol{x} \in \tilde{\mathcal{S}} \\ \boldsymbol{x} \neq \boldsymbol{0}}} R(\boldsymbol{x}) \le \lambda_k$ and equality in (6.11)
lyche-numerical-linear-al F02058	157	■1.000	`\mathcal{S}=\tilde{\mathcal{S}}`	$\mathcal{S}=\tilde{\mathcal{S}}$	follows for $\mathcal{S} = \tilde{\mathcal{S}}$. Moreover, $R(\boldsymbol{u}_k) = \lambda_k$.
lyche-numerical-linear-al F02059	157	■1.000	`R\left(\boldsymbol{u}_{k}\right)=\lambda_{k}`	$R\left(\boldsymbol{u}_{k}\right)=\lambda_{k}$	follows for $\mathcal{S} = \tilde{\mathcal{S}}$. Moreover, $R(\boldsymbol{u}_k) = \lambda_k$.
lyche-numerical-linear-al F02060	158	■1.000	`\mathcal{S}=\tilde{\mathcal{S}}:=\operatorname{span} \left(\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{k} \right)`	$\mathcal{S}=\tilde{\mathcal{S}}:=\operatorname{span} \left(\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{k} \right)$	*with equality for* $\mathcal{S} = \tilde{\mathcal{S}} := \operatorname{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k)$ *and* $\boldsymbol{x} = \boldsymbol{u}_k$. *Here* $(\lambda_j, \boldsymbol{u}_j)$, $1 \le j \le$
lyche-numerical-linear-al F02061	158	■1.000	`\left(\lambda_{j}, \boldsymbol{u}_{j}\right), 1 \leq j \leq`	$\left(\lambda_{j}, \boldsymbol{u}_{j}\right), 1 \leq j \leq$	*with equality for* $\mathcal{S} = \tilde{\mathcal{S}} := \operatorname{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k)$ *and* $\boldsymbol{x} = \boldsymbol{u}_k$. *Here* $(\lambda_j, \boldsymbol{u}_j)$, $1 \le j \le$
lyche-numerical-linear-al F02062	158	■1.000	`\operatorname{span}\left(\boldsymbol{u}_{k}, \ldots, \boldsymbol{u}_{n}\right)`	$\operatorname{span}\left(\boldsymbol{u}_{k}, \ldots, \boldsymbol{u}_{n}\right)$	$\operatorname{span}(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n)$ and show that $R(\boldsymbol{y}) \le \lambda_k$ for some $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}'$. It is easy to see
lyche-numerical-linear-al F02063	158	■1.000	`R(\boldsymbol{y}) \leq \lambda_{k}`	$R(\boldsymbol{y}) \leq \lambda_{k}$	$\operatorname{span}(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n)$ and show that $R(\boldsymbol{y}) \le \lambda_k$ for some $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}'$. It is easy to see
lyche-numerical-linear-al F02064	158	■1.000	`\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}^{\prime}`	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}^{\prime}$	$\operatorname{span}(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n)$ and show that $R(\boldsymbol{y}) \le \lambda_k$ for some $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}'$. It is easy to see
lyche-numerical-linear-al F02065	158	■1.000	`\boldsymbol{y} \in \tilde{\mathcal{S}}`	$\boldsymbol{y} \in \tilde{\mathcal{S}}$	that $R(\boldsymbol{y}) \ge \lambda_k$ for any $\boldsymbol{y} \in \tilde{\mathcal{S}}$.
lyche-numerical-linear-al F02066	158	■0.995	`\boldsymbol{A}, \boldsymbol{B} \in`	$\boldsymbol{A}, \boldsymbol{B} \in$	**Theorem 6.13 (Eigenvalue Perturbation for Hermitian Matrices)** *Let* $\boldsymbol{A}, \boldsymbol{B} \in$
lyche-numerical-linear-al F02067	158	■0.998	`\alpha_{1} \geq \alpha_{2} \geq \cdots \geq \alpha_{n}`	$\alpha_{1} \geq \alpha_{2} \geq \cdots \geq \alpha_{n}$	$\mathbb{C}^{n \times n}$ *be Hermitian with eigenvalues* $\alpha_1 \ge \alpha_2 \ge \cdots \ge \alpha_n$ *and* $\beta_1 \ge \beta_2 \ge \cdots \ge \beta_n$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02068	158	0.998	`\beta_{1} \geq \beta_{2} \geq \cdots \geq \beta_{n}`	$\beta_{1} \geq \beta_{2} \geq \cdots \geq \beta_{n}$	$\mathbb{C}^{n\times n}$ *be Hermitian with eigenvalues* $\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n$ *and* $\beta_1 \geq \beta_2 \geq \cdots \geq \beta_n$.
lyche-numerical-linear-algebra F02069	158	1.000	`\varepsilon_{1} \geq \varepsilon_{2} \geq \cdots \geq \varepsilon_{n}`	$\varepsilon_{1} \geq \varepsilon_{2} \geq \cdots \geq \varepsilon_{n}$	*where* $\varepsilon_1 \geq \varepsilon_2 \geq \cdots \geq \varepsilon_n$ *are the eigenvalues of* $\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$.
lyche-numerical-linear-algebra F02070	158	1.000	`\boldsymbol{E}:=\boldsymbol{B}-\boldsymbol{A}`	$\boldsymbol{E}:=\boldsymbol{B}-\boldsymbol{A}$	*where* $\varepsilon_1 \geq \varepsilon_2 \geq \cdots \geq \varepsilon_n$ *are the eigenvalues of* $\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$.
lyche-numerical-linear-algebra F02071	158	0.997	`\left(\alpha_{j}, \boldsymbol{u}_{j}\right), j=1, \ldots, n`	$\left(\alpha_{j}, \boldsymbol{u}_{j}\right), j=1, \ldots, n$	eigenvalues are real. Let $(\alpha_j, \boldsymbol{u}_j)$, $j = 1, \ldots, n$ be orthonormal eigenpairs for $\boldsymbol{A}$
lyche-numerical-linear-algebra F02072	158	1.000	`\mathcal{S}:=\operatorname{span}\left\{ \boldsymbol{u}_{k}, \ldots, \boldsymbol{u}_{n}\right \}`	$\mathcal{S}:=\operatorname{span}\left\{\boldsymbol{u}_{k}, \ldots, \boldsymbol{u}_{n}\right\}$	and let $\mathcal{S} := \operatorname{span}\{\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n\}$. By Theorem 6.11 we obtain
lyche-numerical-linear-algebra F02073	158	1.000	`\boldsymbol{D}:=-\boldsymbol{E}`	$\boldsymbol{D}:=-\boldsymbol{E}$	and this proves the upper inequality. For the lower one we define $\boldsymbol{D} := -\boldsymbol{E}$ and
lyche-numerical-linear-algebra F02074	158	1.000	`-\varepsilon_{n}`	$-\varepsilon_{n}$	observe that $-\varepsilon_n$ is the largest eigenvalue of $\boldsymbol{D}$. Since $\boldsymbol{A} = \boldsymbol{B}+\boldsymbol{D}$ it follows from the
lyche-numerical-linear-algebra F02075	158	1.000	`\boldsymbol{A}=\boldsymbol{B}+\boldsymbol{D}`	$\boldsymbol{A}=\boldsymbol{B}+\boldsymbol{D}$	observe that $-\varepsilon_n$ is the largest eigenvalue of $\boldsymbol{D}$. Since $\boldsymbol{A} = \boldsymbol{B}+\boldsymbol{D}$ it follows from the
lyche-numerical-linear-algebra F02076	158	1.000	`\alpha_{k} \leq \beta_{k}-\varepsilon_{n}`	$\alpha_{k} \leq \beta_{k}-\varepsilon_{n}$	result just proved that $\alpha_k \leq \beta_k - \varepsilon_n$, which is the same as the lower inequality.
lyche-numerical-linear-algebra F02077	159	1.000	`\mu_{1}, \ldots, \mu_{n}`	$\mu_{1}, \ldots, \mu_{n}$	*normal matrices with eigenvalues* $\lambda_1, \ldots, \lambda_n$ *and* $\mu_1, \ldots, \mu_n$*, respectively. Then*
lyche-numerical-linear-algebra F02078	159	0.655	`\boldsymbol{y} \in \mathbb{C}^{n}`	$\boldsymbol{y} \in \mathbb{C}^{n}$	$\lambda \in \mathbb{C}$ and $\boldsymbol{y} \in \mathbb{C}^n$ is nonzero. We say that $(\lambda, \boldsymbol{y})$ is a **left eigenpair** for $\boldsymbol{A}$ if
lyche-numerical-linear-algebra F02079	159	0.655	`\lambda, \boldsymbol{y}`	$\lambda, \boldsymbol{y}$	$\lambda \in \mathbb{C}$ and $\boldsymbol{y} \in \mathbb{C}^n$ is nonzero. We say that $(\lambda, \boldsymbol{y})$ is a **left eigenpair** for $\boldsymbol{A}$ if
lyche-numerical-linear-algebra F02080	159	0.979	`\boldsymbol{y}^{*} \boldsymbol{A}=\lambda \boldsymbol{y}^{*}`	$\boldsymbol{y}^{*} \boldsymbol{A}=\lambda \boldsymbol{y}^{*}$	$\boldsymbol{y}^*\boldsymbol{A} = \lambda\boldsymbol{y}^*$ or equivalently $\boldsymbol{A}^*\boldsymbol{y} = \bar{\lambda}\boldsymbol{y}$. We say that $(\lambda, \boldsymbol{y})$ is a **right eigenpair** for
lyche-numerical-linear-algebra F02081	159	0.979	`\boldsymbol{A}^{*} \boldsymbol{y}=\bar{\lambda} \boldsymbol{y}`	$\boldsymbol{A}^{*} \boldsymbol{y}=\bar{\lambda} \boldsymbol{y}$	$\boldsymbol{y}^*\boldsymbol{A} = \lambda\boldsymbol{y}^*$ or equivalently $\boldsymbol{A}^*\boldsymbol{y} = \bar{\lambda}\boldsymbol{y}$. We say that $(\lambda, \boldsymbol{y})$ is a **right eigenpair** for
lyche-numerical-linear-algebra F02082	159	0.988	`\boldsymbol{A} \boldsymbol{y}=\lambda \boldsymbol{y}`	$\boldsymbol{A} \boldsymbol{y}=\lambda \boldsymbol{y}$	$\boldsymbol{A}$ if $\boldsymbol{A}\boldsymbol{y} = \lambda\boldsymbol{y}$. If $(\lambda, \boldsymbol{y})$ is a left eigenpair then λ is called a **left eigenvalue** and
lyche-numerical-linear-algebra F02083	159	0.988	`(\lambda, \boldsymbol{y})`	$(\lambda, \boldsymbol{y})$	$\boldsymbol{A}$ if $\boldsymbol{A}\boldsymbol{y} = \lambda\boldsymbol{y}$. If $(\lambda, \boldsymbol{y})$ is a left eigenpair then λ is called a **left eigenvalue** and
lyche-numerical-linear-algebra F02084	159	0.779	`(\lambda, y)`	(λ, y)	$\boldsymbol{y}$ a **left eigenvector**. Similarly if $(\lambda, \boldsymbol{y})$ is a right eigenpair then λ is called a **right**
lyche-numerical-linear-algebra F02085	159	0.999	`\bar{\lambda}`	$\bar{\lambda}$	is an eigenvector of $\boldsymbol{A}^*$. If λ is a left eigenvalue of $\boldsymbol{A}$ then $\bar{\lambda}$ is an eigenvalue of $\boldsymbol{A}^*$
lyche-numerical-linear-algebra F02086	159	1.000	`\left\{\boldsymbol{y}_{1}, \ldots, \boldsymbol{y}_{n} \right\}`	$\left\{\boldsymbol{y}_{1}, \ldots, \boldsymbol{y}_{n}\right\}$	*eigenvectors* $\{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n\}$ *with* $\boldsymbol{y}_i^*\boldsymbol{x}_j = \delta_{i,j}$*. Conversely, if* $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ *has*
lyche-numerical-linear-algebra F02087	159	1.000	`\boldsymbol{y}_{i}^{*} \boldsymbol{x}_{j}=\delta_{i, j}`	$\boldsymbol{y}_{i}^{*} \boldsymbol{x}_{j}=\delta_{i, j}$	*eigenvectors* $\{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n\}$ *with* $\boldsymbol{y}_i^*\boldsymbol{x}_j = \delta_{i,j}$*. Conversely, if* $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ *has*
lyche-numerical-linear-algebra F02088	160	1.000	`\left(\lambda_{1}, \boldsymbol{x}_{1}\right), \ldots, \left(\lambda_{n}, \boldsymbol{x}_{n}\right)`	$\left(\lambda_{1}, \boldsymbol{x}_{1}\right), \ldots,\left(\lambda_{n}, \boldsymbol{x}_{n}\right)$	***Proof*** For any right eigenpairs $(\lambda_1, \boldsymbol{x}_1), \ldots, (\lambda_n, \boldsymbol{x}_n)$ and left eigenpairs
lyche-numerical-linear-algebra F02089	160	0.998	`\left(\lambda_{1}, \boldsymbol{y}_{1}\right), \ldots, \left(\lambda_{n}, \boldsymbol{y}_{n}\right)`	$\left(\lambda_{1}, \boldsymbol{y}_{1}\right), \ldots,\left(\lambda_{n}, \boldsymbol{y}_{n}\right)$	$(\lambda_1, \boldsymbol{y}_1), \ldots, (\lambda_n, \boldsymbol{y}_n)$ of $\boldsymbol{A}$ we have $\boldsymbol{A}\boldsymbol{X} = \boldsymbol{X}\boldsymbol{D}$, $\boldsymbol{Y}^*\boldsymbol{A} = \boldsymbol{D}\boldsymbol{Y}^*$, where
lyche-numerical-linear-algebra F02090	160	0.998	`\boldsymbol{A} \boldsymbol{X}=\boldsymbol{X} \boldsymbol{D}, \boldsymbol{Y}^{*} \boldsymbol{A}= \boldsymbol{D} \boldsymbol{Y}^{*}`	$\boldsymbol{A} \boldsymbol{X}=\boldsymbol{X} \boldsymbol{D}, \boldsymbol{Y}^{*} \boldsymbol{A}=\boldsymbol{D} \boldsymbol{Y}^{*}$	$(\lambda_1, \boldsymbol{y}_1), \ldots, (\lambda_n, \boldsymbol{y}_n)$ of $\boldsymbol{A}$ we have $\boldsymbol{A}\boldsymbol{X} = \boldsymbol{X}\boldsymbol{D}$, $\boldsymbol{Y}^*\boldsymbol{A} = \boldsymbol{D}\boldsymbol{Y}^*$, where
lyche-numerical-linear-algebra F02091	160	1.000	`\boldsymbol{A} \boldsymbol{X}=\boldsymbol{X} \boldsymbol{D} \Longrightarrow \boldsymbol{A}= \boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1} \Longrightarrow \boldsymbol{X}^{-1} \boldsymbol{A}=`	$\boldsymbol{A} \boldsymbol{X}=\boldsymbol{X} \boldsymbol{D} \Longrightarrow \boldsymbol{A}=\boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1} \Longrightarrow \boldsymbol{X}^{-1} \boldsymbol{A}=$	Suppose $\boldsymbol{X}$ is nonsingular. Then $\boldsymbol{A}\boldsymbol{X} = \boldsymbol{X}\boldsymbol{D} \implies \boldsymbol{A} = \boldsymbol{X}\boldsymbol{D}\boldsymbol{X}^{-1} \implies \boldsymbol{X}^{-1}\boldsymbol{A} =$
lyche-numerical-linear-algebra F02092	160	1.000	`\boldsymbol{D} \boldsymbol{X}^{-1}`	$\boldsymbol{D} \boldsymbol{X}^{-1}$	$\boldsymbol{D}\boldsymbol{X}^{-1}$ and it follows that $\boldsymbol{Y}^* := \boldsymbol{X}^{-1}$ contains a collection of left eigenvectors such

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F02093	160	■1.000	\boldsymbol{Y}^{*}:=\boldsymbol{X}^{-1}	$\boldsymbol{Y}^{*}:=\boldsymbol{X}^{-1}$	$\boldsymbol{D}\boldsymbol{X}^{-1}$ and it follows that $\boldsymbol{Y}^* := \boldsymbol{X}^{-1}$ contains a collection of left eigenvectors such
lyche-numerical-linear-algebra_F02094	160	■1.000	\boldsymbol{Y}^{*} \boldsymbol{X}=\boldsymbol{I}	$\boldsymbol{Y}^{*} \boldsymbol{X}=\boldsymbol{I}$	that $\boldsymbol{Y}^*\boldsymbol{X} = \boldsymbol{I}$. Thus the columns of $\boldsymbol{Y}$ are linearly independent and $\boldsymbol{y}_i^*\boldsymbol{x}_j = \delta_{i,j}$.
lyche-numerical-linear-algebra_F02095	160	■1.000	\boldsymbol{Y}	$\boldsymbol{Y}$	that $\boldsymbol{Y}^*\boldsymbol{X} = \boldsymbol{I}$. Thus the columns of $\boldsymbol{Y}$ are linearly independent and $\boldsymbol{y}_i^*\boldsymbol{x}_j = \delta_{i,j}$.
lyche-numerical-linear-algebra_F02096	160	■1.000	\boldsymbol{A} \boldsymbol{Y}^{-*}=\boldsymbol{Y}^{-*} \boldsymbol{D}	$\boldsymbol{A} \boldsymbol{Y}^{-*}=\boldsymbol{Y}^{-*} \boldsymbol{D}$	Similarly, if $\boldsymbol{Y}$ is nonsingular then $\boldsymbol{A}\boldsymbol{Y}^{-*} = \boldsymbol{Y}^{-*}\boldsymbol{D}$ and it follows that $\boldsymbol{X} := \boldsymbol{Y}^{-*}$
lyche-numerical-linear-algebra_F02097	160	■1.000	\boldsymbol{X}:=\boldsymbol{Y}^{-*}	$\boldsymbol{X}:=\boldsymbol{Y}^{-*}$	Similarly, if $\boldsymbol{Y}$ is nonsingular then $\boldsymbol{A}\boldsymbol{Y}^{-*} = \boldsymbol{Y}^{-*}\boldsymbol{D}$ and it follows that $\boldsymbol{X} := \boldsymbol{Y}^{-*}$
lyche-numerical-linear-algebra_F02098	160	■0.996	\boldsymbol{v}=\sum_{j=1}^{n} c_{j} \boldsymbol{x}_{j}	$\boldsymbol{v}=\sum_{j=1}^{n} c_{j} \boldsymbol{x}_{j}$	If $\boldsymbol{v} = \sum_{j=1}^n c_j\boldsymbol{x}_j$ then $\boldsymbol{y}_i^*\boldsymbol{v} = \sum_{j=1}^n c_j\boldsymbol{y}_i^*\boldsymbol{x}_j = c_i\boldsymbol{y}_i^*\boldsymbol{x}_i$, so $c_i = \boldsymbol{y}_i^*\boldsymbol{v}/\boldsymbol{y}_i^*\boldsymbol{x}_i$ for
lyche-numerical-linear-algebra_F02099	160	■0.996	\boldsymbol{y}_{i}^{*} \boldsymbol{v}=\sum_{j=1}^{n} c_{j} \boldsymbol{y}_{i}^{*} \boldsymbol{x}_{j}=c_{i} \boldsymbol{y}_{i}^{*} \boldsymbol{x}_{i}	$\boldsymbol{y}_{i}^{*} \boldsymbol{v}=\sum_{j=1}^{n} c_{j} \boldsymbol{y}_{i}^{*} \boldsymbol{x}_{j}=c_{i} \boldsymbol{y}_{i}^{*} \boldsymbol{x}_{i}$	If $\boldsymbol{v} = \sum_{j=1}^n c_j\boldsymbol{x}_j$ then $\boldsymbol{y}_i^*\boldsymbol{v} = \sum_{j=1}^n c_j\boldsymbol{y}_i^*\boldsymbol{x}_j = c_i\boldsymbol{y}_i^*\boldsymbol{x}_i$, so $c_i = \boldsymbol{y}_i^*\boldsymbol{v}/\boldsymbol{y}_i^*\boldsymbol{x}_i$ for
lyche-numerical-linear-algebra_F02100	160	■0.996	c_{i}=\boldsymbol{y}_{i}^{*} \boldsymbol{v} / \boldsymbol{y}_{i}^{*} \boldsymbol{x}_{i}	$c_{i}=\boldsymbol{y}_{i}^{*} \boldsymbol{v} / \boldsymbol{y}_{i}^{*} \boldsymbol{x}_{i}$	If $\boldsymbol{v} = \sum_{j=1}^n c_j\boldsymbol{x}_j$ then $\boldsymbol{y}_i^*\boldsymbol{v} = \sum_{j=1}^n c_j\boldsymbol{y}_i^*\boldsymbol{x}_j = c_i\boldsymbol{y}_i^*\boldsymbol{x}_i$, so $c_i = \boldsymbol{y}_i^*\boldsymbol{v}/\boldsymbol{y}_i^*\boldsymbol{x}_i$ for
lyche-numerical-linear-algebra_F02101	160	■0.995	\mu, \boldsymbol{y}	$\mu, \boldsymbol{y}$	**Theorem 6.16 (Biorthogonality)** *Suppose* $(\mu, \boldsymbol{y})$ *and* $(\lambda, \boldsymbol{x})$ *are left and right*
lyche-numerical-linear-algebra_F02102	160	■1.000	\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{y}^{*} \boldsymbol{x}=\mu \boldsymbol{y}^{*} \boldsymbol{x}	$\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{y}^{*} \boldsymbol{x}=\mu \boldsymbol{y}^{*} \boldsymbol{x}$	***Proof*** Using the eigenpair relation in two ways we obtain $\boldsymbol{y}^*\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{y}^*\boldsymbol{x} = \mu\boldsymbol{y}^*\boldsymbol{x}$
lyche-numerical-linear-algebra_F02103	160	■1.000	\boldsymbol{y}^{*} \boldsymbol{x} \neq 0	$\boldsymbol{y}^{*} \boldsymbol{x} \neq 0$	*eigenpairs of* $\boldsymbol{A} \in \mathbb{C}^{n\times n}$. *If* λ *has algebraic multiplicity one then* $\boldsymbol{y}^*\boldsymbol{x} \neq 0$.
lyche-numerical-linear-algebra_F02104	160	■1.000	\\|\boldsymbol{x}\\|_{2}=1	$\\|\boldsymbol{x}\\|_{2}=1$	***Proof*** Assume that $\\|\boldsymbol{x}\\|_2 = 1$. We have (cf. (6.8))
lyche-numerical-linear-algebra_F02105	161	■1.000	\boldsymbol{V} \boldsymbol{e}_{1}=\boldsymbol{x}	$\boldsymbol{V} \boldsymbol{e}_{1}=\boldsymbol{x}$	where $\boldsymbol{V}$ is unitary and $\boldsymbol{V}\boldsymbol{e}_1 = \boldsymbol{x}$. We show that if $\boldsymbol{y}^*\boldsymbol{x} = 0$ then λ is also an
lyche-numerical-linear-algebra_F02106	161	■1.000	\boldsymbol{u}:=\boldsymbol{V}^{*} \boldsymbol{y}	$\boldsymbol{u}:=\boldsymbol{V}^{*} \boldsymbol{y}$	eigenvalue of $\boldsymbol{M}$ contradicting the multiplicity assumption of λ. Let $\boldsymbol{u} := \boldsymbol{V}^*\boldsymbol{y}$.
lyche-numerical-linear-algebra_F02107	161	■1.000	(\bar{\lambda}, \boldsymbol{u})	$(\bar{\lambda}, \boldsymbol{u})$	so $(\bar{\lambda}, \boldsymbol{u})$ is an eigenpair of $\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{V}$. But then $\boldsymbol{y}^*\boldsymbol{x} = \boldsymbol{u}^*\boldsymbol{V}^*\boldsymbol{V}\boldsymbol{e}_1 = \boldsymbol{u}^*\boldsymbol{e}_1$. Suppose
lyche-numerical-linear-algebra_F02108	161	■1.000	\boldsymbol{V}^{*} \boldsymbol{A}^{*} \boldsymbol{V}	$\boldsymbol{V}^{*} \boldsymbol{A}^{*} \boldsymbol{V}$	so $(\bar{\lambda}, \boldsymbol{u})$ is an eigenpair of $\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{V}$. But then $\boldsymbol{y}^*\boldsymbol{x} = \boldsymbol{u}^*\boldsymbol{V}^*\boldsymbol{V}\boldsymbol{e}_1 = \boldsymbol{u}^*\boldsymbol{e}_1$. Suppose
lyche-numerical-linear-algebra_F02109	161	■1.000	\boldsymbol{y}^{*} \boldsymbol{x}=\boldsymbol{u}^{*} \boldsymbol{V}^{*} \boldsymbol{V} \boldsymbol{e}_{1}=\boldsymbol{u}^{*} \boldsymbol{e}_{1}	$\boldsymbol{y}^{*} \boldsymbol{x}=\boldsymbol{u}^{*} \boldsymbol{V}^{*} \boldsymbol{V} \boldsymbol{e}_{1}=\boldsymbol{u}^{*} \boldsymbol{e}_{1}$	so $(\bar{\lambda}, \boldsymbol{u})$ is an eigenpair of $\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{V}$. But then $\boldsymbol{y}^*\boldsymbol{x} = \boldsymbol{u}^*\boldsymbol{V}^*\boldsymbol{V}\boldsymbol{e}_1 = \boldsymbol{u}^*\boldsymbol{e}_1$. Suppose
lyche-numerical-linear-algebra_F02110	161	■0.596	\boldsymbol{u}^{*} \boldsymbol{e}_{1}=0	$\boldsymbol{u}^{*} \boldsymbol{e}_{1}=0$	that $\boldsymbol{u}^*\boldsymbol{e}_1 = 0$, i.e., $\boldsymbol{u} = \left[\begin{smallmatrix}0\\ \boldsymbol{v}\end{smallmatrix}\right]$ for some nonzero $\boldsymbol{v} \in \mathbb{C}^{n-1}$. Then
lyche-numerical-linear-algebra_F02111	161	■0.596	\boldsymbol{u}=\left[\begin{array}{l}0 \\\\ \boldsymbol{v}\end{array}\right]	$\boldsymbol{u}=\left[\begin{array}{l}0 \\\\ \boldsymbol{v}\end{array}\right]$	that $\boldsymbol{u}^*\boldsymbol{e}_1 = 0$, i.e., $\boldsymbol{u} = \left[\begin{smallmatrix}0\\ \boldsymbol{v}\end{smallmatrix}\right]$ for some nonzero $\boldsymbol{v} \in \mathbb{C}^{n-1}$. Then
lyche-numerical-linear-algebra_F02112	161	■0.596	\boldsymbol{v} \in \mathbb{C}^{n-1}	$\boldsymbol{v} \in \mathbb{C}^{n-1}$	that $\boldsymbol{u}^*\boldsymbol{e}_1 = 0$, i.e., $\boldsymbol{u} = \left[\begin{smallmatrix}0\\ \boldsymbol{v}\end{smallmatrix}\right]$ for some nonzero $\boldsymbol{v} \in \mathbb{C}^{n-1}$. Then
lyche-numerical-linear-algebra_F02113	161	■1.000	\boldsymbol{A}:=\left[\begin{array}{ll}1 & 1 \\\\ 0 & 1\end{array}\right]	$\boldsymbol{A}:=\left[\begin{array}{ll}1 & 1 \\\\ 0 & 1\end{array}\right]$	$\boldsymbol{A} := \left[\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right]$ has one eigenvalue $\lambda = 1$ of algebraic multiplicity two, one right
lyche-numerical-linear-algebra_F02114	161	■1.000	\boldsymbol{y}=\boldsymbol{e}_{2}	$\boldsymbol{y}=\boldsymbol{e}_{2}$	eigenvector $\boldsymbol{x} = \boldsymbol{e}_1$ and one left eigenvector $\boldsymbol{y} = \boldsymbol{e}_2$. Thus $\boldsymbol{x}$ and $\boldsymbol{y}$ are orthogonal.
lyche-numerical-linear-algebra_F02115	161	■1.000	\operatorname{det}\left(\boldsymbol{B}^{T}\right)=	$\operatorname{det}\left(\boldsymbol{B}^{T}\right)=$	**Exercise 6.2 (Characteristic Polynomial of Transpose)** We have $\det(\boldsymbol{B}^T) =$
lyche-numerical-linear-algebra_F02116	161	■1.000	\operatorname{det}(\boldsymbol{B})	$\operatorname{det}(\boldsymbol{B})$	$\det(\boldsymbol{B})$ and $\det(\overline{\boldsymbol{B}}) = \overline{\det(\boldsymbol{B})}$for any square matrix $\boldsymbol{B}$. Use this to show that
lyche-numerical-linear-algebra_F02117	161	■1.000	\operatorname{det}(\overline{\boldsymbol{B}})=\overline{ \operatorname{det}(\boldsymbol{B})}	$\operatorname{det}(\overline{\boldsymbol{B}})=\overline{ \operatorname{det}(\boldsymbol{B})}$	$\det(\boldsymbol{B})$ and $\det(\overline{\boldsymbol{B}}) = \overline{\det(\boldsymbol{B})}$for any square matrix $\boldsymbol{B}$. Use this to show that

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F02118	161	0.991	`\pi_{\boldsymbol{A}^{T}}=\pi_{\boldsymbol{A}}`	$\pi_{\boldsymbol{A}^{T}}=\pi_{\boldsymbol{A}}$	**a)** $\pi_{\boldsymbol{A}^T} = \pi_{\boldsymbol{A}}$,
lyche-numerical-linear-algebra_F02119	161	1.000	`\pi_{A^{*}}(\bar{\lambda})=\overline{\pi_{A}(\lambda)}`	$\pi_{A^{*}}(\bar{\lambda})=\overline{\pi_{A}(\lambda)}$	**b)** $\pi_{\boldsymbol{A}^*}(\bar{\lambda}) = \overline{\pi_{\boldsymbol{A}}(\lambda)}$.
lyche-numerical-linear-algebra_F02120	161	1.000	`\left(\lambda^{-1}, \boldsymbol{x}\right)`	$\left(\lambda^{-1}, \boldsymbol{x}\right)$	**a)** If $\boldsymbol{A}$ is nonsingular then $(\lambda^{-1}, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^{-1}$.
lyche-numerical-linear-algebra_F02121	161	0.562	`\left(\lambda^{k}, \boldsymbol{x}\right)`	$\left(\lambda^{k}, \boldsymbol{x}\right)$	**b)** $(\lambda^k, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^k$ for $k \in \mathbb{Z}$.
lyche-numerical-linear-algebra_F02122	161	0.562	`\boldsymbol{A}^{k}`	$\boldsymbol{A}^{k}$	**b)** $(\lambda^k, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^k$ for $k \in \mathbb{Z}$.
lyche-numerical-linear-algebra_F02123	161	0.562	`k \in \mathbb{Z}`	$k \in \mathbb{Z}$	**b)** $(\lambda^k, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^k$ for $k \in \mathbb{Z}$.
lyche-numerical-linear-algebra_F02124	162	1.000	`\left(\lambda_{j}, \boldsymbol{x}_{j}\right), j=1, \ldots, n`	$\left(\lambda_{j}, \boldsymbol{x}_{j}\right), j=1, \ldots, n$	is nondefective with eigenpairs $(\lambda_j, \boldsymbol{x}_j)$, $j = 1, \ldots, n$ then for any $\boldsymbol{x} \in \mathbb{C}^n$ and
lyche-numerical-linear-algebra_F02125	162	1.000	`k \in \mathbb{N}`	$k \in \mathbb{N}$	$k \in \mathbb{N}$
lyche-numerical-linear-algebra_F02126	162	1.000	`\boldsymbol{A}^{2}=`	$\boldsymbol{A}^{2}=$	**Exercise 6.5 (Eigenvalues of an Idempotent Matrix)** Let $\lambda \in \sigma(\boldsymbol{A})$ where $\boldsymbol{A}^2 =$
lyche-numerical-linear-algebra_F02127	162	1.000	`\lambda=0`	$\lambda=0$	$\boldsymbol{A} \in \mathbb{C}^{n\times n}$. Show that $\lambda = 0$ or $\lambda = 1$. (A matrix is called **idempotent** if $\boldsymbol{A}^2 = \boldsymbol{A}$).
lyche-numerical-linear-algebra_F02128	162	1.000	`\boldsymbol{A}^{2}=\boldsymbol{A}`	$\boldsymbol{A}^{2}=\boldsymbol{A}$	$\boldsymbol{A} \in \mathbb{C}^{n\times n}$. Show that $\lambda = 0$ or $\lambda = 1$. (A matrix is called **idempotent** if $\boldsymbol{A}^2 = \boldsymbol{A}$).
lyche-numerical-linear-algebra_F02129	162	1.000	`\boldsymbol{A}^{k}=0`	$\boldsymbol{A}^{k}=0$	**Exercise 6.6 (Eigenvalues of a Nilpotent Matrix)** Let $\lambda \in \sigma(\boldsymbol{A})$ where $\boldsymbol{A}^k = 0$
lyche-numerical-linear-algebra_F02130	162	1.000	`\boldsymbol{A}^{*} \boldsymbol{A}=\boldsymbol{I}`	$\boldsymbol{A}^{*} \boldsymbol{A}=\boldsymbol{I}$	**Exercise 6.7 (Eigenvalues of a Unitary Matrix)** Let $\lambda \in \sigma(\boldsymbol{A})$, where $\boldsymbol{A}^*\boldsymbol{A} = \boldsymbol{I}$.
lyche-numerical-linear-algebra_F02131	162	1.000	`|\lambda|=1`	$\lvert\lambda\rvert=1$	Show that $\lvert\lambda\rvert = 1$.
lyche-numerical-linear-algebra_F02132	162	1.000	`\epsilon_{0}>0`	$\epsilon_{0}>0$	$\mathbb{C}^{n\times n}$ is singular. Then we can find $\epsilon_0 > 0$ such that $\boldsymbol{A} + \epsilon\boldsymbol{I}$ is nonsingular for all
lyche-numerical-linear-algebra_F02133	162	1.000	`\boldsymbol{A}+\epsilon \boldsymbol{I}`	$\boldsymbol{A}+\epsilon \boldsymbol{I}$	$\mathbb{C}^{n\times n}$ is singular. Then we can find $\epsilon_0 > 0$ such that $\boldsymbol{A} + \epsilon\boldsymbol{I}$ is nonsingular for all
lyche-numerical-linear-algebra_F02134	162	0.997	`\epsilon \in \mathbb{C}`	$\epsilon \in \mathbb{C}$	$\epsilon \in \mathbb{C}$ with $\lvert\epsilon\rvert < \epsilon_0$. Hint: $\det(\boldsymbol{A}) = \lambda_1\lambda_2\cdots\lambda_n$, where λ_i are the eigenvalues of $\boldsymbol{A}$.
lyche-numerical-linear-algebra_F02135	162	0.997	`|\epsilon|<\epsilon_{0}`	$\lvert\epsilon\rvert<\epsilon_{0}$	$\epsilon \in \mathbb{C}$ with $\lvert\epsilon\rvert < \epsilon_0$. Hint: $\det(\boldsymbol{A}) = \lambda_1\lambda_2\cdots\lambda_n$, where λ_i are the eigenvalues of $\boldsymbol{A}$.
lyche-numerical-linear-algebra_F02136	162	0.997	`\operatorname{det}(\boldsymbol{A})=\lambda_{1} \lambda_{2} \cdots \lambda_{n}`	$\operatorname{det}(\boldsymbol{A})=\lambda_{1} \lambda_{2} \cdots \lambda_{n}$	$\epsilon \in \mathbb{C}$ with $\lvert\epsilon\rvert < \epsilon_0$. Hint: $\det(\boldsymbol{A}) = \lambda_1\lambda_2\cdots\lambda_n$, where λ_i are the eigenvalues of $\boldsymbol{A}$.
lyche-numerical-linear-algebra_F02137	162	1.000	`q_{0}, \ldots, q_{n-1} \in \mathbb{C}`	$q_{0}, \ldots, q_{n-1} \in \mathbb{C}$	**Exercise 6.9 (Companion Matrix)** For $q_0, \ldots, q_{n-1} \in \mathbb{C}$ let $p(\lambda) = \lambda^n +$
lyche-numerical-linear-algebra_F02138	162	1.000	`p(\lambda)=\lambda^{n}+`	$p(\lambda)=\lambda^{n}+$	**Exercise 6.9 (Companion Matrix)** For $q_0, \ldots, q_{n-1} \in \mathbb{C}$ let $p(\lambda) = \lambda^n +$
lyche-numerical-linear-algebra_F02139	162	1.000	`q_{n-1} \lambda^{n-1}+\cdots+q_{0}`	$q_{n-1} \lambda^{n-1}+\cdots+q_{0}$	$q_{n-1}\lambda^{n-1} + \cdots + q_0$ be a polynomial of degree n in λ. We derive two matrices
lyche-numerical-linear-algebra_F02140	162	1.000	`(-1)^{n} p`	$(-1)^{n} p$	that have $(-1)^n p$ as its characteristic polynomial.
lyche-numerical-linear-algebra_F02141	162	0.998	`p=(-1)^{n} \pi_{\boldsymbol{A}}`	$p=(-1)^{n} \pi_{\boldsymbol{A}}$	**a)** Show that $p = (-1)^n\pi_{\boldsymbol{A}}$ where
lyche-numerical-linear-algebra_F02142	162	1.000	`p=(-1)^{n} \pi_{\boldsymbol{B}}`	$p=(-1)^{n} \pi_{\boldsymbol{B}}$	**b)** Show that $p = (-1)^n\pi_{\boldsymbol{B}}$ where
lyche-numerical-linear-algebra_F02143	163	0.998	`\boldsymbol{A}=\left[\begin{array}{lll}1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2\end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{lll}1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2\end{array}\right]$	$\boldsymbol{A} = \left[\begin{array}{lll}1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2\end{array}\right]$. Is $\boldsymbol{A}$ defective?

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F02144	163	1.000	`\boldsymbol{Y} \in \mathbb{R}^{2 \times 2}`	$\boldsymbol{Y} \in \mathbb{R}^{2 \times 2}$	where $a \in \mathbb{R}$. There is a nonsingular matrix $\boldsymbol{Y} \in \mathbb{R}^{2\times 2}$ and a diagonal matrix
lyche-numerical-linear-al F02145	163	1.000	`\boldsymbol{D} \in \mathbb{R}^{2 \times 2}`	$\boldsymbol{D} \in \mathbb{R}^{2 \times 2}$	$\boldsymbol{D} \in \mathbb{R}^{2\times 2}$ such that $\boldsymbol{A} = \boldsymbol{Y}\boldsymbol{D}\boldsymbol{Y}^{-1}$?
lyche-numerical-linear-al F02146	163	1.000	`\boldsymbol{A}=\boldsymbol{Y} \boldsymbol{D} \boldsymbol{Y}^{-1}`	$\boldsymbol{A}=\boldsymbol{Y} \boldsymbol{D} \boldsymbol{Y}^{-1}$	$\boldsymbol{D} \in \mathbb{R}^{2\times 2}$ such that $\boldsymbol{A} = \boldsymbol{Y}\boldsymbol{D}\boldsymbol{Y}^{-1}$?
lyche-numerical-linear-al F02147	163	1.000	`\boldsymbol{A} \in \mathbb{R}^{n, n}`	$\boldsymbol{A} \in \mathbb{R}^{n, n}$	$\boldsymbol{A} \in \mathbb{R}^{n,n}$ be tridiagonal (i.e. $a_{ij} = 0$ when $\lvert i - j \rvert > 1$) and suppose also that
lyche-numerical-linear-al F02148	163	1.000	`a_{i+1, i} a_{i, i+1}>0`	$a_{i+1, i} a_{i, i+1}>0$	$a_{i+1,i}a_{i,i+1} > 0$ for $i = 1, \ldots, n-1$. Show that the eigenvalues of $\boldsymbol{A}$ are real.[2]
lyche-numerical-linear-al F02149	163	1.000	`\boldsymbol{A}=\left[\begin{array}{rrr}3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1\end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{rrr}3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1\end{array}\right]$	For the Jordan factorization of the matrix $\boldsymbol{A} = \left[\begin{smallmatrix}3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1\end{smallmatrix}\right]$ we have $\boldsymbol{J} = \left[\begin{smallmatrix}1&1&0\\0&1&0\\0&0&1\end{smallmatrix}\right]$.
lyche-numerical-linear-al F02150	163	1.000	`\boldsymbol{J}=\left[\begin{array}{lll}1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1\end{array}\right]`	$\boldsymbol{J}=\left[\begin{array}{lll}1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1\end{array}\right]$	For the Jordan factorization of the matrix $\boldsymbol{A} = \left[\begin{smallmatrix}3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1\end{smallmatrix}\right]$ we have $\boldsymbol{J} = \left[\begin{smallmatrix}1&1&0\\0&1&0\\0&0&1\end{smallmatrix}\right]$.
lyche-numerical-linear-al F02151	163	1.000	`\left(\boldsymbol{J}_{m}(\lambda)-\lambda \boldsymbol{I}\right)^{r}=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{I}_{m-r} \\ \mathbf{0} & \mathbf{0}\end{array}\right]`	$\left(\boldsymbol{J}_{m}(\lambda)-\lambda \boldsymbol{I}\right)^{r}=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{I}_{m-r} \\ \mathbf{0} & \mathbf{0}\end{array}\right]$	**Exercise 6.14 (A Nilpotent Matrix)** Show that $(\boldsymbol{J}_m(\lambda) - \lambda\boldsymbol{I})^r = \left[\begin{smallmatrix}\mathbf{0} & \boldsymbol{I}_{m-r} \\ \mathbf{0} & \mathbf{0}\end{smallmatrix}\right]$ for
lyche-numerical-linear-al F02152	163	0.998	`1 \leq r \leq m-1`	$1 \leq r \leq m-1$	$1 \le r \le m-1$ and conclude that $(\boldsymbol{J}_m(\lambda) - \lambda\boldsymbol{I})^m = 0$.
lyche-numerical-linear-al F02153	163	0.998	`\left(\boldsymbol{J}_{m}(\lambda)-\lambda \boldsymbol{I}\right)^{m}=0`	$\left(\boldsymbol{J}_{m}(\lambda)-\lambda \boldsymbol{I}\right)^{m}=0$	$1 \le r \le m-1$ and conclude that $(\boldsymbol{J}_m(\lambda) - \lambda\boldsymbol{I})^m = 0$.
lyche-numerical-linear-al F02154	163	1.000	`r=0,1,2, \ldots, m=2,3, \ldots`	$r=0,1,2, \ldots, m=2,3, \ldots$	Then for $r = 0, 1, 2, \ldots, m = 2, 3, \ldots$, and any $\lambda \in \mathbb{C}$
lyche-numerical-linear-al F02155	163	0.998	`\boldsymbol{A}^{r}=\boldsymbol{S} \boldsymbol{J}^{r} \boldsymbol{S}^{-1}`	$\boldsymbol{A}^{r}=\boldsymbol{S} \boldsymbol{J}^{r} \boldsymbol{S}^{-1}$	**a)** $\boldsymbol{A}^r = \boldsymbol{S}\boldsymbol{J}^r\boldsymbol{S}^{-1}$,
lyche-numerical-linear-al F02156	163	1.000	`\boldsymbol{J}^{r}=\operatorname{diag}\left( \boldsymbol{U}_{1}^{r}, \ldots, \boldsymbol{U}_{k}^{r} \right)`	$\boldsymbol{J}^{r}=\operatorname{diag}\left(\boldsymbol{U}_{1}^{r}, \ldots, \boldsymbol{U}_{k}^{r} \right)$	**b)** $\boldsymbol{J}^r = \operatorname{diag}(\boldsymbol{U}_1^r, \ldots, \boldsymbol{U}_k^r)$,
lyche-numerical-linear-al F02157	163	1.000	`\boldsymbol{D}^{-1} \boldsymbol{A} \boldsymbol{D}`	$\boldsymbol{D}^{-1} \boldsymbol{A} \boldsymbol{D}$	[2]Hint: show that there is a diagonal matrix $\boldsymbol{D}$ such that $\boldsymbol{D}^{-1}\boldsymbol{A}\boldsymbol{D}$ is symmetric.
lyche-numerical-linear-al F02158	164	0.985	`\boldsymbol{U}_{i}^{r}=\operatorname{diag}\left( \boldsymbol{J}_{m_{i, 1}}\left(\lambda_{i}\right)^{r}, \ldots, \boldsymbol{J}_{m_{i, g_{i}}}\left(\lambda_{i} \right)^{r}\right)`	$\boldsymbol{U}_{i}^{r}=\operatorname{diag}\left(\boldsymbol{J}_{m_{i, 1}}\left(\lambda_{i}\right)^{r}, \ldots, \boldsymbol{J}_{m_{i, g_{i}}}\left(\lambda_{i} \right)^{r}\right)$	**c)** $\boldsymbol{U}_i^r = \operatorname{diag}(\boldsymbol{J}_{m_{i,1}}(\lambda_i)^r, \ldots, \boldsymbol{J}_{m_{i,g_i}}(\lambda_i)^r)$,
lyche-numerical-linear-al F02159	164	0.968	`\boldsymbol{J}_{m}(\lambda)^{r}=\left(\boldsymbol{E}_{m}+ \lambda \boldsymbol{I}_{m}\right)^{r}=\sum_{k=0}^{ \min \{r, m-1\}}\binom{r}{k} \lambda^{r-k} \boldsymbol{E}_{m}^{k}`	$\boldsymbol{J}_{m}(\lambda)^{r}=\left(\boldsymbol{E}_{m}+ \lambda \boldsymbol{I}_{m}\right)^{r}=\sum_{k=0}^{ \min \{r, m-1\}}\binom{r}{k} \lambda^{r-k} \boldsymbol{E}_{m}^{k}$	**d)** $\boldsymbol{J}_m(\lambda)^r = (\boldsymbol{E}_m + \lambda\boldsymbol{I}_m)^r = \sum_{k=0}^{\min\{r,m-1\}} \binom{r}{k}\lambda^{r-k}\boldsymbol{E}_m^k$.
lyche-numerical-linear-al F02160	164	1.000	`\boldsymbol{J}^{100}`	$\boldsymbol{J}^{100}$	**Exercise 6.16 (Powers of a Jordan Block)** Find $\boldsymbol{J}^{100}$ and $\boldsymbol{A}^{100}$ for the matrix in
lyche-numerical-linear-al F02161	164	1.000	`\boldsymbol{A}^{100}`	$\boldsymbol{A}^{100}$	**Exercise 6.16 (Powers of a Jordan Block)** Find $\boldsymbol{J}^{100}$ and $\boldsymbol{A}^{100}$ for the matrix in
lyche-numerical-linear-al F02162	164	1.000	`\mu_{\boldsymbol{A}}(\boldsymbol{A})`	$\mu_{\boldsymbol{A}}(\boldsymbol{A})$	is called the **minimal polynomial** of $\boldsymbol{A}$. We define the matrix polynomial $\mu_{\boldsymbol{A}}(\boldsymbol{A})$
lyche-numerical-linear-al F02163	164	1.000	`\lambda_{i}-\lambda`	$\lambda_{i}-\lambda$	by replacing the factors $\lambda_i - \lambda$ by $\lambda_i\boldsymbol{I} - \boldsymbol{A}$.
lyche-numerical-linear-al F02164	164	1.000	`\lambda_{i} \boldsymbol{I}-\boldsymbol{A}`	$\lambda_{i} \boldsymbol{I}-\boldsymbol{A}$	by replacing the factors $\lambda_i - \lambda$ by $\lambda_i\boldsymbol{I} - \boldsymbol{A}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02165	164	1.000	\pi_{\boldsymbol{A}}(\lambda)=\prod_{i=1}^{k} \prod_{j=1}^{g_{i}}\left(\lambda_{i}-\lambda \right)^{m_{i, j}}	$\pi_{\boldsymbol{A}}(\lambda)=\prod_{i=1}^{k} \prod_{j=1}^{g_{i}}\left(\lambda_{i}-\lambda \right)^{m_{i, j}}$	**a)** We have $\pi_{\boldsymbol{A}}(\lambda) = \prod_{i=1}^{k} \prod_{j=1}^{g_i} (\lambda_i - \lambda)^{m_{i,j}}$. Use this to show that the minimal
lyche-numerical-linear-algebra F02166	164	0.629	\pi_{\boldsymbol{A}}=\mu_{\boldsymbol{A}} \nu_{ \boldsymbol{A}}	$\pi_{\boldsymbol{A}}=\mu_{\boldsymbol{A}} \nu_{\boldsymbol{A}}$	polynomial divides the characteristic polynomial, i.e., $\pi_{\boldsymbol{A}} = \mu_{\boldsymbol{A}} \nu_{\boldsymbol{A}}$ for some
lyche-numerical-linear-algebra F02167	164	0.953	\nu_{\boldsymbol{A}}	$\nu_{\boldsymbol{A}}$	polynomial $\nu_{\boldsymbol{A}}$.
lyche-numerical-linear-algebra F02168	164	0.986	\mu_{\boldsymbol{A}}(\boldsymbol{A})=\mathbf{0} \Longleftrightarrow \mu_{\boldsymbol{A}}(\boldsymbol{J})= \mathbf{0}	$\mu_{\boldsymbol{A}}(\boldsymbol{A})=\mathbf{0} \Longleftrightarrow \mu_{\boldsymbol{A}}(\boldsymbol{J})=\mathbf{0}$	**b)** Show that $\mu_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0} \iff \mu_{\boldsymbol{A}}(\boldsymbol{J}) = \mathbf{0}$.
lyche-numerical-linear-algebra F02169	164	1.000	m_{i}	m_{i}	**c)** (can be difficult) Use Exercises 6.14, 6.15 and the maximality of m_i to show that
lyche-numerical-linear-algebra F02170	164	1.000	\mu_{\boldsymbol{A}}(\boldsymbol{A})=0	$\mu_{\boldsymbol{A}}(\boldsymbol{A})=0$	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = 0$. Thus a matrix satisfies its minimal equation. Finally show that the
lyche-numerical-linear-algebra F02171	164	1.000	p(\boldsymbol{A})=\mathbf{0}	$p(\boldsymbol{A})=\mathbf{0}$	degree of any polynomial p such that $p(\boldsymbol{A}) = \mathbf{0}$ is at least as large as the degree
lyche-numerical-linear-algebra F02172	164	1.000	\pi_{\boldsymbol{A}}(\boldsymbol{A})=\mathbf{0}	$\pi_{\boldsymbol{A}}(\boldsymbol{A})=\mathbf{0}$	its characteristic equation $\pi_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0}$.
lyche-numerical-linear-algebra F02173	164	1.000	p(t):=\sum_{j=0}^{r} b_{j} t^{j}	$p(t):=\sum_{j=0}^{r} b_{j} t^{j}$	is a polynomial given by $p(t) := \sum_{j=0}^{r} b_j t^j$, where $b_j \in \mathbb{C}$ and $\boldsymbol{A} \in \mathbb{C}^{n\times n}$. We
lyche-numerical-linear-algebra F02174	164	1.000	b_{j} \in \mathbb{C}	$b_{j} \in \mathbb{C}$	is a polynomial given by $p(t) := \sum_{j=0}^{r} b_j t^j$, where $b_j \in \mathbb{C}$ and $\boldsymbol{A} \in \mathbb{C}^{n\times n}$. We
lyche-numerical-linear-algebra F02175	164	1.000	p(\boldsymbol{A}) \in \mathbb{C}^{n \times n}	$p(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$	define the matrix $p(\boldsymbol{A}) \in \mathbb{C}^{n\times n}$ by
lyche-numerical-linear-algebra F02176	164	1.000	\boldsymbol{A}^{0}:=\boldsymbol{I}	$\boldsymbol{A}^{0}:=\boldsymbol{I}$	where $\boldsymbol{A}^0 := \boldsymbol{I}$. From this it follows that if $p(t) := (t-\alpha_1)\cdots(t-\alpha_r)$ for some
lyche-numerical-linear-algebra F02177	164	1.000	p(t):=\left(t-\alpha_{1}\right) \cdots\left(t-\alpha_{r} \right)	$p(t):=\left(t-\alpha_{1}\right) \cdots\left(t-\alpha_{r}\right)$	where $\boldsymbol{A}^0 := \boldsymbol{I}$. From this it follows that if $p(t) := (t-\alpha_1)\cdots(t-\alpha_r)$ for some
lyche-numerical-linear-algebra F02178	164	1.000	\alpha_{0}, \ldots, \alpha_{r} \in \mathbb{C}	$\alpha_{0}, \ldots, \alpha_{r} \in \mathbb{C}$	$\alpha_0, \ldots, \alpha_r \in \mathbb{C}$ then $p(\boldsymbol{A}) = (\boldsymbol{A}-\alpha_1)\cdots(\boldsymbol{A}-\alpha_r)$. We accept this without proof.
lyche-numerical-linear-algebra F02179	164	1.000	p(\boldsymbol{A})=\left(\boldsymbol{A}-\alpha_{1}\right) \cdots\left(\boldsymbol{A}-\alpha_{r}\right)	$p(\boldsymbol{A})=\left(\boldsymbol{A}-\alpha_{1}\right) \cdots\left(\boldsymbol{A}-\alpha_{r}\right)$	$\alpha_0, \ldots, \alpha_r \in \mathbb{C}$ then $p(\boldsymbol{A}) = (\boldsymbol{A}-\alpha_1)\cdots(\boldsymbol{A}-\alpha_r)$. We accept this without proof.
lyche-numerical-linear-algebra F02180	164	1.000	\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}= \boldsymbol{T}	$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}=\boldsymbol{T}$	Let $\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \boldsymbol{T}$, where $\boldsymbol{U}$ is unitary and $\boldsymbol{T}$ upper triangular with the eigenvalues
lyche-numerical-linear-algebra F02181	164	0.997	\left[\begin{array}{rr}2 & 1 \\ -1 & 4\end{array}\right]	$\left[\begin{array}{rr}2 & 1 \\ -1 & 4\end{array}\right]$	**a)** Find the characteristic polynomial $\pi_{\boldsymbol{A}}$ to $\left[\begin{array}{rr}2 & 1 \\ -1 & 4\end{array}\right]$. Show that $\pi(\boldsymbol{A}) = \mathbf{0}$.
lyche-numerical-linear-algebra F02182	164	0.997	\pi(\boldsymbol{A})=\mathbf{0}	$\pi(\boldsymbol{A})=\mathbf{0}$	**a)** Find the characteristic polynomial $\pi_{\boldsymbol{A}}$ to $\left[\begin{array}{rr}2 & 1 \\ -1 & 4\end{array}\right]$. Show that $\pi(\boldsymbol{A}) = \mathbf{0}$.
lyche-numerical-linear-algebra F02183	164	1.000	p(\boldsymbol{A})=	$p(\boldsymbol{A})=$	**b)** Let now $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ be arbitrary. For any polynomial p show that $p(\boldsymbol{A}) =$
lyche-numerical-linear-algebra F02184	164	0.877	\boldsymbol{U} p(\boldsymbol{T}) \boldsymbol{U}^{*}	$\boldsymbol{U} p(\boldsymbol{T}) \boldsymbol{U}^{*}$	$\boldsymbol{U} p(\boldsymbol{T}) \boldsymbol{U}^*$.
lyche-numerical-linear-algebra F02185	164	1.000	n, k \in \mathbb{N}	$n, k \in \mathbb{N}$	**c)** Let $n, k \in \mathbb{N}$ with $1 \le k < n$. Let $\boldsymbol{C}, \boldsymbol{D} \in \mathbb{C}^{n\times n}$ be upper triangular. Moreover,
lyche-numerical-linear-algebra F02186	164	1.000	\boldsymbol{C}, \boldsymbol{D} \in \mathbb{C}^{n \times n}	$\boldsymbol{C}, \boldsymbol{D} \in \mathbb{C}^{n \times n}$	**c)** Let $n, k \in \mathbb{N}$ with $1 \le k < n$. Let $\boldsymbol{C}, \boldsymbol{D} \in \mathbb{C}^{n\times n}$ be upper triangular. Moreover,
lyche-numerical-linear-algebra F02187	164	1.000	c_{i, j}=0	$c_{i, j}=0$	$c_{i,j} = 0$ for $i, j \le k$ and $d_{k+1,k+1} = 0$. Define $\boldsymbol{E} := \boldsymbol{C}\boldsymbol{D}$ and show that $e_{i,j} = 0$
lyche-numerical-linear-algebra F02188	164	1.000	i, j \leq k	$i, j \leq k$	$c_{i,j} = 0$ for $i, j \le k$ and $d_{k+1,k+1} = 0$. Define $\boldsymbol{E} := \boldsymbol{C}\boldsymbol{D}$ and show that $e_{i,j} = 0$
lyche-numerical-linear-algebra F02189	164	1.000	d_{k+1, k+1}=0	$d_{k+1, k+1}=0$	$c_{i,j} = 0$ for $i, j \le k$ and $d_{k+1,k+1} = 0$. Define $\boldsymbol{E} := \boldsymbol{C}\boldsymbol{D}$ and show that $e_{i,j} = 0$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F02190	164	1.000	`\boldsymbol{E}:=\boldsymbol{C} \boldsymbol{D}`	$\boldsymbol{E}:=\boldsymbol{C} \boldsymbol{D}$	$c_{i,j}=0$ for $i, j \leq k$ and $d_{k+1,k+1}=0$. Define $\boldsymbol{E}:=\boldsymbol{C}\boldsymbol{D}$ and show that $e_{i,j}=0$
lyche-numerical-linear-al F02191	164	1.000	`e_{i, j}=0`	$e_{i, j}=0$	$c_{i,j}=0$ for $i, j \leq k$ and $d_{k+1,k+1}=0$. Define $\boldsymbol{E}:=\boldsymbol{C}\boldsymbol{D}$ and show that $e_{i,j}=0$
lyche-numerical-linear-al F02192	164	1.000	`i, j \leq k+1`	$i, j \leq k+1$	for $i, j \leq k+1$.
lyche-numerical-linear-al F02193	165	0.773	`p:=\pi_{\boldsymbol{A}}`	$p:=\pi_{\boldsymbol{A}}$	**d)** Now let $p:=\pi_{\boldsymbol{A}}$ be the characteristic polynomial of $\boldsymbol{A}$. Show that $p(\boldsymbol{T})=\mathbf{0}$.[3]
lyche-numerical-linear-al F02194	165	0.773	`p(\boldsymbol{T})=\mathbf{0} .^{3}`	$p(\boldsymbol{T})=\mathbf{0} .^{3}$	**d)** Now let $p:=\pi_{\boldsymbol{A}}$ be the characteristic polynomial of $\boldsymbol{A}$. Show that $p(\boldsymbol{T})=\mathbf{0}$.[3]
lyche-numerical-linear-al F02195	165	1.000	`\boldsymbol{A}=\left[\begin{array}{ll}1 & 2 \\ 3 & 2 \end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{ll}1 & 2 \\ 3 & 2 \end{array}\right]$	$\boldsymbol{A}=\left[\begin{smallmatrix}1 & 2 \\ 3 & 2\end{smallmatrix}\right]$ is $\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U}=\left[\begin{smallmatrix}-1 & -1 \\ 0 & 4\end{smallmatrix}\right]$, where $\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{smallmatrix}1 & 1 \\ -1 & 1\end{smallmatrix}\right]$.
lyche-numerical-linear-al F02196	165	1.000	`\boldsymbol{U}^{T} \boldsymbol{A} \boldsymbol{U}=\left[ \begin{array}{cc}-1 & -1 \\ 0 & 4\end{array}\right]`	$\boldsymbol{U}^{T} \boldsymbol{A} \boldsymbol{U}=\left[\begin{array}{cc}-1 & -1 \\ 0 & 4\end{array}\right]$	$\boldsymbol{A}=\left[\begin{smallmatrix}1 & 2 \\ 3 & 2\end{smallmatrix}\right]$ is $\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U}=\left[\begin{smallmatrix}-1 & -1 \\ 0 & 4\end{smallmatrix}\right]$, where $\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{smallmatrix}1 & 1 \\ -1 & 1\end{smallmatrix}\right]$.
lyche-numerical-linear-al F02197	165	1.000	`\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc}1 & 1 \\ -1 & 1\end{array}\right]`	$\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc}1 & 1 \\ -1 & 1\end{array}\right]$	$\boldsymbol{A}=\left[\begin{smallmatrix}1 & 2 \\ 3 & 2\end{smallmatrix}\right]$ is $\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U}=\left[\begin{smallmatrix}-1 & -1 \\ 0 & 4\end{smallmatrix}\right]$, where $\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{smallmatrix}1 & 1 \\ -1 & 1\end{smallmatrix}\right]$.
lyche-numerical-linear-al F02198	165	0.998	`\boldsymbol{C}=\boldsymbol{A}+i \boldsymbol{B}`	$\boldsymbol{C}=\boldsymbol{A}+i \boldsymbol{B}$	**Exercise 6.20 (Skew-Hermitian Matrix)** Suppose $\boldsymbol{C}=\boldsymbol{A}+i\boldsymbol{B}$, where $\boldsymbol{A}, \boldsymbol{B} \in$
lyche-numerical-linear-al F02199	165	1.000	`\boldsymbol{A}^{T}=-\boldsymbol{A}`	$\boldsymbol{A}^{T}=-\boldsymbol{A}$	$\mathbb{R}^{n\times n}$. Show that $\boldsymbol{C}$ is skew-Hermitian if and only if $\boldsymbol{A}^T=-\boldsymbol{A}$ and $\boldsymbol{B}^T=\boldsymbol{B}$.
lyche-numerical-linear-al F02200	165	1.000	`\boldsymbol{B}^{T}=\boldsymbol{B}`	$\boldsymbol{B}^{T}=\boldsymbol{B}$	$\mathbb{R}^{n\times n}$. Show that $\boldsymbol{C}$ is skew-Hermitian if and only if $\boldsymbol{A}^T=-\boldsymbol{A}$ and $\boldsymbol{B}^T=\boldsymbol{B}$.
lyche-numerical-linear-al F02201	165	1.000	`\left(\lambda_{1}, \boldsymbol{u}_{1}\right), \ldots, \left(\lambda_{n}, \boldsymbol{u}_{n}\right)`	$\left(\lambda_{1}, \boldsymbol{u}_{1}\right), \ldots, \left(\lambda_{n}, \boldsymbol{u}_{n}\right)$	that if the eigenpairs $(\lambda_1, \boldsymbol{u}_1), \ldots, (\lambda_n, \boldsymbol{u}_n)$ of $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ are orthogonal, i.e.,
lyche-numerical-linear-al F02202	165	0.999	`\boldsymbol{u}_{j}^{*} \boldsymbol{u}_{k}=0`	$\boldsymbol{u}_{j}^{*} \boldsymbol{u}_{k}=0$	$\boldsymbol{u}_j^*\boldsymbol{u}_k=0$ for $j \neq k$ then the eigenvector expansions of $\boldsymbol{x}$ and $\boldsymbol{A}\boldsymbol{x} \in \mathbb{C}^n$ take
lyche-numerical-linear-al F02203	165	0.999	`j \neq k`	$j \neq k$	$\boldsymbol{u}_j^*\boldsymbol{u}_k=0$ for $j \neq k$ then the eigenvector expansions of $\boldsymbol{x}$ and $\boldsymbol{A}\boldsymbol{x} \in \mathbb{C}^n$ take
lyche-numerical-linear-al F02204	165	0.999	`\boldsymbol{A} \boldsymbol{x} \in \mathbb{C}^{n}`	$\boldsymbol{A} \boldsymbol{x} \in \mathbb{C}^{n}$	$\boldsymbol{u}_j^*\boldsymbol{u}_k=0$ for $j \neq k$ then the eigenvector expansions of $\boldsymbol{x}$ and $\boldsymbol{A}\boldsymbol{x} \in \mathbb{C}^n$ take
lyche-numerical-linear-al F02205	165	0.996	`\lambda_{\text {min }}`	$\lambda_{\text {min }}$	where $\lambda_{\min}$ is the smallest eigenvalue of $\boldsymbol{A}$, and $R(\boldsymbol{x})$ is the Rayleigh quotient
lyche-numerical-linear-al F02206	165	0.996	`R(\boldsymbol{x})`	$R(\boldsymbol{x})$	where $\lambda_{\min}$ is the smallest eigenvalue of $\boldsymbol{A}$, and $R(\boldsymbol{x})$ is the Rayleigh quotient
lyche-numerical-linear-al F02207	166	0.997	`\boldsymbol{y} \neq 0`	$\boldsymbol{y} \neq 0$	**b)** Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ such that $\lVert\boldsymbol{x}\rVert_2=1$ and $\boldsymbol{y} \neq 0$. Show that
lyche-numerical-linear-al F02208	166	1.000	`t>0`	$t>0$	where $t>0$ is small.[4]
lyche-numerical-linear-al F02209	166	1.000	`\boldsymbol{x}^{0} \in B_{1}`	$\boldsymbol{x}^{0} \in B_{1}$	Assume that $\boldsymbol{x}^0 \in B_1$ satisfies $\boldsymbol{A}\boldsymbol{x}^0-R(\boldsymbol{x}^0)\boldsymbol{x}^0 \neq 0$, i.e., $(R(\boldsymbol{x}^0), \boldsymbol{x}^0)$ is not an
lyche-numerical-linear-al F02210	166	1.000	`\boldsymbol{A} \boldsymbol{x}^{0}-R\left( \boldsymbol{x}^{0}\right) \boldsymbol{x}^{0} \neq 0`	$\boldsymbol{A} \boldsymbol{x}^{0}-R\left(\boldsymbol{x}^{0}\right) \boldsymbol{x}^{0} \neq 0$	Assume that $\boldsymbol{x}^0 \in B_1$ satisfies $\boldsymbol{A}\boldsymbol{x}^0-R(\boldsymbol{x}^0)\boldsymbol{x}^0 \neq 0$, i.e., $(R(\boldsymbol{x}^0), \boldsymbol{x}^0)$ is not an
lyche-numerical-linear-al F02211	166	1.000	`\left(R\left(\boldsymbol{x}^{0}\right), \boldsymbol{x}^{0}\right)`	$\left(R\left(\boldsymbol{x}^{0}\right), \boldsymbol{x}^{0}\right)$	Assume that $\boldsymbol{x}^0 \in B_1$ satisfies $\boldsymbol{A}\boldsymbol{x}^0-R(\boldsymbol{x}^0)\boldsymbol{x}^0 \neq 0$, i.e., $(R(\boldsymbol{x}^0), \boldsymbol{x}^0)$ is not an
lyche-numerical-linear-al F02212	166	1.000	`\boldsymbol{x}^{1} \in B_{1}`	$\boldsymbol{x}^{1} \in B_{1}$	eigenpair for $\boldsymbol{A}$. Explain how we can find a vector $\boldsymbol{x}^1 \in B_1$ such that $R(\boldsymbol{x}^1) <$
lyche-numerical-linear-al F02213	166	1.000	`R\left(\boldsymbol{x}^{1}\right)<`	$R\left(\boldsymbol{x}^{1}\right)<$	eigenpair for $\boldsymbol{A}$. Explain how we can find a vector $\boldsymbol{x}^1 \in B_1$ such that $R(\boldsymbol{x}^1) <$
lyche-numerical-linear-al F02214	166	0.993	`R\left(\boldsymbol{x}^{0}\right)`	$R\left(\boldsymbol{x}^{0}\right)$	$R(\boldsymbol{x}^0)$.
lyche-numerical-linear-al F02215	166	1.000	`\beta_{i} \geq \alpha_{i}`	$\beta_{i} \geq \alpha_{i}$	Theorem 6.13, if $\boldsymbol{E}$ is symmetric positive semidefinite then $\beta_i \geq \alpha_i$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02216	166	■ 1.000	\boldsymbol{A}:=\left[\begin{array}{ll}0 & 0 \\ 0 & 4 \end{array}\right]	$\boldsymbol{A}:=\left[\begin{array}{ll}0 & 0 \\ 0 & 4 \end{array}\right]$	ces $A := \left[\begin{smallmatrix}0&0\\0&4\end{smallmatrix}\right]$ and $B := \left[\begin{smallmatrix}-1&-1\\1&1\end{smallmatrix}\right]$. Why does this not contradict the Hoffman-
lyche-numerical-linear-algebra F02217	166	■ 1.000	\boldsymbol{B}:=\left[\begin{array}{cc}-1 & -1 \\ 1 & 1 \end{array}\right]	$\boldsymbol{B}:=\left[\begin{array}{cc}-1 & -1 \\ 1 & 1 \end{array}\right]$	ces $A := \left[\begin{smallmatrix}0&0\\0&4\end{smallmatrix}\right]$ and $B := \left[\begin{smallmatrix}-1&-1\\1&1\end{smallmatrix}\right]$. Why does this not contradict the Hoffman-
lyche-numerical-linear-algebra F02218	166	■ 1.000	\boldsymbol{A}:=\left[\begin{array}{ll}3 & 1 \\ 2 & 2 \end{array}\right]	$\boldsymbol{A}:=\left[\begin{array}{ll}3 & 1 \\ 2 & 2 \end{array}\right]$	the matrix $A := \left[\begin{smallmatrix}3&1\\2&2\end{smallmatrix}\right]$ and the two expansions in (6.16) for any $v \in \mathbb{R}^2$.
lyche-numerical-linear-algebra F02219	166	■ 1.000	\boldsymbol{v} \in \mathbb{R}^{2}	$\boldsymbol{v} \in \mathbb{R}^{2}$	the matrix $A := \left[\begin{smallmatrix}3&1\\2&2\end{smallmatrix}\right]$ and the two expansions in (6.16) for any $v \in \mathbb{R}^2$.
lyche-numerical-linear-algebra F02220	166	■ 1.000	\boldsymbol{y}, \boldsymbol{x} \in	$\boldsymbol{y}, \boldsymbol{x} \in$	**Exercise 6.27 (Generalized Rayleigh Quotient)** For $A \in \mathbb{C}^{n\times n}$ and any $y, x \in$
lyche-numerical-linear-algebra F02221	166	■ 1.000	R(\boldsymbol{y}, \boldsymbol{x})=R_{\boldsymbol{A}}(\boldsymbol{y}, \boldsymbol{x}):=\frac{\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{y}^{*} \boldsymbol{x}}	$R(\boldsymbol{y}, \boldsymbol{x})=R_{\boldsymbol{A}}(\boldsymbol{y}, \boldsymbol{x}):=\frac{\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{y}^{*} \boldsymbol{x}}$	$\mathbb{C}^n$ with $y^*x \neq 0$ the quantity $R(y,x) = R_A(y,x) := \frac{y^*Ax}{y^*x}$ is called a **generalized**
lyche-numerical-linear-algebra F02222	166	■ 1.000	R(\boldsymbol{y}, \boldsymbol{x})=\lambda	$R(\boldsymbol{y}, \boldsymbol{x})=\lambda$	$R(y,x) = \lambda$ for any y with $y^*x \neq 0$. Also show that if (λ, y) is a left eigenpair for
lyche-numerical-linear-algebra F02223	166	■ 1.000	\boldsymbol{A}, \boldsymbol{A}^{T}	$\boldsymbol{A}, \boldsymbol{A}^{T}$	6.7.1 Does A, A^T and A^* have the same eigenvalues? What about A^*A and
lyche-numerical-linear-algebra F02224	166	■ 1.000	\boldsymbol{A} \boldsymbol{A}^{*}	$\boldsymbol{A} \boldsymbol{A}^{*}$	AA^*?
lyche-numerical-linear-algebra F02225	166	■ 0.993	f(t)=R(\boldsymbol{x}-t \boldsymbol{y})	$f(t)=R(\boldsymbol{x}-t \boldsymbol{y})$	[4]Hint: Use Taylor's theorem for the function $f(t) = R(x - ty)$.
lyche-numerical-linear-algebra F02226	168	■ 1.000	\boldsymbol{U}:=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{n}\right] \in \mathbb{C}^{n \times n}	$\boldsymbol{U}:=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{n}\right] \in \mathbb{C}^{n \times n}$	eigenpairs $(\lambda_1, u_1), \ldots, (\lambda_n, u_n)$. Letting $U := [u_1, \ldots, u_n] \in \mathbb{C}^{n\times n}$ and $D :=$
lyche-numerical-linear-algebra F02227	168	■ 1.000	\boldsymbol{D}:=	$\boldsymbol{D}:=$	eigenpairs $(\lambda_1, u_1), \ldots, (\lambda_n, u_n)$. Letting $U := [u_1, \ldots, u_n] \in \mathbb{C}^{n\times n}$ and $D :=$
lyche-numerical-linear-algebra F02228	168	■ 1.000	\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n} \right)	$\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)$	$\operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ we have the spectral decomposition
lyche-numerical-linear-algebra F02229	168	■ 1.000	\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*}	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*}$	$A = U\Sigma V^*$, where U and V are unitary, and Σ is a nonnegative diagonal matrix,
lyche-numerical-linear-algebra F02230	168	■ 1.000	\boldsymbol{\Sigma}	$\boldsymbol{\Sigma}$	$A = U\Sigma V^*$, where U and V are unitary, and Σ is a nonnegative diagonal matrix,
lyche-numerical-linear-algebra F02231	168	■ 1.000	\Sigma_{i j}=0	$\Sigma_{i j}=0$	i.e., $\Sigma_{ij} = 0$ for all $i \neq j$ and $\Sigma_{ii} \geq 0$ for all i. The diagonal elements $\sigma_i := \Sigma_{ii}$ are
lyche-numerical-linear-algebra F02232	168	■ 1.000	\Sigma_{i i} \geq 0	$\Sigma_{i i} \geq 0$	i.e., $\Sigma_{ij} = 0$ for all $i \neq j$ and $\Sigma_{ii} \geq 0$ for all i. The diagonal elements $\sigma_i := \Sigma_{ii}$ are
lyche-numerical-linear-algebra F02233	168	■ 1.000	\sigma_{i}:=\Sigma_{i i}	$\sigma_{i}:=\Sigma_{i i}$	i.e., $\Sigma_{ij} = 0$ for all $i \neq j$ and $\Sigma_{ii} \geq 0$ for all i. The diagonal elements $\sigma_i := \Sigma_{ii}$ are
lyche-numerical-linear-algebra F02234	168	■ 1.000	\sigma_{i} \geq \sigma_{i+1}	$\sigma_{i} \geq \sigma_{i+1}$	$\sigma_i \geq \sigma_{i+1}$ for all i.
lyche-numerical-linear-algebra F02235	169	■ 1.000	\sigma_{1}=2	$\sigma_{1}=2$	mal, and Σ is a nonnegative diagonal matrix with singular values $\sigma_1 = 2$ and
lyche-numerical-linear-algebra F02236	169	■ 1.000	\sigma_{2}=1	$\sigma_{2}=1$	$\sigma_2 = 1$.
lyche-numerical-linear-algebra F02237	169	■ 1.000	\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A})	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A})$	To start we show that bases for the four fundamental subspaces $\mathcal{R}(A)$, $\mathcal{N}(A)$,
lyche-numerical-linear-algebra F02238	169	■ 1.000	\mathcal{R}\left(\boldsymbol{A}^{*}\right)	$\mathcal{R}\left(\boldsymbol{A}^{*}\right)$	$\mathcal{R}(A^*)$ and $\mathcal{N}(A^*)$ of a matrix A can be determined from the eigenpairs of A^*A
lyche-numerical-linear-algebra F02239	169	■ 1.000	\mathcal{N}\left(\boldsymbol{A}^{*}\right)	$\mathcal{N}\left(\boldsymbol{A}^{*}\right)$	$\mathcal{R}(A^*)$ and $\mathcal{N}(A^*)$ of a matrix A can be determined from the eigenpairs of A^*A
lyche-numerical-linear-algebra F02240	169	■ 0.978	\left.\boldsymbol{A}^{*} \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^{*}\right)	$\left.\boldsymbol{A}^{*} \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^{*}\right)$	**Theorem 7.1 (The Matrices A^*A, AA^*)** *Suppose $m, n \in \mathbb{N}$ and $A \in \mathbb{C}^{m\times n}$.*
lyche-numerical-linear-algebra F02241	169	■ 1.000	\boldsymbol{A}^{*} \boldsymbol{A} \in \mathbb{C}^{n \times n}	$\boldsymbol{A}^{*} \boldsymbol{A} \in \mathbb{C}^{n \times n}$	*1. The matrices $A^*A \in \mathbb{C}^{n\times n}$ and $AA^* \in \mathbb{C}^{m\times m}$ have the same nonzero eigenval-*

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02242	169	1.000	`\boldsymbol{A} \boldsymbol{A}^{*} \in \mathbb{C}^{m \times m}`	$\boldsymbol{A} \boldsymbol{A}^{*} \in \mathbb{C}^{m \times m}$	*1. The matrices $\boldsymbol{A}^*\boldsymbol{A} \in \mathbb{C}^{n\times n}$ and $\boldsymbol{A}\boldsymbol{A}^* \in \mathbb{C}^{m\times m}$ have the same nonzero eigenval-*
lyche-numerical-linear-algebra F02243	169	0.992	`\left(\lambda_{j}, \boldsymbol{v}_{j}\right)`	$\left(\lambda_{j}, \boldsymbol{v}_{j}\right)$	*3. Let $(\lambda_j, \boldsymbol{v}_j)$ be orthonormal eigenpairs for $\boldsymbol{A}^*\boldsymbol{A}$ with*
lyche-numerical-linear-algebra F02244	169	1.000	`\left\{\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{r}\right\}`	$\left\{\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{r}\right\}$	*Then $\{\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r\}$ is an orthogonal basis for the column space $\mathcal{R}(\boldsymbol{A}) :=$*
lyche-numerical-linear-algebra F02245	169	1.000	`\mathcal{R}(\boldsymbol{A}):=`	$\mathcal{R}(\boldsymbol{A}):=$	*Then $\{\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r\}$ is an orthogonal basis for the column space $\mathcal{R}(\boldsymbol{A}) :=$*
lyche-numerical-linear-algebra F02246	169	1.000	`\left\{\boldsymbol{A} \boldsymbol{y} \in \mathbb{C}^{m}: \boldsymbol{y} \in \mathbb{C}^{n}\right\}`	$\left\{\boldsymbol{A} \boldsymbol{y} \in \mathbb{C}^{m}: \boldsymbol{y} \in \mathbb{C}^{n}\right\}$	*$\{\boldsymbol{A}\boldsymbol{y} \in \mathbb{C}^m : \boldsymbol{y} \in \mathbb{C}^n\}$ and $\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\}$ is an orthonormal basis for the*
lyche-numerical-linear-algebra F02247	169	1.000	`\left\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_{n}\right\}`	$\left\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_{n}\right\}$	*$\{\boldsymbol{A}\boldsymbol{y} \in \mathbb{C}^m : \boldsymbol{y} \in \mathbb{C}^n\}$ and $\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\}$ is an orthonormal basis for the*
lyche-numerical-linear-algebra F02248	169	0.899	`\mathcal{N}(\boldsymbol{A}):=\left\{\boldsymbol{y} \in \mathbb{C}^{n}: \boldsymbol{A} \boldsymbol{y}=\mathbf{0}\right\}`	$\mathcal{N}(\boldsymbol{A}):=\left\{\boldsymbol{y} \in \mathbb{C}^{n}: \boldsymbol{A} \boldsymbol{y}=\mathbf{0}\right\}$	*nullspace $\mathcal{N}(\boldsymbol{A}) := \{\boldsymbol{y} \in \mathbb{C}^n : \boldsymbol{A}\boldsymbol{y} = \mathbf{0}\}$.*
lyche-numerical-linear-algebra F02249	169	0.996	`\left(\lambda_{j}, \boldsymbol{u}_{j}\right)`	$\left(\lambda_{j}, \boldsymbol{u}_{j}\right)$	*4. Let $(\lambda_j, \boldsymbol{u}_j)$ be orthonormal eigenpairs for $\boldsymbol{A}\boldsymbol{A}^*$. If $\lambda_j > 0$, $j = 1, \ldots, r$ and*
lyche-numerical-linear-algebra F02250	169	0.996	`\lambda_{j}>0, j=1, \ldots, r`	$\lambda_{j}>0, j=1, \ldots, r$	*4. Let $(\lambda_j, \boldsymbol{u}_j)$ be orthonormal eigenpairs for $\boldsymbol{A}\boldsymbol{A}^*$. If $\lambda_j > 0$, $j = 1, \ldots, r$ and*
lyche-numerical-linear-algebra F02251	169	1.000	`\lambda_{j}=0, j=r+1, \ldots, m`	$\lambda_{j}=0, j=r+1, \ldots, m$	*$\lambda_j = 0$, $j = r+1, \ldots, m$ then $\{\boldsymbol{A}^*\boldsymbol{u}_1, \ldots, \boldsymbol{A}^*\boldsymbol{u}_r\}$ is an orthogonal basis for*
lyche-numerical-linear-algebra F02252	169	1.000	`\left\{\boldsymbol{A}^{*} \boldsymbol{u}_{1}, \ldots, \boldsymbol{A}^{*} \boldsymbol{u}_{r}\right\}`	$\left\{\boldsymbol{A}^{*} \boldsymbol{u}_{1}, \ldots, \boldsymbol{A}^{*} \boldsymbol{u}_{r}\right\}$	*$\lambda_j = 0$, $j = r+1, \ldots, m$ then $\{\boldsymbol{A}^*\boldsymbol{u}_1, \ldots, \boldsymbol{A}^*\boldsymbol{u}_r\}$ is an orthogonal basis for*
lyche-numerical-linear-algebra F02253	169	1.000	`\left\{\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_{m}\right\}`	$\left\{\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_{m}\right\}$	*the column space $\mathcal{R}(\boldsymbol{A}^*)$ and $\{\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m\}$ is an orthonormal basis for the*
lyche-numerical-linear-algebra F02254	169	1.000	`\pi_{\boldsymbol{A}^{*} \boldsymbol{A}}`	$\pi_{\boldsymbol{A}^{*} \boldsymbol{A}}$	1. Consider the characteristic polynomials $\pi_{\boldsymbol{A}^*\boldsymbol{A}}$ and $\pi_{\boldsymbol{A}\boldsymbol{A}^*}$. By (6.1) we have
lyche-numerical-linear-algebra F02255	169	1.000	`\pi_{\boldsymbol{A} \boldsymbol{A}^{*}}`	$\pi_{\boldsymbol{A} \boldsymbol{A}^{*}}$	1. Consider the characteristic polynomials $\pi_{\boldsymbol{A}^*\boldsymbol{A}}$ and $\pi_{\boldsymbol{A}\boldsymbol{A}^*}$. By (6.1) we have
lyche-numerical-linear-algebra F02256	170	1.000	`\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{v}=\lambda \boldsymbol{v}`	$\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{v}=\lambda \boldsymbol{v}$	$\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}$ with $\boldsymbol{v} \neq \mathbf{0}$, then
lyche-numerical-linear-algebra F02257	170	1.000	`\boldsymbol{v} \neq \mathbf{0}`	$\boldsymbol{v} \neq \mathbf{0}$	$\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}$ with $\boldsymbol{v} \neq \mathbf{0}$, then
lyche-numerical-linear-algebra F02258	170	1.000	`\left(\boldsymbol{A} \boldsymbol{v}_{j}\right)^{*} \boldsymbol{A} \boldsymbol{v}_{k}=\boldsymbol{v}_{j}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{v}_{k}= \lambda_{k} \boldsymbol{v}_{j}^{*} \boldsymbol{v}_{k}`	$\left(\boldsymbol{A} \boldsymbol{v}_{j}\right)^{*} \boldsymbol{A} \boldsymbol{v}_{k}=\boldsymbol{v}_{j}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{v}_{k}= \lambda_{k} \boldsymbol{v}_{j}^{*} \boldsymbol{v}_{k}$	3. By orthonormality of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ we have $(\boldsymbol{A}\boldsymbol{v}_j)^*\boldsymbol{A}\boldsymbol{v}_k = \boldsymbol{v}_j^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}_k = \lambda_k\boldsymbol{v}_j^*\boldsymbol{v}_k$
lyche-numerical-linear-algebra F02259	170	0.578	`\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{n}`	$\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{n}$	$= 0$ for $j \neq k$, showing that $\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_n$ are orthogonal vectors. Moreover,
lyche-numerical-linear-algebra F02260	170	1.000	`\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{r}`	$\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{r}$	(7.3) implies that $\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r$ are nonzero and $\boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$ for $j = r+$
lyche-numerical-linear-algebra F02261	170	1.000	`\boldsymbol{A} \boldsymbol{v}_{j}=\mathbf{0}`	$\boldsymbol{A} \boldsymbol{v}_{j}=\mathbf{0}$	(7.3) implies that $\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r$ are nonzero and $\boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$ for $j = r+$
lyche-numerical-linear-algebra F02262	170	1.000	`j=r+`	$j=r+$	(7.3) implies that $\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r$ are nonzero and $\boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$ for $j = r+$
lyche-numerical-linear-algebra F02263	170	1.000	`\mathcal{N}(\boldsymbol{A})`	$\mathcal{N}(\boldsymbol{A})$	linearly independent vectors in $\mathcal{R}(\boldsymbol{A})$ and $\mathcal{N}(\boldsymbol{A})$, respectively. The proof will
lyche-numerical-linear-algebra F02264	170	1.000	`\mathcal{R}(\boldsymbol{A}) \subset \operatorname{span}\left(\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{r}\right)`	$\mathcal{R}(\boldsymbol{A}) \subset \operatorname{span}\left(\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{r}\right)$	be complete once it is shown that $\mathcal{R}(\boldsymbol{A}) \subset \operatorname{span}(\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r)$ and $\mathcal{N}(\boldsymbol{A}) \subset$
lyche-numerical-linear-algebra F02265	170	1.000	`\mathcal{N}(\boldsymbol{A}) \subset`	$\mathcal{N}(\boldsymbol{A}) \subset$	be complete once it is shown that $\mathcal{R}(\boldsymbol{A}) \subset \operatorname{span}(\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r)$ and $\mathcal{N}(\boldsymbol{A}) \subset$
lyche-numerical-linear-algebra F02266	170	0.999	`\operatorname{span}\left(\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_{n}\right)`	$\operatorname{span}\left(\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_{n}\right)$	$\operatorname{span}(\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n)$. Suppose $\boldsymbol{x} \in \mathcal{R}(\boldsymbol{A})$. Then $\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y}$ for some $\boldsymbol{y} \in \mathbb{C}^n$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02267	170	0.999	`\boldsymbol{x} \in \mathcal{R}(\boldsymbol{A})`	$\boldsymbol{x} \in \mathcal{R}(\boldsymbol{A})$	span$(\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n)$. Suppose $\boldsymbol{x} \in \mathcal{R}(\boldsymbol{A})$. Then $\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y}$ for some $\boldsymbol{y} \in \mathbb{C}^n$,
lyche-numerical-linear-algebra F02268	170	0.999	`\boldsymbol{x}=\boldsymbol{A} \boldsymbol{y}`	$\boldsymbol{x}=\boldsymbol{A}\boldsymbol{y}$	span$(\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n)$. Suppose $\boldsymbol{x} \in \mathcal{R}(\boldsymbol{A})$. Then $\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y}$ for some $\boldsymbol{y} \in \mathbb{C}^n$,
lyche-numerical-linear-algebra F02269	170	1.000	`\boldsymbol{y}=\sum_{j=1}^{n} c_{j} \boldsymbol{v}_{j}`	$\boldsymbol{y}=\sum_{j=1}^{n} c_{j} \boldsymbol{v}_{j}$	Let $\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{v}_j$ be an eigenvector expansion of $\boldsymbol{y}$. Since $\boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$ for
lyche-numerical-linear-algebra F02270	170	1.000	`\boldsymbol{x}=\boldsymbol{A} \boldsymbol{y}= \sum_{j=1}^{n} c_{j} \boldsymbol{A} \boldsymbol{v}_{j}= \sum_{j=1}^{r} c_{j} \boldsymbol{A} \boldsymbol{v}_{j} \in`	$\boldsymbol{x}=\boldsymbol{A} \boldsymbol{y}= \sum_{j=1}^{n} c_{j} \boldsymbol{A} \boldsymbol{v}_{j}= \sum_{j=1}^{r} c_{j} \boldsymbol{A} \boldsymbol{v}_{j} \in$	$j = r+1, \ldots, n$ we obtain $\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{A}\boldsymbol{v}_j = \sum_{j=1}^r c_j \boldsymbol{A}\boldsymbol{v}_j \in$
lyche-numerical-linear-algebra F02271	170	1.000	`\operatorname{span}\left(\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{r}\right)`	$\operatorname{span}\left(\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{r}\right)$	span$(\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r)$. Finally, if $\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{v}_j \in \mathcal{N}(\boldsymbol{A})$, then we have
lyche-numerical-linear-algebra F02272	170	1.000	`\boldsymbol{y}=\sum_{j=1}^{n} c_{j} \boldsymbol{v}_{j} \in \mathcal{N}(\boldsymbol{A})`	$\boldsymbol{y}=\sum_{j=1}^{n} c_{j} \boldsymbol{v}_{j} \in \mathcal{N}(\boldsymbol{A})$	span$(\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r)$. Finally, if $\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{v}_j \in \mathcal{N}(\boldsymbol{A})$, then we have
lyche-numerical-linear-algebra F02273	170	1.000	`\boldsymbol{A} \boldsymbol{y}=\sum_{j=1}^{r} c_{j} \boldsymbol{A} \boldsymbol{v}_{j}=\mathbf{0}`	$\boldsymbol{A} \boldsymbol{y}=\sum_{j=1}^{r} c_{j} \boldsymbol{A} \boldsymbol{v}_{j}=\mathbf{0}$	$\boldsymbol{A}\boldsymbol{y} = \sum_{j=1}^r c_j \boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$, and $c_1 = \cdots = c_r = 0$ since $\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r$ are
lyche-numerical-linear-algebra F02274	170	1.000	`c_{1}=\cdots=c_{r}=0`	$c_{1}=\cdots=c_{r}=0$	$\boldsymbol{A}\boldsymbol{y} = \sum_{j=1}^r c_j \boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$, and $c_1 = \cdots = c_r = 0$ since $\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r$ are
lyche-numerical-linear-algebra F02275	170	1.000	`\boldsymbol{y}=\sum_{j=r+1}^{n} c_{j} \boldsymbol{v}_{j} \in \operatorname{span}\left(\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_{n}\right)`	$\boldsymbol{y}=\sum_{j=r+1}^{n} c_{j} \boldsymbol{v}_{j} \in \operatorname{span}\left(\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_{n}\right)$	linearly independent. But then $\boldsymbol{y} = \sum_{j=r+1}^n c_j \boldsymbol{v}_j \in$ span$(\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n)$.
lyche-numerical-linear-algebra F02276	170	1.000	`\boldsymbol{A} \boldsymbol{A}^{*}=\boldsymbol{B}^{*} \boldsymbol{B}`	$\boldsymbol{A} \boldsymbol{A}^{*}=\boldsymbol{B}^{*} \boldsymbol{B}$	4. Since $\boldsymbol{A}\boldsymbol{A}^* = \boldsymbol{B}^*\boldsymbol{B}$ with $\boldsymbol{B} := \boldsymbol{A}^*$ this follows from part 3 with $\boldsymbol{A} = \boldsymbol{B}$.
lyche-numerical-linear-algebra F02277	170	1.000	`\boldsymbol{B}:=\boldsymbol{A}^{*}`	$\boldsymbol{B}:=\boldsymbol{A}^{*}$	4. Since $\boldsymbol{A}\boldsymbol{A}^* = \boldsymbol{B}^*\boldsymbol{B}$ with $\boldsymbol{B} := \boldsymbol{A}^*$ this follows from part 3 with $\boldsymbol{A} = \boldsymbol{B}$.
lyche-numerical-linear-algebra F02278	170	1.000	`\boldsymbol{A}=\boldsymbol{B}`	$\boldsymbol{A}=\boldsymbol{B}$	4. Since $\boldsymbol{A}\boldsymbol{A}^* = \boldsymbol{B}^*\boldsymbol{B}$ with $\boldsymbol{B} := \boldsymbol{A}^*$ this follows from part 3 with $\boldsymbol{A} = \boldsymbol{B}$.
lyche-numerical-linear-algebra F02279	170	1.000	`2 \boldsymbol{A}^{*} \boldsymbol{A}`	$2 \boldsymbol{A}^{*} \boldsymbol{A}$	5. By part 1 and 2 $\boldsymbol{A}^*\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^*$ have the same number r of positive eigenvalues
lyche-numerical-linear-algebra F02280	170	1.000	`4 r`	$4 r$	and by part 3 and 4 r is the rank of $\boldsymbol{A}$.
lyche-numerical-linear-algebra F02281	170	1.000	`m, n, r \in \mathbb{N}`	$m, n, r \in \mathbb{N}$	**Theorem 7.2 (Existence of SVD)** *Suppose for $m, n, r \in \mathbb{N}$ that $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ has*
lyche-numerical-linear-algebra F02282	170	1.000	`\lambda_{1} \geq \cdots \geq`	$\lambda_{1} \geq \cdots \geq$	*rank r, and that $(\lambda_j, \boldsymbol{v}_j)$ are orthonormal eigenpairs for $\boldsymbol{A}^*\boldsymbol{A}$ with $\lambda_1 \geq \cdots \geq$*
lyche-numerical-linear-algebra F02283	170	1.000	`\lambda_{r}>0=\lambda_{r+1}=\cdots=\lambda_{n}`	$\lambda_{r}>0=\lambda_{r+1}=\cdots=\lambda_{n}$	$\lambda_r > 0 = \lambda_{r+1} = \cdots = \lambda_n$. *Define*
lyche-numerical-linear-algebra F02284	170	1.000	`\boldsymbol{V}:=\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right] \in \mathbb{C}^{n \times n}`	$\boldsymbol{V}:=\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right] \in \mathbb{C}^{n \times n}$	*1. $\boldsymbol{V} := [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n] \in \mathbb{C}^{n\times n}$,*
lyche-numerical-linear-algebra F02285	170	1.000	`\Sigma \in \mathbb{R}^{m \times n}`	$\Sigma \in \mathbb{R}^{m \times n}$	*2. $\Sigma \in \mathbb{R}^{m\times n}$ is a diagonal matrix with diagonal elements $\sigma_j := \sqrt{\lambda_j}$ for $j =$*
lyche-numerical-linear-algebra F02286	170	1.000	`\sigma_{j}:=\sqrt{\lambda_{j}}`	$\sigma_{j}:=\sqrt{\lambda_{j}}$	*2. $\Sigma \in \mathbb{R}^{m\times n}$ is a diagonal matrix with diagonal elements $\sigma_j := \sqrt{\lambda_j}$ for $j =$*
lyche-numerical-linear-algebra F02287	170	1.000	`j=`	$j=$	*2. $\Sigma \in \mathbb{R}^{m\times n}$ is a diagonal matrix with diagonal elements $\sigma_j := \sqrt{\lambda_j}$ for $j =$*
lyche-numerical-linear-algebra F02288	170	1.000	`1, \ldots, \min (m, n)`	$1, \ldots, \min (m, n)$	$1, \ldots, \min(m, n)$,
lyche-numerical-linear-algebra F02289	170	0.999	`\boldsymbol{U}:=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{m}\right] \in \mathbb{C}^{m \times m}`	$\boldsymbol{U}:=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{m}\right] \in \mathbb{C}^{m \times m}$	*3. $\boldsymbol{U} := [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m] \in \mathbb{C}^{m\times m}$, where $\boldsymbol{u}_j = \sigma_j^{-1}\boldsymbol{A}\boldsymbol{v}_j$ for $j = 1, \ldots, r$ and*
lyche-numerical-linear-algebra F02290	170	0.999	`\boldsymbol{u}_{j}=\sigma_{j}^{-1} \boldsymbol{A} \boldsymbol{v}_{j}`	$\boldsymbol{u}_{j}=\sigma_{j}^{-1} \boldsymbol{A} \boldsymbol{v}_{j}$	*3. $\boldsymbol{U} := [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m] \in \mathbb{C}^{m\times m}$, where $\boldsymbol{u}_j = \sigma_j^{-1}\boldsymbol{A}\boldsymbol{v}_j$ for $j = 1, \ldots, r$ and*

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F02291	170	0.999	`j=1, \ldots, r`	$j=1, \ldots, r$	*3.* $\boldsymbol{U} := [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m] \in \mathbb{C}^{m\times m}$, *where* $\boldsymbol{u}_j = \sigma_j^{-1}\boldsymbol{A}\boldsymbol{v}_j$ *for* $j = 1, \ldots, r$ *and*
lyche-numerical-linear-algebra_F02292	170	1.000	`\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_{m}`	$\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_{m}$	$\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m$ *is any extension of* $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r$ *to an orthonormal basis* $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m$
lyche-numerical-linear-algebra_F02293	170	1.000	`\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{r}`	$\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{r}$	$\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m$ *is any extension of* $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r$ *to an orthonormal basis* $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m$
lyche-numerical-linear-algebra_F02294	170	1.000	`\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{m}`	$\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{m}$	$\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m$ *is any extension of* $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r$ *to an orthonormal basis* $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m$
lyche-numerical-linear-algebra_F02295	170	1.000	`\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}`	$\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$	***Proof*** Let $\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$ be as in the theorem. The vectors $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r$ are orthonormal
lyche-numerical-linear-algebra_F02296	170	1.000	`\sigma_{j}=\left|\boldsymbol{A} \boldsymbol{v}_{j}\right |_{2}>0, j=1, \ldots, r`	$\sigma_{j}=\left\|\boldsymbol{A} \boldsymbol{v}_{j}\right\|_{2}>0, j=1, \ldots, r$	since $\boldsymbol{A}\boldsymbol{v}_1, \ldots, \boldsymbol{A}\boldsymbol{v}_r$ are orthogonal and $\sigma_j = \\|\boldsymbol{A}\boldsymbol{v}_j\\|_2 > 0$, $j = 1, \ldots, r$ by (7.3).
lyche-numerical-linear-algebra_F02297	171	1.000	`\boldsymbol{U} \boldsymbol{\Sigma}=\boldsymbol{A} \boldsymbol{V}`	$\boldsymbol{U} \boldsymbol{\Sigma}=\boldsymbol{A} \boldsymbol{V}$	Thus $\boldsymbol{U}\boldsymbol{\Sigma} = \boldsymbol{A}\boldsymbol{V}$ and since $\boldsymbol{V}$ is square and unitary we find $\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^* = \boldsymbol{A}\boldsymbol{V}\boldsymbol{V}^* = \boldsymbol{A}$
lyche-numerical-linear-algebra_F02298	171	1.000	`\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*}= \boldsymbol{A} \boldsymbol{V} \boldsymbol{V}^{*}= \boldsymbol{A}`	$\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*}=\boldsymbol{A} \boldsymbol{V} \boldsymbol{V}^{*}=\boldsymbol{A}$	Thus $\boldsymbol{U}\boldsymbol{\Sigma} = \boldsymbol{A}\boldsymbol{V}$ and since $\boldsymbol{V}$ is square and unitary we find $\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^* = \boldsymbol{A}\boldsymbol{V}\boldsymbol{V}^* = \boldsymbol{A}$
lyche-numerical-linear-algebra_F02299	171	1.000	`\sigma_{1} \geq \sigma_{2} \geq \cdots \geq \sigma_{r}`	$\sigma_{1} \geq \sigma_{2} \geq \cdots \geq \sigma_{r}$	and we have an SVD of $\boldsymbol{A}$ with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$.
lyche-numerical-linear-algebra_F02300	171	1.000	`\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc}14 & 2 \\ 4 & 22 \\ 16 & 13\end{array}\right]`	$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc}14 & 2 \\ 4 & 22 \\ 16 & 13\end{array}\right]$	*Example 7.2 (Find SVD)* To derive the SVD in (7.2) where $\boldsymbol{A} = \frac{1}{15}\begin{bmatrix}14 & 2\\ 4 & 22\\ 16 & 13\end{bmatrix}$, we
lyche-numerical-linear-algebra_F02301	171	0.990	`\sigma_{1}=2, \sigma_{2}=1`	$\sigma_{1}=2, \sigma_{2}=1$	Thus $\sigma_1 = 2$, $\sigma_2 = 1$, and $\boldsymbol{V} = \frac{1}{5}\begin{bmatrix}3 & 4\\ 4 & -3\end{bmatrix}$. Now $\boldsymbol{u}_1 = \boldsymbol{A}\boldsymbol{v}_1/\sigma_1 = [1, 2, 2]^T/3$,
lyche-numerical-linear-algebra_F02302	171	0.990	`\boldsymbol{V}=\frac{1}{5}\left[\begin{array}{rr}3 & 4 \\ 4 & -3\end{array}\right]`	$\boldsymbol{V}=\frac{1}{5}\left[\begin{array}{rr}3 & 4 \\ 4 & -3\end{array}\right]$	Thus $\sigma_1 = 2$, $\sigma_2 = 1$, and $\boldsymbol{V} = \frac{1}{5}\begin{bmatrix}3 & 4\\ 4 & -3\end{bmatrix}$. Now $\boldsymbol{u}_1 = \boldsymbol{A}\boldsymbol{v}_1/\sigma_1 = [1, 2, 2]^T/3$,
lyche-numerical-linear-algebra_F02303	171	0.990	`\boldsymbol{u}_{1}=\boldsymbol{A} \boldsymbol{v}_{1} / \sigma_{1}=[1,2,2]^{T} / 3`	$\boldsymbol{u}_{1}=\boldsymbol{A} \boldsymbol{v}_{1} / \sigma_{1}=[1,2,2]^{T} / 3$	Thus $\sigma_1 = 2$, $\sigma_2 = 1$, and $\boldsymbol{V} = \frac{1}{5}\begin{bmatrix}3 & 4\\ 4 & -3\end{bmatrix}$. Now $\boldsymbol{u}_1 = \boldsymbol{A}\boldsymbol{v}_1/\sigma_1 = [1, 2, 2]^T/3$,
lyche-numerical-linear-algebra_F02304	171	0.998	`\boldsymbol{u}_{2}=\boldsymbol{A} \boldsymbol{v}_{2} / \sigma_{2}=[2,-2,1]^{T} / 3`	$\boldsymbol{u}_{2}=\boldsymbol{A} \boldsymbol{v}_{2} / \sigma_{2}=[2,-2,1]^{T} / 3$	$\boldsymbol{u}_2 = \boldsymbol{A}\boldsymbol{v}_2/\sigma_2 = [2, -2, 1]^T/3$. For an SVD we also need $\boldsymbol{u}_3$ which is any vector
lyche-numerical-linear-algebra_F02305	171	0.998	`\boldsymbol{u}_{3}`	$\boldsymbol{u}_{3}$	$\boldsymbol{u}_2 = \boldsymbol{A}\boldsymbol{v}_2/\sigma_2 = [2, -2, 1]^T/3$. For an SVD we also need $\boldsymbol{u}_3$ which is any vector
lyche-numerical-linear-algebra_F02306	171	1.000	`\boldsymbol{u}_{1}`	$\boldsymbol{u}_{1}$	of length one orthogonal to $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$. $\boldsymbol{u}_3 = [2, 1, -2]^T/3$ is such a vector and we
lyche-numerical-linear-algebra_F02307	171	1.000	`\boldsymbol{u}_{2} . \boldsymbol{u}_{3}=[2,1,-2]^{T} / 3`	$\boldsymbol{u}_{2} . \boldsymbol{u}_{3}=[2,1,-2]^{T} / 3$	of length one orthogonal to $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$. $\boldsymbol{u}_3 = [2, 1, -2]^T/3$ is such a vector and we
lyche-numerical-linear-algebra_F02308	171	1.000	`\Sigma_{1}`	Σ_{1}	Thus Σ_1 contains the r positive singular values on the diagonal and for $k, l \geq 0$ the
lyche-numerical-linear-algebra_F02309	171	1.000	`k, l \geq 0`	$k, l \geq 0$	Thus Σ_1 contains the r positive singular values on the diagonal and for $k, l \geq 0$ the
lyche-numerical-linear-algebra_F02310	171	1.000	`\mathbf{0}_{k, l}=[]`	$\mathbf{0}_{k, l}=[]$	symbol $\mathbf{0}_{k,l} = [\,]$ denotes the empty matrix if $k = 0$ or $l = 0$, and the zero matrix
lyche-numerical-linear-algebra_F02311	171	1.000	`k=0`	$k=0$	symbol $\mathbf{0}_{k,l} = [\,]$ denotes the empty matrix if $k = 0$ or $l = 0$, and the zero matrix
lyche-numerical-linear-algebra_F02312	171	1.000	`l=0`	$l=0$	symbol $\mathbf{0}_{k,l} = [\,]$ denotes the empty matrix if $k = 0$ or $l = 0$, and the zero matrix
lyche-numerical-linear-algebra_F02313	172	1.000	`l`	l	with k rows and l columns otherwise. We obtain by block multiplication a reduced
lyche-numerical-linear-algebra_F02314	172	1.000	`\boldsymbol{A}=\boldsymbol{U}_{1} \boldsymbol{\Sigma}_{1} \boldsymbol{V}_{1}^{*}`	$\boldsymbol{A}=\boldsymbol{U}_{1} \boldsymbol{\Sigma}_{1} \boldsymbol{V}_{1}^{*}$	factorization of $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ of the form $\boldsymbol{A} = \boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*$, where $\boldsymbol{U}_1 \in \mathbb{C}^{m\times r}$ and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02315	172	1.000	`\boldsymbol{U}_{1} \in \mathbb{C}^{m \times r}`	$\boldsymbol{U}_{1} \in \mathbb{C}^{m \times r}$	factorization of $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ of the form $\boldsymbol{A} = \boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*$, where $\boldsymbol{U}_1 \in \mathbb{C}^{m\times r}$ and
lyche-numerical-linear-algebra F02316	172	1.000	`\boldsymbol{V}_{1} \in \mathbb{C}^{n \times r}`	$\boldsymbol{V}_{1} \in \mathbb{C}^{n \times r}$	$\boldsymbol{V}_1 \in \mathbb{C}^{n\times r}$ have orthonormal columns, and $\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r\times r}$ is a diagonal matrix with
lyche-numerical-linear-algebra F02317	172	1.000	`\boldsymbol{\Sigma}_{1} \in \mathbb{R}^{r \times r}`	$\boldsymbol{\Sigma}_{1} \in \mathbb{R}^{r \times r}$	$\boldsymbol{V}_1 \in \mathbb{C}^{n\times r}$ have orthonormal columns, and $\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r\times r}$ is a diagonal matrix with
lyche-numerical-linear-algebra F02318	172	1.000	`\sigma_{1} \geq \cdots \geq \sigma_{r}>0`	$\sigma_{1} \geq \cdots \geq \sigma_{r}>0$	$\sigma_1 \geq \cdots \geq \sigma_r > 0$.
lyche-numerical-linear-algebra F02319	172	1.000	`\boldsymbol{U}_{1}, \boldsymbol{V}_{1}`	$\boldsymbol{U}_{1}, \boldsymbol{V}_{1}$	$\boldsymbol{U}_1$, $\boldsymbol{V}_1$ contain the first r columns of $\boldsymbol{U}$, $\boldsymbol{V}$ respectively, and $\Sigma_1 \in \mathbb{R}^{r\times r}$ is a
lyche-numerical-linear-algebra F02320	172	1.000	`\boldsymbol{U}, \boldsymbol{V}`	$\boldsymbol{U}, \boldsymbol{V}$	$\boldsymbol{U}_1$, $\boldsymbol{V}_1$ contain the first r columns of $\boldsymbol{U}$, $\boldsymbol{V}$ respectively, and $\Sigma_1 \in \mathbb{R}^{r\times r}$ is a
lyche-numerical-linear-algebra F02321	172	1.000	`\Sigma_{1} \in \mathbb{R}^{r \times r}`	$\Sigma_{1} \in \mathbb{R}^{r \times r}$	$\boldsymbol{U}_1$, $\boldsymbol{V}_1$ contain the first r columns of $\boldsymbol{U}$, $\boldsymbol{V}$ respectively, and $\Sigma_1 \in \mathbb{R}^{r\times r}$ is a
lyche-numerical-linear-algebra F02322	172	1.000	`\boldsymbol{U}_{1}`	$\boldsymbol{U}_{1}$	$\boldsymbol{U}_1$ and $\boldsymbol{V}_1$ in any way to unitary matrices $\boldsymbol{U} \in \mathbb{C}^{m\times m}$ and $\boldsymbol{V} \in \mathbb{C}^{n\times n}$, and let $\boldsymbol{\Sigma}$
lyche-numerical-linear-algebra F02323	172	1.000	`\boldsymbol{V}_{1}`	$\boldsymbol{V}_{1}$	$\boldsymbol{U}_1$ and $\boldsymbol{V}_1$ in any way to unitary matrices $\boldsymbol{U} \in \mathbb{C}^{m\times m}$ and $\boldsymbol{V} \in \mathbb{C}^{n\times n}$, and let $\boldsymbol{\Sigma}$
lyche-numerical-linear-algebra F02324	172	1.000	`\boldsymbol{U} \in \mathbb{C}^{m \times m}`	$\boldsymbol{U} \in \mathbb{C}^{m \times m}$	$\boldsymbol{U}_1$ and $\boldsymbol{V}_1$ in any way to unitary matrices $\boldsymbol{U} \in \mathbb{C}^{m\times m}$ and $\boldsymbol{V} \in \mathbb{C}^{n\times n}$, and let $\boldsymbol{\Sigma}$
lyche-numerical-linear-algebra F02325	172	1.000	`\boldsymbol{V} \in \mathbb{C}^{n \times n}`	$\boldsymbol{V} \in \mathbb{C}^{n \times n}$	$\boldsymbol{U}_1$ and $\boldsymbol{V}_1$ in any way to unitary matrices $\boldsymbol{U} \in \mathbb{C}^{m\times m}$ and $\boldsymbol{V} \in \mathbb{C}^{n\times n}$, and let $\boldsymbol{\Sigma}$
lyche-numerical-linear-algebra F02326	172	1.000	`\boldsymbol{A}=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{r}\right] \operatorname{diag} \left(\sigma_{1}, \ldots, \sigma_{r}\right)\left[ \boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{r}\right]^{*}`	$\boldsymbol{A}=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{r}\right] \operatorname{diag} \left(\sigma_{1}, \ldots, \sigma_{r}\right)\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{r}\right]^{*}$	3. If $\boldsymbol{A} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r]\operatorname{diag}(\sigma_1, \ldots, \sigma_r)[\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r]^*$ is a singular value factoriza-
lyche-numerical-linear-algebra F02327	172	1.000	`7.3(r<n<m)`	$7.3(r<n<m)$	*Example 7.3 ($r < n < m$)* To find the SVF and SVD of
lyche-numerical-linear-algebra F02328	173	1.000	`\sigma_{1}=2, \sigma_{2}=0`	$\sigma_{1}=2, \sigma_{2}=0$	and we find $\sigma_1 = 2, \sigma_2 = 0$, Thus $r = 1, m = 3, n = 2$ and
lyche-numerical-linear-algebra F02329	173	1.000	`r=1, m=3, n=2`	$r=1, m=3, n=2$	and we find $\sigma_1 = 2, \sigma_2 = 0$, Thus $r = 1, m = 3, n = 2$ and
lyche-numerical-linear-algebra F02330	173	1.000	`\boldsymbol{u}_{1}=\boldsymbol{A} \boldsymbol{v}_{1} / \sigma_{1}=\boldsymbol{s}_{1} / \sqrt{2}`	$\boldsymbol{u}_{1}=\boldsymbol{A} \boldsymbol{v}_{1} / \sigma_{1}=\boldsymbol{s}_{1} / \sqrt{2}$	We find $\boldsymbol{u}_1 = \boldsymbol{A}\boldsymbol{v}_1/\sigma_1 = \boldsymbol{s}_1/\sqrt{2}$, where $\boldsymbol{s}_1 = [1, 1, 0]^T$, and the SVF of $\boldsymbol{A}$ is given
lyche-numerical-linear-algebra F02331	173	1.000	`\boldsymbol{s}_{1}=[1,1,0]^{T}`	$\boldsymbol{s}_{1}=[1,1,0]^{T}$	We find $\boldsymbol{u}_1 = \boldsymbol{A}\boldsymbol{v}_1/\sigma_1 = \boldsymbol{s}_1/\sqrt{2}$, where $\boldsymbol{s}_1 = [1, 1, 0]^T$, and the SVF of $\boldsymbol{A}$ is given
lyche-numerical-linear-algebra F02332	173	1.000	`\boldsymbol{s}_{1}`	$\boldsymbol{s}_{1}$	$\boldsymbol{s}_1$ to a basis $\{\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3\}$ for $\mathbb{R}^3$, apply the Gram-Schmidt orthogonalization process
lyche-numerical-linear-algebra F02333	173	1.000	`\left\{\boldsymbol{s}_{1}, \boldsymbol{s}_{2}, \boldsymbol{s}_{3}\right\}`	$\left\{\boldsymbol{s}_{1}, \boldsymbol{s}_{2}, \boldsymbol{s}_{3}\right\}$	$\boldsymbol{s}_1$ to a basis $\{\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3\}$ for $\mathbb{R}^3$, apply the Gram-Schmidt orthogonalization process
lyche-numerical-linear-algebra F02334	173	0.979	`\boldsymbol{w}_{1}=\boldsymbol{s}_{1}, \boldsymbol{w}_{2}=\boldsymbol{s}_{2}-\frac{ \boldsymbol{s}_{2}^{T} \boldsymbol{w}_{1}}{ \boldsymbol{w}_{1}^{T} \boldsymbol{w}_{1}} \boldsymbol{w}_{1}=\left[\begin{array}{c}-1 / 2 \\ \ 1 / 2 \\ 0\end{array}\right], \boldsymbol{w}_{3}= \boldsymbol{s}_{3}-`	$\boldsymbol{w}_{1}=\boldsymbol{s}_{1}, \boldsymbol{w}_{2}=\boldsymbol{s}_{2}-\frac{ \boldsymbol{s}_{2}^{T} \boldsymbol{w}_{1}}{ \boldsymbol{w}_{1}^{T} \boldsymbol{w}_{1}} \boldsymbol{w}_{1}=\left[\begin{array}{c}-1 / 2 \\ \ 1 / 2 \\ 0\end{array}\right], \boldsymbol{w}_{3}= \boldsymbol{s}_{3}-$	we find from (5.8) $\boldsymbol{w}_1 = \boldsymbol{s}_1$, $\boldsymbol{w}_2 = \boldsymbol{s}_2 - \frac{\boldsymbol{s}_2^T\boldsymbol{w}_1}{\boldsymbol{w}_1^T\boldsymbol{w}_1}\boldsymbol{w}_1 = \begin{bmatrix}-1/2\\1/2\\0\end{bmatrix}$, $\boldsymbol{w}_3 = \boldsymbol{s}_3 -$
lyche-numerical-linear-algebra F02335	173	0.763	`\frac{\boldsymbol{s}_{3}^{T} \boldsymbol{w}_{1}}{ \boldsymbol{w}_{1}^{T} \boldsymbol{w}_{1}} \boldsymbol{w}_{1}-\frac{\boldsymbol{s}_{3}^{T} \boldsymbol{w}_{2}}{\boldsymbol{w}_{2}^{T} \boldsymbol{w}_{2}} \boldsymbol{w}_{2}=\left[ \begin{array}{l}0 \\ 0 \\ 1\end{array}\right]`	$\frac{\boldsymbol{s}_{3}^{T} \boldsymbol{w}_{1}}{ \boldsymbol{w}_{1}^{T} \boldsymbol{w}_{1}} \boldsymbol{w}_{1}-\frac{\boldsymbol{s}_{3}^{T} \boldsymbol{w}_{2}}{\boldsymbol{w}_{2}^{T} \boldsymbol{w}_{2}} \boldsymbol{w}_{2}=\left[\begin{array}{l}0 \\ 0 \\ 1\end{array}\right]$	$\frac{\boldsymbol{s}_3^T\boldsymbol{w}_1}{\boldsymbol{w}_1^T\boldsymbol{w}_1}\boldsymbol{w}_1 - \frac{\boldsymbol{s}_3^T\boldsymbol{w}_2}{\boldsymbol{w}_2^T\boldsymbol{w}_2}\boldsymbol{w}_2 = \begin{bmatrix}0\\0\\1\end{bmatrix}$. Normalizing the $\boldsymbol{w}_i$'s we obtain $\boldsymbol{u}_1 = \boldsymbol{w}_1/\|\boldsymbol{w}_1\|_2 =$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02336	173	0.763	`\boldsymbol{w}_{i}`	$\boldsymbol{w}_{i}$	$\frac{s_3^T w_1}{w_1^T w_1} w_1 - \frac{s_3^T w_2}{w_2^T w_2} w_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Normalizing the w_i's we obtain $u_1 = w_1/\|w_1\|_2 =$
lyche-numerical-linear-algebra F02337	173	0.763	`\boldsymbol{u}_{1}=\boldsymbol{w}_{1} /\left| \boldsymbol{w}_{1}\right|_{2}=`	$\boldsymbol{u}_{1}=\boldsymbol{w}_{1} /\left\| \boldsymbol{w}_{1}\right\|_{2}=$	$\frac{s_3^T w_1}{w_1^T w_1} w_1 - \frac{s_3^T w_2}{w_2^T w_2} w_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Normalizing the w_i's we obtain $u_1 = w_1/\|w_1\|_2 =$
lyche-numerical-linear-algebra F02338	173	0.999	`[1 / \sqrt{2}, 1 / \sqrt{2}, 0]^{T}, \boldsymbol{u}_{2}= \boldsymbol{w}_{2} /\left|\boldsymbol{w}_{2}\right |_{2}=[-1 / \sqrt{2}, 1 / \sqrt{2}, 0]^{T}`	$[1 / \sqrt{2}, 1 / \sqrt{2}, 0]^{T}, \boldsymbol{u}_{2}= \boldsymbol{w}_{2} /\left\|\boldsymbol{w}_{2}\right\|_{2}=[-1 / \sqrt{2}, 1 / \sqrt{2}, 0]^{T}$	$[1/\sqrt{2}, 1/\sqrt{2}, 0]^T$, $u_2 = w_2/\|w_2\|_2 = [-1/\sqrt{2}, 1/\sqrt{2}, 0]^T$, and $u_3 = s_3/\|s_3\|_2 =$
lyche-numerical-linear-algebra F02339	173	0.999	`\boldsymbol{u}_{3}=\boldsymbol{s}_{3} /\left| \boldsymbol{s}_{3}\right|_{2}=`	$\boldsymbol{u}_{3}=\boldsymbol{s}_{3} /\left\| \boldsymbol{s}_{3}\right\|_{2}=$	$[1/\sqrt{2}, 1/\sqrt{2}, 0]^T$, $u_2 = w_2/\|w_2\|_2 = [-1/\sqrt{2}, 1/\sqrt{2}, 0]^T$, and $u_3 = s_3/\|s_3\|_2 =$
lyche-numerical-linear-algebra F02340	173	1.000	`[0,0,1]^{T}`	$[0,0,1]^{T}$	$[0, 0, 1]^T$. Therefore, $A = U \Sigma V^T$, is an SVD, where
lyche-numerical-linear-algebra F02341	173	1.000	`\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{T}`	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{T}$	$[0, 0, 1]^T$. Therefore, $A = U \Sigma V^T$, is an SVD, where
lyche-numerical-linear-algebra F02342	174	1.000	`\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}), \mathcal{R}\left(\boldsymbol{A}^{*}\right)`	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}), \mathcal{R}\left(\boldsymbol{A}^{*}\right)$	$\mathcal{R}(A)$, $\mathcal{N}(A)$, $\mathcal{R}(A^*)$, and $\mathcal{N}(A^*)$.
lyche-numerical-linear-algebra F02343	174	1.000	`m, n`	m, n	*m, n let $A \in \mathbb{C}^{m\times n}$ have rank r and a singular value decomposition $A =$*
lyche-numerical-linear-algebra F02344	174	0.985	`\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{m} \right] \boldsymbol{\Sigma}\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right]^{*}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*}`	$\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{m} \right] \boldsymbol{\Sigma}\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right]^{*}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*}$	$[u_1, \ldots, u_m]\Sigma[v_1, \ldots, v_n]^* = U \Sigma V^*$. *Then the singular vectors satisfy*
lyche-numerical-linear-algebra F02345	174	1.000	`\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{r} \right\}`	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{r} \right\}$	1. $\{u_1, \ldots, u_r\}$ *is an orthonormal basis for* $\mathcal{R}(A)$,
lyche-numerical-linear-algebra F02346	174	1.000	`\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{r} \right\}`	$\left\{\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{r} \right\}$	3. $\{v_1, \ldots, v_r\}$ *is an orthonormal basis for* $\mathcal{R}(A^*)$,
lyche-numerical-linear-algebra F02347	174	1.000	`\boldsymbol{A} \boldsymbol{V}=\boldsymbol{U} \boldsymbol{\Sigma}`	$\boldsymbol{A} \boldsymbol{V}=\boldsymbol{U} \boldsymbol{\Sigma}$	***Proof*** If $A = U \Sigma V^*$ then $AV = U \Sigma$, or in terms of the block partition (7.4)
lyche-numerical-linear-algebra F02348	174	0.980	`\boldsymbol{A}\left[\boldsymbol{V}_{1}, \boldsymbol{V}_{2}\right]=\left[\boldsymbol{U}_{1}, \boldsymbol{U}_{2}\right]\left[\begin{array}{cc} \Sigma_{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{array}\right]`	$\boldsymbol{A}\left[\boldsymbol{V}_{1}, \boldsymbol{V}_{2}\right]=\left[\boldsymbol{U}_{1}, \boldsymbol{U}_{2}\right]\left[\begin{array}{cc} \Sigma_{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{array}\right]$	$A[V_1, V_2] = [U_1, U_2]\begin{bmatrix} \Sigma_1 & 0 \\ 0 & 0 \end{bmatrix}$. But then $AV_1 = U_1\Sigma_1$, $AV_2 = 0$, and this
lyche-numerical-linear-algebra F02349	174	0.980	`\boldsymbol{A} \boldsymbol{V}_{1}=\boldsymbol{U}_{1} \boldsymbol{\Sigma}_{1}, \boldsymbol{A} \boldsymbol{V}_{2}=\mathbf{0}`	$\boldsymbol{A} \boldsymbol{V}_{1}=\boldsymbol{U}_{1} \boldsymbol{\Sigma}_{1}, \boldsymbol{A} \boldsymbol{V}_{2}=\mathbf{0}$	$A[V_1, V_2] = [U_1, U_2]\begin{bmatrix} \Sigma_1 & 0 \\ 0 & 0 \end{bmatrix}$. But then $AV_1 = U_1\Sigma_1$, $AV_2 = 0$, and this
lyche-numerical-linear-algebra F02350	174	1.000	`\boldsymbol{A}^{*}=\boldsymbol{V} \boldsymbol{\Sigma}^{*} \boldsymbol{U}^{*}`	$\boldsymbol{A}^{*}=\boldsymbol{V} \boldsymbol{\Sigma}^{*} \boldsymbol{U}^{*}$	$A^* = V \Sigma^* U^*$ or $A^*U = V \Sigma^*$. Using the block partition as before we obtain the
lyche-numerical-linear-algebra F02351	174	1.000	`\boldsymbol{A}^{*} \boldsymbol{U}=\boldsymbol{V} \boldsymbol{\Sigma}^{*}`	$\boldsymbol{A}^{*} \boldsymbol{U}=\boldsymbol{V} \boldsymbol{\Sigma}^{*}$	$A^* = V \Sigma^* U^*$ or $A^*U = V \Sigma^*$. Using the block partition as before we obtain the
lyche-numerical-linear-algebra F02352	174	1.000	`\mathcal{R}(\boldsymbol{A}),\left\{\boldsymbol{A}^{*} \boldsymbol{u}_{1}, \ldots, \boldsymbol{A}^{*} \boldsymbol{u}_{r}\right\}`	$\mathcal{R}(\boldsymbol{A}),\left\{\boldsymbol{A}^{*} \boldsymbol{u}_{1}, \ldots, \boldsymbol{A}^{*} \boldsymbol{u}_{r}\right\}$	$\mathcal{R}(A)$, $\{A^*u_1, \ldots, A^*u_r\}$ is an orthogonal basis for $\mathcal{R}(A^*)$, $\{v_{r+1}, \ldots, v_m\}$ is an
lyche-numerical-linear-algebra F02353	174	1.000	`\mathcal{R}\left(\boldsymbol{A}^{*}\right),\left\{ \boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_{m}\right\}`	$\mathcal{R}\left(\boldsymbol{A}^{*}\right),\left\{ \boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_{m}\right\}$	$\mathcal{R}(A)$, $\{A^*u_1, \ldots, A^*u_r\}$ is an orthogonal basis for $\mathcal{R}(A^*)$, $\{v_{r+1}, \ldots, v_m\}$ is an
lyche-numerical-linear-algebra F02354	174	1.000	`\boldsymbol{T}: \mathbb{R}^{n} \rightarrow \mathbb{R}^{m}`	$\boldsymbol{T}: \mathbb{R}^{n} \rightarrow \mathbb{R}^{m}$	a linear transformation. Consider the linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ given by
lyche-numerical-linear-algebra F02355	174	0.825	`\boldsymbol{T} \boldsymbol{z}:=\boldsymbol{A} \boldsymbol{z}`	$\boldsymbol{T} \boldsymbol{z}:=\boldsymbol{A} \boldsymbol{z}$	$Tz := Az$ where $A \in \mathbb{R}^{m\times n}$. Assume that rank$(A) = n$. In the following theorem
lyche-numerical-linear-algebra F02356	174	0.825	`(\boldsymbol{A})=n`	$(\boldsymbol{A})=n$	$Tz := Az$ where $A \in \mathbb{R}^{m\times n}$. Assume that rank$(A) = n$. In the following theorem
lyche-numerical-linear-algebra F02357	174	0.672	`\mathcal{S}:=\left\{z \in \mathbb{R}^{n}\right.`	$\mathcal{S}:=\left\{z \in \mathbb{R}^{n}\right.$	we show that the function T maps the unit sphere in $\mathbb{R}^n$ given by $\mathcal{S} := \{z \in \mathbb{R}^n :$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02358	174	0.943	`\left.|\boldsymbol{z}|_{2}=1\right\}`	$\left.\|\boldsymbol{z}\|_{2}=1\right\}$	$\|z\|_2 = 1\}$ onto an ellipsoid $\mathcal{E} := \boldsymbol{A}\mathcal{S} = \{\boldsymbol{A}z : z \in \mathcal{S}\}$ in $\mathbb{R}^m$.
lyche-numerical-linear-algebra F02359	174	0.943	`\mathcal{E}:=\boldsymbol{A} \mathcal{S}=\{\boldsymbol{A} \boldsymbol{z}: \boldsymbol{z} \in \mathcal{S}\}`	$\mathcal{E}:=\boldsymbol{A} \mathcal{S}=\{\boldsymbol{A} \boldsymbol{z}: \boldsymbol{z} \in \mathcal{S}\}$	$\|z\|_2 = 1\}$ onto an ellipsoid $\mathcal{E} := \boldsymbol{A}\mathcal{S} = \{\boldsymbol{A}z : z \in \mathcal{S}\}$ in $\mathbb{R}^m$.
lyche-numerical-linear-algebra F02360	174	1.000	`r=n`	$r=n$	**Theorem 7.4 (SVF Ellipse)** *Suppose* $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ *has rank* $r = n$, *and let* $\boldsymbol{A} =$
lyche-numerical-linear-algebra F02361	174	0.977	`\boldsymbol{U}_{1} \Sigma_{1} \boldsymbol{V}_{1}^{T}`	$\boldsymbol{U}_{1} \Sigma_{1} \boldsymbol{V}_{1}^{T}$	$\boldsymbol{U}_1\Sigma_1\boldsymbol{V}_1^T$ *be a singular value factorization of* $\boldsymbol{A}$. *Then*
lyche-numerical-linear-algebra F02362	175	0.407	`z \in \mathcal{S}`	$z \in \mathcal{S}$	***Proof*** Suppose $z \in \mathcal{S}$. Now $\boldsymbol{A}z = \boldsymbol{U}_1\Sigma_1\boldsymbol{V}_1^T z = \boldsymbol{U}_1\boldsymbol{y}$, where $\boldsymbol{y} := \Sigma_1\boldsymbol{V}_1^T z$.
lyche-numerical-linear-algebra F02363	175	0.407	`\boldsymbol{A} z=\boldsymbol{U}_{1} \Sigma_{1} \boldsymbol{V}_{1}^{T} z=\boldsymbol{U}_{1} \boldsymbol{y}`	$\boldsymbol{A} z=\boldsymbol{U}_{1} \Sigma_{1} \boldsymbol{V}_{1}^{T} z=\boldsymbol{U}_{1} \boldsymbol{y}$	***Proof*** Suppose $z \in \mathcal{S}$. Now $\boldsymbol{A}z = \boldsymbol{U}_1\Sigma_1\boldsymbol{V}_1^T z = \boldsymbol{U}_1\boldsymbol{y}$, where $\boldsymbol{y} := \Sigma_1\boldsymbol{V}_1^T z$.
lyche-numerical-linear-algebra F02364	175	0.407	`\boldsymbol{y}:=\Sigma_{1} \boldsymbol{V}_{1}^{T} \boldsymbol{z}`	$\boldsymbol{y}:=\Sigma_{1} \boldsymbol{V}_{1}^{T} \boldsymbol{z}$	***Proof*** Suppose $z \in \mathcal{S}$. Now $\boldsymbol{A}z = \boldsymbol{U}_1\Sigma_1\boldsymbol{V}_1^T z = \boldsymbol{U}_1\boldsymbol{y}$, where $\boldsymbol{y} := \Sigma_1\boldsymbol{V}_1^T z$.
lyche-numerical-linear-algebra F02365	175	0.997	`\operatorname{rank}(\boldsymbol{A})=n`	$\operatorname{rank}(\boldsymbol{A})=n$	Since $\operatorname{rank}(\boldsymbol{A}) = n$ it follows that $\boldsymbol{V}_1 = \boldsymbol{V}$ is square so that $\boldsymbol{V}_1\boldsymbol{V}_1^T = \boldsymbol{I}$. But then
lyche-numerical-linear-algebra F02366	175	0.997	`\boldsymbol{V}_{1}=\boldsymbol{V}`	$\boldsymbol{V}_{1}=\boldsymbol{V}$	Since $\operatorname{rank}(\boldsymbol{A}) = n$ it follows that $\boldsymbol{V}_1 = \boldsymbol{V}$ is square so that $\boldsymbol{V}_1\boldsymbol{V}_1^T = \boldsymbol{I}$. But then
lyche-numerical-linear-algebra F02367	175	0.997	`\boldsymbol{V}_{1} \boldsymbol{V}_{1}^{T}=\boldsymbol{I}`	$\boldsymbol{V}_{1} \boldsymbol{V}_{1}^{T}=\boldsymbol{I}$	Since $\operatorname{rank}(\boldsymbol{A}) = n$ it follows that $\boldsymbol{V}_1 = \boldsymbol{V}$ is square so that $\boldsymbol{V}_1\boldsymbol{V}_1^T = \boldsymbol{I}$. But then
lyche-numerical-linear-algebra F02368	175	0.984	`\boldsymbol{V}_{1} \boldsymbol{\Sigma}_{1}^{-1} \boldsymbol{y}=\boldsymbol{z}`	$\boldsymbol{V}_{1} \boldsymbol{\Sigma}_{1}^{-1} \boldsymbol{y}=\boldsymbol{z}$	$\boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{y} = z$ and we obtain
lyche-numerical-linear-algebra F02369	175	1.000	`\boldsymbol{y} \in \tilde{\mathcal{E}}`	$\boldsymbol{y} \in \tilde{\mathcal{E}}$	This implies that $\boldsymbol{y} \in \tilde{\mathcal{E}}$. Finally, $\boldsymbol{x} = \boldsymbol{A}z = \boldsymbol{U}_1\Sigma_1\boldsymbol{V}_1^T z = \boldsymbol{U}_1\boldsymbol{y}$, where $\boldsymbol{y} \in \tilde{\mathcal{E}}$
lyche-numerical-linear-algebra F02370	175	1.000	`\boldsymbol{x}=\boldsymbol{A} \boldsymbol{z}= \boldsymbol{U}_{1} \Sigma_{1} \boldsymbol{V}_{1}^{T} \boldsymbol{z}=\boldsymbol{U}_{1} \boldsymbol{y}`	$\boldsymbol{x}=\boldsymbol{A} \boldsymbol{z}= \boldsymbol{U}_{1} \Sigma_{1} \boldsymbol{V}_{1}^{T} \boldsymbol{z}=\boldsymbol{U}_{1} \boldsymbol{y}$	This implies that $\boldsymbol{y} \in \tilde{\mathcal{E}}$. Finally, $\boldsymbol{x} = \boldsymbol{A}z = \boldsymbol{U}_1\Sigma_1\boldsymbol{V}_1^T z = \boldsymbol{U}_1\boldsymbol{y}$, where $\boldsymbol{y} \in \tilde{\mathcal{E}}$
lyche-numerical-linear-algebra F02371	175	1.000	`\mathcal{E}=\boldsymbol{U}_{1} \tilde{\mathcal{E}}`	$\mathcal{E}=\boldsymbol{U}_{1} \tilde{\mathcal{E}}$	implies that $\mathcal{E} = \boldsymbol{U}_1\tilde{\mathcal{E}}$.
lyche-numerical-linear-algebra F02372	175	1.000	`1=\frac{y_{1}^{2}}{\sigma_{1}^{2}}+\cdots+ \frac{y_{n}^{2}}{\sigma_{n}^{2}}`	$1=\frac{y_{1}^{2}}{\sigma_{1}^{2}}+\cdots+ \frac{y_{n}^{2}}{\sigma_{n}^{2}}$	The equation $1 = \frac{y_1^2}{\sigma_1^2} + \cdots + \frac{y_n^2}{\sigma_n^2}$ describes an ellipsoid in $\mathbb{R}^n$ with semiaxes
lyche-numerical-linear-algebra F02373	175	1.000	`\sigma_{j}`	σ_{j}	of length σ_j along the unit vectors $\boldsymbol{e}_j$ for $j = 1, \ldots, n$. Since the orthonormal
lyche-numerical-linear-algebra F02374	175	1.000	`\boldsymbol{e}_{j}`	$\boldsymbol{e}_{j}$	of length σ_j along the unit vectors $\boldsymbol{e}_j$ for $j = 1, \ldots, n$. Since the orthonormal
lyche-numerical-linear-algebra F02375	175	0.992	`\boldsymbol{U}_{1} \boldsymbol{y} \rightarrow \boldsymbol{x}`	$\boldsymbol{U}_{1} \boldsymbol{y} \rightarrow \boldsymbol{x}$	transformation $\boldsymbol{U}_1\boldsymbol{y} \rightarrow \boldsymbol{x}$ preserves length, the image $\mathcal{E} = \boldsymbol{A}\mathcal{S}$ is a rotated ellipsoid
lyche-numerical-linear-algebra F02376	175	0.992	`\mathcal{E}=\boldsymbol{A} \mathcal{S}`	$\mathcal{E}=\boldsymbol{A} \mathcal{S}$	transformation $\boldsymbol{U}_1\boldsymbol{y} \rightarrow \boldsymbol{x}$ preserves length, the image $\mathcal{E} = \boldsymbol{A}\mathcal{S}$ is a rotated ellipsoid
lyche-numerical-linear-algebra F02377	175	1.000	`\boldsymbol{u}_{j}=\boldsymbol{U} \boldsymbol{e}_{j}`	$\boldsymbol{u}_{j}=\boldsymbol{U} \boldsymbol{e}_{j}$	with semiaxes along the left singular vectors $\boldsymbol{u}_j = \boldsymbol{U}\boldsymbol{e}_j$, of length σ_j, $j = 1, \ldots, n$.
lyche-numerical-linear-algebra F02378	175	1.000	`\sigma_{j}, j=1, \ldots, n`	$\sigma_{j}, j=1, \ldots, n$	with semiaxes along the left singular vectors $\boldsymbol{u}_j = \boldsymbol{U}\boldsymbol{e}_j$, of length σ_j, $j = 1, \ldots, n$.
lyche-numerical-linear-algebra F02379	175	1.000	`\boldsymbol{A} \boldsymbol{v}_{j}=\sigma_{j} \boldsymbol{u}_{j}`	$\boldsymbol{A} \boldsymbol{v}_{j}=\sigma_{j} \boldsymbol{u}_{j}$	Since $\boldsymbol{A}\boldsymbol{v}_j = \sigma_j\boldsymbol{u}_j$, for $j = 1, \ldots, n$ the right singular vectors defines points in $\mathcal{S}$
lyche-numerical-linear-algebra F02380	175	1.000	`\mathcal{E}`	$\mathcal{E}$	that are mapped onto the semiaxes of $\mathcal{E}$.
lyche-numerical-linear-algebra F02381	175	1.000	`\boldsymbol{A}: \mathbb{R}^{2} \rightarrow \mathbb{R}^{2}`	$\boldsymbol{A}: \mathbb{R}^{2} \rightarrow \mathbb{R}^{2}$	*Example 7.4 (Ellipse)* Consider the transformation $\boldsymbol{A} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by the
lyche-numerical-linear-algebra F02382	175	1.000	`\sigma_{1}=3, \sigma_{2}=1, \boldsymbol{u}_{1}=[3,4]^{T} / 5`	$\sigma_{1}=3, \sigma_{2}=1, \boldsymbol{u}_{1}=[3,4]^{T} / 5$	in Example 7.8. Recall that $\sigma_1 = 3$, $\sigma_2 = 1$, $\boldsymbol{u}_1 = [3,4]^T/5$ and $\boldsymbol{u}_2 = [-4,3]^T/5$.

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ F02383</td><td>175</td><td>1.000</td><td>\boldsymbol{u}_{2}=[-4,3]^{T} / 5</td><td>$\boldsymbol{u}_{2}=[-4,3]^{T} / 5$</td><td>in Example 7.8. Recall that $\sigma_1=3$, $\sigma_2=1$, $\boldsymbol{u}_1=[3,4]^T/5$ and $\boldsymbol{u}_2=[-4,3]^T/5$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02384</td><td>175</td><td>1.000</td><td>y_{1}^{2} / \sigma_{1}^{2}+y_{2}^{2} / \sigma_{2}^{2}=1</td><td>$y_{1}^{2} / \sigma_{1}^{2}+y_{2}^{2} / \sigma_{2}^{2}=1$</td><td>The ellipses $y_1^2/\sigma_1^2+y_2^2/\sigma_2^2=1$ and $\mathcal{E}=\boldsymbol{A}\mathcal{S}=\boldsymbol{U}_1\tilde{\mathcal{E}}$ are shown in Fig. 7.1. Since</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02385</td><td>175</td><td>1.000</td><td>\mathcal{E}=\boldsymbol{A} \mathcal{S}=\boldsymbol{U}_{1} \tilde{\mathcal{E}}</td><td>$\mathcal{E}=\boldsymbol{A} \mathcal{S}=\boldsymbol{U}_{1} \tilde{\mathcal{E}}$</td><td>The ellipses $y_1^2/\sigma_1^2+y_2^2/\sigma_2^2=1$ and $\mathcal{E}=\boldsymbol{A}\mathcal{S}=\boldsymbol{U}_1\tilde{\mathcal{E}}$ are shown in Fig. 7.1. Since</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02386</td><td>175</td><td>1.000</td><td>\boldsymbol{A} \mathcal{S}</td><td>$\boldsymbol{A} \mathcal{S}$</td><td>Fig. 7.1 The ellipse $y_1^2/9+y_2^2=1$ (left) and the rotated ellipse $\boldsymbol{A}\mathcal{S}$ (right)</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02387</td><td>176</td><td>0.997</td><td>\boldsymbol{y}=\boldsymbol{U}_{1}^{T} \boldsymbol{x}= \left[3 / 5 x_{1}+4 / 5 x_{2},-4 / 5 x_{1}+3 / 5 x_{2} \right]^{T}</td><td>$\boldsymbol{y}=\boldsymbol{U}_{1}^{T} \boldsymbol{x}= \left[3 / 5 x_{1}+4 / 5 x_{2},-4 / 5 x_{1}+3 / 5 x_{2} \right]^{T}$</td><td>$\boldsymbol{y}=\boldsymbol{U}_1^T\boldsymbol{x}=[3/5x_1+4/5x_2,-4/5x_1+3/5x_2]^T$, the equation for the ellipse on the</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02388</td><td>176</td><td>1.000</td><td>\left\|\boldsymbol{A}^{*}\right\|_{F}=\|\boldsymbol{A} \|_{F}</td><td>$\left\|\boldsymbol{A}^{*}\right\|_{F}=\|\boldsymbol{A} \|_{F}$</td><td>1. $\|\boldsymbol{A}^*\|_F=\|\boldsymbol{A}\|_F$,</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02389</td><td>176</td><td>1.000</td><td>\|\boldsymbol{A}\|_{F}^{2}=\sum_{j=1}^{n}\left\| \boldsymbol{a}_{: j}\right\|_{2}^{2}</td><td>$\|\boldsymbol{A}\|_{F}^{2}=\sum_{j=1}^{n}\left\| \boldsymbol{a}_{: j}\right\|_{2}^{2}$</td><td>2. $\|\boldsymbol{A}\|_F^2=\sum_{j=1}^n\|\boldsymbol{a}_{:j}\|_2^2$,</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02390</td><td>177</td><td>1.000</td><td>\|\boldsymbol{U} \boldsymbol{A}\|_{F}=\|\boldsymbol{A} \boldsymbol{V}\|_{F}=\|\boldsymbol{A}\|_{F}</td><td>$\|\boldsymbol{U} \boldsymbol{A}\|_{F}=\|\boldsymbol{A} \boldsymbol{V}\|_{F}=\|\boldsymbol{A}\|_{F}$</td><td>3. $\|\boldsymbol{U}\boldsymbol{A}\|_F=\|\boldsymbol{A}\boldsymbol{V}\|_F=\|\boldsymbol{A}\|_F$ for any unitary matrices $\boldsymbol{U}\in\mathbb{C}^{m\times m}$ and $\boldsymbol{V}\in$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02391</td><td>177</td><td>1.000</td><td>\boldsymbol{V} \in</td><td>$\boldsymbol{V} \in$</td><td>3. $\|\boldsymbol{U}\boldsymbol{A}\|_F=\|\boldsymbol{A}\boldsymbol{V}\|_F=\|\boldsymbol{A}\|_F$ for any unitary matrices $\boldsymbol{U}\in\mathbb{C}^{m\times m}$ and $\boldsymbol{V}\in$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02392</td><td>177</td><td>0.999</td><td>\|\boldsymbol{A} \boldsymbol{B}\|_{F} \leq\| \boldsymbol{A}\|_{F}\|\boldsymbol{B}\|_{F}</td><td>$\|\boldsymbol{A} \boldsymbol{B}\|_{F} \leq\| \boldsymbol{A}\|_{F}\|\boldsymbol{B}\|_{F}$</td><td>4. $\|\boldsymbol{A}\boldsymbol{B}\|_F\le\|\boldsymbol{A}\|_F\|\boldsymbol{B}\|_F$ for any $\boldsymbol{B}\in\mathbb{C}^{n,k},\quad k\in\mathbb{N}$,</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02393</td><td>177</td><td>0.999</td><td>\boldsymbol{B} \in \mathbb{C}^{n, k}, \quad k \in \mathbb{N}</td><td>$\boldsymbol{B} \in \mathbb{C}^{n, k}, \quad k \in \mathbb{N}$</td><td>4. $\|\boldsymbol{A}\boldsymbol{B}\|_F\le\|\boldsymbol{A}\|_F\|\boldsymbol{B}\|_F$ for any $\boldsymbol{B}\in\mathbb{C}^{n,k},\quad k\in\mathbb{N}$,</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02394</td><td>177</td><td>0.980</td><td>\|\boldsymbol{A} \boldsymbol{x}\|_{2} \leq\| \boldsymbol{A}\|_{F}\|\boldsymbol{x}\|_{2}</td><td>$\|\boldsymbol{A} \boldsymbol{x}\|_{2} \leq\| \boldsymbol{A}\|_{F}\|\boldsymbol{x}\|_{2}$</td><td>5. $\|\boldsymbol{A}\boldsymbol{x}\|_2\le\|\boldsymbol{A}\|_F\|\boldsymbol{x}\|_2$, for all $\boldsymbol{x}\in\mathbb{C}^n$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02395</td><td>177</td><td>1.000</td><td>\left\|\boldsymbol{A}^{*}\right\|_{F}^{2}=\sum_{j=1}^{n} \sum_{i=1}^{m}\left|\bar{a}_{i j}\right|^{2}= \sum_{i=1}^{m} \sum_{j=1}^{n}\left|a_{i j}\right|^{2}=\| \boldsymbol{A}\|_{F}^{2}</td><td>$\left\|\boldsymbol{A}^{*}\right\|_{F}^{2}=\sum_{j=1}^{n} \sum_{i=1}^{m}\left|\bar{a}_{i j}\right|^{2}= \sum_{i=1}^{m} \sum_{j=1}^{n}\left|a_{i j}\right|^{2}=\| \boldsymbol{A}\|_{F}^{2}$</td><td>1. $\|\boldsymbol{A}^*\|_F^2=\sum_{j=1}^n\sum_{i=1}^m|\bar{a}_{ij}|^2=\sum_{i=1}^m\sum_{j=1}^n|a_{ij}|^2=\|\boldsymbol{A}\|_F^2$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02396</td><td>177</td><td>1.000</td><td>\|\boldsymbol{A}\|_{F}:=\|\operatorname{vec}(\boldsymbol{A})\|_{2}</td><td>$\|\boldsymbol{A}\|_{F}:=\|\operatorname{vec}(\boldsymbol{A})\|_{2}$</td><td>$\|\boldsymbol{A}\|_F:=\|\operatorname{vec}(\boldsymbol{A})\|_2$, where $\operatorname{vec}(\boldsymbol{A})\in\mathbb{C}^{mn}$ is the vector obtained by stacking</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02397</td><td>177</td><td>1.000</td><td>\operatorname{vec}(\boldsymbol{A}) \in \mathbb{C}^{m n}</td><td>$\operatorname{vec}(\boldsymbol{A}) \in \mathbb{C}^{m n}$</td><td>$\|\boldsymbol{A}\|_F:=\|\operatorname{vec}(\boldsymbol{A})\|_2$, where $\operatorname{vec}(\boldsymbol{A})\in\mathbb{C}^{mn}$ is the vector obtained by stacking</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02398</td><td>177</td><td>1.000</td><td>\boldsymbol{U}^{*} \boldsymbol{U}=I</td><td>$\boldsymbol{U}^{*} \boldsymbol{U}=I$</td><td>3. Recall that if $\boldsymbol{U}^*\boldsymbol{U}=\boldsymbol{I}$ then $\|\boldsymbol{U}\boldsymbol{x}\|_2=\|\boldsymbol{x}\|_2$ for all $\boldsymbol{x}\in\mathbb{C}^n$. Applying this to each</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02399</td><td>177</td><td>1.000</td><td>\|\boldsymbol{U} \boldsymbol{x}\|_{2}=\|\boldsymbol{x} \|_{2}</td><td>$\|\boldsymbol{U} \boldsymbol{x}\|_{2}=\|\boldsymbol{x} \|_{2}$</td><td>3. Recall that if $\boldsymbol{U}^*\boldsymbol{U}=\boldsymbol{I}$ then $\|\boldsymbol{U}\boldsymbol{x}\|_2=\|\boldsymbol{x}\|_2$ for all $\boldsymbol{x}\in\mathbb{C}^n$. Applying this to each</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02400</td><td>177</td><td>0.997</td><td>\|\boldsymbol{U} \boldsymbol{A}\|_{F}^{2} \stackrel{2 .}{=} \sum_{j=1}^{n}\left\|\boldsymbol{U} \boldsymbol{a}_{: j}\right\|_{2}^{2}=\sum_{j=1}^{n} \left\|\boldsymbol{a}_{: j}\right\|_{2}^{2} \stackrel{2 .}{=}\|\boldsymbol{A}\|_{F}^{2}</td><td>$\|\boldsymbol{U} \boldsymbol{A}\|_{F}^{2} \stackrel{2 .}{=} \sum_{j=1}^{n}\left\|\boldsymbol{U} \boldsymbol{a}_{: j}\right\|_{2}^{2}=\sum_{j=1}^{n} \left\|\boldsymbol{a}_{: j}\right\|_{2}^{2} \stackrel{2 .}{=}\|\boldsymbol{A}\|_{F}^{2}$</td><td>column $\boldsymbol{a}_{:j}$ of $\boldsymbol{A}$ we find $\|\boldsymbol{U}\boldsymbol{A}\|_F^2\stackrel{2.}{=}\sum_{j=1}^n\|\boldsymbol{U}\boldsymbol{a}_{:j}\|_2^2=\sum_{j=1}^n\|\boldsymbol{a}_{:j}\|_2^2\stackrel{2.}{=}\|\boldsymbol{A}\|_F^2$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02401</td><td>177</td><td>0.933</td><td>\boldsymbol{V} \boldsymbol{V}^{*}=I</td><td>$\boldsymbol{V} \boldsymbol{V}^{*}=I$</td><td>Similarly, since $\boldsymbol{V}\boldsymbol{V}^*=\boldsymbol{I}$ we find $\|\boldsymbol{A}\boldsymbol{V}\|_F\stackrel{1.}{=}\|\boldsymbol{V}^*\boldsymbol{A}^*\|_F=\|\boldsymbol{A}^*\|_F\stackrel{1.}{=}\|\boldsymbol{A}\|_F$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02402</td><td>177</td><td>0.933</td><td>\|\boldsymbol{A} \boldsymbol{V}\|_{F} \stackrel{1 .}{=} \left\|\boldsymbol{V}^{*} \boldsymbol{A}^{*}\right\|_{F}= \left\|\boldsymbol{A}^{*}\right\|_{F} \stackrel{1 .}{=}\| \boldsymbol{A}\|_{F}</td><td>$\|\boldsymbol{A} \boldsymbol{V}\|_{F} \stackrel{1 .}{=} \left\|\boldsymbol{V}^{*} \boldsymbol{A}^{*}\right\|_{F}= \left\|\boldsymbol{A}^{*}\right\|_{F} \stackrel{1 .}{=}\| \boldsymbol{A}\|_{F}$</td><td>Similarly, since $\boldsymbol{V}\boldsymbol{V}^*=\boldsymbol{I}$ we find $\|\boldsymbol{A}\boldsymbol{V}\|_F\stackrel{1.}{=}\|\boldsymbol{V}^*\boldsymbol{A}^*\|_F=\|\boldsymbol{A}^*\|_F\stackrel{1.}{=}\|\boldsymbol{A}\|_F$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02403</td><td>177</td><td>0.986</td><td>\|\boldsymbol{v}\|_{F}=\|\boldsymbol{v}\|_{2}</td><td>$\|\boldsymbol{v}\|_{F}=\|\boldsymbol{v}\|_{2}$</td><td>5. Since $\|\boldsymbol{v}\|_F=\|\boldsymbol{v}\|_2$ for a vector this follows by taking $k=1$ and $\boldsymbol{B}=\boldsymbol{x}$ in 4.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02404</td><td>177</td><td>0.986</td><td>\boldsymbol{B}=\boldsymbol{x}</td><td>$\boldsymbol{B}=\boldsymbol{x}$</td><td>5. Since $\|\boldsymbol{v}\|_F=\|\boldsymbol{v}\|_2$ for a vector this follows by taking $k=1$ and $\boldsymbol{B}=\boldsymbol{x}$ in 4.</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F02405	177	0.996	`|\boldsymbol{A}|_{F}=`	$\|\boldsymbol{A}\|_{F}=$	**Theorem 7.5 (Frobenius Norm and Singular Values)** *We have* $\|\boldsymbol{A}\|_F =$
lyche-numerical-linear-algebra_F02406	177	0.999	`\sqrt{\sigma_{1}^{2}+\cdots+\sigma_{n}^{2}}`	$\sqrt{\sigma_{1}^{2}+\cdots+\sigma_{n}^{2}}$	$\sqrt{\sigma_1^2+\cdots+\sigma_n^2}$, *where* $\sigma_1,\ldots,\sigma_n$ *are the singular values of* $\boldsymbol{A}$.
lyche-numerical-linear-algebra_F02407	177	0.999	`\sigma_{1}, \ldots, \sigma_{n}`	$\sigma_{1}, \ldots, \sigma_{n}$	$\sqrt{\sigma_1^2+\cdots+\sigma_n^2}$, *where* $\sigma_1,\ldots,\sigma_n$ *are the singular values of* $\boldsymbol{A}$.
lyche-numerical-linear-algebra_F02408	177	1.000	`m \geq n \geq 1`	$m \geq n \geq 1$	Suppose $m \geq n \geq 1$ and $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ has a singular value decomposition $\boldsymbol{A} =$
lyche-numerical-linear-algebra_F02409	177	1.000	`\boldsymbol{U}\left[\begin{array}{c}\boldsymbol{D} \\ \mathbf{0}\end{array}\right] \boldsymbol{V}^{*}`	$\boldsymbol{U}\left[\begin{array}{c}\boldsymbol{D} \\ \mathbf{0}\end{array}\right] \boldsymbol{V}^{*}$	$\boldsymbol{U}\left[\begin{smallmatrix}\boldsymbol{D}\\ \mathbf{0}\end{smallmatrix}\right]\boldsymbol{V}^*$, where $\boldsymbol{D}=\operatorname{diag}(\sigma_1,\ldots,\sigma_n)$. We choose $\epsilon>0$ and let $1\leq r\leq n$ be
lyche-numerical-linear-algebra_F02410	177	1.000	`\boldsymbol{D}=\operatorname{diag}\left(\sigma_{1}, \ldots, \sigma_{n}\right)`	$\boldsymbol{D}=\operatorname{diag}\left(\sigma_{1}, \ldots, \sigma_{n}\right)$	$\boldsymbol{U}\left[\begin{smallmatrix}\boldsymbol{D}\\ \mathbf{0}\end{smallmatrix}\right]\boldsymbol{V}^*$, where $\boldsymbol{D}=\operatorname{diag}(\sigma_1,\ldots,\sigma_n)$. We choose $\epsilon>0$ and let $1\leq r\leq n$ be
lyche-numerical-linear-algebra_F02411	177	1.000	`\epsilon>0`	$\epsilon>0$	$\boldsymbol{U}\left[\begin{smallmatrix}\boldsymbol{D}\\ \mathbf{0}\end{smallmatrix}\right]\boldsymbol{V}^*$, where $\boldsymbol{D}=\operatorname{diag}(\sigma_1,\ldots,\sigma_n)$. We choose $\epsilon>0$ and let $1\leq r\leq n$ be
lyche-numerical-linear-algebra_F02412	177	1.000	`1 \leq r \leq n`	$1 \leq r \leq n$	$\boldsymbol{U}\left[\begin{smallmatrix}\boldsymbol{D}\\ \mathbf{0}\end{smallmatrix}\right]\boldsymbol{V}^*$, where $\boldsymbol{D}=\operatorname{diag}(\sigma_1,\ldots,\sigma_n)$. We choose $\epsilon>0$ and let $1\leq r\leq n$ be
lyche-numerical-linear-algebra_F02413	177	1.000	`\sigma_{r+1}^{2}+\cdots+\sigma_{n}^{2}<\epsilon^{2}`	$\sigma_{r+1}^{2}+\cdots+\sigma_{n}^{2}<\epsilon^{2}$	the smallest integer such that $\sigma_{r+1}^2+\cdots+\sigma_n^2<\epsilon^2$. Define $\boldsymbol{A}':=\boldsymbol{U}\left[\begin{smallmatrix}\boldsymbol{D}'\\ \mathbf{0}\end{smallmatrix}\right]\boldsymbol{V}^*$, where
lyche-numerical-linear-algebra_F02414	177	1.000	`\boldsymbol{A}^{\prime}:=\boldsymbol{U}\left[ \begin{array}{c}\boldsymbol{D}^{\prime} \\ \mathbf{0} \end{array}\right] \boldsymbol{V}^{*}`	$\boldsymbol{A}^{\prime}:=\boldsymbol{U}\left[\begin{array}{c}\boldsymbol{D}^{\prime} \\ \mathbf{0} \end{array}\right] \boldsymbol{V}^{*}$	the smallest integer such that $\sigma_{r+1}^2+\cdots+\sigma_n^2<\epsilon^2$. Define $\boldsymbol{A}':=\boldsymbol{U}\left[\begin{smallmatrix}\boldsymbol{D}'\\ \mathbf{0}\end{smallmatrix}\right]\boldsymbol{V}^*$, where
lyche-numerical-linear-algebra_F02415	177	1.000	`\boldsymbol{D}^{\prime}:=\operatorname{diag}\left( \sigma_{1}, \ldots, \sigma_{r}, 0, \ldots, 0\right) \in \mathbb{R}^{n \times n}`	$\boldsymbol{D}^{\prime}:=\operatorname{diag}\left(\sigma_{1}, \ldots, \sigma_{r}, 0, \ldots, 0\right) \in \mathbb{R}^{n \times n}$	$\boldsymbol{D}':=\operatorname{diag}(\sigma_1,\ldots,\sigma_r,0,\ldots,0)\in\mathbb{R}^{n\times n}$. By Lemma 7.1
lyche-numerical-linear-algebra_F02416	178	1.000	`\boldsymbol{A}^{\prime}`	$\boldsymbol{A}^{\prime}$	Thus, if ϵ is small then $\boldsymbol{A}$ is near a matrix $\boldsymbol{A}'$ of rank r. This can be used to determine
lyche-numerical-linear-algebra_F02417	178	1.000	`\sqrt{\sigma_{r+1}^{2}+\cdots+\sigma_{n}^{2}}`	$\sqrt{\sigma_{r+1}^{2}+\cdots+\sigma_{n}^{2}}$	rank numerically. We choose an r such that $\sqrt{\sigma_{r+1}^2+\cdots+\sigma_n^2}$ is "small". Then we
lyche-numerical-linear-algebra_F02418	178	0.972	`\operatorname{rank}(\boldsymbol{A})=r`	$\operatorname{rank}(\boldsymbol{A})=r$	postulate that $\operatorname{rank}(\boldsymbol{A})=r$ since $\boldsymbol{A}$ is close to a matrix of rank r.
lyche-numerical-linear-algebra_F02419	178	1.000	`r, \boldsymbol{A}^{\prime}`	$r, \boldsymbol{A}^{\prime}$	The following theorem shows that of all $m\times n$ matrices of rank r, $\boldsymbol{A}'$ is closest
lyche-numerical-linear-algebra_F02420	178	1.000	`\sigma_{1} \geq \cdots \geq \sigma_{n} \geq 0`	$\sigma_{1} \geq \cdots \geq \sigma_{n} \geq 0$	*values* $\sigma_1\geq\cdots\geq\sigma_n\geq 0$. *For any* $r\leq\operatorname{rank}(\boldsymbol{A})$ *we have*
lyche-numerical-linear-algebra_F02421	178	1.000	`r \leq \operatorname{rank}(\boldsymbol{A})`	$r \leq \operatorname{rank}(\boldsymbol{A})$	*values* $\sigma_1\geq\cdots\geq\sigma_n\geq 0$. *For any* $r\leq\operatorname{rank}(\boldsymbol{A})$ *we have*
lyche-numerical-linear-algebra_F02422	178	1.000	`\boldsymbol{A}=\left[\begin{array}{l}3 \\ 4\end{array} \right]`	$\boldsymbol{A}=\left[\begin{array}{l}3 \\ 4\end{array} \right]$	**a)** $\boldsymbol{A}=\begin{bmatrix}3\\4\end{bmatrix}$.
lyche-numerical-linear-algebra_F02423	178	1.000	`\boldsymbol{A}=\left[\begin{array}{ll}1 & 1 \\ 2 & 2 \\ 2 & 2\end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{ll}1 & 1 \\ 2 & 2 \\ 2 & 2\end{array}\right]$	**b)** $\boldsymbol{A}=\begin{bmatrix}1&1\\2&2\\2&2\end{bmatrix}$.
lyche-numerical-linear-algebra_F02424	179	1.000	`\boldsymbol{A}=\boldsymbol{e}_{1}`	$\boldsymbol{A}=\boldsymbol{e}_{1}$	**a)** $\boldsymbol{A}=\boldsymbol{e}_1$ the first unit vector in $\mathbb{R}^m$.
lyche-numerical-linear-algebra_F02425	179	1.000	`\boldsymbol{A}=\boldsymbol{e}_{n}^{T}`	$\boldsymbol{A}=\boldsymbol{e}_{n}^{T}$	**b)** $\boldsymbol{A}=\boldsymbol{e}_n^T$ the last unit vector in $\mathbb{R}^n$.
lyche-numerical-linear-algebra_F02426	179	1.000	`\boldsymbol{A}=\left[\begin{array}{rr}-1 & 0 \\ 0 & 3 \end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{rr}-1 & 0 \\ 0 & 3 \end{array}\right]$	**c)** $\boldsymbol{A}=\left[\begin{smallmatrix}-1&0\\0&3\end{smallmatrix}\right]$.
lyche-numerical-linear-algebra_F02427	179	1.000	`\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}`	$\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}$	$\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}$ is a singular value decomposition of $\boldsymbol{A}\in\mathbb{C}^{m\times n}$ then

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02428	179	1.000	`\boldsymbol{A}^{*} \boldsymbol{A}=\boldsymbol{V} \operatorname{diag}\left(\sigma_{1}^{2}, \ldots, \sigma_{n}^{2}\right) \boldsymbol{V}^{*}`	$\boldsymbol{A}^{*} \boldsymbol{A}=\boldsymbol{V} \operatorname{diag}\left(\sigma_{1}^{2}, \ldots, \sigma_{n}^{2}\right) \boldsymbol{V}^{*}$	**a)** $\boldsymbol{A}^*\boldsymbol{A} = \boldsymbol{V}\operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)\boldsymbol{V}^*$ is a spectral decomposition of $\boldsymbol{A}^*\boldsymbol{A}$.
lyche-numerical-linear-algebra F02429	179	1.000	`\boldsymbol{A} \boldsymbol{A}^{*}=\boldsymbol{U} \operatorname{diag}\left(\sigma_{1}^{2}, \ldots, \sigma_{m}^{2}\right) \boldsymbol{U}^{*}`	$\boldsymbol{A} \boldsymbol{A}^{*}=\boldsymbol{U} \operatorname{diag}\left(\sigma_{1}^{2}, \ldots, \sigma_{m}^{2}\right) \boldsymbol{U}^{*}$	**b)** $\boldsymbol{A}\boldsymbol{A}^* = \boldsymbol{U}\operatorname{diag}(\sigma_1^2, \ldots, \sigma_m^2)\boldsymbol{U}^*$ is a spectral decomposition of $\boldsymbol{A}\boldsymbol{A}^*$.
lyche-numerical-linear-algebra F02430	179	0.803	`\left(\sigma_{1}, \ldots, \sigma_{n}\right)`	$\left(\sigma_{1}, \ldots, \sigma_{n}\right)$	"If $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ has singular values $(\sigma_1, \ldots, \sigma_n)$ then $\boldsymbol{A}^2$ has singular values
lyche-numerical-linear-algebra F02431	179	0.803	`\boldsymbol{A}^{2}`	$\boldsymbol{A}^{2}$	"If $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ has singular values $(\sigma_1, \ldots, \sigma_n)$ then $\boldsymbol{A}^2$ has singular values
lyche-numerical-linear-algebra F02432	179	1.000	`\left(\sigma_{1}^{2}, \ldots, \sigma_{n}^{2}\right)`	$\left(\sigma_{1}^{2}, \ldots, \sigma_{n}^{2}\right)$	$(\sigma_1^2, \ldots, \sigma_n^2)$". Find a class of matrices for which the statement is true. Show that
lyche-numerical-linear-algebra F02433	179	0.884	`\boldsymbol{A}=\frac{1}{25}\left[\begin{array}{ll}11 & 48 \\ 48 & 39\end{array}\right]`	$\boldsymbol{A}=\frac{1}{25}\left[\begin{array}{ll}11 & 48 \\ 48 & 39\end{array}\right]$	$\boldsymbol{A} = \frac{1}{25}\begin{bmatrix}11 & 48\\ 48 & 39\end{bmatrix}$. Also, using possibly a computer, find its spectral decomposition
lyche-numerical-linear-algebra F02434	179	0.998	`\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^{T}`	$\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^{T}$	$\boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^T$. The matrix $\boldsymbol{A}$ is normal, but the spectral decomposition is not an SVD.
lyche-numerical-linear-algebra F02435	179	1.000	`\boldsymbol{A}=\frac{1}{5}\left[\begin{array}{cc}3 & -4 \\ 4 & 3\end{array}\right]\left[\begin{array}{ll}3 & 0 \\ 0 & 1\end{array}\right] \frac{1}{5}\left[ \begin{array}{cc}3 & 4 \\ 4 & -3\end{array}\right]`	$\boldsymbol{A}=\frac{1}{5}\left[\begin{array}{cc}3 & -4 \\ 4 & 3\end{array}\right]\left[\begin{array}{ll}3 & 0 \\ 0 & 1\end{array}\right] \frac{1}{5}\left[\begin{array}{cc}3 & 4 \\ 4 & -3\end{array}\right]$	[1]Answer: $\boldsymbol{A} = \frac{1}{5}\begin{bmatrix}3 & -4\\ 4 & 3\end{bmatrix}\begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}\frac{1}{5}\begin{bmatrix}3 & 4\\ 4 & -3\end{bmatrix}$.
lyche-numerical-linear-algebra F02436	180	1.000	`\operatorname{rank}(\boldsymbol{A})=\operatorname{rank} \left(\boldsymbol{A}^{*}\right)`	$\operatorname{rank}(\boldsymbol{A})=\operatorname{rank} \left(\boldsymbol{A}^{*}\right)$	**a)** $\operatorname{rank}(\boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A}^*)$.
lyche-numerical-linear-algebra F02437	180	1.000	`\operatorname{rank}(\boldsymbol{A})+\operatorname{null}( \boldsymbol{A})=n`	$\operatorname{rank}(\boldsymbol{A})+\operatorname{null}(\boldsymbol{A})=n$	**b)** $\operatorname{rank}(\boldsymbol{A}) + \operatorname{null}(\boldsymbol{A}) = n$,
lyche-numerical-linear-algebra F02438	180	1.000	`\operatorname{rank}(\boldsymbol{A})+\operatorname{null} \left(\boldsymbol{A}^{*}\right)=m`	$\operatorname{rank}(\boldsymbol{A})+\operatorname{null} \left(\boldsymbol{A}^{*}\right)=m$	**c)** $\operatorname{rank}(\boldsymbol{A}) + \operatorname{null}(\boldsymbol{A}^*) = m$,
lyche-numerical-linear-algebra F02439	180	1.000	`\operatorname{null}(\boldsymbol{A})`	$\operatorname{null}(\boldsymbol{A})$	where $\operatorname{null}(\boldsymbol{A})$ is defined as the dimension of $\mathcal{N}(\boldsymbol{A})$.
lyche-numerical-linear-algebra F02440	180	1.000	`\operatorname{rank} \boldsymbol{A}=\operatorname{rank} \left(\boldsymbol{A}^{*} \boldsymbol{A}\right)= \operatorname{rank}\left(\boldsymbol{A} \boldsymbol{A}^{*}\right)`	$\operatorname{rank} \boldsymbol{A}=\operatorname{rank} \left(\boldsymbol{A}^{*} \boldsymbol{A}\right)= \operatorname{rank}\left(\boldsymbol{A} \boldsymbol{A}^{*}\right)$	**a)** $\operatorname{rank}\boldsymbol{A} = \operatorname{rank}(\boldsymbol{A}^*\boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A}\boldsymbol{A}^*)$,
lyche-numerical-linear-algebra F02441	180	0.997	`\operatorname{null}\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)=\operatorname{null} \boldsymbol{A}`	$\operatorname{null}\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)=\operatorname{null} \boldsymbol{A}$	**b)** $\operatorname{null}(\boldsymbol{A}^*\boldsymbol{A}) = \operatorname{null}\boldsymbol{A}$, and $\operatorname{null}(\boldsymbol{A}\boldsymbol{A}^*) = \operatorname{null}(\boldsymbol{A}^*)$.
lyche-numerical-linear-algebra F02442	180	0.997	`\operatorname{null}\left(\boldsymbol{A} \boldsymbol{A}^{*}\right)=\operatorname{null}\left( \boldsymbol{A}^{*}\right)`	$\operatorname{null}\left(\boldsymbol{A} \boldsymbol{A}^{*}\right)=\operatorname{null}\left(\boldsymbol{A}^{*}\right)$	**b)** $\operatorname{null}(\boldsymbol{A}^*\boldsymbol{A}) = \operatorname{null}\boldsymbol{A}$, and $\operatorname{null}(\boldsymbol{A}\boldsymbol{A}^*) = \operatorname{null}(\boldsymbol{A}^*)$.
lyche-numerical-linear-algebra F02443	180	1.000	`\mathcal{R}(\boldsymbol{B})`	$\mathcal{R}(\boldsymbol{B})$	Give orthonormal bases for $\mathcal{R}(\boldsymbol{B})$ and $\mathcal{N}(\boldsymbol{B})$.
lyche-numerical-linear-algebra F02444	180	1.000	`\mathcal{N}(\boldsymbol{B})`	$\mathcal{N}(\boldsymbol{B})$	Give orthonormal bases for $\mathcal{R}(\boldsymbol{B})$ and $\mathcal{N}(\boldsymbol{B})$.
lyche-numerical-linear-algebra F02445	180	1.000	`\mathcal{R}\left(\boldsymbol{A}^{*} \boldsymbol{A} \right)=`	$\mathcal{R}\left(\boldsymbol{A}^{*} \boldsymbol{A} \right)=$	**Exercise 7.13 (Some Spanning Sets)** Show for any $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ that $\mathcal{R}(\boldsymbol{A}^*\boldsymbol{A}) =$
lyche-numerical-linear-algebra F02446	180	1.000	`\mathcal{R}\left(\boldsymbol{V}_{1}\right)=\mathcal{R} \left(\boldsymbol{A}^{*}\right)`	$\mathcal{R}\left(\boldsymbol{V}_{1}\right)=\mathcal{R} \left(\boldsymbol{A}^{*}\right)$	$\mathcal{R}(\boldsymbol{V}_1) = \mathcal{R}(\boldsymbol{A}^*)$
lyche-numerical-linear-algebra F02447	180	1.000	`\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{m} \in \mathbb{C}^{m}`	$\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{m} \in \mathbb{C}^{m}$	$\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m \in \mathbb{C}^m$, and right singular vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{C}^n$. Show that the matrix
lyche-numerical-linear-algebra F02448	180	1.000	`\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n} \in \mathbb{C}^{n}`	$\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n} \in \mathbb{C}^{n}$	$\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m \in \mathbb{C}^m$, and right singular vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{C}^n$. Show that the matrix
lyche-numerical-linear-algebra F02449	181	1.000	`\boldsymbol{B}:=`	$\boldsymbol{B}:=$	**c)** Use the singular value decomposition of $\boldsymbol{A}$ to give a suitable definition of $\boldsymbol{B} :=$
lyche-numerical-linear-algebra F02450	181	0.981	`\sqrt{\boldsymbol{A}^{T} \boldsymbol{A}}`	$\sqrt{\boldsymbol{A}^{T} \boldsymbol{A}}$	$\sqrt{\boldsymbol{A}^T\boldsymbol{A}}$ so that $\boldsymbol{P} = \boldsymbol{B}$.

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-al F02451</td><td>181</td><td>0.981</td><td>\boldsymbol{P}=\boldsymbol{B}</td><td>$\boldsymbol{P}=\boldsymbol{B}$</td><td>$\sqrt{A^T A}$ so that $\boldsymbol{P}=\boldsymbol{B}$.</td></tr>
<tr><td>lyche-numerical-linear-al F02452</td><td>181</td><td>1.000</td><td>\boldsymbol{X}_{k}=\boldsymbol{U} \boldsymbol{\Sigma}_{k} \boldsymbol{V}^{T}</td><td>$\boldsymbol{X}_{k}=\boldsymbol{U} \boldsymbol{\Sigma}_{k} \boldsymbol{V}^{T}$</td><td>d) Show that the iteration (7.14) is well defined by showing that $\boldsymbol{X}_k = \boldsymbol{U}\boldsymbol{\Sigma}_k\boldsymbol{V}^T$,</td></tr>
<tr><td>lyche-numerical-linear-al F02453</td><td>181</td><td>0.985</td><td>\Sigma_{k}</td><td>Σ_{k}</td><td>for a diagonal matrix Σ_k with positive diagonal elements, $k = 0, 1, 2, \ldots$.</td></tr>
<tr><td>lyche-numerical-linear-al F02454</td><td>181</td><td>0.985</td><td>k=0,1,2, \ldots</td><td>$k=0,1,2, \ldots$</td><td>for a diagonal matrix Σ_k with positive diagonal elements, $k = 0, 1, 2, \ldots$.</td></tr>
<tr><td>lyche-numerical-linear-al F02455</td><td>181</td><td>0.749</td><td>[\mathrm{Q}, \mathrm{P}, \mathrm{k}]=</td><td>$[\mathrm{Q}, \mathrm{P}, \mathrm{k}]=$</td><td>function [Q,P,k] = polardecomp(A,tol,K) to carry out the iter-</td></tr>
<tr><td>lyche-numerical-linear-al F02456</td><td>181</td><td>0.749</td><td>(\mathrm{A}, \mathrm{tol}, \mathrm{K})</td><td>$(\mathrm{A}, \mathrm{tol}, \mathrm{K})$</td><td>function [Q,P,k] = polardecomp(A,tol,K) to carry out the iter-</td></tr>
<tr><td>lyche-numerical-linear-al F02457</td><td>181</td><td>1.000</td><td>\boldsymbol{P}=\boldsymbol{Q}^{T} \boldsymbol{A}</td><td>$\boldsymbol{P}=\boldsymbol{Q}^{T} \boldsymbol{A}$</td><td>ation in (7.14). The output is approximations $\boldsymbol{Q}$ and $\boldsymbol{P} = \boldsymbol{Q}^T\boldsymbol{A}$ to the polar</td></tr>
<tr><td>lyche-numerical-linear-al F02458</td><td>181</td><td>0.977</td><td>\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{P}</td><td>$\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{P}$</td><td>decomposition $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{P}$ of $\boldsymbol{A}$ and the number of iterations k such that</td></tr>
<tr><td>lyche-numerical-linear-al F02459</td><td>181</td><td>0.919</td><td>\left\|\boldsymbol{X}_{k+1}-\boldsymbol{X}_{k}\right \|_{F}<</td><td>$\left\|\boldsymbol{X}_{k+1}-\boldsymbol{X}_{k}\right\|_{F}<$</td><td>$\|\boldsymbol{X}_{k+1} - \boldsymbol{X}_k\|_F < tol * \|\boldsymbol{X}_{k+1}\|_F$. Set $k = K + 1$ if convergence is not achieved</td></tr>
<tr><td>lyche-numerical-linear-al F02460</td><td>181</td><td>0.919</td><td>*\left\|\boldsymbol{X}_{k+1}\right\|_{F}</td><td>$*\left\|\boldsymbol{X}_{k+1}\right\|_{F}$</td><td>$\|\boldsymbol{X}_{k+1} - \boldsymbol{X}_k\|_F < tol * \|\boldsymbol{X}_{k+1}\|_F$. Set $k = K + 1$ if convergence is not achieved</td></tr>
<tr><td>lyche-numerical-linear-al F02461</td><td>181</td><td>0.919</td><td>k=K+1</td><td>$k=K+1$</td><td>$\|\boldsymbol{X}_{k+1} - \boldsymbol{X}_k\|_F < tol * \|\boldsymbol{X}_{k+1}\|_F$. Set $k = K + 1$ if convergence is not achieved</td></tr>
<tr><td>lyche-numerical-linear-al F02462</td><td>181</td><td>1.000</td><td>\|\boldsymbol{A}\|_{1}</td><td>$\|\boldsymbol{A}\|_{1}$</td><td>Compute $\|\boldsymbol{A}\|_1$ and $\|\boldsymbol{A}\|_\infty$.</td></tr>
<tr><td>lyche-numerical-linear-al F02463</td><td>181</td><td>1.000</td><td>\|\boldsymbol{A}\|_{\infty}</td><td>$\|\boldsymbol{A}\|_{\infty}$</td><td>Compute $\|\boldsymbol{A}\|_1$ and $\|\boldsymbol{A}\|_\infty$.</td></tr>
<tr><td>lyche-numerical-linear-al F02464</td><td>181</td><td>0.972</td><td>\left(\boldsymbol{B}^{T}\right)</td><td>$\left(\boldsymbol{B}^{T}\right)$</td><td>Find the spaces $\operatorname{span}(\boldsymbol{B}^T)$ and $\ker(\boldsymbol{B})$.</td></tr>
<tr><td>lyche-numerical-linear-al F02465</td><td>181</td><td>0.972</td><td>\operatorname{ker}(\boldsymbol{B})</td><td>$\operatorname{ker}(\boldsymbol{B})$</td><td>Find the spaces $\operatorname{span}(\boldsymbol{B}^T)$ and $\ker(\boldsymbol{B})$.</td></tr>
<tr><td>lyche-numerical-linear-al F02466</td><td>182</td><td>1.000</td><td>\boldsymbol{x} \in \mathbb{R}^{3}</td><td>$\boldsymbol{x} \in \mathbb{R}^{3}$</td><td>Find the solution $\boldsymbol{x} \in \mathbb{R}^3$ with $\|\boldsymbol{x}\|_2$ as small as possible.</td></tr>
<tr><td>lyche-numerical-linear-al F02467</td><td>182</td><td>1.000</td><td>\boldsymbol{b} \in \mathbb{R}^{m}</td><td>$\boldsymbol{b} \in \mathbb{R}^{m}$</td><td>d) Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a matrix with linearly independent columns, and $\boldsymbol{b} \in \mathbb{R}^m$</td></tr>
<tr><td>lyche-numerical-linear-al F02468</td><td>182</td><td>1.000</td><td>\boldsymbol{A}^{T} \boldsymbol{A} \boldsymbol{x}= \boldsymbol{A}^{T} \boldsymbol{b}</td><td>$\boldsymbol{A}^{T} \boldsymbol{A} \boldsymbol{x}= \boldsymbol{A}^{T} \boldsymbol{b}$</td><td>the normal equations $\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}^T\boldsymbol{b}$. Will the method converge? Justify your</td></tr>
<tr><td>lyche-numerical-linear-al F02469</td><td>182</td><td>1.000</td><td>\mathcal{R}(\boldsymbol{A}), \mathcal{R}\left(\boldsymbol{A}^{T}\right), \mathcal{N}(\boldsymbol{A})</td><td>$\mathcal{R}(\boldsymbol{A}), \mathcal{R}\left(\boldsymbol{A}^{T}\right), \mathcal{N}(\boldsymbol{A})$</td><td>a) Give orthonormal bases for $\mathcal{R}(\boldsymbol{A})$, $\mathcal{R}(\boldsymbol{A}^T)$, $\mathcal{N}(\boldsymbol{A})$ and $\mathcal{N}(\boldsymbol{A}^T)$.</td></tr>
<tr><td>lyche-numerical-linear-al F02470</td><td>182</td><td>1.000</td><td>\mathcal{N}\left(\boldsymbol{A}^{T}\right)</td><td>$\mathcal{N}\left(\boldsymbol{A}^{T}\right)$</td><td>a) Give orthonormal bases for $\mathcal{R}(\boldsymbol{A})$, $\mathcal{R}(\boldsymbol{A}^T)$, $\mathcal{N}(\boldsymbol{A})$ and $\mathcal{N}(\boldsymbol{A}^T)$.</td></tr>
<tr><td>lyche-numerical-linear-al F02471</td><td>182</td><td>0.999</td><td>\boldsymbol{B} \in \mathbb{R}^{4,3}</td><td>$\boldsymbol{B} \in \mathbb{R}^{4,3}$</td><td>b) Explain why for all matrices $\boldsymbol{B} \in \mathbb{R}^{4,3}$ of rank one we have $\|\boldsymbol{A} - \boldsymbol{B}\|_F \geq 6$.</td></tr>
<tr><td>lyche-numerical-linear-al F02472</td><td>182</td><td>0.999</td><td>\|\boldsymbol{A}-\boldsymbol{B}\|_{F} \geq 6</td><td>$\|\boldsymbol{A}-\boldsymbol{B}\|_{F} \geq 6$</td><td>b) Explain why for all matrices $\boldsymbol{B} \in \mathbb{R}^{4,3}$ of rank one we have $\|\boldsymbol{A} - \boldsymbol{B}\|_F \geq 6$.</td></tr>
<tr><td>lyche-numerical-linear-al F02473</td><td>182</td><td>1.000</td><td>\left\|\boldsymbol{A}-\boldsymbol{A}_{1}\right\|_{F}=6</td><td>$\left\|\boldsymbol{A}-\boldsymbol{A}_{1}\right\|_{F}=6$</td><td>c) Give a matrix $\boldsymbol{A}_1$ of rank one such that $\|\boldsymbol{A} - \boldsymbol{A}_1\|_F = 6$.</td></tr>
<tr><td>lyche-numerical-linear-al F02474</td><td>182</td><td>0.980</td><td>(n, 1)</td><td>$(n, 1)$</td><td>diagonal equal to -1. Let $\boldsymbol{B}$ be the matrix obtained from $\boldsymbol{A}$ by changing the $(n, 1)$</td></tr>
<tr><td>lyche-numerical-linear-al F02475</td><td>182</td><td>1.000</td><td>-2^{2-n}</td><td>-2^{2-n}</td><td>element from zero to -2^{2-n}.</td></tr>
<tr><td>lyche-numerical-linear-al F02476</td><td>182</td><td>1.000</td><td>\boldsymbol{x}:=\left[2^{n-2}, 2^{n-3}, \ldots, 2^{0}, 1 \right]^{T}</td><td>$\boldsymbol{x}:=\left[2^{n-2}, 2^{n-3}, \ldots, 2^{0}, 1 \right]^{T}$</td><td>a) Show that $\boldsymbol{B}\boldsymbol{x} = \boldsymbol{0}$, where $\boldsymbol{x} := [2^{n-2}, 2^{n-3}, \ldots, 2^0, 1]^T$. Conclude that $\boldsymbol{B}$ is</td></tr>
<tr><td>lyche-numerical-linear-al F02477</td><td>182</td><td>1.000</td><td>\operatorname{det}(\boldsymbol{A})=1</td><td>$\operatorname{det}(\boldsymbol{A})=1$</td><td>singular, $\det(\boldsymbol{A}) = 1$, and $\|\boldsymbol{A} - \boldsymbol{B}\|_F = 2^{2-n}$. Thus even if $\det(\boldsymbol{A})$ is not small</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra-F02478</td><td>182</td><td>1.000</td><td><code>\|\boldsymbol{A}-\boldsymbol{B}\|_{F}=2^{2-n}</code></td><td>$\|\boldsymbol{A}-\boldsymbol{B}\|_{F}=2^{2-n}$</td><td>singular, $\det(\boldsymbol{A})=1$, and $\|\boldsymbol{A}-\boldsymbol{B}\|_F=2^{2-n}$. Thus even if $\det(\boldsymbol{A})$ is not small</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02479</td><td>182</td><td>1.000</td><td><code>\boldsymbol{A}-\boldsymbol{B}</code></td><td>$\boldsymbol{A}-\boldsymbol{B}$</td><td>the Frobenius norm of $\boldsymbol{A}-\boldsymbol{B}$ is small for large n, and the matrix $\boldsymbol{A}$ is very close</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02480</td><td>182</td><td>1.000</td><td><code>\sigma_{n}</code></td><td>σ_{n}</td><td>b) Use Theorem 7.6 to show that the smallest singular vale σ_n of $\boldsymbol{A}$ is bounded</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02481</td><td>182</td><td>1.000</td><td><code>2^{2-n}</code></td><td>2^{2-n}</td><td>above by 2^{2-n}.</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02482</td><td>183</td><td>1.000</td><td><code>\|\boldsymbol{A}\|_{1},\|\boldsymbol{A}\|_{\infty}</code></td><td>$\|\boldsymbol{A}\|_{1},\|\boldsymbol{A}\|_{\infty}$</td><td>Compute $\|\boldsymbol{A}\|_1$, $\|\boldsymbol{A}\|_\infty$ and $\|\boldsymbol{A}\|_F$.</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02483</td><td>183</td><td>1.000</td><td><code>\|\boldsymbol{A}\|_{F}</code></td><td>$\|\boldsymbol{A}\|_{F}$</td><td>Compute $\|\boldsymbol{A}\|_1$, $\|\boldsymbol{A}\|_\infty$ and $\|\boldsymbol{A}\|_F$.</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02484</td><td>183</td><td>1.000</td><td><code>\boldsymbol{T}=\boldsymbol{L} \boldsymbol{L}^{T}</code></td><td>$\boldsymbol{T}=\boldsymbol{L}\boldsymbol{L}^{T}$</td><td>$\boldsymbol{T}=\boldsymbol{L}\boldsymbol{L}^T$ of $\boldsymbol{T}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02485</td><td>183</td><td>1.000</td><td><code>\boldsymbol{A}^{\prime}=\sum_{i=1}^{r} \sigma_{i} \boldsymbol{u}_{i} \boldsymbol{v}_{i}^{*}</code></td><td>$\boldsymbol{A}^{\prime}=\sum_{i=1}^{r}\sigma_{i}\boldsymbol{u}_{i}\boldsymbol{v}_{i}^{*}$</td><td>$m\ge n$, and let $\boldsymbol{A}'=\sum_{i=1}^r\sigma_i\boldsymbol{u}_i\boldsymbol{v}_i^*$, where $1\le r\le n$, σ_i are the singular values</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02486</td><td>183</td><td>1.000</td><td><code>1 \leq r \leq n, \sigma_{i}</code></td><td>$1\leq r\leq n,\sigma_{i}$</td><td>$m\ge n$, and let $\boldsymbol{A}'=\sum_{i=1}^r\sigma_i\boldsymbol{u}_i\boldsymbol{v}_i^*$, where $1\le r\le n$, σ_i are the singular values</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02487</td><td>183</td><td>1.000</td><td><code>\boldsymbol{u}_{i}, \boldsymbol{v}_{i}</code></td><td>$\boldsymbol{u}_{i},\boldsymbol{v}_{i}$</td><td>of $\boldsymbol{A}$, and where $\boldsymbol{u}_i$, $\boldsymbol{v}_i$ are the columns of $\boldsymbol{U}$ and $\boldsymbol{V}$. Prove that</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02488</td><td>185</td><td>1.000</td><td><code>\|\cdot\|: \mathcal{V} \rightarrow \mathbb{R}</code></td><td>$\|\cdot\|:\mathcal{V}\rightarrow\mathbb{R}$</td><td>space $\mathcal{V}$ is a function $\|\cdot\|:\mathcal{V}\rightarrow\mathbb{R}$ that satisfies for all $\boldsymbol{x}$, $\boldsymbol{y}$ in $\mathcal{V}$ and all a in $\mathbb{R}$</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02489</td><td>185</td><td>0.728</td><td><code>(\mathcal{V}, \mathbb{R},\|\cdot\|)</code></td><td>$(\mathcal{V},\mathbb{R},\|\cdot\|)$</td><td>The triple $(\mathcal{V},\mathbb{R},\|\cdot\|)$ (resp. $(\mathcal{V},\mathbb{C},\|\cdot\|)$) is called a normed vector space and the</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02490</td><td>185</td><td>0.728</td><td><code>(\mathcal{V}, \mathbb{C},\|\cdot\|)</code></td><td>$(\mathcal{V},\mathbb{C},\|\cdot\|)$</td><td>The triple $(\mathcal{V},\mathbb{R},\|\cdot\|)$ (resp. $(\mathcal{V},\mathbb{C},\|\cdot\|)$) is called a normed vector space and the</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02491</td><td>185</td><td>1.000</td><td><code>\mathbb{R}^{n}, \mathbb{C}^{n}</code></td><td>$\mathbb{R}^{n},\mathbb{C}^{n}$</td><td>In this book the vector space will be one of $\mathbb{R}^n$, $\mathbb{C}^n$ or one of the matrix spaces</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02492</td><td>185</td><td>1.000</td><td><code>\mathcal{V}=\mathbb{C}^{n}</code></td><td>$\mathcal{V}=\mathbb{C}^{n}$</td><td>book we will use the following family of vector norms on $\mathcal{V}=\mathbb{C}^n$ and $\mathcal{V}=\mathbb{R}^n$.</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02493</td><td>185</td><td>1.000</td><td><code>\mathcal{V}=\mathbb{R}^{n}</code></td><td>$\mathcal{V}=\mathbb{R}^{n}$</td><td>book we will use the following family of vector norms on $\mathcal{V}=\mathbb{C}^n$ and $\mathcal{V}=\mathbb{R}^n$.</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02494</td><td>186</td><td>0.987</td><td><code>p \geq 1</code></td><td>$p\geq 1$</td><td>Definition 8.2 (Vector p-Norms) We define for $p\ge 1$ and $\boldsymbol{x}\in\mathbb{R}^n$ or $\boldsymbol{x}\in\mathbb{C}^n$ the</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02495</td><td>186</td><td>1.000</td><td><code>p=1,2, \infty</code></td><td>$p=1,2,\infty$</td><td>The most important cases are $p=1,2,\infty$:</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02496</td><td>186</td><td>1.000</td><td><code>\|\boldsymbol{x}\|_{1}:=\sum_{j=1}^{n}\left|x_{j}\right|</code></td><td>$\|\boldsymbol{x}\|_{1}:=\sum_{j=1}^{n}\left|x_{j}\right|$</td><td>1. $\|\boldsymbol{x}\|_1:=\sum_{j=1}^n|x_j|$,</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02497</td><td>186</td><td>1.000</td><td><code>\|\boldsymbol{x}\|_{2}:=\sqrt{\sum_{j=1}^{n}\left|x_{j}\right|^{2}}</code></td><td>$\|\boldsymbol{x}\|_{2}:=\sqrt{\sum_{j=1}^{n}\left|x_{j}\right|^{2}}$</td><td>2. $\|\boldsymbol{x}\|_2:=\sqrt{\sum_{j=1}^n|x_j|^2}$,</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02498</td><td>186</td><td>1.000</td><td><code>l_{2}</code></td><td>l_{2}</td><td>(the two-norm, l_2-norm, or Euclidian norm)</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02499</td><td>186</td><td>1.000</td><td><code>\|\boldsymbol{x}\|_{\infty}:=\max _{1 \leq j \leq n} \left|x_{j}\right|</code></td><td>$\|\boldsymbol{x}\|_{\infty}:=\max_{1\leq j\leq n}\left|x_{j}\right|$</td><td>3. $\|\boldsymbol{x}\|_\infty:=\max_{1\le j\le n}|x_j|$,</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02500</td><td>186</td><td>1.000</td><td><code>l_{\infty}</code></td><td>l_{∞}</td><td>(the infinity-norm, l_∞-norm, or max norm)</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02501</td><td>186</td><td>1.000</td><td><code>1 \leq p \leq \infty</code></td><td>$1\leq p\leq\infty$</td><td>1. In Sect. 8.4, we show that the p-norms are vector norms for $1\le p\le\infty$.</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02502</td><td>186</td><td>1.000</td><td><code>\|\boldsymbol{x}+\boldsymbol{y}\|_{p} \leq\|\boldsymbol{x}\|_{p}+\|\boldsymbol{y}\|_{p}</code></td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|_{p}\leq\|\boldsymbol{x}\|_{p}+\|\boldsymbol{y}\|_{p}$</td><td>2. The triangle inequality $\|\boldsymbol{x}+\boldsymbol{y}\|_p\le\|\boldsymbol{x}\|_p+\|\boldsymbol{y}\|_p$ is called Minkowski's</td></tr>
<tr><td>lyche-numerical-linear-algebra-F02503</td><td>186</td><td>1.000</td><td><code>\frac{1}{p}+\frac{1}{q}=1</code></td><td>$\frac{1}{p}+\frac{1}{q}=1$</td><td>The relation $\frac{1}{p}+\frac{1}{q}=1$ means that if $p=1$ then $q=\infty$ and vice versa.</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_F02504</td><td>186</td><td>1.000</td><td>p=1</td><td>$p=1$</td><td>The relation $\frac{1}{p}+\frac{1}{q}=1$ means that if $p=1$ then $q=\infty$ and vice versa.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02505</td><td>186</td><td>1.000</td><td>q=\infty</td><td>$q=\infty$</td><td>The relation $\frac{1}{p}+\frac{1}{q}=1$ means that if $p=1$ then $q=\infty$ and vice versa.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02506</td><td>186</td><td>1.000</td><td>p=2</td><td>$p=2$</td><td>Theorem 5.1) for the Euclidian norm $p=2$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02507</td><td>186</td><td>1.000</td><td>\lim _{p \rightarrow \infty} n^{1 / p}=1</td><td>$\lim _{p \rightarrow \infty} n^{1 / p}=1$</td><td>Since $\lim_{p\rightarrow\infty} n^{1/p}=1$ for any fixed $n\in\mathbb{N}$, we see that (8.4) follows.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02508</td><td>187</td><td>0.987</td><td>\|\cdot\|</td><td>$\|\cdot\|$</td><td>Definition 8.3 (Equivalent Norms) We say that two norms $\|\cdot\|$ and $\|\cdot\|'$ on $\mathcal{V}$ are</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02509</td><td>187</td><td>0.987</td><td>\|\cdot\|^{\prime}</td><td>$\|\cdot\|^{\prime}$</td><td>Definition 8.3 (Equivalent Norms) We say that two norms $\|\cdot\|$ and $\|\cdot\|'$ on $\mathcal{V}$ are</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02510</td><td>187</td><td>1.000</td><td>M</td><td>M</td><td>equivalent if there are positive constants m and M such that for all vectors $\boldsymbol{x}\in\mathcal{V}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02511</td><td>187</td><td>1.000</td><td>\boldsymbol{x} \in \mathcal{V}</td><td>$\boldsymbol{x} \in \mathcal{V}$</td><td>equivalent if there are positive constants m and M such that for all vectors $\boldsymbol{x}\in\mathcal{V}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02512</td><td>187</td><td>1.000</td><td>\infty</td><td>∞</td><td>By (8.5) the p- and ∞-norms are equivalent for any $p\geq 1$. This result is</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02513</td><td>187</td><td>1.000</td><td>\|\boldsymbol{x}-\boldsymbol{y}\| \geq|\|\boldsymbol{x}\|-\|\boldsymbol{y}\||</td><td>$\|\boldsymbol{x}-\boldsymbol{y}\| \geq|\|\boldsymbol{x}\|-\|\boldsymbol{y}\||$</td><td>1. $\|\boldsymbol{x}-\boldsymbol{y}\|\geq|\,\|\boldsymbol{x}\|-\|\boldsymbol{y}\|\,|$, for all $\boldsymbol{x},\boldsymbol{y}\in\mathbb{C}^n$ (inverse triangle inequality).</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02514</td><td>187</td><td>1.000</td><td>\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n} \quad</td><td>$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n}$</td><td>1. $\|\boldsymbol{x}-\boldsymbol{y}\|\geq|\,\|\boldsymbol{x}\|-\|\boldsymbol{y}\|\,|$, for all $\boldsymbol{x},\boldsymbol{y}\in\mathbb{C}^n$ (inverse triangle inequality).</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02515</td><td>187</td><td>1.000</td><td>\mathcal{V} \rightarrow \mathbb{R}</td><td>$\mathcal{V} \rightarrow \mathbb{R}$</td><td>2. The vector norm is a continuous function $\mathcal{V}\rightarrow\mathbb{R}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02516</td><td>187</td><td>1.000</td><td>\|\boldsymbol{x}\|=\|\boldsymbol{x}-\boldsymbol{y}+\boldsymbol{y}\| \leq\|\boldsymbol{x}-\boldsymbol{y}\|+\|\boldsymbol{y}\|</td><td>$\|\boldsymbol{x}\|=\|\boldsymbol{x}-\boldsymbol{y}+\boldsymbol{y}\| \leq\|\boldsymbol{x}-\boldsymbol{y}\|+\|\boldsymbol{y}\|$</td><td>1. Since $\|\boldsymbol{x}\|=\|\boldsymbol{x}-\boldsymbol{y}+\boldsymbol{y}\|\leq\|\boldsymbol{x}-\boldsymbol{y}\|+\|\boldsymbol{y}\|$ we obtain $\|\boldsymbol{x}-\boldsymbol{y}\|\geq\|\boldsymbol{x}\|-\|\boldsymbol{y}\|$. By</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02517</td><td>187</td><td>1.000</td><td>\|\boldsymbol{x}-\boldsymbol{y}\| \geq\|\boldsymbol{x}\|-\|\boldsymbol{y}\|</td><td>$\|\boldsymbol{x}-\boldsymbol{y}\| \geq\|\boldsymbol{x}\|-\|\boldsymbol{y}\|$</td><td>1. Since $\|\boldsymbol{x}\|=\|\boldsymbol{x}-\boldsymbol{y}+\boldsymbol{y}\|\leq\|\boldsymbol{x}-\boldsymbol{y}\|+\|\boldsymbol{y}\|$ we obtain $\|\boldsymbol{x}-\boldsymbol{y}\|\geq\|\boldsymbol{x}\|-\|\boldsymbol{y}\|$. By</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02518</td><td>187</td><td>1.000</td><td>\|\boldsymbol{x}-\boldsymbol{y}\|=\|\boldsymbol{y}-\boldsymbol{x}\| \geq\|\boldsymbol{y}\|-\|\boldsymbol{x}\|</td><td>$\|\boldsymbol{x}-\boldsymbol{y}\|=\|\boldsymbol{y}-\boldsymbol{x}\| \geq\|\boldsymbol{y}\|-\|\boldsymbol{x}\|$</td><td>symmetry $\|\boldsymbol{x}-\boldsymbol{y}\|=\|\boldsymbol{y}-\boldsymbol{x}\|\geq\|\boldsymbol{y}\|-\|\boldsymbol{x}\|$ and we obtain the inverse triangle</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02519</td><td>187</td><td>1.000</td><td>f: \mathcal{S} \rightarrow \mathbb{R}</td><td>$f: \mathcal{S} \rightarrow \mathbb{R}$</td><td>The set $\mathcal{S}$ is a closed and bounded set and the function $f:\mathcal{S}\rightarrow\mathbb{R}$ given by</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02520</td><td>187</td><td>1.000</td><td>f(\boldsymbol{y})=\|\boldsymbol{y}\|</td><td>$f(\boldsymbol{y})=\|\boldsymbol{y}\|$</td><td>$f(\boldsymbol{y})=\|\boldsymbol{y}\|$ is continuous by what we just showed. Therefore f attains its</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02521</td><td>187</td><td>1.000</td><td>\boldsymbol{y}:=\boldsymbol{x} /\|\boldsymbol{x}\|^{\prime} \in \mathcal{S}</td><td>$\boldsymbol{y}:=\boldsymbol{x} /\|\boldsymbol{x}\|^{\prime} \in \mathcal{S}$</td><td>For any $\boldsymbol{x}\in\mathcal{V}$ we have $\boldsymbol{y}:=\boldsymbol{x}/\|\boldsymbol{x}\|'\in\mathcal{S}$, and (8.7) follows if we apply (8.8) to</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02522</td><td>188</td><td>0.997</td><td>\left(\mathbb{C}^{m \times n}, \mathbb{C}\right)</td><td>$\left(\mathbb{C}^{m \times n}, \mathbb{C}\right)$</td><td>For simplicity we consider only norms on the vector space $(\mathbb{C}^{m\times n},\mathbb{C})$. All results</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02523</td><td>188</td><td>1.000</td><td>\left(\mathbb{R}^{m \times n}, \mathbb{R}\right)</td><td>$\left(\mathbb{R}^{m \times n}, \mathbb{R}\right)$</td><td>also holds for $(\mathbb{R}^{m\times n},\mathbb{R})$. A matrix norm $\|\ \|:\mathbb{C}^{m\times n},\rightarrow\mathbb{R}$ is simply a vector norm</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02524</td><td>188</td><td>1.000</td><td>\left\|\|: \mathbb{C}^{m \times n}, \rightarrow \mathbb{R}\right.</td><td>$\left\|\|: \mathbb{C}^{m \times n}, \rightarrow \mathbb{R}\right.$</td><td>also holds for $(\mathbb{R}^{m\times n},\mathbb{R})$. A matrix norm $\|\ \|:\mathbb{C}^{m\times n},\rightarrow\mathbb{R}$ is simply a vector norm</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02525</td><td>188</td><td>1.000</td><td>m n</td><td>$m n$</td><td>is a matrix norm. Indeed, writing all elements in $\boldsymbol{A}$ in a string of length mn we see</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02526</td><td>188</td><td>1.000</td><td>\mathbb{C}^{m n}</td><td>$\mathbb{C}^{m n}$</td><td>that the Frobenius norm is the Euclidian norm on the space $\mathbb{C}^{mn}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02527</td><td>188</td><td>1.000</td><td>\|\cdot\|_{V}</td><td>$\|\cdot\|_{V}$</td><td>Any vector norm $\|\cdot\|_V$ on $\mathbb{C}^{mn}$ defines a matrix norm on $\mathbb{C}^{m\times n}$ given by $\|\boldsymbol{A}\|:=$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02528</td><td>188</td><td>1.000</td><td>\|\boldsymbol{A}\|:=</td><td>$\|\boldsymbol{A}\|:=$</td><td>Any vector norm $\|\cdot\|_V$ on $\mathbb{C}^{mn}$ defines a matrix norm on $\mathbb{C}^{m\times n}$ given by $\|\boldsymbol{A}\|:=$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02529</td><td>188</td><td>1.000</td><td>\|\operatorname{vec}(\boldsymbol{A})\|_{V}</td><td>$\|\operatorname{vec}(\boldsymbol{A})\|_{V}$</td><td>$\|\mathrm{vec}(\boldsymbol{A})\|_V$, where $\mathrm{vec}(\boldsymbol{A})\in\mathbb{C}^{mn}$ is the vector obtained by stacking the columns</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_F02530</td><td>189</td><td>1.000</td><td>\|\boldsymbol{A} \boldsymbol{B}\| \leq\|\boldsymbol{A}\| \|\boldsymbol{B}\|</td><td>$\|\boldsymbol{A} \boldsymbol{B}\| \leq\|\boldsymbol{A}\| \|\boldsymbol{B}\|$</td><td>4. $\|\boldsymbol{A}\boldsymbol{B}\| \leq \|\boldsymbol{A}\|\,\|\boldsymbol{B}\|$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02531</td><td>189</td><td>0.998</td><td>\|\|</td><td>$\|\|$</td><td>Definition 8.5 (Subordinate Matrix Norms) Suppose $m, n \in \mathbb{N}$ are given, let $\|\ \|$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02532</td><td>189</td><td>1.000</td><td>\left\|\|_{\beta}\right.</td><td>$\left\|\|_{\beta}\right.$</td><td>on $\mathbb{C}^m$ and $\|\ \|_\beta$ on $\mathbb{C}^n$ be vector norms, and let $\|\ \|$ be a matrix norm on $\mathbb{C}^{m\times n}$. We</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02533</td><td>189</td><td>1.000</td><td>\| \|</td><td>$\| \|$</td><td>on $\mathbb{C}^m$ and $\|\ \|_\beta$ on $\mathbb{C}^n$ be vector norms, and let $\|\ \|$ be a matrix norm on $\mathbb{C}^{m\times n}$. We</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02534</td><td>189</td><td>0.999</td><td>\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\| \|\boldsymbol{x}\|_{\beta}</td><td>$\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\| \|\boldsymbol{x}\|_{\beta}$</td><td>$\|\boldsymbol{A}\boldsymbol{x}\| \leq \|\boldsymbol{A}\|\,\|\boldsymbol{x}\|_\beta$ for all $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ and all $\boldsymbol{x} \in \mathbb{C}^n$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02535</td><td>189</td><td>1.000</td><td>m, n \in \mathbb{N}, \boldsymbol{A} \in \mathbb{C}^{m \times n}</td><td>$m, n \in \mathbb{N}, \boldsymbol{A} \in \mathbb{C}^{m \times n}$</td><td>Proposition 8.1 For $m, n \in \mathbb{N}$, $\boldsymbol{A} \in \mathbb{C}^{m\times n}$, all $\boldsymbol{x} \in \mathbb{C}^n$ and any consistent matrix</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02536</td><td>189</td><td>1.000</td><td>\boldsymbol{B}:=\boldsymbol{x} \in \mathbb{C}^{n \times 1}</td><td>$\boldsymbol{B}:=\boldsymbol{x} \in \mathbb{C}^{n \times 1}$</td><td>consistency implies that (8.11) holds for $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ and $\boldsymbol{B} := \boldsymbol{x} \in \mathbb{C}^{n\times 1}$. The last</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02537</td><td>190</td><td>0.969</td><td>n \in \mathbb{N} .^{1}</td><td>$n \in \mathbb{N} .^{1}$</td><td>defined on $\mathbb{C}^n$ for all $n \in \mathbb{N}$.[1]</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02538</td><td>190</td><td>1.000</td><td>\left\|\|\right.</td><td>$\left\|\|\right.$</td><td>Definition 8.6 (Operator Norm) Let $\|\ \|$ be a vector norm defined on $\mathbb{C}^n$ for all</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02539</td><td>190</td><td>1.000</td><td>\|\|: \mathcal{S} \rightarrow \mathbb{R}</td><td>$\|\|: \mathcal{S} \rightarrow \mathbb{R}$</td><td>norm $\|\ \| : \mathcal{S} \rightarrow \mathbb{R}$ is a continuous function, it follows that the function</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02540</td><td>190</td><td>0.995</td><td>f(\boldsymbol{x})=\|\boldsymbol{A} \boldsymbol{x}\|</td><td>$f(\boldsymbol{x})=\|\boldsymbol{A} \boldsymbol{x}\|$</td><td>$f : \mathcal{S} \rightarrow \mathbb{R}$ given by $f(\boldsymbol{x}) = \|\boldsymbol{A}\boldsymbol{x}\|$ is continuous. But then f attains its max and</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02541</td><td>190</td><td>1.000</td><td>\| \|_{\beta}</td><td>$\| \|_{\beta}$</td><td>[1]In the case of one vector norm $\|\ \|$ on $\mathbb{C}^m$ and another vector norm $\|\ \|_\beta$ on $\mathbb{C}^n$ we would define</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02542</td><td>190</td><td>0.587</td><td>\|\boldsymbol{A}\|:=\max _{\boldsymbol{x} \neq 0} \frac{ \|\boldsymbol{A} \boldsymbol{x}\|}{\|\boldsymbol{x}\| \beta}</td><td>$\|\boldsymbol{A}\|:=\max _{\boldsymbol{x} \neq 0} \frac{ \|\boldsymbol{A} \boldsymbol{x}\|}{\|\boldsymbol{x}\| \beta}$</td><td>$\|\boldsymbol{A}\| := \max_{\boldsymbol{x}\neq 0} \frac{\|\boldsymbol{A}\boldsymbol{x}\|}{\|\boldsymbol{x}\|_\beta}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02543</td><td>191</td><td>1.000</td><td>\|\boldsymbol{I}\|=1</td><td>$\|\boldsymbol{I}\|=1$</td><td>consistent matrix norm. Moreover, $\|\boldsymbol{I}\| = 1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02544</td><td>191</td><td>1.000</td><td>\|\boldsymbol{A}\|=0</td><td>$\|\boldsymbol{A}\|=0$</td><td>1. Nonnegativity is obvious. If $\|\boldsymbol{A}\| = 0$ then $\|\boldsymbol{A}\boldsymbol{y}\| = 0$ for each $\boldsymbol{y} \in \mathbb{C}^n$. In</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02545</td><td>191</td><td>1.000</td><td>\|\boldsymbol{A} \boldsymbol{y}\|=0</td><td>$\|\boldsymbol{A} \boldsymbol{y}\|=0$</td><td>1. Nonnegativity is obvious. If $\|\boldsymbol{A}\| = 0$ then $\|\boldsymbol{A}\boldsymbol{y}\| = 0$ for each $\boldsymbol{y} \in \mathbb{C}^n$. In</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02546</td><td>191</td><td>1.000</td><td>\boldsymbol{A}=0</td><td>$\boldsymbol{A}=0$</td><td>particular, each column $\boldsymbol{A}\boldsymbol{e}_j$ in $\boldsymbol{A}$ is zero. Hence $\boldsymbol{A} = 0$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02547</td><td>191</td><td>1.000</td><td>\|c \boldsymbol{A}\|=\max _{\boldsymbol{x}}\|c \boldsymbol{A} \boldsymbol{x}\|=\max _{\boldsymbol{x}}|c| \|\boldsymbol{A} \boldsymbol{x}\|=|c|\|\boldsymbol{A}\|</td><td>$\|c \boldsymbol{A}\|=\max _{\boldsymbol{x}}\|c \boldsymbol{A} \boldsymbol{x}\|=\max _{\boldsymbol{x}}|c| \|\boldsymbol{A} \boldsymbol{x}\|=|c|\|\boldsymbol{A}\|$</td><td>2. $\|c\boldsymbol{A}\| = \max_{\boldsymbol{x}} \|c\boldsymbol{A}\boldsymbol{x}\| = \max_{\boldsymbol{x}} |c|\,\|\boldsymbol{A}\boldsymbol{x}\| = |c|\,\|\boldsymbol{A}\|$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02548</td><td>191</td><td>1.000</td><td>\|\boldsymbol{A}+\boldsymbol{B}\|=\max _{\boldsymbol{x}} \|(\boldsymbol{A}+\boldsymbol{B}) \boldsymbol{x}\| \leq \max _{\boldsymbol{x}}\|\boldsymbol{A} \boldsymbol{x}\|+ \max _{\boldsymbol{x}}\|\boldsymbol{B} \boldsymbol{x}\|= \|\boldsymbol{A}\|+\|\boldsymbol{B}\|</td><td>$\|\boldsymbol{A}+\boldsymbol{B}\|=\max _{\boldsymbol{x}} \|(\boldsymbol{A}+\boldsymbol{B}) \boldsymbol{x}\| \leq \max _{\boldsymbol{x}}\|\boldsymbol{A} \boldsymbol{x}\|+ \max _{\boldsymbol{x}}\|\boldsymbol{B} \boldsymbol{x}\|= \|\boldsymbol{A}\|+\|\boldsymbol{B}\|$</td><td>3. $\|\boldsymbol{A}+\boldsymbol{B}\| = \max_{\boldsymbol{x}} \|(\boldsymbol{A}+\boldsymbol{B})\boldsymbol{x}\| \leq \max_{\boldsymbol{x}} \|\boldsymbol{A}\boldsymbol{x}\| + \max_{\boldsymbol{x}} \|\boldsymbol{B}\boldsymbol{x}\| = \|\boldsymbol{A}\| + \|\boldsymbol{B}\|$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02549</td><td>191</td><td>1.000</td><td>\|\boldsymbol{I}\|_{F}=\sqrt{n}</td><td>$\|\boldsymbol{I}\|_{F}=\sqrt{n}$</td><td>Since $\|\boldsymbol{I}\|_F = \sqrt{n}$, we see that the Frobenius norm is not an operator norm for</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02550</td><td>191</td><td>1.000</td><td>n>1</td><td>$n>1$</td><td>$n > 1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02551</td><td>191</td><td>0.985</td><td>\boldsymbol{p}</td><td>$\boldsymbol{p}$</td><td>8.2.3 The Operator p-Norms</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02552</td><td>191</td><td>1.000</td><td>\ell_{p}</td><td>ℓ_{p}</td><td>Recall that the p or ℓ_p vector norms (8.1) are given by</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02553</td><td>191</td><td>1.000</td><td>\left\|\|_{p}\right.</td><td>$\left\|\|_{p}\right.$</td><td>The operator norms $\|\ \|_p$ defined from these p-vector norms are used quite</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02554</td><td>192</td><td>1.000</td><td>\|\boldsymbol{A}\|_{2}</td><td>$\|\boldsymbol{A}\|_{2}$</td><td>The two-norm $\|\boldsymbol{A}\|_2$ is also called the spectral norm of $\boldsymbol{A}$.</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-al F02555</td><td>192</td><td>■ 0.998</td><td>K_{p}</td><td>K_{p}</td><td>(a) We derive a constant K_p such that $\|Ax\|_p \leq K_p$ for any $x \in \mathbb{C}^n$ with $\|x\|_p = 1$.</td></tr>
<tr><td>lyche-numerical-linear-al F02556</td><td>192</td><td>■ 0.998</td><td>\|\boldsymbol{A} \boldsymbol{x}\|_{p} \leq K_{p}</td><td>$\|\boldsymbol{A} \boldsymbol{x}\|_{p} \leq K_{p}$</td><td>(a) We derive a constant K_p such that $\|Ax\|_p \leq K_p$ for any $x \in \mathbb{C}^n$ with $\|x\|_p = 1$.</td></tr>
<tr><td>lyche-numerical-linear-al F02557</td><td>192</td><td>■ 0.998</td><td>\|\boldsymbol{x}\|_{p}=1</td><td>$\|\boldsymbol{x}\|_{p}=1$</td><td>(a) We derive a constant K_p such that $\|Ax\|_p \leq K_p$ for any $x \in \mathbb{C}^n$ with $\|x\|_p = 1$.</td></tr>
<tr><td>lyche-numerical-linear-al F02558</td><td>192</td><td>■ 1.000</td><td>\boldsymbol{y}_{e} \in \mathbb{C}^{n}</td><td>$\boldsymbol{y}_{e} \in \mathbb{C}^{n}$</td><td>(b) We give an extremal vector $y_e \in \mathbb{C}^n$ with $\|y_e\|_p = 1$ so that $\|Ay_e\|_p = K_p$.</td></tr>
<tr><td>lyche-numerical-linear-al F02559</td><td>192</td><td>■ 1.000</td><td>\left\|\boldsymbol{y}_{e}\right\|_{p}=1</td><td>$\left\|\boldsymbol{y}_{e}\right\|_{p}=1$</td><td>(b) We give an extremal vector $y_e \in \mathbb{C}^n$ with $\|y_e\|_p = 1$ so that $\|Ay_e\|_p = K_p$.</td></tr>
<tr><td>lyche-numerical-linear-al F02560</td><td>192</td><td>■ 1.000</td><td>\left\|\boldsymbol{A} \boldsymbol{y}_{e}\right \|_{p}=K_{p}</td><td>$\left\|\boldsymbol{A} \boldsymbol{y}_{e}\right\|_{p}=K_{p}$</td><td>(b) We give an extremal vector $y_e \in \mathbb{C}^n$ with $\|y_e\|_p = 1$ so that $\|Ay_e\|_p = K_p$.</td></tr>
<tr><td>lyche-numerical-linear-al F02561</td><td>192</td><td>■ 1.000</td><td>\|\boldsymbol{A}\|_{p}=\left\|\boldsymbol{A} \boldsymbol{y}_{e}\right\|_{p}=K_{p}</td><td>$\|\boldsymbol{A}\|_{p}=\left\|\boldsymbol{A} \boldsymbol{y}_{e}\right\|_{p}=K_{p}$</td><td>It then follows from (8.17) that $\|A\|_p = \|Ay_e\|_p = K_p$.</td></tr>
<tr><td>lyche-numerical-linear-al F02562</td><td>192</td><td>■ 1.000</td><td>K_{2}=</td><td>$K_{2}=$</td><td>2-norm: Let $A = U\Sigma V^*$ be a singular value decomposition of A, define $K_2 =$</td></tr>
<tr><td>lyche-numerical-linear-al F02563</td><td>192</td><td>■ 0.998</td><td>\sigma_{1}, \boldsymbol{c}:=\boldsymbol{V}^{*} \boldsymbol{x}</td><td>$\sigma_{1}, \boldsymbol{c}:=\boldsymbol{V}^{*} \boldsymbol{x}$</td><td>$\sigma_1$, $c := V^*x$, and $y_e = v_1$ the singular vector corresponding to σ_1. Then $x =$</td></tr>
<tr><td>lyche-numerical-linear-al F02564</td><td>192</td><td>■ 0.998</td><td>\boldsymbol{y}_{e}=\boldsymbol{v}_{1}</td><td>$\boldsymbol{y}_{e}=\boldsymbol{v}_{1}$</td><td>$\sigma_1$, $c := V^*x$, and $y_e = v_1$ the singular vector corresponding to σ_1. Then $x =$</td></tr>
<tr><td>lyche-numerical-linear-al F02565</td><td>192</td><td>■ 0.998</td><td>\sigma_{1}</td><td>σ_{1}</td><td>σ_1, $c := V^*x$, and $y_e = v_1$ the singular vector corresponding to σ_1. Then $x =$</td></tr>
<tr><td>lyche-numerical-linear-al F02566</td><td>192</td><td>■ 1.000</td><td>\boldsymbol{V} \boldsymbol{c},\|\boldsymbol{c}\|_{2}=\|\boldsymbol{x}\|_{2}=1</td><td>$\boldsymbol{V} \boldsymbol{c},\|\boldsymbol{c}\|_{2}=\|\boldsymbol{x}\|_{2}=1$</td><td>$Vc$, $\|c\|_2 = \|x\|_2 = 1$, and using (7.7) in (b) we find</td></tr>
<tr><td>lyche-numerical-linear-al F02567</td><td>192</td><td>■ 0.998</td><td>\|\boldsymbol{A} \boldsymbol{x}\|_{2}^{2}=\left\|\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*} \boldsymbol{x}\right\|_{2}^{2}=\|\boldsymbol{\Sigma} \boldsymbol{c}\|_{2}^{2}=\sum_{j=1}^{n} \sigma_{j}^{2} \left|c_{j}\right|^{2} \leq \sigma_{1}^{2} \sum_{j=1}^{n} \left|c_{j}\right|^{2}=\sigma_{1}^{2}</td><td>$\|\boldsymbol{A} \boldsymbol{x}\|_{2}^{2}=\left\|\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*} \boldsymbol{x}\right\|_{2}^{2}=\|\boldsymbol{\Sigma} \boldsymbol{c}\|_{2}^{2}=\sum_{j=1}^{n} \sigma_{j}^{2} \left|c_{j}\right|^{2} \leq \sigma_{1}^{2} \sum_{j=1}^{n} \left|c_{j}\right|^{2}=\sigma_{1}^{2}$</td><td>(a) $\|Ax\|_2^2 = \|U\Sigma V^*x\|_2^2 = \|\Sigma c\|_2^2 = \sum_{j=1}^n \sigma_j^2|c_j|^2 \leq \sigma_1^2 \sum_{j=1}^n |c_j|^2 = \sigma_1^2$.</td></tr>
<tr><td>lyche-numerical-linear-al F02568</td><td>192</td><td>■ 1.000</td><td>\left\|\boldsymbol{A} \boldsymbol{v}_{1}\right\|_{2}= \left\|\sigma_{1} \boldsymbol{u}_{1}\right\|_{2}= \sigma_{1}</td><td>$\left\|\boldsymbol{A} \boldsymbol{v}_{1}\right\|_{2}=\left\|\sigma_{1} \boldsymbol{u}_{1}\right\|_{2}=\sigma_{1}$</td><td>(b) $\|Av_1\|_2 = \|\sigma_1 u_1\|_2 = \sigma_1$.</td></tr>
<tr><td>lyche-numerical-linear-al F02569</td><td>192</td><td>■ 1.000</td><td>K_{1}, c</td><td>K_{1}, c</td><td>1-norm: Define K_1, c and y_e by $K_1 := \|Ae_c\|_1 = \max_{1\leq j\leq n}\|Ae_j\|_1$ and $y_e :=$</td></tr>
<tr><td>lyche-numerical-linear-al F02570</td><td>192</td><td>■ 1.000</td><td>\boldsymbol{y}_{e}</td><td>$\boldsymbol{y}_{e}$</td><td>1-norm: Define K_1, c and y_e by $K_1 := \|Ae_c\|_1 = \max_{1\leq j\leq n}\|Ae_j\|_1$ and $y_e :=$</td></tr>
<tr><td>lyche-numerical-linear-al F02571</td><td>192</td><td>■ 1.000</td><td>K_{1}:=\left\|\boldsymbol{A} \boldsymbol{e}_{c}\right \|_{1}=\max _{1 \leq j \leq n}\left\|\boldsymbol{A} \boldsymbol{e}_{j}\right\|_{1}</td><td>$K_{1}:=\left\|\boldsymbol{A} \boldsymbol{e}_{c}\right\|_{1}=\max _{1 \leq j \leq n}\left\|\boldsymbol{A} \boldsymbol{e}_{j}\right\|_{1}$</td><td>1-norm: Define K_1, c and y_e by $K_1 := \|Ae_c\|_1 = \max_{1\leq j\leq n}\|Ae_j\|_1$ and $y_e :=$</td></tr>
<tr><td>lyche-numerical-linear-al F02572</td><td>192</td><td>■ 1.000</td><td>\boldsymbol{y}_{e}:=</td><td>$\boldsymbol{y}_{e}:=$</td><td>1-norm: Define K_1, c and y_e by $K_1 := \|Ae_c\|_1 = \max_{1\leq j\leq n}\|Ae_j\|_1$ and $y_e :=$</td></tr>
<tr><td>lyche-numerical-linear-al F02573</td><td>192</td><td>■ 1.000</td><td>\boldsymbol{e}_{c}</td><td>$\boldsymbol{e}_{c}$</td><td>e_c, a unit vector. Then $\|y_e\|_1 = 1$ and we obtain</td></tr>
<tr><td>lyche-numerical-linear-al F02574</td><td>192</td><td>■ 1.000</td><td>\left\|\boldsymbol{y}_{e}\right\|_{1}=1</td><td>$\left\|\boldsymbol{y}_{e}\right\|_{1}=1$</td><td>$e_c$, a unit vector. Then $\|y_e\|_1 = 1$ and we obtain</td></tr>
<tr><td>lyche-numerical-linear-al F02575</td><td>192</td><td>■ 1.000</td><td>\left\|\boldsymbol{A} \boldsymbol{y}_{e}\right \|_{1}=K_{1}</td><td>$\left\|\boldsymbol{A} \boldsymbol{y}_{e}\right\|_{1}=K_{1}$</td><td>(b) $\|Ay_e\|_1 = K_1$.</td></tr>
<tr><td>lyche-numerical-linear-al F02576</td><td>192</td><td>■ 0.997</td><td>K_{\infty}, r</td><td>K_{∞}, r</td><td>∞-norm: Define K_∞, r and y_e by $K_\infty := \|e_r^T A\|_1 = \max_{1\leq k\leq m}\|e_k^T A\|_1$ and</td></tr>
<tr><td>lyche-numerical-linear-al F02577</td><td>192</td><td>■ 0.997</td><td>K_{\infty}:=\left\|\boldsymbol{e}_{r}^{T} \boldsymbol{A} \right\|_{1}=\max _{1 \leq k \leq m}\left\| \boldsymbol{e}_{k}^{T} \boldsymbol{A}\right\|_{1}</td><td>$K_{\infty}:=\left\|\boldsymbol{e}_{r}^{T} \boldsymbol{A}\right\|_{1}=\max _{1 \leq k \leq m}\left\|\boldsymbol{e}_{k}^{T} \boldsymbol{A}\right\|_{1}$</td><td>$\infty$-norm: Define K_∞, r and y_e by $K_\infty := \|e_r^T A\|_1 = \max_{1\leq k\leq m}\|e_k^T A\|_1$ and</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02578	192	0.998	`\boldsymbol{y}_{e}:=\left[e^{-i \theta_{1}}, \ldots, e^{-i \theta_{n}}\right]^{T}`	$\boldsymbol{y}_{e}:=\left[e^{-i \theta_{1}}, \ldots, e^{-i \theta_{n}}\right]^{T}$	$\boldsymbol{y}_e := [e^{-i\theta_1}, \ldots, e^{-i\theta_n}]^T$, where $a_{rj} = \lvert a_{rj}\rvert e^{i\theta_j}$ for $j = 1, \ldots, n$.
lyche-numerical-linear-algebra F02579	192	0.998	`a_{r j}=\left|a_{r j}\right| e^{i \theta_{j}}`	$a_{r j}=\left\lvert a_{r j}\right\rvert e^{i \theta_{j}}$	$\boldsymbol{y}_e := [e^{-i\theta_1}, \ldots, e^{-i\theta_n}]^T$, where $a_{rj} = \lvert a_{rj}\rvert e^{i\theta_j}$ for $j = 1, \ldots, n$.
lyche-numerical-linear-algebra F02580	192	1.000	`\left\\|\boldsymbol{A} \boldsymbol{y}^{*}\right\\|_{\infty}=\max _{1 \leq k \leq m}\left|\sum_{j=1}^{n} a_{k j} e^{-i \theta_{j}}\right|=K_{\infty}`	$\left\lVert\boldsymbol{A} \boldsymbol{y}^{*}\right\rVert_{\infty}=\max _{1 \leq k \leq m}\left\lvert\sum_{j=1}^{n} a_{k j} e^{-i \theta_{j}}\right\rvert=K_{\infty}$	**(b)** $\lVert \boldsymbol{A}\boldsymbol{y}^*\rVert_\infty = \max_{1\le k\le m} \left\lvert \sum_{j=1}^n a_{kj}e^{-i\theta_j}\right\rvert = K_\infty$.
lyche-numerical-linear-algebra F02581	192	1.000	`\left|\sum_{j=1}^{n} a_{k j} e^{-i \theta_{j}}\right| \leq \sum_{j=1}^{n}\left|a_{k j}\right| \leq K_{\infty}`	$\left\lvert\sum_{j=1}^{n} a_{k j} e^{-i \theta_{j}}\right\rvert \leq \sum_{j=1}^{n}\left\lvert a_{k j}\right\rvert \leq K_{\infty}$	The last equality is correct because $\left\lvert \sum_{j=1}^n a_{kj}e^{-i\theta_j}\right\rvert \le \sum_{j=1}^n \lvert a_{kj}\rvert \le K_\infty$ with
lyche-numerical-linear-algebra F02582	192	1.000	`k=r`	$k=r$	equality for $k = r$.
lyche-numerical-linear-algebra F02583	192	0.999	`\boldsymbol{A}:=\frac{1}{15}\left[\begin{array}{ccc}14 & 4 & 16 \\ 2 & 22 & 13\end{array}\right]`	$\boldsymbol{A}:=\frac{1}{15}\left[\begin{array}{ccc}14 & 4 & 16 \\ 2 & 22 & 13\end{array}\right]$	matrix $\boldsymbol{A} := \frac{1}{15}\left[\begin{smallmatrix}14 & 4 & 16\\ 2 & 22 & 13\end{smallmatrix}\right]$, is $\sigma_1 = 2$ (cf. Example 7.9). We find
lyche-numerical-linear-algebra F02584	193	1.000	`\sigma_{1} \geq \sigma_{2} \geq`	$\sigma_{1} \geq \sigma_{2} \geq$	**Theorem 8.4 (Spectral Norm)** *Suppose* $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ *has singular values* $\sigma_1 \ge \sigma_2 \ge$
lyche-numerical-linear-algebra F02585	193	1.000	`\cdots \geq \sigma_{n}`	$\cdots \geq \sigma_{n}$	$\cdots \ge \sigma_n$ *and eigenvalues* $\lvert\lambda_1\rvert \ge \lvert\lambda_2\rvert \ge \cdots \ge \lvert\lambda_n\rvert$. *Then*
lyche-numerical-linear-algebra F02586	193	1.000	`\left|\lambda_{1}\right| \geq\left|\lambda_{2}\right| \geq \cdots \geq\left|\lambda_{n}\right|`	$\left\lvert\lambda_{1}\right\rvert \geq\left\lvert\lambda_{2}\right\rvert \geq \cdots \geq\left\lvert\lambda_{n}\right\rvert$	$\cdots \ge \sigma_n$ *and eigenvalues* $\lvert\lambda_1\rvert \ge \lvert\lambda_2\rvert \ge \cdots \ge \lvert\lambda_n\rvert$. *Then*
lyche-numerical-linear-algebra F02587	193	1.000	`1 / \sigma_{n}`	$1 / \sigma_{n}$	***Proof*** Since $1/\sigma_n$ is the largest singular value of $\boldsymbol{A}^{-1}$, (8.19) follows. By Exer-
lyche-numerical-linear-algebra F02588	193	1.000	`\\|\boldsymbol{A}\\|_{2}^{2} \leq`	$\lVert\boldsymbol{A}\rVert_{2}^{2} \leq$	**Theorem 8.5 (Spectral Norm Bound)** *For any* $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ *we have* $\lVert\boldsymbol{A}\rVert_2^2 \le$
lyche-numerical-linear-algebra F02589	193	0.994	`\\|\boldsymbol{A}\\|_{1}\\|\boldsymbol{A}\\|_{\infty}`	$\lVert\boldsymbol{A}\rVert_{1}\lVert\boldsymbol{A}\rVert_{\infty}$	$\lVert\boldsymbol{A}\rVert_1\lVert\boldsymbol{A}\rVert_\infty$.
lyche-numerical-linear-algebra F02590	193	0.994	`\left(\sigma^{2}, \boldsymbol{v}\right)`	$\left(\sigma^{2}, \boldsymbol{v}\right)$	***Proof*** Let $(\sigma^2, \boldsymbol{v})$ be an eigenpair for $\boldsymbol{A}^*\boldsymbol{A}$ corresponding to the largest singular
lyche-numerical-linear-algebra F02591	193	1.000	`\left\\|\boldsymbol{A}^{*}\right\\|_{1}=\\|\boldsymbol{A}\\|_{\infty}`	$\left\lVert\boldsymbol{A}^{*}\right\rVert_{1}=\lVert\boldsymbol{A}\rVert_{\infty}$	Observing that $\lVert\boldsymbol{A}^*\rVert_1 = \lVert\boldsymbol{A}\rVert_\infty$ by Theorem 8.3 and canceling $\lVert\boldsymbol{v}\rVert_1$ proves the
lyche-numerical-linear-algebra F02592	193	1.000	`\\|\boldsymbol{v}\\|_{1}`	$\lVert\boldsymbol{v}\rVert_{1}$	Observing that $\lVert\boldsymbol{A}^*\rVert_1 = \lVert\boldsymbol{A}\rVert_\infty$ by Theorem 8.3 and canceling $\lVert\boldsymbol{v}\rVert_1$ proves the
lyche-numerical-linear-algebra F02593	193	1.000	`\\|\boldsymbol{U} \boldsymbol{A} \boldsymbol{V}\\|=\\|\boldsymbol{A}\\|`	$\lVert\boldsymbol{U} \boldsymbol{A} \boldsymbol{V}\rVert=\lVert\boldsymbol{A}\rVert$	**unitary invariant** if $\lVert\boldsymbol{U}\boldsymbol{A}\boldsymbol{V}\rVert = \lVert\boldsymbol{A}\rVert$ for any $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ and any unitary matrices
lyche-numerical-linear-algebra F02594	193	1.000	`\boldsymbol{U}(\boldsymbol{A}+`	$\boldsymbol{U}(\boldsymbol{A}+$	increased by a unitary transformation. Thus if $\boldsymbol{U}$ and $\boldsymbol{V}$ are unitary then $\boldsymbol{U}(\boldsymbol{A} +$
lyche-numerical-linear-algebra F02595	193	1.000	`\boldsymbol{E}) \boldsymbol{V}=\boldsymbol{U} \boldsymbol{A} \boldsymbol{V}+\boldsymbol{F}`	$\boldsymbol{E}) \boldsymbol{V}=\boldsymbol{U} \boldsymbol{A} \boldsymbol{V}+\boldsymbol{F}$	$\boldsymbol{E})\boldsymbol{V} = \boldsymbol{U}\boldsymbol{A}\boldsymbol{V} + \boldsymbol{F}$, where $\lVert\boldsymbol{F}\rVert = \lVert\boldsymbol{E}\rVert$.
lyche-numerical-linear-algebra F02596	193	1.000	`\\|\boldsymbol{F}\\|=\\|\boldsymbol{E}\\|`	$\lVert\boldsymbol{F}\rVert=\lVert\boldsymbol{E}\rVert$	$\boldsymbol{E})\boldsymbol{V} = \boldsymbol{U}\boldsymbol{A}\boldsymbol{V} + \boldsymbol{F}$, where $\lVert\boldsymbol{F}\rVert = \lVert\boldsymbol{E}\rVert$.
lyche-numerical-linear-algebra F02597	194	1.000	`\left\\|\boldsymbol{A}^{*}\right\\|_{2}=\\|\boldsymbol{A}\\|_{2}`	$\left\lVert\boldsymbol{A}^{*}\right\rVert_{2}=\lVert\boldsymbol{A}\rVert_{2}$	Theorem 8.3 that $\lVert\boldsymbol{A}^*\rVert_2 = \lVert\boldsymbol{A}\rVert_2$. Moreover $\boldsymbol{V}^*$ is unitary. Using these facts we
lyche-numerical-linear-algebra F02598	194	1.000	`\boldsymbol{V}^{*}`	$\boldsymbol{V}^{*}$	Theorem 8.3 that $\lVert\boldsymbol{A}^*\rVert_2 = \lVert\boldsymbol{A}\rVert_2$. Moreover $\boldsymbol{V}^*$ is unitary. Using these facts we
lyche-numerical-linear-algebra F02599	194	1.000	`\\|\boldsymbol{x}\\|=\\||\boldsymbol{x}|\\|`	$\lVert\boldsymbol{x}\rVert=\lVert\lvert\boldsymbol{x}\rvert\rVert$	A vector norm on $\mathbb{C}^n$ is an **absolute norm** if $\lVert\boldsymbol{x}\rVert = \lVert\,\lvert\boldsymbol{x}\rvert\,\rVert$ for all $\boldsymbol{x} \in \mathbb{C}^n$. Here
lyche-numerical-linear-algebra F02600	194	1.000	`|\boldsymbol{x}|:=\left[\left|x_{1}\right|, \ldots, \left|x_{n}\right|\right]^{T}`	$\lvert\boldsymbol{x}\rvert:=\left[\left\lvert x_{1}\right\rvert, \ldots, \left\lvert x_{n}\right\rvert\right]^{T}$	$\lvert\boldsymbol{x}\rvert := [\lvert x_1\rvert, \ldots, \lvert x_n\rvert]^T$, the absolute values of the components of $\boldsymbol{x}$. Clearly the
lyche-numerical-linear-algebra F02601	195	1.000	`x_{1}=x_{2}=10`	$x_{1}=x_{2}=10$	whose exact solution is $x_1 = x_2 = 10$. If we replace the second equation by
lyche-numerical-linear-algebra F02602	195	1.000	`x_{1}=30, x_{2}=-10`	$x_{1}=30, x_{2}=-10$	the exact solution changes to $x_1 = 30$, $x_2 = -10$. Here a small change in one of
lyche-numerical-linear-algebra F02603	195	1.000	`1-10^{-16}`	$1-10^{-16}$	the coefficients, from $1 - 10^{-16}$ to $1 + 10^{-16}$, changed the exact solution by a large

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02604	195	1.000	`1+10^{-16}`	$1+10^{-16}$	the coefficients, from $1-10^{-16}$ to $1+10^{-16}$, changed the exact solution by a large
lyche-numerical-linear-algebra F02605	195	0.994	`\boldsymbol{A}, \boldsymbol{b}`	$\boldsymbol{A}, \boldsymbol{b}$	$\boldsymbol{A}$,$\boldsymbol{b}$ has on the inverse of $\boldsymbol{A}$ and on the solution $\boldsymbol{x}$ of a linear system $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$. To
lyche-numerical-linear-algebra F02606	195	1.000	`\boldsymbol{I} \in \mathbb{R}^{n \times n}`	$\boldsymbol{I} \in \mathbb{R}^{n \times n}$	vector norm. This follows from Lemma 7.1. We recall that if $\boldsymbol{I} \in \mathbb{R}^{n\times n}$ then $\lVert\boldsymbol{I}\rVert=1$
lyche-numerical-linear-algebra F02607	195	0.996	`(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{y}= \boldsymbol{b}+\boldsymbol{e}`	$(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{y}= \boldsymbol{b}+\boldsymbol{e}$	Suppose $\boldsymbol{x}$, $\boldsymbol{y}$ solve $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$ and $(\boldsymbol{A}+\boldsymbol{E})\boldsymbol{y}=\boldsymbol{b}+\boldsymbol{e}$, respectively. where $\boldsymbol{A}$, $\boldsymbol{A}+\boldsymbol{E}\in$
lyche-numerical-linear-algebra F02608	195	0.996	`\boldsymbol{A}, \boldsymbol{A}+\boldsymbol{E} \in`	$\boldsymbol{A}, \boldsymbol{A}+\boldsymbol{E} \in$	Suppose $\boldsymbol{x}$, $\boldsymbol{y}$ solve $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$ and $(\boldsymbol{A}+\boldsymbol{E})\boldsymbol{y}=\boldsymbol{b}+\boldsymbol{e}$, respectively. where $\boldsymbol{A}$, $\boldsymbol{A}+\boldsymbol{E}\in$
lyche-numerical-linear-algebra F02609	195	1.000	`\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^{n}`	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^{n}$	$\mathbb{C}^{n\times n}$ are nonsingular and $\boldsymbol{b}, \boldsymbol{e}\in\mathbb{C}^n$. How large can $\boldsymbol{y}-\boldsymbol{x}$ be? The difference $\lVert\boldsymbol{y}-\boldsymbol{x}\rVert$
lyche-numerical-linear-algebra F02610	195	1.000	`\boldsymbol{y}-\boldsymbol{x}`	$\boldsymbol{y}-\boldsymbol{x}$	$\mathbb{C}^{n\times n}$ are nonsingular and $\boldsymbol{b}, \boldsymbol{e}\in\mathbb{C}^n$. How large can $\boldsymbol{y}-\boldsymbol{x}$ be? The difference $\lVert\boldsymbol{y}-\boldsymbol{x}\rVert$
lyche-numerical-linear-algebra F02611	195	1.000	`|\boldsymbol{y}-\boldsymbol{x}|`	$\lVert\boldsymbol{y}-\boldsymbol{x}\rVert$	$\mathbb{C}^{n\times n}$ are nonsingular and $\boldsymbol{b}, \boldsymbol{e}\in\mathbb{C}^n$. How large can $\boldsymbol{y}-\boldsymbol{x}$ be? The difference $\lVert\boldsymbol{y}-\boldsymbol{x}\rVert$
lyche-numerical-linear-algebra F02612	195	1.000	`|\boldsymbol{y}-\boldsymbol{x}| /|\boldsymbol{x}|`	$\lVert\boldsymbol{y}-\boldsymbol{x}\rVert /\lVert\boldsymbol{x}\rVert$	measures the **absolute error** in $\boldsymbol{y}$ as an approximation to $\boldsymbol{x}$, while $\lVert\boldsymbol{y}-\boldsymbol{x}\rVert/\lVert\boldsymbol{x}\rVert$ and
lyche-numerical-linear-algebra F02613	195	1.000	`|\boldsymbol{y}-\boldsymbol{x}| /|\boldsymbol{y}|`	$\lVert\boldsymbol{y}-\boldsymbol{x}\rVert /\lVert\boldsymbol{y}\rVert$	$\lVert\boldsymbol{y}-\boldsymbol{x}\rVert/\lVert\boldsymbol{y}\rVert$ are measures for the **relative error**.
lyche-numerical-linear-algebra F02614	195	0.546	`\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^{n}, \boldsymbol{b} \neq \mathbf{0}`	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^{n}, \boldsymbol{b} \neq \mathbf{0}$	*nonsingular,* $\boldsymbol{b}, \boldsymbol{e}\in\mathbb{C}^n$, $\boldsymbol{b}\neq\mathbf{0}$ *and* $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$, $\boldsymbol{A}\boldsymbol{y}=\boldsymbol{b}+\boldsymbol{e}$. *Then*
lyche-numerical-linear-algebra F02615	195	0.546	`\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}, \boldsymbol{A} \boldsymbol{y}=\boldsymbol{b}+ \boldsymbol{e}`	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}, \boldsymbol{A} \boldsymbol{y}=\boldsymbol{b}+ \boldsymbol{e}$	*nonsingular,* $\boldsymbol{b}, \boldsymbol{e}\in\mathbb{C}^n$, $\boldsymbol{b}\neq\mathbf{0}$ *and* $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$, $\boldsymbol{A}\boldsymbol{y}=\boldsymbol{b}+\boldsymbol{e}$. *Then*
lyche-numerical-linear-algebra F02616	195	1.000	`K(\boldsymbol{A})=|\boldsymbol{A}|\left| \boldsymbol{A}^{-1}\right|`	$K(\boldsymbol{A})=\lVert\boldsymbol{A}\rVert\left\lVert \boldsymbol{A}^{-1}\right\rVert$	*where* $K(\boldsymbol{A})=\lVert\boldsymbol{A}\rVert\,\lVert\boldsymbol{A}^{-1}\rVert$ *is the condition number of* $\boldsymbol{A}$.
lyche-numerical-linear-algebra F02617	195	1.000	`\boldsymbol{A} \boldsymbol{y}=\boldsymbol{b}+ \boldsymbol{e}`	$\boldsymbol{A} \boldsymbol{y}=\boldsymbol{b}+ \boldsymbol{e}$	***Proof*** Subtracting $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$ from $\boldsymbol{A}\boldsymbol{y}=\boldsymbol{b}+\boldsymbol{e}$ we have $\boldsymbol{A}(\boldsymbol{y}-\boldsymbol{x})=\boldsymbol{e}$ or $\boldsymbol{y}-\boldsymbol{x}=\boldsymbol{A}^{-1}\boldsymbol{e}$.
lyche-numerical-linear-algebra F02618	195	1.000	`\boldsymbol{A}(\boldsymbol{y}-\boldsymbol{x})= \boldsymbol{e}`	$\boldsymbol{A}(\boldsymbol{y}-\boldsymbol{x})= \boldsymbol{e}$	***Proof*** Subtracting $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$ from $\boldsymbol{A}\boldsymbol{y}=\boldsymbol{b}+\boldsymbol{e}$ we have $\boldsymbol{A}(\boldsymbol{y}-\boldsymbol{x})=\boldsymbol{e}$ or $\boldsymbol{y}-\boldsymbol{x}=\boldsymbol{A}^{-1}\boldsymbol{e}$.
lyche-numerical-linear-algebra F02619	195	1.000	`\boldsymbol{y}-\boldsymbol{x}=\boldsymbol{A}^{-1} \boldsymbol{e}`	$\boldsymbol{y}-\boldsymbol{x}=\boldsymbol{A}^{-1} \boldsymbol{e}$	***Proof*** Subtracting $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$ from $\boldsymbol{A}\boldsymbol{y}=\boldsymbol{b}+\boldsymbol{e}$ we have $\boldsymbol{A}(\boldsymbol{y}-\boldsymbol{x})=\boldsymbol{e}$ or $\boldsymbol{y}-\boldsymbol{x}=\boldsymbol{A}^{-1}\boldsymbol{e}$.
lyche-numerical-linear-algebra F02620	195	1.000	`|\boldsymbol{y}-\boldsymbol{x}|=\left| \boldsymbol{A}^{-1} \boldsymbol{e}\right| \leq\left |\boldsymbol{A}^{-1}\right||\boldsymbol{e}|`	$\lVert\boldsymbol{y}-\boldsymbol{x}\rVert=\left\lVert \boldsymbol{A}^{-1} \boldsymbol{e}\right\rVert \leq\left\lVert\boldsymbol{A}^{-1}\right\rVert\lVert\boldsymbol{e}\rVert$	Combining $\lVert\boldsymbol{y}-\boldsymbol{x}\rVert=\lVert\boldsymbol{A}^{-1}\boldsymbol{e}\rVert\leq\lVert\boldsymbol{A}^{-1}\rVert\,\lVert\boldsymbol{e}\rVert$ and $\lVert\boldsymbol{b}\rVert=\lVert\boldsymbol{A}\boldsymbol{x}\rVert\leq\lVert\boldsymbol{A}\rVert\,\lVert\boldsymbol{x}\rVert$ we
lyche-numerical-linear-algebra F02621	195	1.000	`|\boldsymbol{b}|=|\boldsymbol{A} \boldsymbol{x}| \leq |\boldsymbol{A}||\boldsymbol{x}|`	$\lVert\boldsymbol{b}\rVert=\lVert\boldsymbol{A} \boldsymbol{x}\rVert \leq \lVert\boldsymbol{A}\rVert\lVert\boldsymbol{x}\rVert$	Combining $\lVert\boldsymbol{y}-\boldsymbol{x}\rVert=\lVert\boldsymbol{A}^{-1}\boldsymbol{e}\rVert\leq\lVert\boldsymbol{A}^{-1}\rVert\,\lVert\boldsymbol{e}\rVert$ and $\lVert\boldsymbol{b}\rVert=\lVert\boldsymbol{A}\boldsymbol{x}\rVert\leq\lVert\boldsymbol{A}\rVert\,\lVert\boldsymbol{x}\rVert$ we
lyche-numerical-linear-algebra F02622	195	1.000	`|\boldsymbol{e}| \leq|\boldsymbol{A}|| \boldsymbol{y}-\boldsymbol{x}|`	$\lVert\boldsymbol{e}\rVert \leq\lVert\boldsymbol{A}\rVert\lVert \boldsymbol{y}-\boldsymbol{x}\rVert$	obtain the upper bound in (8.22). Combining $\lVert\boldsymbol{e}\rVert\leq\lVert\boldsymbol{A}\rVert\,\lVert\boldsymbol{y}-\boldsymbol{x}\rVert$ and $\lVert\boldsymbol{x}\rVert\leq$
lyche-numerical-linear-algebra F02623	195	1.000	`|\boldsymbol{x}| \leq`	$\lVert\boldsymbol{x}\rVert \leq$	obtain the upper bound in (8.22). Combining $\lVert\boldsymbol{e}\rVert\leq\lVert\boldsymbol{A}\rVert\,\lVert\boldsymbol{y}-\boldsymbol{x}\rVert$ and $\lVert\boldsymbol{x}\rVert\leq$
lyche-numerical-linear-algebra F02624	195	1.000	`\left|\boldsymbol{A}^{-1}\right||\boldsymbol{b}|`	$\left\lVert\boldsymbol{A}^{-1}\right\rVert\lVert\boldsymbol{b}\rVert$	$\lVert\boldsymbol{A}^{-1}\rVert\,\lVert\boldsymbol{b}\rVert$ we obtain the lower bound.
lyche-numerical-linear-algebra F02625	196	1.000	`|\boldsymbol{e}| /|\boldsymbol{b}|`	$\lVert\boldsymbol{e}\rVert /\lVert\boldsymbol{b}\rVert$	Consider (8.22). $\lVert\boldsymbol{e}\rVert/\lVert\boldsymbol{b}\rVert$ is a measure of the size of the perturbation $\boldsymbol{e}$ relative
lyche-numerical-linear-algebra F02626	196	1.000	`\boldsymbol{e}`	$\boldsymbol{e}$	Consider (8.22). $\lVert\boldsymbol{e}\rVert/\lVert\boldsymbol{b}\rVert$ is a measure of the size of the perturbation $\boldsymbol{e}$ relative
lyche-numerical-linear-algebra F02627	196	0.999	`K(\boldsymbol{A})`	$K(\boldsymbol{A})$	$K(\boldsymbol{A})$ times as large as $\lVert\boldsymbol{e}\rVert/\lVert\boldsymbol{b}\rVert$.
lyche-numerical-linear-algebra F02628	196	1.000	`K(\boldsymbol{A}) \geq 1`	$K(\boldsymbol{A}) \geq 1$	**well-conditioned** (with respect to inversion). We always have $K(\boldsymbol{A})\geq 1$. For since
lyche-numerical-linear-algebra F02629	196	1.000	`|\boldsymbol{x}|=|\boldsymbol{I} \boldsymbol{x}| \leq |\boldsymbol{I}||\boldsymbol{x}|`	$\lVert\boldsymbol{x}\rVert=\lVert\boldsymbol{I} \boldsymbol{x}\rVert \leq \lVert\boldsymbol{I}\rVert\lVert\boldsymbol{x}\rVert$	$\lVert\boldsymbol{x}\rVert=\lVert\boldsymbol{I}\boldsymbol{x}\rVert\leq\lVert\boldsymbol{I}\rVert\lVert\boldsymbol{x}\rVert$ for any $\boldsymbol{x}$ we have $\lVert\boldsymbol{I}\rVert\geq 1$ and therefore $\lVert\boldsymbol{A}\rVert\,\lVert\boldsymbol{A}^{-1}\rVert\geq$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02630	196	1.000	`|\boldsymbol{I}| \geq 1`	$\\|\boldsymbol{I}\\| \geq 1$	$\\|\boldsymbol{x}\\| = \\|\boldsymbol{I}\boldsymbol{x}\\| \leq \\|\boldsymbol{I}\\|\\|\boldsymbol{x}\\|$ for any $\boldsymbol{x}$ we have $\\|\boldsymbol{I}\\| \geq 1$ and therefore $\\|\boldsymbol{A}\\|\\|\boldsymbol{A}^{-1}\\| \geq$
lyche-numerical-linear-algebra F02631	196	1.000	`|\boldsymbol{A}|\left|\boldsymbol{A}^{-1}\right| \geq`	$\\|\boldsymbol{A}\\|\left\\|\boldsymbol{A}^{-1}\right\\| \geq$	$\\|\boldsymbol{x}\\| = \\|\boldsymbol{I}\boldsymbol{x}\\| \leq \\|\boldsymbol{I}\\|\\|\boldsymbol{x}\\|$ for any $\boldsymbol{x}$ we have $\\|\boldsymbol{I}\\| \geq 1$ and therefore $\\|\boldsymbol{A}\\|\\|\boldsymbol{A}^{-1}\\| \geq$
lyche-numerical-linear-algebra F02632	196	1.000	`\left|\boldsymbol{A} \boldsymbol{A}^{-1}\right|=|\boldsymbol{I}| \geq 1`	$\left\\|\boldsymbol{A} \boldsymbol{A}^{-1}\right\\|=\\|\boldsymbol{I}\\| \geq 1$	$\\|\boldsymbol{A}\boldsymbol{A}^{-1}\\| = \\|\boldsymbol{I}\\| \geq 1$.
lyche-numerical-linear-algebra F02633	196	1.000	`\ell_{1}, \ell_{\infty}`	ℓ_{1}, ℓ_{∞}	when discussing properties of the condition number, and the ℓ_1, ℓ_∞, or Frobenius
lyche-numerical-linear-algebra F02634	196	0.874	`\boldsymbol{A x}=\boldsymbol{b}`	$\boldsymbol{A x}=\boldsymbol{b}$	Suppose we have computed an approximate solution $\boldsymbol{y}$ to $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. The vector
lyche-numerical-linear-algebra F02635	196	1.000	`\boldsymbol{r}(\boldsymbol{y}):=\boldsymbol{A} \boldsymbol{y}-\boldsymbol{b}`	$\boldsymbol{r}(\boldsymbol{y}):=\boldsymbol{A} \boldsymbol{y}-\boldsymbol{b}$	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}$ is called the **residual vector**, or just the residual. We can bound
lyche-numerical-linear-algebra F02636	196	1.000	`\boldsymbol{r}`	$\boldsymbol{r}$	$\boldsymbol{x} - \boldsymbol{y}$ in terms of $\boldsymbol{r}$.
lyche-numerical-linear-algebra F02637	196	0.987	`\boldsymbol{b} \neq \mathbf{0}`	$\boldsymbol{b} \neq \mathbf{0}$	*nonsingular and* $\boldsymbol{b} \neq \mathbf{0}$. *Let* $\boldsymbol{r}(\boldsymbol{y}) = \boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}$ *for* $\boldsymbol{y} \in \mathbb{C}^n$. *If* $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ *then*
lyche-numerical-linear-algebra F02638	196	0.987	`\boldsymbol{r}(\boldsymbol{y})=\boldsymbol{A} \boldsymbol{y}-\boldsymbol{b}`	$\boldsymbol{r}(\boldsymbol{y})=\boldsymbol{A} \boldsymbol{y}-\boldsymbol{b}$	*nonsingular and* $\boldsymbol{b} \neq \mathbf{0}$. *Let* $\boldsymbol{r}(\boldsymbol{y}) = \boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}$ *for* $\boldsymbol{y} \in \mathbb{C}^n$. *If* $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ *then*
lyche-numerical-linear-algebra F02639	196	1.000	`\boldsymbol{e}=\boldsymbol{r}(\boldsymbol{y})`	$\boldsymbol{e}=\boldsymbol{r}(\boldsymbol{y})$	***Proof*** We simply take $\boldsymbol{e} = \boldsymbol{r}(\boldsymbol{y})$ in Theorem 8.7.
lyche-numerical-linear-algebra F02640	196	0.999	`\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}`	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n\times n}$ with $\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E}$ nonsingular. We like to compare the solution $\boldsymbol{x}$ and $\boldsymbol{y}$
lyche-numerical-linear-algebra F02641	196	0.999	`\boldsymbol{A}, \boldsymbol{A}+\boldsymbol{E}`	$\boldsymbol{A}, \boldsymbol{A}+\boldsymbol{E}$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n\times n}$ with $\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E}$ nonsingular. We like to compare the solution $\boldsymbol{x}$ and $\boldsymbol{y}$
lyche-numerical-linear-algebra F02642	196	0.999	`(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{y}=\boldsymbol{b}`	$(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{y}=\boldsymbol{b}$	of the systems $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ and $(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b}$.
lyche-numerical-linear-algebra F02643	196	0.615	`\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^{n}`	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^{n}$	**Theorem 8.9 (Perturbation in Matrix)** *Suppose* $\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n\times n}$, $\boldsymbol{b} \in \mathbb{C}^n$ *with* $\boldsymbol{A}$
lyche-numerical-linear-algebra F02644	196	0.999	`r:=\left|\boldsymbol{A}^{-1} \boldsymbol{E}\right|<1`	$r:=\left\\|\boldsymbol{A}^{-1} \boldsymbol{E}\right\\|<1$	*nonsingular and* $\boldsymbol{b} \neq \mathbf{0}$. *If* $r := \\|\boldsymbol{A}^{-1}\boldsymbol{E}\\| < 1$ *then* $\boldsymbol{A} + \boldsymbol{E}$ *is nonsingular. If* $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$
lyche-numerical-linear-algebra F02645	196	0.999	`\boldsymbol{A}+\boldsymbol{E}`	$\boldsymbol{A}+\boldsymbol{E}$	*nonsingular and* $\boldsymbol{b} \neq \mathbf{0}$. *If* $r := \\|\boldsymbol{A}^{-1}\boldsymbol{E}\\| < 1$ *then* $\boldsymbol{A} + \boldsymbol{E}$ *is nonsingular. If* $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$
lyche-numerical-linear-algebra F02646	196	1.000	`r \geq 1`	$r \geq 1$	***Proof*** We show $\boldsymbol{A} + \boldsymbol{E}$ singular implies $r \geq 1$. Suppose $\boldsymbol{A} + \boldsymbol{E}$ is singular. Then
lyche-numerical-linear-algebra F02647	196	1.000	`(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{x}=\mathbf{0}`	$(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{x}=\mathbf{0}$	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mathbf{0}$ for some nonzero $\boldsymbol{x} \in \mathbb{C}^n$. Multiplying by $\boldsymbol{A}^{-1}$ it follows that
lyche-numerical-linear-algebra F02648	196	1.000	`\left(\boldsymbol{I}+\boldsymbol{A}^{-1} \boldsymbol{E}\right) \boldsymbol{x}=\mathbf{0}`	$\left(\boldsymbol{I}+\boldsymbol{A}^{-1} \boldsymbol{E}\right) \boldsymbol{x}=\mathbf{0}$	$(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})\boldsymbol{x} = \mathbf{0}$ and this implies that $\\|\boldsymbol{x}\\| = \\|\boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{x}\\| \leq r\\|\boldsymbol{x}\\|$. But then $r \geq 1$.
lyche-numerical-linear-algebra F02649	196	1.000	`|\boldsymbol{x}|=\left|\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{x}\right| \leq r|\boldsymbol{x}|`	$\\|\boldsymbol{x}\\|=\left\\|\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{x}\right\\| \leq r\\|\boldsymbol{x}\\|$	$(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})\boldsymbol{x} = \mathbf{0}$ and this implies that $\\|\boldsymbol{x}\\| = \\|\boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{x}\\| \leq r\\|\boldsymbol{x}\\|$. But then $r \geq 1$.
lyche-numerical-linear-algebra F02650	196	0.967	`\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\boldsymbol{E} \boldsymbol{y}`	$\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\boldsymbol{E} \boldsymbol{y}$	Subtracting $(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b}$ from $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ gives $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{E}\boldsymbol{y}$ or
lyche-numerical-linear-algebra F02651	196	1.000	`\boldsymbol{x}-\boldsymbol{y}=\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{y}`	$\boldsymbol{x}-\boldsymbol{y}=\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{y}$	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$. Taking norms and dividing by $\\|\boldsymbol{y}\\|$ proves (8.24). Solving
lyche-numerical-linear-algebra F02652	196	1.000	`|\boldsymbol{y}|`	$\\|\boldsymbol{y}\\|$	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$. Taking norms and dividing by $\\|\boldsymbol{y}\\|$ proves (8.24). Solving
lyche-numerical-linear-algebra F02653	197	1.000	`\boldsymbol{y}=\left(\boldsymbol{I}+\boldsymbol{A}^{-1} \boldsymbol{E}\right)^{-1} \boldsymbol{x}`	$\boldsymbol{y}=\left(\boldsymbol{I}+\boldsymbol{A}^{-1} \boldsymbol{E}\right)^{-1} \boldsymbol{x}$	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$ for $\boldsymbol{y}$ we obtain $\boldsymbol{y} = (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\boldsymbol{x}$. By Theorem 12.14 we
lyche-numerical-linear-algebra F02654	197	0.995	`|\boldsymbol{y}| \leq\left|\left(\boldsymbol{I}+\boldsymbol{A}^{-1} \boldsymbol{E}\right)^{-1}\right||\boldsymbol{x}| \leq \frac{|\boldsymbol{x}|}{1-r}`	$\\|\boldsymbol{y}\\| \leq\left\\|\left(\boldsymbol{I}+\boldsymbol{A}^{-1} \boldsymbol{E}\right)^{-1}\right\\|\\|\boldsymbol{x}\\| \leq \frac{\\|\boldsymbol{x}\\|}{1-r}$	have $\\|\boldsymbol{y}\\| \leq \\|(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\\|\\|\boldsymbol{x}\\| \leq \frac{\\|\boldsymbol{x}\\|}{1-r}$ and (8.24) implies $\\|\boldsymbol{y} - \boldsymbol{x}\\| \leq r\\|\boldsymbol{y}\\| \leq$
lyche-numerical-linear-algebra F02655	197	0.995	`|\boldsymbol{y}-\boldsymbol{x}| \leq r|\boldsymbol{y}| \leq`	$\\|\boldsymbol{y}-\boldsymbol{x}\\| \leq r\\|\boldsymbol{y}\\| \leq$	have $\\|\boldsymbol{y}\\| \leq \\|(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\\|\\|\boldsymbol{x}\\| \leq \frac{\\|\boldsymbol{x}\\|}{1-r}$ and (8.24) implies $\\|\boldsymbol{y} - \boldsymbol{x}\\| \leq r\\|\boldsymbol{y}\\| \leq$

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-al F02656</td><td>197</td><td>■ 0.669</td><td>\frac{r}{1-r}\|\boldsymbol{x}\| \leq \frac{K(\boldsymbol{A})}{1-r}\|\boldsymbol{E}\|\|\boldsymbol{A}\|</td><td>$\frac{r}{1-r}\|\boldsymbol{x}\| \leq \frac{K(\boldsymbol{A})}{1-r}\|\boldsymbol{E}\|\|\boldsymbol{A}\|$</td><td>$\frac{r}{1-r}\|\boldsymbol{x}\| \le \frac{K(\boldsymbol{A})}{1-r}\frac{\|\boldsymbol{E}\|}{\|\boldsymbol{A}\|}\|\boldsymbol{x}\|$. Dividing by $\|\boldsymbol{x}\|$ gives (8.25).</td></tr>
<tr><td>lyche-numerical-linear-al F02657</td><td>197</td><td>■ 0.669</td><td>\|\boldsymbol{x}\|</td><td>$\|\boldsymbol{x}\|$</td><td>$\frac{r}{1-r}\|\boldsymbol{x}\| \le \frac{K(\boldsymbol{A})}{1-r}\frac{\|\boldsymbol{E}\|}{\|\boldsymbol{A}\|}\|\boldsymbol{x}\|$. Dividing by $\|\boldsymbol{x}\|$ gives (8.25).</td></tr>
<tr><td>lyche-numerical-linear-al F02658</td><td>197</td><td>■ 0.986</td><td>\boldsymbol{x} .\|\boldsymbol{E}\| /\|\boldsymbol{A}\|</td><td>$\boldsymbol{x} .\|\boldsymbol{E}\| /\|\boldsymbol{A}\|$</td><td>to $\boldsymbol{y}$ and the relative error in $\boldsymbol{y}$ as an approximation to $\boldsymbol{x}$. $\|\boldsymbol{E}\|/\|\boldsymbol{A}\|$ is a measure for</td></tr>
<tr><td>lyche-numerical-linear-al F02659</td><td>197</td><td>■ 1.000</td><td>\|\boldsymbol{E}\| /\|\boldsymbol{A}\|</td><td>$\|\boldsymbol{E}\| /\|\boldsymbol{A}\|$</td><td>again plays a crucial role. $\|\boldsymbol{y}-\boldsymbol{x}\|/\|\boldsymbol{y}\|$ can be as large as $K(\boldsymbol{A})$ times $\|\boldsymbol{E}\|/\|\boldsymbol{A}\|$. It</td></tr>
<tr><td>lyche-numerical-linear-al F02660</td><td>197</td><td>■ 1.000</td><td>\left\|\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{y}\right\| \leq\left\|\boldsymbol{A}^{-1}\right\|\|\boldsymbol{E}\|\|\boldsymbol{y}\|</td><td>$\left\|\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{y}\right\| \leq\left\|\boldsymbol{A}^{-1}\right\|\|\boldsymbol{E}\|\|\boldsymbol{y}\|$</td><td>the upper bound we used the inequality $\|\boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}\| \le \|\boldsymbol{A}^{-1}\|\,\|\boldsymbol{E}\|\,\|\boldsymbol{y}\|$. For a more</td></tr>
<tr><td>lyche-numerical-linear-al F02661</td><td>197</td><td>■ 1.000</td><td>\left\|\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{y}\right\|</td><td>$\left\|\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{y}\right\|$</td><td>or less random perturbation $\boldsymbol{E}$ this is not a severe overestimate for $\|\boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}\|$. In the</td></tr>
<tr><td>lyche-numerical-linear-al F02662</td><td>197</td><td>■ 1.000</td><td>\sigma_{1} \geq \sigma_{2} \geq \cdots \geq \sigma_{n}>0</td><td>$\sigma_{1} \geq \sigma_{2} \geq \cdots \geq \sigma_{n}>0$</td><td>with singular values $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_n > 0$ and eigenvalues $|\lambda_1| \ge |\lambda_2| \ge \cdots \ge$</td></tr>
<tr><td>lyche-numerical-linear-al F02663</td><td>197</td><td>■ 1.000</td><td>\left|\lambda_{1}\right| \geq\left|\lambda_{2}\right| \geq \cdots \geq</td><td>$\left|\lambda_{1}\right| \geq\left|\lambda_{2}\right| \geq \cdots \geq$</td><td>with singular values $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_n > 0$ and eigenvalues $|\lambda_1| \ge |\lambda_2| \ge \cdots \ge$</td></tr>
<tr><td>lyche-numerical-linear-al F02664</td><td>197</td><td>■ 1.000</td><td>\left|\lambda_{n}\right|>0</td><td>$\left|\lambda_{n}\right|>0$</td><td>$|\lambda_n| > 0$. Then</td></tr>
<tr><td>lyche-numerical-linear-al F02665</td><td>197</td><td>■ 0.999</td><td>\sigma_{1} / \sigma_{n}</td><td>σ_{1} / σ_{n}</td><td>It follows that $\boldsymbol{A}$ is ill-conditioned with respect to inversion if and only if σ_1/σ_n</td></tr>
<tr><td>lyche-numerical-linear-al F02666</td><td>197</td><td>■ 1.000</td><td>\lambda_{1} / \lambda_{n}</td><td>$\lambda_{1} / \lambda_{n}$</td><td>is large, or λ_1/λ_n is large when $\boldsymbol{A}$ is positive definite.</td></tr>
<tr><td>lyche-numerical-linear-al F02667</td><td>197</td><td>■ 1.000</td><td>\|\boldsymbol{y}-\boldsymbol{x}\| /\|\boldsymbol{x}\| \approx\|\boldsymbol{r}(\boldsymbol{y})\| /\|\boldsymbol{b}\|</td><td>$\|\boldsymbol{y}-\boldsymbol{x}\| /\|\boldsymbol{x}\| \approx\|\boldsymbol{r}(\boldsymbol{y})\| /\|\boldsymbol{b}\|$</td><td>If $\boldsymbol{A}$ is well-conditioned, (8.23) says that $\|\boldsymbol{y}-\boldsymbol{x}\|/\|\boldsymbol{x}\| \approx \|\boldsymbol{r}(\boldsymbol{y})\|/\|\boldsymbol{b}\|$. In other</td></tr>
<tr><td>lyche-numerical-linear-al F02668</td><td>197</td><td>■ 1.000</td><td>\|\boldsymbol{b}\| \approx 1</td><td>$\|\boldsymbol{b}\| \approx 1$</td><td>long as $\|\boldsymbol{b}\| \approx 1$. If $\boldsymbol{A}$ is ill-conditioned, anything can happen. We can for example</td></tr>
<tr><td>lyche-numerical-linear-al F02669</td><td>197</td><td>■ 1.000</td><td>\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{E}</td><td>$\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{E}$</td><td>Suppose $\boldsymbol{A}$ is nonsingular and $\boldsymbol{E}$ a perturbation of $\boldsymbol{A}$. We expect $\boldsymbol{B} := \boldsymbol{A}+\boldsymbol{E}$ to be</td></tr>
<tr><td>lyche-numerical-linear-al F02670</td><td>197</td><td>■ 1.000</td><td>\left\|\boldsymbol{B}^{-1}\right\|</td><td>$\left\|\boldsymbol{B}^{-1}\right\|$</td><td>to have bounds on $\|\boldsymbol{B}^{-1}\|$ in terms of $\|\boldsymbol{A}^{-1}\|$ and the difference $\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\|$.</td></tr>
<tr><td>lyche-numerical-linear-al F02671</td><td>197</td><td>■ 1.000</td><td>\left\|\boldsymbol{A}^{-1}\right\|</td><td>$\left\|\boldsymbol{A}^{-1}\right\|$</td><td>to have bounds on $\|\boldsymbol{B}^{-1}\|$ in terms of $\|\boldsymbol{A}^{-1}\|$ and the difference $\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\|$.</td></tr>
<tr><td>lyche-numerical-linear-al F02672</td><td>197</td><td>■ 1.000</td><td>\left\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\|</td><td>$\left\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\|$</td><td>to have bounds on $\|\boldsymbol{B}^{-1}\|$ in terms of $\|\boldsymbol{A}^{-1}\|$ and the difference $\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\|$.</td></tr>
<tr><td>lyche-numerical-linear-al F02673</td><td>197</td><td>■ 1.000</td><td>\left\|\boldsymbol{B}^{-1}\right\| /\left\|\boldsymbol{A}^{-1}\right\|,\left\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\| /\left\|\boldsymbol{B}^{-1}\right\|</td><td>$\left\|\boldsymbol{B}^{-1}\right\| /\left\|\boldsymbol{A}^{-1}\right\|,\left\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\| /\left\|\boldsymbol{B}^{-1}\right\|$</td><td>We consider the relative errors $\|\boldsymbol{B}^{-1}\|/\|\boldsymbol{A}^{-1}\|$, $\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\|/\|\boldsymbol{B}^{-1}\|$ and $\|\boldsymbol{B}^{-1}-$</td></tr>
<tr><td>lyche-numerical-linear-al F02674</td><td>197</td><td>■ 1.000</td><td>\| \boldsymbol{B}^{-1}-</td><td>$\| \boldsymbol{B}^{-1}-$</td><td>We consider the relative errors $\|\boldsymbol{B}^{-1}\|/\|\boldsymbol{A}^{-1}\|$, $\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\|/\|\boldsymbol{B}^{-1}\|$ and $\|\boldsymbol{B}^{-1}-$</td></tr>
<tr><td>lyche-numerical-linear-al F02675</td><td>197</td><td>■ 1.000</td><td>\boldsymbol{A}^{-1}\|/\| \boldsymbol{A}^{-1} \|</td><td>$\boldsymbol{A}^{-1}\|/\| \boldsymbol{A}^{-1} \|$</td><td>$\boldsymbol{A}^{-1}\|/\|\boldsymbol{A}^{-1}\|$.</td></tr>
<tr><td>lyche-numerical-linear-al F02676</td><td>198</td><td>■ 1.000</td><td>\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{E} \in \mathbb{C}^{n \times n}</td><td>$\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{E} \in \mathbb{C}^{n \times n}$</td><td>gular and let $\boldsymbol{B} := \boldsymbol{A}+\boldsymbol{E} \in \mathbb{C}^{n\times n}$ be nonsingular. For any consistent matrix norm</td></tr>
<tr><td>lyche-numerical-linear-al F02677</td><td>198</td><td>■ 0.998</td><td>K(\boldsymbol{A}):=\|\boldsymbol{A}\|\left\|\boldsymbol{A}^{-1}\right\|</td><td>$K(\boldsymbol{A}):=\|\boldsymbol{A}\|\left\|\boldsymbol{A}^{-1}\right\|$</td><td>where $K(\boldsymbol{A}) := \|\boldsymbol{A}\|\,\|\boldsymbol{A}^{-1}\|$. If $r := \|\boldsymbol{A}^{-1}\boldsymbol{E}\| < 1$ then $\boldsymbol{B}$ is nonsingular and</td></tr>
<tr><td>lyche-numerical-linear-al F02678</td><td>198</td><td>■ 0.922</td><td>\left\|\boldsymbol{A}^{-1} \boldsymbol{E}\right\|</td><td>$\left\|\boldsymbol{A}^{-1} \boldsymbol{E}\right\|$</td><td>We can replace $\|\boldsymbol{A}^{-1}\boldsymbol{E}\|$ by $\|\boldsymbol{E}\boldsymbol{A}^{-1}\|$ everywhere.</td></tr>
<tr><td>lyche-numerical-linear-al F02679</td><td>198</td><td>■ 0.922</td><td>\left\|\boldsymbol{E} \boldsymbol{A}^{-1}\right\|</td><td>$\left\|\boldsymbol{E} \boldsymbol{A}^{-1}\right\|$</td><td>We can replace $\|\boldsymbol{A}^{-1}\boldsymbol{E}\|$ by $\|\boldsymbol{E}\boldsymbol{A}^{-1}\|$ everywhere.</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_F02680</td><td>198</td><td>0.999</td><td>r<1</td><td>$r < 1$</td><td>Proof That $\boldsymbol{B}$ is nonsingular if $r < 1$ follows from Theorem 8.9. We have $-\boldsymbol{E}=$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02681</td><td>198</td><td>0.999</td><td>-\boldsymbol{E}=</td><td>$-\boldsymbol{E}=$</td><td>Proof That $\boldsymbol{B}$ is nonsingular if $r < 1$ follows from Theorem 8.9. We have $-\boldsymbol{E}=$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02682</td><td>198</td><td>1.000</td><td>\boldsymbol{A}-\boldsymbol{B}=\boldsymbol{A}\left(\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right) \boldsymbol{B}=\boldsymbol{B}\left(\boldsymbol{B}^{-1}- \boldsymbol{A}^{-1}\right) \boldsymbol{A}</td><td>$\boldsymbol{A}-\boldsymbol{B}=\boldsymbol{A}\left(\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right) \boldsymbol{B}=\boldsymbol{B}\left(\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right) \boldsymbol{A}$</td><td>$\boldsymbol{A}-\boldsymbol{B}=\boldsymbol{A}(\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1})\boldsymbol{B}=\boldsymbol{B}(\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1})\boldsymbol{A}$ so that</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02683</td><td>198</td><td>1.000</td><td>\left\|\boldsymbol{A}^{-1}\right\| \leq\left\| \boldsymbol{B}^{-1}\right\|+r\left\|\boldsymbol{B}^{-1} \right\|</td><td>$\left\|\boldsymbol{A}^{-1}\right\| \leq\left\|\boldsymbol{B}^{-1}\right\|+r\left\|\boldsymbol{B}^{-1}\right\|$</td><td>Similarly we obtain the lower bound in (8.28) from $\|\boldsymbol{A}^{-1}\| \le \|\boldsymbol{B}^{-1}\|+r\|\boldsymbol{B}^{-1}\|$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02684</td><td>198</td><td>1.000</td><td>\left\|\boldsymbol{B}^{-1}\right\| /\left\| \boldsymbol{A}^{-1}\right\|</td><td>$\left\|\boldsymbol{B}^{-1}\right\| /\left\|\boldsymbol{A}^{-1}\right\|$</td><td>The bound in (8.29) follows by multiplying (8.27) by $\|\boldsymbol{B}^{-1}\|/\|\boldsymbol{A}^{-1}\|$ and using</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02685</td><td>199</td><td>1.000</td><td>1 \leq p \leq \infty, \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n}</td><td>$1 \leq p \leq \infty, \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n}$</td><td>Then for all $1 \le p \le \infty$, $\boldsymbol{x},\boldsymbol{y} \in \mathbb{C}^n$ and all $a \in \mathbb{C}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02686</td><td>199</td><td>1.000</td><td>\|\boldsymbol{x}\|_{p} \geq 0</td><td>$\|\boldsymbol{x}\|_{p} \geq 0$</td><td>1. $\|\boldsymbol{x}\|_p \ge 0$ with equality if and only if $\boldsymbol{x}=\boldsymbol{0}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02687</td><td>199</td><td>1.000</td><td>\|a \boldsymbol{x}\|_{p}=|a|\|\boldsymbol{x}\|_{p}</td><td>$\|a \boldsymbol{x}\|_{p}=|a|\|\boldsymbol{x}\|_{p}$</td><td>2. $\|a\boldsymbol{x}\|_p=|a|\,\|\boldsymbol{x}\|_p$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02688</td><td>199</td><td>1.000</td><td>I \subset \mathbb{R}</td><td>$I \subset \mathbb{R}$</td><td>Definition 8.8 (Convex Function) Let $I \subset \mathbb{R}$ be an interval. A function $f: I \rightarrow \mathbb{R}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02689</td><td>199</td><td>1.000</td><td>f: I \rightarrow \mathbb{R}</td><td>$f: I \rightarrow \mathbb{R}$</td><td>Definition 8.8 (Convex Function) Let $I \subset \mathbb{R}$ be an interval. A function $f: I \rightarrow \mathbb{R}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02690</td><td>199</td><td>1.000</td><td>x_{1}, x_{2} \in I</td><td>$x_{1}, x_{2} \in I$</td><td>for all $x_1,x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in [0,1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02691</td><td>199</td><td>1.000</td><td>x_{1}<x_{2}</td><td>$x_{1} < x_{2}$</td><td>for all $x_1,x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in [0,1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02692</td><td>199</td><td>1.000</td><td>\lambda \in[0,1]</td><td>$\lambda \in[0,1]$</td><td>for all $x_1,x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in [0,1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02693</td><td>199</td><td>1.000</td><td>\sum_{j=1}^{n} \lambda_{j} x_{j}</td><td>$\sum_{j=1}^{n} \lambda_{j} x_{j}$</td><td>for all $x_1,x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in [0,1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02694</td><td>199</td><td>1.000</td><td>x_{1}, \ldots, x_{n}</td><td>$x_{1}, \ldots, x_{n}$</td><td>convex combination of $x_1,\ldots,x_n$ if $\lambda_j \ge 0$ for $j=1,\ldots,n$ and $\sum_{j=1}^n \lambda_j=1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02695</td><td>199</td><td>1.000</td><td>\lambda_{j} \geq 0</td><td>$\lambda_{j} \geq 0$</td><td>convex combination of $x_1,\ldots,x_n$ if $\lambda_j \ge 0$ for $j=1,\ldots,n$ and $\sum_{j=1}^n \lambda_j=1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02696</td><td>199</td><td>1.000</td><td>\sum_{j=1}^{n} \lambda_{j}=1</td><td>$\sum_{j=1}^{n} \lambda_{j}=1$</td><td>convex combination of $x_1,\ldots,x_n$ if $\lambda_j \ge 0$ for $j=1,\ldots,n$ and $\sum_{j=1}^n \lambda_j=1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02697</td><td>200</td><td>1.000</td><td>f \in C^{2}[a, b]</td><td>$f \in C^{2}[a, b]$</td><td>Lemma 8.2 (A Sufficient Condition for Convexity) If $f \in C^2[a,b]$ and $f''(x) \ge$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02698</td><td>200</td><td>1.000</td><td>f^{\prime \prime}(x) \geq</td><td>$f^{\prime \prime}(x) \geq$</td><td>Lemma 8.2 (A Sufficient Condition for Convexity) If $f \in C^2[a,b]$ and $f''(x) \ge$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02699</td><td>200</td><td>1.000</td><td>a \leq x_{1} \leq x \leq x_{2} \leq b</td><td>$a \leq x_{1} \leq x \leq x_{2} \leq b$</td><td>numerical methods) For any $a \le x_1 \le x \le x_2 \le b$ there is a $c \in [x_1,x_2]$ such that</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02700</td><td>200</td><td>1.000</td><td>c \in\left[x_{1}, x_{2}\right]</td><td>$c \in\left[x_{1}, x_{2}\right]$</td><td>numerical methods) For any $a \le x_1 \le x \le x_2 \le b$ there is a $c \in [x_1,x_2]$ such that</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02701</td><td>200</td><td>1.000</td><td>f(x) \leq(1-\lambda) f\left(x_{1}\right)+\lambda f \left(x_{2}\right)</td><td>$f(x) \leq(1-\lambda) f\left(x_{1}\right)+\lambda f\left(x_{2}\right)$</td><td>Since $\lambda \in [0,1]$ we have $f(x) \le (1-\lambda)f(x_1)+\lambda f(x_2)$. Moreover,</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02702</td><td>200</td><td>0.999</td><td>I \in \mathbb{R}</td><td>$I \in \mathbb{R}$</td><td>Theorem 8.13 (Jensen's Inequality) Suppose $I \in \mathbb{R}$ is an interval and $f: I \rightarrow \mathbb{R}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02703</td><td>200</td><td>1.000</td><td>z_{1}, \ldots, z_{n} \in I</td><td>$z_{1}, \ldots, z_{n} \in I$</td><td>$\sum_{j=1}^n \lambda_j=1$, and all $z_1,\ldots,z_n \in I$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02704</td><td>200</td><td>1.000</td><td>n \geq 2</td><td>$n \geq 2$</td><td>Proof We use induction on n. The result is trivial for $n=1$. Let $n \ge 2$, assume</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F02705	200	0.999	`\lambda_{j}, z_{j}`	λ_{j}, z_{j}	the inequality holds for $n-1$, and let λ_j, z_j for $j = 1, \ldots, n$ be given as in the
lyche-numerical-linear-al F02706	200	1.000	`\lambda_{i}<1`	$\lambda_{i}<1$	theorem. Since $n \geq 2$ we have $\lambda_i < 1$ for at least one i so assume without loss of
lyche-numerical-linear-al F02707	200	1.000	`\lambda_{1}<1`	$\lambda_{1}<1$	generality that $\lambda_1 < 1$, and define $u := \sum_{j=2}^{n} \frac{\lambda_j}{1-\lambda_1} z_j$. Since $\sum_{j=2}^{n} \lambda_j = 1 - \lambda_1$
lyche-numerical-linear-al F02708	200	1.000	`u:=\sum_{j=2}^{n} \frac{\lambda_{j}}{1-\lambda_{1}} z_{j}`	$u:=\sum_{j=2}^{n} \frac{\lambda_{j}}{1-\lambda_{1}} z_{j}$	generality that $\lambda_1 < 1$, and define $u := \sum_{j=2}^{n} \frac{\lambda_j}{1-\lambda_1} z_j$. Since $\sum_{j=2}^{n} \lambda_j = 1 - \lambda_1$
lyche-numerical-linear-al F02709	200	1.000	`\sum_{j=2}^{n} \lambda_{j}=1-\lambda_{1}`	$\sum_{j=2}^{n} \lambda_{j}=1-\lambda_{1}$	generality that $\lambda_1 < 1$, and define $u := \sum_{j=2}^{n} \frac{\lambda_j}{1-\lambda_1} z_j$. Since $\sum_{j=2}^{n} \lambda_j = 1 - \lambda_1$
lyche-numerical-linear-al F02710	200	1.000	`f(u) \leq \sum_{j=2}^{n} \frac{\lambda_{j}}{1-\lambda_{1}} f\left(z_{j}\right)`	$f(u) \leq \sum_{j=2}^{n} \frac{\lambda_{j}}{1-\lambda_{1}} f\left(z_{j}\right)$	that $f(u) \leq \sum_{j=2}^{n} \frac{\lambda_j}{1-\lambda_1} f(z_j)$. But then by the convexity of f
lyche-numerical-linear-al F02711	201	1.000	`\sum_{j=1}^{n} \lambda_{j} a_{j}`	$\sum_{j=1}^{n} \lambda_{j} a_{j}$	$\sum_{j=1}^{n} \lambda_j a_j$ *is a convex combination of nonnegative numbers* $a_1, \ldots, a_n$. *Then*
lyche-numerical-linear-al F02712	201	1.000	`a_{1}, \ldots, a_{n}`	$a_{1}, \ldots, a_{n}$	$\sum_{j=1}^{n} \lambda_j a_j$ *is a convex combination of nonnegative numbers* $a_1, \ldots, a_n$. *Then*
lyche-numerical-linear-al F02713	201	1.000	`0^{0}:=0`	$0^{0}:=0$	*where* $0^0 := 0$.
lyche-numerical-linear-al F02714	201	1.000	`a_{j}`	a_{j}	***Proof*** The result is trivial if one or more of the a_j's are zero so assume $a_j > 0$
lyche-numerical-linear-al F02715	201	1.000	`a_{j}>0`	$a_{j}>0$	***Proof*** The result is trivial if one or more of the a_j's are zero so assume $a_j > 0$
lyche-numerical-linear-al F02716	201	1.000	`f:(0, \infty) \rightarrow \mathbb{R}`	$f:(0, \infty) \rightarrow \mathbb{R}$	for all j. Consider the function $f : (0, \infty) \to \mathbb{R}$ given by $f(x) = -\log x$. Since
lyche-numerical-linear-al F02717	201	1.000	`f(x)=-\log x`	$f(x)=-\log x$	for all j. Consider the function $f : (0, \infty) \to \mathbb{R}$ given by $f(x) = -\log x$. Since
lyche-numerical-linear-al F02718	201	1.000	`f^{\prime \prime}(x)=1 / x^{2}>0`	$f^{\prime \prime}(x)=1 / x^{2}>0$	$f''(x) = 1/x^2 > 0$ for $x \in (0, \infty)$, it follows from Lemma 8.2 that this function is
lyche-numerical-linear-al F02719	201	1.000	`x \in(0, \infty)`	$x \in(0, \infty)$	$f''(x) = 1/x^2 > 0$ for $x \in (0, \infty)$, it follows from Lemma 8.2 that this function is
lyche-numerical-linear-al F02720	201	1.000	`\log \left(a_{1}^{\lambda_{1}} \cdots a_{n}^{\lambda_{n}}\right) \leq \log \left(\sum_{j=1}^{n} \lambda_{j} a_{j}\right)`	$\log \left(a_{1}^{\lambda_{1}} \cdots a_{n}^{\lambda_{n}}\right) \leq \log \left(\sum_{j=1}^{n} \lambda_{j} a_{j}\right)$	or $\log\left(a_1^{\lambda_1} \cdots a_n^{\lambda_n}\right) \leq \log\left(\sum_{j=1}^{n} \lambda_j a_j\right)$. The inequality follows since $\exp(\log x)$
lyche-numerical-linear-al F02721	201	1.000	`\exp (\log x)`	$\exp (\log x)$	or $\log\left(a_1^{\lambda_1} \cdots a_n^{\lambda_n}\right) \leq \log\left(\sum_{j=1}^{n} \lambda_j a_j\right)$. The inequality follows since $\exp(\log x)$
lyche-numerical-linear-al F02722	201	1.000	`=x`	$=x$	$= x$ for $x > 0$ and the exponential function is monotone increasing.
lyche-numerical-linear-al F02723	201	1.000	`x>0`	$x>0$	$= x$ for $x > 0$ and the exponential function is monotone increasing.
lyche-numerical-linear-al F02724	201	1.000	`\lambda_{j}=\frac{1}{n}`	$\lambda_{j}=\frac{1}{n}$	Taking $\lambda_j = \frac{1}{n}$ for all j in (8.32) we obtain the classical **geometric/arithmetic**
lyche-numerical-linear-al F02725	201	1.000	`p=\infty`	$p=\infty$	***Proof*** We leave the proof for $p = 1$ and $p = \infty$ as an exercise so assume $1 < p <$
lyche-numerical-linear-al F02726	201	1.000	`1<p<`	$1<p<$	***Proof*** We leave the proof for $p = 1$ and $p = \infty$ as an exercise so assume $1 < p <$
lyche-numerical-linear-al F02727	201	1.000	`a, b \geq 0`	$a, b \geq 0$	∞. For any $a, b \geq 0$ the weighted arithmetic/geometric mean inequality implies
lyche-numerical-linear-al F02728	202	1.000	`(p-1) q=p`	$(p-1) q=p$	relation $(p-1)q = p$ the result is
lyche-numerical-linear-al F02729	202	1.000	`a= \pm 1`	$a= \pm 1$	***Proof*** We set $a = \pm 1$ in (5.5) and add the two equations.

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ F02730</td><td>202</td><td>1.000</td><td>\langle\boldsymbol{x}, \boldsymbol{x}\rangle=\| \boldsymbol{x}\|^{2}</td><td>$\langle \boldsymbol{x}, \boldsymbol{x}\rangle=\|\boldsymbol{x}\|^{2}$</td><td>$\langle \boldsymbol{x}, \boldsymbol{x}\rangle = \|\boldsymbol{x}\|^2$ if and only if the parallelogram identity (8.35) holds for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02731</td><td>202</td><td>1.000</td><td>\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}</td><td>$\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$</td><td>$\langle \boldsymbol{x}, \boldsymbol{x}\rangle = \|\boldsymbol{x}\|^2$ if and only if the parallelogram identity (8.35) holds for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02732</td><td>203</td><td>0.670</td><td>\langle\boldsymbol{y}, \boldsymbol{z}\rangle=0</td><td>$\langle\boldsymbol{y}, \boldsymbol{z}\rangle=0$</td><td>In particular, since $\boldsymbol{y} = \boldsymbol{0}$ implies $\langle \boldsymbol{y}, \boldsymbol{z}\rangle = 0$ we obtain $\langle \boldsymbol{x}, \boldsymbol{z}\rangle = 2\langle \frac{\boldsymbol{x}}{2}, \boldsymbol{z}\rangle$ for all</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02733</td><td>203</td><td>0.670</td><td>\langle\boldsymbol{x}, \boldsymbol{z}\rangle=2\left \langle\frac{\boldsymbol{x}}{2}, \boldsymbol{z}\right \rangle</td><td>$\langle\boldsymbol{x}, \boldsymbol{z}\rangle=2\left\langle\frac{\boldsymbol{x}}{2}, \boldsymbol{z}\right\rangle$</td><td>In particular, since $\boldsymbol{y} = \boldsymbol{0}$ implies $\langle \boldsymbol{y}, \boldsymbol{z}\rangle = 0$ we obtain $\langle \boldsymbol{x}, \boldsymbol{z}\rangle = 2\langle \frac{\boldsymbol{x}}{2}, \boldsymbol{z}\rangle$ for all</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02734</td><td>203</td><td>1.000</td><td>\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}</td><td>$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$</td><td>$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$. This means that $2\langle \frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z}\rangle = \langle \boldsymbol{x}+\boldsymbol{y}, \boldsymbol{z}\rangle$ for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and (8.37)</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02735</td><td>203</td><td>1.000</td><td>2\left\langle\frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z}\right\rangle=\langle\boldsymbol{x}+ \boldsymbol{y}, \boldsymbol{z}\rangle</td><td>$2\left\langle\frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z}\right\rangle=\langle\boldsymbol{x}+\boldsymbol{y}, \boldsymbol{z}\rangle$</td><td>$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$. This means that $2\langle \frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z}\rangle = \langle \boldsymbol{x}+\boldsymbol{y}, \boldsymbol{z}\rangle$ for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and (8.37)</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02736</td><td>203</td><td>1.000</td><td>a=n</td><td>$a=n$</td><td>We first show (8.38) when $a = n$ is a positive integer. By induction</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02737</td><td>203</td><td>1.000</td><td>a>0</td><td>$a>0$</td><td>Now if $a > 0$ there is a sequence $\{a_n\}$ of positive rational numbers converging to a.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02738</td><td>203</td><td>1.000</td><td>\left\{a_{n}\right\}</td><td>$\left\{a_{n}\right\}$</td><td>Now if $a > 0$ there is a sequence $\{a_n\}$ of positive rational numbers converging to a.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02739</td><td>204</td><td>1.000</td><td>a\langle\boldsymbol{x}, \boldsymbol{y}\rangle=\langle a \boldsymbol{x}, \boldsymbol{y}\rangle</td><td>$a\langle\boldsymbol{x}, \boldsymbol{y}\rangle=\langle a\boldsymbol{x}, \boldsymbol{y}\rangle$</td><td>Taking limits and using continuity of norms we obtain $a\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \langle a\boldsymbol{x}, \boldsymbol{y}\rangle$. This also</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02740</td><td>204</td><td>0.651</td><td>a<0</td><td>$a<0$</td><td>holds for $a = 0$. Finally, if $a < 0$ then $(-a) > 0$ and from what we just showed</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02741</td><td>204</td><td>0.651</td><td>(-a)>0</td><td>$(-a)>0$</td><td>holds for $a = 0$. Finally, if $a < 0$ then $(-a) > 0$ and from what we just showed</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02742</td><td>204</td><td>1.000</td><td>\mathcal{V}=\mathbb{C}^{n}, 1 \leq p \leq \infty, n \geq 2</td><td>$\mathcal{V}=\mathbb{C}^{n}, 1 \leq p \leq \infty, n \geq 2$</td><td>on $\mathcal{V} = \mathbb{R}^n$ or $\mathcal{V} = \mathbb{C}^n$, $1 \leq p \leq \infty$, $n \geq 2$, there is an inner product on $\mathcal{V}$ such that</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02743</td><td>204</td><td>1.000</td><td>\langle\boldsymbol{x}, \boldsymbol{x}\rangle=\| \boldsymbol{x}\|_{p}^{2}</td><td>$\langle\boldsymbol{x}, \boldsymbol{x}\rangle=\|\boldsymbol{x}\|_{p}^{2}$</td><td>$\langle \boldsymbol{x}, \boldsymbol{x}\rangle = \|\boldsymbol{x}\|_p^2$ for all $\boldsymbol{x} \in \mathcal{V}$ if and only if $p = 2$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02744</td><td>204</td><td>1.000</td><td>p \neq 2</td><td>$p \neq 2$</td><td>standard inner product. If $p \neq 2$ then the parallelogram identity (8.35) does not</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02745</td><td>204</td><td>1.000</td><td>\boldsymbol{x}:=\boldsymbol{e}_{1}</td><td>$\boldsymbol{x}:=\boldsymbol{e}_{1}$</td><td>hold for say $\boldsymbol{x} := \boldsymbol{e}_1$ and $\boldsymbol{y} := \boldsymbol{e}_2$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02746</td><td>204</td><td>1.000</td><td>\boldsymbol{y}:=\boldsymbol{e}_{2}</td><td>$\boldsymbol{y}:=\boldsymbol{e}_{2}$</td><td>hold for say $\boldsymbol{x} := \boldsymbol{e}_1$ and $\boldsymbol{y} := \boldsymbol{e}_2$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02747</td><td>204</td><td>1.000</td><td>0<\lambda_{n} \leq \cdots \leq \lambda_{1}</td><td>$0<\lambda_{n} \leq \cdots \leq \lambda_{1}$</td><td>metric positive definite matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ with eigenvalues $0 < \lambda_n \leq \cdots \leq \lambda_1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02748</td><td>204</td><td>1.000</td><td>\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{n}</td><td>$\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{n}$</td><td>symmetric and positive definite matrix and assume $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n$ is a basis for $\mathbb{R}^n$. We</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02749</td><td>204</td><td>1.000</td><td>\boldsymbol{B}_{k}:=\left[\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{k}\right] \in \mathbb{R}^{n \times k}</td><td>$\boldsymbol{B}_{k}:=\left[\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{k}\right] \in \mathbb{R}^{n \times k}$</td><td>define $\boldsymbol{B}_k := [\boldsymbol{b}_1, \ldots, \boldsymbol{b}_k] \in \mathbb{R}^{n\times k}$ for $k = 1, \ldots, n$. We consider in this exercise the</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02750</td><td>204</td><td>0.973</td><td>\langle\boldsymbol{x}, \boldsymbol{y}\rangle:= \boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{y}</td><td>$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{y}$</td><td>inner product $\langle\cdot,\cdot\rangle$ defined by $\langle \boldsymbol{x}, \boldsymbol{y}\rangle := \boldsymbol{x}^T\boldsymbol{A}\boldsymbol{y}$ for $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ and the corresponding</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02751</td><td>204</td><td>1.000</td><td>\|\boldsymbol{x}\|_{\boldsymbol{A}}:=\langle \boldsymbol{x}, \boldsymbol{x}\rangle^{1 / 2}</td><td>$\|\boldsymbol{x}\|_{\boldsymbol{A}}:=\langle\boldsymbol{x}, \boldsymbol{x}\rangle^{1 / 2}$</td><td>norm $\|\boldsymbol{x}\|_{\boldsymbol{A}} := \langle \boldsymbol{x}, \boldsymbol{x}\rangle^{1/2}$. We define $\tilde{\boldsymbol{b}}_1 := \boldsymbol{b}_1$ and</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02752</td><td>204</td><td>1.000</td><td>\tilde{\boldsymbol{b}}_{1}:=\boldsymbol{b}_{1}</td><td>$\tilde{\boldsymbol{b}}_{1}:=\boldsymbol{b}_{1}$</td><td>norm $\|\boldsymbol{x}\|_{\boldsymbol{A}} := \langle \boldsymbol{x}, \boldsymbol{x}\rangle^{1/2}$. We define $\tilde{\boldsymbol{b}}_1 := \boldsymbol{b}_1$ and</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02753</td><td>204</td><td>1.000</td><td>\boldsymbol{B}_{k}^{T} \boldsymbol{A} \boldsymbol{B}_{k}</td><td>$\boldsymbol{B}_{k}^{T} \boldsymbol{A} \boldsymbol{B}_{k}$</td><td>a) Show that $\boldsymbol{B}_k^T\boldsymbol{A}\boldsymbol{B}_k$ is positive definite for $k = 1, \ldots, n$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02754</td><td>204</td><td>0.999</td><td>\left\langle\tilde{\boldsymbol{b}}_{k}, \boldsymbol{b}_{j}\right\rangle=0</td><td>$\left\langle\tilde{\boldsymbol{b}}_{k}, \boldsymbol{b}_{j}\right\rangle=0$</td><td>b) Show that for $k = 2, \ldots, n$ we have (i) $\langle \tilde{\boldsymbol{b}}_k, \boldsymbol{b}_j\rangle = 0$ for $j = 1, \ldots, k-1$ and</td></tr>
<tr><td>lyche-numerical-linear-algebra_ F02755</td><td>204</td><td>0.999</td><td>j=1, \ldots, k-1</td><td>$j=1, \ldots, k-1$</td><td>b) Show that for $k = 2, \ldots, n$ we have (i) $\langle \tilde{\boldsymbol{b}}_k, \boldsymbol{b}_j\rangle = 0$ for $j = 1, \ldots, k-1$ and</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_F02756</td><td>204</td><td>1.000</td><td>\tilde{\boldsymbol{b}}_{k}-\boldsymbol{b}_{k} \in \operatorname{span}\left(\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{k-1}\right)</td><td>$\tilde{\boldsymbol{b}}_{k}-\boldsymbol{b}_{k} \in \operatorname{span}\left(\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{k-1}\right)$</td><td>(ii) $\tilde{\boldsymbol{b}}_k - \boldsymbol{b}_k \in \operatorname{span}(\boldsymbol{b}_1, \ldots, \boldsymbol{b}_{k-1})$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02757</td><td>205</td><td>0.985</td><td>\tilde{\boldsymbol{b}}_{1}, \ldots, \tilde{ \boldsymbol{b}}_{n}</td><td>$\tilde{\boldsymbol{b}}_{1}, \ldots, \tilde{\boldsymbol{b}}_{n}$</td><td>c) Explain why $\tilde{\boldsymbol{b}}_1, \ldots, \tilde{\boldsymbol{b}}_n$ is a basis for $\mathbb{R}^n$ which in addition is $\boldsymbol{A}$-orthogonal, i.e.,</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02758</td><td>205</td><td>0.984</td><td>\left\langle\tilde{\boldsymbol{b}}_{i}, \tilde{ \boldsymbol{b}}_{j}\right\rangle=0</td><td>$\left\langle\tilde{\boldsymbol{b}}_{i}, \tilde{\boldsymbol{b}}_{j}\right\rangle=0$</td><td>$\langle\tilde{\boldsymbol{b}}_i, \tilde{\boldsymbol{b}}_j\rangle = 0$ for all $i, j \le n, i \ne j$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02759</td><td>205</td><td>0.984</td><td>i, j \leq n, i \neq j</td><td>$i, j \leq n, i \neq j$</td><td>$\langle\tilde{\boldsymbol{b}}_i, \tilde{\boldsymbol{b}}_j\rangle = 0$ for all $i, j \le n, i \ne j$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02760</td><td>205</td><td>1.000</td><td>\tilde{\boldsymbol{B}}_{n}:=\left[\tilde{ \boldsymbol{b}}_{1}, \ldots, \tilde{\boldsymbol{b}}_{n} \right]</td><td>$\tilde{\boldsymbol{B}}_{n}:=\left[\tilde{\boldsymbol{b}}_{1}, \ldots, \tilde{\boldsymbol{b}}_{n}\right]$</td><td>d) Define $\tilde{\boldsymbol{B}}_n := [\tilde{\boldsymbol{b}}_1, \ldots, \tilde{\boldsymbol{b}}_n]$. Show that there is an upper triangular matrix $\boldsymbol{T} \in$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02761</td><td>205</td><td>1.000</td><td>\boldsymbol{T} \in</td><td>$\boldsymbol{T} \in$</td><td>d) Define $\tilde{\boldsymbol{B}}_n := [\tilde{\boldsymbol{b}}_1, \ldots, \tilde{\boldsymbol{b}}_n]$. Show that there is an upper triangular matrix $\boldsymbol{T} \in$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02762</td><td>205</td><td>1.000</td><td>\boldsymbol{B}_{n}=\tilde{\boldsymbol{B}}_{n} \boldsymbol{T}</td><td>$\boldsymbol{B}_{n}=\tilde{\boldsymbol{B}}_{n} \boldsymbol{T}$</td><td>$\mathbb{R}^{n\times n}$ with ones on the diagonal and satisfies $\boldsymbol{B}_n = \tilde{\boldsymbol{B}}_n\boldsymbol{T}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02763</td><td>205</td><td>1.000</td><td>\left|t_{i j}\right| \leq \frac{1}{2}</td><td>$\left|t_{i j}\right| \leq \frac{1}{2}$</td><td>e) Assume that the matrix $\boldsymbol{T}$ in d) is such that $|t_{ij}| \le \frac{1}{2}$ for all $i, j \le n, i \ne j$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02764</td><td>205</td><td>0.947</td><td>\left\|\tilde{\boldsymbol{b}}_{k}\right\|_{ \boldsymbol{A}}^{2} \leq 2\left\|\tilde{ \boldsymbol{b}}_{k+1}\right\|_{\boldsymbol{A}}^{2}</td><td>$\left\|\tilde{\boldsymbol{b}}_{k}\right\|_{\boldsymbol{A}}^{2} \leq 2\left\|\tilde{\boldsymbol{b}}_{k+1}\right\|_{\boldsymbol{A}}^{2}$</td><td>Assume also that $\|\tilde{\boldsymbol{b}}_k\|_{\boldsymbol{A}}^2 \le 2\|\tilde{\boldsymbol{b}}_{k+1}\|_{\boldsymbol{A}}^2$ for $k = 1, \ldots, n-1$ and that $\det(\boldsymbol{B}_n) = 1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02765</td><td>205</td><td>0.947</td><td>\operatorname{det}\left(\boldsymbol{B}_{n}\right)=1</td><td>$\operatorname{det}\left(\boldsymbol{B}_{n}\right)=1$</td><td>Assume also that $\|\tilde{\boldsymbol{b}}_k\|_{\boldsymbol{A}}^2 \le 2\|\tilde{\boldsymbol{b}}_{k+1}\|_{\boldsymbol{A}}^2$ for $k = 1, \ldots, n-1$ and that $\det(\boldsymbol{B}_n) = 1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02766</td><td>205</td><td>0.948</td><td>\left[\begin{array}{ll}1 & 1 \\ 1 & 1\end{array}\right]</td><td>$\left[\begin{array}{ll}1 & 1 \\ 1 & 1\end{array}\right]$</td><td>consistent by considering $\left[\begin{smallmatrix}1 & 1\\ 1 & 1\end{smallmatrix}\right]$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02767</td><td>205</td><td>1.000</td><td>\sqrt{m n}</td><td>$\sqrt{m n}$</td><td>b) Show that the constant $\sqrt{mn}$ can be replaced by m and by n.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02768</td><td>205</td><td>0.985</td><td>\|\boldsymbol{A} \boldsymbol{x}\|_{\infty} \leq</td><td>$\|\boldsymbol{A} \boldsymbol{x}\|_{\infty} \leq$</td><td>a) Show that the max norm is subordinate to the ∞ and 1 norm, i.e., $\|\boldsymbol{A}\boldsymbol{x}\|_\infty \le$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02769</td><td>205</td><td>1.000</td><td>\|\boldsymbol{A}\|_{M}\|\boldsymbol{x}\|_{1}</td><td>$\|\boldsymbol{A}\|_{M}\|\boldsymbol{x}\|_{1}$</td><td>$\|\boldsymbol{A}\|_M\|\boldsymbol{x}\|_1$ holds for all $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ and all $\boldsymbol{x} \in \mathbb{C}^n$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02770</td><td>205</td><td>1.000</td><td>\|\boldsymbol{A}\|_{M}=\left|a_{k l}\right|</td><td>$\|\boldsymbol{A}\|_{M}=\left|a_{k l}\right|$</td><td>b) Show that if $\|\boldsymbol{A}\|_M = |a_{kl}|$, then $\|\boldsymbol{A}\boldsymbol{e}_l\|_\infty = \|\boldsymbol{A}\|_M\|\boldsymbol{e}_l\|_1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02771</td><td>205</td><td>1.000</td><td>\left\|\boldsymbol{A} \boldsymbol{e}_{l}\right\|_{ \infty}=\|\boldsymbol{A}\|_{M}\left\|\boldsymbol{e}_{l} \right\|_{1}</td><td>$\left\|\boldsymbol{A} \boldsymbol{e}_{l}\right\|_{\infty}=\|\boldsymbol{A}\|_{M}\left\|\boldsymbol{e}_{l}\right\|_{1}$</td><td>b) Show that if $\|\boldsymbol{A}\|_M = |a_{kl}|$, then $\|\boldsymbol{A}\boldsymbol{e}_l\|_\infty = \|\boldsymbol{A}\|_M\|\boldsymbol{e}_l\|_1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02772</td><td>205</td><td>1.000</td><td>\|\boldsymbol{A}\|_{M}=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{A} \boldsymbol{x}\|_{ \infty}}{\|\boldsymbol{x}\|_{1}}</td><td>$\|\boldsymbol{A}\|_{M}=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{A} \boldsymbol{x}\|_{\infty}}{\|\boldsymbol{x}\|_{1}}$</td><td>c) Show that $\|\boldsymbol{A}\|_M = \max_{\boldsymbol{x}\ne\mathbf{0}} \frac{\|\boldsymbol{A}\boldsymbol{x}\|_\infty}{\|\boldsymbol{x}\|_1}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02773</td><td>205</td><td>0.480</td><td>\left\|\tilde{\boldsymbol{b}}_{1}\right\|_{ \boldsymbol{A}}^{2} \cdots\left\|\tilde{ \boldsymbol{b}}_{n}\right\|_{\boldsymbol{A}}^{2}= \operatorname{det}(\boldsymbol{A})</td><td>$\left\|\tilde{\boldsymbol{b}}_{1}\right\|_{\boldsymbol{A}}^{2} \cdots\left\|\tilde{\boldsymbol{b}}_{n}\right\|_{\boldsymbol{A}}^{2}=\operatorname{det}(\boldsymbol{A})$</td><td>[2]Hint: Show that $\|\tilde{\boldsymbol{b}}_1\|_{\boldsymbol{A}}^2 \cdots \|\tilde{\boldsymbol{b}}_n\|_{\boldsymbol{A}}^2 = \det(\boldsymbol{A})$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02774</td><td>206</td><td>0.991</td><td>\|\boldsymbol{A} \boldsymbol{x}\|_{2} \geq \sigma_{n}</td><td>$\|\boldsymbol{A} \boldsymbol{x}\|_{2} \geq \sigma_{n}$</td><td>Show that $\|\boldsymbol{A}\boldsymbol{x}\|_2 \ge \sigma_n$ for all $\boldsymbol{x} \in \mathbb{C}^n$ with $\|\boldsymbol{x}\|_2 = 1$. Show that</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02775</td><td>206</td><td>1.000</td><td>\|\boldsymbol{A}\|_{p}</td><td>$\|\boldsymbol{A}\|_{p}$</td><td>Compute $\|\boldsymbol{A}\|_p$ and $\|\boldsymbol{A}^{-1}\|_p$ for $p = 1, 2, \infty$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02776</td><td>206</td><td>1.000</td><td>\left\|\boldsymbol{A}^{-1}\right\|_{p}</td><td>$\left\|\boldsymbol{A}^{-1}\right\|_{p}$</td><td>Compute $\|\boldsymbol{A}\|_p$ and $\|\boldsymbol{A}^{-1}\|_p$ for $p = 1, 2, \infty$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02777</td><td>206</td><td>1.000</td><td>\|\boldsymbol{V} \boldsymbol{A}\|_{2}=</td><td>$\|\boldsymbol{V} \boldsymbol{A}\|_{2}=$</td><td>Exercise 8.11 (Unitary Invariance of the Spectral Norm) Show that $\|\boldsymbol{V}\boldsymbol{A}\|_2 =$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02778</td><td>206</td><td>1.000</td><td>\boldsymbol{V}^{*} \boldsymbol{V}=\boldsymbol{I}</td><td>$\boldsymbol{V}^{*} \boldsymbol{V}=\boldsymbol{I}$</td><td>$\|\boldsymbol{A}\|_2$ holds even for a rectangular $\boldsymbol{V}$ as long as $\boldsymbol{V}^*\boldsymbol{V} = \boldsymbol{I}$.</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-al F02779</td><td>206</td><td>1.000</td><td>8.12\left(\|\boldsymbol{A} \boldsymbol{U}\|_{2}\right.</td><td>$8.12\left(\|\boldsymbol{A} \boldsymbol{U}\|_{2}\right.$</td><td>Exercise 8.12 ($\|\boldsymbol{AU}\|_2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2\times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2\times 1}$ with</td></tr>
<tr><td>lyche-numerical-linear-al F02780</td><td>206</td><td>1.000</td><td>\left.\boldsymbol{A}\right)</td><td>$\left.\boldsymbol{A}\right)$</td><td>Exercise 8.12 ($\|\boldsymbol{AU}\|_2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2\times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2\times 1}$ with</td></tr>
<tr><td>lyche-numerical-linear-al F02781</td><td>206</td><td>1.000</td><td>\boldsymbol{A} \in \mathbb{R}^{2 \times 2}</td><td>$\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$</td><td>Exercise 8.12 ($\|\boldsymbol{AU}\|_2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2\times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2\times 1}$ with</td></tr>
<tr><td>lyche-numerical-linear-al F02782</td><td>206</td><td>1.000</td><td>\boldsymbol{U} \in \mathbb{R}^{2 \times 1}</td><td>$\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$</td><td>Exercise 8.12 ($\|\boldsymbol{AU}\|_2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2\times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2\times 1}$ with</td></tr>
<tr><td>lyche-numerical-linear-al F02783</td><td>206</td><td>1.000</td><td>\|\boldsymbol{A} \boldsymbol{U}\|_{2}<\|\boldsymbol{A} \|_{2}</td><td>$\|\boldsymbol{A} \boldsymbol{U}\|_{2}<\|\boldsymbol{A} \|_{2}$</td><td>$\boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}$ such that $\|\boldsymbol{AU}\|_2 < \|\boldsymbol{A}\|_2$. Thus, in general, $\|\boldsymbol{AU}\|_2 = \|\boldsymbol{A}\|_2$ does</td></tr>
<tr><td>lyche-numerical-linear-al F02784</td><td>206</td><td>1.000</td><td>\|\boldsymbol{A} \boldsymbol{U}\|_{2}=\|\boldsymbol{A} \|_{2}</td><td>$\|\boldsymbol{A} \boldsymbol{U}\|_{2}=\|\boldsymbol{A} \|_{2}$</td><td>$\boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}$ such that $\|\boldsymbol{AU}\|_2 < \|\boldsymbol{A}\|_2$. Thus, in general, $\|\boldsymbol{AU}\|_2 = \|\boldsymbol{A}\|_2$ does</td></tr>
<tr><td>lyche-numerical-linear-al F02785</td><td>206</td><td>0.999</td><td>\|\boldsymbol{A}\|_{p}=\rho(\boldsymbol{A}):=</td><td>$\|\boldsymbol{A}\|_{p}=\rho(\boldsymbol{A}):=$</td><td>Exercise 8.13 ($p$-Norm of Diagonal Matrix) Show that $\|\boldsymbol{A}\|_p = \rho(\boldsymbol{A}) :=$</td></tr>
<tr><td>lyche-numerical-linear-al F02786</td><td>206</td><td>0.972</td><td>\max \left|\lambda_{i}\right|</td><td>$\max \left|\lambda_{i}\right|$</td><td>$\max|\lambda_i|$ (the largest eigenvalue of $\boldsymbol{A}$), $1 \le p \le \infty$, when $\boldsymbol{A}$ is a diagonal matrix.</td></tr>
<tr><td>lyche-numerical-linear-al F02787</td><td>206</td><td>1.000</td><td>\boldsymbol{a} \in \mathbb{C}^{m}</td><td>$\boldsymbol{a} \in \mathbb{C}^{m}$</td><td>Exercise 8.14 (Spectral Norm of a Column Vector) A vector $\boldsymbol{a} \in \mathbb{C}^m$ can also be</td></tr>
<tr><td>lyche-numerical-linear-al F02788</td><td>206</td><td>1.000</td><td>\boldsymbol{A} \in \mathbb{C}^{m, 1}</td><td>$\boldsymbol{A} \in \mathbb{C}^{m, 1}$</td><td>considered as a matrix $\boldsymbol{A} \in \mathbb{C}^{m,1}$.</td></tr>
<tr><td>lyche-numerical-linear-al F02789</td><td>206</td><td>1.000</td><td>\|\boldsymbol{A}\|_{p}=\|\boldsymbol{a}\|_{p}</td><td>$\|\boldsymbol{A}\|_{p}=\|\boldsymbol{a}\|_{p}$</td><td>b) Show that $\|\boldsymbol{A}\|_p = \|\boldsymbol{a}\|_p$ for $1 \le p \le \infty$.</td></tr>
<tr><td>lyche-numerical-linear-al F02790</td><td>206</td><td>1.000</td><td>|\boldsymbol{A}| \in \mathbb{R}^{m \times n}</td><td>$|\boldsymbol{A}| \in \mathbb{R}^{m \times n}$</td><td>let $|\boldsymbol{A}| \in \mathbb{R}^{m\times n}$ be the matrix with elements $|a_{ij}|$.</td></tr>
<tr><td>lyche-numerical-linear-al F02791</td><td>206</td><td>1.000</td><td>\left|a_{i j}\right|</td><td>$\left|a_{i j}\right|$</td><td>let $|\boldsymbol{A}| \in \mathbb{R}^{m\times n}$ be the matrix with elements $|a_{ij}|$.</td></tr>
<tr><td>lyche-numerical-linear-al F02792</td><td>206</td><td>0.999</td><td>|\boldsymbol{A}|</td><td>$|\boldsymbol{A}|$</td><td>a) Compute $|\boldsymbol{A}|$ if $\boldsymbol{A} = \begin{bmatrix} 1+i & -2 \\ 1 & 1-i \end{bmatrix}, \quad i = \sqrt{-1}$.</td></tr>
<tr><td>lyche-numerical-linear-al F02793</td><td>206</td><td>0.999</td><td>\boldsymbol{A}=\left[\begin{array}{cc}1+i & -2 \\ 1 & 1-i \end{array}\right], \quad i=\sqrt{-1}</td><td>$\boldsymbol{A}=\left[\begin{array}{cc}1+i & -2 \\ 1 & 1-i \end{array}\right], \quad i=\sqrt{-1}$</td><td>a) Compute $|\boldsymbol{A}|$ if $\boldsymbol{A} = \begin{bmatrix} 1+i & -2 \\ 1 & 1-i \end{bmatrix}, \quad i = \sqrt{-1}$.</td></tr>
<tr><td>lyche-numerical-linear-al F02794</td><td>206</td><td>1.000</td><td>\boldsymbol{A} \in \mathbb{C}^{m \times n}\| \boldsymbol{A}\|_{F}=\||\boldsymbol{A}|\|_{F},\| \boldsymbol{A}\|_{p}=\||\boldsymbol{A}|\|_{p}</td><td>$\boldsymbol{A} \in \mathbb{C}^{m \times n}\| \boldsymbol{A}\|_{F}=\||\boldsymbol{A}|\|_{F},\| \boldsymbol{A}\|_{p}=\||\boldsymbol{A}|\|_{p}$</td><td>b) Show that for any $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ $\|\boldsymbol{A}\|_F = \|\,|\boldsymbol{A}|\,\|_F$, $\|\boldsymbol{A}\|_p = \|\,|\boldsymbol{A}|\,\|_p$ for $p = 1, \infty$.</td></tr>
<tr><td>lyche-numerical-linear-al F02795</td><td>206</td><td>1.000</td><td>p=1, \infty</td><td>$p=1, \infty$</td><td>b) Show that for any $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ $\|\boldsymbol{A}\|_F = \|\,|\boldsymbol{A}|\,\|_F$, $\|\boldsymbol{A}\|_p = \|\,|\boldsymbol{A}|\,\|_p$ for $p = 1, \infty$.</td></tr>
<tr><td>lyche-numerical-linear-al F02796</td><td>206</td><td>1.000</td><td>\boldsymbol{A} \in \mathbb{C}^{m \times n}\| \boldsymbol{A}\|_{2} \leq\||\boldsymbol{A}|\|_{2}</td><td>$\boldsymbol{A} \in \mathbb{C}^{m \times n}\| \boldsymbol{A}\|_{2} \leq\||\boldsymbol{A}|\|_{2}$</td><td>c) Show that for any $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ $\|\boldsymbol{A}\|_2 \le \|\,|\boldsymbol{A}|\,\|_2$.</td></tr>
<tr><td>lyche-numerical-linear-al F02797</td><td>206</td><td>0.806</td><td>\|\boldsymbol{A}\|_{2}<\||\boldsymbol{A}|\|_{2}</td><td>$\|\boldsymbol{A}\|_{2}<\||\boldsymbol{A}|\|_{2}$</td><td>d) Find a real symmetric 2×2 matrix $\boldsymbol{A}$ such that $\|\boldsymbol{A}\|_2 < \|\,|\boldsymbol{A}|\,\|_2$.</td></tr>
<tr><td>lyche-numerical-linear-al F02798</td><td>206</td><td>1.000</td><td>\left\{\lambda_{1}, \lambda_{2}, \ldots, \lambda_{m} \right\}</td><td>$\left\{\lambda_{1}, \lambda_{2}, \ldots, \lambda_{m} \right\}$</td><td>$\{\lambda_1, \lambda_2, \ldots, \lambda_m\}$, in no particular order. We have that $m \le n$, since some of the</td></tr>
<tr><td>lyche-numerical-linear-al F02799</td><td>206</td><td>1.000</td><td>\boldsymbol{x}_{0} \in \mathbb{C}^{n}</td><td>$\boldsymbol{x}_{0} \in \mathbb{C}^{n}$</td><td>eigenvalues may have multiplicity larger than one. Given $\boldsymbol{x}_0 \in \mathbb{C}^n$, and $k \ge 0$, we</td></tr>
<tr><td>lyche-numerical-linear-al F02800</td><td>207</td><td>1.000</td><td>\left\{\boldsymbol{x}_{k}\right\}_{k=0}^{m-1}</td><td>$\left\{\boldsymbol{x}_{k}\right\}_{k=0}^{m-1}$</td><td>define the sequence $\{\boldsymbol{x}_k\}_{k=0}^{m-1}$ by</td></tr>
<tr><td>lyche-numerical-linear-al F02801</td><td>207</td><td>1.000</td><td>c_{i k}</td><td>$c_{i k}$</td><td>a) Let the coefficients c_{ik} be defined by</td></tr>
<tr><td>lyche-numerical-linear-al F02802</td><td>207</td><td>1.000</td><td>\left\{\left(\sigma_{i}, \boldsymbol{u}_{i}\right)\right \}_{i=1}^{n}</td><td>$\left\{\left(\sigma_{i}, \boldsymbol{u}_{i}\right)\right \}_{i=1}^{n}$</td><td>where $\{(\sigma_i, \boldsymbol{u}_i)\}_{i=1}^n$ are the eigenpairs of $\boldsymbol{A}$. Show that</td></tr>
<tr><td>lyche-numerical-linear-al F02803</td><td>207</td><td>1.000</td><td>l \leq m</td><td>$l \leq m$</td><td>b) Show that for some $l \le m$, we have that $\boldsymbol{x}_l = \boldsymbol{x}_{l+1} = \cdots = \boldsymbol{x}_m = \boldsymbol{x}$, where</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_F02804</td><td>207</td><td>1.000</td><td>\boldsymbol{x}_{l}=\boldsymbol{x}_{l+1}=\cdots= \boldsymbol{x}_{m}=\boldsymbol{x}</td><td>$\boldsymbol{x}_{l}=\boldsymbol{x}_{l+1}=\cdots=\boldsymbol{x}_{m}=\boldsymbol{x}$</td><td>b) Show that for some $l \leq m$, we have that $\boldsymbol{x}_l = \boldsymbol{x}_{l+1} = \cdots = \boldsymbol{x}_m = \boldsymbol{x}$, where</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02805</td><td>207</td><td>1.000</td><td>\boldsymbol{T}=\operatorname{tridiag}(c, d, c)</td><td>$\boldsymbol{T}=\operatorname{tridiag}(c, d, c)$</td><td>c) Consider this iteration for the $n \times n$ matrix $\boldsymbol{T} = \operatorname{tridiag}(c, d, c)$, where d and c</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02806</td><td>207</td><td>1.000</td><td>d>2 c</td><td>$d>2 c$</td><td>are positive real numbers and $d > 2c$. The eigenvalues of $\boldsymbol{T}$ are</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02807</td><td>207</td><td>1.000</td><td>\boldsymbol{T} \boldsymbol{x}=\boldsymbol{b}</td><td>$\boldsymbol{T} \boldsymbol{x}=\boldsymbol{b}$</td><td>What is the operation count for solving $\boldsymbol{T}\boldsymbol{x} = \boldsymbol{b}$ using the iterative algorithm</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02808</td><td>207</td><td>1.000</td><td>b_{i j}=0</td><td>$b_{i j}=0$</td><td>$b_{ij} = 0$ if $|i-j| \leq 1$. Set $\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$, where $\boldsymbol{T}$ is the tridiagonal matrix above.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02809</td><td>207</td><td>1.000</td><td>|i-j| \leq 1</td><td>$|i-j| \leq 1$</td><td>$b_{ij} = 0$ if $|i-j| \leq 1$. Set $\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$, where $\boldsymbol{T}$ is the tridiagonal matrix above.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02810</td><td>207</td><td>1.000</td><td>\boldsymbol{A}=\boldsymbol{T}+\boldsymbol{B}</td><td>$\boldsymbol{A}=\boldsymbol{T}+\boldsymbol{B}$</td><td>$b_{ij} = 0$ if $|i-j| \leq 1$. Set $\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$, where $\boldsymbol{T}$ is the tridiagonal matrix above.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02811</td><td>207</td><td>0.991</td><td>\rho(\boldsymbol{E}):=\max _{i}\left|\lambda_{i}\right|</td><td>$\rho(\boldsymbol{E}):=\max _{i}\left|\lambda_{i}\right|$</td><td>Recall that if $\boldsymbol{E} \in \mathbb{R}^{n \times n}$ has eigenvalues $\lambda_1, \ldots, \lambda_n$ then $\rho(\boldsymbol{E}) := \max_i |\lambda_i|$ is</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02812</td><td>207</td><td>1.000</td><td>\rho\left(\boldsymbol{T}^{-1} \boldsymbol{B}\right) \leq \rho\left(\boldsymbol{T}^{-1}\right) \rho(\boldsymbol{B})</td><td>$\rho\left(\boldsymbol{T}^{-1} \boldsymbol{B}\right) \leq \rho\left(\boldsymbol{T}^{-1}\right) \rho(\boldsymbol{B})$</td><td>the spectral radius of $\boldsymbol{E}$. Show that $\rho(\boldsymbol{T}^{-1}\boldsymbol{B}) \leq \rho(\boldsymbol{T}^{-1})\rho(\boldsymbol{B})$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02813</td><td>208</td><td>1.000</td><td>\delta:=\|\boldsymbol{x}-\boldsymbol{y}\|_{2} /\| \boldsymbol{x}\|_{2}</td><td>$\delta:=\|\boldsymbol{x}-\boldsymbol{y}\|_{2} /\| \boldsymbol{x}\|_{2}$</td><td>We define $\delta := \|\boldsymbol{x} - \boldsymbol{y}\|_2 / \|\boldsymbol{x}\|_2$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02814</td><td>208</td><td>1.000</td><td>K_{2}(\boldsymbol{A})=\|\boldsymbol{A}\|_{2}\left\| \boldsymbol{A}^{-1}\right\|_{2}</td><td>$K_{2}(\boldsymbol{A})=\|\boldsymbol{A}\|_{2}\left\| \boldsymbol{A}^{-1}\right\|_{2}$</td><td>a) Determine $K_2(\boldsymbol{A}) = \|\boldsymbol{A}\|_2 \|\boldsymbol{A}^{-1}\|_2$. Give an upper bound and a positive lower</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02815</td><td>208</td><td>1.000</td><td>\delta</td><td>δ</td><td>bound for δ without computing $\boldsymbol{x}$ and $\boldsymbol{y}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02816</td><td>208</td><td>1.000</td><td>b_{2}=2.0</td><td>$b_{2}=2.0$</td><td>b) Suppose as before that $b_2 = 2.0$ and $\|\boldsymbol{e}\|_2 = 0.1$. Determine b_1 and $\boldsymbol{e}$ which</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02817</td><td>208</td><td>1.000</td><td>\|\boldsymbol{e}\|_{2}=0.1</td><td>$\|\boldsymbol{e}\|_{2}=0.1$</td><td>b) Suppose as before that $b_2 = 2.0$ and $\|\boldsymbol{e}\|_2 = 0.1$. Determine b_1 and $\boldsymbol{e}$ which</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02818</td><td>208</td><td>1.000</td><td>b_{1}</td><td>b_{1}</td><td>b) Suppose as before that $b_2 = 2.0$ and $\|\boldsymbol{e}\|_2 = 0.1$. Determine b_1 and $\boldsymbol{e}$ which</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02819</td><td>208</td><td>1.000</td><td>\boldsymbol{y}_{\boldsymbol{A}}</td><td>$\boldsymbol{y}_{\boldsymbol{A}}$</td><td>special choices of $\boldsymbol{b}$. Suppose $\boldsymbol{y}_A$ and $\boldsymbol{y}_{A^{-1}}$ are vectors with $\|\boldsymbol{y}_A\| = \|\boldsymbol{y}_{A^{-1}}\| = 1$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02820</td><td>208</td><td>1.000</td><td>\boldsymbol{y}_{\boldsymbol{A}^{-1}}</td><td>$\boldsymbol{y}_{\boldsymbol{A}^{-1}}$</td><td>special choices of $\boldsymbol{b}$. Suppose $\boldsymbol{y}_A$ and $\boldsymbol{y}_{A^{-1}}$ are vectors with $\|\boldsymbol{y}_A\| = \|\boldsymbol{y}_{A^{-1}}\| = 1$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02821</td><td>208</td><td>1.000</td><td>\left\|\boldsymbol{y}_{\boldsymbol{A}}\right\|=\left\| \boldsymbol{y}_{\boldsymbol{A}^{-1}}\right\|=1</td><td>$\left\|\boldsymbol{y}_{\boldsymbol{A}}\right\|=\left\| \boldsymbol{y}_{\boldsymbol{A}^{-1}}\right\|=1$</td><td>special choices of $\boldsymbol{b}$. Suppose $\boldsymbol{y}_A$ and $\boldsymbol{y}_{A^{-1}}$ are vectors with $\|\boldsymbol{y}_A\| = \|\boldsymbol{y}_{A^{-1}}\| = 1$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02822</td><td>208</td><td>0.968</td><td>\|\boldsymbol{A}\|=\left\|\boldsymbol{A} \boldsymbol{y}_{ \boldsymbol{A}}\right\|</td><td>$\|\boldsymbol{A}\|=\left\|\boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}\right\|$</td><td>and $\|\boldsymbol{A}\| = \|\boldsymbol{A}\boldsymbol{y}_A\|$ and $\|\boldsymbol{A}^{-1}\| = \|\boldsymbol{A}^{-1}\boldsymbol{y}_{A^{-1}}\|$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02823</td><td>208</td><td>0.968</td><td>\left\|\boldsymbol{A}^{-1}\right\|=\left\| \boldsymbol{A}^{-1} \boldsymbol{y}_{\boldsymbol{A}^{-1}} \right\|</td><td>$\left\|\boldsymbol{A}^{-1}\right\|=\left\|\boldsymbol{A}^{-1} \boldsymbol{y}_{\boldsymbol{A}^{-1}}\right\|$</td><td>and $\|\boldsymbol{A}\| = \|\boldsymbol{A}\boldsymbol{y}_A\|$ and $\|\boldsymbol{A}^{-1}\| = \|\boldsymbol{A}^{-1}\boldsymbol{y}_{A^{-1}}\|$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02824</td><td>208</td><td>1.000</td><td>\boldsymbol{b}=\boldsymbol{A} \boldsymbol{y}_{ \boldsymbol{A}}</td><td>$\boldsymbol{b}=\boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}$</td><td>a) Show that the upper bound in (8.22) is attained if $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{y}_A$ and $\boldsymbol{e} = \boldsymbol{y}_{A^{-1}}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02825</td><td>208</td><td>1.000</td><td>\boldsymbol{e}=\boldsymbol{y}_{\boldsymbol{A}^{-1}}</td><td>$\boldsymbol{e}=\boldsymbol{y}_{\boldsymbol{A}^{-1}}$</td><td>a) Show that the upper bound in (8.22) is attained if $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{y}_A$ and $\boldsymbol{e} = \boldsymbol{y}_{A^{-1}}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02826</td><td>208</td><td>0.980</td><td>\boldsymbol{b}=\boldsymbol{y}_{\boldsymbol{A}^{-1}}</td><td>$\boldsymbol{b}=\boldsymbol{y}_{\boldsymbol{A}^{-1}}$</td><td>b) Show that the lower bound is attained if $\boldsymbol{b} = \boldsymbol{y}_{A^{-1}}$ and $\boldsymbol{e} = \boldsymbol{A}\boldsymbol{y}_A$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02827</td><td>208</td><td>0.980</td><td>\boldsymbol{e}=\boldsymbol{A} \boldsymbol{y}_{ \boldsymbol{A}}</td><td>$\boldsymbol{e}=\boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}$</td><td>b) Show that the lower bound is attained if $\boldsymbol{b} = \boldsymbol{y}_{A^{-1}}$ and $\boldsymbol{e} = \boldsymbol{A}\boldsymbol{y}_A$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02828</td><td>208</td><td>1.000</td><td>m \geq 1</td><td>$m \geq 1$</td><td>will show that for $m \geq 1$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02829</td><td>208</td><td>1.000</td><td>\boldsymbol{T}:=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}</td><td>$\boldsymbol{T}:=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}$</td><td>where $\boldsymbol{T} := \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$ and $\operatorname{cond}_p(\boldsymbol{T}) := \|\boldsymbol{T}\|_p \|\boldsymbol{T}^{-1}\|_p$ is the p-</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_F02830</td><td>208</td><td>1.000</td><td>\operatorname{cond}_{p}(\boldsymbol{T}):=\|\boldsymbol{T}\|_{p}\left\|\boldsymbol{T}^{-1}\right\|_{p}</td><td>$\operatorname{cond}_{p}(\boldsymbol{T}):=\|\boldsymbol{T}\|_{p}\left\|\boldsymbol{T}^{-1}\right\|_{p}$</td><td>where $\boldsymbol{T}:=\operatorname{tridiag}(-1,2,-1)\in\mathbb{R}^{m\times m}$ and $\operatorname{cond}_p(\boldsymbol{T}):=\|\boldsymbol{T}\|_p\|\boldsymbol{T}^{-1}\|_p$ is the p-</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02831</td><td>208</td><td>1.000</td><td>m \geq 3</td><td>$m \geq 3$</td><td>a) Show that for $m\geq 3$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02832</td><td>208</td><td>1.000</td><td>\operatorname{cond}_{1}(\boldsymbol{T})= \operatorname{cond}_{\infty}(\boldsymbol{T})=3</td><td>$\operatorname{cond}_{1}(\boldsymbol{T})=\operatorname{cond}_{\infty}(\boldsymbol{T})=3$</td><td>and that $\operatorname{cond}_1(\boldsymbol{T})=\operatorname{cond}_\infty(\boldsymbol{T})=3$ for $m=2$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02833</td><td>208</td><td>1.000</td><td>m=2</td><td>$m=2$</td><td>and that $\operatorname{cond}_1(\boldsymbol{T})=\operatorname{cond}_\infty(\boldsymbol{T})=3$ for $m=2$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02834</td><td>209</td><td>1.000</td><td>\tan x>x</td><td>$\tan x>x$</td><td>Hint: For the upper bound use the inequality $\tan x>x$ valid for $0<x<\pi/2$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02835</td><td>209</td><td>1.000</td><td>0<x<\pi / 2</td><td>$0<x<\pi/2$</td><td>Hint: For the upper bound use the inequality $\tan x>x$ valid for $0<x<\pi/2$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02836</td><td>209</td><td>1.000</td><td>\cot ^{2} x>\frac{1}{x^{2}}-\frac{2}{3}</td><td>$\cot^{2} x>\frac{1}{x^{2}}-\frac{2}{3}$</td><td>For the lower bound we use (without proof) the inequality $\cot^2 x>\frac{1}{x^2}-\frac{2}{3}$ for</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02837</td><td>209</td><td>1.000</td><td>\boldsymbol{I}-\boldsymbol{E}</td><td>$\boldsymbol{I}-\boldsymbol{E}$</td><td>a) Show that if $\boldsymbol{I}-\boldsymbol{E}$ is nonsingular then</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02838</td><td>209</td><td>1.000</td><td>\|\boldsymbol{E}\|<1</td><td>$\|\boldsymbol{E}\|<1$</td><td>b) If $\|\boldsymbol{E}\|<1$ then $(\boldsymbol{I}-\boldsymbol{E})^{-1}$ is nonsingular by exists and</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02839</td><td>209</td><td>1.000</td><td>(\boldsymbol{I}-\boldsymbol{E})^{-1}</td><td>$(\boldsymbol{I}-\boldsymbol{E})^{-1}$</td><td>b) If $\|\boldsymbol{E}\|<1$ then $(\boldsymbol{I}-\boldsymbol{E})^{-1}$ is nonsingular by exists and</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02840</td><td>209</td><td>1.000</td><td>\boldsymbol{x}=\left[x_{0}, \ldots, x_{n}\right]^{T} \in \mathbb{R}^{n+1}</td><td>$\boldsymbol{x}=\left[x_{0},\ldots,x_{n}\right]^{T}\in\mathbb{R}^{n+1}$</td><td>components of $\boldsymbol{x}=[x_0,\ldots,x_n]^T\in\mathbb{R}^{n+1}$ define a partition of the interval $[a,b]$,</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02841</td><td>210</td><td>0.999</td><td>\boldsymbol{y}:=\left[y_{0}, \ldots, y_{n}\right]^{T} \in \mathbb{R}^{n+1}</td><td>$\boldsymbol{y}:=\left[y_{0},\ldots,y_{n}\right]^{T}\in\mathbb{R}^{n+1}$</td><td>and given a dataset $\boldsymbol{y}:=[y_0,\ldots,y_n]^T\in\mathbb{R}^{n+1}$, where we assume $y_0=y_n$. The</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02842</td><td>210</td><td>0.999</td><td>y_{0}=y_{n}</td><td>$y_{0}=y_{n}$</td><td>and given a dataset $\boldsymbol{y}:=[y_0,\ldots,y_n]^T\in\mathbb{R}^{n+1}$, where we assume $y_0=y_n$. The</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02843</td><td>210</td><td>0.902</td><td>\left(x_{i-1}, x_{i}\right)</td><td>$\left(x_{i-1},x_{i}\right)$</td><td>(Recall that g is a cubic polynomial on each interval (x_{i-1},x_i), for $i=1,\ldots,n$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02844</td><td>210</td><td>1.000</td><td>C^{2}[a, b]</td><td>$C^{2}[a,b]$</td><td>with smoothness $C^2[a,b]$.)</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02845</td><td>210</td><td>0.664</td><td>s_{i}:=g^{\prime}\left(x_{i}\right), i=0, \ldots, n</td><td>$s_{i}:=g^{\prime}\left(x_{i}\right), i=0,\ldots,n$</td><td>We define $s_i:=g'(x_i)$, $i=0,\ldots,n$. It can be shown that the vector $\boldsymbol{s}:=$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02846</td><td>210</td><td>1.000</td><td>\left[s_{1}, \ldots, s_{n}\right]^{T}</td><td>$\left[s_{1},\ldots,s_{n}\right]^{T}$</td><td>$[s_1,\ldots,s_n]^T$ is determined from a linear system</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02847</td><td>211</td><td>1.000</td><td>\left\|\boldsymbol{A}^{-1}\right\|_{\infty} \leq 1</td><td>$\left\|\boldsymbol{A}^{-1}\right\|_{\infty}\leq 1$</td><td>b) Show that $\|\boldsymbol{A}^{-1}\|_\infty\leq 1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02848</td><td>211</td><td>1.000</td><td>\left\|\boldsymbol{A}^{-1}\right\|_{1} \leq \frac{3}{2}</td><td>$\left\|\boldsymbol{A}^{-1}\right\|_{1}\leq\frac{3}{2}$</td><td>c) Show that $\|\boldsymbol{A}^{-1}\|_1\leq\frac{3}{2}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02849</td><td>211</td><td>1.000</td><td>\boldsymbol{e} \in \mathbb{R}^{n}</td><td>$\boldsymbol{e}\in\mathbb{R}^{n}$</td><td>d) Let $\boldsymbol{s}$ and $\boldsymbol{b}$ be as in (8.44), where we assume $\boldsymbol{b}\neq\boldsymbol{0}$. Let $\boldsymbol{e}\in\mathbb{R}^n$ be such that</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02850</td><td>211</td><td>1.000</td><td>\|\boldsymbol{e}\|_{p} /\|\boldsymbol{b}\|_{p} \leq 0.01</td><td>$\|\boldsymbol{e}\|_{p}/\|\boldsymbol{b}\|_{p}\leq 0.01$</td><td>$\|\boldsymbol{e}\|_p/\|\boldsymbol{b}\|_p\leq 0.01$. Suppose $\hat{\boldsymbol{s}}$ satisfies</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02851</td><td>211</td><td>1.000</td><td>\hat{\boldsymbol{s}}</td><td>$\hat{\boldsymbol{s}}$</td><td>$\|\boldsymbol{e}\|_p/\|\boldsymbol{b}\|_p\leq 0.01$. Suppose $\hat{\boldsymbol{s}}$ satisfies</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02852</td><td>211</td><td>1.000</td><td>\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{b} \in \mathbb{R}^{m}</td><td>$\boldsymbol{A}\in\mathbb{R}^{m\times n},\boldsymbol{b}\in\mathbb{R}^{m}$</td><td>$\boldsymbol{A}\in\mathbb{R}^{m\times n}$, $\boldsymbol{b}\in\mathbb{R}^m$, where $\boldsymbol{A}$ has rank n and let $\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ be a singular</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02853</td><td>211</td><td>1.000</td><td>\boldsymbol{U} \in \mathbb{R}^{m \times n}</td><td>$\boldsymbol{U}\in\mathbb{R}^{m\times n}$</td><td>value factorization of $\boldsymbol{A}$. Thus $\boldsymbol{U}\in\mathbb{R}^{m\times n}$ and $\boldsymbol{\Sigma},\boldsymbol{V}\in\mathbb{R}^{n\times n}$. Write a MATLAB</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02854</td><td>211</td><td>1.000</td><td>\boldsymbol{\Sigma}, \boldsymbol{V} \in \mathbb{R}^{n \times n}</td><td>$\boldsymbol{\Sigma},\boldsymbol{V}\in\mathbb{R}^{n\times n}$</td><td>value factorization of $\boldsymbol{A}$. Thus $\boldsymbol{U}\in\mathbb{R}^{m\times n}$ and $\boldsymbol{\Sigma},\boldsymbol{V}\in\mathbb{R}^{n\times n}$. Write a MATLAB</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02855</td><td>211</td><td>0.932</td><td>[\mathrm{x}, \mathrm{K}]=l \mathrm{sq}(\mathrm{A}, \mathrm{b})</td><td>$[\mathrm{x},\mathrm{K}]=l\mathrm{sq}(\mathrm{A},\mathrm{b})$</td><td><code>function [x,K]=lsq(A,b)</code> that uses the singular value factorization of $\boldsymbol{A}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F02856</td><td>211</td><td>1.000</td><td>\boldsymbol{x}=\boldsymbol{V} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}^{T} \boldsymbol{b}</td><td>$\boldsymbol{x}=\boldsymbol{V}\boldsymbol{\Sigma}^{-1}\boldsymbol{U}^{T}\boldsymbol{b}$</td><td>to calculate a least squares solution $\boldsymbol{x}=\boldsymbol{V}\boldsymbol{\Sigma}^{-1}\boldsymbol{U}^T\boldsymbol{b}$ to the system $\boldsymbol{A}\boldsymbol{x}=$</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F02857	211	1.000	`\boldsymbol{A} \boldsymbol{x}=`	$\boldsymbol{A} \boldsymbol{x}=$	to calculate a least squares solution $\boldsymbol{x} = \boldsymbol{V}\boldsymbol{\Sigma}^{-1}\boldsymbol{U}^T\boldsymbol{b}$ to the system $\boldsymbol{A}\boldsymbol{x} =$
lyche-numerical-linear-algebra_F02858	211	0.909	`[\mathrm{U}, \mathrm{Sigma}, \mathrm{V}]= \operatorname{svd}(\mathrm{A}, 0)`	$[\mathrm{U}, \mathrm{Sigma}, \mathrm{V}]= \operatorname{svd}(\mathrm{A}, 0)$	`[U,Sigma,V]=svd(A,0)` computes the singular value factorization of $\boldsymbol{A}$.
lyche-numerical-linear-algebra_F02859	211	0.998	`|\cdot| p`	$\|\cdot\| p$	**Exercise 8.25** (**p Norm for $p = 1$ and $p = \infty$**) Show that $\|\cdot\|p$ is a vector norm
lyche-numerical-linear-algebra_F02860	211	1.000	`p=1, p=\infty`	$p=1, p=\infty$	in $\mathbb{R}^n$ for $p = 1$, $p = \infty$.
lyche-numerical-linear-algebra_F02861	211	0.992	`\langle\boldsymbol{x}, \boldsymbol{y}\rangle=s( \boldsymbol{x}, \boldsymbol{y})+i s(\boldsymbol{x}, i \boldsymbol{y})`	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle=s(\boldsymbol{x}, \boldsymbol{y})+i s(\boldsymbol{x}, i \boldsymbol{y})$	[4]Hint: We have $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = s(\boldsymbol{x}, \boldsymbol{y}) + is(\boldsymbol{x}, i\boldsymbol{y})$, where $s(\boldsymbol{x}, \boldsymbol{y}) := \frac{1}{4}(\|\boldsymbol{x}+\boldsymbol{y}\|^2 - \|\boldsymbol{x}-\boldsymbol{y}\|^2)$.
lyche-numerical-linear-algebra_F02862	211	0.992	`s(\boldsymbol{x}, \boldsymbol{y}):=\frac{1}{4}\left(| \boldsymbol{x}+\boldsymbol{y}|^{2}-|\boldsymbol{x}- \boldsymbol{y}|^{2}\right)`	$s(\boldsymbol{x}, \boldsymbol{y}):=\frac{1}{4}\left(\|\boldsymbol{x}+\boldsymbol{y}\|^{2}-\|\boldsymbol{x}-\boldsymbol{y}\|^{2}\right)$	[4]Hint: We have $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = s(\boldsymbol{x}, \boldsymbol{y}) + is(\boldsymbol{x}, i\boldsymbol{y})$, where $s(\boldsymbol{x}, \boldsymbol{y}) := \frac{1}{4}(\|\boldsymbol{x}+\boldsymbol{y}\|^2 - \|\boldsymbol{x}-\boldsymbol{y}\|^2)$.
lyche-numerical-linear-algebra_F02863	212	1.000	`S_{p}`	S_{p}	is called the unit sphere in $\mathbb{R}^n$ with respect to p. Draw S_p for $p = 1, 2, \infty$ for $n = 2$.
lyche-numerical-linear-algebra_F02864	212	1.000	`|\boldsymbol{x}|_{\infty} \leq|\boldsymbol{x}|_{p} \leq n^{1 / p}|\boldsymbol{x}|_{\infty}`	$\|\boldsymbol{x}\|_{\infty} \leq\|\boldsymbol{x}\|_{p} \leq n^{1 / p}\|\boldsymbol{x}\|_{\infty}$	have $\|\boldsymbol{x}\|_\infty \le \|\boldsymbol{x}\|_p \le n^{1/p}\|\boldsymbol{x}\|_\infty$ (cf. (8.5)).
lyche-numerical-linear-algebra_F02865	212	1.000	`\boldsymbol{x}_{l}`	$\boldsymbol{x}_{l}$	Produce a vector $\boldsymbol{x}_l$ such that $\|\boldsymbol{x}_l\|_\infty = \|\boldsymbol{x}_l\|_p$ and another vector $\boldsymbol{x}_u$ such that
lyche-numerical-linear-algebra_F02866	212	1.000	`\left|\boldsymbol{x}_{l}\right|_{\infty}=\left| \boldsymbol{x}_{l}\right|_{p}`	$\left\|\boldsymbol{x}_{l}\right\|_{\infty}=\left\|\boldsymbol{x}_{l}\right\|_{p}$	Produce a vector $\boldsymbol{x}_l$ such that $\|\boldsymbol{x}_l\|_\infty = \|\boldsymbol{x}_l\|_p$ and another vector $\boldsymbol{x}_u$ such that
lyche-numerical-linear-algebra_F02867	212	1.000	`\boldsymbol{x}_{u}`	$\boldsymbol{x}_{u}$	Produce a vector $\boldsymbol{x}_l$ such that $\|\boldsymbol{x}_l\|_\infty = \|\boldsymbol{x}_l\|_p$ and another vector $\boldsymbol{x}_u$ such that
lyche-numerical-linear-algebra_F02868	212	1.000	`\left|\boldsymbol{x}_{u}\right|_{p}=n^{1 / p}\left| \boldsymbol{x}_{u}\right|_{\infty}`	$\left\|\boldsymbol{x}_{u}\right\|_{p}=n^{1 / p}\left\|\boldsymbol{x}_{u}\right\|_{\infty}$	$\|\boldsymbol{x}_u\|_p = n^{1/p}\|\boldsymbol{x}_u\|_\infty$. Thus, these inequalities are sharp.
lyche-numerical-linear-algebra_F02869	212	1.000	`1 \leq q \leq p \leq \infty`	$1 \leq q \leq p \leq \infty$	**Exercise 8.28** (**p-Norm Inequalities for Arbitrary p**) If $1 \le q \le p \le \infty$ then
lyche-numerical-linear-algebra_F02870	212	1.000	`f(z)=z^{p / q}`	$f(z)=z^{p / q}$	$f(z) = z^{p/q}$ and $z_i = \|x_i\|^q$. For the left inequality consider first $y_i = x_i/\|\boldsymbol{x}\|_\infty$,
lyche-numerical-linear-algebra_F02871	212	1.000	`z_{i}=\left|x_{i}\right|^{q}`	$z_{i}=\left\|x_{i}\right\|^{q}$	$f(z) = z^{p/q}$ and $z_i = \|x_i\|^q$. For the left inequality consider first $y_i = x_i/\|\boldsymbol{x}\|_\infty$,
lyche-numerical-linear-algebra_F02872	212	1.000	`y_{i}=x_{i} /|\boldsymbol{x}|_{\infty}`	$y_{i}=x_{i} /\|\boldsymbol{x}\|_{\infty}$	$f(z) = z^{p/q}$ and $z_i = \|x_i\|^q$. For the left inequality consider first $y_i = x_i/\|\boldsymbol{x}\|_\infty$,
lyche-numerical-linear-algebra_F02873	212	0.551	`| \boldsymbol{A} \boldsymbol{x}|\leq| \boldsymbol{A} \left|_{F}\right| \boldsymbol{x} |`	$\| \boldsymbol{A} \boldsymbol{x}\|\leq\| \boldsymbol{A} \left\|_{F}\right\| \boldsymbol{x} \|$	8.6.2 Does there exist a vector norm $\|\ \|$ such that $\|\boldsymbol{A}\boldsymbol{x}\| \le \|\boldsymbol{A}\|_F\|\boldsymbol{x}\|$ for all $\boldsymbol{A} \in$
lyche-numerical-linear-algebra_F02874	212	0.980	`\mathbb{C}^{n \times n}, \boldsymbol{x} \in \mathbb{C}^{n}, m, n \in \mathbb{N}`	$\mathbb{C}^{n \times n}, \boldsymbol{x} \in \mathbb{C}^{n}, m, n \in \mathbb{N}$	$\mathbb{C}^{n\times n}, \boldsymbol{x} \in \mathbb{C}^n, m, n \in \mathbb{N}$?
lyche-numerical-linear-algebra_F02875	212	1.000	`|\boldsymbol{A}|_{2} \leq|\boldsymbol{A}|_{F}`	$\|\boldsymbol{A}\|_{2} \leq\|\boldsymbol{A}\|_{F}$	8.6.3 Why is $\|\boldsymbol{A}\|_2 \le \|\boldsymbol{A}\|_F$ for any matrix $\boldsymbol{A}$?
lyche-numerical-linear-algebra_F02876	213	1.000	`\boldsymbol{b} \notin \mathcal{R}(\boldsymbol{A})`	$\boldsymbol{b} \notin \mathcal{R}(\boldsymbol{A})$	either case the system can only be solved approximately if $\boldsymbol{b} \notin \mathcal{R}(\boldsymbol{A})$, the column
lyche-numerical-linear-algebra_F02877	213	0.999	`|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}|`	$\|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\|$	$\|\cdot\|$, say a p-norm, and look for $\boldsymbol{x} \in \mathbb{C}^n$ which minimizes $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|$. The use of
lyche-numerical-linear-algebra_F02878	213	1.000	`\boldsymbol{b} \in \mathbb{C}^{m}`	$\boldsymbol{b} \in \mathbb{C}^{m}$	and $\boldsymbol{b} \in \mathbb{C}^m$. To find $\boldsymbol{x} \in \mathbb{C}^n$ that minimizes $E : \mathbb{C}^n \to \mathbb{R}$ given by
lyche-numerical-linear-algebra_F02879	213	1.000	`E: \mathbb{C}^{n} \rightarrow \mathbb{R}`	$E: \mathbb{C}^{n} \rightarrow \mathbb{R}$	and $\boldsymbol{b} \in \mathbb{C}^m$. To find $\boldsymbol{x} \in \mathbb{C}^n$ that minimizes $E : \mathbb{C}^n \to \mathbb{R}$ given by
lyche-numerical-linear-algebra_F02880	213	1.000	`E(\boldsymbol{x})`	$E(\boldsymbol{x})$	Since the square root function is monotone, minimizing $E(\boldsymbol{x})$ or $\sqrt{E(\boldsymbol{x})}$ is
lyche-numerical-linear-algebra_F02881	213	1.000	`\sqrt{E(\boldsymbol{x})}`	$\sqrt{E(\boldsymbol{x})}$	Since the square root function is monotone, minimizing $E(\boldsymbol{x})$ or $\sqrt{E(\boldsymbol{x})}$ is
lyche-numerical-linear-algebra_F02882	214	1.000	`6 x_{1}-8=0`	$6 x_{1}-8=0$	Setting the first derivative with respect to x_1 equal to zero we obtain $6x_1 - 8 = 0$ or

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02883	214	■ 1.000	`x_{1}=4 / 3`	$x_{1}=4 / 3$	$x_1 = 4/3$, the average of b_1, b_2, b_3. The second derivative is positive and $x_1 = 4/3$
lyche-numerical-linear-algebra F02884	214	■ 1.000	`b_{1}, b_{2}, b_{3}`	b_{1}, b_{2}, b_{3}	$x_1 = 4/3$, the average of b_1, b_2, b_3. The second derivative is positive and $x_1 = 4/3$
lyche-numerical-linear-algebra F02885	214	■ 1.000	`\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}= \boldsymbol{A}^{*} \boldsymbol{b}`	$\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}= \boldsymbol{A}^{*} \boldsymbol{b}$	*if and only if* $\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}^*\boldsymbol{b}$.
lyche-numerical-linear-algebra F02886	214	■ 1.000	`p(t)=x_{1}+x_{2} t`	$p(t)=x_{1}+x_{2} t$	*Example 9.2 (Linear Regression)* We want to fit a straight line $p(t) = x_1 + x_2 t$ to
lyche-numerical-linear-algebra F02887	214	■ 1.000	`\left(t_{k}, y_{k}\right) \in \mathbb{R}^{2}, k=1, \ldots, m`	$\left(t_{k}, y_{k}\right) \in \mathbb{R}^{2}, k=1, \ldots, m$	$m \geq 2$ given data $(t_k, y_k) \in \mathbb{R}^2$, $k = 1, \ldots, m$. This is part of the linear regression
lyche-numerical-linear-algebra F02888	214	■ 1.000	`m>2`	$m>2$	This is square for $m = 2$ and overdetermined for $m > 2$. The matrix $\boldsymbol{A}$ has linearly
lyche-numerical-linear-algebra F02889	214	■ 1.000	`\left\{t_{1}, \ldots, t_{m}\right\}`	$\left\{t_{1}, \ldots, t_{m}\right\}$	independent columns if and only if the set $\{t_1, \ldots, t_m\}$ of sites contains at least two
lyche-numerical-linear-algebra F02890	214	■ 1.000	`t_{i} \neq t_{j}`	$t_{i} \neq t_{j}$	distinct elements. For if say $t_i \neq t_j$ then
lyche-numerical-linear-algebra F02891	215	■ 1.000	`t_{1}=\cdots=t_{m}`	$t_{1}=\cdots=t_{m}$	Conversely, if $t_1 = \cdots = t_m$ then the columns of $\boldsymbol{A}$ are linearly dependent. The
lyche-numerical-linear-algebra F02892	215	■ 0.970	`t_{1} \neq t_{2}`	$t_{1} \neq t_{2}$	$m = 2$ and $t_1 \neq t_2$ then both systems $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ and $\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}^*\boldsymbol{b}$ are square, and p
lyche-numerical-linear-algebra F02893	215	■ 1.000	`p\left(t_{k}\right)=y_{k}, k=1,2`	$p\left(t_{k}\right)=y_{k}, k=1,2$	is the linear interpolant to the data. Indeed, p is linear and $p(t_k) = y_k$, $k = 1, 2$.
lyche-numerical-linear-algebra F02894	215	■ 1.000	`\left[\begin{array}{cc}4 & 10 \\ 10 & 30\end{array} \right]\left[\begin{array}{l}x_{1} \\ x_{2}\end{array} \right]=\left[\begin{array}{c}6 \\ 10.1\end{array}\right]`	$\left[\begin{array}{cc}4 & 10 \\ 10 & 30\end{array} \right]\left[\begin{array}{l}x_{1} \\ x_{2}\end{array} \right]=\left[\begin{array}{c}6 \\ 10.1\end{array}\right]$	the normal equations become $\begin{bmatrix}4 & 10\\ 10 & 30\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} = \begin{bmatrix}6\\ 10.1\end{bmatrix}$. The data and the least
lyche-numerical-linear-algebra F02895	215	■ 1.000	`p(t)=x_{1}+x_{2} t=3.95-0.98 t`	$p(t)=x_{1}+x_{2} t=3.95-0.98 t$	squares polynomial $p(t) = x_1 + x_2 t = 3.95 - 0.98t$ are shown in Fig. 9.1.
lyche-numerical-linear-algebra F02896	215	■ 1.000	`y \in \mathbb{R}`	$y \in \mathbb{R}$	To every input $\boldsymbol{u} \in \mathbb{R}^n$ we obtain an output $y \in \mathbb{R}$. Assuming we have a linear
lyche-numerical-linear-algebra F02897	216	■ 0.994	`\min \\|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b} \\|_{2}^{2}`	$\min \\|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\\|_{2}^{2}$	We can estimate $\boldsymbol{x}$ by solving the least squares problem $\min \\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\\|_2^2$.
lyche-numerical-linear-algebra F02898	216	■ 1.000	`1 \leq n \leq m`	$1 \leq n \leq m$	• size: $1 \leq n \leq m$,
lyche-numerical-linear-algebra F02899	216	■ 1.000	`\mathcal{S}:=\left\{t_{1}, t_{2}, \ldots, t_{m}\right\} \subset[a, b]`	$\mathcal{S}:=\left\{t_{1}, t_{2}, \ldots, t_{m}\right\} \subset[a, b]$	• sites: $\mathcal{S} := \{t_1, t_2, \ldots, t_m\} \subset [a, b]$,
lyche-numerical-linear-algebra F02900	216	■ 1.000	`\boldsymbol{y}=\left[y_{1}, y_{2}, \ldots, y_{m} \right]^{*} \in \mathbb{R}^{m}`	$\boldsymbol{y}=\left[y_{1}, y_{2}, \ldots, y_{m} \right]^{*} \in \mathbb{R}^{m}$	• y-values: $\boldsymbol{y} = [y_1, y_2, \ldots, y_m]^* \in \mathbb{R}^m$,
lyche-numerical-linear-algebra F02901	216	■ 1.000	`\phi_{j}:[a, b] \rightarrow \mathbb{R}, j=1, \ldots, n`	$\phi_{j}:[a, b] \rightarrow \mathbb{R}, j=1, \ldots, n$	• functions: $\phi_j : [a, b] \to \mathbb{R}$, $j = 1, \ldots, n$.
lyche-numerical-linear-algebra F02902	216	■ 0.999	`p:[a, b] \rightarrow \mathbb{R}`	$p:[a, b] \rightarrow \mathbb{R}$	Find a function (curve fit) $p : [a, b] \to \mathbb{R}$ given by $p := \sum_{j=1}^n x_j \phi_j$ such that
lyche-numerical-linear-algebra F02903	216	■ 0.999	`p:=\sum_{j=1}^{n} x_{j} \phi_{j}`	$p:=\sum_{j=1}^{n} x_{j} \phi_{j}$	Find a function (curve fit) $p : [a, b] \to \mathbb{R}$ given by $p := \sum_{j=1}^n x_j \phi_j$ such that
lyche-numerical-linear-algebra F02904	216	■ 1.000	`p\left(t_{k}\right) \approx y_{k}`	$p\left(t_{k}\right) \approx y_{k}$	$p(t_k) \approx y_k$ for $k = 1, \ldots, m$.
lyche-numerical-linear-algebra F02905	217	■ 1.000	`\phi_{j}`	ϕ_{j}	Typical examples of functions ϕ_j are polynomials, trigonometric functions, expo-
lyche-numerical-linear-algebra F02906	217	■ 1.000	`w_{k}>0`	$w_{k}>0$	In (9.2) one can also include **weights** $w_k > 0$ for $k = 1, \ldots, m$ and minimize
lyche-numerical-linear-algebra F02907	217	■ 1.000	`y_{k}`	y_{k}	If y_k is an accurate observation, we can choose a large weight w_k. This will force
lyche-numerical-linear-algebra F02908	217	■ 1.000	`w_{k}`	w_{k}	If y_k is an accurate observation, we can choose a large weight w_k. This will force

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ F02909	217	1.000	`p\left(t_{k}\right)-y_{k}`	$p\left(t_{k}\right)-y_{k}$	$p(t_k) - y_k$ to be small. Similarly, a small w_k will allow $p(t_k) - y_k$ to be large. If
lyche-numerical-linear-algebra_ F02910	217	1.000	`\delta y_{k}`	δy_{k}	an estimate for the standard deviation δy_k in y_k is known for each k, we can choose
lyche-numerical-linear-algebra_ F02911	217	1.000	`w_{k}=1 /\left(\delta y_{k}\right)^{2}, k=1,2, \ldots, m`	$w_{k}=1 /\left(\delta y_{k}\right)^{2}, k=1,2, \ldots, m$	$w_k = 1/(\delta y_k)^2$, $k = 1, 2, \ldots, m$. For simplicity we will assume in the following
lyche-numerical-linear-algebra_ F02912	217	1.000	`w_{k}=1`	$w_{k}=1$	that $w_k = 1$ for all k.
lyche-numerical-linear-algebra_ F02913	217	1.000	`4.2 \boldsymbol{A}^{*} \boldsymbol{A}`	$4.2 \boldsymbol{A}^{*} \boldsymbol{A}$	***Proof*** By Lemma 4.2 $\boldsymbol{A}^*\boldsymbol{A}$ is positive definite if and only if $\boldsymbol{A}$ has linearly
lyche-numerical-linear-algebra_ F02914	217	0.989	`(\boldsymbol{A} \boldsymbol{x})_{k}=\sum_{j=1}^{n} x_{j} \phi_{j}\left(t_{k}\right), k=1, \ldots, m`	$(\boldsymbol{A} \boldsymbol{x})_{k}=\sum_{j=1}^{n} x_{j} \phi_{j}\left(t_{k}\right), k=1, \ldots, m$	independent columns. Since $(\boldsymbol{A}\boldsymbol{x})_k = \sum_{j=1}^n x_j\phi_j(t_k)$, $k = 1, \ldots, m$ this is
lyche-numerical-linear-algebra_ F02915	217	1.000	`\phi_{j}(t):=t^{j-1}`	$\phi_{j}(t):=t^{j-1}$	mials $\phi_j(t) := t^{j-1}$, for $j = 1, \ldots, n$ and equidistant sites $t_k = (k-1)/(m-1)$
lyche-numerical-linear-algebra_ F02916	217	1.000	`t_{k}=(k-1) /(m-1)`	$t_{k}=(k-1) /(m-1)$	mials $\phi_j(t) := t^{j-1}$, for $j = 1, \ldots, n$ and equidistant sites $t_k = (k-1)/(m-1)$
lyche-numerical-linear-algebra_ F02917	217	1.000	`\boldsymbol{B}_{n} \boldsymbol{x}=\boldsymbol{c}_{n}`	$\boldsymbol{B}_{n} \boldsymbol{x}=\boldsymbol{c}_{n}$	for $k = 1, \ldots, m$. The normal equations are $\boldsymbol{B}_n\boldsymbol{x} = \boldsymbol{c}_n$, where for $n = 3$
lyche-numerical-linear-algebra_ F02918	217	1.000	`\boldsymbol{B}_{n}`	$\boldsymbol{B}_{n}$	$\boldsymbol{B}_n$ is positive definite if at least n of the t's are distinct. However $\boldsymbol{B}_n$ is extremely
lyche-numerical-linear-algebra_ F02919	217	1.000	`t`	t	$\boldsymbol{B}_n$ is positive definite if at least n of the t's are distinct. However $\boldsymbol{B}_n$ is extremely
lyche-numerical-linear-algebra_ F02920	217	1.000	`\frac{1}{m} \boldsymbol{B}_{n} \approx \boldsymbol{H}_{n}`	$\frac{1}{m} \boldsymbol{B}_{n} \approx \boldsymbol{H}_{n}$	ill-conditioned even for moderate n. Indeed, $\frac{1}{m}\boldsymbol{B}_n \approx \boldsymbol{H}_n$, where $\boldsymbol{H}_n \in \mathbb{R}^{n\times n}$ is the
lyche-numerical-linear-algebra_ F02921	217	1.000	`\boldsymbol{H}_{n} \in \mathbb{R}^{n \times n}`	$\boldsymbol{H}_{n} \in \mathbb{R}^{n \times n}$	ill-conditioned even for moderate n. Indeed, $\frac{1}{m}\boldsymbol{B}_n \approx \boldsymbol{H}_n$, where $\boldsymbol{H}_n \in \mathbb{R}^{n\times n}$ is the
lyche-numerical-linear-algebra_ F02922	217	1.000	`1 /(i+j-1)`	$1 /(i+j-1)$	**Hilbert Matrix** with i, j element $1/(i+j-1)$. Thus, for $n = 3$
lyche-numerical-linear-algebra_ F02923	218	1.000	`\frac{1}{m} \boldsymbol{B}_{n}`	$\frac{1}{m} \boldsymbol{B}_{n}$	The elements of $\frac{1}{m}\boldsymbol{B}_n$ are Riemann sums approximations to the elements of $\boldsymbol{H}_n$. In
lyche-numerical-linear-algebra_ F02924	218	1.000	`\boldsymbol{H}_{n}`	$\boldsymbol{H}_{n}$	The elements of $\frac{1}{m}\boldsymbol{B}_n$ are Riemann sums approximations to the elements of $\boldsymbol{H}_n$. In
lyche-numerical-linear-algebra_ F02925	218	1.000	`\boldsymbol{B}_{n}=\left[b_{i, j}\right]_{i, j=1}^{n}`	$\boldsymbol{B}_{n}=\left[b_{i, j}\right]_{i, j=1}^{n}$	fact, if $\boldsymbol{B}_n = [b_{i,j}]_{i,j=1}^n$ then
lyche-numerical-linear-algebra_ F02926	218	1.000	`\boldsymbol{H}_{n}^{-1}`	$\boldsymbol{H}_{n}^{-1}$	The elements of $\boldsymbol{H}_n^{-1}$ are determined in Exercise 1.13. We find $K_1(\boldsymbol{H}_6)$:=
lyche-numerical-linear-algebra_ F02927	218	1.000	`K_{1}\left(\boldsymbol{H}_{6}\right):=`	$K_{1}\left(\boldsymbol{H}_{6}\right):=$	The elements of $\boldsymbol{H}_n^{-1}$ are determined in Exercise 1.13. We find $K_1(\boldsymbol{H}_6)$:=
lyche-numerical-linear-algebra_ F02928	218	0.998	`\left|\boldsymbol{H}_{6}\right|_{1}\left|\boldsymbol{H}_{6}^{-1}\right|_{1} \approx 3 \cdot 10^{7}`	$\left\|\boldsymbol{H}_{6}\right\|_{1}\left\|\boldsymbol{H}_{6}^{-1}\right\|_{1} \approx 3 \cdot 10^{7}$	$\\|\boldsymbol{H}_6\\|_1\\|\boldsymbol{H}_6^{-1}\\|_1 \approx 3 \cdot 10^7$. It appears that $\frac{1}{m}\boldsymbol{B}_n$ and hence $\boldsymbol{B}_n$ is ill-conditioned
lyche-numerical-linear-algebra_ F02929	218	1.000	`(t-\tilde{t})^{j-1}, j=1, \ldots, n`	$(t-\tilde{t})^{j-1}, j=1, \ldots, n$	possibility is to use the shifted power basis $(t-\tilde{t})^{j-1}$, $j = 1, \ldots, n$, for a suitable $\tilde{t}$.
lyche-numerical-linear-algebra_ F02930	218	1.000	`\tilde{t}`	$\tilde{t}$	possibility is to use the shifted power basis $(t-\tilde{t})^{j-1}$, $j = 1, \ldots, n$, for a suitable $\tilde{t}$.
lyche-numerical-linear-algebra_ F02931	218	1.000	`\langle\boldsymbol{x}, \boldsymbol{y}\rangle= \boldsymbol{y}^{*} \boldsymbol{x}`	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle= \boldsymbol{y}^{*} \boldsymbol{x}$	With the usual inner product $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \boldsymbol{y}^*\boldsymbol{x}$, orthogonal sums and projections we
lyche-numerical-linear-algebra_ F02932	218	1.000	`\mathcal{S}:=\mathcal{R}(\boldsymbol{A})`	$\mathcal{S}:=\mathcal{R}(\boldsymbol{A})$	problems. For $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ we consider the column space $\mathcal{S} := \mathcal{R}(\boldsymbol{A})$ of $\boldsymbol{A}$ and the
lyche-numerical-linear-algebra_ F02933	218	1.000	`\mathcal{T}:=\mathcal{N}\left(\boldsymbol{A}^{*}\right)`	$\mathcal{T}:=\mathcal{N}\left(\boldsymbol{A}^{*}\right)$	null space $\mathcal{T} := \mathcal{N}(\boldsymbol{A}^*)$ of $\boldsymbol{A}^*$. These are subspaces of $\mathbb{C}^m$ and by Theorem 7.3 we
lyche-numerical-linear-algebra_ F02934	218	1.000	`\boldsymbol{b}=\boldsymbol{b}_{1}+\boldsymbol{b}_{2}`	$\boldsymbol{b}=\boldsymbol{b}_{1}+\boldsymbol{b}_{2}$	uniquely as $\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2$, where $\boldsymbol{b}_1$ is the orthogonal projection of $\boldsymbol{b}$ into $\mathcal{R}(\boldsymbol{A})$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02935	218	1.000	`\boldsymbol{b}_{1}`	$\boldsymbol{b}_{1}$	uniquely as $\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2$, where $\boldsymbol{b}_1$ is the orthogonal projection of $\boldsymbol{b}$ into $\mathcal{R}(\boldsymbol{A})$
lyche-numerical-linear-algebra F02936	218	1.000	`\boldsymbol{b}_{2}`	$\boldsymbol{b}_{2}$	and $\boldsymbol{b}_2$ is the orthogonal projection of $\boldsymbol{b}$ into $\mathcal{N}(\boldsymbol{A}^*)$. Suppose $\boldsymbol{x} \in \mathbb{C}^n$. Clearly
lyche-numerical-linear-algebra F02937	218	1.000	`\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}_{1} \in \mathcal{R}(\boldsymbol{A})`	$\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}_{1} \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{Ax}-\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$ since it is a subspace and $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. But then $\langle \boldsymbol{Ax}-\boldsymbol{b}_1, \boldsymbol{b}_2\rangle = 0$
lyche-numerical-linear-algebra F02938	218	1.000	`\boldsymbol{b}_{2} \in \mathcal{N}\left( \boldsymbol{A}^{*}\right)`	$\boldsymbol{b}_{2} \in \mathcal{N}\left(\boldsymbol{A}^{*}\right)$	$\boldsymbol{Ax}-\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$ since it is a subspace and $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. But then $\langle \boldsymbol{Ax}-\boldsymbol{b}_1, \boldsymbol{b}_2\rangle = 0$
lyche-numerical-linear-algebra F02939	218	1.000	`\left\langle\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}_{1}, \boldsymbol{b}_{2}\right\rangle=0`	$\left\langle\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}_{1}, \boldsymbol{b}_{2}\right\rangle=0$	$\boldsymbol{Ax}-\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$ since it is a subspace and $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. But then $\langle \boldsymbol{Ax}-\boldsymbol{b}_1, \boldsymbol{b}_2\rangle = 0$
lyche-numerical-linear-algebra F02940	219	1.000	`\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}_{1}`	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}_{1}$	with equality if and only if $\boldsymbol{Ax} = \boldsymbol{b}_1$. It follows that the set of all least squares
lyche-numerical-linear-algebra F02941	219	1.000	`\boldsymbol{b}_{1} \in \mathcal{R}(\boldsymbol{A})`	$\boldsymbol{b}_{1} \in \mathcal{R}(\boldsymbol{A})$	This set is nonempty since $\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$.
lyche-numerical-linear-algebra F02942	219	1.000	`\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}_{1}= \mathbf{0}`	$\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}_{1}= \mathbf{0}$	***Proof of Theorem 9.3*** If $\boldsymbol{x}$ solves the least squares problem then $\boldsymbol{Ax} - \boldsymbol{b}_1 = \mathbf{0}$ and
lyche-numerical-linear-algebra F02943	219	0.699	`\boldsymbol{A}^{*}(\boldsymbol{A x}-\boldsymbol{b})= \boldsymbol{A}^{*}\left(\boldsymbol{A x}- \boldsymbol{b}_{1}\right)=\mathbf{0}`	$\boldsymbol{A}^{*}(\boldsymbol{A x}-\boldsymbol{b})= \boldsymbol{A}^{*}\left(\boldsymbol{A x}- \boldsymbol{b}_{1}\right)=\mathbf{0}$	it follows that $\boldsymbol{A}^*(\boldsymbol{Ax}-\boldsymbol{b}) = \boldsymbol{A}^*(\boldsymbol{Ax}-\boldsymbol{b}_1) = \mathbf{0}$ since $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. This shows that
lyche-numerical-linear-algebra F02944	219	1.000	`\boldsymbol{A}^{*} \boldsymbol{b}_{2}=\mathbf{0}`	$\boldsymbol{A}^{*} \boldsymbol{b}_{2}=\mathbf{0}$	the normal equations hold. Conversely, if $\boldsymbol{A}^*\boldsymbol{Ax} = \boldsymbol{A}^*\boldsymbol{b}$ then $\boldsymbol{A}^*\boldsymbol{b}_2 = \mathbf{0}$ implies that
lyche-numerical-linear-algebra F02945	219	0.995	`\boldsymbol{A}^{*}\left(\boldsymbol{A} \boldsymbol{x}- \boldsymbol{b}_{1}\right)=\mathbf{0}`	$\boldsymbol{A}^{*}\left(\boldsymbol{A} \boldsymbol{x}- \boldsymbol{b}_{1}\right)=\mathbf{0}$	$\boldsymbol{A}^*(\boldsymbol{Ax}-\boldsymbol{b}_1) = \mathbf{0}$. But then $\boldsymbol{Ax}-\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$ showing that $\boldsymbol{Ax}-\boldsymbol{b}_1 = \mathbf{0}$,
lyche-numerical-linear-algebra F02946	219	0.995	`\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}_{1} \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}\left( \boldsymbol{A}^{*}\right)`	$\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}_{1} \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}\left(\boldsymbol{A}^{*}\right)$	$\boldsymbol{A}^*(\boldsymbol{Ax}-\boldsymbol{b}_1) = \mathbf{0}$. But then $\boldsymbol{Ax}-\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$ showing that $\boldsymbol{Ax}-\boldsymbol{b}_1 = \mathbf{0}$,
lyche-numerical-linear-algebra F02947	219	1.000	`m \geq n, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^{m}`	$m \geq n, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^{m}$	We assume that $m \geq n$, $\boldsymbol{A} \in \mathbb{C}^{m\times n}$, $\boldsymbol{b} \in \mathbb{C}^m$. We consider numerical methods based
lyche-numerical-linear-algebra F02948	219	1.000	`\boldsymbol{B}:=\boldsymbol{A}^{*} \boldsymbol{A}`	$\boldsymbol{B}:=\boldsymbol{A}^{*} \boldsymbol{A}$	coefficient matrix $\boldsymbol{B} := \boldsymbol{A}^*\boldsymbol{A}$ in the normal equations is positive definite, and we
lyche-numerical-linear-algebra F02949	219	0.998	`\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)_{i, j}= \sum_{k=1}^{m} \bar{a}_{k, i} a_{k, j}, i, j=1, \ldots, n`	$\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)_{i, j}= \sum_{k=1}^{m} \bar{a}_{k, i} a_{k, j}, i, j=1, \ldots, n$	1. inner product: $(\boldsymbol{A}^*\boldsymbol{A})_{i,j} = \sum_{k=1}^m \bar{a}_{k,i}a_{k,j}, i, j = 1, \ldots, n$,
lyche-numerical-linear-algebra F02950	219	1.000	`\boldsymbol{A}^{*} \boldsymbol{A}=\sum_{k=1}^{m}\left[\begin{array}{c}\bar{a}_{k, 1} \\ \vdots \\ \bar{a}_{k, n}\end{array}\right]\left[a_{k 1} \cdots a_{k n}\right], \boldsymbol{A}^{*} \boldsymbol{b}=\sum_{k=1}^{m}\left[\begin{array}{c}\bar{a}_{k, 1} \\ \vdots \\ \bar{a}_{k, n}\end{array}\right] b_{k}`	$\boldsymbol{A}^{*} \boldsymbol{A}=\sum_{k=1}^{m}\left[\begin{array}{c}\bar{a}_{k, 1} \\ \vdots \\ \bar{a}_{k, n}\end{array}\right]\left[a_{k 1} \cdots a_{k n}\right], \boldsymbol{A}^{*} \boldsymbol{b}=\sum_{k=1}^{m}\left[\begin{array}{c}\bar{a}_{k, 1} \\ \vdots \\ \bar{a}_{k, n}\end{array}\right] b_{k}$	2. outer product: $\boldsymbol{A}^*\boldsymbol{A} = \sum_{k=1}^m \begin{bmatrix}\bar{a}_{k,1}\\ \vdots \\ \bar{a}_{k,n}\end{bmatrix} [a_{k1} \cdots a_{kn}]$, $\boldsymbol{A}^*\boldsymbol{b} = \sum_{k=1}^m \begin{bmatrix}\bar{a}_{k,1}\\ \vdots \\ \bar{a}_{k,n}\end{bmatrix} b_k$.
lyche-numerical-linear-algebra F02951	219	1.000	`n(n+1) / 2`	$n(n+1) / 2$	we only need to compute $n(n+1)/2$ such inner products. It follows that $\boldsymbol{B}$ can be
lyche-numerical-linear-algebra F02952	219	1.000	`m n^{2}`	$m n^{2}$	computed in approximately mn^2 arithmetic operations. In conclusion the number
lyche-numerical-linear-algebra F02953	220	1.000	`\boldsymbol{B}, 2 m n`	$\boldsymbol{B}, 2 m n$	of operations are mn^2 to find $\boldsymbol{B}$, $2mn$ to find $\boldsymbol{c} := \boldsymbol{A}^*\boldsymbol{b}$, $n^3/3$ to find $\boldsymbol{L}$ such that
lyche-numerical-linear-algebra F02954	220	1.000	`\boldsymbol{c}:=\boldsymbol{A}^{*} \boldsymbol{b}, n^{3} / 3`	$\boldsymbol{c}:=\boldsymbol{A}^{*} \boldsymbol{b}, n^{3} / 3$	of operations are mn^2 to find $\boldsymbol{B}$, $2mn$ to find $\boldsymbol{c} := \boldsymbol{A}^*\boldsymbol{b}$, $n^3/3$ to find $\boldsymbol{L}$ such that
lyche-numerical-linear-algebra F02955	220	1.000	`\boldsymbol{B}=\boldsymbol{L} \boldsymbol{L}^{*}, n^{2}`	$\boldsymbol{B}=\boldsymbol{L} \boldsymbol{L}^{*}, n^{2}$	$\boldsymbol{B} = \boldsymbol{LL}^*$, n^2 to solve $\boldsymbol{Ly} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-algebra F02956	220	1.000	`\boldsymbol{L} \boldsymbol{y}=\boldsymbol{c}`	$\boldsymbol{L} \boldsymbol{y}=\boldsymbol{c}$	$\boldsymbol{B} = \boldsymbol{LL}^*$, n^2 to solve $\boldsymbol{Ly} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-algebra F02957	220	1.000	`\boldsymbol{L}^{*} \boldsymbol{x}=\boldsymbol{y}`	$\boldsymbol{L}^{*} \boldsymbol{x}=\boldsymbol{y}$	$\boldsymbol{B} = \boldsymbol{LL}^*$, n^2 to solve $\boldsymbol{Ly} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F02958	220	1.000	`m \approx n`	$m \approx n$	$\boldsymbol{B} = \boldsymbol{L}\boldsymbol{L}^*$, n^2 to solve $\boldsymbol{L}\boldsymbol{y} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-al F02959	220	1.000	`\frac{4}{3} n^{3}=`	$\frac{4}{3} n^{3}=$	$\boldsymbol{B} = \boldsymbol{L}\boldsymbol{L}^*$, n^2 to solve $\boldsymbol{L}\boldsymbol{y} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-al F02960	220	1.000	`\lambda_{1} \geq \cdots \geq \lambda_{n}>0`	$\lambda_{1} \geq \cdots \geq \lambda_{n}>0$	*where* $\lambda_1 \geq \cdots \geq \lambda_n > 0$ *are the eigenvalues of* $\boldsymbol{A}^*\boldsymbol{A}$, *and* $\sigma_1 \geq \cdots \geq \sigma_n > 0$ *are*
lyche-numerical-linear-al F02961	220	1.000	`\sigma_{1} \geq \cdots \geq \sigma_{n}>0`	$\sigma_{1} \geq \cdots \geq \sigma_{n}>0$	*where* $\lambda_1 \geq \cdots \geq \lambda_n > 0$ *are the eigenvalues of* $\boldsymbol{A}^*\boldsymbol{A}$, *and* $\sigma_1 \geq \cdots \geq \sigma_n > 0$ *are*
lyche-numerical-linear-al F02962	220	1.000	`K_{2}(\boldsymbol{A})=\frac{\sigma_{1}}{\sigma_{n}}`	$K_{2}(\boldsymbol{A})=\frac{\sigma_{1}}{\sigma_{n}}$	***Proof*** Since $\boldsymbol{A}^*\boldsymbol{A}$ is Hermitian it follows from Theorem 8.10 that $K_2(\boldsymbol{A}) = \frac{\sigma_1}{\sigma_n}$ and
lyche-numerical-linear-al F02963	220	0.999	`K_{2}\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)= \frac{\lambda_{1}}{\lambda_{n}}`	$K_{2}\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)= \frac{\lambda_{1}}{\lambda_{n}}$	$K_2(\boldsymbol{A}^*\boldsymbol{A}) = \frac{\lambda_1}{\lambda_n}$. But $\lambda_i = \sigma_i^2$ by Theorem 7.2 and the proof is complete.
lyche-numerical-linear-al F02964	220	0.999	`\lambda_{i}=\sigma_{i}^{2}`	$\lambda_{i}=\sigma_{i}^{2}$	$K_2(\boldsymbol{A}^*\boldsymbol{A}) = \frac{\lambda_1}{\lambda_n}$. But $\lambda_i = \sigma_i^2$ by Theorem 7.2 and the proof is complete.
lyche-numerical-linear-al F02965	220	1.000	`\boldsymbol{R}_{1}^{*}`	$\boldsymbol{R}_{1}^{*}$	Since $\boldsymbol{A}$ has rank n the matrix $\boldsymbol{R}_1^*$ is nonsingular and can be canceled. Thus
lyche-numerical-linear-al F02966	220	1.000	`\boldsymbol{c}_{1}`	$\boldsymbol{c}_{1}$	We can use Householder transformations or Givens rotations to find $\boldsymbol{R}_1$ and $\boldsymbol{c}_1$.
lyche-numerical-linear-al F02967	220	0.997	`\boldsymbol{R}=\boldsymbol{Q}^{*} \boldsymbol{A}`	$\boldsymbol{R}=\boldsymbol{Q}^{*} \boldsymbol{A}$	$\boldsymbol{R} = \boldsymbol{Q}^*\boldsymbol{A}$ and $\boldsymbol{c} = \boldsymbol{Q}^*\boldsymbol{b}$, where $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ is the QR decomposition of $\boldsymbol{A}$.
lyche-numerical-linear-al F02968	220	0.997	`\boldsymbol{c}=\boldsymbol{Q}^{*} \boldsymbol{b}`	$\boldsymbol{c}=\boldsymbol{Q}^{*} \boldsymbol{b}$	$\boldsymbol{R} = \boldsymbol{Q}^*\boldsymbol{A}$ and $\boldsymbol{c} = \boldsymbol{Q}^*\boldsymbol{b}$, where $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ is the QR decomposition of $\boldsymbol{A}$.
lyche-numerical-linear-al F02969	220	1.000	`\boldsymbol{c}`	$\boldsymbol{c}$	The matrices $\boldsymbol{R}_1$ and $\boldsymbol{c}_1$ are located in the first n rows of $\boldsymbol{R}$ and $\boldsymbol{c}$. Using also
lyche-numerical-linear-al F02970	220	0.769	`[\mathrm{R}, \mathrm{c}]=`	$[\mathrm{R}, \mathrm{c}]=$	1. `[R,c]=housetriang(A,b).`
lyche-numerical-linear-al F02971	221	0.958	`\boldsymbol{x}=[1,0,0]^{*}`	$\boldsymbol{x}=[1,0,0]^{*}$	and we find $\boldsymbol{x} = [1, 0, 0]^*$.
lyche-numerical-linear-al F02972	221	1.000	`\boldsymbol{R}_{1} \boldsymbol{x}=\boldsymbol{c}_{1}`	$\boldsymbol{R}_{1} \boldsymbol{x}=\boldsymbol{c}_{1}$	problems, see [2]. The 2 norm condition number for the system $\boldsymbol{R}_1\boldsymbol{x} = \boldsymbol{c}_1$ is
lyche-numerical-linear-al F02973	221	1.000	`K_{2}\left(\boldsymbol{R}_{1}\right)=K_{2}\left( \boldsymbol{Q}_{1} \boldsymbol{R}_{1}\right)=K_{2}( \boldsymbol{A})`	$K_{2}\left(\boldsymbol{R}_{1}\right)=K_{2}\left(\boldsymbol{Q}_{1} \boldsymbol{R}_{1}\right)=K_{2}(\boldsymbol{A})$	$K_2(\boldsymbol{R}_1) = K_2(\boldsymbol{Q}_1\boldsymbol{R}_1) = K_2(\boldsymbol{A})$, and as discussed in the previous section this
lyche-numerical-linear-al F02974	221	1.000	`K_{2}\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)`	$K_{2}\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)$	is the square root of $K_2(\boldsymbol{A}^*\boldsymbol{A})$, the condition number for the normal equations.
lyche-numerical-linear-al F02975	221	1.000	`2 m n^{2}-`	$2 m n^{2}-$	the number of arithmetic operations in Algorithm 5.2 is approximately $2mn^2 -$
lyche-numerical-linear-al F02976	221	1.000	`\mathrm{x}=\mathrm{A} \backslash \mathrm{b}`	$\mathrm{x}=\mathrm{A} \backslash \mathrm{b}$	Using MATLAB a least squares solution can be found using `x=A\b` if $\boldsymbol{A}$ has
lyche-numerical-linear-al F02977	221	0.996	`\mathrm{x}=\operatorname{lscov}(\mathrm{A}, \mathrm{b})`	$\mathrm{x}=\operatorname{lscov}(\mathrm{A}, \mathrm{b})$	full rank.For rank deficient problems the function `x=lscov(A,b)` finds a least
lyche-numerical-linear-al F02978	222	1.000	`\boldsymbol{U}_{2}`	$\boldsymbol{U}_{2}$	• the set of columns of $\boldsymbol{U}_2$ is an orthonormal basis for $\mathcal{N}(\boldsymbol{A}^*)$,
lyche-numerical-linear-al F02979	222	1.000	`\boldsymbol{V}_{2}`	$\boldsymbol{V}_{2}$	• the set of columns of $\boldsymbol{V}_2$ is an orthonormal basis for $\mathcal{N}(\boldsymbol{A})$.
lyche-numerical-linear-al F02980	222	1.000	`\boldsymbol{A}^{\dagger} \in \mathbb{C}^{n \times m}`	$\boldsymbol{A}^{\dagger} \in \mathbb{C}^{n \times m}$	*there is a unique matrix* $\boldsymbol{A}^\dagger \in \mathbb{C}^{n\times m}$ *such that*
lyche-numerical-linear-al F02981	222	1.000	`\boldsymbol{U}_{1} \boldsymbol{\Sigma}_{1} \boldsymbol{V}_{1}^{*}`	$\boldsymbol{U}_{1} \boldsymbol{\Sigma}_{1} \boldsymbol{V}_{1}^{*}$	*If* $\boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*$ *is a singular value factorization of* $\boldsymbol{A}$ *then*
lyche-numerical-linear-al F02982	222	1.000	`\left(\boldsymbol{A}^{\dagger} \boldsymbol{A}\right)^{*}= \boldsymbol{V}_{1} \boldsymbol{V}_{1}^{*}`	$\left(\boldsymbol{A}^{\dagger} \boldsymbol{A}\right)^{*}= \boldsymbol{V}_{1} \boldsymbol{V}_{1}^{*}$	A similar calculation shows that $(\boldsymbol{A}^\dagger\boldsymbol{A})^* = \boldsymbol{V}_1\boldsymbol{V}_1^*$ and $(\boldsymbol{A}\boldsymbol{A}^\dagger)^* = \boldsymbol{U}_1\boldsymbol{U}_1^*$ showing
lyche-numerical-linear-al F02983	222	1.000	`\left(\boldsymbol{A} \boldsymbol{A}^{\dagger}\right)^{*}= \boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*}`	$\left(\boldsymbol{A} \boldsymbol{A}^{\dagger}\right)^{*}= \boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*}$	A similar calculation shows that $(\boldsymbol{A}^\dagger\boldsymbol{A})^* = \boldsymbol{V}_1\boldsymbol{V}_1^*$ and $(\boldsymbol{A}\boldsymbol{A}^\dagger)^* = \boldsymbol{U}_1\boldsymbol{U}_1^*$ showing

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F02984	222	■ 1.000	`\boldsymbol{A}^{\dagger} \boldsymbol{A}`	$\boldsymbol{A}^{\dagger} \boldsymbol{A}$	that $\boldsymbol{A}^{\dagger}\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^{\dagger}$ are Hermitian. Moreover, by (9.10)
lyche-numerical-linear-algebra F02985	222	■ 1.000	`\boldsymbol{A} \boldsymbol{A}^{\dagger}`	$\boldsymbol{A} \boldsymbol{A}^{\dagger}$	that $\boldsymbol{A}^{\dagger}\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^{\dagger}$ are Hermitian. Moreover, by (9.10)
lyche-numerical-linear-algebra F02986	223	■ 1.000	`\boldsymbol{A}^{\dagger}`	$\boldsymbol{A}^{\dagger}$	The matrix $\boldsymbol{A}^{\dagger}$ is called the **generalized inverse** of $\boldsymbol{A}$. We note that
lyche-numerical-linear-algebra F02987	223	■ 0.999	`\boldsymbol{A}^{-1}=\boldsymbol{A}^{\dagger}`	$\boldsymbol{A}^{-1}=\boldsymbol{A}^{\dagger}$	1. If $\boldsymbol{A}$ is square and nonsingular then $\boldsymbol{A}^{-1}$ satisfies (9.8) so that $\boldsymbol{A}^{-1}=\boldsymbol{A}^{\dagger}$. Indeed,
lyche-numerical-linear-algebra F02988	223	■ 1.000	`\boldsymbol{A}^{-1} \boldsymbol{A}`	$\boldsymbol{A}^{-1} \boldsymbol{A}$	$\boldsymbol{A}^{-1}\boldsymbol{A}=\boldsymbol{A}\boldsymbol{A}^{-1}=\boldsymbol{I}$ implies that $\boldsymbol{A}^{-1}\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^{-1}$ are Hermitian. Moreover,
lyche-numerical-linear-algebra F02989	223	■ 1.000	`\boldsymbol{A} \boldsymbol{A}^{-1}`	$\boldsymbol{A} \boldsymbol{A}^{-1}$	$\boldsymbol{A}^{-1}\boldsymbol{A}=\boldsymbol{A}\boldsymbol{A}^{-1}=\boldsymbol{I}$ implies that $\boldsymbol{A}^{-1}\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^{-1}$ are Hermitian. Moreover,
lyche-numerical-linear-algebra F02990	223	■ 1.000	`\boldsymbol{A} \boldsymbol{A}^{-1} \boldsymbol{A}= \boldsymbol{A}, \boldsymbol{A}^{-1} \boldsymbol{A} \boldsymbol{A}^{-1}=\boldsymbol{A}^{-1}`	$\boldsymbol{A} \boldsymbol{A}^{-1} \boldsymbol{A}= \boldsymbol{A}, \boldsymbol{A}^{-1} \boldsymbol{A} \boldsymbol{A}^{-1}=\boldsymbol{A}^{-1}$	$\boldsymbol{A}\boldsymbol{A}^{-1}\boldsymbol{A}=\boldsymbol{A}$, $\boldsymbol{A}^{-1}\boldsymbol{A}\boldsymbol{A}^{-1}=\boldsymbol{A}^{-1}$. By uniqueness $\boldsymbol{A}^{-1}=\boldsymbol{A}^{\dagger}$.
lyche-numerical-linear-algebra F02991	223	■ 1.000	`\mathcal{R}(\boldsymbol{A}), \mathcal{N}\left( \boldsymbol{A}^{*}\right)`	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}\left(\boldsymbol{A}^{*}\right)$	*and let $\mathcal{S}$ be one of the subspaces $\mathcal{R}(\boldsymbol{A})$, $\mathcal{N}(\boldsymbol{A}^*)$. The orthogonal projection of*
lyche-numerical-linear-algebra F02992	223	■ 1.000	`\boldsymbol{v} \in \mathbb{C}^{m}`	$\boldsymbol{v} \in \mathbb{C}^{m}$	$\boldsymbol{v}\in\mathbb{C}^m$ *into $\mathcal{S}$ can be written as a matrix $\boldsymbol{P}_{\mathcal{S}}$ times the vector $\boldsymbol{v}$ in the form $\boldsymbol{P}_{\mathcal{S}}\boldsymbol{v}$,*
lyche-numerical-linear-algebra F02993	223	■ 1.000	`\boldsymbol{P}_{\mathcal{S}}`	$\boldsymbol{P}_{\mathcal{S}}$	$\boldsymbol{v}\in\mathbb{C}^m$ *into $\mathcal{S}$ can be written as a matrix $\boldsymbol{P}_{\mathcal{S}}$ times the vector $\boldsymbol{v}$ in the form $\boldsymbol{P}_{\mathcal{S}}\boldsymbol{v}$,*
lyche-numerical-linear-algebra F02994	223	■ 1.000	`\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}`	$\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$	$\boldsymbol{v}\in\mathbb{C}^m$ *into $\mathcal{S}$ can be written as a matrix $\boldsymbol{P}_{\mathcal{S}}$ times the vector $\boldsymbol{v}$ in the form $\boldsymbol{P}_{\mathcal{S}}\boldsymbol{v}$,*
lyche-numerical-linear-algebra F02995	223	■ 0.998	`\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*} \in \mathbb{C}^{m \times n}`	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*} \in \mathbb{C}^{m \times n}$	*where $\boldsymbol{A}^{\dagger}$ is the generalized inverse of $\boldsymbol{A}$, and $\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*\in\mathbb{C}^{m\times n}$ is a singular*
lyche-numerical-linear-algebra F02996	223	■ 1.000	`\boldsymbol{s}=\boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*} \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})`	$\boldsymbol{s}=\boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*} \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$	where $\boldsymbol{s}=\boldsymbol{U}_1\boldsymbol{U}_1^*\boldsymbol{v}\in\mathcal{R}(\boldsymbol{A})$ and $\boldsymbol{t}=\boldsymbol{U}_2\boldsymbol{U}_2^*\boldsymbol{v}\in\mathcal{N}(\boldsymbol{A}^*)$. By uniqueness and (9.10)
lyche-numerical-linear-algebra F02997	223	■ 1.000	`\boldsymbol{t}=\boldsymbol{U}_{2} \boldsymbol{U}_{2}^{*} \boldsymbol{v} \in \mathcal{N}\left(\boldsymbol{A}^{*} \right)`	$\boldsymbol{t}=\boldsymbol{U}_{2} \boldsymbol{U}_{2}^{*} \boldsymbol{v} \in \mathcal{N}\left(\boldsymbol{A}^{*} \right)$	where $\boldsymbol{s}=\boldsymbol{U}_1\boldsymbol{U}_1^*\boldsymbol{v}\in\mathcal{R}(\boldsymbol{A})$ and $\boldsymbol{t}=\boldsymbol{U}_2\boldsymbol{U}_2^*\boldsymbol{v}\in\mathcal{N}(\boldsymbol{A}^*)$. By uniqueness and (9.10)
lyche-numerical-linear-algebra F02998	223	■ 1.000	`\boldsymbol{v}=\left(\boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*}+\boldsymbol{U}_{2} \boldsymbol{U}_{2}^{*}\right) \boldsymbol{v}`	$\boldsymbol{v}=\left(\boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*}+\boldsymbol{U}_{2} \boldsymbol{U}_{2}^{*}\right) \boldsymbol{v}$	we obtain the first equation in (9.12). Since $\boldsymbol{v}=(\boldsymbol{U}_1\boldsymbol{U}_1^*+\boldsymbol{U}_2\boldsymbol{U}_2^*)\boldsymbol{v}$ for any $\boldsymbol{v}\in\mathbb{C}^m$
lyche-numerical-linear-algebra F02999	223	■ 0.999	`\boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*}+ \boldsymbol{U}_{2} \boldsymbol{U}_{2}^{*}=\boldsymbol{I}`	$\boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*}+ \boldsymbol{U}_{2} \boldsymbol{U}_{2}^{*}=\boldsymbol{I}$	we have $\boldsymbol{U}_1\boldsymbol{U}_1^*+\boldsymbol{U}_2\boldsymbol{U}_2^*=\boldsymbol{I}$, and hence $\boldsymbol{U}_2\boldsymbol{U}_2^*=\boldsymbol{I}-\boldsymbol{U}_1\boldsymbol{U}_1^*=\boldsymbol{I}-\boldsymbol{A}\boldsymbol{A}^{\dagger}$ and the
lyche-numerical-linear-algebra F03000	223	■ 0.999	`\boldsymbol{U}_{2} \boldsymbol{U}_{2}^{*}=\boldsymbol{I}- \boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*}=\boldsymbol{I}- \boldsymbol{A} \boldsymbol{A}^{\dagger}`	$\boldsymbol{U}_{2} \boldsymbol{U}_{2}^{*}=\boldsymbol{I}- \boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*}=\boldsymbol{I}- \boldsymbol{A} \boldsymbol{A}^{\dagger}$	we have $\boldsymbol{U}_1\boldsymbol{U}_1^*+\boldsymbol{U}_2\boldsymbol{U}_2^*=\boldsymbol{I}$, and hence $\boldsymbol{U}_2\boldsymbol{U}_2^*=\boldsymbol{I}-\boldsymbol{U}_1\boldsymbol{U}_1^*=\boldsymbol{I}-\boldsymbol{A}\boldsymbol{A}^{\dagger}$ and the
lyche-numerical-linear-algebra F03001	223	■ 0.848	`\min _{x}|\boldsymbol{A} \boldsymbol{x}-b|_{2}^{2}`	$\min _{x}\|\boldsymbol{A} \boldsymbol{x}-b\|_{2}^{2}$	*solves the least squares problem* $\min_x\|\boldsymbol{A}\boldsymbol{x}-b\|_2^2$ *if and only if* $\boldsymbol{x}=\boldsymbol{A}^{\dagger}\boldsymbol{b}+\boldsymbol{z}$,
lyche-numerical-linear-algebra F03002	223	■ 0.848	`\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{b}+ \boldsymbol{z}`	$\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{b}+ \boldsymbol{z}$	*solves the least squares problem* $\min_x\|\boldsymbol{A}\boldsymbol{x}-b\|_2^2$ *if and only if* $\boldsymbol{x}=\boldsymbol{A}^{\dagger}\boldsymbol{b}+\boldsymbol{z}$,
lyche-numerical-linear-algebra F03003	223	■ 0.893	`\boldsymbol{z} \in \mathcal{N}(\boldsymbol{A})`	$\boldsymbol{z} \in \mathcal{N}(\boldsymbol{A})$	*where $\boldsymbol{A}^{\dagger}$ is the generalized inverse of $\boldsymbol{A}$ and $\boldsymbol{z}\in\mathcal{N}(\boldsymbol{A})$.*
lyche-numerical-linear-algebra F03004	224	■ 1.000	`\boldsymbol{b}_{1}:=\boldsymbol{A} \boldsymbol{A}^{ \dagger} \boldsymbol{b}`	$\boldsymbol{b}_{1}:=\boldsymbol{A} \boldsymbol{A}^{ \dagger} \boldsymbol{b}$	***Proof*** It follows from Theorem 9.6 that $\boldsymbol{b}_1:=\boldsymbol{A}\boldsymbol{A}^{\dagger}\boldsymbol{b}$ is the orthogonal projection
lyche-numerical-linear-algebra F03005	224	■ 0.998	`\boldsymbol{z}:=\boldsymbol{x}-\boldsymbol{A}^{\dagger} \boldsymbol{b}`	$\boldsymbol{z}:=\boldsymbol{x}-\boldsymbol{A}^{\dagger} \boldsymbol{b}$	Let $\boldsymbol{x}$ be a least squares solution, i.e., $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}_1$. If $\boldsymbol{z}:=\boldsymbol{x}-\boldsymbol{A}^{\dagger}\boldsymbol{b}$ then $\boldsymbol{A}\boldsymbol{z}=$
lyche-numerical-linear-algebra F03006	224	■ 0.998	`\boldsymbol{A} \boldsymbol{z}=`	$\boldsymbol{A} \boldsymbol{z}=$	Let $\boldsymbol{x}$ be a least squares solution, i.e., $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}_1$. If $\boldsymbol{z}:=\boldsymbol{x}-\boldsymbol{A}^{\dagger}\boldsymbol{b}$ then $\boldsymbol{A}\boldsymbol{z}=$
lyche-numerical-linear-algebra F03007	224	■ 1.000	`\boldsymbol{A} \boldsymbol{x}-\boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b}= \boldsymbol{b}_{1}-\boldsymbol{b}_{1}=\mathbf{0}`	$\boldsymbol{A} \boldsymbol{x}-\boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b}= \boldsymbol{b}_{1}-\boldsymbol{b}_{1}=\mathbf{0}$	$\boldsymbol{A}\boldsymbol{x}-\boldsymbol{A}\boldsymbol{A}^{\dagger}\boldsymbol{b}=\boldsymbol{b}_1-\boldsymbol{b}_1=\boldsymbol{0}$ and $\boldsymbol{x}=\boldsymbol{A}^{\dagger}\boldsymbol{b}+\boldsymbol{z}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F03008	224	0.997	`\boldsymbol{A} \boldsymbol{z}=\mathbf{0}`	$\boldsymbol{A} \boldsymbol{z}=\mathbf{0}$	Conversely, if $\boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{b} + \boldsymbol{z}$ with $\boldsymbol{A}\boldsymbol{z} = \mathbf{0}$ then $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{A}^{\dagger}\boldsymbol{b} + \boldsymbol{z}) = \boldsymbol{b}_1$ and $\boldsymbol{x}$ is
lyche-numerical-linear-al F03009	224	0.997	`\boldsymbol{A} \boldsymbol{x}=\boldsymbol{A}\left( \boldsymbol{A}^{\dagger} \boldsymbol{b}+\boldsymbol{z} \right)=\boldsymbol{b}_{1}`	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{A}\left(\boldsymbol{A}^{\dagger} \boldsymbol{b}+\boldsymbol{z} \right)=\boldsymbol{b}_{1}$	Conversely, if $\boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{b} + \boldsymbol{z}$ with $\boldsymbol{A}\boldsymbol{z} = \mathbf{0}$ then $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{A}^{\dagger}\boldsymbol{b} + \boldsymbol{z}) = \boldsymbol{b}_1$ and $\boldsymbol{x}$ is
lyche-numerical-linear-al F03010	224	1.000	`\boldsymbol{A}^{\dagger} \boldsymbol{b}`	$\boldsymbol{A}^{\dagger} \boldsymbol{b}$	The least squares solution $\boldsymbol{A}^{\dagger}\boldsymbol{b}$ has an interesting property.
lyche-numerical-linear-al F03011	224	1.000	`\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{b}`	$\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{b}$	*Euclidian norm is* $\boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{b}$ *corresponding to* $\boldsymbol{z} = \mathbf{0}$.
lyche-numerical-linear-al F03012	224	0.999	`\boldsymbol{z}=\boldsymbol{V}_{2} \boldsymbol{y}`	$\boldsymbol{z}=\boldsymbol{V}_{2} \boldsymbol{y}$	$\mathcal{N}(\boldsymbol{A})$ we have $\boldsymbol{z} = \boldsymbol{V}_2\boldsymbol{y}$ for some $\boldsymbol{y}$. Moreover, $\boldsymbol{V}_2^*\boldsymbol{V}_1 = \mathbf{0}$ since $\boldsymbol{V}$ has orthonormal
lyche-numerical-linear-al F03013	224	0.999	`\boldsymbol{V}_{2}^{*} \boldsymbol{V}_{1}=\mathbf{0}`	$\boldsymbol{V}_{2}^{*} \boldsymbol{V}_{1}=\mathbf{0}$	$\mathcal{N}(\boldsymbol{A})$ we have $\boldsymbol{z} = \boldsymbol{V}_2\boldsymbol{y}$ for some $\boldsymbol{y}$. Moreover, $\boldsymbol{V}_2^*\boldsymbol{V}_1 = \mathbf{0}$ since $\boldsymbol{V}$ has orthonormal
lyche-numerical-linear-al F03014	224	1.000	`\boldsymbol{z}^{*} \boldsymbol{A}^{\dagger} \boldsymbol{b}=\boldsymbol{y}^{*} \boldsymbol{V}_{2}^{*} \boldsymbol{V}_{1} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_{1}^{*} \boldsymbol{b}=\mathbf{0}`	$\boldsymbol{z}^{*} \boldsymbol{A}^{\dagger} \boldsymbol{b}=\boldsymbol{y}^{*} \boldsymbol{V}_{2}^{*} \boldsymbol{V}_{1} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_{1}^{*} \boldsymbol{b}=\mathbf{0}$	columns. But then $\boldsymbol{z}^*\boldsymbol{A}^{\dagger}\boldsymbol{b} = \boldsymbol{y}^*\boldsymbol{V}_2^*\boldsymbol{V}_1\boldsymbol{\Sigma}^{-1}\boldsymbol{U}_1^*\boldsymbol{b} = \mathbf{0}$. Thus $\boldsymbol{z}$ and $\boldsymbol{A}^{\dagger}\boldsymbol{b}$ are orthogonal
lyche-numerical-linear-al F03015	224	1.000	`|\boldsymbol{x}|_{2}^{2}=\left|\boldsymbol{A}^{ \dagger} \boldsymbol{b}+\boldsymbol{z}\right|_{2}^{2}= \left|\boldsymbol{A}^{\dagger} \boldsymbol{b}\right |_{2}^{2}+|\boldsymbol{z}|_{2}^{2} \geq\left| \boldsymbol{A}^{\dagger} \boldsymbol{b}\right|_{2}^{2}`	$\|\boldsymbol{x}\|_{2}^{2}=\left\|\boldsymbol{A}^{\dagger} \boldsymbol{b}+\boldsymbol{z}\right\|_{2}^{2}=\left\|\boldsymbol{A}^{\dagger} \boldsymbol{b}\right\|_{2}^{2}+\|\boldsymbol{z}\|_{2}^{2} \geq\left\|\boldsymbol{A}^{\dagger} \boldsymbol{b}\right\|_{2}^{2}$	so that by Pythagoras $\|\boldsymbol{x}\|_2^2 = \|\boldsymbol{A}^{\dagger}\boldsymbol{b} + \boldsymbol{z}\|_2^2 = \|\boldsymbol{A}^{\dagger}\boldsymbol{b}\|_2^2 + \|\boldsymbol{z}\|_2^2 \geq \|\boldsymbol{A}^{\dagger}\boldsymbol{b}\|_2^2$ with
lyche-numerical-linear-al F03016	224	0.997	`\boldsymbol{A}=\left[\begin{array}{ll}1 & 1 \\ 1 & 1 \end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{ll}1 & 1 \\ 1 & 1 \end{array}\right]$	problem with $\boldsymbol{A} = \left[\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right]$ and $\boldsymbol{b} = [1, 1]^*$. The singular value factorization, $\boldsymbol{A}^{\dagger}$ and
lyche-numerical-linear-al F03017	224	0.997	`\boldsymbol{b}=[1,1]^{*}`	$\boldsymbol{b}=[1,1]^{*}$	problem with $\boldsymbol{A} = \left[\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right]$ and $\boldsymbol{b} = [1, 1]^*$. The singular value factorization, $\boldsymbol{A}^{\dagger}$ and
lyche-numerical-linear-al F03018	224	1.000	`[1 / 2,1 / 2]+[a,-a]`	$[1 / 2,1 / 2]+[a,-a]$	Using Corollary 9.1 we find the general solution $[1/2, 1/2] + [a, -a]$ for any $a \in \mathbb{C}$.
lyche-numerical-linear-al F03019	224	0.653	`a=1 / 2`	$a=1 / 2$	The MATLAB function `lscov` gives the solution $[1, 0]^*$ corresponding to $a = 1/2$,
lyche-numerical-linear-al F03020	224	0.761	`[1 / 2,1 / 2]`	$[1 / 2,1 / 2]$	while the minimal norm solution is $[1/2, 1/2]$ obtained for $a = 0$.
lyche-numerical-linear-al F03021	224	0.933	`\min |\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}|_{2}`	$\min \|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\|_{2}$	solution $\boldsymbol{x}$ of the least squares problem $\min\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2$.
lyche-numerical-linear-al F03022	225	1.000	`\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^{m}`	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^{m}$	*independent columns, and let* $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^m$. *Let* $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ *be the solutions of*
lyche-numerical-linear-al F03023	225	0.994	`\min |\boldsymbol{A} \boldsymbol{y}-\boldsymbol{b}-\boldsymbol{e}|_{2}`	$\min \|\boldsymbol{A} \boldsymbol{y}-\boldsymbol{b}-\boldsymbol{e}\|_{2}$	$\min\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2$ *and* $\min\|\boldsymbol{A}\boldsymbol{y} - \boldsymbol{b} - \boldsymbol{e}\|_2$. *Finally, let* $\boldsymbol{b}_1$, $\boldsymbol{e}_1$ *be the orthogonal*
lyche-numerical-linear-al F03024	225	0.994	`\boldsymbol{b}_{1}, \boldsymbol{e}_{1}`	$\boldsymbol{b}_{1}, \boldsymbol{e}_{1}$	$\min\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2$ *and* $\min\|\boldsymbol{A}\boldsymbol{y} - \boldsymbol{b} - \boldsymbol{e}\|_2$. *Finally, let* $\boldsymbol{b}_1$, $\boldsymbol{e}_1$ *be the orthogonal*
lyche-numerical-linear-al F03025	225	1.000	`\boldsymbol{b}_{1} \neq \mathbf{0}`	$\boldsymbol{b}_{1} \neq \mathbf{0}$	*projections of* $\boldsymbol{b}$ *and* $\boldsymbol{e}$ *into* $\mathcal{R}(\boldsymbol{A})$. *If* $\boldsymbol{b}_1 \neq \mathbf{0}$, *we have for any operator norm*
lyche-numerical-linear-al F03026	225	1.000	`\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{b}_{1}`	$\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{b}_{1}$	***Proof*** Subtracting $\boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{b}_1$ from $\boldsymbol{y} = \boldsymbol{A}^{\dagger}\boldsymbol{b}_1 + \boldsymbol{A}^{\dagger}\boldsymbol{e}_1$ we have $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{e}_1$.
lyche-numerical-linear-al F03027	225	1.000	`\boldsymbol{y}=\boldsymbol{A}^{\dagger} \boldsymbol{b}_{1}+\boldsymbol{A}^{\dagger} \boldsymbol{e}_{1}`	$\boldsymbol{y}=\boldsymbol{A}^{\dagger} \boldsymbol{b}_{1}+\boldsymbol{A}^{\dagger} \boldsymbol{e}_{1}$	***Proof*** Subtracting $\boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{b}_1$ from $\boldsymbol{y} = \boldsymbol{A}^{\dagger}\boldsymbol{b}_1 + \boldsymbol{A}^{\dagger}\boldsymbol{e}_1$ we have $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{e}_1$.
lyche-numerical-linear-al F03028	225	1.000	`\boldsymbol{y}-\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{e}_{1}`	$\boldsymbol{y}-\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{e}_{1}$	***Proof*** Subtracting $\boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{b}_1$ from $\boldsymbol{y} = \boldsymbol{A}^{\dagger}\boldsymbol{b}_1 + \boldsymbol{A}^{\dagger}\boldsymbol{e}_1$ we have $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{e}_1$.
lyche-numerical-linear-al F03029	225	1.000	`|\boldsymbol{y}-\boldsymbol{x}|=\left|\boldsymbol{A}^{ \dagger} \boldsymbol{e}_{1}\right| \leq\left| \boldsymbol{A}^{\dagger}\right|\left|\boldsymbol{e}_{1} \right|`	$\|\boldsymbol{y}-\boldsymbol{x}\|=\left\|\boldsymbol{A}^{\dagger} \boldsymbol{e}_{1}\right\| \leq\left\|\boldsymbol{A}^{\dagger}\right\|\left\|\boldsymbol{e}_{1}\right\|$	Thus $\|\boldsymbol{y} - \boldsymbol{x}\| = \|\boldsymbol{A}^{\dagger}\boldsymbol{e}_1\| \leq \|\boldsymbol{A}^{\dagger}\|\|\boldsymbol{e}_1\|$. Moreover, $\|\boldsymbol{b}_1\| = \|\boldsymbol{A}\boldsymbol{x}\| \leq \|\boldsymbol{A}\|\|\boldsymbol{x}\|$.

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra F03030</td><td>225</td><td>1.000</td><td><code>\left\|\boldsymbol{b}_{1}\right\|=\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\|\|\boldsymbol{x}\|</code></td><td>$\left\|\boldsymbol{b}_{1}\right\|=\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\|\|\boldsymbol{x}\|$</td><td>Thus $\|\boldsymbol{y}-\boldsymbol{x}\| = \|\boldsymbol{A}^{\dagger}\boldsymbol{e}_1\| \leq \|\boldsymbol{A}^{\dagger}\|\|\boldsymbol{e}_1\|$. Moreover, $\|\boldsymbol{b}_1\| = \|\boldsymbol{A}\boldsymbol{x}\| \leq \|\boldsymbol{A}\|\|\boldsymbol{x}\|$.</td></tr>
<tr><td>lyche-numerical-linear-algebra F03031</td><td>225</td><td>1.000</td><td><code>\|\boldsymbol{y}-\boldsymbol{x}\| /\|\boldsymbol{x}\| \leq\|\boldsymbol{A}\|\left\|\boldsymbol{A}^{\dagger} \right\|\left\|\boldsymbol{e}_{1}\right\| /\left\| \boldsymbol{b}_{1}\right\|</code></td><td>$\|\boldsymbol{y}-\boldsymbol{x}\| /\|\boldsymbol{x}\| \leq\|\boldsymbol{A}\|\left\|\boldsymbol{A}^{\dagger} \right\|\left\|\boldsymbol{e}_{1}\right\| /\left\| \boldsymbol{b}_{1}\right\|$</td><td>Therefore $\|\boldsymbol{y}-\boldsymbol{x}\|/\|\boldsymbol{x}\| \leq \|\boldsymbol{A}\|\|\boldsymbol{A}^{\dagger}\|\|\boldsymbol{e}_1\|/\|\boldsymbol{b}_1\|$ proving the rightmost inequality.</td></tr>
<tr><td>lyche-numerical-linear-algebra F03032</td><td>225</td><td>1.000</td><td><code>\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})= \boldsymbol{e}_{1}</code></td><td>$\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})= \boldsymbol{e}_{1}$</td><td>From $\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y}) = \boldsymbol{e}_1$ and $\boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{b}_1$ we obtain the leftmost inequality.</td></tr>
<tr><td>lyche-numerical-linear-algebra F03033</td><td>225</td><td>1.000</td><td><code>K(\boldsymbol{A})=\|\boldsymbol{A}\|\left\| \boldsymbol{A}^{\dagger}\right\|</code></td><td>$K(\boldsymbol{A})=\|\boldsymbol{A}\|\left\| \boldsymbol{A}^{\dagger}\right\|$</td><td>the number $K(\boldsymbol{A}) = \|\boldsymbol{A}\|\|\boldsymbol{A}^{\dagger}\|$ generalizes the condition number $\|\boldsymbol{A}\|\|\boldsymbol{A}^{-1}\|$ for</td></tr>
<tr><td>lyche-numerical-linear-algebra F03034</td><td>225</td><td>1.000</td><td><code>\|\boldsymbol{A}\|\left\|\boldsymbol{A}^{-1}\right\|</code></td><td>$\|\boldsymbol{A}\|\left\|\boldsymbol{A}^{-1}\right\|$</td><td>the number $K(\boldsymbol{A}) = \|\boldsymbol{A}\|\|\boldsymbol{A}^{\dagger}\|$ generalizes the condition number $\|\boldsymbol{A}\|\|\boldsymbol{A}^{-1}\|$ for</td></tr>
<tr><td>lyche-numerical-linear-algebra F03035</td><td>225</td><td>0.997</td><td><code>\left\|\boldsymbol{e}_{1}\right\| /\left\| \boldsymbol{b}_{1}\right\|</code></td><td>$\left\|\boldsymbol{e}_{1}\right\| /\left\| \boldsymbol{b}_{1}\right\|$</td><td>$\|\boldsymbol{e}\|/\|\boldsymbol{b}\|$ in (8.22) has been replaced by $\|\boldsymbol{e}_1\|/\|\boldsymbol{b}_1\|$, the orthogonal projections of</td></tr>
<tr><td>lyche-numerical-linear-algebra F03036</td><td>225</td><td>1.000</td><td><code>\|\boldsymbol{b}\| /\left\|\boldsymbol{b}_{1}\right\|</code></td><td>$\|\boldsymbol{b}\| /\left\|\boldsymbol{b}_{1}\right\|$</td><td>$\boldsymbol{e}$ and $\boldsymbol{b}$ into $\mathcal{R}(\boldsymbol{A})$. If $\boldsymbol{b}$ lies almost entirely in $\mathcal{N}(\boldsymbol{A}^*)$, i.e. $\|\boldsymbol{b}\|/\|\boldsymbol{b}_1\|$ is large, then</td></tr>
<tr><td>lyche-numerical-linear-algebra F03037</td><td>226</td><td>1.000</td><td><code>K_{\infty}(\boldsymbol{A})=\|\boldsymbol{A}\|_{\infty} \left\|\boldsymbol{A}^{\dagger}\right\|_{\infty}=2 \cdot 2=4</code></td><td>$K_{\infty}(\boldsymbol{A})=\|\boldsymbol{A}\|_{\infty} \left\|\boldsymbol{A}^{\dagger}\right\|_{\infty}=2 \cdot 2=4$</td><td>Thus $K_\infty(\boldsymbol{A}) = \|\boldsymbol{A}\|_\infty\|\boldsymbol{A}^{\dagger}\|_\infty = 2\cdot 2 = 4$ is quite small.</td></tr>
<tr><td>lyche-numerical-linear-algebra F03038</td><td>226</td><td>0.999</td><td><code>\boldsymbol{A} \boldsymbol{A}^{\dagger}=\left[\begin{array}{lll}1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]</code></td><td>$\boldsymbol{A} \boldsymbol{A}^{\dagger}=\left[\begin{array}{lll}1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$</td><td>Consider now the projections $\boldsymbol{b}_1$ and $\boldsymbol{e}_1$. We find $\boldsymbol{A}\boldsymbol{A}^{\dagger} = \left[\begin{smallmatrix}1&0&0\\0&1&0\\0&0&0\end{smallmatrix}\right]$. Hence</td></tr>
<tr><td>lyche-numerical-linear-algebra F03039</td><td>226</td><td>1.000</td><td><code>\left\|\boldsymbol{e}_{1}\right\|_{\infty} /\left\| \boldsymbol{b}_{1}\right\|_{\infty}=10^{-2}</code></td><td>$\left\|\boldsymbol{e}_{1}\right\|_{\infty} /\left\| \boldsymbol{b}_{1}\right\|_{\infty}=10^{-2}$</td><td>Thus $\|\boldsymbol{e}_1\|_\infty/\|\boldsymbol{b}_1\|_\infty = 10^{-2}$ and (9.13) takes the form</td></tr>
<tr><td>lyche-numerical-linear-algebra F03040</td><td>226</td><td>1.000</td><td><code>\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{b}= \left[10^{-4}, 0\right]^{*}</code></td><td>$\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{b}= \left[10^{-4}, 0\right]^{*}$</td><td>To verify the bounds we compute the solutions as $\boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{b} = [10^{-4}, 0]^*$ and</td></tr>
<tr><td>lyche-numerical-linear-algebra F03041</td><td>226</td><td>0.999</td><td><code>\boldsymbol{y}=\boldsymbol{A}^{\dagger}(\boldsymbol{b}+ \boldsymbol{e})=\left[10^{-4}+10^{-6}, 0\right]^{*}</code></td><td>$\boldsymbol{y}=\boldsymbol{A}^{\dagger}(\boldsymbol{b}+ \boldsymbol{e})=\left[10^{-4}+10^{-6}, 0\right]^{*}$</td><td>$\boldsymbol{y} = \boldsymbol{A}^{\dagger}(\boldsymbol{b}+\boldsymbol{e}) = [10^{-4}+10^{-6}, 0]^*$. Hence</td></tr>
<tr><td>lyche-numerical-linear-algebra F03042</td><td>226</td><td>1.000</td><td><code>\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{m \times n}, m>n</code></td><td>$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{m \times n}, m>n$</td><td>Theorem 9.9 (Perturbing the Matrix) Suppose $\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{m\times n}$, $m > n$, where $\boldsymbol{A}$</td></tr>
<tr><td>lyche-numerical-linear-algebra F03043</td><td>226</td><td>1.000</td><td><code>\alpha:=1-\|\boldsymbol{E}\|_{2}\left\|\boldsymbol{A}^{ \dagger}\right\|_{2}>0</code></td><td>$\alpha:=1-\|\boldsymbol{E}\|_{2}\left\|\boldsymbol{A}^{ \dagger}\right\|_{2}>0$</td><td>has linearly independent columns and $\alpha := 1 - \|\boldsymbol{E}\|_2\|\boldsymbol{A}^{\dagger}\|_2 > 0$. Then $\boldsymbol{A}+\boldsymbol{E}$</td></tr>
<tr><td>lyche-numerical-linear-algebra F03044</td><td>226</td><td>1.000</td><td><code>\boldsymbol{b}=\boldsymbol{b}_{1}+\boldsymbol{b}_{2} \in \mathbb{C}^{m}</code></td><td>$\boldsymbol{b}=\boldsymbol{b}_{1}+\boldsymbol{b}_{2} \in \mathbb{C}^{m}$</td><td>has linearly independent columns. Let $\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2 \in \mathbb{C}^m$ where $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ are the</td></tr>
<tr><td>lyche-numerical-linear-algebra F03045</td><td>226</td><td>0.755</td><td><code>\min \|(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{y}- \boldsymbol{b}\|_{2}</code></td><td>$\min \|(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{y}- \boldsymbol{b}\|_{2}$</td><td>and $\boldsymbol{y}$ be the solutions of $\min\|\boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\|_2$ and $\min\|(\boldsymbol{A}+\boldsymbol{E})\boldsymbol{y}-\boldsymbol{b}\|_2$. Then</td></tr>
<tr><td>lyche-numerical-linear-algebra F03046</td><td>226</td><td>1.000</td><td><code>K(1+\beta K) / \alpha</code></td><td>$K(1+\beta K) / \alpha$</td><td>most $K(1+\beta K)/\alpha$ times as large as the size $\|\boldsymbol{E}\|_2/\|\boldsymbol{A}\|_2$ of the relative perturbation</td></tr>
<tr><td>lyche-numerical-linear-algebra F03047</td><td>226</td><td>1.000</td><td><code>\|\boldsymbol{E}\|_{2} /\|\boldsymbol{A}\|_{2}</code></td><td>$\|\boldsymbol{E}\|_{2} /\|\boldsymbol{A}\|_{2}$</td><td>most $K(1+\beta K)/\alpha$ times as large as the size $\|\boldsymbol{E}\|_2/\|\boldsymbol{A}\|_2$ of the relative perturbation</td></tr>
<tr><td>lyche-numerical-linear-algebra F03048</td><td>227</td><td>0.993</td><td><code>\boldsymbol{A} . \beta</code></td><td>$\boldsymbol{A} . \beta$</td><td>in $\boldsymbol{A}$. β will be small if $\boldsymbol{b}$ lies almost entirely in $\mathcal{R}(\boldsymbol{A})$, and we have approximately</td></tr>
<tr><td>lyche-numerical-linear-algebra F03049</td><td>227</td><td>1.000</td><td><code>\rho \leq \frac{1}{\alpha} K\|\boldsymbol{E}\|_{2} /\| \boldsymbol{A}\|_{2}</code></td><td>$\rho \leq \frac{1}{\alpha} K\|\boldsymbol{E}\|_{2} /\| \boldsymbol{A}\|_{2}$</td><td>$\rho \leq \frac{1}{\alpha}K\|\boldsymbol{E}\|_2/\|\boldsymbol{A}\|_2$. This corresponds to the estimate (8.25) for linear systems.</td></tr>
<tr><td>lyche-numerical-linear-algebra F03050</td><td>227</td><td>1.000</td><td><code>\beta</code></td><td>β</td><td>If β is not small, the term $\frac{1}{\alpha}K^2\beta\|\boldsymbol{E}\|_2/\|\boldsymbol{A}\|_2$ will dominate. In other words, the</td></tr>
<tr><td>lyche-numerical-linear-algebra F03051</td><td>227</td><td>1.000</td><td><code>\frac{1}{\alpha} K^{2} \beta\|\boldsymbol{E}\|_{2} /\| \boldsymbol{A}\|_{2}</code></td><td>$\frac{1}{\alpha} K^{2} \beta\|\boldsymbol{E}\|_{2} /\| \boldsymbol{A}\|_{2}$</td><td>If β is not small, the term $\frac{1}{\alpha}K^2\beta\|\boldsymbol{E}\|_2/\|\boldsymbol{A}\|_2$ will dominate. In other words, the</td></tr>
<tr><td>lyche-numerical-linear-algebra F03052</td><td>227</td><td>1.000</td><td><code>K(\boldsymbol{A})^{2} \beta</code></td><td>$K(\boldsymbol{A})^{2} \beta$</td><td>condition number is roughly $K(\boldsymbol{A})$ if β is small and $K(\boldsymbol{A})^2\beta$ if β is not small. Note</td></tr>
<tr><td>lyche-numerical-linear-algebra F03053</td><td>227</td><td>0.999</td><td><code>\boldsymbol{b}_{2}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}</code></td><td>$\boldsymbol{b}_{2}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}$</td><td>that β is large if $\boldsymbol{b}$ is almost orthogonal to $\mathcal{R}(\boldsymbol{A})$ and that $\boldsymbol{b}_2 = \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}$ is the residual</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-al F03054</td><td>227</td><td>0.999</td><td><code>\sigma_{1}, \sigma_{2}, \ldots, \sigma_{n}</code></td><td>$\sigma_{1}, \sigma_{2}, \ldots, \sigma_{n}$</td><td>$\boldsymbol{A} \in \mathbb{C}^{m \times n}$ has singular values $\sigma_1, \sigma_2, \ldots, \sigma_n$ ordered so that $\sigma_1 \geq \cdots \geq \sigma_n$. Then</td></tr>
<tr><td>lyche-numerical-linear-al F03055</td><td>227</td><td>0.999</td><td><code>\sigma_{1} \geq \cdots \geq \sigma_{n}</code></td><td>$\sigma_{1} \geq \cdots \geq \sigma_{n}$</td><td>$\boldsymbol{A} \in \mathbb{C}^{m \times n}$ has singular values $\sigma_1, \sigma_2, \ldots, \sigma_n$ ordered so that $\sigma_1 \geq \cdots \geq \sigma_n$. Then</td></tr>
<tr><td>lyche-numerical-linear-al F03056</td><td>228</td><td>1.000</td><td><code>\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m \times n}</code></td><td>$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m \times n}$</td><td>Theorem 9.11 (Perturbation of Singular Values) Let $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m \times n}$ be rectangu-</td></tr>
<tr><td>lyche-numerical-linear-al F03057</td><td>228</td><td>1.000</td><td><code>n-j+1</code></td><td>$n-j+1$</td><td>Proof Fix j and let $\mathcal{S}$ be the $n-j+1$ dimensional subspace for which the minimum</td></tr>
<tr><td>lyche-numerical-linear-al F03058</td><td>228</td><td>1.000</td><td><code>\beta_{j} \leq \alpha_{j}+\|\boldsymbol{A}-\boldsymbol{B}\|_{2}</code></td><td>$\beta_{j} \leq \alpha_{j}+\|\boldsymbol{A}-\boldsymbol{B}\|_{2}$</td><td>By symmetry we obtain $\beta_j \leq \alpha_j + \|\boldsymbol{A}-\boldsymbol{B}\|_2$ and the proof is complete.</td></tr>
<tr><td>lyche-numerical-linear-al F03059</td><td>228</td><td>1.000</td><td><code>\boldsymbol{A}, \boldsymbol{E} \in</code></td><td>$\boldsymbol{A}, \boldsymbol{E} \in$</td><td>Theorem 9.12 (Generalized Inverse When Perturbing the Matrix) Let $\boldsymbol{A}, \boldsymbol{E} \in$</td></tr>
<tr><td>lyche-numerical-linear-al F03060</td><td>228</td><td>1.000</td><td><code>\alpha_{1} \geq \cdots \geq \alpha_{n}</code></td><td>$\alpha_{1} \geq \cdots \geq \alpha_{n}$</td><td>$\mathbb{R}^{m \times n}$ have singular values $\alpha_1 \geq \cdots \geq \alpha_n$ and $\epsilon_1 \geq \cdots \geq \epsilon_n$. If $\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E}) \leq$</td></tr>
<tr><td>lyche-numerical-linear-al F03061</td><td>228</td><td>1.000</td><td><code>\epsilon_{1} \geq \cdots \geq \epsilon_{n}</code></td><td>$\epsilon_{1} \geq \cdots \geq \epsilon_{n}$</td><td>$\mathbb{R}^{m \times n}$ have singular values $\alpha_1 \geq \cdots \geq \alpha_n$ and $\epsilon_1 \geq \cdots \geq \epsilon_n$. If $\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E}) \leq$</td></tr>
<tr><td>lyche-numerical-linear-al F03062</td><td>228</td><td>1.000</td><td><code>\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E}) \leq</code></td><td>$\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E}) \leq$</td><td>$\mathbb{R}^{m \times n}$ have singular values $\alpha_1 \geq \cdots \geq \alpha_n$ and $\epsilon_1 \geq \cdots \geq \epsilon_n$. If $\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E}) \leq$</td></tr>
<tr><td>lyche-numerical-linear-al F03063</td><td>228</td><td>0.893</td><td><code>\left\|\boldsymbol{A}^{\dagger}\right\|_{2}\|\boldsymbol{E}\|_{2}<1</code></td><td>$\left\|\boldsymbol{A}^{\dagger}\right\|_{2}\|\boldsymbol{E}\|_{2}<1$</td><td>$\operatorname{rank}(\boldsymbol{A}) = r$ and $\|\boldsymbol{A}^\dagger\|_2\|\boldsymbol{E}\|_2 < 1$ then</td></tr>
<tr><td>lyche-numerical-linear-al F03064</td><td>228</td><td>1.000</td><td><code>\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E})= \operatorname{rank}(\boldsymbol{A})</code></td><td>$\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E})=\operatorname{rank}(\boldsymbol{A})$</td><td>1. $\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E}) = \operatorname{rank}(\boldsymbol{A})$,</td></tr>
<tr><td>lyche-numerical-linear-al F03065</td><td>228</td><td>1.000</td><td><code>\left\|(\boldsymbol{A}+\boldsymbol{E})^{\dagger}\right \|_{2} \leq \frac{\left\|\boldsymbol{A}^{\dagger}\right \|_{2}}{1-\left\|\boldsymbol{A}^{\dagger}\right\|_{2}\| \boldsymbol{E}\|_{2}}=\frac{1}{\alpha_{r}-\epsilon_{1}}</code></td><td>$\left\|(\boldsymbol{A}+\boldsymbol{E})^{\dagger}\right\|_{2} \leq \frac{\left\|\boldsymbol{A}^{\dagger}\right\|_{2}}{1-\left\|\boldsymbol{A}^{\dagger}\right\|_{2}\|\boldsymbol{E}\|_{2}}=\frac{1}{\alpha_{r}-\epsilon_{1}}$</td><td>2. $\|(\boldsymbol{A}+\boldsymbol{E})^\dagger\|_2 \leq \frac{\|\boldsymbol{A}^\dagger\|_2}{1-\|\boldsymbol{A}^\dagger\|_2\|\boldsymbol{E}\|_2} = \frac{1}{\alpha_r-\epsilon_1}$.</td></tr>
<tr><td>lyche-numerical-linear-al F03066</td><td>228</td><td>1.000</td><td><code>\beta_{1} \geq</code></td><td>$\beta_{1} \geq$</td><td>Proof Suppose $\boldsymbol{A}$ has rank r and let $\boldsymbol{B} := \boldsymbol{A}+\boldsymbol{E}$ have singular values $\beta_1 \geq$</td></tr>
<tr><td>lyche-numerical-linear-al F03067</td><td>228</td><td>1.000</td><td><code>\cdots \geq \beta_{n}</code></td><td>$\cdots \geq \beta_{n}$</td><td>$\cdots \geq \beta_n$. In terms of singular values the inequality $\|\boldsymbol{A}^\dagger\|_2\|\boldsymbol{E}\|_2 < 1$ can be written</td></tr>
<tr><td>lyche-numerical-linear-al F03068</td><td>228</td><td>1.000</td><td><code>\epsilon_{1} / \alpha_{r}<1</code></td><td>$\epsilon_{1} / \alpha_{r}<1$</td><td>$\epsilon_1/\alpha_r < 1$ or $\alpha_r > \epsilon_1$. By Theorem 9.11 we have $\alpha_r - \beta_r \leq \|\boldsymbol{E}\|_2 = \epsilon_1$, which</td></tr>
<tr><td>lyche-numerical-linear-al F03069</td><td>228</td><td>1.000</td><td><code>\alpha_{r}>\epsilon_{1}</code></td><td>$\alpha_{r}>\epsilon_{1}$</td><td>$\epsilon_1/\alpha_r < 1$ or $\alpha_r > \epsilon_1$. By Theorem 9.11 we have $\alpha_r - \beta_r \leq \|\boldsymbol{E}\|_2 = \epsilon_1$, which</td></tr>
<tr><td>lyche-numerical-linear-al F03070</td><td>228</td><td>1.000</td><td><code>\alpha_{r}-\beta_{r} \leq\|\boldsymbol{E}\|_{2}= \epsilon_{1}</code></td><td>$\alpha_{r}-\beta_{r} \leq\|\boldsymbol{E}\|_{2}=\epsilon_{1}$</td><td>$\epsilon_1/\alpha_r < 1$ or $\alpha_r > \epsilon_1$. By Theorem 9.11 we have $\alpha_r - \beta_r \leq \|\boldsymbol{E}\|_2 = \epsilon_1$, which</td></tr>
<tr><td>lyche-numerical-linear-al F03071</td><td>228</td><td>0.703</td><td><code>\beta_{r} \geq \alpha_{r}-\epsilon_{1}>0</code></td><td>$\beta_{r} \geq \alpha_{r}-\epsilon_{1}>0$</td><td>implies $\beta_r \geq \alpha_r - \epsilon_1 > 0$, and this shows that $\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E}) \geq r$. Thus 1. follows.</td></tr>
<tr><td>lyche-numerical-linear-al F03072</td><td>228</td><td>0.703</td><td><code>\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E}) \geq r</code></td><td>$\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E}) \geq r$</td><td>implies $\beta_r \geq \alpha_r - \epsilon_1 > 0$, and this shows that $\operatorname{rank}(\boldsymbol{A}+\boldsymbol{E}) \geq r$. Thus 1. follows.</td></tr>
<tr><td>lyche-numerical-linear-al F03073</td><td>228</td><td>0.960</td><td><code>\beta_{r} \geq \alpha_{r}-\epsilon_{1}</code></td><td>$\beta_{r} \geq \alpha_{r}-\epsilon_{1}$</td><td>To prove 2., the inequality $\beta_r \geq \alpha_r - \epsilon_1$ implies that</td></tr>
<tr><td>lyche-numerical-linear-al F03074</td><td>228</td><td>1.000</td><td><code>\mu_{1} \geq \cdots \geq \mu_{n}</code></td><td>$\mu_{1} \geq \cdots \geq \mu_{n}$</td><td>and $\mu_1 \geq \cdots \geq \mu_n$, respectively.</td></tr>
<tr><td>lyche-numerical-linear-al F03075</td><td>229</td><td>1.000</td><td><code>\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{m \times n}</code></td><td>$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{m \times n}$</td><td>$m, n \in \mathbb{N}$ and $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{m \times n}$ we have</td></tr>
<tr><td>lyche-numerical-linear-al F03076</td><td>229</td><td>1.000</td><td><code>\beta_{1} \geq \cdots \geq \beta_{n}</code></td><td>$\beta_{1} \geq \cdots \geq \beta_{n}$</td><td>where $\alpha_1 \geq \cdots \geq \alpha_n$ and $\beta_1 \geq \cdots \geq \beta_n$ are the singular values of $\boldsymbol{A}$ and $\boldsymbol{B}$,</td></tr>
<tr><td>lyche-numerical-linear-al F03077</td><td>229</td><td>1.000</td><td><code>\lambda_{1} \geq \cdots \geq \lambda_{m+n}</code></td><td>$\lambda_{1} \geq \cdots \geq \lambda_{m+n}$</td><td>If $\boldsymbol{C}$ and $\boldsymbol{D}$ have eigenvalues $\lambda_1 \geq \cdots \geq \lambda_{m+n}$ and $\mu_1 \geq \cdots \geq \mu_{m+n}$, respectively</td></tr>
<tr><td>lyche-numerical-linear-al F03078</td><td>229</td><td>1.000</td><td><code>\mu_{1} \geq \cdots \geq \mu_{m+n}</code></td><td>$\mu_{1} \geq \cdots \geq \mu_{m+n}$</td><td>If $\boldsymbol{C}$ and $\boldsymbol{D}$ have eigenvalues $\lambda_1 \geq \cdots \geq \lambda_{m+n}$ and $\mu_1 \geq \cdots \geq \mu_{m+n}$, respectively</td></tr>
<tr><td>lyche-numerical-linear-al F03079</td><td>229</td><td>0.651</td><td><code>\operatorname{SVD}\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{m}\right] \boldsymbol{\Sigma}\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right]^{*}</code></td><td>$\operatorname{SVD}\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{m}\right] \boldsymbol{\Sigma}\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{n}\right]^{*}$</td><td>Suppose $\boldsymbol{A}$ has rank r and SVD $[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m]\boldsymbol{\Sigma}[\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n]^*$. We use (7.7) and</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F03080	229	1.000	2 r	$2r$	Thus $\boldsymbol{C}$ has the $2r$ eigenvalues $\alpha_1, -\alpha_1, \ldots, \alpha_r, -\alpha_r$ and $m+n-2r$ additional
lyche-numerical-linear-algebra F03081	229	1.000	\alpha_{1},-\alpha_{1}, \ldots, \alpha_{r},-\alpha_{r}	$\alpha_{1},-\alpha_{1}, \ldots, \alpha_{r},-\alpha_{r}$	Thus $\boldsymbol{C}$ has the $2r$ eigenvalues $\alpha_1, -\alpha_1, \ldots, \alpha_r, -\alpha_r$ and $m+n-2r$ additional
lyche-numerical-linear-algebra F03082	229	1.000	m+n-2 r	$m+n-2r$	Thus $\boldsymbol{C}$ has the $2r$ eigenvalues $\alpha_1, -\alpha_1, \ldots, \alpha_r, -\alpha_r$ and $m+n-2r$ additional
lyche-numerical-linear-algebra F03083	229	1.000	2 s	$2s$	zero eigenvalues. Similarly, if $\boldsymbol{B}$ has rank s then $\boldsymbol{D}$ has the $2s$ eigenvalues
lyche-numerical-linear-algebra F03084	229	0.999	\beta_{1},-\beta_{1}, \ldots, \beta_{s},-\beta_{s}	$\beta_{1},-\beta_{1}, \ldots, \beta_{s},-\beta_{s}$	$\beta_1, -\beta_1, \ldots, \beta_s, -\beta_s$ and $m+n-2s$ additional zero eigenvalues. Let
lyche-numerical-linear-algebra F03085	229	0.999	m+n-2 s	$m+n-2s$	$\beta_1, -\beta_1, \ldots, \beta_s, -\beta_s$ and $m+n-2s$ additional zero eigenvalues. Let
lyche-numerical-linear-algebra F03086	230	1.000	\sum_{i=1}^{t}\left|\alpha_{i}-\beta_{i}\right|^{2} \leq \\|\boldsymbol{B}-\boldsymbol{A}\\|_{F}^{2}	$\sum_{i=1}^{t}\left\lvert\alpha_{i}-\beta_{i}\right\rvert^{2} \leq \lVert\boldsymbol{B}-\boldsymbol{A}\rVert_{F}^{2}$	But then (9.21) implies $\sum_{i=1}^{t}\lvert\alpha_i-\beta_i\rvert^2 \leq \lVert\boldsymbol{B}-\boldsymbol{A}\rVert_F^2$. Since $t \leq n$ and $\alpha_i = \beta_i = 0$
lyche-numerical-linear-algebra F03087	230	1.000	t \leq n	$t \leq n$	But then (9.21) implies $\sum_{i=1}^{t}\lvert\alpha_i-\beta_i\rvert^2 \leq \lVert\boldsymbol{B}-\boldsymbol{A}\rVert_F^2$. Since $t \leq n$ and $\alpha_i = \beta_i = 0$
lyche-numerical-linear-algebra F03088	230	1.000	\alpha_{i}=\beta_{i}=0	$\alpha_{i}=\beta_{i}=0$	But then (9.21) implies $\sum_{i=1}^{t}\lvert\alpha_i-\beta_i\rvert^2 \leq \lVert\boldsymbol{B}-\boldsymbol{A}\rVert_F^2$. Since $t \leq n$ and $\alpha_i = \beta_i = 0$
lyche-numerical-linear-algebra F03089	230	1.000	i=t+1, \ldots, n	$i=t+1, \ldots, n$	for $i = t+1, \ldots, n$ we obtain (9.20).
lyche-numerical-linear-algebra F03090	230	1.000	\left(t-c_{1}\right)^{2}+\left(y-c_{2}\right)^{2}=r^{2}	$\left(t-c_{1}\right)^{2}+\left(y-c_{2}\right)^{2}=r^{2}$	to fit a circle $(t-c_1)^2 + (y-c_2)^2 = r^2$ to $m \geq 3$ given points $(t_i, y_i)_{i=1}^m$ in the
lyche-numerical-linear-algebra F03091	230	1.000	\left(t_{i}, y_{i}\right)_{i=1}^{m}	$\left(t_{i}, y_{i}\right)_{i=1}^{m}$	to fit a circle $(t-c_1)^2 + (y-c_2)^2 = r^2$ to $m \geq 3$ given points $(t_i, y_i)_{i=1}^m$ in the
lyche-numerical-linear-algebra F03092	230	0.999	(t, y)	(t, y)	(t, y)-plane. We obtain the overdetermined system
lyche-numerical-linear-algebra F03093	230	1.000	c_{1}, c_{2}	c_{1}, c_{2}	of m equations in the three unknowns c_1, c_2 and r. This system is nonlinear, but it
lyche-numerical-linear-algebra F03094	230	1.000	c_{1}=x_{1} / 2, c_{2}=x_{2} / 2	$c_{1}=x_{1} / 2, c_{2}=x_{2} / 2$	and then setting $c_1 = x_1/2$, $c_2 = x_2/2$ and $r^2 = c_1^2 + c_2^2 + x_3$.
lyche-numerical-linear-algebra F03095	230	1.000	r^{2}=c_{1}^{2}+c_{2}^{2}+x_{3}	$r^{2}=c_{1}^{2}+c_{2}^{2}+x_{3}$	and then setting $c_1 = x_1/2$, $c_2 = x_2/2$ and $r^2 = c_1^2 + c_2^2 + x_3$.
lyche-numerical-linear-algebra F03096	230	1.000	c_{1}, c_{2}, r	c_{1}, c_{2}, r	**a)** Derive (9.23) from (9.22). Explain how we can find c_1, c_2, r once $[x_1, x_2, x_3]$ is
lyche-numerical-linear-algebra F03097	230	1.000	\left[x_{1}, x_{2}, x_{3}\right]	$\left[x_{1}, x_{2}, x_{3}\right]$	**a)** Derive (9.23) from (9.22). Explain how we can find c_1, c_2, r once $[x_1, x_2, x_3]$ is
lyche-numerical-linear-algebra F03098	230	0.593	(1,4),(3,2),(1,0)	$(1,4),(3,2),(1,0)$	**d)** Use (9.23) to find the circle passing through the three points $(1, 4)$, $(3, 2)$, $(1, 0)$.
lyche-numerical-linear-algebra F03099	231	1.000	\left[\begin{array}{cc}\sqrt{2} & \sqrt{2} \\\\ 0 & \sqrt{3}\end{array}\right]	$\left[\begin{array}{cc}\sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3}\end{array}\right]$	**a)** Let $\boldsymbol{A}$ be the matrix $\begin{bmatrix}\sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3}\end{bmatrix}$. Find the singular values of $\boldsymbol{A}$, and compute
lyche-numerical-linear-algebra F03100	231	0.993	\boldsymbol{A}=\left[\begin{array}{cc}3 & \alpha \\\\ \alpha & 1\end{array}\right]	$\boldsymbol{A}=\left[\begin{array}{cc}3 & \alpha \\ \alpha & 1\end{array}\right]$	**b)** Consider the matrix $\boldsymbol{A} = \begin{bmatrix}3 & \alpha \\ \alpha & 1\end{bmatrix}$, where α is a real number. For which values of
lyche-numerical-linear-algebra F03101	231	1.000	\boldsymbol{p}_{1}=(0,1), \boldsymbol{p}_{2}=(1,0), \boldsymbol{p}_{3}=(2,1)	$\boldsymbol{p}_{1}=(0,1), \boldsymbol{p}_{2}=(1,0), \boldsymbol{p}_{3}=(2,1)$	**c)** We would like to fit the points $\boldsymbol{p}_1 = (0, 1)$, $\boldsymbol{p}_2 = (1, 0)$, $\boldsymbol{p}_3 = (2, 1)$ to a straight
lyche-numerical-linear-algebra F03102	231	1.000	p(x)=m x+b	$p(x)=m x+b$	line in the plane. Find a line $p(x) = mx + b$ which minimizes
lyche-numerical-linear-algebra F03103	231	1.000	\boldsymbol{p}_{i}=\left(x_{i}, y_{i}\right)	$\boldsymbol{p}_{i}=\left(x_{i}, y_{i}\right)$	where $\boldsymbol{p}_i = (x_i, y_i)$. Is this solution unique?
lyche-numerical-linear-algebra F03104	231	1.000	F: \mathbb{R}^{n} \rightarrow \mathbb{R}	$F: \mathbb{R}^{n} \rightarrow \mathbb{R}$	$\mathbb{R}^{m\times n}$ and let $\boldsymbol{I} \in \mathbb{R}^{n\times n}$ be the identity matrix. We define $F : \mathbb{R}^n \to \mathbb{R}$ by

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F03105	231	1.000	`\boldsymbol{B}:=\boldsymbol{I}+\boldsymbol{A}^{T} \boldsymbol{A}`	$\boldsymbol{B}:=\boldsymbol{I}+\boldsymbol{A}^{T}\boldsymbol{A}$	**a)** Show that the matrix $\boldsymbol{B} := \boldsymbol{I} + \boldsymbol{A}^T\boldsymbol{A}$ is symmetric and positive definite.
lyche-numerical-linear-al F03106	231	1.000	`\left(\boldsymbol{I}+\boldsymbol{A}^{T} \boldsymbol{A} \right) \boldsymbol{x}=\boldsymbol{A}^{T} \boldsymbol{b}`	$\left(\boldsymbol{I}+\boldsymbol{A}^{T}\boldsymbol{A}\right)\boldsymbol{x}=\boldsymbol{A}^{T}\boldsymbol{b}$	is the unique solution of the linear system $(\boldsymbol{I} + \boldsymbol{A}^T\boldsymbol{A})\boldsymbol{x} = \boldsymbol{A}^T\boldsymbol{b}$.
lyche-numerical-linear-al F03107	231	1.000	`\operatorname{diag}\left(d_{1}, d_{2}, \ldots, d_{m} \right) \in \mathbb{R}^{m \times m}`	$\operatorname{diag}\left(d_{1}, d_{2}, \ldots, d_{m}\right) \in \mathbb{R}^{m \times m}$	$\operatorname{diag}(d_1, d_2, \ldots, d_m) \in \mathbb{R}^{m\times m}$, where $d_i > 0, i = 1, 2, \ldots, m$. We want to minimize
lyche-numerical-linear-al F03108	231	1.000	`d_{i}>0, i=1,2, \ldots, m`	$d_{i}>0, i=1,2, \ldots, m$	$\operatorname{diag}(d_1, d_2, \ldots, d_m) \in \mathbb{R}^{m\times m}$, where $d_i > 0, i = 1, 2, \ldots, m$. We want to minimize
lyche-numerical-linear-al F03109	232	1.000	`r_{i}=r_{i}(\boldsymbol{x}), i=1,2, \ldots, m`	$r_{i}=r_{i}(\boldsymbol{x}), i=1,2, \ldots, m$	where $r_i = r_i(\boldsymbol{x}), i = 1, 2, \ldots, m$ are the components of the vector
lyche-numerical-linear-al F03110	232	1.000	`|\boldsymbol{r}(\boldsymbol{x})|_{D}^{2}`	$\lVert\boldsymbol{r}(\boldsymbol{x})\rVert_{D}^{2}$	**a)** Show that $\lVert\boldsymbol{r}(\boldsymbol{x})\rVert_D^2$ in (9.24) obtains a unique minimum when $\boldsymbol{x} = \boldsymbol{x}_{min}$ is the
lyche-numerical-linear-al F03111	232	1.000	`\boldsymbol{x}=\boldsymbol{x}_{\text {min }}`	$\boldsymbol{x}=\boldsymbol{x}_{\text {min }}$	**a)** Show that $\lVert\boldsymbol{r}(\boldsymbol{x})\rVert_D^2$ in (9.24) obtains a unique minimum when $\boldsymbol{x} = \boldsymbol{x}_{min}$ is the
lyche-numerical-linear-al F03112	232	1.000	`K_{2}(\boldsymbol{B}):=|\boldsymbol{B}|_{2}\left| \boldsymbol{B}^{-1}\right|_{2}`	$K_{2}(\boldsymbol{B}):=\lVert\boldsymbol{B}\rVert_{2}\left\lVert\boldsymbol{B}^{-1}\right\rVert_{2}$	where for any nonsingular matrix $K_2(\boldsymbol{B}) := \lVert\boldsymbol{B}\rVert_2\lVert\boldsymbol{B}^{-1}\rVert_2$.
lyche-numerical-linear-al F03113	232	0.999	`\boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times m}`	$\boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times m}$	pose $\boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n\times m}$ satisfy
lyche-numerical-linear-al F03114	232	1.000	`\boldsymbol{B}=\boldsymbol{C}`	$\boldsymbol{B}=\boldsymbol{C}$	Verify the following proof that $\boldsymbol{B} = \boldsymbol{C}$.
lyche-numerical-linear-al F03115	232	1.000	`\boldsymbol{A}=\left[\begin{array}{ll}1 & 1 \\ 1 & 1 \\ 0 & 0\end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{ll}1 & 1 \\ 1 & 1 \\ 0 & 0\end{array}\right]$	generalized inverse of $\boldsymbol{A} = \left[\begin{smallmatrix}1&1\\1&1\\0&0\end{smallmatrix}\right]$ is $\boldsymbol{A}^\dagger = \frac{1}{4}\left[\begin{smallmatrix}1&1&0\\1&1&0\end{smallmatrix}\right]$ without using the singular
lyche-numerical-linear-al F03116	232	1.000	`\boldsymbol{A}^{\dagger}=\frac{1}{4}\left[ \begin{array}{lll}1 & 1 & 0 \\ 1 & 1 & 0\end{array} \right]`	$\boldsymbol{A}^{\dagger}=\frac{1}{4}\left[\begin{array}{lll}1 & 1 & 0 \\ 1 & 1 & 0\end{array}\right]$	generalized inverse of $\boldsymbol{A} = \left[\begin{smallmatrix}1&1\\1&1\\0&0\end{smallmatrix}\right]$ is $\boldsymbol{A}^\dagger = \frac{1}{4}\left[\begin{smallmatrix}1&1&0\\1&1&0\end{smallmatrix}\right]$ without using the singular
lyche-numerical-linear-al F03117	232	1.000	`\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1} \boldsymbol{A}^{*}`	$\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^{*}\boldsymbol{A}\right)^{-1}\boldsymbol{A}^{*}$	and $\boldsymbol{A}^\dagger = (\boldsymbol{A}^*\boldsymbol{A})^{-1}\boldsymbol{A}^*$. If $\boldsymbol{A}$ has linearly independent rows, then show that $\boldsymbol{A}\boldsymbol{A}^*$ is
lyche-numerical-linear-al F03118	232	1.000	`\boldsymbol{A}^{\dagger}=\boldsymbol{A}^{*}\left( \boldsymbol{A} \boldsymbol{A}^{*}\right)^{-1}`	$\boldsymbol{A}^{\dagger}=\boldsymbol{A}^{*}\left(\boldsymbol{A}\boldsymbol{A}^{*}\right)^{-1}$	nonsingular and $\boldsymbol{A}^\dagger = \boldsymbol{A}^*(\boldsymbol{A}\boldsymbol{A}^*)^{-1}$.
lyche-numerical-linear-al F03119	233	1.000	`\mathcal{R}\left(\boldsymbol{A}^{*}\right), \mathcal{N}( \boldsymbol{A})`	$\mathcal{R}\left(\boldsymbol{A}^{*}\right), \mathcal{N}(\boldsymbol{A})$	rank r, and let $\mathcal{S}$ be one of the subspaces $\mathcal{R}(\boldsymbol{A}^*), \mathcal{N}(\boldsymbol{A})$. Show that the orthogonal
lyche-numerical-linear-al F03120	233	0.996	`\quad \mathbb{C}^{n}=\mathcal{R}\left(\boldsymbol{A}^{*} \right) \stackrel{\perp}{\oplus} \mathcal{N}( \boldsymbol{A})`	$\quad \mathbb{C}^{n}=\mathcal{R}\left(\boldsymbol{A}^{*}\right) \stackrel{\perp}{\oplus} \mathcal{N}(\boldsymbol{A})$	orthogonal sum $\quad \mathbb{C}^n = \mathcal{R}(\boldsymbol{A}^*)\stackrel{\perp}{\oplus}\mathcal{N}(\boldsymbol{A})$.
lyche-numerical-linear-al F03121	233	1.000	`\boldsymbol{u}^{\dagger}=\left(\boldsymbol{u}^{*} \boldsymbol{u}\right)^{-1} \boldsymbol{u}^{*}`	$\boldsymbol{u}^{\dagger}=\left(\boldsymbol{u}^{*}\boldsymbol{u}\right)^{-1}\boldsymbol{u}^{*}$	**Exercise 9.9 (The Generalized Inverse of a Vector)** Show that $\boldsymbol{u}^\dagger = (\boldsymbol{u}^*\boldsymbol{u})^{-1}\boldsymbol{u}^*$
lyche-numerical-linear-al F03122	233	1.000	`\boldsymbol{u} \in \mathbb{C}^{n, 1}`	$\boldsymbol{u} \in \mathbb{C}^{n, 1}$	if $\boldsymbol{u} \in \mathbb{C}^{n,1}$ is nonzero.
lyche-numerical-linear-al F03123	233	1.000	`\boldsymbol{A}=\boldsymbol{u} \boldsymbol{v}^{*}`	$\boldsymbol{A}=\boldsymbol{u}\boldsymbol{v}^{*}$	**Exercise 9.10 (The Generalized Inverse of an Outer Product)** If $\boldsymbol{A} = \boldsymbol{u}\boldsymbol{v}^*$ where
lyche-numerical-linear-al F03124	233	1.000	`\boldsymbol{u} \in \mathbb{C}^{m}, \boldsymbol{v} \in \mathbb{C}^{n}`	$\boldsymbol{u} \in \mathbb{C}^{m}, \boldsymbol{v} \in \mathbb{C}^{n}$	$\boldsymbol{u} \in \mathbb{C}^m$, $\boldsymbol{v} \in \mathbb{C}^n$ are nonzero, show that
lyche-numerical-linear-al F03125	233	0.990	`\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n} \right)^{\dagger}=\operatorname{diag}\left(\lambda_{1}^{ \dagger}, \ldots, \lambda_{n}^{\dagger}\right)`	$\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)^{\dagger}=\operatorname{diag}\left(\lambda_{1}^{\dagger}, \ldots, \lambda_{n}^{\dagger}\right)$	$\operatorname{diag}(\lambda_1, \ldots, \lambda_n)^\dagger = \operatorname{diag}(\lambda_1^\dagger, \ldots, \lambda_n^\dagger)$ where
lyche-numerical-linear-al F03126	233	1.000	`\left(\boldsymbol{A}^{*}\right)^{\dagger}=\left( \boldsymbol{A}^{\dagger}\right)^{*}`	$\left(\boldsymbol{A}^{*}\right)^{\dagger}=\left(\boldsymbol{A}^{\dagger}\right)^{*}$	**a)** $(\boldsymbol{A}^*)^\dagger = (\boldsymbol{A}^\dagger)^*$.
lyche-numerical-linear-al F03127	233	0.999	`\left(\boldsymbol{A}^{\dagger}\right)^{\dagger}= \boldsymbol{A}`	$\left(\boldsymbol{A}^{\dagger}\right)^{\dagger}=\boldsymbol{A}$	**b)** $(\boldsymbol{A}^\dagger)^\dagger = \boldsymbol{A}$.
lyche-numerical-linear-al F03128	233	0.995	`(\alpha \boldsymbol{A})^{\dagger}=\frac{1}{\alpha} \boldsymbol{A}^{\dagger}, \quad \alpha \neq 0`	$(\alpha \boldsymbol{A})^{\dagger}=\frac{1}{\alpha}\boldsymbol{A}^{\dagger}, \quad \alpha \neq 0$	**c)** $(\alpha\boldsymbol{A})^\dagger = \frac{1}{\alpha}\boldsymbol{A}^\dagger, \quad \alpha \neq 0$.

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-al
F03129</td><td>233</td><td>■1.000</td><td>k, m, n \in \mathbb{N}, \boldsymbol{A} \in</td><td>$k, m, n \in \mathbb{N}, \boldsymbol{A} \in$</td><td>**Exercise 9.13 (The Generalized Inverse of a Product)** Suppose $k, m, n \in \mathbb{N}$, $\boldsymbol{A} \in$</td></tr>
<tr><td>lyche-numerical-linear-al
F03130</td><td>233</td><td>■0.623</td><td>\mathbb{C}^{m \times n}, \boldsymbol{B} \in \mathbb{C}^{n \times k}</td><td>$\mathbb{C}^{m \times n}, \boldsymbol{B} \in \mathbb{C}^{n \times k}$</td><td>$\mathbb{C}^{m\times n}$, $\boldsymbol{B} \in \mathbb{C}^{n\times k}$. Suppose $\boldsymbol{A}$ has linearly independent columns and $\boldsymbol{B}$ has linearly</td></tr>
<tr><td>lyche-numerical-linear-al
F03131</td><td>233</td><td>■0.993</td><td>(\boldsymbol{A} \boldsymbol{B})^{\dagger}= \boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}</td><td>$(\boldsymbol{A} \boldsymbol{B})^{\dagger}= \boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}$</td><td>**a)** Show that $(\boldsymbol{A}\boldsymbol{B})^{\dagger} = \boldsymbol{B}^{\dagger}\boldsymbol{A}^{\dagger}$.</td></tr>
<tr><td>lyche-numerical-linear-al
F03132</td><td>233</td><td>■1.000</td><td>\boldsymbol{E}=\boldsymbol{A} \boldsymbol{B}, \boldsymbol{F}=\boldsymbol{B}^{\dagger} \boldsymbol{A}^{ \dagger}</td><td>$\boldsymbol{E}=\boldsymbol{A} \boldsymbol{B}, \boldsymbol{F}=\boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}$</td><td>Hint: Let $\boldsymbol{E} = \boldsymbol{A}\boldsymbol{B}$, $\boldsymbol{F} = \boldsymbol{B}^{\dagger}\boldsymbol{A}^{\dagger}$. Show by using $\boldsymbol{A}^{\dagger}\boldsymbol{A} = \boldsymbol{B}\boldsymbol{B}^{\dagger} = \boldsymbol{I}$ that $\boldsymbol{F}$ is the</td></tr>
<tr><td>lyche-numerical-linear-al
F03133</td><td>233</td><td>■1.000</td><td>\boldsymbol{A}^{\dagger} \boldsymbol{A}=\boldsymbol{B} \boldsymbol{B}^{\dagger}=\boldsymbol{I}</td><td>$\boldsymbol{A}^{\dagger} \boldsymbol{A}=\boldsymbol{B} \boldsymbol{B}^{\dagger}=\boldsymbol{I}$</td><td>Hint: Let $\boldsymbol{E} = \boldsymbol{A}\boldsymbol{B}$, $\boldsymbol{F} = \boldsymbol{B}^{\dagger}\boldsymbol{A}^{\dagger}$. Show by using $\boldsymbol{A}^{\dagger}\boldsymbol{A} = \boldsymbol{B}\boldsymbol{B}^{\dagger} = \boldsymbol{I}$ that $\boldsymbol{F}$ is the</td></tr>
<tr><td>lyche-numerical-linear-al
F03134</td><td>233</td><td>■0.998</td><td>\boldsymbol{A} \in \mathbb{R}^{1,2}, \boldsymbol{B} \in \mathbb{R}^{2,1}</td><td>$\boldsymbol{A} \in \mathbb{R}^{1,2}, \boldsymbol{B} \in \mathbb{R}^{2,1}$</td><td>**b)** Find $\boldsymbol{A} \in \mathbb{R}^{1,2}$, $\boldsymbol{B} \in \mathbb{R}^{2,1}$ such that $(\boldsymbol{A}\boldsymbol{B})^{\dagger} \neq \boldsymbol{B}^{\dagger}\boldsymbol{A}^{\dagger}$.</td></tr>
<tr><td>lyche-numerical-linear-al
F03135</td><td>233</td><td>■0.998</td><td>(\boldsymbol{A} \boldsymbol{B})^{\dagger} \neq \boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}</td><td>$(\boldsymbol{A} \boldsymbol{B})^{\dagger} \neq \boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}$</td><td>**b)** Find $\boldsymbol{A} \in \mathbb{R}^{1,2}$, $\boldsymbol{B} \in \mathbb{R}^{2,1}$ such that $(\boldsymbol{A}\boldsymbol{B})^{\dagger} \neq \boldsymbol{B}^{\dagger}\boldsymbol{A}^{\dagger}$.</td></tr>
<tr><td>lyche-numerical-linear-al
F03136</td><td>234</td><td>■1.000</td><td>\boldsymbol{A}^{*}=\boldsymbol{A}^{\dagger}</td><td>$\boldsymbol{A}^{*}=\boldsymbol{A}^{\dagger}$</td><td>$\boldsymbol{A}^* = \boldsymbol{A}^{\dagger}$ if and only if all singular values of $\boldsymbol{A}$ are either zero or one.</td></tr>
<tr><td>lyche-numerical-linear-al
F03137</td><td>234</td><td>■0.998</td><td>\boldsymbol{A}\left(\boldsymbol{A}^{*} \boldsymbol{A} \right)^{-1} \boldsymbol{A}^{*} \boldsymbol{b}</td><td>$\boldsymbol{A}\left(\boldsymbol{A}^{*} \boldsymbol{A} \right)^{-1} \boldsymbol{A}^{*} \boldsymbol{b}$</td><td>$\boldsymbol{A}(\boldsymbol{A}^*\boldsymbol{A})^{-1}\boldsymbol{A}^*\boldsymbol{b}$ is the projection of $\boldsymbol{b}$ into $\mathcal{R}(\boldsymbol{A})$. (Cf. Exercise 9.8.)</td></tr>
<tr><td>lyche-numerical-linear-al
F03138</td><td>234</td><td>■1.000</td><td>r>0</td><td>$r>0$</td><td>system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ where $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ has rank $r > 0$ and $\boldsymbol{b} \in \mathbb{C}^n$. Let</td></tr>
<tr><td>lyche-numerical-linear-al
F03139</td><td>234</td><td>■1.000</td><td>\boldsymbol{c}=\left[c_{1}, \ldots, c_{n}\right]^{*}= \boldsymbol{U}^{*} \boldsymbol{b}</td><td>$\boldsymbol{c}=\left[c_{1}, \ldots, c_{n}\right]^{*}= \boldsymbol{U}^{*} \boldsymbol{b}$</td><td>**a)** Let $\boldsymbol{c} = [c_1,\ldots,c_n]^* = \boldsymbol{U}^*\boldsymbol{b}$ and $\boldsymbol{y} = [y_1,\ldots,y_n]^* = \boldsymbol{V}^*\boldsymbol{x}$. Show that $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$</td></tr>
<tr><td>lyche-numerical-linear-al
F03140</td><td>234</td><td>■1.000</td><td>\boldsymbol{y}=\left[y_{1}, \ldots, y_{n}\right]^{*}= \boldsymbol{V}^{*} \boldsymbol{x}</td><td>$\boldsymbol{y}=\left[y_{1}, \ldots, y_{n}\right]^{*}= \boldsymbol{V}^{*} \boldsymbol{x}$</td><td>**a)** Let $\boldsymbol{c} = [c_1,\ldots,c_n]^* = \boldsymbol{U}^*\boldsymbol{b}$ and $\boldsymbol{y} = [y_1,\ldots,y_n]^* = \boldsymbol{V}^*\boldsymbol{x}$. Show that $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$</td></tr>
<tr><td>lyche-numerical-linear-al
F03141</td><td>234</td><td>■0.999</td><td>c_{r+1}=\cdots=c_{n}=0</td><td>$c_{r+1}=\cdots=c_{n}=0$</td><td>**b)** Show that $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ has a solution $\boldsymbol{x}$ if and only if $c_{r+1} = \cdots = c_n = 0$.</td></tr>
<tr><td>lyche-numerical-linear-al
F03142</td><td>234</td><td>■1.000</td><td>\boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^{n}</td><td>$\boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^{n}$</td><td>**Exercise 9.17 (Fredholm's Alternative)** For any $\boldsymbol{A} \in \mathbb{C}^{m\times n}$, $\boldsymbol{b} \in \mathbb{C}^n$ show that</td></tr>
<tr><td>lyche-numerical-linear-al
F03143</td><td>234</td><td>■1.000</td><td>\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})</td><td>$\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$</td><td>In other words either $\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$, or we can find $\boldsymbol{y} \in \mathcal{N}(\boldsymbol{A}^*)$ such that $\boldsymbol{y}^*\boldsymbol{b} \neq 0$. This</td></tr>
<tr><td>lyche-numerical-linear-al
F03144</td><td>234</td><td>■1.000</td><td>\boldsymbol{y} \in \mathcal{N}\left(\boldsymbol{A}^{*} \right)</td><td>$\boldsymbol{y} \in \mathcal{N}\left(\boldsymbol{A}^{*} \right)$</td><td>In other words either $\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$, or we can find $\boldsymbol{y} \in \mathcal{N}(\boldsymbol{A}^*)$ such that $\boldsymbol{y}^*\boldsymbol{b} \neq 0$. This</td></tr>
<tr><td>lyche-numerical-linear-al
F03145</td><td>234</td><td>■1.000</td><td>\boldsymbol{y}^{*} \boldsymbol{b} \neq 0</td><td>$\boldsymbol{y}^{*} \boldsymbol{b} \neq 0$</td><td>In other words either $\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$, or we can find $\boldsymbol{y} \in \mathcal{N}(\boldsymbol{A}^*)$ such that $\boldsymbol{y}^*\boldsymbol{b} \neq 0$. This</td></tr>
<tr><td>lyche-numerical-linear-al
F03146</td><td>234</td><td>■1.000</td><td>\mathbb{C}^{(m-n) \times n}</td><td>$\mathbb{C}^{(m-n) \times n}$</td><td>where $\boldsymbol{B}$ is a non-singular $n \times n$ matrix and $\boldsymbol{C}$ is in $\mathbb{C}^{(m-n)\times n}$. Let $\boldsymbol{A}^{\dagger}$ denote the</td></tr>
<tr><td>lyche-numerical-linear-al
F03147</td><td>234</td><td>■1.000</td><td>\left\|\boldsymbol{A}^{\dagger}\right\|_{2} \leq\left\| \boldsymbol{B}^{-1}\right\|_{2}</td><td>$\left\|\boldsymbol{A}^{\dagger}\right\|_{2} \leq\left\| \boldsymbol{B}^{-1}\right\|_{2}$</td><td>pseudoinverse of $\boldsymbol{A}$. Show that $\|\boldsymbol{A}^{\dagger}\|_2 \leq \|\boldsymbol{B}^{-1}\|_2$.</td></tr>
<tr><td>lyche-numerical-linear-al
F03148</td><td>235</td><td>■1.000</td><td>K(\boldsymbol{A})=\|\boldsymbol{A}\|_{2}\left\| \boldsymbol{A}^{\dagger}\right\|_{2}</td><td>$K(\boldsymbol{A})=\|\boldsymbol{A}\|_{2}\left\| \boldsymbol{A}^{\dagger}\right\|_{2}$</td><td>**b)** Compute $K(\boldsymbol{A}) = \|\boldsymbol{A}\|_2\|\boldsymbol{A}^{\dagger}\|_2$.</td></tr>
<tr><td>lyche-numerical-linear-al
F03149</td><td>235</td><td>■0.999</td><td>\boldsymbol{y}_{A^{\dagger}}</td><td>$\boldsymbol{y}_{A^{\dagger}}$</td><td>and $\boldsymbol{y}_{A^{\dagger}}$ are vectors with $\|\boldsymbol{y}_A\| = \|\boldsymbol{y}_{A^{\dagger}}\| = 1$ and $\|\boldsymbol{A}\| = \|\boldsymbol{A}\boldsymbol{y}_A\|$ and $\|\boldsymbol{A}^{\dagger}\| =$</td></tr>
<tr><td>lyche-numerical-linear-al
F03150</td><td>235</td><td>■0.999</td><td>\left\|\boldsymbol{y}_{\boldsymbol{A}}\right\|=\left\| \boldsymbol{y}_{A^{\dagger}}\right\|=1</td><td>$\left\|\boldsymbol{y}_{\boldsymbol{A}}\right\|=\left\| \boldsymbol{y}_{A^{\dagger}}\right\|=1$</td><td>and $\boldsymbol{y}_{A^{\dagger}}$ are vectors with $\|\boldsymbol{y}_A\| = \|\boldsymbol{y}_{A^{\dagger}}\| = 1$ and $\|\boldsymbol{A}\| = \|\boldsymbol{A}\boldsymbol{y}_A\|$ and $\|\boldsymbol{A}^{\dagger}\| =$</td></tr>
<tr><td>lyche-numerical-linear-al
F03151</td><td>235</td><td>■0.999</td><td>\left\|\boldsymbol{A}^{\dagger}\right\|=</td><td>$\left\|\boldsymbol{A}^{\dagger}\right\|=$</td><td>and $\boldsymbol{y}_{A^{\dagger}}$ are vectors with $\|\boldsymbol{y}_A\| = \|\boldsymbol{y}_{A^{\dagger}}\| = 1$ and $\|\boldsymbol{A}\| = \|\boldsymbol{A}\boldsymbol{y}_A\|$ and $\|\boldsymbol{A}^{\dagger}\| =$</td></tr>
<tr><td>lyche-numerical-linear-al
F03152</td><td>235</td><td>■0.999</td><td>\left\|\boldsymbol{A}^{\dagger} \boldsymbol{y}_{A^{ \dagger}}\right\|</td><td>$\left\|\boldsymbol{A}^{\dagger} \boldsymbol{y}_{A^{\dagger}}\right\|$</td><td>$\|\boldsymbol{A}^{\dagger}\boldsymbol{y}_{A^{\dagger}}\|$.</td></tr>
<tr><td>lyche-numerical-linear-al
F03153</td><td>235</td><td>■1.000</td><td>\boldsymbol{b}=\boldsymbol{A} \boldsymbol{y}_{A}, \boldsymbol{e}_{1}=\boldsymbol{y}_{A^{\dagger}}</td><td>$\boldsymbol{b}=\boldsymbol{A} \boldsymbol{y}_{A}, \boldsymbol{e}_{1}=\boldsymbol{y}_{A^{\dagger}}$</td><td>**a)** Show that we have equality to the right in (9.13) if $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{y}_A$, $\boldsymbol{e}_1 = \boldsymbol{y}_{A^{\dagger}}$.</td></tr>
<tr><td>lyche-numerical-linear-al
F03154</td><td>235</td><td>■1.000</td><td>\boldsymbol{A}^{\dagger}=\boldsymbol{A}^{-1}</td><td>$\boldsymbol{A}^{\dagger}=\boldsymbol{A}^{-1}$</td><td>nonsingular then $\boldsymbol{A}^{\dagger} = \boldsymbol{A}^{-1}$ and $\boldsymbol{e}_1 = \boldsymbol{e}$.</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-a F03155	235	■1.000	`\boldsymbol{e}_{1}=\boldsymbol{e}`	$\boldsymbol{e}_{1}=\boldsymbol{e}$	nonsingular then $\boldsymbol{A}^{\dagger}=\boldsymbol{A}^{-1}$ and $\boldsymbol{e}_1=\boldsymbol{e}$.
lyche-numerical-linear-a F03156	235	■1.000	`(2,2)`	$(2,2)$	**b)** Suppose ϵ is small and we replace the $(2,2)$ entry $3+2\epsilon+\epsilon^2$ in $\boldsymbol{A}^*\boldsymbol{A}$ by $3+2\epsilon$.
lyche-numerical-linear-a F03157	235	■1.000	`3+2 \epsilon+\epsilon^{2}`	$3+2 \epsilon+\epsilon^{2}$	**b)** Suppose ϵ is small and we replace the $(2,2)$ entry $3+2\epsilon+\epsilon^2$ in $\boldsymbol{A}^*\boldsymbol{A}$ by $3+2\epsilon$.
lyche-numerical-linear-a F03158	235	■1.000	`3+2 \epsilon`	$3+2 \epsilon$	**b)** Suppose ϵ is small and we replace the $(2,2)$ entry $3+2\epsilon+\epsilon^2$ in $\boldsymbol{A}^*\boldsymbol{A}$ by $3+2\epsilon$.
lyche-numerical-linear-a F03159	235	■1.000	`\epsilon<\sqrt{u}, u`	$\epsilon<\sqrt{u}, u$	(This will be done in a computer if $\epsilon<\sqrt{u}$, u being the round-off unit). For
lyche-numerical-linear-a F03160	235	■1.000	`u=10^{-16}`	$u=10^{-16}$	example, if $u=10^{-16}$ then $\sqrt{u}=10^{-8}$. Solve $\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x}=\boldsymbol{A}^*\boldsymbol{b}$ for $\boldsymbol{x}$ and compare
lyche-numerical-linear-a F03161	235	■1.000	`\sqrt{u}=10^{-8}`	$\sqrt{u}=10^{-8}$	example, if $u=10^{-16}$ then $\sqrt{u}=10^{-8}$. Solve $\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x}=\boldsymbol{A}^*\boldsymbol{b}$ for $\boldsymbol{x}$ and compare
lyche-numerical-linear-a F03162	235	■1.000	`\boldsymbol{A}(\epsilon) \in \mathbb{R}^{n \times n}`	$\boldsymbol{A}(\epsilon) \in \mathbb{R}^{n \times n}$	$\boldsymbol{A}(\epsilon)\in\mathbb{R}^{n\times n}$ be bidiagonal with $a_{i,j}=0$ for $i,j=1,\ldots,n$ and $j\neq i,i+1$.
lyche-numerical-linear-a F03163	235	■1.000	`j \neq i, i+1`	$j \neq i, i+1$	$\boldsymbol{A}(\epsilon)\in\mathbb{R}^{n\times n}$ be bidiagonal with $a_{i,j}=0$ for $i,j=1,\ldots,n$ and $j\neq i,i+1$.
lyche-numerical-linear-a F03164	235	■1.000	`a_{k, k+1}=\epsilon \in \mathbb{R}`	$a_{k, k+1}=\epsilon \in \mathbb{R}$	Moreover, for some $1\le k\le n-1$ we have $a_{k,k+1}=\epsilon\in\mathbb{R}$. Show that
lyche-numerical-linear-a F03165	235	■1.000	`\sigma_{i}(\epsilon), i=1, \ldots, n`	$\sigma_{i}(\epsilon), i=1, \ldots, n$	where $\sigma_i(\epsilon)$, $i=1,\ldots,n$ are the singular values of $\boldsymbol{A}(\epsilon)$.
lyche-numerical-linear-a F03166	235	■1.000	`\boldsymbol{A}(\epsilon)`	$\boldsymbol{A}(\epsilon)$	where $\sigma_i(\epsilon)$, $i=1,\ldots,n$ are the singular values of $\boldsymbol{A}(\epsilon)$.
lyche-numerical-linear-a F03167	238	■1.000	`\Omega:=(0,1)^{2}=\{(x, y): 0<x, y<1\}`	$\Omega:=(0,1)^{2}=\{(x, y): 0<x, y<1\}$	Let $\Omega:=(0,1)^2=\{(x,y):0<x,y<1\}$ be the open unit square with boundary
lyche-numerical-linear-a F03168	238	■1.000	`\partial \Omega`	$\partial \Omega$	$\partial\Omega$. Consider the problem
lyche-numerical-linear-a F03169	238	■1.000	`u=`	$u=$	Here the function f is given and continuous on Ω, and we seek a function $u=$
lyche-numerical-linear-a F03170	238	■1.000	`u(x, y)`	$u(x, y)$	$u(x,y)$ such that (10.1) holds and which is zero on $\partial\Omega$.
lyche-numerical-linear-a F03171	238	■1.000	`v_{j, k} \approx u(j h, k h)`	$v_{j, k} \approx u(j h, k h)$	approximations $v_{j,k}\approx u(jh,kh)$ on a grid of points given by
lyche-numerical-linear-a F03172	238	■1.000	`\Omega_{h}:=\{(j h, k h): j, k=1, \ldots, m\}`	$\Omega_{h}:=\{(j h, k h): j, k=1, \ldots, m\}$	The points $\Omega_h:=\{(jh,kh):\ j,k=1,\ldots,m\}$ are called the interior points, while
lyche-numerical-linear-a F03173	238	■1.000	`\bar{\Omega}_{h} \backslash \Omega_{h}`	$\bar{\Omega}_{h} \backslash \Omega_{h}$	$\overline{\Omega}_h\setminus\Omega_h$ are the boundary points. The solution is zero at the boundary points. Using
lyche-numerical-linear-a F03174	239	■1.000	`f_{j, k}:=f(j h, k h)`	$f_{j, k}:=f(j h, k h)$	where $f_{j,k}:=f(jh,kh)$ and
lyche-numerical-linear-a F03175	239	■1.000	`n:=m^{2}`	$n:=m^{2}$	Let us take a closer look at (10.2). It consists of $n:=m^2$ linear equations. Since the
lyche-numerical-linear-a F03176	239	■1.000	`v_{j, k}`	$v_{j, k}$	values at the boundary points are known, the unknowns are the n numbers $v_{j,k}$ at
lyche-numerical-linear-a F03177	239	■1.000	`\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}`	$\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}$	where $\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1)\in\mathbb{R}^{m\times m}$ is the second derivative matrix given by
lyche-numerical-linear-a F03178	239	■0.913	`j, k`	j, k	Indeed, the (j,k) element in $\boldsymbol{T}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{T}$ is given by
lyche-numerical-linear-a F03179	239	■0.913	`\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V T}`	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V T}$	Indeed, the (j,k) element in $\boldsymbol{T}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{T}$ is given by
lyche-numerical-linear-a F03180	239	■1.000	`\boldsymbol{B} \in \mathbb{R}^{m \times n}`	$\boldsymbol{B} \in \mathbb{R}^{m \times n}$	**Definition 10.1 (vec Operation)** For any $\boldsymbol{B}\in\mathbb{R}^{m\times n}$ we define the vector
lyche-numerical-linear-a F03181	240	■1.000	`\boldsymbol{x}:=\operatorname{vec}(\boldsymbol{V}) \in \mathbb{R}^{n}`	$\boldsymbol{x}:=\operatorname{vec}(\boldsymbol{V}) \in \mathbb{R}^{n}$	Let $\boldsymbol{x}:=vec(\boldsymbol{V})\in\mathbb{R}^n$, where $n=m^2$. Note that forming $\boldsymbol{x}$ by stacking the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F03182	240	1.000	`n=m^{2}`	$n=m^{2}$	Let $\boldsymbol{x} := vec(\boldsymbol{V}) \in \mathbb{R}^n$, where $n = m^2$. Note that forming $\boldsymbol{x}$ by stacking the
lyche-numerical-linear-algebra F03183	240	1.000	`x_{i}=v_{j, k}`	$x_{i}=v_{j, k}$	where $x_i = v_{j,k}$ and $b_i = h^2 f_{j,k}$. This must be modified close to the boundary. We
lyche-numerical-linear-algebra F03184	240	1.000	`b_{i}=h^{2} f_{j, k}`	$b_{i}=h^{2} f_{j, k}$	where $x_i = v_{j,k}$ and $b_i = h^2 f_{j,k}$. This must be modified close to the boundary. We
lyche-numerical-linear-algebra F03185	240	1.000	`\boldsymbol{x}=\operatorname{vec}(\boldsymbol{V}), \boldsymbol{b}=h^{2} \operatorname{vec}(\boldsymbol{F})`	$\boldsymbol{x}=\operatorname{vec}(\boldsymbol{V}), \boldsymbol{b}=h^{2} \operatorname{vec}(\boldsymbol{F})$	where $\boldsymbol{x} = vec(\boldsymbol{V})$, $\boldsymbol{b} = h^2 vec(\boldsymbol{F})$ and $\boldsymbol{A}$ is the **Poisson matrix** given by
lyche-numerical-linear-algebra F03186	241	0.905	`\boldsymbol{T}_{1} \in \mathbb{R}^{m \times m}`	$\boldsymbol{T}_{1} \in \mathbb{R}^{m \times m}$	In Sect. 2.4 we encountered the 1-dimensional test matrix $\boldsymbol{T}_1 \in \mathbb{R}^{m\times m}$ defined for
lyche-numerical-linear-algebra F03187	241	1.000	`a, d`	a, d	any real numbers a, d by
lyche-numerical-linear-algebra F03188	241	1.000	`\boldsymbol{T}_{2}=\left[a_{i j}\right] \in`	$\boldsymbol{T}_{2}=\left[a_{i j}\right] \in$	The (2-dimensional) Poisson matrix is a special case of the matrix $\boldsymbol{T}_2 = [a_{ij}] \in$
lyche-numerical-linear-algebra F03189	242	0.956	`\boldsymbol{T}_{2}`	$\boldsymbol{T}_{2}$	The partition into 3×3 sub matrices shows that $\boldsymbol{T}_2$ is block tridiagonal.
lyche-numerical-linear-algebra F03190	242	1.000	`p, q, r, s`	p, q, r, s	**Definition 10.2 (Kronecker Product)** For any positive integers p, q, r, s we
lyche-numerical-linear-algebra F03191	242	1.000	`\boldsymbol{A} \in \mathbb{R}^{p \times q}`	$\boldsymbol{A} \in \mathbb{R}^{p \times q}$	define the **Kronecker product** of two matrices $\boldsymbol{A} \in \mathbb{R}^{p\times q}$ and $\boldsymbol{B} \in \mathbb{R}^{r\times s}$ as a
lyche-numerical-linear-algebra F03192	242	1.000	`\boldsymbol{B} \in \mathbb{R}^{r \times s}`	$\boldsymbol{B} \in \mathbb{R}^{r \times s}$	define the **Kronecker product** of two matrices $\boldsymbol{A} \in \mathbb{R}^{p\times q}$ and $\boldsymbol{B} \in \mathbb{R}^{r\times s}$ as a
lyche-numerical-linear-algebra F03193	242	1.000	`\boldsymbol{C} \in \mathbb{R}^{p r \times q s}`	$\boldsymbol{C} \in \mathbb{R}^{p r \times q s}$	matrix $\boldsymbol{C} \in \mathbb{R}^{pr\times qs}$ given in block form as
lyche-numerical-linear-algebra F03194	242	1.000	`\boldsymbol{C}=\boldsymbol{A} \otimes \boldsymbol{B}`	$\boldsymbol{C}=\boldsymbol{A} \otimes \boldsymbol{B}$	We denote the Kronecker product of $\boldsymbol{A}$ and $\boldsymbol{B}$ by $\boldsymbol{C} = \boldsymbol{A} \otimes \boldsymbol{B}$.
lyche-numerical-linear-algebra F03195	242	1.000	`\boldsymbol{B} \otimes \boldsymbol{A}`	$\boldsymbol{B} \otimes \boldsymbol{A}$	which in our notation is given by $\boldsymbol{B} \otimes \boldsymbol{A}$.
lyche-numerical-linear-algebra F03196	242	1.000	`\boldsymbol{u} \otimes \boldsymbol{v}=\left[ \boldsymbol{u}^{T} v_{1}, \ldots, \boldsymbol{u}^{T} v_{r}\right]^{T}`	$\boldsymbol{u} \otimes \boldsymbol{v}=\left[\boldsymbol{u}^{T} v_{1}, \ldots, \boldsymbol{u}^{T} v_{r}\right]^{T}$	The Kronecker product $\boldsymbol{u} \otimes \boldsymbol{v} = \left[\boldsymbol{u}^T v_1, \ldots, \boldsymbol{u}^T v_r\right]^T$ of two column vectors
lyche-numerical-linear-algebra F03197	242	1.000	`\boldsymbol{u} \in \mathbb{R}^{p}`	$\boldsymbol{u} \in \mathbb{R}^{p}$	$\boldsymbol{u} \in \mathbb{R}^p$ and $\boldsymbol{v} \in \mathbb{R}^r$ is a column vector of length $p \cdot r$.
lyche-numerical-linear-algebra F03198	242	1.000	`\boldsymbol{v} \in \mathbb{R}^{r}`	$\boldsymbol{v} \in \mathbb{R}^{r}$	$\boldsymbol{u} \in \mathbb{R}^p$ and $\boldsymbol{v} \in \mathbb{R}^r$ is a column vector of length $p \cdot r$.
lyche-numerical-linear-algebra F03199	242	1.000	`p \cdot r`	$p \cdot r$	$\boldsymbol{u} \in \mathbb{R}^p$ and $\boldsymbol{v} \in \mathbb{R}^r$ is a column vector of length $p \cdot r$.
lyche-numerical-linear-algebra F03200	243	1.000	`r, s, k`	r, s, k	**Definition 10.3 (Kronecker Sum)** For positive integers r, s, k, let $\boldsymbol{A} \in \mathbb{R}^{r\times r}$, $\boldsymbol{B} \in$
lyche-numerical-linear-algebra F03201	243	1.000	`\boldsymbol{A} \in \mathbb{R}^{r \times r}, \boldsymbol{B} \in`	$\boldsymbol{A} \in \mathbb{R}^{r \times r}, \boldsymbol{B} \in$	**Definition 10.3 (Kronecker Sum)** For positive integers r, s, k, let $\boldsymbol{A} \in \mathbb{R}^{r\times r}$, $\boldsymbol{B} \in$
lyche-numerical-linear-algebra F03202	243	1.000	`\mathbb{R}^{s \times s}`	$\mathbb{R}^{s \times s}$	$\mathbb{R}^{s\times s}$, and $\boldsymbol{I}_k$ be the identity matrix of order k. The sum $\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}$ is known
lyche-numerical-linear-algebra F03203	243	1.000	`\boldsymbol{I}_{k}`	$\boldsymbol{I}_{k}$	$\mathbb{R}^{s\times s}$, and $\boldsymbol{I}_k$ be the identity matrix of order k. The sum $\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}$ is known
lyche-numerical-linear-algebra F03204	243	1.000	`\boldsymbol{A} \otimes \boldsymbol{I}_{s}+\boldsymbol{I}_{r} \otimes \boldsymbol{B}`	$\boldsymbol{A} \otimes \boldsymbol{I}_{s}+\boldsymbol{I}_{r} \otimes \boldsymbol{B}$	$\mathbb{R}^{s\times s}$, and $\boldsymbol{I}_k$ be the identity matrix of order k. The sum $\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}$ is known
lyche-numerical-linear-algebra F03205	243	1.000	`\lambda, \mu`	λ, μ	λ, μ and matrices $\boldsymbol{A}, \boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{B}, \boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{C}$ of dimensions such that the operations
lyche-numerical-linear-algebra F03206	243	1.000	`\boldsymbol{A}, \boldsymbol{A}_{1}, \boldsymbol{A}_{2}, \boldsymbol{B}, \boldsymbol{B}_{1}, \boldsymbol{B}_{2}, \boldsymbol{C}`	$\boldsymbol{A}, \boldsymbol{A}_{1}, \boldsymbol{A}_{2}, \boldsymbol{B}, \boldsymbol{B}_{1}, \boldsymbol{B}_{2}, \boldsymbol{C}$	λ, μ and matrices $\boldsymbol{A}, \boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{B}, \boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{C}$ of dimensions such that the operations
lyche-numerical-linear-algebra F03207	243	1.000	`\boldsymbol{A} \otimes \boldsymbol{B} \neq \boldsymbol{B} \otimes \boldsymbol{A}`	$\boldsymbol{A} \otimes \boldsymbol{B} \neq \boldsymbol{B} \otimes \boldsymbol{A}$	Note however that in general we have $\boldsymbol{A} \otimes \boldsymbol{B} \neq \boldsymbol{B} \otimes \boldsymbol{A}$, but it can be shown that

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F03208	243	1.000	`\boldsymbol{P}, \boldsymbol{Q}`	$\boldsymbol{P}, \boldsymbol{Q}$	there are permutation matrices $\boldsymbol{P}$, $\boldsymbol{Q}$ such that $\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$, see [9].
lyche-numerical-linear-algebra F03209	243	1.000	`\boldsymbol{B} \otimes \boldsymbol{A}=\boldsymbol{P}( \boldsymbol{A} \otimes \boldsymbol{B}) \boldsymbol{Q}`	$\boldsymbol{B} \otimes \boldsymbol{A}=\boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B}) \boldsymbol{Q}$	there are permutation matrices $\boldsymbol{P}$, $\boldsymbol{Q}$ such that $\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$, see [9].
lyche-numerical-linear-algebra F03210	243	0.989	`\boldsymbol{B} \boldsymbol{D}`	$\boldsymbol{B} \boldsymbol{D}$	*ces with dimensions so that the products* $\boldsymbol{AC}$ *and* $\boldsymbol{BD}$ *are well defined. Then the*
lyche-numerical-linear-algebra F03211	243	1.000	`(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})`	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})$	*product* $(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})$ *is defined and*
lyche-numerical-linear-algebra F03212	243	1.000	`\boldsymbol{B} \in \mathbb{R}^{r, t}`	$\boldsymbol{B} \in \mathbb{R}^{r, t}$	***Proof*** If $\boldsymbol{B} \in \mathbb{R}^{r,t}$ and $\boldsymbol{D} \in \mathbb{R}^{t,s}$ for some integers r, s, t, then
lyche-numerical-linear-algebra F03213	243	1.000	`\boldsymbol{D} \in \mathbb{R}^{t, s}`	$\boldsymbol{D} \in \mathbb{R}^{t, s}$	***Proof*** If $\boldsymbol{B} \in \mathbb{R}^{r,t}$ and $\boldsymbol{D} \in \mathbb{R}^{t,s}$ for some integers r, s, t, then
lyche-numerical-linear-algebra F03214	243	1.000	`r, s, t`	r, s, t	***Proof*** If $\boldsymbol{B} \in \mathbb{R}^{r,t}$ and $\boldsymbol{D} \in \mathbb{R}^{t,s}$ for some integers r, s, t, then
lyche-numerical-linear-algebra F03215	244	1.000	`r, s \in \mathbb{N}`	$r, s \in \mathbb{N}$	**Theorem 10.1 (Properties of Kronecker Products)** *Suppose for* $r, s \in \mathbb{N}$ *that*
lyche-numerical-linear-algebra F03216	244	1.000	`\boldsymbol{A} \in \mathbb{R}^{r \times r}`	$\boldsymbol{A} \in \mathbb{R}^{r \times r}$	$\boldsymbol{A} \in \mathbb{R}^{r\times r}$ *and* $\boldsymbol{B} \in \mathbb{R}^{s\times s}$ *are square matrices with eigenpairs* $(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \ldots, r$
lyche-numerical-linear-algebra F03217	244	1.000	`\boldsymbol{B} \in \mathbb{R}^{s \times s}`	$\boldsymbol{B} \in \mathbb{R}^{s \times s}$	$\boldsymbol{A} \in \mathbb{R}^{r\times r}$ *and* $\boldsymbol{B} \in \mathbb{R}^{s\times s}$ *are square matrices with eigenpairs* $(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \ldots, r$
lyche-numerical-linear-algebra F03218	244	1.000	`\left(\lambda_{i}, \boldsymbol{u}_{i}\right) i=1, \ldots, r`	$\left(\lambda_{i}, \boldsymbol{u}_{i}\right) i=1, \ldots, r$	$\boldsymbol{A} \in \mathbb{R}^{r\times r}$ *and* $\boldsymbol{B} \in \mathbb{R}^{s\times s}$ *are square matrices with eigenpairs* $(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \ldots, r$
lyche-numerical-linear-algebra F03219	244	1.000	`\left(\mu_{j}, \boldsymbol{v}_{j}\right), j=1, \ldots, s`	$\left(\mu_{j}, \boldsymbol{v}_{j}\right), j=1, \ldots, s$	*and* $(\mu_j, \boldsymbol{v}_j)$, $j = 1, \ldots, s$. *Moreover, let* $\boldsymbol{F}, \boldsymbol{V} \in \mathbb{R}^{r\times s}$. *Then*
lyche-numerical-linear-algebra F03220	244	1.000	`\boldsymbol{F}, \boldsymbol{V} \in \mathbb{R}^{r \times s}`	$\boldsymbol{F}, \boldsymbol{V} \in \mathbb{R}^{r \times s}$	*and* $(\mu_j, \boldsymbol{v}_j)$, $j = 1, \ldots, s$. *Moreover, let* $\boldsymbol{F}, \boldsymbol{V} \in \mathbb{R}^{r\times s}$. *Then*
lyche-numerical-linear-algebra F03221	244	0.999	`(\boldsymbol{A} \otimes \boldsymbol{B})^{T}= \boldsymbol{A}^{T} \otimes \boldsymbol{B}^{T}`	$(\boldsymbol{A} \otimes \boldsymbol{B})^{T}= \boldsymbol{A}^{T} \otimes \boldsymbol{B}^{T}$	*1.* $(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T$, *(this also holds for rectangular matrices).*
lyche-numerical-linear-algebra F03222	244	0.999	`\boldsymbol{A} \otimes \boldsymbol{B}`	$\boldsymbol{A} \otimes \boldsymbol{B}$	*2. If* $\boldsymbol{A}$ *and* $\boldsymbol{B}$ *are nonsingular then* $\boldsymbol{A} \otimes \boldsymbol{B}$ *is nonsingular. with* $(\boldsymbol{A} \otimes \boldsymbol{B})^{-1} =$
lyche-numerical-linear-algebra F03223	244	0.999	`(\boldsymbol{A} \otimes \boldsymbol{B})^{-1}=`	$(\boldsymbol{A} \otimes \boldsymbol{B})^{-1}=$	*2. If* $\boldsymbol{A}$ *and* $\boldsymbol{B}$ *are nonsingular then* $\boldsymbol{A} \otimes \boldsymbol{B}$ *is nonsingular. with* $(\boldsymbol{A} \otimes \boldsymbol{B})^{-1} =$
lyche-numerical-linear-algebra F03224	244	1.000	`\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}`	$\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}$	$\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}$.
lyche-numerical-linear-algebra F03225	244	1.000	`(\boldsymbol{A} \otimes \boldsymbol{B})\left( \boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)= \lambda_{i} \mu_{j}\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right), \quad i=1, \ldots, r, \quad j=1, \ldots, s`	$(\boldsymbol{A} \otimes \boldsymbol{B})\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)= \lambda_{i} \mu_{j}\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right), \quad i=1, \ldots, r, \quad j=1, \ldots, s$	*4.* $(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i\mu_j(\boldsymbol{u}_i \otimes \boldsymbol{v}_j), \quad i = 1, \ldots, r, \quad j = 1, \ldots, s,$
lyche-numerical-linear-algebra F03226	244	0.999	`\left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}+ \boldsymbol{I}_{r} \otimes \boldsymbol{B}\right)\left( \boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)= \left(\lambda_{i}+\mu_{j}\right)\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right), \quad i=1, \ldots, r, \quad j=1, \ldots, s`	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}+ \boldsymbol{I}_{r} \otimes \boldsymbol{B}\right)\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)= \left(\lambda_{i}+\mu_{j}\right)\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right), \quad i=1, \ldots, r, \quad j=1, \ldots, s$	*5.* $(\boldsymbol{A}\otimes\boldsymbol{I}_s+\boldsymbol{I}_r\otimes\boldsymbol{B})(\boldsymbol{u}_i\otimes\boldsymbol{v}_j) = (\lambda_i+\mu_j)(\boldsymbol{u}_i\otimes\boldsymbol{v}_j), \quad i = 1, \ldots, r, \quad j = 1, \ldots, s.,$
lyche-numerical-linear-algebra F03227	244	1.000	`\boldsymbol{A} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{B}`	$\boldsymbol{A} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{B}$	$\boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$ *is positive definite.*
lyche-numerical-linear-algebra F03228	244	0.689	`\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^{T}= \boldsymbol{F} \quad \Leftrightarrow \quad(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}( \boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})`	$\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^{T}= \boldsymbol{F} \quad \Leftrightarrow \quad(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$	*7.* $\boldsymbol{AVB}^T = \boldsymbol{F} \quad \Leftrightarrow \quad (\boldsymbol{A} \otimes \boldsymbol{B})\operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F}),$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F03229	244	1.000	`\boldsymbol{A} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{B}^{T}=\boldsymbol{F} \quad \Leftrightarrow \quad\left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}+ \boldsymbol{I}_{r} \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}( \boldsymbol{F})`	$\boldsymbol{A} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{B}^{T}=\boldsymbol{F} \quad \Leftrightarrow \quad\left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}+ \boldsymbol{I}_{r} \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$	8. $\boldsymbol{A V}+\boldsymbol{V B}^{T}=\boldsymbol{F} \quad \Leftrightarrow \quad (\boldsymbol{A} \otimes \boldsymbol{I}_{s}+\boldsymbol{I}_{r} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$.
lyche-numerical-linear-algebra F03230	244	1.000	`\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+ \boldsymbol{I} \otimes \boldsymbol{T}`	$\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+ \boldsymbol{I} \otimes \boldsymbol{T}$	Poisson matrix $\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}$ is also positive definite.
lyche-numerical-linear-algebra F03231	244	1.000	`\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^{T}= \boldsymbol{F}`	$\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^{T}= \boldsymbol{F}$	4. The system $\boldsymbol{A V B}^{T}=\boldsymbol{F}$ in part 7 can be solved by first finding $\boldsymbol{W}$ from $\boldsymbol{A W}=$
lyche-numerical-linear-algebra F03232	244	1.000	`\boldsymbol{A} \boldsymbol{W}=`	$\boldsymbol{A} \boldsymbol{W}=$	4. The system $\boldsymbol{A V B}^{T}=\boldsymbol{F}$ in part 7 can be solved by first finding $\boldsymbol{W}$ from $\boldsymbol{A W}=$
lyche-numerical-linear-algebra F03233	244	1.000	`\boldsymbol{B} \boldsymbol{V}^{T}=\boldsymbol{W}^{T}`	$\boldsymbol{B} \boldsymbol{V}^{T}=\boldsymbol{W}^{T}$	$\boldsymbol{F}$, and then finding $\boldsymbol{V}$ from $\boldsymbol{B V}^{T}=\boldsymbol{W}^{T}$. This is preferable to solving the much
lyche-numerical-linear-algebra F03234	244	0.544	`(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}( \boldsymbol{F})`	$(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$	larger linear system $(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$.
lyche-numerical-linear-algebra F03235	244	0.965	`\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V T}= \boldsymbol{F}`	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V T}= \boldsymbol{F}$	5. A fast way to solve the 2D Poisson problem in the form $\boldsymbol{T V}+\boldsymbol{V T}=\boldsymbol{F}$ will be
lyche-numerical-linear-algebra F03236	245	0.998	`(\boldsymbol{A} \otimes \boldsymbol{B})\left( \boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}\right)= \left(\boldsymbol{A} \boldsymbol{A}^{-1}\right) \otimes \left(\boldsymbol{B} \boldsymbol{B}^{-1}\right)=`	$(\boldsymbol{A} \otimes \boldsymbol{B})\left(\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}\right)= \left(\boldsymbol{A} \boldsymbol{A}^{-1}\right) \otimes \left(\boldsymbol{B} \boldsymbol{B}^{-1}\right)=$	2. By the mixed product rule $(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1})=(\boldsymbol{A A}^{-1}) \otimes(\boldsymbol{B B}^{-1})=$
lyche-numerical-linear-algebra F03237	245	0.997	`\boldsymbol{I}_{r} \otimes \boldsymbol{I}_{s}= \boldsymbol{I}_{r s}`	$\boldsymbol{I}_{r} \otimes \boldsymbol{I}_{s}= \boldsymbol{I}_{r s}$	$\boldsymbol{I}_{r} \otimes \boldsymbol{I}_{s}=\boldsymbol{I}_{r s}$.Thus $(\boldsymbol{A} \otimes \boldsymbol{B})$ is nonsingular with the indicated inverse.
lyche-numerical-linear-algebra F03238	245	0.997	`(\boldsymbol{A} \otimes \boldsymbol{B})`	$(\boldsymbol{A} \otimes \boldsymbol{B})$	$\boldsymbol{I}_{r} \otimes \boldsymbol{I}_{s}=\boldsymbol{I}_{r s}$.Thus $(\boldsymbol{A} \otimes \boldsymbol{B})$ is nonsingular with the indicated inverse.
lyche-numerical-linear-algebra F03239	245	1.000	`1,(\boldsymbol{A} \otimes \boldsymbol{B})^{T}= \boldsymbol{A}^{T} \otimes \boldsymbol{B}^{T}= \boldsymbol{A} \otimes \boldsymbol{B}`	$1,(\boldsymbol{A} \otimes \boldsymbol{B})^{T}= \boldsymbol{A}^{T} \otimes \boldsymbol{B}^{T}= \boldsymbol{A} \otimes \boldsymbol{B}$	3. By 1, $(\boldsymbol{A} \otimes \boldsymbol{B})^{T}=\boldsymbol{A}^{T} \otimes \boldsymbol{B}^{T}=\boldsymbol{A} \otimes \boldsymbol{B}$. Moreover, since then $\boldsymbol{A} \otimes \boldsymbol{I}$ and $\boldsymbol{I} \otimes \boldsymbol{B}$
lyche-numerical-linear-algebra F03240	245	1.000	`\boldsymbol{A} \otimes \boldsymbol{I}`	$\boldsymbol{A} \otimes \boldsymbol{I}$	3. By 1, $(\boldsymbol{A} \otimes \boldsymbol{B})^{T}=\boldsymbol{A}^{T} \otimes \boldsymbol{B}^{T}=\boldsymbol{A} \otimes \boldsymbol{B}$. Moreover, since then $\boldsymbol{A} \otimes \boldsymbol{I}$ and $\boldsymbol{I} \otimes \boldsymbol{B}$
lyche-numerical-linear-algebra F03241	245	1.000	`\boldsymbol{I} \otimes \boldsymbol{B}`	$\boldsymbol{I} \otimes \boldsymbol{B}$	3. By 1, $(\boldsymbol{A} \otimes \boldsymbol{B})^{T}=\boldsymbol{A}^{T} \otimes \boldsymbol{B}^{T}=\boldsymbol{A} \otimes \boldsymbol{B}$. Moreover, since then $\boldsymbol{A} \otimes \boldsymbol{I}$ and $\boldsymbol{I} \otimes \boldsymbol{B}$
lyche-numerical-linear-algebra F03242	245	1.000	`(\boldsymbol{A} \otimes \boldsymbol{B})\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)= \left(\boldsymbol{A} \boldsymbol{u}_{i}\right) \otimes \left(\boldsymbol{B} \boldsymbol{v}_{j}\right)=\left(\lambda_{i} \boldsymbol{u}_{i}\right) \otimes\left(\mu_{j} \boldsymbol{v}_{j}\right)=\left(\lambda_{i} \mu_{j}\right)\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)`	$(\boldsymbol{A} \otimes \boldsymbol{B})\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)= \left(\boldsymbol{A} \boldsymbol{u}_{i}\right) \otimes \left(\boldsymbol{B} \boldsymbol{v}_{j}\right)=\left(\lambda_{i} \boldsymbol{u}_{i}\right) \otimes\left(\mu_{j} \boldsymbol{v}_{j}\right)=\left(\lambda_{i} \mu_{j}\right)\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)$	4. $(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j})=(\boldsymbol{A} \boldsymbol{u}_{i}) \otimes(\boldsymbol{B} \boldsymbol{v}_{j})=(\lambda_{i} \boldsymbol{u}_{i}) \otimes(\mu_{j} \boldsymbol{v}_{j})=(\lambda_{i} \mu_{j})(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j})$, for
lyche-numerical-linear-algebra F03243	245	0.475	`\left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}\right) \left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j} \right)=\lambda_{i}\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right), \quad`	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}\right) \left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j} \right)=\lambda_{i}\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right), \quad$	5. $(\boldsymbol{A} \otimes \boldsymbol{I}_{s})(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j})=\lambda_{i}(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}), \quad$ and $\quad (\boldsymbol{I}_{r} \otimes \boldsymbol{B})(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j})=\mu_{j}(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j})$.
lyche-numerical-linear-algebra F03244	245	0.475	`\quad\left(\boldsymbol{I}_{r} \otimes \boldsymbol{B} \right)\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)=\mu_{j}\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)`	$\quad\left(\boldsymbol{I}_{r} \otimes \boldsymbol{B} \right)\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)=\mu_{j}\left(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}\right)$	5. $(\boldsymbol{A} \otimes \boldsymbol{I}_{s})(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j})=\lambda_{i}(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j}), \quad$ and $\quad (\boldsymbol{I}_{r} \otimes \boldsymbol{B})(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j})=\mu_{j}(\boldsymbol{u}_{i} \otimes \boldsymbol{v}_{j})$.
lyche-numerical-linear-algebra F03245	245	0.892	`1, \boldsymbol{A} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{B}`	$1, \boldsymbol{A} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{B}$	6. By 1, $\boldsymbol{A} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{B}$ is symmetric. Moreover, the eigenvalues $\lambda_{i}+\mu_{j}$ are positive
lyche-numerical-linear-algebra F03246	245	0.892	`\lambda_{i}+\mu_{j}`	$\lambda_{i}+\mu_{j}$	6. By 1, $\boldsymbol{A} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{B}$ is symmetric. Moreover, the eigenvalues $\lambda_{i}+\mu_{j}$ are positive
lyche-numerical-linear-algebra F03247	245	0.500	`\mu_{j}`	μ_{j}	since for all i, j, both λ_{i} and μ_{j} are nonnegative and one of them is positive. It

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F03248	245	1.000	\boldsymbol{V}, \boldsymbol{F}	$\boldsymbol{V}, \boldsymbol{F}$	7. We partition $\boldsymbol{V}$, $\boldsymbol{F}$, and $\boldsymbol{B}^T$ by columns as $\boldsymbol{V} = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_s]$, $\boldsymbol{F} = [\boldsymbol{f}_1, \ldots, \boldsymbol{f}_s]$
lyche-numerical-linear-algebra_F03249	245	1.000	\boldsymbol{B}^{T}	$\boldsymbol{B}^{T}$	7. We partition $\boldsymbol{V}$, $\boldsymbol{F}$, and $\boldsymbol{B}^T$ by columns as $\boldsymbol{V} = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_s]$, $\boldsymbol{F} = [\boldsymbol{f}_1, \ldots, \boldsymbol{f}_s]$
lyche-numerical-linear-algebra_F03250	245	1.000	\boldsymbol{V}=\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{s}\right], \boldsymbol{F}=\left[\boldsymbol{f}_{1}, \ldots, \boldsymbol{f}_{s}\right]	$\boldsymbol{V}=\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{s}\right], \boldsymbol{F}=\left[\boldsymbol{f}_{1}, \ldots, \boldsymbol{f}_{s}\right]$	7. We partition $\boldsymbol{V}$, $\boldsymbol{F}$, and $\boldsymbol{B}^T$ by columns as $\boldsymbol{V} = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_s]$, $\boldsymbol{F} = [\boldsymbol{f}_1, \ldots, \boldsymbol{f}_s]$
lyche-numerical-linear-algebra_F03251	245	1.000	\boldsymbol{B}^{T}=\left[\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{s}\right]	$\boldsymbol{B}^{T}=\left[\boldsymbol{b}_{1}, \ldots, \boldsymbol{b}_{s}\right]$	and $\boldsymbol{B}^T = [\boldsymbol{b}_1, \ldots, \boldsymbol{b}_s]$. Then we have
lyche-numerical-linear-algebra_F03252	245	0.890	\left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}\right) \operatorname{vec}(\boldsymbol{V})=\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_{s}^{T}	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}\right) \operatorname{vec}(\boldsymbol{V})=\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_{s}^{T}$	8. From the proof of 7. we have $(\boldsymbol{A} \otimes \boldsymbol{I}_s) \operatorname{vec}(\boldsymbol{V}) = \boldsymbol{A}\boldsymbol{V}\boldsymbol{I}_s^T$ and $(\boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) =$
lyche-numerical-linear-algebra_F03253	245	0.890	\left(\boldsymbol{I}_{r} \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V})=	$\left(\boldsymbol{I}_{r} \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V})=$	8. From the proof of 7. we have $(\boldsymbol{A} \otimes \boldsymbol{I}_s) \operatorname{vec}(\boldsymbol{V}) = \boldsymbol{A}\boldsymbol{V}\boldsymbol{I}_s^T$ and $(\boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) =$
lyche-numerical-linear-algebra_F03254	245	1.000	\boldsymbol{I}_{r} \boldsymbol{V} \boldsymbol{B}^{T}	$\boldsymbol{I}_{r} \boldsymbol{V} \boldsymbol{B}^{T}$	$\boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T$. But then
lyche-numerical-linear-algebra_F03255	246	0.527	\left(\lambda_{j}, \boldsymbol{s}_{j}\right)	$\left(\lambda_{j}, \boldsymbol{s}_{j}\right)$	*where* $(\lambda_j, \boldsymbol{s}_j)$ *are the eigenpairs of* $\boldsymbol{T}_1$ *given by* (10.15) *and* (10.16). *The*
lyche-numerical-linear-algebra_F03256	246	0.975	\boldsymbol{s}_{j} \otimes \boldsymbol{s}_{k}	$\boldsymbol{s}_{j} \otimes \boldsymbol{s}_{k}$	*eigenvectors* $\boldsymbol{s}_j \otimes \boldsymbol{s}_k$ *are orthogonal*
lyche-numerical-linear-algebra_F03257	246	1.000	d>0	$d>0$	*and* $\boldsymbol{T}_2$ *is positive definite if* $d > 0$ *and* $d \geq 2\|a\|$.
lyche-numerical-linear-algebra_F03258	246	1.000	d \geq 2|a|	$d \geq 2\|a\|$	*and* $\boldsymbol{T}_2$ *is positive definite if* $d > 0$ *and* $d \geq 2\|a\|$.
lyche-numerical-linear-algebra_F03259	246	1.000	j, k, p, q=1, \ldots, m	$j, k, p, q=1, \ldots, m$	rule, the mixed product rule and (2.32) we find for $j, k, p, q = 1, \ldots, m$
lyche-numerical-linear-algebra_F03260	246	1.000	\lambda_{j}	λ_{j}	and (10.19) follows. Since $\boldsymbol{T}_2$ is symmetric, $\boldsymbol{T}_2$ is positive definite if the λ_j given
lyche-numerical-linear-algebra_F03261	246	0.990	\mathbb{R}^{m^{2} \times m^{2}}	$\mathbb{R}^{m^{2} \times m^{2}}$	$\mathbb{R}^{m^2 \times m^2}$ *given by* (10.8) *is given by*
lyche-numerical-linear-algebra_F03262	246	1.000	d=2, a=-1	$d=2, a=-1$	***Proof*** Recall that by (10.15) with $d = 2$, $a = -1$, and (10.18), the eigenvalues $\lambda_{j,k}$
lyche-numerical-linear-algebra_F03263	246	1.000	\lambda_{j, k}	$\lambda_{j, k}$	***Proof*** Recall that by (10.15) with $d = 2$, $a = -1$, and (10.18), the eigenvalues $\lambda_{j,k}$
lyche-numerical-linear-algebra_F03264	247	0.967	\\|\boldsymbol{A}\\|_{2}\left\\|\boldsymbol{A}^{-1}\right \\|_{2}=\frac{\lambda_{\text {max }}}{\lambda_{\text {min }}}	$\\|\boldsymbol{A}\\|_{2}\left\\|\boldsymbol{A}^{-1}\right\\|_{2}=\frac{\lambda_{\text {max }}}{\lambda_{\text {min }}}$	$\\|\boldsymbol{A}\\|_2 \\|\boldsymbol{A}^{-1}\\|_2 = \frac{\lambda_{max}}{\lambda_{min}}$ and (10.20) follows.
lyche-numerical-linear-algebra_F03265	247	1.000	\lambda=2 a+2 d	$\lambda=2 a+2 d$	Show that $\lambda = 2a + 2d$ is an eigenvalue corresponding to the eigenvector $\boldsymbol{x} =$
lyche-numerical-linear-algebra_F03266	247	1.000	[1,1,1,1]^{T}	$[1,1,1,1]^{T}$	$[1, 1, 1, 1]^T$. Verify that apart from a scaling of the eigenvector this agrees with
lyche-numerical-linear-algebra_F03267	247	1.000	j=k=1	$j=k=1$	(10.15) and (10.16) for $j = k = 1$ and $m = 2$.
lyche-numerical-linear-algebra_F03268	247	1.000	-\Delta u=f, u=0	$-\Delta u=f, u=0$	9 point difference approximation to the Poisson problem $-\Delta u = f$, $u = 0$ on the
lyche-numerical-linear-algebra_F03269	248	1.000	-\left(\square_{h} v\right)_{j, k}=(\mu f)_{j, k}	$-\left(\square_{h} v\right)_{j, k}=(\mu f)_{j, k}$	(a) $-(\square_h v)_{j,k} = (\mu f)_{j,k}$
lyche-numerical-linear-algebra_F03270	248	1.000	j, k=1, \ldots, m	$j, k=1, \ldots, m$	$j, k = 1, \ldots, m$
lyche-numerical-linear-algebra_F03271	248	1.000	0=v_{0, k}=v_{m+1, k}=v_{j, 0}=v_{j, m+1}, j, k=0,1, \ldots, m+1	$0=v_{0, k}=v_{m+1, k}=v_{j, 0}=v_{j, m+1}, j, k=0,1, \ldots, m+1$	$0 = v_{0,k} = v_{m+1,k} = v_{j,0} = v_{j,m+1}$, $j, k = 0, 1, \ldots, m+1$,
lyche-numerical-linear-algebra_F03272	248	1.000	-\left(\square_{h} v\right)_{j, k}=\left[20 v_{j, k}-4 v_{j-1, k}-4 v_{j, k-1}-4 v_{j+1, k}-4 v_{j, k+1}\right.	$-\left(\square_{h} v\right)_{j, k}=\left[20 v_{j, k}-4 v_{j-1, k}-4 v_{j, k-1}-4 v_{j+1, k}-4 v_{j, k+1}\right.$	$-(\square_h v)_{j,k} = [20v_{j,k} - 4v_{j-1,k} - 4v_{j,k-1} - 4v_{j+1,k} - 4v_{j,k+1}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F03273	248	1.000	`\left.-v_{j-1, k-1}-v_{j+1, k-1}-v_{j-1, k+1}-v_{j+1, k+1}\right] /\left(6 h^{2}\right)`	$\left.-v_{j-1, k-1}-v_{j+1, k-1}-v_{j-1, k+1}-v_{j+1, k+1}\right] /\left(6 h^{2}\right)$	$-v_{j-1,k-1}-v_{j+1,k-1}-v_{j-1,k+1}-v_{j+1,k+1}]/(6h^2)$,
lyche-numerical-linear-algebra F03274	248	0.877	`(\mu f)_{j, k}=\left[8 f_{j, k}+f_{j-1, k}+f_{j, k-1}+f_{j+1, k}+f_{j, k+1}\right] / 12`	$(\mu f)_{j, k}=\left[8 f_{j, k}+f_{j-1, k}+f_{j, k-1}+f_{j+1, k}+f_{j, k+1}\right] / 12$	$(\mu f)_{j,k} = [8f_{j,k}+f_{j-1,k}+f_{j,k-1}+f_{j+1,k}+f_{j,k+1}]/12.$
lyche-numerical-linear-algebra F03275	248	1.000	`j, k=1,2`	$j, k=1,2$	**b)** Find $v_{j,k}$ for $j,k = 1,2$, if $f(x,y) = 2\pi^2\sin(\pi x)\sin(\pi y)$ and $m = 2$.
lyche-numerical-linear-algebra F03276	248	1.000	`f(x, y)=2 \pi^{2} \sin (\pi x) \sin (\pi y)`	$f(x, y)=2 \pi^{2} \sin (\pi x) \sin (\pi y)$	**b)** Find $v_{j,k}$ for $j,k = 1,2$, if $f(x,y) = 2\pi^2\sin(\pi x)\sin(\pi y)$ and $m = 2$.
lyche-numerical-linear-algebra F03277	248	1.000	`v_{j, k}=5 \pi^{2} / 66`	$v_{j, k}=5 \pi^{2} / 66$	Answer: $v_{j,k} = 5\pi^2/66$.
lyche-numerical-linear-algebra F03278	248	1.000	`O\left(h^{4}\right)`	$O\left(h^{4}\right)$	It can be shown that (10.21) defines an $O(h^4)$ approximation to (10.1).
lyche-numerical-linear-algebra F03279	248	0.997	`\mu \boldsymbol{F}`	$\mu \boldsymbol{F}$	Here $\mu\boldsymbol{F}$ has elements $(\mu f)_{j,k}$ given by (10.21d) and $\boldsymbol{T} = \text{tridiag}(-1,2,-1)$.
lyche-numerical-linear-algebra F03280	248	0.997	`(\mu f)_{j, k}`	$(\mu f)_{j, k}$	Here $\mu\boldsymbol{F}$ has elements $(\mu f)_{j,k}$ given by (10.21d) and $\boldsymbol{T} = \text{tridiag}(-1,2,-1)$.
lyche-numerical-linear-algebra F03281	248	1.000	`\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \otimes \boldsymbol{T}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{V})`	$\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \otimes \boldsymbol{T}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{V})$	$\boldsymbol{A} = \boldsymbol{T}\otimes\boldsymbol{I}+\boldsymbol{I}\otimes\boldsymbol{T}-\frac{1}{6}\boldsymbol{T}\otimes\boldsymbol{T}$, $\boldsymbol{x} = \text{vec}(\boldsymbol{V})$, and $\boldsymbol{b} = h^2\text{vec}(\mu\boldsymbol{F})$.
lyche-numerical-linear-algebra F03282	248	1.000	`\boldsymbol{b}=h^{2} \operatorname{vec}(\mu \boldsymbol{F})`	$\boldsymbol{b}=h^{2} \operatorname{vec}(\mu \boldsymbol{F})$	$\boldsymbol{A} = \boldsymbol{T}\otimes\boldsymbol{I}+\boldsymbol{I}\otimes\boldsymbol{T}-\frac{1}{6}\boldsymbol{T}\otimes\boldsymbol{T}$, $\boldsymbol{x} = \text{vec}(\boldsymbol{V})$, and $\boldsymbol{b} = h^2\text{vec}(\mu\boldsymbol{F})$.
lyche-numerical-linear-algebra F03283	248	1.000	`\Delta u=0`	$\Delta u=0$	Here Ω is the open unit square. The condition $\Delta u = 0$ is called the *Navier boundary*
lyche-numerical-linear-algebra F03284	248	0.994	`\Delta^{2} u=u_{x x x x}+2 u_{x x y y}+u_{y y y y}`	$\Delta^{2} u=u_{x x x x}+2 u_{x x y y}+u_{y y y y}$	*condition*. Moreover, $\Delta^2 u = u_{xxxx}+2u_{xxyy}+u_{yyyy}$.
lyche-numerical-linear-algebra F03285	248	1.000	`v=-\Delta u`	$v=-\Delta u$	**a)** Let $v = -\Delta u$. Show that (10.23) can be written as a system
lyche-numerical-linear-algebra F03286	248	1.000	`\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}, h=1 /(m+`	$\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}, h=1 /(m+$	**b)** Discretizing, using (10.4), with $\boldsymbol{T} = \text{tridiag}(-1,2,-1) \in \mathbb{R}^{m\times m}$, $h = 1/(m+$
lyche-numerical-linear-algebra F03287	248	0.584	`\boldsymbol{F}=(f(j h, k h))_{j, k=1}^{m}`	$\boldsymbol{F}=(f(j h, k h))_{j, k=1}^{m}$	1), and $\boldsymbol{F} = \big(f(jh,kh)\big)_{j,k=1}^m$ we get two matrix equations
lyche-numerical-linear-algebra F03288	249	1.000	`\boldsymbol{A}=(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T})^{2}`	$\boldsymbol{A}=(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T})^{2}$	and hence $\boldsymbol{A} = (\boldsymbol{T}\otimes\boldsymbol{I}+\boldsymbol{I}\otimes\boldsymbol{T})^2$ is the matrix for the standard form of the
lyche-numerical-linear-algebra F03289	249	0.999	`\boldsymbol{A}=\boldsymbol{T}^{2} \otimes \boldsymbol{I}+2 \boldsymbol{T} \otimes \boldsymbol{T}+\boldsymbol{I} \otimes \boldsymbol{T}^{2} \in \mathbb{R}^{n \times n}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{U})`	$\boldsymbol{A}=\boldsymbol{T}^{2} \otimes \boldsymbol{I}+2 \boldsymbol{T} \otimes \boldsymbol{T}+\boldsymbol{I} \otimes \boldsymbol{T}^{2} \in \mathbb{R}^{n \times n}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{U})$	where $\boldsymbol{A} = \boldsymbol{T}^2\otimes\boldsymbol{I}+2\boldsymbol{T}\otimes\boldsymbol{T}+\boldsymbol{I}\otimes\boldsymbol{T}^2 \in \mathbb{R}^{n\times n}$, $\boldsymbol{x} = \text{vec}(\boldsymbol{U})$, and $\boldsymbol{b} = h^4\,\text{vec}(\boldsymbol{F})$.
lyche-numerical-linear-algebra F03290	249	0.999	`\boldsymbol{b}=h^{4} \operatorname{vec}(\boldsymbol{F})`	$\boldsymbol{b}=h^{4} \operatorname{vec}(\boldsymbol{F})$	where $\boldsymbol{A} = \boldsymbol{T}^2\otimes\boldsymbol{I}+2\boldsymbol{T}\otimes\boldsymbol{T}+\boldsymbol{I}\otimes\boldsymbol{T}^2 \in \mathbb{R}^{n\times n}$, $\boldsymbol{x} = \text{vec}(\boldsymbol{U})$, and $\boldsymbol{b} = h^4\,\text{vec}(\boldsymbol{F})$.
lyche-numerical-linear-algebra F03291	249	0.831	`\mathbf{b}`	$\mathbf{b}$	representations for the matrix $\boldsymbol{A}$, the product one in **b)** and the one in **c)**. Which
lyche-numerical-linear-algebra F03292	249	0.831	`\mathbf{c}`	$\mathbf{c}$	representations for the matrix $\boldsymbol{A}$, the product one in **b)** and the one in **c)**. Which
lyche-numerical-linear-algebra F03293	249	1.000	`\boldsymbol{T}_{1}:=\operatorname{tridiagonal}(a, d, a)`	$\boldsymbol{T}_{1}:=\operatorname{tridiagonal}(a, d, a)$	10.5.2 What are the eigenpairs of $\boldsymbol{T}_1 := \text{tridiagonal}(a,d,a)$?
lyche-numerical-linear-algebra F03294	249	1.000	`(\boldsymbol{A} \otimes`	$(\boldsymbol{A} \otimes$	• give an economical general way to solve the linear system $(\boldsymbol{A}\otimes$
lyche-numerical-linear-algebra F03295	249	1.000	`\boldsymbol{B}) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})`	$\boldsymbol{B}) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$	$\boldsymbol{B})\,\text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$?

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F03296	249	■ 0.999	`\left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}+ \boldsymbol{I}_{r} \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}( \boldsymbol{F})`	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}+\boldsymbol{I}_{r} \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$	• Same for $(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$.
lyche-numerical-linear-algebra_F03297	250	■ 1.000	`n=9`	$n=9$	matrix (10.8). Thus, for $n = 9$
lyche-numerical-linear-algebra_F03298	251	■ 1.000	`d=\sqrt{n}`	$d=\sqrt{n}$	lower triangular, to solve $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. Since $\boldsymbol{A}$ and $\boldsymbol{L}$ has the same bandwidth $d = \sqrt{n}$
lyche-numerical-linear-algebra_F03299	251	■ 1.000	`O\left(n d^{2}\right)=O\left(n^{2}\right)`	$O\left(n d^{2}\right)=O\left(n^{2}\right)$	the complexity of this factorization is $O(nd^2) = O(n^2)$, cf. Algorithm 4.2. We need
lyche-numerical-linear-algebra_F03300	251	■ 0.929	`(n=100)`	$(n=100)$	**Fig. 11.1** Fill-inn in the Cholesky factor of the Poisson matrix $(n = 100)$
lyche-numerical-linear-algebra_F03301	252	■ 1.000	`\boldsymbol{D}_{1}, \ldots, \boldsymbol{D}_{m}`	$\boldsymbol{D}_{1}, \ldots, \boldsymbol{D}_{m}$	Here $\boldsymbol{D}_1, \ldots, \boldsymbol{D}_m$ and $\boldsymbol{U}_1, \ldots, \boldsymbol{U}_m$ are square matrices while $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1$,
lyche-numerical-linear-algebra_F03302	252	■ 1.000	`\boldsymbol{U}_{1}, \ldots, \boldsymbol{U}_{m}`	$\boldsymbol{U}_{1}, \ldots, \boldsymbol{U}_{m}$	Here $\boldsymbol{D}_1, \ldots, \boldsymbol{D}_m$ and $\boldsymbol{U}_1, \ldots, \boldsymbol{U}_m$ are square matrices while $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1$,
lyche-numerical-linear-algebra_F03303	252	■ 1.000	`\boldsymbol{A}_{1}, \ldots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_{1}`	$\boldsymbol{A}_{1}, \ldots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_{1}$	Here $\boldsymbol{D}_1, \ldots, \boldsymbol{D}_m$ and $\boldsymbol{U}_1, \ldots, \boldsymbol{U}_m$ are square matrices while $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1$,
lyche-numerical-linear-algebra_F03304	252	■ 1.000	`\ldots, \boldsymbol{L}_{m-1}`	$\ldots, \boldsymbol{L}_{m-1}$	$\ldots, \boldsymbol{L}_{m-1}$ and $\boldsymbol{C}_1, \ldots, \boldsymbol{C}_{m-1}$ can be rectangular.
lyche-numerical-linear-algebra_F03305	252	■ 1.000	`\boldsymbol{C}_{1}, \ldots, \boldsymbol{C}_{m-1}`	$\boldsymbol{C}_{1}, \ldots, \boldsymbol{C}_{m-1}$	$\ldots, \boldsymbol{L}_{m-1}$ and $\boldsymbol{C}_1, \ldots, \boldsymbol{C}_{m-1}$ can be rectangular.
lyche-numerical-linear-algebra_F03306	252	■ 0.995	`\boldsymbol{b}^{T}=`	$\boldsymbol{b}^{T}=$	To solve the system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ we partition $\boldsymbol{b}$ conformally with $\boldsymbol{A}$ in the form $\boldsymbol{b}^T =$
lyche-numerical-linear-algebra_F03307	252	■ 1.000	`\left[\boldsymbol{b}_{1}^{T}, \ldots, \boldsymbol{b}_{m}^{T}\right]`	$\left[\boldsymbol{b}_{1}^{T}, \ldots, \boldsymbol{b}_{m}^{T}\right]$	$[\boldsymbol{b}_1^T, \ldots, \boldsymbol{b}_m^T]$. The formulas for solving $\boldsymbol{L}\boldsymbol{y} = \boldsymbol{b}$ and $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{y}$ are as follows:
lyche-numerical-linear-algebra_F03308	252	■ 1.000	`\boldsymbol{x}^{T}=\left[\boldsymbol{x}_{1}^{T}, \ldots, \boldsymbol{x}_{m}^{T}\right]`	$\boldsymbol{x}^{T}=\left[\boldsymbol{x}_{1}^{T}, \ldots, \boldsymbol{x}_{m}^{T}\right]$	The solution is then $\boldsymbol{x}^T = [\boldsymbol{x}_1^T, \ldots, \boldsymbol{x}_m^T]$. To find $\boldsymbol{L}_k$ in (11.2) we solve the linear
lyche-numerical-linear-algebra_F03309	252	■ 1.000	`\boldsymbol{L}_{k}`	$\boldsymbol{L}_{k}$	The solution is then $\boldsymbol{x}^T = [\boldsymbol{x}_1^T, \ldots, \boldsymbol{x}_m^T]$. To find $\boldsymbol{L}_k$ in (11.2) we solve the linear
lyche-numerical-linear-algebra_F03310	252	■ 1.000	`\boldsymbol{L}_{k} \boldsymbol{U}_{k}=\boldsymbol{A}_{k}`	$\boldsymbol{L}_{k} \boldsymbol{U}_{k}=\boldsymbol{A}_{k}$	systems $\boldsymbol{L}_k\boldsymbol{U}_k = \boldsymbol{A}_k$. Similarly we need to solve a linear system to find $\boldsymbol{x}_k$ in (11.3).
lyche-numerical-linear-algebra_F03311	252	■ 1.000	`\boldsymbol{x}_{k}`	$\boldsymbol{x}_{k}$	systems $\boldsymbol{L}_k\boldsymbol{U}_k = \boldsymbol{A}_k$. Similarly we need to solve a linear system to find $\boldsymbol{x}_k$ in (11.3).
lyche-numerical-linear-algebra_F03312	252	■ 1.000	`O\left(n^{3 / 2}\right)`	$O\left(n^{3 / 2}\right)$	The algorithm we now derive will only require $O(n^{3/2})$ arithmetic operations and
lyche-numerical-linear-algebra_F03313	252	■ 1.000	`O(n \log n)`	$O(n \log n)$	number of arithmetic operations can be reduced further to $O(n \log n)$.
lyche-numerical-linear-algebra_F03314	252	■ 1.000	`\boldsymbol{V}=`	$\boldsymbol{V}=$	where $\boldsymbol{T} = \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{m\times m}$ is the second derivative matrix, $\boldsymbol{V} =$
lyche-numerical-linear-algebra_F03315	252	■ 1.000	`\left(v_{j, k}\right) \in \mathbb{R}^{m \times m}`	$\left(v_{j, k}\right) \in \mathbb{R}^{m \times m}$	$(v_{j,k}) \in \mathbb{R}^{m\times m}$ are the unknowns, and $\boldsymbol{F} = (f_{j,k}) = (f(jh, kh)) \in \mathbb{R}^{m\times m}$ contains
lyche-numerical-linear-algebra_F03316	252	■ 1.000	`\boldsymbol{F}=\left(f_{j, k}\right)=(f(j h, k h)) \in \mathbb{R}^{m \times m}`	$\boldsymbol{F}=\left(f_{j, k}\right)=(f(j h, k h)) \in \mathbb{R}^{m \times m}$	$(v_{j,k}) \in \mathbb{R}^{m\times m}$ are the unknowns, and $\boldsymbol{F} = (f_{j,k}) = (f(jh, kh)) \in \mathbb{R}^{m\times m}$ contains
lyche-numerical-linear-algebra_F03317	253	■ 0.999	`\boldsymbol{X} \in \mathbb{R}^{m \times m}`	$\boldsymbol{X} \in \mathbb{R}^{m \times m}$	Define $\boldsymbol{X} \in \mathbb{R}^{m\times m}$ by $\boldsymbol{V} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$, where $\boldsymbol{V}$ is the solution of $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}$.
lyche-numerical-linear-algebra_F03318	253	■ 0.999	`\boldsymbol{V}=\boldsymbol{S} \boldsymbol{X} \boldsymbol{S}`	$\boldsymbol{V}=\boldsymbol{S} \boldsymbol{X} \boldsymbol{S}$	Define $\boldsymbol{X} \in \mathbb{R}^{m\times m}$ by $\boldsymbol{V} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$, where $\boldsymbol{V}$ is the solution of $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}$.
lyche-numerical-linear-algebra_F03319	253	■ 0.999	`\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^{2} \boldsymbol{F}`	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^{2} \boldsymbol{F}$	Define $\boldsymbol{X} \in \mathbb{R}^{m\times m}$ by $\boldsymbol{V} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$, where $\boldsymbol{V}$ is the solution of $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}$.
lyche-numerical-linear-algebra_F03320	253	■ 1.000	`\boldsymbol{D} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D}=4 h^{4} \boldsymbol{G}`	$\boldsymbol{D} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D}=4 h^{4} \boldsymbol{G}$	Since $\boldsymbol{D}$ is diagonal, the equation $\boldsymbol{D}\boldsymbol{X} + \boldsymbol{X}\boldsymbol{D} = 4h^4\boldsymbol{G}$, is easy to solve. For the j, k
lyche-numerical-linear-algebra_F03321	254	■ 1.000	`\boldsymbol{G}=\boldsymbol{S} \boldsymbol{F} \boldsymbol{S}`	$\boldsymbol{G}=\boldsymbol{S} \boldsymbol{F} \boldsymbol{S}$	1. $\boldsymbol{G} = \boldsymbol{S}\boldsymbol{F}\boldsymbol{S}$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al… F03322	254	■1.000	`x_{j, k}=h^{4} g_{j, k} /\left(\sigma_{j}+\sigma_{k}\right), \quad j, k=1, \ldots, m`	$x_{j, k}=h^{4} g_{j, k} /\left(\sigma_{j}+\sigma_{k}\right), \quad j, k=1, \ldots, m$	2. $x_{j,k} = h^4 g_{j,k}/(\sigma_j + \sigma_k), \quad j,k = 1,\ldots,m,$
lyche-numerical-linear-al… F03323	254	■1.000	`\boldsymbol{X}, \boldsymbol{S}`	$\boldsymbol{X}, \boldsymbol{S}$	We can compute $\boldsymbol{X}, \boldsymbol{S}$ and the σ's without using loops. Using outer products,
lyche-numerical-linear-al… F03324	254	■0.999	`-\Delta u=f`	$-\Delta u=f$	$-\Delta u = f$ on $\Omega = (0,1)^2$ and $u = 0$ on $\partial\Omega$ using the 5-point scheme, i.e., let $m \in$
lyche-numerical-linear-al… F03325	254	■0.999	`\Omega=(0,1)^{2}`	$\Omega=(0,1)^{2}$	$-\Delta u = f$ on $\Omega = (0,1)^2$ and $u = 0$ on $\partial\Omega$ using the 5-point scheme, i.e., let $m \in$
lyche-numerical-linear-al… F03326	254	■0.999	`m \in`	$m \in$	$-\Delta u = f$ on $\Omega = (0,1)^2$ and $u = 0$ on $\partial\Omega$ using the 5-point scheme, i.e., let $m \in$
lyche-numerical-linear-al… F03327	254	■1.000	`\mathbb{N}, h=1 /(m+1)`	$\mathbb{N}, h=1 /(m+1)$	$\mathbb{N}$, $h = 1/(m+1)$, and $\boldsymbol{F} = (f(jh,kh)) \in \mathbb{R}^{m\times m}$. We compute $\boldsymbol{V} \in \mathbb{R}^{(m+2)\times(m+2)}$
lyche-numerical-linear-al… F03328	254	■1.000	`\boldsymbol{F}=(f(j h, k h)) \in \mathbb{R}^{m \times m}`	$\boldsymbol{F}=(f(j h, k h)) \in \mathbb{R}^{m \times m}$	$\mathbb{N}$, $h = 1/(m+1)$, and $\boldsymbol{F} = (f(jh,kh)) \in \mathbb{R}^{m\times m}$. We compute $\boldsymbol{V} \in \mathbb{R}^{(m+2)\times(m+2)}$
lyche-numerical-linear-al… F03329	254	■1.000	`\boldsymbol{V} \in \mathbb{R}^{(m+2) \times(m+2)}`	$\boldsymbol{V} \in \mathbb{R}^{(m+2) \times(m+2)}$	$\mathbb{N}$, $h = 1/(m+1)$, and $\boldsymbol{F} = (f(jh,kh)) \in \mathbb{R}^{m\times m}$. We compute $\boldsymbol{V} \in \mathbb{R}^{(m+2)\times(m+2)}$
lyche-numerical-linear-al… F03330	254	■1.000	`O\left(m^{3}\right)`	$O\left(m^{3}\right)$	requires $O(m^3)$ arithmetic operations, the complexity of this algorithm is for large
lyche-numerical-linear-al… F03331	254	■1.000	`O\left(4 \times 2 m^{3}\right)=`	$O\left(4 \times 2 m^{3}\right)=$	m determined by 4 m-by-m matrix multiplications and is given by $O(4 \times 2m^3) =$
lyche-numerical-linear-al… F03332	254	■0.943	`O\left(8 n^{3 / 2}\right) .{ }^{1}`	$O\left(8 n^{3 / 2}\right) .{ }^{1}$	$O(8n^{3/2})$.[1] The method is very fast and will be used as a preconditioner for a more
lyche-numerical-linear-al… F03333	254	■1.000	`10^{6}`	10^{6}	find the 10^6 unknowns $v_{j,k}$ on a 1000×1000 grid.
lyche-numerical-linear-al… F03334	254	■1.000	`1000 \times 1000`	1000×1000	find the 10^6 unknowns $v_{j,k}$ on a 1000×1000 grid.
lyche-numerical-linear-al… F03335	254	■1.000	`O\left(4 n^{3 / 2}\right)`	$O\left(4 n^{3 / 2}\right)$	to $O(4n^{3/2})$. This is detailed in Problem 11.4.
lyche-numerical-linear-al… F03336	255	■1.000	`\boldsymbol{A} \in \mathbb{R}^{m \times m}`	$\boldsymbol{A} \in \mathbb{R}^{m \times m}$	given by (11.4) and a matrix $\boldsymbol{A} \in \mathbb{R}^{m\times m}$. Since the matrices are m-by-m this will
lyche-numerical-linear-al… F03337	255	■1.000	`\boldsymbol{S} \boldsymbol{A}`	$\boldsymbol{S} \boldsymbol{A}$	calculate the products $\boldsymbol{SA}$ and $\boldsymbol{AS}$ in $O(m^2 \log_2 m)$ operations.
lyche-numerical-linear-al… F03338	255	■1.000	`\boldsymbol{A} \boldsymbol{S}`	$\boldsymbol{A} \boldsymbol{S}$	calculate the products $\boldsymbol{SA}$ and $\boldsymbol{AS}$ in $O(m^2 \log_2 m)$ operations.
lyche-numerical-linear-al… F03339	255	■1.000	`O\left(m^{2} \log _{2} m\right)`	$O\left(m^{2} \log _{2} m\right)$	calculate the products $\boldsymbol{SA}$ and $\boldsymbol{AS}$ in $O(m^2 \log_2 m)$ operations.
lyche-numerical-linear-al… F03340	255	■1.000	`\boldsymbol{v}=\left[v_{1}, \ldots, v_{m}\right]^{T} \in \mathbb{R}^{m}`	$\boldsymbol{v}=\left[v_{1}, \ldots, v_{m}\right]^{T} \in \mathbb{R}^{m}$	Given $\boldsymbol{v} = [v_1,\ldots,v_m]^T \in \mathbb{R}^m$ we say that the vector $\boldsymbol{w} = [w_1,\ldots,w_m]^T$ given
lyche-numerical-linear-al… F03341	255	■1.000	`\boldsymbol{w}=\left[w_{1}, \ldots, w_{m}\right]^{T}`	$\boldsymbol{w}=\left[w_{1}, \ldots, w_{m}\right]^{T}$	Given $\boldsymbol{v} = [v_1,\ldots,v_m]^T \in \mathbb{R}^m$ we say that the vector $\boldsymbol{w} = [w_1,\ldots,w_m]^T$ given
lyche-numerical-linear-al… F03342	255	■1.000	`\boldsymbol{w}=\boldsymbol{S} \boldsymbol{v}`	$\boldsymbol{w}=\boldsymbol{S} \boldsymbol{v}$	as the matrix times vector $\boldsymbol{w} = \boldsymbol{Sv}$, where $\boldsymbol{S}$ is the sine matrix given by (11.4). We
lyche-numerical-linear-al… F03343	255	■1.000	`\boldsymbol{B}=\boldsymbol{S} \boldsymbol{A}`	$\boldsymbol{B}=\boldsymbol{S} \boldsymbol{A}$	can then identify the matrix $\boldsymbol{B} = \boldsymbol{SA}$ as the DST of $\boldsymbol{A} \in \mathbb{R}^{m,n}$, i.e. as the DST of
lyche-numerical-linear-al… F03344	255	■1.000	`\boldsymbol{A} \in \mathbb{R}^{m, n}`	$\boldsymbol{A} \in \mathbb{R}^{m, n}$	can then identify the matrix $\boldsymbol{B} = \boldsymbol{SA}$ as the DST of $\boldsymbol{A} \in \mathbb{R}^{m,n}$, i.e. as the DST of
lyche-numerical-linear-al… F03345	255	■1.000	`\boldsymbol{B}=\boldsymbol{A} \boldsymbol{S}`	$\boldsymbol{B}=\boldsymbol{A} \boldsymbol{S}$	the columns of $\boldsymbol{A}$. The product $\boldsymbol{B} = \boldsymbol{AS}$ can also be interpreted as a DST. Indeed,
lyche-numerical-linear-al… F03346	255	■1.000	`\boldsymbol{B}=\left(\boldsymbol{S} \boldsymbol{A}^{T}\right)^{T}`	$\boldsymbol{B}=\left(\boldsymbol{S} \boldsymbol{A}^{T}\right)^{T}$	since $\boldsymbol{S}$ is symmetric we have $\boldsymbol{B} = (\boldsymbol{SA}^T)^T$ which means that $\boldsymbol{B}$ is the transpose
lyche-numerical-linear-al… F03347	255	■1.000	`N \in \mathbb{N}`	$N \in \mathbb{N}$	as the fast Fourier transform (FFT). To define the DFT let for $N \in \mathbb{N}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F03348	256	■1.000	`\boldsymbol{y}=\left[y_{1}, \ldots, y_{N}\right]^{T} \in \mathbb{R}^{N}`	$\boldsymbol{y}=\left[y_{1}, \ldots, y_{N}\right]^{T} \in \mathbb{R}^{N}$	where $i=\sqrt{-1}$ is the imaginary unit. Given $\boldsymbol{y}=[y_1,\ldots,y_N]^T \in \mathbb{R}^N$ we say that
lyche-numerical-linear-al F03349	256	■1.000	`z=\left[z_{1}, \ldots, z_{N}\right]^{T}`	$z=\left[z_{1}, \ldots, z_{N}\right]^{T}$	$\boldsymbol{z}=[z_1,\ldots,z_N]^T$ given by
lyche-numerical-linear-al F03350	256	■0.991	`\boldsymbol{z}=\boldsymbol{F}_{N} \boldsymbol{y}`	$\boldsymbol{z}=\boldsymbol{F}_{N} \boldsymbol{y}$	vector product $\boldsymbol{z}=\boldsymbol{F}_N\boldsymbol{y}$, where the **Fourier matrix** $\boldsymbol{F}_N \in \mathbb{C}^{N\times N}$ has elements
lyche-numerical-linear-al F03351	256	■0.991	`\boldsymbol{F}_{N} \in \mathbb{C}^{N \times N}`	$\boldsymbol{F}_{N} \in \mathbb{C}^{N \times N}$	vector product $\boldsymbol{z}=\boldsymbol{F}_N\boldsymbol{y}$, where the **Fourier matrix** $\boldsymbol{F}_N \in \mathbb{C}^{N\times N}$ has elements
lyche-numerical-linear-al F03352	256	■1.000	`\omega_{N}^{j k}, j, k=0,1, \ldots, N-1`	$\omega_{N}^{j k}, j, k=0,1, \ldots, N-1$	ω_N^{jk}, $j,k=0,1,\ldots,N-1$. For a matrix we say that $\boldsymbol{B}=\boldsymbol{F}_N\boldsymbol{A}$ is the DFT of $\boldsymbol{A}$.
lyche-numerical-linear-al F03353	256	■1.000	`\boldsymbol{B}=\boldsymbol{F}_{N} \boldsymbol{A}`	$\boldsymbol{B}=\boldsymbol{F}_{N} \boldsymbol{A}$	ω_N^{jk}, $j,k=0,1,\ldots,N-1$. For a matrix we say that $\boldsymbol{B}=\boldsymbol{F}_N\boldsymbol{A}$ is the DFT of $\boldsymbol{A}$.
lyche-numerical-linear-al F03354	256	■1.000	`\omega_{4}^{2}=(-i)^{2}=-1, \omega_{4}^{3}=(-i)(-1)=i, \omega_{4}^{4}=(-1)^{2}=1, \omega_{4}^{6}=i^{2}=-1`	$\omega_{4}^{2}=(-i)^{2}=-1, \omega_{4}^{3}=(-i)(-1)=i, \omega_{4}^{4}=(-1)^{2}=1, \omega_{4}^{6}=i^{2}=-1$	we find $\omega_4^2=(-i)^2=-1$, $\omega_4^3=(-i)(-1)=i$, $\omega_4^4=(-1)^2=1$, $\omega_4^6=i^2=-1$,
lyche-numerical-linear-al F03355	256	■1.000	`\omega_{4}^{9}=i^{3}=-i`	$\omega_{4}^{9}=i^{3}=-i$	$\omega_4^9=i^3=-i$, and so
lyche-numerical-linear-al F03356	256	■1.000	`2 m+2`	$2 m+2$	computed from the discrete Fourier transform of order $2m+2$. We recall that for
lyche-numerical-linear-al F03357	256	■1.000	`w`	w	any complex number w
lyche-numerical-linear-al F03358	256	■1.000	`\boldsymbol{S} \boldsymbol{x}`	$\boldsymbol{S} \boldsymbol{x}$	*and a vector* $\boldsymbol{x}\in\mathbb{R}^m$. *Component* k *of* $\boldsymbol{Sx}$ *is equal to* $i/2$ *times component* $k+1$ *of*
lyche-numerical-linear-al F03359	256	■1.000	`i / 2`	$i / 2$	*and a vector* $\boldsymbol{x}\in\mathbb{R}^m$. *Component* k *of* $\boldsymbol{Sx}$ *is equal to* $i/2$ *times component* $k+1$ *of*
lyche-numerical-linear-al F03360	256	■1.000	`\boldsymbol{F}_{2 m+2} \boldsymbol{z}`	$\boldsymbol{F}_{2 m+2} \boldsymbol{z}$	$\boldsymbol{F}_{2m+2}\boldsymbol{z}$ *where*
lyche-numerical-linear-al F03361	256	■1.000	`\omega=\omega_{2 m+2}=e^{-2 \pi i /(2 m+2)}=e^{-\pi i /(m+1)}`	$\omega=\omega_{2 m+2}=e^{-2 \pi i /(2 m+2)}=e^{-\pi i /(m+1)}$	***Proof*** Let $\omega=\omega_{2m+2}=e^{-2\pi i/(2m+2)}=e^{-\pi i/(m+1)}$. We note that
lyche-numerical-linear-al F03362	257	■1.000	`-2 i`	$-2 i$	Dividing both sides by $-2i$ and noting $-1/(2i)=-i/(2i^2)=i/2$, proves the
lyche-numerical-linear-al F03363	257	■1.000	`-1 /(2 i)=-i /\left(2 i^{2}\right)=i / 2`	$-1 /(2 i)=-i /\left(2 i^{2}\right)=i / 2$	Dividing both sides by $-2i$ and noting $-1/(2i)=-i/(2i^2)=i/2$, proves the
lyche-numerical-linear-al F03364	257	■1.000	`N=2 m+2`	$N=2 m+2$	from the DFT of length $N=2m+2$.
lyche-numerical-linear-al F03365	257	■1.000	`\boldsymbol{F}_{N} \boldsymbol{y}`	$\boldsymbol{F}_{N} \boldsymbol{y}$	the matrix- vector product $\boldsymbol{F}_N\boldsymbol{y}$. Suppose N is even. The key to the FFT is a
lyche-numerical-linear-al F03366	257	■1.000	`\boldsymbol{F}_{N}`	$\boldsymbol{F}_{N}$	connection between $\boldsymbol{F}_N$ and $\boldsymbol{F}_{N/2}$ which makes it possible to compute the FFT
lyche-numerical-linear-al F03367	257	■1.000	`\boldsymbol{F}_{N / 2}`	$\boldsymbol{F}_{N / 2}$	connection between $\boldsymbol{F}_N$ and $\boldsymbol{F}_{N/2}$ which makes it possible to compute the FFT
lyche-numerical-linear-al F03368	257	■1.000	`N / 2`	$N / 2$	of order N as two FFT's of order $N/2$. By repeating this process we can reduce the
lyche-numerical-linear-al F03369	257	■1.000	`O\left(N^{2}\right)`	$O\left(N^{2}\right)$	number of arithmetic operations to compute a DFT from $O(N^2)$ to $O(N\log_2 N)$.
lyche-numerical-linear-al F03370	257	■1.000	`O\left(N \log _{2} N\right)`	$O\left(N \log _{2} N\right)$	number of arithmetic operations to compute a DFT from $O(N^2)$ to $O(N\log_2 N)$.
lyche-numerical-linear-al F03371	257	■1.000	`\boldsymbol{P}_{N} \in \mathbb{R}^{N \times N}`	$\boldsymbol{P}_{N} \in \mathbb{R}^{N \times N}$	tion matrix $\boldsymbol{P}_N\in\mathbb{R}^{N\times N}$ given by
lyche-numerical-linear-al F03372	257	■0.998	`\boldsymbol{e}_{k}=\left(\delta_{j, k}\right)`	$\boldsymbol{e}_{k}=\left(\delta_{j, k}\right)$	where the $\boldsymbol{e}_k=(\delta_{j,k})$ are unit vectors. If $\boldsymbol{A}$ is a matrix with N columns $[\boldsymbol{a}_1,\ldots,\boldsymbol{a}_N]$
lyche-numerical-linear-al F03373	257	■0.998	`\left[\boldsymbol{a}_{1}, \ldots, \boldsymbol{a}_{N}\right]`	$\left[\boldsymbol{a}_{1}, \ldots, \boldsymbol{a}_{N}\right]$	where the $\boldsymbol{e}_k=(\delta_{j,k})$ are unit vectors. If $\boldsymbol{A}$ is a matrix with N columns $[\boldsymbol{a}_1,\ldots,\boldsymbol{a}_N]$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F03374	257	0.965	`\boldsymbol{P}_{N}`	$\boldsymbol{P}_{N}$	i.e. post multiplying $\boldsymbol{A}$ by $\boldsymbol{P}_N$ permutes the columns of $\boldsymbol{A}$ so that all the odd-indexed
lyche-numerical-linear-al F03375	258	1.000	`\boldsymbol{F}_{4} \boldsymbol{P}_{4}`	$\boldsymbol{F}_{4} \boldsymbol{P}_{4}$	where we have indicated a certain block structure of $\boldsymbol{F}_4\boldsymbol{P}_4$. These blocks can be
lyche-numerical-linear-al F03376	258	1.000	`\boldsymbol{F}_{2}`	$\boldsymbol{F}_{2}$	related to the 2-by-2 matrix $\boldsymbol{F}_2$. We define the diagonal scaling matrix $\boldsymbol{D}_2$ by
lyche-numerical-linear-al F03377	258	1.000	`\omega_{2}=\exp ^{-2 \pi i / 2}=-1`	$\omega_{2}=\exp ^{-2 \pi i / 2}=-1$	Since $\omega_2 = \exp^{-2\pi i/2} = -1$ we find
lyche-numerical-linear-al F03378	258	1.000	`N=2 m`	$N=2 m$	**Theorem 11.1 (Fast Fourier Transform)** *If $N = 2m$ is even then*
lyche-numerical-linear-al F03379	258	1.000	`p, q`	p, q	***Proof*** Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p-1$ and $k := q-1$.
lyche-numerical-linear-al F03380	258	1.000	`1 \leq p, q \leq m`	$1 \leq p, q \leq m$	***Proof*** Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p-1$ and $k := q-1$.
lyche-numerical-linear-al F03381	258	1.000	`j:=p-1`	$j:=p-1$	***Proof*** Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p-1$ and $k := q-1$.
lyche-numerical-linear-al F03382	258	1.000	`k:=q-1`	$k:=q-1$	***Proof*** Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p-1$ and $k := q-1$.
lyche-numerical-linear-al F03383	258	1.000	`\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}`	$\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}$	It follows that the four m-by-m blocks of $\boldsymbol{F}_{2m}\boldsymbol{P}_{2m}$ have the required structure.
lyche-numerical-linear-al F03384	258	1.000	`\boldsymbol{y} \in \mathbb{R}^{2 m}`	$\boldsymbol{y} \in \mathbb{R}^{2 m}$	$\boldsymbol{y} \in \mathbb{R}^{2m}$ and set $\boldsymbol{w} = \boldsymbol{P}_{2m}^T\boldsymbol{y} = [\boldsymbol{w}_1^T, \boldsymbol{w}_2^T]^T$, where
lyche-numerical-linear-al F03385	258	1.000	`\boldsymbol{w}=\boldsymbol{P}_{2 m}^{T} \boldsymbol{y}= \left[\boldsymbol{w}_{1}^{T}, \boldsymbol{w}_{2}^{T} \right]^{T}`	$\boldsymbol{w}=\boldsymbol{P}_{2 m}^{T} \boldsymbol{y}= \left[\boldsymbol{w}_{1}^{T}, \boldsymbol{w}_{2}^{T} \right]^{T}$	$\boldsymbol{y} \in \mathbb{R}^{2m}$ and set $\boldsymbol{w} = \boldsymbol{P}_{2m}^T\boldsymbol{y} = [\boldsymbol{w}_1^T, \boldsymbol{w}_2^T]^T$, where
lyche-numerical-linear-al F03386	259	1.000	`\boldsymbol{F}_{2 m} \boldsymbol{y}`	$\boldsymbol{F}_{2 m} \boldsymbol{y}$	In order to compute $\boldsymbol{F}_{2m}\boldsymbol{y}$ we need to compute $\boldsymbol{F}_m\boldsymbol{w}_1$ and $\boldsymbol{F}_m\boldsymbol{w}_2$. Thus, by
lyche-numerical-linear-al F03387	259	1.000	`\boldsymbol{F}_{m} \boldsymbol{w}_{1}`	$\boldsymbol{F}_{m} \boldsymbol{w}_{1}$	In order to compute $\boldsymbol{F}_{2m}\boldsymbol{y}$ we need to compute $\boldsymbol{F}_m\boldsymbol{w}_1$ and $\boldsymbol{F}_m\boldsymbol{w}_2$. Thus, by
lyche-numerical-linear-al F03388	259	1.000	`\boldsymbol{F}_{m} \boldsymbol{w}_{2}`	$\boldsymbol{F}_{m} \boldsymbol{w}_{2}$	In order to compute $\boldsymbol{F}_{2m}\boldsymbol{y}$ we need to compute $\boldsymbol{F}_m\boldsymbol{w}_1$ and $\boldsymbol{F}_m\boldsymbol{w}_2$. Thus, by
lyche-numerical-linear-al F03389	259	1.000	`n=2^{k}`	$n=2^{k}$	combining two FFT's of order m we obtain an FFT of order $2m$. If $n = 2^k$ then
lyche-numerical-linear-al F03390	259	1.000	`\boldsymbol{y} \in \mathbb{R}^{n}`	$\boldsymbol{y} \in \mathbb{R}^{n}$	Statement 3 is included so that the input $\boldsymbol{y} \in \mathbb{R}^n$ can be either a row or column
lyche-numerical-linear-al F03391	259	1.000	`\gamma N \log _{2} N`	$\gamma N \log _{2} N$	The complexity of the FFT is given by $\gamma N \log_2 N$ for some constant γ
lyche-numerical-linear-al F03392	259	1.000	`\gamma`	γ	The complexity of the FFT is given by $\gamma N \log_2 N$ for some constant γ
lyche-numerical-linear-al F03393	259	1.000	`N=2^{k}`	$N=2^{k}$	let x_k be the complexity (the number of arithmetic operations) when $N = 2^k$.
lyche-numerical-linear-al F03394	259	1.000	`N / 2=2^{k-1}`	$N / 2=2^{k-1}$	Since we need two FFT's of order $N/2 = 2^{k-1}$ and a multiplication with the
lyche-numerical-linear-al F03395	259	1.000	`\boldsymbol{D}_{N / 2}`	$\boldsymbol{D}_{N / 2}$	diagonal matrix $\boldsymbol{D}_{N/2}$, it is reasonable to assume that $x_k = 2x_{k-1} + \gamma 2^k$ for
lyche-numerical-linear-al F03396	259	1.000	`x_{k}=2 x_{k-1}+\gamma 2^{k}`	$x_{k}=2 x_{k-1}+\gamma 2^{k}$	diagonal matrix $\boldsymbol{D}_{N/2}$, it is reasonable to assume that $x_k = 2x_{k-1} + \gamma 2^k$ for
lyche-numerical-linear-al F03397	259	1.000	`x_{0}=0`	$x_{0}=0$	some constant γ independent of k. Since $x_0 = 0$ we obtain by induction on k
lyche-numerical-linear-al F03398	259	1.000	`x_{k}=\gamma k 2^{k}`	$x_{k}=\gamma k 2^{k}$	that $x_k = \gamma k 2^k$. Indeed, this holds for $k = 0$ and if $x_{k-1} = \gamma(k-1)2^{k-1}$ then
lyche-numerical-linear-al F03399	259	1.000	`x_{k-1}=\gamma(k-1) 2^{k-1}`	$x_{k-1}=\gamma(k-1) 2^{k-1}$	that $x_k = \gamma k 2^k$. Indeed, this holds for $k = 0$ and if $x_{k-1} = \gamma(k-1)2^{k-1}$ then

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F03400	259	■ 1.000	`x_{k}=2 x_{k-1}+\gamma 2^{k}=2 \gamma(k-1) 2^{k-1}+\gamma 2^{k}=\gamma k 2^{k}`	$x_{k}=2 x_{k-1}+\gamma 2^{k}=2 \gamma(k-1) 2^{k-1}+\gamma 2^{k}=\gamma k 2^{k}$	$x_k = 2x_{k-1} + \gamma 2^k = 2\gamma(k-1)2^{k-1} + \gamma 2^k = \gamma k 2^k$. Reasonable implementations
lyche-numerical-linear-algebra_F03401	259	■ 1.000	`\gamma \approx 5`	$\gamma \approx 5$	of FFT typically have $\gamma \approx 5$, see [14].
lyche-numerical-linear-algebra_F03402	259	■ 1.000	`O\left(8 n^{2}\right)`	$O\left(8 n^{2}\right)$	large N. The direct multiplication $\boldsymbol{F}_N \boldsymbol{y}$ requires $O(8n^2)$ arithmetic operations since
lyche-numerical-linear-algebra_F03403	260	■ 1.000	`5 N \log _{2} N`	$5 N \log _{2} N$	complex arithmetic is involved. Assuming that the FFT uses $5N \log_2 N$ arithmetic
lyche-numerical-linear-algebra_F03404	260	■ 1.000	`N=2^{20} \approx 10^{6}`	$N=2^{20} \approx 10^{6}$	operations we find for $N = 2^{20} \approx 10^6$ the ratio
lyche-numerical-linear-algebra_F03405	260	■ 1.000	`\boldsymbol{H}=\boldsymbol{S} \boldsymbol{F}`	$\boldsymbol{H}=\boldsymbol{S} \boldsymbol{F}$	Algorithm 11.1. We first compute $\boldsymbol{H} = \boldsymbol{SF}$ using Lemma 11.1 and m FFT's, one
lyche-numerical-linear-algebra_F03406	260	■ 0.998	`\boldsymbol{G}=\boldsymbol{H} \boldsymbol{S}`	$\boldsymbol{G}=\boldsymbol{H} \boldsymbol{S}$	for each of the m columns of $\boldsymbol{F}$. We then compute $\boldsymbol{G} = \boldsymbol{HS}$ by m FFT's, one for
lyche-numerical-linear-algebra_F03407	260	■ 1.000	`\boldsymbol{Z}=\boldsymbol{S} \boldsymbol{X}`	$\boldsymbol{Z}=\boldsymbol{S} \boldsymbol{X}$	each of the rows of $\boldsymbol{H}$. After $\boldsymbol{X}$ is determined we compute $\boldsymbol{Z} = \boldsymbol{SX}$ and $\boldsymbol{V} = \boldsymbol{ZS}$
lyche-numerical-linear-algebra_F03408	260	■ 1.000	`\boldsymbol{V}=\boldsymbol{Z} \boldsymbol{S}`	$\boldsymbol{V}=\boldsymbol{Z} \boldsymbol{S}$	each of the rows of $\boldsymbol{H}$. After $\boldsymbol{X}$ is determined we compute $\boldsymbol{Z} = \boldsymbol{SX}$ and $\boldsymbol{V} = \boldsymbol{ZS}$
lyche-numerical-linear-algebra_F03409	260	■ 1.000	`O\left(\gamma(2 m+2) \log _{2}(2 m+2)\right)`	$O\left(\gamma(2 m+2) \log _{2}(2 m+2)\right)$	one FFT requires $O(\gamma(2m+2)\log_2(2m+2))$ arithmetic operations the $4m$ FFT's
lyche-numerical-linear-algebra_F03410	260	■ 1.000	`O\left(8 n^{3 / 2}\right)`	$O\left(8 n^{3 / 2}\right)$	if Cholesky factorization was used. This should be compared to the $O(8n^{3/2})$
lyche-numerical-linear-algebra_F03411	260	■ 1.000	`\boldsymbol{F}_{4}`	$\boldsymbol{F}_{4}$	**Exercise 11.1 (Fourier Matrix)** Show that the Fourier matrix $\boldsymbol{F}_4$ is symmetric,
lyche-numerical-linear-algebra_F03412	261	■ 1.000	`\left(v_{i, j}\right)_{i, j=1}^{m}`	$\left(v_{i, j}\right)_{i, j=1}^{m}$	$(v_{i,j})_{i,j=1}^m$, where
lyche-numerical-linear-algebra_F03413	261	■ 0.608	`\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V T}=h^{2} \boldsymbol{F}`	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V T}=h^{2} \boldsymbol{F}$	in $\boldsymbol{TV} + \boldsymbol{VT} = h^2\boldsymbol{F}$. Let $\boldsymbol{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$, where $\lambda_j = 4\sin^2(j\pi h/2)$,
lyche-numerical-linear-algebra_F03414	261	■ 0.608	`\boldsymbol{D}=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{m}\right)`	$\boldsymbol{D}=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{m}\right)$	in $\boldsymbol{TV} + \boldsymbol{VT} = h^2\boldsymbol{F}$. Let $\boldsymbol{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$, where $\lambda_j = 4\sin^2(j\pi h/2)$,
lyche-numerical-linear-algebra_F03415	261	■ 0.608	`\lambda_{j}=4 \sin ^{2}(j \pi h / 2)`	$\lambda_{j}=4 \sin ^{2}(j \pi h / 2)$	in $\boldsymbol{TV} + \boldsymbol{VT} = h^2\boldsymbol{F}$. Let $\boldsymbol{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$, where $\lambda_j = 4\sin^2(j\pi h/2)$,
lyche-numerical-linear-algebra_F03416	261	■ 1.000	`\boldsymbol{X}=\left[\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{m}\right]`	$\boldsymbol{X}=\left[\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{m}\right]$	where $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m]$ and $\boldsymbol{C} = [\boldsymbol{c}_1, \ldots, \boldsymbol{c}_m]$. Thus we can find $\boldsymbol{X}$ by solving
lyche-numerical-linear-algebra_F03417	261	■ 1.000	`\boldsymbol{C}=\left[\boldsymbol{c}_{1}, \ldots, \boldsymbol{c}_{m}\right]`	$\boldsymbol{C}=\left[\boldsymbol{c}_{1}, \ldots, \boldsymbol{c}_{m}\right]$	where $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m]$ and $\boldsymbol{C} = [\boldsymbol{c}_1, \ldots, \boldsymbol{c}_m]$. Thus we can find $\boldsymbol{X}$ by solving
lyche-numerical-linear-algebra_F03418	261	■ 1.000	`8 m-7`	$8 m-7$	$m \times m$ system can be solved by Algorithms 2.1 and 2.2 in $8m - 7$ arithmetic
lyche-numerical-linear-algebra_F03419	261	■ 1.000	`O\left(\delta m^{2}\right)`	$O\left(\delta m^{2}\right)$	operations. Give an algorithm to find $\boldsymbol{X}$ which only requires $O(\delta m^2)$ arithmetic
lyche-numerical-linear-algebra_F03420	261	■ 1.000	`O\left(4 m^{3}\right)=O\left(4 n^{3 / 2}\right)`	$O\left(4 m^{3}\right)=O\left(4 n^{3 / 2}\right)$	**c)** Describe a method to compute $\boldsymbol{V}$ which only requires $O(4m^3) = O(4n^{3/2})$
lyche-numerical-linear-algebra_F03421	261	■ 1.000	`O\left(2 \gamma n \log _{2} n\right)`	$O\left(2 \gamma n \log _{2} n\right)$	$O(2\gamma n \log_2 n)$ where γ is the same constant as mentioned at the end of the
lyche-numerical-linear-algebra_F03422	261	■ 1.000	`\boldsymbol{V}=\boldsymbol{S} \boldsymbol{X} \boldsymbol{S}=\left(x_{j, k}\right)`	$\boldsymbol{V}=\boldsymbol{S} \boldsymbol{X} \boldsymbol{S}=\left(x_{j, k}\right)$	$\boldsymbol{V} = \boldsymbol{SXS} = (x_{j,k})$ where $\boldsymbol{V}$ is the solution of (10.22). Show that
lyche-numerical-linear-algebra_F03423	262	■ 1.000	`\lambda_{j}=4 \sin ^{2}(j \pi h / 2), j=1, \ldots, m`	$\lambda_{j}=4 \sin ^{2}(j \pi h / 2), j=1, \ldots, m$	where $\boldsymbol{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$, with $\lambda_j = 4\sin^2(j\pi h/2)$, $j = 1, \ldots, m$, and that
lyche-numerical-linear-algebra_F03424	262	■ 1.000	`\sigma_{j}+\sigma_{k}-\frac{2}{3} \sigma_{j} \sigma_{k}>0`	$\sigma_{j}+\sigma_{k}-\frac{2}{3} \sigma_{j} \sigma_{k}>0$	Show that $\sigma_j + \sigma_k - \frac{2}{3}\sigma_j\sigma_k > 0$ for $j, k = 1, 2, \ldots, m$. Conclude that the matrix
lyche-numerical-linear-algebra_F03425	262	■ 1.000	`j, k=1,2, \ldots, m`	$j, k=1,2, \ldots, m$	Show that $\sigma_j + \sigma_k - \frac{2}{3}\sigma_j\sigma_k > 0$ for $j, k = 1, 2, \ldots, m$. Conclude that the matrix

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-a F03426	262	1.000	`\boldsymbol{X}=\left(x_{j, k}\right)`	$\boldsymbol{X}=\left(x_{j, k}\right)$	Define the matrix $\boldsymbol{X} = (x_{j,k})$ by $\boldsymbol{U} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$ where $\boldsymbol{U}$ is the solution of (10.25).
lyche-numerical-linear-a F03427	262	1.000	`\boldsymbol{U}=\boldsymbol{S} \boldsymbol{X} \boldsymbol{S}`	$\boldsymbol{U}=\boldsymbol{S} \boldsymbol{X} \boldsymbol{S}$	Define the matrix $\boldsymbol{X} = (x_{j,k})$ by $\boldsymbol{U} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$ where $\boldsymbol{U}$ is the solution of (10.25).
lyche-numerical-linear-a F03428	262	1.000	`O\left(\delta n^{3 / 2}\right)`	$O\left(\delta n^{3 / 2}\right)$	which requires only $O(\delta n^{3/2})$ operations to find $\boldsymbol{U}$ in Problem 10.9. Here δ is some
lyche-numerical-linear-a F03429	262	0.713	`\boldsymbol{F}=`	$\boldsymbol{F}=$	Exercise 11.8 compute the solution $\boldsymbol{U}$ corresponding to $\boldsymbol{F}$ = `ones(m,m)`. For
lyche-numerical-linear-a F03430	262	1.000	`\boldsymbol{x}=\boldsymbol{A} \backslash \boldsymbol{b}`	$\boldsymbol{x}=\boldsymbol{A} \backslash \boldsymbol{b}$	standard form $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ in (10.25). You can use $\boldsymbol{x} = \boldsymbol{A}\backslash\boldsymbol{b}$ for solving $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.
lyche-numerical-linear-a F03431	262	0.581	`\boldsymbol{x} \in \mathbb{R}^{m^{2}}`	$\boldsymbol{x} \in \mathbb{R}^{m^{2}}$	Use `F(:)` to vectorize a matrix and `reshape(x,m,m)` to turn a vector $\boldsymbol{x} \in \mathbb{R}^{m^2}$
lyche-numerical-linear-a F03432	262	1.000	`U`	U	into an $m \times m$ matrix. Use the MATLAB command `surf(U)` for plotting U for, say,
lyche-numerical-linear-a F03433	262	1.000	`m=50`	$m=50$	$m = 50$. Compare the result with Exercise 11.8 by plotting the difference between
lyche-numerical-linear-a F03434	264	1.000	`\boldsymbol{x}_{0}`	$\boldsymbol{x}_{0}$	approximation $\boldsymbol{x}_0$ to the exact solution $\boldsymbol{x}$ and then compute a sequence $\{\boldsymbol{x}_k\}$ such that
lyche-numerical-linear-a F03435	264	1.000	`\left\{\boldsymbol{x}_{k}\right\}`	$\left\{\boldsymbol{x}_{k}\right\}$	approximation $\boldsymbol{x}_0$ to the exact solution $\boldsymbol{x}$ and then compute a sequence $\{\boldsymbol{x}_k\}$ such that
lyche-numerical-linear-a F03436	264	1.000	`\boldsymbol{x}_{k} \rightarrow \boldsymbol{x}`	$\boldsymbol{x}_{k} \rightarrow \boldsymbol{x}$	hopefully $\boldsymbol{x}_k \rightarrow \boldsymbol{x}$. Iterative methods are mainly used for large sparse systems, i.e.,
lyche-numerical-linear-a F03437	265	1.000	`\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right] \left[\begin{array}{l}y \\ z\end{array}\right]=\left[ \begin{array}{l}1 \\ 1\end{array}\right]`	$\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right] \left[\begin{array}{l}y \\ z\end{array}\right]=\left[\begin{array}{l}1 \\ 1\end{array}\right]$	onal elements, the linear system $\left[\begin{smallmatrix}2 & -1 \\ -1 & 2\end{smallmatrix}\right]\left[\begin{smallmatrix}y \\ z\end{smallmatrix}\right] = \left[\begin{smallmatrix}1 \\ 1\end{smallmatrix}\right]$ can be written in component
lyche-numerical-linear-a F03438	265	1.000	`y=(z+1) / 2`	$y=(z+1) / 2$	form as $y = (z+1)/2$ and $z = (y+1)/2$. Starting with y_0, z_0 we generate two
lyche-numerical-linear-a F03439	265	1.000	`z=(y+1) / 2`	$z=(y+1) / 2$	form as $y = (z+1)/2$ and $z = (y+1)/2$. Starting with y_0, z_0 we generate two
lyche-numerical-linear-a F03440	265	1.000	`y_{0}, z_{0}`	y_{0}, z_{0}	form as $y = (z+1)/2$ and $z = (y+1)/2$. Starting with y_0, z_0 we generate two
lyche-numerical-linear-a F03441	265	0.996	`\left\{y_{k}\right\}`	$\left\{y_{k}\right\}$	sequences $\{y_k\}$ and $\{z_k\}$ using the difference equations $y_{k+1} = (z_k+1)/2$ and
lyche-numerical-linear-a F03442	265	0.996	`\left\{z_{k}\right\}`	$\left\{z_{k}\right\}$	sequences $\{y_k\}$ and $\{z_k\}$ using the difference equations $y_{k+1} = (z_k+1)/2$ and
lyche-numerical-linear-a F03443	265	0.996	`y_{k+1}=\left(z_{k}+1\right) / 2`	$y_{k+1}=\left(z_{k}+1\right) / 2$	sequences $\{y_k\}$ and $\{z_k\}$ using the difference equations $y_{k+1} = (z_k+1)/2$ and
lyche-numerical-linear-a F03444	265	1.000	`z_{k+1}=\left(y_{k}+1\right) / 2`	$z_{k+1}=\left(y_{k}+1\right) / 2$	$z_{k+1} = (y_k+1)/2$. This is an example of Jacobi's method. If $y_0 = z_0 = 0$ then
lyche-numerical-linear-a F03445	265	1.000	`y_{0}=z_{0}=0`	$y_{0}=z_{0}=0$	$z_{k+1} = (y_k+1)/2$. This is an example of Jacobi's method. If $y_0 = z_0 = 0$ then
lyche-numerical-linear-a F03446	265	1.000	`y_{1}=z_{1}=1 / 2`	$y_{1}=z_{1}=1 / 2$	we find $y_1 = z_1 = 1/2$ and in general $y_k = z_k = 1 - 2^{-k}$ for $k = 0, 1, 2, 3, \ldots$.
lyche-numerical-linear-a F03447	265	1.000	`y_{k}=z_{k}=1-2^{-k}`	$y_{k}=z_{k}=1-2^{-k}$	we find $y_1 = z_1 = 1/2$ and in general $y_k = z_k = 1 - 2^{-k}$ for $k = 0, 1, 2, 3, \ldots$.
lyche-numerical-linear-a F03448	265	1.000	`k=0,1,2,3, \ldots`	$k=0,1,2,3, \ldots$	we find $y_1 = z_1 = 1/2$ and in general $y_k = z_k = 1 - 2^{-k}$ for $k = 0, 1, 2, 3, \ldots$.
lyche-numerical-linear-a F03449	265	1.000	`[1,1]^{T}`	$[1,1]^{T}$	The iteration converges to the exact solution $[1,1]^T$, and the error is halved in each
lyche-numerical-linear-a F03450	266	1.000	`z_{k+1}=\left(y_{k+1}+1\right) / 2`	$z_{k+1}=\left(y_{k+1}+1\right) / 2$	$z_{k+1} = (y_{k+1}+1)/2$. If $y_0 = z_0 = 0$ then we find $y_1 = 1/2$, $z_1 = 3/4$, $y_2 = 7/8$,
lyche-numerical-linear-a F03451	266	1.000	`y_{1}=1 / 2, z_{1}=3 / 4, y_{2}=7 / 8`	$y_{1}=1 / 2, z_{1}=3 / 4, y_{2}=7 / 8$	$z_{k+1} = (y_{k+1}+1)/2$. If $y_0 = z_0 = 0$ then we find $y_1 = 1/2$, $z_1 = 3/4$, $y_2 = 7/8$,
lyche-numerical-linear-a F03452	266	1.000	`z_{2}=15 / 16`	$z_{2}=15 / 16$	$z_2 = 15/16$, and in general $y_k = 1 - 2 \cdot 4^{-k}$ and $z_k = 1 - 4^{-k}$ for $k = 1, 2, 3, \ldots$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F03453	266	1.000	`y_{k}=1-2 \cdot 4^{-k}`	$y_{k}=1-2 \cdot 4^{-k}$	$z_2 = 15/16$, and in general $y_k = 1 - 2 \cdot 4^{-k}$ and $z_k = 1 - 4^{-k}$ for $k = 1, 2, 3, \ldots$.
lyche-numerical-linear-al F03454	266	1.000	`z_{k}=1-4^{-k}`	$z_{k}=1-4^{-k}$	$z_2 = 15/16$, and in general $y_k = 1 - 2 \cdot 4^{-k}$ and $z_k = 1 - 4^{-k}$ for $k = 1, 2, 3, \ldots$.
lyche-numerical-linear-al F03455	266	1.000	`k=1,2,3, \ldots`	$k=1,2,3, \ldots$	$z_2 = 15/16$, and in general $y_k = 1 - 2 \cdot 4^{-k}$ and $z_k = 1 - 4^{-k}$ for $k = 1, 2, 3, \ldots$.
lyche-numerical-linear-al F03456	266	1.000	`\boldsymbol{x}_{k}=\left[\boldsymbol{x}_{k}(1), \ldots, \boldsymbol{x}_{k}(n)\right]^{T}`	$\boldsymbol{x}_{k}=\left[\boldsymbol{x}_{k}(1), \ldots, \boldsymbol{x}_{k}(n)\right]^{T}$	Suppose we know an approximation $\boldsymbol{x}_k = [\boldsymbol{x}_k(1), \ldots, \boldsymbol{x}_k(n)]^T$ to the exact solution
lyche-numerical-linear-al F03457	266	1.000	`\boldsymbol{x}(i)`	$\boldsymbol{x}(i)$	diagonal elements. Solving the ith equation of $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ for $\boldsymbol{x}(i)$, we obtain a **fixed-**
lyche-numerical-linear-al F03458	266	1.000	`\boldsymbol{x}_{k+1}(i)`	$\boldsymbol{x}_{k+1}(i)$	where we use the new $\boldsymbol{x}_{k+1}(i)$ immediately after it has been computed.
lyche-numerical-linear-al F03459	266	1.000	`0<\omega<2`	$0<\omega<2$	ducing an acceleration parameter $0 < \omega < 2$ in the GS method. We write
lyche-numerical-linear-al F03460	266	1.000	`\boldsymbol{x}(i)=\omega \boldsymbol{x}(i)+(1-\omega) \boldsymbol{x}(i)`	$\boldsymbol{x}(i)=\omega \boldsymbol{x}(i)+(1-\omega) \boldsymbol{x}(i)$	$\boldsymbol{x}(i) = \omega\boldsymbol{x}(i) + (1-\omega)\boldsymbol{x}(i)$ and this leads to the method
lyche-numerical-linear-al F03461	266	1.000	`\omega=1`	$\omega=1$	The SOR method reduces to the Gauss-Seidel method for $\omega = 1$. Denoting the
lyche-numerical-linear-al F03462	266	1.000	`\boldsymbol{x}_{k+1}^{g s}`	$\boldsymbol{x}_{k+1}^{g s}$	right hand side of (12.3) by $\boldsymbol{x}_{k+1}^{gs}$ we can write (12.4) as $\boldsymbol{x}_{k+1} = \omega\boldsymbol{x}_{k+1}^{gs} + (1 -$
lyche-numerical-linear-al F03463	266	1.000	`\boldsymbol{x}_{k+1}=\omega \boldsymbol{x}_{k+1}^{g s}+(1-`	$\boldsymbol{x}_{k+1}=\omega \boldsymbol{x}_{k+1}^{g s}+(1-$	right hand side of (12.3) by $\boldsymbol{x}_{k+1}^{gs}$ we can write (12.4) as $\boldsymbol{x}_{k+1} = \omega\boldsymbol{x}_{k+1}^{gs} + (1 -$
lyche-numerical-linear-al F03464	266	1.000	`\omega) \boldsymbol{x}_{k}`	$\omega) \boldsymbol{x}_{k}$	$\omega)\boldsymbol{x}_k$, and we see that $\boldsymbol{x}_{k+1}$ is located on the straight line passing through the
lyche-numerical-linear-al F03465	266	1.000	`\boldsymbol{x}_{k+1}`	$\boldsymbol{x}_{k+1}$	$\omega)\boldsymbol{x}_k$, and we see that $\boldsymbol{x}_{k+1}$ is located on the straight line passing through the
lyche-numerical-linear-al F03466	267	1.000	`1 \leq \omega<2`	$1 \leq \omega<2$	parameter ω in the range $1 \leq \omega < 2$ and then $\boldsymbol{x}_{k+1}$ is computed by linear
lyche-numerical-linear-al F03467	267	1.000	`\boldsymbol{x}_{k+1 / 2}`	$\boldsymbol{x}_{k+1 / 2}$	(12.4), computing an approximation denoted $\boldsymbol{x}_{k+1/2}$ instead of $\boldsymbol{x}_{k+1}$, is followed
lyche-numerical-linear-al F03468	267	1.000	`i=n, n-1, \ldots 1`	$i=n, n-1, \ldots 1$	in the order $i = n, n-1, \ldots 1$. The method is slower and more complicated
lyche-numerical-linear-al F03469	267	0.999	`\boldsymbol{v}(i, j)`	$\boldsymbol{v}(i, j)$	with homogenous boundary conditions also given in (10.4). Solving for $\boldsymbol{v}(i, j)$ we
lyche-numerical-linear-al F03470	267	0.570	`e_{i, j}:=f_{i, j} /(m+1)^{2}`	$e_{i, j}:=f_{i, j} /(m+1)^{2}$	where $e_{i,j} := f_{i,j}/(m+1)^2$. The J, GS , and SOR methods take the form
lyche-numerical-linear-al F03471	268	0.651	`k_{100}`	k_{100}	k_{100}, k_{2500}, $k_{10\,000}$, $k_{40\,000}$, $k_{160\,000}$; J: 385, 8386; GS: 194, 4194; SOR: 35, 164, 324, 645, 1286
lyche-numerical-linear-al F03472	268	0.651	`k_{2500}`	k_{2500}	k_{100}, k_{2500}, $k_{10\,000}$, $k_{40\,000}$, $k_{160\,000}$; J: 385, 8386; GS: 194, 4194; SOR: 35, 164, 324, 645, 1286
lyche-numerical-linear-al F03473	268	0.651	`k_{10} 000`	$k_{10}000$	k_{100}, k_{2500}, $k_{10\,000}$, $k_{40\,000}$, $k_{160\,000}$; J: 385, 8386; GS: 194, 4194; SOR: 35, 164, 324, 645, 1286

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-a F03474</td><td>268</td><td>0.651</td><td><code>k_{40000}</code></td><td>k_{40000}</td><td><table><tr><th></th><th>k_{100}</th><th>k_{2500}</th><th>$k_{10\,000}$</th><th>$k_{40\,000}$</th><th>$k_{160\,000}$</th></tr><tr><td>J</td><td>385</td><td>8386</td><td></td><td></td><td></td></tr><tr><td>GS</td><td>194</td><td>4194</td><td></td><td></td><td></td></tr><tr><td>SOR</td><td>35</td><td>164</td><td>324</td><td>645</td><td>1286</td></tr></table></td></tr>
<tr><td>lyche-numerical-linear-a F03475</td><td>268</td><td>0.651</td><td><code>k_{160} 000</code></td><td>$k_{160}000$</td><td><table><tr><th></th><th>k_{100}</th><th>k_{2500}</th><th>$k_{10\,000}$</th><th>$k_{40\,000}$</th><th>$k_{160\,000}$</th></tr><tr><td>J</td><td>385</td><td>8386</td><td></td><td></td><td></td></tr><tr><td>GS</td><td>194</td><td>4194</td><td></td><td></td><td></td></tr><tr><td>SOR</td><td>35</td><td>164</td><td>324</td><td>645</td><td>1286</td></tr></table></td></tr>
<tr><td>lyche-numerical-linear-a F03476</td><td>268</td><td>0.995</td><td><code>\left(i_{1}, j_{1}\right)<</code></td><td>$\left(i_{1}, j_{1}\right) <$</td><td>We note that for GS and SOR we have used the natural ordering, i.e., $(i_1, j_1) <$</td></tr>
<tr><td>lyche-numerical-linear-a F03477</td><td>268</td><td>0.960</td><td><code>i_{2}, j_{2}</code></td><td>i_{2}, j_{2}</td><td>(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering</td></tr>
<tr><td>lyche-numerical-linear-a F03478</td><td>268</td><td>0.960</td><td><code>j_{1} \leq j_{2}</code></td><td>$j_{1} \leq j_{2}$</td><td>(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering</td></tr>
<tr><td>lyche-numerical-linear-a F03479</td><td>268</td><td>0.960</td><td><code>i_{1}<i_{2}</code></td><td>$i_{1} < i_{2}$</td><td>(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering</td></tr>
<tr><td>lyche-numerical-linear-a F03480</td><td>268</td><td>0.960</td><td><code>j_{1}=j_{2}</code></td><td>$j_{1}=j_{2}$</td><td>(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering</td></tr>
<tr><td>lyche-numerical-linear-a F03481</td><td>268</td><td>1.000</td><td><code>\boldsymbol{F}=\left(f_{i, j}\right) \in \mathbb{R}^{m \times m}</code></td><td>$\boldsymbol{F}=\left(f_{i, j}\right) \in \mathbb{R}^{m \times m}$</td><td>on the linear system (12.6) with $\boldsymbol{F} = (f_{i,j}) \in \mathbb{R}^{m\times m}$, starting with $\boldsymbol{V}_0 = \mathbf{0} \in$</td></tr>
<tr><td>lyche-numerical-linear-a F03482</td><td>268</td><td>1.000</td><td><code>\boldsymbol{V}_{0}=\mathbf{0} \in</code></td><td>$\boldsymbol{V}_{0}=\mathbf{0} \in$</td><td>on the linear system (12.6) with $\boldsymbol{F} = (f_{i,j}) \in \mathbb{R}^{m\times m}$, starting with $\boldsymbol{V}_0 = \mathbf{0} \in$</td></tr>
<tr><td>lyche-numerical-linear-a F03483</td><td>268</td><td>1.000</td><td><code>\mathbb{R}^{(m+2) \times(m+2)}</code></td><td>$\mathbb{R}^{(m+2) \times(m+2)}$</td><td>$\mathbb{R}^{(m+2)\times(m+2)}$. The output is the number of iterations k, to obtain $\|\boldsymbol{V}^{(k)} - \boldsymbol{U}\|_M :=$</td></tr>
<tr><td>lyche-numerical-linear-a F03484</td><td>268</td><td>1.000</td><td><code>\left\|\boldsymbol{V}^{(k)}-\boldsymbol{U}\right\|_{M}:=</code></td><td>$\left\|\boldsymbol{V}^{(k)}-\boldsymbol{U}\right\|_{M}:=$</td><td>$\mathbb{R}^{(m+2)\times(m+2)}$. The output is the number of iterations k, to obtain $\|\boldsymbol{V}^{(k)} - \boldsymbol{U}\|_M :=$</td></tr>
<tr><td>lyche-numerical-linear-a F03485</td><td>268</td><td>0.911</td><td><code>\max _{i, j}\left|v_{i j}-u_{i j}\right|<</code></td><td>$\max _{i, j}\left|v_{i j}-u_{i j}\right| <$</td><td>$\max_{i,j}|v_{ij} - u_{ij}| < tol$. Here $[u_{ij}] \in \mathbb{R}^{(m+2)\times(m+2)}$ is the ”exact" solution of (12.6)</td></tr>
<tr><td>lyche-numerical-linear-a F03486</td><td>268</td><td>0.911</td><td><code>\left[u_{i j}\right] \in \mathbb{R}^{(m+2) \times(m+2)}</code></td><td>$\left[u_{i j}\right] \in \mathbb{R}^{(m+2) \times(m+2)}$</td><td>$\max_{i,j}|v_{ij} - u_{ij}| < tol$. Here $[u_{ij}] \in \mathbb{R}^{(m+2)\times(m+2)}$ is the ”exact" solution of (12.6)</td></tr>
<tr><td>lyche-numerical-linear-a F03487</td><td>268</td><td>0.843</td><td><code>k=k_{n}</code></td><td>$k=k_{n}$</td><td>$k = k_n$ from this algorithm using $\boldsymbol{F} = \texttt{ones}(m, m)$ for $m = 10, 50$, $K = 10^4$, and</td></tr>
<tr><td>lyche-numerical-linear-a F03488</td><td>268</td><td>0.843</td><td><code>\boldsymbol{F}=\operatorname{ones}(m, m)</code></td><td>$\boldsymbol{F}=\operatorname{ones}(m, m)$</td><td>$k = k_n$ from this algorithm using $\boldsymbol{F} = \texttt{ones}(m, m)$ for $m = 10, 50$, $K = 10^4$, and</td></tr>
<tr><td>lyche-numerical-linear-a F03489</td><td>268</td><td>0.843</td><td><code>m=10,50, K=10^{4}</code></td><td>$m=10,50, K=10^{4}$</td><td>$k = k_n$ from this algorithm using $\boldsymbol{F} = \texttt{ones}(m, m)$ for $m = 10, 50$, $K = 10^4$, and</td></tr>
<tr><td>lyche-numerical-linear-a F03490</td><td>268</td><td>0.825</td><td><code>=10^{-8}</code></td><td>$=10^{-8}$</td><td>$tol = 10^{-8}$. We also show the number of iterations for Gauss-Seidel and SOR with</td></tr>
<tr><td>lyche-numerical-linear-a F03491</td><td>268</td><td>1.000</td><td><code>\omega^{*}:=2 /\left(1+\sin \left(\frac{\pi}{m+1}\right)\right)</code></td><td>$\omega^{*}:=2 /\left(1+\sin \left(\frac{\pi}{m+1}\right)\right)$</td><td>a value of ω known as the optimal acceleration parameter $\omega^* := 2/\big(1 + \sin(\frac{\pi}{m+1})\big)$.</td></tr>
<tr><td>lyche-numerical-linear-a F03492</td><td>269</td><td>1.000</td><td><code>0<w<2</code></td><td>$0 < w < 2$</td><td>is an acceleration parameter with $0 < w < 2$. For $w = 1$ we obtain Gauss-Seidel’s</td></tr>
<tr><td>lyche-numerical-linear-a F03493</td><td>269</td><td>1.000</td><td><code>w=1</code></td><td>$w=1$</td><td>is an acceleration parameter with $0 < w < 2$. For $w = 1$ we obtain Gauss-Seidel’s</td></tr>
<tr><td>lyche-numerical-linear-a F03494</td><td>269</td><td>1.000</td><td><code>k_{n}=O(n)</code></td><td>$k_{n}=O(n)$</td><td>$k_n = O(n)$ for the J and GS method and $k_n = O(\sqrt{n})$ for SOR with optimal ω.</td></tr>
<tr><td>lyche-numerical-linear-a F03495</td><td>269</td><td>1.000</td><td><code>k_{n}=O(\sqrt{n})</code></td><td>$k_{n}=O(\sqrt{n})$</td><td>$k_n = O(n)$ for the J and GS method and $k_n = O(\sqrt{n})$ for SOR with optimal ω.</td></tr>
<tr><td>lyche-numerical-linear-a F03496</td><td>269</td><td>1.000</td><td><code>\sqrt{n}</code></td><td>$\sqrt{n}$</td><td>The choice of tol will only influence the constants multiplying n or $\sqrt{n}$.</td></tr>
<tr><td>lyche-numerical-linear-a F03497</td><td>269</td><td>1.000</td><td><code>O\left(k_{n} \times n\right)</code></td><td>$O\left(k_{n} \times n\right)$</td><td>$O(k_n \times n)$. Therefore the number of arithmetic operations for the J and GS</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F03498	270	1.000	`\boldsymbol{G} \in \mathbb{C}^{n \times n}`	$\boldsymbol{G} \in \mathbb{C}^{n \times n}$	$\{\boldsymbol{x}_k\}$ of vectors in $\mathbb{C}^n$. For a general $\boldsymbol{G} \in \mathbb{C}^{n\times n}$ and $\boldsymbol{c} \in \mathbb{C}^n$ a solution of $\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$
lyche-numerical-linear-algebra_F03499	270	1.000	`\boldsymbol{x}=\boldsymbol{G} \boldsymbol{x}+ \boldsymbol{c}`	$\boldsymbol{x}=\boldsymbol{G} \boldsymbol{x}+\boldsymbol{c}$	$\{\boldsymbol{x}_k\}$ of vectors in $\mathbb{C}^n$. For a general $\boldsymbol{G} \in \mathbb{C}^{n\times n}$ and $\boldsymbol{c} \in \mathbb{C}^n$ a solution of $\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$
lyche-numerical-linear-algebra_F03500	270	1.000	`\boldsymbol{I}-\boldsymbol{G}`	$\boldsymbol{I}-\boldsymbol{G}$	is called a **fixed-point**. The fixed-point is unique if $\boldsymbol{I} - \boldsymbol{G}$ is nonsingular.
lyche-numerical-linear-algebra_F03501	270	1.000	`\lim _{k \rightarrow \infty} \boldsymbol{x}_{k}= \boldsymbol{x}`	$\lim _{k \rightarrow \infty} \boldsymbol{x}_{k}=\boldsymbol{x}$	If $\lim_{k\to\infty} \boldsymbol{x}_k = \boldsymbol{x}$ for some $\boldsymbol{x} \in \mathbb{C}^n$ then $\boldsymbol{x}$ is a fixed point since
lyche-numerical-linear-algebra_F03502	270	1.000	`\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x}= \boldsymbol{M}^{-1} \boldsymbol{b}`	$\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{M}^{-1} \boldsymbol{b}$	$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ in the equivalent form $\boldsymbol{M}^{-1}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{M}^{-1}\boldsymbol{b}$ or $\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{M}^{-1}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{M}^{-1}\boldsymbol{b}$.
lyche-numerical-linear-algebra_F03503	270	1.000	`\boldsymbol{x}=\boldsymbol{x}-\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x}+\boldsymbol{M}^{-1} \boldsymbol{b}`	$\boldsymbol{x}=\boldsymbol{x}-\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x}+\boldsymbol{M}^{-1} \boldsymbol{b}$	$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ in the equivalent form $\boldsymbol{M}^{-1}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{M}^{-1}\boldsymbol{b}$ or $\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{M}^{-1}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{M}^{-1}\boldsymbol{b}$.
lyche-numerical-linear-algebra_F03504	270	1.000	`\boldsymbol{A}=\boldsymbol{D}-\boldsymbol{A}_{ \boldsymbol{L}}-\boldsymbol{A}_{\boldsymbol{R}}`	$\boldsymbol{A}=\boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}-\boldsymbol{A}_{\boldsymbol{R}}$	of three matrices, $\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{A}_{\boldsymbol{L}} - \boldsymbol{A}_{\boldsymbol{R}}$, where $-\boldsymbol{A}_{\boldsymbol{L}}$, $\boldsymbol{D}$, and $-\boldsymbol{A}_{\boldsymbol{R}}$ are the lower,
lyche-numerical-linear-algebra_F03505	270	1.000	`-\boldsymbol{A}_{\boldsymbol{L}}, \boldsymbol{D}`	$-\boldsymbol{A}_{\boldsymbol{L}}, \boldsymbol{D}$	of three matrices, $\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{A}_{\boldsymbol{L}} - \boldsymbol{A}_{\boldsymbol{R}}$, where $-\boldsymbol{A}_{\boldsymbol{L}}$, $\boldsymbol{D}$, and $-\boldsymbol{A}_{\boldsymbol{R}}$ are the lower,
lyche-numerical-linear-algebra_F03506	270	1.000	`-\boldsymbol{A}_{\boldsymbol{R}}`	$-\boldsymbol{A}_{\boldsymbol{R}}$	of three matrices, $\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{A}_{\boldsymbol{L}} - \boldsymbol{A}_{\boldsymbol{R}}$, where $-\boldsymbol{A}_{\boldsymbol{L}}$, $\boldsymbol{D}$, and $-\boldsymbol{A}_{\boldsymbol{R}}$ are the lower,
lyche-numerical-linear-algebra_F03507	270	1.000	`\boldsymbol{D}:=\operatorname{diag}\left(a_{11}, \ldots, a_{n n}\right)`	$\boldsymbol{D}:=\operatorname{diag}\left(a_{11}, \ldots, a_{n n}\right)$	diagonal, and upper part of $\boldsymbol{A}$, respectively. Thus $\boldsymbol{D} := \operatorname{diag}(a_{11}, \ldots, a_{nn})$,
lyche-numerical-linear-algebra_F03508	271	0.753	`\boldsymbol{M}_{J}, \boldsymbol{M}_{1}`	$\boldsymbol{M}_{J}, \boldsymbol{M}_{1}$	*$\boldsymbol{M}_J$, $\boldsymbol{M}_1$ and $\boldsymbol{M}_\omega$ for the J, GS and SOR methods are given by*
lyche-numerical-linear-algebra_F03509	271	0.753	`\boldsymbol{M}_{\omega}`	$\boldsymbol{M}_{\omega}$	*$\boldsymbol{M}_J$, $\boldsymbol{M}_1$ and $\boldsymbol{M}_\omega$ for the J, GS and SOR methods are given by*
lyche-numerical-linear-algebra_F03510	271	0.753	`J, G S`	$J, G S$	*$\boldsymbol{M}_J$, $\boldsymbol{M}_1$ and $\boldsymbol{M}_\omega$ for the J, GS and SOR methods are given by*
lyche-numerical-linear-algebra_F03511	271	0.955	`\boldsymbol{D} \boldsymbol{x}_{k+1}=(\boldsymbol{D}- \boldsymbol{A}) \boldsymbol{x}_{k}+\boldsymbol{b}`	$\boldsymbol{D} \boldsymbol{x}_{k+1}=(\boldsymbol{D}-\boldsymbol{A}) \boldsymbol{x}_{k}+\boldsymbol{b}$	obtain the matrix form $\boldsymbol{D}\boldsymbol{x}_{k+1} = (\boldsymbol{D} - \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$ showing that $\boldsymbol{M}_J = \boldsymbol{D}$.
lyche-numerical-linear-algebra_F03512	271	0.955	`\boldsymbol{M}_{J}=\boldsymbol{D}`	$\boldsymbol{M}_{J}=\boldsymbol{D}$	obtain the matrix form $\boldsymbol{D}\boldsymbol{x}_{k+1} = (\boldsymbol{D} - \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$ showing that $\boldsymbol{M}_J = \boldsymbol{D}$.
lyche-numerical-linear-algebra_F03513	271	1.000	`\boldsymbol{A}_{L} \boldsymbol{x}_{k+1}`	$\boldsymbol{A}_{L} \boldsymbol{x}_{k+1}$	Dividing both sides by ω and moving $\boldsymbol{A}_L\boldsymbol{x}_{k+1}$ to the left hand side this takes the
lyche-numerical-linear-algebra_F03514	271	1.000	`\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{L} \right) \boldsymbol{x}_{k+1}=\boldsymbol{A}_{R} \boldsymbol{x}_{k}+\boldsymbol{b}+\left(\omega^{-1}-1 \right) \boldsymbol{D} \boldsymbol{x}_{k}`	$\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{L}\right) \boldsymbol{x}_{k+1}=\boldsymbol{A}_{R} \boldsymbol{x}_{k}+\boldsymbol{b}+\left(\omega^{-1}-1\right) \boldsymbol{D} \boldsymbol{x}_{k}$	form $(\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)\boldsymbol{x}_{k+1} = \boldsymbol{A}_R\boldsymbol{x}_k + \boldsymbol{b} + (\omega^{-1} - 1)\boldsymbol{D}\boldsymbol{x}_k$ showing that $\boldsymbol{M}_\omega =$
lyche-numerical-linear-algebra_F03515	271	1.000	`\boldsymbol{M}_{\omega}=`	$\boldsymbol{M}_{\omega}=$	form $(\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)\boldsymbol{x}_{k+1} = \boldsymbol{A}_R\boldsymbol{x}_k + \boldsymbol{b} + (\omega^{-1} - 1)\boldsymbol{D}\boldsymbol{x}_k$ showing that $\boldsymbol{M}_\omega =$
lyche-numerical-linear-algebra_F03516	271	1.000	`\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{L}`	$\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{L}$	$\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L$. We obtain $\boldsymbol{M}_1$ by letting $\omega = 1$ in $\boldsymbol{M}_\omega$.
lyche-numerical-linear-algebra_F03517	271	1.000	`\boldsymbol{M}_{1}`	$\boldsymbol{M}_{1}$	$\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L$. We obtain $\boldsymbol{M}_1$ by letting $\omega = 1$ in $\boldsymbol{M}_\omega$.
lyche-numerical-linear-algebra_F03518	271	1.000	`\boldsymbol{G}_{\omega}=\boldsymbol{I}-\boldsymbol{M}_{ \omega}^{-1} \boldsymbol{A}`	$\boldsymbol{G}_{\omega}=\boldsymbol{I}-\boldsymbol{M}_{\omega}^{-1} \boldsymbol{A}$	The iteration matrix $\boldsymbol{G}_\omega = \boldsymbol{I} - \boldsymbol{M}_\omega^{-1}\boldsymbol{A}$ is given by
lyche-numerical-linear-algebra_F03519	272	1.000	`\boldsymbol{x}_{k+1}(1)`	$\boldsymbol{x}_{k+1}(1)$	Substituting the value of $\boldsymbol{x}_{k+1}(1)$ from the first equation into the second equation
lyche-numerical-linear-algebra_F03520	272	1.000	`\boldsymbol{x}_{k+1}:=\boldsymbol{G} \boldsymbol{x}_{k}+ \boldsymbol{c}`	$\boldsymbol{x}_{k+1}:=\boldsymbol{G} \boldsymbol{x}_{k}+\boldsymbol{c}$	for methods of the form $\boldsymbol{x}_{k+1} := \boldsymbol{G}\boldsymbol{x}_k + \boldsymbol{c}$.
lyche-numerical-linear-algebra_F03521	272	1.000	`\boldsymbol{G} \in \mathbb{C}^{n \times n}, \boldsymbol{c} \in \mathbb{C}^{n}`	$\boldsymbol{G} \in \mathbb{C}^{n \times n}, \boldsymbol{c} \in \mathbb{C}^{n}$	In the following we assume that $\boldsymbol{G} \in \mathbb{C}^{n\times n}$, $\boldsymbol{c} \in \mathbb{C}^n$ and $\boldsymbol{I} - \boldsymbol{G}$ is nonsingular.
lyche-numerical-linear-algebra_F03522	272	1.000	`\lim _{k \rightarrow \infty} \boldsymbol{G}^{k}= \mathbf{0}`	$\lim _{k \rightarrow \infty} \boldsymbol{G}^{k}=\mathbf{0}$	$\boldsymbol{x}_{k+1} := \boldsymbol{G}\boldsymbol{x}_k + \boldsymbol{c}$ *converges if and only if* $\lim_{k\to\infty} \boldsymbol{G}^k = \mathbf{0}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F03523	272	■0.991	\boldsymbol{x}_{k+1}=\boldsymbol{G} \boldsymbol{x}_{k}+ \boldsymbol{c}	$\boldsymbol{x}_{k+1}=\boldsymbol{G} \boldsymbol{x}_{k}+\boldsymbol{c}$	$\boldsymbol{x}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}$ from $\boldsymbol{x}_{k+1}=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c}$. The vector $\boldsymbol{c}$ cancels and we obtain $\boldsymbol{x}_{k+1}-\boldsymbol{x}=$
lyche-numerical-linear-algebra F03524	272	■0.991	\boldsymbol{x}_{k+1}-\boldsymbol{x}=	$\boldsymbol{x}_{k+1}-\boldsymbol{x}=$	$\boldsymbol{x}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}$ from $\boldsymbol{x}_{k+1}=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c}$. The vector $\boldsymbol{c}$ cancels and we obtain $\boldsymbol{x}_{k+1}-\boldsymbol{x}=$
lyche-numerical-linear-algebra F03525	272	■1.000	\boldsymbol{G}\left(\boldsymbol{x}_{k}-\boldsymbol{x} \right)	$\boldsymbol{G}\left(\boldsymbol{x}_{k}-\boldsymbol{x}\right)$	$\boldsymbol{G}(\boldsymbol{x}_k-\boldsymbol{x})$. By induction on k
lyche-numerical-linear-algebra F03526	272	■0.999	\boldsymbol{x}_{k}-\boldsymbol{x} \rightarrow \mathbf{0}	$\boldsymbol{x}_{k}-\boldsymbol{x} \rightarrow \mathbf{0}$	Clearly $\boldsymbol{x}_k-\boldsymbol{x}\rightarrow\mathbf{0}$ if $\boldsymbol{G}^k\rightarrow\mathbf{0}$ and the method converges. For the converse we let
lyche-numerical-linear-algebra F03527	272	■0.999	\boldsymbol{G}^{k} \rightarrow \mathbf{0}	$\boldsymbol{G}^{k} \rightarrow \mathbf{0}$	Clearly $\boldsymbol{x}_k-\boldsymbol{x}\rightarrow\mathbf{0}$ if $\boldsymbol{G}^k\rightarrow\mathbf{0}$ and the method converges. For the converse we let
lyche-numerical-linear-algebra F03528	272	■1.000	\boldsymbol{x}_{0}-\boldsymbol{x}=\boldsymbol{e}_{j}	$\boldsymbol{x}_{0}-\boldsymbol{x}=\boldsymbol{e}_{j}$	$\boldsymbol{x}$ be as before and choose $\boldsymbol{x}_0-\boldsymbol{x}=\boldsymbol{e}_j$, the jth unit vector for $j=1,\ldots,n$. Since
lyche-numerical-linear-algebra F03529	272	■1.000	\boldsymbol{G}^{k} \boldsymbol{e}_{j} \rightarrow \mathbf{0}	$\boldsymbol{G}^{k} \boldsymbol{e}_{j} \rightarrow \mathbf{0}$	$\boldsymbol{x}_k-\boldsymbol{x}\rightarrow\mathbf{0}$ for any $\boldsymbol{x}_0$ we have $\boldsymbol{G}^k\boldsymbol{e}_j\rightarrow\mathbf{0}$ for $j=1,\ldots,n$ which implies that
lyche-numerical-linear-algebra F03530	272	■1.000	\\|\boldsymbol{G}\\|<1	$\\|\boldsymbol{G}\\|<1$	**Theorem 12.3 (Sufficient Condition for Convergence)** *If* $\\|\boldsymbol{G}\\|<1$ *then the*
lyche-numerical-linear-algebra F03531	273	■1.000	\rho(\boldsymbol{G})<1	$\rho(\boldsymbol{G})<1$	*and only if* $\rho(\boldsymbol{G})<1$.
lyche-numerical-linear-algebra F03532	273	■1.000	\boldsymbol{r}_{k}:=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k}	$\boldsymbol{r}_{k}:=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_{k}$	adding a multiple of the residual vector $\boldsymbol{r}_k:=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$. Note that we do not need
lyche-numerical-linear-algebra F03533	273	■0.992	\lambda_{1} \geq \lambda_{2} \geq \cdots \geq \lambda_{n}>0	$\lambda_{1} \geq \lambda_{2} \geq \cdots \geq \lambda_{n}>0$	*values* $\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_n>0$ *then the R method given by* $\boldsymbol{x}_{k+1}=\big(\boldsymbol{I}-\alpha\boldsymbol{A}\big)\boldsymbol{x}_k+\boldsymbol{b}$
lyche-numerical-linear-algebra F03534	273	■0.992	\boldsymbol{x}_{k+1}=(\boldsymbol{I}-\alpha \boldsymbol{A}) \boldsymbol{x}_{k}+\boldsymbol{b}	$\boldsymbol{x}_{k+1}=(\boldsymbol{I}-\alpha\boldsymbol{A})\boldsymbol{x}_{k}+\boldsymbol{b}$	*values* $\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_n>0$ *then the R method given by* $\boldsymbol{x}_{k+1}=\big(\boldsymbol{I}-\alpha\boldsymbol{A}\big)\boldsymbol{x}_k+\boldsymbol{b}$
lyche-numerical-linear-algebra F03535	273	■1.000	0<\alpha<2 / \lambda_{1}	$0<\alpha<2/\lambda_{1}$	*converges if and only if* $0<\alpha<2/\lambda_1$. *Moreover,*
lyche-numerical-linear-algebra F03536	274	■1.000	\boldsymbol{I}-\alpha \boldsymbol{A}	$\boldsymbol{I}-\alpha\boldsymbol{A}$	***Proof*** The eigenvalues of $\boldsymbol{I}-\alpha\boldsymbol{A}$ are $1-\alpha\lambda_j,\quad j=1,\ldots,n$. We have
lyche-numerical-linear-algebra F03537	274	■1.000	1-\alpha \lambda_{j}, \quad j=1, \ldots, n	$1-\alpha\lambda_{j},\quad j=1,\ldots,n$	***Proof*** The eigenvalues of $\boldsymbol{I}-\alpha\boldsymbol{A}$ are $1-\alpha\lambda_j,\quad j=1,\ldots,n$. We have
lyche-numerical-linear-algebra F03538	274	■1.000	1-\alpha \lambda_{n}=\alpha \lambda_{1}-1	$1-\alpha\lambda_{n}=\alpha\lambda_{1}-1$	see Fig. 12.1. Clearly $1-\alpha\lambda_n=\alpha\lambda_1-1$ for $\alpha=\alpha_o$ and
lyche-numerical-linear-algebra F03539	274	■1.000	\alpha=\alpha_{o}	$\alpha=\alpha_{o}$	see Fig. 12.1. Clearly $1-\alpha\lambda_n=\alpha\lambda_1-1$ for $\alpha=\alpha_o$ and
lyche-numerical-linear-algebra F03540	274	■1.000	\rho_{\alpha}<1	$\rho_{\alpha}<1$	We have $\rho_\alpha<1$ if and only if $\alpha>0$ and $\alpha\lambda_1-1<1$ showing convergence if
lyche-numerical-linear-algebra F03541	274	■1.000	\alpha \lambda_{1}-1<1	$\alpha\lambda_{1}-1<1$	We have $\rho_\alpha<1$ if and only if $\alpha>0$ and $\alpha\lambda_1-1<1$ showing convergence if
lyche-numerical-linear-algebra F03542	274	■1.000	\rho_{\alpha}>\rho_{\alpha_{o}}	$\rho_{\alpha}>\rho_{\alpha_{o}}$	and only if $0<\alpha<2/\lambda_1$ and $\rho_\alpha>\rho_{\alpha_o}$ for $\alpha\leq 0$ and $\alpha\geq 2/\lambda_1$. Finally, if
lyche-numerical-linear-algebra F03543	274	■1.000	\alpha \leq 0	$\alpha\leq 0$	and only if $0<\alpha<2/\lambda_1$ and $\rho_\alpha>\rho_{\alpha_o}$ for $\alpha\leq 0$ and $\alpha\geq 2/\lambda_1$. Finally, if
lyche-numerical-linear-algebra F03544	274	■1.000	\alpha \geq 2 / \lambda_{1}	$\alpha\geq 2/\lambda_{1}$	and only if $0<\alpha<2/\lambda_1$ and $\rho_\alpha>\rho_{\alpha_o}$ for $\alpha\leq 0$ and $\alpha\geq 2/\lambda_1$. Finally, if
lyche-numerical-linear-algebra F03545	274	■1.000	0<\alpha<\alpha_{o}	$0<\alpha<\alpha_{o}$	$0<\alpha<\alpha_o$ then $\rho_\alpha=1-\alpha\lambda_n>1-\alpha_o\lambda_n=\rho_{\alpha_o}$ and if $\alpha_o<\alpha<2/\lambda_1$ then
lyche-numerical-linear-algebra F03546	274	■1.000	\rho_{\alpha}=1-\alpha \lambda_{n}>1-\alpha_{o} \lambda_{n}=\rho_{\alpha_{o}}	$\rho_{\alpha}=1-\alpha\lambda_{n}>1-\alpha_{o}\lambda_{n}=\rho_{\alpha_{o}}$	$0<\alpha<\alpha_o$ then $\rho_\alpha=1-\alpha\lambda_n>1-\alpha_o\lambda_n=\rho_{\alpha_o}$ and if $\alpha_o<\alpha<2/\lambda_1$ then
lyche-numerical-linear-algebra F03547	274	■1.000	\alpha_{o}<\alpha<2 / \lambda_{1}	$\alpha_{o}<\alpha<2/\lambda_{1}$	$0<\alpha<\alpha_o$ then $\rho_\alpha=1-\alpha\lambda_n>1-\alpha_o\lambda_n=\rho_{\alpha_o}$ and if $\alpha_o<\alpha<2/\lambda_1$ then
lyche-numerical-linear-algebra F03548	274	■0.998	\rho_{\alpha}=\alpha \lambda_{1}-1>\alpha_{o} \lambda_{1}-1=\rho_{\alpha_{o}}	$\rho_{\alpha}=\alpha\lambda_{1}-1>\alpha_{o}\lambda_{1}-1=\rho_{\alpha_{o}}$	$\rho_\alpha=\alpha\lambda_1-1>\alpha_o\lambda_1-1=\rho_{\alpha_o}$.
lyche-numerical-linear-algebra F03549	274	■0.964	\mathbf{R}	$\mathbf{R}$	**Corollary 12.1 (Rate of Convergence for the R Method)** *Suppose A is positive*

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra- F03550</td><td>274</td><td>■ 0.991</td><td>\lambda_{\text {max }}</td><td>λ_{max}</td><td>definite with largest and smallest eigenvalue λ_{max} and λ_{min}, respectively. Richard-</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03551</td><td>274</td><td>■ 0.999</td><td>0<\alpha<2 / \lambda_{\text {max }}</td><td>$0<\alpha<2/\lambda_{\text{max}}$</td><td>son's method $\boldsymbol{x}_{k+1} = (\boldsymbol{I}-\alpha\boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$ converges if and only if $0<\alpha<2/\lambda_{max}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03552</td><td>274</td><td>■ 1.000</td><td>\alpha=\alpha_{o}:=\frac{2}{\lambda_{\text {max }}+ \lambda_{\text {min }}}</td><td>$\alpha=\alpha_{o}:=\frac{2}{\lambda_{\text{max}}+\lambda_{\text{min}}}$</td><td>With $\alpha=\alpha_o := \frac{2}{\lambda_{max}+\lambda_{min}}$ we have the error estimate</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03553</td><td>274</td><td>■ 0.960</td><td>\kappa:=\lambda_{\text {max }} / \lambda_{\text {min }}</td><td>$\kappa:=\lambda_{\text{max}}/\lambda_{\text{min}}$</td><td>where $\kappa := \lambda_{max}/\lambda_{min}$ is the spectral condition number of $\boldsymbol{A}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03554</td><td>274</td><td>■ 0.988</td><td>\left\|\|_{2}\right.</td><td>$\|\|_{2}$</td><td>Proof The spectral norm $\| \|_2$ is consistent and therefore $\|\boldsymbol{x}_k - \boldsymbol{x}\|_2 \le \|\boldsymbol{I} -$</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03555</td><td>274</td><td>■ 0.988</td><td>\| \boldsymbol{x}_{k}-\boldsymbol{x}\left\|_{2} \leq \right\| \boldsymbol{I}-</td><td>$\|\boldsymbol{x}_{k}-\boldsymbol{x}\|_{2} \leq \| \boldsymbol{I}-$</td><td>Proof The spectral norm $\| \|_2$ is consistent and therefore $\|\boldsymbol{x}_k - \boldsymbol{x}\|_2 \le \|\boldsymbol{I} -$</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03556</td><td>274</td><td>■ 0.994</td><td>\alpha_{o} \boldsymbol{A}\left\|_{2}^{k}\right\| \boldsymbol{x}_{0}-\boldsymbol{x} \|_{2}</td><td>$\alpha_{o} \boldsymbol{A}\|_{2}^{k}\| \boldsymbol{x}_{0}-\boldsymbol{x} \|_{2}$</td><td>$\alpha_o \boldsymbol{A}\|_2^k\|\boldsymbol{x}_0 - \boldsymbol{x}\|_2$. But for a positive definite matrix the spectral norm is equal to</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03557</td><td>275</td><td>■ 1.000</td><td>\omega \in(0,2)</td><td>$\omega \in(0,2)$</td><td>The condition $\omega \in (0,2)$ is necessary for convergence of the SOR method.</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03558</td><td>275</td><td>■ 1.000</td><td>\boldsymbol{D} \boldsymbol{x}_{k+1}=\omega \left(\boldsymbol{A}_{L} \boldsymbol{x}_{k+1}+ \boldsymbol{A}_{R} \boldsymbol{x}_{k}+\boldsymbol{b} \right)+(1-\omega) \boldsymbol{D} \boldsymbol{x}_{k}</td><td>$\boldsymbol{D} \boldsymbol{x}_{k+1}=\omega\left(\boldsymbol{A}_{L} \boldsymbol{x}_{k+1}+\boldsymbol{A}_{R} \boldsymbol{x}_{k}+\boldsymbol{b}\right)+(1-\omega) \boldsymbol{D} \boldsymbol{x}_{k}$</td><td>Proof We have (cf. (12.13)) $\boldsymbol{D}\boldsymbol{x}_{k+1} = \omega(\boldsymbol{A}_L\boldsymbol{x}_{k+1}+\boldsymbol{A}_R\boldsymbol{x}_k+\boldsymbol{b})+(1-\omega)\boldsymbol{D}\boldsymbol{x}_k$</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03559</td><td>275</td><td>■ 1.000</td><td>\boldsymbol{x}_{k+1}=\omega\left(\boldsymbol{L} \boldsymbol{x}_{k+1}+\boldsymbol{R} \boldsymbol{x}_{k}+ \boldsymbol{D}^{-1} \boldsymbol{b}\right)+(1-\omega) \boldsymbol{x}_{k}</td><td>$\boldsymbol{x}_{k+1}=\omega\left(\boldsymbol{L} \boldsymbol{x}_{k+1}+\boldsymbol{R} \boldsymbol{x}_{k}+\boldsymbol{D}^{-1} \boldsymbol{b}\right)+(1-\omega) \boldsymbol{x}_{k}$</td><td>or $\boldsymbol{x}_{k+1} = \omega(\boldsymbol{L}\boldsymbol{x}_{k+1}+\boldsymbol{R}\boldsymbol{x}_k+\boldsymbol{D}^{-1}\boldsymbol{b})+(1-\omega)\boldsymbol{x}_k$, where $\boldsymbol{L} := \boldsymbol{D}^{-1}\boldsymbol{A}_L$ and</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03560</td><td>275</td><td>■ 1.000</td><td>\boldsymbol{L}:=\boldsymbol{D}^{-1} \boldsymbol{A}_{ \boldsymbol{L}}</td><td>$\boldsymbol{L}:=\boldsymbol{D}^{-1} \boldsymbol{A}_{\boldsymbol{L}}$</td><td>or $\boldsymbol{x}_{k+1} = \omega(\boldsymbol{L}\boldsymbol{x}_{k+1}+\boldsymbol{R}\boldsymbol{x}_k+\boldsymbol{D}^{-1}\boldsymbol{b})+(1-\omega)\boldsymbol{x}_k$, where $\boldsymbol{L} := \boldsymbol{D}^{-1}\boldsymbol{A}_L$ and</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03561</td><td>275</td><td>■ 1.000</td><td>\boldsymbol{R}:=\boldsymbol{D}^{-1} \boldsymbol{A}_{ \boldsymbol{R}}</td><td>$\boldsymbol{R}:=\boldsymbol{D}^{-1} \boldsymbol{A}_{\boldsymbol{R}}$</td><td>$\boldsymbol{R} := \boldsymbol{D}^{-1}\boldsymbol{A}_{\boldsymbol{R}}$. Thus $(\boldsymbol{I}-\omega\boldsymbol{L})\boldsymbol{x}_{k+1} = (\omega\boldsymbol{R}+(1-\omega)\boldsymbol{I})\boldsymbol{x}_k + \omega\boldsymbol{D}^{-1}\boldsymbol{b}$ so the</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03562</td><td>275</td><td>■ 1.000</td><td>(\boldsymbol{I}-\omega \boldsymbol{L}) \boldsymbol{x}_{k+1}=(\omega \boldsymbol{R}+(1- \omega) \boldsymbol{I}) \boldsymbol{x}_{k}+\omega \boldsymbol{D}^{-1} \boldsymbol{b}</td><td>$(\boldsymbol{I}-\omega \boldsymbol{L}) \boldsymbol{x}_{k+1}=(\omega \boldsymbol{R}+(1-\omega) \boldsymbol{I}) \boldsymbol{x}_{k}+\omega \boldsymbol{D}^{-1} \boldsymbol{b}$</td><td>$\boldsymbol{R} := \boldsymbol{D}^{-1}\boldsymbol{A}_{\boldsymbol{R}}$. Thus $(\boldsymbol{I}-\omega\boldsymbol{L})\boldsymbol{x}_{k+1} = (\omega\boldsymbol{R}+(1-\omega)\boldsymbol{I})\boldsymbol{x}_k + \omega\boldsymbol{D}^{-1}\boldsymbol{b}$ so the</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03563</td><td>275</td><td>■ 1.000</td><td>\boldsymbol{G}_{\omega}</td><td>$\boldsymbol{G}_{\omega}$</td><td>We next compute the determinant of $\boldsymbol{G}_\omega$. Since $\boldsymbol{I}-\omega\boldsymbol{L}$ is lower triangular with</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03564</td><td>275</td><td>■ 1.000</td><td>\boldsymbol{I}-\omega \boldsymbol{L}</td><td>$\boldsymbol{I}-\omega \boldsymbol{L}$</td><td>We next compute the determinant of $\boldsymbol{G}_\omega$. Since $\boldsymbol{I}-\omega\boldsymbol{L}$ is lower triangular with</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03565</td><td>275</td><td>■ 1.000</td><td>\omega \boldsymbol{R}+(1-\omega) \boldsymbol{I}</td><td>$\omega \boldsymbol{R}+(1-\omega) \boldsymbol{I}$</td><td>the determinant of this matrix is equal to one. The matrix $\omega\boldsymbol{R}+(1-\omega)\boldsymbol{I}$ is upper</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03566</td><td>275</td><td>■ 1.000</td><td>1-\omega</td><td>$1-\omega$</td><td>triangular with $1-\omega$ on the diagonal and therefore its determinant equals $(1-\omega)^n$.</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03567</td><td>275</td><td>■ 1.000</td><td>(1-\omega)^{n}</td><td>$(1-\omega)^{n}$</td><td>triangular with $1-\omega$ on the diagonal and therefore its determinant equals $(1-\omega)^n$.</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03568</td><td>275</td><td>■ 0.998</td><td>\operatorname{det}\left(\boldsymbol{G}_{\omega} \right)=(1-\omega)^{n}</td><td>$\operatorname{det}\left(\boldsymbol{G}_{\omega}\right)=(1-\omega)^{n}$</td><td>It follows that $\det(\boldsymbol{G}_\omega) = (1-\omega)^n$. Since the determinant of a matrix equals the</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03569</td><td>275</td><td>■ 1.000</td><td>|\lambda| \geq|1-\omega|</td><td>$|\lambda| \geq|1-\omega|$</td><td>product of its eigenvalues we must have $|\lambda| \ge |1-\omega|$ for at least one eigenvalue λ</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03570</td><td>275</td><td>■ 1.000</td><td>\rho\left(\boldsymbol{G}_{\omega}\right) \geq|\omega-1|</td><td>$\rho\left(\boldsymbol{G}_{\omega}\right) \geq|\omega-1|$</td><td>of $\boldsymbol{G}_\omega$ and we conclude that $\rho(\boldsymbol{G}_\omega) \ge |\omega-1|$. But then $\rho(\boldsymbol{G}_\omega) \ge 1$ if ω is not in</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03571</td><td>275</td><td>■ 1.000</td><td>\rho\left(\boldsymbol{G}_{\omega}\right) \geq 1</td><td>$\rho\left(\boldsymbol{G}_{\omega}\right) \geq 1$</td><td>of $\boldsymbol{G}_\omega$ and we conclude that $\rho(\boldsymbol{G}_\omega) \ge |\omega-1|$. But then $\rho(\boldsymbol{G}_\omega) \ge 1$ if ω is not in</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03572</td><td>275</td><td>■ 1.000</td><td>|\lambda|<1</td><td>$|\lambda|<1$</td><td>to show that $|\lambda| < 1$. The following identity will be shown below:</td></tr>
<tr><td>lyche-numerical-linear-algebra- F03573</td><td>275</td><td>■ 1.000</td><td>\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}</td><td>$\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}$</td><td>where $\boldsymbol{D} := \operatorname{diag}(a_{11},\dots,a_{nn})$. Now $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}$ and $\boldsymbol{x}^*\boldsymbol{D}\boldsymbol{x}$ are positive for all nonzero</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F03574	275	■ 1.000	`\boldsymbol{x}^{*} \boldsymbol{D} \boldsymbol{x}`	$\boldsymbol{x}^{*} \boldsymbol{D} \boldsymbol{x}$	where $\boldsymbol{D} := \operatorname{diag}(a_{11}, \ldots, a_{nn})$. Now $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}$ and $\boldsymbol{x}^*\boldsymbol{D}\boldsymbol{x}$ are positive for all nonzero
lyche-numerical-linear-algebra_F03575	275	■ 1.000	`x \in \mathbb{C}^{n}`	$x \in \mathbb{C}^{n}$	$x \in \mathbb{C}^n$ since a positive definite matrix has positive diagonal elements $a_{ii} =$
lyche-numerical-linear-algebra_F03576	275	■ 1.000	`a_{i i}=`	$a_{i i}=$	$x \in \mathbb{C}^n$ since a positive definite matrix has positive diagonal elements $a_{ii} =$
lyche-numerical-linear-algebra_F03577	275	■ 1.000	`\boldsymbol{e}_{i}^{T} \boldsymbol{A} \boldsymbol{e}_{i}>0`	$\boldsymbol{e}_{i}^{T} \boldsymbol{A} \boldsymbol{e}_{i}>0$	$\boldsymbol{e}_i^T\boldsymbol{A}\boldsymbol{e}_i > 0$. It follows that the left hand side of (12.22) is nonnegative and then
lyche-numerical-linear-algebra_F03578	275	■ 1.000	`|\lambda| \leq 1`	$\lvert\lambda\rvert \leq 1$	the right hand side must be nonnegative as well. This implies $\lvert\lambda\rvert \le 1$. If $\lvert\lambda\rvert = 1$
lyche-numerical-linear-algebra_F03579	276	■ 1.000	`\boldsymbol{G}_{\omega} \boldsymbol{x}=\lambda \boldsymbol{x}`	$\boldsymbol{G}_{\omega} \boldsymbol{x}=\lambda \boldsymbol{x}$	and the eigenpair equation $\boldsymbol{G}_\omega \boldsymbol{x} = \lambda \boldsymbol{x}$ can be written $\boldsymbol{x} - (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_{\boldsymbol{L}})^{-1}\boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{x}$
lyche-numerical-linear-algebra_F03580	276	■ 1.000	`\boldsymbol{x}-\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}\right)^{-1} \boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{x}`	$\boldsymbol{x}-\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}\right)^{-1} \boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{x}$	and the eigenpair equation $\boldsymbol{G}_\omega \boldsymbol{x} = \lambda \boldsymbol{x}$ can be written $\boldsymbol{x} - (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_{\boldsymbol{L}})^{-1}\boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{x}$
lyche-numerical-linear-algebra_F03581	276	■ 0.976	`\boldsymbol{A} \boldsymbol{x} \neq \mathbf{0}`	$\boldsymbol{A} \boldsymbol{x} \neq \mathbf{0}$	Now $\boldsymbol{A}\boldsymbol{x} \neq \mathbf{0}$ implies that $\lambda \neq 1$.
lyche-numerical-linear-algebra_F03582	276	■ 0.976	`\lambda \neq 1`	$\lambda \neq 1$	Now $\boldsymbol{A}\boldsymbol{x} \neq \mathbf{0}$ implies that $\lambda \neq 1$.
lyche-numerical-linear-algebra_F03583	276	■ 1.000	`(\boldsymbol{A} \boldsymbol{x})^{*} \boldsymbol{y}=\boldsymbol{y}^{*}\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}\right)^{*} \boldsymbol{y}=`	$(\boldsymbol{A} \boldsymbol{x})^{*} \boldsymbol{y}=\boldsymbol{y}^{*}\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}\right)^{*} \boldsymbol{y}=$	Again using (12.23), (12.24) and adding $(\boldsymbol{A}\boldsymbol{x})^*\boldsymbol{y} = \boldsymbol{y}^*(\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_{\boldsymbol{L}})^*\boldsymbol{y} =$
lyche-numerical-linear-algebra_F03584	276	■ 1.000	`\boldsymbol{y}^{*}\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{R}}\right) \boldsymbol{y}`	$\boldsymbol{y}^{*}\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{R}}\right) \boldsymbol{y}$	$\boldsymbol{y}^*(\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_{\boldsymbol{R}})\boldsymbol{y}$ and $\boldsymbol{y}^*(\lambda\boldsymbol{A}\boldsymbol{x}) = \boldsymbol{y}^*\boldsymbol{E}\boldsymbol{y} = \boldsymbol{y}^*(\omega^{-1}\boldsymbol{D} + \boldsymbol{A}_{\boldsymbol{R}} - \boldsymbol{D})\boldsymbol{y}$ we find
lyche-numerical-linear-algebra_F03585	276	■ 1.000	`\boldsymbol{y}^{*}(\lambda \boldsymbol{A} \boldsymbol{x})=\boldsymbol{y}^{*} \boldsymbol{E} \boldsymbol{y}=\boldsymbol{y}^{*}\left(\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_{\boldsymbol{R}}-\boldsymbol{D}\right) \boldsymbol{y}`	$\boldsymbol{y}^{*}(\lambda \boldsymbol{A} \boldsymbol{x})=\boldsymbol{y}^{*} \boldsymbol{E} \boldsymbol{y}=\boldsymbol{y}^{*}\left(\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_{\boldsymbol{R}}-\boldsymbol{D}\right) \boldsymbol{y}$	$\boldsymbol{y}^*(\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_{\boldsymbol{R}})\boldsymbol{y}$ and $\boldsymbol{y}^*(\lambda\boldsymbol{A}\boldsymbol{x}) = \boldsymbol{y}^*\boldsymbol{E}\boldsymbol{y} = \boldsymbol{y}^*(\omega^{-1}\boldsymbol{D} + \boldsymbol{A}_{\boldsymbol{R}} - \boldsymbol{D})\boldsymbol{y}$ we find
lyche-numerical-linear-algebra_F03586	276	■ 1.000	`(\boldsymbol{A} \boldsymbol{x})^{*}=\boldsymbol{x}^{*} \boldsymbol{A}^{*}=\boldsymbol{x}^{*} \boldsymbol{A}, \boldsymbol{y}:=(1-\lambda) \boldsymbol{x}`	$(\boldsymbol{A} \boldsymbol{x})^{*}=\boldsymbol{x}^{*} \boldsymbol{A}^{*}=\boldsymbol{x}^{*} \boldsymbol{A}, \boldsymbol{y}:=(1-\lambda) \boldsymbol{x}$	Since $(\boldsymbol{A}\boldsymbol{x})^* = \boldsymbol{x}^*\boldsymbol{A}^* = \boldsymbol{x}^*\boldsymbol{A}$, $\boldsymbol{y} := (1-\lambda)\boldsymbol{x}$ and $\boldsymbol{y}^* = (1-\bar{\lambda})\boldsymbol{x}^*$ this also equals
lyche-numerical-linear-algebra_F03587	276	■ 1.000	`\boldsymbol{y}^{*}=(1-\bar{\lambda}) \boldsymbol{x}^{*}`	$\boldsymbol{y}^{*}=(1-\bar{\lambda}) \boldsymbol{x}^{*}$	Since $(\boldsymbol{A}\boldsymbol{x})^* = \boldsymbol{x}^*\boldsymbol{A}^* = \boldsymbol{x}^*\boldsymbol{A}$, $\boldsymbol{y} := (1-\lambda)\boldsymbol{x}$ and $\boldsymbol{y}^* = (1-\bar{\lambda})\boldsymbol{x}^*$ this also equals
lyche-numerical-linear-algebra_F03588	276	■ 0.997	`\boldsymbol{G}_{J}=\boldsymbol{I}-\boldsymbol{D}^{-1} \boldsymbol{A}=\boldsymbol{I}-\boldsymbol{A} / 4`	$\boldsymbol{G}_{J}=\boldsymbol{I}-\boldsymbol{D}^{-1} \boldsymbol{A}=\boldsymbol{I}-\boldsymbol{A} / 4$	Consider first the J method. The matrix $\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1}\boldsymbol{A} = \boldsymbol{I} - \boldsymbol{A}/4$ has eigenvalues
lyche-numerical-linear-algebra_F03589	277	■ 1.000	`\rho\left(\boldsymbol{G}_{J}\right)=\cos (\pi h)<1`	$\rho\left(\boldsymbol{G}_{J}\right)=\cos (\pi h)<1$	It follows that $\rho(\boldsymbol{G}_J) = \cos(\pi h) < 1$. Since $\boldsymbol{G}_J$ is symmetric it is normal, and the
lyche-numerical-linear-algebra_F03590	277	■ 1.000	`\boldsymbol{G}_{J}`	$\boldsymbol{G}_{J}$	It follows that $\rho(\boldsymbol{G}_J) = \cos(\pi h) < 1$. Since $\boldsymbol{G}_J$ is symmetric it is normal, and the
lyche-numerical-linear-algebra_F03591	277	■ 0.932	`\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\alpha \boldsymbol{r}_{k}`	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\alpha \boldsymbol{r}_{k}$	The R method given by $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha\boldsymbol{r}_k$ with $\alpha = 2/(\lambda_{max} + \lambda_{min}) = 1/4$
lyche-numerical-linear-algebra_F03592	277	■ 0.932	`\alpha=2 /\left(\lambda_{\text {max }}+\lambda_{\text {min }}\right)=1 / 4`	$\alpha=2 /\left(\lambda_{\text {max }}+\lambda_{\text {min }}\right)=1 / 4$	The R method given by $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha\boldsymbol{r}_k$ with $\alpha = 2/(\lambda_{max} + \lambda_{min}) = 1/4$
lyche-numerical-linear-algebra_F03593	277	■ 1.000	`\kappa`	κ	from Corollary 12.1 with κ given by (10.20).
lyche-numerical-linear-algebra_F03594	277	■ 1.000	`\rho\left(\boldsymbol{G}_{\omega}\right)`	$\rho\left(\boldsymbol{G}_{\omega}\right)$	For the SOR method it is possible to explicitly determine $\rho(\boldsymbol{G}_\omega)$ for any $\omega \in$
lyche-numerical-linear-algebra_F03595	277	■ 1.000	`\omega \in`	$\omega \in$	For the SOR method it is possible to explicitly determine $\rho(\boldsymbol{G}_\omega)$ for any $\omega \in$
lyche-numerical-linear-algebra_F03596	277	■ 1.000	`\beta:=\rho\left(\boldsymbol{G}_{J}\right)=\cos (\pi h)`	$\beta:=\rho\left(\boldsymbol{G}_{J}\right)=\cos (\pi h)$	*where* $\beta := \rho(\boldsymbol{G}_J) = \cos(\pi h)$ *and*
lyche-numerical-linear-algebra_F03597	277	■ 1.000	`\omega^{*}`	ω^{*}	decreases monotonically to the minimum ω^*. Then it increases linearly to the value

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F03598	277	1.000	`\omega=2`	$\omega=2$	one for $\omega = 2$. We call ω^* the **optimal relaxation parameter**.
lyche-numerical-linear-algebra_F03599	277	1.000	`\beta=\cos (\pi h)`	$\beta=\cos (\pi h)$	For the discrete Poisson problem we have $\beta = \cos(\pi h)$ and it follows from
lyche-numerical-linear-algebra_F03600	277	0.998	`\rho\left(\boldsymbol{G}_{1}\right)=\beta^{2}=\rho\left(\boldsymbol{G}_{J}\right)^{2}=\cos ^{2}(\pi h)`	$\rho\left(\boldsymbol{G}_{1}\right)=\beta^{2}=\rho\left(\boldsymbol{G}_{J}\right)^{2}=\cos ^{2}(\pi h)$	Letting $\omega = 1$ in (12.27) we find $\rho(\boldsymbol{G}_1) = \beta^2 = \rho(\boldsymbol{G}_J)^2 = \cos^2(\pi h)$ for the GS
lyche-numerical-linear-algebra_F03601	277	0.999	`\rho\left(\boldsymbol{G}_{J}\right), \rho\left(\boldsymbol{G}_{1}\right)`	$\rho\left(\boldsymbol{G}_{J}\right), \rho\left(\boldsymbol{G}_{1}\right)$	The values of $\rho(\boldsymbol{G}_J)$, $\rho(\boldsymbol{G}_1)$, and $\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1$ are shown in Table 12.2 for
lyche-numerical-linear-algebra_F03602	277	0.999	`\rho\left(\boldsymbol{G}_{\omega^{*}}\right)=\omega^{*}-1`	$\rho\left(\boldsymbol{G}_{\omega^{*}}\right)=\omega^{*}-1$	The values of $\rho(\boldsymbol{G}_J)$, $\rho(\boldsymbol{G}_1)$, and $\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1$ are shown in Table 12.2 for
lyche-numerical-linear-algebra_F03603	277	1.000	`\rho(\boldsymbol{G})^{k_{n}} \leq`	$\rho(\boldsymbol{G})^{k_{n}} \leq$	$n = 100$ and $n = 2500$. We also show the smallest integer k_n such that $\rho(\boldsymbol{G})^{k_n} \leq$
lyche-numerical-linear-algebra_F03604	278	1.000	`\mathrm{n}=100`	$\mathrm{n}=100$	— n=100 n = 2500 k_{100} k_{2500} J 0.959493 0.998103 446 9703 GS 0.920627 0.99621 223 4852 SOR 0.56039 0.88402 32 150
lyche-numerical-linear-algebra_F03605	278	1.000	`\mathrm{n}=2500`	$\mathrm{n}=2500$	— n=100 n = 2500 k_{100} k_{2500} J 0.959493 0.998103 446 9703 GS 0.920627 0.99621 223 4852 SOR 0.56039 0.88402 32 150
lyche-numerical-linear-algebra_F03606	278	1.000	`\boldsymbol{x}_{k}-\boldsymbol{x}=\boldsymbol{G}^{k}\left(\boldsymbol{x}_{0}-\boldsymbol{x}\right)`	$\boldsymbol{x}_{k}-\boldsymbol{x}=\boldsymbol{G}^{k}\left(\boldsymbol{x}_{0}-\boldsymbol{x}\right)$	know how fast the iterative method converges. Recall that $\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k(\boldsymbol{x}_0 - \boldsymbol{x})$.
lyche-numerical-linear-algebra_F03607	278	1.000	`\lim _{k \rightarrow \infty}\left|\boldsymbol{G}^{k}\right|^{1 / k}=`	$\lim _{k \rightarrow \infty}\left\|\boldsymbol{G}^{k}\right\|^{1 / k}=$	For the last formula we apply Theorem 12.13 which says that $\lim_{k\to\infty}\|\boldsymbol{G}^k\|^{1/k} =$
lyche-numerical-linear-algebra_F03608	278	1.000	`\rho(\boldsymbol{G})`	$\rho(\boldsymbol{G})$	$\rho(\boldsymbol{G})$. For Jacobi's method and the spectral norm we have $\|\boldsymbol{G}_J^k\|_2 = \rho(\boldsymbol{G}_J)^k$ (cf.
lyche-numerical-linear-algebra_F03609	278	1.000	`\left|\boldsymbol{G}_{J}^{k}\right|_{2}=\rho\left(\boldsymbol{G}_{J}\right)^{k}`	$\left\|\boldsymbol{G}_{J}^{k}\right\|_{2}=\rho\left(\boldsymbol{G}_{J}\right)^{k}$	$\rho(\boldsymbol{G})$. For Jacobi's method and the spectral norm we have $\|\boldsymbol{G}_J^k\|_2 = \rho(\boldsymbol{G}_J)^k$ (cf.
lyche-numerical-linear-algebra_F03610	278	1.000	`\rho(\boldsymbol{G})=1-\eta`	$\rho(\boldsymbol{G})=1-\eta$	**Lemma 12.1 (Number of Iterations)** *Suppose* $\rho(\boldsymbol{G}) = 1 - \eta$ *for some* $0 < \eta < 1$
lyche-numerical-linear-algebra_F03611	278	1.000	`0<\eta<1`	$0<\eta<1$	**Lemma 12.1 (Number of Iterations)** *Suppose* $\rho(\boldsymbol{G}) = 1 - \eta$ *for some* $0 < \eta < 1$
lyche-numerical-linear-algebra_F03612	278	1.000	`s \in \mathbb{N}`	$s \in \mathbb{N}$	*and let* $s \in \mathbb{N}$. *Then*
lyche-numerical-linear-algebra_F03613	278	1.000	`\rho(\boldsymbol{G})^{k} \leq 10^{-s}`	$\rho(\boldsymbol{G})^{k} \leq 10^{-s}$	*is an estimate for the smallest number of iterations k so that* $\rho(\boldsymbol{G})^k \leq 10^{-s}$.
lyche-numerical-linear-algebra_F03614	279	1.000	`\tilde{k}`	$\tilde{k}$	***Proof*** The estimate $\tilde{k}$ is an approximate solution of the equation $\rho(\boldsymbol{G})^k = 10^{-s}$.
lyche-numerical-linear-algebra_F03615	279	1.000	`\rho(\boldsymbol{G})^{k}=10^{-s}`	$\rho(\boldsymbol{G})^{k}=10^{-s}$	***Proof*** The estimate $\tilde{k}$ is an approximate solution of the equation $\rho(\boldsymbol{G})^k = 10^{-s}$.
lyche-numerical-linear-algebra_F03616	279	1.000	`-\log (1-\eta) \approx \eta`	$-\log (1-\eta) \approx \eta$	Thus, since $-\log(1 - \eta) \approx \eta$ when η is small
lyche-numerical-linear-algebra_F03617	279	1.000	`\eta`	η	Thus, since $-\log(1 - \eta) \approx \eta$ when η is small
lyche-numerical-linear-algebra_F03618	279	0.996	`\rho\left(\boldsymbol{G}_{J}\right)=\cos (\pi h)=1-\eta, \eta=1-\cos (\pi h)=\frac{1}{2} \pi^{2} h^{2}+O \left(h^{4}\right)=`	$\rho\left(\boldsymbol{G}_{J}\right)=\cos (\pi h)=1-\eta, \eta=1-\cos (\pi h)=\frac{1}{2} \pi^{2} h^{2}+O \left(h^{4}\right)=$	• R and J: $\rho(\boldsymbol{G}_J) = \cos(\pi h) = 1 - \eta$, $\eta = 1 - \cos(\pi h) = \frac{1}{2}\pi^2 h^2 + O(h^4) =$
lyche-numerical-linear-algebra_F03619	279	1.000	`\frac{\pi^{2}}{2} / n+O\left(n^{-2}\right)`	$\frac{\pi^{2}}{2} / n+O\left(n^{-2}\right)$	$\frac{\pi^2}{2}/n + O(n^{-2})$. Thus,
lyche-numerical-linear-algebra_F03620	279	0.512	`\operatorname{GS}: \rho\left(\boldsymbol{G}_{1}\right)=\cos ^{2}(\pi h)=1-\eta, \eta=1-\cos ^{2}(\pi h)=\sin ^{2} \pi h=\pi^{2} h^{2}+`	$\operatorname{GS}: \rho\left(\boldsymbol{G}_{1}\right)=\cos ^{2}(\pi h)=1-\eta, \eta=1-\cos ^{2}(\pi h)=\sin ^{2} \pi h=\pi^{2} h^{2}+$	• GS: $\rho(\boldsymbol{G}_1) = \cos^2(\pi h) = 1 - \eta$, $\eta = 1 - \cos^2(\pi h) = \sin^2 \pi h = \pi^2 h^2 +$

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra F03621</td><td>279</td><td>1.000</td><td>O\left(h^{4}\right)=\pi^{2} / n+O\left(n^{-2}\right)</td><td>$O\left(h^{4}\right)=\pi^{2} / n+O\left(n^{-2}\right)$</td><td>$O(h^4) = \pi^2/n + O(n^{-2})$. Thus,</td></tr>
<tr><td>lyche-numerical-linear-algebra F03622</td><td>279</td><td>0.906</td><td>\rho\left(\boldsymbol{G}_{\omega^{*}}\right)=\frac{1-\sin (\pi h)}{1+\sin (\pi h)}=1-2 \pi h+O\left(h^{2}\right)</td><td>$\rho\left(\boldsymbol{G}_{\omega^{*}}\right)=\frac{1-\sin (\pi h)}{1+\sin (\pi h)}=1-2 \pi h+O\left(h^{2}\right)$</td><td>• SOR: $\rho(\boldsymbol{G}_{\omega^*}) = \frac{1-\sin(\pi h)}{1+\sin(\pi h)} = 1 - 2\pi h + O(h^2)$. Thus,</td></tr>
<tr><td>lyche-numerical-linear-algebra F03623</td><td>279</td><td>1.000</td><td>\boldsymbol{G}^{k}</td><td>$\boldsymbol{G}^{k}$</td><td>1. The convergence depends on the behavior of the powers $\boldsymbol{G}^k$ as k increases.</td></tr>
<tr><td>lyche-numerical-linear-algebra F03624</td><td>279</td><td>0.998</td><td>\lim _{k \rightarrow \infty}\left\|\boldsymbol{G}^{k} \right\|^{1 / k}=\rho(\boldsymbol{G})</td><td>$\lim _{k \rightarrow \infty}\left\|\boldsymbol{G}^{k} \right\|^{1 / k}=\rho(\boldsymbol{G})$</td><td>2. The convergence $\lim_{k\to\infty}\|\boldsymbol{G}^k\|^{1/k} = \rho(\boldsymbol{G})$ can be quite slow (cf. Exer-</td></tr>
<tr><td>lyche-numerical-linear-algebra F03625</td><td>279</td><td>1.000</td><td>\left\|\boldsymbol{x}_{k+1}-\boldsymbol{x}_{k}\right\|</td><td>$\left\|\boldsymbol{x}_{k+1}-\boldsymbol{x}_{k}\right\|$</td><td>One possibility is to choose a vector norm, keep track of $\|\boldsymbol{x}_{k+1} - \boldsymbol{x}_k\|$, and stop</td></tr>
<tr><td>lyche-numerical-linear-algebra F03626</td><td>280</td><td>1.000</td><td>\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\|</td><td>$\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\|$</td><td>when this number is sufficiently small. The following result indicates that $\|\boldsymbol{x}_k - \boldsymbol{x}\|$</td></tr>
<tr><td>lyche-numerical-linear-algebra F03627</td><td>280</td><td>1.000</td><td>\|\boldsymbol{G}\|</td><td>$\|\boldsymbol{G}\|$</td><td>can be quite large if $\|\boldsymbol{G}\|$ is close to one.</td></tr>
<tr><td>lyche-numerical-linear-algebra F03628</td><td>280</td><td>1.000</td><td>\boldsymbol{x}_{k}=\boldsymbol{G} \boldsymbol{x}_{k-1}+ \boldsymbol{c}, \boldsymbol{x}=\boldsymbol{G} \boldsymbol{x}+\boldsymbol{c}</td><td>$\boldsymbol{x}_{k}=\boldsymbol{G} \boldsymbol{x}_{k-1}+ \boldsymbol{c}, \boldsymbol{x}=\boldsymbol{G} \boldsymbol{x}+\boldsymbol{c}$</td><td>Lemma 12.2 (Be Careful When Stopping) If $\boldsymbol{x}_k = \boldsymbol{G}\boldsymbol{x}_{k-1} + \boldsymbol{c}$, $\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$ and</td></tr>
<tr><td>lyche-numerical-linear-algebra F03629</td><td>280</td><td>0.998</td><td>(1-\|\boldsymbol{G}\|)\left\|\boldsymbol{x}_{k}- \boldsymbol{x}\right\| \leq\|\boldsymbol{G}\|\left\| \boldsymbol{x}_{k-1}-\boldsymbol{x}_{k}\right\|</td><td>$(1-\|\boldsymbol{G}\|)\left\|\boldsymbol{x}_{k}- \boldsymbol{x}\right\| \leq\|\boldsymbol{G}\|\left\| \boldsymbol{x}_{k-1}-\boldsymbol{x}_{k}\right\|$</td><td>Thus $(1 - \|\boldsymbol{G}\|)\|\boldsymbol{x}_k - \boldsymbol{x}\| \le \|\boldsymbol{G}\|\|\boldsymbol{x}_{k-1} - \boldsymbol{x}_k\|$ which implies (12.32).</td></tr>
<tr><td>lyche-numerical-linear-algebra F03630</td><td>280</td><td>0.997</td><td>\boldsymbol{M} \boldsymbol{x}_{k+1}=\boldsymbol{M} \boldsymbol{x}_{k}-\boldsymbol{A} \boldsymbol{x}_{k}+ \boldsymbol{b}</td><td>$\boldsymbol{M} \boldsymbol{x}_{k+1}=\boldsymbol{M} \boldsymbol{x}_{k}-\boldsymbol{A} \boldsymbol{x}_{k}+ \boldsymbol{b}$</td><td>splitting matrix of the method then by (12.9) we have $\boldsymbol{M}\boldsymbol{x}_{k+1} = \boldsymbol{M}\boldsymbol{x}_k - \boldsymbol{A}\boldsymbol{x}_k + \boldsymbol{b}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra F03631</td><td>280</td><td>1.000</td><td>\left\{\boldsymbol{A}^{k}\right\}</td><td>$\left\{\boldsymbol{A}^{k}\right\}$</td><td>sequence $\{\boldsymbol{A}^k\}$ of powers of $\boldsymbol{A}$. We want to know when this sequence converges to</td></tr>
<tr><td>lyche-numerical-linear-algebra F03632</td><td>281</td><td>1.000</td><td>\lim _{k \rightarrow \infty} \boldsymbol{A}^{k}= \mathbf{0}</td><td>$\lim _{k \rightarrow \infty} \boldsymbol{A}^{k}= \mathbf{0}$</td><td>Theorem 12.10 (When Is $\lim_{k\to\infty} \boldsymbol{A}^k = \mathbf{0}$?) For any $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ we have</td></tr>
<tr><td>lyche-numerical-linear-algebra F03633</td><td>281</td><td>1.000</td><td>\rho(\boldsymbol{A})</td><td>$\rho(\boldsymbol{A})$</td><td>where $\rho(\boldsymbol{A})$ is the spectral radius of $\boldsymbol{A}$ given by (12.17).</td></tr>
<tr><td>lyche-numerical-linear-algebra F03634</td><td>281</td><td>0.913</td><td>\rho(\boldsymbol{A})<1</td><td>$\rho(\boldsymbol{A})<1$</td><td>Clearly $\rho(\boldsymbol{A}) < 1$ is a necessary condition for $\lim_{k\to\infty} \boldsymbol{A}^k = \mathbf{0}$. For if $(\lambda, \boldsymbol{x})$ is</td></tr>
<tr><td>lyche-numerical-linear-algebra F03635</td><td>281</td><td>1.000</td><td>|\lambda| \geq 1</td><td>$|\lambda| \geq 1$</td><td>an eigenpair of $\boldsymbol{A}$ with $|\lambda| \ge 1$ and $\|\boldsymbol{x}\|_2 = 1$ then $\boldsymbol{A}^k\boldsymbol{x} = \lambda^k\boldsymbol{x}$, and this implies</td></tr>
<tr><td>lyche-numerical-linear-algebra F03636</td><td>281</td><td>1.000</td><td>\boldsymbol{A}^{k} \boldsymbol{x}=\lambda^{k} \boldsymbol{x}</td><td>$\boldsymbol{A}^{k} \boldsymbol{x}=\lambda^{k} \boldsymbol{x}$</td><td>an eigenpair of $\boldsymbol{A}$ with $|\lambda| \ge 1$ and $\|\boldsymbol{x}\|_2 = 1$ then $\boldsymbol{A}^k\boldsymbol{x} = \lambda^k\boldsymbol{x}$, and this implies</td></tr>
<tr><td>lyche-numerical-linear-algebra F03637</td><td>281</td><td>1.000</td><td>\left\|\boldsymbol{A}^{k}\right\|_{2} \geq\left\| \boldsymbol{A}^{k} \boldsymbol{x}\right\|_{2}=\left\| \lambda^{k} \boldsymbol{x}\right\|_{2}=|\lambda|^{k}</td><td>$\left\|\boldsymbol{A}^{k}\right\|_{2} \geq\left\| \boldsymbol{A}^{k} \boldsymbol{x}\right\|_{2}=\left\| \lambda^{k} \boldsymbol{x}\right\|_{2}=|\lambda|^{k}$</td><td>$\|\boldsymbol{A}^k\|_2 \ge \|\boldsymbol{A}^k\boldsymbol{x}\|_2 = \|\lambda^k\boldsymbol{x}\|_2 = |\lambda|^k$, and it follows that $\boldsymbol{A}^k$ does not tend to zero.</td></tr>
<tr><td>lyche-numerical-linear-algebra F03638</td><td>281</td><td>1.000</td><td>\|\boldsymbol{A}\|<1</td><td>$\|\boldsymbol{A}\|<1$</td><td>norm on $\mathbb{C}^{n\times n}$ such that $\|\boldsymbol{A}\| < 1$ and then use Theorems 12.2 and 12.3.</td></tr>
<tr><td>lyche-numerical-linear-algebra F03639</td><td>281</td><td>1.000</td><td>\rho(\boldsymbol{A}) \leq</td><td>$\rho(\boldsymbol{A}) \leq$</td><td>matrix norm $\|\cdot\|$ that is consistent on $\mathbb{C}^{n\times n}$ and any $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ we have $\rho(\boldsymbol{A}) \le$</td></tr>
<tr><td>lyche-numerical-linear-algebra F03640</td><td>281</td><td>1.000</td><td>\|\boldsymbol{A}\|</td><td>$\|\boldsymbol{A}\|$</td><td>$\|\boldsymbol{A}\|$.</td></tr>
<tr><td>lyche-numerical-linear-algebra F03641</td><td>281</td><td>0.679</td><td>\boldsymbol{A},\| \|</td><td>$\boldsymbol{A},\| \|$</td><td>Proof Let $(\lambda, \boldsymbol{x})$ be an eigenpair for $\boldsymbol{A}$, $\| \ \|$ a consistent matrix norm on $\mathbb{C}^{n\times n}$</td></tr>
<tr><td>lyche-numerical-linear-algebra F03642</td><td>281</td><td>1.000</td><td>\boldsymbol{X}:=[\boldsymbol{x}, \ldots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}</td><td>$\boldsymbol{X}:=[\boldsymbol{x}, \ldots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}$</td><td>and define $\boldsymbol{X} := [\boldsymbol{x}, \ldots, \boldsymbol{x}] \in \mathbb{C}^{n\times n}$. Then $\lambda\boldsymbol{X} = \boldsymbol{A}\boldsymbol{X}$, which implies $|\lambda|\,\|\boldsymbol{X}\| =$</td></tr>
<tr><td>lyche-numerical-linear-algebra F03643</td><td>281</td><td>1.000</td><td>\lambda \boldsymbol{X}=\boldsymbol{A} \boldsymbol{X}</td><td>$\lambda \boldsymbol{X}=\boldsymbol{A} \boldsymbol{X}$</td><td>and define $\boldsymbol{X} := [\boldsymbol{x}, \ldots, \boldsymbol{x}] \in \mathbb{C}^{n\times n}$. Then $\lambda\boldsymbol{X} = \boldsymbol{A}\boldsymbol{X}$, which implies $|\lambda|\,\|\boldsymbol{X}\| =$</td></tr>
<tr><td>lyche-numerical-linear-algebra F03644</td><td>281</td><td>1.000</td><td>|\lambda|\|\boldsymbol{X}\|=</td><td>$|\lambda|\|\boldsymbol{X}\|=$</td><td>and define $\boldsymbol{X} := [\boldsymbol{x}, \ldots, \boldsymbol{x}] \in \mathbb{C}^{n\times n}$. Then $\lambda\boldsymbol{X} = \boldsymbol{A}\boldsymbol{X}$, which implies $|\lambda|\,\|\boldsymbol{X}\| =$</td></tr>
<tr><td>lyche-numerical-linear-algebra F03645</td><td>281</td><td>0.999</td><td>\|\lambda \boldsymbol{X}\|=\|\boldsymbol{A} \boldsymbol{X}\| \leq\|\boldsymbol{A}\|\|\boldsymbol{X}\|</td><td>$\|\lambda \boldsymbol{X}\|=\|\boldsymbol{A} \boldsymbol{X}\| \leq\|\boldsymbol{A}\|\|\boldsymbol{X}\|$</td><td>$\|\lambda\boldsymbol{X}\| = \|\boldsymbol{A}\boldsymbol{X}\| \le \|\boldsymbol{A}\|\,\|\boldsymbol{X}\|$. Since $\|\boldsymbol{X}\| \ne 0$ we obtain $|\lambda| \le \|\boldsymbol{A}\|$.</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-al F03646</td><td>281</td><td>0.999</td><td>\|\boldsymbol{X}\| \neq 0</td><td>$\|\boldsymbol{X}\| \neq 0$</td><td>$\|\lambda \boldsymbol{X}\| = \|\boldsymbol{A}\boldsymbol{X}\| \leq \|\boldsymbol{A}\|\,\|\boldsymbol{X}\|$. Since $\|\boldsymbol{X}\| \neq 0$ we obtain $|\lambda| \leq \|\boldsymbol{A}\|$.</td></tr>
<tr><td>lyche-numerical-linear-al F03647</td><td>281</td><td>0.999</td><td>|\lambda| \leq\|\boldsymbol{A}\|</td><td>$|\lambda| \leq \|\boldsymbol{A}\|$</td><td>$\|\lambda \boldsymbol{X}\| = \|\boldsymbol{A}\boldsymbol{X}\| \leq \|\boldsymbol{A}\|\,\|\boldsymbol{X}\|$. Since $\|\boldsymbol{X}\| \neq 0$ we obtain $|\lambda| \leq \|\boldsymbol{A}\|$.</td></tr>
<tr><td>lyche-numerical-linear-al F03648</td><td>281</td><td>1.000</td><td>\rho(\boldsymbol{A}) \leq\|\boldsymbol{A}\| \leq \rho(\boldsymbol{A})+\epsilon</td><td>$\rho(\boldsymbol{A}) \leq \|\boldsymbol{A}\| \leq \rho(\boldsymbol{A})+\epsilon$</td><td>that $\rho(\boldsymbol{A}) \leq \|\boldsymbol{A}\| \leq \rho(\boldsymbol{A})+\epsilon$.</td></tr>
<tr><td>lyche-numerical-linear-al F03649</td><td>281</td><td>1.000</td><td>\boldsymbol{R}=\left[r_{i j}\right]</td><td>$\boldsymbol{R}=\left[r_{i j}\right]$</td><td>there is a unitary matrix $\boldsymbol{U}$ and an upper triangular matrix $\boldsymbol{R} = [r_{ij}]$ such that</td></tr>
<tr><td>lyche-numerical-linear-al F03650</td><td>281</td><td>1.000</td><td>\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}= \boldsymbol{R}</td><td>$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}= \boldsymbol{R}$</td><td>$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \boldsymbol{R}$. For $t > 0$ we define $\boldsymbol{D}_t := \operatorname{diag}(t, t^2, \ldots, t^n) \in \mathbb{R}^{n\times n}$, and note that</td></tr>
<tr><td>lyche-numerical-linear-al F03651</td><td>281</td><td>1.000</td><td>\boldsymbol{D}_{t}:=\operatorname{diag}\left(t, t^{2}, \ldots, t^{n}\right) \in \mathbb{R}^{n \times n}</td><td>$\boldsymbol{D}_{t}:=\operatorname{diag}\left(t, t^{2}, \ldots, t^{n}\right) \in \mathbb{R}^{n \times n}$</td><td>$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \boldsymbol{R}$. For $t > 0$ we define $\boldsymbol{D}_t := \operatorname{diag}(t, t^2, \ldots, t^n) \in \mathbb{R}^{n\times n}$, and note that</td></tr>
<tr><td>lyche-numerical-linear-al F03652</td><td>281</td><td>0.718</td><td>(i, j)</td><td>(i, j)</td><td>the (i, j) element in $\boldsymbol{D}_t\boldsymbol{R}\boldsymbol{D}_t^{-1}$ is given by $t^{i-j}r_{ij}$ for all i, j. For $n = 3$</td></tr>
<tr><td>lyche-numerical-linear-al F03653</td><td>281</td><td>0.718</td><td>\boldsymbol{D}_{t} \boldsymbol{R} \boldsymbol{D}_{t}^{-1}</td><td>$\boldsymbol{D}_{t} \boldsymbol{R} \boldsymbol{D}_{t}^{-1}$</td><td>the (i, j) element in $\boldsymbol{D}_t\boldsymbol{R}\boldsymbol{D}_t^{-1}$ is given by $t^{i-j}r_{ij}$ for all i, j. For $n = 3$</td></tr>
<tr><td>lyche-numerical-linear-al F03654</td><td>281</td><td>0.718</td><td>t^{i-j} r_{i j}</td><td>$t^{i-j} r_{i j}$</td><td>the (i, j) element in $\boldsymbol{D}_t\boldsymbol{R}\boldsymbol{D}_t^{-1}$ is given by $t^{i-j}r_{ij}$ for all i, j. For $n = 3$</td></tr>
<tr><td>lyche-numerical-linear-al F03655</td><td>281</td><td>0.877</td><td>\|\boldsymbol{B}\|_{t}:=\left\|\boldsymbol{D}_{t} \boldsymbol{U}^{*} \boldsymbol{B} \boldsymbol{U} \boldsymbol{D}_{t}^{-1}\right\|_{1}</td><td>$\|\boldsymbol{B}\|_{t}:=\left\|\boldsymbol{D}_{t} \boldsymbol{U}^{*} \boldsymbol{B} \boldsymbol{U} \boldsymbol{D}_{t}^{-1}\right\|_{1}$</td><td>$\|\boldsymbol{B}\|_t := \|\boldsymbol{D}_t\boldsymbol{U}^*\boldsymbol{B}\boldsymbol{U}\boldsymbol{D}_t^{-1}\|_1$. We leave it as an exercise to show that $\|\ \|_t$ is a</td></tr>
<tr><td>lyche-numerical-linear-al F03656</td><td>281</td><td>0.877</td><td>\left\|\|_{t}\right.</td><td>$\left\|\|_{t}\right.$</td><td>$\|\boldsymbol{B}\|_t := \|\boldsymbol{D}_t\boldsymbol{U}^*\boldsymbol{B}\boldsymbol{U}\boldsymbol{D}_t^{-1}\|_1$. We leave it as an exercise to show that $\|\ \|_t$ is a</td></tr>
<tr><td>lyche-numerical-linear-al F03657</td><td>281</td><td>1.000</td><td>\|\boldsymbol{B}\|:=\|\boldsymbol{B}\|_{t}</td><td>$\|\boldsymbol{B}\|:=\|\boldsymbol{B}\|_{t}$</td><td>consistent matrix norm on $\mathbb{C}^{n\times n}$. We define $\|\boldsymbol{B}\| := \|\boldsymbol{B}\|_t$, where t is chosen so</td></tr>
<tr><td>lyche-numerical-linear-al F03658</td><td>282</td><td>1.000</td><td>\lambda^{k}</td><td>λ^{k}</td><td>then λ^k is an eigenvalue of $\boldsymbol{A}^k$ for any $k \in \mathbb{N}$. By Theorem 12.11 we then obtain</td></tr>
<tr><td>lyche-numerical-linear-al F03659</td><td>282</td><td>1.000</td><td>\rho(\boldsymbol{A})^{k}=\rho\left(\boldsymbol{A}^{k}\right) \leq\left\|\boldsymbol{A}^{k}\right\|</td><td>$\rho(\boldsymbol{A})^{k}=\rho\left(\boldsymbol{A}^{k}\right) \leq\left\|\boldsymbol{A}^{k}\right\|$</td><td>$\rho(\boldsymbol{A})^k = \rho(\boldsymbol{A}^k) \leq \|\boldsymbol{A}^k\|$ for any $k \in \mathbb{N}$ so that $\rho(\boldsymbol{A}) \leq \|\boldsymbol{A}^k\|^{1/k}$. Let $\epsilon > 0$ and</td></tr>
<tr><td>lyche-numerical-linear-al F03660</td><td>282</td><td>1.000</td><td>\rho(\boldsymbol{A}) \leq\left\|\boldsymbol{A}^{k}\right\|^{1 / k}</td><td>$\rho(\boldsymbol{A}) \leq\left\|\boldsymbol{A}^{k}\right\|^{1 / k}$</td><td>$\rho(\boldsymbol{A})^k = \rho(\boldsymbol{A}^k) \leq \|\boldsymbol{A}^k\|$ for any $k \in \mathbb{N}$ so that $\rho(\boldsymbol{A}) \leq \|\boldsymbol{A}^k\|^{1/k}$. Let $\epsilon > 0$ and</td></tr>
<tr><td>lyche-numerical-linear-al F03661</td><td>282</td><td>1.000</td><td>\boldsymbol{B}:=(\rho(\boldsymbol{A})+\epsilon)^{-1} \boldsymbol{A}</td><td>$\boldsymbol{B}:=(\rho(\boldsymbol{A})+\epsilon)^{-1} \boldsymbol{A}$</td><td>consider the matrix $\boldsymbol{B} := (\rho(\boldsymbol{A})+\epsilon)^{-1}\boldsymbol{A}$. Then $\rho(\boldsymbol{B}) = \rho(\boldsymbol{A})/(\rho(\boldsymbol{A})+\epsilon) < 1$</td></tr>
<tr><td>lyche-numerical-linear-al F03662</td><td>282</td><td>1.000</td><td>\rho(\boldsymbol{B})=\rho(\boldsymbol{A}) /(\rho(\boldsymbol{A})+\epsilon)<1</td><td>$\rho(\boldsymbol{B})=\rho(\boldsymbol{A}) /(\rho(\boldsymbol{A})+\epsilon)<1$</td><td>consider the matrix $\boldsymbol{B} := (\rho(\boldsymbol{A})+\epsilon)^{-1}\boldsymbol{A}$. Then $\rho(\boldsymbol{B}) = \rho(\boldsymbol{A})/(\rho(\boldsymbol{A})+\epsilon) < 1$</td></tr>
<tr><td>lyche-numerical-linear-al F03663</td><td>282</td><td>1.000</td><td>\left\|\boldsymbol{B}^{k}\right\| \rightarrow 0</td><td>$\left\|\boldsymbol{B}^{k}\right\| \rightarrow 0$</td><td>and $\|\boldsymbol{B}^k\| \to 0$ by Theorem 12.10 as $k \to \infty$. Choose $N \in \mathbb{N}$ such that $\|\boldsymbol{B}^k\| < 1$</td></tr>
<tr><td>lyche-numerical-linear-al F03664</td><td>282</td><td>1.000</td><td>k \rightarrow \infty</td><td>$k \rightarrow \infty$</td><td>and $\|\boldsymbol{B}^k\| \to 0$ by Theorem 12.10 as $k \to \infty$. Choose $N \in \mathbb{N}$ such that $\|\boldsymbol{B}^k\| < 1$</td></tr>
<tr><td>lyche-numerical-linear-al F03665</td><td>282</td><td>1.000</td><td>\left\|\boldsymbol{B}^{k}\right\|<1</td><td>$\left\|\boldsymbol{B}^{k}\right\|<1$</td><td>and $\|\boldsymbol{B}^k\| \to 0$ by Theorem 12.10 as $k \to \infty$. Choose $N \in \mathbb{N}$ such that $\|\boldsymbol{B}^k\| < 1$</td></tr>
<tr><td>lyche-numerical-linear-al F03666</td><td>282</td><td>1.000</td><td>k \geq N</td><td>$k \geq N$</td><td>for all $k \geq N$. Then for $k \geq N$</td></tr>
<tr><td>lyche-numerical-linear-al F03667</td><td>282</td><td>1.000</td><td>\rho(\boldsymbol{A}) \leq\left\|\boldsymbol{A}^{k}\right\|^{1 / k} \leq \rho(\boldsymbol{A})+\epsilon</td><td>$\rho(\boldsymbol{A}) \leq\left\|\boldsymbol{A}^{k}\right\|^{1 / k} \leq \rho(\boldsymbol{A})+\epsilon$</td><td>We have shown that $\rho(\boldsymbol{A}) \leq \|\boldsymbol{A}^k\|^{1/k} \leq \rho(\boldsymbol{A})+\epsilon$ for $k \geq N$. Since ϵ is arbitrary</td></tr>
<tr><td>lyche-numerical-linear-al F03668</td><td>282</td><td>1.000</td><td>\sum_{k=0}^{\infty} \boldsymbol{A}_{k}</td><td>$\sum_{k=0}^{\infty} \boldsymbol{A}_{k}$</td><td>Consider an infinite series $\sum_{k=0}^{\infty} \boldsymbol{A}_k$ of matrices in $\mathbb{C}^{n\times n}$. We say that the series</td></tr>
<tr><td>lyche-numerical-linear-al F03669</td><td>282</td><td>1.000</td><td>\left\{\boldsymbol{S}_{m}\right\}</td><td>$\left\{\boldsymbol{S}_{m}\right\}$</td><td>converges if the sequence of partial sums $\{\boldsymbol{S}_m\}$ given by $\boldsymbol{S}_m = \sum_{k=0}^{m} \boldsymbol{A}_k$ converges.</td></tr>
<tr><td>lyche-numerical-linear-al F03670</td><td>282</td><td>1.000</td><td>\boldsymbol{S}_{m}=\sum_{k=0}^{m} \boldsymbol{A}_{k}</td><td>$\boldsymbol{S}_{m}=\sum_{k=0}^{m} \boldsymbol{A}_{k}$</td><td>converges if the sequence of partial sums $\{\boldsymbol{S}_m\}$ given by $\boldsymbol{S}_m = \sum_{k=0}^{m} \boldsymbol{A}_k$ converges.</td></tr>
<tr><td>lyche-numerical-linear-al F03671</td><td>282</td><td>1.000</td><td>\left\|\boldsymbol{S}_{l}-\boldsymbol{S}_{m}\right\|<\epsilon</td><td>$\left\|\boldsymbol{S}_{l}-\boldsymbol{S}_{m}\right\|<\epsilon$</td><td>there exists an integer N so that $\|\boldsymbol{S}_l - \boldsymbol{S}_m\| < \epsilon$ for all $l > m \geq N$.</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-al F03672</td><td>282</td><td>1.000</td><td>l>m \geq N</td><td>$l>m \geq N$</td><td>there exists an integer N so that $\|\boldsymbol{S}_l-\boldsymbol{S}_m\|<\epsilon$ for all $l>m\geq N$.</td></tr>
<tr><td>lyche-numerical-linear-al F03673</td><td>282</td><td>1.000</td><td>\sum_{k=0}^{\infty} \boldsymbol{B}^{k}</td><td>$\sum_{k=0}^{\infty} \boldsymbol{B}^{k}$</td><td>1. The series $\sum_{k=0}^{\infty}\boldsymbol{B}^k$ converges if and only if $\rho(\boldsymbol{B})<1$.</td></tr>
<tr><td>lyche-numerical-linear-al F03674</td><td>282</td><td>1.000</td><td>\rho(\boldsymbol{B})<1</td><td>$\rho(\boldsymbol{B})<1$</td><td>1. The series $\sum_{k=0}^{\infty}\boldsymbol{B}^k$ converges if and only if $\rho(\boldsymbol{B})<1$.</td></tr>
<tr><td>lyche-numerical-linear-al F03675</td><td>282</td><td>1.000</td><td>(\boldsymbol{I}-\boldsymbol{B})</td><td>$(\boldsymbol{I}-\boldsymbol{B})$</td><td>2. If $\rho(\boldsymbol{B})<1$ then $(\boldsymbol{I}-\boldsymbol{B})$ is nonsingular and $(\boldsymbol{I}-\boldsymbol{B})^{-1}=\sum_{k=0}^{\infty}\boldsymbol{B}^k$.</td></tr>
<tr><td>lyche-numerical-linear-al F03676</td><td>282</td><td>1.000</td><td>(\boldsymbol{I}-\boldsymbol{B})^{-1}=\sum_{k=0}^{\infty} \boldsymbol{B}^{k}</td><td>$(\boldsymbol{I}-\boldsymbol{B})^{-1}=\sum_{k=0}^{\infty} \boldsymbol{B}^{k}$</td><td>2. If $\rho(\boldsymbol{B})<1$ then $(\boldsymbol{I}-\boldsymbol{B})$ is nonsingular and $(\boldsymbol{I}-\boldsymbol{B})^{-1}=\sum_{k=0}^{\infty}\boldsymbol{B}^k$.</td></tr>
<tr><td>lyche-numerical-linear-al F03677</td><td>282</td><td>1.000</td><td>\|\boldsymbol{B}\|<1</td><td>$\|\boldsymbol{B}\|<1$</td><td>3. If $\|\boldsymbol{B}\|<1$ for some consistent matrix norm $\|\cdot\|$ on $\mathbb{C}^{n\times n}$ then</td></tr>
<tr><td>lyche-numerical-linear-al F03678</td><td>283</td><td>1.000</td><td>\boldsymbol{S}_{m}:=\sum_{k=0}^{m} \boldsymbol{B}^{k}</td><td>$\boldsymbol{S}_{m}:=\sum_{k=0}^{m} \boldsymbol{B}^{k}$</td><td>1. Suppose $\rho(\boldsymbol{B})<1$. We show that $\boldsymbol{S}_m:=\sum_{k=0}^{m}\boldsymbol{B}^k$ is a Cauchy sequence and</td></tr>
<tr><td>lyche-numerical-linear-al F03679</td><td>283</td><td>1.000</td><td>l>m</td><td>$l>m$</td><td>norm $\|\cdot\|$ on $\mathbb{C}^{n\times n}$ such that $\|\boldsymbol{B}\|<1$. Then for $l>m$</td></tr>
<tr><td>lyche-numerical-linear-al F03680</td><td>283</td><td>1.000</td><td>\frac{\|\boldsymbol{B}\|^{N+1}}{1-\|\boldsymbol{B}\|}< \epsilon</td><td>$\frac{\|\boldsymbol{B}\|^{N+1}}{1-\|\boldsymbol{B}\|}< \epsilon$</td><td>But then $\{\boldsymbol{S}_m\}$ is a Cauchy sequence provided N is such that $\frac{\|\boldsymbol{B}\|^{N+1}}{1-\|\boldsymbol{B}\|}<\epsilon$.</td></tr>
<tr><td>lyche-numerical-linear-al F03681</td><td>283</td><td>0.876</td><td>\boldsymbol{S}_{m} \boldsymbol{x}=</td><td>$\boldsymbol{S}_{m} \boldsymbol{x}=$</td><td>Conversely, suppose $(\lambda,\boldsymbol{x})$ is an eigenpair for $\boldsymbol{B}$ with $|\lambda|\geq 1$. We find $\boldsymbol{S}_m\boldsymbol{x}=$</td></tr>
<tr><td>lyche-numerical-linear-al F03682</td><td>283</td><td>1.000</td><td>\sum_{k=0}^{m} \boldsymbol{B}^{k} \boldsymbol{x}=\left(\sum_{k=0}^{m} \lambda^{k}\right) \boldsymbol{x}</td><td>$\sum_{k=0}^{m} \boldsymbol{B}^{k} \boldsymbol{x}=\left(\sum_{k=0}^{m} \lambda^{k}\right) \boldsymbol{x}$</td><td>$\sum_{k=0}^{m}\boldsymbol{B}^k\boldsymbol{x}=\big(\sum_{k=0}^{m}\lambda^k\big)\boldsymbol{x}$. Since λ^k does not tend to zero the series $\sum_{k=0}^{\infty}\lambda^k$ is</td></tr>
<tr><td>lyche-numerical-linear-al F03683</td><td>283</td><td>1.000</td><td>\sum_{k=0}^{\infty} \lambda^{k}</td><td>$\sum_{k=0}^{\infty} \lambda^{k}$</td><td>$\sum_{k=0}^{m}\boldsymbol{B}^k\boldsymbol{x}=\big(\sum_{k=0}^{m}\lambda^k\big)\boldsymbol{x}$. Since λ^k does not tend to zero the series $\sum_{k=0}^{\infty}\lambda^k$ is</td></tr>
<tr><td>lyche-numerical-linear-al F03684</td><td>283</td><td>1.000</td><td>\left\{\boldsymbol{S}_{m} \boldsymbol{x}\right\}</td><td>$\left\{\boldsymbol{S}_{m} \boldsymbol{x}\right\}$</td><td>not convergent and therefore $\{\boldsymbol{S}_m\boldsymbol{x}\}$ and hence $\{\boldsymbol{S}_m\}$ does not converge.</td></tr>
<tr><td>lyche-numerical-linear-al F03685</td><td>283</td><td>1.000</td><td>\boldsymbol{B}^{m+1} \rightarrow 0</td><td>$\boldsymbol{B}^{m+1} \rightarrow 0$</td><td>Since $\rho(\boldsymbol{B})<1$ we conclude that $\boldsymbol{B}^{m+1}\rightarrow 0$ and hence taking limits in (12.36)</td></tr>
<tr><td>lyche-numerical-linear-al F03686</td><td>283</td><td>0.947</td><td>\left(\sum_{k=0}^{\infty} \boldsymbol{B}^{k}\right)(\boldsymbol{I}-\boldsymbol{B})=\boldsymbol{I}</td><td>$\left(\sum_{k=0}^{\infty} \boldsymbol{B}^{k}\right)(\boldsymbol{I}-\boldsymbol{B})=\boldsymbol{I}$</td><td>we obtain $\big(\sum_{k=0}^{\infty}\boldsymbol{B}^k\big)(\boldsymbol{I}-\boldsymbol{B})=\boldsymbol{I}$ which completes the proof of 2.</td></tr>
<tr><td>lyche-numerical-linear-al F03687</td><td>283</td><td>0.999</td><td>\left\|(\boldsymbol{I}-\boldsymbol{B})^{-1}\right\|=\left \|\sum_{k=0}^{\infty} \boldsymbol{B}^{k}\right\| \leq \sum_{k=0}^{\infty}\|\boldsymbol{B}\|^{k}=\frac{1}{1-\| \boldsymbol{B}\|}</td><td>$\left\|(\boldsymbol{I}-\boldsymbol{B})^{-1}\right\|=\left \|\sum_{k=0}^{\infty} \boldsymbol{B}^{k}\right\| \leq \sum_{k=0}^{\infty}\|\boldsymbol{B}\|^{k}=\frac{1}{1-\| \boldsymbol{B}\|}$</td><td>3. By 2: $\|(\boldsymbol{I}-\boldsymbol{B})^{-1}\|=\|\sum_{k=0}^{\infty}\boldsymbol{B}^k\|\leq\sum_{k=0}^{\infty}\|\boldsymbol{B}\|^k=\frac{1}{1-\|\boldsymbol{B}\|}$.</td></tr>
<tr><td>lyche-numerical-linear-al F03688</td><td>283</td><td>1.000</td><td>\boldsymbol{\omega}</td><td>$\boldsymbol{\omega}$</td><td>12.5 The Optimal SOR Parameter $\boldsymbol{\omega}$</td></tr>
<tr><td>lyche-numerical-linear-al F03689</td><td>283</td><td>0.546</td><td>\boldsymbol{G}_{J} \boldsymbol{v}=\mu \boldsymbol{v}</td><td>$\boldsymbol{G}_{J} \boldsymbol{v}=\mu \boldsymbol{v}$</td><td>write these equations using the matrix formulation $\boldsymbol{T}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{T}=h^2\boldsymbol{F}$. If $\boldsymbol{G}_J\boldsymbol{v}=\mu\boldsymbol{v}$</td></tr>
<tr><td>lyche-numerical-linear-al F03690</td><td>283</td><td>1.000</td><td>\boldsymbol{v}:=\operatorname{vec}(\boldsymbol{V}) \in \mathbb{R}^{m^{2}}</td><td>$\boldsymbol{v}:=\operatorname{vec}(\boldsymbol{V}) \in \mathbb{R}^{m^{2}}$</td><td>where $\boldsymbol{v}:=\operatorname{vec}(\boldsymbol{V})\in\mathbb{R}^{m^2}$ and $v_{i,j}=0$ if $i\in\{0,m+1\}$ or $j\in\{0,m+1\}$.</td></tr>
<tr><td>lyche-numerical-linear-al F03691</td><td>283</td><td>1.000</td><td>v_{i, j}=0</td><td>$v_{i, j}=0$</td><td>where $\boldsymbol{v}:=\operatorname{vec}(\boldsymbol{V})\in\mathbb{R}^{m^2}$ and $v_{i,j}=0$ if $i\in\{0,m+1\}$ or $j\in\{0,m+1\}$.</td></tr>
<tr><td>lyche-numerical-linear-al F03692</td><td>283</td><td>1.000</td><td>i \in\{0, m+1\}</td><td>$i \in\{0, m+1\}$</td><td>where $\boldsymbol{v}:=\operatorname{vec}(\boldsymbol{V})\in\mathbb{R}^{m^2}$ and $v_{i,j}=0$ if $i\in\{0,m+1\}$ or $j\in\{0,m+1\}$.</td></tr>
<tr><td>lyche-numerical-linear-al F03693</td><td>283</td><td>1.000</td><td>j \in\{0, m+1\}</td><td>$j \in\{0, m+1\}$</td><td>where $\boldsymbol{v}:=\operatorname{vec}(\boldsymbol{V})\in\mathbb{R}^{m^2}$ and $v_{i,j}=0$ if $i\in\{0,m+1\}$ or $j\in\{0,m+1\}$.</td></tr>
<tr><td>lyche-numerical-linear-al F03694</td><td>283</td><td>0.588</td><td>(\lambda, \boldsymbol{w})</td><td>$(\lambda, \boldsymbol{w})$</td><td>Suppose $(\lambda,\boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_\omega$. By (12.21) $(\boldsymbol{I}-\omega\boldsymbol{L})^{-1}(\omega\boldsymbol{R}+(1-$</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F03695	283	0.588	`(12.21)(\boldsymbol{I}-\omega \boldsymbol{L})^{-1}(\omega \boldsymbol{R}+(1-`	$(12.21)(\boldsymbol{I}-\omega \boldsymbol{L})^{-1}(\omega \boldsymbol{R}+(1-$	Suppose $(\lambda, \boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_\omega$. By (12.21) $(\boldsymbol{I}-\omega \boldsymbol{L})^{-1}(\omega \boldsymbol{R}+(1-$
lyche-numerical-linear-al F03696	283	1.000	`\omega) \boldsymbol{I}) \boldsymbol{w}=\lambda \boldsymbol{w}`	$\omega) \boldsymbol{I}) \boldsymbol{w}=\lambda \boldsymbol{w}$	$\omega)\boldsymbol{I})\boldsymbol{w}=\lambda\boldsymbol{w}$ or
lyche-numerical-linear-al F03697	284	1.000	`\boldsymbol{w}=\operatorname{vec}(\boldsymbol{W})`	$\boldsymbol{w}=\operatorname{vec}(\boldsymbol{W})$	Let $\boldsymbol{w}=\operatorname{vec}(\boldsymbol{W})$, where $\boldsymbol{W} \in \mathbb{C}^{m \times m}$. Then (12.38) can be written
lyche-numerical-linear-al F03698	284	1.000	`\boldsymbol{W} \in \mathbb{C}^{m \times m}`	$\boldsymbol{W} \in \mathbb{C}^{m \times m}$	Let $\boldsymbol{w}=\operatorname{vec}(\boldsymbol{W})$, where $\boldsymbol{W} \in \mathbb{C}^{m \times m}$. Then (12.38) can be written
lyche-numerical-linear-al F03699	284	1.000	`w_{i, j}=0`	$w_{i, j}=0$	where $w_{i,j}=0$ if $i \in \{0, m+1\}$ or $j \in \{0, m+1\}$.
lyche-numerical-linear-al F03700	284	0.430	`(\mu, \boldsymbol{v})`	$(\mu, \boldsymbol{v})$	***Proof*** Suppose $(\lambda, \boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_\omega$. We claim that $(\mu, \boldsymbol{v})$ is an eigenpair
lyche-numerical-linear-al F03701	284	1.000	`\boldsymbol{v}=(\boldsymbol{V})`	$\boldsymbol{v}=(\boldsymbol{V})$	for $\boldsymbol{G}_J$, where μ is given by (12.40) and $\boldsymbol{v}=(\boldsymbol{V})$ with $v_{i,j}:=\lambda^{-(i+j)/2}w_{i,j}$.
lyche-numerical-linear-al F03702	284	1.000	`v_{i, j}:=\lambda^{-(i+j) / 2} w_{i, j}`	$v_{i, j}:=\lambda^{-(i+j) / 2} w_{i, j}$	for $\boldsymbol{G}_J$, where μ is given by (12.40) and $\boldsymbol{v}=(\boldsymbol{V})$ with $v_{i,j}:=\lambda^{-(i+j)/2}w_{i,j}$.
lyche-numerical-linear-al F03703	284	1.000	`w_{i, j}`	$w_{i, j}$	Indeed, replacing $w_{i,j}$ by $\lambda^{(i+j)/2}v_{i,j}$ in (12.39) and cancelling the common factor
lyche-numerical-linear-al F03704	284	1.000	`\lambda^{(i+j) / 2} v_{i, j}`	$\lambda^{(i+j) / 2} v_{i, j}$	Indeed, replacing $w_{i,j}$ by $\lambda^{(i+j)/2}v_{i,j}$ in (12.39) and cancelling the common factor
lyche-numerical-linear-al F03705	284	1.000	`\lambda^{(i+j) / 2}`	$\lambda^{(i+j) / 2}$	$\lambda^{(i+j)/2}$ we obtain
lyche-numerical-linear-al F03706	284	0.999	`\boldsymbol{v}=: \operatorname{vec}(\boldsymbol{V}), \boldsymbol{W}=\operatorname{vec}(\boldsymbol{W})`	$\boldsymbol{v}=: \operatorname{vec}(\boldsymbol{V}), \boldsymbol{W}=\operatorname{vec}(\boldsymbol{W})$	(12.41). We define as before $\boldsymbol{v}=:\operatorname{vec}(\boldsymbol{V})$, $\boldsymbol{W}=\operatorname{vec}(\boldsymbol{W})$ with $w_{i,j}:=\lambda^{(i+j)/2}v_{i,j}$.
lyche-numerical-linear-al F03707	284	0.999	`w_{i, j}:=\lambda^{(i+j) / 2} v_{i, j}`	$w_{i, j}:=\lambda^{(i+j) / 2} v_{i, j}$	(12.41). We define as before $\boldsymbol{v}=:\operatorname{vec}(\boldsymbol{V})$, $\boldsymbol{W}=\operatorname{vec}(\boldsymbol{W})$ with $w_{i,j}:=\lambda^{(i+j)/2}v_{i,j}$.
lyche-numerical-linear-al F03708	284	1.000	`\lambda^{-(i+j) / 2}`	$\lambda^{-(i+j) / 2}$	Inserting this in (12.37) and canceling $\lambda^{-(i+j)/2}$ we obtain
lyche-numerical-linear-al F03709	284	1.000	`\omega \lambda^{1 / 2}`	$\omega \lambda^{1 / 2}$	Multiplying by $\omega\lambda^{1/2}$ we obtain
lyche-numerical-linear-al F03710	284	0.952	`\omega \mu \lambda^{1 / 2}=\lambda+\omega-1`	$\omega \mu \lambda^{1 / 2}=\lambda+\omega-1$	Thus, if $\omega\mu\lambda^{1/2}=\lambda+\omega-1$ then by (12.39) $(\lambda, \boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_\omega$.
lyche-numerical-linear-al F03711	284	0.952	`\lambda, \boldsymbol{w}`	$\lambda, \boldsymbol{w}$	Thus, if $\omega\mu\lambda^{1/2}=\lambda+\omega-1$ then by (12.39) $(\lambda, \boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_\omega$.
lyche-numerical-linear-al F03712	285	1.000	`\rho\left(\boldsymbol{G}_{\omega}\right)=|\lambda(\mu)|`	$\rho\left(\boldsymbol{G}_{\omega}\right)=\lvert\lambda(\mu)\rvert$	that $\rho(\boldsymbol{G}_\omega)=\lvert\lambda(\mu)\rvert$, where $\lambda(\mu)$ is an eigenvalue of $\boldsymbol{G}_\omega$ satisfying (12.41) for
lyche-numerical-linear-al F03713	285	1.000	`\lambda(\mu)`	$\lambda(\mu)$	that $\rho(\boldsymbol{G}_\omega)=\lvert\lambda(\mu)\rvert$, where $\lambda(\mu)$ is an eigenvalue of $\boldsymbol{G}_\omega$ satisfying (12.41) for
lyche-numerical-linear-al F03714	285	1.000	`\frac{1}{2} \cos (j \pi h)+\frac{1}{2} \cos (k \pi h)`	$\frac{1}{2} \cos (j \pi h)+\frac{1}{2} \cos (k \pi h)$	some eigenvalue μ of $\boldsymbol{G}_J$. The eigenvalues of $\boldsymbol{G}_J$ are $\frac{1}{2}\cos(j\pi h)+\frac{1}{2}\cos(k\pi h)$,
lyche-numerical-linear-al F03715	285	1.000	`-\mu`	$-\mu$	$j, k=1, \ldots, m$, so μ is real and both μ and $-\mu$ are eigenvalues. Thus, to compute
lyche-numerical-linear-al F03716	285	1.000	`\lambda=\lambda(\mu)`	$\lambda=\lambda(\mu)$	(12.41) for $\lambda=\lambda(\mu)$ gives
lyche-numerical-linear-al F03717	285	0.998	`d(0)=4>0`	$d(0)=4>0$	Moreover $d(0)=4>0$ and $d(2)=4\mu^2-4<0$. As a function of ω, $\lambda(\mu)$ changes
lyche-numerical-linear-al F03718	285	0.998	`d(2)=4 \mu^{2}-4<0`	$d(2)=4 \mu^{2}-4<0$	Moreover $d(0)=4>0$ and $d(2)=4\mu^2-4<0$. As a function of ω, $\lambda(\mu)$ changes
lyche-numerical-linear-al F03719	285	0.998	`\omega, \lambda(\mu)`	$\omega, \lambda(\mu)$	Moreover $d(0)=4>0$ and $d(2)=4\mu^2-4<0$. As a function of ω, $\lambda(\mu)$ changes
lyche-numerical-linear-al F03720	285	0.618	`d(\omega)=0`	$d(\omega)=0$	from real to complex when $d(\omega)=0$. The root in $(0,2)$ is

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F03721	285	1.000	`\tilde{\omega}(\mu)`	$\tilde{\omega}(\mu)$	Both $\lambda(\mu)$ and $\tilde{\omega}(\mu)$ are strictly increasing as functions of μ. It follows that $\lvert\lambda(\mu)\rvert$
lyche-numerical-linear-al F03722	285	1.000	`|\lambda(\mu)|`	$\lvert\lambda(\mu)\rvert$	Both $\lambda(\mu)$ and $\tilde{\omega}(\mu)$ are strictly increasing as functions of μ. It follows that $\lvert\lambda(\mu)\rvert$
lyche-numerical-linear-al F03723	285	1.000	`\mu=\rho\left(\boldsymbol{G}_{J}\right)=: \beta`	$\mu=\rho\left(\boldsymbol{G}_{J}\right)=: \beta$	is maximized for $\mu = \rho(\boldsymbol{G}_J) =: \beta$ and for this value of μ we obtain (12.27) for
lyche-numerical-linear-al F03724	285	0.995	`0<\omega \leq \tilde{\omega}(\beta)=\omega^{*}`	$0<\omega \leq \tilde{\omega}(\beta)=\omega^{*}$	$0 < \omega \leq \tilde{\omega}(\beta) = \omega^*$.
lyche-numerical-linear-al F03725	285	1.000	`\rho\left(\boldsymbol{G}_{\omega}\right)=\omega-1`	$\rho\left(\boldsymbol{G}_{\omega}\right)=\omega-1$	Evidently $\rho(\boldsymbol{G}_\omega) = \omega - 1$ is strictly increasing in $\omega^* < \omega < 2$. Equation (12.29)
lyche-numerical-linear-al F03726	285	1.000	`\omega^{*}<\omega<2`	$\omega^{*}<\omega<2$	Evidently $\rho(\boldsymbol{G}_\omega) = \omega - 1$ is strictly increasing in $\omega^* < \omega < 2$. Equation (12.29)
lyche-numerical-linear-al F03727	285	1.000	`0<\omega<\omega^{*}`	$0<\omega<\omega^{*}$	will follow if we can show that $\rho(\boldsymbol{G}_\omega)$ is strictly decreasing in $0 < \omega < \omega^*$. By
lyche-numerical-linear-al F03728	285	1.000	`\beta^{2}\left(\omega^{2} \beta^{2}-4 \omega+4\right)< \left(2-\omega \beta^{2}\right)^{2}`	$\beta^{2}\left(\omega^{2} \beta^{2}-4 \omega+4\right)<\left(2-\omega \beta^{2}\right)^{2}$	Since $\beta^2(\omega^2\beta^2 - 4\omega + 4) < (2 - \omega\beta^2)^2$ the numerator is negative and the strict
lyche-numerical-linear-al F03729	286	1.000	`a_{i i}=d \neq 0`	$a_{i i}=d \neq 0$	**Exercise 12.1 (Richardson and Jacobi)** Show that if $a_{ii} = d \neq 0$ for all i then
lyche-numerical-linear-al F03730	286	1.000	`\alpha:=1 / d`	$\alpha:=1 / d$	Richardson's method with $\alpha := 1/d$ is the same as Jacobi's method.
lyche-numerical-linear-al F03731	286	1.000	`u_{j}`	u_{j}	all eigenvalues λ_j of $\boldsymbol{A}$ have positive real parts u_j for $j = 1, \ldots, n$ and that α is
lyche-numerical-linear-al F03732	286	0.766	`0<\alpha<\min _{j}\left(2 u_{j} /\left|\lambda_{j} \right|^{2}\right)`	$0<\alpha<\min _{j}\left(2 u_{j} /\left\lvert\lambda_{j}\right\rvert^{2}\right)$	real. Show that the R method converges if and only if $0 < \alpha < \min_j(2u_j/\lvert\lambda_j\rvert^2)$.
lyche-numerical-linear-al F03733	286	1.000	`\boldsymbol{A}=\left[\begin{array}{ll}1 & 2 \\ 3 & 4 \end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{ll}1 & 2 \\ 3 & 4\end{array}\right]$	method and Gauss-Seidel's method diverge for $\boldsymbol{A} = \left[\begin{smallmatrix}1 & 2\\ 3 & 4\end{smallmatrix}\right]$.
lyche-numerical-linear-al F03734	286	0.945	`\boldsymbol{A}:=\left[\begin{array}{ll}a_{11} & a_{12} \\ a_{21} & a_{22}\end{array}\right] \in \mathbb{R}^{2 \times 2}`	$\boldsymbol{A}:=\left[\begin{array}{ll}a_{11} & a_{12} \\ a_{21} & a_{22}\end{array}\right] \in \mathbb{R}^{2 \times 2}$	and only if Jacobi converges for a 2 by 2 matrix $\boldsymbol{A} := \left[\begin{smallmatrix}a_{11} & a_{12}\\ a_{21} & a_{22}\end{smallmatrix}\right] \in \mathbb{R}^{2\times 2}$.
lyche-numerical-linear-al F03735	286	1.000	`\left[\begin{array}{ccc}1 & a & a \\ a & 1 & a \\ a & a & 1\end{array}\right]`	$\left[\begin{array}{ccc}1 & a & a \\ a & 1 & a \\ a & a & 1\end{array}\right]$	values) that the matrix $\left[\begin{smallmatrix}1 & a & a\\ a & 1 & a\\ a & a & 1\end{smallmatrix}\right]$ is positive definite for $-1/2 < a < 1$. Thus, GS
lyche-numerical-linear-al F03736	286	1.000	`-1 / 2<a<1`	$-1 / 2<a<1$	values) that the matrix $\left[\begin{smallmatrix}1 & a & a\\ a & 1 & a\\ a & a & 1\end{smallmatrix}\right]$ is positive definite for $-1/2 < a < 1$. Thus, GS
lyche-numerical-linear-al F03737	286	1.000	`1 / 2<a<1`	$1 / 2<a<1$	$1/2 < a < 1$.
lyche-numerical-linear-al F03738	286	0.995	`\boldsymbol{G}_{1}`	$\boldsymbol{G}_{1}$	**Exercise 12.6 (Example: GS Diverges, J Converges)** Let $\boldsymbol{G}_J$ and $\boldsymbol{G}_1$ be the
lyche-numerical-linear-al F03739	286	0.924	`\boldsymbol{A}:=\left[\begin{array}{ccc}1 & 0 & 1 / 2 \\ 1 & 1 & 0 \\ -1 & 1 & 1\end{array}\right] .1`	$\boldsymbol{A}:=\left[\begin{array}{ccc}1 & 0 & 1 / 2 \\ 1 & 1 & 0 \\ -1 & 1 & 1\end{array}\right] .1$	$\boldsymbol{A} := \left[\begin{smallmatrix}1 & 0 & 1/2\\ 1 & 1 & 0\\ -1 & 1 & 1\end{smallmatrix}\right]$.[1]
lyche-numerical-linear-al F03740	286	0.996	`\boldsymbol{G}_{1}:=\left[\begin{array}{ccc}0 & 0 & -1 / 2 \\ 0 & 0 & 1 / 2 \\ 0 & 0 & -1\end{array}\right]`	$\boldsymbol{G}_{1}:=\left[\begin{array}{ccc}0 & 0 & -1 / 2 \\ 0 & 0 & 1 / 2 \\ 0 & 0 & -1\end{array}\right]$	**a)** Show that $\boldsymbol{G}_1 := \left[\begin{smallmatrix}0 & 0 & -1/2\\ 0 & 0 & 1/2\\ 0 & 0 & -1\end{smallmatrix}\right]$ and conclude that GS diverges.
lyche-numerical-linear-al F03741	286	1.000	`p(\lambda):=\operatorname{det}\left(\lambda \boldsymbol{I}-\boldsymbol{G}_{J}\right)=\lambda^{3}+ \frac{1}{2} \lambda+\frac{1}{2}`	$p(\lambda):=\operatorname{det}\left(\lambda \boldsymbol{I}-\boldsymbol{G}_{J}\right)=\lambda^{3}+\frac{1}{2} \lambda+\frac{1}{2}$	**b)** Show that $p(\lambda) := \det(\lambda\boldsymbol{I} - \boldsymbol{G}_J) = \lambda^3 + \frac{1}{2}\lambda + \frac{1}{2}$.
lyche-numerical-linear-al F03742	286	0.997	`p(\lambda) \neq 0`	$p(\lambda) \neq 0$	**c)** Show that if $\lvert\lambda\rvert \geq 1$ then $p(\lambda) \neq 0$. Conclude that J converges.
lyche-numerical-linear-al F03743	286	1.000	`\left|a_{i i}\right|>\sum_{j \neq i}\left|a_{i j}\right|`	$\left\lvert a_{i i}\right\rvert>\sum_{j \neq i}\left\lvert a_{i j}\right\rvert$	method converges if $\lvert a_{ii}\rvert > \sum_{j\neq i}\lvert a_{ij}\rvert$ for $i = 1, \ldots, n$.
lyche-numerical-linear-al F03744	287	1.000	`r:=\max _{i} r_{i}<1`	$r:=\max _{i} r_{i}<1$	GS method. Suppose $r := \max_i r_i < 1$, where $r_i = \sum_{j\neq i}\frac{\lvert a_{ij}\rvert}{\lvert a_{ii}\rvert}$. Show using

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-al F03745</td><td>287</td><td>1.000</td><td>r_{i}=\sum_{j \neq i} \frac{\left|a_{i j}\right|}{\left|a_{i i}\right|}</td><td>$r_{i}=\sum_{j \neq i} \frac{\left|a_{i j}\right|}{\left|a_{i i}\right|}$</td><td>GS method. Suppose $r := \max_i r_i < 1$, where $r_i = \sum_{j\neq i} \frac{|a_{ij}|}{|a_{ii}|}$. Show using</td></tr>
<tr><td>lyche-numerical-linear-al F03746</td><td>287</td><td>1.000</td><td>\left|\boldsymbol{\epsilon}_{k+1}(j)\right| \leq r\left\|\boldsymbol{\epsilon}_{k}\right\|_{\infty}</td><td>$\left|\boldsymbol{\epsilon}_{k+1}(j)\right| \leq r\left\|\boldsymbol{\epsilon}_{k}\right\|_{\infty}$</td><td>induction on i that $|\boldsymbol{\epsilon}_{k+1}(j)| \le r\|\boldsymbol{\epsilon}_k\|_\infty$ for $j = 1,\ldots,i$. Conclude that Gauss-</td></tr>
<tr><td>lyche-numerical-linear-al F03747</td><td>287</td><td>1.000</td><td>j=1, \ldots, i</td><td>$j=1, \ldots, i$</td><td>induction on i that $|\boldsymbol{\epsilon}_{k+1}(j)| \le r\|\boldsymbol{\epsilon}_k\|_\infty$ for $j = 1,\ldots,i$. Conclude that Gauss-</td></tr>
<tr><td>lyche-numerical-linear-al F03748</td><td>287</td><td>1.000</td><td>a \in</td><td>$a \in$</td><td>Exercise 12.9 (Convergence Example for Fix Point Iteration) Consider for $a \in$</td></tr>
<tr><td>lyche-numerical-linear-al F03749</td><td>287</td><td>1.000</td><td>\boldsymbol{x}_{0}=\mathbf{0}</td><td>$\boldsymbol{x}_{0}=\mathbf{0}$</td><td>Starting with $\boldsymbol{x}_0 = \mathbf{0}$ show by induction</td></tr>
<tr><td>lyche-numerical-linear-al F03750</td><td>287</td><td>1.000</td><td>\boldsymbol{x}=[1,1]^{T}</td><td>$\boldsymbol{x}=[1,1]^{T}$</td><td>and conclude that the iteration converges to the fixed-point $\boldsymbol{x} = [1, 1]^T$ for $|a| < 1$</td></tr>
<tr><td>lyche-numerical-linear-al F03751</td><td>287</td><td>1.000</td><td>|a|<1</td><td>$|a|<1$</td><td>and conclude that the iteration converges to the fixed-point $\boldsymbol{x} = [1, 1]^T$ for $|a| < 1$</td></tr>
<tr><td>lyche-numerical-linear-al F03752</td><td>287</td><td>1.000</td><td>|a|>1</td><td>$|a|>1$</td><td>and diverges for $|a| > 1$. Show that $\rho(\boldsymbol{G}) = 1 - \eta$ with $\eta = 1 - |a|$. Compute the</td></tr>
<tr><td>lyche-numerical-linear-al F03753</td><td>287</td><td>1.000</td><td>\eta=1-|a|</td><td>$\eta=1-|a|$</td><td>and diverges for $|a| > 1$. Show that $\rho(\boldsymbol{G}) = 1 - \eta$ with $\eta = 1 - |a|$. Compute the</td></tr>
<tr><td>lyche-numerical-linear-al F03754</td><td>287</td><td>0.999</td><td>a=0.9</td><td>$a=0.9$</td><td>estimate (12.31) for the rate of convergence for $a = 0.9$ and $s = 16$ and compare</td></tr>
<tr><td>lyche-numerical-linear-al F03755</td><td>287</td><td>0.999</td><td>s=16</td><td>$s=16$</td><td>estimate (12.31) for the rate of convergence for $a = 0.9$ and $s = 16$ and compare</td></tr>
<tr><td>lyche-numerical-linear-al F03756</td><td>287</td><td>1.000</td><td>|a|^{k} \leq 10^{-16}</td><td>$|a|^{k} \leq 10^{-16}$</td><td>with the true number of iterations determined from $|a|^k \le 10^{-16}$.</td></tr>
<tr><td>lyche-numerical-linear-al F03757</td><td>287</td><td>1.000</td><td>\rho\left(\boldsymbol{G}_{J}\right)=1 / 2</td><td>$\rho\left(\boldsymbol{G}_{J}\right)=1 / 2$</td><td>in Example 12.2. Show that $\rho(\boldsymbol{G}_J) = 1/2$. Then show that $\boldsymbol{x}_k(1) = \boldsymbol{x}_k(2) = 1-2^{-k}$</td></tr>
<tr><td>lyche-numerical-linear-al F03758</td><td>287</td><td>1.000</td><td>\boldsymbol{x}_{k}(1)=\boldsymbol{x}_{k}(2)=1-2^{-k}</td><td>$\boldsymbol{x}_{k}(1)=\boldsymbol{x}_{k}(2)=1-2^{-k}$</td><td>in Example 12.2. Show that $\rho(\boldsymbol{G}_J) = 1/2$. Then show that $\boldsymbol{x}_k(1) = \boldsymbol{x}_k(2) = 1-2^{-k}$</td></tr>
<tr><td>lyche-numerical-linear-al F03759</td><td>287</td><td>1.000</td><td>\boldsymbol{A}=\boldsymbol{I}-\boldsymbol{L}-\boldsymbol{L}^{T}</td><td>$\boldsymbol{A}=\boldsymbol{I}-\boldsymbol{L}-\boldsymbol{L}^{T}$</td><td>be written $\boldsymbol{A} = \boldsymbol{I} - \boldsymbol{L} - \boldsymbol{L}^T$, where $\boldsymbol{L}$ is lower triangular with zero's on the diagonal,</td></tr>
<tr><td>lyche-numerical-linear-al F03760</td><td>287</td><td>1.000</td><td>l_{i, j}=0</td><td>$l_{i, j}=0$</td><td>$l_{i,j} = 0$, when $j >= i$. The method is defined by</td></tr>
<tr><td>lyche-numerical-linear-al F03761</td><td>287</td><td>1.000</td><td>j>=i</td><td>$j>=i$</td><td>$l_{i,j} = 0$, when $j >= i$. The method is defined by</td></tr>
<tr><td>lyche-numerical-linear-al F03762</td><td>287</td><td>1.000</td><td>\boldsymbol{N}</td><td>$\boldsymbol{N}$</td><td>where $\boldsymbol{M}$ and $\boldsymbol{N}$ are given by the splitting</td></tr>
<tr><td>lyche-numerical-linear-al F03763</td><td>287</td><td>1.000</td><td>\boldsymbol{M}^{-1} \boldsymbol{N}</td><td>$\boldsymbol{M}^{-1} \boldsymbol{N}$</td><td>a) Let $\boldsymbol{x} \neq \mathbf{0}$ be an eigenvector of $\boldsymbol{M}^{-1}\boldsymbol{N}$ with eigenvalue λ. Show that</td></tr>
<tr><td>lyche-numerical-linear-al F03764</td><td>288</td><td>1.000</td><td>\boldsymbol{x}=[x(1), x(2), \ldots, x(n)]^{T}</td><td>$\boldsymbol{x}=[x(1), x(2), \ldots, x(n)]^{T}$</td><td>1. Choose $\boldsymbol{x} = [x(1), x(2), \ldots, x(n)]^T$.</td></tr>
<tr><td>lyche-numerical-linear-al F03765</td><td>288</td><td>0.996</td><td>i=1,2, \ldots, n-1, n, n, n-1, n-2, \ldots, 1</td><td>$i=1,2, \ldots, n-1, n, n, n-1, n-2, \ldots, 1$</td><td>for $i = 1, 2, \ldots, n-1, n, n, n-1, n-2, \ldots, 1$</td></tr>
<tr><td>lyche-numerical-linear-al F03766</td><td>288</td><td>0.999</td><td>\boldsymbol{b}:=[1,9,-2]^{T}</td><td>$\boldsymbol{b}:=[1,9,-2]^{T}$</td><td>and $\boldsymbol{b} := [1, 9, -2]^T$.</td></tr>
<tr><td>lyche-numerical-linear-al F03767</td><td>288</td><td>1.000</td><td>\boldsymbol{x}_{0}=[1,1,1]^{t}</td><td>$\boldsymbol{x}_{0}=[1,1,1]^{t}$</td><td>a) With $\boldsymbol{x}_0 = [1, 1, 1]^t$, carry out one iteration of the Gauss-Seidel method to find</td></tr>
<tr><td>lyche-numerical-linear-al F03768</td><td>288</td><td>1.000</td><td>\boldsymbol{x}_{1} \in \mathbb{R}^{3}</td><td>$\boldsymbol{x}_{1} \in \mathbb{R}^{3}$</td><td>$\boldsymbol{x}_1 \in \mathbb{R}^3$.</td></tr>
<tr><td>lyche-numerical-linear-al F03769</td><td>288</td><td>1.000</td><td>\boldsymbol{E} \in \mathbb{C}^{n \times n}</td><td>$\boldsymbol{E} \in \mathbb{C}^{n \times n}$</td><td>$\boldsymbol{E} \in \mathbb{C}^{n\times n}$. Show that $\boldsymbol{A} + \boldsymbol{E}$ is nonsingular if $\rho(\boldsymbol{A}^{-1}\boldsymbol{E}) < 1$.</td></tr>
<tr><td>lyche-numerical-linear-al F03770</td><td>288</td><td>1.000</td><td>\rho\left(\boldsymbol{A}^{-1} \boldsymbol{E}\right)<1</td><td>$\rho\left(\boldsymbol{A}^{-1} \boldsymbol{E}\right)<1$</td><td>$\boldsymbol{E} \in \mathbb{C}^{n\times n}$. Show that $\boldsymbol{A} + \boldsymbol{E}$ is nonsingular if $\rho(\boldsymbol{A}^{-1}\boldsymbol{E}) < 1$.</td></tr>
<tr><td>lyche-numerical-linear-al F03771</td><td>289</td><td>0.998</td><td>\lim _{k \rightarrow \infty} \|</td><td>$\lim _{k \rightarrow \infty} \|$</td><td>Exercise 12.15 (Slow Spectral Radius Convergence) The convergence $\lim_{k\to\infty}\|$</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F03772	289	1.000	`\boldsymbol{A}^{k} |^{1 / k}=\rho(\boldsymbol{A})`	$\boldsymbol{A}^{k} \|^{1 / k}=\rho(\boldsymbol{A})$	$\boldsymbol{A}^k\|^{1/k} = \rho(\boldsymbol{A})$ can be quite slow. Consider
lyche-numerical-linear-al F03773	289	1.000	`|\lambda|=\rho(\boldsymbol{A})<1`	$\|\lambda\|=\rho(\boldsymbol{A})<1$	If $\|\lambda\| = \rho(\boldsymbol{A}) < 1$ then $\lim_{k\to\infty} \boldsymbol{A}^k = \mathbf{0}$ for any $a \in \mathbb{R}$. We show below that the
lyche-numerical-linear-al F03774	289	1.000	`(1, n)`	$(1, n)$	$(1, n)$ element of $\boldsymbol{A}^k$ is given by $f(k) := \binom{k}{n-1} a^{n-1}\lambda^{k-n+1}$ for $k \geq n-1$.
lyche-numerical-linear-al F03775	289	1.000	`f(k):=\binom{k}{n-1} a^{n-1} \lambda^{k-n+1}`	$f(k):=\binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$	$(1, n)$ element of $\boldsymbol{A}^k$ is given by $f(k) := \binom{k}{n-1} a^{n-1}\lambda^{k-n+1}$ for $k \geq n-1$.
lyche-numerical-linear-al F03776	289	1.000	`k \geq n-1`	$k \geq n-1$	$(1, n)$ element of $\boldsymbol{A}^k$ is given by $f(k) := \binom{k}{n-1} a^{n-1}\lambda^{k-n+1}$ for $k \geq n-1$.
lyche-numerical-linear-al F03777	289	1.000	`n=5`	$n=5$	**a)** Pick an n, e.g. $n = 5$, and make a plot of $f(k)$ for $\lambda = 0.9$, $a = 10$, and
lyche-numerical-linear-al F03778	289	1.000	`f(k)`	$f(k)$	**a)** Pick an n, e.g. $n = 5$, and make a plot of $f(k)$ for $\lambda = 0.9$, $a = 10$, and
lyche-numerical-linear-al F03779	289	1.000	`\lambda=0.9, a=10`	$\lambda=0.9, a=10$	**a)** Pick an n, e.g. $n = 5$, and make a plot of $f(k)$ for $\lambda = 0.9$, $a = 10$, and
lyche-numerical-linear-al F03780	289	1.000	`n-1 \leq k \leq 200`	$n-1 \leq k \leq 200$	$n-1 \leq k \leq 200$. Your program should also compute $\max_k f(k)$. Use your
lyche-numerical-linear-al F03781	289	1.000	`\max _{k} f(k)`	$\max _{k} f(k)$	$n-1 \leq k \leq 200$. Your program should also compute $\max_k f(k)$. Use your
lyche-numerical-linear-al F03782	289	1.000	`f(k)<10^{-8}`	$f(k)<10^{-8}$	program to determine how large k must be before $f(k) < 10^{-8}$.
lyche-numerical-linear-al F03783	289	1.000	`\boldsymbol{E}:=(\boldsymbol{A}-\lambda \boldsymbol{I}) / a`	$\boldsymbol{E}:=(\boldsymbol{A}-\lambda \boldsymbol{I}) / a$	**b)** We can determine the elements of $\boldsymbol{A}^k$ explicitly for any k. Let $\boldsymbol{E} := (\boldsymbol{A} - \lambda\boldsymbol{I})/a$.
lyche-numerical-linear-al F03784	289	1.000	`\boldsymbol{E}^{k}=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{I}_{n-k} \\ \mathbf{0} & \mathbf{0} \end{array}\right]`	$\boldsymbol{E}^{k}=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{I}_{n-k} \\ \mathbf{0} & \mathbf{0}\end{array}\right]$	Show by induction that $\boldsymbol{E}^k = \left[\begin{smallmatrix}\mathbf{0} & \boldsymbol{I}_{n-k} \\ \mathbf{0} & \mathbf{0}\end{smallmatrix}\right]$ for $1 \leq k \leq n-1$ and that $\boldsymbol{E}^n = \mathbf{0}$.
lyche-numerical-linear-al F03785	289	1.000	`\boldsymbol{E}^{n}=\mathbf{0}`	$\boldsymbol{E}^{n}=\mathbf{0}$	Show by induction that $\boldsymbol{E}^k = \left[\begin{smallmatrix}\mathbf{0} & \boldsymbol{I}_{n-k} \\ \mathbf{0} & \mathbf{0}\end{smallmatrix}\right]$ for $1 \leq k \leq n-1$ and that $\boldsymbol{E}^n = \mathbf{0}$.
lyche-numerical-linear-al F03786	289	0.997	`\boldsymbol{A}^{k}=(a \boldsymbol{E}+\lambda \boldsymbol{I})^{k}=\sum_{j=0}^{\min \{k, n-1\}} \binom{k}{j} a^{j} \lambda^{k-j} \boldsymbol{E}^{j}`	$\boldsymbol{A}^{k}=(a \boldsymbol{E}+\lambda \boldsymbol{I})^{k}=\sum_{j=0}^{\min \{k, n-1\}} \binom{k}{j} a^{j} \lambda^{k-j} \boldsymbol{E}^{j}$	**c)** We have $\boldsymbol{A}^k = (a\boldsymbol{E} + \lambda\boldsymbol{I})^k = \sum_{j=0}^{\min\{k,n-1\}} \binom{k}{j} a^j \lambda^{k-j} \boldsymbol{E}^j$ and conclude that the
lyche-numerical-linear-al F03787	289	1.000	`\boldsymbol{A}^{n} \rightarrow \mathbf{0}`	$\boldsymbol{A}^{n} \rightarrow \mathbf{0}$	• Give a necessary and sufficient condition that $\boldsymbol{A}^n \rightarrow \mathbf{0}$.
lyche-numerical-linear-al F03788	289	1.000	`| \boldsymbol{A} |<\rho(\boldsymbol{A})`	$\\| \boldsymbol{A} \\|<\rho(\boldsymbol{A})$	• Is there a matrix norm $\\| \ \\|$ consistent on $\mathbb{C}^{n\times n}$ such that $\\|\boldsymbol{A}\\| < \rho(\boldsymbol{A})$?
lyche-numerical-linear-al F03789	290	0.969	`\kappa:=\frac{\lambda_{\text {max }}}{\lambda_{\text {min }}}`	$\kappa:=\frac{\lambda_{\text {max }}}{\lambda_{\text {min }}}$	condition number of $\boldsymbol{A}$ is given by $\kappa := \frac{\lambda_{max}}{\lambda_{min}}$, where λ_{max} and λ_{min} are the largest
lyche-numerical-linear-al F03790	290	1.000	`\langle\boldsymbol{x}, \boldsymbol{y}\rangle= \boldsymbol{x}^{T} \boldsymbol{y}`	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle= \boldsymbol{x}^{T} \boldsymbol{y}$	on $\mathbb{R}^n$, the usual inner product $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \boldsymbol{x}^T\boldsymbol{y}$ with the associated Euclidian norm
lyche-numerical-linear-al F03791	290	1.000	`|\boldsymbol{x}|_{2}=\sqrt{\boldsymbol{x}^{T} \boldsymbol{x}}`	$\\|\boldsymbol{x}\\|_{2}=\sqrt{\boldsymbol{x}^{T} \boldsymbol{x}}$	$\\|\boldsymbol{x}\\|_2 = \sqrt{\boldsymbol{x}^T\boldsymbol{x}}$, and the **$\boldsymbol{A}$-inner product** and the corresponding **$\boldsymbol{A}$-norm** given by
lyche-numerical-linear-al F03792	291	1.000	`\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in`	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in$	We note that the $\boldsymbol{A}$-inner product is an inner product on $\mathbb{R}^n$. Indeed, for any $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in$
lyche-numerical-linear-al F03793	291	1.000	`\langle\boldsymbol{x}, \boldsymbol{x}\rangle_{ \boldsymbol{A}}=\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x} \geq 0`	$\langle\boldsymbol{x}, \boldsymbol{x}\rangle_{\boldsymbol{A}}=\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x} \geq 0$	1. $\langle \boldsymbol{x}, \boldsymbol{x}\rangle_{\boldsymbol{A}} = \boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} \geq 0$ and $\langle \boldsymbol{x}, \boldsymbol{x}\rangle_{\boldsymbol{A}} = 0$ if and only if $\boldsymbol{x} = \mathbf{0}$, since $\boldsymbol{A}$ is positive
lyche-numerical-linear-al F03794	291	1.000	`\langle\boldsymbol{x}, \boldsymbol{x}\rangle_{ \boldsymbol{A}}=0`	$\langle\boldsymbol{x}, \boldsymbol{x}\rangle_{\boldsymbol{A}}=0$	1. $\langle \boldsymbol{x}, \boldsymbol{x}\rangle_{\boldsymbol{A}} = \boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} \geq 0$ and $\langle \boldsymbol{x}, \boldsymbol{x}\rangle_{\boldsymbol{A}} = 0$ if and only if $\boldsymbol{x} = \mathbf{0}$, since $\boldsymbol{A}$ is positive

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F03795	291	0.983	`\langle\boldsymbol{x}, \boldsymbol{y}\rangle_{ \boldsymbol{A}}:=\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{y}=\left(\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{y}\right)^{T}=\boldsymbol{y}^{T} \boldsymbol{A}^{T} \boldsymbol{x}=\boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{x}=\langle\boldsymbol{y}, \boldsymbol{x}\rangle_{\boldsymbol{A}}`	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle_{ \boldsymbol{A}}:=\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{y}=\left(\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{y}\right)^{T}=\boldsymbol{y}^{T} \boldsymbol{A}^{T} \boldsymbol{x}=\boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{x}=\langle\boldsymbol{y}, \boldsymbol{x}\rangle_{\boldsymbol{A}}$	2. $\langle \boldsymbol{x}, \boldsymbol{y}\rangle_A := \boldsymbol{x}^T \boldsymbol{A}\boldsymbol{y} = (\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{y})^T = \boldsymbol{y}^T\boldsymbol{A}^T\boldsymbol{x} = \boldsymbol{y}^T\boldsymbol{A}\boldsymbol{x} = \langle\boldsymbol{y},\boldsymbol{x}\rangle_A$ by symmetry of
lyche-numerical-linear-algebra F03796	291	1.000	`\langle\boldsymbol{x}+\boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}}:=\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{z}+\boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{z}=\langle\boldsymbol{x}, \boldsymbol{z}\rangle_{\boldsymbol{A}}+\langle \boldsymbol{y}, \boldsymbol{z}\rangle_{\boldsymbol{A}}`	$\langle\boldsymbol{x}+\boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}}:=\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{z}+\boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{z}=\langle\boldsymbol{x}, \boldsymbol{z}\rangle_{\boldsymbol{A}}+\langle \boldsymbol{y}, \boldsymbol{z}\rangle_{\boldsymbol{A}}$	3. $\langle\boldsymbol{x}+\boldsymbol{y},\boldsymbol{z}\rangle_A := \boldsymbol{x}^T\boldsymbol{A}\boldsymbol{z} + \boldsymbol{y}^T\boldsymbol{A}\boldsymbol{z} = \langle\boldsymbol{x},\boldsymbol{z}\rangle_A + \langle\boldsymbol{y},\boldsymbol{z}\rangle_A$, true for any $\boldsymbol{A}$.
lyche-numerical-linear-algebra F03797	291	1.000	`\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{b} \in \mathbb{R}^{n}`	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{b} \in \mathbb{R}^{n}$	Consider for a positive definite $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\boldsymbol{b} \in \mathbb{R}^n$ and $c \in \mathbb{R}$ the quadratic
lyche-numerical-linear-algebra F03798	291	1.000	`c \in \mathbb{R}`	$c \in \mathbb{R}$	Consider for a positive definite $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\boldsymbol{b} \in \mathbb{R}^n$ and $c \in \mathbb{R}$ the quadratic
lyche-numerical-linear-algebra F03799	291	1.000	`Q: \mathbb{R}^{n} \rightarrow \mathbb{R}`	$Q: \mathbb{R}^{n} \rightarrow \mathbb{R}$	function $Q : \mathbb{R}^n \to \mathbb{R}$ given by
lyche-numerical-linear-algebra F03800	291	1.000	`\boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^{n}`	$\boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^{n}$	The following expansion will be used repeatedly. For $\boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^n$ and $\varepsilon \in \mathbb{R}$
lyche-numerical-linear-algebra F03801	291	1.000	`\varepsilon \in \mathbb{R}`	$\varepsilon \in \mathbb{R}$	The following expansion will be used repeatedly. For $\boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^n$ and $\varepsilon \in \mathbb{R}$
lyche-numerical-linear-algebra F03802	291	0.996	`\boldsymbol{r}(\boldsymbol{y}):=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{y}`	$\boldsymbol{r}(\boldsymbol{y}):=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{y}$	*if and only if* $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. *Moreover, the residual* $\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{b} - \boldsymbol{A}\boldsymbol{y}$ *for any* $\boldsymbol{y} \in \mathbb{R}^n$ *is*
lyche-numerical-linear-algebra F03803	291	1.000	`\boldsymbol{r}(\boldsymbol{y})=-\nabla Q(\boldsymbol{y})`	$\boldsymbol{r}(\boldsymbol{y})=-\nabla Q(\boldsymbol{y})$	*equal to the negative gradient, i.e.,* $\boldsymbol{r}(\boldsymbol{y}) = -\nabla Q(\boldsymbol{y})$, *where* $\nabla := \left[\frac{\partial}{\partial y_1}, \ldots, \frac{\partial}{\partial y_n}\right]^T$.
lyche-numerical-linear-algebra F03804	291	1.000	`\nabla:=\left[\frac{\partial}{\partial y_{1}}, \ldots, \frac{\partial}{\partial y_{n}}\right]^{T}`	$\nabla:=\left[\frac{\partial}{\partial y_{1}}, \ldots, \frac{\partial}{\partial y_{n}}\right]^{T}$	*equal to the negative gradient, i.e.,* $\boldsymbol{r}(\boldsymbol{y}) = -\nabla Q(\boldsymbol{y})$, *where* $\nabla := \left[\frac{\partial}{\partial y_1}, \ldots, \frac{\partial}{\partial y_n}\right]^T$.
lyche-numerical-linear-algebra F03805	291	0.998	`\boldsymbol{y}=\boldsymbol{x}, \varepsilon=1`	$\boldsymbol{y}=\boldsymbol{x}, \varepsilon=1$	***Proof*** If $\boldsymbol{y} = \boldsymbol{x}$, $\varepsilon = 1$, and $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, then (13.5) simplifies to $Q(\boldsymbol{x}+\boldsymbol{h}) =$
lyche-numerical-linear-algebra F03806	291	0.998	`Q(\boldsymbol{x}+\boldsymbol{h})=`	$Q(\boldsymbol{x}+\boldsymbol{h})=$	***Proof*** If $\boldsymbol{y} = \boldsymbol{x}$, $\varepsilon = 1$, and $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, then (13.5) simplifies to $Q(\boldsymbol{x}+\boldsymbol{h}) =$
lyche-numerical-linear-algebra F03807	291	1.000	`Q(\boldsymbol{x})+\frac{1}{2} \boldsymbol{h}^{T} \boldsymbol{A} \boldsymbol{h}`	$Q(\boldsymbol{x})+\frac{1}{2} \boldsymbol{h}^{T} \boldsymbol{A} \boldsymbol{h}$	$Q(\boldsymbol{x}) + \frac{1}{2}\boldsymbol{h}^T\boldsymbol{A}\boldsymbol{h}$, and since $\boldsymbol{A}$ is positive definite $Q(\boldsymbol{x}+\boldsymbol{h}) > Q(\boldsymbol{x})$ for all nonzero
lyche-numerical-linear-algebra F03808	291	1.000	`Q(\boldsymbol{x}+\boldsymbol{h})>Q(\boldsymbol{x})`	$Q(\boldsymbol{x}+\boldsymbol{h})>Q(\boldsymbol{x})$	$Q(\boldsymbol{x}) + \frac{1}{2}\boldsymbol{h}^T\boldsymbol{A}\boldsymbol{h}$, and since $\boldsymbol{A}$ is positive definite $Q(\boldsymbol{x}+\boldsymbol{h}) > Q(\boldsymbol{x})$ for all nonzero
lyche-numerical-linear-algebra F03809	291	0.982	`\boldsymbol{h} \in \mathbb{R}^{n}`	$\boldsymbol{h} \in \mathbb{R}^{n}$	$\boldsymbol{h} \in \mathbb{R}^n$. It follows that $\boldsymbol{x}$ is the unique minimum of Q. Conversely, if $\boldsymbol{A}\boldsymbol{x} \neq \boldsymbol{b}$ and
lyche-numerical-linear-algebra F03810	291	0.982	`\boldsymbol{A} \boldsymbol{x} \neq \boldsymbol{b}`	$\boldsymbol{A} \boldsymbol{x} \neq \boldsymbol{b}$	$\boldsymbol{h} \in \mathbb{R}^n$. It follows that $\boldsymbol{x}$ is the unique minimum of Q. Conversely, if $\boldsymbol{A}\boldsymbol{x} \neq \boldsymbol{b}$ and
lyche-numerical-linear-algebra F03811	292	1.000	`\boldsymbol{h}:=\boldsymbol{r}(\boldsymbol{x})`	$\boldsymbol{h}:=\boldsymbol{r}(\boldsymbol{x})$	$\boldsymbol{h} := \boldsymbol{r}(\boldsymbol{x})$, then by (13.5), $Q(\boldsymbol{x}+\varepsilon\boldsymbol{h}) - Q(\boldsymbol{x}) = -\varepsilon(\boldsymbol{h}^T\boldsymbol{r}(x) - \frac{1}{2}\varepsilon\boldsymbol{h}^T\boldsymbol{A}\boldsymbol{h}) < 0$ for
lyche-numerical-linear-algebra F03812	292	1.000	`Q(\boldsymbol{x}+\varepsilon \boldsymbol{h})-Q( \boldsymbol{x})=-\varepsilon\left(\boldsymbol{h}^{T} \boldsymbol{r}(x)-\frac{1}{2} \varepsilon \boldsymbol{h}^{T} \boldsymbol{A} \boldsymbol{h}\right)<0`	$Q(\boldsymbol{x}+\varepsilon \boldsymbol{h})-Q(\boldsymbol{x})=-\varepsilon\left(\boldsymbol{h}^{T} \boldsymbol{r}(x)-\frac{1}{2} \varepsilon \boldsymbol{h}^{T} \boldsymbol{A} \boldsymbol{h}\right)<0$	$\boldsymbol{h} := \boldsymbol{r}(\boldsymbol{x})$, then by (13.5), $Q(\boldsymbol{x}+\varepsilon\boldsymbol{h}) - Q(\boldsymbol{x}) = -\varepsilon(\boldsymbol{h}^T\boldsymbol{r}(x) - \frac{1}{2}\varepsilon\boldsymbol{h}^T\boldsymbol{A}\boldsymbol{h}) < 0$ for
lyche-numerical-linear-algebra F03813	292	1.000	`\boldsymbol{x}_{0} \in \mathbb{R}^{n}`	$\boldsymbol{x}_{0} \in \mathbb{R}^{n}$	1. Choose $\boldsymbol{x}_0 \in \mathbb{R}^n$.
lyche-numerical-linear-algebra F03814	292	1.000	`\alpha_{k}`	α_{k}	Choose a "step length" α_k,
lyche-numerical-linear-algebra F03815	292	1.000	`\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}`	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}$	Compute $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k\boldsymbol{p}_k$.

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_F03816</td><td>293</td><td>0.999</td><td>Q\left(\boldsymbol{x}_{k+1}\right)</td><td>$Q\left(\boldsymbol{x}_{k+1}\right)$</td><td>For a fixed direction $\boldsymbol{p}_k$ we say that α_k is **optimal** if $Q(\boldsymbol{x}_{k+1})$ is as small as</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03817</td><td>293</td><td>1.000</td><td>Q\left(\boldsymbol{x}_{k}+\alpha \boldsymbol{p}_{k} \right)=Q\left(\boldsymbol{x}_{k}\right)-\alpha \boldsymbol{p}_{k}^{T} \boldsymbol{r}_{k}+\frac{1}{2} \alpha^{2} \boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}</td><td>$Q\left(\boldsymbol{x}_{k}+\alpha \boldsymbol{p}_{k} \right)=Q\left(\boldsymbol{x}_{k}\right)-\alpha \boldsymbol{p}_{k}^{T} \boldsymbol{r}_{k}+\frac{1}{2} \alpha^{2} \boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}$</td><td>By (13.5) we have $Q(\boldsymbol{x}_k+\alpha\boldsymbol{p}_k)=Q(\boldsymbol{x}_k)-\alpha\boldsymbol{p}_k^T\boldsymbol{r}_k+\frac{1}{2}\alpha^2\boldsymbol{p}_k^T\boldsymbol{A}\boldsymbol{p}_k$, where $\boldsymbol{r}_k :=$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03818</td><td>293</td><td>1.000</td><td>\boldsymbol{r}_{k}:=</td><td>$\boldsymbol{r}_{k}:=$</td><td>By (13.5) we have $Q(\boldsymbol{x}_k+\alpha\boldsymbol{p}_k)=Q(\boldsymbol{x}_k)-\alpha\boldsymbol{p}_k^T\boldsymbol{r}_k+\frac{1}{2}\alpha^2\boldsymbol{p}_k^T\boldsymbol{A}\boldsymbol{p}_k$, where $\boldsymbol{r}_k :=$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03819</td><td>293</td><td>1.000</td><td>\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k}</td><td>$\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k}$</td><td>$\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$. Since $\boldsymbol{p}_k^T\boldsymbol{A}\boldsymbol{p}_k\geq 0$ we find a minimum α_k by solving $\frac{\partial}{\partial\alpha}Q(\boldsymbol{x}_k+\alpha\boldsymbol{p}_k)=0$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03820</td><td>293</td><td>1.000</td><td>\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k} \geq 0</td><td>$\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k} \geq 0$</td><td>$\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$. Since $\boldsymbol{p}_k^T\boldsymbol{A}\boldsymbol{p}_k\geq 0$ we find a minimum α_k by solving $\frac{\partial}{\partial\alpha}Q(\boldsymbol{x}_k+\alpha\boldsymbol{p}_k)=0$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03821</td><td>293</td><td>1.000</td><td>\frac{\partial}{\partial \alpha} Q\left(\boldsymbol{x}_{k}+\alpha \boldsymbol{p}_{k}\right)=0</td><td>$\frac{\partial}{\partial \alpha} Q\left(\boldsymbol{x}_{k}+\alpha \boldsymbol{p}_{k}\right)=0$</td><td>$\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$. Since $\boldsymbol{p}_k^T\boldsymbol{A}\boldsymbol{p}_k\geq 0$ we find a minimum α_k by solving $\frac{\partial}{\partial\alpha}Q(\boldsymbol{x}_k+\alpha\boldsymbol{p}_k)=0$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03822</td><td>293</td><td>1.000</td><td>\boldsymbol{p}_{k}=\boldsymbol{r}_{k}</td><td>$\boldsymbol{p}_{k}=\boldsymbol{r}_{k}$</td><td>choose $\boldsymbol{p}_k=\boldsymbol{r}_k$ the negative gradient, and the optimal α_k. Starting from $\boldsymbol{x}_0$ we</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03823</td><td>293</td><td>1.000</td><td>k=0,1,2 \ldots</td><td>$k=0,1,2 \ldots$</td><td>compute for $k=0,1,2\ldots$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03824</td><td>293</td><td>1.000</td><td>\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{k}=\left(\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{r}_{k}\right)^{T} \boldsymbol{r}_{k}=0</td><td>$\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{k}=\left(\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{r}_{k}\right)^{T} \boldsymbol{r}_{k}=0$</td><td>are orthogonal. Indeed, by (13.10), $\boldsymbol{r}_{k+1}^T\boldsymbol{r}_k=(\boldsymbol{r}_k-\alpha_k\boldsymbol{A}\boldsymbol{r}_k)^T\boldsymbol{r}_k=0$ since $\alpha_k=$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03825</td><td>293</td><td>1.000</td><td>\alpha_{k}=</td><td>$\alpha_{k}=$</td><td>are orthogonal. Indeed, by (13.10), $\boldsymbol{r}_{k+1}^T\boldsymbol{r}_k=(\boldsymbol{r}_k-\alpha_k\boldsymbol{A}\boldsymbol{r}_k)^T\boldsymbol{r}_k=0$ since $\alpha_k=$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03826</td><td>293</td><td>0.887</td><td>\frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{ \boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}}</td><td>$\frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{ \boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}}$</td><td>$\frac{\boldsymbol{r}_k^T\boldsymbol{r}_k}{\boldsymbol{r}_k^T\boldsymbol{A}\boldsymbol{r}_k}$ and $\boldsymbol{A}$ is symmetric.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03827</td><td>293</td><td>1.000</td><td>\boldsymbol{x}_{0}=[-1,-1 / 2]^{T}</td><td>$\boldsymbol{x}_{0}=[-1,-1 / 2]^{T}$</td><td>Starting with $\boldsymbol{x}_0=[-1,-1/2]^T$ and $\boldsymbol{r}_0=-\boldsymbol{A}\boldsymbol{x}_0=[3/2,0]^T$ we find</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03828</td><td>293</td><td>1.000</td><td>\boldsymbol{r}_{0}=-\boldsymbol{A} \boldsymbol{x}_{0}=[3 / 2,0]^{T}</td><td>$\boldsymbol{r}_{0}=-\boldsymbol{A} \boldsymbol{x}_{0}=[3 / 2,0]^{T}$</td><td>Starting with $\boldsymbol{x}_0=[-1,-1/2]^T$ and $\boldsymbol{r}_0=-\boldsymbol{A}\boldsymbol{x}_0=[3/2,0]^T$ we find</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03829</td><td>294</td><td>1.000</td><td>k \geq 1</td><td>$k \geq 1$</td><td>and in general for $k\geq 1$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03830</td><td>294</td><td>1.000</td><td>\alpha_{k}=1 / 2</td><td>$\alpha_{k}=1 / 2$</td><td>Since $\alpha_k=1/2$ is constant for all k the methods of Richardson, Jacobi and steepest</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03831</td><td>294</td><td>1.000</td><td>\left\|\boldsymbol{x}_{j+1}\right\|_{2} /\left\| \boldsymbol{x}_{j}\right\|=\left\|\boldsymbol{r}_{j+1} \right\|_{2} /\left\|\boldsymbol{r}_{j}\right\|_{2}=1 / 2</td><td>$\left\|\boldsymbol{x}_{j+1}\right\|_{2} /\left\| \boldsymbol{x}_{j}\right\|=\left\|\boldsymbol{r}_{j+1} \right\|_{2} /\left\|\boldsymbol{r}_{j}\right\|_{2}=1 / 2$</td><td>of convergence is determined from $\|\boldsymbol{x}_{j+1}\|_2/\|\boldsymbol{x}_j\| = \|\boldsymbol{r}_{j+1}\|_2/\|\boldsymbol{r}_j\|_2 = 1/2$ for</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03832</td><td>294</td><td>1.000</td><td>\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{j}=0</td><td>$\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{j}=0$</td><td>**$\boldsymbol{A}$-orthogonal search directions** i.e., $\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_j=0$ for all $i\neq j$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03833</td><td>294</td><td>1.000</td><td>\boldsymbol{r}_{0}=</td><td>$\boldsymbol{r}_{0}=$</td><td>As in the steepest descent method we choose a starting vector $\boldsymbol{x}_0\in\mathbb{R}^n$. If $\boldsymbol{r}_0=$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03834</td><td>294</td><td>1.000</td><td>\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{0}= \mathbf{0}</td><td>$\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{0}= \mathbf{0}$</td><td>$\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_0=\boldsymbol{0}$ then $\boldsymbol{x}_0$ is the exact solution and we are finished, otherwise we initially</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03835</td><td>294</td><td>1.000</td><td>\boldsymbol{r}_{1}^{T} \boldsymbol{r}_{0}=0</td><td>$\boldsymbol{r}_{1}^{T} \boldsymbol{r}_{0}=0$</td><td>make a steepest descent step. It follows that $\boldsymbol{r}_1^T\boldsymbol{r}_0=0$ and $\boldsymbol{p}_0:=\boldsymbol{r}_0$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03836</td><td>294</td><td>1.000</td><td>\boldsymbol{p}_{0}:=\boldsymbol{r}_{0}</td><td>$\boldsymbol{p}_{0}:=\boldsymbol{r}_{0}$</td><td>make a steepest descent step. It follows that $\boldsymbol{r}_1^T\boldsymbol{r}_0=0$ and $\boldsymbol{p}_0:=\boldsymbol{r}_0$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03837</td><td>294</td><td>1.000</td><td>j \geq 0</td><td>$j \geq 0$</td><td>For the general case we define for $j\geq 0$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F03838</td><td>294</td><td>1.000</td><td>\boldsymbol{p}_{j}</td><td>$\boldsymbol{p}_{j}$</td><td>1. $\boldsymbol{p}_j$ is computed by the Gram-Schmidt orthogonalization process applied to the</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F03839	294	0.999	`\boldsymbol{r}_{0}, \ldots, \boldsymbol{r}_{j}`	$\boldsymbol{r}_{0}, \ldots, \boldsymbol{r}_{j}$	residuals $\boldsymbol{r}_0, \ldots, \boldsymbol{r}_j$ using the $\boldsymbol{A}$-inner product. The search directions are there-
lyche-numerical-linear-al F03840	294	1.000	`\alpha_{j}`	α_{j}	3. It can be shown that the step length α_j is optimal for all j (cf. Exercise 13.7).
lyche-numerical-linear-al F03841	295	1.000	`\boldsymbol{x}_{j}`	$\boldsymbol{x}_{j}$	$\boldsymbol{x}_j$ *is well defined,* $\boldsymbol{r}_j \neq 0$, *and* $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ *for* $i, j = 0, 1, \ldots, k$, $i \neq j$. *Then* $\boldsymbol{x}_{k+1}$
lyche-numerical-linear-al F03842	295	1.000	`\boldsymbol{r}_{j} \neq 0`	$\boldsymbol{r}_{j} \neq 0$	$\boldsymbol{x}_j$ *is well defined,* $\boldsymbol{r}_j \neq 0$, *and* $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ *for* $i, j = 0, 1, \ldots, k$, $i \neq j$. *Then* $\boldsymbol{x}_{k+1}$
lyche-numerical-linear-al F03843	295	1.000	`\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{j}=0`	$\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{j}=0$	$\boldsymbol{x}_j$ *is well defined,* $\boldsymbol{r}_j \neq 0$, *and* $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ *for* $i, j = 0, 1, \ldots, k$, $i \neq j$. *Then* $\boldsymbol{x}_{k+1}$
lyche-numerical-linear-al F03844	295	1.000	`i, j=0,1, \ldots, k, i \neq j`	$i, j=0,1, \ldots, k, i \neq j$	$\boldsymbol{x}_j$ *is well defined,* $\boldsymbol{r}_j \neq 0$, *and* $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ *for* $i, j = 0, 1, \ldots, k$, $i \neq j$. *Then* $\boldsymbol{x}_{k+1}$
lyche-numerical-linear-al F03845	295	1.000	`\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{j}=0`	$\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{j}=0$	*is well defined and* $\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$ *for* $j = 0, 1, \ldots, k$.
lyche-numerical-linear-al F03846	295	1.000	`j=0,1, \ldots, k`	$j=0,1, \ldots, k$	*is well defined and* $\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$ *for* $j = 0, 1, \ldots, k$.
lyche-numerical-linear-al F03847	295	1.000	`\boldsymbol{r}_{j}`	$\boldsymbol{r}_{j}$	***Proof*** Since the residuals $\boldsymbol{r}_j$ are orthogonal and nonzero for $j \leq k$, they are linearly
lyche-numerical-linear-al F03848	295	1.000	`j \leq k`	$j \leq k$	***Proof*** Since the residuals $\boldsymbol{r}_j$ are orthogonal and nonzero for $j \leq k$, they are linearly
lyche-numerical-linear-al F03849	295	1.000	`\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{i}=0`	$\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{i}=0$	and $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$ for $i < k$. But then $\boldsymbol{x}_{k+1}$ and $\boldsymbol{r}_{k+1}$ are well defined. Now
lyche-numerical-linear-al F03850	295	1.000	`i<k`	$i<k$	and $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$ for $i < k$. But then $\boldsymbol{x}_{k+1}$ and $\boldsymbol{r}_{k+1}$ are well defined. Now
lyche-numerical-linear-al F03851	295	1.000	`\boldsymbol{r}_{k+1}`	$\boldsymbol{r}_{k+1}$	and $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$ for $i < k$. But then $\boldsymbol{x}_{k+1}$ and $\boldsymbol{r}_{k+1}$ are well defined. Now
lyche-numerical-linear-al F03852	295	1.000	`j<k`	$j<k$	orthogonality for $j < k$ and by the definition of α_k for $j = k$. This completes
lyche-numerical-linear-al F03853	295	1.000	`\operatorname{dim} \mathbb{R}^{n}=n`	$\operatorname{dim} \mathbb{R}^{n}=n$	orthogonal and therefore linearly independent if they are nonzero. Since $\dim \mathbb{R}^n = n$
lyche-numerical-linear-al F03854	295	1.000	`\boldsymbol{r}_{0}, \ldots, \boldsymbol{r}_{n}`	$\boldsymbol{r}_{0}, \ldots, \boldsymbol{r}_{n}$	the $n + 1$ residuals $\boldsymbol{r}_0, \ldots, \boldsymbol{r}_n$ cannot all be nonzero and we must have $\boldsymbol{r}_k = 0$ for
lyche-numerical-linear-al F03855	295	1.000	`\boldsymbol{r}_{k}=0`	$\boldsymbol{r}_{k}=0$	the $n + 1$ residuals $\boldsymbol{r}_0, \ldots, \boldsymbol{r}_n$ cannot all be nonzero and we must have $\boldsymbol{r}_k = 0$ for
lyche-numerical-linear-al F03856	295	1.000	`j=k+1`	$j=k+1$	With $j = k + 1$ (13.11) therefore takes the simple form $\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$ and
lyche-numerical-linear-al F03857	295	1.000	`\boldsymbol{p}_{k+1}=\boldsymbol{r}_{k+1}+\beta_{k} \boldsymbol{p}_{k}`	$\boldsymbol{p}_{k+1}=\boldsymbol{r}_{k+1}+\beta_{k} \boldsymbol{p}_{k}$	With $j = k + 1$ (13.11) therefore takes the simple form $\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$ and
lyche-numerical-linear-al F03858	295	1.000	`\boldsymbol{x}_{0}, \boldsymbol{p}_{0}= \boldsymbol{r}_{0}=`	$\boldsymbol{x}_{0}, \boldsymbol{p}_{0}= \boldsymbol{r}_{0}=$	To summarize, in the **conjugate gradient method** we start with $\boldsymbol{x}_0$, $\boldsymbol{p}_0 = \boldsymbol{r}_0 =$
lyche-numerical-linear-al F03859	295	1.000	`\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{0}`	$\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{0}$	$\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_0$ and then generate a sequence of vectors $\{\boldsymbol{x}_k\}$ as follows:
lyche-numerical-linear-al F03860	296	1.000	`\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right] \left[\begin{array}{l}x_{1} \\ x_{2}\end{array}\right]= \left[\begin{array}{l}0 \\ 0\end{array}\right]`	$\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right] \left[\begin{array}{l}x_{1} \\ x_{2}\end{array}\right]= \left[\begin{array}{l}0 \\ 0\end{array}\right]$	itive definite linear system $\left[\begin{smallmatrix}2 & -1 \\ -1 & 2\end{smallmatrix}\right]\left[\begin{smallmatrix}x_1 \\ x_2\end{smallmatrix}\right] = \left[\begin{smallmatrix}0 \\ 0\end{smallmatrix}\right]$. Starting as in Example 13.1 with
lyche-numerical-linear-al F03861	296	1.000	`\boldsymbol{x}_{0}=\left[\begin{array}{c}-1 \\ -1 / 2 \end{array}\right]`	$\boldsymbol{x}_{0}=\left[\begin{array}{c}-1 \\ -1 / 2 \end{array}\right]$	$\boldsymbol{x}_0 = \left[\begin{smallmatrix}-1 \\ -1/2\end{smallmatrix}\right]$ we find $\boldsymbol{p}_0 = \boldsymbol{r}_0 = \left[\begin{smallmatrix}3/2 \\ 0\end{smallmatrix}\right]$ and then
lyche-numerical-linear-al F03862	296	1.000	`\boldsymbol{p}_{0}=\boldsymbol{r}_{0}=\left[ \begin{array}{c}3 / 2 \\ 0\end{array}\right]`	$\boldsymbol{p}_{0}=\boldsymbol{r}_{0}=\left[\begin{array}{c}3 / 2 \\ 0\end{array}\right]$	$\boldsymbol{x}_0 = \left[\begin{smallmatrix}-1 \\ -1/2\end{smallmatrix}\right]$ we find $\boldsymbol{p}_0 = \boldsymbol{r}_0 = \left[\begin{smallmatrix}3/2 \\ 0\end{smallmatrix}\right]$ and then
lyche-numerical-linear-al F03863	296	1.000	`\boldsymbol{x}_{2}`	$\boldsymbol{x}_{2}$	Thus $\boldsymbol{x}_2$ is the exact solution as illustrated in the right part of Fig. 13.1.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F03864	296	0.969	`\left|\boldsymbol{r}_{k}\right|_{2} /|\boldsymbol{b}|_{2} \leq`	$\lVert \boldsymbol{r}_k \rVert_2 / \lVert \boldsymbol{b} \rVert_2 \leq$	starting vector for the iteration. The iteration is stopped when $\lVert \boldsymbol{r}_k \rVert_2 / \lVert \boldsymbol{b} \rVert_2 \leq$ tol or
lyche-numerical-linear-al F03865	297	0.803	`k>`	$k >$	$k >$ itmax. K is the number of iterations used:
lyche-numerical-linear-al F03866	297	0.996	`(\boldsymbol{t}=\boldsymbol{A} \boldsymbol{p})`	$(\boldsymbol{t}=\boldsymbol{A} \boldsymbol{p})$	1. one matrix times vector ($\boldsymbol{t} = \boldsymbol{A}\boldsymbol{p}$),
lyche-numerical-linear-al F03867	297	1.000	`\left(\boldsymbol{p}^{T} \boldsymbol{t}\right.`	$\left(\boldsymbol{p}^{T} \boldsymbol{t}\right.$	2. two inner products (($\boldsymbol{p}^T\boldsymbol{t}$ and $\boldsymbol{r}^T\boldsymbol{r}$),
lyche-numerical-linear-al F03868	297	1.000	`\left.\boldsymbol{r}^{T} \boldsymbol{r}\right)`	$\left.\boldsymbol{r}^{T} \boldsymbol{r}\right)$	2. two inner products (($\boldsymbol{p}^T\boldsymbol{t}$ and $\boldsymbol{r}^T\boldsymbol{r}$),
lyche-numerical-linear-al F03869	297	0.758	`\boldsymbol{x}=\boldsymbol{x}+a \boldsymbol{p}, \boldsymbol{r}=\boldsymbol{r}-a \boldsymbol{t}`	$\boldsymbol{x}=\boldsymbol{x}+a \boldsymbol{p}, \boldsymbol{r}=\boldsymbol{r}-a \boldsymbol{t}$	3. three vector-plus-scalar-times-vector ($\boldsymbol{x} = \boldsymbol{x} + a\boldsymbol{p}$, $\boldsymbol{r} = \boldsymbol{r} - a\boldsymbol{t}$ and $\boldsymbol{p} = \boldsymbol{r} +$
lyche-numerical-linear-al F03870	297	0.758	`\boldsymbol{p}=\boldsymbol{r}+`	$\boldsymbol{p}=\boldsymbol{r}+$	3. three vector-plus-scalar-times-vector ($\boldsymbol{x} = \boldsymbol{x} + a\boldsymbol{p}$, $\boldsymbol{r} = \boldsymbol{r} - a\boldsymbol{t}$ and $\boldsymbol{p} = \boldsymbol{r} +$
lyche-numerical-linear-al F03871	297	1.000	`\boldsymbol{t}=\boldsymbol{A} \boldsymbol{p}`	$\boldsymbol{t}=\boldsymbol{A} \boldsymbol{p}$	The dominating part is the computation of $\boldsymbol{t} = \boldsymbol{A}\boldsymbol{p}$.
lyche-numerical-linear-al F03872	297	1.000	`\boldsymbol{T}_{2}:=\boldsymbol{T}_{1} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}_{1}`	$\boldsymbol{T}_{2}:=\boldsymbol{T}_{1} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}_{1}$	Kronecker sum $\boldsymbol{T}_2 := \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1$, where $\boldsymbol{T}_1 := \operatorname{tridiag}_m(a,d,a) \in \mathbb{R}^{m\times m}$
lyche-numerical-linear-al F03873	297	1.000	`\boldsymbol{T}_{1}:=\operatorname{tridiag}_{m}(a, d, a) \in \mathbb{R}^{m \times m}`	$\boldsymbol{T}_{1}:=\operatorname{tridiag}_{m}(a, d, a) \in \mathbb{R}^{m \times m}$	Kronecker sum $\boldsymbol{T}_2 := \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1$, where $\boldsymbol{T}_1 := \operatorname{tridiag}_m(a,d,a) \in \mathbb{R}^{m\times m}$
lyche-numerical-linear-al F03874	297	1.000	`\boldsymbol{f}=[1, \ldots, 1]^{T} \in \mathbb{R}^{n}`	$\boldsymbol{f}=[1, \ldots, 1]^{T} \in \mathbb{R}^{n}$	(cf. Theorem 10.2). We set $h = 1/(m+1)$ and $\boldsymbol{f} = [1,\ldots,1]^T \in \mathbb{R}^n$.
lyche-numerical-linear-al F03875	297	1.000	`a=1 / 9, d=5 / 18`	$a=1 / 9, d=5 / 18$	1. $a = 1/9$, $d = 5/18$, the Averaging matrix.
lyche-numerical-linear-al F03876	297	1.000	`a=-1, d=2`	$a=-1, d=2$	2. $a = -1$, $d = 2$, the Poisson matrix.
lyche-numerical-linear-al F03877	297	1.000	`O(5 n)`	$O(5 n)$	Note that for our test problems $\boldsymbol{T}_2$ only has $O(5n)$ nonzero elements. Therefore,
lyche-numerical-linear-al F03878	297	1.000	`\boldsymbol{t}`	$\boldsymbol{t}$	taking advantage of the sparseness of $\boldsymbol{T}_2$ we can compute $\boldsymbol{t}$ in Algorithm 13.1
lyche-numerical-linear-al F03879	298	1.000	`\boldsymbol{V}, \boldsymbol{R}, \boldsymbol{P}, \boldsymbol{B}, \boldsymbol{T} \in`	$\boldsymbol{V}, \boldsymbol{R}, \boldsymbol{P}, \boldsymbol{B}, \boldsymbol{T} \in$	tageous to use a matrix equation formulation. We define matrices $\boldsymbol{V}, \boldsymbol{R}, \boldsymbol{P}, \boldsymbol{B}, \boldsymbol{T} \in$
lyche-numerical-linear-al F03880	298	1.000	`\boldsymbol{x}=\operatorname{vec}(\boldsymbol{V}), \boldsymbol{r}=\operatorname{vec}(\boldsymbol{R}), \boldsymbol{p}=\operatorname{vec}(\boldsymbol{P}), \boldsymbol{t}=\operatorname{vec}(\boldsymbol{T})`	$\boldsymbol{x}=\operatorname{vec}(\boldsymbol{V}), \boldsymbol{r}=\operatorname{vec}(\boldsymbol{R}), \boldsymbol{p}=\operatorname{vec}(\boldsymbol{P}), \boldsymbol{t}=\operatorname{vec}(\boldsymbol{T})$	$\mathbb{R}^{m\times m}$ by $\boldsymbol{x} = \operatorname{vec}(\boldsymbol{V})$, $\boldsymbol{r} = \operatorname{vec}(\boldsymbol{R})$, $\boldsymbol{p} = \operatorname{vec}(\boldsymbol{P})$, $\boldsymbol{t} = \operatorname{vec}(\boldsymbol{T})$, and $h^2\boldsymbol{f} = \operatorname{vec}(\boldsymbol{B})$.
lyche-numerical-linear-al F03881	298	1.000	`h^{2} \boldsymbol{f}=\operatorname{vec}(\boldsymbol{B})`	$h^{2} \boldsymbol{f}=\operatorname{vec}(\boldsymbol{B})$	$\mathbb{R}^{m\times m}$ by $\boldsymbol{x} = \operatorname{vec}(\boldsymbol{V})$, $\boldsymbol{r} = \operatorname{vec}(\boldsymbol{R})$, $\boldsymbol{p} = \operatorname{vec}(\boldsymbol{P})$, $\boldsymbol{t} = \operatorname{vec}(\boldsymbol{T})$, and $h^2\boldsymbol{f} = \operatorname{vec}(\boldsymbol{B})$.
lyche-numerical-linear-al F03882	298	1.000	`\boldsymbol{T}_{2} \boldsymbol{x}=h^{2} \boldsymbol{f} \Longleftrightarrow \boldsymbol{T}_{1} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}_{1}=\boldsymbol{B}`	$\boldsymbol{T}_{2} \boldsymbol{x}=h^{2} \boldsymbol{f} \Longleftrightarrow \boldsymbol{T}_{1} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}_{1}=\boldsymbol{B}$	Then $\boldsymbol{T}_2\boldsymbol{x} = h^2\boldsymbol{f} \Longleftrightarrow \boldsymbol{T}_1\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T}_1 = \boldsymbol{B}$, and $\boldsymbol{t} = \boldsymbol{T}_2\boldsymbol{p} \Longleftrightarrow \boldsymbol{T} = \boldsymbol{T}_1\boldsymbol{P} + \boldsymbol{P}\boldsymbol{T}_1$.
lyche-numerical-linear-al F03883	298	1.000	`\boldsymbol{t}=\boldsymbol{T}_{2} \boldsymbol{p} \Longleftrightarrow \boldsymbol{T}=\boldsymbol{T}_{1} \boldsymbol{P}+\boldsymbol{P} \boldsymbol{T}_{1}`	$\boldsymbol{t}=\boldsymbol{T}_{2} \boldsymbol{p} \Longleftrightarrow \boldsymbol{T}=\boldsymbol{T}_{1} \boldsymbol{P}+\boldsymbol{P} \boldsymbol{T}_{1}$	Then $\boldsymbol{T}_2\boldsymbol{x} = h^2\boldsymbol{f} \Longleftrightarrow \boldsymbol{T}_1\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T}_1 = \boldsymbol{B}$, and $\boldsymbol{t} = \boldsymbol{T}_2\boldsymbol{p} \Longleftrightarrow \boldsymbol{T} = \boldsymbol{T}_1\boldsymbol{P} + \boldsymbol{P}\boldsymbol{T}_1$.
lyche-numerical-linear-al F03884	298	0.958	`t o l=10^{-8}`	$tol=10^{-8}$	For both the averaging- and Poison matrix we use $tol = 10^{-8}$.
lyche-numerical-linear-al F03885	298	1.000	`K / \sqrt{n}`	$K / \sqrt{n}$	of iterations, and $K/\sqrt{n}$.
lyche-numerical-linear-al F03886	299	1.000	`|\boldsymbol{x}|_{\boldsymbol{A}}:=\sqrt{\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}}`	$\lVert\boldsymbol{x}\rVert_{\boldsymbol{A}}:=\sqrt{\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}}$	Recall that the $\boldsymbol{A}$-norm of a vector $\boldsymbol{x} \in \mathbb{R}^n$ is given by $\lVert\boldsymbol{x}\rVert_{\boldsymbol{A}} := \sqrt{\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}}$. The
lyche-numerical-linear-al F03887	299	0.995	`\kappa=\operatorname{cond}_{2}(\boldsymbol{A}):=\lambda_{\text {max }} / \lambda_{\text {min }}`	$\kappa=\operatorname{cond}_{2}(\boldsymbol{A}):=\lambda_{\text {max }} / \lambda_{\text {min }}$	*Here $\kappa = cond_2(\boldsymbol{A}) := \lambda_{max}/\lambda_{min}$ is the spectral condition number of $\boldsymbol{A}$, where*
lyche-numerical-linear-al F03888	299	1.000	`\frac{\kappa-1}{\kappa+1}<1`	$\frac{\kappa-1}{\kappa+1}<1$	1. Since $\frac{\kappa-1}{\kappa+1} < 1$ the steepest descent method always converges for a positive

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F03889	299	■1.000	\frac{\kappa-1}{\kappa+1}	$\frac{\kappa-1}{\kappa+1}$	definite matrix. The convergence can be slow when $\frac{\kappa-1}{\kappa+1}$ is close to one, and this
lyche-numerical-linear-algebra F03890	300	■1.000	d=5 / 18, a=1 / 9	$d=5/18, a=1/9$	For the averaging problem given by $d = 5/18$, $a = 1/9$, the largest and smallest
lyche-numerical-linear-algebra F03891	300	■0.999	\lambda_{\text {max }}=\frac{5}{9}+\frac{4}{9} \cos (\pi h)	$\lambda_{\text{max }}=\frac{5}{9}+\frac{4}{9}\cos(\pi h)$	eigenvalue of $\boldsymbol{T}_2$ are given by $\lambda_{max} = \frac{5}{9} + \frac{4}{9}\cos(\pi h)$ and $\lambda_{min} = \frac{5}{9} - \frac{4}{9}\cos(\pi h)$.
lyche-numerical-linear-algebra F03892	300	■0.999	\lambda_{\text {min }}=\frac{5}{9}-\frac{4}{9} \cos (\pi h)	$\lambda_{\text{min }}=\frac{5}{9}-\frac{4}{9}\cos(\pi h)$	eigenvalue of $\boldsymbol{T}_2$ are given by $\lambda_{max} = \frac{5}{9} + \frac{4}{9}\cos(\pi h)$ and $\lambda_{min} = \frac{5}{9} - \frac{4}{9}\cos(\pi h)$.
lyche-numerical-linear-algebra F03893	300	■1.000	\mathbb{W}_{0}=\{\mathbf{0}\}	$\mathbb{W}_{0}=\{\mathbf{0}\}$	The Krylov spaces are defined by $\mathbb{W}_0 = \{\mathbf{0}\}$ and
lyche-numerical-linear-algebra F03894	301	■1.000	\operatorname{dim}\left(\mathbb{W}_{k}\right) \leq k	$\operatorname{dim}\left(\mathbb{W}_{k}\right) \leq k$	with $\dim(\mathbb{W}_k) \leq k$ for all $k \geq 0$. Moreover, If $\boldsymbol{v} \in \mathbb{W}_k$ then $\boldsymbol{A}\boldsymbol{v} \in \mathbb{W}_{k+1}$.
lyche-numerical-linear-algebra F03895	301	■1.000	\boldsymbol{v} \in \mathbb{W}_{k}	$\boldsymbol{v} \in \mathbb{W}_{k}$	with $\dim(\mathbb{W}_k) \leq k$ for all $k \geq 0$. Moreover, If $\boldsymbol{v} \in \mathbb{W}_k$ then $\boldsymbol{A}\boldsymbol{v} \in \mathbb{W}_{k+1}$.
lyche-numerical-linear-algebra F03896	301	■1.000	\boldsymbol{A} \boldsymbol{v} \in \mathbb{W}_{k+1}	$\boldsymbol{A} \boldsymbol{v} \in \mathbb{W}_{k+1}$	with $\dim(\mathbb{W}_k) \leq k$ for all $k \geq 0$. Moreover, If $\boldsymbol{v} \in \mathbb{W}_k$ then $\boldsymbol{A}\boldsymbol{v} \in \mathbb{W}_{k+1}$.
lyche-numerical-linear-algebra F03897	301	■1.000	\boldsymbol{p}_{0}=\boldsymbol{r}_{0}	$\boldsymbol{p}_{0}=\boldsymbol{r}_{0}$	***Proof*** Equation (13.22) clearly holds for $k = 0$ since $\boldsymbol{p}_0 = \boldsymbol{r}_0$. Suppose it holds
lyche-numerical-linear-algebra F03898	301	■1.000	\boldsymbol{r}_{k+1}=\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k} \in \mathbb{W}_{k+2}, \boldsymbol{p}_{k+1}=\boldsymbol{r}_{k+1}+\beta_{k} \boldsymbol{p}_{k} \in	$\boldsymbol{r}_{k+1}=\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k} \in \mathbb{W}_{k+2}, \boldsymbol{p}_{k+1}=\boldsymbol{r}_{k+1}+\beta_{k} \boldsymbol{p}_{k} \in$	for some $k \geq 0$. Then $\boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{A}\boldsymbol{p}_k \in \mathbb{W}_{k+2}$, $\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k \in$
lyche-numerical-linear-algebra F03899	301	■0.995	\mathbb{W}_{k+2}	$\mathbb{W}_{k+2}$	$\mathbb{W}_{k+2}$ and $\boldsymbol{x}_{k+1} - \boldsymbol{x}_0 \stackrel{(13.12)}{=} \boldsymbol{x}_k - \boldsymbol{x}_0 + \alpha_k \boldsymbol{p}_k \in \mathbb{W}_{k+1}$. Thus (13.22) follows by
lyche-numerical-linear-algebra F03900	301	■0.995	\boldsymbol{x}_{k+1}-\boldsymbol{x}_{0} \stackrel{\text { (13.12) }}{=} \boldsymbol{x}_{k}-\boldsymbol{x}_{0}+ \alpha_{k} \boldsymbol{p}_{k} \in \mathbb{W}_{k+1}	$\boldsymbol{x}_{k+1}-\boldsymbol{x}_{0} \stackrel{\text{ (13.12) }}{=} \boldsymbol{x}_{k}-\boldsymbol{x}_{0}+\alpha_{k} \boldsymbol{p}_{k} \in \mathbb{W}_{k+1}$	$\mathbb{W}_{k+2}$ and $\boldsymbol{x}_{k+1} - \boldsymbol{x}_0 \stackrel{(13.12)}{=} \boldsymbol{x}_k - \boldsymbol{x}_0 + \alpha_k \boldsymbol{p}_k \in \mathbb{W}_{k+1}$. Thus (13.22) follows by
lyche-numerical-linear-algebra F03901	301	■0.984	\boldsymbol{w} \in \mathbb{W}_{k}	$\boldsymbol{w} \in \mathbb{W}_{k}$	induction. The equation (13.23) follows since any $\boldsymbol{w} \in \mathbb{W}_k$ is a linear combination
lyche-numerical-linear-algebra F03902	301	■1.000	\left\{\boldsymbol{r}_{0}, \boldsymbol{r}_{1}, \ldots, \boldsymbol{r}_{k-1}\right\}	$\left\{\boldsymbol{r}_{0}, \boldsymbol{r}_{1}, \ldots, \boldsymbol{r}_{k-1}\right\}$	of $\{\boldsymbol{r}_0, \boldsymbol{r}_1, \ldots, \boldsymbol{r}_{k-1}\}$ and also $\{\boldsymbol{p}_0, \boldsymbol{p}_1, \ldots, \boldsymbol{p}_{k-1}\}$.
lyche-numerical-linear-algebra F03903	301	■1.000	\left\{\boldsymbol{p}_{0}, \boldsymbol{p}_{1}, \ldots, \boldsymbol{p}_{k-1}\right\}	$\left\{\boldsymbol{p}_{0}, \boldsymbol{p}_{1}, \ldots, \boldsymbol{p}_{k-1}\right\}$	of $\{\boldsymbol{r}_0, \boldsymbol{r}_1, \ldots, \boldsymbol{r}_{k-1}\}$ and also $\{\boldsymbol{p}_0, \boldsymbol{p}_1, \ldots, \boldsymbol{p}_{k-1}\}$.
lyche-numerical-linear-algebra F03904	301	■0.995	\boldsymbol{u}:=\boldsymbol{x}_{k}-\boldsymbol{x}_{0}- \boldsymbol{w}	$\boldsymbol{u}:=\boldsymbol{x}_{k}-\boldsymbol{x}_{0}-\boldsymbol{w}$	***Proof*** Fix k, let $\boldsymbol{w} \in \mathbb{W}_k$ and $\boldsymbol{u} := \boldsymbol{x}_k - \boldsymbol{x}_0 - \boldsymbol{w}$. By (13.22) $\boldsymbol{u} \in \mathbb{W}_k$ and then (13.23)
lyche-numerical-linear-algebra F03905	301	■0.995	\boldsymbol{u} \in \mathbb{W}_{k}	$\boldsymbol{u} \in \mathbb{W}_{k}$	***Proof*** Fix k, let $\boldsymbol{w} \in \mathbb{W}_k$ and $\boldsymbol{u} := \boldsymbol{x}_k - \boldsymbol{x}_0 - \boldsymbol{w}$. By (13.22) $\boldsymbol{u} \in \mathbb{W}_k$ and then (13.23)
lyche-numerical-linear-algebra F03906	301	■1.000	\left\langle\boldsymbol{x}-\boldsymbol{x}_{k}, \boldsymbol{u}\right\rangle=\boldsymbol{r}_{k}^{T} \boldsymbol{u}=0	$\left\langle\boldsymbol{x}-\boldsymbol{x}_{k}, \boldsymbol{u}\right\rangle=\boldsymbol{r}_{k}^{T} \boldsymbol{u}=0$	implies that $\langle \boldsymbol{x} - \boldsymbol{x}_k, \boldsymbol{u} \rangle = \boldsymbol{r}_k^T \boldsymbol{u} = 0$. Using Corollary 5.2 we obtain
lyche-numerical-linear-algebra F03907	301	■1.000	\boldsymbol{u}=\mathbf{0}	$\boldsymbol{u}=\mathbf{0}$	with equality for $\boldsymbol{u} = \mathbf{0}$.
lyche-numerical-linear-algebra F03908	301	■1.000	\boldsymbol{x}_{0} \neq \mathbf{0}	$\boldsymbol{x}_{0} \neq \mathbf{0}$	solution $\boldsymbol{x}$ in the $\boldsymbol{A}$-norm. More generally, if $\boldsymbol{x}_0 \neq \mathbf{0}$ then $\boldsymbol{x} - \boldsymbol{x}_k = (\boldsymbol{x} - \boldsymbol{x}_0) -$
lyche-numerical-linear-algebra F03909	301	■1.000	\boldsymbol{x}-\boldsymbol{x}_{k}=\left(\boldsymbol{x}- \boldsymbol{x}_{0}\right)-	$\boldsymbol{x}-\boldsymbol{x}_{k}=\left(\boldsymbol{x}-\boldsymbol{x}_{0}\right)-$	solution $\boldsymbol{x}$ in the $\boldsymbol{A}$-norm. More generally, if $\boldsymbol{x}_0 \neq \mathbf{0}$ then $\boldsymbol{x} - \boldsymbol{x}_k = (\boldsymbol{x} - \boldsymbol{x}_0) -$
lyche-numerical-linear-algebra F03910	301	■1.000	\left(\boldsymbol{x}_{k}-\boldsymbol{x}_{0}\right)	$\left(\boldsymbol{x}_{k}-\boldsymbol{x}_{0}\right)$	$(\boldsymbol{x}_k - \boldsymbol{x}_0)$ and $\boldsymbol{x}_k - \boldsymbol{x}_0$ is the element in $\mathbb{W}_k$ that is closest to $\boldsymbol{x} - \boldsymbol{x}_0$ in the $\boldsymbol{A}$-norm.
lyche-numerical-linear-algebra F03911	301	■1.000	\boldsymbol{x}_{k}-\boldsymbol{x}_{0}	$\boldsymbol{x}_{k}-\boldsymbol{x}_{0}$	$(\boldsymbol{x}_k - \boldsymbol{x}_0)$ and $\boldsymbol{x}_k - \boldsymbol{x}_0$ is the element in $\mathbb{W}_k$ that is closest to $\boldsymbol{x} - \boldsymbol{x}_0$ in the $\boldsymbol{A}$-norm.
lyche-numerical-linear-algebra F03912	301	■1.000	p(t):=\sum_{j=0}^{m} a_{j} t^{m}	$p(t):=\sum_{j=0}^{m} a_{j} t^{m}$	Recall that to each polynomial $p(t) := \sum_{j=0}^{m} a_j t^m$ there corresponds a matrix

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F03913	301	0.808	p(\boldsymbol{A}):=a_{0} \boldsymbol{I}+a_{1} \boldsymbol{A}+\cdots+a_{m} \boldsymbol{A}^{m}	$p(\boldsymbol{A}):=a_{0} \boldsymbol{I}+a_{1} \boldsymbol{A}+\cdots+a_{m} \boldsymbol{A}^{m}$	polynomial $p(\boldsymbol{A}) := a_0\boldsymbol{I}+a_1\boldsymbol{A}+\cdots+a_m\boldsymbol{A}^m$. Moreover, if $(\lambda_j, \boldsymbol{u}_j)$ are eigenpairs
lyche-numerical-linear-algebra F03914	301	0.979	\left(p\left(\lambda_{j}\right), \boldsymbol{u}_{j} \right)	$\left(p\left(\lambda_{j}\right), \boldsymbol{u}_{j} \right)$	of $\boldsymbol{A}$ then $(p(\lambda_j), \boldsymbol{u}_j)$ are eigenpairs of $p(\boldsymbol{A})$ for $j = 1, \ldots, n$.
lyche-numerical-linear-algebra F03915	301	0.979	p(\boldsymbol{A})	$p(\boldsymbol{A})$	of $\boldsymbol{A}$ then $(p(\lambda_j), \boldsymbol{u}_j)$ are eigenpairs of $p(\boldsymbol{A})$ for $j = 1, \ldots, n$.
lyche-numerical-linear-algebra F03916	301	0.991	\left(\lambda_{j}, \boldsymbol{u}_{j}\right), j=1,2, \ldots, n	$\left(\lambda_{j}, \boldsymbol{u}_{j}\right), j=1,2, \ldots, n$	$\mathbb{R}^{n\times n}$ *is positive definite with orthonormal eigenpairs* $(\lambda_j, \boldsymbol{u}_j)$, $j = 1, 2, \ldots, n$,
lyche-numerical-linear-algebra F03917	301	1.000	\boldsymbol{r}_{0}:=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{0}	$\boldsymbol{r}_{0}:=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{0}$	*and let* $\boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_0$ *for some* $\boldsymbol{x}_0 \in \mathbb{R}^n$. *To each* $\boldsymbol{w} \in \mathbb{W}_k$ *there corresponds*
lyche-numerical-linear-algebra F03918	302	1.000	P(t):=\sum_{j=0}^{k-1} a_{j} t^{k-1}	$P(t):=\sum_{j=0}^{k-1} a_{j} t^{k-1}$	*a polynomial* $P(t) := \sum_{j=0}^{k-1} a_j t^{k-1}$ *such that* $\boldsymbol{w} = P(\boldsymbol{A})\boldsymbol{r}_0$. *Moreover, if* $\boldsymbol{r}_0 =$
lyche-numerical-linear-algebra F03919	302	1.000	\boldsymbol{w}=P(\boldsymbol{A}) \boldsymbol{r}_{0}	$\boldsymbol{w}=P(\boldsymbol{A}) \boldsymbol{r}_{0}$	*a polynomial* $P(t) := \sum_{j=0}^{k-1} a_j t^{k-1}$ *such that* $\boldsymbol{w} = P(\boldsymbol{A})\boldsymbol{r}_0$. *Moreover, if* $\boldsymbol{r}_0 =$
lyche-numerical-linear-algebra F03920	302	1.000	\sum_{j=1}^{n} \sigma_{j} \boldsymbol{u}_{j}	$\sum_{j=1}^{n} \sigma_{j} \boldsymbol{u}_{j}$	$\sum_{j=1}^n \sigma_j \boldsymbol{u}_j$ *then*
lyche-numerical-linear-algebra F03921	302	1.000	\boldsymbol{w}=a_{0} \boldsymbol{r}_{0}+a_{1} \boldsymbol{A} \boldsymbol{r}_{0}+\cdots+a_{k-1} \boldsymbol{A}^{k-1} \boldsymbol{r}_{0}	$\boldsymbol{w}=a_{0} \boldsymbol{r}_{0}+a_{1} \boldsymbol{A} \boldsymbol{r}_{0}+\cdots+a_{k-1} \boldsymbol{A}^{k-1} \boldsymbol{r}_{0}$	***Proof*** If $\boldsymbol{w} \in \mathbb{W}_k$ then $\boldsymbol{w} = a_0\boldsymbol{r}_0 + a_1\boldsymbol{A}\boldsymbol{r}_0 + \cdots + a_{k-1}\boldsymbol{A}^{k-1}\boldsymbol{r}_0$ for some scalars
lyche-numerical-linear-algebra F03922	302	1.000	a_{0}, \ldots, a_{k-1}	$a_{0}, \ldots, a_{k-1}$	$a_0, \ldots, a_{k-1}$. But then $\boldsymbol{w} = P(\boldsymbol{A})\boldsymbol{r}_0$. We find $\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A})\boldsymbol{r}_0 = \boldsymbol{A}^{-1}(\boldsymbol{r}_0 -$
lyche-numerical-linear-algebra F03923	302	1.000	\boldsymbol{x}-\boldsymbol{x}_{0}-P(\boldsymbol{A}) \boldsymbol{r}_{0}=\boldsymbol{A}^{-1}\left(\boldsymbol{r}_{0}-\right.	$\boldsymbol{x}-\boldsymbol{x}_{0}-P(\boldsymbol{A}) \boldsymbol{r}_{0}=\boldsymbol{A}^{-1}\left(\boldsymbol{r}_{0}-\right.$	$a_0, \ldots, a_{k-1}$. But then $\boldsymbol{w} = P(\boldsymbol{A})\boldsymbol{r}_0$. We find $\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A})\boldsymbol{r}_0 = \boldsymbol{A}^{-1}(\boldsymbol{r}_0 -$
lyche-numerical-linear-algebra F03924	302	1.000	\boldsymbol{A} P(\boldsymbol{A})) \boldsymbol{r}_{0}= \boldsymbol{A}^{-1} Q(\boldsymbol{A}) \boldsymbol{r}_{0}	$\boldsymbol{A} P(\boldsymbol{A})) \boldsymbol{r}_{0}= \boldsymbol{A}^{-1} Q(\boldsymbol{A}) \boldsymbol{r}_{0}$	$\boldsymbol{A}P(\boldsymbol{A}))\boldsymbol{r}_0 = \boldsymbol{A}^{-1}Q(\boldsymbol{A})\boldsymbol{r}_0$ and $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A})\boldsymbol{r}_0) = Q(\boldsymbol{A})\boldsymbol{r}_0$. Therefore,
lyche-numerical-linear-algebra F03925	302	1.000	\boldsymbol{A}\left(\boldsymbol{x}-\boldsymbol{x}_{0}-P(\boldsymbol{A}) \boldsymbol{r}_{0}\right)=Q(\boldsymbol{A}) \boldsymbol{r}_{0}	$\boldsymbol{A}\left(\boldsymbol{x}-\boldsymbol{x}_{0}-P(\boldsymbol{A}) \boldsymbol{r}_{0}\right)=Q(\boldsymbol{A}) \boldsymbol{r}_{0}$	$\boldsymbol{A}P(\boldsymbol{A}))\boldsymbol{r}_0 = \boldsymbol{A}^{-1}Q(\boldsymbol{A})\boldsymbol{r}_0$ and $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A})\boldsymbol{r}_0) = Q(\boldsymbol{A})\boldsymbol{r}_0$. Therefore,
lyche-numerical-linear-algebra F03926	302	1.000	\boldsymbol{r}_{0}	$\boldsymbol{r}_{0}$	Using the eigenvector expansion for $\boldsymbol{r}_0$ we obtain
lyche-numerical-linear-algebra F03927	303	0.978	0<a<b	$0<a<b$	$0 < a < b$ *is an interval containing all the eigenvalues of* $\boldsymbol{A}$. *Then in the conjugate*
lyche-numerical-linear-algebra F03928	303	0.978	A	A	$0 < a < b$ *is an interval containing all the eigenvalues of* $\boldsymbol{A}$. *Then in the conjugate*
lyche-numerical-linear-algebra F03929	303	1.000	\Pi_{k}	Π_{k}	*where* Π_k *denotes the class of univariate polynomials of degree* $\leq k$ *with real*
lyche-numerical-linear-algebra F03930	303	1.000	\leq k	$\leq k$	*where* Π_k *denotes the class of univariate polynomials of degree* $\leq k$ *with real*
lyche-numerical-linear-algebra F03931	303	0.998	Q(t)=1	$Q(t)=1$	***Proof*** By (13.25) with $Q(t) = 1$ (corresponding to $P(\boldsymbol{A}) = \mathbf{0}$) we find $\|\boldsymbol{x} -$
lyche-numerical-linear-algebra F03932	303	0.998	P(\boldsymbol{A})=\mathbf{0}	$P(\boldsymbol{A})=\mathbf{0}$	***Proof*** By (13.25) with $Q(t) = 1$ (corresponding to $P(\boldsymbol{A}) = \mathbf{0}$) we find $\|\boldsymbol{x} -$
lyche-numerical-linear-algebra F03933	303	0.998	\\| \boldsymbol{x}-	$\\| \boldsymbol{x}-$	***Proof*** By (13.25) with $Q(t) = 1$ (corresponding to $P(\boldsymbol{A}) = \mathbf{0}$) we find $\|\boldsymbol{x} -$
lyche-numerical-linear-algebra F03934	303	0.969	\boldsymbol{x}_{0} \\|_{\boldsymbol{A}}^{2}=\sum_{j=1}^{n} \frac{\sigma_{j}^{2}}{\lambda_{j}}	$\boldsymbol{x}_{0} \\|_{\boldsymbol{A}}^{2}=\sum_{j=1}^{n} \frac{\sigma_{j}^{2}}{\lambda_{j}}$	$\boldsymbol{x}_0\|_{\boldsymbol{A}}^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}$. Therefore, by the best approximation property Theorem 13.4
lyche-numerical-linear-algebra F03935	303	1.000	Q \in \Pi_{k}	$Q \in \Pi_{k}$	where $Q \in \Pi_k$ and $Q(0) = 1$. Minimizing over such polynomials Q and taking
lyche-numerical-linear-algebra F03936	303	1.000	Q(0)=1	$Q(0)=1$	where $Q \in \Pi_k$ and $Q(0) = 1$. Minimizing over such polynomials Q and taking
lyche-numerical-linear-algebra F03937	303	0.999	\kappa=\lambda_{\text {max }} / \lambda_{\text {min }}	$\kappa=\lambda_{\text {max }} / \lambda_{\text {min }}$	where $\kappa = \lambda_{max}/\lambda_{min}$ is the spectral condition number of $\boldsymbol{A}$. Ignoring the second

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-al F03938	303	1.000	`y=1 / x`	$y=1 / x$	$y=1/x$, using $2^k/k!=2$, $k=1,2$, and $2^k/k!<2$ for $k>2$, we find
lyche-numerical-linear-al F03939	303	1.000	`2^{k} / k!=2, k=1,2`	$2^{k} / k!=2, k=1,2$	$y=1/x$, using $2^k/k!=2$, $k=1,2$, and $2^k/k!<2$ for $k>2$, we find
lyche-numerical-linear-al F03940	303	1.000	`2^{k} / k!<2`	$2^{k} / k!<2$	$y=1/x$, using $2^k/k!=2$, $k=1,2$, and $2^k/k!<2$ for $k>2$, we find
lyche-numerical-linear-al F03941	303	1.000	`k>2`	$k>2$	$y=1/x$, using $2^k/k!=2$, $k=1,2$, and $2^k/k!<2$ for $k>2$, we find
lyche-numerical-linear-al F03942	304	1.000	`c \notin[a, b]`	$c \notin[a, b]$	$c\notin[a,b]$ and $k\in\mathbb{N}$. Consider the set $\mathcal{S}_k$ of all polynomials Q of degree $\le k$ such
lyche-numerical-linear-al F03943	304	1.000	`\mathcal{S}_{k}`	$\mathcal{S}_{k}$	$c\notin[a,b]$ and $k\in\mathbb{N}$. Consider the set $\mathcal{S}_k$ of all polynomials Q of degree $\le k$ such
lyche-numerical-linear-al F03944	304	1.000	`Q(c)=1`	$Q(c)=1$	that $Q(c)=1$. For any continuous function f on $[a,b]$ we define
lyche-numerical-linear-al F03945	304	1.000	`Q^{*} \in \mathcal{S}_{k}`	$Q^{*} \in \mathcal{S}_{k}$	We want to find a polynomial $Q^*\in\mathcal{S}_k$ such that
lyche-numerical-linear-al F03946	304	1.000	`Q^{*}`	Q^{*}	We will show that Q^* is uniquely given as a suitably shifted and normalized
lyche-numerical-linear-al F03947	304	1.000	`T_{n}`	T_{n}	version of the **Chebyshev polynomial**. The Chebyshev polynomial T_n of degree n
lyche-numerical-linear-al F03948	304	0.999	`T_{0}(t)=1`	$T_{0}(t)=1$	starting with $T_0(t)=1$ and $T_1(t)=t$. Thus $T_2(t)=2t^2-1$, $T_3(t)=4t^3-3t$ etc.
lyche-numerical-linear-al F03949	304	0.999	`T_{1}(t)=t`	$T_{1}(t)=t$	starting with $T_0(t)=1$ and $T_1(t)=t$. Thus $T_2(t)=2t^2-1$, $T_3(t)=4t^3-3t$ etc.
lyche-numerical-linear-al F03950	304	0.999	`T_{2}(t)=2 t^{2}-1, T_{3}(t)=4 t^{3}-3 t`	$T_{2}(t)=2 t^{2}-1, T_{3}(t)=4 t^{3}-3 t$	starting with $T_0(t)=1$ and $T_1(t)=t$. Thus $T_2(t)=2t^2-1$, $T_3(t)=4t^3-3t$ etc.
lyche-numerical-linear-al F03951	304	1.000	`T_{n}(t)=\cos (n \arccos t)`	$T_{n}(t)=\cos (n \arccos t)$	*1.* $T_n(t)=\cos(n\arccos t)$ *for* $t\in[-1,1]$,
lyche-numerical-linear-al F03952	304	1.000	`t \in[-1,1]`	$t \in[-1,1]$	*1.* $T_n(t)=\cos(n\arccos t)$ *for* $t\in[-1,1]$,
lyche-numerical-linear-al F03953	304	1.000	`T_{n}(t)=\frac{1}{2}\left[\left(t+\sqrt{t^{2}-1}\right)^{n}+\left(t+\sqrt{t^{2}-1}\right)^{-n}\right]`	$T_{n}(t)=\frac{1}{2}\left[\left(t+\sqrt{t^{2}-1}\right)^{n}+\left(t+\sqrt{t^{2}-1}\right)^{-n}\right]$	*2.* $T_n(t)=\frac{1}{2}\left[\left(t+\sqrt{t^2-1}\right)^n+\left(t+\sqrt{t^2-1}\right)^{-n}\right]$ *for* $\lvert t\rvert\ge 1$.
lyche-numerical-linear-al F03954	304	1.000	`|t| \geq 1`	$\lvert t\rvert \geq 1$	*2.* $T_n(t)=\frac{1}{2}\left[\left(t+\sqrt{t^2-1}\right)^n+\left(t+\sqrt{t^2-1}\right)^{-n}\right]$ *for* $\lvert t\rvert\ge 1$.
lyche-numerical-linear-al F03955	304	1.000	`P_{n}(t)=\cos (n \arccos t)`	$P_{n}(t)=\cos (n \arccos t)$	1. With $P_n(t)=\cos(n\arccos t)$ we have $P_n(t)=\cos n\phi$, where $t=\cos\phi$.
lyche-numerical-linear-al F03956	304	1.000	`P_{n}(t)=\cos n \phi`	$P_{n}(t)=\cos n \phi$	1. With $P_n(t)=\cos(n\arccos t)$ we have $P_n(t)=\cos n\phi$, where $t=\cos\phi$.
lyche-numerical-linear-al F03957	304	1.000	`t=\cos \phi`	$t=\cos \phi$	1. With $P_n(t)=\cos(n\arccos t)$ we have $P_n(t)=\cos n\phi$, where $t=\cos\phi$.
lyche-numerical-linear-al F03958	304	1.000	`P_{n}`	P_{n}	and it follows that P_n satisfies the same recurrence relation as T_n. Since $P_0=T_0$
lyche-numerical-linear-al F03959	304	1.000	`P_{0}=T_{0}`	$P_{0}=T_{0}$	and it follows that P_n satisfies the same recurrence relation as T_n. Since $P_0=T_0$
lyche-numerical-linear-al F03960	304	1.000	`P_{1}=T_{1}`	$P_{1}=T_{1}$	and $P_1=T_1$ we have $P_n=T_n$ for all $n\ge 0$.
lyche-numerical-linear-al F03961	304	1.000	`P_{n}=T_{n}`	$P_{n}=T_{n}$	and $P_1=T_1$ we have $P_n=T_n$ for all $n\ge 0$.
lyche-numerical-linear-al F03962	304	1.000	`x_{n}:=T_{n}(t)`	$x_{n}:=T_{n}(t)$	2. Fix t with $\lvert t\rvert\ge 1$ and let $x_n:=T_n(t)$ for $n\ge 0$. The recurrence relation for the
lyche-numerical-linear-al F03963	305	0.711	`x_{n}=z^{n}`	$x_{n}=z^{n}$	To solve this difference equation we insert $x_n=z^n$ into (13.31) and obtain $z^{n+1}-$
lyche-numerical-linear-al F03964	305	0.711	`z^{n+1}-`	$z^{n+1}-$	To solve this difference equation we insert $x_n=z^n$ into (13.31) and obtain $z^{n+1}-$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F03965	305	■ 1.000	`2 t z^{n}+z^{n-1}=0`	$2tz^{n}+z^{n-1}=0$	$2tz^n + z^{n-1} = 0$ or $z^2 - 2tz + 1 = 0$. The roots of this equation are
lyche-numerical-linear-algebra F03966	305	■ 1.000	`z^{2}-2 t z+1=0`	$z^{2}-2tz+1=0$	$2tz^n + z^{n-1} = 0$ or $z^2 - 2tz + 1 = 0$. The roots of this equation are
lyche-numerical-linear-algebra F03967	305	■ 1.000	`z_{1}^{n}, z_{2}^{n}`	z_{1}^{n}, z_{2}^{n}	Now z_1^n, z_2^n and more generally $c_1z_1^n + c_2z_2^n$ are solutions of (13.31) for any
lyche-numerical-linear-algebra F03968	305	■ 1.000	`c_{1} z_{1}^{n}+c_{2} z_{2}^{n}`	$c_{1}z_{1}^{n}+c_{2}z_{2}^{n}$	Now z_1^n, z_2^n and more generally $c_1z_1^n + c_2z_2^n$ are solutions of (13.31) for any
lyche-numerical-linear-algebra F03969	305	■ 1.000	`c_{1}`	c_{1}	constants c_1 and c_2. We find these constants from the initial conditions x_0 =
lyche-numerical-linear-algebra F03970	305	■ 1.000	`c_{2}`	c_{2}	constants c_1 and c_2. We find these constants from the initial conditions x_0 =
lyche-numerical-linear-algebra F03971	305	■ 1.000	`x_{0}=`	$x_{0}=$	constants c_1 and c_2. We find these constants from the initial conditions x_0 =
lyche-numerical-linear-algebra F03972	305	■ 1.000	`c_{1}+c_{2}=1`	$c_{1}+c_{2}=1$	$c_1 + c_2 = 1$ and $x_1 = c_1z_1 + c_2z_2 = t$. Since $z_1 + z_2 = 2t$ the solution is
lyche-numerical-linear-algebra F03973	305	■ 1.000	`x_{1}=c_{1} z_{1}+c_{2} z_{2}=t`	$x_{1}=c_{1}z_{1}+c_{2}z_{2}=t$	$c_1 + c_2 = 1$ and $x_1 = c_1z_1 + c_2z_2 = t$. Since $z_1 + z_2 = 2t$ the solution is
lyche-numerical-linear-algebra F03974	305	■ 1.000	`z_{1}+z_{2}=2 t`	$z_{1}+z_{2}=2t$	$c_1 + c_2 = 1$ and $x_1 = c_1z_1 + c_2z_2 = t$. Since $z_1 + z_2 = 2t$ the solution is
lyche-numerical-linear-algebra F03975	305	■ 1.000	`c_{1}=c_{2}=\frac{1}{2}`	$c_{1}=c_{2}=\frac{1}{2}$	$c_1 = c_2 = \frac{1}{2}$.
lyche-numerical-linear-algebra F03976	305	■ 1.000	`a<b, c \notin[a, b]`	$a<b, c \notin[a, b]$	**Theorem 13.6 (A Minimal Norm Problem)** *Suppose* $a < b$, $c \notin [a,b]$ *and* $k \in$
lyche-numerical-linear-algebra F03977	305	■ 1.000	`k \in`	$k \in$	**Theorem 13.6 (A Minimal Norm Problem)** *Suppose* $a < b$, $c \notin [a,b]$ *and* $k \in$
lyche-numerical-linear-algebra F03978	305	■ 1.000	`\mathbb{N}`	$\mathbb{N}$	$\mathbb{N}$. *If* $Q \in \mathcal{S}_k$ *and* $Q \neq Q^*$ *then* $\|Q\|_\infty > \|Q^*\|_\infty$.
lyche-numerical-linear-algebra F03979	305	■ 1.000	`Q \in \mathcal{S}_{k}`	$Q \in \mathcal{S}_{k}$	$\mathbb{N}$. *If* $Q \in \mathcal{S}_k$ *and* $Q \neq Q^*$ *then* $\|Q\|_\infty > \|Q^*\|_\infty$.
lyche-numerical-linear-algebra F03980	305	■ 1.000	`Q \neq Q^{*}`	$Q \neq Q^{*}$	$\mathbb{N}$. *If* $Q \in \mathcal{S}_k$ *and* $Q \neq Q^*$ *then* $\|Q\|_\infty > \|Q^*\|_\infty$.
lyche-numerical-linear-algebra F03981	305	■ 1.000	`|Q|_{\infty}>\left|Q^{*}\right|_{\infty}`	$\|Q\|_{\infty}>\left\|Q^{*}\right\|_{\infty}$	$\mathbb{N}$. *If* $Q \in \mathcal{S}_k$ *and* $Q \neq Q^*$ *then* $\|Q\|_\infty > \|Q^*\|_\infty$.
lyche-numerical-linear-algebra F03982	305	■ 1.000	`p(z)=p^{\prime}(z)=0`	$p(z)=p^{\prime}(z)=0$	$p(z) = p'(z) = 0$, we say that p has a double zero at z. Counting such a zero as
lyche-numerical-linear-algebra F03983	305	■ 1.000	`\left|Q^{*}\right|`	$\left\|Q^{*}\right\|$	$\|Q^*\|$ takes on its maximum $1/\|T_k(u(c))\|$ at the $k+1$ points $\mu_0, \ldots, \mu_k$ in $[a,b]$
lyche-numerical-linear-algebra F03984	305	■ 1.000	`1 /\left|T_{k}(u(c))\right|`	$1/\left\|T_{k}(u(c))\right\|$	$\|Q^*\|$ takes on its maximum $1/\|T_k(u(c))\|$ at the $k+1$ points $\mu_0, \ldots, \mu_k$ in $[a,b]$
lyche-numerical-linear-algebra F03985	305	■ 1.000	`\mu_{0}, \ldots, \mu_{k}`	$\mu_{0}, \ldots, \mu_{k}$	$\|Q^*\|$ takes on its maximum $1/\|T_k(u(c))\|$ at the $k+1$ points $\mu_0, \ldots, \mu_k$ in $[a,b]$
lyche-numerical-linear-algebra F03986	305	■ 1.000	`u\left(\mu_{i}\right)=\cos (i \pi / k)`	$u\left(\mu_{i}\right)=\cos(i\pi/k)$	such that $u(\mu_i) = \cos(i\pi/k)$ for $i = 0, 1, \ldots, k$. Suppose $Q \in S_k$ and that
lyche-numerical-linear-algebra F03987	305	■ 1.000	`i=0,1, \ldots, k`	$i=0,1,\ldots,k$	such that $u(\mu_i) = \cos(i\pi/k)$ for $i = 0, 1, \ldots, k$. Suppose $Q \in S_k$ and that
lyche-numerical-linear-algebra F03988	305	■ 1.000	`Q \in S_{k}`	$Q \in S_{k}$	such that $u(\mu_i) = \cos(i\pi/k)$ for $i = 0, 1, \ldots, k$. Suppose $Q \in S_k$ and that
lyche-numerical-linear-algebra F03989	305	■ 1.000	`|Q|_{\infty} \leq\left|Q^{*}\right|_{\infty}`	$\|Q\|_{\infty} \leq\left\|Q^{*}\right\|_{\infty}$	$\|Q\|_\infty \leq \|Q^*\|_\infty$. We have to show that $Q \equiv Q^*$. Let $f \equiv Q - Q^*$. We show
lyche-numerical-linear-algebra F03990	305	■ 1.000	`Q \equiv Q^{*}`	$Q \equiv Q^{*}$	$\|Q\|_\infty \leq \|Q^*\|_\infty$. We have to show that $Q \equiv Q^*$. Let $f \equiv Q - Q^*$. We show
lyche-numerical-linear-algebra F03991	305	■ 1.000	`f \equiv Q-Q^{*}`	$f \equiv Q-Q^{*}$	$\|Q\|_\infty \leq \|Q^*\|_\infty$. We have to show that $Q \equiv Q^*$. Let $f \equiv Q - Q^*$. We show

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F03992	305	■ 1.000	`f(c)=0`	$f(c)=0$	$f(c)=0$, this means that $f\equiv 0$ or equivalently $Q\equiv Q^{*}$.
lyche-numerical-linear-algebra_F03993	305	■ 1.000	`f \equiv 0`	$f \equiv 0$	$f(c)=0$, this means that $f\equiv 0$ or equivalently $Q\equiv Q^{*}$.
lyche-numerical-linear-algebra_F03994	305	■ 1.000	`I_{j}=\left[\mu_{j-1}, \mu_{j}\right]`	$I_{j}=\left[\mu_{j-1}, \mu_{j}\right]$	Consider $I_j=[\mu_{j-1},\mu_j]$ for a fixed j. Let
lyche-numerical-linear-algebra_F03995	305	■ 1.000	`\sigma_{j} \leq 0`	$\sigma_{j} \leq 0$	We have $\sigma_j\leq 0$. For if say $Q^*(\mu_j)>0$ then
lyche-numerical-linear-algebra_F03996	305	■ 1.000	`Q^{*}\left(\mu_{j}\right)>0`	$Q^{*}\left(\mu_{j}\right)>0$	We have $\sigma_j\leq 0$. For if say $Q^*(\mu_j)>0$ then
lyche-numerical-linear-algebra_F03997	305	■ 1.000	`f\left(\mu_{j}\right) \leq 0`	$f\left(\mu_{j}\right) \leq 0$	so that $f(\mu_j)\leq 0$. Moreover,
lyche-numerical-linear-algebra_F03998	305	■ 1.000	`f\left(\mu_{j-1}\right) \geq 0`	$f\left(\mu_{j-1}\right) \geq 0$	Thus $f(\mu_{j-1})\geq 0$ and it follows that $\sigma_j\leq 0$. Similarly, $\sigma_j\leq 0$ if $Q^*(\mu_j)<0$.
lyche-numerical-linear-algebra_F03999	305	■ 1.000	`Q^{*}\left(\mu_{j}\right)<0`	$Q^{*}\left(\mu_{j}\right)<0$	Thus $f(\mu_{j-1})\geq 0$ and it follows that $\sigma_j\leq 0$. Similarly, $\sigma_j\leq 0$ if $Q^*(\mu_j)<0$.
lyche-numerical-linear-algebra_F04000	305	■ 1.000	`\sigma_{j}<0, f`	$\sigma_{j}<0, f$	If $\sigma_j<0$, f must have a zero in I_j since it is continuous. Suppose $\sigma_j=0$.
lyche-numerical-linear-algebra_F04001	305	■ 1.000	`I_{j}`	I_{j}	If $\sigma_j<0$, f must have a zero in I_j since it is continuous. Suppose $\sigma_j=0$.
lyche-numerical-linear-algebra_F04002	305	■ 1.000	`\sigma_{j}=0`	$\sigma_{j}=0$	If $\sigma_j<0$, f must have a zero in I_j since it is continuous. Suppose $\sigma_j=0$.
lyche-numerical-linear-algebra_F04003	305	■ 1.000	`f\left(\mu_{j-1}\right)=0`	$f\left(\mu_{j-1}\right)=0$	Then $f(\mu_{j-1})=0$ or $f(\mu_j)=0$. If $f(\mu_j)=0$ then $Q(\mu_j)=Q^*(\mu_j)$. But then
lyche-numerical-linear-algebra_F04004	305	■ 1.000	`f\left(\mu_{j}\right)=0`	$f\left(\mu_{j}\right)=0$	Then $f(\mu_{j-1})=0$ or $f(\mu_j)=0$. If $f(\mu_j)=0$ then $Q(\mu_j)=Q^*(\mu_j)$. But then
lyche-numerical-linear-algebra_F04005	305	■ 1.000	`Q\left(\mu_{j}\right)=Q^{*}\left(\mu_{j}\right)`	$Q\left(\mu_{j}\right)=Q^{*}\left(\mu_{j}\right)$	Then $f(\mu_{j-1})=0$ or $f(\mu_j)=0$. If $f(\mu_j)=0$ then $Q(\mu_j)=Q^*(\mu_j)$. But then
lyche-numerical-linear-algebra_F04006	305	■ 0.998	`\mu_{j} \in(a, b)`	$\mu_{j} \in(a, b)$	μ_j is a maximum or minimum both for Q and Q^*. If $\mu_j\in(a,b)$ then $Q'(\mu_j)=$
lyche-numerical-linear-algebra_F04007	305	■ 0.998	`Q^{\prime}\left(\mu_{j}\right)=`	$Q^{\prime}\left(\mu_{j}\right)=$	μ_j is a maximum or minimum both for Q and Q^*. If $\mu_j\in(a,b)$ then $Q'(\mu_j)=$
lyche-numerical-linear-algebra_F04008	306	■ 1.000	`Q^{* \prime}\left(\mu_{j}\right)=0`	$Q^{* \prime}\left(\mu_{j}\right)=0$	$Q^{*\prime}(\mu_j)=0$. Thus $f(\mu_j)=f'(\mu_j)=0$, and f has a double zero at μ_j. We can
lyche-numerical-linear-algebra_F04009	306	■ 1.000	`f\left(\mu_{j}\right)=f^{\prime}\left(\mu_{j}\right)=0`	$f\left(\mu_{j}\right)=f^{\prime}\left(\mu_{j}\right)=0$	$Q^{*\prime}(\mu_j)=0$. Thus $f(\mu_j)=f'(\mu_j)=0$, and f has a double zero at μ_j. We can
lyche-numerical-linear-algebra_F04010	306	■ 1.000	`I_{j+1}`	I_{j+1}	count this as one zero for I_j and one for I_{j+1}. If $\mu_j=b$, we still have a zero in I_j.
lyche-numerical-linear-algebra_F04011	306	■ 1.000	`\mu_{j}=b`	$\mu_{j}=b$	count this as one zero for I_j and one for I_{j+1}. If $\mu_j=b$, we still have a zero in I_j.
lyche-numerical-linear-algebra_F04012	306	■ 1.000	`\mu_{j-1}`	μ_{j-1}	Similarly, if $f(\mu_{j-1})=0$, a double zero of f at μ_{j-1} appears if $\mu_{j-1}\in(a,b)$. We
lyche-numerical-linear-algebra_F04013	306	■ 1.000	`\mu_{j-1} \in(a, b)`	$\mu_{j-1} \in(a, b)$	Similarly, if $f(\mu_{j-1})=0$, a double zero of f at μ_{j-1} appears if $\mu_{j-1}\in(a,b)$. We
lyche-numerical-linear-algebra_F04014	306	■ 1.000	`I_{j-1}`	I_{j-1}	count this as one zero for I_{j-1} and one for I_j.
lyche-numerical-linear-algebra_F04015	306	■ 1.000	`I_{j}, j=`	$I_{j}, j=$	In this way we associate one zero of f for each of the k intervals I_j, $j=$
lyche-numerical-linear-algebra_F04016	306	■ 0.999	`1,2, \ldots, k`	$1,2, \ldots, k$	$1,2,\ldots,k$. We conclude that f has at least k zeros in $[a,b]$.
lyche-numerical-linear-algebra_F04017	306	■ 1.000	`c=0`	$c=0$	Theorem 13.6 with a, and b, the smallest and largest eigenvalue of $\boldsymbol{A}$, and $c=0$
lyche-numerical-linear-algebra_F04018	306	■ 1.000	`t=(b+a) /(b-a)`	$t=(b+a) /(b-a)$	Moreover with $t=(b+a)/(b-a)$ we have

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_F04019</td><td>307</td><td>1.000</td><td><code>M:=\lambda_{\text {max }}</code></td><td>$M := \lambda_{\text{max}}$</td><td>where $M := \lambda_{max}$ and $m := \lambda_{min}$ are the largest and smallest eigenvalue of $\boldsymbol{A}$,</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04020</td><td>307</td><td>1.000</td><td><code>m:=\lambda_{\text {min }}</code></td><td>$m := \lambda_{\text{min}}$</td><td>where $M := \lambda_{max}$ and $m := \lambda_{min}$ are the largest and smallest eigenvalue of $\boldsymbol{A}$,</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04021</td><td>307</td><td>0.997</td><td><code>\left(\lambda_{j}^{-1}, \boldsymbol{u}_{j}\right)</code></td><td>$\left(\lambda_{j}^{-1}, \boldsymbol{u}_{j}\right)$</td><td>Proof If $(\lambda_j, \boldsymbol{u}_j)$ are orthonormal eigenpairs of $\boldsymbol{A}$ then $(\lambda_j^{-1}, \boldsymbol{u}_j)$ are eigenpairs</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04022</td><td>307</td><td>1.000</td><td><code>\boldsymbol{A}^{-1}, j=1, \ldots, n</code></td><td>$\boldsymbol{A}^{-1}, j=1, \ldots, n$</td><td>for $\boldsymbol{A}^{-1}$, $j = 1, \ldots, n$. Let $\boldsymbol{y} = \sum_{j=1}^{n} c_j \boldsymbol{u}_j$ be the corresponding eigenvector</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04023</td><td>307</td><td>1.000</td><td><code>\boldsymbol{y}=\sum_{j=1}^{n} c_{j} \boldsymbol{u}_{j}</code></td><td>$\boldsymbol{y}=\sum_{j=1}^{n} c_{j} \boldsymbol{u}_{j}$</td><td>for $\boldsymbol{A}^{-1}$, $j = 1, \ldots, n$. Let $\boldsymbol{y} = \sum_{j=1}^{n} c_j \boldsymbol{u}_j$ be the corresponding eigenvector</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04024</td><td>307</td><td>1.000</td><td><code>f:[m c, M c] \rightarrow \mathbb{R}</code></td><td>$f:[m c, M c] \rightarrow \mathbb{R}$</td><td>where $f : [mc, Mc] \rightarrow \mathbb{R}$ is given by $f(x) := x + 1/x$. By (13.38)</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04025</td><td>307</td><td>1.000</td><td><code>f(x):=x+1 / x</code></td><td>$f(x):=x+1 / x$</td><td>where $f : [mc, Mc] \rightarrow \mathbb{R}$ is given by $f(x) := x + 1/x$. By (13.38)</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04026</td><td>307</td><td>1.000</td><td><code>f \in C^{2}</code></td><td>$f \in C^{2}$</td><td>Since $f \in C^2$ and f'' is positive it follows from Lemma 8.2 that f is a convex</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04027</td><td>307</td><td>1.000</td><td><code>f^{\prime \prime}</code></td><td>$f^{\prime \prime}$</td><td>Since $f \in C^2$ and f'' is positive it follows from Lemma 8.2 that f is a convex</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04028</td><td>308</td><td>1.000</td><td><code>c:=1 / \sqrt{m M}</code></td><td>$c:=1 / \sqrt{m M}$</td><td>Choosing $c := 1/\sqrt{mM}$ we find $f(mc) = f(Mc) = \sqrt{\frac{M}{m}} + \sqrt{\frac{m}{M}} = \frac{M+m}{\sqrt{mM}}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04029</td><td>308</td><td>1.000</td><td><code>f(m c)=f(M c)=\sqrt{\frac{M}{m}}+\sqrt{\frac{m}{M}}=\frac{M+m}{\sqrt{m M}}</code></td><td>$f(m c)=f(M c)=\sqrt{\frac{M}{m}}+\sqrt{\frac{m}{M}}=\frac{M+m}{\sqrt{m M}}$</td><td>Choosing $c := 1/\sqrt{mM}$ we find $f(mc) = f(Mc) = \sqrt{\frac{M}{m}} + \sqrt{\frac{m}{M}} = \frac{M+m}{\sqrt{mM}}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04030</td><td>308</td><td>1.000</td><td><code>\boldsymbol{\epsilon}_{j}:=\boldsymbol{x}-\boldsymbol{x}_{j}, j=0,1, \ldots</code></td><td>$\boldsymbol{\epsilon}_{j}:=\boldsymbol{x}-\boldsymbol{x}_{j}, j=0,1, \ldots$</td><td>Proof of (13.19) Let $\boldsymbol{\epsilon}_j := \boldsymbol{x} - \boldsymbol{x}_j$, $j = 0, 1, \ldots$, where $\boldsymbol{Ax} = \boldsymbol{b}$. It is enough to</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04031</td><td>308</td><td>1.000</td><td><code>\left\|\boldsymbol{\epsilon}_{k}\right\|_{\boldsymbol{A}} \leq\left(\frac{\kappa-1}{\kappa+1}\right)\left\|\boldsymbol{\epsilon}_{k-1}\right\| \leq \cdots \leq \left(\frac{\kappa-1}{\kappa+1}\right)^{k}\left\|\boldsymbol{\epsilon}_{0}\right\|</code></td><td>$\left\|\boldsymbol{\epsilon}_{k}\right\|_{\boldsymbol{A}} \leq\left(\frac{\kappa-1}{\kappa+1}\right)\left\|\boldsymbol{\epsilon}_{k-1}\right\| \leq \cdots \leq \left(\frac{\kappa-1}{\kappa+1}\right)^{k}\left\|\boldsymbol{\epsilon}_{0}\right\|$</td><td>for then $\|\boldsymbol{\epsilon}_k\|_{\boldsymbol{A}} \leq \left(\frac{\kappa-1}{\kappa+1}\right) \|\boldsymbol{\epsilon}_{k-1}\| \leq \cdots \leq \left(\frac{\kappa-1}{\kappa+1}\right)^k \|\boldsymbol{\epsilon}_0\|$. It follows from (13.8) that</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04032</td><td>309</td><td>1.000</td><td><code>\boldsymbol{r}_{m+1}=\mathbf{0}</code></td><td>$\boldsymbol{r}_{m+1}=\mathbf{0}$</td><td>gradient method m be the smallest integer such that $\boldsymbol{r}_{m+1} = \mathbf{0}$. For $k \leq m$ we</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04033</td><td>309</td><td>1.000</td><td><code>k \leq m</code></td><td>$k \leq m$</td><td>gradient method m be the smallest integer such that $\boldsymbol{r}_{m+1} = \mathbf{0}$. For $k \leq m$ we</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04034</td><td>309</td><td>1.000</td><td><code>\left\|\boldsymbol{\epsilon}_{k+1}\right\|_{2}<\left\|\boldsymbol{\epsilon}_{k}\right\|_{2}</code></td><td>$\left\|\boldsymbol{\epsilon}_{k+1}\right\|_{2}<\left\|\boldsymbol{\epsilon}_{k}\right\|_{2}$</td><td>have $\|\boldsymbol{\epsilon}_{k+1}\|_2 < \|\boldsymbol{\epsilon}_k\|_2$. More precisely,</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04035</td><td>309</td><td>0.983</td><td><code>\boldsymbol{\epsilon}_{j}=\boldsymbol{x}-\boldsymbol{x}_{j}</code></td><td>$\boldsymbol{\epsilon}_{j}=\boldsymbol{x}-\boldsymbol{x}_{j}$</td><td>where $\boldsymbol{\epsilon}_j = \boldsymbol{x} - \boldsymbol{x}_j$ and $\boldsymbol{Ax} = \boldsymbol{b}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04036</td><td>309</td><td>1.000</td><td><code>j \leq m</code></td><td>$j \leq m$</td><td>Proof For $j \leq m$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04037</td><td>309</td><td>1.000</td><td><code>\beta_{i-1} \cdots \beta_{k}=\left(\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}\right) /\left(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}\right)</code></td><td>$\beta_{i-1} \cdots \beta_{k}=\left(\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}\right) /\left(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}\right)$</td><td>and since $\beta_{i-1} \cdots \beta_k = (\boldsymbol{r}_i^T \boldsymbol{r}_i)/(\boldsymbol{r}_k^T \boldsymbol{r}_k)$ we find</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04038</td><td>310</td><td>1.000</td><td><code>\operatorname{cond}_{2}(\boldsymbol{A})</code></td><td>$\operatorname{cond}_{2}(\boldsymbol{A})$</td><td>For problems $\boldsymbol{Ax} = \boldsymbol{b}$ of size n, where both n and $\operatorname{cond}_2(\boldsymbol{A})$ are large, it is</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04039</td><td>310</td><td>1.000</td><td><code>\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{B} \boldsymbol{b}</code></td><td>$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{B} \boldsymbol{b}$</td><td>equivalent system $\boldsymbol{BAx} = \boldsymbol{Bb}$, where $\boldsymbol{B}$ is nonsingular and $\operatorname{cond}_2(\boldsymbol{BA})$ is smaller</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04040</td><td>310</td><td>1.000</td><td><code>\operatorname{cond}_{2}(\boldsymbol{B} \boldsymbol{A})</code></td><td>$\operatorname{cond}_{2}(\boldsymbol{B} \boldsymbol{A})$</td><td>equivalent system $\boldsymbol{BAx} = \boldsymbol{Bb}$, where $\boldsymbol{B}$ is nonsingular and $\operatorname{cond}_2(\boldsymbol{BA})$ is smaller</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04041</td><td>310</td><td>0.578</td><td><code>\boldsymbol{B}=\boldsymbol{M}^{-1}</code></td><td>$\boldsymbol{B}=\boldsymbol{M}^{-1}$</td><td>$\boldsymbol{B} = \boldsymbol{M}^{-1}$. We cannot use CG on $\boldsymbol{BAx} = \boldsymbol{Bb}$ directly since $\boldsymbol{BA}$ in general is not</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04042</td><td>310</td><td>1.000</td><td><code>\boldsymbol{B}=\boldsymbol{C}^{T} \boldsymbol{C}</code></td><td>$\boldsymbol{B}=\boldsymbol{C}^{T} \boldsymbol{C}$</td><td>such that $\boldsymbol{B} = \boldsymbol{C}^T \boldsymbol{C}$. ($\boldsymbol{C}$ is only needed for the derivation and will not appear in the</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F04043	310	1.000	`\boldsymbol{y}:=\boldsymbol{C}^{-T} \boldsymbol{x}`	$\boldsymbol{y}:=\boldsymbol{C}^{-T} \boldsymbol{x}$	where $\boldsymbol{y} := \boldsymbol{C}^{-T}\boldsymbol{x}$. We have 3 linear systems
lyche-numerical-linear-algebra_F04044	311	1.000	`\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}`	$\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}$	being positive definite the matrix $\boldsymbol{CAC}^T$ is similar to $\boldsymbol{BA}$. Indeed,
lyche-numerical-linear-algebra_F04045	311	1.000	`\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T} \right) \boldsymbol{y}=\boldsymbol{C} \boldsymbol{b}`	$\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T} \right) \boldsymbol{y}=\boldsymbol{C} \boldsymbol{b}$	We apply the conjugate gradient method to $(\boldsymbol{CAC}^T)\boldsymbol{y} = \boldsymbol{Cb}$. Denoting the
lyche-numerical-linear-algebra_F04046	311	0.997	`\boldsymbol{q}_{k}`	$\boldsymbol{q}_{k}$	search direction by $\boldsymbol{q}_k$ and the residual by $\boldsymbol{z}_k := \boldsymbol{Cb} - \boldsymbol{CAC}^T\boldsymbol{y}_k$ we obtain the
lyche-numerical-linear-algebra_F04047	311	0.997	`\boldsymbol{z}_{k}:=\boldsymbol{C} \boldsymbol{b}-\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T} \boldsymbol{y}_{k}`	$\boldsymbol{z}_{k}:=\boldsymbol{C} \boldsymbol{b}-\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T} \boldsymbol{y}_{k}$	search direction by $\boldsymbol{q}_k$ and the residual by $\boldsymbol{z}_k := \boldsymbol{Cb} - \boldsymbol{CAC}^T\boldsymbol{y}_k$ we obtain the
lyche-numerical-linear-algebra_F04048	311	0.998	`\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}, \boldsymbol{r}_{k}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k}`	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}, \boldsymbol{r}_{k}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k}$	Here $\boldsymbol{x}_k$ will be an approximation to the solution $\boldsymbol{x}$ of $\boldsymbol{Ax} = \boldsymbol{b}$, $\boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{Ax}_k$ is
lyche-numerical-linear-algebra_F04049	311	0.573	`\boldsymbol{s}_{k}=\boldsymbol{C}^{T} \boldsymbol{z}_{k}=\boldsymbol{C}^{T}\left(\boldsymbol{C}-\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}\right) \boldsymbol{y}_{k}=\boldsymbol{B} \boldsymbol{b}-`	$\boldsymbol{s}_{k}=\boldsymbol{C}^{T} \boldsymbol{z}_{k}=\boldsymbol{C}^{T}\left(\boldsymbol{C}-\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}\right) \boldsymbol{y}_{k}=\boldsymbol{B} \boldsymbol{b}-$	the residual in the original system, and $\boldsymbol{s}_k = \boldsymbol{C}^T\boldsymbol{z}_k = \boldsymbol{C}^T(\boldsymbol{C} - \boldsymbol{CAC}^T)\boldsymbol{y}_k = \boldsymbol{Bb} -$
lyche-numerical-linear-algebra_F04050	311	1.000	`\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}_{k}`	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}_{k}$	$\boldsymbol{BAx}_k$ is the residual in the preconditioned system. If we set $\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{Ax}_0$, $\boldsymbol{p}_0 =$
lyche-numerical-linear-algebra_F04051	311	1.000	`\boldsymbol{r}_{0}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{0}, \boldsymbol{p}_{0}=`	$\boldsymbol{r}_{0}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{0}, \boldsymbol{p}_{0}=$	$\boldsymbol{BAx}_k$ is the residual in the preconditioned system. If we set $\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{Ax}_0$, $\boldsymbol{p}_0 =$
lyche-numerical-linear-algebra_F04052	311	0.835	`\boldsymbol{s}_{0}=\boldsymbol{B r}_{0}`	$\boldsymbol{s}_{0}=\boldsymbol{B r}_{0}$	$\boldsymbol{s}_0 = \boldsymbol{Br}_0$, we obtain the following preconditioned conjugate gradient algorithm for
lyche-numerical-linear-algebra_F04053	311	0.995	`\mathrm{x}\left(=\boldsymbol{x}_{0}\right)`	$\mathrm{x}\left(=\boldsymbol{x}_{0}\right)$	and $\mathrm{x}(= \boldsymbol{x}_0)$ is the starting iteration.
lyche-numerical-linear-algebra_F04054	312	1.000	`\rho`	ρ	Apart from the calculation of ρ this algorithm is quite similar to Algorithm 13.1.
lyche-numerical-linear-algebra_F04055	312	1.000	`w=B * t`	$w=B * t$	The main additional work is contained in $w = B * t$. We'll discuss this further in
lyche-numerical-linear-algebra_F04056	312	1.000	`\boldsymbol{y}_{k}`	$\boldsymbol{y}_{k}$	where $\boldsymbol{y}_k$ is the conjugate gradient approximation to the solution $\boldsymbol{y}$ of (13.46) and κ
lyche-numerical-linear-algebra_F04057	313	0.998	`\boldsymbol{B} \boldsymbol{x}`	$\boldsymbol{B} \boldsymbol{x}$	2. The evaluation of $\boldsymbol{Bx}$ for a given vector $\boldsymbol{x}$ should not be expensive in storage and
lyche-numerical-linear-algebra_F04058	313	1.000	`u=u(x, y)`	$u=u(x, y)$	and c are given and we seek a function $u = u(x, y)$ such that (13.52) holds. We
lyche-numerical-linear-algebra_F04059	313	1.000	`c(x, y)>0`	$c(x, y)>0$	assume that c and f are defined and continuous on Ω and that $c(x, y) > 0$ for
lyche-numerical-linear-algebra_F04060	313	0.991	`(x, y) \in \Omega`	$(x, y) \in \Omega$	all $(x, y) \in \Omega$. The problem (13.52) reduces to the Poisson problem (10.1) in the
lyche-numerical-linear-algebra_F04061	313	1.000	`c(x, y)=1`	$c(x, y)=1$	special case where $c(x, y) = 1$ for $(x, y) \in \Omega$.
lyche-numerical-linear-algebra_F04062	313	1.000	`m \in \mathbb{N}`	$m \in \mathbb{N}$	To solve (13.52) numerically, we choose $m \in \mathbb{N}$, set $h := 1/(m + 1)$, and define
lyche-numerical-linear-algebra_F04063	313	1.000	`v_{j, k} \approx u\left(x_{j}, y_{k}\right)`	$v_{j, k} \approx u\left(x_{j}, y_{k}\right)$	We compute approximations $v_{j,k} \approx u(x_j, y_k)$ on a grid of points
lyche-numerical-linear-algebra_F04064	314	1.000	`f, g`	f, g	using a finite difference method. For univariate functions f, g we approximate
lyche-numerical-linear-algebra_F04065	314	1.000	`f_{j, k}:=f\left(x_{j}, y_{k}\right),(d v)_{j, k}:= \left(d_{1} v\right)_{j, k}+\left(d_{2} v\right)_{j, k}`	$f_{j, k}:=f\left(x_{j}, y_{k}\right),(d v)_{j, k}:= \left(d_{1} v\right)_{j, k}+\left(d_{2} v\right)_{j, k}$	where $f_{j,k} := f(x_j, y_k)$, $(dv)_{j,k} := (d_1v)_{j,k} + (d_2v)_{j,k}$,
lyche-numerical-linear-algebra_F04066	314	1.000	`c_{p, q}=c(p h, q h)`	$c_{p, q}=c(p h, q h)$	and where $c_{p,q} = c(ph, qh)$ for $p, q \in \mathbb{R}$. The equation (13.53) can be written in
lyche-numerical-linear-algebra_F04067	314	1.000	`p, q \in \mathbb{R}`	$p, q \in \mathbb{R}$	and where $c_{p,q} = c(ph, qh)$ for $p, q \in \mathbb{R}$. The equation (13.53) can be written in

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04068	314	0.997	`h^{2} L_{h} v=h^{2} \boldsymbol{F}`	$h^{2} L_{h} v=h^{2} \boldsymbol{F}$	as unknowns. The system $h^2 L_h v = h^2 \boldsymbol{F}$ can be written in standard form $\boldsymbol{Ax} = \boldsymbol{b}$
lyche-numerical-linear-algebra F04069	315	1.000	`\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m \times m}`	$\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m \times m}$	**Lemma 13.6 (Discrete Inner Product)** *If* $\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m\times m}$ *and* $v_{j,k} = w_{j,k} = 0$ *for*
lyche-numerical-linear-algebra F04070	315	1.000	`v_{j, k}=w_{j, k}=0`	$v_{j, k}=w_{j, k}=0$	**Lemma 13.6 (Discrete Inner Product)** *If* $\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m\times m}$ *and* $v_{j,k} = w_{j,k} = 0$ *for*
lyche-numerical-linear-algebra F04071	315	1.000	`(j, k) \in \partial I_{m}`	$(j, k) \in \partial I_{m}$	$(j, k) \in \partial I_m$, *then*
lyche-numerical-linear-algebra F04072	315	0.994	`m \in \mathbb{N}, a_{i}, b_{i}, c_{i} \in \mathbb{R}`	$m \in \mathbb{N}, a_{i}, b_{i}, c_{i} \in \mathbb{R}$	***Proof*** If $m \in \mathbb{N}, a_i, b_i, c_i \in \mathbb{R}$ for $i = 0, \ldots, m$ and $b_0 = c_0 = b_{m+1} = c_{m+1} = 0$
lyche-numerical-linear-algebra F04073	315	0.994	`i=0, \ldots, m`	$i=0, \ldots, m$	***Proof*** If $m \in \mathbb{N}, a_i, b_i, c_i \in \mathbb{R}$ for $i = 0, \ldots, m$ and $b_0 = c_0 = b_{m+1} = c_{m+1} = 0$
lyche-numerical-linear-algebra F04074	315	0.994	`b_{0}=c_{0}=b_{m+1}=c_{m+1}=0`	$b_{0}=c_{0}=b_{m+1}=c_{m+1}=0$	***Proof*** If $m \in \mathbb{N}, a_i, b_i, c_i \in \mathbb{R}$ for $i = 0, \ldots, m$ and $b_0 = c_0 = b_{m+1} = c_{m+1} = 0$
lyche-numerical-linear-algebra F04075	315	1.000	`\left(d_{1} v\right)_{j, k} w_{j, k}`	$\left(d_{1} v\right)_{j, k} w_{j, k}$	and the right hand side of (13.60) follows. We apply (13.60) to $(d_1 v)_{j,k} w_{j,k}$ and
lyche-numerical-linear-algebra F04076	315	1.000	`\left(d_{2} v\right)_{j, k} w_{j, k}`	$\left(d_{2} v\right)_{j, k} w_{j, k}$	$(d_2 v)_{j,k} w_{j,k}$ given by (13.55) and (13.59) follows.
lyche-numerical-linear-algebra F04077	315	1.000	`\left\langle L_{h} v, \boldsymbol{W}\right\rangle=\left \langle\boldsymbol{W}, L_{h} v\right\rangle`	$\left\langle L_{h} v, \boldsymbol{W}\right\rangle=\left\langle\boldsymbol{W}, L_{h} v\right\rangle$	***Proof*** By (13.59) $\langle L_h v, \boldsymbol{W}\rangle = \langle \boldsymbol{W}, L_h v\rangle$ and symmetry follows. We take $\boldsymbol{W} =$
lyche-numerical-linear-algebra F04078	315	1.000	`\boldsymbol{W}=`	$\boldsymbol{W}=$	***Proof*** By (13.59) $\langle L_h v, \boldsymbol{W}\rangle = \langle \boldsymbol{W}, L_h v\rangle$ and symmetry follows. We take $\boldsymbol{W} =$
lyche-numerical-linear-algebra F04079	315	1.000	`c_{j+\frac{1}{2}, k}`	$c_{j+\frac{1}{2}, k}$	$\boldsymbol{V}$ and obtain quadratic factors in (13.59). Since $c_{j+\frac{1}{2},k}$ and $c_{j,k+\frac{1}{2}}$ correspond to
lyche-numerical-linear-algebra F04080	315	1.000	`c_{j, k+\frac{1}{2}}`	$c_{j, k+\frac{1}{2}}$	$\boldsymbol{V}$ and obtain quadratic factors in (13.59). Since $c_{j+\frac{1}{2},k}$ and $c_{j,k+\frac{1}{2}}$ correspond to
lyche-numerical-linear-algebra F04081	315	1.000	`\left\langle L_{h} v, \boldsymbol{V}\right\rangle \geq 0`	$\left\langle L_{h} v, \boldsymbol{V}\right\rangle \geq 0$	and $\langle L_h v, \boldsymbol{V}\rangle \geq 0$ for all $\boldsymbol{V} \in \mathbb{R}^{m\times m}$. If $\langle L_h v, \boldsymbol{V}\rangle = 0$ then all the quadratic factors
lyche-numerical-linear-algebra F04082	315	1.000	`\boldsymbol{V} \in \mathbb{R}^{m \times m}`	$\boldsymbol{V} \in \mathbb{R}^{m \times m}$	and $\langle L_h v, \boldsymbol{V}\rangle \geq 0$ for all $\boldsymbol{V} \in \mathbb{R}^{m\times m}$. If $\langle L_h v, \boldsymbol{V}\rangle = 0$ then all the quadratic factors
lyche-numerical-linear-algebra F04083	315	1.000	`\left\langle L_{h} v, \boldsymbol{V}\right\rangle=0`	$\left\langle L_{h} v, \boldsymbol{V}\right\rangle=0$	and $\langle L_h v, \boldsymbol{V}\rangle \geq 0$ for all $\boldsymbol{V} \in \mathbb{R}^{m\times m}$. If $\langle L_h v, \boldsymbol{V}\rangle = 0$ then all the quadratic factors
lyche-numerical-linear-algebra F04084	315	1.000	`v_{j, k+1}=v_{j, k}`	$v_{j, k+1}=v_{j, k}$	must be zero, and $v_{j,k+1} = v_{j,k}$ for $k = 0, 1, \ldots, m$ and $j = 1, \ldots, m$. Now
lyche-numerical-linear-algebra F04085	315	1.000	`k=0,1, \ldots, m`	$k=0,1, \ldots, m$	must be zero, and $v_{j,k+1} = v_{j,k}$ for $k = 0, 1, \ldots, m$ and $j = 1, \ldots, m$. Now
lyche-numerical-linear-algebra F04086	315	1.000	`v_{j, 0}=v_{j, m+1}=0`	$v_{j, 0}=v_{j, m+1}=0$	$v_{j,0} = v_{j,m+1} = 0$ implies that $\boldsymbol{V} = \mathbf{0}$. It follows that the linear system (13.56) is
lyche-numerical-linear-algebra F04087	315	1.000	`\boldsymbol{V}=\mathbf{0}`	$\boldsymbol{V}=\mathbf{0}$	$v_{j,0} = v_{j,m+1} = 0$ implies that $\boldsymbol{V} = \mathbf{0}$. It follows that the linear system (13.56) is
lyche-numerical-linear-algebra F04088	316	1.000	`b \in \mathbb{R}^{n}`	$b \in \mathbb{R}^{n}$	Consider solving $\boldsymbol{Ax} = \boldsymbol{b}$, where $\boldsymbol{A}$ is given by (13.57) and $b \in \mathbb{R}^n$. Since $\boldsymbol{A}$ is
lyche-numerical-linear-algebra F04089	316	1.000	`x \in \mathbb{R}^{n}`	$x \in \mathbb{R}^{n}$	positive definite it is nonsingular and the system has a unique solution $x \in \mathbb{R}^n$.
lyche-numerical-linear-algebra F04090	316	1.000	`m=\sqrt{n}`	$m=\sqrt{n}$	to find $\boldsymbol{x}$. Since the bandwidth of $\boldsymbol{A}$ is $m = \sqrt{n}$ both of these methods require $O(n^2)$
lyche-numerical-linear-algebra F04091	316	1.000	`c(x, y) \equiv 1`	$c(x, y) \equiv 1$	If we choose $c(x, y) \equiv 1$ in (13.52), we get the Poisson problem. With this in
lyche-numerical-linear-algebra F04092	316	1.000	`\boldsymbol{A}_{p}`	$\boldsymbol{A}_{p}$	mind, we may think of the coefficient matrix $\boldsymbol{A}_p$ arising from the discretization of
lyche-numerical-linear-algebra F04093	316	1.000	`\boldsymbol{B}=\boldsymbol{A}_{p}^{-1}`	$\boldsymbol{B}=\boldsymbol{A}_{p}^{-1}$	$\boldsymbol{B} = \boldsymbol{A}_p^{-1}$, the inverse of the discrete Poisson matrix as a preconditioner for the
lyche-numerical-linear-algebra F04094	316	0.928	`\boldsymbol{w}=\boldsymbol{B} \boldsymbol{t}`	$\boldsymbol{w}=\boldsymbol{B} \boldsymbol{t}$	Consider Algorithm 13.3. With this preconditioner the calculation $\boldsymbol{w} = \boldsymbol{Bt}$ takes

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04095	316	1.000	`\boldsymbol{A}_{p} \boldsymbol{w}_{k}=\boldsymbol{t}_{k}`	$\boldsymbol{A}_{p} \boldsymbol{w}_{k}=\boldsymbol{t}_{k}$	the form $\boldsymbol{A}_p \boldsymbol{w}_k = \boldsymbol{t}_k$.
lyche-numerical-linear-algebra F04096	316	1.000	`\boldsymbol{A}_{p} \boldsymbol{w}=\boldsymbol{t}`	$\boldsymbol{A}_{p} \boldsymbol{w}=\boldsymbol{t}$	This method can be utilized to solve $\boldsymbol{A}_p \boldsymbol{w} = \boldsymbol{t}$.
lyche-numerical-linear-algebra F04097	316	1.000	`\boldsymbol{x}_{0}=0`	$\boldsymbol{x}_{0}=0$	lem (13.52). We used $\boldsymbol{x}_0 = 0$ and $\epsilon = 10^{-8}$. The results are shown in Table 13.3.
lyche-numerical-linear-algebra F04098	316	1.000	`\epsilon=10^{-8}`	$\epsilon=10^{-8}$	lem (13.52). We used $\boldsymbol{x}_0 = 0$ and $\epsilon = 10^{-8}$. The results are shown in Table 13.3.
lyche-numerical-linear-algebra F04099	316	1.000	`O\left(n \log _{2} n\right)`	$O\left(n \log _{2} n\right)$	without preconditioning to $O(n^{3/2})$ or $O(n \log_2 n)$ with preconditioning. This is not
lyche-numerical-linear-algebra F04100	316	1.000	`K_{\text {pre }}`	$K_{\text {pre }}$	n: 2500, 10000, 22500, 40000, 62500 K: 222, 472, 728, 986, 1246 $K/\sqrt{n}$: 4.44, 4.72, 4.85, 4.93, 4.98 K_{pre}: 22, 23, 23, 23, 23
lyche-numerical-linear-algebra F04101	317	1.000	`0<c_{0} \leq`	$0<c_{0} \leq$	**Theorem 13.12 (Eigenvalues of Preconditioned Matrix)** *Suppose* $0 < c_0 \leq$
lyche-numerical-linear-algebra F04102	317	0.973	`c(x, y) \leq c_{1}`	$c(x, y) \leq c_{1}$	$c(x, y) \leq c_1$ *for all* $(x, y) \in [0, 1]^2$. *For the eigenvalues of the matrix* $\boldsymbol{B}\boldsymbol{A} = \boldsymbol{A}_p^{-1}\boldsymbol{A}$
lyche-numerical-linear-algebra F04103	317	0.973	`(x, y) \in[0,1]^{2}`	$(x, y) \in[0,1]^{2}$	$c(x, y) \leq c_1$ *for all* $(x, y) \in [0, 1]^2$. *For the eigenvalues of the matrix* $\boldsymbol{B}\boldsymbol{A} = \boldsymbol{A}_p^{-1}\boldsymbol{A}$
lyche-numerical-linear-algebra F04104	317	0.973	`\boldsymbol{B} \boldsymbol{A}=\boldsymbol{A}_{p}^{-1} \boldsymbol{A}`	$\boldsymbol{B} \boldsymbol{A}=\boldsymbol{A}_{p}^{-1} \boldsymbol{A}$	$c(x, y) \leq c_1$ *for all* $(x, y) \in [0, 1]^2$. *For the eigenvalues of the matrix* $\boldsymbol{B}\boldsymbol{A} = \boldsymbol{A}_p^{-1}\boldsymbol{A}$
lyche-numerical-linear-algebra F04105	317	1.000	`\boldsymbol{A}_{p}^{-1} \boldsymbol{A} \boldsymbol{x}= \lambda x`	$\boldsymbol{A}_{p}^{-1} \boldsymbol{A} \boldsymbol{x}= \lambda x$	***Proof*** Suppose $\boldsymbol{A}_p^{-1}\boldsymbol{A}\boldsymbol{x} = \lambda x$ for some $\boldsymbol{x} \in \mathbb{R}^n \setminus \{0\}$. Then $\boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{A}_p x$.
lyche-numerical-linear-algebra F04106	317	1.000	`\boldsymbol{x} \in \mathbb{R}^{n} \backslash\{0\}`	$\boldsymbol{x} \in \mathbb{R}^{n} \backslash\{0\}$	***Proof*** Suppose $\boldsymbol{A}_p^{-1}\boldsymbol{A}\boldsymbol{x} = \lambda x$ for some $\boldsymbol{x} \in \mathbb{R}^n \setminus \{0\}$. Then $\boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{A}_p x$.
lyche-numerical-linear-algebra F04107	317	1.000	`\boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{A}_{p} x`	$\boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{A}_{p} x$	***Proof*** Suppose $\boldsymbol{A}_p^{-1}\boldsymbol{A}\boldsymbol{x} = \lambda x$ for some $\boldsymbol{x} \in \mathbb{R}^n \setminus \{0\}$. Then $\boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{A}_p x$.
lyche-numerical-linear-algebra F04108	317	1.000	`\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}`	$\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}$	We computed $\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}$ in (13.59) and we obtain $\boldsymbol{x}^T\boldsymbol{A}_p\boldsymbol{x}$ by setting all the c's there
lyche-numerical-linear-algebra F04109	317	1.000	`\boldsymbol{x}^{T} \boldsymbol{A}_{p} \boldsymbol{x}`	$\boldsymbol{x}^{T} \boldsymbol{A}_{p} \boldsymbol{x}$	We computed $\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}$ in (13.59) and we obtain $\boldsymbol{x}^T\boldsymbol{A}_p\boldsymbol{x}$ by setting all the c's there
lyche-numerical-linear-algebra F04110	317	0.997	`\boldsymbol{x}^{T} \boldsymbol{A}_{p} x>0`	$\boldsymbol{x}^{T} \boldsymbol{A}_{p} x>0$	Thus $\boldsymbol{x}^T\boldsymbol{A}_p x > 0$ and bounding all the c's in (13.59) from below by c_0 and above
lyche-numerical-linear-algebra F04111	317	0.997	`c_{0}`	c_{0}	Thus $\boldsymbol{x}^T\boldsymbol{A}_p x > 0$ and bounding all the c's in (13.59) from below by c_0 and above
lyche-numerical-linear-algebra F04112	317	1.000	`c_{0} \leq \lambda \leq c_{1}`	$c_{0} \leq \lambda \leq c_{1}$	which implies that $c_0 \leq \lambda \leq c_1$ for all eigenvalues λ of $\boldsymbol{B}\boldsymbol{A} = \boldsymbol{A}_p^{-1}\boldsymbol{A}$.
lyche-numerical-linear-algebra F04113	317	0.998	`c(x, y)=e^{-x+y}`	$c(x, y)=e^{-x+y}$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa \leq e^2 \approx$
lyche-numerical-linear-algebra F04114	317	0.998	`c_{0}=e^{-2}`	$c_{0}=e^{-2}$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa \leq e^2 \approx$
lyche-numerical-linear-algebra F04115	317	0.998	`c_{1}=1`	$c_{1}=1$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa \leq e^2 \approx$
lyche-numerical-linear-algebra F04116	317	0.998	`\kappa \leq e^{2} \approx`	$\kappa \leq e^{2} \approx$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa \leq e^2 \approx$
lyche-numerical-linear-algebra F04117	317	1.000	`\boldsymbol{A}=\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^{T}`	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^{T}$	**Exercise 13.2 (Paraboloid)** Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^T$ be the spectral decomposition of $\boldsymbol{A}$,
lyche-numerical-linear-algebra F04118	318	0.966	`\boldsymbol{v}=\left[v_{1}, \ldots, v_{n}\right]^{T}:= \boldsymbol{U}^{T} \boldsymbol{y}`	$\boldsymbol{v}=\left[v_{1}, \ldots, v_{n}\right]^{T}:= \boldsymbol{U}^{T} \boldsymbol{y}$	$\boldsymbol{v} = [v_1, \ldots, v_n]^T := \boldsymbol{U}^T\boldsymbol{y}$, and set $\boldsymbol{c} := \boldsymbol{U}^T\boldsymbol{b} = [c_1, \ldots, c_n]^T$. Show that

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04119	318	0.966	\boldsymbol{c}:=\boldsymbol{U}^{T} \boldsymbol{b}= \left[c_{1}, \ldots, c_{n}\right]^{T}	$\boldsymbol{c}:=\boldsymbol{U}^{T} \boldsymbol{b}=\left[c_{1}, \ldots, c_{n}\right]^{T}$	$\boldsymbol{v}=[v_1,\ldots,v_n]^T:=\boldsymbol{U}^T\boldsymbol{y}$, and set $\boldsymbol{c}:=\boldsymbol{U}^T\boldsymbol{b}=[c_1,\ldots,c_n]^T$. Show that
lyche-numerical-linear-algebra F04120	318	0.999	\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{r}_{k}	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\alpha_{k}\boldsymbol{r}_{k}$	the iteration is $\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\alpha_k\boldsymbol{r}_k$, where $\boldsymbol{r}_k$ is the residual, $\boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$, and
lyche-numerical-linear-algebra F04121	318	0.999	\boldsymbol{r}_{k}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k}	$\boldsymbol{r}_{k}=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_{k}$	the iteration is $\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\alpha_k\boldsymbol{r}_k$, where $\boldsymbol{r}_k$ is the residual, $\boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$, and
lyche-numerical-linear-algebra F04122	318	0.973	\alpha_{k}=\frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{r}_{k}^{T} \boldsymbol{A r}_{k}}	$\alpha_{k}=\frac{\boldsymbol{r}_{k}^{T}\boldsymbol{r}_{k}}{\boldsymbol{r}_{k}^{T}\boldsymbol{A r}_{k}}$	$\alpha_k=\frac{\boldsymbol{r}_k^T\boldsymbol{r}_k}{\boldsymbol{r}_k^T\boldsymbol{A}\boldsymbol{r}_k}$.
lyche-numerical-linear-algebra F04123	318	0.927	\boldsymbol{A}=\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right], \boldsymbol{b}=\left[\begin{array}{ll}1 & 1\end{array}\right]^{T}	$\boldsymbol{A}=\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right], \boldsymbol{b}=\left[\begin{array}{ll}1 & 1\end{array}\right]^{T}$	**a)** Compute $\boldsymbol{x}_1$ if $\boldsymbol{A}=\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$, $\boldsymbol{b}=[1\ 1]^T$ and $\boldsymbol{x}_0=\boldsymbol{0}$.
lyche-numerical-linear-algebra F04124	318	1.000	\boldsymbol{e}_{k}=\boldsymbol{x}_{k}-\boldsymbol{x}	$\boldsymbol{e}_{k}=\boldsymbol{x}_{k}-\boldsymbol{x}$	**b)** If the k-th error, $\boldsymbol{e}_k=\boldsymbol{x}_k-\boldsymbol{x}$, is an eigenvector of $\boldsymbol{A}$, what can you say about
lyche-numerical-linear-algebra F04125	318	0.907	\boldsymbol{x}_{1}=\left(\frac{\boldsymbol{b}^{T} \boldsymbol{b}}{\boldsymbol{b}^{T} \boldsymbol{A} \boldsymbol{b}}\right) \boldsymbol{b}	$\boldsymbol{x}_{1}=\left(\frac{\boldsymbol{b}^{T}\boldsymbol{b}}{\boldsymbol{b}^{T}\boldsymbol{A}\boldsymbol{b}}\right)\boldsymbol{b}$	conjugate gradient method when $\boldsymbol{x}_0=\boldsymbol{0}$. (Answer: $\boldsymbol{x}_1=\left(\frac{\boldsymbol{b}^T\boldsymbol{b}}{\boldsymbol{b}^T\boldsymbol{A}\boldsymbol{b}}\right)\boldsymbol{b}$.)
lyche-numerical-linear-algebra F04126	318	0.990	\boldsymbol{A} \boldsymbol{y}=\boldsymbol{r}_{0}:= \boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{0}	$\boldsymbol{A}\boldsymbol{y}=\boldsymbol{r}_{0}:=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_{0}$	system $\boldsymbol{A}\boldsymbol{y}=\boldsymbol{r}_0:=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_0$ starting with $\boldsymbol{y}_0=\boldsymbol{0}$.[3]
lyche-numerical-linear-algebra F04127	318	0.990	\boldsymbol{y}_{0}=\mathbf{0} .^{3}	$\boldsymbol{y}_{0}=\mathbf{0} .^{3}$	system $\boldsymbol{A}\boldsymbol{y}=\boldsymbol{r}_0:=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_0$ starting with $\boldsymbol{y}_0=\boldsymbol{0}$.[3]
lyche-numerical-linear-algebra F04128	318	1.000	\boldsymbol{p}_{k}=\boldsymbol{r}_{k}+\sum_{j=0}^{k-1} a_{k, j} \boldsymbol{r}_{j}	$\boldsymbol{p}_{k}=\boldsymbol{r}_{k}+\sum_{j=0}^{k-1} a_{k, j}\boldsymbol{r}_{j}$	[2]Hint: use induction on k to show that $\boldsymbol{p}_k=\boldsymbol{r}_k+\sum_{j=0}^{k-1}a_{k,j}\boldsymbol{r}_j$ for some constants $a_{k,j}$.
lyche-numerical-linear-algebra F04129	318	0.881	\boldsymbol{A} \boldsymbol{y}=\boldsymbol{r}_{0}	$\boldsymbol{A}\boldsymbol{y}=\boldsymbol{r}_{0}$	[3]Hint: The conjugate gradient method for $\boldsymbol{A}\boldsymbol{y}=\boldsymbol{r}_0$ can be written $\boldsymbol{y}_{k+1}:=\boldsymbol{y}_k+\gamma_k\boldsymbol{q}_k$, $\gamma_k:=$
lyche-numerical-linear-algebra F04130	318	0.881	\boldsymbol{y}_{k+1}:=\boldsymbol{y}_{k}+\gamma_{k} \boldsymbol{q}_{k}, \gamma_{k}:=	$\boldsymbol{y}_{k+1}:=\boldsymbol{y}_{k}+\gamma_{k}\boldsymbol{q}_{k}, \gamma_{k}:=$	[3]Hint: The conjugate gradient method for $\boldsymbol{A}\boldsymbol{y}=\boldsymbol{r}_0$ can be written $\boldsymbol{y}_{k+1}:=\boldsymbol{y}_k+\gamma_k\boldsymbol{q}_k$, $\gamma_k:=$
lyche-numerical-linear-algebra F04131	318	0.888	\frac{\boldsymbol{s}_{k}^{T} \boldsymbol{s}_{k}}{ \boldsymbol{q}_{k}^{T} \boldsymbol{A} \boldsymbol{q}_{k}}, \boldsymbol{s}_{k+1}:= \boldsymbol{s}_{k}-\gamma_{k} \boldsymbol{A} \boldsymbol{q}_{k}, \boldsymbol{q}_{k+1}:= \boldsymbol{s}_{k+1}+\delta_{k} \boldsymbol{q}_{k}, \delta_{k}:=\frac{\boldsymbol{s}_{k+1}^{T} \boldsymbol{s}_{k+1}}{\boldsymbol{s}_{k}^{T} \boldsymbol{s}_{k}}	$\frac{\boldsymbol{s}_{k}^{T}\boldsymbol{s}_{k}}{\boldsymbol{q}_{k}^{T}\boldsymbol{A}\boldsymbol{q}_{k}}, \boldsymbol{s}_{k+1}:=\boldsymbol{s}_{k}-\gamma_{k}\boldsymbol{A}\boldsymbol{q}_{k}, \boldsymbol{q}_{k+1}:=\boldsymbol{s}_{k+1}+\delta_{k}\boldsymbol{q}_{k}, \delta_{k}:=\frac{\boldsymbol{s}_{k+1}^{T}\boldsymbol{s}_{k+1}}{\boldsymbol{s}_{k}^{T}\boldsymbol{s}_{k}}$	$\frac{\boldsymbol{s}_k^T\boldsymbol{s}_k}{\boldsymbol{q}_k^T\boldsymbol{A}\boldsymbol{q}_k}$, $\boldsymbol{s}_{k+1}:=\boldsymbol{s}_k-\gamma_k\boldsymbol{A}\boldsymbol{q}_k$, $\boldsymbol{q}_{k+1}:=\boldsymbol{s}_{k+1}+\delta_k\boldsymbol{q}_k$, $\delta_k:=\frac{\boldsymbol{s}_{k+1}^T\boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T\boldsymbol{s}_k}$. Show that $\boldsymbol{y}_k=\boldsymbol{x}_k-\boldsymbol{x}_0$,
lyche-numerical-linear-algebra F04132	318	0.888	\boldsymbol{y}_{k}=\boldsymbol{x}_{k}-\boldsymbol{x}_{0}	$\boldsymbol{y}_{k}=\boldsymbol{x}_{k}-\boldsymbol{x}_{0}$	$\frac{\boldsymbol{s}_k^T\boldsymbol{s}_k}{\boldsymbol{q}_k^T\boldsymbol{A}\boldsymbol{q}_k}$, $\boldsymbol{s}_{k+1}:=\boldsymbol{s}_k-\gamma_k\boldsymbol{A}\boldsymbol{q}_k$, $\boldsymbol{q}_{k+1}:=\boldsymbol{s}_{k+1}+\delta_k\boldsymbol{q}_k$, $\delta_k:=\frac{\boldsymbol{s}_{k+1}^T\boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T\boldsymbol{s}_k}$. Show that $\boldsymbol{y}_k=\boldsymbol{x}_k-\boldsymbol{x}_0$,
lyche-numerical-linear-algebra F04133	318	0.977	\boldsymbol{s}_{k}=\boldsymbol{r}_{k}	$\boldsymbol{s}_{k}=\boldsymbol{r}_{k}$	$\boldsymbol{s}_k=\boldsymbol{r}_k$, and $\boldsymbol{q}_k=\boldsymbol{p}_k$, for $k=0,1,2\ldots$.
lyche-numerical-linear-algebra F04134	318	0.977	\boldsymbol{q}_{k}=\boldsymbol{p}_{k}	$\boldsymbol{q}_{k}=\boldsymbol{p}_{k}$	$\boldsymbol{s}_k=\boldsymbol{r}_k$, and $\boldsymbol{q}_k=\boldsymbol{p}_k$, for $k=0,1,2\ldots$.
lyche-numerical-linear-algebra F04135	319	1.000	n=100,400,1600,2500	$n=100,400,1600,2500$	to test for say $n=100,400,1600,2500$, and tabulate K/n instead of $K/\sqrt{n}$ in the
lyche-numerical-linear-algebra F04136	319	1.000	K / n	K / n	to test for say $n=100,400,1600,2500$, and tabulate K/n instead of $K/\sqrt{n}$ in the
lyche-numerical-linear-algebra F04137	319	1.000	\boldsymbol{A} \in \mathbb{R}^{m, n}, \boldsymbol{b} \in \mathbb{R}^{m}	$\boldsymbol{A}\in\mathbb{R}^{m, n}, \boldsymbol{b}\in\mathbb{R}^{m}$	$\boldsymbol{A}\in\mathbb{R}^{m,n}$, $\boldsymbol{b}\in\mathbb{R}^m$ and $\boldsymbol{A}^T\boldsymbol{A}$ is positive definite.[4] Explain why only the following
lyche-numerical-linear-algebra F04138	319	0.996	\mathrm{r}=\mathrm{A}^{\prime}\left(\mathrm{b}- \mathrm{A}^{*} \mathrm{x}\right) ; \mathrm{p}=\mathrm{r}	$\mathrm{r}=\mathrm{A}^{\prime}\left(\mathrm{b}-\mathrm{A}^{*}\mathrm{x}\right) ; \mathrm{p}=\mathrm{r}$	1. r=A'(b-A*x); p=r;
lyche-numerical-linear-algebra F04139	319	0.926	\mathrm{r}=\mathrm{r}-\mathrm{a}^{*} \mathrm{~A}^{*} \mathrm{t}	$\mathrm{r}=\mathrm{r}-\mathrm{a}^{*}\mathrm{~A}^{*}\mathrm{t}$	3. r=r-a*A'*t;

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04140	319	1.000	`\operatorname{cond}_{2}(\boldsymbol{A})^{2}`	$\operatorname{cond}_{2}(\boldsymbol{A})^{2}$	Note that the condition number of the normal equations is $\operatorname{cond}_2(\boldsymbol{A})^2$, the square of
lyche-numerical-linear-algebra F04141	319	1.000	`\left\{\boldsymbol{v}_{i}\right\}_{i=1}^{k}`	$\left\{\boldsymbol{v}_{i}\right\}_{i=1}^{k}$	**a)** Let $\{\boldsymbol{v}_i\}_{i=1}^k$ be a set of linearly independent vectors in $\mathbb{R}^n$, and let $\langle\cdot,\cdot\rangle$ be an
lyche-numerical-linear-algebra F04142	319	1.000	`n_{i j}=\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{j}\right\rangle`	$n_{i j}=\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{j}\right\rangle$	inner product in $\mathbb{R}^n$. Explain that the $k \times k$-matrix $\boldsymbol{N}$ with entries $n_{ij} = \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle$
lyche-numerical-linear-algebra F04143	319	0.997	`\mathbb{W} \subset \mathbb{R}^{n}`	$\mathbb{W} \subset \mathbb{R}^{n}$	**b)** Let $\mathbb{W} \subset \mathbb{R}^n$ be any linear subspace. Show that there is one and only one vector
lyche-numerical-linear-algebra F04144	319	1.000	`\hat{\boldsymbol{x}} \in \mathbb{W}`	$\hat{\boldsymbol{x}} \in \mathbb{W}$	$\hat{\boldsymbol{x}} \in \mathbb{W}$ so that
lyche-numerical-linear-algebra F04145	319	1.000	`\hat{\boldsymbol{x}}`	$\hat{\boldsymbol{x}}$	and that $\hat{\boldsymbol{x}}$ satisfies
lyche-numerical-linear-algebra F04146	319	1.000	`\mathbb{W}`	$\mathbb{W}$	space $\mathbb{W}$ is taken to be the Krylov space
lyche-numerical-linear-algebra F04147	320	1.000	`\boldsymbol{x}_{k} \in \mathbb{W}_{k}`	$\boldsymbol{x}_{k} \in \mathbb{W}_{k}$	$\boldsymbol{x}_k$. Assume that $\boldsymbol{x}_k \in \mathbb{W}_k$ is already determined. In addition, assume that we
lyche-numerical-linear-algebra F04148	320	1.000	`\boldsymbol{p}_{k} \in \mathbb{W}_{k+1}`	$\boldsymbol{p}_{k} \in \mathbb{W}_{k+1}$	already have computed a "search direction" $\boldsymbol{p}_k \in \mathbb{W}_{k+1}$ such that $\lVert \boldsymbol{A}\boldsymbol{p}_k \rVert_2 =$
lyche-numerical-linear-algebra F04149	320	1.000	`\left|\boldsymbol{A} \boldsymbol{p}_{k}\right|_{2}=`	$\left\lVert\boldsymbol{A} \boldsymbol{p}_{k}\right\rVert_{2}=$	already have computed a "search direction" $\boldsymbol{p}_k \in \mathbb{W}_{k+1}$ such that $\lVert \boldsymbol{A}\boldsymbol{p}_k \rVert_2 =$
lyche-numerical-linear-algebra F04150	320	1.000	`\left|\boldsymbol{p}_{k}\right|_{\boldsymbol{A}}=1`	$\left\lVert\boldsymbol{p}_{k}\right\rVert_{\boldsymbol{A}}=1$	$\lVert \boldsymbol{p}_k \rVert_{\boldsymbol{A}} = 1$, and such that
lyche-numerical-linear-algebra F04151	320	1.000	`\alpha_{k} \in \mathbb{R}`	$\alpha_{k} \in \mathbb{R}$	Show that $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k$ for a suitable $\alpha_k \in \mathbb{R}$, and express α_k in terms of
lyche-numerical-linear-algebra F04152	320	1.000	`\boldsymbol{p}_{k-2}, \boldsymbol{p}_{k-1}`	$\boldsymbol{p}_{k-2}, \boldsymbol{p}_{k-1}$	further that the vectors $\boldsymbol{p}_{k-2}$, $\boldsymbol{p}_{k-1}$, and $\boldsymbol{p}_k$ are already known with properties as
lyche-numerical-linear-algebra F04153	320	1.000	`k \leq 3`	$k \leq 3$	**a)** Determine the vectors defining the Krylov spaces for $k \leq 3$ taking as initial
lyche-numerical-linear-algebra F04154	320	1.000	`\left[\boldsymbol{b}, \boldsymbol{A} \boldsymbol{b}, \boldsymbol{A}^{2} \boldsymbol{b}\right]=\left[ \begin{array}{rrr}4 & 8 & 20 \\ 0 & -4 & -16 \\ 0 & 0 & 4\end{array}\right]`	$\left[\boldsymbol{b}, \boldsymbol{A} \boldsymbol{b}, \boldsymbol{A}^{2} \boldsymbol{b}\right]=\left[\begin{array}{rrr}4 & 8 & 20 \\ 0 & -4 & -16 \\ 0 & 0 & 4\end{array}\right]$	approximation $\boldsymbol{x} = \boldsymbol{0}$. Answer: $[\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \boldsymbol{A}^2\boldsymbol{b}] = \begin{bmatrix} 4 & 8 & 20 \\ 0 & -4 & -16 \\ 0 & 0 & 4 \end{bmatrix}$.
lyche-numerical-linear-algebra F04155	321	0.983	`\operatorname{dim}\left(\mathbb{W}_{k}\right)=k`	$\operatorname{dim}\left(\mathbb{W}_{k}\right)=k$	• $\dim(\mathbb{W}_k) = k$ for $k = 0, 1, 2, 3$.
lyche-numerical-linear-algebra F04156	321	0.983	`k=0,1,2,3`	$k=0,1,2,3$	• $\dim(\mathbb{W}_k) = k$ for $k = 0, 1, 2, 3$.
lyche-numerical-linear-algebra F04157	321	0.987	`\boldsymbol{x}_{3}`	$\boldsymbol{x}_{3}$	• $\boldsymbol{x}_3$ is the exact solution of $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.
lyche-numerical-linear-algebra F04158	321	1.000	`\boldsymbol{r}_{0}, \ldots, \boldsymbol{r}_{k-1}`	$\boldsymbol{r}_{0}, \ldots, \boldsymbol{r}_{k-1}$	• $\boldsymbol{r}_0, \ldots, \boldsymbol{r}_{k-1}$ is an orthogonal basis for $\mathbb{W}_k$ for $k = 1, 2, 3$.
lyche-numerical-linear-algebra F04159	321	1.000	`k=1,2,3`	$k=1,2,3$	• $\boldsymbol{r}_0, \ldots, \boldsymbol{r}_{k-1}$ is an orthogonal basis for $\mathbb{W}_k$ for $k = 1, 2, 3$.
lyche-numerical-linear-algebra F04160	321	1.000	`\boldsymbol{p}_{0}, \ldots, \boldsymbol{p}_{k-1}`	$\boldsymbol{p}_{0}, \ldots, \boldsymbol{p}_{k-1}$	• $\boldsymbol{p}_0, \ldots, \boldsymbol{p}_{k-1}$ is an $\boldsymbol{A}$-orthogonal basis for $\mathbb{W}_k$ for $k = 1, 2, 3$.
lyche-numerical-linear-algebra F04161	321	1.000	`\left\{\left|\boldsymbol{r}_{k}\right|\right.`	$\left\{\left\lVert\boldsymbol{r}_{k}\right\rVert\right.$	• $\{\lVert \boldsymbol{r}_k \rVert$ is monotonically decreasing.
lyche-numerical-linear-algebra F04162	321	0.991	`\left\{\left|\boldsymbol{x}_{k}-\boldsymbol{x}\right| \right.`	$\left\{\left\lVert\boldsymbol{x}_{k}-\boldsymbol{x}\right\rVert \right.$	• $\{\lVert \boldsymbol{x}_k - \boldsymbol{x} \rVert$ is monotonically decreasing.
lyche-numerical-linear-algebra F04163	321	1.000	`\boldsymbol{B}^{T}=-\boldsymbol{B}`	$\boldsymbol{B}^{T}=-\boldsymbol{B}$	Let $\boldsymbol{B} \in \mathbb{R}^{n\times n}$ be an antisymmetric matrix, i.e., $\boldsymbol{B}^T = -\boldsymbol{B}$, and let $\boldsymbol{A} := \boldsymbol{I} - \boldsymbol{B}$,
lyche-numerical-linear-algebra F04164	321	1.000	`\boldsymbol{A}:=\boldsymbol{I}-\boldsymbol{B}`	$\boldsymbol{A}:=\boldsymbol{I}-\boldsymbol{B}$	Let $\boldsymbol{B} \in \mathbb{R}^{n\times n}$ be an antisymmetric matrix, i.e., $\boldsymbol{B}^T = -\boldsymbol{B}$, and let $\boldsymbol{A} := \boldsymbol{I} - \boldsymbol{B}$,

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-a F04165</td><td>321</td><td>0.998</td><td>\|\boldsymbol{A} \boldsymbol{x}\|_{2}^{2}=\| \boldsymbol{x}\|_{2}^{2}+\|\boldsymbol{B} \boldsymbol{x}\|_{2}^{2}</td><td>$\|\boldsymbol{A} \boldsymbol{x}\|_{2}^{2}=\| \boldsymbol{x}\|_{2}^{2}+\|\boldsymbol{B} \boldsymbol{x}\|_{2}^{2}$</td><td>b) Show that $\|\boldsymbol{A}\boldsymbol{x}\|_2^2 = \|\boldsymbol{x}\|_2^2 + \|\boldsymbol{B}\boldsymbol{x}\|_2^2$ and that $\|\boldsymbol{A}\|_2 = \sqrt{1+\|\boldsymbol{B}\|_2^2}$.</td></tr>
<tr><td>lyche-numerical-linear-a F04166</td><td>321</td><td>0.998</td><td>\|\boldsymbol{A}\|_{2}=\sqrt{1+\|\boldsymbol{B} \|_{2}^{2}}</td><td>$\|\boldsymbol{A}\|_{2}=\sqrt{1+\|\boldsymbol{B} \|_{2}^{2}}$</td><td>b) Show that $\|\boldsymbol{A}\boldsymbol{x}\|_2^2 = \|\boldsymbol{x}\|_2^2 + \|\boldsymbol{B}\boldsymbol{x}\|_2^2$ and that $\|\boldsymbol{A}\|_2 = \sqrt{1+\|\boldsymbol{B}\|_2^2}$.</td></tr>
<tr><td>lyche-numerical-linear-a F04167</td><td>321</td><td>1.000</td><td>\|\boldsymbol{A}\|_{2} \leq 1</td><td>$\|\boldsymbol{A}\|_{2} \leq 1$</td><td>and $\|\boldsymbol{A}\|_2 \le 1$.</td></tr>
<tr><td>lyche-numerical-linear-a F04168</td><td>321</td><td>1.000</td><td>1 \leq k \leq n, \mathcal{W}=\operatorname{span}\left(\boldsymbol{w}_{1}, \ldots, \boldsymbol{w}_{k}\right)</td><td>$1 \leq k \leq n, \mathcal{W}=\operatorname{span}\left(\boldsymbol{w}_{1}, \ldots, \boldsymbol{w}_{k}\right)$</td><td>d) Let $1 \le k \le n$, $\mathcal{W} = \operatorname{span}(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k)$ a k-dimensional subspace of $\mathbb{R}^n$ and</td></tr>
<tr><td>lyche-numerical-linear-a F04169</td><td>321</td><td>1.000</td><td>\boldsymbol{x} \in \mathcal{W}</td><td>$\boldsymbol{x} \in \mathcal{W}$</td><td>$\boldsymbol{b} \in \mathbb{R}^n$. Show that if $\boldsymbol{x} \in \mathcal{W}$ is such that</td></tr>
<tr><td>lyche-numerical-linear-a F04170</td><td>321</td><td>0.835</td><td>\|\boldsymbol{x}\|_{2} \leq\|\boldsymbol{b}\|_{2}</td><td>$\|\boldsymbol{x}\|_{2} \leq\|\boldsymbol{b}\|_{2}$</td><td>then $\|\boldsymbol{x}\|_2 \le \|\boldsymbol{b}\|_2$.</td></tr>
<tr><td>lyche-numerical-linear-a F04171</td><td>322</td><td>1.000</td><td>\boldsymbol{x}:=\sum_{j=1}^{k} x_{j} \boldsymbol{w}_{j}</td><td>$\boldsymbol{x}:=\sum_{j=1}^{k} x_{j} \boldsymbol{w}_{j}$</td><td>With $\boldsymbol{x} := \sum_{j=1}^{k} x_j \boldsymbol{w}_j$ the problem (13.65) is equivalent to finding real</td></tr>
<tr><td>lyche-numerical-linear-a F04172</td><td>322</td><td>1.000</td><td>x_{1}, \ldots, x_{k}</td><td>$x_{1}, \ldots, x_{k}$</td><td>numbers $x_1, \ldots, x_k$ solving the linear system</td></tr>
<tr><td>lyche-numerical-linear-a F04173</td><td>322</td><td>0.996</td><td>\boldsymbol{x}^{*}:=\boldsymbol{A}^{-1} \boldsymbol{b}</td><td>$\boldsymbol{x}^{*}:=\boldsymbol{A}^{-1} \boldsymbol{b}$</td><td>e) Let $\boldsymbol{x}^* := \boldsymbol{A}^{-1}\boldsymbol{b}$. Show that</td></tr>
<tr><td>lyche-numerical-linear-a F04174</td><td>322</td><td>1.000</td><td>\boldsymbol{x}_{k} \in \mathcal{W}_{k}</td><td>$\boldsymbol{x}_{k} \in \mathcal{W}_{k}$</td><td>For $k = 1, 2, \ldots, n$ we define $\boldsymbol{x}_k \in \mathcal{W}_k$ by</td></tr>
<tr><td>lyche-numerical-linear-a F04175</td><td>322</td><td>0.986</td><td>\boldsymbol{x} *</td><td>$\boldsymbol{x} *$</td><td>“good” approximation to $\boldsymbol{x}*$ follows from (13.67). In this exercise we will derive a</td></tr>
<tr><td>lyche-numerical-linear-a F04176</td><td>322</td><td>1.000</td><td>k=0, \ldots, n</td><td>$k=0, \ldots, n$</td><td>For $k = 0, \ldots, n$ we set</td></tr>
<tr><td>lyche-numerical-linear-a F04177</td><td>322</td><td>1.000</td><td>\omega_{0}, \omega_{1}, \ldots, \omega_{m}</td><td>$\omega_{0}, \omega_{1}, \ldots, \omega_{m}$</td><td>Let $\omega_0, \omega_1, \ldots, \omega_m$ be real numbers defined recursively for $k = 1, 2, \ldots, m$ by</td></tr>
<tr><td>lyche-numerical-linear-a F04178</td><td>322</td><td>1.000</td><td>k=1,2, \ldots, m</td><td>$k=1,2, \ldots, m$</td><td>Let $\omega_0, \omega_1, \ldots, \omega_m$ be real numbers defined recursively for $k = 1, 2, \ldots, m$ by</td></tr>
<tr><td>lyche-numerical-linear-a F04179</td><td>323</td><td>1.000</td><td>k=0,1, \ldots, m-1</td><td>$k=0,1, \ldots, m-1$</td><td>$k = 0, 1, \ldots, m-1$</td></tr>
<tr><td>lyche-numerical-linear-a F04180</td><td>323</td><td>1.000</td><td>\boldsymbol{x}_{0}=\boldsymbol{x}_{-1}=0</td><td>$\boldsymbol{x}_{0}=\boldsymbol{x}_{-1}=0$</td><td>starting with $\boldsymbol{x}_0 = \boldsymbol{x}_{-1} = 0$ and $\boldsymbol{r}_0 = \boldsymbol{r}_{-1} = \boldsymbol{b}$.</td></tr>
<tr><td>lyche-numerical-linear-a F04181</td><td>323</td><td>1.000</td><td>\boldsymbol{r}_{0}=\boldsymbol{r}_{-1}=\boldsymbol{b}</td><td>$\boldsymbol{r}_{0}=\boldsymbol{r}_{-1}=\boldsymbol{b}$</td><td>starting with $\boldsymbol{x}_0 = \boldsymbol{x}_{-1} = 0$ and $\boldsymbol{r}_0 = \boldsymbol{r}_{-1} = \boldsymbol{b}$.</td></tr>
<tr><td>lyche-numerical-linear-a F04182</td><td>323</td><td>1.000</td><td>0<\omega_{k}<1</td><td>$0<\omega_{k}<1$</td><td>a) Show that $0 < \omega_k < 1$ for $k = 1, 2, \ldots, m$.</td></tr>
<tr><td>lyche-numerical-linear-a F04183</td><td>323</td><td>1.000</td><td>\left\langle\boldsymbol{r}_{k}, \boldsymbol{r}_{j}\right \rangle=0</td><td>$\left\langle\boldsymbol{r}_{k}, \boldsymbol{r}_{j}\right \rangle=0$</td><td>c) Show that $\langle \boldsymbol{r}_k, \boldsymbol{r}_j \rangle = 0$ for $j = 0, 1, \ldots, k-1$.</td></tr>
<tr><td>lyche-numerical-linear-a F04184</td><td>323</td><td>1.000</td><td>j=0,1, \ldots, k-1</td><td>$j=0,1, \ldots, k-1$</td><td>c) Show that $\langle \boldsymbol{r}_k, \boldsymbol{r}_j \rangle = 0$ for $j = 0, 1, \ldots, k-1$.</td></tr>
<tr><td>lyche-numerical-linear-a F04185</td><td>323</td><td>1.000</td><td>k \leq m+1</td><td>$k \leq m+1$</td><td>d) Show that if $k \le m+1$ then $\mathcal{W}_k = \operatorname{span}(\boldsymbol{r}_0, \boldsymbol{r}_1, \ldots, \boldsymbol{r}_{k-1})$ and $\dim \mathcal{W}_k = k$.</td></tr>
<tr><td>lyche-numerical-linear-a F04186</td><td>323</td><td>1.000</td><td>\mathcal{W}_{k}=\operatorname{span}\left(\boldsymbol{r}_{0}, \boldsymbol{r}_{1}, \ldots, \boldsymbol{r}_{k-1}\right)</td><td>$\mathcal{W}_{k}=\operatorname{span}\left(\boldsymbol{r}_{0}, \boldsymbol{r}_{1}, \ldots, \boldsymbol{r}_{k-1}\right)$</td><td>d) Show that if $k \le m+1$ then $\mathcal{W}_k = \operatorname{span}(\boldsymbol{r}_0, \boldsymbol{r}_1, \ldots, \boldsymbol{r}_{k-1})$ and $\dim \mathcal{W}_k = k$.</td></tr>
<tr><td>lyche-numerical-linear-a F04187</td><td>323</td><td>1.000</td><td>\operatorname{dim} \mathcal{W}_{k}=k</td><td>$\operatorname{dim} \mathcal{W}_{k}=k$</td><td>d) Show that if $k \le m+1$ then $\mathcal{W}_k = \operatorname{span}(\boldsymbol{r}_0, \boldsymbol{r}_1, \ldots, \boldsymbol{r}_{k-1})$ and $\dim \mathcal{W}_k = k$.</td></tr>
<tr><td>lyche-numerical-linear-a F04188</td><td>323</td><td>1.000</td><td>1 \leq k \leq m-1</td><td>$1 \leq k \leq m-1$</td><td>e) Show that if $1 \le k \le m-1$ then</td></tr>
<tr><td>lyche-numerical-linear-a F04189</td><td>323</td><td>0.605</td><td>\alpha_{k}:=\left\langle\boldsymbol{B r}_{k}, \boldsymbol{r}_{k+1}\right\rangle / \rho_{k+1}</td><td>$\alpha_{k}:=\left\langle\boldsymbol{B r}_{k}, \boldsymbol{r}_{k+1}\right\rangle / \rho_{k+1}$</td><td>where $\alpha_k := \langle \boldsymbol{B}\boldsymbol{r}_k, \boldsymbol{r}_{k+1} \rangle / \rho_{k+1}$ and $\beta_k := \langle \boldsymbol{B}\boldsymbol{r}_k, \boldsymbol{r}_{k-1} \rangle / \rho_{k-1}$.</td></tr>
<tr><td>lyche-numerical-linear-a F04190</td><td>323</td><td>0.605</td><td>\beta_{k}:=\left\langle\boldsymbol{B r}_{k}, \boldsymbol{r}_{k-1}\right\rangle / \rho_{k-1}</td><td>$\beta_{k}:=\left\langle\boldsymbol{B r}_{k}, \boldsymbol{r}_{k-1}\right\rangle / \rho_{k-1}$</td><td>where $\alpha_k := \langle \boldsymbol{B}\boldsymbol{r}_k, \boldsymbol{r}_{k+1} \rangle / \rho_{k+1}$ and $\beta_k := \langle \boldsymbol{B}\boldsymbol{r}_k, \boldsymbol{r}_{k-1} \rangle / \rho_{k-1}$.</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04191	323	0.865	`\alpha_{0}:=\left\langle\boldsymbol{B r}_{0}, \boldsymbol{r}_{1}\right\rangle / \rho_{1}`	$\alpha_{0}:=\left\langle\boldsymbol{B r}_{0}, \boldsymbol{r}_{1}\right\rangle / \rho_{1}$	**f)** Define $\alpha_0 := \langle \boldsymbol{B r}_0, \boldsymbol{r}_1\rangle/\rho_1$ and show that $\alpha_0 = 1$.
lyche-numerical-linear-algebra F04192	323	0.865	`\alpha_{0}=1`	$\alpha_{0}=1$	**f)** Define $\alpha_0 := \langle \boldsymbol{B r}_0, \boldsymbol{r}_1\rangle/\rho_1$ and show that $\alpha_0 = 1$.
lyche-numerical-linear-algebra F04193	323	1.000	`\beta_{k}=-\alpha_{k-1} \rho_{k} / \rho_{k-1}`	$\beta_{k}=-\alpha_{k-1} \rho_{k} / \rho_{k-1}$	**g)** Show that if $1 \leq k \leq m-1$ then $\beta_k = -\alpha_{k-1}\rho_k/\rho_{k-1}$.
lyche-numerical-linear-algebra F04194	323	1.000	`\alpha_{k}+\beta_{k}=1`	$\alpha_{k}+\beta_{k}=1$	**i)** Use (13.71) and (13.72) to show that $\alpha_k + \beta_k = 1$ for $k = 1, 2, \ldots, m-1$.
lyche-numerical-linear-algebra F04195	323	1.000	`k=1,2, \ldots, m-1`	$k=1,2, \ldots, m-1$	**i)** Use (13.71) and (13.72) to show that $\alpha_k + \beta_k = 1$ for $k = 1, 2, \ldots, m-1$.
lyche-numerical-linear-algebra F04196	323	1.000	`\alpha_{k} \geq 1`	$\alpha_{k} \geq 1$	**j)** Show that $\alpha_k \geq 1$ for $k = 1, 2, \ldots, m-1$.
lyche-numerical-linear-algebra F04197	323	1.000	`\boldsymbol{x}_{k}, \boldsymbol{r}_{k}`	$\boldsymbol{x}_{k}, \boldsymbol{r}_{k}$	**k)** Show that $\boldsymbol{x}_k$, $\boldsymbol{r}_k$ and $\boldsymbol{\omega}_k$ satisfy the recurrence relations (13.68), (13.69)
lyche-numerical-linear-algebra F04198	323	1.000	`\boldsymbol{\omega}_{k}`	$\boldsymbol{\omega}_{k}$	**k)** Show that $\boldsymbol{x}_k$, $\boldsymbol{r}_k$ and $\boldsymbol{\omega}_k$ satisfy the recurrence relations (13.68), (13.69)
lyche-numerical-linear-algebra F04199	323	0.887	`\operatorname{arccosh}`	$\operatorname{arccosh}$	where arccosh is the inverse function of $\cosh x := (e^x + e^{-x})/2$.
lyche-numerical-linear-algebra F04200	323	0.887	`\cosh x:=\left(e^{x}+e^{-x}\right) / 2`	$\cosh x:=\left(e^{x}+e^{-x}\right) / 2$	where arccosh is the inverse function of $\cosh x := (e^x + e^{-x})/2$.
lyche-numerical-linear-algebra F04201	323	1.000	`\boldsymbol{A}^{-1}\left(\boldsymbol{r}_{k+1}-\boldsymbol{r}_{j}\right) \in \mathcal{W}_{k+1}`	$\boldsymbol{A}^{-1}\left(\boldsymbol{r}_{k+1}-\boldsymbol{r}_{j}\right) \in \mathcal{W}_{k+1}$	[5]Hint: Show that $\boldsymbol{A}^{-1}(\boldsymbol{r}_{k+1} - \boldsymbol{r}_j) \in \mathcal{W}_{k+1}$.
lyche-numerical-linear-algebra F04202	324	1.000	`f:[a, b] \rightarrow \mathbb{R}`	$f:[a, b] \rightarrow \mathbb{R}$	**Exercise 13.16 (Maximum of a Convex Function)** Show that if $f : [a, b] \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra F04203	324	1.000	`\max _{a \leq x \leq b} f(x) \leq \max \{f(a), f(b)\}`	$\max _{a \leq x \leq b} f(x) \leq \max \{f(a), f(b)\}$	is convex then $\max_{a \leq x \leq b} f(x) \leq \max\{f(a), f(b)\}$.
lyche-numerical-linear-algebra F04204	326	1.000	`\boldsymbol{A} \boldsymbol{e}_{i}=a_{i i} \boldsymbol{e}_{i}`	$\boldsymbol{A} \boldsymbol{e}_{i}=a_{i i} \boldsymbol{e}_{i}$	**Diagonal Matrices** The eigenpairs are easily determined. Since $\boldsymbol{A}\boldsymbol{e}_i = a_{ii}\boldsymbol{e}_i$
lyche-numerical-linear-algebra F04205	326	0.965	`\left(\lambda_{i}, \boldsymbol{e}_{i}\right)`	$\left(\lambda_{i}, \boldsymbol{e}_{i}\right)$	the eigenpairs are $(\lambda_i, \boldsymbol{e}_i)$, where $\lambda_i = a_{ii}$ for $i = 1, \ldots, n$. Moreover, the
lyche-numerical-linear-algebra F04206	326	0.965	`\lambda_{i}=a_{i i}`	$\lambda_{i}=a_{i i}$	the eigenpairs are $(\lambda_i, \boldsymbol{e}_i)$, where $\lambda_i = a_{ii}$ for $i = 1, \ldots, n$. Moreover, the
lyche-numerical-linear-algebra F04207	326	1.000	`\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\prod_{i=1}^{n}\left(a_{i i}-\lambda \right)`	$\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\prod_{i=1}^{n}\left(a_{i i}-\lambda \right)$	the eigenvalues Since $\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = \prod_{i=1}^{n}(a_{ii} - \lambda)$ the eigenvalues are $\lambda_i = a_{ii}$
lyche-numerical-linear-algebra F04208	326	1.000	`\boldsymbol{A}_{i} \boldsymbol{X}_{i}=\boldsymbol{X}_{i} \boldsymbol{D}_{i}`	$\boldsymbol{A}_{i} \boldsymbol{X}_{i}=\boldsymbol{X}_{i} \boldsymbol{D}_{i}$	Here the eigenpair problem reduces to r smaller problems. Let $\boldsymbol{A}_i\boldsymbol{X}_i = \boldsymbol{X}_i\boldsymbol{D}_i$
lyche-numerical-linear-algebra F04209	326	1.000	`\boldsymbol{A}_{i}`	$\boldsymbol{A}_{i}$	define the eigenpairs of $\boldsymbol{A}_i$ for $i = 1, \ldots, r$ and let $\boldsymbol{X} := \operatorname{diag}(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_r)$,
lyche-numerical-linear-algebra F04210	326	1.000	`\boldsymbol{X}:=\operatorname{diag}\left( \boldsymbol{X}_{1}, \ldots, \boldsymbol{X}_{r}\right)`	$\boldsymbol{X}:=\operatorname{diag}\left(\boldsymbol{X}_{1}, \ldots, \boldsymbol{X}_{r}\right)$	define the eigenpairs of $\boldsymbol{A}_i$ for $i = 1, \ldots, r$ and let $\boldsymbol{X} := \operatorname{diag}(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_r)$,
lyche-numerical-linear-algebra F04211	326	1.000	`\boldsymbol{D}:=\operatorname{diag}\left( \boldsymbol{D}_{1}, \ldots, \boldsymbol{D}_{r}\right)`	$\boldsymbol{D}:=\operatorname{diag}\left(\boldsymbol{D}_{1}, \ldots, \boldsymbol{D}_{r}\right)$	$\boldsymbol{D} := \operatorname{diag}(\boldsymbol{D}_1, \ldots, \boldsymbol{D}_r)$. Then the eigenpairs for $\boldsymbol{A}$ are given by
lyche-numerical-linear-algebra F04212	327	0.988	`\boldsymbol{A}_{11}, \boldsymbol{A}_{22}, \ldots, \boldsymbol{A}_{r r}`	$\boldsymbol{A}_{11}, \boldsymbol{A}_{22}, \ldots, \boldsymbol{A}_{r r}$	$\boldsymbol{A}_{11}, \boldsymbol{A}_{22}, \ldots, \boldsymbol{A}_{rr}$ be the diagonal blocks of $\boldsymbol{A}$. By Property 8. of determinants
lyche-numerical-linear-algebra F04213	327	1.000	`\pi_{\boldsymbol{A}}(\lambda)`	$\pi_{\boldsymbol{A}}(\lambda)$	a small change in one of the coefficients of $\pi_{\boldsymbol{A}}(\lambda)$ can lead to a large change in the
lyche-numerical-linear-algebra F04214	327	0.405	`\left.\pi_{\boldsymbol{A}}(\lambda):\right)=\lambda^{16}`	$\left.\pi_{\boldsymbol{A}}(\lambda):\right)=\lambda^{16}$	roots of the polynomial. For example, if $\pi_{\boldsymbol{A}}(\lambda :) = \lambda^{16}$ and $q(\lambda) = \lambda^{16} - 10^{-16}$ then
lyche-numerical-linear-algebra F04215	327	0.405	`q(\lambda)=\lambda^{16}-10^{-16}`	$q(\lambda)=\lambda^{16}-10^{-16}$	roots of the polynomial. For example, if $\pi_{\boldsymbol{A}}(\lambda :) = \lambda^{16}$ and $q(\lambda) = \lambda^{16} - 10^{-16}$ then
lyche-numerical-linear-algebra F04216	327	0.999	`\lambda_{j}=10^{-1} e^{2 \pi i j / 16}`	$\lambda_{j}=10^{-1} e^{2 \pi i j / 16}$	the roots of $\pi_{\boldsymbol{A}}$ are all equal to zero, while the roots of q are $\lambda_j = 10^{-1}e^{2\pi ij/16}$,
lyche-numerical-linear-algebra F04217	327	1.000	`j=1, \ldots, 16`	$j=1, \ldots, 16$	$j = 1, \ldots, 16$. The roots of q have absolute value 0.1 and a perturbation in one

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04218	327	1.000	`10^{-16}`	10^{-16}	of the polynomial coefficients of magnitude 10^{-16} has led to an error in the roots
lyche-numerical-linear-algebra F04219	328	1.000	`R \cap C`	$R \cap C$	*Then any eigenvalue of* $\boldsymbol{A}$ *lies in* $R \cap C$ *where* $R = R_1 \cup R_2 \cup \cdots \cup R_n$ *and*
lyche-numerical-linear-algebra F04220	328	1.000	`R=R_{1} \cup R_{2} \cup \cdots \cup R_{n}`	$R=R_{1} \cup R_{2} \cup \cdots \cup R_{n}$	*Then any eigenvalue of* $\boldsymbol{A}$ *lies in* $R \cap C$ *where* $R = R_1 \cup R_2 \cup \cdots \cup R_n$ *and*
lyche-numerical-linear-algebra F04221	328	1.000	`C=C_{1} \cup C_{2} \cup \cdots \cup C_{n}`	$C=C_{1} \cup C_{2} \cup \cdots \cup C_{n}$	$C = C_1 \cup C_2 \cup \cdots \cup C_n$.
lyche-numerical-linear-algebra F04222	328	0.621	`\lambda \in R_{i}`	$\lambda \in R_{i}$	***Proof*** Suppose $(\lambda, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}$. We claim that $\lambda \in R_i$, where i is such
lyche-numerical-linear-algebra F04223	328	0.997	`\left|x_{i}\right|=\\|\boldsymbol{x}\\|_{\infty}`	$\left\|x_{i}\right\|=\\|\boldsymbol{x}\\|_{\infty}$	that $\|x_i\| = \\|\boldsymbol{x}\\|_\infty$. Indeed, $\boldsymbol{Ax} = \lambda\boldsymbol{x}$ implies that $\sum_j a_{ij}x_j = \lambda x_i$ or $(\lambda - a_{ii})x_i =$
lyche-numerical-linear-algebra F04224	328	0.997	`\sum_{j} a_{i j} x_{j}=\lambda x_{i}`	$\sum_{j} a_{i j} x_{j}=\lambda x_{i}$	that $\|x_i\| = \\|\boldsymbol{x}\\|_\infty$. Indeed, $\boldsymbol{Ax} = \lambda\boldsymbol{x}$ implies that $\sum_j a_{ij}x_j = \lambda x_i$ or $(\lambda - a_{ii})x_i =$
lyche-numerical-linear-algebra F04225	328	0.997	`\left(\lambda-a_{i i}\right) x_{i}=`	$\left(\lambda-a_{i i}\right) x_{i}=$	that $\|x_i\| = \\|\boldsymbol{x}\\|_\infty$. Indeed, $\boldsymbol{Ax} = \lambda\boldsymbol{x}$ implies that $\sum_j a_{ij}x_j = \lambda x_i$ or $(\lambda - a_{ii})x_i =$
lyche-numerical-linear-algebra F04226	328	1.000	`\sum_{j \neq i} a_{i j} x_{j}`	$\sum_{j \neq i} a_{i j} x_{j}$	$\sum_{j\neq i} a_{ij}x_j$. Dividing by x_i and taking absolute values we find
lyche-numerical-linear-algebra F04227	328	1.000	`\left|x_{j} / x_{i}\right| \leq 1`	$\left\|x_{j} / x_{i}\right\| \leq 1$	since $\|x_j/x_i\| \leq 1$ for all j. Thus $\lambda \in R_i$.
lyche-numerical-linear-algebra F04228	328	1.000	`C_{j}`	C_{j}	But these are the column disks C_j of $\boldsymbol{A}$. Hence $\lambda \in C_j$ for some j.
lyche-numerical-linear-algebra F04229	328	1.000	`\lambda \in C_{j}`	$\lambda \in C_{j}$	But these are the column disks C_j of $\boldsymbol{A}$. Hence $\lambda \in C_j$ for some j.
lyche-numerical-linear-algebra F04230	328	1.000	`r_{i}`	r_{i}	with center at a_{ii} and radius r_i, c.f. Fig. 14.1. R_i is called a (Gerschgorin) row disk.
lyche-numerical-linear-algebra F04231	328	1.000	`R_{1}, \ldots, R_{n}`	$R_{1}, \ldots, R_{n}$	An eigenvalue λ lies in the union of the row disks $R_1, \ldots, R_n$ and also in the
lyche-numerical-linear-algebra F04232	328	1.000	`C_{1}, \ldots, C_{n}`	$C_{1}, \ldots, C_{n}$	union of the column disks $C_1, \ldots, C_n$. If $\boldsymbol{A}$ is Hermitian then $R_i = C_i$ for $i =$
lyche-numerical-linear-algebra F04233	328	1.000	`R_{i}=C_{i}`	$R_{i}=C_{i}$	union of the column disks $C_1, \ldots, C_n$. If $\boldsymbol{A}$ is Hermitian then $R_i = C_i$ for $i =$
lyche-numerical-linear-algebra F04234	328	1.000	`R_{1}=R_{m}=\{z \in \mathbb{R}:|z-2| \leq 1\}`	$R_{1}=R_{m}=\{z \in \mathbb{R}:\|z-2\| \leq 1\}$	eigenvalues are real. We find $R_1 = R_m = \{z \in \mathbb{R} : \|z-2\| \leq 1\}$ and
lyche-numerical-linear-algebra F04235	328	1.000	`\lambda \in[0,4]`	$\lambda \in[0,4]$	We conclude that $\lambda \in [0,4]$ for any eigenvalue λ of $\boldsymbol{T}$. To check this, we recall that
lyche-numerical-linear-algebra F04236	329	1.000	`4\left[\sin \frac{\pi}{2(m+1)}\right]^{2}`	$4\left[\sin \frac{\pi}{2(m+1)}\right]^{2}$	When m is large the smallest eigenvalue $4\left[\sin \frac{\pi}{2(m+1)}\right]^2$ is very close to zero and
lyche-numerical-linear-algebra F04237	329	0.969	`4\left[\sin \frac{m \pi}{2(m+1)}\right]^{2}`	$4\left[\sin \frac{m \pi}{2(m+1)}\right]^{2}$	the largest eigenvalue $4\left[\sin \frac{m\pi}{2(m+1)}\right]^2$ is very close to 4. Thus Gerschgorin's theorem
lyche-numerical-linear-algebra F04238	329	1.000	`\boldsymbol{A}(0)=\boldsymbol{D}`	$\boldsymbol{A}(0)=\boldsymbol{D}$	We have $\boldsymbol{A}(0) = \boldsymbol{D}$ and $\boldsymbol{A}(1) = \boldsymbol{A}$. As a function of t, every eigenvalue of $\boldsymbol{A}(t)$ is
lyche-numerical-linear-algebra F04239	329	1.000	`\boldsymbol{A}(1)=\boldsymbol{A}`	$\boldsymbol{A}(1)=\boldsymbol{A}$	We have $\boldsymbol{A}(0) = \boldsymbol{D}$ and $\boldsymbol{A}(1) = \boldsymbol{A}$. As a function of t, every eigenvalue of $\boldsymbol{A}(t)$ is
lyche-numerical-linear-algebra F04240	329	1.000	`\boldsymbol{A}(t)`	$\boldsymbol{A}(t)$	We have $\boldsymbol{A}(0) = \boldsymbol{D}$ and $\boldsymbol{A}(1) = \boldsymbol{A}$. As a function of t, every eigenvalue of $\boldsymbol{A}(t)$ is
lyche-numerical-linear-algebra F04241	329	1.000	`R_{i}(t)`	$R_{i}(t)$	row disks $R_i(t)$ of $\boldsymbol{A}(t)$ have radius proportional to t, indeed
lyche-numerical-linear-algebra F04242	329	1.000	`0 \leq t_{1}<t_{2} \leq 1`	$0 \leq t_{1}<t_{2} \leq 1$	Clearly $0 \leq t_1 < t_2 \leq 1$ implies $R_i(t_1) \subset R_i(t_2)$ and $R_i(1)$ is a row disk of
lyche-numerical-linear-algebra F04243	329	1.000	`R_{i}\left(t_{1}\right) \subset R_{i}\left(t_{2}\right)`	$R_{i}\left(t_{1}\right) \subset R_{i}\left(t_{2}\right)$	Clearly $0 \leq t_1 < t_2 \leq 1$ implies $R_i(t_1) \subset R_i(t_2)$ and $R_i(1)$ is a row disk of

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-a F04244</td><td>329</td><td>1.000</td><td>R_{i}(1)</td><td>$R_i(1)$</td><td>Clearly $0 \leq t_1 < t_2 \leq 1$ implies $R_i(t_1) \subset R_i(t_2)$ and $R_i(1)$ is a row disk of</td></tr>
<tr><td>lyche-numerical-linear-a F04245</td><td>329</td><td>0.992</td><td>\bigcup_{k=1}^{p} R_{i_{k}}(1)</td><td>$\bigcup_{k=1}^{p} R_{i_k}(1)$</td><td>$\boldsymbol{A}$ for all i. Suppose $\bigcup_{k=1}^{p} R_{i_k}(1)$ are disjoint from the other disks of $\boldsymbol{A}$ and set</td></tr>
<tr><td>lyche-numerical-linear-a F04246</td><td>329</td><td>0.992</td><td>R^{p}(t):=\bigcup_{k=1}^{p} R_{i_{k}}(t)</td><td>$R^p(t) := \bigcup_{k=1}^{p} R_{i_k}(t)$</td><td>$R^p(t) := \bigcup_{k=1}^{p} R_{i_k}(t)$ for $t \in [0,1]$. Now $R^p(0)$ contains only the p eigenvalues</td></tr>
<tr><td>lyche-numerical-linear-a F04247</td><td>329</td><td>0.992</td><td>t \in[0,1]</td><td>$t \in [0,1]$</td><td>$R^p(t) := \bigcup_{k=1}^{p} R_{i_k}(t)$ for $t \in [0,1]$. Now $R^p(0)$ contains only the p eigenvalues</td></tr>
<tr><td>lyche-numerical-linear-a F04248</td><td>329</td><td>0.992</td><td>R^{p}(0)</td><td>$R^p(0)$</td><td>$R^p(t) := \bigcup_{k=1}^{p} R_{i_k}(t)$ for $t \in [0,1]$. Now $R^p(0)$ contains only the p eigenvalues</td></tr>
<tr><td>lyche-numerical-linear-a F04249</td><td>329</td><td>1.000</td><td>a_{i_{1}, i_{1}}, \ldots, a_{i_{p}, i_{p}}</td><td>$a_{i_1,i_1}, \ldots, a_{i_p,i_p}$</td><td>$a_{i_1,i_1}, \ldots, a_{i_p,i_p}$ of $\boldsymbol{A}(0) = \boldsymbol{D}$. As t increases from zero to one the set $R^p(t)$ is</td></tr>
<tr><td>lyche-numerical-linear-a F04250</td><td>329</td><td>1.000</td><td>R^{p}(t)</td><td>$R^p(t)$</td><td>$a_{i_1,i_1}, \ldots, a_{i_p,i_p}$ of $\boldsymbol{A}(0) = \boldsymbol{D}$. As t increases from zero to one the set $R^p(t)$ is</td></tr>
<tr><td>lyche-numerical-linear-a F04251</td><td>329</td><td>1.000</td><td>R^{p}(1)</td><td>$R^p(1)$</td><td>cannot loose or gain eigenvalues. It follows that $R^p(1)$ must contain p eigenvalues</td></tr>
<tr><td>lyche-numerical-linear-a F04252</td><td>329</td><td>1.000</td><td>\boldsymbol{A}=\left[\begin{array}{ccc}1 & \epsilon_{1} & \epsilon_{2} \\ \epsilon_{3} & 2 & \epsilon_{4} \\ \epsilon_{5} & \epsilon_{6} & 3\end{array}\right]</td><td>$\boldsymbol{A}=\left[\begin{array}{ccc}1 & \epsilon_{1} & \epsilon_{2} \\ \epsilon_{3} & 2 & \epsilon_{4} \\ \epsilon_{5} & \epsilon_{6} & 3\end{array}\right]$</td><td>Example 14.2 Consider the matrix $\boldsymbol{A} = \left[\begin{smallmatrix}1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3\end{smallmatrix}\right]$, where $|\epsilon_i| \leq 10^{-15}$ all i. By</td></tr>
<tr><td>lyche-numerical-linear-a F04253</td><td>329</td><td>1.000</td><td>\left|\epsilon_{i}\right| \leq 10^{-15}</td><td>$|\epsilon_i| \leq 10^{-15}$</td><td>Example 14.2 Consider the matrix $\boldsymbol{A} = \left[\begin{smallmatrix}1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3\end{smallmatrix}\right]$, where $|\epsilon_i| \leq 10^{-15}$ all i. By</td></tr>
<tr><td>lyche-numerical-linear-a F04254</td><td>329</td><td>1.000</td><td>\lambda_{1}, \lambda_{2}, \lambda_{3}</td><td>$\lambda_1, \lambda_2, \lambda_3$</td><td>Corollary 14.1 the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of $\boldsymbol{A}$ are distinct and satisfy $|\lambda_j - j| \leq$</td></tr>
<tr><td>lyche-numerical-linear-a F04255</td><td>329</td><td>1.000</td><td>\left|\lambda_{j}-j\right| \leq</td><td>$|\lambda_j - j| \leq$</td><td>Corollary 14.1 the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of $\boldsymbol{A}$ are distinct and satisfy $|\lambda_j - j| \leq$</td></tr>
<tr><td>lyche-numerical-linear-a F04256</td><td>329</td><td>1.000</td><td>2 \times 10^{-15}</td><td>2×10^{-15}</td><td>2×10^{-15} for $j = 1, 2, 3$.</td></tr>
<tr><td>lyche-numerical-linear-a F04257</td><td>329</td><td>1.000</td><td>j=1,2,3</td><td>$j = 1, 2, 3$</td><td>2×10^{-15} for $j = 1, 2, 3$.</td></tr>
<tr><td>lyche-numerical-linear-a F04258</td><td>329</td><td>0.979</td><td>\boldsymbol{A}+</td><td>$\boldsymbol{A}+$</td><td>we think of $\boldsymbol{E}$ as a perturbation of $\boldsymbol{A}$. By how much do the eigenvalues of $\boldsymbol{A}$ and $\boldsymbol{A}+$</td></tr>
<tr><td>lyche-numerical-linear-a F04259</td><td>329</td><td>1.000</td><td>\boldsymbol{A}_{0}:=\mathbf{0}</td><td>$\boldsymbol{A}_0 := \mathbf{0}$</td><td>We illustrate this by considering two examples. Suppose $\boldsymbol{A}_0 := \mathbf{0}$ is the zero</td></tr>
<tr><td>lyche-numerical-linear-a F04260</td><td>329</td><td>0.998</td><td>\lambda \in \sigma\left(\boldsymbol{A}_{0}+\boldsymbol{E}\right)=\sigma(\boldsymbol{E})</td><td>$\lambda \in \sigma(\boldsymbol{A}_0 + \boldsymbol{E}) = \sigma(\boldsymbol{E})$</td><td>matrix. If $\lambda \in \sigma(\boldsymbol{A}_0 + \boldsymbol{E}) = \sigma(\boldsymbol{E})$, then $|\lambda| \leq \|\boldsymbol{E}\|_\infty$ by Theorem 12.11, and any</td></tr>
<tr><td>lyche-numerical-linear-a F04261</td><td>329</td><td>0.998</td><td>|\lambda| \leq\|\boldsymbol{E}\|_{\infty}</td><td>$|\lambda| \leq \|\boldsymbol{E}\|_\infty$</td><td>matrix. If $\lambda \in \sigma(\boldsymbol{A}_0 + \boldsymbol{E}) = \sigma(\boldsymbol{E})$, then $|\lambda| \leq \|\boldsymbol{E}\|_\infty$ by Theorem 12.11, and any</td></tr>
<tr><td>lyche-numerical-linear-a F04262</td><td>330</td><td>0.998</td><td>\boldsymbol{A}_{0}</td><td>$\boldsymbol{A}_0$</td><td>zero eigenvalue of $\boldsymbol{A}_0$ is perturbed by at most $\|\boldsymbol{E}\|_\infty$. On the other hand consider</td></tr>
<tr><td>lyche-numerical-linear-a F04263</td><td>330</td><td>0.998</td><td>\|\boldsymbol{E}\|_{\infty}</td><td>$\|\boldsymbol{E}\|_\infty$</td><td>zero eigenvalue of $\boldsymbol{A}_0$ is perturbed by at most $\|\boldsymbol{E}\|_\infty$. On the other hand consider</td></tr>
<tr><td>lyche-numerical-linear-a F04264</td><td>330</td><td>1.000</td><td>\boldsymbol{A}_{1}+\boldsymbol{E}</td><td>$\boldsymbol{A}_1 + \boldsymbol{E}$</td><td>The characteristic polynomial of $\boldsymbol{A}_1 + \boldsymbol{E}$ is $\pi(\lambda) := (-1)^n(\lambda^n - \epsilon)$, and the zero</td></tr>
<tr><td>lyche-numerical-linear-a F04265</td><td>330</td><td>1.000</td><td>\pi(\lambda):=(-1)^{n}\left(\lambda^{n}-\epsilon\right)</td><td>$\pi(\lambda) := (-1)^n (\lambda^n - \epsilon)$</td><td>The characteristic polynomial of $\boldsymbol{A}_1 + \boldsymbol{E}$ is $\pi(\lambda) := (-1)^n(\lambda^n - \epsilon)$, and the zero</td></tr>
<tr><td>lyche-numerical-linear-a F04266</td><td>330</td><td>1.000</td><td>|\lambda|=\|\boldsymbol{E}\|_{\infty}^{1 / n}</td><td>$|\lambda| = \|\boldsymbol{E}\|_\infty^{1/n}$</td><td>eigenvalues of $\boldsymbol{A}_1$ are perturbed by the amount $|\lambda| = \|\boldsymbol{E}\|_\infty^{1/n}$. Thus, for $n = 16$, a</td></tr>
<tr><td>lyche-numerical-linear-a F04267</td><td>330</td><td>1.000</td><td>n=16</td><td>$n = 16$</td><td>eigenvalues of $\boldsymbol{A}_1$ are perturbed by the amount $|\lambda| = \|\boldsymbol{E}\|_\infty^{1/n}$. Thus, for $n = 16$, a</td></tr>
<tr><td>lyche-numerical-linear-a F04268</td><td>330</td><td>0.978</td><td>\epsilon=10^{-16}</td><td>$\epsilon = 10^{-16}$</td><td>perturbation of say $\epsilon = 10^{-16}$ gives a change in eigenvalue of 0.1.</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F04269	330	1.000	|\boldsymbol{E}|_{\infty}^{1 / n}	$\|\boldsymbol{E}\|_{\infty}^{1 / n}$	The following theorem shows that a dependence $\|\boldsymbol{E}\|_{\infty}^{1/n}$ is the worst that can
lyche-numerical-linear-algebra_F04270	330	1.000	\mu \in	$\mu \in$	**Theorem 14.2 (Elsner's Theorem (1985))** *Suppose* $\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$. *To every* $\mu \in$
lyche-numerical-linear-algebra_F04271	330	1.000	\sigma(\boldsymbol{A}+\boldsymbol{E})	$\sigma(\boldsymbol{A}+\boldsymbol{E})$	$\sigma(\boldsymbol{A}+\boldsymbol{E})$ *there is a* $\lambda \in \sigma(\boldsymbol{A})$ *such that*
lyche-numerical-linear-algebra_F04272	330	1.000	\left|\boldsymbol{u}_{1}\right|_{2}=1	$\left\|\boldsymbol{u}_{1}\right\|_{2}=1$	μ. Let $\boldsymbol{u}_1$ with $\|\boldsymbol{u}_1\|_2 = 1$ be an eigenvector corresponding to μ, and extend $\boldsymbol{u}_1$ to
lyche-numerical-linear-algebra_F04273	330	1.000	|\boldsymbol{E}|_{2}^{1 / n}	$\|\boldsymbol{E}\|_{2}^{1 / n}$	elements of the matrix. The factor $\|\boldsymbol{E}\|_2^{1/n}$ shows that this dependence is almost,
lyche-numerical-linear-algebra_F04274	330	1.000	\left[\begin{array}{cc}1 & 1 \\ \epsilon & 1\end{array}\right]	$\left[\begin{array}{cc}1 & 1 \\ \epsilon & 1\end{array}\right]$	but not quite, differentiable. As an example, the eigenvalues of the matrix $\left[\begin{smallmatrix}1 & 1 \\ \epsilon & 1\end{smallmatrix}\right]$ are
lyche-numerical-linear-algebra_F04275	330	1.000	1 \pm \sqrt{\epsilon}	$1 \pm \sqrt{\epsilon}$	$1 \pm \sqrt{\epsilon}$ and these expressions are not differentiable at $\epsilon = 0$.
lyche-numerical-linear-algebra_F04276	330	1.000	\epsilon=0	$\epsilon=0$	$1 \pm \sqrt{\epsilon}$ and these expressions are not differentiable at $\epsilon = 0$.
lyche-numerical-linear-algebra_F04277	331	1.000	1 / n	$1 / n$	nondefective matrices we can get rid of the annoying exponent $1/n$ in $\|\boldsymbol{E}\|_2$
lyche-numerical-linear-algebra_F04278	331	1.000	|\boldsymbol{E}|_{2}	$\|\boldsymbol{E}\|_{2}$	nondefective matrices we can get rid of the annoying exponent $1/n$ in $\|\boldsymbol{E}\|_2$
lyche-numerical-linear-algebra_F04279	331	1.000	\mu \in \mathbb{C}	$\mu \in \mathbb{C}$	*any* $\mu \in \mathbb{C}$ *and* $\boldsymbol{x} \in \mathbb{C}^n$ *with* $\|\boldsymbol{x}\|_p = 1$ *we can find an eigenvalue* λ *of* $\boldsymbol{A}$ *such that*
lyche-numerical-linear-algebra_F04280	331	0.999	\boldsymbol{r}:=\boldsymbol{A} \boldsymbol{x}-\mu \boldsymbol{x}	$\boldsymbol{r}:=\boldsymbol{A} \boldsymbol{x}-\mu \boldsymbol{x}$	*where* $\boldsymbol{r} := \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x}$ *and* $K_p(\boldsymbol{X}) := \|\boldsymbol{X}\|_p \|\boldsymbol{X}^{-1}\|_p$. *If for some* $\boldsymbol{E} \in \mathbb{C}^{n\times n}$ *it holds*
lyche-numerical-linear-algebra_F04281	331	0.999	K_{p}(\boldsymbol{X}):=|\boldsymbol{X}|_{p}\left|\boldsymbol{X}^{-1}\right|_{p}	$K_{p}(\boldsymbol{X}):=\|\boldsymbol{X}\|_{p}\left\|\boldsymbol{X}^{-1}\right\|_{p}$	*where* $\boldsymbol{r} := \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x}$ *and* $K_p(\boldsymbol{X}) := \|\boldsymbol{X}\|_p \|\boldsymbol{X}^{-1}\|_p$. *If for some* $\boldsymbol{E} \in \mathbb{C}^{n\times n}$ *it holds*
lyche-numerical-linear-algebra_F04282	331	0.902	(\mu, \boldsymbol{x})	$(\mu, \boldsymbol{x})$	*that* $(\mu, \boldsymbol{x})$ *is an eigenpair for* $\boldsymbol{A}+\boldsymbol{E}$, *then we can find an eigenvalue* λ *of* $\boldsymbol{A}$ *such*
lyche-numerical-linear-algebra_F04283	331	1.000	\mu \in \sigma(\boldsymbol{A})	$\mu \in \sigma(\boldsymbol{A})$	***Proof*** If $\mu \in \sigma(\boldsymbol{A})$ then we can take $\lambda = \mu$ and (14.2), (14.3) hold trivially. So
lyche-numerical-linear-algebra_F04284	331	1.000	\lambda=\mu	$\lambda=\mu$	***Proof*** If $\mu \in \sigma(\boldsymbol{A})$ then we can take $\lambda = \mu$ and (14.2), (14.3) hold trivially. So
lyche-numerical-linear-algebra_F04285	331	1.000	\mu \notin \sigma(\boldsymbol{A})	$\mu \notin \sigma(\boldsymbol{A})$	assume $\mu \notin \sigma(\boldsymbol{A})$. Since $\boldsymbol{A}$ is nondefective it can be diagonalized, we have $\boldsymbol{A} =$
lyche-numerical-linear-algebra_F04286	331	1.000	\boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1}	$\boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1}$	$\boldsymbol{X}\boldsymbol{D}\boldsymbol{X}^{-1}$, where $\boldsymbol{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $(\lambda_j, \boldsymbol{x}_j)$ are the eigenpairs of $\boldsymbol{A}$ for $j =$
lyche-numerical-linear-algebra_F04287	331	1.000	\left(\lambda_{j}, \boldsymbol{x}_{j}\right)	$\left(\lambda_{j}, \boldsymbol{x}_{j}\right)$	$\boldsymbol{X}\boldsymbol{D}\boldsymbol{X}^{-1}$, where $\boldsymbol{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $(\lambda_j, \boldsymbol{x}_j)$ are the eigenpairs of $\boldsymbol{A}$ for $j =$
lyche-numerical-linear-algebra_F04288	331	1.000	\boldsymbol{D}_{1}:=\boldsymbol{D}-\mu \boldsymbol{I}	$\boldsymbol{D}_{1}:=\boldsymbol{D}-\mu \boldsymbol{I}$	$1, \ldots, n$. Define $\boldsymbol{D}_1 := \boldsymbol{D} - \mu\boldsymbol{I}$. Then $\boldsymbol{D}_1^{-1} = \operatorname{diag}\left((\lambda_1-\mu)^{-1}, \ldots, (\lambda_n-\mu)^{-1}\right)$
lyche-numerical-linear-algebra_F04289	331	1.000	\boldsymbol{D}_{1}^{-1}=\operatorname{diag}\left(\left(\lambda_{1}-\mu\right)^{-1}, \ldots,\left(\lambda_{n}-\mu\right)^{-1}\right)	$\boldsymbol{D}_{1}^{-1}=\operatorname{diag}\left(\left(\lambda_{1}-\mu\right)^{-1}, \ldots,\left(\lambda_{n}-\mu\right)^{-1}\right)$	$1, \ldots, n$. Define $\boldsymbol{D}_1 := \boldsymbol{D} - \mu\boldsymbol{I}$. Then $\boldsymbol{D}_1^{-1} = \operatorname{diag}\left((\lambda_1-\mu)^{-1}, \ldots, (\lambda_n-\mu)^{-1}\right)$
lyche-numerical-linear-algebra_F04290	331	0.658	(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{x}=\mu \boldsymbol{x}	$(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{x}=\mu \boldsymbol{x}$	But then (14.2) follows. If $(\boldsymbol{A}+\boldsymbol{E})\boldsymbol{x} = \mu\boldsymbol{x}$ then $\boldsymbol{0} = \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x} + \boldsymbol{E}\boldsymbol{x} = \boldsymbol{r} + \boldsymbol{E}\boldsymbol{x}$.
lyche-numerical-linear-algebra_F04291	331	0.658	\mathbf{0}=\boldsymbol{A} \boldsymbol{x}-\mu \boldsymbol{x}+\boldsymbol{E} \boldsymbol{x}=\boldsymbol{r}+\boldsymbol{E} \boldsymbol{x}	$\mathbf{0}=\boldsymbol{A} \boldsymbol{x}-\mu \boldsymbol{x}+\boldsymbol{E} \boldsymbol{x}=\boldsymbol{r}+\boldsymbol{E} \boldsymbol{x}$	But then (14.2) follows. If $(\boldsymbol{A}+\boldsymbol{E})\boldsymbol{x} = \mu\boldsymbol{x}$ then $\boldsymbol{0} = \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x} + \boldsymbol{E}\boldsymbol{x} = \boldsymbol{r} + \boldsymbol{E}\boldsymbol{x}$.
lyche-numerical-linear-algebra_F04292	331	0.954	|\boldsymbol{r}|_{p}=|-\boldsymbol{E} \boldsymbol{x}|_{p} \leq|\boldsymbol{E}|_{p}	$\|\boldsymbol{r}\|_{p}=\|-\boldsymbol{E} \boldsymbol{x}\|_{p} \leq\|\boldsymbol{E}\|_{p}$	But then $\|\boldsymbol{r}\|_p = \|-\boldsymbol{E}\boldsymbol{x}\|_p \leq \|\boldsymbol{E}\|_p$. Inserting this in (14.2) proves (14.3).
lyche-numerical-linear-algebra_F04293	331	1.000	K_{p}(\boldsymbol{X})	$K_{p}(\boldsymbol{X})$	be magnified by at most $K_p(\boldsymbol{X})$, the condition number of the eigenvector matrix
lyche-numerical-linear-algebra_F04294	332	0.593	K_{2}(\boldsymbol{X})=1	$K_{2}(\boldsymbol{X})=1$	we restrict attention to the 2-norm then $K_2(\boldsymbol{X}) = 1$ and (14.3) implies the following

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-a F04295	332	1.000	`|\lambda-\mu| \leq|\boldsymbol{E}|_{2}`	$\lvert\lambda-\mu\rvert \leq\lVert\boldsymbol{E}\rVert_{2}$	*eigenvalue* λ *of* $\boldsymbol{A}$ *such that* $\lvert\lambda-\mu\rvert \leq \lVert\boldsymbol{E}\rVert_2$.
lyche-numerical-linear-a F04296	332	0.995	`\boldsymbol{A}=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)`	$\boldsymbol{A}=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)$	**Lemma 14.1** (*p*-**Norm of a Diagonal Matrix**) *If* $\boldsymbol{A} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ *is a*
lyche-numerical-linear-a F04297	332	1.000	`|\boldsymbol{A}|_{p}=\rho(\boldsymbol{A})`	$\lVert\boldsymbol{A}\rVert_{p}=\rho(\boldsymbol{A})$	*diagonal matrix then* $\lVert\boldsymbol{A}\rVert_p = \rho(\boldsymbol{A})$ *for* $1 \leq p \leq \infty$.
lyche-numerical-linear-a F04298	332	1.000	`p<\infty`	$p<\infty$	***Proof*** For $p = \infty$ the proof is left as an exercise. For any $\boldsymbol{x} \in \mathbb{C}^n$ and $p < \infty$ we
lyche-numerical-linear-a F04299	332	0.874	`|\boldsymbol{A}|_{p}=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{|\boldsymbol{A}|_{p}}{|\boldsymbol{x}|_{p}} \leq \rho(\boldsymbol{A})`	$\lVert\boldsymbol{A}\rVert_{p}=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\lVert\boldsymbol{A}\rVert_{p}}{\lVert\boldsymbol{x}\rVert_{p}} \leq \rho(\boldsymbol{A})$	Thus $\lVert\boldsymbol{A}\rVert_p = \max_{\boldsymbol{x}\neq\mathbf{0}} \frac{\lVert\boldsymbol{Ax}\rVert_p}{\lVert\boldsymbol{x}\rVert_p} \leq \rho(\boldsymbol{A})$. But from Theorem 12.11 we have $\rho(\boldsymbol{A}) \leq$
lyche-numerical-linear-a F04300	332	0.784	`\mu, \boldsymbol{x}`	$\mu, \boldsymbol{x}$	*where* $\boldsymbol{r} := \boldsymbol{Ax} - \mu\boldsymbol{x}$. *If for some* $\boldsymbol{E} \in \mathbb{C}^{n\times n}$ *it holds that* $(\mu, \boldsymbol{x})$ *is an eigenpair for*
lyche-numerical-linear-a F04301	333	1.000	`\boldsymbol{B}:=\mu \boldsymbol{A}^{-1}`	$\boldsymbol{B}:=\mu \boldsymbol{A}^{-1}$	and (14.4) follows from (14.2). To prove (14.5) we define the matrices $\boldsymbol{B} := \mu\boldsymbol{A}^{-1}$
lyche-numerical-linear-a F04302	333	0.858	`\boldsymbol{F}:=-\boldsymbol{A}^{-1} \boldsymbol{E}`	$\boldsymbol{F}:=-\boldsymbol{A}^{-1} \boldsymbol{E}$	and $\boldsymbol{F} := -\boldsymbol{A}^{-1}\boldsymbol{E}$. If $(\lambda_j, \boldsymbol{x})$ are the eigenpairs for $\boldsymbol{A}$ then $\left(\frac{\mu}{\lambda_j}, \boldsymbol{x}\right)$ are the eigenpairs
lyche-numerical-linear-a F04303	333	0.858	`\left(\lambda_{j}, \boldsymbol{x}\right)`	$\left(\lambda_{j}, \boldsymbol{x}\right)$	and $\boldsymbol{F} := -\boldsymbol{A}^{-1}\boldsymbol{E}$. If $(\lambda_j, \boldsymbol{x})$ are the eigenpairs for $\boldsymbol{A}$ then $\left(\frac{\mu}{\lambda_j}, \boldsymbol{x}\right)$ are the eigenpairs
lyche-numerical-linear-a F04304	333	0.858	`\left(\frac{\mu}{\lambda_{j}}, \boldsymbol{x}\right)`	$\left(\frac{\mu}{\lambda_{j}}, \boldsymbol{x}\right)$	and $\boldsymbol{F} := -\boldsymbol{A}^{-1}\boldsymbol{E}$. If $(\lambda_j, \boldsymbol{x})$ are the eigenpairs for $\boldsymbol{A}$ then $\left(\frac{\mu}{\lambda_j}, \boldsymbol{x}\right)$ are the eigenpairs
lyche-numerical-linear-a F04305	333	0.999	`(1, \boldsymbol{x})`	$(1, \boldsymbol{x})$	Thus $(1, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{B} + \boldsymbol{F}$. Applying Theorem 14.3 to this eigenvalue
lyche-numerical-linear-a F04306	333	0.999	`\boldsymbol{B}+\boldsymbol{F}`	$\boldsymbol{B}+\boldsymbol{F}$	Thus $(1, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{B} + \boldsymbol{F}$. Applying Theorem 14.3 to this eigenvalue
lyche-numerical-linear-a F04307	333	1.000	`\left|\frac{\mu}{\lambda}-1\right| \leq K_{p}(\boldsymbol{X})|\boldsymbol{F}|_{p}=K_{p}(\boldsymbol{X})\left|\boldsymbol{A}^{-1} \boldsymbol{E}\right|_{p}`	$\left\lvert\frac{\mu}{\lambda}-1\right\rvert \leq K_{p}(\boldsymbol{X})\lVert\boldsymbol{F}\rVert_{p}=K_{p}(\boldsymbol{X})\left\lVert\boldsymbol{A}^{-1} \boldsymbol{E}\right\rVert_{p}$	we can find $\lambda \in \sigma(\boldsymbol{A})$ such that $\lvert\frac{\mu}{\lambda} - 1\rvert \leq K_p(\boldsymbol{X})\lVert\boldsymbol{F}\rVert_p = K_p(\boldsymbol{X})\lVert\boldsymbol{A}^{-1}\boldsymbol{E}\rVert_p$ which
lyche-numerical-linear-a F04308	333	1.000	`1,2, \ldots, i-2, i=3,4, \ldots, n`	$1,2, \ldots, i-2, i=3,4, \ldots, n$	$1, 2, \ldots, i-2, i = 3, 4, \ldots, n$. We will reduce $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ to upper Hessenberg form
lyche-numerical-linear-a F04309	333	1.000	`\boldsymbol{A}_{1}=\boldsymbol{A}`	$\boldsymbol{A}_{1}=\boldsymbol{A}$	by unitary similarity transformations. Let $\boldsymbol{A}_1 = \boldsymbol{A}$ and define $\boldsymbol{A}_{k+1} = \boldsymbol{H}_k\boldsymbol{A}_k\boldsymbol{H}_k$ for
lyche-numerical-linear-a F04310	333	1.000	`\boldsymbol{A}_{k+1}=\boldsymbol{H}_{k} \boldsymbol{A}_{k} \boldsymbol{H}_{k}`	$\boldsymbol{A}_{k+1}=\boldsymbol{H}_{k} \boldsymbol{A}_{k} \boldsymbol{H}_{k}$	by unitary similarity transformations. Let $\boldsymbol{A}_1 = \boldsymbol{A}$ and define $\boldsymbol{A}_{k+1} = \boldsymbol{H}_k\boldsymbol{A}_k\boldsymbol{H}_k$ for
lyche-numerical-linear-a F04311	333	1.000	`k=1,2, \ldots, n-2`	$k=1,2, \ldots, n-2$	$k = 1, 2, \ldots, n-2$. Here $\boldsymbol{H}_k$ is a Householder transformation chosen to introduce
lyche-numerical-linear-a F04312	333	1.000	`\boldsymbol{H}_{k}`	$\boldsymbol{H}_{k}$	$k = 1, 2, \ldots, n-2$. Here $\boldsymbol{H}_k$ is a Householder transformation chosen to introduce
lyche-numerical-linear-a F04313	333	1.000	`\boldsymbol{A}_{n-1}`	$\boldsymbol{A}_{n-1}$	$\boldsymbol{A}_{n-1}$ will be upper Hessenberg. Householder transformations were used in Chap. 5
lyche-numerical-linear-a F04314	333	1.000	`\boldsymbol{A}_{k}^{*}=\boldsymbol{A}_{k}`	$\boldsymbol{A}_{k}^{*}=\boldsymbol{A}_{k}$	if $\boldsymbol{A}_k^* = \boldsymbol{A}_k$ then
lyche-numerical-linear-a F04315	334	1.000	`\boldsymbol{B}_{k} \in \mathbb{C}^{k, k}`	$\boldsymbol{B}_{k} \in \mathbb{C}^{k, k}$	Suppose $\boldsymbol{B}_k \in \mathbb{C}^{k,k}$ is upper Hessenberg, and the first $k-1$ columns of $\boldsymbol{D}_k \in$
lyche-numerical-linear-a F04316	334	1.000	`\boldsymbol{D}_{k} \in`	$\boldsymbol{D}_{k} \in$	Suppose $\boldsymbol{B}_k \in \mathbb{C}^{k,k}$ is upper Hessenberg, and the first $k-1$ columns of $\boldsymbol{D}_k \in$
lyche-numerical-linear-a F04317	334	0.999	`\mathbb{C}^{n-k, k}`	$\mathbb{C}^{n-k, k}$	$\mathbb{C}^{n-k,k}$ are zero, i.e. $\boldsymbol{D}_k = [\mathbf{0}, \mathbf{0}, \ldots, \mathbf{0}, \boldsymbol{d}_k]$. Let $\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k\boldsymbol{v}_k^* \in \mathbb{C}^{n-k,n-k}$ be a
lyche-numerical-linear-a F04318	334	0.999	`\boldsymbol{D}_{k}=\left[\mathbf{0}, \mathbf{0}, \ldots, \mathbf{0}, \boldsymbol{d}_{k}\right]`	$\boldsymbol{D}_{k}=\left[\mathbf{0}, \mathbf{0}, \ldots, \mathbf{0}, \boldsymbol{d}_{k}\right]$	$\mathbb{C}^{n-k,k}$ are zero, i.e. $\boldsymbol{D}_k = [\mathbf{0}, \mathbf{0}, \ldots, \mathbf{0}, \boldsymbol{d}_k]$. Let $\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k\boldsymbol{v}_k^* \in \mathbb{C}^{n-k,n-k}$ be a
lyche-numerical-linear-a F04319	334	0.999	`\boldsymbol{V}_{k}=\boldsymbol{I}-\boldsymbol{v}_{k} \boldsymbol{v}_{k}^{*} \in \mathbb{C}^{n-k, n-k}`	$\boldsymbol{V}_{k}=\boldsymbol{I}-\boldsymbol{v}_{k} \boldsymbol{v}_{k}^{*} \in \mathbb{C}^{n-k, n-k}$	$\mathbb{C}^{n-k,k}$ are zero, i.e. $\boldsymbol{D}_k = [\mathbf{0}, \mathbf{0}, \ldots, \mathbf{0}, \boldsymbol{d}_k]$. Let $\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k\boldsymbol{v}_k^* \in \mathbb{C}^{n-k,n-k}$ be a

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-a F04320	334	0.999	`\boldsymbol{V}_{k} \boldsymbol{d}_{k}=\alpha_{k} \boldsymbol{e}_{1}`	$\boldsymbol{V}_{k} \boldsymbol{d}_{k}=\alpha_{k} \boldsymbol{e}_{1}$	Householder transformation such that $\boldsymbol{V}_k \boldsymbol{d}_k = \alpha_k \boldsymbol{e}_1$. Define
lyche-numerical-linear-a F04321	334	1.000	`\boldsymbol{V}_{k} \boldsymbol{D}_{k}=\left[ \boldsymbol{V}_{k} \mathbf{0}, \ldots, \boldsymbol{V}_{k} \mathbf{0}, \boldsymbol{V}_{k} \boldsymbol{d}_{k} \right]=\left(\mathbf{0}, \ldots, \mathbf{0}, \alpha_{k} \boldsymbol{e}_{1}\right)`	$\boldsymbol{V}_{k} \boldsymbol{D}_{k}=\left[\boldsymbol{V}_{k} \mathbf{0}, \ldots, \boldsymbol{V}_{k} \mathbf{0}, \boldsymbol{V}_{k} \boldsymbol{d}_{k} \right]=\left(\mathbf{0}, \ldots, \mathbf{0}, \alpha_{k} \boldsymbol{e}_{1}\right)$	Now $\boldsymbol{V}_k \boldsymbol{D}_k = [\boldsymbol{V}_k \mathbf{0}, \ldots, \boldsymbol{V}_k \mathbf{0}, \boldsymbol{V}_k \boldsymbol{d}_k] = (\mathbf{0}, \ldots, \mathbf{0}, \alpha_k \boldsymbol{e}_1)$. Moreover, the matrix
lyche-numerical-linear-a F04322	334	1.000	`(k+1) \times(k+1)`	$(k+1) \times(k+1)$	$\boldsymbol{B}_k$ is not affected by the $\boldsymbol{H}_k$ transformation. Therefore the upper left $(k+1)\times(k+1)$
lyche-numerical-linear-a F04323	334	1.000	`\boldsymbol{v}_{k}`	$\boldsymbol{v}_{k}$	To find $\boldsymbol{A}_{k+1}$ we use Algorithm 5.1 to find $\boldsymbol{v}_k$ and α_k. We store $\boldsymbol{v}_k$ in the kth
lyche-numerical-linear-a F04324	334	1.000	`\boldsymbol{L}(k+1: n, k)=\boldsymbol{v}_{k}`	$\boldsymbol{L}(k+1: n, k)=\boldsymbol{v}_{k}$	column of a matrix $\boldsymbol{L}$ as $\boldsymbol{L}(k+1:n,k) = \boldsymbol{v}_k$. This leads to the following algorithm
lyche-numerical-linear-a F04325	334	0.994	`\boldsymbol{B} . \boldsymbol{B}`	$\boldsymbol{B} . \boldsymbol{B}$	transformations. The algorithm returns the reduced matrix $\boldsymbol{B}$. $\boldsymbol{B}$ is tridiagonal if $\boldsymbol{A}$
lyche-numerical-linear-a F04326	334	1.000	`\boldsymbol{B}=\boldsymbol{Q}^{*} \boldsymbol{A} \boldsymbol{Q}`	$\boldsymbol{B}=\boldsymbol{Q}^{*} \boldsymbol{A} \boldsymbol{Q}$	unitary matrix $\boldsymbol{Q}$ such that $\boldsymbol{B} = \boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$. Algorithm 5.1 is used in each step of the
lyche-numerical-linear-a F04327	335	1.000	`\boldsymbol{Q}^{*} \boldsymbol{A} \boldsymbol{Q}`	$\boldsymbol{Q}^{*} \boldsymbol{A} \boldsymbol{Q}$	$\boldsymbol{Q}$ is orthogonal and $\boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$ is upper Hessenberg. We need to compute the product
lyche-numerical-linear-a F04328	335	0.950	`\boldsymbol{Q}=\boldsymbol{H}_{1} \boldsymbol{H}_{2} \cdots \boldsymbol{H}_{n-2}`	$\boldsymbol{Q}=\boldsymbol{H}_{1} \boldsymbol{H}_{2} \cdots \boldsymbol{H}_{n-2}$	$\boldsymbol{Q} = \boldsymbol{H}_1 \boldsymbol{H}_2 \cdots \boldsymbol{H}_{n-2}$, where $\boldsymbol{H}_k = \left[\begin{smallmatrix}\boldsymbol{I} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{I}-\boldsymbol{v}_k\boldsymbol{v}_k^T\end{smallmatrix}\right]$ and $\boldsymbol{v}_k \in \mathbb{R}^{n-k}$. Since $\boldsymbol{v}_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-a F04329	335	0.950	`\boldsymbol{H}_{k}=\left[\begin{array}{cc}\boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}-\boldsymbol{v}_{k} \boldsymbol{v}_{k}^{T}\end{array}\right]`	$\boldsymbol{H}_{k}=\left[\begin{array}{cc}\boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}-\boldsymbol{v}_{k} \boldsymbol{v}_{k}^{T}\end{array}\right]$	$\boldsymbol{Q} = \boldsymbol{H}_1 \boldsymbol{H}_2 \cdots \boldsymbol{H}_{n-2}$, where $\boldsymbol{H}_k = \left[\begin{smallmatrix}\boldsymbol{I} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{I}-\boldsymbol{v}_k\boldsymbol{v}_k^T\end{smallmatrix}\right]$ and $\boldsymbol{v}_k \in \mathbb{R}^{n-k}$. Since $\boldsymbol{v}_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-a F04330	335	0.950	`\boldsymbol{v}_{k} \in \mathbb{R}^{n-k}`	$\boldsymbol{v}_{k} \in \mathbb{R}^{n-k}$	$\boldsymbol{Q} = \boldsymbol{H}_1 \boldsymbol{H}_2 \cdots \boldsymbol{H}_{n-2}$, where $\boldsymbol{H}_k = \left[\begin{smallmatrix}\boldsymbol{I} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{I}-\boldsymbol{v}_k\boldsymbol{v}_k^T\end{smallmatrix}\right]$ and $\boldsymbol{v}_k \in \mathbb{R}^{n-k}$. Since $\boldsymbol{v}_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-a F04331	335	0.950	`\boldsymbol{v}_{1} \in \mathbb{R}^{n-1}`	$\boldsymbol{v}_{1} \in \mathbb{R}^{n-1}$	$\boldsymbol{Q} = \boldsymbol{H}_1 \boldsymbol{H}_2 \cdots \boldsymbol{H}_{n-2}$, where $\boldsymbol{H}_k = \left[\begin{smallmatrix}\boldsymbol{I} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{I}-\boldsymbol{v}_k\boldsymbol{v}_k^T\end{smallmatrix}\right]$ and $\boldsymbol{v}_k \in \mathbb{R}^{n-k}$. Since $\boldsymbol{v}_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-a F04332	335	1.000	`\boldsymbol{v}_{n-2} \in \mathbb{R}^{2}`	$\boldsymbol{v}_{n-2} \in \mathbb{R}^{2}$	and $\boldsymbol{v}_{n-2} \in \mathbb{R}^2$ it is most economical to assemble the product from right to left. We
lyche-numerical-linear-a F04333	335	1.000	`\boldsymbol{Q}_{k+1}`	$\boldsymbol{Q}_{k+1}$	Suppose $\boldsymbol{Q}_{k+1}$ has the form $\left[\begin{smallmatrix}\boldsymbol{I}_k & \mathbf{0}\\ \mathbf{0} & \boldsymbol{U}_k\end{smallmatrix}\right]$, where $\boldsymbol{U}_k \in \mathbb{R}^{n-k,n-k}$. Then
lyche-numerical-linear-a F04334	335	1.000	`\left[\begin{array}{cc}\boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_{k}\end{array}\right]`	$\left[\begin{array}{cc}\boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_{k}\end{array}\right]$	Suppose $\boldsymbol{Q}_{k+1}$ has the form $\left[\begin{smallmatrix}\boldsymbol{I}_k & \mathbf{0}\\ \mathbf{0} & \boldsymbol{U}_k\end{smallmatrix}\right]$, where $\boldsymbol{U}_k \in \mathbb{R}^{n-k,n-k}$. Then
lyche-numerical-linear-a F04335	335	1.000	`\boldsymbol{U}_{k} \in \mathbb{R}^{n-k, n-k}`	$\boldsymbol{U}_{k} \in \mathbb{R}^{n-k, n-k}$	Suppose $\boldsymbol{Q}_{k+1}$ has the form $\left[\begin{smallmatrix}\boldsymbol{I}_k & \mathbf{0}\\ \mathbf{0} & \boldsymbol{U}_k\end{smallmatrix}\right]$, where $\boldsymbol{U}_k \in \mathbb{R}^{n-k,n-k}$. Then
lyche-numerical-linear-a F04336	335	1.000	`\lambda_{1} \geq \lambda_{2} \geq \cdots \geq \lambda_{n}`	$\lambda_{1} \geq \lambda_{2} \geq \cdots \geq \lambda_{n}$	Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. In this section
lyche-numerical-linear-a F04337	335	1.000	`1 \leq m \leq n`	$1 \leq m \leq n$	some $1 \leq m \leq n$. Using Householder similarity transformations as outlined in the
lyche-numerical-linear-a F04338	336	1.000	`c_{i}=0`	$c_{i}=0$	Suppose one of the off-diagonal elements is equal to zero, say $c_i = 0$. We then have
lyche-numerical-linear-a F04339	336	1.000	`\boldsymbol{A}=\left[\begin{array}{cc}\boldsymbol{A}_{1} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{A}_{2} \end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{cc}\boldsymbol{A}_{1} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{A}_{2} \end{array}\right]$	$\boldsymbol{A} = \left[\begin{smallmatrix}A_1 & \mathbf{0}\\ \mathbf{0} & A_2\end{smallmatrix}\right]$, where
lyche-numerical-linear-a F04340	336	1.000	`c_{i} \neq 0`	$c_{i} \neq 0$	so that $\boldsymbol{A}$ is irreducible, i.e., $c_i \neq 0$ for $i = 1, \ldots, n-1$.
lyche-numerical-linear-a F04341	336	1.000	`x \in \mathbb{R}`	$x \in \mathbb{R}$	for $x \in \mathbb{R}$ the polynomial $p_k(x) := \det(x\boldsymbol{I}_k - \boldsymbol{A}_k)$ for $k = 1, \ldots, n$, where $\boldsymbol{A}_k$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-a F04342	336	■ 1.000	p_{k}(x):=\operatorname{det}\left(x \boldsymbol{I}_{k}-\boldsymbol{A}_{k}\right)	$p_{k}(x):=\operatorname{det}\left(x \boldsymbol{I}_{k}-\boldsymbol{A}_{k}\right)$	for $x \in \mathbb{R}$ the polynomial $p_k(x) := \det(x\boldsymbol{I}_k - \boldsymbol{A}_k)$ for $k = 1, \ldots, n$, where $\boldsymbol{A}_k$
lyche-numerical-linear-a F04343	336	■ 1.000	p_{n}	p_{n}	The eigenvalues of $\boldsymbol{A}$ are the roots of the polynomial p_n. Using the last column to
lyche-numerical-linear-a F04344	336	■ 1.000	k \geq 2	$k \geq 2$	expand for $k \geq 2$ the determinant $p_{k+1}(x)$ we find
lyche-numerical-linear-a F04345	336	■ 1.000	p_{k+1}(x)	$p_{k+1}(x)$	expand for $k \geq 2$ the determinant $p_{k+1}(x)$ we find
lyche-numerical-linear-a F04346	336	■ 0.990	p_{1}(x)=x-d_{1}	$p_{1}(x)=x-d_{1}$	Since $p_1(x) = x - d_1$ and $p_2(x) = (x - d_2)(x - d_1) - c_1^2$ this also holds for $k = 0, 1$
lyche-numerical-linear-a F04347	336	■ 0.990	p_{2}(x)=\left(x-d_{2}\right)\left(x-d_{1}\right)-c_{1}^{2}	$p_{2}(x)=\left(x-d_{2}\right)\left(x-d_{1}\right)-c_{1}^{2}$	Since $p_1(x) = x - d_1$ and $p_2(x) = (x - d_2)(x - d_1) - c_1^2$ this also holds for $k = 0, 1$
lyche-numerical-linear-a F04348	336	■ 0.990	k=0,1	$k=0,1$	Since $p_1(x) = x - d_1$ and $p_2(x) = (x - d_2)(x - d_1) - c_1^2$ this also holds for $k = 0, 1$
lyche-numerical-linear-a F04349	336	■ 1.000	p_{-1}(x)=0	$p_{-1}(x)=0$	if we define $p_{-1}(x) = 0$ and $p_0(x) = 1$. For M sufficiently large we have
lyche-numerical-linear-a F04350	336	■ 1.000	p_{0}(x)=1	$p_{0}(x)=1$	if we define $p_{-1}(x) = 0$ and $p_0(x) = 1$. For M sufficiently large we have
lyche-numerical-linear-a F04351	336	■ 1.000	y_{1} \in\left(-M, d_{1}\right)	$y_{1} \in\left(-M, d_{1}\right)$	Since p_2 is continuous there are $y_1 \in (-M, d_1)$ and $y_2 \in (d_1, M)$ such that
lyche-numerical-linear-a F04352	336	■ 1.000	y_{2} \in\left(d_{1}, M\right)	$y_{2} \in\left(d_{1}, M\right)$	Since p_2 is continuous there are $y_1 \in (-M, d_1)$ and $y_2 \in (d_1, M)$ such that
lyche-numerical-linear-a F04353	336	■ 1.000	p_{2}\left(y_{1}\right)=p_{2}\left(y_{2}\right)=0	$p_{2}\left(y_{1}\right)=p_{2}\left(y_{2}\right)=0$	$p_2(y_1) = p_2(y_2) = 0$. It follows that the root d_1 of p_1 separates the roots of p_2, so
lyche-numerical-linear-a F04354	336	■ 1.000	y_{1}	y_{1}	y_1 and y_2 must be distinct. Consider next
lyche-numerical-linear-a F04355	336	■ 1.000	y_{2}	y_{2}	y_1 and y_2 must be distinct. Consider next
lyche-numerical-linear-a F04356	337	■ 1.000	y_{1}<d_{1}<y_{2}	$y_{1}<d_{1}<y_{2}$	Since $y_1 < d_1 < y_2$ we have for M sufficiently large
lyche-numerical-linear-a F04357	337	■ 1.000	x_{1}, x_{2}, x_{3}	x_{1}, x_{2}, x_{3}	Thus the roots x_1, x_2, x_3 of p_3 are separated by the roots y_1, y_2 of p_2. In the general
lyche-numerical-linear-a F04358	337	■ 1.000	p_{3}	p_{3}	Thus the roots x_1, x_2, x_3 of p_3 are separated by the roots y_1, y_2 of p_2. In the general
lyche-numerical-linear-a F04359	337	■ 1.000	z_{1}, \ldots, z_{k-1}	$z_{1}, \ldots, z_{k-1}$	case suppose for $k \geq 2$ that the roots $z_1, \ldots, z_{k-1}$ of p_{k-1} separate the roots
lyche-numerical-linear-a F04360	337	■ 1.000	p_{k-1}	p_{k-1}	case suppose for $k \geq 2$ that the roots $z_1, \ldots, z_{k-1}$ of p_{k-1} separate the roots
lyche-numerical-linear-a F04361	337	■ 1.000	y_{1}, \ldots, y_{k}	$y_{1}, \ldots, y_{k}$	$y_1, \ldots, y_k$ of p_k. Choose M so that $y_0 := -M < y_1$, $y_{k+1} := M > y_k$. Then
lyche-numerical-linear-a F04362	337	■ 1.000	p_{k}	p_{k}	$y_1, \ldots, y_k$ of p_k. Choose M so that $y_0 := -M < y_1$, $y_{k+1} := M > y_k$. Then
lyche-numerical-linear-a F04363	337	■ 1.000	y_{0}:=-M<y_{1}, y_{k+1}:=M>y_{k}	$y_{0}:=-M<y_{1}, y_{k+1}:=M>y_{k}$	$y_1, \ldots, y_k$ of p_k. Choose M so that $y_0 := -M < y_1$, $y_{k+1} := M > y_k$. Then
lyche-numerical-linear-a F04364	337	■ 1.000	j=0, k+1	$j=0, k+1$	This holds for $j = 0, k + 1$, and for $j = 1, \ldots, k$ since
lyche-numerical-linear-a F04365	337	■ 0.832	x_{1}, \ldots, x_{k+1}	$x_{1}, \ldots, x_{k+1}$	It follows that the roots $x_1, \ldots, x_{k+1}$ are separated by the roots $y_1, \ldots, y_k$ of p_k and
lyche-numerical-linear-a F04366	337	■ 1.000	\boldsymbol{A}=\boldsymbol{E}^{*} \boldsymbol{B} \boldsymbol{E}	$\boldsymbol{A}=\boldsymbol{E}^{*} \boldsymbol{B} \boldsymbol{E}$	We say that two matrices $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n\times n}$ are **congruent** if $\boldsymbol{A} = \boldsymbol{E}^*\boldsymbol{B}\boldsymbol{E}$ for some
lyche-numerical-linear-a F04367	337	■ 1.000	\boldsymbol{D}, \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}=\boldsymbol{D}	$\boldsymbol{D}, \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}=\boldsymbol{D}$	congruent and similar to a diagonal matrix $\boldsymbol{D}$, $\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \boldsymbol{D}$ where $\boldsymbol{U}$ is unitary. The
lyche-numerical-linear-a F04368	337	■ 1.000	\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})	$\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})$	eigenvalues of $\boldsymbol{A}$ are the diagonal elements of $\boldsymbol{D}$. Let $\pi(\boldsymbol{A})$, $\zeta(\boldsymbol{A})$ and $\upsilon(\boldsymbol{A})$ denote

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F04369	337	1.000	`v(\boldsymbol{A})`	$v(\boldsymbol{A})$	eigenvalues of $\boldsymbol{A}$ are the diagonal elements of $\boldsymbol{D}$. Let $\pi(\boldsymbol{A})$, $\zeta(\boldsymbol{A})$ and $\upsilon(\boldsymbol{A})$ denote
lyche-numerical-linear-algebra_F04370	337	1.000	`\pi(\boldsymbol{A})+\zeta(\boldsymbol{A})+v(\boldsymbol{A})=n`	$\pi(\boldsymbol{A})+\zeta(\boldsymbol{A})+v(\boldsymbol{A})=n$	all eigenvalues are real and $\pi(\boldsymbol{A})+\zeta(\boldsymbol{A})+\upsilon(\boldsymbol{A})=n$.
lyche-numerical-linear-algebra_F04371	337	1.000	`\pi(\boldsymbol{A})=\pi(\boldsymbol{B}), \zeta(\boldsymbol{A})=\zeta(\boldsymbol{B})`	$\pi(\boldsymbol{A})=\pi(\boldsymbol{B}), \zeta(\boldsymbol{A})=\zeta(\boldsymbol{B})$	*congruent then* $\pi(\boldsymbol{A})=\pi(\boldsymbol{B})$, $\zeta(\boldsymbol{A})=\zeta(\boldsymbol{B})$ *and* $\upsilon(\boldsymbol{A})=\upsilon(\boldsymbol{B})$.
lyche-numerical-linear-algebra_F04372	337	1.000	`v(\boldsymbol{A})=v(\boldsymbol{B})`	$v(\boldsymbol{A})=v(\boldsymbol{B})$	*congruent then* $\pi(\boldsymbol{A})=\pi(\boldsymbol{B})$, $\zeta(\boldsymbol{A})=\zeta(\boldsymbol{B})$ *and* $\upsilon(\boldsymbol{A})=\upsilon(\boldsymbol{B})$.
lyche-numerical-linear-algebra_F04373	337	1.000	`\pi(\boldsymbol{A})=k`	$\pi(\boldsymbol{A})=k$	are diagonal matrices. Suppose $\pi(\boldsymbol{A})=k$ and $\pi(\boldsymbol{B})=m<k$. We shall show that
lyche-numerical-linear-algebra_F04374	337	1.000	`\pi(\boldsymbol{B})=m<k`	$\pi(\boldsymbol{B})=m<k$	are diagonal matrices. Suppose $\pi(\boldsymbol{A})=k$ and $\pi(\boldsymbol{B})=m<k$. We shall show that
lyche-numerical-linear-algebra_F04375	337	1.000	`\boldsymbol{E}_{1}`	$\boldsymbol{E}_{1}$	this leads to a contradiction. Let $\boldsymbol{E}_1$ be the upper left $m\times k$ corner of $\boldsymbol{E}$. Since
lyche-numerical-linear-algebra_F04376	337	1.000	`m \times k`	$m \times k$	this leads to a contradiction. Let $\boldsymbol{E}_1$ be the upper left $m\times k$ corner of $\boldsymbol{E}$. Since
lyche-numerical-linear-algebra_F04377	337	1.000	`m<k`	$m<k$	$m<k$, we can find a nonzero $\boldsymbol{x}$ such that $\boldsymbol{E}_1\boldsymbol{x}=\mathbf{0}$ (cf. Lemma 1.3). Let $\boldsymbol{y}^T=$
lyche-numerical-linear-algebra_F04378	337	1.000	`\boldsymbol{E}_{1} \boldsymbol{x}=\mathbf{0}`	$\boldsymbol{E}_{1} \boldsymbol{x}=\mathbf{0}$	$m<k$, we can find a nonzero $\boldsymbol{x}$ such that $\boldsymbol{E}_1\boldsymbol{x}=\mathbf{0}$ (cf. Lemma 1.3). Let $\boldsymbol{y}^T=$
lyche-numerical-linear-algebra_F04379	337	1.000	`\boldsymbol{y}^{T}=`	$\boldsymbol{y}^{T}=$	$m<k$, we can find a nonzero $\boldsymbol{x}$ such that $\boldsymbol{E}_1\boldsymbol{x}=\mathbf{0}$ (cf. Lemma 1.3). Let $\boldsymbol{y}^T=$
lyche-numerical-linear-algebra_F04380	337	0.996	`\left[\boldsymbol{x}^{T}, \mathbf{0}^{T}\right] \in \mathbb{C}^{n}`	$\left[\boldsymbol{x}^{T}, \mathbf{0}^{T}\right] \in \mathbb{C}^{n}$	$[\boldsymbol{x}^T,\mathbf{0}^T]\in\mathbb{C}^n$, and $\boldsymbol{z}=[z_1,\ldots,z_n]^T=\boldsymbol{E}\boldsymbol{y}$. Then $z_i=0$ for $i=1,2,\ldots,m$.
lyche-numerical-linear-algebra_F04381	337	0.996	`\boldsymbol{z}=\left[z_{1}, \ldots, z_{n}\right]^{T}=\boldsymbol{E} \boldsymbol{y}`	$\boldsymbol{z}=\left[z_{1}, \ldots, z_{n}\right]^{T}=\boldsymbol{E} \boldsymbol{y}$	$[\boldsymbol{x}^T,\mathbf{0}^T]\in\mathbb{C}^n$, and $\boldsymbol{z}=[z_1,\ldots,z_n]^T=\boldsymbol{E}\boldsymbol{y}$. Then $z_i=0$ for $i=1,2,\ldots,m$.
lyche-numerical-linear-algebra_F04382	337	0.996	`z_{i}=0`	$z_{i}=0$	$[\boldsymbol{x}^T,\mathbf{0}^T]\in\mathbb{C}^n$, and $\boldsymbol{z}=[z_1,\ldots,z_n]^T=\boldsymbol{E}\boldsymbol{y}$. Then $z_i=0$ for $i=1,2,\ldots,m$.
lyche-numerical-linear-algebra_F04383	337	0.996	`i=1,2, \ldots, m`	$i=1,2, \ldots, m$	$[\boldsymbol{x}^T,\mathbf{0}^T]\in\mathbb{C}^n$, and $\boldsymbol{z}=[z_1,\ldots,z_n]^T=\boldsymbol{E}\boldsymbol{y}$. Then $z_i=0$ for $i=1,2,\ldots,m$.
lyche-numerical-linear-algebra_F04384	337	1.000	`\mu_{i} \leq 0`	$\mu_{i} \leq 0$	$\mu_i\le 0$ for $i\ge m+1$ then
lyche-numerical-linear-algebra_F04385	337	1.000	`i \geq m+1`	$i \geq m+1$	$\mu_i\le 0$ for $i\ge m+1$ then
lyche-numerical-linear-algebra_F04386	338	1.000	`\pi(\boldsymbol{A})=\pi(\boldsymbol{B})`	$\pi(\boldsymbol{A})=\pi(\boldsymbol{B})$	We conclude that $\pi(\boldsymbol{A})=\pi(\boldsymbol{B})$ if $\boldsymbol{A}$ and $\boldsymbol{B}$ are diagonal. Moreover, $\upsilon(\boldsymbol{A})=$
lyche-numerical-linear-algebra_F04387	338	1.000	`v(\boldsymbol{A})=`	$v(\boldsymbol{A})=$	We conclude that $\pi(\boldsymbol{A})=\pi(\boldsymbol{B})$ if $\boldsymbol{A}$ and $\boldsymbol{B}$ are diagonal. Moreover, $\upsilon(\boldsymbol{A})=$
lyche-numerical-linear-algebra_F04388	338	1.000	`\pi(-\boldsymbol{A})=\pi(-\boldsymbol{B})=v(\boldsymbol{B})`	$\pi(-\boldsymbol{A})=\pi(-\boldsymbol{B})=v(\boldsymbol{B})$	$\pi(-\boldsymbol{A})=\pi(-\boldsymbol{B})=\upsilon(\boldsymbol{B})$ and $\zeta(\boldsymbol{A})=n-\pi(\boldsymbol{A})-\upsilon(\boldsymbol{A})=n-\pi(\boldsymbol{B})-\upsilon(\boldsymbol{B})=$
lyche-numerical-linear-algebra_F04389	338	1.000	`\zeta(\boldsymbol{A})=n-\pi(\boldsymbol{A})-v(\boldsymbol{A})=n-\pi(\boldsymbol{B})-v(\boldsymbol{B})=`	$\zeta(\boldsymbol{A})=n-\pi(\boldsymbol{A})-v(\boldsymbol{A})=n-\pi(\boldsymbol{B})-v(\boldsymbol{B})=$	$\pi(-\boldsymbol{A})=\pi(-\boldsymbol{B})=\upsilon(\boldsymbol{B})$ and $\zeta(\boldsymbol{A})=n-\pi(\boldsymbol{A})-\upsilon(\boldsymbol{A})=n-\pi(\boldsymbol{B})-\upsilon(\boldsymbol{B})=$
lyche-numerical-linear-algebra_F04390	338	1.000	`\zeta(\boldsymbol{B})`	$\zeta(\boldsymbol{B})$	$\zeta(\boldsymbol{B})$. This completes the proof for diagonal matrices.
lyche-numerical-linear-algebra_F04391	338	1.000	`\boldsymbol{U}_{1}^{*} \boldsymbol{A} \boldsymbol{U}_{1}=\boldsymbol{D}_{1}`	$\boldsymbol{U}_{1}^{*} \boldsymbol{A} \boldsymbol{U}_{1}=\boldsymbol{D}_{1}$	Let in the general case $\boldsymbol{U}_1$ and $\boldsymbol{U}_2$ be unitary matrices such that $\boldsymbol{U}_1^*\boldsymbol{A}\boldsymbol{U}_1=\boldsymbol{D}_1$
lyche-numerical-linear-algebra_F04392	338	0.999	`\boldsymbol{U}_{2}^{*} \boldsymbol{B} \boldsymbol{U}_{2}=\boldsymbol{D}_{2}`	$\boldsymbol{U}_{2}^{*} \boldsymbol{B} \boldsymbol{U}_{2}=\boldsymbol{D}_{2}$	and $\boldsymbol{U}_2^*\boldsymbol{B}\boldsymbol{U}_2=\boldsymbol{D}_2$ where $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$ are diagonal matrices. Since $\boldsymbol{A}=\boldsymbol{E}^*\boldsymbol{B}\boldsymbol{E}$,
lyche-numerical-linear-algebra_F04393	338	0.999	`\boldsymbol{D}_{1}=\boldsymbol{F}^{*} \boldsymbol{D}_{2} \boldsymbol{F}`	$\boldsymbol{D}_{1}=\boldsymbol{F}^{*} \boldsymbol{D}_{2} \boldsymbol{F}$	we find $\boldsymbol{D}_1=\boldsymbol{F}^*\boldsymbol{D}_2\boldsymbol{F}$ where $\boldsymbol{F}=\boldsymbol{U}_2^*\boldsymbol{E}\boldsymbol{U}_1$ is nonsingular. Thus $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$
lyche-numerical-linear-algebra_F04394	338	0.999	`\boldsymbol{F}=\boldsymbol{U}_{2}^{*} \boldsymbol{E} \boldsymbol{U}_{1}`	$\boldsymbol{F}=\boldsymbol{U}_{2}^{*} \boldsymbol{E} \boldsymbol{U}_{1}$	we find $\boldsymbol{D}_1=\boldsymbol{F}^*\boldsymbol{D}_2\boldsymbol{F}$ where $\boldsymbol{F}=\boldsymbol{U}_2^*\boldsymbol{E}\boldsymbol{U}_1$ is nonsingular. Thus $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$
lyche-numerical-linear-algebra_F04395	338	1.000	`\boldsymbol{D}_{1}, \boldsymbol{B}`	$\boldsymbol{D}_{1}, \boldsymbol{B}$	are congruent diagonal matrices. But since $\boldsymbol{A}$ and $\boldsymbol{D}_1$, $\boldsymbol{B}$ and $\boldsymbol{D}_2$ have the same

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04396	338	■ 1.000	`\pi(\boldsymbol{A})=\pi\left(\boldsymbol{D}_{1}\right)=\pi\left(\boldsymbol{D}_{2}\right)=\pi(\boldsymbol{B})`	$\pi(\boldsymbol{A})=\pi\left(\boldsymbol{D}_{1}\right)=\pi\left(\boldsymbol{D}_{2}\right)=\pi(\boldsymbol{B})$	eigenvalues, we find $\pi(\boldsymbol{A})=\pi(\boldsymbol{D}_1)=\pi(\boldsymbol{D}_2)=\pi(\boldsymbol{B})$. Similar results hold for ζ
lyche-numerical-linear-algebra F04397	338	■ 1.000	`\zeta`	ζ	eigenvalues, we find $\pi(\boldsymbol{A})=\pi(\boldsymbol{D}_1)=\pi(\boldsymbol{D}_2)=\pi(\boldsymbol{B})$. Similar results hold for ζ
lyche-numerical-linear-algebra F04398	338	■ 1.000	`\boldsymbol{A}=\operatorname{tridiag}\left(c_{i}, d_{i}, c_{i}\right) \in \mathbb{R}^{n \times n}`	$\boldsymbol{A}=\operatorname{tridiag}\left(c_{i}, d_{i}, c_{i}\right) \in \mathbb{R}^{n \times n}$	$\boldsymbol{A}=\operatorname{tridiag}(c_i,d_i,c_i)\in\mathbb{R}^{n\times n}$ *is symmetric and that* $\alpha\in\mathbb{R}$ *is such that* $\boldsymbol{A}-\alpha\boldsymbol{I}$
lyche-numerical-linear-algebra F04399	338	■ 1.000	`\alpha \in \mathbb{R}`	$\alpha \in \mathbb{R}$	$\boldsymbol{A}=\operatorname{tridiag}(c_i,d_i,c_i)\in\mathbb{R}^{n\times n}$ *is symmetric and that* $\alpha\in\mathbb{R}$ *is such that* $\boldsymbol{A}-\alpha\boldsymbol{I}$
lyche-numerical-linear-algebra F04400	338	■ 1.000	`\boldsymbol{A}-\alpha \boldsymbol{I}`	$\boldsymbol{A}-\alpha \boldsymbol{I}$	$\boldsymbol{A}=\operatorname{tridiag}(c_i,d_i,c_i)\in\mathbb{R}^{n\times n}$ *is symmetric and that* $\alpha\in\mathbb{R}$ *is such that* $\boldsymbol{A}-\alpha\boldsymbol{I}$
lyche-numerical-linear-algebra F04401	338	■ 0.999	`\boldsymbol{A}-\alpha \boldsymbol{I}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^{T}`	$\boldsymbol{A}-\alpha \boldsymbol{I}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^{T}$	*has an symmetric LU factorization, i.e.* $\boldsymbol{A}-\alpha\boldsymbol{I}=\boldsymbol{LDL}^T$ *where* $\boldsymbol{L}$ *is unit lower*
lyche-numerical-linear-algebra F04402	338	■ 1.000	`d_{1}(\alpha), \ldots, d_{n}(\alpha)`	$d_{1}(\alpha), \ldots, d_{n}(\alpha)$	$d_1(\alpha),\ldots,d_n(\alpha)$ *in* $\boldsymbol{D}$ *can be computed recursively as follows*
lyche-numerical-linear-algebra F04403	338	■ 1.000	`\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^{T}`	$\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^{T}$	elements in $\boldsymbol{D}$ in an $\boldsymbol{LDL}^T$ factorization we see that the formulas in (14.8) follows
lyche-numerical-linear-algebra F04404	338	■ 1.000	`v(\boldsymbol{A}-\alpha \boldsymbol{I})=v(\boldsymbol{D})`	$v(\boldsymbol{A}-\alpha \boldsymbol{I})=v(\boldsymbol{D})$	By the previous theorem $\upsilon(\boldsymbol{A}-\alpha\boldsymbol{I})=\upsilon(\boldsymbol{D})$, the number of negative diagonal
lyche-numerical-linear-algebra F04405	338	■ 0.988	`(\boldsymbol{A}-\alpha \boldsymbol{I}) \boldsymbol{x}=(\lambda-\alpha) \boldsymbol{x}`	$(\boldsymbol{A}-\alpha \boldsymbol{I}) \boldsymbol{x}=(\lambda-\alpha) \boldsymbol{x}$	elements in $\boldsymbol{D}$. If $\boldsymbol{Ax}=\lambda\boldsymbol{x}$ then $(\boldsymbol{A}-\alpha\boldsymbol{I})\boldsymbol{x}=(\lambda-\alpha)\boldsymbol{x}$, and $\lambda-\alpha$ is an eigenvalue
lyche-numerical-linear-algebra F04406	338	■ 0.988	`\lambda-\alpha`	$\lambda-\alpha$	elements in $\boldsymbol{D}$. If $\boldsymbol{Ax}=\lambda\boldsymbol{x}$ then $(\boldsymbol{A}-\alpha\boldsymbol{I})\boldsymbol{x}=(\lambda-\alpha)\boldsymbol{x}$, and $\lambda-\alpha$ is an eigenvalue
lyche-numerical-linear-algebra F04407	338	■ 1.000	`v(\boldsymbol{A}-\alpha \boldsymbol{I})`	$v(\boldsymbol{A}-\alpha \boldsymbol{I})$	of $\boldsymbol{A}-\alpha\boldsymbol{I}$. But then $\upsilon(\boldsymbol{A}-\alpha\boldsymbol{I})$ equals the number of eigenvalues of $\boldsymbol{A}$ which are
lyche-numerical-linear-algebra F04408	338	■ 1.000	`\lambda_{1} \geq \lambda_{2} \geq`	$\lambda_{1} \geq \lambda_{2} \geq$	Corollary 14.2 can be used to determine the mth eigenvalue of $\boldsymbol{A}$, where $\lambda_1\geq\lambda_2\geq$
lyche-numerical-linear-algebra F04409	338	■ 0.999	`\cdots \geq \lambda_{n}`	$\cdots \geq \lambda_{n}$	$\cdots\geq\lambda_n$. Using Gerschgorin's theorem we first find an interval $[a,b]$, such that
lyche-numerical-linear-algebra F04410	338	■ 0.999	`(a, b)`	(a, b)	(a,b) contains the eigenvalues of $\boldsymbol{A}$. Let for $x\in[a,b]$
lyche-numerical-linear-algebra F04411	339	■ 1.000	`\rho(a)=`	$\rho(a)=$	be the number of eigenvalues of $\boldsymbol{A}$ which are strictly greater than x. Clearly $\rho(a)=$
lyche-numerical-linear-algebra F04412	339	■ 1.000	`n, \rho(b)=0`	$n, \rho(b)=0$	n, $\rho(b)=0$. Choosing a tolerance ϵ and using bisection we proceed as follows:
lyche-numerical-linear-algebra F04413	339	■ 1.000	`\left\{\left[a_{j}, b_{j}\right]\right\}`	$\left\{\left[a_{j}, b_{j}\right]\right\}$	We generate a sequence $\{[a_j,b_j]\}$ of intervals, each containing λ_m and $b_j-a_j=$
lyche-numerical-linear-algebra F04414	339	■ 1.000	`b_{j}-a_{j}=`	$b_{j}-a_{j}=$	We generate a sequence $\{[a_j,b_j]\}$ of intervals, each containing λ_m and $b_j-a_j=$
lyche-numerical-linear-algebra F04415	339	■ 1.000	`2^{-j}(b-a)`	$2^{-j}(b-a)$	$2^{-j}(b-a)$.
lyche-numerical-linear-algebra F04416	339	■ 1.000	`d_{k}(\alpha)`	$d_{k}(\alpha)$	As it stands this method will fail if in (14.8) one of the $d_k(\alpha)$ is zero. One
lyche-numerical-linear-algebra F04417	339	■ 1.000	`\delta_{k}=`	$\delta_{k}=$	possibility is to replace such a $d_k(\alpha)$ by a suitable small number, say $\delta_k=$
lyche-numerical-linear-algebra F04418	339	■ 1.000	`c_{k} \epsilon_{M}`	$c_{k} \epsilon_{M}$	$c_k\epsilon_M$, where ϵ_M is the Machine epsilon, typically 2×10^{-16} for MATLAB. This
lyche-numerical-linear-algebra F04419	339	■ 1.000	`\epsilon_{M}`	ϵ_{M}	$c_k\epsilon_M$, where ϵ_M is the Machine epsilon, typically 2×10^{-16} for MATLAB. This
lyche-numerical-linear-algebra F04420	339	■ 1.000	`2 \times 10^{-16}`	2×10^{-16}	$c_k\epsilon_M$, where ϵ_M is the Machine epsilon, typically 2×10^{-16} for MATLAB. This
lyche-numerical-linear-algebra F04421	339	■ 1.000	`\left|d_{k}(\alpha)\right|<\left|\delta_{k}\right|`	$\left\lvert d_{k}(\alpha)\right\rvert<\left\lvert\delta_{k}\right\rvert$	replacement is done if $\lvert d_k(\alpha)\rvert<\lvert\delta_k\rvert$.
lyche-numerical-linear-algebra F04422	340	■ 0.998	`\boldsymbol{A}\left(\left|a_{i, i}\right|>\right.`	$\boldsymbol{A}\left(\left\lvert a_{i, i}\right\rvert>\right.$	Gershgorin's circle theorem that a strictly diagonally dominant matrix $\boldsymbol{A}$ $(\lvert a_{i,i}\rvert>$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F04423	340	■ 1.000	\sum_{j \neq i}\left|a_{i, j}\right|	$\sum_{j \neq i}\left\|a_{i, j}\right\|$	$\sum_{j\neq i}\|a_{i,j}\|$ for all i) is nonsingular.
lyche-numerical-linear-algebra_F04424	340	■ 1.000	R:=R_{1} \cup \cdots \cup R_{n}	$R:=R_{1} \cup \cdots \cup R_{n}$	$\boldsymbol{A} \in \mathbb{R}^{n,n}$ lie inside $R \cap C$, where $R := R_1 \cup \cdots \cup R_n$ is the union of the row disks
lyche-numerical-linear-algebra_F04425	340	■ 1.000	C=C_{1} \cup \cdots \cup C_{n}	$C=C_{1} \cup \cdots \cup C_{n}$	R_i of $\boldsymbol{A}$, and $C = C_1 \cup \cdots \cup C_n$ is the union of the column disks C_j. You do not
lyche-numerical-linear-algebra_F04426	340	■ 0.987	\boldsymbol{s}=\left[s_{1}, \ldots, s_{n}\right] \in \mathbb{R}^{n}	$\boldsymbol{s}=\left[s_{1}, \ldots, s_{n}\right] \in \mathbb{R}^{n}$	computes the centres $\boldsymbol{s} = [s_1, \ldots, s_n] \in \mathbb{R}^n$ of the row and column disks, and their
lyche-numerical-linear-algebra_F04427	340	■ 1.000	\boldsymbol{r}=\left[r_{1}, \ldots, r_{n}\right] \in \mathbb{R}^{n}	$\boldsymbol{r}=\left[r_{1}, \ldots, r_{n}\right] \in \mathbb{R}^{n}$	radii $\boldsymbol{r} = [r_1, \ldots, r_n] \in \mathbb{R}^n$ and $\boldsymbol{c} = [c_1, \ldots, c_n] \in \mathbb{R}^n$, respectively.
lyche-numerical-linear-algebra_F04428	340	■ 1.000	\boldsymbol{c}=\left[c_{1}, \ldots, c_{n}\right] \in \mathbb{R}^{n}	$\boldsymbol{c}=\left[c_{1}, \ldots, c_{n}\right] \in \mathbb{R}^{n}$	radii $\boldsymbol{r} = [r_1, \ldots, r_n] \in \mathbb{R}^n$ and $\boldsymbol{c} = [c_1, \ldots, c_n] \in \mathbb{R}^n$, respectively.
lyche-numerical-linear-algebra_F04429	340	■ 1.000	\boldsymbol{A}\left(t_{2}\right)	$\boldsymbol{A}\left(t_{2}\right)$	$0 \le t_1 < t_2 \le 1$ and that μ is an eigenvalue of $\boldsymbol{A}(t_2)$. Show, using Theorem 14.2
lyche-numerical-linear-algebra_F04430	340	■ 1.000	\boldsymbol{A}=\boldsymbol{A}\left(t_{1}\right)	$\boldsymbol{A}=\boldsymbol{A}\left(t_{1}\right)$	with $\boldsymbol{A} = \boldsymbol{A}(t_1)$ and $\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$, that $\boldsymbol{A}(t_1)$ has an eigenvalue λ such that
lyche-numerical-linear-algebra_F04431	340	■ 1.000	\boldsymbol{E}=\boldsymbol{A}\left(t_{2}\right)-\boldsymbol{A}\left(t_{1}\right)	$\boldsymbol{E}=\boldsymbol{A}\left(t_{2}\right)-\boldsymbol{A}\left(t_{1}\right)$	with $\boldsymbol{A} = \boldsymbol{A}(t_1)$ and $\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$, that $\boldsymbol{A}(t_1)$ has an eigenvalue λ such that
lyche-numerical-linear-algebra_F04432	340	■ 1.000	\boldsymbol{A}\left(t_{1}\right)	$\boldsymbol{A}\left(t_{1}\right)$	with $\boldsymbol{A} = \boldsymbol{A}(t_1)$ and $\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$, that $\boldsymbol{A}(t_1)$ has an eigenvalue λ such that
lyche-numerical-linear-algebra_F04433	340	■ 1.000	\\|\boldsymbol{A}\\|_{\infty}=	$\\|\boldsymbol{A}\\|_{\infty}=$	**Exercise 14.6 (∞-Norm of a Diagonal Matrix)** Give a direct proof that $\\|\boldsymbol{A}\\|_\infty =$
lyche-numerical-linear-algebra_F04434	340	■ 1.000	\left[a_{k j}\right], \boldsymbol{E}=\left[e_{k j}\right]	$\left[a_{k j}\right], \boldsymbol{E}=\left[e_{k j}\right]$	$[a_{kj}]$, $\boldsymbol{E} = [e_{kj}]$, and $\boldsymbol{B} = [b_{kj}]$ be matrices in $\mathbb{R}^{n,n}$ with
lyche-numerical-linear-algebra_F04435	340	■ 1.000	\boldsymbol{B}=\left[b_{k j}\right]	$\boldsymbol{B}=\left[b_{k j}\right]$	$[a_{kj}]$, $\boldsymbol{E} = [e_{kj}]$, and $\boldsymbol{B} = [b_{kj}]$ be matrices in $\mathbb{R}^{n,n}$ with
lyche-numerical-linear-algebra_F04436	340	■ 1.000	\mathbb{R}^{n, n}	$\mathbb{R}^{n, n}$	$[a_{kj}]$, $\boldsymbol{E} = [e_{kj}]$, and $\boldsymbol{B} = [b_{kj}]$ be matrices in $\mathbb{R}^{n,n}$ with
lyche-numerical-linear-algebra_F04437	341	■ 1.000	\boldsymbol{B}=\boldsymbol{A}+\boldsymbol{E}	$\boldsymbol{B}=\boldsymbol{A}+\boldsymbol{E}$	and $\boldsymbol{B} = \boldsymbol{A} + \boldsymbol{E}$, where $0 < \epsilon < 1$. Thus for $n = 4$,
lyche-numerical-linear-algebra_F04438	341	■ 1.000	\\|\boldsymbol{A}\\|_{2}=\\|\boldsymbol{B}\\|_{2}=1	$\\|\boldsymbol{A}\\|_{2}=\\|\boldsymbol{B}\\|_{2}=1$	**b)** Show that $\\|\boldsymbol{A}\\|_2 = \\|\boldsymbol{B}\\|_2 = 1$ for arbitrary $n \in \mathbb{N}$.
lyche-numerical-linear-algebra_F04439	341	■ 1.000	\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{B}	$\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{B}$	**c)** Recall Elsner's Theorem (Theorem 14.2). Let $\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{B}$ be given by (14.10).
lyche-numerical-linear-algebra_F04440	341	■ 1.000	\mu=\epsilon^{1 / n}	$\mu=\epsilon^{1 / n}$	$\mu = \epsilon^{1/n}$ of $\boldsymbol{B}$? How sharp is this upper bound?
lyche-numerical-linear-algebra_F04441	341	■ 1.000	\frac{10}{3} n^{3}=5 G_{n}	$\frac{10}{3} n^{3}=5 G_{n}$	Show that the number of arithmetic operations for Algorithm 14.1 is $\frac{10}{3}n^3 = 5G_n$.
lyche-numerical-linear-algebra_F04442	341	■ 1.000	\frac{4}{3} n^{3}=2 G_{n}	$\frac{4}{3} n^{3}=2 G_{n}$	of arithmetic operations required by Algorithm 14.2 is $\frac{4}{3}n^3 = 2G_n$.
lyche-numerical-linear-algebra_F04443	341	■ 1.000	\boldsymbol{V}_{k} \boldsymbol{E}_{k} \boldsymbol{V}_{k}	$\boldsymbol{V}_{k} \boldsymbol{E}_{k} \boldsymbol{V}_{k}$	compute $\boldsymbol{V}_k\boldsymbol{E}_k\boldsymbol{V}_k$ where $\boldsymbol{E}_k$ is symmetric. Dropping subscripts we have to compute
lyche-numerical-linear-algebra_F04444	341	■ 1.000	\boldsymbol{E}_{k}	$\boldsymbol{E}_{k}$	compute $\boldsymbol{V}_k\boldsymbol{E}_k\boldsymbol{V}_k$ where $\boldsymbol{E}_k$ is symmetric. Dropping subscripts we have to compute
lyche-numerical-linear-algebra_F04445	341	■ 1.000	\boldsymbol{G}=\left(\boldsymbol{I}-\boldsymbol{v} \boldsymbol{v}^{T}\right) \boldsymbol{E}\left(\boldsymbol{I}-\boldsymbol{v} \boldsymbol{v}^{T}\right)	$\boldsymbol{G}=\left(\boldsymbol{I}-\boldsymbol{v} \boldsymbol{v}^{T}\right) \boldsymbol{E}\left(\boldsymbol{I}-\boldsymbol{v} \boldsymbol{v}^{T}\right)$	a product of the form $\boldsymbol{G} = (\boldsymbol{I} - \boldsymbol{v}\boldsymbol{v}^T)\boldsymbol{E}(\boldsymbol{I} - \boldsymbol{v}\boldsymbol{v}^T)$. Let $\boldsymbol{w} := \boldsymbol{E}\boldsymbol{v}$, $\beta := \frac{1}{2}\boldsymbol{v}^T\boldsymbol{w}$ and
lyche-numerical-linear-algebra_F04446	341	■ 1.000	\boldsymbol{w}:=\boldsymbol{E} \boldsymbol{v}, \beta:= \frac{1}{2} \boldsymbol{v}^{T} \boldsymbol{w}	$\boldsymbol{w}:=\boldsymbol{E} \boldsymbol{v}, \beta:= \frac{1}{2} \boldsymbol{v}^{T} \boldsymbol{w}$	a product of the form $\boldsymbol{G} = (\boldsymbol{I} - \boldsymbol{v}\boldsymbol{v}^T)\boldsymbol{E}(\boldsymbol{I} - \boldsymbol{v}\boldsymbol{v}^T)$. Let $\boldsymbol{w} := \boldsymbol{E}\boldsymbol{v}$, $\beta := \frac{1}{2}\boldsymbol{v}^T\boldsymbol{w}$ and
lyche-numerical-linear-algebra_F04447	341	■ 0.999	\boldsymbol{z}:=\boldsymbol{w}-\beta \boldsymbol{v}	$\boldsymbol{z}:=\boldsymbol{w}-\beta \boldsymbol{v}$	$\boldsymbol{z} := \boldsymbol{w} - \beta\boldsymbol{v}$. Show that $\boldsymbol{G} = \boldsymbol{E} - \boldsymbol{v}\boldsymbol{z}^T - \boldsymbol{z}\boldsymbol{v}^T$. Since $\boldsymbol{G}$ is symmetric, only the sub- or
lyche-numerical-linear-algebra_F04448	341	■ 0.999	\boldsymbol{G}=\boldsymbol{E}-\boldsymbol{v} \boldsymbol{z}^{T}-\boldsymbol{z} \boldsymbol{v}^{T}	$\boldsymbol{G}=\boldsymbol{E}-\boldsymbol{v} \boldsymbol{z}^{T}-\boldsymbol{z} \boldsymbol{v}^{T}$	$\boldsymbol{z} := \boldsymbol{w} - \beta\boldsymbol{v}$. Show that $\boldsymbol{G} = \boldsymbol{E} - \boldsymbol{v}\boldsymbol{z}^T - \boldsymbol{z}\boldsymbol{v}^T$. Since $\boldsymbol{G}$ is symmetric, only the sub- or

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04449	341	1.000	`O\left(4 n^{3} / 3\right)`	$O\left(4 n^{3} / 3\right)$	be shown that we need $O(4n^3/3)$ operations to tridiagonalize a symmetric matrix
lyche-numerical-linear-algebra F04450	342	0.999	`d_{k}`	d_{k}	**a)** Let d_k be the diagonal elements of $\boldsymbol{D}$ in an LDL* factorization of $\boldsymbol{A}_n$. Show that
lyche-numerical-linear-algebra F04451	342	0.999	`\boldsymbol{A}_{n}`	$\boldsymbol{A}_{n}$	**a)** Let d_k be the diagonal elements of $\boldsymbol{D}$ in an LDL* factorization of $\boldsymbol{A}_n$. Show that
lyche-numerical-linear-algebra F04452	342	1.000	`5+\sqrt{24}<d_{k} \leq 10, k=1,2, \ldots, n`	$5+\sqrt{24}<d_{k} \leq 10, k=1,2, \ldots, n$	$5+\sqrt{24} < d_k \leq 10, k=1,2,\ldots,n$.
lyche-numerical-linear-algebra F04453	342	1.000	`D_{n}:=\operatorname{det}\left(\boldsymbol{A}_{n} \right)>(5+\sqrt{24})^{n}`	$D_{n}:=\operatorname{det}\left(\boldsymbol{A}_{n}\right)>(5+\sqrt{24})^{n}$	**b)** Show that $D_n := \det(\boldsymbol{A}_n) > (5+\sqrt{24})^n$. Give $n_0 \in \mathbb{N}$ such that your computer
lyche-numerical-linear-algebra F04454	342	1.000	`n_{0} \in \mathbb{N}`	$n_{0} \in \mathbb{N}$	**b)** Show that $D_n := \det(\boldsymbol{A}_n) > (5+\sqrt{24})^n$. Give $n_0 \in \mathbb{N}$ such that your computer
lyche-numerical-linear-algebra F04455	342	1.000	`D_{n_{0}}`	$D_{n_{0}}$	gives an overflow when D_{n_0} is computed in floating point arithmetic.
lyche-numerical-linear-algebra F04456	342	0.935	`\boldsymbol{B}=\boldsymbol{U}^{T} \boldsymbol{D} \boldsymbol{U}`	$\boldsymbol{B}=\boldsymbol{U}^{T} \boldsymbol{D} \boldsymbol{U}$	where $\boldsymbol{A}^T = \boldsymbol{A}$ and $\boldsymbol{B}$ is symmetric positive definite. Then $\boldsymbol{B} = \boldsymbol{U}^T\boldsymbol{D}\boldsymbol{U}$ where
lyche-numerical-linear-algebra F04457	342	1.000	`\hat{\boldsymbol{A}}=\boldsymbol{D}^{-1 / 2} \boldsymbol{U} \boldsymbol{A} \boldsymbol{U}^{T} \boldsymbol{D}^{-1 / 2}`	$\hat{\boldsymbol{A}}=\boldsymbol{D}^{-1 / 2} \boldsymbol{U} \boldsymbol{A} \boldsymbol{U}^{T} \boldsymbol{D}^{-1 / 2}$	$\hat{\boldsymbol{A}} = \boldsymbol{D}^{-1/2}\boldsymbol{U}\boldsymbol{A}\boldsymbol{U}^T\boldsymbol{D}^{-1/2}$ where
lyche-numerical-linear-algebra F04458	342	1.000	`\hat{\boldsymbol{A}}`	$\hat{\boldsymbol{A}}$	**a)** Show that $\hat{\boldsymbol{A}}$ is symmetric.
lyche-numerical-linear-algebra F04459	342	1.000	`\hat{\boldsymbol{A}}=\hat{\boldsymbol{U}}^{T} \hat{ \boldsymbol{D}} \hat{\boldsymbol{U}}`	$\hat{\boldsymbol{A}}=\hat{\boldsymbol{U}}^{T} \hat{\boldsymbol{D}} \hat{\boldsymbol{U}}$	Let $\hat{\boldsymbol{A}} = \hat{\boldsymbol{U}}^T\hat{\boldsymbol{D}}\hat{\boldsymbol{U}}$ where $\hat{\boldsymbol{U}}$ is orthonormal and $\hat{\boldsymbol{D}}$ is diagonal. Set $\boldsymbol{E} =$
lyche-numerical-linear-algebra F04460	342	1.000	`\hat{\boldsymbol{D}}`	$\hat{\boldsymbol{D}}$	Let $\hat{\boldsymbol{A}} = \hat{\boldsymbol{U}}^T\hat{\boldsymbol{D}}\hat{\boldsymbol{U}}$ where $\hat{\boldsymbol{U}}$ is orthonormal and $\hat{\boldsymbol{D}}$ is diagonal. Set $\boldsymbol{E} =$
lyche-numerical-linear-algebra F04461	342	1.000	`\boldsymbol{E}=`	$\boldsymbol{E}=$	Let $\hat{\boldsymbol{A}} = \hat{\boldsymbol{U}}^T\hat{\boldsymbol{D}}\hat{\boldsymbol{U}}$ where $\hat{\boldsymbol{U}}$ is orthonormal and $\hat{\boldsymbol{D}}$ is diagonal. Set $\boldsymbol{E} =$
lyche-numerical-linear-algebra F04462	342	1.000	`\boldsymbol{U}^{T} \boldsymbol{D}^{-1 / 2} \hat{ \boldsymbol{U}}^{T}`	$\boldsymbol{U}^{T} \boldsymbol{D}^{-1 / 2} \hat{\boldsymbol{U}}^{T}$	$\boldsymbol{U}^T\boldsymbol{D}^{-1/2}\hat{\boldsymbol{U}}^T$.
lyche-numerical-linear-algebra F04463	342	0.999	`\boldsymbol{E}^{T} \boldsymbol{A} \boldsymbol{E}=\hat{ \boldsymbol{D}}, \boldsymbol{E}^{T} \boldsymbol{B} \boldsymbol{E}=\boldsymbol{I}`	$\boldsymbol{E}^{T} \boldsymbol{A} \boldsymbol{E}=\hat{\boldsymbol{D}}, \boldsymbol{E}^{T} \boldsymbol{B} \boldsymbol{E}=\boldsymbol{I}$	**b)** Show that $\boldsymbol{E}$ is nonsingular and that $\boldsymbol{E}^T\boldsymbol{A}\boldsymbol{E} = \hat{\boldsymbol{D}}, \boldsymbol{E}^T\boldsymbol{B}\boldsymbol{E} = \boldsymbol{I}$.
lyche-numerical-linear-algebra F04464	342	1.000	`\boldsymbol{A}=\operatorname{tridiag}(\boldsymbol{c}, \boldsymbol{d}, \boldsymbol{c})`	$\boldsymbol{A}=\operatorname{tridiag}(\boldsymbol{c}, \boldsymbol{d}, \boldsymbol{c})$	**Exercise 14.14 (Program Code for One Eigenvalue)** Suppose $\boldsymbol{A} = \operatorname{tridiag}(\boldsymbol{c},\boldsymbol{d},\boldsymbol{c})$
lyche-numerical-linear-algebra F04465	342	1.000	`d_{1}, \ldots, d_{n}`	$d_{1}, \ldots, d_{n}$	is symmetric and tridiagonal with elements $d_1,\ldots,d_n$ on the diagonal and
lyche-numerical-linear-algebra F04466	342	1.000	`c_{1}, \ldots, c_{n-1}`	$c_{1}, \ldots, c_{n-1}$	$c_1,\ldots,c_{n-1}$ on the neighboring subdiagonals. Let $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ be the
lyche-numerical-linear-algebra F04467	342	0.998	`\mathrm{k}=\operatorname{counting}(\mathrm{c}, \mathrm{d}, \mathrm{x})`	$\mathrm{k}=\operatorname{counting}(\mathrm{c}, \mathrm{d}, \mathrm{x})$	**a)** Write a `function k=counting(c,d,x)` which for given x counts the
lyche-numerical-linear-algebra F04468	342	1.000	`d_{j}(x)`	$d_{j}(x)$	above if one of the $d_j(x)$ is close to zero.
lyche-numerical-linear-algebra F04469	342	0.907	`(a, b]`	$(a, b]$	interval $(a,b]$ containing all eigenvalues of $\boldsymbol{A}$ and then generates a sequence
lyche-numerical-linear-algebra F04470	342	0.901	`\left\{\left(a_{j}, b_{j}\right]\right\}`	$\left\{\left(a_{j}, b_{j}\right]\right\}$	$\{(a_j,b_j]\}$ of intervals each containing λ_m. Iterate until $b_j - a_j \leq (b-a)\epsilon_M$,
lyche-numerical-linear-algebra F04471	342	0.901	`b_{j}-a_{j} \leq(b-a) \epsilon_{M}`	$b_{j}-a_{j} \leq(b-a) \epsilon_{M}$	$\{(a_j,b_j]\}$ of intervals each containing λ_m. Iterate until $b_j - a_j \leq (b-a)\epsilon_M$,
lyche-numerical-linear-algebra F04472	342	1.000	`\epsilon_{M} \approx 2.22 \times 10^{-16}`	$\epsilon_{M} \approx 2.22 \times 10^{-16}$	where ϵ_M is MATLAB's machine epsilon `eps`. Typically $\epsilon_M \approx 2.22 \times 10^{-16}$.
lyche-numerical-linear-algebra F04473	342	0.795	`\boldsymbol{T}:=\operatorname{tridiag}(-1,2,-1)`	$\boldsymbol{T}:=\operatorname{tridiag}(-1,2,-1)$	**c)** Test the program on $\boldsymbol{T} := \operatorname{tridiag}(-1,2,-1)$ of size 100. Compare the exact
lyche-numerical-linear-algebra F04474	342	1.000	`\lambda_{5}`	λ_{5}	value of λ_5 with your result and the result obtained by using MATLAB's built-in

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04475	343	1.000	`x \in \mathbb{C}`	$x \in \mathbb{C}$	is upper Hessenberg and $x \in \mathbb{C}$. We will study two algorithms to compute $f(x) =$
lyche-numerical-linear-algebra F04476	343	1.000	`f(x)=`	$f(x)=$	is upper Hessenberg and $x \in \mathbb{C}$. We will study two algorithms to compute $f(x) =$
lyche-numerical-linear-algebra F04477	343	1.000	`\operatorname{det}(\boldsymbol{A}-x \boldsymbol{I})`	$\operatorname{det}(\boldsymbol{A}-x \boldsymbol{I})$	$\det(\boldsymbol{A} - x\boldsymbol{I})$.
lyche-numerical-linear-algebra F04478	343	1.000	`\mu \in \sigma(\boldsymbol{A}+\boldsymbol{E})`	$\mu \in \sigma(\boldsymbol{A}+\boldsymbol{E})$	14.7.1 Suppose $\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n\times n}$. To every $\mu \in \sigma(\boldsymbol{A}+\boldsymbol{E})$ there is a $\lambda \in \sigma(\boldsymbol{A})$ which
lyche-numerical-linear-algebra F04479	345	0.604	`2 z_{k} / 3^{k}`	$2 z_{k} / 3^{k}$	It follows that $2z_k/3^k$ converges to $[1, -1]$, an eigenvector corresponding to the
lyche-numerical-linear-algebra F04480	345	0.604	`[1,-1]`	$[1,-1]$	It follows that $2z_k/3^k$ converges to $[1, -1]$, an eigenvector corresponding to the
lyche-numerical-linear-algebra F04481	345	0.954	`\lambda=3`	$\lambda=3$	dominant eigenvalue $\lambda = 3$. The sequence of Rayleigh quotients $\{z_k^T \boldsymbol{A} z_k / z_k^T z_k\}$
lyche-numerical-linear-algebra F04482	345	0.954	`\left\{z_{k}^{T} \boldsymbol{A} z_{k} / z_{k}^{T} z_{k} \right\}`	$\left\{z_{k}^{T} \boldsymbol{A} z_{k} / z_{k}^{T} z_{k} \right\}$	dominant eigenvalue $\lambda = 3$. The sequence of Rayleigh quotients $\{z_k^T \boldsymbol{A} z_k / z_k^T z_k\}$
lyche-numerical-linear-algebra F04483	345	0.628	`\boldsymbol{z}_{0}`	$\boldsymbol{z}_{0}$	To understand better what happens we expand z_0 in terms of the eigenvectors
lyche-numerical-linear-algebra F04484	345	1.000	`\left(\lambda_{j}^{k}, \boldsymbol{v}_{j}\right), j=1,2`	$\left(\lambda_{j}^{k}, \boldsymbol{v}_{j}\right), j=1,2$	Since $\boldsymbol{A}^k$ has eigenpairs $(\lambda_j^k, \boldsymbol{v}_j)$, $j = 1, 2$ we find
lyche-numerical-linear-algebra F04485	345	0.749	`3^{-k} z_{k}=c_{1} \boldsymbol{v}_{1}+3^{-k} c_{2} \boldsymbol{v}_{2} \rightarrow c_{1} \boldsymbol{v}_{1}`	$3^{-k} z_{k}=c_{1} \boldsymbol{v}_{1}+3^{-k} c_{2} \boldsymbol{v}_{2} \rightarrow c_{1} \boldsymbol{v}_{1}$	Thus $3^{-k}z_k = c_1\boldsymbol{v}_1 + 3^{-k}c_2\boldsymbol{v}_2 \rightarrow c_1\boldsymbol{v}_1$. Since $c_1 \neq 0$ we obtain convergence to the
lyche-numerical-linear-algebra F04486	345	0.749	`c_{1} \neq 0`	$c_{1} \neq 0$	Thus $3^{-k}z_k = c_1\boldsymbol{v}_1 + 3^{-k}c_2\boldsymbol{v}_2 \rightarrow c_1\boldsymbol{v}_1$. Since $c_1 \neq 0$ we obtain convergence to the
lyche-numerical-linear-algebra F04487	345	1.000	`\left(\lambda_{j}, \boldsymbol{v}_{j}\right), j=1, \ldots, n`	$\left(\lambda_{j}, \boldsymbol{v}_{j}\right), j=1, \ldots, n$	Let $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ have eigenpairs $(\lambda_j, \boldsymbol{v}_j)$, $j = 1, \ldots, n$ with $\lvert\lambda_1\rvert > \lvert\lambda_2\rvert \geq \cdots \geq$
lyche-numerical-linear-algebra F04488	345	1.000	`\left|\lambda_{1}\right|>\left|\lambda_{2}\right| \geq \cdots \geq`	$\left\lvert\lambda_{1}\right\rvert>\left\lvert\lambda_{2}\right\rvert \geq \cdots \geq$	Let $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ have eigenpairs $(\lambda_j, \boldsymbol{v}_j)$, $j = 1, \ldots, n$ with $\lvert\lambda_1\rvert > \lvert\lambda_2\rvert \geq \cdots \geq$
lyche-numerical-linear-algebra F04489	345	1.000	`\left|\lambda_{n}\right|`	$\left\lvert\lambda_{n}\right\rvert$	$\lvert\lambda_n\rvert$.
lyche-numerical-linear-algebra F04490	345	0.905	`z_{0} \in \mathbb{C}^{n}`	$z_{0} \in \mathbb{C}^{n}$	Given $z_0 \in \mathbb{C}^n$ we assume that
lyche-numerical-linear-algebra F04491	345	0.999	`z_{0}`	z_{0}	multiplicity one. The second assumption says that z_0 has a component in the
lyche-numerical-linear-algebra F04492	345	0.522	`z_{0}=c_{1} \boldsymbol{v}_{1}+c_{2} \boldsymbol{v}_{2}+ \cdots+c_{n} \boldsymbol{v}_{n}`	$z_{0}=c_{1} \boldsymbol{v}_{1}+c_{2} \boldsymbol{v}_{2}+ \cdots+c_{n} \boldsymbol{v}_{n}$	To see what happens let $z_0 = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_n\boldsymbol{v}_n$, where by assumption
lyche-numerical-linear-algebra F04493	345	1.000	`\boldsymbol{A}^{k} \boldsymbol{v}_{j}=\lambda_{j}^{k} \boldsymbol{v}_{j}`	$\boldsymbol{A}^{k} \boldsymbol{v}_{j}=\lambda_{j}^{k} \boldsymbol{v}_{j}$	(ii) of (15.2) we have $c_1 \neq 0$. Since $\boldsymbol{A}^k\boldsymbol{v}_j = \lambda_j^k\boldsymbol{v}_j$ for all j we see that
lyche-numerical-linear-algebra F04494	345	1.000	`\lambda_{1}^{k}`	λ_{1}^{k}	Dividing by λ_1^k we find
lyche-numerical-linear-algebra F04495	346	0.772	`\left(\lambda_{j} / \lambda_{1}\right)^{k} \rightarrow 0`	$\left(\lambda_{j} / \lambda_{1}\right)^{k} \rightarrow 0$	Assumption (i) of (15.2) implies that $(\lambda_j/\lambda_1)^k \rightarrow 0$ as $k \rightarrow \infty$ for all $j \geq 2$ and
lyche-numerical-linear-algebra F04496	346	0.534	`(i)`	(i)	matrices as long as (i) and (ii) of (15.2) hold, see for example page 58 of [18].
lyche-numerical-linear-algebra F04497	346	0.989	`z_{k}`	z_{k}	In practice we need to scale the iterates z_k somehow and we normally do not
lyche-numerical-linear-algebra F04498	346	0.766	`\boldsymbol{x}_{0}=z_{0} /\left|z_{0}\right|`	$\boldsymbol{x}_{0}=z_{0} /\left\lVert z_{0}\right\rVert$	know λ_1. Instead we choose a norm on $\mathbb{C}^n$, set $\boldsymbol{x}_0 = z_0/\lVert z_0\rVert$ and generate for
lyche-numerical-linear-algebra F04499	346	1.000	`k=1,2, \ldots`	$k=1,2, \ldots$	$k = 1, 2, \ldots$ unit vectors as follows:
lyche-numerical-linear-algebra F04500	346	1.000	`\lambda_{1}>0`	$\lambda_{1}>0$	*In particular, if* $\lambda_1 > 0$ *and* $c_1 > 0$ *then the sequence* $\{\boldsymbol{x}_k\}$ *will converge to the*
lyche-numerical-linear-algebra F04501	346	1.000	`c_{1}>0`	$c_{1}>0$	*In particular, if* $\lambda_1 > 0$ *and* $c_1 > 0$ *then the sequence* $\{\boldsymbol{x}_k\}$ *will converge to the*

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra_F04502</td><td>346</td><td>1.000</td><td>\boldsymbol{u}_{1}:=\boldsymbol{v}_{1} /\left\| \boldsymbol{v}_{1}\right\|</td><td>$\boldsymbol{u}_{1}:=\boldsymbol{v}_{1} /\left\| \boldsymbol{v}_{1}\right\|$</td><td>eigenvector $\boldsymbol{u}_1 := \boldsymbol{v}_1/\|\boldsymbol{v}_1\|$ of unit length.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04503</td><td>346</td><td>0.980</td><td>\boldsymbol{x}_{k}=\boldsymbol{z}_{k} /\left\| \boldsymbol{z}_{k}\right\|</td><td>$\boldsymbol{x}_{k}=\boldsymbol{z}_{k} /\left\| \boldsymbol{z}_{k}\right\|$</td><td>Proof By induction on k it follows that $\boldsymbol{x}_k = \boldsymbol{z}_k/\|\boldsymbol{z}_k\|$ for all $k \ge 0$, where $\boldsymbol{z}_k =$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04504</td><td>346</td><td>0.980</td><td>\boldsymbol{z}_{k}=</td><td>$\boldsymbol{z}_{k}=$</td><td>Proof By induction on k it follows that $\boldsymbol{x}_k = \boldsymbol{z}_k/\|\boldsymbol{z}_k\|$ for all $k \ge 0$, where $\boldsymbol{z}_k =$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04505</td><td>346</td><td>0.999</td><td>\boldsymbol{A}^{k} \boldsymbol{z}_{0}</td><td>$\boldsymbol{A}^{k} \boldsymbol{z}_{0}$</td><td>$\boldsymbol{A}^k\boldsymbol{z}_0$. Indeed, this holds for $k = 1$, and if it holds for $k-1$ then $\boldsymbol{y}_k = \boldsymbol{A}\boldsymbol{x}_{k-1} =$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04506</td><td>346</td><td>0.999</td><td>\boldsymbol{y}_{k}=\boldsymbol{A} \boldsymbol{x}_{k-1}=</td><td>$\boldsymbol{y}_{k}=\boldsymbol{A} \boldsymbol{x}_{k-1}=$</td><td>$\boldsymbol{A}^k\boldsymbol{z}_0$. Indeed, this holds for $k = 1$, and if it holds for $k-1$ then $\boldsymbol{y}_k = \boldsymbol{A}\boldsymbol{x}_{k-1} =$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04507</td><td>346</td><td>1.000</td><td>\boldsymbol{A} \boldsymbol{z}_{k-1} /\left\| \boldsymbol{z}_{k-1}\right\|=\boldsymbol{z}_{k} /\left\| \boldsymbol{z}_{k-1}\right\|</td><td>$\boldsymbol{A} \boldsymbol{z}_{k-1} /\left\| \boldsymbol{z}_{k-1}\right\|=\boldsymbol{z}_{k} /\left\| \boldsymbol{z}_{k-1}\right\|$</td><td>$\boldsymbol{A}\boldsymbol{z}_{k-1}/\|\boldsymbol{z}_{k-1}\| = \boldsymbol{z}_k/\|\boldsymbol{z}_{k-1}\|$ and $\boldsymbol{x}_k = (\boldsymbol{z}_k/\|\boldsymbol{z}_{k-1}\|)(\|\boldsymbol{z}_{k-1}\|/\|\boldsymbol{z}_k\|) = \boldsymbol{z}_k/\|\boldsymbol{z}_k\|$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04508</td><td>346</td><td>1.000</td><td>\boldsymbol{x}_{k}=\left(\boldsymbol{z}_{k} /\left \|\boldsymbol{z}_{k-1}\right\|\right)\left(\left\| \boldsymbol{z}_{k-1}\right\| /\left\|\boldsymbol{z}_{k} \right\|\right)=\boldsymbol{z}_{k} /\left\| \boldsymbol{z}_{k}\right\|</td><td>$\boldsymbol{x}_{k}=\left(\boldsymbol{z}_{k} /\left \|\boldsymbol{z}_{k-1}\right\|\right)\left(\left\| \boldsymbol{z}_{k-1}\right\| /\left\|\boldsymbol{z}_{k} \right\|\right)=\boldsymbol{z}_{k} /\left\| \boldsymbol{z}_{k}\right\|$</td><td>$\boldsymbol{A}\boldsymbol{z}_{k-1}/\|\boldsymbol{z}_{k-1}\| = \boldsymbol{z}_k/\|\boldsymbol{z}_{k-1}\|$ and $\boldsymbol{x}_k = (\boldsymbol{z}_k/\|\boldsymbol{z}_{k-1}\|)(\|\boldsymbol{z}_{k-1}\|/\|\boldsymbol{z}_k\|) = \boldsymbol{z}_k/\|\boldsymbol{z}_k\|$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04509</td><td>346</td><td>1.000</td><td>r(\lambda):=\boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u}</td><td>$r(\lambda):=\boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u}$</td><td>residual $r(\lambda) := \boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04510</td><td>346</td><td>1.000</td><td>\boldsymbol{u} \in \mathbb{C}^{n} \backslash\{\mathbf{0} \}</td><td>$\boldsymbol{u} \in \mathbb{C}^{n} \backslash\{\mathbf{0} \}$</td><td>$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$, and let $\rho : \mathbb{C} \to \mathbb{R}$ be given by $\rho(\lambda) = \|\boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}\|_2$. Then ρ is</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04511</td><td>346</td><td>1.000</td><td>\rho: \mathbb{C} \rightarrow \mathbb{R}</td><td>$\rho: \mathbb{C} \rightarrow \mathbb{R}$</td><td>$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$, and let $\rho : \mathbb{C} \to \mathbb{R}$ be given by $\rho(\lambda) = \|\boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}\|_2$. Then ρ is</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04512</td><td>346</td><td>1.000</td><td>\rho(\lambda)=\|\boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u}\|_{2}</td><td>$\rho(\lambda)=\|\boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u}\|_{2}$</td><td>$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$, and let $\rho : \mathbb{C} \to \mathbb{R}$ be given by $\rho(\lambda) = \|\boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}\|_2$. Then ρ is</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04513</td><td>346</td><td>0.906</td><td>\lambda:=\frac{\boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^{*} \boldsymbol{u}}</td><td>$\lambda:=\frac{\boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^{*} \boldsymbol{u}}$</td><td>minimized when $\lambda := \frac{\boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}}{\boldsymbol{u}^*\boldsymbol{u}}$, the Rayleigh quotient for $\boldsymbol{A}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04514</td><td>347</td><td>1.000</td><td>\{\boldsymbol{u}, \boldsymbol{U}\}</td><td>$\{\boldsymbol{u}, \boldsymbol{U}\}$</td><td>Proof Assume $\boldsymbol{u}^*\boldsymbol{u} = 1$ and extend $\boldsymbol{u}$ to an orthonormal basis $\{\boldsymbol{u}, \boldsymbol{U}\}$ for $\mathbb{C}^n$. Then</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04515</td><td>347</td><td>1.000</td><td>\boldsymbol{U}^{*} \boldsymbol{u}=\mathbf{0}</td><td>$\boldsymbol{U}^{*} \boldsymbol{u}=\mathbf{0}$</td><td>$\boldsymbol{U}^*\boldsymbol{u} = \mathbf{0}$ and</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04516</td><td>347</td><td>1.000</td><td>\lambda=\boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}</td><td>$\lambda=\boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}$</td><td>and ρ has a global minimum at $\lambda = \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04517</td><td>347</td><td>0.909</td><td>(\mu, \boldsymbol{u})</td><td>$(\mu, \boldsymbol{u})$</td><td>$(\mu, \boldsymbol{u})$ is an approximate eigenpair with $\|\boldsymbol{u}\|_2 = 1$, then by (14.4) we can find an</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04518</td><td>347</td><td>0.909</td><td>\|\boldsymbol{u}\|_{2}=1</td><td>$\|\boldsymbol{u}\|_{2}=1$</td><td>$(\mu, \boldsymbol{u})$ is an approximate eigenpair with $\|\boldsymbol{u}\|_2 = 1$, then by (14.4) we can find an</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04519</td><td>347</td><td>0.860</td><td>l, \boldsymbol{x}</td><td>$l, \boldsymbol{x}$</td><td>estimate for the eigenvalue is used to compute a dominant eigenpair $(l, \boldsymbol{x})$ of $\boldsymbol{A}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04520</td><td>347</td><td>0.767</td><td>\|\boldsymbol{A} \boldsymbol{x}-l \boldsymbol{x}\|_{2} / \|\boldsymbol{A}\|_{F}<t o l</td><td>$\|\boldsymbol{A} \boldsymbol{x}-l \boldsymbol{x}\|_{2} / \|\boldsymbol{A}\|_{F} < t o l$</td><td>$\|\boldsymbol{A}\boldsymbol{x} - l\boldsymbol{x}\|_2/\|\boldsymbol{A}\|_F < tol$. If no such eigenpair is found in K iterations the value</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04521</td><td>347</td><td>0.877</td><td>=K+1</td><td>$=K+1$</td><td>$it = K + 1$ is returned.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04522</td><td>348</td><td>0.900</td><td>\boldsymbol{z}=[0.6602,0.3420]</td><td>$\boldsymbol{z}=[0.6602,0.3420]$</td><td>In each case we start with the random vector $\boldsymbol{z} = [0.6602, 0.3420]$ and $tol = 10^{-6}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04523</td><td>348</td><td>0.900</td><td>=10^{-6}</td><td>$=10^{-6}$</td><td>In each case we start with the random vector $\boldsymbol{z} = [0.6602, 0.3420]$ and $tol = 10^{-6}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04524</td><td>348</td><td>1.000</td><td>\frac{\left|\lambda_{2}\right|}{\left|\lambda_{1}\right|}</td><td>$\frac{\left|\lambda_{2}\right|}{\left|\lambda_{1}\right|}$</td><td>convergence depends on the ratio $\frac{|\lambda_2|}{|\lambda_1|}$. If this ratio is small then the convergence</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04525</td><td>348</td><td>1.000</td><td>\lambda_{1}=5.3723</td><td>$\lambda_{1}=5.3723$</td><td>$\boldsymbol{A}_1$ are $\lambda_1 = 5.3723$ and $\lambda_2 = -0.3723$ giving a quite small ratio of 0.07 and</td></tr>
<tr><td>lyche-numerical-linear-algebra_F04526</td><td>348</td><td>1.000</td><td>\lambda_{2}=-0.3723</td><td>$\lambda_{2}=-0.3723$</td><td>$\boldsymbol{A}_1$ are $\lambda_1 = 5.3723$ and $\lambda_2 = -0.3723$ giving a quite small ratio of 0.07 and</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>lyche-numerical-linear-algebra F04527</td><td>348</td><td>■ 1.000</td><td>\lambda_{2}=1.9</td><td>$\lambda_2 = 1.9$</td><td>$\lambda_2 = 1.9$ and the corresponding ratio is 0.95 resulting in slow convergence.</td></tr>
<tr><td>lyche-numerical-linear-algebra F04528</td><td>348</td><td>■ 1.000</td><td>\boldsymbol{A}-s \boldsymbol{I}</td><td>$\boldsymbol{A}-s\boldsymbol{I}$</td><td>choose a number s and apply the power method to the matrix $\boldsymbol{A}-s\boldsymbol{I}$. The number</td></tr>
<tr><td>lyche-numerical-linear-algebra F04529</td><td>348</td><td>■ 1.000</td><td>\lambda-s</td><td>$\lambda-s$</td><td>s is called a shift since it shifts an eigenvalue λ of $\boldsymbol{A}$ to $\lambda-s$ of $\boldsymbol{A}-s\boldsymbol{I}$. Sometimes</td></tr>
<tr><td>lyche-numerical-linear-algebra F04530</td><td>348</td><td>■ 1.000</td><td>(\boldsymbol{A}-s \boldsymbol{I})^{-1}</td><td>$(\boldsymbol{A}-s\boldsymbol{I})^{-1}$</td><td>method to the inverse matrix $(\boldsymbol{A}-s\boldsymbol{I})^{-1}$, where s is a shift. If $\boldsymbol{A}$ has eigenvalues</td></tr>
<tr><td>lyche-numerical-linear-algebra F04531</td><td>348</td><td>■ 1.000</td><td>\lim _{s \rightarrow \lambda_{1}}\left|\mu_{1}(s)\right|=\infty</td><td>$\lim_{s\rightarrow\lambda_1}\left|\mu_1(s)\right|=\infty$</td><td>Suppose λ_1 is a simple eigenvalue of $\boldsymbol{A}$. Then $\lim_{s\to\lambda_1}|\mu_1(s)| = \infty$, while</td></tr>
<tr><td>lyche-numerical-linear-algebra F04532</td><td>348</td><td>■ 1.000</td><td>\lim _{s \rightarrow \lambda_{1}} \mu_{j}(s)=\left(\lambda_{j}-\lambda_{1}\right)^{-1}<\infty</td><td>$\lim_{s\rightarrow\lambda_1}\mu_j(s)=\left(\lambda_j-\lambda_1\right)^{-1}<\infty$</td><td>$\lim_{s\to\lambda_1}\mu_j(s) = (\lambda_j-\lambda_1)^{-1} < \infty$ for $j = 2,\ldots,n$. Hence, by choosing s</td></tr>
<tr><td>lyche-numerical-linear-algebra F04533</td><td>348</td><td>■ 1.000</td><td>j=2, \ldots, n</td><td>$j=2,\ldots,n$</td><td>$\lim_{s\to\lambda_1}\mu_j(s) = (\lambda_j-\lambda_1)^{-1} < \infty$ for $j = 2,\ldots,n$. Hence, by choosing s</td></tr>
<tr><td>lyche-numerical-linear-algebra F04534</td><td>349</td><td>■ 0.987</td><td>s_{k-1}</td><td>s_{k-1}</td><td>previous Rayleigh quotient s_{k-1} as the current shift. In each iteration we need to</td></tr>
<tr><td>lyche-numerical-linear-algebra F04535</td><td>349</td><td>■ 1.000</td><td>\boldsymbol{A} \boldsymbol{x}_{k}</td><td>$\boldsymbol{A}\boldsymbol{x}_k$</td><td>We can avoid the calculation of $\boldsymbol{A}\boldsymbol{x}_k$ in (iii) and (iv). Let</td></tr>
<tr><td>lyche-numerical-linear-algebra F04536</td><td>350</td><td>■ 0.913</td><td>i i</td><td>ii</td><td>this error disappears when we normalize $\boldsymbol{y}_k$ in (ii). Miraculously, the normalized</td></tr>
<tr><td>lyche-numerical-linear-algebra F04537</td><td>350</td><td>■ 0.995</td><td>(s, \boldsymbol{x})</td><td>$(s,\boldsymbol{x})$</td><td>Given an approximation $(s,\boldsymbol{x})$ to an eigenpair $(\lambda,\boldsymbol{v})$ of a matrix $\boldsymbol{A}\in\mathbb{C}^{n\times n}$. The</td></tr>
<tr><td>lyche-numerical-linear-algebra F04538</td><td>350</td><td>■ 0.995</td><td>(\lambda, \boldsymbol{v})</td><td>$(\lambda,\boldsymbol{v})$</td><td>Given an approximation $(s,\boldsymbol{x})$ to an eigenpair $(\lambda,\boldsymbol{v})$ of a matrix $\boldsymbol{A}\in\mathbb{C}^{n\times n}$. The</td></tr>
<tr><td>lyche-numerical-linear-algebra F04539</td><td>350</td><td>■ 1.000</td><td>n r</td><td>nr</td><td>one Rayleigh quotient iteration. The length nr of the new residual is also returned.</td></tr>
<tr><td>lyche-numerical-linear-algebra F04540</td><td>350</td><td>■ 1.000</td><td>\lambda_{1}=(5-\sqrt{33}) / 2 \approx-0.37</td><td>$\lambda_1=(5-\sqrt{33})/2\approx-0.37$</td><td>$\boldsymbol{A} = \left[\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right]$ is $\lambda_1 = (5-\sqrt{33})/2 \approx -0.37$. Starting with $\boldsymbol{x} = [1,1]^T$ and $s =$</td></tr>
<tr><td>lyche-numerical-linear-algebra F04541</td><td>350</td><td>■ 1.000</td><td>s=</td><td>$s=$</td><td>$\boldsymbol{A} = \left[\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right]$ is $\lambda_1 = (5-\sqrt{33})/2 \approx -0.37$. Starting with $\boldsymbol{x} = [1,1]^T$ and $s =$</td></tr>
<tr><td>lyche-numerical-linear-algebra F04542</td><td>350</td><td>■ 0.798</td><td>\|\boldsymbol{r}\|_{2}</td><td>$\Vert\boldsymbol{r}\Vert_2$</td><td>$k$ 1 2 3 4 5
$\Vert\boldsymbol{r}\Vert_2$ 1.0e+000 7.7e−002 1.6e−004 8.2e−010 2.0e−020
$\lvert s-\lambda_1\rvert$ 3.7e−001 −1.2e−002 −2.9e−005 −1.4e−010 −2.2e−016</td></tr>
<tr><td>lyche-numerical-linear-algebra F04543</td><td>350</td><td>■ 0.798</td><td>1.0 \mathrm{e}+000</td><td>$1.0\mathrm{e}+000$</td><td>k 1 2 3 4 5
$\Vert\boldsymbol{r}\Vert_2$ 1.0e+000 7.7e−002 1.6e−004 8.2e−010 2.0e−020
$\lvert s-\lambda_1\rvert$ 3.7e−001 −1.2e−002 −2.9e−005 −1.4e−010 −2.2e−016</td></tr>
<tr><td>lyche-numerical-linear-algebra F04544</td><td>350</td><td>■ 0.798</td><td>\left|s-\lambda_{1}\right|</td><td>$\left\lvert s-\lambda_1\right\rvert$</td><td>$k$ 1 2 3 4 5
$\Vert\boldsymbol{r}\Vert_2$ 1.0e+000 7.7e−002 1.6e−004 8.2e−010 2.0e−020
$\lvert s-\lambda_1\rvert$ 3.7e−001 −1.2e−002 −2.9e−005 −1.4e−010 −2.2e−016</td></tr>
<tr><td>lyche-numerical-linear-algebra F04545</td><td>351</td><td>■ 0.960</td><td>\boldsymbol{Q} \in \mathbb{C}^{n \times n}</td><td>$\boldsymbol{Q}\in\mathbb{C}^{n\times n}$</td><td>same. If $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ is a QR factorization then $\boldsymbol{Q}\in\mathbb{C}^{n\times n}$ is unitary, $\boldsymbol{Q}^*\boldsymbol{Q} = \boldsymbol{I}$ and</td></tr>
<tr><td>lyche-numerical-linear-algebra F04546</td><td>351</td><td>■ 1.000</td><td>\boldsymbol{R} \in \mathbb{C}^{n \times n}</td><td>$\boldsymbol{R}\in\mathbb{C}^{n\times n}$</td><td>$\boldsymbol{R}\in\mathbb{C}^{n\times n}$ is upper triangular.</td></tr>
<tr><td>lyche-numerical-linear-algebra F04547</td><td>351</td><td>■ 1.000</td><td>\boldsymbol{R}_{k} \boldsymbol{Q}_{k}</td><td>$\boldsymbol{R}_k\boldsymbol{Q}_k$</td><td>The determination of the QR factorization of $\boldsymbol{A}_k$ and the computation of $\boldsymbol{R}_k\boldsymbol{Q}_k$</td></tr>
<tr><td>lyche-numerical-linear-algebra F04548</td><td>351</td><td>■ 1.000</td><td>\boldsymbol{R}_{k}=\boldsymbol{Q}_{k}^{*} \boldsymbol{A}_{k}</td><td>$\boldsymbol{R}_k=\boldsymbol{Q}_k^*\boldsymbol{A}_k$</td><td>point, since $\boldsymbol{R}_k = \boldsymbol{Q}_k^*\boldsymbol{A}_k$ we find</td></tr>
<tr><td>lyche-numerical-linear-algebra F04549</td><td>352</td><td>■ 1.000</td><td>\boldsymbol{A}_{10}</td><td>$\boldsymbol{A}_{10}$</td><td>$\boldsymbol{A}_{10}$ is almost diagonal and contains approximations to the eigenvalues $\lambda_1 = 3$ and</td></tr>
<tr><td>lyche-numerical-linear-algebra F04550</td><td>352</td><td>■ 1.000</td><td>\lambda_{1}=3</td><td>$\lambda_1=3$</td><td>$\boldsymbol{A}_{10}$ is almost diagonal and contains approximations to the eigenvalues $\lambda_1 = 3$ and</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-a F04551	352	0.999	`\lambda_{2}=1`	$\lambda_{2}=1$	$\lambda_2 = 1$ on the diagonal.
lyche-numerical-linear-a F04552	352	1.000	`\lambda_{1}, \ldots, \lambda_{4}`	$\lambda_{1}, \ldots, \lambda_{4}$	This matrix is almost quasi-triangular and estimates for the eigenvalues $\lambda_1, \ldots, \lambda_4$
lyche-numerical-linear-a F04553	352	0.835	`\boldsymbol{A}_{14}`	$\boldsymbol{A}_{14}$	of $\boldsymbol{A}$ can now easily be determined from the diagonal blocks of $\boldsymbol{A}_{14}$. The 1×1
lyche-numerical-linear-a F04554	352	1.000	`\lambda_{1} \approx 2.323`	$\lambda_{1} \approx 2.323$	blocks give us two real eigenvalues $\lambda_1 \approx 2.323$ and $\lambda_4 \approx 0.2275$. The middle
lyche-numerical-linear-a F04555	352	1.000	`\lambda_{4} \approx 0.2275`	$\lambda_{4} \approx 0.2275$	blocks give us two real eigenvalues $\lambda_1 \approx 2.323$ and $\lambda_4 \approx 0.2275$. The middle
lyche-numerical-linear-a F04556	352	0.813	`\lambda_{2} \approx 0.0914+0.4586 i`	$\lambda_{2} \approx 0.0914+0.4586 i$	2×2 block has complex eigenvalues resulting in $\lambda_2 \approx 0.0914 + 0.4586i$ and $\lambda_3 \approx$
lyche-numerical-linear-a F04557	352	0.813	`\lambda_{3} \approx`	$\lambda_{3} \approx$	2×2 block has complex eigenvalues resulting in $\lambda_2 \approx 0.0914 + 0.4586i$ and $\lambda_3 \approx$
lyche-numerical-linear-a F04558	352	1.000	`\left(\boldsymbol{A}_{k}\right)_{k}`	$\left(\boldsymbol{A}_{k}\right)_{k}$	$(\boldsymbol{A}_k)_k$ converges to the triangular Schur form (Cf. Theorem 6.5) if all the eigenvalues
lyche-numerical-linear-a F04559	352	0.674	`k=1,2,3 \ldots`	$k=1,2,3 \ldots$	**Theorem 15.3 (QR and Power)** *For* $k = 1, 2, 3, \ldots$, *the QR factorization of* A^k *is*
lyche-numerical-linear-a F04560	352	0.674	`A^{k}`	A^{k}	**Theorem 15.3 (QR and Power)** *For* $k = 1, 2, 3, \ldots$, *the QR factorization of* A^k *is*
lyche-numerical-linear-a F04561	352	1.000	`\boldsymbol{A}^{k}=\tilde{\boldsymbol{Q}}_{k} \tilde{ \boldsymbol{R}}_{k}`	$\boldsymbol{A}^{k}=\tilde{\boldsymbol{Q}}_{k} \tilde{\boldsymbol{R}}_{k}$	$\boldsymbol{A}^k = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k$, *where*
lyche-numerical-linear-a F04562	352	1.000	`\boldsymbol{Q}_{1}, \ldots, \boldsymbol{Q}_{k}, \boldsymbol{R}_{1}, \ldots, \boldsymbol{R}_{k}`	$\boldsymbol{Q}_{1}, \ldots, \boldsymbol{Q}_{k}, \boldsymbol{R}_{1}, \ldots, \boldsymbol{R}_{k}$	*and* $\boldsymbol{Q}_1, \ldots, \boldsymbol{Q}_k$, $\boldsymbol{R}_1, \ldots, \boldsymbol{R}_k$ *are the matrices generated by the basic QR algo-*
lyche-numerical-linear-a F04563	353	1.000	`\tilde{\boldsymbol{Q}}_{1} \tilde{\boldsymbol{R}}_{1}= \boldsymbol{Q}_{1} \boldsymbol{R}_{1}=\boldsymbol{A}_{1}`	$\tilde{\boldsymbol{Q}}_{1} \tilde{\boldsymbol{R}}_{1}=\boldsymbol{Q}_{1} \boldsymbol{R}_{1}=\boldsymbol{A}_{1}$	The proof is by induction on k. Clearly $\tilde{\boldsymbol{Q}}_1 \tilde{\boldsymbol{R}}_1 = \boldsymbol{Q}_1 \boldsymbol{R}_1 = \boldsymbol{A}_1$. Suppose
lyche-numerical-linear-a F04564	353	1.000	`\tilde{\boldsymbol{Q}}_{k-1} \tilde{ \boldsymbol{R}}_{k-1}=\boldsymbol{A}^{k-1}`	$\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1}=\boldsymbol{A}^{k-1}$	$\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1} = \boldsymbol{A}^{k-1}$ for some $k \geq 2$. Since $\boldsymbol{Q}_k \boldsymbol{R}_k = \boldsymbol{A}_k$ and using (15.11)
lyche-numerical-linear-a F04565	353	1.000	`\boldsymbol{Q}_{k} \boldsymbol{R}_{k}=\boldsymbol{A}_{k}`	$\boldsymbol{Q}_{k} \boldsymbol{R}_{k}=\boldsymbol{A}_{k}$	$\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1} = \boldsymbol{A}^{k-1}$ for some $k \geq 2$. Since $\boldsymbol{Q}_k \boldsymbol{R}_k = \boldsymbol{A}_k$ and using (15.11)
lyche-numerical-linear-a F04566	353	1.000	`\tilde{\boldsymbol{R}}_{k}`	$\tilde{\boldsymbol{R}}_{k}$	Since $\tilde{\boldsymbol{R}}_k$ is upper triangular, its first column is a multiple of $\boldsymbol{e}_1$ so that
lyche-numerical-linear-a F04567	353	1.000	`\left|\tilde{\boldsymbol{q}}_{1}^{(k)}\right|_{2}=1`	$\left\|\tilde{\boldsymbol{q}}_{1}^{(k)}\right\|_{2}=1$	Since $\\|\tilde{\boldsymbol{q}}_1^{(k)}\\|_2 = 1$ the first column of $\tilde{\boldsymbol{Q}}_k$ is the result of applying the normalized
lyche-numerical-linear-a F04568	353	1.000	`\tilde{\boldsymbol{Q}}_{k}`	$\tilde{\boldsymbol{Q}}_{k}$	Since $\\|\tilde{\boldsymbol{q}}_1^{(k)}\\|_2 = 1$ the first column of $\tilde{\boldsymbol{Q}}_k$ is the result of applying the normalized
lyche-numerical-linear-a F04569	353	1.000	`\boldsymbol{x}_{0}=\boldsymbol{e}_{1}`	$\boldsymbol{x}_{0}=\boldsymbol{e}_{1}$	power iteration (15.6) to the starting vector $\boldsymbol{x}_0 = \boldsymbol{e}_1$. If this iteration converges we
lyche-numerical-linear-a F04570	353	1.000	`\lambda_{1} \boldsymbol{e}_{1}`	$\lambda_{1} \boldsymbol{e}_{1}$	It can be shown that the first column of $\boldsymbol{A}_k$ must then converge to $\lambda_1 \boldsymbol{e}_1$, where λ_1 is a
lyche-numerical-linear-a F04571	353	1.000	`\left(\tilde{\boldsymbol{Q}}_{k}^{*} \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k}\right)_{k}`	$\left(\tilde{\boldsymbol{Q}}_{k}^{*} \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k}\right)_{k}$	Indeed, what is observed in practice is that the sequence $(\tilde{\boldsymbol{Q}}_k^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_k)_k$ converges to a
lyche-numerical-linear-a F04572	353	1.000	`\boldsymbol{H}_{1}`	$\boldsymbol{H}_{1}$	an initial reduction of $\boldsymbol{A}$ to upper Hessenberg form $\boldsymbol{H}_1$ using Algorithm 14.1, the
lyche-numerical-linear-a F04573	353	1.000	`\boldsymbol{P}_{i, i+1}, i=1, \ldots, n-1`	$\boldsymbol{P}_{i, i+1}, i=1, \ldots, n-1$	determine plane rotations $\boldsymbol{P}_{i,i+1}$, $i = 1, \ldots, n-1$ so that $\boldsymbol{P}_{n-1,n} \cdots \boldsymbol{P}_{1,2} \boldsymbol{H}_1 = \boldsymbol{R}_1$
lyche-numerical-linear-a F04574	353	1.000	`\boldsymbol{P}_{n-1, n} \cdots \boldsymbol{P}_{1,2} \boldsymbol{H}_{1}=\boldsymbol{R}_{1}`	$\boldsymbol{P}_{n-1, n} \cdots \boldsymbol{P}_{1,2} \boldsymbol{H}_{1}=\boldsymbol{R}_{1}$	determine plane rotations $\boldsymbol{P}_{i,i+1}$, $i = 1, \ldots, n-1$ so that $\boldsymbol{P}_{n-1,n} \cdots \boldsymbol{P}_{1,2} \boldsymbol{H}_1 = \boldsymbol{R}_1$
lyche-numerical-linear-a F04575	353	1.000	`\boldsymbol{H}_{1}=\boldsymbol{Q}_{1} \boldsymbol{R}_{1}`	$\boldsymbol{H}_{1}=\boldsymbol{Q}_{1} \boldsymbol{R}_{1}$	is upper triangular. The details were described in Sect. 5.6. Thus $\boldsymbol{H}_1 = \boldsymbol{Q}_1 \boldsymbol{R}_1$,
lyche-numerical-linear-a F04576	353	0.999	`\boldsymbol{Q}_{1}=\boldsymbol{P}_{1,2}^{*} \cdots \boldsymbol{P}_{n-1, n}^{*}`	$\boldsymbol{Q}_{1}=\boldsymbol{P}_{1,2}^{*} \cdots \boldsymbol{P}_{n-1, n}^{*}$	where $\boldsymbol{Q}_1 = \boldsymbol{P}_{1,2}^* \cdots \boldsymbol{P}_{n-1,n}^*$ is a QR factorization of $\boldsymbol{H}_1$. To finish the QR step we

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-a F04577	353	0.996	`\boldsymbol{R}_{1} \boldsymbol{Q}_{1}=\boldsymbol{R}_{1} \boldsymbol{P}_{1,2}^{*} \cdots \boldsymbol{P}_{n-1, n}^{*}`	$\boldsymbol{R}_{1} \boldsymbol{Q}_{1}=\boldsymbol{R}_{1} \boldsymbol{P}_{1,2}^{*} \cdots \boldsymbol{P}_{n-1, n}^{*}$	compute $\boldsymbol{R}_1\boldsymbol{Q}_1 = \boldsymbol{R}_1\boldsymbol{P}^*_{1,2}\cdots \boldsymbol{P}^*_{n-1,n}$. This postmultiplication step is illustrated by
lyche-numerical-linear-a F04578	353	0.999	`(i+1, i)`	$(i+1, i)$	The postmultiplication by $\boldsymbol{P}_{i,i+1}$ introduces a nonzero in position $(i+1,i)$
lyche-numerical-linear-a F04579	354	0.993	`\boldsymbol{R} \boldsymbol{P}_{1,2}^{*} \cdots \boldsymbol{P}_{n-1, n}^{*}`	$\boldsymbol{R} \boldsymbol{P}_{1,2}^{*} \cdots \boldsymbol{P}_{n-1, n}^{*}$	matrix $\boldsymbol{R}\boldsymbol{P}^*_{1,2}\cdots \boldsymbol{P}^*_{n-1,n}$ is upper Hessenberg and a QR step leaves the Hessenberg
lyche-numerical-linear-a F04580	354	1.000	`a_{i+1, i}`	$a_{i+1, i}$	If a subdiagonal element $a_{i+1,i}$ of an upper Hessenberg matrix $\boldsymbol{A}$ is equal to zero,
lyche-numerical-linear-a F04581	354	1.000	`A(1: i, 1: i)`	$A(1: i, 1: i)$	matrices $A(1:i,1:i)$ and $A(i+1:n,i+1:n)$. Thus, if during the iteration
lyche-numerical-linear-a F04582	354	1.000	`A(i+1: n, i+1: n)`	$A(i+1: n, i+1: n)$	matrices $A(1:i,1:i)$ and $A(i+1:n,i+1:n)$. Thus, if during the iteration
lyche-numerical-linear-a F04583	354	1.000	`\left|a_{i+1, i}^{(k)}\right| \leq \epsilon`	$\left\vert a_{i+1, i}^{(k)}\right\vert \leq \epsilon$	To see what effect this can have on the eigenvalues of $\boldsymbol{A}$ suppose $\vert a^{(k)}_{i+1,i}\vert \le \epsilon$. Let
lyche-numerical-linear-a F04584	354	0.992	`\hat{\boldsymbol{A}}_{k}:=\boldsymbol{A}_{k}-a_{i+1, i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_{i}^{T}`	$\hat{\boldsymbol{A}}_{k}:=\boldsymbol{A}_{k}-a_{i+1, i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_{i}^{T}$	$\hat{\boldsymbol{A}}_k := \boldsymbol{A}_k - a^{(k)}_{i+1,i}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T$ be the matrix obtained from $\boldsymbol{A}_k$ by setting the $(i+1,i)$
lyche-numerical-linear-a F04585	354	1.000	`\boldsymbol{A}_{k}=\tilde{\boldsymbol{Q}}_{k-1}^{*} \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1}`	$\boldsymbol{A}_{k}=\tilde{\boldsymbol{Q}}_{k-1}^{*} \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1}$	element equal to zero. Since $\boldsymbol{A}_k = \tilde{\boldsymbol{Q}}^*_{k-1}\boldsymbol{A}\tilde{\boldsymbol{Q}}_{k-1}$ we have
lyche-numerical-linear-a F04586	354	1.000	`\tilde{\boldsymbol{Q}}_{k-1}`	$\tilde{\boldsymbol{Q}}_{k-1}$	Since $\tilde{\boldsymbol{Q}}_{k-1}$ is unitary, $\Vert\boldsymbol{E}\Vert_F = \Vert a^{(k)}_{i+1,i}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T\Vert_F = \vert a^{(k)}_{i+1,i}\vert \le \epsilon$ and setting
lyche-numerical-linear-a F04587	354	1.000	`\\|\boldsymbol{E}\\|_{F}=\left\\|a_{i+1, i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_{i}^{T}\right\\|_{F}=\left|a_{i+1, i}^{(k)}\right| \leq \epsilon`	$\Vert\boldsymbol{E}\Vert_{F}=\left\Vert a_{i+1, i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_{i}^{T}\right\Vert_{F}=\left\vert a_{i+1, i}^{(k)}\right\vert \leq \epsilon$	Since $\tilde{\boldsymbol{Q}}_{k-1}$ is unitary, $\Vert\boldsymbol{E}\Vert_F = \Vert a^{(k)}_{i+1,i}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T\Vert_F = \vert a^{(k)}_{i+1,i}\vert \le \epsilon$ and setting
lyche-numerical-linear-a F04588	354	1.000	`a_{i+1, i}^{(k)}=0`	$a_{i+1, i}^{(k)}=0$	$a^{(k)}_{i+1,i} = 0$ amounts to a perturbation in the original $\boldsymbol{A}$ of at most ϵ. For how to
lyche-numerical-linear-a F04589	355	1.000	`\boldsymbol{R}_{k}=\boldsymbol{Q}_{k}^{*}\left(\boldsymbol{A}_{k}-s_{k} \boldsymbol{I}\right)`	$\boldsymbol{R}_{k}=\boldsymbol{Q}_{k}^{*}\left(\boldsymbol{A}_{k}-s_{k} \boldsymbol{I}\right)$	Since $\boldsymbol{R}_k = \boldsymbol{Q}^*_k(\boldsymbol{A}_k - s_k\boldsymbol{I})$ we find
lyche-numerical-linear-a F04590	355	1.000	`\boldsymbol{Q}_{k}`	$\boldsymbol{Q}_{k}$	Theorem 15.3 and also the inverse power method. In fact the last column of $\boldsymbol{Q}_k$
lyche-numerical-linear-a F04591	355	0.998	`\boldsymbol{A}-s_{k} \boldsymbol{I}=\boldsymbol{Q}_{k} \boldsymbol{R}_{k}`	$\boldsymbol{A}-s_{k} \boldsymbol{I}=\boldsymbol{Q}_{k} \boldsymbol{R}_{k}$	since $\boldsymbol{A} - s_k\boldsymbol{I} = \boldsymbol{Q}_k\boldsymbol{R}_k$ we have $(\boldsymbol{A}-s_k\boldsymbol{I})^* = \boldsymbol{R}^*_k\boldsymbol{Q}^*_k$ and $(\boldsymbol{A}-s_k\boldsymbol{I})^*\boldsymbol{Q}_k = \boldsymbol{R}^*_k$.
lyche-numerical-linear-a F04592	355	0.998	`\left(\boldsymbol{A}-s_{k} \boldsymbol{I}\right)^{*}=\boldsymbol{R}_{k}^{*} \boldsymbol{Q}_{k}^{*}`	$\left(\boldsymbol{A}-s_{k} \boldsymbol{I}\right)^{*}=\boldsymbol{R}_{k}^{*} \boldsymbol{Q}_{k}^{*}$	since $\boldsymbol{A} - s_k\boldsymbol{I} = \boldsymbol{Q}_k\boldsymbol{R}_k$ we have $(\boldsymbol{A}-s_k\boldsymbol{I})^* = \boldsymbol{R}^*_k\boldsymbol{Q}^*_k$ and $(\boldsymbol{A}-s_k\boldsymbol{I})^*\boldsymbol{Q}_k = \boldsymbol{R}^*_k$.
lyche-numerical-linear-a F04593	355	0.998	`\left(\boldsymbol{A}-s_{k} \boldsymbol{I}\right)^{*} \boldsymbol{Q}_{k}=\boldsymbol{R}_{k}^{*}`	$\left(\boldsymbol{A}-s_{k} \boldsymbol{I}\right)^{*} \boldsymbol{Q}_{k}=\boldsymbol{R}_{k}^{*}$	since $\boldsymbol{A} - s_k\boldsymbol{I} = \boldsymbol{Q}_k\boldsymbol{R}_k$ we have $(\boldsymbol{A}-s_k\boldsymbol{I})^* = \boldsymbol{R}^*_k\boldsymbol{Q}^*_k$ and $(\boldsymbol{A}-s_k\boldsymbol{I})^*\boldsymbol{Q}_k = \boldsymbol{R}^*_k$.
lyche-numerical-linear-a F04594	355	1.000	`\boldsymbol{R}_{k}^{*}`	$\boldsymbol{R}_{k}^{*}$	Thus, since $\boldsymbol{R}^*_k$ is lower triangular with n,n element $\bar{r}^{(k)}_{nn}$ we find $(\boldsymbol{A}-s_k\boldsymbol{I})^*\boldsymbol{Q}_k\boldsymbol{e}_n =$
lyche-numerical-linear-a F04595	355	1.000	`n, n`	n, n	Thus, since $\boldsymbol{R}^*_k$ is lower triangular with n,n element $\bar{r}^{(k)}_{nn}$ we find $(\boldsymbol{A}-s_k\boldsymbol{I})^*\boldsymbol{Q}_k\boldsymbol{e}_n =$
lyche-numerical-linear-a F04596	355	1.000	`\bar{r}_{n n}^{(k)}`	$\bar{r}_{n n}^{(k)}$	Thus, since $\boldsymbol{R}^*_k$ is lower triangular with n,n element $\bar{r}^{(k)}_{nn}$ we find $(\boldsymbol{A}-s_k\boldsymbol{I})^*\boldsymbol{Q}_k\boldsymbol{e}_n =$
lyche-numerical-linear-a F04597	355	1.000	`\left(\boldsymbol{A}-s_{k} \boldsymbol{I}\right)^{*} \boldsymbol{Q}_{k} \boldsymbol{e}_{n}=`	$\left(\boldsymbol{A}-s_{k} \boldsymbol{I}\right)^{*} \boldsymbol{Q}_{k} \boldsymbol{e}_{n}=$	Thus, since $\boldsymbol{R}^*_k$ is lower triangular with n,n element $\bar{r}^{(k)}_{nn}$ we find $(\boldsymbol{A}-s_k\boldsymbol{I})^*\boldsymbol{Q}_k\boldsymbol{e}_n =$
lyche-numerical-linear-a F04598	355	1.000	`\boldsymbol{R}_{k}^{*} \boldsymbol{e}_{n}=\bar{r}_{n n}^{(k)} \boldsymbol{e}_{n}`	$\boldsymbol{R}_{k}^{*} \boldsymbol{e}_{n}=\bar{r}_{n n}^{(k)} \boldsymbol{e}_{n}$	$\boldsymbol{R}^*_k\boldsymbol{e}_n = \bar{r}^{(k)}_{nn}\boldsymbol{e}_n$ from which the conclusion follows.
lyche-numerical-linear-a F04599	355	1.000	`s_{k}:=\boldsymbol{e}_{n}^{T} \boldsymbol{A}_{k} \boldsymbol{e}_{n}`	$s_{k}:=\boldsymbol{e}_{n}^{T} \boldsymbol{A}_{k} \boldsymbol{e}_{n}$	The shift $s_k := \boldsymbol{e}_n^T\boldsymbol{A}_k\boldsymbol{e}_n$ is called the **Rayleigh quotient shift**, while the
lyche-numerical-linear-a F04600	355	0.999	`\boldsymbol{A}:=?\left[\begin{array}{ll}0 & 1 \\ 1 & 0 \end{array}\right]`	$\boldsymbol{A}:=?\left[\begin{array}{ll}0 & 1 \\ 1 & 0 \end{array}\right]$	is given by $\boldsymbol{A} :=? \left[\begin{smallmatrix}0 & 1\\ 1 & 0\end{smallmatrix}\right]$. We obtain $\boldsymbol{A}_k = \boldsymbol{A}$ for all $k \in \mathbb{N}$. For convergence of the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04601	355	0.999	`\boldsymbol{A}_{k}=\boldsymbol{A}`	$\boldsymbol{A}_{k}=\boldsymbol{A}$	is given by $\boldsymbol{A} :=?\left[\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right]$. We obtain $\boldsymbol{A}_k = \boldsymbol{A}$ for all $k \in \mathbb{N}$. For convergence of the
lyche-numerical-linear-algebra F04602	355	1.000	`\boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u}`	$\boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u}$	**Exercise 15.1 (Orthogonal Vectors)** Show that $\boldsymbol{u}$ and $\boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}$ are orthogonal
lyche-numerical-linear-algebra F04603	355	1.000	`\lambda=\frac{\boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^{*} \boldsymbol{u}}`	$\lambda=\frac{\boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^{*} \boldsymbol{u}}$	when $\lambda = \frac{\boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}}{\boldsymbol{u}^*\boldsymbol{u}}$.
lyche-numerical-linear-algebra F04604	358	1.000	`f: \mathbb{R}^{n} \rightarrow \mathbb{R}`	$f: \mathbb{R}^{n} \rightarrow \mathbb{R}$	For any sufficiently differentiable $f : \mathbb{R}^n \to \mathbb{R}$ we recall that the partial derivative
lyche-numerical-linear-algebra F04605	358	1.000	`\nabla f(\boldsymbol{x}) \in \mathbb{R}^{n}`	$\nabla f(\boldsymbol{x}) \in \mathbb{R}^{n}$	$\nabla f(\boldsymbol{x}) \in \mathbb{R}^n$, and the **hessian** $\boldsymbol{H} f = \nabla\nabla^T f(\boldsymbol{x}) \in \mathbb{R}^{n\times n}$ of f by
lyche-numerical-linear-algebra F04606	358	1.000	`\boldsymbol{H} f=\nabla \nabla^{T} f(\boldsymbol{x}) \in \mathbb{R}^{n \times n}`	$\boldsymbol{H} f=\nabla \nabla^{T} f(\boldsymbol{x}) \in \mathbb{R}^{n \times n}$	$\nabla f(\boldsymbol{x}) \in \mathbb{R}^n$, and the **hessian** $\boldsymbol{H} f = \nabla\nabla^T f(\boldsymbol{x}) \in \mathbb{R}^{n\times n}$ of f by
lyche-numerical-linear-algebra F04607	358	1.000	`\nabla^{T} f:=(\nabla f)^{T}`	$\nabla^{T} f:=(\nabla f)^{T}$	where $\nabla^T f := (\nabla f)^T$ is the row vector gradient. The operators $\nabla\nabla^T$ and $\nabla^T\nabla$ are
lyche-numerical-linear-algebra F04608	358	1.000	`\nabla \nabla^{T}`	$\nabla \nabla^{T}$	where $\nabla^T f := (\nabla f)^T$ is the row vector gradient. The operators $\nabla\nabla^T$ and $\nabla^T\nabla$ are
lyche-numerical-linear-algebra F04609	358	1.000	`\nabla^{T} \nabla`	$\nabla^{T} \nabla$	where $\nabla^T f := (\nabla f)^T$ is the row vector gradient. The operators $\nabla\nabla^T$ and $\nabla^T\nabla$ are
lyche-numerical-linear-algebra F04610	358	1.000	`\nabla^{T} \nabla f=D_{1}^{2} f+\cdots+D_{n}^{2} f=: \nabla^{2}`	$\nabla^{T} \nabla f=D_{1}^{2} f+\cdots+D_{n}^{2} f=: \nabla^{2}$	quite different. Indeed, $\nabla^T\nabla f = D_1^2 f + \cdots + D_n^2 f =: \nabla^2$ the **Laplacian** of f,
lyche-numerical-linear-algebra F04611	358	1.000	`f, g: \mathbb{R}^{n} \rightarrow \mathbb{R}`	$f, g: \mathbb{R}^{n} \rightarrow \mathbb{R}$	**Lemma 16.1 (Product Rules)** *For* $f, g : \mathbb{R}^n \to \mathbb{R}$ *we have the product rules*
lyche-numerical-linear-algebra F04612	358	0.982	`\nabla(f g)=f \nabla g+g \nabla f, \quad \nabla^{T}(f g)=f \nabla^{T} g+g \nabla^{T} f`	$\nabla(f g)=f \nabla g+g \nabla f, \quad \nabla^{T}(f g)=f \nabla^{T} g+g \nabla^{T} f$	*1.* $\nabla(fg) = f\nabla g + g\nabla f, \quad \nabla^T(fg) = f\nabla^T g + g\nabla^T f,$
lyche-numerical-linear-algebra F04613	358	1.000	`\nabla \nabla^{T}(f g)=\nabla f \nabla^{T} g+\nabla g \nabla^{T} f+f \nabla \nabla^{T} g+g \nabla \nabla^{T} f`	$\nabla \nabla^{T}(f g)=\nabla f \nabla^{T} g+\nabla g \nabla^{T} f+f \nabla \nabla^{T} g+g \nabla \nabla^{T} f$	*2.* $\nabla\nabla^T(fg) = \nabla f\nabla^T g + \nabla g\nabla^T f + f\nabla\nabla^T g + g\nabla\nabla^T f.$
lyche-numerical-linear-algebra F04614	358	1.000	`\nabla^{2}(f g)=2 \nabla^{T} f \nabla g+f \nabla^{2} g+g \nabla^{2} f`	$\nabla^{2}(f g)=2 \nabla^{T} f \nabla g+f \nabla^{2} g+g \nabla^{2} f$	*3.* $\nabla^2(fg) = 2\nabla^T f\nabla g + f\nabla^2 g + g\nabla^2 f.$
lyche-numerical-linear-algebra F04615	359	1.000	`\boldsymbol{f}=\left[f_{1}, \ldots f_{m}\right]^{T}: \mathbb{R}^{n} \rightarrow \mathbb{R}^{m}`	$\boldsymbol{f}=\left[f_{1}, \ldots f_{m}\right]^{T}: \mathbb{R}^{n} \rightarrow \mathbb{R}^{m}$	We define the **Jacobian** of a vector function $\boldsymbol{f} = [f_1, \ldots f_m]^T : \mathbb{R}^n \to \mathbb{R}^m$ as
lyche-numerical-linear-algebra F04616	359	1.000	`f(\boldsymbol{x})=f(x, y)=x^{2}-x y+y^{2}`	$f(\boldsymbol{x})=f(x, y)=x^{2}-x y+y^{2}$	As an example, if $f(\boldsymbol{x}) = f(x, y) = x^2 - xy + y^2$ and $\boldsymbol{g}(x, y) := [f(x, y),\ x - y]^T$
lyche-numerical-linear-algebra F04617	359	1.000	`\boldsymbol{g}(x, y):=[f(x, y), x-y]^{T}`	$\boldsymbol{g}(x, y):=[f(x, y), x-y]^{T}$	As an example, if $f(\boldsymbol{x}) = f(x, y) = x^2 - xy + y^2$ and $\boldsymbol{g}(x, y) := [f(x, y),\ x - y]^T$
lyche-numerical-linear-algebra F04618	359	1.000	`f \in C^{2}(\Omega)`	$f \in C^{2}(\Omega)$	**Lemma 16.2 (Second Order Taylor Expansion)** *Suppose* $f \in C^2(\Omega)$, *where*
lyche-numerical-linear-algebra F04619	359	0.633	`\Omega \in \mathbb{R}^{n}`	$\Omega \in \mathbb{R}^{n}$	$\Omega \in \mathbb{R}^n$ *contains two points* $\boldsymbol{x}, \boldsymbol{x}+\boldsymbol{h} \in \Omega$, *such that the line segment* $L := \{\boldsymbol{x}+t\boldsymbol{h} :$
lyche-numerical-linear-algebra F04620	359	0.633	`\boldsymbol{x}, \boldsymbol{x}+\boldsymbol{h} \in \Omega`	$\boldsymbol{x}, \boldsymbol{x}+\boldsymbol{h} \in \Omega$	$\Omega \in \mathbb{R}^n$ *contains two points* $\boldsymbol{x}, \boldsymbol{x}+\boldsymbol{h} \in \Omega$, *such that the line segment* $L := \{\boldsymbol{x}+t\boldsymbol{h} :$
lyche-numerical-linear-algebra F04621	359	0.633	`L:=\{\boldsymbol{x}+t \boldsymbol{h}:`	$L:=\{\boldsymbol{x}+t \boldsymbol{h}:$	$\Omega \in \mathbb{R}^n$ *contains two points* $\boldsymbol{x}, \boldsymbol{x}+\boldsymbol{h} \in \Omega$, *such that the line segment* $L := \{\boldsymbol{x}+t\boldsymbol{h} :$
lyche-numerical-linear-algebra F04622	359	1.000	`t \in(0,1)\} \subset \Omega`	$t \in(0,1)\} \subset \Omega$	$t \in (0, 1)\} \subset \Omega$. *Then*
lyche-numerical-linear-algebra F04623	359	1.000	`g:[0,1] \rightarrow \mathbb{R}`	$g:[0,1] \rightarrow \mathbb{R}$	***Proof*** Let $g : [0, 1] \to \mathbb{R}$ be defined by $g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$. Then $g \in C^2[0, 1]$ and
lyche-numerical-linear-algebra F04624	359	1.000	`g(t):=f(\boldsymbol{x}+t \boldsymbol{h})`	$g(t):=f(\boldsymbol{x}+t \boldsymbol{h})$	***Proof*** Let $g : [0, 1] \to \mathbb{R}$ be defined by $g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$. Then $g \in C^2[0, 1]$ and
lyche-numerical-linear-algebra F04625	359	1.000	`g \in C^{2}[0,1]`	$g \in C^{2}[0,1]$	***Proof*** Let $g : [0, 1] \to \mathbb{R}$ be defined by $g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$. Then $g \in C^2[0, 1]$ and
lyche-numerical-linear-algebra F04626	359	1.000	`\boldsymbol{c}=\boldsymbol{x}+u \boldsymbol{h}`	$\boldsymbol{c}=\boldsymbol{x}+u \boldsymbol{h}$	we obtain (16.2) with $\boldsymbol{c} = \boldsymbol{x} + u\boldsymbol{h}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F04627	360	1.000	`m, n \in \mathbb{N}, \boldsymbol{B} \in \mathbb{R}^{n \times n}, \boldsymbol{C} \in`	$m, n \in \mathbb{N}, \boldsymbol{B} \in \mathbb{R}^{n \times n}, \boldsymbol{C} \in$	**Lemma 16.3 (Functions Involving Matrices)** *For any* $m, n \in \mathbb{N}$, $\boldsymbol{B} \in \mathbb{R}^{n \times n}$, $\boldsymbol{C} \in$
lyche-numerical-linear-algebra_F04628	360	1.000	`\boldsymbol{x} \in \mathbb{R}^{n}, \boldsymbol{y} \in \mathbb{R}^{m}`	$\boldsymbol{x} \in \mathbb{R}^{n}, \boldsymbol{y} \in \mathbb{R}^{m}$	$\mathbb{R}^{m \times n}$, *and* $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{y} \in \mathbb{R}^m$ *we have*
lyche-numerical-linear-algebra_F04629	360	1.000	`\nabla\left(\boldsymbol{y}^{T} \boldsymbol{C}\right)=\nabla^{T}(\boldsymbol{C} \boldsymbol{x})=\boldsymbol{C}`	$\nabla\left(\boldsymbol{y}^{T} \boldsymbol{C}\right)=\nabla^{T}(\boldsymbol{C} \boldsymbol{x})=\boldsymbol{C}$	*1.* $\nabla(\boldsymbol{y}^T\boldsymbol{C}) = \nabla^T(\boldsymbol{C}\boldsymbol{x}) = \boldsymbol{C}$,
lyche-numerical-linear-algebra_F04630	360	1.000	`\nabla\left(\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}\right)=\left(\boldsymbol{B}+\boldsymbol{B}^{T}\right) \boldsymbol{x}, \quad \nabla^{T}\left(\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}\right)=\boldsymbol{x}^{T}\left(\boldsymbol{B}+\boldsymbol{B}^{T}\right)`	$\nabla\left(\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}\right)=\left(\boldsymbol{B}+\boldsymbol{B}^{T}\right) \boldsymbol{x}, \quad \nabla^{T}\left(\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}\right)=\boldsymbol{x}^{T}\left(\boldsymbol{B}+\boldsymbol{B}^{T}\right)$	*2.* $\nabla(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}) = (\boldsymbol{B}+\boldsymbol{B}^T)\boldsymbol{x}, \quad \nabla^T(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}) = \boldsymbol{x}^T(\boldsymbol{B}+\boldsymbol{B}^T)$,
lyche-numerical-linear-algebra_F04631	360	1.000	`\nabla \nabla^{T}\left(\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}\right)=\boldsymbol{B}+\boldsymbol{B}^{T}`	$\nabla \nabla^{T}\left(\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}\right)=\boldsymbol{B}+\boldsymbol{B}^{T}$	*3.* $\nabla\nabla^T(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}) = \boldsymbol{B}+\boldsymbol{B}^T$.
lyche-numerical-linear-algebra_F04632	360	1.000	`D_{i}\left(\boldsymbol{y}^{T} \boldsymbol{C}\right)=\lim _{h \rightarrow 0} \frac{1}{h}\left(\left(\boldsymbol{y}+h \boldsymbol{e}_{i}\right)^{T} \boldsymbol{C}-\boldsymbol{y}^{T} \boldsymbol{C}\right)=\boldsymbol{e}_{i}^{T} \boldsymbol{C}`	$D_{i}\left(\boldsymbol{y}^{T} \boldsymbol{C}\right)=\lim _{h \rightarrow 0} \frac{1}{h}\left(\left(\boldsymbol{y}+h \boldsymbol{e}_{i}\right)^{T} \boldsymbol{C}-\boldsymbol{y}^{T} \boldsymbol{C}\right)=\boldsymbol{e}_{i}^{T} \boldsymbol{C}$	1. We find $D_i(\boldsymbol{y}^T\boldsymbol{C}) = \lim_{h\to 0}\frac{1}{h}\big((\boldsymbol{y}+h\boldsymbol{e}_i)^T\boldsymbol{C} - \boldsymbol{y}^T\boldsymbol{C}\big) = \boldsymbol{e}_i^T\boldsymbol{C}$ and $D_i(\boldsymbol{C}\boldsymbol{x}) =$
lyche-numerical-linear-algebra_F04633	360	1.000	`D_{i}(\boldsymbol{C} \boldsymbol{x})=`	$D_{i}(\boldsymbol{C} \boldsymbol{x})=$	1. We find $D_i(\boldsymbol{y}^T\boldsymbol{C}) = \lim_{h\to 0}\frac{1}{h}\big((\boldsymbol{y}+h\boldsymbol{e}_i)^T\boldsymbol{C} - \boldsymbol{y}^T\boldsymbol{C}\big) = \boldsymbol{e}_i^T\boldsymbol{C}$ and $D_i(\boldsymbol{C}\boldsymbol{x}) =$
lyche-numerical-linear-algebra_F04634	360	1.000	`\lim _{h \rightarrow 0} \frac{1}{h}\left(\boldsymbol{C} \left(\boldsymbol{x}+h \boldsymbol{e}_{i}\right)-\boldsymbol{C} \boldsymbol{x}\right)=\boldsymbol{C} \boldsymbol{e}_{i}`	$\lim _{h \rightarrow 0} \frac{1}{h}\left(\boldsymbol{C} \left(\boldsymbol{x}+h \boldsymbol{e}_{i}\right)-\boldsymbol{C} \boldsymbol{x}\right)=\boldsymbol{C} \boldsymbol{e}_{i}$	$\lim_{h\to 0}\frac{1}{h}(\boldsymbol{C}(\boldsymbol{x}+h\boldsymbol{e}_i) - \boldsymbol{C}\boldsymbol{x}) = \boldsymbol{C}\boldsymbol{e}_i$ and 1. follows.
lyche-numerical-linear-algebra_F04635	363	0.783	`\boldsymbol{x}^{T} \boldsymbol{A y}`	$\boldsymbol{x}^{T} \boldsymbol{A y}$	1979-3, $\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{y}$ inequality, 119
lyche-numerical-linear-algebra_F04636	363	1.000	`\boldsymbol{T} \boldsymbol{H} \boldsymbol{x}=\boldsymbol{b}, 79`	$\boldsymbol{T} \boldsymbol{H} \boldsymbol{x}=\boldsymbol{b}, 79$	1981-3, solving $\boldsymbol{T}\boldsymbol{H}\boldsymbol{x} = \boldsymbol{b}$, 79
lyche-numerical-linear-algebra_F04637	365	0.995	`\boldsymbol{A}^{*} \boldsymbol{A}, 86`	$\boldsymbol{A}^{*} \boldsymbol{A}, 86$	$\boldsymbol{A}^*\boldsymbol{A}$, 86
lyche-numerical-linear-algebra_F04638	374	1.000	`X X`	$X X$	*Engineering XX.*